



pyCoastal

A Python toolbox for coastal, port and ocean engineering

User manual and reference, version 0.2.0

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Preface

pyCoastal started as the culmination of my academic journey in numerical modeling, fluid dynamics, and hydraulics. Students often believe that numerical models are overwhelmingly complex tools; however, their structure follows recurrent patterns common to many modeling frameworks. When properly configured and validated, these models can reproduce real-world phenomena, provide insight, and let us explore scenarios beyond direct observation. Some researchers prefer to work in the analytical domain, which has its own advantages and limitations, and I admire those who can capture the complexity of natural processes through concise formulations that, even if simplified, reflect the essence of how the world operates.

Since the first edition of this manual the package has grown in a different direction as well. Next to the numerical framework it now carries a complete design chain: a wave record becomes a design condition, the design condition sizes a structure, and the structure comes out as a dimensioned drawing with its quantities and its specification notes. Every relation names its source, states its range of validity, and says so when it is pushed outside that range. A browser version of the design modules, the Coastal Design Bench, puts each result next to what the peer-reviewed literature says about the ground it stands on, through an extract of the CoastalWiki knowledge base.

This manual documents all of it. Part I gets the package installed and running. Parts II and III describe the numerical framework and the standalone engineering formulae. Part IV is the core of the book: one chapter per design application, each with its theory, its interface, a worked example with the output the code actually prints, and the limits of the method. Part V covers drawings and deliverables, Part VI the browser app, and Part VII the test suite that keeps all of it honest. Part VIII reproduces the complete CoastalWiki extract that ships with the app, topic by topic, and the appendices hold the full API reference and the source of every example.

I do not expect pyCoastal to be the tool behind groundbreaking discoveries, although I hope it may contribute to them, or at least serve as a starting point for someone.

Stefano Biondi

Part I. Getting started

1 Introduction

Coastal zones concentrate people, infrastructure, and ecosystems in environments that are energetic and constantly changing. Waves, tides, surges, and currents interact with complex shorelines to drive erosion, flooding, sediment transport, and loads on structures. Anticipating these processes is essential for safe navigation, coastal protection, habitat conservation, and climate adaptation.

pyCoastal is a Python toolbox for coastal, port, and ocean engineering. It has two faces that share one code base:

1. **A numerical framework** of uniform grids, finite-difference operators, explicit time integrators, boundary handlers, Poisson solvers, and physics building blocks (shallow water, incompressible Navier-Stokes, eddy-viscosity closures). It favors clarity over ultimate efficiency so that students and practitioners can see and modify every numerical brick.
2. **A design chain** of scenario-level applications that take an engineering design and report the quantities a project actually turns on: a 100-year wave with its confidence band, an armor stone size and a crest level, a seawall base width, a scour depth over a tidal cycle, a dredge level, a fender size, a flood extent, and the drawing sheet that carries all of it.

1.1 The design chain

A real coastal scheme runs as a chain, and pyCoastal is organized the same way ([Table 1.1](#)). Each step hands its result to the next as an object, not as a number you retype, so a drawing cannot drift out of step with the calculation behind it.

Table 1.1: The design chain and the modules that implement each step.

Step	Module	What it gives you
Design condition	<code>applications.extremes</code>	100-year wave or water level, with a confidence band
Bed and soil	<code>applications.sediment</code>	grain mobility, fall velocity, earth pressure
Nearshore	<code>applications.port</code>	phase-resolved diffraction into a harbor, berth agitation
Structure	<code>applications.structures</code> , <code>applications.seawall</code>	armor size, crest level, stability checks
Loads	<code>applications.piles</code> , <code>applications.berthing</code>	Morison loads through the wave cycle, berthing energy and fenders
Scour	<code>applications.scour</code> , <code>applications.river</code>	contraction, pier and abutment scour over the tidal cycle; the flow split that feeds it
Navigation	<code>applications.channel</code>	dredge level, channel width, dredge volume
Coastline	<code>applications.nourishment</code> , <code>applications.surge</code>	fill life and profile, water level budget and a connected flood map
Deliverable	<code>drafting</code> , <code>applications.sections</code>	a dimensioned drawing sheet and a DXF

1.2 Principles

Three rules run through the whole package, and they explain most of the design decisions you will meet in the later chapters.

Every relation names its source. Module docstrings list their sources (Van der Meer 1988, EurOtop 2018, Goda 2010, PIANC 2014, HEC-18, and so on), and each function says which equation it implements.

Nothing silently extrapolates. Where a formula has a documented range of validity, the functions compute outside it if asked, because engineering judgement outside a range is the engineer's call, but they say so through a warnings list, a flag such as `within_range`, `impulsive`, or `screening_only`, or a note in the returned dictionary.

Assumptions are reported, not buried. Width components of a channel that you did not specify are taken at their most benign class and reported as assumed. The +1 safety term in Froehlich's abutment equation is reported as a separate `safety_margin`. The steady-current floor added to the Sumer and Fredsoe scour relation is flagged as `current_governs`. A reviewer should always be able to see which assumption to argue with.

1.3 Before the math

A few terms recur throughout the numerical parts of this manual. A *field* is any variable defined over the domain, such as water depth, velocity, or pressure. A *grid* divides the domain into cells where these variables are stored; in `pyCoastal` the values live at cell centers. A *stencil* is the small pattern of neighboring cells used to approximate derivatives through finite differences, and these approximations form the spatial *operators*. Once the spatial terms are discretized, a *time integrator* updates the solution step by step, using a right-hand side that gathers fluxes, advection, diffusion, pressure gradients, and sources. *Boundary conditions* specify what happens at the domain edges.

On the design side, a *design condition* is the wave and water level a structure is sized against, a *design object* (for example `SeawallDesign` or `ChannelDesign`) is the fully dimensioned result of a design function, and a *sheet* is a drawing at a true, stated scale with a border, a title block, and specification notes.

1.4 How this manual is organized

- **Part I** covers installation, the repository layout, conventions, and a quick start.
- **Part II** describes the numerical framework and the governing equations, with the simulation examples that exercise them.
- **Part III** documents the standalone engineering formulae in `pyCoastal.tools`.
- **Part IV** has one chapter per design application.
- **Part V** covers the drafting layer: sections, sheets, and DXF export.
- **Part VI** describes the Coastal Design Bench browser app.
- **Part VII** describes the test suite and the verification of the browser engine against the Python.
- **Part VIII** reproduces the CoastalWiki knowledge base shipped with the app.
- **Appendices** give the complete API reference, the full source of every example, a list of symbols, and the reference lists.

2 Installation and first steps

2.1 Requirements

pyCoastal needs Python 3.10 or newer. The runtime dependencies are kept to two, so the design modules install anywhere:

- `numpy` ≥ 1.21 for arrays, gradients, and linear algebra;
- `PyYAML` ≥ 5.4 for YAML case files.

Everything else is optional and imported lazily, only by the functions that need it (Table 2.1).

Table 2.1: Optional dependency groups declared in `pyproject.toml`.

Extra	Installs	Needed for
<code>sparse</code>	<code>scipy >= 1.7</code>	sparse direct solver in <code>physics.poisson</code>
<code>vtk</code>	<code>vtk >= 9.0</code>	<code>io.write_vtk</code>
<code>plots</code>	<code>matplotlib >= 3.4</code>	drafting, <code>applications.sections</code> , plotting, and the examples
<code>dev</code>	<code>pytest, scipy, matplotlib</code>	running the test suite
<code>docs</code>	<code>sphinx, sphinx-rtd-theme</code>	the Sphinx API pages in <code>docs/</code>

2.2 Installing

From PyPI:

```
pip install pyCoastal
pip install "pyCoastal[plots]" # with drawings and plotting
```

From a clone, in editable mode, which is the right choice for development and for running the examples:

```
git clone https://github.com/stebiondi/pyCoastal.git
cd pyCoastal
python -m venv .venv
source .venv/bin/activate # on Windows: .venv\Scripts\activate
pip install -e ".[dev]"
```

Check the installation and run the tests:

```
python -c "import pyCoastal; print(pyCoastal.__version__)"
python -m pytest tests/
```

Continuous integration runs the same suite on Ubuntu and Windows under Python 3.10, 3.11, and 3.13 on every push to `main` and on every pull request (`.github/workflows/tests.yml`). A separate workflow builds and publishes the distribution to PyPI (`.github/workflows/publish.yml`).

2.3 Repository layout

```
pyCoastal/
├── pyCoastal/           the package
│   ├── numerics/       grids, operators, time integrators, BCs, solver
│   ├── physics/        shallow water, Navier-Stokes, Poisson, turbulence
│   ├── tools/          standalone formulae: waves, sediment, structures,
│   │                   one-line shoreline, morphodynamics
│   └── applications/   design modules (extremes, structures, seawall,
│                       port, piles, scour, river, channel, berthing,
│                       nourishment, surge, sediment, sections)
├── drafting.py         drawing primitives, sheets, title blocks, DXF
├── plotting.py         color maps and helpers for the examples
├── config.py, io.py    case-file reading and VTK output
└── examples/          worked examples and their YAML configs
```

└─ tests/	the pytest suite
└─ webapp/	Coastal Design Bench (browser app)
└─ media/	figures, animations, drawing sheets, DXF files
└─ docs/	Sphinx sources and this manual (docs/manual)
└─ paper/	JOSS-style paper
└─ pyCoastal manual.pdf	this manual

2.4 Conventions

The same conventions hold in every module. Where a module departs from them its docstring says so.

Coordinates. x and y span the horizontal plane and z is elevation, positive upward. In every 2D array x is axis 0 and y is axis 1, matching `UniformGrid`, which builds its coordinates with `indexing="ij"` and flattens as $i*ny + j$. In cross-shore work x is the chainage and in alongshore work (the one-line model) x is alongshore and y is the cross-shore shoreline offset, positive seaward.

Units. SI throughout: meters, seconds, kilograms, newtons. Forces on walls and walls' weights are kN per meter run; overtopping discharges are liters per second per meter (l/s/m), the unit of the tolerability tables; ship speeds are in knots where the formula is written in knots and every such function says so. Grain sizes are meters, not millimeters. Angles are degrees where a geotechnical or navigational input is conventionally stated in degrees, and radians in wave-direction and slope-angle arguments of the low-level tools; each function states which.

Levels. Levels are meters relative to chart datum (m CD) and increase upward. Depths and draughts are positive numbers. Pier scour drawings use the initial bed as datum, because every dimension there is a depth below it.

Wave parameters. Design wave heights are spectral significant heights H_{m0} at the toe of the structure. `EurOtop` relations use the spectral period $T_{m-1,0}$; if only the peak period is known, `DesignConditions.from_peak_period` converts with $T_{m-1,0} = T_p/1.1$. Goda's pressure method is written around the significant period, so the seawall module passes T_p to it.

Slopes. Slopes are given as $\cot\alpha$, the horizontal run per unit rise, because that is how they appear on drawings.

2.5 Quick start

A seawall from a design condition to a drawing sheet, in eight lines:

```
from pyCoastal.applications.structures import DesignConditions
from pyCoastal.applications.seawall import design_seawall
from pyCoastal.applications.sections import seawall_sheet

conditions = DesignConditions.from_peak_period(Hm0=2.8, Tp=9.5, depth=8.5)
wall = design_seawall(conditions, still_water_level=2.9, seabed_level=-5.6,
                      tolerable_use="trained_staff")
print(wall.summary())
sheet = seawall_sheet(wall, project="Bayfront promenade protection", size="A3")
sheet.save("seawall.png")      # 600 dpi, at a true scale stated on the sheet
sheet.to_dxf("seawall.dxf")    # the geometry on named layers, for CAD
```

The printed summary is the design report, in the order an engineer checks it (crest, founding level, base, wave force, safety factors, bearing). The full worked example is in [Section 10](#).

Every example in `examples/` runs from the repository root:

```
python examples/seawall_section.py
python examples/breakwater_design.py
python examples/design_wave.py
```

and writes its figures to `media/`. Appendix B lists the full source of each one.

2.6 Reading case files

The numerical examples read their parameters from YAML files in `examples/configs/`. Two helpers parse case files into dictionaries:

- `pyCoastal.config.load_config(path)` reads `.yaml/.yml` (PyYAML), `.json`, and `.ini/.cfg` files;

- `pyCoastal.io.read_data(path)` does the same for any path-like object, and `pyCoastal.io.write_vtk(grid, filename)` writes a VTK grid object to a legacy `.vtk` file (needs the `vtk` extra).

`pyCoastal.numerics.domain.Domain(cfg)` builds a `Mesh1D` or `Mesh2D` from a domain: block with dimension, `x0`, `x1`, `nx` (and `y0`, `y1`, `ny` in 2D).

Part II. Numerical framework

3 Grids, operators, and time integration

Computational coastal engineering connects governing equations with the numerical choices that approximate them. `pyCoastal.numerics` exposes those choices one brick at a time: a grid, the finite-difference operators that act on it, the integrators that march it in time, and the boundary conditions that close it. Every choice is visible in the code and can be changed one at a time.

3.1 Grids

3.1.1 UniformGrid

Most numerical examples use a uniform Cartesian mesh, the simplest setting in which to illustrate discretization ideas. Two storage layouts are common: cell-centered (unknowns at cell centers) and node-centered (unknowns at grid intersections). `pyCoastal` adopts the cell-centered layout, which keeps stencils symmetric and boundary handling simple on rectangular domains.

```
from pyCoastal.numerics.grid import UniformGrid

grid = UniformGrid(shape=(200, 100), spacing=(2.0, 2.0), origin=(0.0, 0.0))
grid.Xc      # cell-center coordinates, a list [X, Y] built with indexing="ij"
grid.Xf      # face coordinates
grid.spacing  # (dx, dy)
grid.n_cells, grid.cell_volume
grid.boundary_indices["west"] # flat indices of the west boundary cells
```

`UniformGrid(shape, spacing, origin=None)` works in one and two dimensions (three-dimensional coordinates are built, but boundary indexing is 1D/2D only). Cell centers sit at $x_i = x_0 + (i + \frac{1}{2})\Delta x$. In 2D the flat index of cell (i, j) is $i n_y + j$, so that x is axis 0 and y is axis 1 in every array. The grid precomputes, for each side ("west", "east", "south", "north"), the flat indices of the boundary cells in `boundary_indices` and a copy in `sponge_indices`, which the boundary conditions of [Section 3.4](#) use. `neumann_indices(side)` returns the pair of boundary and first-interior indices needed to impose a gradient.

3.1.2 Mesh1D, Mesh2D, and Domain

`pyCoastal.numerics.domain` provides simple structured meshes built from a case file. `Mesh1D(x0, x1, nx)` stores cell centers x and spacing dx ; `Mesh2D(x0, x1, nx, y0, y1, ny)` stores 2D center arrays x, y and spacings dx, dy . `Domain(cfg)` reads a domain: block with dimension: 1 or 2 and instantiates the right mesh; `Domain.info()` prints a summary.

Indexing and memory layout. Arrays follow NumPy C order, so the last index is contiguous. Spacings $\Delta x, \Delta y$ are constant.

3.2 Finite differences

Finite differences approximate derivatives with algebraic expressions built from nearby grid values. On a uniform grid, a first derivative at a cell center can be estimated by a centered difference, which samples one point on either side to keep symmetry and second-order accuracy. Higher derivatives and one-sided formulas follow the same idea. The approach turns differential equations into systems of algebraic equations, which is why it remains a common choice for structured-grid problems in fluid mechanics and wave modeling.

3.2.1 Centered operators

Centered operators combine values symmetrically around the point of interest, which keeps the approximation free of directional bias and gives higher accuracy for smooth solutions. For first derivatives and the Laplacian:

$$\left. \frac{\partial f}{\partial x} \right|_{i,j} \approx \frac{f_{i+1,j} - f_{i-1,j}}{2\Delta x}, \quad \left. \frac{\partial f}{\partial y} \right|_{i,j} \approx \frac{f_{i,j+1} - f_{i,j-1}}{2\Delta y}, \quad (3.1)$$

$$\nabla^2 f|_{i,j} \approx \frac{f_{i+1,j} - 2f_{i,j} + f_{i-1,j}}{\Delta x^2} + \frac{f_{i,j+1} - 2f_{i,j} + f_{i,j-1}}{\Delta y^2}. \quad (3.2)$$

3.2.2 Upwind differences for advection

For an advection term with velocity u the solution at a point is determined by what arrives from upstream, so the stencil is taken from the upstream side:

$$\left. \frac{\partial f}{\partial x} \right|_i \approx \begin{cases} \frac{f_i - f_{i-1}}{\Delta x}, & u > 0, \\ \frac{f_{i+1} - f_i}{\Delta x}, & u < 0. \end{cases} \quad (3.3)$$

The one-sided stencil introduces numerical diffusion but is stable for hyperbolic problems and prevents non-physical oscillations.

3.2.3 The operator library

`pyCoastal.numerics.operators` implements these stencils on a `UniformGrid` (Table 3.1). The operators use periodic (roll) indexing, so boundary values must be imposed afterwards by the boundary conditions.

Table 3.1: Operators in `pyCoastal.numerics.operators`.

Function	Operation
<code>laplacian(field, grid)</code>	5-point Laplacian, Equation 3.2
<code>gradient(field, grid)</code>	centered gradient, returns (fx, fy)
<code>grad_x, grad_y</code>	centered first derivatives
<code>upwind_x(field, u, grid), upwind_y(field, v, grid)</code>	first-order upwind derivatives
<code>divergence(ux, uy, grid)</code>	centered $\nabla \cdot \mathbf{u}$
<code>curl_z(u, v, grid)</code>	$\partial v / \partial x - \partial u / \partial y$
<code>biharmonic(field, grid)</code>	$\nabla^4 f$ as the Laplacian of the Laplacian
<code>mixed_xy(field, grid)</code>	$\partial^2 f / \partial x \partial y$
<code>advect(u, v, field, grid, scheme="upwind")</code>	$u \partial_x f + v \partial_y f$
<code>smooth3(field, grid)</code>	5-point smoothing filter, weights summing to one

`pyCoastal.numerics.scheme` adds axis-generic versions, `central_difference(phi, spacing, axis)` and `upwind(phi, vel, spacing, axis)`.

3.3 Time integration

Once space is discretized, the semi-discrete variable $u(t)$ evolves under a right-hand side that gathers fluxes and sources, $du/dt = \text{RHS}(u, t)$. Explicit schemes compute the new state from information at the current level only, which makes them simple to apply and to check. The choice of integrator controls stability, accuracy, and how fast information moves through the grid.

Forward Euler (first order) treats the rate as constant over the step:

$$u^{n+1} = u^n + \Delta t \text{RHS}(u^n).$$

Heun / RK2 (second order) averages the slopes at the start and the end of the step:

$$k_1 = \text{RHS}(u^n, t), \quad k_2 = \text{RHS}(u^n + \Delta t k_1, t + \Delta t), \quad u^{n+1} = u^n + \frac{\Delta t}{2}(k_1 + k_2).$$

Classical RK4 (fourth order):

$$u^{n+1} = u^n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

SSP RK3 (third order, strong stability preserving) blends each stage with the earlier state, which keeps monotonicity for advection-dominated problems under a CFL limit:

$$\begin{aligned}
u^{(1)} &= u^n + \Delta t \text{ RHS}(u^n), \\
u^{(2)} &= \frac{3}{4}u^n + \frac{1}{4}\left[u^{(1)} + \Delta t \text{ RHS}(u^{(1)})\right], \\
u^{n+1} &= \frac{1}{3}u^n + \frac{2}{3}\left[u^{(2)} + \Delta t \text{ RHS}(u^{(2)})\right].
\end{aligned} \tag{3.4}$$

SSP RK3 is the recommended all-rounder for wave and transport demonstrations.

Adams-Bashforth 2 (second order, multistep) reuses the previous right-hand side, and needs one starting step from another method:

$$u^{n+1} = u^n + \frac{\Delta t}{2} (3 \text{ RHS}(u^n) - \text{RHS}(u^{n-1})).$$

These live in `pyCoastal.numerics.time_intg` as array functions, `euler_step(u, t, dt, rhs)`, `rk2_step`, `rk4_step`, `rk3_ssp_step`, and `ab2_step(u, u_prev, t, dt, rhs)`, where `rhs(u, t)` returns an array of the same shape as `u`.

For multi-field states, `pyCoastal.numerics.scheme` provides integrator classes that act on a dictionary of arrays: `EulerIntegrator(dt)` and `SSPRK2Integrator(dt)` (Shu-Osher second-order TVD), both with `step(state, rhs, t, **kwargs)` returning `(new_state, new_time)`, where `rhs(state, t, **kwargs)` returns a dictionary of tendencies.

3.3.1 The solver driver

`pyCoastal.numerics.solver.Solver(grid, physics, bc, integrator=None)` ties the pieces together: `physics` provides `initialize_state(grid)`, `rhs(state, t, ...)` and a `dt`; `bc` provides `apply(state, t)`; the integrator defaults to forward Euler. `Solver.run(t0, t_end, callback=None)` applies the boundary conditions, takes one step, and calls `callback(state, t)` after every step for output or plotting.

3.3.2 Stability

The CFL condition states that information carried by advection cannot move farther than one cell per step. For an advective speed c :

$$\text{CFL} = \max\left(\frac{c \Delta t}{\Delta x}, \frac{c \Delta t}{\Delta y}\right) < 1. \tag{3.5}$$

Diffusion imposes its own, often more severe, limit:

$$\Delta t \leq \frac{1}{2\nu} \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right)^{-1}. \tag{3.6}$$

The time step must satisfy both. The design applications apply the same rule internally: the port solver steps at a Courant number of 0.35 by default, and the one-line shoreline model at 0.9 of its diffusive limit (`tools.shoreline.suggest_dt`).

3.4 Boundary conditions

Boundary conditions close the discrete system. They fix a value, prescribe a gradient, let waves or flow enter or leave, or enforce a wall, and they control how signals reflect and how energy enters or exits the domain.

3.4.1 Ghost-cell formulas

For a cell-centered scheme the classical formulas at $x = 0$ are:

- **Dirichlet** (value), $q|_{\Gamma} = q_D(x, t)$: $q_0 = 2q_b - q_1$ with $q_b = q_D$.
- **Neumann** (gradient), $\partial q / \partial n|_{\Gamma} = g_N$: $q_0 = q_1 - \Delta x g_N$.
- **Solid wall**: $u_n = 0$, mirroring the normal component antisymmetrically; free slip mirrors the tangential velocity symmetrically, no slip antisymmetrically.
- **Sponge layer** (wave absorption): a damping zone relaxes the solution toward a reference state,

$$\frac{\partial q}{\partial t} = -\sigma(x) [q - q_{\text{ref}}], \quad \sigma(s) = \sigma_{\max} \sin^2\left(\frac{\pi s}{2}\right), \quad s = \frac{x - x_0}{L_{\text{sp}}} \in [0, 1]. \tag{3.7}$$

A sponge of $L_{\text{sp}} \approx 1$ to 3 wavelengths with $\sigma_{\max} \sim \kappa c / L_{\text{sp}}$, $\kappa \approx 3$ to 6, absorbs well; monitor reflections with gauges inside the domain.

3.4.2 The boundary classes

`pyCoastal.numerics.boundary` implements boundary conditions that act directly on the boundary cells, using the flat indices precomputed by the grid. Every class takes a side name and a list of field keys and implements `apply(fields, grid, t)`:

- `DirichletBC(location, var_names, value)`, where `value` is a number or a function of time;
- `NeumannBC(location, var_names, gradient=0.0)`, which copies the interior value plus `gradient * dx` onto the boundary;
- `WallBC(location, var_names)`, which sets the named fields (normally the normal velocity or flux) to zero;
- `SpongeBC(location, var_names, damping)`, which multiplies the boundary strip by a constant factor or by `damping(t, k)` evaluated per flat index `k`, so a spatial ramp can be built from the index;
- `BoundaryManager(bcs=())` holds a list of conditions; `add(bc)` registers one and `apply_all(fields, grid, t)` applies them all, in order, at time `t`.

```
import numpy as np
from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.numerics.boundary import (
    DirichletBC, NeumannBC, WallBC, SpongeBC, BoundaryManager,
)

grid = UniformGrid(shape=(200, 200), spacing=(2.0, 2.0))
fields = {"eta": np.zeros(grid.shape), "u": np.zeros(grid.shape),
          "v": np.zeros(grid.shape)}

bcs = BoundaryManager()
bcs.add(DirichletBC("south", ["eta"], lambda t: 0.5 * np.sin(2 * np.pi * t / 3.0)))
bcs.add(DirichletBC("west", ["u"], value=0.2))           # inflow
bcs.add(NeumannBC("east", ["u", "eta", "v"]))           # zero-gradient outflow
bcs.add(WallBC("north", ["v"]))                         # no normal flow
bcs.add(SpongeBC("north", ["eta"], damping=0.9))        # damp the strip
bcs.apply_all(fields, grid, t=0.0)
```

The complete version of this demonstration is `examples/numerics/2D_irr_turb.py` (Appendix B).

4 Governing equations and closures

`pyCoastal.physics` holds the governing equations as building blocks: flux functions and source terms that a solver assembles into a right-hand side, a pressure-projection step, Poisson solvers, and eddy-viscosity closures.

4.1 Shallow water equations

The shallow water equations describe a fluid layer whose horizontal scales are much larger than its depth. They track the total depth and the depth-averaged velocity, with the pressure hydrostatic. They capture wave propagation, flooding and draining, and large-scale currents:

$$\frac{\partial h}{\partial t} + \frac{\partial(hu)}{\partial x} + \frac{\partial(hv)}{\partial y} = 0, \quad (4.1)$$

$$\frac{\partial(hu)}{\partial t} + \frac{\partial}{\partial x} \left(hu^2 + \frac{1}{2}gh^2 \right) + \frac{\partial(huv)}{\partial y} = S_x, \quad (4.2)$$

$$\frac{\partial(hv)}{\partial t} + \frac{\partial(huv)}{\partial x} + \frac{\partial}{\partial y} \left(hv^2 + \frac{1}{2}gh^2 \right) = S_y. \quad (4.3)$$

Here h is the total depth, u and v the depth-averaged velocities, and S_x, S_y collect bed slope, bottom friction, and Coriolis. When adding sources, preserve the lake-at-rest balance.

`ShallowWater2D(g=9.81)` provides `fluxes(h, hu, hv)`, which returns the three flux pairs (F_h, G_h) , (F_{hu}, G_{hu}) , (F_{hv}, G_{hv}) , and `source_bed_slope(h, zb, grid=None)`, which returns $S_x = -gh \partial z_b / \partial x$ and $S_y = -gh \partial z_b / \partial y$. Pass the grid so the derivatives use the real spacing; without it the spacing is taken as one.

4.2 Incompressible Navier-Stokes

For incompressible viscous flow a popular explicit strategy is Chorin's projection (fractional step) method: predict the velocity with advection and diffusion, solve a Poisson equation for the pressure that removes the divergence, then correct:

$$\tilde{\mathbf{u}} = \mathbf{u}^n + \Delta t \left(N(\mathbf{u}^n) + \nu \nabla^2 \mathbf{u}^n \right), \quad (4.4)$$

$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \tilde{\mathbf{u}}, \quad (4.5)$$

$$\mathbf{u}^{n+1} = \tilde{\mathbf{u}} - \Delta t \nabla p^{n+1}, \quad (4.6)$$

with $\mathbf{u} = (u, v)$, p the kinematic pressure, and ν the viscosity. `pyCoastal.physics.navier_stokes` provides `initialize_state(grid)`, which returns zero u, v, p fields, and `rhs(state, t, grid, bc_mgr, v=0.001)`, which returns the convection and diffusion tendencies for u and v (the predictor of Equation 4.4), with the velocity boundary conditions enforced on the result.

4.3 Poisson solver

A Poisson solver computes a field whose Laplacian matches a given source. It appears in the pressure projection, in potential flow, and in diffusion steady states:

$$\nabla^2 \phi = b \text{ in } \Omega, \quad \phi|_{\Gamma_D} = \phi_D, \quad \frac{\partial \phi}{\partial n} \Big|_{\Gamma_N} = g_N,$$

with the compatibility condition $\int_{\Omega} b \, d\Omega + \int_{\Gamma_N} g_N \, d\Gamma = 0$ for a pure Neumann problem. The five-point Laplacian of Equation 3.2 gives a sparse linear system. The Jacobi iteration, with $h = \Delta x = \Delta y$, reads

$$\phi_{i,j}^{(k+1)} = \frac{1}{4} \left(\phi_{i+1,j}^{(k)} + \phi_{i-1,j}^{(k)} + \phi_{i,j+1}^{(k)} + \phi_{i,j-1}^{(k)} - h^2 b_{i,j} \right),$$

stopping when $\|b - A\phi^{(k)}\|_2 / \|b\|_2 < \varepsilon$.

`pyCoastal.physics.poisson` offers:

- `build_laplacian(nx, ny, dx, dy)`, the sparse CSR operator with zero Dirichlet boundaries (needs SciPy);
- `solve_direct(rhs, dx, dy, bc_mask=None, bc_values=None)`, a sparse direct solve;

- `solve_jacobi(rhs, dx, dy, bc_mask=None, bc_values=None, tol=1e-6, maxiter=5000);`
- `solve_poisson(rhs, dx, dy, ..., method="auto")`, which picks the direct solver when SciPy is available and Jacobi otherwise;
- `apply_dirichlet_bc(phi, bc_mask, bc_values)`, which imposes Dirichlet values in place.

4.4 Turbulence closures

When the grid cannot resolve all scales, eddy-viscosity closures emulate the transport by unresolved motions. The closures here are educational scaffolds: production RANS or LES needs careful calibration and near-wall treatment.

Smagorinsky (LES). The eddy viscosity is proportional to the local strain rate and the filter width:

$$\nu_t = (C_s \Delta)^2 |S|, \quad |S| = \sqrt{2 S_{ij} S_{ij}}, \quad S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right),$$

with $C_s \approx 0.1$ to 0.2 . `SmagorinskyModel(Cs=0.17, filter_width=1.0)` computes `eddy_viscosity(u, v, grid)`.

k-ε (RANS). Two transport equations for the turbulent kinetic energy k and its dissipation ε , with $\nu_t = C_\mu k^2 / \varepsilon$:

$$\begin{aligned} \partial_t k + U_j \partial_{x_j} k &= P - \varepsilon + \partial_{x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \partial_{x_j} k \right], \\ \partial_t \varepsilon + U_j \partial_{x_j} \varepsilon &= C_{\varepsilon 1} \frac{\varepsilon}{k} P - C_{\varepsilon 2} \frac{\varepsilon^2}{k} + \partial_{x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_\varepsilon} \right) \partial_{x_j} \varepsilon \right], \end{aligned}$$

with $P = 2\nu_t S_{ij} S_{ij}$. `KEpsilonModel(Cmu=0.09, sigma_k=1.0, sigma_e=1.3, C1=1.44, C2=1.92)` returns the eddy viscosity from k and ε ; the transport equations belong to the solver.

k-ω (RANS). Pairs k with the specific dissipation $\omega = \varepsilon / k$, which behaves better near walls, with $\nu_t = k / \omega$:

$$\begin{aligned} \partial_t k + U_j \partial_{x_j} k &= P - \beta^* k \omega + \partial_{x_j} \left[(\nu + \sigma_k^* \nu_t) \partial_{x_j} k \right], \\ \partial_t \omega + U_j \partial_{x_j} \omega &= \alpha \frac{\omega}{k} P - \beta \omega^2 + \partial_{x_j} \left[(\nu + \sigma_\omega \nu_t) \partial_{x_j} \omega \right]. \end{aligned}$$

`KOmegaModel(sigma_k=2.0, sigma_omega=2.0, beta_star=0.09, beta=0.075, gamma=0.52)` returns k / ω from `eddy_viscosity(k, ω)`.

5 Simulation examples

Each numerical example connects one governing equation, one set of numerics, and one diagnostic to a single learning objective. The scripts read their parameters from YAML files in `examples/configs/`, so an experiment can be repeated or swept by editing a few lines of text. They animate their fields with matplotlib; run them from the repository root.

```
python examples/numerics/water_drop.py
python examples/waves2D.py
python examples/wave2D_irregular.py
python examples/current.py
python examples/pollutant.py
python examples/numerics/viscous_fluid.py
python examples/numerics/2D_irr_turb.py
python examples/equilibrium_shoreline.py
```

5.1 Case files

The YAML keys used by the examples are collected in [Table 5.1](#). Grid keys give cell counts and either spacings or domain lengths; physics keys the parameters of the equation; solver keys the time step (directly or through a CFL number) and the run length; output keys the gauge locations as grid indices.

Table 5.1: YAML keys used by the numerical examples.

Config	grid	physics	solver / forcing	output
water_drop.yaml	Nx, Ny, Lx, Ly	wave_speed, sigma	CFL, t_final	
waves2D.yaml	nx, ny, dx, dy	gravity, depth	amplitude, period; cfl, t_final	obs_points
waves2D_irregular.yaml	nx, ny, dx, dy	gravity, depth	type (pm or jonswap), gamma, Hs, Tp; dt, duration	gauge
current.yaml	nx, ny, dx, dy	speed	cfl, t_final	gauge
pollutant.yaml	Lx, Ly, Nx, Ny	diffusivity, U0, R	t_final, c0_sigma	
viscous_fluid.yaml	Nx, Ny, Lx, Ly	viscosity, background_speed, vortex_sigma	dt, t_final	
equilibrium_shoreline.yaml	Lshore, Gb, Lb1, Lb2, Yi		n_points, sag_side_factor	

5.2 Water drop: the 2D wave equation

Objective. Verify the accuracy of the Laplacian and the stability of the time stepping on a simple hyperbolic system,

$$\frac{\partial^2 \eta}{\partial t^2} = c^2 \nabla^2 \eta.$$

A Gaussian bump of width `sigma` is released at the center of the domain with zero Dirichlet boundaries, and the circular front should travel at `c`. Refining the grid and the time step should converge the solution at the expected order.

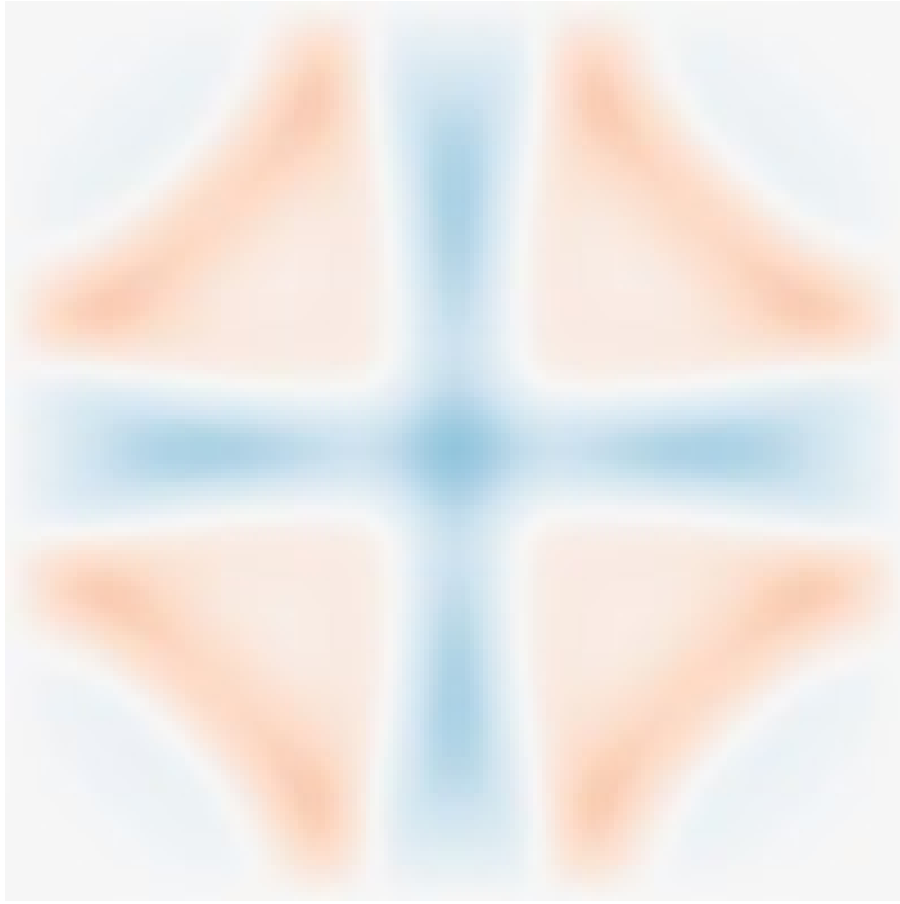


Figure 5.1: Circular wave propagation from a central disturbance (examples/numerics/water_drop.py).

5.3 Regular waves: waves2D.py

A depth-averaged wave field in uniform depth, forced by a sinusoid of amplitude `amplitude` and period `period` at the $y = 0$ boundary and periodic in x . The script solves linear advection along y column by column with an upwind stencil and forward Euler, animates η , and records the time series at the observation points. The wave speed comes from the linear dispersion relation for the configured depth and period.

5.4 Irregular waves: a numerical flume

Objective. Generate a target spectrum and manage reflections. The script synthesizes a surface elevation record from a Pierson-Moskowitz or JONSWAP spectrum (Section 6.1), imposes it as a Dirichlet condition at the inflow boundary, and records a gauge inside the domain, so the input and interior spectra (H_s , T_p) can be compared. A sponge at the outflow absorbs the outgoing waves.

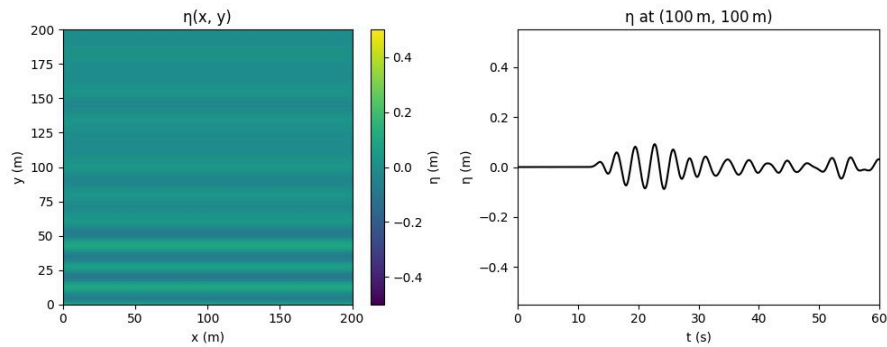


Figure 5.2: Irregular wave field at the end of the run (examples/wave2D_irregular.py).

5.5 Current and pollutant: advection and diffusion

Objective. Observe upwind smearing and diffusive spreading. `current.py` advects a passive field with a uniform current, $u_t + c u_x = 0$, with a Dirichlet inlet on the west and a Neumann outlet on the east, and records a gauge. `pollutant.py` solves the full advection-diffusion equation for a Gaussian patch in a vortex flow in a shallow pond,

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D \nabla^2 C,$$

with upwind advection, centered diffusion (`operators.gradient` and `operators.laplacian`), and zero-flux boundaries. For pure diffusion the variance of the patch grows as $2Dt$, which the run can be checked against.

5.6 Viscous fluid: velocity decay

Objective. Check the diffusive time scale and operator accuracy. `viscous_fluid.py` evolves two counter-rotating vortices on a periodic domain with central differences and forward Euler. The diffusion part can be checked against the decaying mode

$$u(x, y, t) = U_0 \sin(k_x x) \sin(k_y y) \exp[-v(k_x^2 + k_y^2)t].$$

5.7 Boundary-condition test: 2D_irr_turb.py

A pure boundary-condition exercise on a 400 m by 400 m grid: a sinusoidal Dirichlet wave maker on the south side, a spatially ramped sponge on the north, a Dirichlet inflow velocity and a zero-gradient surface on the west, zero-gradient outflow on the east, and walls for the normal velocity. No physics is advanced; the script applies the `BoundaryManager` every step and records a gauge, which isolates what the boundary classes do.

5.8 Equilibrium shoreline behind detached breakwaters

`equilibrium_shoreline.py` builds a schematic equilibrium planform behind a pair of detached breakwaters from three circular arcs (a left lateral arc, a central salient, and a right lateral arc), following the Hsu and Evans style of parabolic bay construction. The geometry comes from `equilibrium_shoreline.yaml` (shoreline length, gap between breakwaters, their lengths, and offshore distance) and the construction uses `pyCoastal.tools.morphodynamics` (Section 6.5), which also reports the retreat and the accretion percentage.

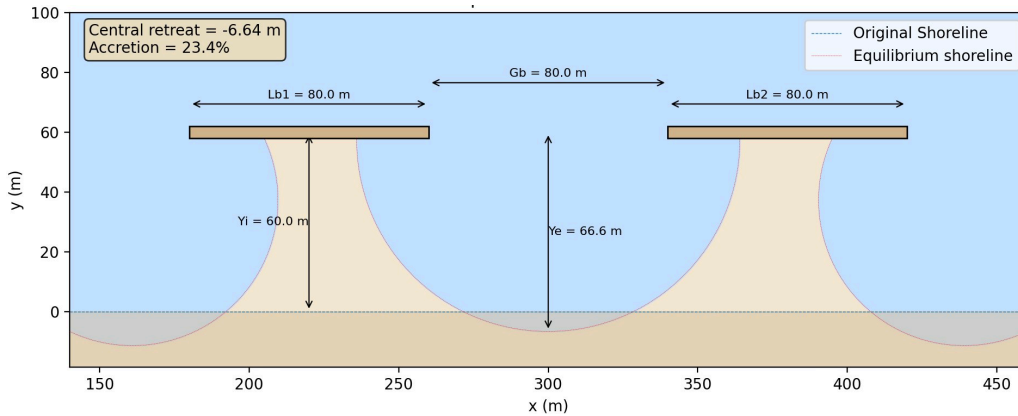


Figure 5.3: Equilibrium shoreline behind two detached breakwaters (`examples/equilibrium_shoreline.py`).

5.9 Diagnostics, verification, and validation

Numerical experiments should be accompanied by simple, repeatable checks:

- **CFL monitor.** Compute and report the Courant number each step, and adapt Δt if needed.
- **Divergence.** After a projection step, $\|\nabla \cdot \mathbf{u}\|_2$ should sit near the Poisson solver tolerance.
- **Energy and variance.** Monitor $\int \eta^2$ or the kinetic energy to spot drift or spurious reflections.
- **Convergence.** Refine Δx , Δy , and Δt and verify the expected order.
- **Spectral targets.** Compare H_s and T_p at interior gauges with the input, and estimate reflection by separating incident and reflected waves.

Save arrays as NumPy files and figures with matplotlib; export animations as `.gif` or `.mp4`. Store the run metadata (git hash, YAML, random seeds) beside the outputs so a figure can be regenerated exactly.

5.10 Extending the framework

To add physics or numerics:

1. implement a right-hand side `rhs(u, t)` and pass it to an integrator;
2. provide boundary handlers consistent with the stencils;
3. add YAML keys with sensible defaults;
4. add a small example and a docstring that states the equations, the variables, the numerics, and the stability limits.

5.11 Limitations of the numerical framework

- Uniform Cartesian grids only; no curvilinear or unstructured meshes.
- Explicit time stepping, which is restrictive for stiff sources such as the ε or ω transport.
- The turbulence closures are educational scaffolds, not production RANS or LES.
- No wetting and drying or moving shorelines in the 2D solvers. (The design applications handle shoreline change with the one-line model and flooding with connectivity-aware mapping; see [Section 17](#) and [Section 18](#).)

Part III. Engineering tools

6 Standalone formulae: pyCoastal.tools

pyCoastal.tools collects standalone coastal engineering formulae. Every function takes its parameters explicitly, with no hidden hard-coded values, so defaults can be set or overridden from a case file. These are the first checks an engineer runs; the design applications of Part IV build full design chains on top of the same relations, with validity checks and reporting. Where a tool and an application overlap, prefer the application for design.

6.1 Waves: tools.wave

6.1.1 Linear dispersion and derived numbers

Under linear (small-amplitude) theory, frequency, wavenumber, and depth are linked by the dispersion relation, which sets the phase and group speeds and underpins most coastal scalings:

$$\omega^2 = gk \tanh(kh), \quad c = \frac{\omega}{k}, \quad c_g = \frac{c}{2} \left(1 + \frac{2kh}{\sinh 2kh} \right). \quad (6.1)$$

dispersion(T, h, g=9.81, tol=1e-12) solves it for the wavelength L by Newton iteration and raises on a non-positive period or depth; wave_number(T, h) returns $k = 2\pi/L$. The deep-water wavelength is $L_0 = gT^2/(2\pi)$.

Two nondimensional numbers classify the waves:

- **Iribarren (surf similarity)**, $\xi_0 = \tan\beta/\sqrt{H_0/L_0}$: surf_similarity(alpha, H, T) returns it and breaker_type(alpha, H, T) names the breaker, spilling for $\xi < 0.4$, plunging for $0.4 \leq \xi \leq 2$, and collapsing or surging above 2.
- **Ursell**, $U = HL^2/h^3$: ursell_number(H, T, h) returns the value and an interpretation, linear theory for $U < 32$ and a nonlinear theory (Stokes, Boussinesq) above.

wave_setup(Hb, gamma=0.8) gives the shoreline setup $\bar{\eta} = \frac{5}{16}\gamma H_b$.

6.1.2 Spectral synthesis

A wave spectrum describes how the energy of a sea state is distributed over frequency. To drive a numerical flume with irregular seas, a surface elevation record is synthesized by random-phase superposition consistent with a target spectrum:

$$\eta(t) = \sum_{i=1}^N \sqrt{2S(f_i)\Delta f} \cos(2\pi f_i t + \varphi_i), \quad \varphi_i \sim U[0, 2\pi]. \quad (6.2)$$

The spectral moments $m_n = \int_0^\infty f^n S(f) df$ give $H_s \approx 4\sqrt{m_0}$ and $T_z \approx \sqrt{m_0/m_2}$. generate_irregular_wave(Hs, Tp, duration, dt, spectrum="pm", gamma=3.3) returns (t, eta) for a Pierson-Moskowitz (fully developed) or JONSWAP (fetch-limited, peak enhancement gamma) spectrum scaled to the requested H_s .

6.2 Sediment transport: tools.sediment_transport

The Shields parameter gauges bed mobility under a bed shear stress $\tau_b = \rho u_*^2$:

$$\theta = \frac{\tau_b}{(\rho_s - \rho) g d}, \quad (6.3)$$

with a critical value $\theta_c \approx 0.03$ to 0.06 for sand (the design modules compute it from the grain size, [Section 8](#)). The transport formulae in [Table 6.1](#) convert mobility into rates.

Table 6.1: Transport relations in pyCoastal.tools.sediment_transport.

Function	Relation
shields_parameter(tau_b, rho_s, rho, d)	Equation 6.3
van_rijn_bedload(Ue, h, d50, rho_s, rho)	$q_b = 0.015 \rho_s U_e h (d_{50}/h)^{1.2} M_e^{1.5}$, $M_e = (U_e - U_{cr})/\sqrt{(s-1)gd_{50}}$ (Van Rijn 1984)
van_rijn_suspended(Ue, h, d50, rho_s, rho)	$q_s = 0.008 \rho_s U_e d_{50} M_e^{2.4} D_*^{-0.6}$
bijker_bedload(tau_wave, tau_current, rho_s, rho, d50)	$\propto \sqrt{\tau_{\text{total}}}$, waves and current combined (Bijker 1971)

Function	Relation
cerc_transport(Eb, angle, K=0.39)	$Q = K E_b / (\rho g) \sin \varphi_b \cos \varphi_b$ (CERC)
bagnold_sediment(H, c, rho_s)	$(\rho_s / \rho) H^2 c$, energetics estimate
izbash_current(rho_s, rho, d)	$u_c = 1.7 \sqrt{\Delta g d}$, stone stability in current
einstein_bedload(tau_star, d, s)	$q_b = 8 \sqrt{g(s-1)d^3 \tau_*^{1.5}}$

The critical velocity in the Van Rijn functions is $U_{cr} = 0.19 d_{50}^{0.1} \log(12h/3d_{50})$ and $s = \rho_s / \rho$.

6.3 Structures and runup: tools.structural

- hudson_dn50(Hs, Delta, theta, Kd=3.0): $H_s / (\Delta D_{n50}) = (K_D \cot \theta)^{1/3} / 1.27$ (SPM 1984).
- vandermeer_dn50(Hs, Delta, P, N, alpha, xi_m, damage=2.0, safety=1.0): Van der Meer (1988), plunging and surging branches (Equation 9.1 and Equation 9.2 in Section 9), with the wave count saturating at 7500. The surf similarity must be supplied, because it depends on the period.
- hunt_runup(beta, H, L): $R \approx H \tan \beta / \sqrt{H/L}$ (Hunt 1959).
- stockdon_runup(H, L, beta): the Stockdon et al. (2006) 2% runup.
- goda_wave_force(H, T, h, beta): a screening estimate of the Goda horizontal force on a vertical face, in N/m, taking $H_{\text{design}} = 1.8H$. The full Goda distribution with uplift and stability checks is applications.seawall.goda_pressures (Section 10).
- iribarren_stability(H, alpha, rho_s, rho_w, mu, N): the Iribarren stone weight, $W \geq N \rho_s g H^3 / [\Delta^3 (\mu \cos \alpha - \sin \alpha)^3]$.

6.4 One-line shoreline model: tools.shoreline

The one-line model treats the shoreline position $y(x, t)$ as the only morphological variable and conserves sediment alongshore:

$$\frac{\partial y}{\partial t} = -\frac{1}{(1-p)D} \frac{\partial Q_{ls}}{\partial x}, \quad Q_{ls} = K_{\text{cerc}} H_b^{5/2} \sin(2\alpha_b), \quad \alpha_b = \alpha_0 - \frac{\partial y}{\partial x}, \quad (6.4)$$

with $H_b = K_d K_t H_{\text{free}}$, where K_d is a diffraction shadow factor and K_t a transmission factor. The explicit scheme is stable for

$$\Delta t \leq 0.45 \frac{\Delta x^2}{\max G}, \quad G = \frac{2K_{\text{cerc}} H_b^{5/2}}{(1-p)D}. \quad (6.5)$$

OneLineParams(D=8, p=0.4, Kcerc=0.4, Hfree=1.0, alpha0=10 deg, Kt=1.0, beta0=-30 deg, morfac=1.0) holds the parameters; any of Hfree, alpha0, Kt can be arrays over x . The building blocks are make_grid(Lx, dx), breakwaters_geometry(x, tips, y_tip, crest_freeboard) for any number of detached breakwaters, kd_diffraction_field(x, geom, beta0) for a simple shadow factor $K_d = \frac{1}{2}(1 + \cos \theta)$ from each tip, compute_flux_Qls, rhs_y_t, suggest_dt, and apply_bcs(y, bc="fixed_ends" | "zero_slope"). run_one_line_model() runs a default two-breakwater case. The nourishment application (Section 17) drives this solver with a fill planform.

6.5 Morphodynamics helpers: tools.morphodynamics

- bruuns_rule(S, beta, L=None, h=None, B=None): shoreline retreat under sea-level rise S , $R = SL/(h + B)$ when the profile length, closure depth, and berm height are given, and $R = S/\tan \beta$ otherwise.
- exner_change(qs_dx, porosity=0.64): bed change rate from the transport divergence, $\partial \eta / \partial t = -q_{s,x} / (1 - n)$.
- linear_slope(grid, north_level, south_level): a plane beach on a grid.
- circle_through_points, two_point_arc, and build_tombolo_arcs(Lshore, Gb, Lb1, Lb2, yb, ...): the three-arc equilibrium planform behind a pair of detached breakwaters used by examples/equilibrium_shoreline.py, with retreat $Y_e = 1.204 Y_i - 0.07 G_b$ below the breakwater line; accretion_metrics(x_eq, y_eq, geom) returns the retreat and the accretion percentage.

Part IV. Design applications

7 Design conditions from a record

Module: `pyCoastal.applications.extremes`. *Example:* `examples/design_wave.py`. *Browser:* Return values.

Every structure in this package is sized against a design condition, and that condition has to come from somewhere. This module is where it comes from: a measured or hindcast record in, a return value with an honest confidence band out.

7.1 Two routes

Peaks over threshold (POT). Keep the independent storm peaks above a threshold, fit a generalized Pareto distribution (GPD) to the excesses, and combine it with the rate at which peaks arrive. It uses the record efficiently, but the answer depends on the threshold, so the threshold is chosen with the diagnostics below rather than by eye.

Block maxima. Keep the largest value in each year and fit a generalized extreme value (GEV) distribution. It wastes data but is hard to get wrong, and it is the right check on a POT answer.

Both are fitted by **L-moments** (Hosking and Wallis 1997), which are close to unbiased for the sample sizes a coastal record offers, are not troubled by the flat likelihood surfaces that defeat maximum likelihood on short records, and need no optimizer. Shape parameters use the extreme-value convention: a positive shape is a heavy tail with no upper bound, a negative shape a finite upper limit. Hosking writes $k = -\text{shape}$; the conversion is done internally and never exposed.

7.2 Theory

Declustering. `decluster(values, threshold, separation)` walks the record and keeps only the largest value in each run above the threshold. Two runs separated by fewer than `separation` samples below the threshold are one storm. For a 3-hourly record, 24 samples is a three-day separation, the usual choice for storm waves.

GPD fit. With survivor function $1 - F(x) = (1 + \xi x/\sigma)^{-1/\xi}$ (exponential as $\xi \rightarrow 0$), the L-moment estimators from the mean λ_1 and L-CV $\tau = \lambda_2/\lambda_1$ of the excesses are

$$\xi = 2 - \frac{1}{\tau}, \quad \sigma = \lambda_1(1 - \xi). \quad (7.1)$$

A shape at or above one means the fitted mean does not exist; the fit is returned and flagged rather than clipped.

POT return value. With λ peaks per year, the T -year value is

$$x_T = u + \frac{\sigma}{\xi} [(\lambda T)^\xi - 1], \quad (7.2)$$

with the logarithmic form in the exponential limit. A return period shorter than $1/\lambda$ is outside the model.

GEV fit. With $F(x) = \exp\{-[1 + \xi(x - \mu)/\sigma]^{-1/\xi}\}$, the parameters come from Hosking's L-moment estimators, whose polynomial approximation for k from the L-skewness is accurate to better than 10^{-4} for $-0.5 < k < 0.5$.

Plotting positions. The empirical return periods use the Gringorten position $(i - a)/(n + 1 - 2a)$ with $a = 0.44$, close to unbiased for the Gumbel and GEV families.

Threshold diagnostics. Above a threshold where the GPD holds, the mean excess is linear in the threshold (`mean_residual_life`), and both the fitted shape and the *modified* scale $\sigma^* = \sigma - \xi u$ are flat (`threshold_stability`). The raw scale is not threshold-invariant, and reading it as if it were is a common way to pick a threshold badly.

Confidence. `ExtremeFit.confidence` is a percentile bootstrap over the fitted peaks. It captures sampling error in the parameters and nothing else: not measurement error, not the choice of threshold, and not whether the climate that produced the record is the one the structure will see. Those are usually the larger uncertainties.

7.3 Using the module

```
from pyCoastal.applications.extremes import (
    fit_pot, fit_block_maxima, mean_residual_life, threshold_stability,
)

mr1 = mean_residual_life(record, thresholds)          # pick the threshold properly
stab = threshold_stability(record, thresholds, separation=24)
```

```

fit = fit_pot(record, threshold=2.5, separation=24, samples_per_year=2920)

fit.return_value(100)          # 100-year Hm0
fit.confidence(100, level=0.90) # and how much it could have been
fit.extrapolation_note(100)    # how far past the record that reaches
print(fit.summary())

check = fit_block_maxima(record, block=2920) # annual maxima cross-check

```

`fit_pot` and `fit_block_maxima` return an `ExtremeFit` with kind ("pot" or "gev"), parameters, the peaks or maxima in data, their positions, the rate and record length years, the bootstrap samples, and a warnings list. The fit warns when there are fewer than one or two peaks per year or fewer than about thirty in total.

7.4 Worked example

`examples/design_wave.py` builds a 40-year synthetic 3-hourly hindcast of H_{m0} with known parameters, chooses the threshold with the diagnostics, fits POT and annual maxima, and hands the 100-year wave to the seawall designer. Because the record is synthetic, the right answer is known, so the example also does what a real study cannot: it repeats the whole analysis on 60 fresh 40-year records and measures the true sampling spread of the 100-year estimate, then checks the bootstrap band against it.

Output of `examples/design_wave.py`

```

Record: 40 years, 116800 samples, max 5.21 m, 240 storms (6.0/yr)
Threshold 2.50 m chosen from the diagnostics (the record was built with 2.50 m)

Peaks over threshold
Model          peaks over threshold
Record         40.0 years, 240 peaks
Threshold      2.50
Rate           6.00 peaks per year
Scale, shape   0.560, -0.071

Return period  value    90% band
   10 yr       4.49    4.19 to 4.83
   50 yr       5.12    4.61 to 5.78
  100 yr       5.38    4.74 to 6.19
the record was built with scale 0.550, shape -0.080, rate 6.0/yr

Annual maxima, as a cross-check
Model          annual maxima
Record         40.0 years, 40 maxima
Location       3.489
Scale, shape   0.484, -0.058

Return period  value    90% band
   10 yr       4.51    4.22 to 4.74
   50 yr       5.18    4.63 to 5.58
  100 yr       5.45    4.77 to 5.95

100-year Hm0 = 5.38 m (4.74 to 6.19 m, 90% bootstrap)
true value for this climate: 5.25 m
2.5 times the 40 year record. Treat the confidence band as a lower bound on the real uncertainty.

Repeating the study on 60 fresh 40-year records
100-year estimate: mean 5.23 m, sd 0.44 m
5th to 95th percentile: 4.61 to 5.94 m (width 1.32 m)
bootstrap band from the single record: width 1.45 m
this record sits at the 70th percentile of that spread
The bootstrap band matches the real sampling spread.

Design wave at the toe 4.68 m (depth limited)
Seawall crest +13.13 m CD, base 20.83 m, concrete 45.0 m3/m

Wrote media/design_wave.png

```

Design wave from a 40-year record: 100-year $H_{m0} = 5.38$ m (4.74 to 6.19 m), true value 5.25 m

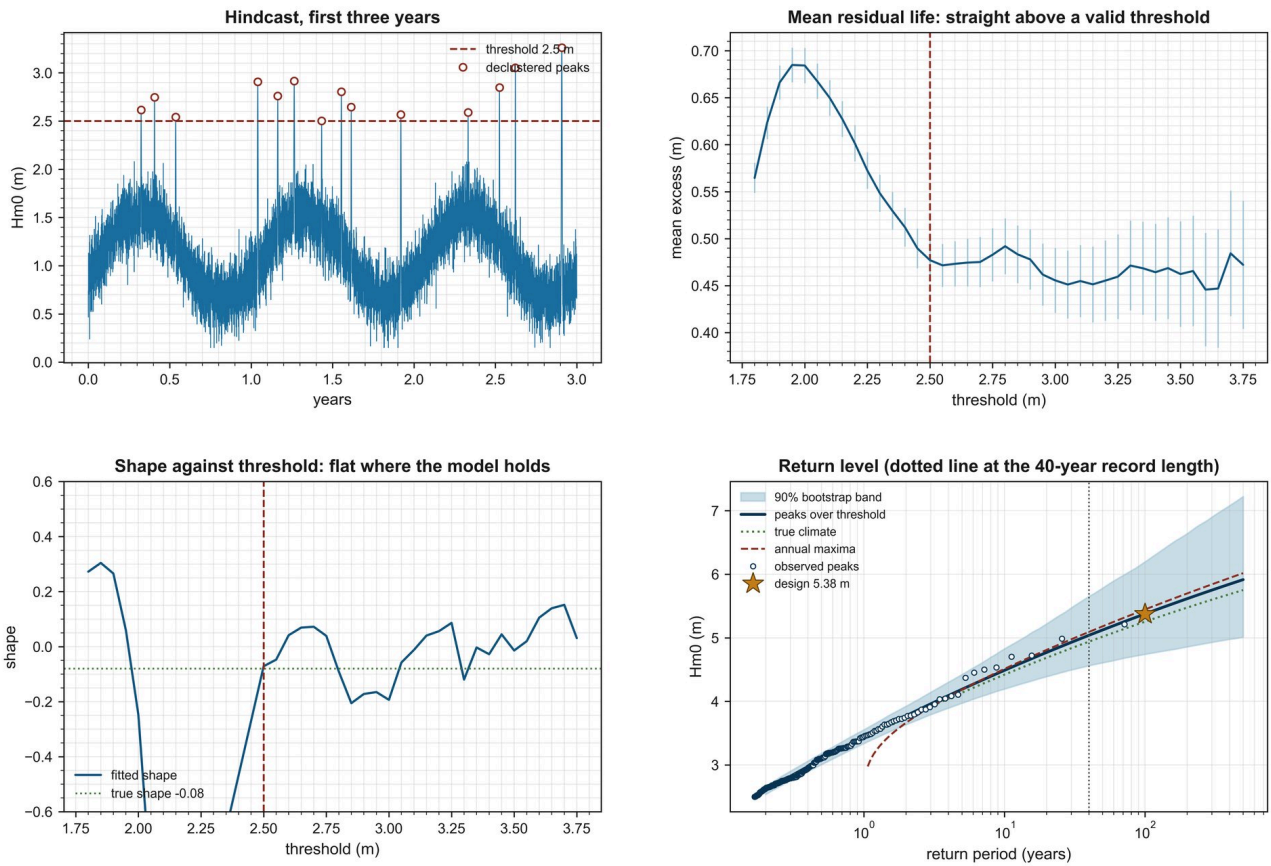


Figure 7.1: Extreme value analysis of a 40-year record: threshold diagnostics, return level plot with bootstrap band, and the sampling spread of the 100-year estimate.

The bootstrap band from the single record (width 1.45 m) matches the real sampling spread across records (1.32 m), and the true 100-year value of 5.25 m sits inside it. The 100-year wave is 2.5 times the record length, so the band should be treated as a lower bound on the real uncertainty, which is exactly what `extrapolation_note` says.

7.5 Limits

- Stationarity is assumed: no trend or climate change in the record.
- The band is sampling error only.
- Very short records, or thresholds that leave too few peaks, are flagged but still fitted.

8 Sediment and soil

Module: `pyCoastal.applications.sediment`.

Every structure in this package sits on, in, or behind something. This module makes that something a material with properties that propagate: the grain size that decides whether the bed moves at all, the friction angle that decides how hard the backfill pushes on a wall, and the unit weight that decides how much of that push is soil and how much is water. The seawall, breakwater, channel, pile, scour, and nourishment modules all take a `Sediment` or a catalogue key.

Two families of relation live here. **Mobility** (grain size, fall velocity, the Shields threshold) gates scour: a bed that never reaches its threshold does not scour, whatever the wave height. **Earth pressure** (Rankine and Coulomb coefficients, and the pressure diagram with the water table and surcharge in the right places) loads the back of a wall.

8.1 The material catalogue

`Sediment(name, d50, specific_gravity=2.65, porosity=0.4, friction_angle=32, cohesion=0, phi_sorting=0.6, d90=None, description="")` stores a material, with derived `relative_density` ($s - 1$), `dry_unit_weight`, `saturated_unit_weight`, `submerged_unit_weight`, `grading` (d_{90}/d_{50} , with d_{90} defaulting to $2.5 d_{50}$), and `cohesive`. `sediment(key)` looks up the catalogue (Table 8.1) and names the options on a bad key.

Table 8.1: The built-in material catalogue.

Key	d_{50} (mm)	ϕ' (deg)	Description
soft_clay	cohesive	22	normally consolidated, undrained strength governs
stiff_clay	cohesive	26	overconsolidated
silt	0.03	28	mobile at almost any wave
very_fine_sand	0.09	29	suspends readily, high siltation
fine_sand	0.19	31	the usual beach and nearshore sand
medium_sand	0.38	33	typical dredged fill
coarse_sand	0.75	35	good drained backfill
fine_gravel	6	38	free draining
coarse_gravel	30	40	shingle beach
rock_fill	150	42	engineered granular backfill

8.2 Mobility

The dimensionless grain size, the critical Shields parameter (Soulsby and Whitehouse 1997), and the settling velocity (Soulsby 1997) are

$$D_* = d_{50} \left[\frac{g(s-1)}{v^2} \right]^{1/3}, \quad \theta_{cr} = \frac{0.30}{1 + 1.2D_*} + 0.055[1 - e^{-0.020D_*}], \quad w_s = \frac{v}{d_{50}} \left[\sqrt{10.36^2 + 1.049D_*^3} - 10.36 \right]. \quad (8.1)$$

The fall velocity relation is valid from silt to gravel, which is why it is preferred to Stokes at one end and a drag law at the other.

Under waves, `wave_orbital_velocity(Hs, T, depth)` gives the near-bed orbital amplitude from linear theory and `wave_shields(material, Hs, T, depth)` the Shields parameter, with the Swart friction factor for a rough turbulent bed and a Nikuradse roughness $k_s = 2.5 d_{50}$:

$$f_w = \exp \left[5.213 \left(\frac{A}{k_s} \right)^{-0.194} - 5.977 \right], \quad f_w \leq 0.3. \quad (8.2)$$

`bed_mobility(material, Hs, T, depth)` is the gate the scour functions use. It returns whether the bed is live, the regime (live bed, clear water, or cohesive), and by how much the threshold is exceeded. Scour relations are almost all live-bed results, and applying one to a bed that never reaches its threshold predicts a hole that will not form.

8.3 Earth pressure

`earth_pressure_coefficient(friction_angle, kind="active", wall_friction=0, backslope=0)` returns the Rankine coefficient for "active", "at_rest" ($K_0 = 1 - \sin\phi'$), or "passive"; a non-zero wall friction switches the active case to Coulomb, which is the honest choice for a rough concrete face and gives a smaller force. Which coefficient applies is a question about movement, not about soil: a wall that cannot move the millimeter or two that mobilizes the active state carries at-rest pressure, roughly half as large again. Choosing "active" because it is smaller is the most common way a retaining structure is under-designed.

`lateral_earth_pressure(material, height, water_table=0, surcharge=0, kind="active")` builds the pressure diagram with effective stress and pore water apart; `lateral_earth_force(...)` integrates it into soil, water, and total forces with the lever arm of the total. Below the water table the soil pushes with its submerged unit weight and the water separately with its full hydrostatic gradient, $K = 1$. The sum is larger than dry soil alone would give, which is why drainage behind a seawall is a structural matter and not a detail. The soil force can be argued down with a better backfill; the water force can only be drained away.

```
from pyCoastal.applications.sediment import (
    sediment, critical_shields, fall_velocity, wave_shields, lateral_earth_force,
)

sand = sediment("medium_sand")
critical_shields(sand), fall_velocity(sand)          # about 0.035 and 0.05 m/s
wave_shields(sand, Hs=2.0, T=9.0, depth=8.0)["mobile"]
lateral_earth_force(sand, height=10.0, water_table=0.0, surcharge=10.0,
                    kind="at_rest")["total"]         # kN/m
```

9 Rubble-mound breakwaters

Module: pyCoastal.applications.structures. *Example:* examples/breakwater_design.py. *Browser:* Breakwater.

Armor sizing and wave overtopping, with every relation traced to its source: Van der Meer (1988) and Hudson (SPM 1984) for stability, the Rock Manual (2007) for layer geometry, and EurOtop (2018) for overtopping and the tolerable discharge limits. The module is aimed at design work rather than illustration: every relation is named, its validity range stated, and the functions compute outside a range when asked but say so.

9.1 The design condition

`DesignConditions(Hm0, Tm10, depth=15.0, storm_duration=6*3600)` carries the wave and water level at the toe. `DesignConditions.from_peak_period(Hm0, Tp, ...)` converts with $T_{m-1,0} = T_p/1.1$ for a single-peaked spectrum. Derived properties are the wave count $N = t_{\text{storm}}/T_m$, the deep and local wavelengths, and the surf similarity

$$\xi_{m-1,0} = \frac{\tan\alpha}{\sqrt{H_{m0}/L_{m-1,0}}}, \quad L_{m-1,0} = \frac{gT_{m-1,0}^2}{2\pi}.$$

9.2 Armor stability

Van der Meer (1988). Two regimes, selected by the surf similarity against a critical value:

$$\frac{H_s}{\Delta D_{n50}} = 6.2 P^{0.18} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \xi^{-0.5} \quad (\xi < \xi_{cr}, \text{ plunging}), \quad (9.1)$$

$$\frac{H_s}{\Delta D_{n50}} = 1.0 P^{-0.13} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \sqrt{\cot\alpha} \xi^P \quad (\xi \geq \xi_{cr}, \text{ surging}), \quad (9.2)$$

$$\xi_{cr} = \left(6.2 P^{0.31} \sqrt{\tan\alpha} \right)^{1/(P+0.5)}.$$

$\Delta = \rho_s/\rho_w - 1$ (1.585 for 2650 kg/m³ rock in seawater), P is the notional permeability (0.1 impermeable core with a filter, 0.4 permeable core, 0.5 homogeneous, 0.6 very permeable), and S the damage level (DAMAGE_LEVELS: 2 is start of damage and the usual design value, 8 and 12 to 17 failure). `rock_armour_vandermeer(conditions, cot_alpha, Delta=1.585, permeability=0.4, damage=2.0, safety_factor=1.0)` returns D_{n50} , $M50$, the governing regime, ξ , and ξ_{cr} . It is valid for non-depth-limited waves at the toe; the shallow-water modification is not applied.

Hudson (SPM 1984). $H_s/(\Delta D_{n50}) = (K_D \cot\alpha)^{1/3}/1.27$, where 1.27 converts the SPM's $H_{1/10}$ basis to H_s . `rock_armour_hudson(conditions, cot_alpha, Delta, Kd=4.0)` is a first estimate and cross-check: it has no dependence on period, storm duration, permeability, or damage.

Layer geometry (Rock Manual). `armour_layer(Dn50, n_layers=2, layer_coefficient=1.0, porosity=0.37)` returns the layer thickness $t = nk_t D_{n50}$ and the number of stones per square meter.

9.3 Overtopping

For a sloping structure, EurOtop (2018) takes the lesser of the breaking and the maximum expressions:

$$\frac{q}{\sqrt{gH_{m0}^3}} = \frac{0.023}{\sqrt{\tan\alpha}} \gamma_b \xi \exp \left[- \left(2.7 \frac{R_c}{\xi H_{m0} \gamma_b \gamma_f \gamma_\beta \gamma_v} \right)^{1.3} \right], \quad (9.3)$$

$$\frac{q}{\sqrt{gH_{m0}^3}} = 0.09 \exp \left[- \left(1.5 \frac{R_c}{H_{m0} \gamma_f \gamma_\beta} \right)^{1.3} \right]. \quad (9.4)$$

For a plain vertical wall in non-impulsive conditions (EurOtop eq. 7.1):

$$\frac{q}{\sqrt{gH_{m0}^3}} = 0.047 \exp \left[- \left(2.35 \frac{R_c}{H_{m0} \gamma_\beta} \right)^{1.3} \right], \quad (9.5)$$

valid when the impulsiveness parameter $h_* = 1.35 (h/H_{m0})(2\pi h/gT_{m-1,0}^2)$ exceeds about 0.23; `overtopping_vertical` reports `h_star` and `impulsive`.

The factors are roughness γ_f (ROUGHNESS_FACTORS: 1.0 for smooth concrete, 0.55 and 0.40 for impermeable and permeable two-layer rock, 0.38 to 0.49 for concrete units), obliquity γ_β (`obliquity_factor(beta_degrees)`), berm γ_b , and wave wall γ_v . `EurOtop` reports roughly a factor-of-three scatter about the mean discharge; `overtopping_with_uncertainty(result, factor=3.0)` returns the band, and `required_crest_freeboard(conditions, q_allowable, cot_alpha, ...)` inverts the relation by bisection. `assess_overtopping(q)` lists which uses a discharge is tolerable for (Table 9.1).

Table 9.1: Tolerable mean discharges (TOLERABLE_DISCHARGE, `EurOtop` 2018).

Key	q (l/s/m)	Use
<code>pedestrians_unaware</code>	0.03	unaware pedestrians, narrow walkway, clear view of the sea
<code>pedestrians_aware</code>	0.1	aware pedestrians, able to see and avoid the hazard
<code>harbour_quay_equipment</code>	0.4	equipment set back 5 to 10 m from the crest
<code>trained_staff</code>	1	trained staff, well shod and protected, wide walkway
<code>buildings_structural</code>	1	structural damage to buildings behind the defense
<code>vehicles_low_speed</code>	10	vehicles at low speed on a road behind the crest
<code>vehicles_moderate_speed</code>	50	driving at moderate speed, pulsating overtopping
<code>embankment_seaward</code>	50	damage to a maintained grass or armored embankment

9.4 Designing a section

`design_rubble_mound(conditions, cot_alpha=2.0, armour="rock_two_layer_permeable", damage=2.0, permeability=0.4, Delta=1.585, tolerable_use="trained_staff", safety_factor=1.0, scatter_factor=3.0)` sizes the stone with Van der Meer and sets the crest so that the **upper** confidence bound on overtopping, not the mean, meets the limit. Designing to the mean would be exceeded about half the time. It returns a `BreakwaterDesign` with `Dn50`, `M50`, `regime`, `crest_freeboard`, `q_mean`, `q_upper`, `layer`, `governing_limit`, `section`, and `summary()`.

The rest of the section is built from the design:

- **Toe scour.** `reflection_coefficient(cot_alpha, xi, permeable=True)` (Seelig and Ahrens 1981, $K_r = a\xi^2/(b + \xi^2)$) with $a = 0.6$, $b = 6.6$ for rubble) and `toe_scour(bed, Hs, T, depth, reflection)`, which scales Xie's standing-wave result by K_r : $S/H_s = 0.4 K_r / \sinh(kh)^{1.35}$. This scaling is an engineering assumption with the right limits, flagged `screening_only`. `breakwater_toe_scour(design, depth, bed)` applies it to a design, gated by `bed_mobility`.
- **Foundation.** `mound_foundation(design, depth, bed="medium_sand", bedding_material="coarse_sand", settlement_allowance=0.0)` returns the bedding blanket on a geotextile, carried past each toe by the larger of the scour apron and twice the scour depth (at least 3 m), and a toe berm of filter stone, which sits low where the orbital velocities are smaller than at the waterline.
- **Crown wall.** `crown_wall(design, still_water_level, deck_width=7.5, parapet_width=2.0, ...)` proportions a stepped concrete crown block and its concrete volume. It is proportioned, not designed: its sliding and overturning under impact and uplift are a separate calculation.

- **Roundhead.** A head is attacked from more directions and its armor gets less support, so practice uses a lower stability coefficient. With a ratio $r = K_{D, \text{head}}/K_{D, \text{trunk}}$ from `roundhead_kd_ratio(armour, cot_alpha)` (Table 9.2), $D_{n50, \text{head}} = D_{n50, \text{trunk}}/r^{1/3}$ and $M_{50, \text{head}} = M_{50, \text{trunk}}/r$. `roundhead(design, kd_ratio=None, raise_crest=0.0)` returns the head design.

Table 9.2: Head to trunk stability coefficient ratios (ROUNDHEAD_KD_RATIO).

Armor family	slope 1:1.5	1:2	1:3
rock	0.95	0.80	0.65
cubes	0.85	0.75	0.60
tetrapod	0.72	0.64	0.50
accropode	0.80	0.75	0.65
dolos	0.65	0.58	0.45

```
from pyCoastal.applications.structures import (
    DesignConditions, design_rubble_mound, roundhead, mound_foundation,
)

conditions = DesignConditions.from_peak_period(
    Hm0=4.0, Tp=11.0, depth=12.0, storm_duration=6 * 3600,
)
design = design_rubble_mound(conditions, cot_alpha=2.0, tolerable_use="trained_staff")
print(design.summary())
head = roundhead(design) # heavier armor at the head
base = mound_foundation(design, depth=12.0, bed="medium_sand")
```

9.5 Worked example

Output of `examples/breakwater_design.py`

```
Design condition  Hm0 = 4.00 m, Tm-1,0 = 10.00 s, depth = 12.0 m
Storm            6.0 h, N = 2160 waves
Section          trunk
Slope            1 : 2
Armour           rock_two_layer_permeable, Dn50 = 1.59 m, M50 = 10.7 t (plunging)
Layer            3.18 m thick, 0.50 stones/m2
Crest freeboard  Rc = 5.69 m (Rc/Hm0 = 1.42)
Overtopping      mean 0.333 l/s/m, upper bound 1 l/s/m
Governing limit  1 l/s/m, Trained staff, well shod and protected, wide walkway

Tolerability of the mean discharge:
NO    0.03 l/s/m  pedestrians_unaware
NO    0.10 l/s/m  pedestrians_aware
ok    0.40 l/s/m  harbour_quay_equipment
ok    1.00 l/s/m  trained_staff
ok    1.00 l/s/m  buildings_structural
ok    10.00 l/s/m vehicles_low_speed
ok    50.00 l/s/m vehicles_moderate_speed
ok    50.00 l/s/m embankment_seaward

Wrote media/breakwater_design.png
Wrote media/breakwater_section.png, media/breakwater_sheet.png and media/breakwater_section.dxf
```

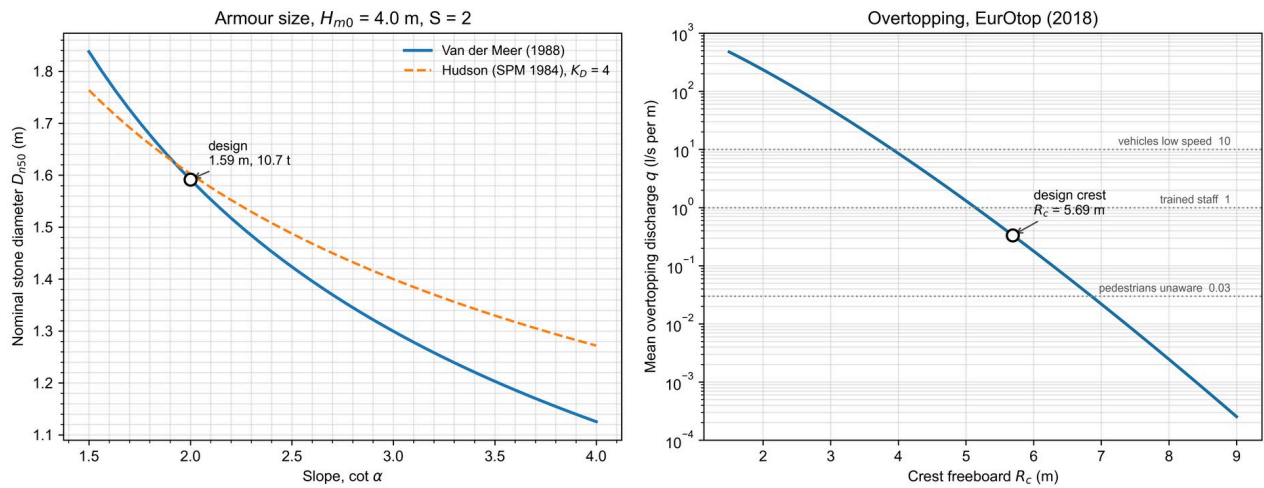



Figure 9.1: Breakwater design curves: armor size against slope, and overtopping against crest freeboard with the EurOtop scatter band.

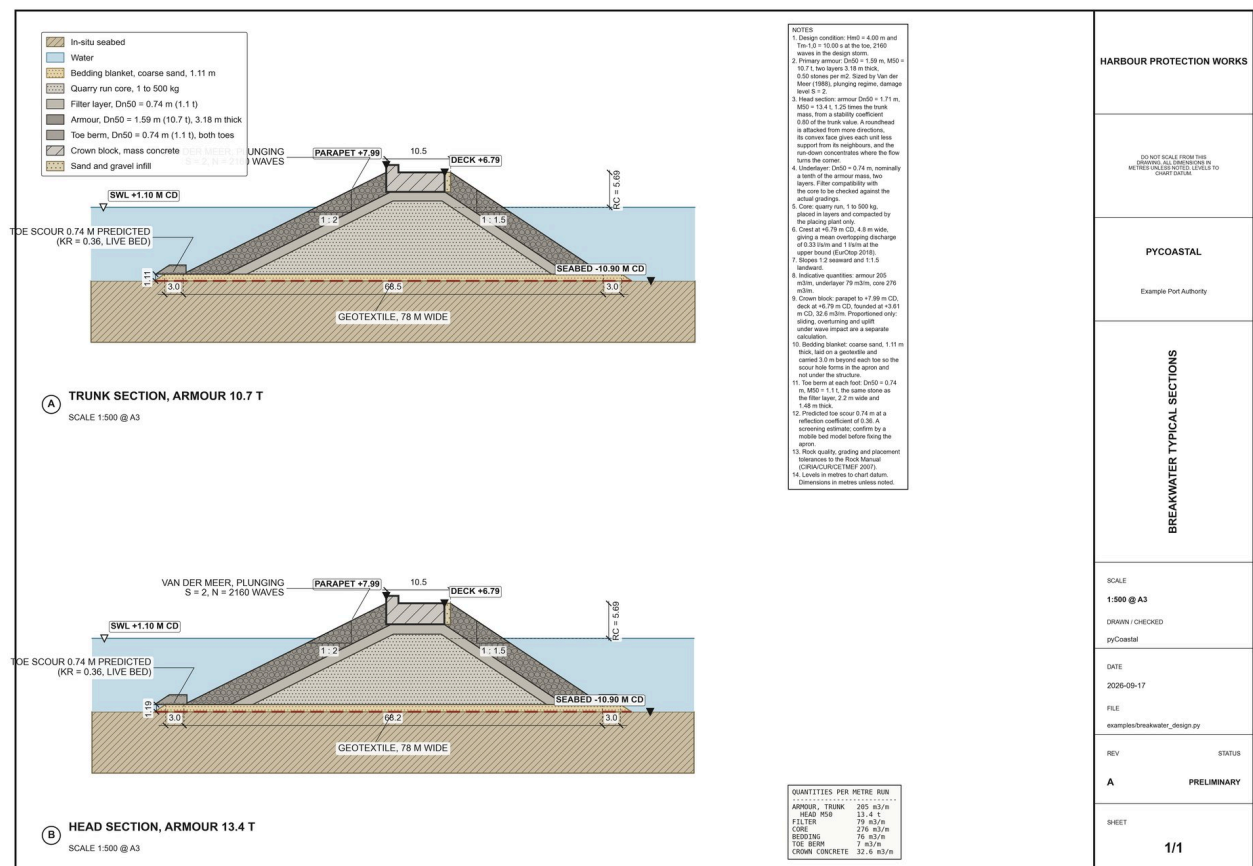


Figure 9.2: The same design issued as a drawing sheet: trunk and head sections with the armor drawn as individual stones at the computed D_{n50} , the bedding blanket, toe berms, crown wall, and generated specification notes.

10 Vertical seawalls

Module: pyCoastal.applications.seawall. *Example:* examples/seawall_section.py. *Browser:* Seawall.

The full chain for an L-shaped gravity seawall: crest level from EurOtop, founding level from scour, Goda wave pressures on the wetted face, earth and water pressure from the backfill, then the base width grown until sliding, overturning, and the base pressure all pass under every load case. applications.sections turns the result into a drawing and a bill of quantities (Section 19).

10.1 Goda pressures

Goda's method (1974, 2010) gives a trapezoidal pressure distribution on the face of a vertical wall. With the design wave $H_{\max} = 1.8H_{m0}$, limited to $\gamma_b d$ in shallow water (breaker_index=0.78), the pressure at the still water level, the elevation of the distribution above it, and the pressure at the base of the upright are

$$p_1 = \frac{1}{2}(1 + \cos\beta)(\alpha_1 + \alpha_2 \cos^2\beta)\rho g H_{\max}, \quad \eta^* = 0.75(1 + \cos\beta)H_{\max}, \quad p_3 = \alpha_3 p_1, \quad (10.1)$$

with the pressure at the crest p_4 interpolated where the crest is below η^* , and a triangular uplift $p_u = \frac{1}{2}(1 + \cos\beta)\alpha_1\alpha_3\rho g H_{\max}$ at the seaward edge. The coefficients are

$$\alpha_1 = 0.6 + \frac{1}{2} \left(\frac{4\pi h/L}{\sinh 4\pi h/L} \right)^2, \quad \alpha_2 = \min \left[\frac{h_b - d}{3h_b} \left(\frac{H_{\max}}{d} \right)^2, \frac{2d}{H_{\max}} \right], \quad \alpha_3 = 1 - \frac{h'}{h} \left[1 - \frac{1}{\cosh 2\pi h/L} \right],$$

where h_b is the depth five wave heights seaward (from the seabed slope), d the depth over the berm, and h' the depth to the underside of the upright. goda_pressures(Hm0, T, depth, wall_toe_depth, berm_depth=None, crest_freeboard=5.0, beta_degrees=0.0, slope=1/30, Hmax_factor=1.8, breaker_index=0.78) returns the pressures, eta_star, the force F, its lever arm, and the uplift. This is the standard non-impulsive distribution; Takahashi's impulsive coefficient is not applied, so a wall on a high mound needs a separate check.

10.2 Scour, toe, and stability relations

- scour_depth_vertical_wall(Hm0, T, depth, coefficient=0.4): Xie (1981), $S/H = 0.4/\sinh(kh)^{1.35}$, the equilibrium depth a quarter wavelength from the wall, used as a scour allowance for the founding level.
- toe_stone_size(Hm0, toe_depth, water_depth, Delta=1.585, damage=0.5): Van der Meer's toe formula, $H_s/(\Delta D_{n50}) = (2 + 6.2(h_t/h)^{2.7})N_{od}^{0.15}$, calibrated for $0.4 < h_t/h < 0.9$ and flagged outside it.
- sliding_safety(F, weight, uplift, friction=0.6): FoS = $\mu(W - U)/F$, zero when uplift exceeds weight (flotation).
- overturning_safety(F, arm, restoring_moment, uplift, base_width): about the rear heel, with the triangular uplift centroid at two thirds of the base width.
- bearing_pressures(normal, base_width, net_moment): $p_{\max}$, $p_{\min}$, the eccentricity, and middle_third; outside the middle third the no-tension triangular distribution is used.
- hydrostatic_force(depth): $\frac{1}{2}\gamma h^2$ at $h/3$, for the water standing in front of the wall when the trough passes.

10.3 Designing a wall

design_seawall(conditions, still_water_level, seabed_level, tolerable_use="pedestrians_aware", ...) follows the chain a design office follows:

- Crest level** from EurOtop, set so the upper bound of the overtopping scatter meets the tolerable limit.
- Founding level** from the scour allowance, bracketed by minimum_embedment and maximum_embedment (deeper is a piling question, not a gravity-wall one).
- Goda pressures** on the wetted face, from the still water level down to the seabed.
- Earth pressure** from the backfill (backfill="medium_sand", water_table, surcharge, earth_pressure_driving="at_rest"), with the pore water apart.
- Base width** grown in step increments until sliding, overturning, and, with require_middle_third, the bearing check pass under both load cases: the wave crest pushing landward, and the trough with the saturated backfill pushing seaward against

only the water left in front (the drawdown case, `drawdown_level`).

6. **Toe stone** from the Van der Meer toe formula, iterated because the berm thickness changes the depth over the berm.

Full Goda uplift is applied under the base even when it is embedded, which is deliberately on the safe side. The result is a `SeawallDesign` with every level and dimension, the pressures and forces, both load cases, `governing_case`, the overtopping, `warnings`, `summary()`, and `quantities()` (concrete, backfill, toe rock, excavation per meter run).

10.4 Worked example

Output of `examples/seawall_section.py`

```
Design condition      Hm0 = 2.80 m, Tm-1,0 = 8.64 s
Water level           SWL +2.90 m CD, depth at wall 8.50 m

Crest level           +8.59 m CD (Rc = 5.69 m, Rc/Hm0 = 2.03)
Promenade level       +7.59 m CD
Founding level        -7.11 m CD (embedment 1.51 m, scour 1.51 m)
Wall height           15.70 m
Base                  15.05 m wide x 1.20 m thick, stem 1.00 m
Sizing                73 iterations on base width

Goda Hmax             5.04 m
Pressures             p1 = 45.8 kPa, p3 = 37.5 kPa, pu = 36.8 kPa
Wave force            517 kN/m at 8.08 m above the base
Weight, uplift        2788 kN/m, 277 kN/m

Backfill              Medium sand, phi' = 33 deg, retained 14.70 m
Water table           0.00 m below the surface, saturated to the top
Earth pressure        1643 kN/m at 5.00 m (K = 0.455, 557 soil + 1087 water)

Load case 1, wave crest pushing landward
  sliding             inf
  overturning         inf
Load case 2, trough with the backfill pushing seaward
  front water         253 kN/m at +1.50 m CD
  net seaward         1390 kN/m
  sliding             1.20
  overturning         2.60
Governing case        drawdown
Bearing               p_max = 245 kPa, e = -1.18 m

Toe protection         Dn50 = 0.27 m, M50 = 0.05 t, berm 2.0 m wide x 0.55 m thick
Overtopping           mean 0.333 l/s/m, upper bound 1 l/s/m
Governing limit       1 l/s/m, trained staff, well shod and protected, wide walkway

Concrete              32.56 m3/m (78.2 t/m)
Backfill              189.71 m3/m
Toe rock              1.82 t/m
Excavation            22.74 m3/m

Warnings
- Toe depth ratio ht/h = 0.94 is outside the 0.4 to 0.9 calibration range of the Van der Meer toe formula.
- Pore water is 66% of the pressure on the back of the wall. Drainage is a structural matter here, not a detail: a working drain would cut the total from 1643 to about 854 kN/m.
- The drawdown case governs, not the wave. The wall is sized by the saturated backfill pushing it seaward at the trough, so the backfill grading and the drainage detail matter more than the design wave.

Overtopping tolerability at the design discharge
no  0.03 l/s/m  pedestrians_unaware
no  0.1 l/s/m   pedestrians_aware
no  0.4 l/s/m   harbour_quay_equipment
ok  1 l/s/m     trained_staff
ok  1 l/s/m     buildings_structural
ok  10 l/s/m    vehicles_low_speed
ok  50 l/s/m    vehicles_moderate_speed
ok  50 l/s/m    embankment_seaward

Wrote media/seawall_section.png and media/seawall_section.dxf
Wrote media/seawall_sheet.png
```

Two warnings in this output are the point of the example. Pore water is 66% of the pressure on the back of the wall, and the drawdown case governs, not the wave: the wall is sized by the saturated backfill pushing it seaward at the trough, so the backfill grading and the drainage detail matter more than the design wave. The toe depth ratio is also just outside the calibration range

of the toe formula, and the output says so.

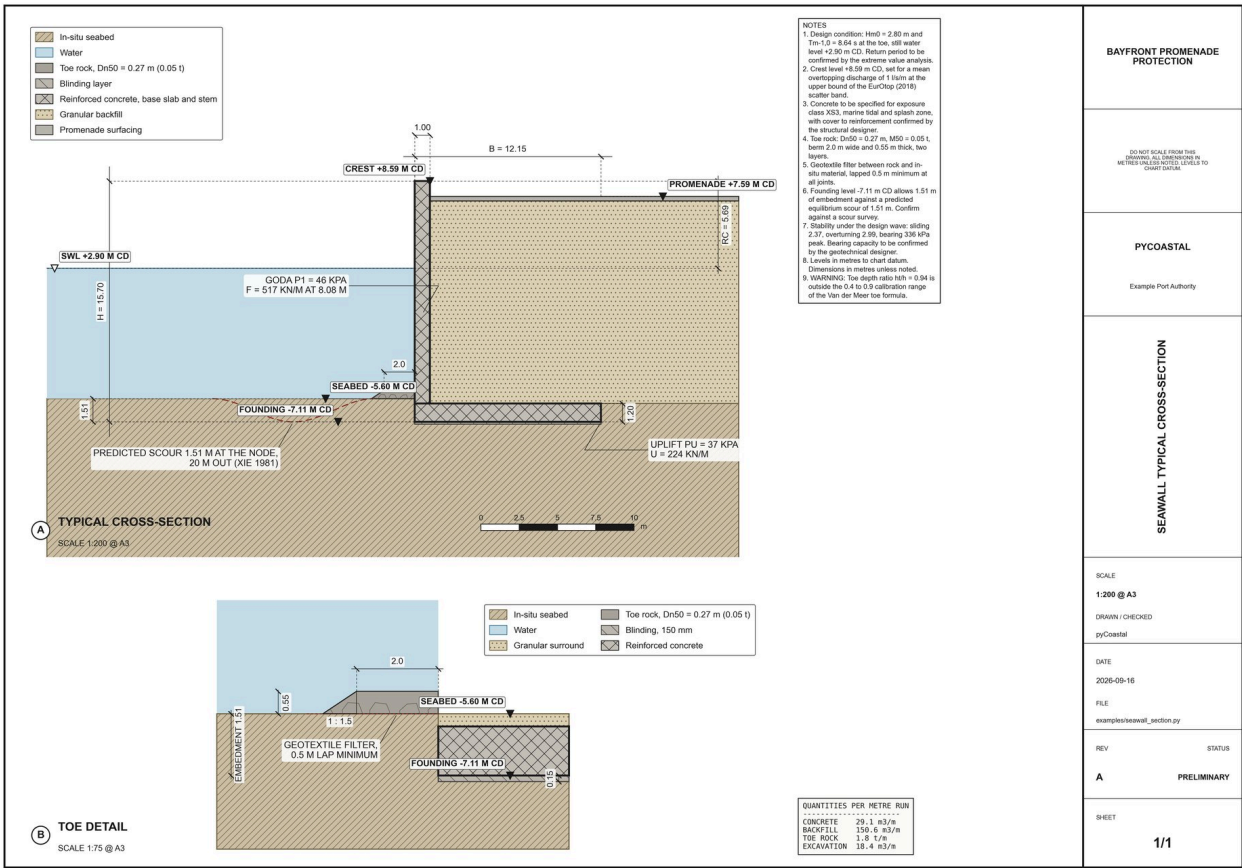


Figure 10.1: The seawall drawing sheet: typical section at a true stated scale, an enlarged toe detail, the design parameter block, and specification notes generated from the design.

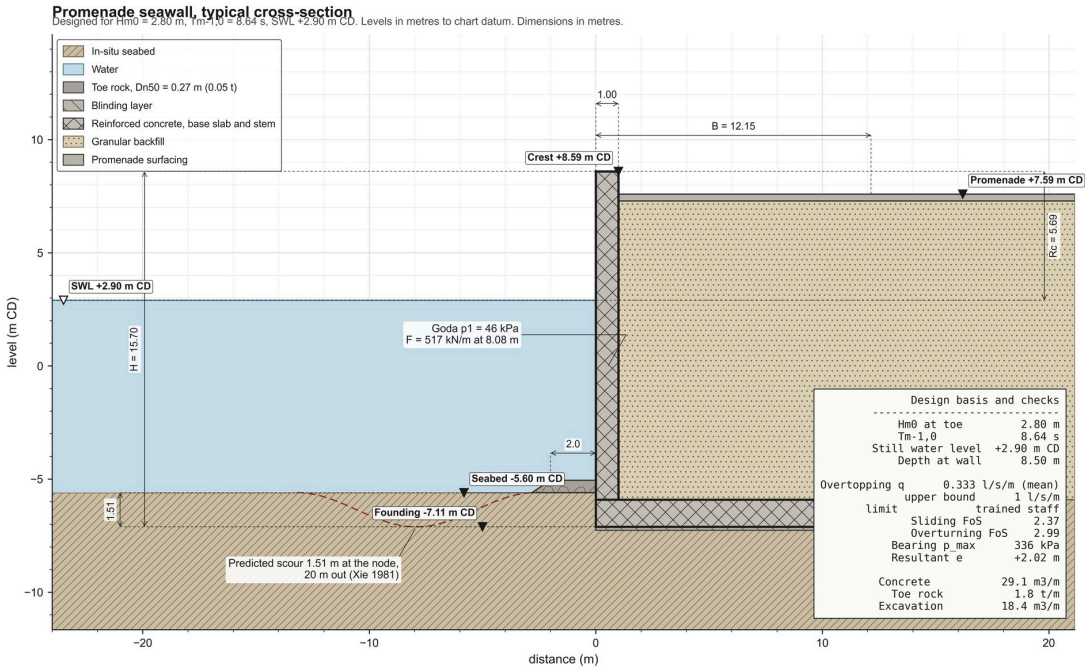


Figure 10.2: Standalone seawall section with plot axes, for a report.

11 Harbor agitation and port layout

Module: `pyCoastal.applications.port`. *Examples:* `examples/port_diffraction.py`, `examples/port_layout_comparison.py`.

Phase-resolved wave propagation into a harbor, reported as the wave energy that reaches the berths. Breakwaters are rasterized onto the grid as reflecting walls with a settable absorption, waves enter from a soft source line, and the open boundaries are damped by sponge layers so that outgoing and reflected energy leaves the domain instead of ringing around it.

11.1 The solver

The model integrates the second-order wave equation in flux form,

$$\frac{\partial^2 \eta}{\partial t^2} = \nabla \cdot (c^2 \nabla \eta), \quad (11.1)$$

with zero flux on every face touching a breakwater cell, by an explicit leapfrog step at a Courant number of 0.35. The celerity comes from the linear dispersion relation (Equation 6.1) at the local depth and the run period, so the wavelength is correct in intermediate water rather than the shallow-water $\sqrt{gh}$. The model is therefore accurate for the single frequency it is run at. It is not a spectral model and does not represent breaking, wave-current interaction, refraction over a varying bed, or nonlinear transfers.

Harbor performance is the disturbance coefficient

$$K_d = \frac{H_{\text{local}}}{H_{\text{reference}}},$$

with the reference height measured at a probe in open water outside the harbor, which makes the result independent of how the source is calibrated. Against a semi-infinite breakwater the solver gives a deep quiet shadow, K_d near 0.5 on the geometric shadow boundary, and Fresnel fringes in the illuminated field.

11.2 Geometry

`Breakwater(points, width=20.0, name="breakwater", absorption=0.0)` is a polyline of a given thickness; cells within `width/2` of the centerline become land. `absorption` is a damping collar in the water next to the structure, a model knob rather than the reflection coefficient itself. `measure_reflection(absorption, period, depth)` fires a normally incident wave at a full-width wall and infers $K_r = (H_{\text{max}} - H_{\text{min}})/(H_{\text{max}} + H_{\text{min}})$ from the standing-wave envelope (Table 11.1). The mapping is not monotonic: past about 0.75 the collar becomes an impedance step and reflects again. Use 0.3 to 0.4 for a rubble mound (K_r near 0.4) and 0 for a vertical caisson.

Table 11.1: Reflection coefficient against the absorption setting.

absorption	0.00	0.25	0.50	0.75	1.00
K_r	0.95	0.48	0.22	0.19	0.25

`PortLayout(Lx=1200, Ly=900, dx=4.0, depth=10.0, breakwaters=[])` holds the domain, grid spacing, constant depth, and structures. Five layouts are built in (`LAYOUTS`):

- `harbour_layout ("two_arm")`: two shore-normal arms with a gap entrance, and optionally a back wall (drop "east" from the sponge sides so the quay reflects);
- `offset_entrance_layout ("offset_entrance")`: overlapping arms making a dog-leg entrance, so waves must diffract twice;
- `hooked_breakwater_layout ("hooked")`: a main breakwater with a shore-parallel hook, so the entrance faces along the coast;
- `detached_breakwater_layout ("detached_screen")`: two arms with a detached screen off the gap;
- `marina_layout ("marina")`: an outer harbor protecting an inner basin through an offset second opening.

Wave direction. `IncidentWave(height=1.5, period=8.0, direction=0.0)` supports an oblique direction, but a phased source line only lights a parallelogram of the domain (`illuminated_mask`), and `simulate_port` warns when a structure falls outside it. The accurate way to study direction is to turn the harbor into the waves with `rotate_layout(layout, degrees)` and drive it shore-normal, since only the relative angle matters.

Absorption matters. Left fully reflecting, a detached screen forms a pocket with the arms that rings, and basin agitation goes up. For a 150 m gap with 280 m arms at $T = 9$ s, mean basin K_d went from 0.35 to 0.51 with a reflecting screen, and from 0.20 to 0.04 once the structures were armored.

11.3 Running a simulation

```
import numpy as np
from pyCoastal.applications.port import IncidentWave, harbour_layout, simulate_port

layout = harbour_layout(gap=130, arm_length=300, back_wall=True, absorption=0.35)
wave = IncidentWave(height=1.5, period=9.0, direction=np.deg2rad(0))

result = simulate_port(layout, wave, sponge_sides=("west", "north", "south"))

result.disturbance_coefficient      # Kd over the whole basin
result.probe((1120, 450))          # Kd at one point
result.berth_report({"quay": (1120, 450)})
result.operable_fraction({"quay": (1120, 450)}, limit=0.5)
```

`simulate_port(layout, wave, duration=None, cfl=0.35, sponge_thickness=None, sponge_strength=2.0, sponge_sides=..., source_x=None, ramp_periods=3.0, analysis_periods=6.0, n_snapshots=120, store_from=None, reference_point=None)` returns a `PortResult` with the coordinate axes, stored surface snapshots, the land mask, the analysis-window `wave_height`, and the `reference_height`. The default duration lets the wave cross the domain twice plus the ramp and analysis windows; the sponge defaults to 1.5 wavelengths.

11.4 Worked examples

`examples/port_diffraction.py` drives a 9 s swell at 25 degrees into a two-arm harbor (by rotating the layout), writes an animation of the instantaneous surface, and reports K_d at four berths:

Output of `examples/port_diffraction.py`

```
Wavelength      : 81.7 m (20 cells per wave)
Celerity        : 9.08 m/s
Entrance gap    : 130 m = 1.59 wavelengths
Approach angle  : 25 deg off shore-normal
Incident height : 1.50 m

    berth      Kd      Hs (m)
entrance      0.12      0.18
outer basin   0.33      0.49
north berth   0.55      0.82
south berth   0.13      0.20

Berths within a 0.5 m operational limit:
entrance      yes
outer basin   yes
north berth   no
south berth   yes

Wrote media/port_disturbance.png
Wrote media/port_diffraction.gif
```

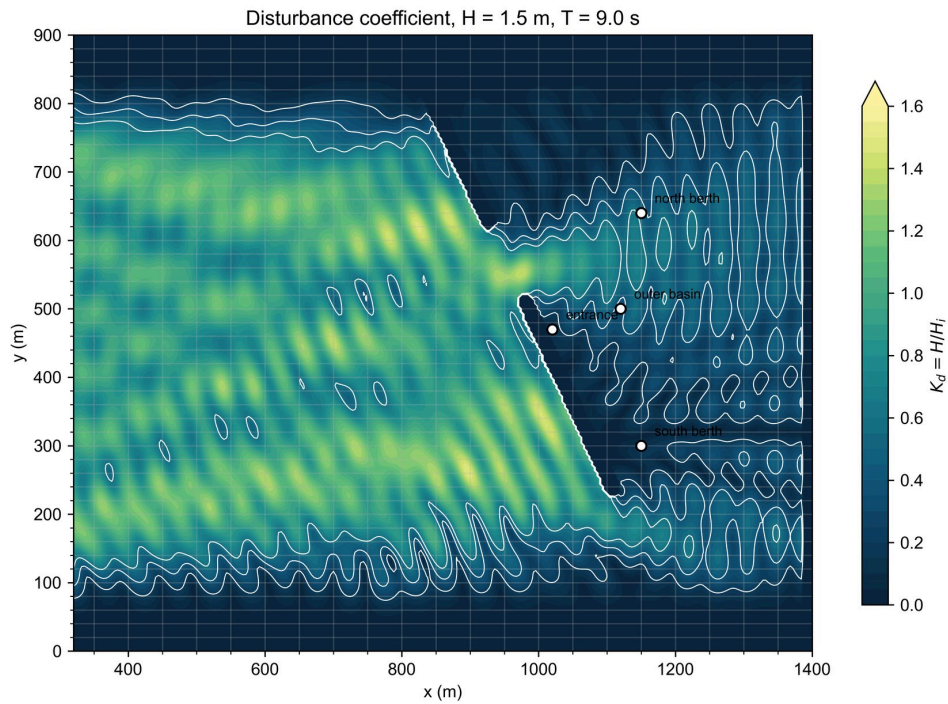


Figure 11.1: Disturbance coefficient in the harbor. The north berth sits in the diffracted field from the entrance and exceeds the 0.5 m limit.

`examples/port_layout_comparison.py` runs all five layouts under the same design wave and ranks them by basin agitation:

Output of `examples/port_layout_comparison.py`

```
ran two_arm
ran offset_entrance
ran hooked
ran detached_screen
ran marina
```

layout	mean K_d	95th pct K_d
Outer harbour and inner basin	0.007	0.022
Overlapping arms, dog-leg	0.008	0.029
Detached screen off the gap	0.039	0.067
Two arms, straight gap	0.146	0.370
Hooked main breakwater	0.180	0.768

Quietest basin: Outer harbour and inner basin (0.007 mean K_d)

Noisiest basin: Hooked main breakwater (0.180 mean K_d)

Wrote `media/port_layouts.png`

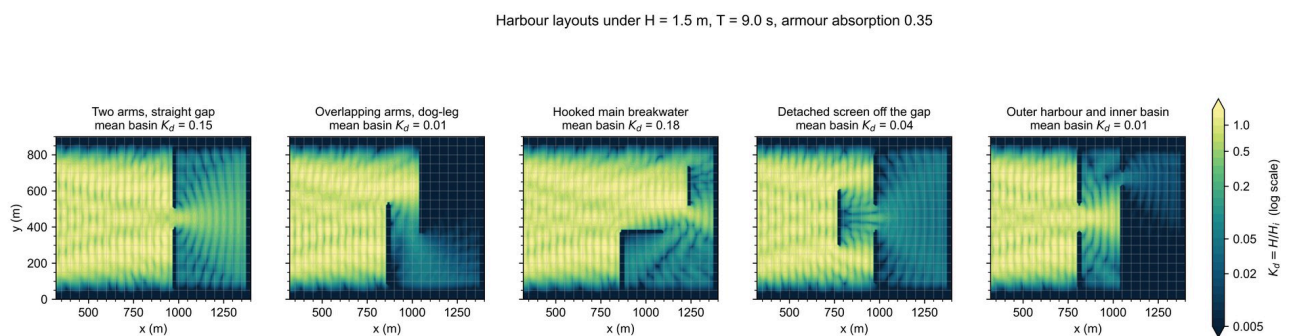


Figure 11.2: The five built-in layouts under the same design wave.

12 Wave loads on piles

Module: `pyCoastal.applications.piles`. Example: `examples/pile_wave_loads.py`. Browser: Monopile.

The load on a pile is not one number. It is a distribution up the pile that changes through the wave cycle, and the two things a designer needs from it, the base shear and the overturning moment at the mudline, peak at different phases. This module computes the distribution, integrates it, and sweeps the phase, rather than evaluating a formula at the crest.

12.1 The Morison equation

The Morison equation (Morison et al. 1950) splits the load per unit length on a slender member into a drag term in phase with the velocity and an inertia term in phase with the acceleration:

$$f(z) = \frac{1}{2}\rho C_d D u|u| + \rho C_m \frac{\pi D^2}{4} \frac{\partial u}{\partial t}. \quad (12.1)$$

Slender means D/L below about 0.2; above that the wave diffracts around the member and the functions warn rather than return a number.

Kinematics. `wave_kinematics(H, T, depth, z, phase=0.0, stretching="wheeler")` gives linear-theory velocity and acceleration. Linear theory says nothing about the water between the still water level and the crest, which is exactly where the load is largest, and extrapolating the cosh profile there overstates the velocity badly. Wheeler (1970) stretching maps the still-water profile onto the instantaneous water column and is the default; "none" and "extrapolate" are available for comparison.

Regime and coefficients. The Keulegan-Carpenter number $KC = u_{\max}T/D$ sets which term dominates: below about 3 the load is almost all inertia, above about 20 drag governs. `drag_inertia_coefficients(KC, rough=True)` returns post-critical values from DNV-RP-C205: $C_d = 1.05$ for a rough cylinder (any pile with marine growth) and 0.65 for a clean one, with C_m falling from the potential-flow value of 2.0 as KC rises. These are concept-design starting values.

Integration and sweep. `morison_load_profile` evaluates Equation 12.1 up the pile, `integrate_load(z, load, mudline)` integrates the force and the mudline moment, and `morison_pile_load(diameter, H, T, depth, phase=0.0, ...)` returns a `PileLoad` at one phase. `phase_sweep(diameter, H, T, depth, phases=181)` sweeps the cycle and reports where each peaks. `design_monopile(diameter, H, T, depth, rough=True, stretching="wheeler", phases=181, bed=None)` returns the worst-phase load, the scour, and the ratio `crest_underestimate` by which a crest-phase evaluation would understate the moment.

12.2 Scour at a pile

Sumer, Fredsoe and Christiansen (1992), for waves:

$$\frac{S}{D} = 1.3 \left[1 - e^{-0.03(KC-6)} \right] \quad (KC > 6), \quad (12.2)$$

with no scour below $KC = 6$, where the horseshoe vortex does not form. Under a steady current $S/D = 1.3$ with a standard deviation of 0.7, which is the `current_only` branch and the limit of the wave relation. `scour_depth_pile(diameter, KC, current_only=False, live_bed=True, bed=None, ...)` returns the equilibrium depth; pass a bed to gate it by mobility.

```
from pyCoastal.applications.piles import design_monopile

result = design_monopile(diameter=8.0, H=12.0, T=13.0, depth=30.0)
result["load"].force / 1e3      # kN, at the worst phase
result["load"].moment / 1e6     # MNm about the mudline
result["load"].inertia_fraction # share carried by the inertia term
result["crest_underestimate"]   # what a crest-phase check would miss
```

12.3 Worked example

Output of `examples/pile_wave_loads.py`

```
Pile      8.00 m diameter in 30.0 m of water
Wave      H = 12.00 m, T = 13.0 s
KC, regime 6.3, mixed
Cd, Cm    1.05, 1.87
```


D / L 0.041

Worst phase 56 degrees (crest at 0), surface at +3.36 m
Base shear 4,297 kN
Mudline moment 77.2 MNm
Effective arm 17.96 m above the mudline
Inertia share 90% of the load

Warnings

- The worst moment is 62% larger than the moment at the crest phase. Designing on the crest alone would have understated it.

At the crest phase 1,437 kN, 29.6 MNm
At the worst phase 4,297 kN, 77.2 MNm
Designing on the crest would miss 62% of the moment

Max base shear at 70 degrees
Max mudline moment at 56 degrees

Scour 0.10 m (0.01 D) at KC = 6.3
under a steady current: 10.40 m (1.3 D, standard deviation 5.60 m)
waves alone barely scour a pile this large, because KC is small; current is what governs.

How the load splits with pile diameter

D (m)	KC	regime	inertia share	moment (MNm)
1.0	50.7	drag dominated	5%	3.8
2.0	25.3	drag dominated	15%	8.1
4.0	12.7	mixed	50%	20.7
8.0	6.3	mixed	90%	77.2
12.0	4.2	mixed	96%	176.0

Wrote media/pile_wave_loads.png

A large monopile is inertia dominated, so the load follows the fluid acceleration, which peaks a quarter cycle before the crest. Evaluating the load under the crest, because that is where the water is highest, understates the overturning moment by 62% here. That is what the phase sweep exists to catch. The diameter table shows the transition from drag to inertia dominance as the pile grows, and the scour result shows that waves alone barely scour a pile this large because KC is small; current is what governs.

Monopile 8 m in 30 m: $H = 12$ m, $T = 13$ s, 4,297 kN and 77 MNm, 90% inertia

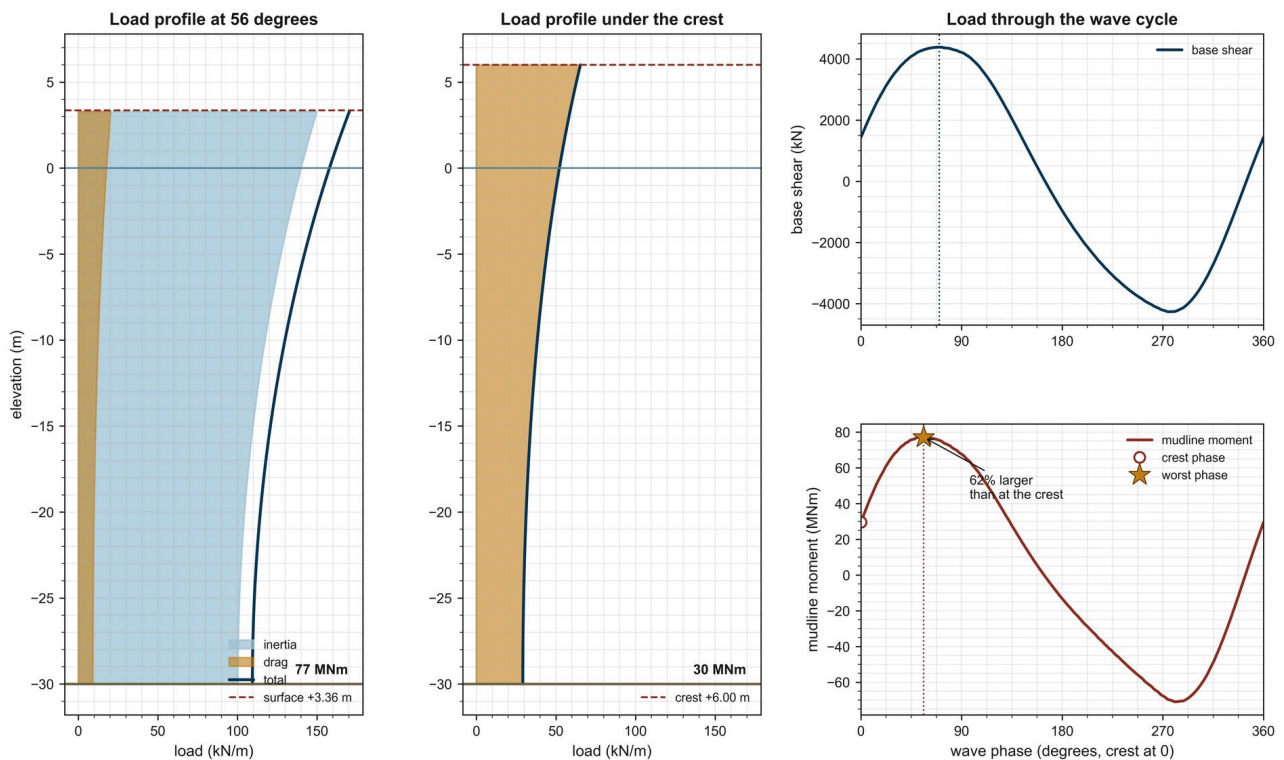


Figure 12.1: Morison loads on an 8 m monopile: load profiles at the crest and worst phases, and force and moment through the wave cycle.

13 Scour at piers and bridges

Module: `pyCoastal.applications.scour`. *Examples:* `examples/pier_scour.py`, `examples/bridge_scour.py`.

The hard part of estuarine scour is not any single relation. It is that the drivers do not peak together and do not even point the same way. The tide reverses twice a day, the river does not, the waves come and go with the wind, and the water depth changes under all of it. A calculation done at “the design current” and “the design wave” without asking whether they can occur at the same moment is either optimistic or absurd, and it is usually not obvious which. So this module works a tidal cycle rather than a load case: it evaluates the scour at every phase and reports the envelope, which phase governs, and by how much.

13.1 Pier scour in combined waves and current

Geometry. A pier is a stem (`Pier(diameter, shape="circular", length=None)`) on a base (`PierBase(width, length, height, top_level, skew)`), because that is what piers are, and the base is where the interesting failure lives. A buried pile cap does nothing until the scour hole reaches it; then it is exposed, it is wider than the stem, and the scour deepens because of it, which exposes more of it. The obstacle is reduced to one width by weighting each element over the depth of flow it occupies:

$$D_e = \frac{D_{\text{base}} h_{\text{base}} + D_{\text{stem}} (h - h_{\text{base}})}{h}, \quad (13.1)$$

where h_{base} is how much of the base stands proud of the *scoured* bed. Each element carries the HEC-18 shape factor K_1 (`SHAPE_FACTOR`) and the alignment factor

$$K_2 = \left(\cos\theta + \frac{L}{a} \sin\theta \right)^{0.65}, \quad L/a \leq 12. \quad (13.2)$$

A base set square to the ebb is skewed to the flood (the flood skew is taken as the supplement), which is what makes a tidal estuary awkward.

Combined waves and current. Sumer and Fredsoe (2001):

$$\frac{S}{D} = 1.3 \{ 1 - \exp[-A(KC - B)] \}, \quad A = 0.03 + \frac{3}{4} U_{cw}^{2.6}, \quad B = 6 \exp(-4.7 U_{cw}), \quad (13.3)$$

with $KC = U_m T / D$ on the *near-bed* orbital velocity and $U_{cw} = U_c / (U_c + U_m)$. At $U_{cw} = 0$ it is exactly the waves-only relation (Equation 12.2); at $U_{cw} = 1$ the threshold vanishes and $S/D \rightarrow 1.3$ as KC grows.

The steady-current floor. KC is built on the wave orbital velocity, so a strong current under small waves has a high U_{cw} and a low KC at the same time, and the bare formula returns almost no scour: a pier in a 1.5 m/s current would scour less than in still water with a ripple on it. A steady current is the $KC \rightarrow \infty$ limit of an oscillatory flow, not the $KC \rightarrow 0$ one, and a current that moves the bed digs its own horseshoe vortex. So when the approach current alone is live-bed, the ratio is floored at the steady-current value. This is an addition to the published relation and is flagged as `current_governs` (`scour_ratio_combined(KC, Ucw, current_live_bed=...)`; `None` disables it).

Depth limitation and time scale. `depth_limitation(depth, diameter)` applies $\tanh(h/D)$ (Breusers et al. 1977), worth nothing above $h/D = 3$ and a quarter below $h/D = 1$. The skin-friction Shields parameter under the current uses $u_* = U\kappa / \ln(11h/k_s)$ with $k_s = 2.5d_{50}$, and the time scale (Sumer, Christiansen and Fredsoe 1992) is

$$T^* = \frac{1}{2000} \frac{h}{D} \theta^{-2.2}, \quad T = T^* \frac{D^2}{\sqrt{g(s-1)d_{50}^3}}, \quad S(t) = S_{\text{eq}} [1 - e^{-t/T}]. \quad (13.4)$$

For a pier a couple of meters across T is typically most of a day, while the tide reverses every six hours.

Feedback. `equilibrium_scour(pier, base, depth, current, Um, period, ...)` solves the coupling between scour and effective diameter for its *deepest* self-consistent root by bisection, because a scour hole does not refill and the shallow root is not an outcome the pier can stay at.

Protection. `riprap_size(velocity)` follows HEC-23 guideline 12, $d_{50} = 0.692(KV)^2 / [2g(s-1)]$ with $K = 1.5$ for round-nosed and 1.7 for rectangular piers, and `scour_protection(pier, base, velocity, scour)` sizes a falling apron reaching two obstacle widths from the face, at least three stones thick, with a launch allowance: an apron relocates the scour to its edge rather than removing it.

The tidal cycle. EstuaryConditions(mean_depth, tidal_amplitude, tidal_current, river_current, Hs, Tp, tidal_period=M2, current_phase=90, bed="medium_sand", wave_follows_tide=False) uses positive for ebb: the river adds to the tide on the ebb and opposes it on the flood, which is why estuarine scour is usually an ebb problem. current_phase sets the lead of the current over the elevation (90 degrees for a standing tide, 0 for a progressive one). design_pier_scour(pier, conditions, base, samples=73, protection_factor=1.0) evaluates every phase and returns a PierScourDesign with the equilibrium scour (for protection and the long-term foundation check), the tidal_limited scour one half cycle can cut, base_exposed, undermined, the envelope, and the protection.

```
from pyCoastal.applications.scour import (
    Pier, PierBase, EstuaryConditions, design_pier_scour,
)

pier = Pier(diameter=2.5, shape="circular")
base = PierBase(width=7.0, length=12.0, height=2.5, top_level=-1.5, skew=20.0)
estuary = EstuaryConditions(mean_depth=9.0, tidal_amplitude=2.2,
                             tidal_current=1.1, river_current=0.4,
                             Hs=1.2, Tp=5.5, bed="medium_sand")
design = design_pier_scour(pier, estuary, base)
design.equilibrium, design.tidal_limited, design.undermined, design.protection
```

13.1.1 Worked example: pier scour

Output of examples/pier_scour.py

Pier 2.5 m on a 7 x 12 m base, top -1.5 m to the bed
Peak ebb 1.50 m/s, peak flood 0.70 m/s

Governing phase	145 deg (ebb)
current	0.78 m/s (peak is 1.50)
depth	7.20 m
effective diameter	5.15 m (stem is 2.50)
equilibrium scour	5.93 m
one tidal half	1.76 m (time scale 17.7 h)
base	UNDERMINED

- Governing phase 145 deg, on the ebb: 0.78 m/s current and 0.47 m/s near-bed orbital velocity in 7.20 m of water.
- $U_{cw} = 0.62$, so the waves and current both matter. $KC = 0.5$ against a threshold of 0.32.
- Equilibrium scour 5.93 m, 1.30 times the effective diameter of 5.15 m.
- Shallow water takes 12% off the unlimited depth: $h/D = 1.4$.
- The hole exposes the base completely (2.5 m) and reaches its underside. The foundation is undermined, not merely exposed, and the bearing check has to be redone on the scoured section.
- Flow misalignment multiplies the base width by 1.32 at 20 deg of skew. A base set square to one direction of a reversing tide is skewed to the other.
- Scour time scale 17.7 hours against a 6.2 hour tidal half cycle, so one half cycle reaches 1.76 m, 30% of equilibrium. The hole does not refill on the reverse flow, so successive tides work it deeper towards the equilibrium. Size protection on the equilibrium either way.
- Sumer and Fredsoe's relations carry a standard deviation near 0.7 diameters in steady current. Treat the number as a mean, not a bound, and design protection with margin on it.

Wrote media/pier_scour.png, media/pier_scour_sheet.png and media/pier_scour_tide.png

13.2 Total bridge scour

Local scour at a pier is the component everyone computes, and on a contracted crossing it is routinely the smallest of the three. HEC-18 splits total scour into contraction, local, and abutment scour; all three are here and all three are worked over the tidal cycle.

The opening. `BridgeOpening(approach_width, opening_width, pier_blockage=0, abutment_length=25, abutment_shape="spill_through", abutment_skew=90, slope=5e-4, flow_fraction=1.0, abutment_depth_fraction=0.4)` gives the contraction ratio W_1/W_2 with the piers subtracted from the gross opening. `flow_fraction` is Q_2/Q_1 , best computed with `river.flow_distribution` (Section 14). `abutment_depth_fraction` sets Froehlich's y_a , the depth where the embankment actually stands; feeding it a mid-channel depth in a deep estuary returns a hole deeper than the water.

Contraction scour. Two modes, switched by the critical velocity $V_c = 6.19 y^{1/6} D_{50}^{1/3}$ (HEC-18 eq. 6.1). Live bed, Laursen (1960):

$$\frac{y_2}{y_1} = \left(\frac{Q_2}{Q_1} \right)^{6/7} \left(\frac{W_1}{W_2} \right)^{k_1}, \quad (13.5)$$

with $k_1 = 0.59, 0.64$, or 0.69 as V^*/w is below 0.5 , between 0.5 and 2 , or above 2 (`transport_exponent`). Clear water (HEC-18 eq. 6.4):

$$y_2 = \left[\frac{0.025 Q_2^2}{D_m^{2/3} W_2^2} \right]^{3/7}. \quad (13.6)$$

The live-bed relation is a sediment balance: faster water carries more but delivers more, and the two cancel, leaving the width ratio. The two relations were fitted separately and do not meet at the threshold; `contraction_scour(..., regime="live" | "clear")` computes either on demand so the step can be seen.

Abutment scour. Froehlich (1989) for $L'/y < 25$ and HIRE for longer abutments:

$$\frac{y_s}{y_a} = 2.27 K_1 K_2 \left(\frac{L'}{y_a} \right)^{0.43} Fr^{0.61} + 1, \quad \frac{y_s}{y_a} = 4 Fr^{0.33} \frac{K_1}{0.55} K_2, \quad (13.7)$$

with K_1 from `ABUTMENT_SHAPE` (1.0 vertical, 0.82 wing wall, 0.55 spill through). The $+1$ in Froehlich is a factor of safety HEC-18 added, not physics, and it is often more than half the answer; it is reported as `safety_margin`. These are the fourth-edition HEC-18 relations, the edition the pier scour also comes from.

Chained, not parallel. `bridge_scour_state` computes the components in the order the water meets them: contraction from the approach flow, then local scour from the flow *in the contracted opening*, which is faster and deeper, and abutment scour from the approach flow it turns. They are added only where they coexist: contraction plus pier at a pier, contraction plus abutment at an abutment. `design_bridge_scour(opening, conditions, pier, base)` envelopes each component over the cycle and returns a `BridgeScourDesign` with `contraction`, `pier_local`, `abutment`, `total_at_pier`, `total_at_abutment`, `component(name)`, and `notes`. When a total exceeds the water depth the result says so: that is a signal the relations have been pushed past their fitted range, not a foundation level.

```
from pyCoastal.applications.scour import BridgeOpening, design_bridge_scour

opening = BridgeOpening(approach_width=140, opening_width=80, pier_blockage=5,
                        abutment_length=22, abutment_shape="spill_through")
design = design_bridge_scour(opening, estuary, pier, base)
design.total_at_pier, design.total_at_abutment, design.component("pier")
```

13.2.1 Worked example: bridge scour

Output of `examples/bridge_scour.py`

```
Waterway 140 m, opening 80 m gross less 5 m of piers
Contraction 1.87 to 1

Governing phase 35 deg (ebb)
  approach          1.24 m/s in 10.80 m
  through the opening 1.51 m/s in 16.62 m

  contraction      6.03 m
```


pier local 5.93 m
 abutment 8.37 m

 at a pier 10.72 m
 at abutment 14.09 m <- governs

- Opening 80 m gross less 5.0 m of piers in a 140 m waterway: a 1.87 to 1 contraction.
- Governing phase 35 deg on the ebb: 1.24 m/s approaching in 10.80 m, becoming 1.51 m/s in 16.62 m through the opening.
- Live-bed contraction scour: the approach at 1.24 m/s is 1.9 times the 0.67 m/s threshold, so the opening is fed from upstream and scours only until it can pass what arrives. Note that this mode does not depend on how fast the water goes: Laursen's relation is a sediment balance, and above the threshold it turns on the width ratio alone.
- Components: contraction 6.03 m, pier local 5.93 m, abutment 8.37 m. Each is the worst over the whole cycle; they do not all peak at the same phase.
- Total 10.72 m at a pier and 14.09 m at an abutment. The abutment governs. These are not added together: they are depths at two different places, and combining them would invent a hole that exists nowhere.
- Abutment scour by Froehlich, of which 4.32 m (52%) is the +1 flow depth HEC-18 adds as a factor of safety rather than as physics. Worth knowing before it is treated as a measurement.
- Local scour is computed on the flow in the contracted opening, not on the approach, since the contraction has already accelerated and deepened it. Taking approach values here is the usual way a pier gets understated.
- A total of 14.09 m exceeds the 10.80 m of water at the governing phase. That is a signal the relations have been pushed past where they were fitted, not a foundation level. Check the abutment depth and the contraction ratio, and get a physical model or a morphodynamic run before committing to it.
- Long-term degradation or aggradation of the reach is not included and is a separate study. HEC-18 adds it to these; on a river with a moving bed it can exceed all three.

Wrote media/bridge_scour.png, media/bridge_scour_sheet.png and media/bridge_scour_components.png

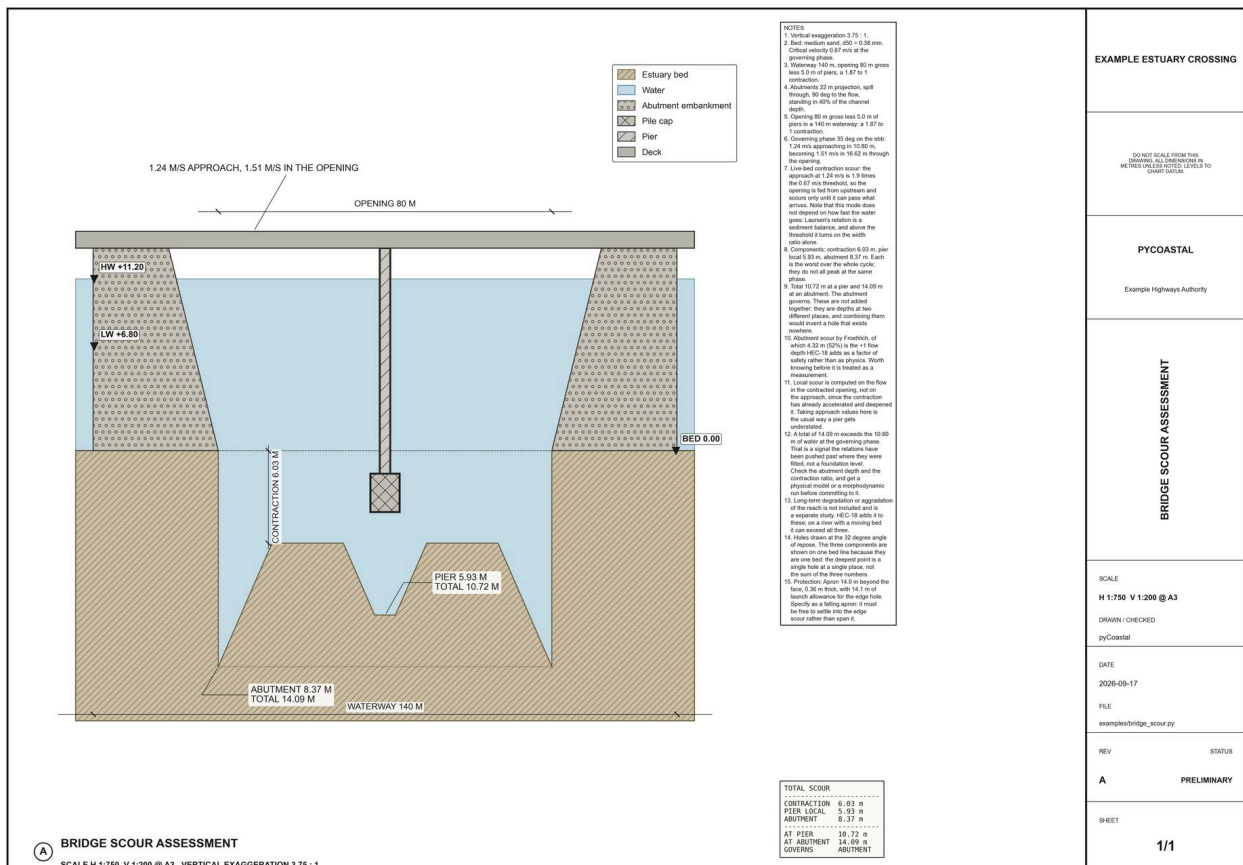


Figure 13.3: Bridge scour assessment sheet. The three components are drawn on one bed line so that the deepest point reads as one hole at one place, not the sum of three numbers.

Total scour at a crossing: three components, three timings, and two contraction regimes that disagree at the threshold between them

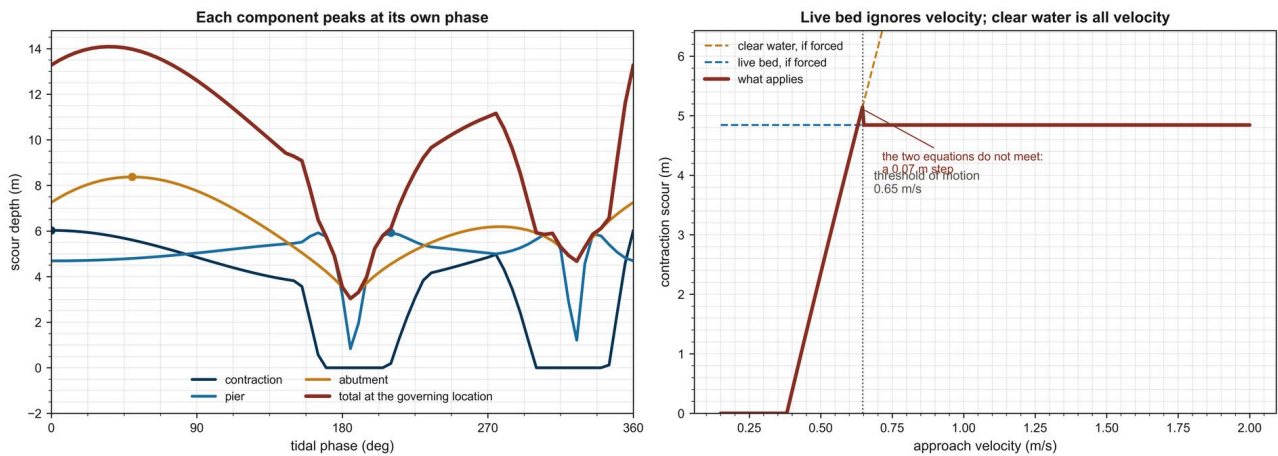


Figure 13.4: Scour components through the tidal cycle. They peak at different phases, so each has its own envelope.

13.3 What is not modeled

A pile cap at bed level can act as a collar and reduce scour; crediting it would be unconservative, so it is not credited. Cohesive beds are refused rather than guessed at. Long-term degradation of the reach and channel migration are separate studies that HEC-18 adds to these totals.

14 Open-channel hydraulics and backwater

Module: `pyCoastal.applications.river`. Example: `examples/backwater.py`.

The foundation the fluvial side of a crossing stands on. Everything a bridge does to a river it does by changing the depth: it squeezes the flow, the water backs up, and that backwater reaches upstream for a distance nobody guesses correctly by eye. The module also exists for one number in the scour chain: the fraction of the discharge that actually goes through a bridge opening, which is a conveyance calculation, not a guess.

14.1 The channel

`Channel(width=30.0, side_slope=2.0, roughness="natural_clean")` is a trapezoidal prismatic section (zero width is triangular, zero side slope rectangular). Roughness is a Manning n or a key into `MANNING`: 0.013 and 0.017 for smooth and rough concrete, 0.022 and 0.030 for clean and weedy earth, 0.028 gravel, 0.035 cobbles, 0.030 and 0.045 for clean and weedy natural channels, and 0.035, 0.07, 0.10 for pasture, brush, and wooded floodplains. The section provides area, perimeter, top_width, hydraulic_radius, velocity, and the conveyance

$$K = \frac{AR^{2/3}}{n}, \quad Q = K\sqrt{S}. \quad (14.1)$$

14.2 Two depths

Normal depth is where friction balances the bed slope, $Q = K(y_n)\sqrt{S_0}$ (normal_depth, which raises on a horizontal or adverse slope, where uniform flow does not exist). **Critical depth** is where the specific energy $E = y + V^2/2g$ is least and the Froude number is one, $Q^2T/(gA^3) = 1$ (critical_depth), with $Fr = V/\sqrt{gA/T}$ (froude_number), the form that is right for any section. `classify_slope(channel, discharge, slope)` compares them: mild, steep, critical, horizontal, or adverse. The classification belongs to the channel and the discharge together, so the same reach can be mild in flood and steep at low flow. `profile_type(depth, classification)` names the profile in Chow's notation: M1 behind a bridge on a mild reach, M2 a drawdown to a free overfall, S1 behind a structure on a steep reach.

14.3 Gradually varied flow

`gvf_profile(channel, discharge, slope, control_depth, steps=200, approach=0.99)` integrates the direct step method,

$$\Delta x = \frac{\Delta E}{S_0 - S_f}, \quad S_f = \left(\frac{Qn}{AR^{2/3}} \right)^2, \quad (14.2)$$

away from the control. The depths are chosen and the distances computed, which is exact for a prismatic channel and never iterates. The integration stops as the depth approaches normal depth, because it gets there only asymptotically: a backwater curve has no end, and the extent of backwater is always a convention. `approach` sets the one used (0.99 of the way to normal depth by default; 0.95 gives a noticeably shorter and equally defensible reach), and the notes of the returned `BackwaterResult` say which. The result carries distance, depth, water_surface, bed_level, profile, reach, and depth_at(distance).

14.4 Afflux at bridge piers

Yarnell (1934):

$$\Delta H = K(K + 5Fr^2 - 0.6)(a + 15a^4)Fr^2 y, \quad (14.3)$$

with a the fraction of the area blocked by the piers, Fr the downstream Froude number, and K the pier shape coefficient (`PIER_SHAPE`: 0.9 semicircular nose to 2.5 for a ten-pile trestle). The fourth power on the blockage is what makes it bite. Yarnell's tests were in a rectangular flume at blockages up to about 0.4 with the flow class unchanged; where a bridge chokes the flow to critical, use the HEC-RAS energy or momentum methods. `yarnell_afflux(channel, discharge, downstream_depth, blockage, shape="semicircular_nose")` returns the afflux and the upstream depth.

14.5 Flow distribution

`flow_distribution(main, main_depth, slope, floodplains=[(Channel, depth), ...])` splits the discharge by conveyance,

$$\frac{Q_i}{Q} = \frac{K_i}{\sum_j K_j} \quad (14.4)$$

This is the number `scour.BridgeOpening` calls `flow_fraction`. A wooded floodplain can be half the width of the section and carry a tenth of the flow; guessing the fraction is the largest assumption in a contraction scour calculation, and it hides behind the most precise-looking number in the output.

```
from pyCoastal.applications.river import (
    Channel, classify_slope, flow_distribution, gvf_profile, yarnell_afflux,
)

channel = Channel(width=30.0, side_slope=2.0, roughness="natural_clean")
plain = Channel(width=120.0, side_slope=3.0, roughness="floodplain_trees")

state = classify_slope(channel, 250.0, 0.0008) # normal, critical, mild
afflux = yarnell_afflux(channel, 250.0, state["normal"], blockage=0.18)
profile = gvf_profile(channel, 250.0, 0.0008, afflux["upstream_depth"])
profile.profile, profile.reach, profile.depth_at(1500.0)

split = flow_distribution(channel, state["normal"], 0.0008, [(plain, 1.2)])
split["main_fraction"]
```

14.6 Worked example

Output of `examples/backwater.py`

```
250 m3/s on a 0.0008 bed, natural_clean
normal depth 3.54 m
critical depth 1.84 m
reach is mild, so a bridge backs water up upstream

Piers blocking 18% of the section, Froude 0.35
afflux 0.071 m, so the control sits at 3.61 m
M1 profile reaching 5.14 km
- M1 profile on a mild reach. Control at 3.61 m against normal 3.54 m and critical 1.84 m.
- Backwater reaches 5.14 km to get 99% of the way to normal depth. The curve is asymptotic, so that figure is a convention
and moves a long way with it: quoting an extent without quoting the criterion is meaningless.

In flood, 1.2 m over a wooded floodplain:
the channel is 26% of the width
and carries 84% of the flow
contraction scour 1.08 m with the computed split, 1.81 m if it is assumed to be all of it

Wrote media/backwater.png
```

Backwater at a bridge: the profile is physics, the extent quoted for it is a convention

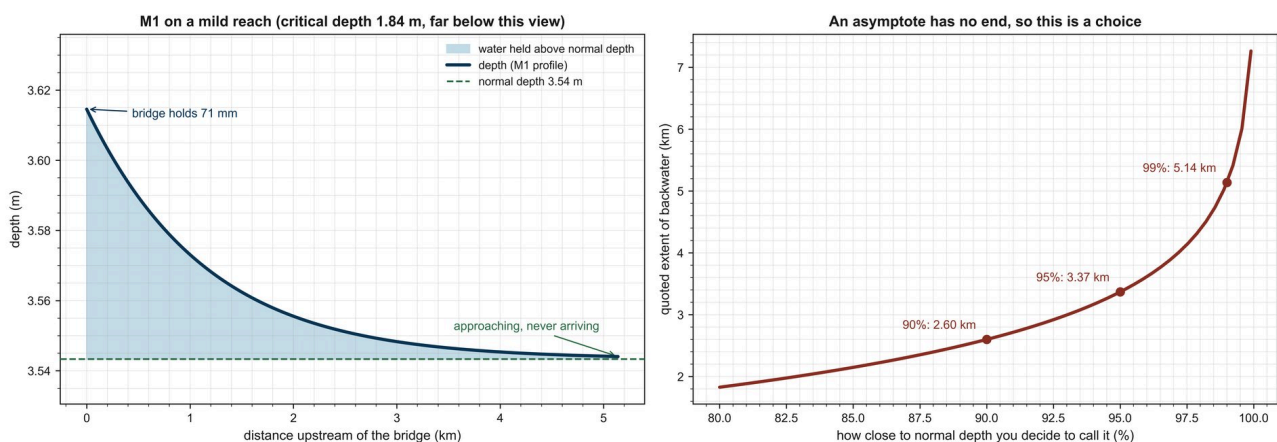


Figure 14.1: The backwater profile and the extent convention. Depth, not level, is plotted, because the bed rises 4 m over the reach while the depth changes by 7 cm.

In the example the channel is 26% of the section and carries 84% of the flow, because conveyance goes as the roughness and the hydraulic radius, not the area. Feeding the computed fraction into `contraction_scour` gives 1.08 m, where assuming all the flow goes through the opening gives 1.81 m.

15 Navigation channels

Module: pyCoastal.applications.channel. *Example:* examples/navigation_channel.py. *Browser:* Channel.

How deep and how wide does the approach channel have to be? The answer is a stack of allowances, each small, each defensible, and together often several meters. This module builds that stack explicitly so it can be argued over line by line, which is how a dredging budget actually gets agreed. It follows the concept-design method of PIANC (2014), Report 121. The width and lane tables are exposed as editable dictionaries rather than buried, because the governing edition of the guideline, and the pilots on the day, decide the numbers on a real scheme; a real channel is confirmed by simulation.

15.1 The vessel

`Vessel(name, length, beam, draught, block_coefficient=0.75, length_pp=None)` is the design ship. L_{pp} defaults to $0.96L_{oa}$. `displaced_volume` is $C_b L_{pp} B T$ and displacement is in tonnes at 1025 kg/m³. The same class is used by the berthing module.

15.2 The depth chain

Start at the design water level, subtract the static draught, then in turn the squat as the ship moves, its vertical response to waves, the net clearance over the bed, and the tolerances the dredger and the survey cannot beat. What is left is the dredge level.

Squat. ICORELS (1980), as given by PIANC:

$$S = C_s \frac{\nabla}{L_{pp}^2} \frac{F_{nh}^2}{\sqrt{1 - F_{nh}^2}}, \quad F_{nh} = \frac{V}{\sqrt{gh}}, \quad (15.1)$$

with $C_s = 2.4$ for an open or wide channel. It blows up as $F_{nh} \rightarrow 1$ and is normally kept to $F_{nh} < 0.7$; the function flags speeds where it is unusable. Barrass (1979) is the back-of-envelope check, $S_{\max} = C_b V_k^2 / 100$ in open water and /50 in a confined channel, with the speed in knots (`squat_icorels`, `squat_barrass`).

Wave response. `wave_response_allowance(Hs, factor=0.5, period=None, vessel=None)` takes a fraction of H_s as keel motion: about 0.3 for a long vessel in short head seas up to about 0.7 for a short vessel in long beam seas. With any swell this is the single largest judgement in the depth chain. Given the period and vessel, it also reports the ratio of wavelength to ship length, which governs whether the ship contours or bridges the wave.

Clearance stack. `underkeel_clearance(vessel, squat, wave_allowance, net_clearance=0.6, water_level_allowance=0.0, dredging_tolerance=0.3, survey_tolerance=0.2, siltation_allowance=0.2, density_allowance=0.0)` returns each allowance, the gross total below the keel, and the required depth. Leaving out the dredging and survey tolerances is the most common way a channel ends up shallower than its drawing.

Because the squat depends on the depth and the depth depends on the squat, `design_channel` solves the depth by a short fixed-point iteration.

15.3 The width chain

`channel_width(vessel, manoeuvrability="moderate", section="outer", two_way=False, speed_class="moderate", bank="sloping_channel_edges", conditions=None)` builds the PIANC width: a basic maneuvering lane (MANOEUVRING_LANE: 1.3, 1.5, 1.8 beams for good, moderate, poor maneuverability), plus additional widths from WIDTH_COMPONENTS for vessel speed, crosswind, crosscurrent, longitudinal current, waves, aids to navigation, bottom surface, depth of waterway, and cargo hazard (with separate outer and inner-channel columns), plus a bank clearance each side (BANK_CLEARANCE), plus a passing distance for two-way traffic (PASSING_DISTANCE). **Components you do not specify are taken at their most benign class and reported as assumed**, because a silent default is how a channel ends up too narrow on paper.

15.4 Side slopes, volumes, and turning basins

`dredged_side_slope`(bed, factor=2.0) gives $\cot\beta = f/\tan\phi'$; the factor of 2 covers both the partial factor on a submerged granular slope and the overcut and slumping of dredging, landing near 1:2.6 in gravel and 1:3.8 in silt. `turning_basin_diameter`(vessel, assisted=True, current=False) gives 1.5 L_{oa} with tugs, 2.0 unassisted, plus half a length where a current runs through.

`design_channel`(vessel, speed, design_water_level, Hs=0, Tp=None, wave_factor=0.5, net_clearance=0.6, ..., bed="medium_sand", existing_bed=None, **width_kwargs) returns a ChannelDesign with the dredge_level, required_depth, the clearance stack, squat and wave details, width, top_width, side_slope, dredge_volume(length), warnings, and summary().

```
from pyCoastal.applications.channel import Vessel, design_channel

vessel = Vessel(name="Post-Panamax container ship", length=336, beam=48.2,
                draught=14.5, block_coefficient=0.68)
channel = design_channel(vessel, speed=8.0, design_water_level=1.20,
                        Hs=1.8, Tp=9.0, existing_bed=-11.5, two_way=True,
                        conditions={"crosswind": "moderate", "waves": "moderate"})
channel.dredge_level, channel.width, channel.dredge_volume(1000)
```

15.5 Worked example

Output of examples/navigation_channel.py

```
Design vessel      Post-Panamax container ship
                  336 x 48.2 x 14.5 m, Cb = 0.68, 157,130 t
Speed              8.0 knots (FnH = 0.31)

Depth chain (m below the design water level)
  static draught      14.50
  squat                0.37
  wave response        0.90
  density              0.00
  net clearance        0.80
  water level uncertainty 0.30
  dredging tolerance   0.30
  survey tolerance     0.20
  siltation            0.20
  required depth       17.57

Design water level +1.20 m CD
Dredge level       -16.37 m CD

Channel width      415 m (8.6 beams, 2-way)
  basic manoeuvring lane 72.3
  crosswind (moderate)  14.5
  crosscurrent (low)     9.6
  waves (moderate)       24.1
  aids_to_navigation (good) 9.6
  bottom_surface (smooth_and_soft) 4.8
  depth_of_waterway (shallow) 9.6
  bank clearance, each side 24.1
  passing distance       77.1

Existing bed       -11.50 m CD
Dredge depth       4.87 m
Top width          463 m at 1:5 side slopes
Volume             2,135 thousand m3 per km

Turning basin      672 m (2.0 x Loa, assisted, current running)

Squat, two ways
  6.0 kn  ICORELS 0.20 m  Barrass 0.24 m  FnH 0.24
  8.0 kn  ICORELS 0.37 m  Barrass 0.44 m  FnH 0.31
  10.0 kn ICORELS 0.59 m  Barrass 0.68 m  FnH 0.39
  12.0 kn ICORELS 0.89 m  Barrass 0.98 m  FnH 0.47

What moves the dredge level
  wave factor 0.3 rather than 0.5  -0.35 m  -162 k m3/km
  wave factor 0.7 rather than 0.5  +0.35 m  +164 k m3/km
  speed 12 kn rather than 8 kn      +0.49 m  +229 k m3/km
  net clearance 1.5 m rather than 0.8 m +0.69 m  +320 k m3/km
  no siltation allowance            -0.20 m  -90 k m3/km
```

Wrote media/navigation_channel.png, media/navigation_channel_sheet.png and media/navigation_channel.dxf
Wrote media/channel_depth_chain.png

The sensitivity table at the end is the lesson: one judgement call, the fraction of the wave height taken as vertical vessel motion, moves the dredge level by 0.35 m either way, more than the dredging and survey tolerances together.

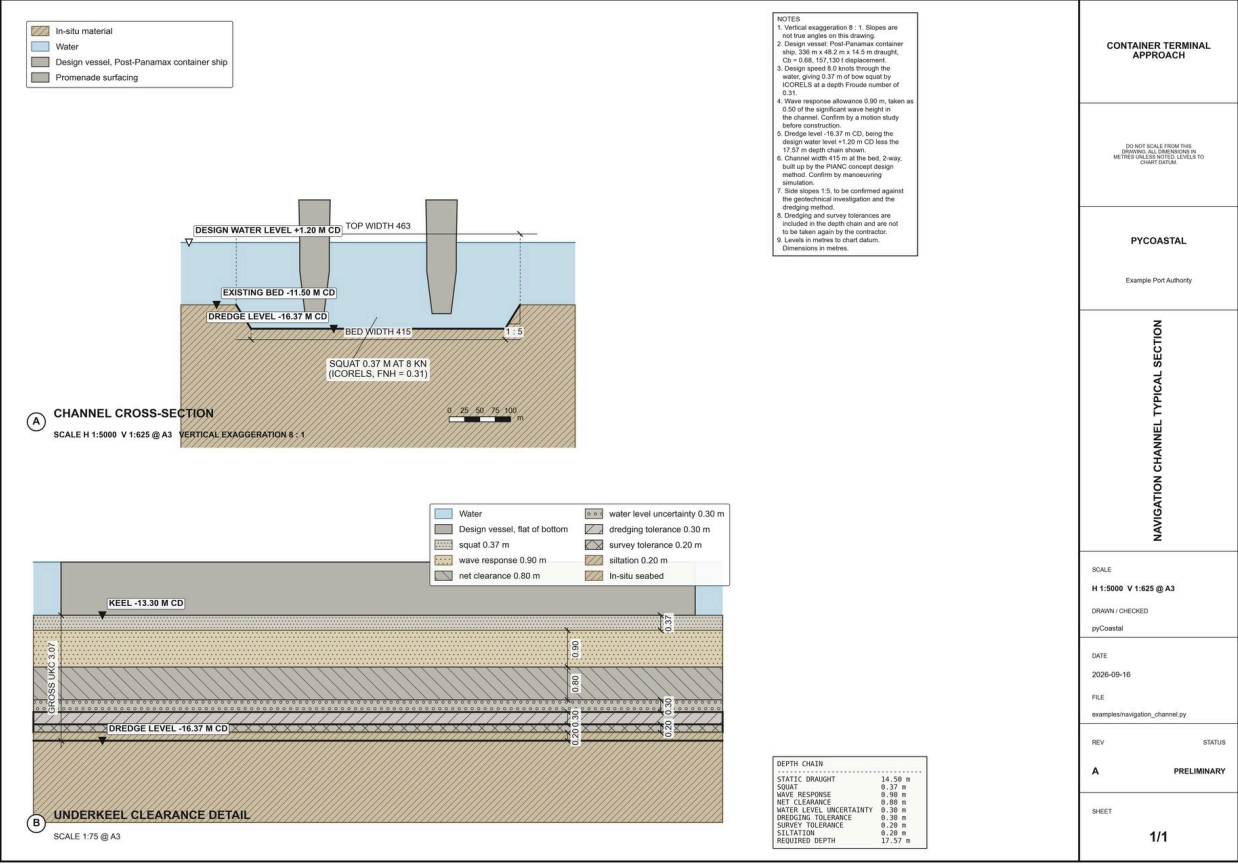


Figure 15.1: Navigation channel drawing sheet: the whole channel with a stated vertical exaggeration, and the underkeel clearance at a true scale where the allowances read as real thicknesses.

Approach channel for Post-Panamax container ship: dredge level -16.37 m CD, 415 m wide

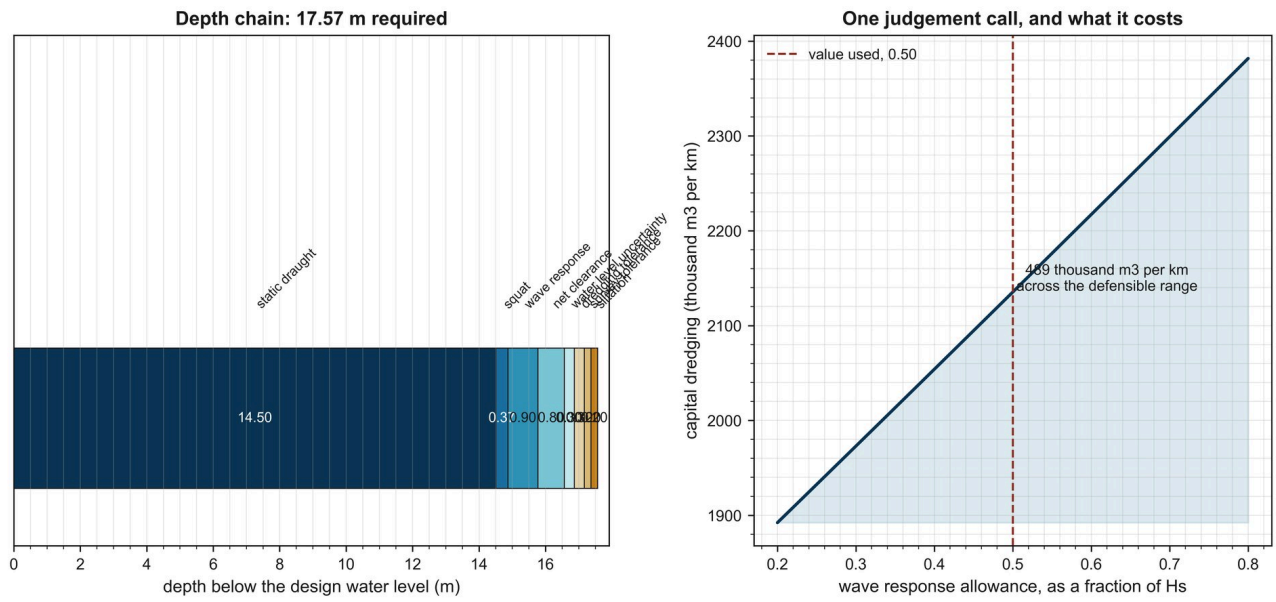


Figure 15.2: The depth chain and its sensitivity to each judgement.

16 Berthing energy and fenders

Module: `pyCoastal.applications.berthing`. Example: `examples/berth_fenders.py`.

A ship comes alongside carrying kinetic energy, and something has to absorb it. A berth fails in three different ways and only one of them is about energy. The fender can be too small, and the ship reaches the quay. The panel can be too small, and the reaction dents the side shell even though the energy was absorbed perfectly. The fenders can be too far apart, and the hull touches the structure between them. All three are checked.

16.1 The energy chain

PIANC (2002, Working Group 33) writes the normal berthing energy as the kinetic energy times four factors:

$$E_N = \frac{1}{2}MV^2 C_m C_e C_s C_c. \quad (16.1)$$

- **Added mass** (Vasco Costa 1964), $C_m = 1 + 2T/B$, the water the ship drags sideways. It is the only factor that makes the energy larger. Below about 1.1 draughts of water depth the form starts to understate it, and `added_mass_factor(vessel, depth)` warns.
- **Eccentricity**, $C_e = (K^2 + R^2 \cos^2 \gamma) / (K^2 + R^2)$ with the radius of gyration $K = (0.19C_b + 0.11)L_{pp}$, R the distance from the contact point to the center of mass along the berth (default $L_{pp}/4$, the quarter point), and γ the angle between the velocity and that line (default 90 degrees). The rotation carries away energy the fender never sees.
- **Softness** C_s : 1.0 for a soft fender, 0.9 where it is stiff enough that the hull deflects with it.
- **Configuration** C_c (BERTH_CONFIGURATION): 0.8 solid quay, 0.9 semi-solid, 1.0 open piled.

The **abnormal factor**, applied afterwards (ABNORMAL_FACTOR: 1.25 for a large tanker up to 2.0 for ro-ro, ferries and tugs, 1.5 for container ships), covers the berthing that goes wrong: a parted tug line, an engine that does not go astern. It is reported separately because it is frequently the largest single term.

Velocity is squared and is the least well known quantity. BERTHING_VELOCITY holds the Brolsma et al. (1977) style table (Table 16.1); `berthing_velocity(vessel, condition)` picks the size class from the displacement.

Table 16.1: Indicative design berthing velocities (m/s), BERTHING_VELOCITY.

Condition	small	medium	large
easy_sheltered	0.20	0.15	0.10
difficult_sheltered	0.30	0.20	0.15
easy_exposed	0.40	0.30	0.20
good_exposed	0.50	0.40	0.25
difficult_exposed	0.80	0.60	0.40

16.2 Fender, panel, and spacing

Fender family. Within one family and rubber grade, performance scales with size: energy with the volume of rubber, reaction with the section,

$$E = E_0 \left(\frac{H}{H_0} \right)^3, \quad R = R_0 \left(\frac{H}{H_0} \right)^2. \quad (16.2)$$

A fender chosen to absorb twice the energy pushes back only about 1.6 times as hard. `FenderFamily` holds a reference size (the default `CONE_FENDER` is a mid-grade cone fender, 1.0 m, 500 kNm, 1000 kN, in standard heights from 0.3 to 2.5 m) and `select_fender(energy, family)` returns the smallest standard size that absorbs the energy and how much of its capacity is used. This is a scaling model, not a catalogue: size the fender you are asking for here, then select against the manufacturer's tested curves.

Hull pressure. `hull_pressure(reaction, panel_width, panel_height, vessel_class)` checks the panel against `HULL_PRESSURE_LIMIT` (150 kN/m² for large tankers, 200 for small tankers and bulk carriers, 400 for container ships and general cargo, 700 for tugs).

Spacing. `fender_spacing(vessel, spacing, projection, bow_radius=None, smallest_vessel=None, clearance=0.15)` applies two checks. Geometric: with the fenders compressed, the standoff must exceed the sagitta of the hull across the gap,

$$s \leq 2 \sqrt{R_b^2 - (R_b - p + c)^2}. \quad (16.3)$$

Practical: the smallest ship using the berth must reach two fenders at once, about $0.15L_{pp}$. Without a bow radius the contact is on the straight parallel midbody, so the geometric check cannot bind and the length rule governs; the rule uses the *smallest* vessel, not the design one.

`design_berth(vessel, velocity=None, condition="easy_sheltered", vessel_class="container", configuration="open_piled", ..., smallest_vessel=None, panel_aspect=(1.2, 2.0), family=CONE_FENDER)` runs the whole chain and returns a `BerthDesign` with normal, abnormal, energy, fender, panel, pressure, spacing, adequate, and notes.

```
from pyCoastal.applications.berthing import design_berth
from pyCoastal.applications.channel import Vessel

ship = Vessel(name="Post-Panamax container", length=366, beam=48.2,
              draught=15.2, block_coefficient=0.68)
feeder = Vessel(name="Feeder", length=140, beam=22.0, draught=8.5,
                block_coefficient=0.68)
design = design_berth(ship, velocity=0.15, vessel_class="container",
                     configuration="open_piled", depth=17.0,
                     smallest_vessel=feeder)
design.energy, design.fender["height"], design.pressure["pressure"], design.adequate
```

16.3 Worked example

Output of `examples/berth_fenders.py`

```
Post-Panamax container: 179 thousand tonnes
Approaching at 0.15 m/s
```

```
kinetic energy      2018 kNm
x Cm 1.631          added water
x Ce 0.478          rotation about the contact
x Cs 1.000          fender softness
x Cc 1.000          berth configuration
normal berthing     1573 kNm
x 1.50             abnormal allowance
design energy        2360 kNm

fender              2.00 m, 4000 kNm rated (59% used)
reaction            4000 kN
hull pressure       325 kN/m2 against 400 allowed
spacing             16.1 m against 20.2 m allowed
ADEQUATE            True
```

- Post-Panamax container: 366 m by 48.2 m, 15.2 m draught, 179 thousand tonnes displacement.
- Berthing at 0.15 m/s, as specified. Velocity is squared in the energy, so it carries more of the answer than anything else here.
- Factors: Cm 1.63 adds the water the ship drags, Ce 0.48 takes away what the ship rotates out, Cs 1.00 and Cc 1.00 take away a little more. Net, the fender sees 78% of the 2018 kNm the ship arrived with.
- Eccentricity is taken at the quarter point, the usual design assumption. A ship that lands flat on its midships rotates about nothing and hands the fender all of it: Ce becomes 1 and the energy rises by 2.09 times.
- Abnormal factor 1.50 for a container, taking the design energy from 1573 to 2360 kNm. This is the allowance for the berthing that goes wrong, and it is the largest single term in this calculation.
- Cone fender, mid grade, 2.00 m: 4000 kNm rated against 2360 kNm required, so it works at 59% of capacity and pushes back 4000 kN.
- Hull pressure 325 kN/m2 over a 2.8 by 4.4 m panel, within the 400 kN/m2 allowed for this class.
- Fenders at 16.1 m centres, inside the 20.2 m limit set by vessel length for Feeder. Contact is on the parallel midbody, which is straight, so only the vessel length rule binds; check the end fenders separately with a bow radius.
- Fender performance here is a scaling model within one family, not a catalogue. Size the fender you are asking for from this, then select against the manufacturer's tested curves at the design temperature, angle and velocity, which is what they guarantee.

Wrote `media/berth_energy.png`

Three of the four PIANC factors reduce the energy and the abnormal allowance puts most of it back: the design energy of 2360 kNm is close to the 2018 kNm the ship arrived with, which makes it easy to conclude wrongly that the factors did not matter.

Berthing energy: three of the four PIANC factors reduce it, and the abnormal allowance puts most of it back

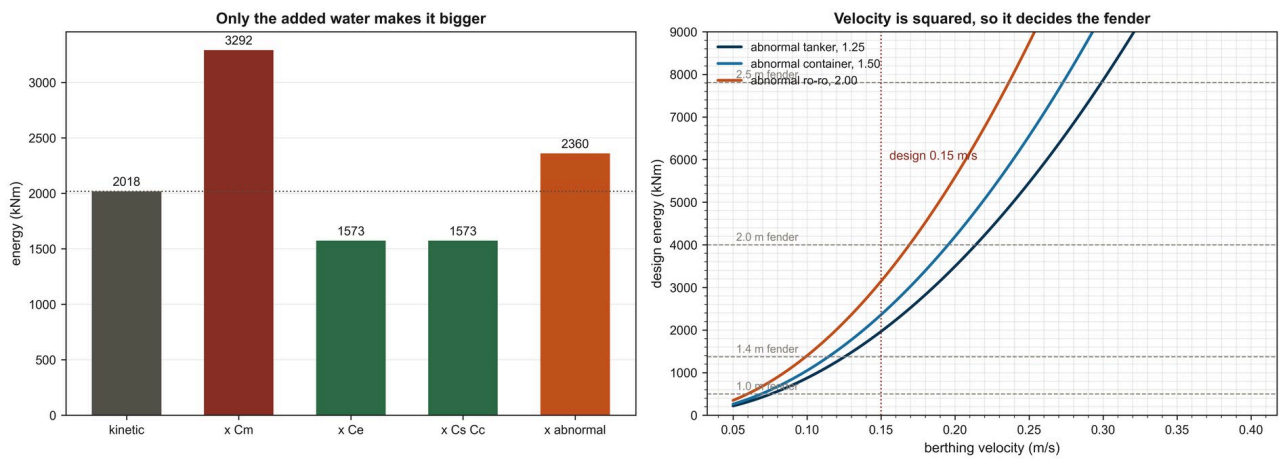


Figure 16.1: The berthing energy chain and the design levers.

17 Beach nourishment

Module: `pyCoastal.applications.nourishment.`
`examples/nourishment_profile.py`. **Browser:** Nourishment.

Examples: `examples/nourishment_design.py`,

A beach fill is designed in two directions at once. Alongshore, it is a perturbation to a straight shoreline that spreads and leaves the project area; the question is how long it lasts. Cross-shore, the beach takes the profile its own grain size can hold; the question is how much dry beach a given volume buys, and how much that depends on where the sand comes from.

17.1 The planform: how long a fill lasts

Under small wave angles the one-line model (Equation 6.4) reduces to a diffusion equation. With the CERC transport in the solid-volume SI form this model needs,

$$Q = K_{\text{cerc}} H_b^{5/2} \sin 2\alpha, \quad K_{\text{cerc}} = \frac{K \sqrt{g/\gamma_b}}{16(s-1)}, \quad (17.1)$$

(`cerc_coefficient(K=0.39, s=2.65, gamma_b=0.78)`; the solver divides by $1-p$ itself, so a bulk-volume coefficient would double-count porosity), linearization gives

$$\frac{\partial y}{\partial t} = \varepsilon \frac{\partial^2 y}{\partial x^2}, \quad \varepsilon = \frac{2K_{\text{cerc}} H_b^{5/2}}{(1-p)(D+B)}, \quad (17.2)$$

(`longshore_diffusivity`). For an initially rectangular fill of width W and length $L = 2a$, Pelnard-Considere (1956) gives

$$y(x, t) = \frac{W}{2} \left[\operatorname{erf} \left(\frac{a-x'}{2\sqrt{\varepsilon t}} \right) + \operatorname{erf} \left(\frac{a+x'}{2\sqrt{\varepsilon t}} \right) \right], \quad (17.3)$$

with x' from the fill center (`pelnard_considere`). The center width $W \operatorname{erf}(a/2\sqrt{\varepsilon t})$ halves when the argument reaches $\operatorname{erf}^{-1}(1/2) = 0.4769$, which defines the spreading half-life, the natural clock of a fill (`sections.spreading_half_life`).

`NourishmentDesign(length=1000, berm_width=30, taper=100, center=None, D=8, B=2, porosity=0.4)` is the fill: full-width length, shoreline advance at placement, linear tapers at each end, closure depth D , and berm height B . `WaveClimate(Hb=1.0, T=8.0, alpha0=0.0, Kcerc=KCERC_DEFAULT)` drives it; a zero angle means the fill diffuses symmetrically without migrating. `simulate_nourishment(design, climate, duration, domain_length=None, dx=10.0, n_outputs=25, bc="fixed_ends", cfl=0.9)` evolves the tapered planform with `tools.shoreline` on a domain five times the fill footprint and returns a `NourishmentResult` with `planforms`, `volume_retained`, `retained_fraction`, `berm_width`, and `design_life(threshold=0.5)`. `renourishment_schedule(design, climate, horizon, threshold=0.5)` returns the interval, the number of renourishments, the placement times, and the total volume over a planning horizon. The analytical solution verifies the solver in the test suite.

Output of `examples/nourishment_design.py`

```
Placed volume      : 960,000 m3
Active profile height: 10.0 m
Diffusivity        : 0.0175 m2/s (551,100 m2/yr)

year  retained %  berm width (m)
0.0   100.0      30.0
4.0   54.0       16.6
8.0   41.5       12.3
12.0  34.9       10.2
16.0  30.7       8.9
20.0  27.8       8.0

Design life (50% retained): 5.0 yr
Renourishment interval    : 4.9 yr
Renourishments in 30 yr   : 6
Total sand over 30 yr      : 6,720,000 m3

Wrote media/nourishment_design.png
```

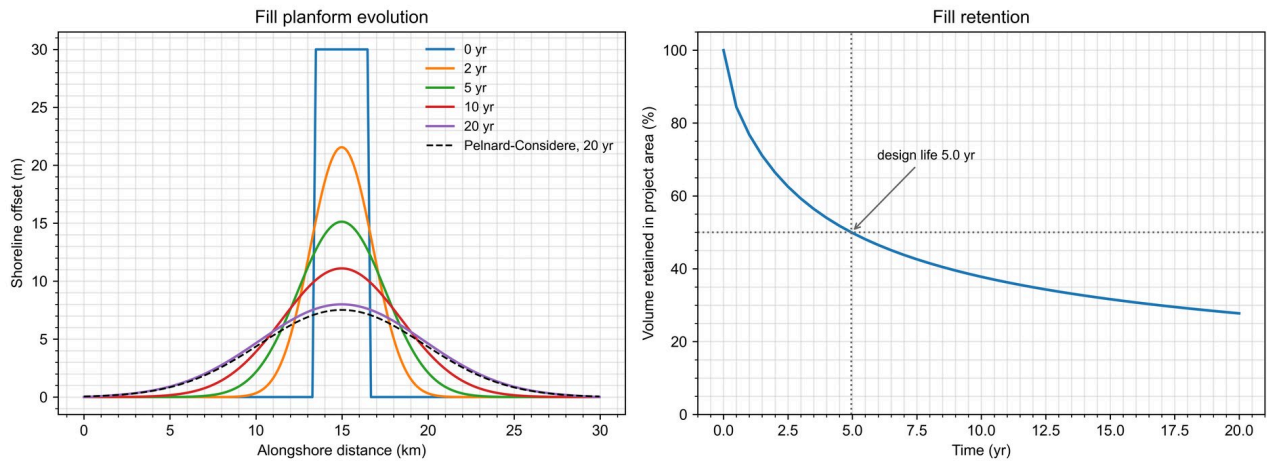


Figure 17.1: Nourishment design study: planform evolution against the Pelnard-Considere solution, retained volume, and the renourishment cycle.

17.2 The profile: what a borrow source is worth

Dean's equilibrium profile is $h = Ay^{2/3}$ (equilibrium_profile), with the scale parameter tied to the fall velocity (Kriebel, Kraus and Larson 1991),

$$A = 0.067 w^{0.44} \quad (w \text{ in cm/s}), \quad (17.4)$$

so a sediment cannot have one settling velocity for scour and another for its profile (dean_scale; fine sand at 0.19 mm gives 0.094 and medium sand at 0.38 mm gives 0.141, against Dean's tabulated 0.10 and 0.14).

A fill with scale A_f placed on a native beach with scale A_n pushes the profile seaward at each depth by $\Delta(h) = a + (h/A_f)^{3/2} - (h/A_n)^{3/2}$, where a is the shoreline advance. Integrating over *depth* (not distance, because the active profile is bounded by the closure contour) from the waterline to closure gives

$$V = Ba + ah_L + 0.4 h_L^{5/2} (A_f^{-3/2} - A_n^{-3/2}), \quad (17.5)$$

where h_L is the closure depth, or the depth where the profiles meet if the fill is coarse enough to intersect first (fill_volume_for_advance). For matched sand it is exactly $V = (B + h_*)a$. Three cases follow: coarser fill **intersects** the native profile and buys more beach per cubic meter; finer fill is **non-intersecting**; and below a **critical volume** (critical_volume) a fine fill buys no dry beach at all, because the whole placement goes into flattening the underwater profile. shoreline_advance(A_{native} , A_{fill} , volume, berm_height, closure_depth) solves for the advance by bisection (one solver that cannot pick the wrong branch) and names the case. profile_overfill_factor(native, borrow, ...) is the ratio of borrow to native volume for the same advance, which is not James's (1975) textural overfill ratio. grain_compatibility(native, borrow) compares the phi means and sortings ($\delta = (\phi_b - \phi_n)/\sigma_n$) and gives a verdict; phi_size and size_from_phi convert grain sizes.

Output of examples/nourishment_profile.py

```
Native beach: Medium sand, d50 = 0.38 mm, A = 0.141
Berm 2.0 m, closure 6.0 m, fill 250 m3 per metre

borrow      A      delta  profile      dry beach  critical
Very fine sand 0.051 +3.78 submerged      0.0 m    2362 m3/m
Fine sand     0.094 +1.82 submerged      0.0 m    566 m3/m
Medium sand   0.141 +0.00 matched       31.2 m     0 m3/m
Coarse sand   0.183 -1.78 intersecting    54.5 m     0 m3/m
Fine gravel   0.307 -7.24 intersecting    66.2 m     0 m3/m

What that costs, per metre of beach, for 40 m of dry sand
Very fine sand 2682 m3/m (8.38 times the native requirement)
Fine sand     886 m3/m (2.77 times the native requirement)
Medium sand   320 m3/m (1.00 times the native requirement)
Coarse sand   164 m3/m (0.51 times the native requirement)
Fine gravel   131 m3/m (0.41 times the native requirement)

Compatibility
Very fine sand much finer than native, poorly suited
Fine sand     much finer than native, poorly suited
```

Medium sand close to native, suitable
 Coarse sand coarser than native, well suited
 Fine gravel coarser than native, well suited

Planform, 1500 m of fill in $H_b = 1.2$ m:

spreading half-life 0.57 yr
 placed volume 719 thousand m³ in place

Wrote media/nourishment_profile.png, media/nourishment_plan.png and media/nourishment_sheet.png
 Wrote media/nourishment_borrow.png

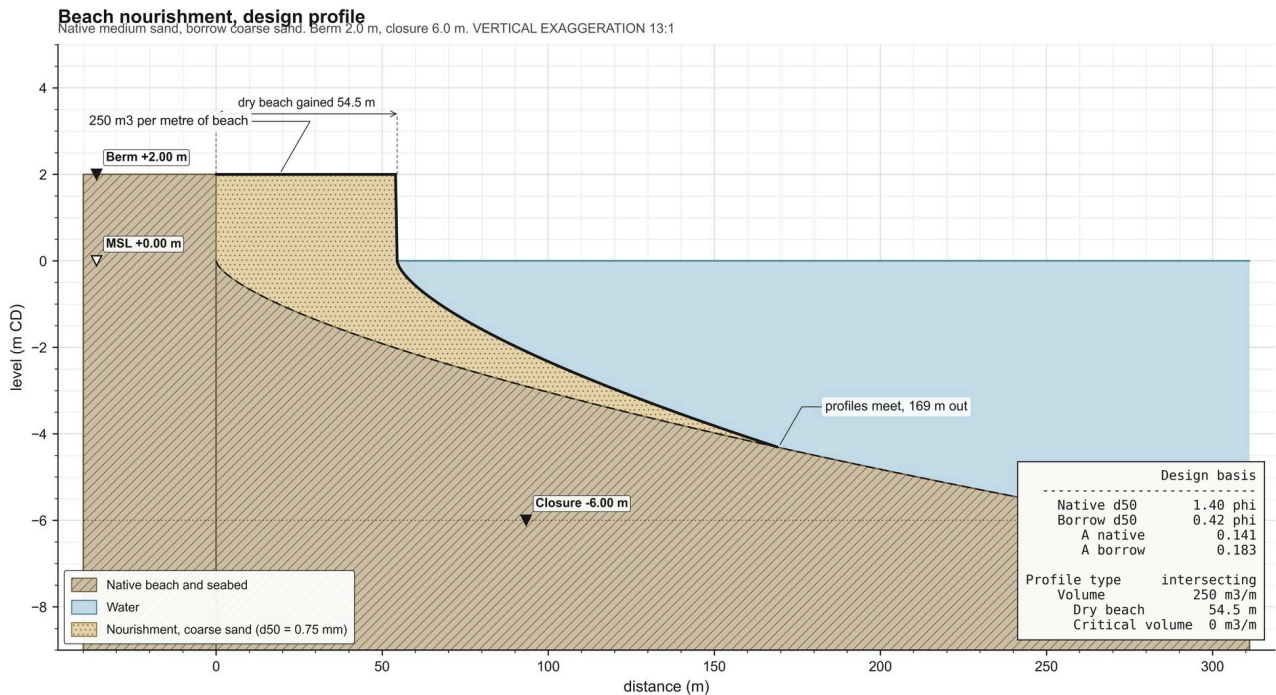


Figure 17.2: Profiles for borrow sources from very fine sand to fine gravel on a medium sand beach.

Nourishment on medium sand: the borrow grain size decides what the sand is worth

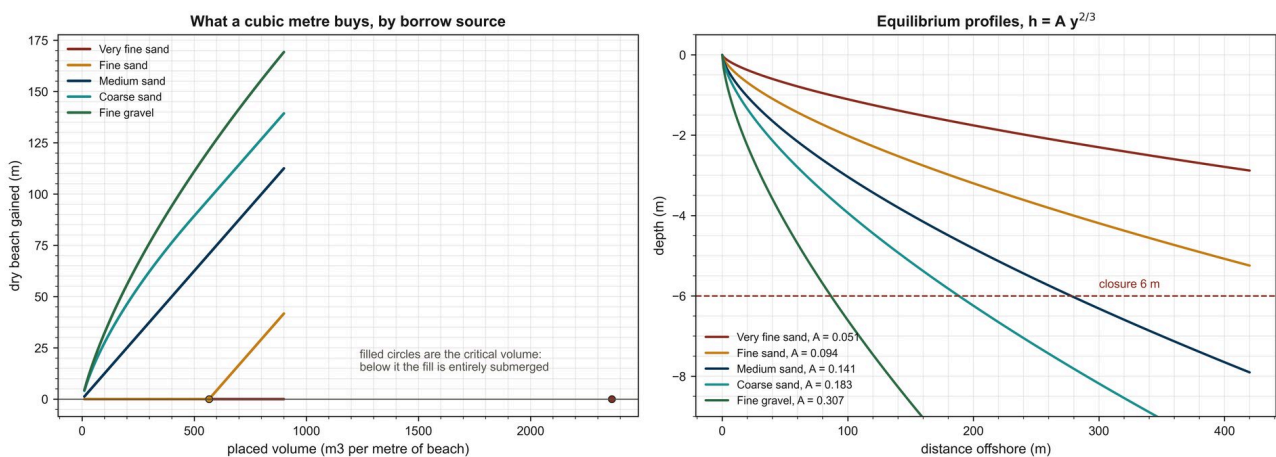


Figure 17.3: The cost of borrow sources: volume per meter of beach for the same dry beach width.

Beach nourishment, planform evolution

Pelinard-Considere (1956). Fill 1500 m long, 54 m wide, diffusivity 1087 thousand m²/yr. Half-life 0.57 yr. CROSS-SHORE EXAGGERATION 25:1

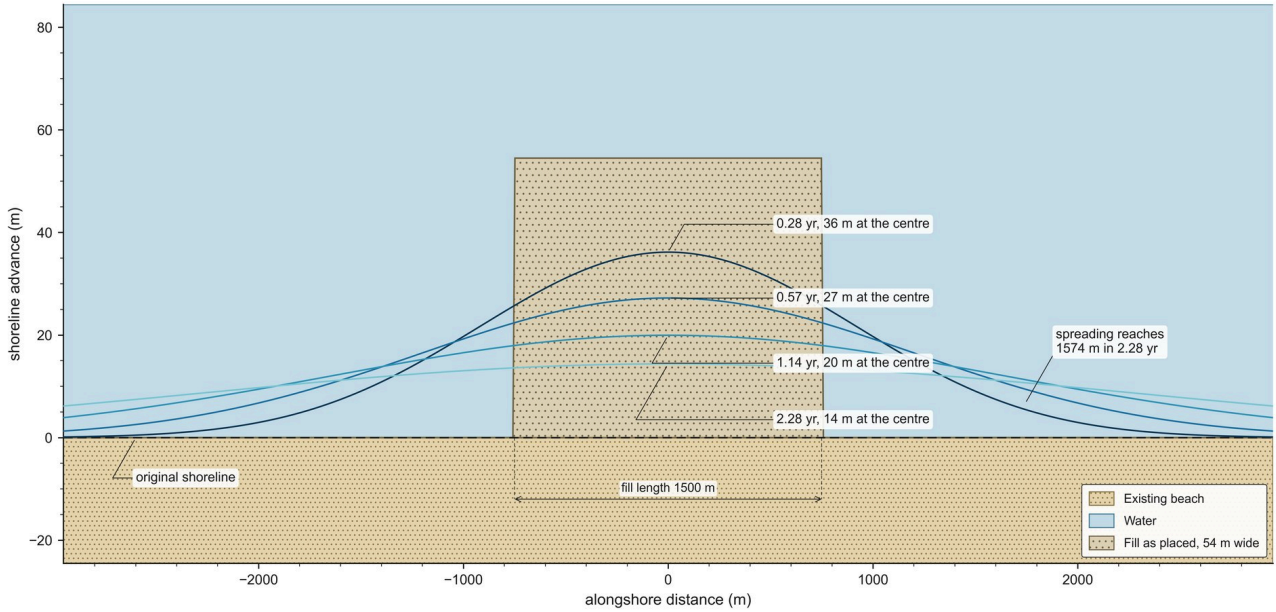


Figure 17.4: Planform spreading alongshore from the analytical solution, at times taken from the fill's own half-life. The cross-shore axis is stretched to fill the sheet and the factor is stated on the drawing.

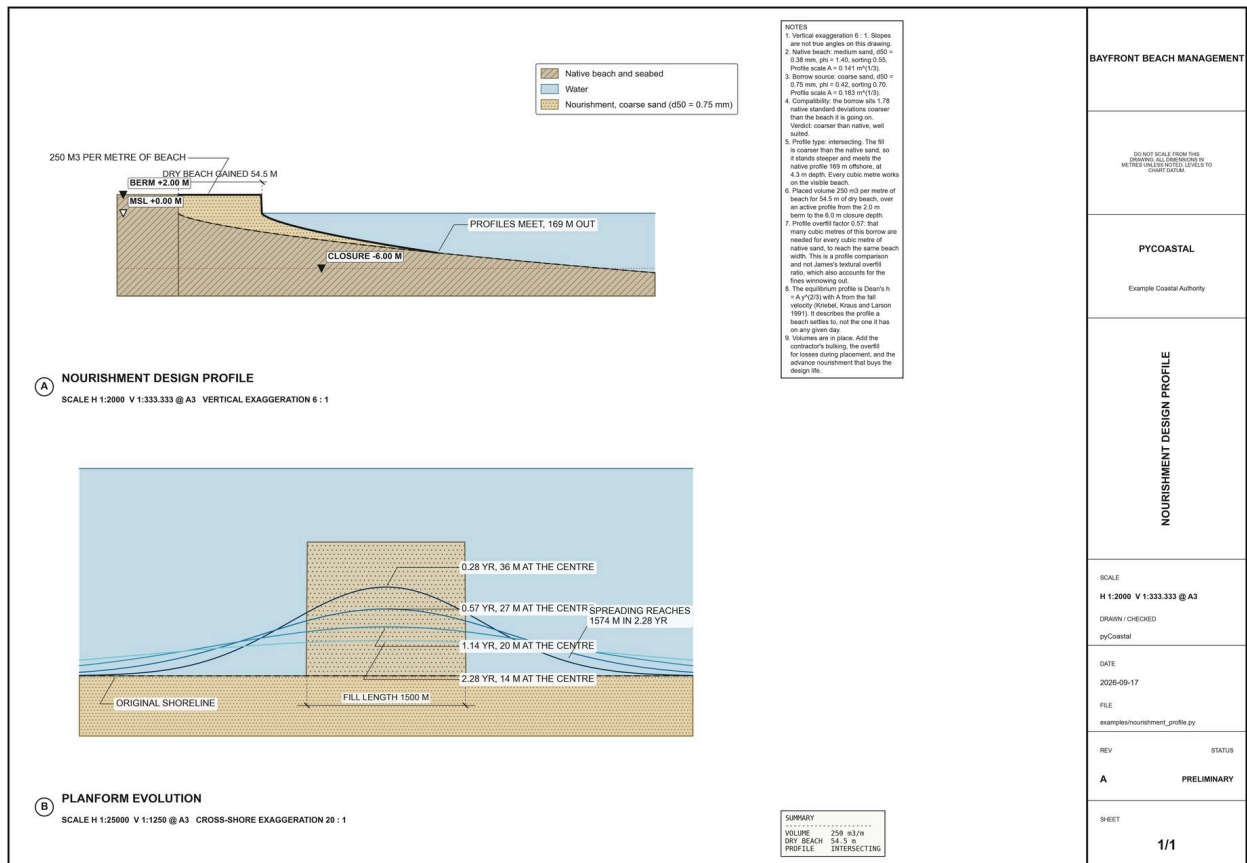


Figure 17.5: Nourishment drawing sheet with the design profile and the planform on one sheet, each at its own standard scale.

17.3 Limits

The analytical planform is for a rectangular fill with constant diffusivity; the numerical solver handles tapers but still uses one wave climate. The profile method assumes both profiles reach equilibrium and ignores winnowing of fines, which the compatibility check addresses separately.

18 Storm surge and coastal flooding

Module: `pyCoastal.applications.surge`. Example: `examples/storm_surge_flooding.py`.

The module builds a total water level from its components and then finds how far a storm floods: along a cross-shore profile, or over a terrain grid with hydraulic connectivity, so that low ground behind a continuous barrier is not flooded by a level that cannot physically reach it.

18.1 The water level budget

The still water level is the sum of the astronomical tide (supplied, not predicted), any sea-level rise, and three setups:

- **Barometric setup**, the inverted-barometer response, $\eta_p = (p_a - p_c)/(\rho g)$, close to 1 cm per hPa (`barometric_setup`). It is the equilibrium response and an upper bound for a fast-moving storm.
- **Wind setup**, from integrating $d\eta/dx = nkW^2 \cos\theta/(gd)$ over the fetch, with $k = 3.3 \times 10^{-6}$ and a return-flow factor n of 1.15 to 1.30. Over a constant depth,

$$\eta_w = \frac{nkW^2 F \cos\theta}{gd}, \quad (18.1)$$

(`wind_setup`). Because setup goes as $1/d$ it is dominated by the shallowest part of the shelf, so `wind_setup_profile(depths, dx, ...)` integrates across a real shelf, carrying the accumulated setup into the local depth $d + \eta$, which damps further setup.

- **Wave setup** in the surf zone, from the breaking height (`surf_zone_setup(breaking_height, gamma=0.8)`).

Runup is added on top for the total water level, from Stockdon et al. (2006):

$$R_{2\%} = 1.1 \left(0.35\beta\sqrt{H_0 L_0} + \frac{\sqrt{H_0 L_0 (0.563\beta^2 + 0.004)}}{2} \right), \quad (18.2)$$

returned as setup and swash separately (`stockdon_runup`). Runup is a swash oscillation, not a sustained level, so flood extent is taken from the still water level and the wave hazard at the shoreline from the total.

`StormConditions(wind_speed=40, central_pressure=95000, tide=0, Hm0=6, Tp=12, fetch=100e3, shelf_depth=20, wind_angle=0, sea_level_rise=0)` is the storm; `total_water_level(storm, beach_slope=0.05, breaker_index=0.78, shelf_depths=None, dx=None, return_flow=1.15)` returns every component, the still water level, and the total water level.

18.2 Flood mapping

A plain elevation threshold floods every low cell, including basins sealed off from the sea by higher ground. `bathtub_flood(terrain, level, seed=None, connectivity=4)` walks outward from the coast instead, so only ground the water can reach is flooded; seed marks where the sea enters (default: any below-level cell on the domain edge) and connectivity is 4 or 8. `isolated_low_ground` returns the cells below the level that the sea cannot reach, `flooded_area(flooded, dx)` and `flood_volume(terrain, flooded, level, dx)` quantify the map, and `flood_depths(z, level)` gives depths regardless of connectivity. On a profile, `inundation_limit(x, z, level)` walks landward and stops at the first ground above the level.

```
import numpy as np
from pyCoastal.applications.surge import (
    StormConditions, total_water_level, bathtub_flood, isolated_low_ground,
)

storm = StormConditions(wind_speed=45, central_pressure=94000, tide=0.6,
                       Hm0=7.0, Tp=13.0, fetch=120e3)
shelf = np.linspace(60.0, 8.0, 200)
levels = total_water_level(storm, shelf_depths=shelf, dx=120e3 / 199)

sea = np.zeros_like(terrain, dtype=bool); sea[0, :] = True
```

```
flooded = bathtub_flood(terrain, levels["still_water_level"], seed=sea)
stranded = isolated_low_ground(terrain, levels["still_water_level"], seed=sea)
```

18.3 Worked example

Output of `examples/storm_surge_flooding.py`

```
Water level budget (m above datum)
      tide      0.60
    sea level rise  0.00
    barometric setup  0.73
      wind setup   3.28
      wave setup   1.75
    still water level  6.36
      runup R2%    2.32
    total water level  8.68

Still water level 6.36 m
threshold-only flooded area : 19.17 km2
connected flooded area      : 18.82 km2
overstated by the threshold : 0.35 km2 (2% of it)
water volume on land        : 113.81 Mm3

with the inlet open : 18.82 km2 flooded
with the inlet sealed : 5.45 km2 flooded
a single breach in the barrier is what floods the bay

inundation limit, intact barrier: 1418 m
inundation limit, at the inlet : 4990 m

Wrote media/storm_surge_flooding.png
```

Connectivity matters: on the synthetic barrier island, a single breach in the barrier is the difference between 5.45 km² and 18.82 km² flooded.

Storm surge flooding: 45 m/s, 940 hPa, Hm0 = 7.0 m, tide 0.6 m

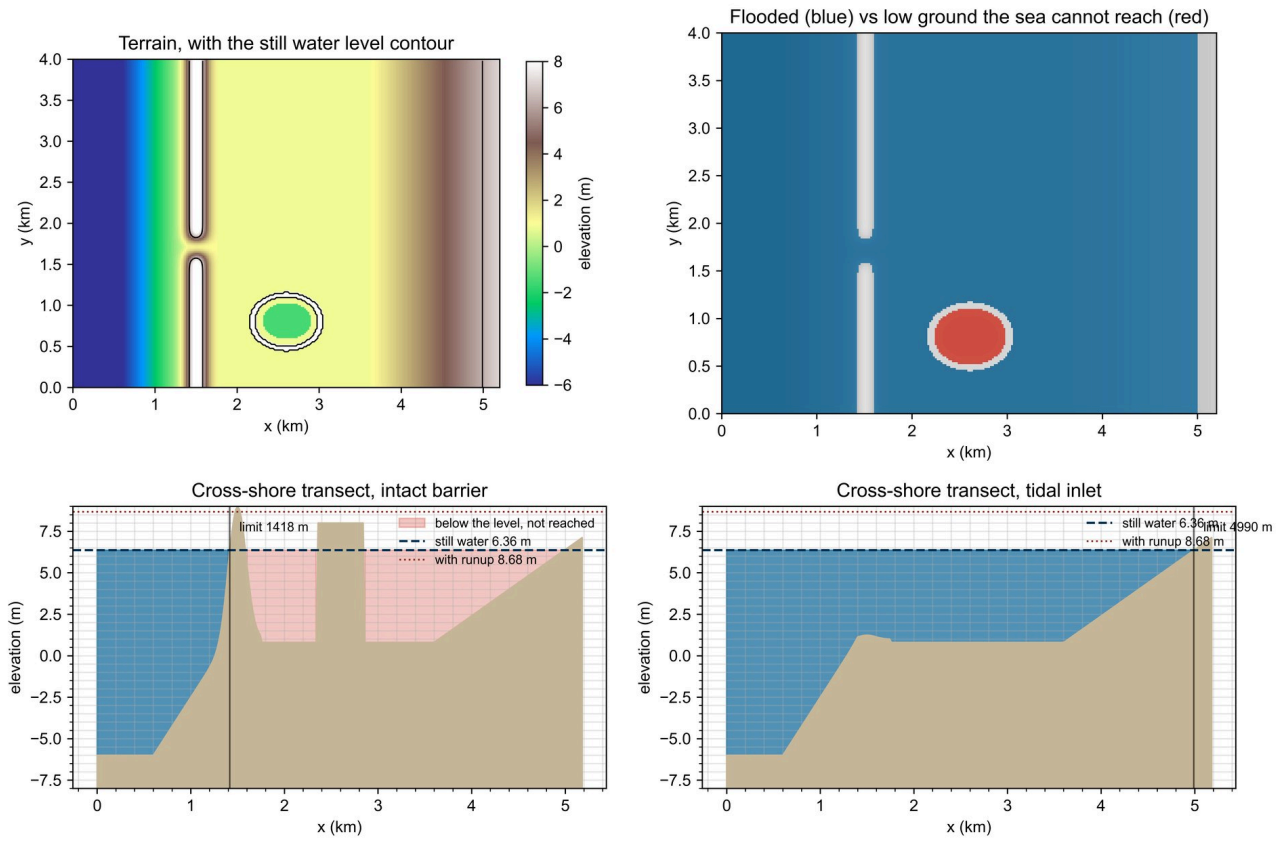


Figure 18.1: Storm surge flood mapping on a barrier-island terrain: threshold-only against connected flooding, with the inlet open and sealed.

Part V. Drawings and deliverables

19 Drawings, sections, and sheets

Modules: pyCoastal.drafting, pyCoastal.applications.sections, pyCoastal.plotting. Needs the plots extra.

A coastal cross-section is not a plot. It is a scaled drawing: the geometry is in real meters, line weights carry meaning, materials are hatched rather than colored by value, and every dimension that governs the design is called out on the paper. pyCoastal.drafting gives the small set of primitives that takes matplotlib from plotting to drafting, and applications.sections turns each design object into a drawing and a take-off.

19.1 Drafting primitives

`Section(title="", subtitle="", figsize=(13, 7), ax=None, caps=False, dim_style="arrow", frame=True, exaggeration=1.0)` is a drawing in real coordinates (x cross-shore, z elevation), locked to equal aspect so a 1:2 slope looks like a 1:2 slope. Text is sized in points and placed with offsets in points, so annotation stays readable when the section is rescaled; only the geometry scales. Its methods are the vocabulary of a drawing office:

- `material(points, kind, label=None)` fills a closed polygon with a material from the library (Table 19.1);
- `line(points, weight="medium", style="-")` draws construction, ground, or hidden lines with the line weights in WEIGHTS (thin 0.5, medium 1.0, heavy 1.8, extra 2.4);
- `water(x0, x1, level, bed=None, label=None)` fills water between a level and the bed, with the level symbol;
- `dim_h(x0, x1, z, text=None, extend_from=None)` and `dim_v(z0, z1, x, text=None, side="right")` draw dimension lines with extension lines and terminators;
- `level(x, z, text=None, side="left", symbol="level")` draws the levelling triangle and elevation callout;
- `slope(apex, cot_alpha, rise, direction="left")` draws the slope triangle, labelled 1: cot α ;
- `note(xy, text, offset=(40, 30))` places a leader line and note;
- `detail_bubble(label, title, scale="")` places the circled detail marker;
- `notes_block(lines, title="NOTES", numbered=True)` places numbered specification notes;
- `table(rows, title="Design parameters")` places the parameter block (a drawing that does not carry its design inputs is not a deliverable);
- `scale_bar(length, divisions=4)` and `key(loc="upper left")` add a checkered scale bar and a material key;
- `fit_scale(xlim, zlim, scales=None, paper="")` sets the view to a true, round drawing scale that fits the extents, and `auto_exaggeration(xlim, zlim, cap=200)` stretches the second axis just enough to fill the viewport when a true scale is unreadable;
- `finish(xlim=None, zlim=None, grid=True), save(path, dpi=600), and to_dxf(path, scale=1.0).`

Table 19.1: The material library in pyCoastal.drafting.

Key	Drawn as
armour, secondary, underlayer, toe	individual stones at the layer's stone size
core	quarry run, dotted hatch
concrete, reinforced	mass and reinforced concrete, diagonal and cross hatch, heavy edge
blinding	blinding layer
rock_fill, granular, sand	fills, circle and dot hatches
subgrade	in-situ seabed, dense diagonal hatch
pavement	promenade surfacing
water	water, thin edge

`Material(name, face, edge, hatch=None, weight="medium", stones=0.0)` defines a new one, and `use_crisp_style()` sets global matplotlib settings for legible technical figures.

`Sheet(titleblock, size="A3", dpi=400, margin=8.0, strip=52.0)` is a drawing sheet with a border and a title block strip down the right edge, in any of PAPER (A0 to A4). `viewport(rect)` adds a Section view within the usable area, `set_scale_from(view)` copies a view's fitted scale into the title block, `save(path)` writes the sheet with no trimming (the border

is the edge), and `to_dxf(path)` exports every view to one DXF. `TitleBlock` holds project, title, organisation, client, scale, date, file, sheet, revision, drawn_by, checked_by, status, and a standard disclaimer.

DXF. `write_dxf(path, entities, scale=1.0)` writes polylines on named layers to a minimal DXF R12 file that any CAD package opens.

19.2 Sections for each design

Every structure in `applications.sections` has three entry points, and all three take a design object rather than loose numbers:

- `draw_*` puts the geometry and annotation into an open `Section` and returns the extents it needs;
- `*_section` is a standalone figure with plot axes, for a report or slide;
- `*_sheet` is a full drawing sheet with border, title block, and notes, at a true stated scale; extra keyword arguments go to the `TitleBlock`.

Table 19.2: Drawing functions in `pyCoastal.applications.sections`.

Structure	draw	section	sheet	notes
Seawall	<code>draw_seawall,</code> <code>draw_seawall_toe_detail</code>	<code>seawall_section</code>	<code>seawall_sheet</code>	<code>seawall_notes</code>
Rubble mound	<code>draw_rubble_mound</code>	<code>rubble_mound_section</code>	<code>rubble_mound_sheet (trunk and head)</code>	<code>rubble_mound_notes</code>
Channel	<code>draw_channel,</code> <code>draw_channel_detail</code>	<code>channel_section</code>	<code>channel_sheet</code>	<code>channel_notes</code>
Nourishment profile	<code>draw_nourishment</code>	<code>nourishment_section</code>	<code>nourishment_sheet</code>	<code>nourishment_notes</code>
Nourishment plan	<code>draw_nourishment_plan</code>	<code>nourishment_plan_section</code>	via <code>nourishment_sheet(plan_design=...)</code>	
Pier scour	<code>draw_pier_scour</code>	<code>pier_scour_section</code>	<code>pier_scour_sheet</code>	<code>pier_scour_notes</code>
Bridge scour	<code>draw_bridge_scour</code>	<code>bridge_scour_section</code>	<code>bridge_scour_sheet</code>	<code>bridge_scour_notes</code>

Supporting geometry: `MoundProfile(x_sea, x_land, crest, cot_sea, cot_land)` describes a trapezoidal mound, with `offset(thickness)` giving the exact perpendicular offset of a layer (the crest edges move by $t(\sqrt{1 + \cot^2 \alpha} - \cot \alpha)$), `band`, and `area`; `mound_layer_volumes(profiles, bed)` takes off the layer quantities. The underlayer follows the Rock Manual rule of a tenth of the armor mass, so its D_{n50} falls by $10^{1/3} \approx 2.15$. `vessel_outline` draws a midship section, `scour_hole_profile` a hole whose faces stand at the angle of repose (`REPOSE_ANGLE = 32` degrees), `bridge_bed_profile` the scoured bed across the whole waterway as one line, and `plan_margin` and `spreading_half_life` set the extents and times of a planform.

19.2.1 Scale and exaggeration

Sections are drawn at a true scale by default, fitted to a round standard scale that the title block then states. Where a true scale is unreadable, as for a channel 600 m wide and 20 m deep or a beach profile hundreds of meters long and a few meters deep, the exaggeration is applied deliberately and printed on the drawing next to the scale. A drawing set then carries a second view at true scale where the detail matters: the channel keel and underkeel allowances, and the seawall toe.

19.3 The drawing sheets

The worked examples of Part IV issue these sheets; they are collected here as the deliverables of the design chain.

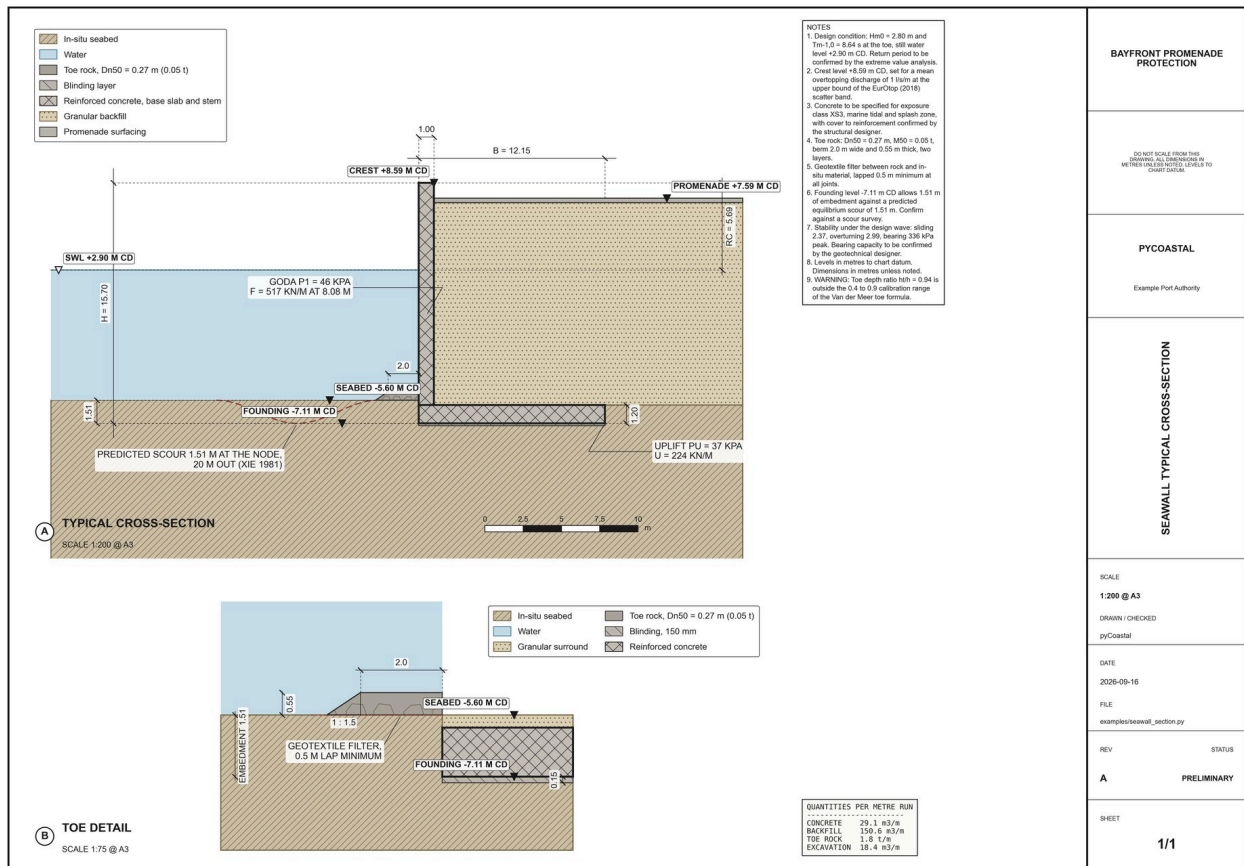


Figure 19.1: Seawall typical cross-section with toe detail (seawa11_sheet).

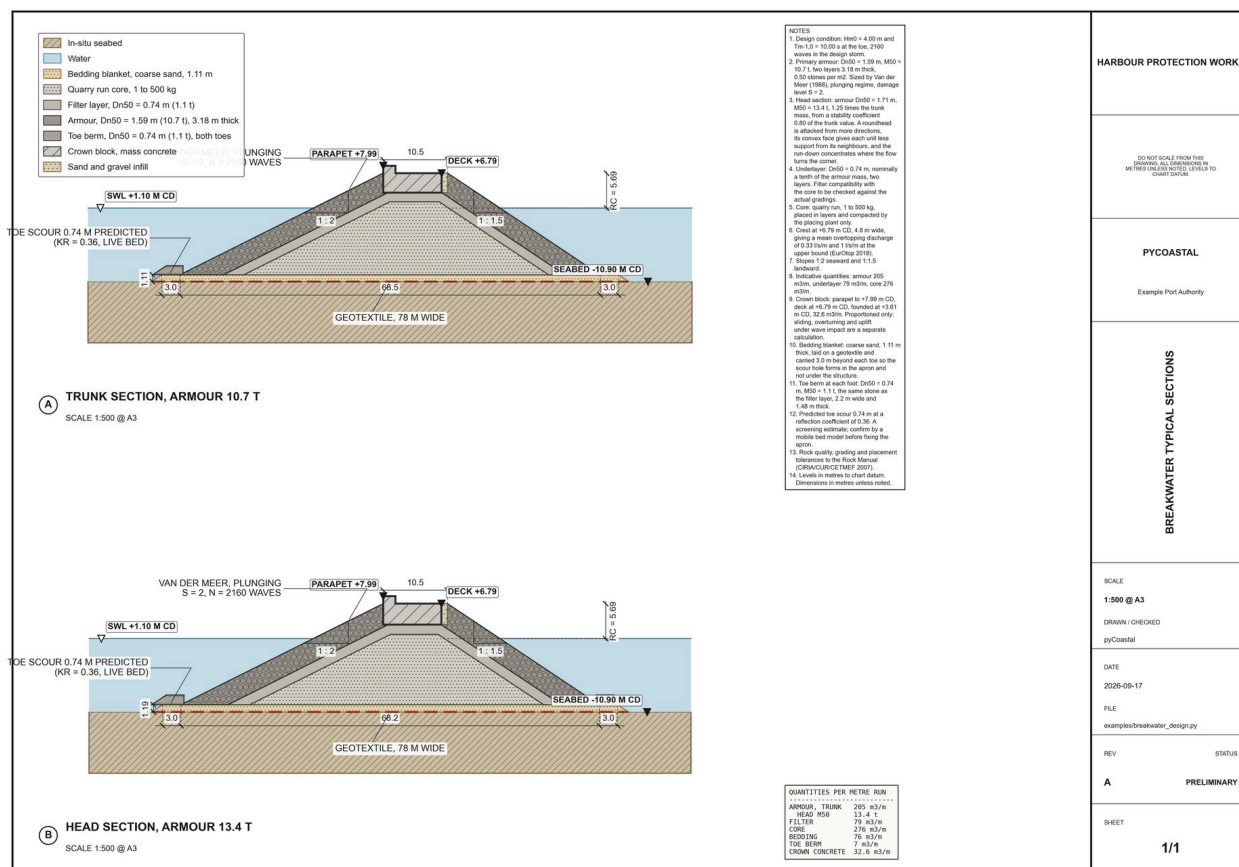


Figure 19.2: Breakwater trunk and head sections (rubb1e_mound_sheet).

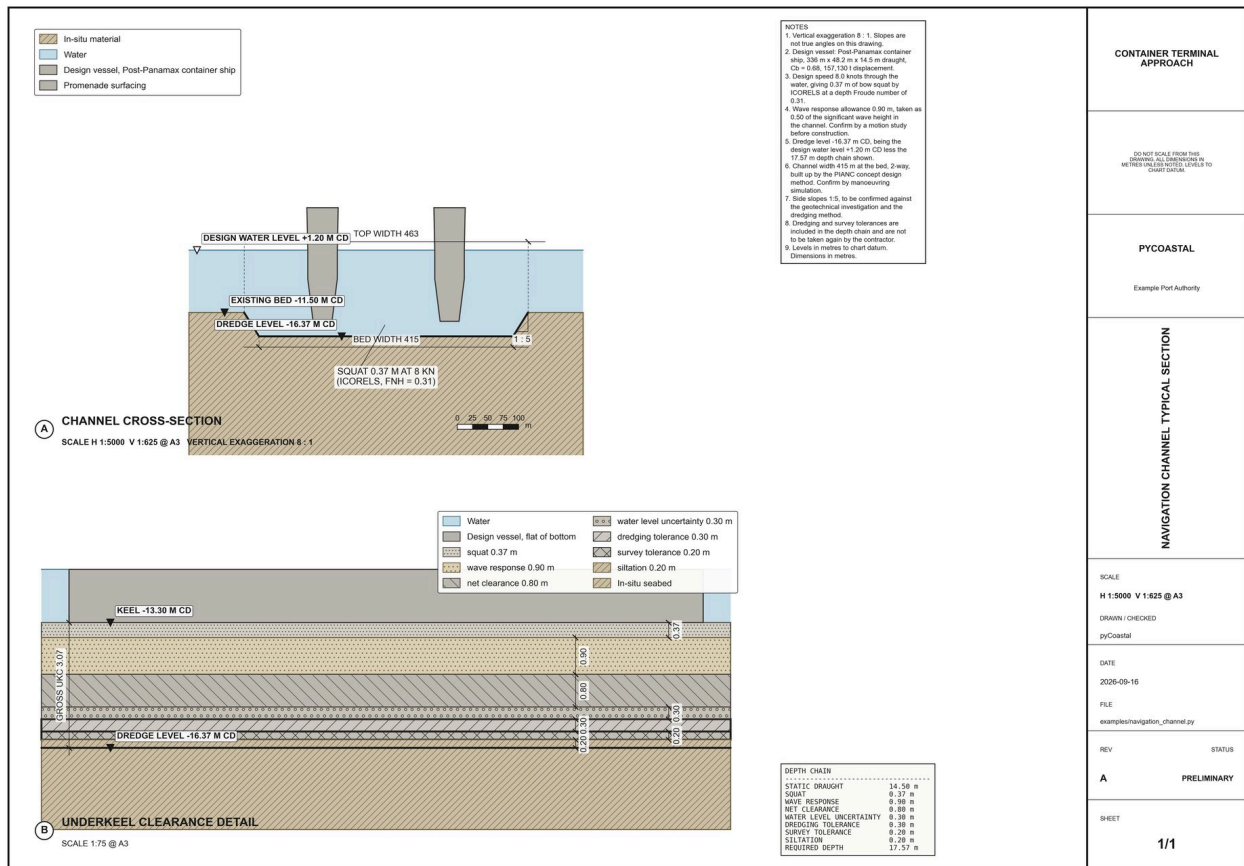


Figure 19.3: Navigation channel with true-scale keel detail (channel1_sheet).

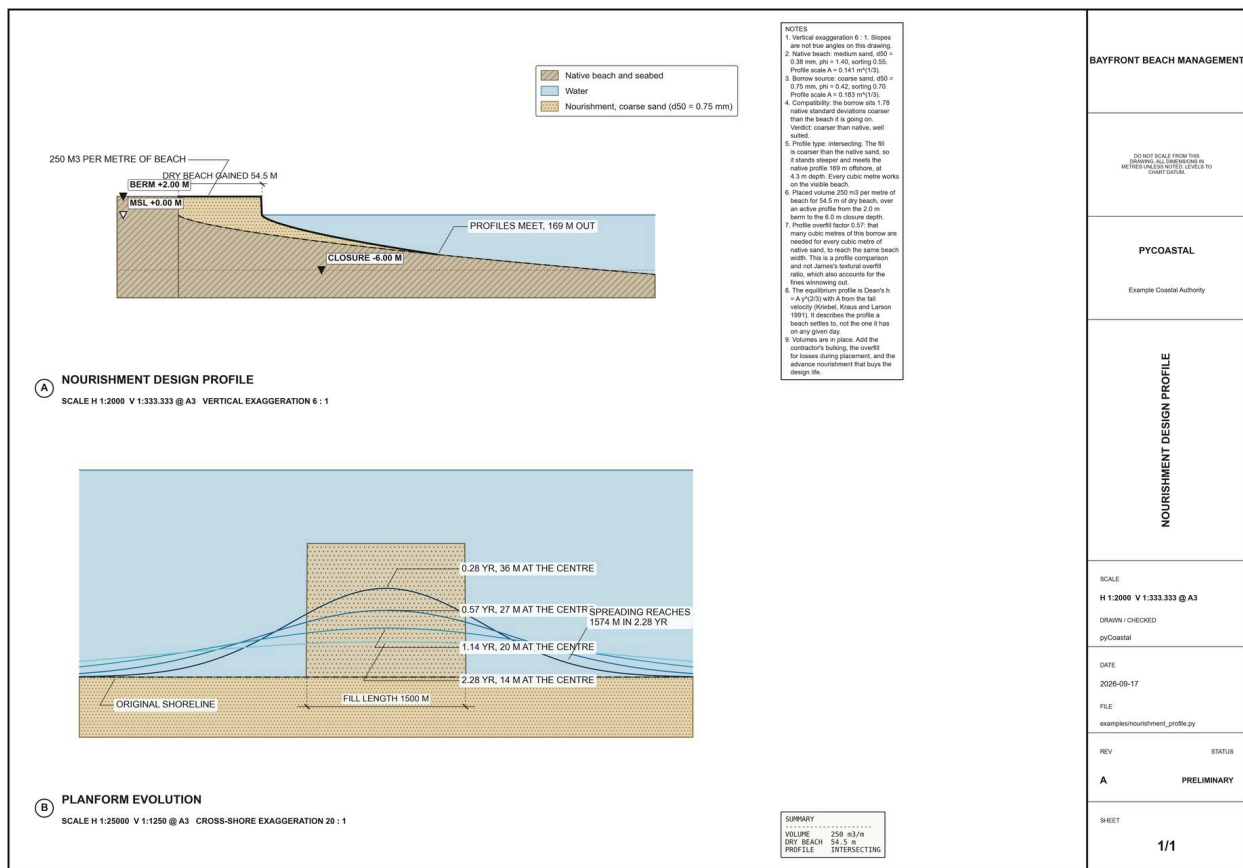
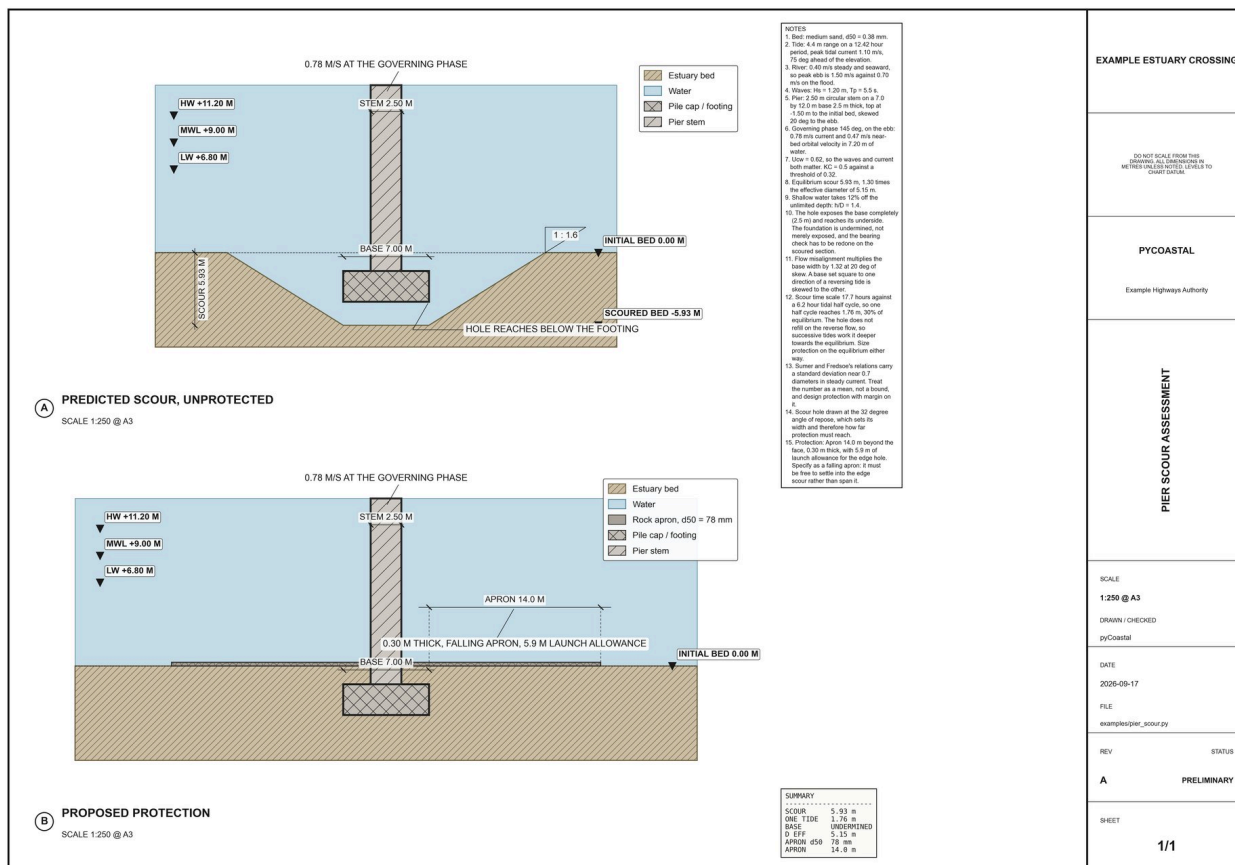


Figure 19.4: Nourishment design profile and planform (nourishment_sheet).



EXAMPLE ESTUARY CROSSING	
DO NOT SCALE FROM THIS DRAWING. ALL DIMENSIONS IN METERS UNLESS NOTED. LEVELS TO LOWEST SURFACE.	
PYCOASTAL	
Example Highways Authority	
PIER SCOUR ASSESSMENT	
SCALE	1:250 @ A3
DRAWN / CHECKED	pyCoastal
DATE	2026-05-17
FILE	examples/pier_scour.py
REV	STATUS
A	PRELIMINARY
SHEET	1/1

Figure 19.5: Pier scour, unprotected and protected (pier_scour_sheet).

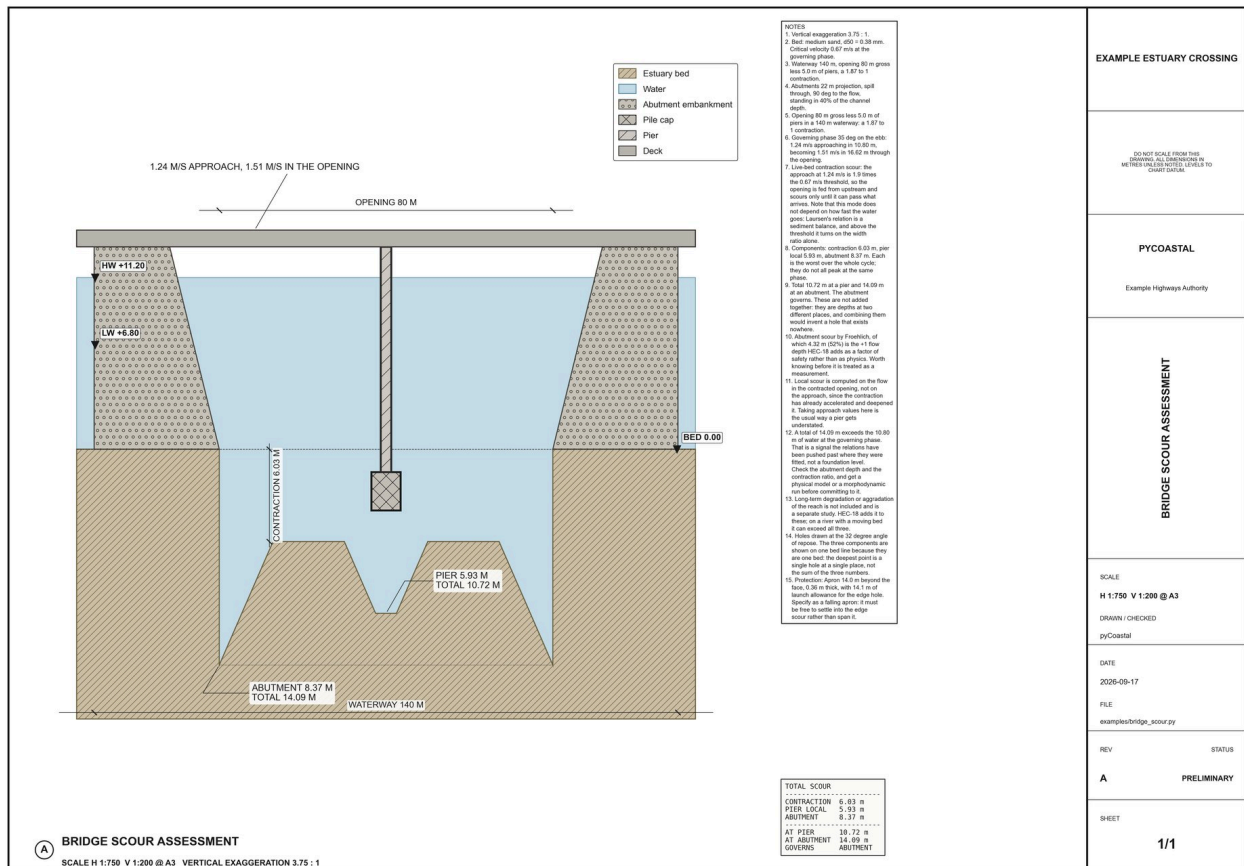


Figure 19.6: Bridge scour, all three components on one bed line (bridge_scor_sheet).

The examples also write DXF files of the geometry (media/seawall_section.dxf, media/breakwater_section.dxf, media/navigation_channel.dxf), with each material on its own layer.

19.4 Plot helpers

pyCoastal.plotting holds the color maps and helpers shared by the examples: water_colormap(), a diverging map for the instantaneous surface built to look like water; agitation_colormap(), a sequential map for wave height or disturbance coefficient; surface_norm(snapshots, percentile=99.5), a symmetric color scale centered on still water; land_overlay(land), a float array that is 1 on structures and NaN on water; and LAND_COLOR. For quantitative fields use agitation_colormap or a standard scientific map, and keep water_colormap for reading the wave pattern at a glance.

Part VI. The Coastal Design Bench

20 The browser app

Directory: webapp/.

The Coastal Design Bench is a static web page that runs the pyCoastal design modules in the browser, draws the result as a sheet, writes the design report, and puts next to it what the peer-reviewed literature says about the ground the relations stand on. Its tagline is the whole idea: size a structure with the pyCoastal relations, then read what the literature says about the ground they stand on.

20.1 How it works

A browser has no Python, so webapp/engine.js carries a JavaScript transliteration of the design modules, function for function: where the Python bisects, the JavaScript bisects; where the Python iterates on the base width, so does the JavaScript. Units follow the Python. A second implementation of a design code is worth having only if somebody checks it, so:

- make_vectors.py dumps reference cases straight out of pyCoastal into vectors.json, deliberately including awkward ones (the surging armor branch, an impulsive wall, a depth-limited Goda wave, drag- and inertia-dominated piles, a resultant outside the middle third);
- verify_engine.py runs engine.js in QuickJS against those cases and fails on any disagreement beyond floating-point noise;
- the page re-runs the same comparison on load and shows the result as a chip in the masthead, so the user sees it too.

app.js defines the modules, their inputs, the run functions, the report, and the checks; draw.js draws each module's sheet as SVG; and index.html is the application frame: a fixed masthead, an input panel, the drawing, the design report, and the Theory and sources panel, each scrolling on its own. Above 1560 px the report sits beside the drawing in its own pane. The page uses the fonts the desktop already has (Segoe UI and Consolas) and fetches no web font, so the first frame is correct.

20.2 Modules and inputs

The app has six modules. Each input is a slider with a range, a select, or a toggle, and the defaults follow the worked examples of Part IV.

Seawall

Gravity wall: crest from EurOtop, base from Goda pressures. (module key seawall)

Input	Label	Unit	Control	Range or options	Default
Hm0	Wave height Hm0	m	slider	0.5 to 8, step 0.1	2.8
Tp	Peak period Tp	s	slider	4 to 18, step 0.1	9.5
swl	Still water level	m CD	slider	-1 to 6, step 0.1	2.9
bed	Seabed level	m CD	slider	-20 to 2, step 0.1	-5.6
use	Tolerable overtopping	-	select	pedestrians_unaware, pedestrians_aware, harbour_quay_equipment, trained_staff, vehicles_low_speed	trained_staff
beta	Wave angle	deg	slider	0 to 60, step 5	0
scour	Scour coefficient	-	slider	0 to 0.6, step 0.05	0.4
backfill	Backfill material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand
table	Water table below surface	m	slider	0 to 16, step 0.5	0

Input	Label	Unit	Control	Range or options	Default
surcharge	Surcharge on the promenade	kPa	slider	0 to 60, step 5	10

Breakwater

Rock armour by Van der Meer, crest by EurOtop. (module key breakwater)

Input	Label	Unit	Control	Range or options	Default
Hm0	Wave height Hm0	m	slider	0.5 to 9, step 0.1	4.0
TP	Peak period TP	s	slider	4 to 18, step 0.1	11.0
depth	Depth at toe	m	slider	2 to 40, step 0.5	12.0
cot	Seaward slope 1 :	-	slider	1.5 to 4, step 0.25	2.0
storm	Storm duration	h	slider	1 to 48, step 1	6
damage	Damage level S	-	slider	2 to 17, step 1	2
use	Tolerable overtopping	-	select	pedestrians_unaware, pedestrians_aware, harbour_quay_equipment, trained_staff, vehicles_low_speed	trained_staff
armour	Armour type	-	select	rock_two_layer_permeable, rock_two_layer_impermeable, cubes_two_layer_random, tetrapod, accropode, xbloc	rock_two_layer_permeable
bed	Seabed material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand
section	Section	-	select	trunk, head	trunk

Channel

PIANC depth chain and width build-up. (module key channel)

Input	Label	Unit	Control	Range or options	Default
length	Vessel length	m	slider	80 to 400, step 5	336
beam	Beam	m	slider	12 to 62, step 0.5	48.2
draught	Draught	m	slider	3 to 20, step 0.1	14.5
cb	Block coefficient	-	slider	0.5 to 0.9, step 0.01	0.68
speed	Speed	kn	slider	3 to 14, step 0.5	8
wl	Design water level	m CD	slider	-1 to 4, step 0.1	1.2
Hs	Wave height in channel	m	slider	0 to 4, step 0.1	1.8
factor	Wave response factor	-	slider	0.2 to 0.8, step 0.05	0.5
net	Net clearance	m	slider	0.3 to 2, step 0.1	0.8
bed	Existing bed	m CD	slider	-25 to -5, step 0.5	-11.5

Input	Label	Unit	Control	Range or options	Default
twoway	Two-way traffic	-	toggle	on / off	true
material	Bed material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand

Monopile

Morison loads through the wave cycle. (module key monopile)

Input	Label	Unit	Control	Range or options	Default
diameter	Pile diameter	m	slider	0.5 to 12, step 0.1	8.0
H	Wave height H	m	slider	1 to 20, step 0.5	12
T	Period T	s	slider	4 to 20, step 0.5	13
depth	Water depth	m	slider	5 to 60, step 1	30
rough	Marine growth	-	toggle	on / off	true
bed	Seabed material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand

Nourishment

What a borrow source is worth, on Dean's equilibrium profile. (module key nourishment)

Input	Label	Unit	Control	Range or options	Default
native	Native beach sand	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel	medium_sand
borrow	Borrow source	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel	coarse_sand
volume	Placed volume	m ³ /m	slider	20 to 1200, step 10	250
berm	Berm height	m	slider	0.5 to 5, step 0.1	2.0
closure	Depth of closure	m	slider	2 to 14, step 0.5	6.0
view	View	-	select	profile, planform	profile
length	Fill length (planform)	m	slider	200 to 6000, step 50	1500
Hb	Breaking wave height (planform)	m	slider	0.3 to 3.0, step 0.1	1.2

Return values

Generalised Pareto tail above a storm threshold. (module key extremes)

Input	Label	Unit	Control	Range or options	Default
threshold	Threshold	m	slider	0.5 to 8, step 0.1	2.5
scale	Scale	m	slider	0.1 to 3, step 0.05	0.55

Input	Label	Unit	Control	Range or options	Default
shape	Shape	-	slider	-0.4 to 0.4, step 0.01	-0.08
rate	Peaks per year	-	slider	0.5 to 30, step 0.5	6
record	Record length	yr	slider	5 to 100, step 1	40
period	Design return period	yr	slider	5 to 500, step 5	100

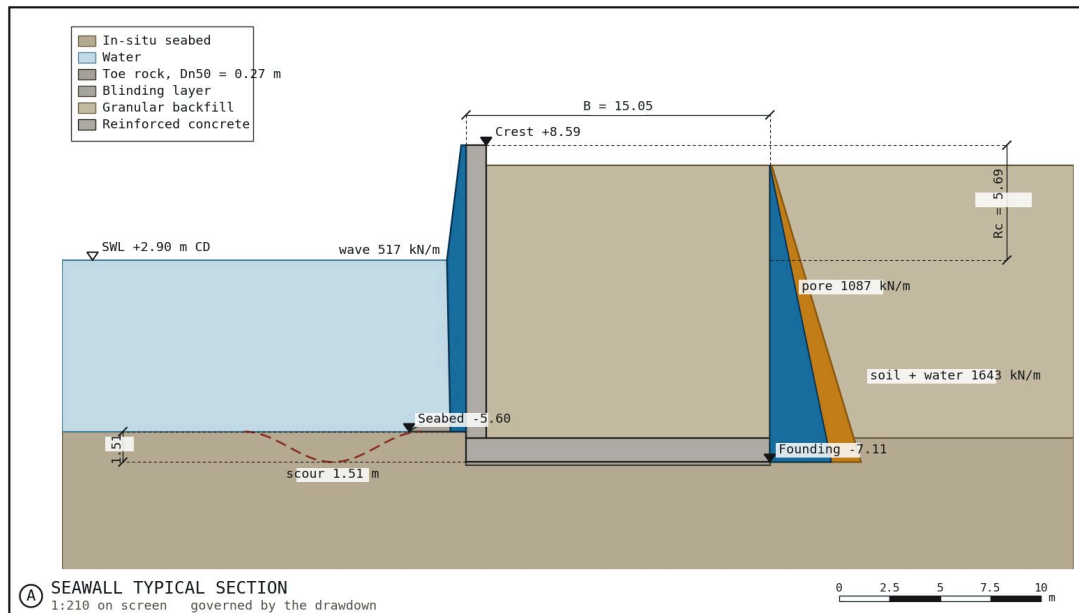


Figure 20.1: Seawall module sheet as rendered by draw.js.

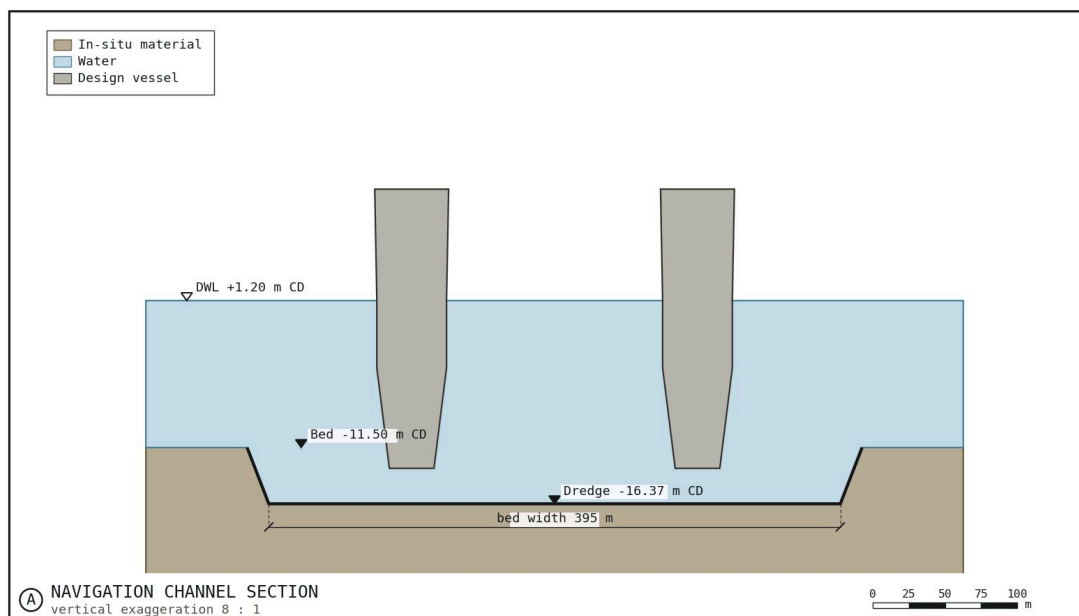


Figure 20.2: Channel module sheet.

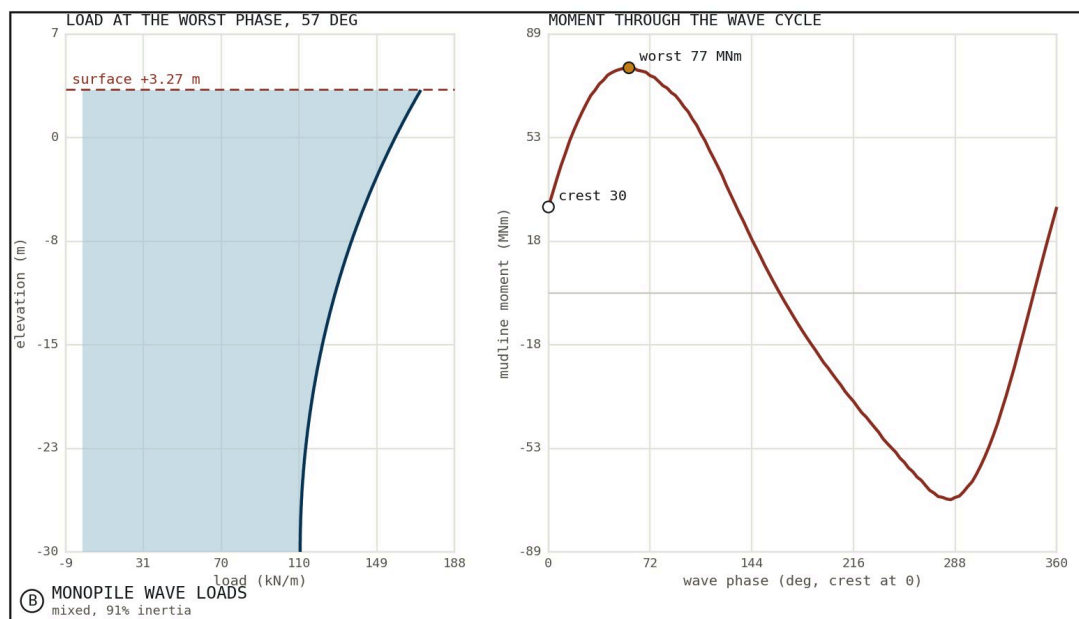


Figure 20.3: Monopile module sheet.

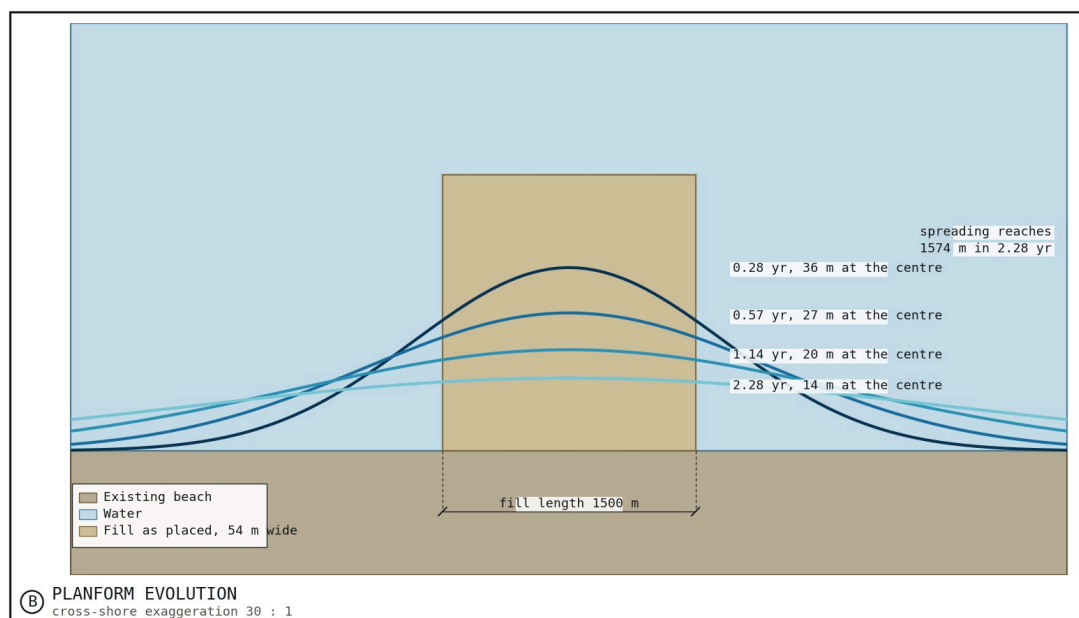


Figure 20.4: Nourishment module, planform view.

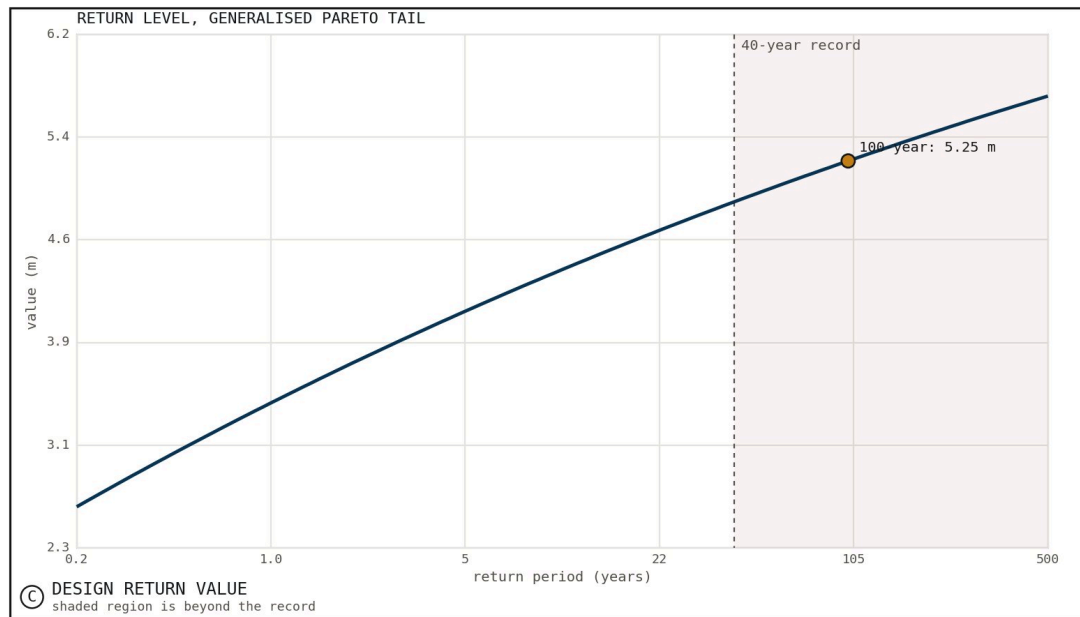


Figure 20.5: Return values module.

20.3 Theory and sources

The Theory and sources panel reads `knowledge.json`, an extract of the CoastalWiki database built by `build_knowledge.py`. The wiki database is 146 MB of SQLite, which no browser will open; the app needs the topic tree, the curated synthesis for each topic, the atomic claims with their regime bounds, the equations, and enough paper metadata to cite and link, which comes to under 2 MB. Each design module is mapped to the wiki topics that bear on it (the mapping is deliberately generous, because a design relation sits where several topics meet), and the panel shows each topic's synthesis, claims, and equations with a DOI behind every piece. Coverage is uneven by design: the wiki records what has been screened, not what exists. Part VIII of this manual reproduces the whole extract.

```
python webapp/build_knowledge.py --db ../2026_coastalwiki/data/coastalwiki.db \
    --out webapp/knowledge.json
```

20.4 Verification and review tools

- `verify_app.py` loads the page's own scripts into QuickJS behind a stub DOM, so the module definitions, run functions, report and check builders, and in-page verification all execute at least once before publishing. It catches the things that break a static page: a typo in a function name, an input key that no run function reads, a check that throws on the defaults.
- `render_preview.py` runs `draw.js` in QuickJS behind a DOM stub that serializes to real SVG, then rasterizes it, because a script that runs can still draw nothing (a sheet that paints its background last passes every check and shows a blank page). Output goes to `media/webapp/`.
- `rasterise.py` draws the app's SVG with matplotlib, since the SVG is deliberately narrow (paths of M, L, H, V, Z plus rects, circles, and text). It is a review tool: is the thing on the page, in the right place, in the right order.

```
pip install quickjs svglib
python webapp/make_vectors.py --out webapp/vectors.json
python webapp/verify_engine.py
python webapp/verify_app.py
python webapp/render_preview.py --out media/webapp
```

To use the app locally, serve the folder (`python -m http.server` inside `webapp/`) and open `index.html`; the page loads `knowledge.json` and `vectors.json` with `fetch`, which needs a server rather than a file URL.

Part VII. Quality assurance

21 The test suite

A design code that nobody checks is a liability, so pyCoastal carries a test suite that grew with every module. It runs with

```
python -m pytest tests/
```

and in continuous integration on Ubuntu and Windows under Python 3.10, 3.11, and 3.13 on every push to `main` and every pull request.

21.1 What the tests check

The tests are not only regression checks. Most of them pin a result to an independent reference, and the kinds of check recur across modules:

- **Published values and limits.** The Van der Meer plunging and surging branches against the published forms; the Sumer and Fredsoe relation collapsing to the waves-only form at $U_{cw} = 0$ and to $S/D = 1.3$ in pure current; the toe scour recovering Xie at $K_r = 1$; Dean's profile scale against his table.
- **Analytical solutions.** The one-line solver against Pelnard-Considere; the port solver against the semi-infinite breakwater shadow; the fill volume and critical volume for matched sand; normal depth against Manning's equation.
- **Physical consistency.** Monotonic responses (more overtopping at lower crest, deeper scour at higher velocity), conservation (volume in the nourishment model), and conventions (axis order, units, sign of levels).
- **Honest failure.** Functions that must refuse (cohesive beds in scour, normal depth on an adverse slope, non-positive freeboard in the overtopping relations) raise, and functions that must warn outside their range do so.
- **Drawings.** Sheets reject unknown paper sizes and report their fitted scale, and DXF export round-trips one line per edge with every layer queued.

21.2 Tests by file

The suite holds 842 test functions in 24 files (parametrized tests expand to more cases when run).

Table 21.1: Test files and what they cover.

File	Tests	Scope
<code>test_applications_berthing.py</code>	49	Berthing energy and fender selection.
<code>test_applications_bridge_scour.py</code>	47	Contraction and abutment scour: the other two thirds of HEC-18.
<code>test_applications_channel.py</code>	46	Navigation channel design: vessel, squat, clearance, width and depth.
<code>test_applications_extremes.py</code>	51	Extreme value analysis: L-moments, fits, return values and diagnostics.
<code>test_applications_nourishment.py</code>	25	Beach nourishment simulator.
<code>test_applications_nourishment_profile.py</code>	30	Cross-shore nourishment: equilibrium profiles and borrow compatibility.
<code>test_applications_piles.py</code>	45	Morison wave loads on piles: kinematics, coefficients, loads and scour.
<code>test_applications_plan_view.py</code>	34	The nourishment planform: the analytical map and the scales it is drawn at.
<code>test_applications_port.py</code>	28	Port layout simulator.
<code>test_applications_port_layouts.py</code>	24	Built-in port layouts and oblique wave incidence.
<code>test_applications_river.py</code>	47	Open channel hydraulics: normal and critical depth, backwater, afflux.
<code>test_applications_scour.py</code>	76	Scour at a pier in combined waves, tide and river current.
<code>test_applications_seawall.py</code>	52	Vertical seawall design: Goda pressures, stability, scour and toe.
<code>test_applications_sections.py</code>	34	Section drawings: geometry, layer offsets, quantities and sheets.
<code>test_applications_sediment.py</code>	36	Sediment and soil: material properties, mobility, and earth pressure.
<code>test_applications_structures.py</code>	70	Armour sizing and wave overtopping.

File	Tests	Scope
test_applications_surge.py	48	Storm surge components and flood mapping.
test_drafting.py	23	Drawing primitives: geometry, scale, and the DXF export.
test_numerics_grid.py	8	Grid geometry and boundary-index bookkeeping.
test_numerics_operators.py	17	Finite-difference operators, checked against analytic derivatives.
test_physics_conventions.py	8	The whole package uses one array layout: x is axis 0, y is axis 1.
test_tools_sediment.py	13	Sediment transport formulae: closed-form checks and physical scaling.
test_tools_structural.py	16	Armour sizing, run-up and wave force formulae.
test_tools_wave.py	15	Wave kinematics and spectra, checked against analytic limits.

21.3 The browser engine

The JavaScript engine of the Coastal Design Bench is checked against the Python by `webapp/verify_engine.py` on the reference cases of `webapp/make_vectors.py`, and the page repeats the comparison on load ([Section 20](#)). `webapp/verify_app.py` executes every module of the page once. These verifiers are not yet wired into continuous integration and are run by hand before publishing the app.

Part VIII. The CoastalWiki knowledge base

22 About the knowledge base

Every design relation in pyCoastal names a source and states where it stops being valid. CoastalWiki holds the same shape of information as queryable data: a provenance-first corpus of peer-reviewed coastal engineering, in which every claim carries the regime it was established in and a DOI behind it, because a relation quoted outside its range is the most expensive kind of mistake. The Coastal Design Bench (Part VI) ships an extract of the wiki and shows it beside each design; this part of the manual reproduces that extract in full.

22.1 What an entry contains

The wiki is organized as a topic tree (for example `structures`, `structures.overtopping`, `structures.overtopping.*`). Each topic carries:

- a **synthesis**: curated statements under fixed headings (what is well established, the governing physics, the dimensionless parameters, the major equations, typical methods, numerical models, experimental datasets, validated ranges, recent advances, disagreements, limitations, open questions, and the seminal papers);
- **claims**: atomic findings, each with the regime it holds in, a confidence class (direct finding, literature review statement, inferred relationship, or proposed hypothesis), the kind of evidence (field, experimental, numerical, analytical, mixed, or review), and the paper it comes from;
- **equations**: named relations in LaTeX with their variables and regime;
- a count of the **papers** screened for the topic.

Citations in the following chapters are given as [number], pointing to the wiki bibliography in Appendix E, where each entry carries its DOI.

22.2 Coverage

The extract was generated on 2026-09-17. The full wiki holds 4,926 papers, 1,751 claims, 163 equations, 156 topics, and 2,071 synthesis statements. The extract shipped with the app, and reproduced here, holds 156 topics, 1,751 claims, 163 equations, and 1,638 cited papers.

Coverage is uneven by design: the wiki records what has been screened, not what exists. A topic with few claims is a topic that has not yet been read closely, not one where nothing is known.

22.3 How the chapters are arranged

The next chapter follows the Coastal Design Bench: for each design module it lists the topics the app shows beside that module. The chapters after it reproduce every topic of the tree, grouped by top-level branch, with its synthesis, equations, and claims. Topics mapped to a design module are reproduced once, in the tree, and cross-referenced from the module chapter.

23 The knowledge base by design module

Each design module of the Coastal Design Bench opens the wiki topics below in its Theory and sources panel. The topics themselves are reproduced in the chapters that follow.

23.1 Design conditions

Return values and their uncertainty, from a record. (module key extremes)

Topic	Papers	Claims	Equations
Extreme wave climates (<code>waves.generation.extremes</code>)	26	25	0
Structural and system reliability (<code>uncertainty.reliability</code>)	5	3	1
Aleatory and epistemic uncertainty (<code>uncertainty.sources</code>)	13	10	0
Probabilistic hazard analysis (<code>risk.probablistic</code>)	20	17	7
Extreme-event observations (<code>field.extreme</code>)	10	7	5

23.2 Seawall design

Crest level, wave pressures, stability, scour and toe. (module key seawall)

Topic	Papers	Claims	Equations
Seawalls and revetments (<code>structures.seawalls</code>)	16	11	0
Overtopping (<code>structures.overtopping</code>)	22	17	0
Structural failure and damage (<code>structures.failure</code>)	4	3	0
Local scour at structures (<code>scour.structures</code>)	14	14	0
Scour protection (<code>scour.protection</code>)	10	10	0
Design conditions and criteria (<code>design.criteria</code>)	5	3	0

23.3 Breakwater design

Rock armour stability and overtopping. (module key structures)

Topic	Papers	Claims	Equations
Breakwaters (<code>structures.breakwaters</code>)	10	7	3
Overtopping (<code>structures.overtopping</code>)	22	17	0
Transmission and reflection (<code>wave_structure.transmission</code>)	16	13	0
Porous structures (<code>wave_structure.porous</code>)	19	19	0
Performance-based design (<code>design.performance</code>)	3	2	1

23.4 Navigation channel

Depth chain, squat and channel width. (module key channel)

Topic	Papers	Claims	Equations
Navigation and vessel response (<code>ports.navigation</code>)	23	19	0
Harbor sedimentation and dredging (<code>ports.sedimentation</code>)	15	13	0
Mooring hydrodynamics (<code>ports.mooring</code>)	10	8	0
Codes and design guidance (<code>design.guidance</code>)	10	7	0

23.5 Pile wave loads

Morison drag and inertia, and pile scour. (module key piles)

Topic	Papers	Claims	Equations
Wave loads (wave_structure.loads)	22	13	7
Impact and impulsive loads (wave_structure.loads.impact)	6	4	2
Local scour at structures (scour.structures)	14	14	0
Pipeline and cable scour (scour.pipelines)	18	18	0

23.6 Storm surge and flooding

Water level budget and inundation extent. (module key surge)

Topic	Papers	Claims	Equations
Surge forcing (storm_surge.forcing)	28	28	0
Wind stress and drag (storm_surge.forcing.wind)	22	28	1
Atmospheric pressure forcing (storm_surge.forcing.pressure)	11	11	0
Wave setup contribution to storm surge (storm_surge.coupling.wave_setup)	5	15	1
Compound flooding (storm_surge.compound)	17	16	8
Wave runoff (waves.transformation.breaking.runup)	12	9	0
Inundation modeling (inundation.models)	23	12	0
Flood-map uncertainty (inundation.uncertainty)	25	21	0

23.7 Harbour agitation

Diffraction into a harbour and berth downtime. (module key port)

Topic	Papers	Claims	Equations
Harbor agitation and resonance (ports.agitation)	14	17	1
Diffraction (waves.transformation.diffraction)	11	5	1
Reflection (waves.transformation.reflection)	5	2	0

23.8 Beach nourishment

One-line shoreline response to a fill. (module key nourishment)

Topic	Papers	Claims	Equations
Beach nourishment (beaches.nourishment)	10	8	2
Longshore currents (nearshore.longshore)	11	9	0
Planform evolution (morphodynamics.planform)	8	4	0
Shoreline change (beaches.change)	20	13	2

24 Wiki: Analytical methods

24.1 Analytical methods: overview

Topic analytical. Closed-form and asymptotic coastal-process analysis.

Papers screened: 18. Claims: 10. Equations: 0.

Well established. Analytical coastal models reduce conservation laws and boundary conditions to exact, approximate, asymptotic, similarity, perturbation, spectral, or reduced-order relations whose assumptions remain visible and testable.

Governing physics. Mass, momentum, energy and sediment conservation are closed with gravity, pressure, dispersion, nonlinearity, friction, turbulence or transport parameterizations and boundary conditions for the bed, shoreline, free surface, currents and structures.

Dimensionless parameters. Regime selection depends on relative depth, wave steepness, Ursell and Froude numbers, surf similarity, current-to-wave speed, relative bathymetric or boundary amplitude, bandwidth, sediment mobility and fall-velocity scaling, and ratios among forcing, dissipation and adjustment scales.

Major equations. Core families include potential-flow and free-surface boundary conditions, shallow-water and Boussinesq-type equations, wave-action and spectral balances, radiation-stress or momentum balances, Exner continuity, transport energetics, eigenvalue/scattering systems and perturbation expansions.

Typical methods. Methods nondimensionalize the governing problem, identify small or large parameters, impose idealized initial and boundary conditions, derive exact or asymptotic solutions, test limiting behavior, and compare with laboratory, field or converged numerical evidence.

Numerical models. Analytical solutions serve as standalone reduced-order models and verification benchmarks for spectral wave, shallow-water, Boussinesq, morphodynamic, scattering, wave-current and fluid-structure solvers.

Experimental datasets. The reviewed evidence connects theory to finite-depth crest and setup measurements, truncated-beach overtopping experiments, estuarine fetch/depth wave observations, and coastal wave or morphology datasets used to constrain asymptotic regimes.

Validated ranges. Examples span weakly nonlinear finite-depth waves, fetch- and depth-limited basins, long waves over gently varying or corrugated beds, idealized planar beaches, partial reflection, deformable boundaries and energetics-based sandy shoreface adjustment.

Recent advances. Recent analytical work increasingly couples currents, complex or compliant boundaries, partial reflections and morphology, and uses observations plus high-fidelity computation to map validity rather than presenting a closed form without an error regime.

Disagreements. Alternative closures can share conservation laws yet predict different asymptotes; exactness under idealized boundaries does not imply field accuracy; fitted empirical coefficients can outperform theory locally while transferring less reliably; formally higher order is not always better outside its scaling regime.

Limitations. Tractability usually requires simplified geometry, weak nonlinearity or dispersion, scale separation, prescribed closures, narrow spectra, uniform sediment or currents, and idealized boundaries; breaking, turbulence, three-dimensionality and stochastic forcing often enter indirectly.

Open questions. Priorities are analytical hierarchies for compound forcing, irregular directional waves, nonlinear breaking, mixed sediments, porous and deformable boundaries, coupled morphology, nonstationarity, uncertainty propagation and explicit transition criteria between model orders.

Seminal papers. Classical linear and nonlinear wave theories, shallow-water characteristics, perturbation and multiple-scale methods, radiation stress, spectral balances and sediment-continuity reductions established the analytical vocabulary used by coastal engineering.

Claims

1. Reanalysis of fully developed wind seas relates asymptotic integral spectral limits to the source-term balances used to validate operational wave models. *Regime: Revisiting the Pierson–Moskowitz Asymptotic Limits for Fully Developed Wind Waves* direct finding, analytical [48]
2. An energetics-based cross-shore sediment-transport formulation yields a mechanistic morphodynamic depth of closure rather than prescribing one solely from wave statistics. *Regime: Exploring shoreface dynamics and a mechanistic explanation for a morphodynamic depth of closure* direct finding, analytical [1057]
3. Second-order three-dimensional finite-depth theory explains setup and surface-elevation and crest statistics when evaluated against controlled experimental data. *Regime: Second-Order Theory and Setup in Surface Gravity Waves: A Comparison with Experimental Data* direct finding, mixed [1413]
4. Fetch- and depth-limited estuarine wind waves can be organized by nondimensional scaling that distinguishes growth constrained by basin geometry and water depth. *Regime: Wind Wave Behavior in Fetch and Depth Limited Estuaries* direct finding, mixed [688]
5. Weakly dispersive surface and internal waves admit reduced analytical descriptions of interacting long-wave structures, with applicability bounded by the asymptotic assumptions. *Regime: Analytical behavior of weakly dispersive surface and internal waves in the ocean* direct finding, analytical [74]
6. Theory and experiments for a truncated planar beach relate an individual incident wave to overtopped volume and identify the limits of idealized run-up representations. *Regime: Overtopping a truncated planar beach* direct finding, mixed [607]
7. An asymptotic solution describes Bragg scattering by variable bathymetry with a uniform current and floating membrane, exposing how current, depth variation, and compliance modify resonance. *Regime: Bragg scattering of gravity waves by a sea bed of varying depth in the presence of uniform current covered by a floating membrane* direct finding, analytical [120]
8. A nonlinear shallow-water formulation coupled to an elastic sheet on a viscoelastic foundation describes wave propagation over a deformable seabed. *Regime: Nonlinear waves propagating over a deformable seafloor* direct finding, analytical [738]

9. A free-long-wave framework with partial reflections resolves nearshore transformation relevant to inundation and shoreline or dune erosion. *Regime: Free Long-Wave Transformation in the Nearshore Zone through Partial Reflections* direct finding, analytical [294]
10. Perturbation analysis shows how a wavy bed modifies currents induced by long waves approaching a beach, including nearshore and swash-zone circulation effects. *Regime: Currents induced by long waves propagating towards a beach over a wavy bed* direct finding, analytical [758]

24.2 Linear wave theory

Topic analytical. Linear. Linearized wave solutions.

Papers screened: 6. Claims: 4. Equations: 0.

Well established. Linear wave theory remains the reference model for small-amplitude surface-gravity-wave dispersion, kinematics, energy flux, shoaling, refraction and diffraction. Its usefulness depends on stating depth, slope, amplitude, directionality and scattering assumptions.

Governing physics. Core physics are gravity-restored free-surface motion, depth-dependent dispersion, conservation of energy or wave action, bathymetric refraction and diffraction, and slope-dependent vertical structure.

Dimensionless parameters. Key regime controls include relative depth kh , H/h , wave steepness, Ursell number, bed slope, directional spread and topographic length relative to wavelength.

Major equations. Common formulations include the Airy dispersion relation, linear velocity-transfer functions, shoaling and group-velocity relations, mild-slope and coupled-mode equations, and spectral wave-action balance.

Typical methods. Methods compare analytical solutions with field velocity profiles, laboratory transformation data, stochastic moment equations, non-hydrostatic solvers and operational spectral hindcast-nowcast-forecast systems.

Numerical models. Models include sloping-bed linear transfer theory, forward-scattering stochastic coupled modes, a two-layer non-hydrostatic solver benchmarked against linear dispersion, and SWAN driven by WAM, WW3 or buoy observations.

Experimental datasets. Reviewed evidence includes storm observations over a 2-degree nearshore slope, three laboratory cases including a fringing reef and barred beach, laboratory topographic-scattering observations, and six regional operational-model applications.

Validated ranges. Explicit abstract-level bounds include $H/h < 0.3$ and Ursell number < 0.6 on a 2-degree slope, and improved two-layer dispersion and shoaling through $kh=4$ compared with a one-layer shallow-water range $kh < 1$.

Recent advances. Recent developments improve dispersion through layered non-hydrostatic pressure structure, retain coherent interference in stochastic coupled modes, and integrate increasingly resolved coastal domains into operational spectral systems.

Disagreements. A model can match bulk wave height while scattering more in setup or second-order statistics, and horizontal-bottom kinematics can fail near a slope even when horizontal velocity remains little affected. Accuracy therefore depends on the diagnostic, not merely the model label.

Limitations. Linearization omits finite-amplitude harmonic coupling, breaking, strongly nonlinear runup and some current interactions. Forward-scattering, hydrostatic structure, spectral closure and abstract-only evidence impose additional source-specific limits.

Open questions. Priorities include quantified switching criteria between linear and nonlinear models, uncertainty propagation from bathymetry and forcing, efficient wide-angle scattering, and common benchmarks spanning kinematics, spectra, setup and extremes.

Seminal papers. The branch rests on classical Airy dispersion and kinematics, energy-flux shoaling, mild-slope transformation and action-balance modeling; the current review slice validates selected descendants rather than establishing a complete historical canon.

Claims

1. For observed nearshore waves on a 2-degree bed slope with H/h below 0.3 and Ursell number below 0.6, sloping-bed linear theory reproduced near-bed vertical-velocity magnitude and phase substantially better than horizontal-bottom theory, while horizontal velocity was comparatively insensitive to slope. *Regime: Near-bed observations on a 2-degree slope for H/h below 0.3 and Ursell number below 0.6 during two storm intervals.* direct finding, mixed [1626]
2. Against linear-wave dispersion and shoaling benchmarks, the two-layer non-hydrostatic model extended useful behavior from the one-layer model's shallow-water range $kh < 1$ to shallow and intermediate depths through $kh=4$, though setup and second-order statistics showed more scatter than bulk and spectral wave height. *Regime: Shallow to intermediate water through $kh=4$, with validation cases including a fringing reef and barred beach.* direct finding, mixed [1190]
3. For the tested two-dimensional topographies, a stochastic coupled-mode model based on forward scattering reproduced coherent interference and the combined wide-angle refraction-diffraction effects in agreement with analytical solutions and laboratory observations within its linear-interaction focus. *Regime: Random directionally spread waves over fully two-dimensional coastal topography under a forward-scattering approximation; the reported paper focuses on linear topographic interaction.* direct finding, mixed [648]
4. Across six ocean and coastal applications, SWAN-based hindcast, nowcast, and forecast schemes provided an effective wave-prediction framework when forced by WAM, WW3, or buoy observations, but the abstract does not establish universal accuracy or isolate linear-theory error. *Regime: SWAN applications from high-resolution coastal domains to quasi-oceanic scales in Portugal, Madeira, Sardinia, and the Black Sea.* review statement, review [1226]

24.3 Nonlinear and asymptotic theory

Topic analytical. nonLinear. Perturbation and nonlinear solutions.

Papers screened: 4. Claims: 2. Equations: 0.

Well established. Nonlinear coastal theory is useful when finite amplitude, shallow depth, moving shorelines, wave asymmetry, harmonic coupling, dispersion, or morphology feedback invalidate linear superposition. Its value depends on explicit ordering assumptions and validation inside the retained regime.

Governing physics. Key mechanisms are finite-amplitude kinematics and dynamic boundary conditions, weak dispersion, quadratic and higher-order harmonic coupling, free-bound interference, nonlinear shoaling and runup, moving wet-dry boundaries, wave-current diffusion, settling lag, and feedback toward or away from morphological equilibrium.

Dimensionless parameters. Approximation choice is organized by wave steepness, relative depth kh , Ursell number, relative amplitude H/h , dispersion and nonlinearity orders, surf similarity, Froude number, Shields parameter, suspension number, settling-to-advection time ratio, and morphology relaxation time.

Major equations. Principal frameworks include nonlinear shallow-water equations, Boussinesq and extended mild-slope equations, Stokes and perturbation expansions, eigenfunction wavemaker solutions, asymptotic convection-diffusion reductions, exact Lagrangian seiche solutions, stochastic half-cycle transport, and dynamic-equilibrium profile laws.

Typical methods. Methods derive ordered reduced equations, exact or semi-analytical solutions, matched or regular perturbation expansions, spectral/eigenfunction decompositions, stochastic averaging, equilibrium stability analysis, and comparisons with laboratory, field, or higher-dimensional numerical solutions.

Numerical models. Reviewed implementations include semi-analytical eigenfunction-FFT wavemakers, higher-order Boussinesq solvers, nonlinear shallow-water inundation models, SWAN phase-decoupled diffraction, quasi-3-D transport, exact seiche benchmarks, stochastic bedload models, and dynamic-equilibrium shoreline models.

Experimental datasets. Theory has been checked against transient-wave flumes, Tainan nonlinear shoaling tests, Synolakis plane-beach and Briggs conical-island runup, Monai valley inundation, current-wave suspended-transport measurements, and multi-year shoreline or inlet observations.

Validated ranges. Examples include Tainan $H=0.05\text{--}0.40$ m and $T=2\text{--}5$ s, plane-beach $H/d=0.021\text{--}0.626$ with breaking above 0.045 on a 1:19.85 slope, conical-island $H/d=0.10$ and 0.20, narrow-band second-order random waves, and source-specific low-to-steep wavemaker tests.

Recent advances. Recent work combines free-bound harmonic separation, nonlinear spectral diagnostics, stochastic transport, benchmark-driven model hierarchies, and dynamic-equilibrium reduced models to retain essential physics while remaining efficient enough for design or forecasting.

Disagreements. A reduced model may solve its parent equations accurately yet omit decisive physics. Linear or weakly nonlinear wavemaker theory fails at higher steepness; nonlinear shallow-water theory cannot reproduce dispersive or post-breaking behavior; local linear wavelengths can misinterpret free-bound spatial beating.

Limitations. Asymptotic truncation, idealized geometry, inviscid or hydrostatic assumptions, weak dispersion, prescribed profile shape, unidirectionality, nonbreaking conditions, friction closure, and scale separation constrain transfer. Mathematical existence alone does not establish coastal relevance.

Open questions. Needs include uniform theories across shoaling, breaking and runup; multidirectional and current-coupled nonlinear interactions; error bounds for truncation and closure; efficient stochastic extremes; sediment and morphology coupling; and principled switching among reduced and process-resolving models.

Seminal papers. Within the reviewed corpus, exact shallow-water shoreline benchmarks, Boussinesq runup validation, mild-slope-derived diffraction, quasi-3-D transport asymptotics, and nonlinear wavemaker theory form reusable analytical foundations across coastal applications.

Claims

1. Sine-Gordon and generalized Kudryashov methods yield soliton solutions of the Boussinesq equation used for weakly nonlinear long waves in harbors and shallow seas. *Regime: Soliton solutions to the Boussinesq equation through sine-Gordon method and Kudryashov method* direct finding, analytical [23]
2. Nonlinear WKB and perturbation analyses show a convex non-reflecting beach generates higher harmonics during wave propagation. *Regime: Nonlinear wave effects at the non-reflecting beach* direct finding, analytical [340]

24.4 Similarity and scaling analysis

Topic analytical.similarity. Dimensional and similarity methods.

Papers screened: 9. Claims: 8. Equations: 0.

Well established. Physical-model similarity requires matching dominant nondimensional controls; geometric resemblance alone does not guarantee dynamic similarity when gravity, viscosity, elasticity, air entrainment, granular drag, structural response, or power conversion compete.

Governing physics. Froude scaling preserves gravity-inertia balance, Reynolds scaling represents viscous effects, Cauchy-type controls represent elastic-fluid interaction, and power similarity governs energy-conversion stages; exact simultaneous similitude is often impossible.

Dimensionless parameters. Reviewed controls include slide Froude number, relative slide thickness and mass, grain Reynolds number, Cauchy and WFSI parameters, fence porosity and relative thickness, and stage-specific power ratios.

Major equations. Core tools are Buckingham-Pi dimensional analysis and model-prototype scale ratios for length, time, velocity, force, pressure, stiffness, power, Froude number, Reynolds number, Cauchy number, and process-specific relative geometry.

Typical methods. Methods combine multi-scale flume tests, systematic parameter variation, prototype-model numerical comparison, discrete-element or hydrodynamic simulation, calibration, uncertainty/error estimation, and mechanism-by-mechanism scaling orchestration.

Numerical models. Numerical support includes discrete-element slide simulations, prototype/model wave-flexible-plate simulations, SWASH fence upscaling, and process-decomposed WEC scaling frameworks.

Experimental datasets. Evidence includes 144 block-slide tests, factor-four granular-slide tests in a 6 m apparatus, plate-impact models up to 40 times smaller, 1:5 wooden-fence tests, and three WEC PTO case studies.

Validated ranges. Reported bounds include wave-height effects up to a factor two from block-slide parameters, granular velocity and runout effects of 35% and 26%, breaking-pressure errors to 132%, displacement errors to 98%, new WFSI deviations below 4.3%, and fence scale bias near 5%.

Recent advances. Recent work uses paired numerical and physical models to quantify rather than merely note scale effects, proposes partial-parameter WFSI scaling, orchestrates similarity across energy-conversion stages, and separates scale bias from design-variable effects.

Disagreements. Rigid landslide blocks do not necessarily generate larger waves than granular slides, and a scaling law adequate for nonbreaking loads can fail badly for breaking pressure or flexible response. Apparent disagreement reflects unmatched secondary physics and response choice.

Limitations. Abstract-level evidence, laboratory confinement, finite scale range, unscaled air or fluid properties, particle stiffness, material substitution, simplified PTO behavior, and incomplete prototype validation bound transfer.

Open questions. Needs include optimal multi-physics scale selection, principled distortion correction, uncertainty propagation to prototypes, shared multi-scale benchmarks, and field validation for breaking impacts, nature-based structures, slides, and energy devices.

Seminal papers. Classical Froude similitude and Buckingham-Pi analysis provide the foundation; the reviewed corpus demonstrates why Reynolds, elasticity, grain-air, porosity, and power constraints must be added for specific coastal processes.

Claims

1. Across 144 Froude-model block-slide tests, slide Froude number, relative thickness, and relative mass changed wave amplitude and height by as much as a factor of two, and rigid blocks did not systematically generate larger waves than deformable granular slides. *Regime: Systematic variation of slide Froude number, relative thickness, and relative mass in physical landslide-tsunami models.* direct finding, experimental [584]
2. For wave impacts on plates modeled at scales down to 1:40, precise Froude scaling removed detectable scale effects, whereas unscaled fluid properties produced breaking-pressure errors up to 132% and traditional Froude scaling underestimated displacement by up to 98%; a new WFSI approach kept validated peak deviations below 4.3%. *Regime: Rigid and flexible plates under breaking and non-breaking regular and solitary waves, with models up to 40 times smaller.* direct finding, mixed [85]
3. A review of wave-energy-converter power-take-off experiments found that credible small-scale reproduction requires explicit setup enhancement, calibration, and error estimation, with some recommendations transferable across technologies despite device-specific scaling choices. *Regime: Small-scale wave-energy-converter PTO experiments across technology-dependent configurations.* review statement, review [496]
4. For oscillating-buoy wave-energy-converter model tests, the proposed orchestration applies Froude similarity to the water-contacting capture stage and power similarity to later conversion stages without requiring their geometric similitude. *Regime: Oscillating-buoy WEC model tests decomposed into capture and downstream conversion stages.* direct finding, analytical [1133]
5. For porosity-0.81 wooden fences, a SWASH model validated against 1:5 tests with 4.7% transmitted-height RMSE predicted model-scale reflection and transmission about 5% above full-scale values, smaller than the effects of fence thickness and porosity. *Regime: Wooden fence porosity 0.81 with varied thickness, comparing 1:5 and full-scale simulations.* direct finding, mixed [317]
6. Across factor-four Froude-scaled dry granular-slide models, nondimensional surface velocity increased up to 35% and runout up to 26% from smallest to largest scale; correlation with grain Reynolds number and air-free DEM comparisons implicated particle-air drag. *Regime: Dry granular slides in a 6 m apparatus across a factor-four scale range.* direct finding, mixed [706]
7. For Froude-scaled dry granular slides in 0.25-1.00 m channels at grain Reynolds numbers of order 10^2 - 10^3 , experiments showed significant velocity and runout scale effects that the discrete-element model did not capture at the smaller scale. *Regime: Dry granular slides in channels 0.25-1.00 m wide with grain Reynolds numbers of order 10^2 - 10^3 .* direct finding, mixed [705]
8. Similarity criteria relate turbulent vertical-displacement time scales to phytoplankton photoadaptation time scales, identifying regimes in which vertical mixing alters light exposure and production response. *Regime: Relationships between vertical mixing and photoadaptation of phytoplankton: similarity criteria* direct finding, analytical [807]

25 Wiki: Beaches and shoreline evolution

25.1 Beaches and shoreline evolution: overview

Topic beaches. Beach state and shoreline change.

Papers screened: 17. Claims: 15. Equations: 0.

Well established. Beaches are open sediment systems whose shoreline and profile evolve through coupled cross-shore and longshore transport, changing water levels, sediment sources and sinks, antecedent morphology and human interventions.

Governing physics. Wave and current stresses move sediment; water level controls where energy is delivered; gradients in longshore transport change planform; cross-shore exchange reshapes profiles; rivers, cliffs, shelf stores, dunes and engineering alter the budget; morphology feeds back on forcing.

Dimensionless parameters. Controls include wave steepness and direction, relative water level, surf similarity, fall-velocity parameter, closure depth, profile slope, sediment size, transport-gradient and recovery time scales, compartment leakage and sediment-input/transport-capacity ratios.

Major equations. Core formulations are Exner sediment continuity, one-line shoreline continuity, equilibrium-profile and disequilibrium response, cross-/longshore transport laws, sediment-budget mass balance, state-space/data assimilation and coupled shelf-shoreline models.

Typical methods. Methods combine repeated topobathymetry, satellite shorelines, lidar/UAS, sediment tracing and budgets, wave/water-level records, trend segmentation, equilibrium and process models, multi-model frameworks, data assimilation and scenario analysis.

Numerical models. Represented tools include shoreline trend clustering, satellite-assimilative dynamic shoreline models, sediment-budget calculators, coupled cross-/longshore and shelf-shoreline models, equilibrium response, mesoscale multi-model frameworks and climate-driver attribution.

Experimental datasets. Evidence includes Australia-wide Landsat coastlines, global satellite shorelines, the Elwha dam-removal coast, West African and Ebro sediment systems, Shoalhaven compartments and modeled shelf-ridge/shoreline coupling.

Validated ranges. The corpus spans sandy open coasts and barriers, adjacent managed compartments, river- and port-modified littoral cells, low-gradient deltas, seasonal through multi-decadal records, national mapping and global climate variability.

Recent advances. Recent work uses decades of satellite shorelines, data assimilation, cluster-based reaches, global climate attribution, intervention natural experiments, coupled shelf-shoreline models and hierarchical management compartments.

Disagreements. A shoreline line alone may indicate retreat while subaerial volume or profile storage behaves differently; equilibrium models are efficient but miss pathways and memory; local trends may reflect basin climate rather than secular forcing; fixed cells may leak sediment.

Limitations. Shoreline proxies and tidal correction introduce error; bathymetric records are sparse; sediment budgets rarely close; recovery may exceed monitoring duration; intervention histories are incomplete; model parameters are nonstationary and scale-dependent.

Open questions. Priorities include profile-planform coupling, shelf exchange, event recovery and memory, mixed sediment, compound water levels, human intervention attribution, dynamic compartment boundaries and probabilistic multi-decadal forecasting.

Seminal papers. Profile and one-line shoreline models established reduced beach evolution, while sediment budgets and littoral-cell concepts established source-pathway-sink management.

Claims

1. Three decades of Landsat observations provide consistent mean-sea-level coastline mapping for measuring Australia-wide dynamic shoreline change. *Regime: Mapping Australia's dynamic coastline at mean sea level using three decades of Landsat imagery* direct finding, mixed [159]
2. Cluster-based segmentation improves shoreline-trend analysis by identifying spatially coherent change regimes rather than imposing arbitrary reaches. *Regime: Understanding coastal change using shoreline trend analysis supported by cluster-based segmentation* direct finding, numerical [217]
3. Removal of a large dam restored fluvial sediment supply and reversed coastal erosion, demonstrating direct sediment-budget control of shoreline trajectory. *Regime: World's largest dam removal reverses coastal erosion* direct finding, field [1489]
4. Assimilating satellite-derived shorelines into a dynamic model enables fine-scale wave- and sea-level-driven predictions across large coastal regions. *Regime: A Model Integrating Satellite-Derived Shoreline Observations for Predicting Fine-Scale Shoreline Response to Waves and Sea-Level Rise Across Large Coastal Regions* direct finding, mixed [1458]
5. A quantitative West African sediment budget separates river regulation, port interception, natural transport and climate effects on a connected coastal barrier. *Regime: A quantitative assessment of human interventions and climate change on the West African sediment budget* direct finding, mixed [497]
6. Integrated analysis of the Ebro Delta links storms, morphology, climate change, river regulation and development to combined flooding and erosion. *Regime: Coastal Flooding and Erosion under a Changing Climate: Implications at a Low-Lying Coast (Ebro Delta)* direct finding, mixed [526]
7. A mesoscale framework combines complementary models to represent interacting shoreline processes for decadal-to-centennial coastal management. *Regime: Simulating mesoscale coastal evolution for decadal coastal management: A new framework integrating multiple, complementary modelling approaches* direct finding, numerical [886]
8. Global satellite shorelines show El Niño reorganizes shoreline-position variability through basin-scale changes in waves and water levels. *Regime: Influence of El Niño on the variability of global shoreline position* direct finding, field [36]
9. Intra-seasonal sea-level anomalies exacerbate beach erosion independently of named storms, showing water-level history modifies morphological response. *Regime: Sea level anomalies exacerbate beach erosion* direct finding, field [1407]
10. A review frames coastal morphodynamics as coupled feedback among waves, water levels, sediment, morphology and climate across scales. *Regime: Coastal Morphodynamics and Climate Change: A Review of Recent Advances* review statement, review [1530]

11. Coastal compartments provide a hierarchical management unit in which sediment supply and local morphodynamics explain differing behavior of adjacent beaches. *Regime: Coastal compartments: the role of sediment supply and morphodynamics in a beach management context* direct finding, mixed [233]
12. A coupled shelf-shoreline model shows waves and tides link kilometer-scale shelf sand ridges to long-term shoreline morphodynamics. *Regime: Long-Term Morphodynamics of a Coupled Shelf-Shoreline System Forced by Waves and Tides, a Model Approach* direct finding, numerical [1030]
13. Nearly five years of Torrey Pines observations show an approximately 40 m seasonal shoreline retreat associated with incident wave-energy changes and support an equilibrium shoreline-response model. *Regime: Equilibrium shoreline response: Observations and modeling* direct finding, mixed [1568]
14. Geologically controlled beaches differ systematically from unconstrained beaches because headlands, rock platforms and coral or bedrock controls set accommodation, wave shadows, rip circulation, sediment-budget closure, rotation and storm response along both cross-shore and longshore gradients. *Regime: Sandy beaches constrained by headlands, rock platforms, coral formations or other basement geology.* review statement, review [475]
15. Multivariate analysis of 92 English and Welsh beaches identifies nine morphodynamic types: dimensionless fall velocity and relative tide range separate groups, but absolute wave-energy flux is additionally needed to distinguish intermediate beaches with three-dimensional bar-rip morphology. *Regime: Gravel, sand, embayed, exposed, macro- and mesotidal beaches across England and Wales under their mean wave, tide, sediment, and morphological conditions.* direct finding, mixed [1272]

25.2 Shoreline change

Topic beaches . change. Retreat, advance, and variability.

Papers screened: 20. Claims: 13. Equations: 2. Shown in the Coastal Design Bench beside: Beach nourishment.

Well established. Beach change is spatially and temporally variable: shoreline indicators respond to waves, water level, sediment supply, morphology and engineering, while erosion at one interval or transect can coexist with accretion nearby or later.

Governing physics. Cross-shore storm erosion and recovery redistribute profile sediment, alongshore transport gradients rotate or translate shorelines, inlets and headlands reorganize cells, and nourishment or construction alters sediment budgets; waterline position also moves instantaneously with tide, runup and setup.

Dimensionless parameters. Controls include shoreline displacement relative to positional uncertainty, observation duration relative to storm and recovery cycles, beach width normalized by wave runup, sediment-budget imbalance, closure-depth ratio, wave obliquity, nourishment volume per shoreline length and erosion/accretion cluster significance.

Major equations. Change metrics fit shoreline position versus time for endpoint or linear-regression rates and uncertainty, compute transect displacement and area or volume budgets, and relate gradients in alongshore transport to shoreline movement through one-line continuity; probabilistic methods characterize indicator and classification error.

Typical methods. Workflows define a repeatable shoreline proxy, harmonize imagery and datums, correct tide or waterline effects, quantify georeferencing and classification uncertainty, cast stable transects, calculate rates and confidence intervals, detect clusters or change points, validate against surveys and interpret results with forcing and interventions.

Numerical models. The reviewed set emphasizes satellite classification and statistical change analysis, including fuzzy c-means and spatial clustering. Hybrid two-dimensional/one-line, moment and climate-response shoreline models remain in the lawful-full-text queue.

Experimental datasets. Reviewed evidence includes 769 Haikou images from 1986-2023, Iztuzu transects and clusters for 2013-2022, national cross-mission remote sensing, northwest Ireland waterlines from 1999-2024 and fuzzy-classified northern Java change from 1994-2000.

Validated ranges. Findings cover selected urban, nourished, storm-affected, dissipative, reflective and deltaic coasts over years to decades. Reported rates and intervention associations are site-, proxy-, resolution- and interval-specific.

Recent advances. Recent advances combine cross-mission optical and SAR imagery, subpixel and probabilistic classification, lidar and video validation, cloud-scale transect analysis, Bayesian change points, physics-informed shoreline models and explicit propagation of waterline and geolocation uncertainty.

Disagreements. Different shoreline proxies—instantaneous waterline, wet/dry line, vegetation line or modeled datum contour—can yield different trends. Linear rates can conceal episodic storms, recovery and interventions, and apparent accretion need not imply positive subaqueous sediment volume.

Limitations. Clouds, pixel size, georegistration, tidal stage, runup, datum mismatch, proxy migration, sparse ground truth, short or uneven sampling, anthropogenic confounding and absent profile volume limit attribution. Random positional error does not capture all systematic bias.

Open questions. Priorities include globally consistent uncertainty-aware shoreline products, separation of waterline and morphology, profile-volume assimilation, intervention causal inference, storm-recovery memory, sediment-cell closure, compound drivers and forecast validation under nonstationary climate and management.

Seminal papers. Repeated beach profiles and aerial-photo transects established quantitative shoreline monitoring; sediment-budget and one-line theory linked transport gradients to movement, while satellite archives and automated extraction enabled dense regional and global time series.

Equations

- *Net shoreline movement* [759]

$$NSM = d_{t_2} - d_{t_1}$$

Regime: Each shore-normal DSAS transect between two georeferenced shoreline dates. Variables: d_{t_2} shoreline position at the youngest date; d_{t_1} shoreline position at the oldest date

- *End point shoreline-change rate* [759]

$$EPR = \frac{d_{t_2} - d_{t_1}}{t_2 - t_1}$$

Regime: Endpoint trend at an Iztuzu DSAS transect; ignores intermediate shorelines. Variables: d_{t_2} youngest shoreline position; d_{t_1} oldest shoreline position; $t_2 - t_1$ elapsed time

Claims

- Between 2013 and 2022, Iztuzu Zones II-IV were predominantly erosional, whereas Zones I and V contained the strongest accretion, with substantial variation among subperiods. *Regime: Manually digitized Google Earth shorelines at 18 dates along Iztuzu Beach; shoreline-position uncertainty from water level and image timing remains.* direct finding, field [759]
- Of 819 Iztuzu shoreline transects, 287 points (35%) belonged to statistically significant erosion or deposition clusters, with erosion clusters concentrated mainly in Zone II and deposition clusters in Zone V. *Regime: 2013-2022 NSM-based hotspot analysis at 5 m transect spacing; results depend on manual shoreline extraction and neighborhood definition.* direct finding, field [759]
- Fuzzy c-means shoreline classification at the northern Java site achieved kappa values of 0.86-0.96 and represented both boundary position and pixel-level change uncertainty, including 739 ha converted to water in 1994-2000. *Regime: The 1994-2015 optical images, $t=0.5$ membership threshold, and northern Java class definitions.* direct finding, field [337]
- Across 769 Haikou images from 1986-2023, artificial-island construction and five nourishment projects were mainly associated with accretion, with a 4.9 m random waterline-position error. *Regime: Haikou Beach, its engineering chronology, and satellite mean-waterline proxy; association is not universal causation.* direct finding, field [618]
- Over 1999-2024, two broad dissipative northwest Ireland beaches remained comparatively stable or accreting under consistent extreme-waterline trends, whereas the narrower reflective beach showed the greatest variability and landward retreat. *Regime: The three undisturbed northwest Ireland beaches and their satellite waterline proxies.* direct finding, field [1187]
- In an 18-year Tairua blind test, 19 process and machine-learning shoreline models lost skill on unseen 2014–2017 data and fast extremes; an ensemble outperformed individual models and exposed architecture uncertainty. *Regime: Blind testing of shoreline evolution models* direct finding, mixed [975]
- Satellite analysis for 1984–2016 estimates sandy beaches comprise 31% of ice-free shoreline; 24% erode faster than 0.5 m/yr, 28% accrete and 48% are stable. *Regime: The State of the World's Beaches* direct finding, mixed [875]
- Mekong delta shoreline analysis links rapid erosion to human-driven reductions and redistribution of river sediment rather than coastal forcing alone. *Regime: Linking rapid erosion of the Mekong River delta to human activities* direct finding, mixed [63]
- A shoreline-prediction review identifies nonstationary forcing, multiscale processes, data limitations and model uncertainty as central barriers to reliable coastal forecasts. *Regime: Challenges and Opportunities in Coastal Shoreline Prediction* review statement, review [1337]
- Multitemporal remote sensing at Sayung documents spatially variable shoreline erosion and accretion suitable for diagnosing coastal change and management priorities. *Regime: Dynamics of shoreline changes in the coastal region of Sayung, Indonesia* direct finding, mixed [338]
- Global long-term satellite observations quantify coastal erosion and accretion patterns and demonstrate that shoreline change is geographically heterogeneous rather than uniformly erosional. *Regime: Global long-term observations of coastal erosion and accretion* direct finding, mixed [957]
- An ensemble Kalman filter improves shoreline modeling by assimilating observations and representing nonstationary future wave climates and associated forecast uncertainty. *Regime: Enhanced Coastal Shoreline Modeling Using an Ensemble Kalman Filter to Include Nonstationarity in Future Wave Climates* direct finding, numerical [628]
- Remote surveys quantify rapid erosion of ice-rich Tuktoyaktuk Island and connect thermal-mechanical degradation to loss of a natural harbor barrier. *Regime: Mechanisms, volumetric assessment, and prognosis for rapid coastal erosion of Tuktoyaktuk Island, an important natural barrier for the harbour and community* direct finding, field [1508]

25.3 Dunes and overwash

Topic beaches . dunes. Dune erosion, overwash, and recovery.

Papers screened: 5. Claims: 8. Equations: 1.

Well established. Storm waves and elevated water levels erode dune faces and redistribute sediment toward the beach and nearshore; whether impact remains collision and scarping or progresses to overwash and inundation depends jointly on forcing intensity and pre-storm berm–dune geometry.

Governing physics. In the reviewed New Jersey study, peak joint wave–water-level forcing was the dominant tested predictor of dune loss, while berm and dune morphology modulated how much forcing reached and removed the dune.

Dimensionless parameters. Transferable regime descriptors include water level and runup relative to dune toe and crest, freeboard normalized by wave height, surf similarity, storm duration relative to profile-adjustment time, berm and dune volume normalized by active-profile scales, and erosion or overwash volume relative to initial dune volume.

Major equations. Dune impact was quantified as 100 times the difference between pre- and post-storm area inside the pre-storm dune footprint divided by the pre-storm area.

Typical methods. One demonstrated workflow combines repeated pre/post-storm beach profiles, hindcast wave and water-level intensity indices, correlation analysis, and classification-tree ensembles with held-out testing.

Numerical models. A 260-tree bootstrap ensemble predicted Minor, Moderate, and Major dune-volume-loss classes with reported within-dataset accuracies of 92%, 84%, and 98%, respectively; this was not external regional validation.

Experimental datasets. The New Jersey Beach Profile Network supported profile-based erosion analysis for eighteen historical storms selected from a 1980-2013 erosion-potential climatology.

Validated ranges. The model was trained and tested for New Jersey profiles with berm volume 1.8-420.1 m³/m, dune volume 1.5-281.7 m³/m, median grain size 0.16-2.19 mm, crest elevation 2.2-7.8 m NAVD, toe elevation 1.0-4.8 m NAVD, SEI 55-3871, and PEI 24.3-163.3.

Recent advances. For this dataset, PEI below 69 generally separated Minor impacts and PEI above 102 generally separated Moderate/Major impacts; morphology was necessary between those thresholds.

Disagreements. Forcing-only thresholds can separate weak and severe impacts in some datasets, while intermediate cases require morphology; process-based profile models explain sediment redistribution but do not by themselves establish universal damage classes. These are complementary scales, not interchangeable predictors.

Limitations. The available reviewed evidence is geographically narrow; its data-driven thresholds and performance should not be transferred outside New Jersey or beyond the reported storm and morphology ranges without retraining or increased uncertainty.

Open questions. Unresolved needs include representing the timing and duration of peak erosion intensity, adding rare severe impacts, testing other beach regimes, and externally validating the model.

Seminal papers. Early storm-surge profile models established offshore redistribution of eroded dune sand under elevated water levels; later impact-regime concepts organized collision, overwash and inundation, and modern data-driven work quantifies how storm intensity and morphology jointly classify loss.

Equations

- Dune volume loss percentage [798]

$$d_{loss}(\%) = \frac{A_{pre} - A_{post}}{A_{pre}} \times 100$$

Regime: Profile-based storm impact classification used in this study; Minor <5%, Moderate 5-40%, Major >40% loss. Variables: A_pre pre-storm cross-sectional area within the pre-storm dune footprint; A_post post-storm cross-sectional area within the pre-storm dune footprint

Claims

1. Among the seven tested predictors, Peak Erosion Intensity was the most important predictor of storm-induced dune damage. *Regime: Compiled New Jersey beach-profile observations for eighteen storms, using the seven reported predictors and PEI range 24.3-163.3.* direct finding, field [798]
2. PEI below 69 generally identified Minor dune damage, whereas PEI above 102 generally identified Moderate or Major damage in the study data. *Regime: New Jersey training observations; damage defined by dune-volume loss (<5% Minor, 5-40% Moderate, >40% Major); high-PEI branch based on only 33 observations.* direct finding, field [798]
3. For intermediate storm intensity, beach-dune morphology was needed in addition to PEI to distinguish dune-damage outcomes. *Regime: New Jersey observations and the reported morphology ranges; derived from the fitted tree and ensemble, not a universal physical threshold.* direct finding, field [798]
4. Larger berm volume was associated with lower percent dune loss under comparable storm intensities in the compiled observations. *Regime: New Jersey profile/storm data, particularly PEI 69-82; association within the fitted data does not alone establish universal causality.* direct finding, field [798]
5. Under the modeled storm-surge process, eroded dune sand was transported offshore and deposited on the beach as the profile adjusted to the elevated water level. *Regime: Sandy beach-dune profiles subjected to the study's storm-surge and random-wave conditions.* direct finding, experimental [1449]
6. Pre- and post-Sandy LiDAR across more than 800 profiles found dune faces steepened 43% without becoming vertical and only half the dune toes followed the assumed foreshore-slope trajectory, challenging common erosion-model assumptions. *Regime: Testing model parameters for wave-induced dune erosion using observations from Hurricane Sandy* direct finding, field [1064]
7. Nine months of hourly Greenwich Dunes monitoring show that moderate 8-12 m/s winds delivered much of the foredune sediment, whereas only 1 of 15 large wind events produced large deposition and the strongest storms instead caused wave scarping; wind angle, fetch, moisture and snow/ice governed event effectiveness. *Regime: Greenwich Dunes, Prince Edward Island, across nine months of wind, beach and seasonal conditions.* direct finding, field [329]
8. A two-step meso-scale dune-supply model improved nine-month sediment-input predictions by first filtering winds that fail supply thresholds and then reducing potential transport for fetch and moisture; predictions matched measured deposition order of magnitude while coarse surveys missed erosion between events. *Regime: Greenwich Dunes, Prince Edward Island, with hourly wind and supply controls over nine months.* direct finding, mixed [330]

25.4 Beach nourishment

Topic beaches.nourishment. Placement, spreading, and performance.

Papers screened: 10. Claims: 8. Equations: 2. Shown in the Coastal Design Bench beside: Beach nourishment.

Well established. Nourishment creates a deliberately nonequilibrium sediment body that begins adjusting cross-shore and alongshore immediately; persistence, protection, and adverse effects depend on grain size, constructed geometry, waves, shoreface and inlet connectivity, structures, and the boundaries used for accounting.

Governing physics. Constructed shoreline gradients drive alongshore diffusion even in calm conditions, wave-energy disequilibrium drives cross-shore profile response, storms accelerate redistribution, and steep shorefaces, canyons, inlets, and groins create sinks or transfers that a local control volume may register as loss.

Dimensionless parameters. Transferable controls include relative grain size, relative fill width and length, profile-scale and equilibrium-shape parameters, wave-energy disequilibrium, relative notch position within the active swash zone, and normalized crest rise under sea-level change.

Major equations. Common reduced descriptions include the Dean equilibrium profile $h = A x^{2/3}$, dynamic shoreline relaxation proportional to wave-energy disequilibrium, one-line shoreline diffusion, empirical linear or exponential volume-loss curves, and sediment control-volume balances.

Typical methods. Robust evaluations combine repeated topographic and bathymetric profiles, shoreline or daily video observations, wave records, UAS-SfM elevation differencing, placed-volume histories, open-boundary sediment budgets, and calibration-validation that checks parameter physics as well as fit.

Numerical models. One-parameter dynamic-equilibrium models can match more parameterized shoreline models at a calibrated site, while one-line volume models can retain more credible parameters than statistically adequate empirical decay curves; neither result removes the need to represent inlets, structures, or open sediment boundaries where relevant.

Experimental datasets. The screened evidence spans four southern California fills monitored for 4-16 years, an 1100 m Florida fill through its first calm and hurricane periods, thirty years at Nice, two years of daily video in Barcelona, nine UAS surveys at Deal, and volume histories at two South Carolina barrier beaches.

Validated ranges. Directly tested ranges include 500-1500 m by about 50 m southern California pads integrated to 8 m depth, an 1100 m Florida project with 45 calm days before strong storms, 558,000 m³ placed along 4.5 km at Nice, two years of daily video at Nova Icaria, nine surveys of eight groins including three notches, and 1-5 m sea-level-rise scenarios at four gravel defenses.

Recent advances. Recent advances use parsimonious dynamic-equilibrium prediction, multi-year regional sediment accounting, UAS-SfM diagnosis of nourishment-structure interaction, and scenario-based crest and volume design under sea-level rise.

Disagreements. Multi-year subaerial persistence in southern California and no net width gain after decades at Nice are conditional outcomes, not a universal contradiction: sediment size, shoreface steepness, canyon proximity, wave exposure, monitoring metric, and boundary flux differ. Likewise, statistical model fit does not establish physically credible lifecycle prediction.

Limitations. Evidence remains site-limited, monitoring often misses subaqueous or cross-boundary flux, storms confound early equilibration, ecological and groundwater impacts are sparse, and reduced models omit inlet, barrier, and structure interactions. Success metrics based only on dry-beach width can therefore be misleading.

Open questions. Priorities are prospective multi-site validation, full littoral-cell sediment budgets, predictive grain-size and profile design, coupled inlet and structure effects, ecological and groundwater thresholds, adaptive monitoring triggers, and lifecycle optimization under uncertain storms and sea-level rise.

Seminal papers. Within this screened branch, the 2006 Upham field study establishes simultaneous immediate planform and profile adjustment, while the 2011 Nice record provides a rare multi-decadal test of nourishment efficiency.

Equations

- *Dean equilibrium beach profile* [404]

$$h = Ax^{2/3}$$

Regime: Idealized profile equilibration of the monitored fill. Variables: h water depth; A sediment-dependent profile scale; x offshore distance

- *Dynamic-equilibrium shoreline response* [649]

$$\frac{dy}{dt} = k(E_{eq} - E)$$

Regime: Medium-to-long-term cross-shore shoreline response at Nova Icaria beach. Variables: y shoreline position; k calibrated response-rate parameter; E_{eq} equilibrium wave energy; E incident wave energy

Claims

- At four southern California nourishments, coarse subaerial pads persisted for years but redistributed across open control-volume boundaries; Torrey Pines lost about 300,000 m³ over 16 years, and Imperial Beach drift contributed to the 2016 Tijuana River mouth closure. *Regime: The four surveyed southern California fills; natural variability and open boundaries limit attribution.* direct finding, field [873]
- At the 1100 m Upham Beach project, planform adjustment began during calm conditions immediately after construction, and a subsequent strong storm caused a substantial portion of total profile equilibration, demonstrating simultaneous longshore and cross-shore adjustment. *Regime: Low-energy west Florida nourishment with strong constructed-width gradients and hurricane-influenced early months.* direct finding, field [404]
- At Nice's 4.5 km gravel beach, 558,000 m³ placed from 1976-2005 produced no significant net width increase across 50 transects and 87 surveys, implying chronic offshore loss promoted by the steep shoreface and seasonal profile flattening. *Regime: Steep reflective, almost tideless gravel beach beside narrow shoreface and canyon heads; offshore pathways were inferred.* direct finding, field [62]
- At Deal, New Jersey, three identically notched groins stabilized adjacent shoreline but only the notch remaining near the seaward swash-zone edge promoted bidirectional bypassing; notches buried by post-nourishment widening behaved like traditional groins. *Regime: The monitored eight-groin field under northward net transport and evolving 2016 nourishment morphology.* direct finding, field [1623]
- At Folly Beach and Hunting Island, four nourishment-loss models all fit observed volumes statistically, but only the wave-linked One-Line model retained physically credible parameters at both sites; inlet and barrier-island processes remained unresolved. *Regime: Two South Carolina barrier-beach projects with measured volume time series and wave information.* direct finding, numerical [1495]
- Nineteen Dutch shoreface nourishments supplied sediment landward for years, while energetic breaking waves and cross-shore transport dominated initial nourishment erosion. *Regime: Observations and Modelling of Shoreface Nourishment Behaviour* direct finding, mixed [622]
- Fine sediment in a beach nourishment briefly elevated local surf-zone fecal indicator bacteria, followed by rapid inactivation and wave-current dispersal. *Regime: Beach Nourishment Impacts on Bacteriological Water Quality and Phytoplankton Bloom Dynamics* direct finding, field [1196]
- Monitoring of the 2014 Dunkirk nourishment shows rapid initial adjustment and predominantly eastward wave-driven redistribution rather than simple loss, benefiting the downdrift recreational beach while erosion weakened during the second year. *Regime: Dredged-sand nourishment fronting the Digue des Allies at Dunkirk on a tide-dominated northern French coast.* direct finding, field [1338]

25.5 Storm-driven beach response

Topic beaches. storms. Erosion and recovery during storms.

Well established. Storm beach response is a redistribution problem governed jointly by hydrodynamic forcing and antecedent/local morphology: elevated waves erode the subaerial beach or dune and move sediment, but the magnitude and even sign of local shoreline change vary with exposure, orientation, reefs, currents, and sediment supply.

Governing physics. Integrated wave energy and duration organize storm demand, while direction controls longshore-current forcing and antecedent width/profile controls available sediment and exposure. Local refraction, wave focusing, rips, river input, and reefs can amplify erosion or produce adjacent deposition.

Dimensionless parameters. The foundational reviewed scaling parameter is relative fall velocity $H/(Tw)$, combining wave height, period, and sediment settling velocity. Correlation, explained skill, and return probability are statistical rather than universal physical similarity parameters and must retain their site/sample definition.

Major equations. Reviewed formulations range from Vellinga's fall-velocity similitude $H/(Tw)$, through Leont'yev's zone-dependent $q=qW+qR$ transport decomposition, to storm indices $SPI=H_s^2 t$ and $TSWE=\int(P_w dt)$, shoreline rates $NSM/EPR/LRR$, and empirical multilinear or extreme-value response models.

Typical methods. Current evidence combines mobile-bed flumes, process-based cross-shore transport models, high-frequency video or profile monitoring, satellite/DSAS shoreline analysis, spectral wave downscaling, regression forecasting, extreme-value analysis, and event-scale 3D morphology mapping.

Numerical models. The reviewed model spectrum spans Vellinga's equilibrium-profile safety method, Leont'yev's cross-shore qW/qR process decomposition, site-specific multilinear forecasting, nested SWAN diagnostics, DSAS shoreline statistics, and empirical extreme-value/amplification curves; these models answer different temporal and spatial questions.

Experimental datasets. Reviewed datasets include random-wave Delta Flume mobile-bed tests up to $H_s=2$ m, 276 monitored Narrabeen-Collaroy storms, 48 years of five Narrabeen profiles plus six high-resolution storm responses, and 18 Iztuzu shorelines at 819 transects paired with a modeled 2013-2022 storm record.

Validated ranges. Quantitative skill is currently site-bound: the 2024 Narrabeen multilinear forecast reports RMSE 3.7-6.4 m over 276 events; the 2026 Narrabeen analysis spans 48 years but only six high-resolution events; Iztuzu uses 18 images over ten years and explicitly reports broad confidence intervals; Vellinga's large-scale waves reached $H_s=2$ m.

Recent advances. Recent advances combine high-frequency morphology with fast site-specific forecasting, direction-resolved satellite/model attribution, and full-beach 3D amplification analysis that exposes erosion maxima missed by conventional sparse transects.

Disagreements. Water level appears minor in the 2024 wave-dominated Narrabeen event regression, whereas storm-surge elevation is fundamental to Vellinga's dune-attack experiments. This is a regime difference, not evidence that water level is generally unimportant. Likewise, positive energy-erosion relations at exposed sectors coexist with deposition at sheltered or sediment-supplied Iztuzu sectors.

Limitations. The evidence is dominated by Dutch, Narrabeen, and Iztuzu cases. Sparse profiles can miss 3D hotspots, sparse images weaken correlation inference, abstract-only model records hide calibration ranges, and site-trained regressions may not extrapolate to surge-dominated, different-sediment, or differently oriented coasts.

Open questions. Major needs are transferable event models with uncertainty, simultaneous waves-water levels-currents-sediment observations, prediction of localized 3D amplification, separation of storm clusters from single events, field validation at data-poor coasts, and recovery-aware forecasting after erosion.

Seminal papers. Vellinga (1982) is the foundational reviewed journal study in this slice, establishing large-scale mobile-bed storm-dune experiments and fall-velocity similitude for Dutch dune safety assessment.

Equations

- Storm power index [759]

$$SPI = H_s^2 t$$

Regime: Modeled Iztuzu storms exceeding the study threshold for at least 12 hours. Variables: H_s storm significant wave height; t storm duration in hours

- Total storm wave energy [759]

$$TSWE = \int_i^n P_w(t) dt$$

Regime: Time-integrated modeled wave-power proxy for each identified storm. Variables: P_w wave energy flux; i, n storm integration bounds

- Deep-water wave energy flux [759]

$$P_w = \frac{\rho g^2}{64\pi} H_s^2 T_e$$

Regime: Wave-power calculation used in the paper's storm-energy metric. Variables: ρ water density; g gravitational acceleration; H_s significant wave height; T_e energy period

Claims

1. At Iztuzu Zones III-V, southern-storm duration, storm power index, and total storm wave energy correlated above 0.8 with end-point shoreline-change rate and above 0.7 with linear-regression rate. *Regime: Five-zone Iztuzu analysis over 2013-2022 using 18 manually digitized images and modeled storms; many 95% confidence intervals were wide because sample size was small.* direct finding, mixed [759]
2. Storms from the southern sector were more strongly associated with erosion of Iztuzu's southwest-facing central zones than storms from the south-southwest sector. *Regime: Iztuzu Zones II-IV, their reported orientations, modeled storm directions, and 2013-2022 shoreline record; sediment transport was inferred rather than measured.* inferred, mixed [759]
3. A stronger storm metric did not imply erosion everywhere at Iztuzu: reef- and river-outlet-influenced sectors showed negative correlations consistent with local wave-energy sheltering or sediment deposition. *Regime: Specific northern and reef-adjacent Iztuzu transects; causal sediment supply and transport were proposed but not directly observed.* inferred, mixed [759]
4. The Iztuzu near-annual correlations often lacked significance at the 95% level because only a small number of high-resolution image intervals were available. *Regime: This study's 18-image, ten-year correlation design; absence of significance does not establish absence of storm influence.* direct finding, mixed [759]
5. Vellinga's large-scale Delta Flume tests used random waves up to 2 m significant height and supported mobile-bed similitude based on the dimensionless fall-velocity parameter $H/(T w)$. *Regime: The paper's sandy mobile-bed 2D/3D program and Delta Flume storm-surge tests; transfer outside the tested sediment and profile regimes is not established here.* direct finding, experimental [1449]
6. Across 276 Narrabeen-Collaroy storm events, cumulative storm wave energy was the dominant erosion predictor, followed by pre-storm beach width and wave direction; water level contributed only weakly at this site. *Regime: Wave-dominated Narrabeen-Collaroy Beach and the observed 276-event predictor domain; the ranking is not universal for surge-dominated coasts.* direct finding, field [629]
7. During the 2016 Narrabeen-Collaroy storm, the measured local maximum erosion reached 235 m³/m above mean sea level and 58.3 m of shoreline retreat. *Regime: June 2016 event and the paper's above-MSL volume/shoreline definitions at Narrabeen-Collaroy.* direct finding, field [573]
8. Localized reef-related wave focusing and a migrating rip current produced erosion hotspots that were not represented by the five fixed historical profile locations. *Regime: The six high-resolution Narrabeen-Collaroy storm datasets and site-specific reef/rip morphology.* direct finding, mixed [573]
9. After the 2013/14 winter, 38 southwest England beaches showed mechanism-dependent recovery: more than 200 m³/m offshore loss at exposed sand beaches, overwash loss at gravel barriers, and rotation at obliquely forced sites. *Regime: The extreme 2013/2014 winter storms: Beach recovery along the southwest coast of England* direct finding, field [1273]
10. The 2013–2014 southwest England storms produced strongly site-dependent erosion and rotation governed by storm track, tide timing, coastal orientation, and embayment. *Regime: The extreme 2013/2014 winter storms: hydrodynamic forcing and coastal response along the southwest coast of England* direct finding, field [929]
11. During consecutive storms, antecedent morphology initially controlled beach response, while tide and surge became more important later and cumulative dune erosion persisted. *Regime: Beach erosion and recovery during consecutive storms at a steep-sloping, meso-tidal beach* direct finding, field [1465]
12. Post-storm beach recovery along Atlantic Europe was site-specific and multi-annual, with energetic winters stalling recovery and moderate winters accelerating it. *Regime: Beach recovery from extreme storm activity during the 2013–14 winter along the Atlantic coast of Europe* direct finding, field [350]
13. LiDAR observations classify extreme-storm beach response by exposure, wave-approach angle, embayment, net volume change, and alongshore variability. *Regime: Classification of beach response to extreme storms* direct finding, field [219]
14. XBeach-G reproduced gravel-beach storm responses from berm building to barrier rollover, and acceleration forces materially improved predictive skill. *Regime: Modelling the morphodynamics of gravel beaches during storms with XBeach-G* direct finding, numerical [943]
15. A storm-hazard matrix jointly classifies coastal flooding and beach erosion so events with different combinations of water-level and morphological impact are not reduced to a single hazard metric. *Regime: A storm hazard matrix combining coastal flooding and beach erosion* direct finding, mixed [779]

26 Wiki: Sea-level rise and climate adaptation

26.1 Sea-level rise and climate adaptation: overview

Topic *Climate*. Long-term change and adaptation of coasts.

Papers screened: 18. Claims: 1. Equations: 0.

Well established. Climate-driven mean sea-level rise shifts the baseline on which tides, waves, surge and river or rainfall flooding act. Small mean changes can strongly amplify threshold exceedance, while exposure, vulnerability, geomorphic response, ecosystems and adaptation determine consequences.

Governing physics. Ocean thermal expansion and land-ice loss raise mean sea level; regional circulation, gravity, vertical land motion and tides alter local change. Storms, waves, runoff, erosion, sediment accretion, wetland migration and reef condition govern how that baseline becomes flooding or shoreline impact.

Dimensionless parameters. Useful controls include sea-level increment relative to local extreme-level scale, flood-frequency amplification, exposed-population fraction, vulnerability indices normalized to 0–1, accommodation-space ratio, defense freeboard relative to projected total water level and adaptation benefit-cost or avoided-loss ratios.

Major equations. Assessment combines sea-level projections with extreme-value or joint-probability distributions, dynamic inundation and erosion models, exposure overlays and risk as a function of hazard, exposure and vulnerability. Carbon accounts quantify stored stock and avoided emissions; adaptation comparisons modify hazards, assets or response capacity.

Typical methods. Workflows combine tide gauges, altimetry, elevation and socioeconomic data with emissions and sea-level scenarios, tide-surge-wave or compound-flood models, dynamic shoreline response, ecosystem and carbon inventories, vulnerability indices and participatory adaptation assessment. Uncertainty should be propagated across all layers.

Numerical models. Models range from probabilistic sea-level and extreme-value ensembles to global or local hydrodynamic flooding, GIS exposure and vulnerability overlays, wetland accommodation models, shoreline equilibrium or erosion models and coupled demographic scenarios. Static bathtub mapping omits important dynamic pathways.

Experimental datasets. Reviewed evidence includes global coastal populations and elevation models, probabilistic extreme-level projections, national US 30 m compound-flood maps, Mediterranean heritage sites, California dynamic exposure, Bangladesh cyclone records, Australian blue-carbon inventories and Grenada reef and shoreline observations.

Validated ranges. Reported evidence spans site, city, national and global scales and scenarios through 2100. Quantitative values are conditional on emissions, elevation, population, protection and process assumptions; the US compound model reports map-validation CSI of 0.69–0.82, not universal predictive skill.

Recent advances. Recent advances correct coastal elevation with machine learning, couple waves, surge, tides, rainfall and rivers, represent erosion and cliff retreat dynamically, quantify blue carbon and wetland accommodation, and compare flexible adaptation pathways with explicit equity and ecosystem services.

Disagreements. Magnitude and timing differ across emissions, ice-sheet, vertical-motion, storm, wave, demographic and adaptation assumptions. Static exposure can seriously understate dynamic flooding, while unconstrained global assessments can overstate realized impacts if protection or migration is ignored.

Limitations. Elevation bias, incomplete hazard dependence, coarse coastal grids, uncertain ice response, socioeconomic pathways, autonomous adaptation, protection failure, ecosystem feedbacks and distributional inequity limit inference. Global rankings identify hotspots but do not replace local design analysis.

Open questions. Priorities include local vertical-motion integration, joint compound extremes, dynamic morphology, cascading infrastructure failure, ecosystem adaptation limits, migration and development feedbacks, equitable financing, adaptation effectiveness and transparent deep-uncertainty decisions.

Seminal papers. Detection and attribution established climate-driven sea-level rise; scenario and extreme-value studies translated it into changing flood frequency, and integrated vulnerability frameworks joined physical exposure with social capacity. Dynamic and nature-based approaches subsequently broadened assessment beyond static inundation.

Claims

1. Coastal climate assessment must connect observed and projected sea-level rise and its causes to physical and societal impacts and feasible response options. *Regime: Cross-cutting global and Japanese sea-level-rise assessment.* review statement, review [965]

26.2 Adaptation pathways

Topic *Climate*. adaptation. Sequenced protection, accommodation, and retreat.

Papers screened: 26. Claims: 15. Equations: 0.

Well established. Coastal adaptation is a continuing risk-management process rather than a one-time project; effective strategies combine protection, accommodation, ecosystem measures and retreat with monitoring, maintenance, finance, governance and triggers for adjustment as hazards and information change.

Governing physics. Relative sea-level rise raises baseline water levels and shifts wave, surge, erosion, groundwater and drainage hazards; compound events and geomorphic or ecosystem response determine local exposure, while intervention performance depends on deterioration, recovery and sediment or ecological feedbacks.

Dimensionless parameters. Useful controls include sea-level rise relative to freeboard, benefit-cost ratio, residual-risk fraction, adaptation lifetime relative to hazard-change timescale, lead-time-to-threshold ratio, avoided-loss fraction, robustness across scenarios and ecological recovery relative to disturbance interval.

Major equations. Decision analysis combines probabilistic hazard and damage functions, discounted life-cycle cost and benefit, dynamic programming or real-options logic, and robustness across scenarios. Adaptive pathways encode actions, thresholds, lead times and transfer points when objectives are no longer met.

Typical methods. Workflows co-produce objectives, map compound hazards and vulnerability, evaluate portfolios under multiple projections, include capital and maintenance costs and distributional impacts, identify thresholds and lead times, stress-test pathways, monitor indicators and pre-authorize adjustment or retreat options.

Numerical models. The set includes scenario-based hazard and pathway frameworks, building-level Monte Carlo damage simulation and dynamic programming. Nature-based evidence emphasizes forward-looking performance, maintenance and adaptive intervention rather than a single fixed design model.

Experimental datasets. Reviewed evidence includes a global coastal-adaptation literature synthesis, Bay County building-level Monte Carlo damage and dynamic-programming analysis under four sea-level projections, and a review of living-shoreline utility under changing climatic conditions.

Validated ranges. Bay County findings apply to its building inventory and four projections; global and living-shoreline reviews support principles rather than universal local benefit-cost rankings. Transfer requires local hazards, institutions, ecosystems, costs and stakeholder objectives.

Recent advances. Recent advances integrate high-resolution exposure, probabilistic projections, adaptation tipping points, building-level digital twins, participatory pathway tools, natural-capital accounting, climate services and evaluation of implementation, maintenance and distributional outcomes.

Disagreements. Protection can reduce near-term risk but induce lock-in and residual catastrophe; retreat avoids exposure but faces social and political barriers. Nature-based measures add co-benefits and adaptability yet have uncertain extreme-event and long-term performance; hybrid portfolios often manage these tradeoffs.

Limitations. Deep climate uncertainty, incomplete compound hazards, uncertain damage and behavior, discounting, omitted distributional effects, institutional fragmentation, funding and maintenance gaps, path dependence and ecological nonstationarity can dominate apparently precise rankings.

Open questions. Priorities include equitable pathway design, endogenous development and migration, compound hazard and cascading infrastructure risk, finance and governance experiments, monitoring thresholds, adaptation effectiveness audits and comparable long-term evidence for hybrid and nature-based portfolios.

Seminal papers. Protect–accommodate–retreat frameworks organized response options; robust decision making, real options and dynamic adaptive policy pathways shifted planning from fixed forecasts toward sequences that remain adjustable under uncertainty.

Claims

1. A 2025 review concludes that durable living-shoreline performance under climate change requires forward-looking design, coordinated holistic planning, and maintenance and adaptive interventions, because present-day success does not establish resistance or recovery under future extremes. *Regime: Conceptual guidance across nature-based and hybrid shoreline types; no universal quantitative threshold is claimed.* review statement, review [345]
2. A global coastal-adaptation review concludes that long-term plans must combine locally relevant sea-level projections with compound hazards, agreed action thresholds, stakeholder priorities, finance, and maintenance capacity; construction alone does not establish durable adaptation. *Regime: Strategic planning across diverse coasts; local design values require site-specific analysis.* review statement, review [535]
3. Australian vegetated coastal ecosystems store an estimated 70–185 Tg C above ground and 1,055–1,540 Tg C in the upper metre of soil; ongoing losses could emit 2.1–3.1 Tg CO₂-e per year. *Regime: Australian vegetated coastal ecosystems as global hotspots for climate change mitigation* direct finding, mixed [1284]
4. Projected global coastal-wetland loss by 2100 ranges from 0 to 30% without added accommodation space; resilience is governed primarily by space for landward migration, which infrastructure can constrain and adaptation can restore. *Regime: Future response of global coastal wetlands to sea-level rise* direct finding, mixed [1268]
5. The Sapelo Island analysis argues that colorblind sea-level-rise adaptation can reproduce environmental racism when it ignores racialized land ownership, employment and barriers to African American participation. *Regime: Racial coastal formation: The environmental injustice of colorblind adaptation planning for sea-level rise* direct finding, mixed [571]
6. At Grenville Bay, six decades of evidence and wave modeling link reef degradation more strongly than historical climate or sea-level change to northern-bay erosion and motivate an engineered reef-restoration pilot for risk reduction. *Regime: Coral reefs for coastal protection: A new methodological approach and engineering case study in Grenada* direct finding, mixed [1172]
7. A review of Indonesian community adaptation identifies infrastructure, mangrove rehabilitation, warning systems, education, livelihood diversification, resettlement, drainage and resource management, emphasizing participatory and locally flexible implementation. *Regime: Adaptation Strategies of Coastal Communities to Sea Level Rise Due to Climate Change* direct finding, mixed [684]
8. Adaptation pathways for six coastal archetypes show options decline as sea level rises and that early low-regret action, monitoring and preparation for transformative choices such as retreat preserve flexibility. *Regime: Generic adaptation pathways for coastal archetypes under uncertain sea-level rise* review statement, review [558]

9. Coastal risk assessments should be iteratively updated for new sea-level scenarios and high-end ice-sheet uncertainty while retaining still-useful prior analysis and testing the robustness of proactive strategies. *Regime: Integrating new sea-level scenarios into coastal risk and adaptation assessments: An ongoing process* review statement, review [1022]
10. The adaptation-tipping-point method identifies when current management no longer meets objectives under sea-level rise, supporting staged Dutch decisions tied to monitored thresholds. *Regime: Using adaptation tipping points to prepare for climate change and sea level rise: a case study in the Netherlands* review statement, review [757]
11. Faced with sea level rise and the intensification of extreme events, human populations living on the coasts are developing responses to address local situations. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, review [781]
12. Sea-level-rise planning must combine protection, accommodation and managed retreat through iterative adaptation pathways because relative rise interacts with subsidence, storms, erosion, salinization and wetland loss, while global sea level rose about 17 cm in the twentieth century and could exceed 1 m in the twenty-first. *Regime: Global coastal settlements and ecosystems, with locally varying relative sea-level change, subsidence, hazards and development.* review statement, review [1020]
13. DIVA simulations for the European Union project that without adaptation, annual coastal-flood exposure in 2100 reaches about 780,000 people under A2 and 200,000 under B1; modeled dike raising and beach nourishment substantially reduce impacts and are economically rational at aggregate scale. *Regime: European Union coast through 2100 under IPCC A2 and B1 socioeconomic/sea-level scenarios.* direct finding, numerical [598]
14. Review of Australian sea-level-rise and coastal-flood policy finds adaptation fragmented without stable national guidance: vertical and cross-jurisdictional consistency, continuity across electoral changes, bipartisan commitment, and sustained resources are necessary for effective local planning. *Regime: Australian national, state, territory, and local planning for sea-level rise and coastal flooding, with international policy implications.* direct finding, review [327]
15. Coastal climate-risk practice should replace stationary, single-hazard assumptions with multidimensional probabilistic frameworks that represent changing hazards, exposure, vulnerability, and uncertainty, and connect risk assessment to flexible, monitoring-supported adaptation pathways. *Regime: Coastal flood, erosion, infrastructure, ecosystem, and socioeconomic risks under long-term climate change, including compound drivers and deep uncertainty beyond 2050.* review statement, review [1417]

26.3 Coastal impacts of climate change

Topic *climate. impacts. Changing hazards and morphology.*

Papers screened: 44. Claims: 29. Equations: 8.

Well established. Coastal climate impacts arise through interacting changes in mean sea level, extremes, waves, rainfall and groundwater, then propagate through shoreline position, overtopping, drainage, habitats, assets, economies, water security and institutions. Adaptation materially changes outcomes but is itself constrained by governance and local context.

Governing physics. Rising mean water level reduces structure freeboard, shifts groundwater upward, increases boundary head on drainage, exposes shorelines to profile adjustment and intensifies coastal squeeze where landward habitat migration is blocked. Coincident tide, waves, rainfall and groundwater make impacts nonlinear and spatially heterogeneous.

Dimensionless parameters. Useful controls include relative sea-level rise to freeboard, relative wave height and steepness, exceedance probability, return-period ratio, groundwater depth normalized by relief, rainfall-to-drainage-capacity ratio, recession normalized by active-profile width, benefit-cost ratio and fractional avoided loss.

Major equations. Core formulations include joint extreme-value probability for waves and water levels, overtopping discharge as a nonlinear function of relative freeboard, two-dimensional shallow-water continuity and momentum, Bruun-rule equilibrium shoreline translation, and discounted impact-adaptation objective functions that combine damage and intervention cost.

Typical methods. Methods span climate-model ensembles, downscaling, joint probability and extreme-value analysis, HEC-RAS 2D inundation, GIS exposure mapping, equilibrium-profile rules, integrated assessment and optimization, systematic reviews, policy robustness assessment and Indigenous community-centered adaptation frameworks.

Numerical models. Extracted models include an integrated offshore-to-overtopping chain using EurOtop, HEC-RAS 5.0.7 2D, GCM ensembles coupled to Bruun-rule/GIS translation, and pyCIAM. Conceptual and review frameworks cover indirect impacts, WAMPUM, habitat-compensation robustness and institutional barriers.

Experimental datasets. This branch is dominated by field-derived and scenario datasets rather than laboratory experiments: the 2007 Walcott flood, Oakland topography/groundwater/rainfall data, 13 Indian port-adjacent beach segments, UK habitat-compensation records, Chinese coastal economic/geographic inputs and reviewed Australian and Tribal adaptation evidence.

Validated ranges. Quantitative findings are bounded to their sources: Walcott 0.35–1 m SLR cases; Oakland 1 m SLR, 1.94 m high tide and 2-year rainfall; Indian RCP4.5/RCP8.5 projections through 2100; and pyCIAM strategies along China's segmented coast. These ranges are scenarios, not universal thresholds.

Recent advances. Recent work couples groundwater emergence with urban rainfall flooding, evaluates flexible adaptation portfolios at national scale, tests ecological compensation against robustness principles, centers Tribal sovereignty and water security, and systematically diagnoses institutional barriers rather than treating adaptation as an unconstrained model choice.

Disagreements. The literature differs less on whether impacts grow than on representation and response: static area accounting versus ecological function, autonomous optimization versus governance feasibility, equilibrium shoreline rules versus process-rich morphology, and generic technical planning versus sovereign community-grounded adaptation.

Limitations. Recurring limitations are climate and socioeconomic scenario uncertainty, static exposure, omitted compound drivers, simplified groundwater or morphology, empirical overtopping, discounting and valuation choices, incomplete monitoring, publication bias, qualitative causal chains and transfer of governance conclusions across communities.

Open questions. Priorities include coupled groundwater-surface-coastal validation, dynamic morphology and habitat succession, correlated extremes, adaptation tipping points, distributional and indirect losses, Indigenous data sovereignty, monitoring of compensation performance, governance implementation and transparent propagation of deep uncertainty.

Seminal papers. The 2012 Walcott analysis demonstrated how modest freeboard loss can radically change overtopping recurrence, while the UK indirect-impact study broadened coastal assessment beyond direct local damage. Together they frame physical amplification and cross-border propagation.

Equations

- *Climate-conditioned coastal impact mapping* [694]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: Peer-reviewed literature on Australian coastal settlements included by the review protocol. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

- *Climate-conditioned coastal impact mapping* [1485]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: pyCIAM coastal segments, modeled protection/retreat/no-adaptation choices and Chinese geographic/economic inputs through 2100. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

- *Climate-conditioned coastal impact mapping* [276]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: Walcott seawall geometry, mesotidal setting, selected future climate projections and 0.35–1.0 m sea-level-rise cases. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

- *Climate-conditioned coastal impact mapping* [197]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: UK Habitats Directive compensation and Shoreline Management Plan context. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

- *Climate-conditioned coastal impact mapping* [1083]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: Selected sandy beach segments near 13 Indian ports, GCM ensemble projections and equilibrium-profile assumptions through 2100. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

- *Climate-conditioned coastal impact mapping* [1021]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: UK exposure to overseas coastal impacts under alternative climate, socioeconomic and adaptation outcomes. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

- *Climate-conditioned coastal impact mapping* [801]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: Eastern US coastal Tribal contexts; principles require Nation-specific governance and knowledge protocols. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

- *Climate-conditioned coastal impact mapping* [1146]

$$I = f(\Delta S, H_s, T, P, G, E, V, A)$$

Regime: Oakland Flatlands topography, prescribed groundwater-inundation footprint, 1.94 m high tide, 1 m SLR and 2-year precipitation. Variables: I coastal impact metric; Delta S sea-level change; H_s wave climate; T tide and surge; P precipitation; G groundwater response; E exposure; V vulnerability; A adaptation

Claims

1. For the Walcott seawall case, the estimated return period of an overtopping event comparable to the November 2007 event fell from 100 years to about 5 years under 0.35 m of sea-level rise and to just over 1 year under 1 m; projected overtopping change was more sensitive to water-level change than to significant-wave-height change. *Regime: Walcott seawall geometry, mesotidal setting, selected future climate projections and 0.35–1.0 m sea-level-rise cases.* direct finding, numerical [276]
2. In the modeled Oakland Flatlands case, combining 1 m sea-level rise, sea-level-driven groundwater rise, high tide and a 2-year precipitation event exceeded drainage capacity and flooded more than 700 acres of built infrastructure, substantially more than scenarios omitting compound groundwater effects. *Regime: Oakland Flatlands topography, prescribed groundwater-inundation footprint, 1.94 m high tide, 1 m SLR and 2-year precipitation.* direct finding, numerical [1146]

3. A United Kingdom assessment found that overseas coastal climate impacts can propagate nationally through supply-chain disruption, migration and security pressures, finance and insurance losses, while also creating export opportunities for coastal-hazard and management expertise; outcomes depend on climate, socioeconomic change and adaptation success. *Regime: UK exposure to overseas coastal impacts under alternative climate, socioeconomic and adaptation outcomes.* review statement, review [1021]
4. The WAMPUM framework argues that sea-level-rise adaptation for eastern coastal Tribal Nations must center Indigenous knowledge, sovereignty, cultural values and relationships to water rather than apply externally imposed technical planning alone. *Regime: Eastern US coastal Tribal contexts; principles require Nation-specific governance and knowledge protocols.* direct finding, mixed [801]
5. Across beach segments near 13 Indian ports, projected absolute sea-level-rise rates were 3.38–5.16 mm/yr under RCP4.5 and 5.36–7.20 mm/yr under RCP8.5; Bruun-rule recession by 2100 was 14.10–29.22 m and 21.05–45.40 m, respectively. *Regime: Selected sandy beach segments near 13 Indian ports, GCM ensemble projections and equilibrium-profile assumptions through 2100.* direct finding, analytical [1083]
6. For United Kingdom shoreline-management habitat compensation, simple area-based balance sheets can overstate adaptation progress because they omit uncertainty and ecological functioning, integrity and coherence; robust compensation requires monitoring, transparency, flexible pathways and scenario-based planning. *Regime: UK Habitats Directive compensation and Shoreline Management Plan context.* review statement, review [197]
7. In a refined pyCIAM assessment of China's coast, locally optimized adaptation reduced cumulative national sea-level-rise losses by 2100 from about US\$4.5 trillion without adaptation to below US\$0.9 trillion; flexible strategies reduced losses by up to 86%, with especially large modeled reductions in Shanghai, Jiangsu and Zhejiang. *Regime: pyCIAM coastal segments, modeled protection/retreat/no-adaptation choices and Chinese geographic/economic inputs through 2100.* direct finding, numerical [1485]
8. A systematic review of Australian coastal-settlement adaptation identified unclear and inconsistent national/state guidance and weak stakeholder coordination as recurring institutional barriers that foster adaptation inertia; it calls for integrated governance, inter-agency communication and coordinated policy. *Regime: Peer-reviewed literature on Australian coastal settlements included by the review protocol.* review statement, review [694]
9. Scenario projections to 2030 and 2060 place the largest absolute low-elevation coastal-zone and 100-year-flood exposure in Asia, while Africa has the highest projected rates of coastal population growth and urbanization. *Regime: Future Coastal Population Growth and Exposure to Sea-Level Rise and Coastal Flooding - A Global Assessment* direct finding, mixed [1017]
10. Using the neural-network-corrected CoastalDEM triples SRTM-based global exposure estimates; under low emissions an estimated 190 million people occupy land below projected 2100 high-tide lines, and under high emissions as many as 630 million occupy land below projected annual-flood levels. *Regime: New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding* direct finding, mixed [751]
11. Extreme-value analysis including waves, tides and surge indicates that 10–20 cm of sea-level rise expected by 2050 can more than double extreme-water-level frequency in many tropical coasts with short-tailed present-day distributions. *Regime: Doubling of coastal flooding frequency within decades due to sea-level rise* direct finding, mixed [1457]
12. A Coastal City Flood Vulnerability Index combines exposure, susceptibility and resilience across hydro-geological, socioeconomic and politico-administrative components on a 0–1 comparative scale and was demonstrated for nine cities. *Regime: A flood vulnerability index for coastal cities and its use in assessing climate change impacts* direct finding, mixed [113]
13. Under mean RCP8.5 with no protection or adaptation, global tide-surge-wave-setup modeling projects that land area, population and assets at risk of episodic flooding increase by 48%, 52% and 46%, respectively, by 2100. *Regime: Projections of global-scale extreme sea levels and resulting episodic coastal flooding over the 21st Century* direct finding, mixed [722]
14. A Bangladesh assessment combines a 119-year cyclone record with storm-surge scenarios and qualitative erosion and backwater analyses to examine climate impacts and adaptation options. *Regime: Climate change impacts and adaptation assessment in Bangladesh* direct finding, mixed [29]
15. A 30 m national US framework jointly models pluvial, fluvial and coastal hazards for current, 2035 and 2050 RCP4.5 conditions; validation against local and FEMA maps produced Critical Success Index values of 0.69–0.82. *Regime: Combined Modeling of US Fluvial, Pluvial, and Coastal Flood Hazard Under Current and Future Climates* direct finding, mixed [128]
16. Of 49 low-lying Mediterranean cultural World Heritage sites, 37 are currently at risk from a 100-year flood and 42 from erosion; aggregate flood and erosion risk may rise 50% and 13% by 2100. *Regime: Mediterranean UNESCO World Heritage at risk from coastal flooding and erosion due to sea-level rise* direct finding, mixed [1173]
17. A Cape May County GIS analysis combines riverine and surge inundation with social vulnerability and shows that sea-level rise and poorly managed development increase exposure of people, property and critical facilities. *Regime: Vulnerability of coastal communities to sea-level rise: a case study of Cape May County, New Jersey, USA* direct finding, mixed [1531]
18. For California, integrating sea-level rise, tides, waves, storms, erosion and cliff retreat yields up to 600,000 people and more than \$150 billion in property exposed by 2100—about three times the population from static sea-level analysis. *Regime: Dynamic flood modeling essential to assess the coastal impacts of climate change* direct finding, mixed [121]
19. Flood-threshold exceedance odds grow exponentially with sea-level rise; the present US 50-year level is projected to be exceeded annually before 2050 along about 70% of the coast and nearly daily at peak tide before 2100 along 90%. *Regime: Sea-level rise exponentially increases coastal flood frequency* direct finding, mixed [1388]
20. A review of open sandy coasts organizes climate impacts through changes in sea level, storms, waves and sediment budgets and emphasizes coupled local shoreline response. *Regime: Assessing climate change impacts on open sandy coasts: A review* review statement, review [1157]
21. The Baltic Sea provides a long-record test case showing how interacting warming and anthropogenic pressures can offset coastal ecosystem-management gains. *Regime: The Baltic Sea as a time machine for the future coastal ocean* review statement, review [1181]
22. A global synthesis links coastal hypoxia susceptibility to stratification, nutrient and organic loading, and restricted exchange, with nonlinear ecosystem consequences. *Regime: Natural and human-induced hypoxia and consequences for coastal areas: synthesis and future development* review statement, review [1586]
23. A review synthesizes climate-change impacts on marine foundation species that structure coastal habitats and ecosystem services. *Regime: Impacts of Climate Change on Marine Foundation Species* review statement, review [1507]
24. Arctic shelf observations constrain subsea-permafrost degradation and gas-migration pathways controlling methane release after marine inundation. *Regime: Current rates and mechanisms of subsea permafrost degradation in the East Siberian Arctic Shelf* direct finding, field [1289]
25. Eelgrass protects sedimentary organic carbon only when canopy density is high: sparse canopies allow greater erosion and carbon resuspension, oscillatory flow releases more particulate carbon than unidirectional flow, and most mobilized upper-sediment carbon can remineralize rapidly. *Regime: Laboratory eelgrass sediments with documented low/high shoot density under progressively increased unidirectional and oscillatory hydrodynamics.* direct finding, experimental [395]

26. Coastal-wetland response to sea-level rise, storms and altered freshwater, sediment and nutrient input is mediated by geomorphic-biological feedbacks and human modifications; resilience depends on whether accretion, migration and ecological adjustment can keep pace with forcing. *Regime: Coastal and estuarine wetlands exposed to sea-level, storm and watershed-input changes under interacting human modification.* review statement, review [325]
27. A national InVEST coastal-exposure index for Ireland shows exposure is locally controlled by shoreline, metocean and climate factors; modeled coastal habitats potentially reduce exposure along 38% of the assessed coast, while dense eastern populations remain highly exposed despite stronger Atlantic forcing in the west. *Regime: Relative national-scale screening of the Republic of Ireland coastline using the documented physical, climatic, habitat and population datasets.* direct finding, numerical [1474]
28. Relative sea-level rise is likely the dominant future climate threat to mangroves where sediment-surface elevation cannot keep pace and landward migration is constrained; resilience can be increased through migration space, catchment and stressor management, rehabilitation, protected-area networks, and standardized regional monitoring. *Regime: Intertidal mangrove systems worldwide, with vulnerability evidence concentrated in the western Pacific and Wider Caribbean and greatest risk where elevation is declining or inland migration is blocked.* review statement, review [501]
29. Circum-Arctic coastal erosion is spatially heterogeneous and cannot be explained by a single mapped factor; wave and storm-surge exposure, sea-ice protection, permafrost and ground ice, sediment properties, and coastal geometry interact locally to control retreat and coastal material fluxes. *Regime: Permafrost-affected Arctic coastlines bordering the Arctic Ocean, including consolidated, unconsolidated, ice-rich, low-lying, and cliffed segments under varying sea-ice and storm exposure.* direct finding, field [769]

26.4 Sea-level projections

Topic *climate.projections. Global-to-local sea-level scenarios.*

Papers screened: 11. Claims: 5. Equations: 0.

Well established. Coastal engineering must treat mean sea level as nonstationary over asset lifetimes. Global trajectories require localization for land motion and regional ocean effects, then combination with tides, surge, waves, morphology, exposure, and vulnerability before they become design criteria.

Governing physics. Thermal expansion and land-ice loss drive global mean rise, while ocean circulation, gravitational fingerprints, tides, vertical land motion, waves, surge, sediment response, and groundwater processes shape local relative level and impact.

Dimensionless parameters. Useful controls include exceedance probability, relative sea-level change normalized by freeboard or crest height, crest-rise amplification, relative wave height and runup, discount and maintenance fractions, and ensemble weights or scenario probabilities.

Major equations. Engineering translations include products of time-varying annual-maximum distributions, scenario crest-rise ratios, runup and barrier-stability relations, depth-damage functions, Monte Carlo expectation, and Bellman recursion for sequential adaptation.

Typical methods. Methods span tide gauges and satellite altimetry, process-based or scenario projections, localization with land motion, nonstationary extreme-value analysis, profile and runup models, GIS exposure inventories, Monte Carlo damage simulation, cost-benefit optimization, and adaptive-pathway planning.

Numerical models. Models include multi-distribution nonstationary extremes, the Shingle Beach Run-up Tool, Barrier Inertia Model, stochastic depth-damage simulation, and dynamic adaptation optimization; they are consequence translators rather than interchangeable sea-level projection generators.

Experimental datasets. The screened engineering evidence uses Adelaide annual maxima, four southeast England beach profiles and a 2014 validation storm, a Bay County building and flood inventory, and more than three decades of multi-mission altimetry; these datasets support different links in the projection-to-decision chain.

Validated ranges. Reported applications include four southeast England beaches under 1-5 m scenarios with crest rise up to 1.26 times SLR, a 200-year joint wave-water-level standard, four SLR trajectories in Bay County, and a location-shift extreme model demonstrated for Adelaide; none constitutes a universal design factor.

Recent advances. Recent engineering work couples multiple SLR trajectories to stochastic damages and revisable actions, while coastal design studies resolve how local wave transformation and morphology can amplify physical requirements beyond the imposed mean rise.

Disagreements. Differences among design outcomes often reflect scenario choice, localization, impact model, asset lifetime, discounting, and adaptation objective rather than disagreement about whether sea level is rising. A deterministic high scenario can stress-test feasibility but is not itself a probabilistic forecast.

Limitations. The branch lacks a full primary projection ensemble and ice-sheet synthesis. Local vertical land motion, changing surge and wave climatology, datum consistency, compound flooding, sediment supply, model structural error, socioeconomic change, and deep uncertainty remain incompletely propagated into most engineering decisions.

Open questions. Priorities include traceable global-to-local probabilistic workflows, nonstationary extremes beyond simple location shifts, compound-hazard projections, observational signposts for adaptive pathways, equitable financing, sediment-aware defense adaptation, and validation of decisions as sea level and exposure evolve.

Seminal papers. Within this screened slice, the 1990 paper is seminal for replacing stationary return-period logic with time-varying annual distributions, providing an early direct bridge from projected mean rise to future design extremes.

Claims

1. The projected global-average 100-year extreme sea level rises 34–76 cm by 2100 under a moderate-mitigation scenario and 58–172 cm under business as usual, making present 100-year levels annual across much of the tropics from about 2050. *Regime: Global probabilistic projections of extreme sea levels show intensification of coastal flood hazard* direct finding, mixed [1467]
2. A comprehensive review links observed and projected sea-level rise to its climatic causes, global and Japanese impacts, and societal response options while identifying unresolved projection and adaptation questions. *Regime: Sea-level rise caused by climate change and its implications for society* direct finding, mixed [965]
3. A sea-level review defines and links local, global, mean, variable and changing sea-level concepts with physical and mathematical terminology intended to prevent ambiguity across research and policy. *Regime: Concepts and Terminology for Sea Level: Mean, Variability and Change, Both Local and Global* review statement, review [532]
4. Global coastal sea-level projections under warming above 2°C show regionally varying relative rise that depends on ocean, ice, gravity and land-motion contributions. *Regime: Coastal sea level rise with warming above 2 °C* direct finding, numerical [653]
5. Emulated multi-model projections estimate that limiting warming to 1.5 degrees Celsius nearly halves median 2100 land-ice sea-level contribution relative to current pledges, while Antarctic uncertainty leaves coastal adaptation facing roughly a factor-of-three range between central low-warming and risk-averse current-policy cases. *Regime: Global land-ice contribution to mean sea level from 2016-2100 under five Shared Socioeconomic Pathways and a current nationally determined contributions scenario.* direct finding, numerical [394]

26.5 Relative sea level and subsidence

Topic *climate.subsidence. Vertical land motion contributions.*

Papers screened: 9. Claims: 8. Equations: 0.

Well established. Relative sea-level change combines ocean-surface change with vertical land motion; subsidence can equal or exceed the climatic ocean component and therefore must be estimated locally.

Governing physics. Coastal land moves through groundwater and hydrocarbon withdrawal, sediment compaction, loading, tectonic and postseismic deformation, glacial isostatic adjustment and shallow infrastructure settlement, while steric and mass changes move the sea surface.

Dimensionless parameters. Useful normalized controls include subsidence-to-eustatic-rate ratio, vertical-motion uncertainty relative to trend, record duration relative to climate oscillations, compaction relative to sediment thickness and projected subsidence relative to defense freeboard.

Major equations. With upward-positive land motion, relative sea-level rate is approximately absolute sea-level rate minus vertical land-motion rate; time-varying analyses add acceleration, periodic variability, datum offsets and correlated error terms.

Typical methods. Studies combine long tide-gauge records with colocated GNSS, satellite altimetry or InSAR, homogenize datum and offsets, estimate nonlinear trends and accelerations, compare independent geodetic products and attribute mechanisms using extraction, geology and earthquake histories.

Numerical models. Methods range from segmented and nonlinear tide-gauge regressions to differencing relative tide-gauge and absolute altimetry signals; hydromechanical aquifer or sediment-compaction models can test causal attribution but are not represented directly in this reviewed set.

Experimental datasets. Reviewed evidence covers Galveston Pier 21 since 1909, five long-record Oceania gauges plus Tuvalu and Adelaide cases, incomplete Chinese gauge records, and 50 Southeast Asian sites with overlapping 1993–2020 tide-gauge and altimetry observations.

Validated ranges. Evidence spans individual gauges and regional networks over multi-decadal periods. Rates are site- and interval-specific and should not be transferred without colocated vertical-motion observations and consistent reference frames.

Recent advances. Recent work fuses tide gauges, GNSS, altimetry and InSAR, resolves nonlinear site histories, maps aquifer and sediment compaction, and incorporates time-varying vertical motion into local sea-level and flood-risk projections.

Disagreements. Linear trends can obscure changing extraction, postseismic transients and compaction, while short records can misidentify climate variability as acceleration. Different geodetic products and reference frames can yield conflicting vertical-motion estimates.

Limitations. Sparse colocated GNSS, datum changes, gauge relocation, altimetry coastal contamination, InSAR reference choices, nonlinear motion, short records and uncertain causal attribution limit projections; abstract-level values also require full-text methodological audit.

Open questions. Priorities include forecasting nonlinear subsidence under changing water management, separating shallow from deep motion, reconciling geodetic frames, propagating local motion into probabilistic extremes and sustaining dense observations in rapidly subsiding deltas and cities.

Seminal papers. Tide-gauge geodesy established that relative records combine ocean and crustal motion; leveling and later continuous GNSS enabled correction, while satellite altimetry and InSAR expanded regional separation and spatial mapping.

Claims

1. A 27-year tide-gauge and satellite-altimetry dataset at 50 Southeast Asian coastal sites revealed highly variable, nonlinear land motion linked to groundwater extraction, postseismic deformation, and sediment compaction. *Regime: The 50 sites with overlapping quality-controlled records; attribution strength varies by location.* direct finding, field [1093]
2. At Galveston Pier 21, estimated subsidence rates changed from 3.53 mm/yr in 1909–1937 to 6.08 mm/yr in 1937–1983 and 3.51 mm/yr thereafter; aquifer compaction contributed as much as 85% of 0.7 m relative sea-level rise since 1909. *Regime: Land subsidence contributions to relative sea level rise at tide gauge Galveston Pier 21, Texas* direct finding, field [845]

3. Across five long-record Oceania tide gauges, the reported mean relative rise was 1.306 mm/yr and mean absolute rise 0.125 mm/yr, with subsidence contributing substantially to local relative trends. *Regime: Relative sea-level rise and land subsidence in Oceania from tide gauge and satellite GPS* direct finding, field [171]
4. For China, short or incomplete tide-gauge records and unassessed instrument vertical motion prevented robust separation of absolute sea-level trend from land motion. *Regime: Sea Level Rise and Land Subsidence Contributions to the Signals from the Tide Gauges of China* direct finding, field [1080]
5. San Francisco Bay analysis shows climate-driven sea-level rise and local land subsidence jointly expand inundation exposure beyond estimates that assume a stable land surface. *Regime: Global climate change and local land subsidence exacerbate inundation risk to the San Francisco Bay Area* direct finding, mixed [1304]
6. Observed vertical land motion reveals hidden Atlantic-coast vulnerability because subsidence increases relative sea-level rise and future inundation beyond ocean-only projections. *Regime: Hidden vulnerability of US Atlantic coast to sea-level rise due to vertical land motion* direct finding, mixed [1042]
7. In many coastal and delta cities, anthropogenic land subsidence exceeds absolute sea-level rise and amplifies flood frequency, depth, duration, and structural damage. *Regime: Sinking coastal cities* review statement, review [413]
8. Coastal fluid extraction can make anthropogenic land subsidence the dominant component of relative sea-level rise: documented extremes approach 30 cm per year and more than 13 m cumulatively, far exceeding typical natural sediment compaction and contemporary global sea-level-rise rates. *Regime: Coastal deltas and aquifer systems experiencing substantial groundwater or hydrocarbon extraction.* review statement, review [635]

27 Wiki: Coastal engineering design

27.1 Coastal engineering design: overview

Topic design. Design criteria, performance, and lifecycle decisions.

Papers screened: 6. Claims: 5. Equations: 0.

Well established. Coastal design converts hazards and functional demands into limit states, dimensions, materials, detailing, construction, monitoring and intervention decisions while balancing safety, serviceability, durability, cost, environmental effects and acceptable risk.

Governing physics. Design actions arise from waves, currents, tides, surge, overtopping, flooding, erosion and geotechnical change; resistance evolves through structural response, soil interaction, fatigue, corrosion, wear, settlement, ecological succession and cumulative damage.

Dimensionless parameters. Important controls include reliability index, utilization and safety factors, return-period and exceedance probability, damage and serviceability indices, relative freeboard, wave steepness and surf similarity, armor stability numbers, corrosion or fatigue life ratios, redundancy, recovery time and benefit-cost ratios.

Major equations. Core formulations include action–resistance limit states, partial-factor or reliability equations, joint-probability load combinations, hydraulic and structural response relations, deterioration and condition-transition models, lifecycle cost or utility, benefit-cost and risk integrals, and adaptation decision rules.

Typical methods. Practice combines standards and consequence classes, deterministic and probabilistic design, physical and numerical modelling, field monitoring, condition assessment, failure-mode analysis, lifecycle optimization, multi-criteria appraisal, stakeholder and ecological assessment, and adaptive pathways.

Numerical models. Tools span spectral and phase-resolving hydrodynamics, overtopping and inundation models, geotechnical and structural finite elements, reliability and Bayesian networks, deterioration and asset-management models, lifecycle optimization, digital asset records and climate-adaptation scenarios.

Experimental datasets. Evidence comes from material and component tests, hydraulic model tests, prototype storm and damage records, long-term defence and structure monitoring, inspection and maintenance histories, failure databases, environmental performance, and infrastructure risk inventories.

Validated ranges. The screened evidence covers reinforced-concrete coastal structures, river and coastal bioengineering, Dutch flood defences, coastal railway exposure and offshore multi-system infrastructure; each transfers only within its asset, hazard, jurisdiction, material and governance regime.

Recent advances. Recent practice integrates probabilistic flood-defence management, climate-risk screening of infrastructure corridors, durable low-carbon materials, ecological engineering, multi-system offshore design, monitoring-informed asset management and flexible adaptation pathways.

Disagreements. Deterministic factors and probabilistic reliability encode uncertainty differently; least first cost can conflict with lifecycle reliability and environmental benefit; hard protection can reduce one failure mode while shifting erosion or ecological impacts; robust near-term actions may differ from expected-value optima.

Limitations. Joint hazard tails, deterioration, hidden defects, failure dependence, construction quality, maintenance behavior, ecosystem trajectories, indirect losses and future climate remain weakly observed; standards may lag nonstationarity and institutional responsibilities can fragment system performance.

Open questions. Priorities include nonstationary joint hazards, cascading and network failure, observation-updated reliability, hybrid and nature-based limit states, embodied carbon and biodiversity, constructability, autonomous inspection, adaptive design triggers, equity and transparent deep-uncertainty decisions.

Seminal papers. Classical coastal design established return-period loading, hydraulic similitude, armor and overtopping formulae, limit-state checks and safety factors; reliability and lifecycle methods later formalized uncertainty, consequences, deterioration and intervention timing.

Claims

1. A review of soil and water bioengineering identifies the need to integrate flood and erosion protection with ecological function in river and coastal projects. *Regime: Ecological challenges* for the use of soil and water bioengineering techniques in river and coastal engineering projects* review statement, review [1165]
2. A review of reinforced-concrete coastal structures connects durability, materials, exposure, structural performance, construction, inspection, repair, and future climate demands. *Regime: New directions for reinforced concrete coastal structures* review statement, review [1032]
3. Offshore renewable-hydrogen infrastructure requires integrated design across support structures, production, storage, transport, operation, safety, reliability, and marine-environment constraints. *Regime: A critical review of challenges and opportunities for the design and operation of offshore structures supporting renewable hydrogen production, storage, and transport* review statement, review [1203]
4. Dutch flood-defence practice links probabilistic reliability assessment, performance monitoring, asset management, intervention planning, and institutional responsibility. *Regime: Developments in the management of flood defences and hydraulic infrastructure in the Netherlands* direct finding, mixed [671]
5. A sea-level-rise risk framework for coastal railways translates hazard, exposure, vulnerability, and adaptation options into infrastructure planning evidence. *Regime: Risk assessment of national railway infrastructure due to sea-level rise: an application of a methodological framework in Italian coastal railways* direct finding, mixed [1189]

27.2 Design conditions and criteria

Topic design.criteria. Selection of design loads and levels.

Papers screened: 5. Claims: 3. Equations: 0. Shown in the Coastal Design Bench beside: Seawall design.

Well established. Coastal design criteria translate probabilistic hazards, performance objectives, consequence classes and physical response into verifiable elevations, loads, geometry and maintenance requirements; compliance with a nominal return period alone does not establish lifecycle reliability.

Governing physics. Waves, water levels, currents, erosion and material or ecosystem response combine nonlinearly, while sea-level rise shifts baseline hazards and deterioration changes capacity. Spatial nonuniformity and weak points can govern failure even when average performance is acceptable.

Dimensionless parameters. Controls include reliability index, return-period-to-design-life ratio, freeboard relative to wave/runup scale, demand-to-capacity ratio, partial safety factors, crest deficit, attenuation or damage threshold and layout-uniformity parameters such as S_r .

Major equations. Criteria use extreme-value quantiles, load and resistance factors, reliability or exceedance probability, freeboard and crest checks and response functions for overtopping, stability or attenuation. Under nonstationarity, annual distributions shift through time and lifecycle failure integrates yearly exceedance.

Typical methods. Workflows define limit states and consequences, estimate joint and nonstationary hazards, propagate uncertainty, select target reliability, evaluate alternatives and failure modes, audit constructed geometry with surveys, test response across representative conditions and specify inspection, maintenance and adaptation triggers.

Numerical models. The set includes a multi-distributional nonstationary extreme-value model, geospatial crest-compliance analysis and a layout-response criterion for vegetation attenuation. Direct intermediate-depth design-wave selection remains pending full-text review.

Experimental datasets. Reviewed evidence includes changing annual sea-level extreme distributions, airborne-LiDAR profiles of federal levees in south Louisiana and four flexible-vegetation layouts tested for spatial wave-attenuation uniformity.

Validated ranges. The levee audit found about 5% of evaluated points met the stated 100-year minimum; the vegetation S_r threshold was derived from four tested layouts; the sea-level method addresses shifting annual distributions. None supplies universal criteria across asset classes.

Recent advances. Recent advances combine nonstationary extremes, Bayesian joint hazards, lidar and digital twins for as-built compliance, real-time condition monitoring, probabilistic fragility and adaptive criteria that trigger staged upgrades as observations and projections change.

Disagreements. Stationary return periods are familiar but can misstate future magnitude under changing baselines. Prescriptive geometry is auditable but may not capture system response, while performance-based criteria are flexible yet depend on models, uncertainty and validation.

Limitations. Hazard nonstationarity, dependence, model and survey error, undocumented weak points, construction tolerance, deterioration, maintenance, climate scenario spread and unmodeled failure modes can undermine nominal compliance. Evidence is heterogeneous and sparse for cross-system transfer.

Open questions. Priorities include joint nonstationary hazards, lifecycle and adaptive reliability targets, system-versus-component criteria, nature-based durability, cascading failure, equity and consequence classes, digital compliance monitoring and transparent treatment of deep uncertainty.

Seminal papers. Return-period design and factor-of-safety methods established prescriptive criteria; reliability-based design introduced probabilistic load–resistance balance, followed by performance-based, lifecycle and adaptive approaches under nonstationary climate.

Claims

1. Airborne-LiDAR crest profiles showed only about 5% of evaluated federal levee crest points in the Atchafalaya study met the USACE minimum height for the 100-year flood. *Regime: The two evaluated federal levees and the contemporaneous USACE height requirement; this is not a structural-failure probability.* direct finding, field [1075]
2. A coupled seepage and deformation analysis relates tidal groundwater fluctuations to the performance of coastal deep foundation pits. *Regime: Seepage Flow Model and Deformation Properties of Coastal Deep Foundation Pit under Tidal Influence* direct finding, numerical [815]
3. A South African recreational-coast case derives transferable planning parameters for balancing wave climate, sediment behavior, structural function, safety and public use in coastal-structure design. *Regime: DESIGN OF COASTAL STRUCTURES FOR RECREATIONAL PURPOSES* review statement, review [1035]

27.3 Codes and design guidance

Topic design.guidance. Evaluation and development of standards.

Papers screened: 10. Claims: 7. Equations: 0. Shown in the Coastal Design Bench beside: Navigation channel.

Well established. Coastal design guidance converts hazard and performance evidence into repeatable decisions. Credible guidance states scope, assumptions, failure modes, uncertainty, safety and serviceability criteria, maintenance, monitoring, and consequences rather than presenting coefficients without provenance.

Governing physics. Guidance must connect waves, water levels, tsunamis, erosion, scour, overtopping, structural resistance, ecological response, human use, deterioration, and adaptation because performance and failure emerge from interacting loads and systems.

Dimensionless parameters. Process-specific controls include relative freeboard, surf similarity, stability and scour numbers, while guidance-level controls include return period, target probability, reliability index, consequence class, safety factor, discount rate, service life, and performance-indicator normalization.

Major equations. Common design frameworks combine joint probability and return levels, limit states and partial factors, runup and overtopping relations, stability and scour formulae, reliability or risk integrals, lifecycle cost, multi-criteria decisions, and adaptive pathways.

Typical methods. Methods synthesize codes, field failures, laboratory and numerical models, expert judgment, probabilistic analysis, consequence assessment, lifecycle and cost-benefit methods, multi-criteria analysis, monitoring, and periodic revision.

Numerical models. Guidance-supporting tools include joint-probability extreme-level models, spectral wave and overtopping models, XBeach with multilevel Monte Carlo, performance-indicator frameworks, and multi-hazard risk models.

Experimental datasets. Reviewed sources draw on post-tsunami surveys, 187 hybrid-measure studies, eight expert interviews, NNBF literature and experts, and global adaptation portfolios; adjacent evidence provides joint sea levels, port overtopping, and hydro-morphodynamic uncertainty.

Validated ranges. This abstract-bounded slice supports no universal coefficient range. Quantitative adjacent evidence includes 40-fold-or-greater MLMC speedups and a 187-study hybrid assessment base; every formula must retain its source-specific validity limits.

Recent advances. Recent guidance emphasizes portfolios, nature-based and hybrid infrastructure, uncertainty-efficient ensembles, integrated earthquake-tsunami assessment, broader performance indicators, and explicit links between structural and evacuation measures.

Disagreements. Hard, nature-based, hybrid, and nonstructural measures are not interchangeable. Structural measures can delay tsunami inundation yet fail without geometry, construction and maintenance; hybrid ambitions often exceed the hazard-focused indicators currently used to evaluate them.

Limitations. Guidance can lag climate, technology, and evidence; transfer across morphology and governance can fail. Sparse extremes, correlated loads, uncertain deterioration, ecological dynamics, maintenance, informal development, and inconsistent indicators constrain standardization.

Open questions. Needs include climate-adjusted and adaptive standards, transparent hybrid indicators, multi-hazard sequences, lifecycle reliability, monitored performance, equitable consequences, update triggers, and machine-readable provenance for every recommendation.

Seminal papers. The branch rests on probabilistic coastal levels, empirical structure manuals, reliability and limit-state design, risk-based planning, and field-forensic learning; current sources extend these toward NNBF, hybrid, and multi-hazard guidance.

Claims

1. A literature-and-expert synthesis concludes that emergent vegetation can be designed with current evidence but requires a risk-based wave-attenuation framework that makes calculation uncertainty and engineering, ecological, and social implementation barriers explicit. *Regime: Emergent vegetation systems used as coastal infrastructure.* review statement, review [1061]
2. Post-tsunami field evidence indicates that seawalls, breakwaters, and canals can reduce tsunami energy and delay inundation, but reliable mitigation depends on geometry, construction quality, maintenance, and integration with evacuation rather than structure presence alone. *Regime: Seawalls, breakwaters, and water-filled canals across major tsunami events including 2011 Tohoku.* review statement, review [73]
3. Across 187 studies and eight expert interviews, hybrid flood-measure assessment was dominated by hydraulic and cost metrics, while long-term adaptation, component interaction, ecological development, maintenance, and governance effectiveness were rarely evaluated together. *Regime: Grey-green-blue hybrid measures across pluvial, fluvial, and coastal flooding.* review statement, review [165]
4. An integrated coastal-structure risk framework treats earthquake and tsunami demands jointly, addressing the vulnerability created when seismic damage changes structural capacity before tsunami loading. *Regime: Coastal structures and communities exposed to sequential or combined earthquake and tsunami hazards.* review statement, review [1175]
5. A global adaptation synthesis finds that flood-risk practice has shifted from structure-only modification toward portfolios that include community-centered and nature-based measures, supporting guidance based on complementary rather than isolated interventions. *Regime: Flood adaptation measures under changing climate risk.* review statement, review [93]
6. A coastal-restoration synthesis identifies research priorities and practitioner motivations needed to align project design, monitoring, ecosystem recovery and stakeholder outcomes. *Regime: Priorities and Motivations of Marine Coastal Restoration Research* review statement, review [132]
7. A historical synthesis traces coastal-defense development and the changing engineering responses to sea-level variation and shoreline retreat. *Regime: Panorama of the History of Coastal Protection* review statement, review [250]

27.4 Lifecycle and maintenance

Topic design. Lifecycle. Durability, inspection, and intervention.

Papers screened: 10. Claims: 10. Equations: 0.

Well established. Lifecycle management links condition inspection, deterioration prediction, safety or reliability targets, maintenance and repair actions, replacement, and discounted whole-life cost rather than treating each decision independently.

Governing physics. Assets transition among condition states as loading and exposure accumulate damage. Inspections observe state imperfectly; maintenance changes condition or transition rate; delayed action increases future failure and repair consequences.

Dimensionless parameters. Decision controls include inspection interval/service life, transition and failure probabilities, false-positive/false-negative rates, intervention effectiveness, discount rate, cost ratios, reliability index and value-of-information ratios.

Major equations. Core tools include Markov transition matrices, partially observable or sequential decision processes, reliability and hazard functions, Bayesian state updating, discounted/average life-cycle cost, and constrained or multiobjective policy optimization.

Typical methods. Methods use periodic condition surveys, deterioration-curve calibration, Markov simulation, inspection-error models, maintenance-action matrices, dynamic or linear programming, reliability constraints, cost-benefit analysis and scenario sensitivity.

Numerical models. Models include homogeneous and time-varying Markov chains, continuous two-stage deterioration, partially observed maintenance decisions, data-driven condition curves, life-cycle-cost calculators and optimal preventive-timing frameworks.

Experimental datasets. The retained abstract evidence uses long-term condition and maintenance records for generic civil infrastructure, concrete bridges and pavements; direct breakwater, seawall and coastal-bridge records remain lawful-access targets.

Validated ranges. Evidence spans component and network decisions, discrete and continuous deterioration, perfect and imperfect inspection, fixed and changing transition rates, preventive and reactive actions, and short inspection intervals through multi-decade horizons.

Recent advances. Recent work uses data-calibrated and time-varying deterioration, explicit two-dimensional inspection error, timing-sensitive preventive maintenance, and integrated cost, environmental and reliability objectives.

Disagreements. Constant transition rates are convenient but may misstate aging and environment changes; fixed inspection schedules are simple but can waste resources; condition thresholds alone may omit consequences, while cost-only policies may violate reliability or sustainability goals.

Limitations. Coastal validation is presently weak in the lawful corpus; storm shocks, armor rearrangement, chloride and corrosion, scour, sea-level rise and adaptation are not captured by generic gradual-deterioration models; action-effect data are sparse.

Open questions. Priorities include coastal condition-state standards, event-driven deterioration, spatial dependence, remote inspection error, climate nonstationarity, adaptation flexibility, residual value, embodied carbon, ecological co-benefits and robust decision-making.

Seminal papers. Markov inspection–maintenance–replacement scheduling established sequential lifecycle decisions; later reliability and partially observable formulations incorporated safety and imperfect information.

Claims

1. Markov-chain deterioration and average-cost sequential decision methods yield optimal inspection, maintenance, and replacement policies. *Regime: Transferable infrastructure lifecycle methodology; coastal calibration required.* direct finding, analytical [727]
2. Condition-based maintenance for two-stage continuous deterioration must account jointly for false condition classification and measurement error in inspections. *Regime: Transferable infrastructure lifecycle methodology; coastal calibration required.* direct finding, analytical [1482]
3. A life-cycle-cost framework links periodic condition diagnosis, intervention planning, safety, and long-term infrastructure budgets. *Regime: Transferable infrastructure lifecycle methodology; coastal calibration required.* direct finding, analytical [1041]
4. Markov deterioration curves calibrated from condition data support life-cycle maintenance-cost estimates for concrete infrastructure. *Regime: Transferable infrastructure lifecycle methodology; coastal calibration required.* direct finding, numerical [1098]
5. Allowing Markov transition probabilities to change over time alters predicted deterioration and optimal maintenance intervention schedules. *Regime: Transferable infrastructure lifecycle methodology; coastal calibration required.* direct finding, numerical [314]
6. Data-driven pavement deterioration and life-cycle cost analysis show preventive maintenance benefit is highly sensitive to intervention timing. *Regime: Transferable infrastructure lifecycle methodology; coastal calibration required.* direct finding, numerical [718]
7. A synthesis evaluates how offshore platforms and pipelines alter seascape ecological connectivity and how decommissioning choices affect those functions. *Regime: Influence of offshore oil and gas structures on seascape ecological connectivity* review statement, review [946]
8. A multi-purpose offshore-infrastructure assessment examines shared energy and aquaculture uses together with environmental pressures in the northern Adriatic. *Regime: Boosting Blue Growth in a Mild Sea: Analysis of the Synergies Produced by a Multi-Purpose Offshore Installation in the Northern Adriatic, Italy* review statement, review [1581]
9. A review identifies flooding, storm surge, and hurricane mechanisms that threaten buried pipelines, tunnels, and culverts and organizes resilience measures. *Regime: Impact of flooding events on buried infrastructures: a review* review statement, review [13]
10. British coast-protection experience attributes important concrete deterioration to abrasion, chemical exposure, reinforcement corrosion and construction quality while finding frost, biological attack and alkali-aggregate reaction secondary in the reviewed cases. *Regime: DURABILITY OF CONCRETE IN COAST PROTECTION WORKS* direct finding, field [33]

27.5 Performance-based design

Topic design.performance. Response and consequence-based objectives.

Papers screened: 3. Claims: 2. Equations: 1. Shown in the Coastal Design Bench beside: Breakwater design.

Well established. Performance-based coastal design begins with explicit functional objectives and limit states, maps uncertain hazards to response and consequences, and verifies that structural, hydraulic, morphological and operational performance remains acceptable over stated conditions and time horizons.

Governing physics. Performance emerges from interaction of waves, water levels, currents, sediment and ecological or structural geometry; changing submergence, depth, exposure, material state and morphology alter loads, attenuation, erosion, overtopping and failure pathways.

Dimensionless parameters. Useful controls include demand-to-capacity and reliability indices, annual exceedance probability, relative crest elevation or submergence, response normalized by allowable limit, adaptation increment relative to sea-level rise, lifecycle utilization and expected loss or downtime ratios.

Major equations. Core formulations combine hazard exceedance with conditional response and limit-state probability, reliability or fragility functions, lifecycle accumulation and consequence integration; system-specific equations relate crest or reef elevation, water depth, wave transformation, sediment volume and load resistance.

Typical methods. Workflows define stakeholders and objectives, identify hazards and failure modes, select measurable performance metrics and thresholds, propagate joint forcing through response models, validate component behavior, quantify epistemic and aleatory uncertainty, compare alternatives and monitor triggers for adaptation.

Numerical models. Current evidence combines scenario-based beach-defense crest and volume design with observed reef wave-attenuation performance; broader response-based, reliability and consequence models are represented only by title-screened records and are not treated as extracted evidence.

Experimental datasets. The reviewed evidence includes four southeast England mixed sand–gravel defenses evaluated under 1–5 m sea-level-rise scenarios and a Virginia restored-oyster field experiment comparing two substrates, reef elevation, water level and marsh-edge response.

Validated ranges. Direct support is limited to four mixed sand–gravel sites and one restored-oyster setting with two substrates. It does not establish transferable performance targets, fragilities, design lives or reliability levels for caissons, levees, pumps or other systems.

Recent advances. Recent advances use environmental contours, stochastic response simulation, time-dependent climate scenarios, lifecycle reliability, adaptive pathways, digital monitoring, coupled consequence assessment and performance objectives for hybrid and nature-based coastal systems.

Disagreements. Prescriptive event-based design is simpler and transparent, whereas response-based probabilistic design better represents nonlinear response and joint uncertainty but requires more data and computation. Nature-based performance also changes biologically and morphologically, challenging fixed resistance assumptions.

Limitations. The evidence slice is small; objectives and acceptable consequences are context dependent; hazard dependence and nonstationarity are difficult to estimate; model and deterioration uncertainty can dominate tails; ecological functions and cascading systems lack common metrics; and monitoring may not match design variables.

Open questions. Priorities include multi-hazard joint probability, response-based versus event-based calibration, time-dependent deterioration and climate adaptation, system fragility and cascading failure, nature-based lifecycle reliability, equity-weighted consequences, observation updating and transferable verification datasets.

Seminal papers. Reliability-based and limit-state structural design established probabilistic performance checks; coastal work extended them to overtopping, sliding, erosion, flood consequences and lifecycle adaptation, while performance-based earthquake engineering supplied hazard–response–damage–loss integration concepts.

Equations

- *Scenario crest-rise amplification* [368]

$$\Delta z_c / \Delta SLR \leq 1.26$$

Regime: Four fixed-position southeast England mixed sand-gravel beach defenses under the tested 1-5 m scenarios. Variables: Δz_c required increase in beach crest elevation; ΔSLR imposed sea-level rise

Claims

1. Across four southeast England mixed sand-gravel defenses under 1-5 m sea-level-rise scenarios, required crest elevation increased by as much as 1.26 times the sea-level rise, with the largest beach-volume adjustments on shallow foreshores and obliquely exposed beaches. *Regime: Fixed-position mixed sand-gravel beaches represented by the Shingle Beach Run-up Tool and Barrier Inertia Model; future wave and sediment changes remain uncertain.* direct finding, numerical [368]
2. A practitioner-oriented catalogue organizes tested eco-engineering designs for breakwaters, revetments, seawalls, piers, and tidal walls to enhance biodiversity and ecosystem services. *Regime: Design catalogue for eco-engineering of coastal artificial structures: a multifunctional approach for stakeholders and end-users* review statement, review [1068]

28 Wiki: Coastal ecosystems and ecohydraulics

28.1 Coastal ecosystems and ecohydraulics: overview

Topic *ecohydraulics*. Coupled hydrodynamic and ecological processes.

Papers screened: 15. Claims: 3. Equations: 0.

Well established. Coastal organisms alter waves, currents, turbulence and sediment transport, while hydroperiod, salinity, bed change and disturbance control organism survival and morphology. These two-way feedbacks create habitat and can provide flood, erosion, water-quality and carbon benefits.

Governing physics. Vegetation drag extracts wave and current momentum; flexibility, buoyancy and reconfiguration change drag. Canopy shear layers and wakes redistribute turbulence and sediment, roots reinforce soil, and tidal inundation controls oxygen, salinity and habitat zonation. Morphodynamic and ecological responses feed back on flow geometry.

Dimensionless parameters. Controls include Reynolds and Froude numbers, Keulegan–Carpenter number, submergence, frontal-area density, solid-volume fraction, Cauchy or flexibility number, wave transmission and attenuation coefficients, Shields stress, Rouse number, hydroperiod fraction and relative sea-level/accretion rate.

Major equations. Models couple mass and momentum conservation to vegetation drag, wave-energy dissipation, turbulent diffusivity, sediment advection–diffusion and erosion/deposition, then link inundation or stress to habitat response. Drag and bulk dissipation closures must retain the scale and morphology assumptions of their calibration.

Typical methods. Evidence combines flumes with live or surrogate plants, field velocity/wave/sediment and vegetation surveys, erosion tests, remote sensing, calibrated hydrodynamic or morphodynamic models and before–after restoration monitoring. Comparative designs vary density, height, flexibility, patch geometry and water level.

Numerical models. Approaches range from cylinder-array drag and diffusivity closures to phase-averaged wave–vegetation, 2D wetland hydrodynamics, inlet circulation and coupled hydro-ecological or morphodynamic models. High-resolution ecological detail is usually limited to small domains; landscape models require aggregation.

Experimental datasets. Reviewed datasets include a 410 ha controlled-tide Ramsar restoration monitored for eight years, real-morphology Typha channel profiles, 1:20 marsh–dyke basin tests, winter marsh erosion flumes, Phragmites wave tests, an Oregon Inlet model and in-situ high-velocity marsh erodibility tests.

Validated ranges. Findings are regime-specific: artificial Phragmites reduced wave height 6–25% at canopy entry; marsh erodibility tests exceeded 0.5 m/s; the restoration covered 410 ha. These values cannot be transferred across species, season, scale or geometry without similarity and uncertainty checks.

Recent advances. Recent work uses real plant morphology, automated eco-hydrological restoration, in-situ failure testing, cross-disciplinary vegetation taxonomies, gray-to-green system comparisons and nested high-resolution/landscape models with more explicit ecological variability.

Disagreements. Aboveground biomass can strongly attenuate waves yet showed no mean winter erosion benefit in one soil-controlled test, indicating that wave attenuation, bed reinforcement and breach erosion are distinct services. Gray–green comparisons also depend on whether success means local velocity reduction, system change or ecological value.

Limitations. Surrogate fidelity, scale effects, plant reconfiguration, seasonal biomass, mortality, patch fragmentation, uncertain drag, limited extreme-event data, evolving topography and mismatch between patch observations and landscape decisions constrain prediction.

Open questions. Priorities include multiscale upscaling, mixed-species and fragmented canopies, failure thresholds under extremes, long-term adaptation to sea-level rise, sediment–vegetation feedback, standardized attenuation metrics, ecological trade-offs and monitored performance of hybrid defenses.

Seminal papers. Foundational canopy-flow and vegetation-drag studies established momentum extraction and wake turbulence; wetland biogeomorphology connected inundation and sediment accretion to habitat persistence. Nature-based engineering later framed these processes as quantifiable protection services.

Claims

1. Ecohydraulic prediction and management require joint physical and biological treatment across plant, patch, gap and landscape scales rather than one-way use of ecology as a model input. *Regime: Fragmented marine canopies across multiple habitat types*. review statement, review [454]
2. Coastal eutrophication response is filtered by system-specific attributes and includes linked changes in oxygen, water clarity, vegetation, biogeochemistry, harmful blooms and food webs rather than a universal nutrient-to-biomass response. *Regime: Our evolving conceptual model of the coastal eutrophication problem* review statement, review [285]
3. A systematic review of 95 papers finds hybrid sea defenses potentially combine robustness and ecological benefit but lack sufficiently consistent quantitative wave-structure performance evidence for confident transfer. *Regime: Advances in understanding the challenges and opportunities of hybrid sea defence approaches for coastal resilience* review statement, review [1554]

28.2 Habitat and restoration

Topic *ecohydraulics.habitat*. Engineering-ecology restoration outcomes.

Papers screened: 24. Claims: 18. Equations: 0.

Well established. Building or protecting habitat does not by itself establish ecological function. Outcomes depend on environmental suitability, hydrodynamics and sediment, connectivity and dispersal, structural complexity, target species and life stage, unresolved stressors, restoration age, maintenance, and the reference and metric used to define success.

Governing physics. Waves, wakes, tides, inundation, sediment moisture and grain size, erosion and accretion, elevation, and structural porosity or complexity set the physical template; colonization, recruitment, trophic subsidy, growth, mortality, dispersal, and succession determine whether that template becomes functioning habitat.

Dimensionless parameters. Useful nondimensional measures include relative elevation and inundation, porosity and relative structural dimensions, habitat or vegetation fraction, standardized effect size, classification agreement, survival or recruitment ratios, carbon additionality and permanence factors, and quality-adjusted habitat units normalized by cost.

Major equations. Common formulations include sediment and habitat-state balances, species-density and occupancy models, dispersal-distance relationships, random-effects standardized mean difference, shoreline-rate comparison, vegetation spectral regressions, carbon accounting, and cost per average annual habitat unit.

Typical methods. Robust assessment uses BACI or restored-natural-disturbed comparisons, replicated physical and biological surveys, shoreline and elevation monitoring, sediment and water measurements, abundance plus demographic and diet metrics, remote sensing, long-term maintenance records, meta-analysis, spatial prioritization, and lifecycle cost and service accounting.

Numerical models. Models include habitat-suitability and colonization regressions, random-effects meta-analysis and moderator tests, spatial conservation prioritization, vegetation biomass regression, point-cloud and habitat classifiers, blue-carbon accounting, and cost-effectiveness per quality-adjusted habitat outcome.

Experimental datasets. Evidence includes a 3.5-year created-wetland colonization series at Mugu Lagoon, four years at three paired hybrid-shoreline sites, six Louisiana oyster reefs revisited after ten years, restored reef and living-shoreline fish observations, 133 Gulf restoration projects, and 42 wetland-fisheries studies with 15 suitable for meta-analysis.

Validated ranges. Observed regimes include 3.5 years of wetland colonization; four-year shoreline trends from roughly 1 m/y advance to nearly 2 m/y retreat; ten-year oyster density falling below 60 individuals/m² from early values above 1000/m²; project extents of 0.04-364 ha; and a 42-study review yielding only 15 quantitatively compatible studies.

Recent advances. Recent work shifts from acreage and simple abundance toward life-stage habitat function, decade-scale persistence, remote high-resolution habitat condition, systematic evidence synthesis, spatial recovery prioritization, blue-carbon finance, and quality-adjusted restoration cost.

Disagreements. Stable or advancing hybrid shorelines and failed decade-scale oyster persistence are conditional, contrasting outcomes rather than a universal verdict on living shorelines. Similar mean fish density in restored and natural sites also does not prove equal fisheries production, because abundance, ontogenetic use, trophic export, and harvest occur at different scales.

Limitations. Most evidence covers few sites, taxa, designs, and years; reference sites can be non-equivalent; direct production and trophic subsidy are rare; restoration area is often substituted for quality; physical protection may outlast biological function or vice versa; and carbon and economic valuation require uncertain boundaries and permanence assumptions.

Open questions. Priorities are function-specific design rules, seascape connectivity and trophic-subsidy measurement, demographic and fisheries-production outcomes, multi-decadal persistence, adaptive maintenance thresholds, cross-habitat tradeoffs, standardized quality-adjusted metrics, and coupled ecological-hydrodynamic-economic optimization under climate change.

Seminal papers. Within this screened branch, the 2004 Mugu Lagoon study demonstrates that habitat suitability and dispersal jointly limit colonization, while the 2018 boulder synthesis frames structural complexity and stability as explicit ecological-engineering design variables.

Claims

1. Adding boulder habitat to coastal ecological engineering can increase structural complexity and alongshore availability for specialist species, but biodiversity benefit depends on boulder stability and usable under-boulder microhabitat. *Regime: The habitat types, taxa, designs, and monitoring periods reviewed in the source.* review statement, review [861]
2. During 3.5 years at a created Mugu Lagoon wetland, horn-snail density often remained below the natural site; sediment moisture and organic content explained 47% of subadult and 55% of adult variation, while adult density also declined with distance from the source wetland. *Regime: The habitat types, taxa, designs, and monitoring periods reviewed in the source.* direct finding, field [79]
3. Across 42 wetland-restoration articles, only 15 supported meta-analysis; restored and undisturbed sites had no significant overall species-density difference (SMD=-0.17 +/- 0.11, p=0.10), with high heterogeneity and little direct evidence of fisheries production. *Regime: The habitat types, taxa, designs, and monitoring periods reviewed in the source.* review statement, review [205]
4. Soft-sediment protection targets can fail as residual reserves when site selection ignores ecological value, pressure, connectivity, and recovery potential; spatial goals therefore require science-based prioritization. *Regime: The habitat types, taxa, designs, and monitoring periods reviewed in the source.* review statement, review [416]
5. Mangrove snapper partitioned restored habitat by life stage: living shorelines showed greater recruitment potential, while restored oyster reefs supported more juveniles and subadults, so restoration success metrics must reflect ontogeny and habitat function. *Regime: The habitat types, taxa, designs, and monitoring periods reviewed in the source.* direct finding, field [862]
6. Coastal blue-carbon accounting can create additional incentives for habitat protection and restoration, but credible use depends on defensible carbon quantification, permanence, additionality, and policy integration. *Regime: Coastal habitat protection and restoration projects with measurable ecosystem-service outcomes.* review statement, review [1377]
7. Across 133 Gulf of Mexico restoration projects, average annual habitat units provided a quality-adjusted outcome denominator for comparing costs, improving on cost-per-acre metrics that ignore habitat quality. *Regime: Coastal habitat protection and restoration projects with measurable ecosystem-service outcomes.* direct finding, field [517]
8. Coastal eutrophication recovery can be accelerated by combining nutrient-input reduction with hydrological flushing, seaweed or mussel harvest, benthic-habitat restoration, sediment sequestration and denitrification, subject to ecosystem-scale safeguards. *Regime: Intervention Options to Accelerate Ecosystem Recovery From Coastal Eutrophication* direct finding, mixed [380]

9. At a 410 ha Australian Ramsar wetland, a calibrated 2D model and automated controls recreated target inundation depth and hydroperiod; eight years of observations showed substantial saltmarsh expansion and achievement of restoration targets. *Regime: Eco-hydrology as a driver for tidal restoration: Observations from a Ramsar wetland in eastern Australia* direct finding, mixed [507]
10. During 19 months of intertidal-sandflat measurements, benthic production and respiration rose with air temperature; their reported Q10 factors were 3.72 and 4.80, respectively. *Regime: In situ production and respiration of the benthic community during emersion on subtropical intertidal sandflats* direct finding, mixed [786]
11. In calibrated Oregon Inlet simulations, channel relocation and island restoration reduced erosive shoreline velocities with minimal system-wide change, whereas gray alternatives caused either little local benefit or, for jetties, large inlet-wide velocity changes. *Regime: Hydrodynamics of a tidal inlet under gray to green coastal protection interventions* direct finding, mixed [1450]
12. A synthesis found 44% of ICES-advice species use coastal habitats and those stocks provide 77% of commercial landings, supporting habitat protection and restoration for population persistence and fishery yield. *Regime: Ecological value of coastal habitats for commercially and ecologically important species* review statement, review [1279]
13. Across 11 Asian cities, urban reefs show compressed depth ranges, low complexity, domed growth forms and city-specific gradients, alongside episodic impacts, recovery and coral colonization of infrastructure. *Regime: Urban coral reefs: Degradation and resilience of hard coral assemblages in coastal cities of East and Southeast Asia* review statement, review [581]
14. An urban-marine-ecology synthesis identifies city infrastructure, pollution, altered habitats and connectivity as interacting drivers of coastal ecosystem pattern and process. *Regime: Towards an urban marine ecology: characterizing the drivers, patterns and processes of marine ecosystems in coastal cities* review statement, review [1411]
15. Modeled tidal-wetland restoration shows tidal amplitude, floodgates, and creek microtopography control inundation, velocity, and hydrologic connectivity. *Regime: Assessing hydrodynamic effects of ecological restoration scenarios for a tidal-dominated wetland in Liaodong Bay (China)* direct finding, mixed [818]
16. Physical coral-canopy experiments identify interception, settlement, and wake accumulation as microplastic-trapping mechanisms governed by flow and canopy density. *Regime: Microplastic trapping efficiency and hydrodynamics in model coral reefs: A physical experimental investigation* direct finding, experimental [953]
17. A northeast Atlantic synthesis identifies threats, evidence gaps, and ecosystem services associated with kelp forests along temperate and polar coasts. *Regime: Threats and knowledge gaps for ecosystem services provided by kelp forests: a northeast Atlantic perspective* review statement, review [1312]
18. Sustainable estuarine restoration requires ecoengineering to recreate or reconnect site-specific hydrographic and sedimentary processes, while adding direct planting or restocking when physical habitat recovery alone cannot restore ecological function. *Regime: Human-modified estuaries and coastal wetlands undergoing managed realignment, tidal reconnection, flow alteration, habitat creation, mangrove or seagrass restoration, or shoreline intervention.* review statement, review [405]

28.3 Mangrove hydrodynamics

Topic ecohydraulics.mangrove. Flow and sediment through mangroves.

Papers screened: 26. Claims: 25. Equations: 0.

Well established. Mangroves dissipate waves and alter currents through form drag, turbulence, blockage, sheltering, and bed friction, while creeks can bypass forest resistance and redistribute water levels.

Governing physics. Controls include submerged frontal area, stem/root density and diameter, porosity, flexibility, vertical leaf distribution, relative water depth, Reynolds regime, forest width, channels, bathymetry, and sediment feedback.

Dimensionless parameters. Key quantities include drag coefficient, vegetation and root Reynolds numbers, relative submergence, frontal-area density, porosity, blockage, forest-width-to-wavelength scale, applied bed stress, and sediment availability.

Major equations. Models use vegetation drag and wave-energy sinks, porous-canopy momentum exchange, turbulence closures, frontal-area and bulk-drag relations, surge-wave coupling, and bio-morphodynamic sediment conservation.

Typical methods. Evidence spans natural-forest velocity arrays, storm and erosion surveys, prototype and reduced flumes, detailed geometry scanning, field-calibrated drag, coupled surge-wave models, and long-term bio-morphodynamics.

Numerical models. CH3D-SWAN, SWAN-type vegetation dissipation, 3-D vegetation-resolving surge models, porous-root flow models, delta-scale hydrodynamics, and bio-morphodynamic models span event to sea-level-rise scales.

Experimental datasets. Accessible cases include Pearl River typhoons, Hurricane Andrew/Ike/Harvey, Typhoon Mangkhut, New Zealand creek systems, Australian temperate mangroves, Mumbai waves, natural pneumatophore turbulence, and *Rhizophora* flumes.

Validated ranges. Case results include about 20% transmission through a 75 m *Rhizophora* forest, 77% reduction over 100 m during Mangkhut, nearly 70% over 100 m temperate forest, high-water gradients of 12-36 cm/km, and drag formulations with explicit Reynolds bounds.

Recent advances. Recent work quantifies prototype drag, geometry-measurement uncertainty, nonlinear low-cover erosion benefits, age-dependent failure, field-calibrated *Rhizophora* coefficients, delta-wide placement effects, and climate-driven morphology.

Disagreements. Mangroves do not uniformly attenuate every hazard: short waves are more consistently reduced than long surge, low-drag channels can eliminate forest-scale attenuation, foreshore breaking can exceed forest contribution, and poorly located patches can amplify water elsewhere.

Limitations. Drag coefficients and frontal areas depend on measurement method, species, age, water depth, damage, flexibility, scale, and event. Many studies remain model- or site-specific, and vertical structure and channel networks are often simplified.

Open questions. Needs include extreme-event drag and failure, flexible and broken vegetation, channel-forest exchange, wave-surge-tide interaction, evolving bathymetry, sediment budgets, species transfer, uncertainty, and coupled ecological persistence.

Seminal papers. The branch develops from mangrove wave-propagation and vegetation-drag theory toward parameterized-tree experiments, vegetation-resolving storm models, field turbulence, drag scaling, and bio-morphodynamic design.

Claims

1. A mangrove-specific wave framework established vegetation resistance as a control on surface-wave propagation through forests. *Regime: Mangrove geometry and resistance alter wave propagation and couple water motion to forest processes.* direct finding, analytical [927]
2. Wetlands attenuate short-period storm waves more reliably than long-period storm surges, whose response depends on event, vegetation, and landscape geometry. *Regime: Wetlands consistently attenuate short-period storm waves, while long-period surge attenuation is weaker or absent depending on storm, wetland, and landscape geometry.* review statement, review [1405]
3. Mangrove protection from tsunamis and cyclones is conditional on forest structure, width, bathymetry, event characteristics, and evidence quality. *Regime: Protection varies with forest width, density, structure, bathymetry, hazard, and evidence quality; mangroves are not an absolute shield.* review statement, review [920]
4. Permeable-dam restoration on mangrove-mud coasts requires adequate sediment supply, system-scale diagnosis, maintenance, and governance rather than structure placement alone. *Regime: Permeable structures can restore sediment balance only when embedded in system understanding, maintenance, governance, and suitable sediment supply.* review statement, review [1523]
5. Engineering with mangroves requires joint assessment of flood attenuation, ecological persistence, uncertainty, and integration with conventional defenses. *Regime: Implementation must jointly evaluate hydraulic functionality, ecosystem persistence, uncertainty, and compatibility with conventional defenses.* review statement, review [500]
6. Pearl River Delta simulations showed that forest width, drag, and location control local and nonlocal typhoon water levels, including possible amplification elsewhere. *Regime: Wide, high-drag forests increased local attenuation; narrow upper-estuary patches could attenuate upstream levels but could amplify water elsewhere if poorly placed.* direct finding, numerical [357]
7. Parameterized Rhizophora tests found about 20% transmission through a 75 m prototype-equivalent forest, with foreshore breaking important and solitary-wave drag coefficient near 1.5. *Regime: Foreshore breaking contributed strongly; a prototype-equivalent 75 m forest transmitted about 20%, and solitary-wave drag was approximately 1.5.* direct finding, experimental [1362]
8. Contrasting field sites showed that channels can bypass forest drag and eliminate measurable long-wave attenuation, making landscape routing decisive. *Regime: One site attenuated long waves, while another showed no measurable flood-tide attenuation because flow bypassed vegetation through low-drag channels.* direct finding, field [976]
9. Prototype-scale Rhizophora tests proposed $CD=0.6+3\times10^{-4}/Re_{DBH}$ for $Re_{DBH} 5\times10^3-1.9\times10^5$, with stated uncertainty 0.69. *Regime: Combined scaled datasets yielded $CD=0.6+3e4/Re_{DBH}$ over $Re_{DBH} 5e3-1.9e5$, with uncertainty about 0.69.* direct finding, experimental [699]
10. Bio-morphodynamic simulations found mangrove accretion can accelerate under sea-level rise, with nonlinear dependence on sediment supply, waves, slope, tidal range, and forest-edge distance. *Regime: Forest accretion could accelerate under SLR, but timing and magnitude varied nonlinearly with coastal conditions and distance from the seaward edge.* direct finding, numerical [1546]
11. Hurricane Harvey plots showed up to 196 m² erosion without mangroves and up to 9 m bank retreat with decreasing cover, with nonlinear benefits at low cover. *Regime: The zero-cover plot lost 196 m² and banks retreated up to 9 m as cover decreased; protection was nonlinear and low cover supplied much of the benefit.* direct finding, field [1094]
12. A modeled intermediate mangrove-root porosity minimized near-bed fluctuations through a vorticity sink and inhibited sediment transport through dominant viscous dissipation. *Regime: An intermediate porosity created a vorticity sink; viscous dissipation exceeded turbulence production and obstructed sediment transport.* direct finding, numerical [692]
13. Dense-array experiments showed that blockage, sheltering, reflection, orientation, and water depth control wave dissipation relevant to mangrove-restoration structures. *Regime: Element orientation, blockage, sheltering, and reflection controlled dissipation; a conceptual model reproduced arrangement effects.* direct finding, experimental [909]
14. Mangrove-creek observations measured high-water attenuation up to 12 cm/km in the creek and 36 cm/km in forest and identified creek-platform exchange as a tidal-asymmetry control. *Regime: High water fell up to 12 cm/km in the creek and 36 cm/km in forest; creek-sheet-flow interaction controlled asymmetry and attenuation.* direct finding, field [611]
15. During Typhoon Mangkhut, a roughly 100 m planted mangrove forest reduced wave height by 77%, avoided uprooting/lethal stem breakage, and showed sedimentation. *Regime: A roughly 100 m, six-year-old forest reduced wave height 77%, survived without uprooting or lethal stem breakage, and accumulated sediment.* direct finding, field [1597]
16. Natural pneumatophore canopies produced Reynolds stress up to $5.3\times10^{-3} \text{ m}^2/\text{s}^2$, eddy viscosity up to $1.9\times10^{-3} \text{ m}^2/\text{s}$, and near-bed drag up to an order above unvegetated expectations. *Regime: Reynolds stresses reached $5.3e-3 \text{ m}^2/\text{s}^2$, eddy viscosity $1.9e-3 \text{ m}^2/\text{s}$, and near-bed drag was up to an order above unvegetated expectation.* direct finding, field [1034]
17. A field-validated Hurricane Andrew model found vegetation removal greatly increased inundation, while buffering depended on vertical wetted and frontal leaf-area profiles. *Regime: Removing vegetation greatly increased inundation volume and area; buffering depended on vertical wetted and frontal leaf-area structure.* direct finding, mixed [1296]
18. For Hurricane Ike, physics-based 3-D vegetation-resolving modelling captured surge dynamics more accurately than a 2-D Manning approach and represented onshore sediment transport. *Regime: A physics-based 3-D vegetation treatment captured complex surge dynamics more accurately than a 2-D Manning representation while resolving onshore sediment transport.* direct finding, numerical [772]
19. Mangrove design modelling found Sonneratia wave attenuation peaked near age 10, while canopy emergence after age 20 shifted drag and increased wind-failure exposure. *Regime: Sonneratia attenuation peaked near age 10; after age 20 canopy emergence shifted attenuation to trunks/roots and increased cyclone wind-failure exposure.* direct finding, numerical [910]
20. A field-calibrated Rhizophora formulation uses root diameter as characteristic length for bulk drag in wave-attenuation models. *Regime: An integrated bulk-drag formulation used root diameter as characteristic length to improve wave-attenuation inputs for complex Rhizophora geometry.* direct finding, mixed [881]
21. Mumbai modelling and measurements found 49-55% attenuation for 0.175 stems/m², 0.3 m diameter, and drag coefficient 0.4-1.5. *Regime: For density 0.175/m², 0.3 m stems, and CD 0.4-1.5, modeled attenuation was 49-55% and agreed reasonably with measurements.* direct finding, mixed [1246]
22. Woody-vegetation frontal-area methods produced inferred bulk drag coefficients from 0.94 to 1.70, demonstrating measurement-driven wave-attenuation uncertainty. *Regime: Alternative frontal-area methods yielded bulk drag coefficients 0.94-1.70; TLS shadowing required correction.* direct finding, mixed [682]
23. An Australian temperate mangrove belt 100 m wide reduced ordinary wave heights by nearly 70%, with attenuation decreasing as water depth increased. *Regime: A 100 m belt reduced ordinary incoming waves by nearly 70%; efficiency decreased with depth and UAV data under-resolved near-bed obstacles.* direct finding, field [1170]
24. Homogenized long-wave analysis represents emergent coastal forests as cylinder arrays and quantifies frictional and turbulence-mediated attenuation. *Regime: Long waves through emergent coastal vegetation* direct finding, analytical [949]
25. Submerged mangrove canopies materially increase regular-wave attenuation, and a hydraulic submerged volume index collapses damping across no-, sparse- and dense-canopy configurations; drag also varies with Reynolds, Keulegan-Carpenter and Ursell regimes. *Regime: Regular waves through the tested 1:10 young/short-mangrove patch with no, sparse and dense canopies over submerged to emergent states.* direct finding, experimental [1599]

28.4 Salt marsh dynamics

Topic *ecohydraulics.marsh*. Flow, sediment, and marsh response.

Papers screened: 35. Claims: 34. Equations: 0.

Well established. Salt marshes are coupled biogeomorphic systems: vegetation changes drag and routing, which alters sediment delivery and channel form, while elevation and morphology feed back on inundation and plant survival.

Governing physics. Controls include tidal prism and asymmetry, surge level, wind waves, creek-edge exchange, sediment concentration and settling, plant frontal area and flexibility, platform elevation, edge geometry, sediment supply, subsidence, and relative sea-level rise.

Dimensionless parameters. Useful controls include relative water level and platform elevation, wave-to-current velocity ratio, relative submergence, drag and frontal-area density, settling-to-advection time, sediment supply relative to accommodation, and SLR relative to accretion.

Major equations. Frameworks combine shallow-water and wave-current flow, vegetation drag, suspended-sediment conservation, erosion/deposition, Exner-type morphology, tidal storage, edge retreat, plant response, and elevation-dependent accretion.

Typical methods. Methods span flux stations, erosion and elevation surveys, channel-network metrics, true-scale surge tests, mown-plot experiments, storm reconstructions, remote topography, and coupled hydro-biogeomorphic modelling.

Numerical models. Three-dimensional routing models, estuary storage models, delta sediment models, decadal edge and channel models, restoration biogeomorphology, pond-growth models, and wave-current plant models address complementary scales.

Experimental datasets. Evidence includes Venice and Massachusetts networks, confined-estuary scenarios, true-scale surge surfaces, Wax Lake floods, restored marshes, microtidal storm delivery, paired vegetation plots, and cross-marsh topographic signatures.

Validated ranges. Reported cases include less than 0.6 cm mean storm-surface lowering, roughly 0.5 cm extra accretion per 60-day flood with stronger tides, doubled marsh flux during a moderate surge, and effects that can reverse sign with sediment supply or scale.

Recent advances. Recent studies resolve competing density effects, surge-enabled sediment delivery, tide-wave import switches, topographic storm signatures, creek-complexity resilience, pond intervention windows, and restoration design controls.

Disagreements. Vegetation does not universally increase deposition: trapping can be offset by flow buffering, small mown plots showed no local effect, and vegetation simultaneously promotes channelization through drag and channel infilling through enhanced platform accretion.

Limitations. Marsh response is scale-, species-, season-, and event-dependent. Sparse storm observations, uncertain belowground and edge processes, simplified plant mechanics, incomplete sediment budgets, and transfer across tidal regimes constrain forecasts.

Open questions. Priorities include storm versus fair-weather budgets, creek-edge pathway partitioning, edge failure distributions, vegetation trait and damage dynamics, pond thresholds, subsidence, sediment diversion, restoration geometry, and uncertainty under accelerating SLR.

Seminal papers. The branch progresses from marsh-surface flux and vegetation-routing concepts to network efficiency, flexible-plant damping, field-calibrated sediment budgets, and coupled restoration and climate-response models.

Claims

1. Three-dimensional modelling showed vegetation organizes unvegetated channels and vegetated levee-basin platforms by shifting flow routing and sedimentation with water level. *Regime: Vegetation routed low-stage flow into unvegetated channels and created levee-basin deposition; overtopping produced more homogeneous sheet-flow sedimentation.* direct finding, numerical [1404]
2. Salt-marsh persistence under climate change depends jointly on vertical accretion, vegetation feedbacks, edge erosion, sediment supply, and accelerating sea-level rise. *Regime: Marsh response reflects interacting mineral and organic accretion, vegetation, edge dynamics, sea-level rise, warming, CO₂, and sediment supply.* review statement, review [448]
3. Cross-system comparisons found vegetation promotes more efficient tidal-channel networks by altering channel branching and meandering. *Regime: Vegetated Massachusetts and Venice marshes had more efficient networks than unvegetated arid systems through vegetation-related branching and meandering.* direct finding, field [695]
4. Salt-marsh budgets require simultaneous high-frequency measurement of water, sediment, organic matter, nutrients, and silica exchange across tidal boundaries. *Regime: Marsh exchange spans tidal water and sediment plus particulate, dissolved, and biogenic compounds, requiring high-frequency, long-duration measurements.* review statement, review [418]
5. Tidal-flat-marsh flood defense is a dynamic biogeomorphic service whose performance changes as morphology and vegetation adjust to forcing and intervention. *Regime: Dynamic morphology and vegetation adjust to forcing and can strengthen or weaken defense; interventions can accelerate favorable natural processes.* review statement, review [1168]
6. Confined-estuary modelling found diminishing attenuation beyond a wetland-size threshold and stronger upstream benefit when the same wetland area was located farther landward. *Regime: Attenuation increased with wetland area to a threshold, and equal-area wetlands farther upstream stored more local flood volume and reduced upstream high water more.* direct finding, numerical [1317]
7. Salt-marsh edge erosion increased with wind-wave exposure, while rare large failures produced long-tailed local erosion even under low-wave-energy conditions. *Regime: Erosion related significantly to wind-wave exposure; low-energy conditions still produced rare large failures and long-tailed local rates.* direct finding, field [803]
8. Deltaic-marsh modelling showed vegetation density has competing trapping and flow-buffering effects, while resuspension extends and homogenizes sediment delivery. *Regime: Vegetation both trapped sediment and buffered delivery; with resuspension sediment penetrated farther and deposited more uniformly.* direct finding, mixed [1551]
9. True-scale storm-surge tests produced less than 0.6 cm mean surface lowering, with organic matter and plant species controlling soil stability and canopy damage. *Regime: Repeated simulated surges lowered surfaces less than 0.6 cm on average; organic matter and species controlled stability and damage.* direct finding, experimental [1333]
10. Biogeomorphic modelling showed restoration inlet geometry can steer marsh development, with larger breach-width contrast increasing habitat diversity. *Regime: Inlet-width asymmetry changed development and greater breach-width contrast increased restored-habitat diversity.* direct finding, numerical [518]
11. A decadal model found sediment supply can reverse wave-driven marsh-edge retreat into expansion, with outcome also controlled by vegetation response time. *Regime: Higher waves lowered and retreated the edge, but sufficient sediment supply could reverse a wave-stressed marsh from retreat to expansion.* direct finding, numerical [1514]

12. Wax Lake simulations found vegetation enhances sedimentation only when trapping exceeds flow buffering, while larger tides added about 0.5 cm accretion per 60-day high-discharge flood. *Regime: Vegetation increased retention only when trapping exceeded flow buffering; larger tides at high discharge added about 0.5 cm accretion per 60-day flood.* direct finding, numerical [1053]
13. Marsh plant-flow prediction requires direct measurement of drag-relevant geometry and mechanics, including frontal area, density, stiffness, and reconfiguration. *Regime: Drag-relevant properties depend on density, frontal area, stiffness, reconfiguration, and measurement method rather than biomass or cover alone.* review statement, review [1406]
14. Restored-marsh observations and modelling linked channel morphology and vegetation to tidal asymmetry, accretion, and possible sea-level-rise amplification of ebb-driven sediment export. *Regime: Ebb dominance and sediment export interact with channels and vegetation and may intensify with sea-level rise, reducing resilience.* direct finding, mixed [449]
15. Wind-wave power controls marsh-margin form, while halophyte colonization reduces lateral erosion and shifts edges toward gentler, irregular morphology. *Regime: Vegetation reduced lateral erosion; high cliffs and smooth edges marked rapid erosion, whereas low-energy shallow-flat margins were gentler and irregular.* direct finding, field [442]
16. Flexible-marsh-plant experiments and modelling showed weak current can reduce wave damping while strong current increases it through reconfiguration and wave-current kinematics. *Regime: Weak currents reduced wave dissipation whereas strong currents increased it through reconfiguration, group velocity, and canopy velocity changes.* direct finding, mixed [1596]
17. Microtidal-bay simulations found wind waves resuspend sediment but surge water levels govern marsh delivery; a moderate surge doubled flux to most marshes. *Regime: Waves controlled flat resuspension, but surge inundation controlled marsh deposition; a moderate surge doubled flux to most marshes and deposition rose linearly with surge intensity.* direct finding, numerical [1621]
18. Vegetation has competing effects on tidal channels—drag promotes channelization while platform accretion reduces tidal prism and infills channels—with dominance set by platform elevation and sediment/SLR regime. *Regime: Vegetation can enhance channelization through drag yet shrink tidal prism through platform accretion and infilling; platform elevation determines which effect dominates.* direct finding, numerical [1285]
19. Field-calibrated models linked pond expansion to sea-level-rise drowning and used a power-law growth relation to identify a finite intervention window before widespread merging. *Regime: Pond growth mechanisms were consistent with SLR-driven drowning and could be summarized by a power law that bounds intervention time before merging.* direct finding, mixed [597]
20. Paired 4 m² marsh plots found no deposition difference between mown and vegetated patches, showing vegetation does not automatically enhance sedimentation at small scales. *Regime: At 4 m² scale mowing did not change flow or deposition, and vegetation presence did not automatically increase local sedimentation.* direct finding, field [1270]
21. Biogeomorphic modelling and real networks showed greater creek-network complexity enhances flow concentration, sediment deposition, and marsh resilience to rising water levels. *Regime: Drainage complexity concentrated flow and increased sediment transport/deposition, improving resilience alongside vegetation traits.* direct finding, mixed [299]
22. Marsh topography preserves a detectable signature of surge-enabled, wave-driven sediment delivery across the tidal-flat-marsh transition. *Regime: Surge water levels enabled wave-driven cross-edge delivery and left a detectable topographic signature consistent across marshes with differing tide and wave regimes.* direct finding, field [1416]
23. Salt-marsh sediment supply can switch between net import and export depending on the joint tide-wave regime and pathway through creeks versus edges. *Regime: Net import or export through creeks and across edges depended on the joint tidal and wave regime and varied among estuaries and marshes.* direct finding, mixed [1372]
24. In 1:20 marsh-dyke basin tests, vegetation reduced waves and overtopping across varied forcing; for a fixed marsh width, taller and denser surrogate marshes produced stronger attenuation. *Regime: Experimental Study of Wave Attenuation Across an Artificial Salt Marsh* direct finding, mixed [106]
25. Winter flume tests found no significant mean-erosion effect of aboveground biomass or difference between two marsh species with comparable root biomass and soil density, suggesting persistent belowground protection in the tested patches. *Regime: Erosion protection by winter state of salt marsh vegetation* direct finding, mixed [1087]
26. A Wadden Sea review concludes that salt-marsh flood and erosion management must integrate site initial conditions, conservation aims, biogeomorphology and climate adaptation; marshes can adapt to sea-level rise under suitable management. *Regime: Management of Wadden Sea Salt Marshes in the Context of Nature Conservation, Coastal Flooding and Erosion Risks: A Review* direct finding, mixed [392]
27. In-situ tests above 0.5 m/s found a mature tidal marsh highly erosion resistant; for comparable soil and vegetation, breach-flow surface erosion may be negligible relative to headcut erosion, but transfer requires further testing. *Regime: Assessment of in-situ tidal marsh erodibility under high flow velocities* direct finding, mixed [142]
28. Hydrodynamic attenuation changes wetting, drying, and inundation depth and can increase modeled coastal-wetland vulnerability to sea-level rise. *Regime: Potential increase in coastal wetland vulnerability to sea-level rise suggested by considering hydrodynamic attenuation effects* direct finding, numerical [1202]
29. Emergent-plant morphology controls vertical velocity and turbulent-kinetic-energy profiles, enabling canopy turbulence prediction from frontal-area structure. *Regime: Measured and Predicted Turbulent Kinetic Energy in Flow Through Emergent Vegetation With Real Plant Morphology* direct finding, experimental [1549]
30. Wider salt marshes and higher foreshores reduced storm wave run-up at Dutch Wadden Sea dikes, but marshes were least likely to form where protection was most needed. *Regime: Using salt marshes for coastal protection: Effective but hard to get where needed most* direct finding, mixed [919]
31. Established marsh topsoil resisted fast-flow erosion better than bare tidal flats, with cohesive layers and fine-root density governing resistance after disturbance. *Regime: The importance of marshes providing soil stabilization to resist fast-flow erosion in case of a dike breach* direct finding, experimental [918]
32. Across eight modeled estuaries, salt marshes reduced storm flooding and damages, with the strongest benefits during high-intensity events. *Regime: Coastal wetlands mitigate storm flooding and associated costs in estuaries* direct finding, numerical [419]
33. Eco-geomorphic scenarios indicate that accelerated sea-level rise can expand newly colonized wetland area while reducing its soil-carbon accumulation opportunity, so vegetation-area conservation does not guarantee maximum carbon storage. *Regime: Accelerated Sea-Level Rise Limits Vegetation Capacity to Sequester Soil Carbon in Coastal Wetlands: A Study Case in Southeastern Australia* direct finding, numerical [1249]
34. Seasonal tests of 418 *Elymus* specimens and resin surrogates spanning 8.45-1708.42 MPa Young's modulus produced bending angles from 0 to 77.4 degrees under 0.4-1.2 m/s currents, supporting Cauchy-scaled plant analogues. *Regime: From seasonal field study to surrogate modeling: Investigating the biomechanical dynamics of Elymus sp. in salt marshes* direct finding, experimental [696]

28.5 Seagrass and aquatic vegetation

Topic *ecohydraulics*. *seagrass*. *Canopy-flow and sediment feedbacks*.

Papers screened: 52. Claims: 50. Equations: 0.

Well established. Seagrass modifies waves, currents, turbulence and sediment through distributed drag, flexible reconfiguration, canopy-interface shear, wave streaming, near-bed sheltering, and particle interception; these physical changes feed back on light, accretion and meadow persistence.

Governing physics. Relative blade–water velocity sets drag; flexibility reduces projected area and changes phase; cumulative drag dissipates waves; canopy shear generates coherent vortices and vertical exchange; sheltering lowers bed stress while wakes add turbulence; sediment–light feedback can create bistability.

Dimensionless parameters. Controls include submergence, canopy height/depth, wave excursion/shoot spacing, Cauchy number, blade stiffness and buoyancy, Reynolds and Keulegan–Carpenter numbers, frontal-area density, meadow length/wavelength, gap/patch size, current/wave velocity ratio, and Shields stress.

Major equations. Models use Morison/drag force with effective frontal area and relative velocity, beam or blade dynamics, wave-energy decay, Reynolds-averaged canopy momentum, turbulent kinetic-energy budgets, advection–diffusion–settling, critical Shields stress, and coupled sediment–light–biomass equations.

Typical methods. Methods span wave/current flumes with flexible mimics or live plants, ADV velocity and turbulence, wave gauges, force and blade-motion imaging, PIV, sediment traps and optical backscatter, field meadow transects and storms, spectral budgets, coupled CFD–structure models, and ecosystem feedback models.

Numerical models. Models include relative-velocity blade dynamics, eelgrass–flow coupling, meadow-scale attenuation, digital twins for surge scenarios, subgrid canopy closures, SPH-finite-element stem interaction, sediment–light bistability, and spatial particle-deposition models.

Experimental datasets. The evidence includes *Posidonia* and *Zostera* field meadows, natural and transplanted canopies, storm before/after surveys, oscillatory and combined-flow flumes, flexible/stiff mimics, fragmented and gapped patches, particle capture/deposition, streaming observations, and spectral wave dissipation.

Validated ranges. The corpus covers low-energy through storm waves, steady, oscillatory and combined flow, submerged and near-emergent canopies, rigid-to-flexible blades, single- and multispecies meadows, intact-to-fragmented patches, and blade-to-barrier-coast scales; detailed numeric ranges remain source-specific.

Recent advances. Recent work directly measures frequency-dependent meadow dissipation, combined-flow resuspension thresholds, turbulence and bedload, carbon-accretion heterogeneity, storm-surge digital twins, transplanted-meadow attenuation, and stem-scale SPH-FE interaction.

Disagreements. Rigid-canopy drag can overpredict attenuation when blades move with water, yet flexible canopies may generate different turbulence rather than uniformly less effect; vegetation can both shelter the bed and add wake stress; shoot density alone is insufficient when flexibility, morphology, patch length and gaps vary.

Limitations. Mimic materials and geometric scaling distort reconfiguration; many studies isolate waves or currents; field morphology is seasonal and heterogeneous; storm breakage, cohesive sediment, root effects, species mixtures, meadow recovery, and full-scale sediment budgets remain underrepresented.

Open questions. Priorities include spectral dissipation across changing morphology, combined wave–current–tide forcing, patch-edge networks, breakage and uprooting, cohesive sediment and carbon, sediment–light thresholds, restoration trajectories, and uncertainty-aware upscaling to coastal protection.

Seminal papers. The branch progresses from eelgrass flow and transport observations and relative-blade-velocity drag to flexible blade models, canopy vortex and streaming measurements, fragmented-patch hydraulics, sediment feedback, spectral budgets, and coupled fluid–structure simulation.

Claims

1. Wave attenuation depends on blade velocity relative to water, so flexible seagrass in low-energy waves experiences less drag than a rigid-canopy assumption predicts. *Regime: Relative velocity of seagrass blades: Implications for wave attenuation in low-energy environments* direct finding, mixed [183]
2. Feedback among seagrass cover, sediment resuspension, and light can create alternative stable vegetated and unvegetated lagoon states. *Regime: Stability and bistability of seagrass ecosystems in shallow coastal lagoons: Role of feedbacks with sediment resuspension and light attenuation* direct finding, numerical [232]
3. Submerged-vegetation wave attenuation varies jointly with organism traits and the magnitude and direction of tidal current. *Regime: Wave attenuation by submerged vegetation: combining the effect of organism traits and tidal current* direct finding, experimental [1084]
4. Vegetation flexibility changes both wave damping and the vertical velocity structure, so stiffness is a primary canopy-response parameter. *Regime: Experimental study on vegetation flexibility as control parameter for wave damping and velocity structure* direct finding, experimental [1445]
5. A storm reorganized flow and suspended-particle distributions within a nearshore seagrass meadow, demonstrating event-scale canopy feedback. *Regime: Flow and particle distributions in a nearshore seagrass meadow before and after a storm* direct finding, field [524]
6. Sediment–light feedback is strongest where resuspension controls turbidity and therefore the light available for seagrass growth. *Regime: Feedback between sediment and light for seagrass: Where is it important?* direct finding, mixed [16]
7. Two-way coupling between eelgrass and flow alters canopy velocity and transport compared with treating vegetation as passive roughness. *Regime: Modeling coupling between eelgrass *Zostera marina* and water flow* direct finding, numerical [9]
8. A blade-dynamics parameterization informed by numerical and laboratory tests improves prediction of flexible-seagrass wave attenuation. *Regime: Improved parameterization of seagrass blade dynamics and wave attenuation based on numerical and laboratory experiments* direct finding, mixed [1583]
9. Eelgrass canopies reduce mean flow and reshape turbulent and scalar transport within and above the meadow. *Regime: Effect of eelgrass *Zostera marina* canopies on flow and transport* direct finding, field [8]
10. Oscillatory flow generates canopy-scale turbulent kinetic energy whose distribution depends on vegetation geometry and wave excursion. *Regime: Turbulent Kinetic Energy in Submerged Model Canopies Under Oscillatory Flow* direct finding, experimental [1589]

11. A digital-twin framework predicts how seagrass nature-based solutions modify storm-surge hydrodynamics under spatially explicit meadow scenarios. *Regime: A Digital Twin modelling framework for the assessment of seagrass Nature Based Solutions against storm surges* direct finding, numerical [1109]
12. Along-meadow development under steady flow changes turbulence and particle deposition, producing spatially varying capture within a submerged canopy. *Regime: Turbulence and Particle Deposition Under Steady Flow Along a Submerged Seagrass Meadow* direct finding, experimental [1593]
13. Field measurements identify wave-induced mean streaming through a submerged *Posidonia* meadow, adding transport not represented by oscillatory velocity alone. *Regime: Field observations of wave-induced streaming through a submerged seagrass (Posidonia oceanica) meadow* direct finding, field [874]
14. Eelgrass modifies bed-load transport and sediment resuspension under oscillatory flow through drag, turbulence, and near-bed sheltering. *Regime: Role of eelgrass on bed-load transport and sediment resuspension under oscillatory flow* direct finding, experimental [917]
15. Particle capture by seagrass under oscillatory flow depends on canopy geometry, particle settling, and the wave-driven exchange through the meadow. *Regime: Particle capture by seagrass canopies under an oscillatory flow* direct finding, experimental [117]
16. Seagrass current reduction cannot be inferred from shoot density alone because blade flexibility changes frontal area and drag. *Regime: Effect of Seagrass on Current Speed: Importance of Flexibility vs. Shoot Density* direct finding, experimental [455]
17. Canopy resistance with complex morphology collapses under a universal scaling based on reconfiguration and effective frontal area. *Regime: Universal scaling-law for flow resistance over canopies with complex morphology* direct finding, mixed [1218]
18. Invasive vegetation and megaherbivore grazing reduce seagrass coastal-protection service by changing canopy structure and wave attenuation. *Regime: Seagrass coastal protection services reduced by invasive species expansion and megaherbivore grazing* direct finding, field [646]
19. Current- and wave-driven velocities attenuate differently within multispecific submerged canopies because species morphology and flexibility interact. *Regime: The attenuation of current- and wave-driven flow within submerged multispecific vegetative canopies* direct finding, experimental [1502]
20. Fragmentation produces patch-edge acceleration and spatially heterogeneous flow that differs from continuous-meadow behavior. *Regime: Interactions between Fragmented Seagrass Canopies and the Local Hydrodynamics* direct finding, experimental [32]
21. Oscillatory canopy flow generates coherent vortices at the canopy interface that control vertical exchange and turbulence. *Regime: Vortex generation in oscillatory canopy flow* direct finding, experimental [495]
22. Anthropogenic gaps reduce wave attenuation through a *Posidonia* meadow by interrupting cumulative canopy drag. *Regime: Impact of anthropogenically created canopy gaps on wave attenuation in a Posidonia oceanica seagrass meadow* direct finding, field [288]
23. Meadow-scale length and patch arrangement can influence within-canopy flow as strongly as local shoot-scale traits. *Regime: Comparison of the influence of patch-scale and meadow-scale characteristics on flow within seagrass meadows: a flume study* direct finding, experimental [17]
24. Seagrass wave damping under combined waves and currents depends on current direction and magnitude through altered blade posture and relative velocity. *Regime: Wave damping by seagrass meadows in combined wave-current conditions* direct finding, experimental [1261]
25. Coastal canopies enhance vertical mixing through shear-layer turbulence while suppressing turbulence deeper inside dense vegetation. *Regime: Vertical mixing in coastal canopies* review statement, review [10]
26. Field observations in a sheltered environment show seagrass meadows reduce wind-wave-driven sediment resuspension relative to unvegetated bed. *Regime: Seagrass Meadows Reduce Wind-Wave Driven Sediment Resuspension in a Sheltered Environment* direct finding, field [1015]
27. Under unidirectional flow, increasing vegetation flexibility changes canopy deflection, effective blockage, velocity reduction, and turbulence. *Regime: Hydrodynamics affected by submerged vegetation with different flexibility under unidirectional flow* direct finding, mixed [1536]
28. Sediment and carbon accretion vary within a meadow in correlation with spatial hydrodynamic intensity. *Regime: Spatial heterogeneity in sediment and carbon accretion rates within a seagrass meadow correlated with the hydrodynamic intensity* direct finding, field [795]
29. An underlying current changes wave propagation over natural seagrass by modifying relative blade motion and the cumulative work of vegetation drag. *Regime: Let it flow: how does an underlying current affect wave propagation over a natural seagrass meadow?* direct finding, field [1085]
30. Marine-canopy edges create distinct oscillatory-flow zones of adjustment, enhanced turbulence, and exchange. *Regime: Local hydrodynamics at edges of marine canopies under oscillatory flows* direct finding, experimental [1283]
31. Submerged vegetation modifies turbulence and bed-load transport through competing sheltering and canopy-generated stress. *Regime: Turbulence and Bedload Transport in Submerged Vegetation Canopies* direct finding, experimental [1609]
32. Combined waves and currents generate a three-dimensional artificial-meadow flow structure not recoverable by superposing wave-only and current-only cases. *Regime: Flow Structure in an Artificial Seagrass Meadow in Combined Wave-Current Conditions* direct finding, experimental [1260]
33. Coupled SPH-finite-element modelling resolves stem-scale fluid-structure interaction and the feedback of blade deformation on wave loading. *Regime: Exploring Wave-Vegetation Interaction at Stem Scale: Analysis of the Coupled Flow-Structure Interactions Using the SPH-Based DualSPHysics Code and the FEA Module of Chrono* direct finding, numerical [1144]
34. Field spectral budgets directly resolve frequency-dependent wave-energy dissipation across a seagrass meadow. *Regime: Spectral Wave Energy Dissipation by a Seagrass Meadow* direct finding, field [1016]
35. Mimic stiffness controls regular-wave transmission and canopy turbulence by changing plant reconfiguration. *Regime: Effect of Mimic Vegetation with Different Stiffness on Regular Wave Propagation and Turbulence* direct finding, experimental [1390]
36. Site-scale assessment links seagrass meadow geometry and local wave climate to barrier-beach coastal protection. *Regime: Assessing the coastal protection role of seagrass meadows along a barrier beach, southern Romanian coast* direct finding, mixed [344]
37. Transplanted seagrass can attenuate fetch-limited waves, with performance governed by meadow maturity, density, and incident conditions. *Regime: Potential of Transplanted Seagrass Meadows on Wave Attenuation in a Fetch-Limited Environment* direct finding, numerical [1307]
38. Combined wave-current forcing shifts the threshold and spatial pattern of sediment resuspension within an artificial seagrass meadow. *Regime: Initiation of Sediment Resuspension by Combined Wave-Current Conditions in an Artificial Seagrass Meadow* direct finding, experimental [1610]
39. Within a real-morphology *Typha* canopy, equilibrium suspended sediment was nearly uniform through the culm region and declined in the distributed-leaf region; a morphology-adapted diffusivity model reproduced profile evolution. *Regime: Suspended Sediment Concentration Profile in a <i>Typha Latifolia</i> Canopy* direct finding, mixed [1550]
40. A cross-canopy review identifies the mismatch between patch/gap measurements and landscape-scale management as the principal upscaling obstacle and calls for more wave, three-dimensional-gap and fuzzy-boundary studies. *Regime: Biophysical Interactions in Fragmented Marine Canopies: Fundamental Processes, Consequences, and Upscaling* direct finding, mixed [454]

41. Artificial emergent *Phragmites* attenuated regular and irregular waves across tested densities; wave height fell 6–25% at canopy entry and continued decreasing through the vegetation field. *Regime: LABORATORY SIMULATIONS OF WAVE ATTENUATION BY AN EMERGENT VEGETATION OF ARTIFICIAL PHRAGMITES AUSTRALIS: AN EXPERIMENTAL STUDY OF AN OPEN-CHANNEL WAVE FLUME* direct finding, mixed [813]
42. A 2000–2024 experimental review recommends jointly reporting wave-energy dissipation, drag, transmission, attenuation, wave-induced velocity and turbulent kinetic energy to compare seagrass wave attenuation consistently. *Regime: Quantifying wave attenuation by seagrass: a comprehensive review of assessment techniques* direct finding, mixed [1555]
43. Wave–vegetation models trade ecological detail and domain scale: high-resolution models resolve morphology and biomechanics over small areas, while ecosystem-scale engineering models simplify those features. *Regime: Numerical Modelling of Wave–Vegetation Interaction: Embracing a Cross-Disciplinary Approach for Bridging Ecology and Engineering for Nature-Inclusive Coastal Defence Systems* direct finding, mixed [1145]
44. Intertidal vegetation stiffness creates a trade-off: stiffer shoots dissipate more wave energy but experience greater drag than flexible vegetation. *Regime: TRADE-OFFS RELATED TO ECOSYSTEM ENGINEERING: A CASE STUDY ON STIFFNESS OF EMERGING MACROPHYTES* direct finding, experimental [176]
45. A review links seagrass wave-current attenuation to sedimentation and ecological processes while identifying major tropical Indonesian evidence gaps. *Regime: Hydrodynamics across seagrass meadows and its impacts on Indonesian coastal ecosystems: A review* review statement, review [1198]
46. A hydrodynamic-ecological synthesis shows that realistic plant morphology, patch dimensions, density, and spacing are necessary for transferable vegetation-flow experiments and models. *Regime: How to build vegetation patches in hydraulic studies: a hydrodynamic-ecological perspective on a biological object* review statement, review [298]
47. Live submerged-vegetation species and patch configuration control the length and strength of downstream sheltering wakes. *Regime: Flow and Wake Length Downstream of Live Submerged Vegetation Patches: How Do Different Species and Patch Configurations Create Sheltering in Stressful Habitats?* direct finding, experimental [297]
48. Vegetated-bed sediment pickup is closely related to turbulent kinetic energy, with canopy geometry modifying entrainment relative to bare beds. *Regime: Sediment Pickup Rate in Bare and Vegetated Channels* direct finding, experimental [1553]
49. Artificial-seagrass meadow experiments quantify downstream shelter distance and translate wake behavior into restoration-layout guidance. *Regime: Wake length of an artificial seagrass meadow: a study of shelter and its feasibility for restoration* direct finding, experimental [1456]
50. Physical modeling shows that surrogate flexural rigidity, buoyancy, and geometry govern the flow and wake generated by individual seagrass blades. *Regime: Flow field and wake structure characteristics imposed by single seagrass blade surrogates* direct finding, experimental [1396]

29 Wiki: Estuaries and tidal inlets

29.1 Estuaries and tidal inlets: overview

Topic estuaries. Hydrodynamics, mixing, and exchange in estuaries and inlets.

Papers screened: 13. Claims: 1. Equations: 0.

Well established. Estuaries exchange freshwater, salt, sediment and tracers under interacting river, tidal and density forcing. Stratification and mixing set salinity intrusion; gravitational circulation, tidal pumping and straining control exchange, while geometry and adjacent-shelf variability can reorganize these balances.

Governing physics. Longitudinal density gradients drive baroclinic exchange, tides advect and mix salinity fronts, and river discharge opposes landward salt transport. Wind and coastal-ocean density anomalies alter boundary conditions; convergence, rotation and evolving bathymetry modify residual circulation and sediment pathways.

Dimensionless parameters. Controls include tidal-excursion-to-intrusion-length, estuarine Richardson, Froude and Simpson numbers, freshwater and tidal-volume ratios, mixing and dispersion parameters, convergence length, stratification, Rossby number, settling and Shields measures.

Major equations. Models solve hydrostatic mass, momentum, salt and sediment conservation with turbulence closure and free-surface forcing. Knudsen bulk relations and isohaline coordinates diagnose exchange; analytical intrusion laws balance advection and dispersion, and morphodynamic systems add bed evolution.

Typical methods. Studies combine salinity, velocity and water-level moorings, hydrographic transects, shelf observations and water sampling with unstructured-grid circulation, HEC-RAS, analytical intrusion, isohaline-budget, coupled groundwater and three-dimensional morphodynamic models.

Numerical models. Reviewed systems include validated unstructured hydrodynamics, isohaline-coordinate diagnostics, Savenije intrusion theory, HEC-RAS salinity transport, coupled surface–subsurface variable-density flow and COAWST-like three-dimensional sediment morphodynamics.

Experimental datasets. Evidence includes the Merrimack and Hudson contrast, low-flow Willapa Bay, the 2014 Changjiang intrusion, 2015 Sebou supermoon observations, lower Meghna irrigation sampling, idealized funnel estuaries and marsh–farmland surge scenarios.

Validated ranges. Examples include 17.6 m/s Changjiang wind and 23-day intake interruption, near-10 cm/s Willapa intrusion, 1.3 dS/m lower-Meghna salinity, and conditional agricultural protection by 200–400 m marsh migration under at least 3 m surge on a 0.1% slope.

Recent advances. Recent advances use isohaline coordinates across contrasting regimes, resolve episodic wind-driven intrusion, couple surface and subsurface salinization, embed baroclinicity in long-term morphology and quantify counterintuitive convergence–mixing interactions.

Disagreements. Classical steady gravitational-circulation scaling works in long estuaries but can fail in short tidally dominated systems. Convergence may enhance residual flow under constant viscosity yet weaken or reverse tidal-straining circulation when mixing varies over the tide.

Limitations. Sparse vertical and lateral sampling, uncertain bathymetry and diffusivity, rapidly changing discharge and coastal boundaries, unresolved turbulence, wetland and groundwater coupling, morphology feedback and limited extreme-event records constrain transfer.

Open questions. Priorities include compound drought–surge–wind intrusion, shelf–estuary exchange, wetland migration, oxygen and sediment feedbacks, data assimilation, climate-driven regime shifts and scalable predictors spanning short and long estuaries.

Seminal papers. Classical salt-balance, Knudsen and gravitational-circulation theories established exchange and intrusion scaling; later tidal-straining, pumping and estuarine-classification frameworks explained time-dependent mixing and residual transport.

Claims

1. For the validated 2006-2012 Delaware Bay hindcast, shelf waves were predominantly remote swell, bathymetric and tidal-current refraction controlled entrance energy, and energetic in-bay waves were mainly locally wind driven with fetch set by wind direction. *Regime: Delaware Bay and adjacent shelf, 2006-2012, validated at three buoys.* direct finding, mixed [748]

29.2 Estuarine exchange flow

Topic estuaries. exchange. Net exchange and flushing.

Papers screened: 35. Claims: 33. Equations: 0.

Well established. Estuarine exchange is compensating inflow and outflow driven by baroclinic pressure but strongly modified by tides, mixing, wind, river discharge, lateral circulation, geometry and shelf conditions; net volume transport alone conceals this renewal.

Governing physics. Density gradients drive gravitational circulation; tidal straining and periodic stratification modulate shear; covariance of eddy viscosity and shear rectifies tides; lateral advection and bathymetry redistribute exchange; wind can reinforce, weaken or reverse flow; mixing entrains salt and transforms water masses.

Dimensionless parameters. Controls include estuarine Froude and Richardson numbers, tidal and freshwater velocity ratio, Simpson and Wedderburn-type straining measures, salinity difference, mixing efficiency, depth/width and convergence, tidal excursion/length, river flow/tidal prism, wind stress/baroclinic pressure, and flushing-time/tidal-period.

Major equations. Core formulations include along-channel momentum and salt balance, Knudsen relations, residual/tidal decomposition, exchange in salinity coordinates (TEF), salinity variance and mixing budgets, eddy-viscosity-shear covariance, box/flushing models, water-age transport, and nondimensional estuarine scaling.

Typical methods. Methods combine moored velocity/salinity sections, ship surveys and tracers, residual harmonic and covariance analysis, primitive-equation models, idealized channels, TEF salinity-coordinate integration, variance/mixing budgets, particle or age tracers, flushing boxes, sensitivity experiments, and shelf-source tracking.

Numerical models. Models include idealized 1DV/2DV and three-dimensional circulation systems, Hudson and urban-channel configurations, TEF diagnostics, salinity-variance budgets, eddy-viscosity-shear covariance, unsteady flushing models, shelf-source particle tracking, plume/coastal forcing experiments, and tracer budgets.

Experimental datasets. The corpus spans Hudson and San Francisco systems, Galveston and Capibaribe bays, Wadden and German Bight estuaries, macrotidal and narrow idealized channels, along-coast coupled estuaries, glacial fjords, and the segmented multi-basin Salish Sea.

Validated ranges. Evidence covers well-mixed to partially and periodically stratified estuaries, weakly to strongly tidal and macrotidal settings, narrow to wide channels, seasonal segmentation, spring-neap variability, local and remote winds, channel deepening, glacial plume forcing, and connected shelf-estuary systems.

Recent advances. Recent advances use salinity-coordinate TEF, variance budgets tying exchange to mixing, shelf-source tracing, segmented-estuary diagnostics, plume-versus-coastal attribution in fjords, multiple closed tracer budgets, and explicit tests of remote-ocean forcing limits.

Disagreements. Classical two-layer gravitational circulation is useful but incomplete: tidal rectification can generate exchange in well-mixed water, lateral advection can rival longitudinal dynamics, wind can reverse the expected pattern, and flushing/residence estimates depend on diagnostic definition and temporal averaging.

Limitations. Sparse sections miss lateral recirculation; residuals depend on averaging; salinity-coordinate transport can hide geography; model mixing is resolution/closure sensitive; residence time is not unique; open-boundary shelf properties and episodic winds are uncertain; constituent transformation complicates tracer closure.

Open questions. Needs include unified TEF and water-age diagnostics, lateral and multi-inlet exchange, event-to-climate variability, morphology/deepening and sea-level effects, compound wind-river-shelf forcing, exchange-biogeochemistry coupling, and observation-based mixing closure.

Seminal papers. The branch evolves from partially mixed momentum/salt balances and tidal rectification toward lateral-advection models, mixing covariance, wind reversal, salinity-variance links, TEF, shelf-source attribution, seasonal segmentation, and multi-tracer budgets.

Claims

1. Estuarine circulation emerges from gravitational, tidal, wind, lateral, and mixing processes and cannot generally be reduced to a steady two-layer gravitational flow. *Regime: The Estuarine Circulation* review statement, review [493]
2. Partially mixed estuary dynamics require coupled salt balance, stratification, vertical mixing, and residual circulation rather than classical steady assumptions alone. *Regime: The Dynamics of a Partially Mixed Estuary** direct finding, analytical [492]
3. Time-dependent salt flux in a partially stratified estuary reflects covariance among tidal velocity, salinity, stratification, and mixing, not only mean exchange. *Regime: Mechanisms Driving the Time-Dependent Salt Flux in a Partially Stratified Estuary** direct finding, mixed [805]
4. Tidal asymmetry in stratification and mixing creates residual estuarine flows even when simple time-mean forcing appears symmetric. *Regime: Creation of residual flows in a partially stratified estuary* direct finding, numerical [1344]
5. Hudson observations and modelling show subtidal salinity and velocity respond jointly to river discharge, tides, and changing stratification. *Regime: Subtidal Salinity and Velocity in the Hudson River Estuary: Observations and Modeling* direct finding, mixed [1151]
6. In tidally energetic idealized estuaries, residual circulation is set by gravitational forcing together with tidal mixing and channel geometry. *Regime: Drivers of Residual Estuarine Circulation in Tidally Energetic Estuaries: Straight and Irrotational Channels with Parabolic Cross Section* direct finding, analytical [211]
7. Lateral advection materially alters residual Hudson estuary circulation, exposing limits of one-dimensional exchange-flow models. *Regime: The Influence of Lateral Advection on the Residual Estuarine Circulation: A Numerical Modeling Study of the Hudson River Estuary* direct finding, numerical [1276]
8. Channel deepening strengthens salinity intrusion, estuarine circulation, and stratification through altered geometry and mixing in an urbanized estuary. *Regime: Response to Channel Deepening of the Salinity Intrusion, Estuarine Circulation, and Stratification in an Urbanized Estuary* direct finding, numerical [1153]
9. Total exchange flow is quantitatively related to mixing through the estuarine salinity-variance budget. *Regime: Estuarine Exchange Flow Is Related to Mixing through the Salinity Variance Budget* direct finding, analytical [887]
10. Local and remote winds can reverse estuarine circulation, with along-coast estuaries enhancing the response. *Regime: Reverse Estuarine Circulation Due to Local and Remote Wind Forcing, Enhanced by the Presence of Along-Coast Estuaries* direct finding, numerical [498]
11. Covariance between eddy viscosity and shear provides a tidal and gravitational mechanism for generating estuarine exchange flow. *Regime: Generation of exchange flows in estuaries by tidal and gravitational eddy viscosity-shear covariance (ESCO)* direct finding, analytical [343]
12. Galveston Bay flushing and transport time scales vary with unsteady river, wind, tide, and exchange forcing rather than a single fixed residence time. *Regime: Time scales in Galveston Bay: An unsteady estuary* direct finding, numerical [1166]
13. Asymmetric tidal mixing generates residual currents in weakly stratified narrow estuaries. *Regime: Residual Currents Induced by Asymmetric Tidal Mixing in Weakly Stratified Narrow Estuaries* direct finding, analytical [267]
14. Tidally periodic stratification creates subtidal circulation through correlations between shear and mixing over the tidal cycle. *Regime: Role of tidally periodic density stratification in the creation of estuarine subtidal circulation* direct finding, mixed [1345]
15. Tidal straining can drive estuarine circulation even in nominally well-mixed estuaries. *Regime: Analysis of Tidal Straining as Driver for Estuarine Circulation in Well-Mixed Estuaries* direct finding, analytical [210]

16. Observed lower Capibaribe circulation governs scalar transport through vertically and laterally structured tidal and residual flow. *Regime: The circulation of the lower Capibaribe Estuary (Brazil) and its implications for the transport of scalars* direct finding, field [1262]
17. The balance between wind straining and gravitational forcing determines exchange-flow direction and magnitude in tidally energetic estuaries. *Regime: The Relative Importance of Wind Straining and Gravitational Forcing in Driving Exchange Flows in Tidally Energetic Estuaries* direct finding, mixed [768]
18. Macrotidal estuary circulation responds distinctly to tidal, riverine, and wind forcing, requiring their joint treatment. *Regime: Tidal, Riverine, and Wind Influences on the Circulation of a Macrotidal Estuary* direct finding, mixed [166]
19. Depth-to-width ratio controls exchange circulation in periodically stratified tidal channels by changing lateral and vertical mixing structure. *Regime: Impact of the Depth-to-Width Ratio of Periodically Stratified Tidal Channels on the Estuarine Circulation* direct finding, numerical [1269]
20. Seasonal segmentation reorganizes estuarine exchange-flow pathways and variability rather than merely scaling a fixed two-layer pattern. *Regime: Estuarine Exchange Flow Variability in a Seasonal, Segmented Estuary* direct finding, mixed [292]
21. Total Exchange Flow analysis resolves the magnitude and salinity classes of exchange across the interconnected Salish Sea. *Regime: Estuarine Exchange Flow in the Salish Sea* direct finding, mixed [888]
22. Estuarine inflow water can originate from distinct shelf source regions, linking exchange properties to shelf circulation and water-mass history. *Regime: The shelf sources of estuarine inflow* direct finding, numerical [187]
23. Flushing-rate analysis separates spring–neap and spatial variations of gravitational circulation from tidal exchange. *Regime: Using flushing rate to investigate spring–neap and spatial variations of gravitational circulation and tidal exchanges in an estuary* direct finding, numerical [1288]
24. Glacial-fjord exchange variability reflects competing plume forcing and coastal-ocean forcing. *Regime: Relative Roles of Plume and Coastal Forcing on Exchange Flow Variability of a Glacial Fjord* direct finding, mixed [1247]
25. Exchange flow materially controls multiple tracer budgets in the Salish Sea and cannot be replaced by net volume transport alone. *Regime: Impact of Estuarine Exchange Flow on Multiple Tracer Budgets in the Salish Sea* direct finding, mixed [1548]
26. Salish Sea exchange has finite sensitivity to remote ocean forcing because internal mixing, geometry, and local forcing constrain the response. *Regime: The Limits of Ocean Forcing on the Exchange Flow of the Salish Sea* direct finding, numerical [1248]
27. Isohaline-coordinate analysis satisfies the Knudsen relation in long and short estuaries and shows Merrimack exchange scales with tidal volume flux and tidal-excursion-to-intrusion-length ratio rather than baroclinic pressure gradient. *Regime: Estuarine Exchange Flow Quantified with Isohaline Coordinates: Contrasting Long and Short Estuaries* direct finding, numerical [258]
28. During low runoff, Willapa Bay salinity, temperature and currents were controlled mainly by shelf upwelling–downwelling variability; gravity-current transmission moved new water landward near 10 cm/s with multi-day inland lags. *Regime: Coupling between the California Current System and a coastal plain estuary in low riverflow conditions* direct finding, field [594]
29. Three-dimensional baroclinic simulations show density-driven subtidal flow has leading-order morphodynamic effects, promoting near-bed flood dominance and sediment import while changing channel concavity, delta volume, bars and asymmetry. *Regime: Effects of Density-Driven Flows on the Long-Term Morphodynamic Evolution of Funnel-Shaped Estuaries* direct finding, numerical [1049]
30. Idealized simulations show estuarine convergence can reduce or reverse tidal-straining circulation, counter to constant-viscosity theory, and thereby weaken up-estuary particulate transport. *Regime: Impact of estuarine convergence on residual circulation in tidally energetic estuaries and inlets* direct finding, numerical [212]
31. Even in the vigorously mixed Marsdiep tidal inlet, slack water with weak wind and high freshwater discharge permits stratification, density-front reversal and mode-one solitary internal waves generated as decelerating ebb flow crosses a transcritical regime; trench lee waves provide a second source. *Regime: The highly energetic, alternating-tidal Marsdiep inlet of the western Wadden Sea during transient stratification around slack tide.* direct finding, field [538]
32. Clustered GPS drifters in a shallow microtidal estuary reveal anisotropic, scale-dependent dispersion dominated by horizontal strain from shear and stretching rather than turbulence alone, with streamwise spreading much stronger than cross-stream spreading. *Regime: Surface dispersion over subtidal-period observations in a small shallow microtidal estuary during two flood and one slack tide.* direct finding, field [1364]
33. A two-layer Estuary Box Model that balances volume, salinity/density and gravitational potential energy converts river discharge, mouth salinity and tidal mixing into estuarine outflow properties; Columbia tests reproduce seasonal salinity and exchange, while global CESM coupling changes local and remote surface salinity relative to direct runoff insertion. *Regime: Estuaries representable by an adjusted two-layer exchange structure in coarse global ocean/Earth-system grids.* direct finding, numerical [1368]

29.3 Tidal inlet dynamics

Topic estuaries. inlets. Hydraulics and stability of inlets.

Papers screened: 21. Claims: 18. Equations: 1.

Well established. A sandy tidal inlet remains open when current-driven sediment export and scour can balance marine or littoral sediment delivery; closure or cross-sectional adjustment occurs when that balance is persistently shifted.

Governing physics. Tidal prism and entrance geometry set exchange velocities, while waves, littoral drift, ebb transport, river flow, storms, and antecedent morphology control sediment import and export; feedback between area, velocity, and transport creates stable, unstable, or closed states.

Dimensionless parameters. No universal dimensionless stability parameter is established by this reviewed slice. Tide-wave dominance, transport-capacity ratios, and geometry-normalized relations are regime descriptors whose coefficients depend on inlet population and units.

Major equations. Common reduced formulations include the O'Brien prism-area relation, the Escoffier area-velocity stability diagram, Bruun sediment-capacity criteria, and nonlinear attractor equations; the 2004 Frisian formulation adds a physical ebb-transport balance to the Escoffier construction.

Typical methods. Methods combine field wave-current-bed observations, coupled morphodynamic simulation, analytical sediment balances, equilibrium/stability diagrams, nonlinear dynamical systems, aerial or chart morphology mapping, hindcast climatologies, and multi-inlet empirical correlation.

Numerical models. Wilson Inlet uses coupled cross-shore/longshore morphodynamics to isolate closure mechanisms; reduced engineering tools include O'Brien, Escoffier, Bruun, and an attractor map, each with distinct sediment and hydrodynamic simplifications.

Experimental datasets. Key reviewed evidence includes the summer 1995 Wilson Inlet field campaign, historical Frisian Inlet adjustment before and after basin reduction, a Snowy River observed/design case, and an 89-inlet United States database paired with 20-year wave hindcasts.

Validated ranges. Evidence is regime-bound: Wilson Inlet is a microtidal wave-dominated seasonal inlet; Frisian results address basin reduction; the attractor comparison uses Snowy River; and morphology correlations span 89 United States inlets but perform poorly in the moderate-wave-exposure group.

Recent advances. The 2012 multi-inlet database quantifies limits of classical morphology classification, while the 2018 attractor comparison unifies long-term tidal and river-controlled states and clarifies how reduced tools should support preliminary design.

Disagreements. Simple prism-area relations imply a compact equilibrium geometry, whereas the empirical morphology study found limited classification improvement from tidal prism alone. These results address different outputs but jointly warn against treating prism as a sufficient universal predictor.

Limitations. The reviewed studies simplify sediment supply, geometry, or transient forcing; two extractions are abstract-limited, field cases are geographically sparse, empirical correlations omit geology and engineering controls, and reduced rules are unsuitable as stand-alone final design models.

Open questions. Priorities include transferable stability criteria under nonstationary sea level and wave climate, transient closure/reopening prediction, multi-inlet interaction, uncertainty-aware sediment budgets, and independent validation across geological and engineered inlet classes.

Seminal papers. Within this screened branch, Wilson Inlet (1999) supplies a process-resolving closure case and the Frisian Inlet study (2004) gives Escoffier stability a sediment-balance basis; earlier O'Brien, Escoffier, and Bruun works remain seminal but were not directly extracted here.

Equations

- O'Brien prism-area check [600]

$$A/P = 6.56 \times 10^{-5}$$

Regime: The paper's stated calibration and units; not a universal dimensionless constant. Variables: A inlet entrance cross-sectional area; P tidal prism

Claims

1. At Wilson Inlet, persistent summer swell drove onshore sediment transport that exceeded removal by weak inlet currents and formed the entrance bar responsible for seasonal closure. *Regime: Wilson Inlet and comparable microtidal, wave-dominated, seasonally open sandy inlets under persistent summer swell.* direct finding, mixed [1156]
2. The influence of summer streamflow and storm events on inlet open duration depended on both event intensity and timing. *Regime: The Wilson Inlet model scenarios; event-response direction is not universal outside the modeled sequence and sediment state.* direct finding, numerical [1156]
3. For the Frisian Inlet framework, annual-mean entrance area is an equilibrium in which ebb currents restore cross-section losses caused by seasonal storm deposition. *Regime: The paper's sandy inlet schematization with annually averaged littoral input and separable entrance/interior sections.* direct finding, analytical [743]
4. An inlet equilibrium is stable only when a perturbation in cross-sectional area drives the section back toward its equilibrium value; the allowable reduction before closure measures its stability margin. *Regime: The extended Escoffier construction applied to basin reduction.* direct finding, analytical [743]
5. O'Brien, Escoffier, Bruun, and attractor methods can provide rapid preliminary stability checks, but the authors advise that final inlet design use more sophisticated models. *Regime: Sandy barrier estuaries and tidal inlets consistent with each reduced method's assumptions.* direct finding, review [600]
6. The attractor formulation generalizes prism-area, Escoffier, and Bruun concepts by representing long-term tidal and river-controlled entrance states within one nonlinear dynamical framework. *Regime: Barrier estuaries with erodible entrances and the model's reduced sediment and hydrodynamic balances.* direct finding, analytical [600]
7. Across 89 United States tidal inlets, the Hayes tide-range/wave-height diagram had limited applicability, and replacing tide range with tidal prism did not materially improve morphology classification. *Regime: The sampled Atlantic, Gulf, and Pacific United States inlets and available morphology classifications.* direct finding, field [1630]
8. In idealized COAWST inlet simulations, ebb-shoal bathymetry was the dominant control on outflow-jet structure, while incident waves increased lateral spreading and limited seaward reach through momentum conversion, friction, and Stokes-drift interaction, with regime-dependent return-current effects. *Regime: Idealized tidal-inlet configurations varying ebb shoal, outflow, incident waves, and channel geometry.* direct finding, numerical [1048]
9. At shallow inlets, maximum ebb and flood balances are dominated by advection, pressure and bottom friction alongstream and centrifugal-pressure balance cross-stream; near slack, local acceleration and pressure dominate. *Regime: Transient Tidal Circulation and Momentum Balances at a Shallow Inlet* direct finding, numerical [585]
10. Stability analysis and morphodynamic modeling relate tidal-basin channel number and spacing to bed slope, Shields parameter, and mean water depth. *Regime: Modeling of channel patterns in short tidal basins* direct finding, analytical [913]
11. Field measurements at a wave-dominated tidal inlet show currents modify wave propagation and create systematic spatial and temporal variations in wave height and water level. *Regime: Wave-current interactions in a wave-dominated tidal inlet* direct finding, field [349]
12. Analysis of short unvegetated tidal flats identifies geometric and hydraulic controls on funnel-shaped tidal-channel equilibrium morphology. *Regime: On funneling of tidal channels* direct finding, analytical [771]
13. Delft3D-SWAN experiments show oblique waves drive tidal-inlet migration through coupled channel, shoal, and alongshore sediment-transport mechanisms. *Regime: Mechanics and rates of tidal inlet migration: Modeling and application to natural examples* direct finding, numerical [1027]
14. Field reconstruction of New Inlet links barrier narrowing, storm surge, astronomical tide, restricted exchange, and tidal phase differences to breach formation and sediment bypassing. *Regime: Inlet Formation and Evolution of the Sediment Bypassing System: New Inlet, Cape Cod, Massachusetts* direct finding, field [446]

15. A Vietnamese inlet study relates seasonal monsoon waves and microtidal forcing to sand-spit evolution and recurring entrance-morphology change. *Regime: Comprehensive Study of the Sand Spit Evolution at Tidal Inlets in the Central Coast of Vietnam* direct finding, mixed [61]
16. Across 35 estuaries and 190 tidal bars, summed bar width approximately follows channel excess width and bar dimensions scale with channel width or tidal prism. *Regime: Topographic forcing of tidal sandbar patterns for irregular estuary planforms* direct finding, mixed [806]
17. In a 110-year Delft3D analogue of Ameland Inlet, increasing relative sea-level rise strengthens flood dominance, erodes the ebb-tidal delta and accretes the basin; tidal flats persist near the 0.2 m scenario but drown under 0.7 m sea-level rise plus subsidence. *Regime: Highly schematized large tidal-inlet/basin morphology representing Ameland Inlet under the specified relative-sea-level scenarios.* direct finding, numerical [346]
18. At Nagatsura-ura Lagoon, tidal-inlet depth controls lagoon water-level response and the magnitude of salinity intrusion: spectral observations and three-dimensional simulations show deeper channels strengthen tidal coherence and salt exchange. *Regime: Nagatsura-ura Lagoon and its shallow connection to Oppa Bay under the observed tidal response and tested inlet-depth cases.* direct finding, mixed [1492]

29.4 River plumes

Topic estuaries.plumes. Buoyant coastal discharge.

Papers screened: 31. Claims: 31. Equations: 0.

Well established. River discharge forms buoyant bulges, fronts and coastal currents whose fate reflects competition among inertia, buoyancy, rotation, tides, wind, waves, mixing and shelf geometry.

Governing physics. Freshwater spreads radially near the mouth, adjusts geostrophically over a deformation radius and commonly turns into a coastally trapped current. Supercritical outflow forms leading fronts; tides strain and mix the plume; wind-driven Ekman transport compresses, deepens or exports it.

Dimensionless parameters. Key controls include internal Froude, Kelvin/source-width-to-deformation-radius, Rossby and Richardson numbers, discharge and buoyancy flux, tidal excursion/source width, wind stress/buoyancy forcing, plume depth/shelf depth and Ekman transport.

Major equations. Models use reduced-gravity shallow-water or hydrostatic primitive equations with salinity buoyancy, Coriolis, turbulence closure and surface/bottom stress. Integral salt, volume and freshwater budgets estimate entrainment, flushing and transport.

Typical methods. Approaches combine ship and ferry salinity transects, gliders, moorings, HF radar, microstructure, drifters, satellite color/temperature/turbidity, Lagrangian inversion, idealized theory and nested three-dimensional coastal models.

Numerical models. The corpus spans reduced-gravity theory, idealized rotating plume models, FVCOM and other primitive-equation shelf models, Lagrangian discharge inversion, front-resolving Rhine simulations and wave-current models for small nearshore outflows.

Experimental datasets. Evidence includes Columbia, Fraser, Changjiang, Rhine, Gironde and Winyah Bay plumes; Great Barrier Reef floods; Oregon and Kochi coastal currents; small Southern California streams; and northern Adriatic flood-plume interactions.

Validated ranges. Studies cover small creeks to major rivers, weakly rotating to rotational plumes, tidal and nontidal shelves, surface- and bottom-trapped structures, downwelling and upwelling winds, flood pulses to four-year records and near-field to regional scales.

Recent advances. Recent work resolves bottom-generated tidal mixing, bottom trapping, wind-driven supercritical export, small-creek sensitivity, high-resolution frontal instability, satellite-Lagrangian discharge inversion and coupled radar/glider/model attribution.

Disagreements. Classical surface-trapped coastal-current theory does not encompass bottom-trapped plumes sustained by tidal mixing, upstream spreading, or wind-driven supercritical export; apparent plume boundaries also differ among salinity, temperature and optical tracers.

Limitations. Fronts and mixing are intermittent and undersampled, discharge data can be uncertain, satellite proxies are not unique salinity measures, turbulence closures differ, nested boundaries affect transport and idealized theories simplify source geometry and ambient currents.

Open questions. Needs include predictive mixing closure, submesoscale frontal instability, wave effects, multiple interacting plumes, sediment and biogeochemical feedback, flood extremes, data assimilation, and robust transfer among source-width, rotation and tidal regimes.

Seminal papers. Foundational theory framed radial spreading versus rotational trapping; long river-mouth records and Columbia/Fraser studies then established tidal pulsing, supercritical fronts, bulges and multi-timescale estuarine forcing.

Claims

1. A simple theory predicts vertical structure and offshore spreading of localized buoyant coastal discharge from competition between radial surface spreading and rotational coastal trapping. *Regime: A Simple Theory for the Fate of Buoyant Coastal Discharges** direct finding, analytical [1564]
2. Landsat turbidity and temperature imagery combined with modeled salinity resolves interacting northern Adriatic river plumes during a major flood. *Regime: High-resolution satellite turbidity and sea surface temperature observations of river plume interactions during a significant flood event* direct finding, mixed [186]
3. Columbia River observations show a sharp tidal-plume front forms when discharge speed exceeds the plume internal-wave speed and expands from the time-dependent source. *Regime: Structure and dynamics of the Columbia River tidal plume front* direct finding, field [712]
4. Bottom-generated tidal mixing sustains a bottom-trapped river plume and buoyant coastal current on an energetic shelf beyond the conventional trapping-depth picture. *Regime: Tidal Mixing Sustains a Bottom-Trapped River Plume and Buoyant Coastal Current on an Energetic Continental Shelf* direct finding, mixed [1532]
5. Repeated Oregon glider sections document seasonally varying buoyancy-driven coastal currents under river input, wind, and shelf circulation. *Regime: Buoyancy-Driven Coastal Currents off Oregon during Fall and Winter* direct finding, field [940]
6. Changjiang simulations show tides transform a surface-attached buoyant coastal current by mixing freshwater vertically and altering freshwater and volume transports. *Regime: Effects of tides on freshwater and volume transports in the Changjiang River plume* direct finding, numerical [811]

7. Four years of high-resolution Fraser plume transects resolve river-flow and tidal control from semidiurnal through annual scales. *Regime: Estuarine forcing of a river plume by river flow and tides* direct finding, field [563]
8. Field and radar observations north of the Rhine show tide and wind jointly control propagation of shallow plume fronts. *Regime: The Influence of Tide and Wind on the Propagation of Fronts in a Shallow River Plume* direct finding, field [1192]
9. Nearshore surface waves modify transport and mixing of small buoyant outflows through wave–current momentum and turbulence interactions. *Regime: Impacts of Nearshore Wave-Current Interaction on Transport and Mixing of Small-Scale Buoyant Plumes* direct finding, numerical [1200]
10. FVCOM simulations off Kochi produce an offshore bulge and coastal current whose seasonal extent and transport vary with discharge and monsoon forcing. *Regime: Seasonal spreading and transport of buoyant plumes in the shelf off Kochi, South west coast of India- A modeling approach* direct finding, numerical [1278]
11. HF radar, hydrography, moorings and imagery partition tidal, wind, and river-discharge control of Fraser plume surface currents. *Regime: Tide, Wind, and River Forcing of the Surface Currents in the Fraser River Plume* direct finding, field [565]
12. Idealized small-mouth Kelvin-number plume experiments show rotation and discharge regulate rapid near-field spreading and net vertical mixing. *Regime: The Effects of Rotation and River Discharge on Net Mixing in Small-Mouth Kelvin Number Plumes* direct finding, numerical [286]
13. Observations and modeling quantify dispersion and dilution of episodic storm-water plumes from small Southern California coastal streams. *Regime: Characterizing storm water dispersion and dilution from small coastal streams* direct finding, mixed [1212]
14. Satellite plume shape combined with Lagrangian modeling provides an inverse estimate of freshwater discharge rate. *Regime: A method for quantifying freshwater discharge rates from satellite observations and Lagrangian numerical modeling of river plumes* direct finding, mixed [1058]
15. Integrated remote sensing and in situ observations describe rapid freshwater-plume evolution and salinity reduction in the Great Barrier Reef lagoon after tropical river floods. *Regime: Evolution and dynamics of tropical river plumes in the Great Barrier Reef: An integrated remote sensing and in situ study* direct finding, field [218]
16. Idealized and process analysis quantifies bottom-generated tidal mixing in tidally pulsed river plumes and its role in tracer fate over the inner shelf. *Regime: The Effect of Bottom – Generated Tidal Mixing on Tidally Pulsed River Plumes* direct finding, numerical [1335]
17. A three-dimensional Rhine ROFI model identifies multiple interacting formation mechanisms for fronts in a shallow, strongly tidal and rotational plume. *Regime: The Evolution of Plume Fronts in the Rhine Region of Freshwater Influence* direct finding, numerical [1193]
18. Theory and simulations explain upstream spreading of bottom-trapped plumes opposite the propagation direction of coastally trapped waves. *Regime: The Upstream Spreading of Bottom-Trapped Plumes* direct finding, analytical [931]
19. In situ observations of a small river plume show wind direction and magnitude rapidly reshape plume position, size and geometry. *Regime: Response of a Small River Plume on Wind Forcing* direct finding, field [1059]
20. Small river and creek plumes, with weak rotation and rapidly depleted inertia, reverse with tides and respond strongly to wind. *Regime: Wind Effects on Small-Scale River and Creek Plumes* direct finding, mixed [123]
21. Four years of ferry transects support steady salt-balance estimates of entrainment and flushing time in the Fraser estuary and plume. *Regime: Entrainment and flushing time in the Fraser River estuary and plume from a steady salt balance analysis* direct finding, field [564]
22. High-resolution observations and analysis reveal frontal instability and mixing processes in the Gironde plume across small spatial and temporal scales. *Regime: Non-Linear Processes in the Gironde River Plume (North-East Atlantic): Instabilities and Mixing* direct finding, field [91]
23. Winyah Bay observations show light upwelling-favorable wind modifies a tidally formed anticyclonic plume bulge and its offshore export. *Regime: Response of a Coastal Plume Formed by Tidally Modulated Estuarine Outflow to Light Upwelling-Favorable Wind* direct finding, field [1565]
24. A supercritical Winyah Bay plume under roughly 1.5 days of moderate upwelling wind spreads offshore rather than remaining coastally trapped. *Regime: Offshore Spreading of a Supercritical Plume Under Upwelling Wind Forcing: A Case Study of the Winyah Bay Outflow* direct finding, field [1566]
25. An idealized wide buoyancy-front model shows downwelling-favorable wind advects and mixes a river-plume coastal current toward the coast and bottom. *Regime: Idealized Two-Dimensional Modeling of a Coastal Buoyancy Front, or River Plume, under Downwelling-Favorable Wind Forcing with Application to the Alaska Coastal Current* direct finding, numerical [1515]
26. Tides alter river-mouth-bar hydrodynamics and morphology, demonstrating that plume and tidal forcing jointly organize sediment deposition at river mouths. *Regime: Effect of tides on mouth bar morphology and hydrodynamics* direct finding, numerical [802]
27. Numerical simulation resolves the spreading and coastal circulation of the Plata River plume along southeastern South America. *Regime: A numerical study of the Plata River plume along the southeastern South American continental shelf* direct finding, numerical [1110]
28. An Ebro River plume model validated against field observations reproduces the coastal buoyant-plume structure and circulation. *Regime: Modelling of the Ebro River plume. Validation with field observations* direct finding, mixed [961]
29. A dry-season process study identifies circulation and exchange mechanisms connecting the Pearl River Estuary plume with the adjacent coastal ocean. *Regime: Process Study of Dry-Season Circulation in the Pearl River Estuary and Adjacent Coastal Waters using a Triple-Nested Coastal Circulation Model* direct finding, mixed [655]
30. Near-field Merrimack River plume observations and simulations show lateral spreading depends on plume density and velocity structure and expose model sensitivity to the represented mixing processes. *Regime: Lateral spreading of a near-field river plume: Observations and numerical simulations* direct finding, mixed [257]
31. At New River Inlet, drifters and models agree best in strongly mixed inlet channels but diverge offshore where observations indicate a buoyant plume and greater lateral spreading. *Regime: Observed and modeled drifters at a tidal inlet* direct finding, mixed [1341]

29.5 Salinity intrusion and mixing

Topic estuaries. salt. Salt transport and stratification.

Papers screened: 33. Claims: 31. Equations: 0.

Well established. Estuarine salinity reflects a dynamic balance between landward ocean salt transport and seaward river flushing, mediated by gravitational exchange, tidal oscillatory transport, stratification-dependent mixing, dispersion and geometry.

Governing physics. Longitudinal density gradients drive landward bottom flow and seaward surface flow; tides alternately strain and mix salinity; river discharge advects salt seaward; wind, waves, ocean salinity, sea level, branching and constrictions alter circulation and mixing.

Dimensionless parameters. Controls include estuarine Froude and Richardson numbers, tidal excursion/estuary length, freshwater Froude, mixing/stratification parameters, Simpson number, discharge/tidal prism, convergence length, drag, X2 position, sea-level/depth ratio and wind stress.

Major equations. The salt budget uses advection–diffusion conservation with resolved or effective dispersion. Exchange decompositions separate mean, tidal oscillatory and shear transports; intrusion models link length to discharge, tidal velocity, convergence and dispersion; tracer-variance decay diagnoses irreversible mixing.

Typical methods. Methods include longitudinal and cross-channel CTD/ADCP surveys, fixed salinity and microstructure time series, tracer variance, box and one-dimensional models, three-dimensional hydrostatic models, empirical intrusion equations, scenario ensembles and global forcing attribution.

Numerical models. Models include UnTRIM, FVCOM, coupled ocean–wave systems, primitive-equation estuary models, time-dependent box models, integrated coastal–inland delta models, effective-dispersion salt-wedge models and global scenario frameworks.

Experimental datasets. Evidence spans Willapa, Puget/Salish, Pearl, Mekong, Sumjin, San Francisco, Hudson, Yarra, Strymon and Rhine–Meuse systems, plus idealized barrier/constriction cases and a global estuary ensemble.

Validated ranges. The corpus covers well-mixed, partially mixed, periodically stratified and salt-wedge regimes; microtidal to macrotidal systems; single and branching channels; normal flow, drought and near-zero discharge; storm barriers, constrictions and future sea level.

Recent advances. Recent work uses tracer-variance mixing, lateral transport theory, sub-estuary networks, drought observations, coupled wind–wave effects, integrated delta models and global attribution of compounding future salt intrusion.

Disagreements. One-dimensional effective dispersion can predict intrusion length but obscures lateral and temporal mechanisms; equilibrium formulas miss transient memory; greater depth from sea-level rise can either enhance intrusion or alter tides/mixing enough to offset it, depending on geometry.

Limitations. Sparse cross-channel observations limit lateral-flux closure; mixing is intermittent; bathymetry and roughness evolve; discharge and ocean salinity boundaries are uncertain; empirical coefficients are site-dependent; global projections simplify local structures and operations.

Open questions. Needs include transient memory and hysteresis, branching deltas, lateral transport, barrier operations, morphology feedback, compound drought–surge–sea-level forcing, ocean salinity change, data assimilation and probabilistic freshwater thresholds.

Seminal papers. Classical adjustment, salt-flux and effective-dispersion studies established river–tide control and gravitational exchange; long field records then revealed unsteadiness, lateral circulation and intermittent stratification.

Claims

1. A review shows sea-level rise modifies estuarine salinity through changes in depth, tides, circulation, mixing, stratification and river–ocean pressure gradients, with strongly geometry-dependent outcomes. *Regime: Sea level rise impacts on estuarine dynamics: A review* review statement, review [711]
2. A validated Salish Sea hindcast resolves hydrography and exchange circulation across connected fjords and straits under realistic river, tide and ocean forcing. *Regime: A Model Study of the Salish Sea Estuarine Circulation** direct finding, numerical [1376]
3. Three years of Willapa Bay observations show salinity and stratification in a macrotidal partially mixed estuary respond to variable river and ocean end members and wind-driven shelf intrusions. *Regime: Dynamics of Willapa Bay, Washington: A Highly Unsteady, Partially Mixed Estuary* direct finding, field [114]
4. A time-dependent Puget Sound box model reproduces seasonal salinity variability and diagnoses residence times and interbasin salt transports. *Regime: Seasonal and Interannual Variability in the Circulation of Puget Sound, Washington: A Box Model Study* direct finding, numerical [99]
5. Coupled ocean–atmosphere–wave–sediment simulations show winds and waves materially alter salt intrusion in the Pearl River Estuary. *Regime: Effect of winds and waves on salt intrusion in the Pearl River estuary* direct finding, numerical [513]
6. Field observations and integrated coastal–inland modelling resolve delta-wide Mekong salt-intrusion pathways threatening freshwater supply. *Regime: Dynamics of salt intrusion in the Mekong Delta: results of field observations and integrated coastal–inland modelling* direct finding, mixed [414]
7. The three-dimensional UnTRIM San Francisco Estuary model reproduces water level, flow and salinity and evaluates X2 and the low-salinity zone. *Regime: Three-Dimensional Modeling of Hydrodynamics and Salinity in the San Francisco Estuary: An Evaluation of Model Accuracy, X2, and the Low-Salinity Zone* direct finding, numerical [895]
8. Intensive Sumjin observations test empirical salt-intrusion models across spring–neap and river-flow conditions and identify the most suitable predictive formulation. *Regime: Comparison of empirical models with intensively observed data for prediction of salt intrusion in the Sumjin River estuary, Korea* direct finding, mixed [1287]
9. Microstructure and velocity transects show lateral straining in a curved tidal inlet creates late-flood stratification and strengthens exchange flow. *Regime: Lateral Circulation Generates Flood Tide Stratification and Estuarine Exchange Flow in a Curved Tidal Inlet* direct finding, field [135]
10. FVCOM simulations show downwelling wind and tides jointly reshape Pearl River estuarine plume circulation, stratification and salt transport. *Regime: Downwelling wind, tides, and estuarine plume dynamics* direct finding, numerical [761]
11. Global projections attribute increasing future estuarine salt intrusion to compounding river-flow, sea-level, tidal and ocean-salinity changes. *Regime: Global increases of salt intrusion in estuaries under future environmental conditions* direct finding, numerical [792]
12. Revised predictive equations link one-dimensional salt-intrusion parameters to observable hydraulic and geometric variables. *Regime: Revised predictive equations for salt intrusion modelling in estuaries* direct finding, analytical [505]
13. A Yarra River model links river flow and salt-wedge dynamics to stratification, residence and oxygen depletion. *Regime: Hydrodynamic controls on oxygen dynamics in a riverine salt wedge estuary, the Yarra River estuary, Australia* direct finding, numerical [198]
14. Tidal oscillatory salt transport in a partially stratified estuary arises from correlated tidal salinity and velocity variations and contributes materially to the subtidal salt balance. *Regime: Mechanisms of Tidal Oscillatory Salt Transport in a Partially Stratified Estuary* direct finding, analytical [1477]
15. Salt intrusion responds nonlinearly across distinct time scales to transient river and ocean forcing rather than following a single equilibrium length. *Regime: Multiscale Temporal Response of Salt Intrusion to Transient River and Ocean Forcing* direct finding, mixed [1089]
16. A three-dimensional model simulates salt-wedge intrusion in the microtidal Strymon mouth under near-zero summer river flow. *Regime: Salt wedge intrusion modeling along the lower reaches of a Mediterranean river* direct finding, numerical [1574]
17. Storm-surge barrier infrastructure changes cross-section, drag, tidal mixing and exchange flow during normal operation in a partially mixed estuary. *Regime: Impacts of Storm Surge Barriers on Drag, Mixing, and Exchange Flow in a Partially Mixed Estuary* direct finding, numerical [1154]

18. Tracer-variance decay identifies where and when salinity mixing varies in the Hudson River Estuary. *Regime: Using Tracer Variance Decay to Quantify Variability of Salinity Mixing in the Hudson River Estuary* direct finding, mixed [1488]
19. A well-mixed sub-estuary connected to a partially mixed main channel exhibits distinct intrusion dynamics governed by branching exchange and forcing. *Regime: Salt intrusion dynamics in a well-mixed sub-estuary connected to a partially to well-mixed main estuary* direct finding, mixed [833]
20. Rhine–Meuse drought observations show branching geometry, low discharge, wind and tides jointly govern prolonged salt-intrusion events. *Regime: Observations of Estuarine Salt Intrusion Dynamics During a Prolonged Drought Event in the Rhine-Meuse Delta* direct finding, field [1498]
21. A constricted microtidal entrance couples freshwater flow and tidal forcing to salinity intrusion and flushing time. *Regime: Estuarine Salinity Intrusion and Flushing Time Response to Freshwater Flows and Tidal Forcing Under the Constricted Entrance* direct finding, numerical [708]
22. Lateral salt transport controls tidally averaged gravitational circulation when tidal mixing redistributes the cross-channel pressure-gradient response. *Regime: Lateral Transport Controls the Tidally Averaged Gravitationally Driven Estuarine Circulation: Tidal Mixing Effects* direct finding, analytical [749]
23. In the Merrimack, salinity intrusion shifts from short and strongly stratified at moderate–high discharge to longer and weakly stratified at low discharge when intrusion length approaches tidal excursion; tidal asymmetry dominates salt flux. *Regime: Structure, variability, and salt flux in a strongly forced salt wedge estuary* direct finding, mixed [1152]
24. Lower Meghna irrigation water measured 1.3 dS/m salinity and 500 ppm chloride in 2011, exceeding earlier estimates and desired levels and coinciding with reported dry-season crop-yield loss. *Regime: Salinity Intrusion in Interior Coast of Bangladesh: Challenges to Agriculture in South-Central Coastal Zone* direct finding, field [127]
25. Coupled surface–subsurface simulations show that 200–400 m marsh migration can protect landward groundwater and crop yield on 0.1% slopes for surges at least 3 m, although wider migration or lower surges did not maintain yield. *Regime: Effects of Marsh Migration on Flooding, Saltwater Intrusion, and Crop Yield in Coastal Agricultural Land Subject to Storm Surge Inundation* direct finding, numerical [550]
26. A nine-day northerly wind reaching 17.6 m/s drove landward net transport and horizontal circulation that overpowered normal Changjiang runoff, interrupting Qingcaosha freshwater supply for 23 days. *Regime: Dynamic mechanism of an extremely severe saltwater intrusion in the Changjiang estuary in February 2014* direct finding, mixed [1619]
27. Field-calibrated Savenije and HEC-RAS models reproduced altered axial salinity profiles and elevated water level in the Sebou estuary during the 2015 supermoon eclipse, supporting rapid intrusion assessment during rare forcing. *Regime: Analytical and numerical study of the salinity intrusion in the Sebou river estuary (Morocco) – effect of the “Super Blood Moon” (total lunar eclipse) of 2015* direct finding, mixed [561]
28. Optimization of pumping in a saltwater-intruded coastal aquifer demonstrates that extraction design can balance water supply objectives against further saline intrusion. *Regime: Pumping optimization in saltwater-intruded coastal aquifers* direct finding, numerical [266]
29. A synthesis describes a coastal groundwater squeeze in which dense development, abstraction, pollution, and marine interaction compress and degrade freshwater resources. *Regime: Science, society, and the coastal groundwater squeeze* review statement, review [962]
30. Three-dimensional Monte Carlo modeling shows that tidal oscillations and hydraulic-conductivity heterogeneity jointly control mixing and spreading at the coastal freshwater–seawater interface. *Regime: Effects of tidal fluctuations and spatial heterogeneity on mixing and spreading in spatially heterogeneous coastal aquifers* direct finding, numerical [1120]
31. Long-duration observations and three-dimensional modeling in the Ohtagawa Estuary show downstream-directed wind enhances salinity intrusion, upstream wind suppresses it, and wind direction controls stratification strength and suspended-sediment transport. *Regime: Ohtagawa Estuary salt intrusion and suspended-sediment response under the documented tides, river flow and opposing wind directions.* direct finding, mixed [691]

30 Wiki: Field measurements

30.1 Field measurements: overview

Topic field. In-situ observations of coastal processes.

Papers screened: 24. Claims: 10. Equations: 0.

Well established. Coastal field measurement requires simultaneous attention to the process, decision variable, spatial and temporal scales, platform response, calibration, datum, sampling design, uncertainty, metadata and survivability; a sensor record is not automatically a representative observation.

Governing physics. Instruments sample coupled waves, currents, water levels, turbulence, sediment, bathymetry and morphology through mechanical, pressure, acoustic, optical, electromagnetic or satellite interactions, each filtered by platform motion, geometry, propagation, scattering and environmental interference.

Dimensionless parameters. Observational adequacy depends on sampling-to-process frequency, record-length-to-event scale, spatial spacing relative to wavelength or morphology, signal-to-noise ratio, platform speed and footprint, relative water depth, wave and current regime, sediment size, calibration range, data return and uncertainty ratios.

Major equations. Core relations include sensor transfer and calibration functions, wave spectra and directional moments, velocity and discharge transformations, acoustic or optical backscatter inversion, sediment and volume budgets, datum and coordinate transformations, sampling and aliasing criteria, and uncertainty propagation or data-assimilation observation operators.

Typical methods. Methods combine fixed gauges and profilers, pressure and velocity arrays, tide gauges, GNSS and survey control, lidar and photogrammetry, multibeam and sidescan acoustics, sediment traps and optical sensors, drifters, autonomous vehicles, satellites, repeat topobathymetry and coordinated multi-platform campaigns.

Numerical models. Field data support retrieval and inversion algorithms, quality-control systems, spectral and directional estimators, observation operators, data assimilation, process-model calibration and validation, uncertainty models, adaptive sampling and digital observing-system design.

Experimental datasets. The reviewed evidence includes a nine-station barred-beach array, multi-site European storm morphology, Willapa Bay wave-current observations, reef-to-island flooding transects, barrier-reef networks, autonomous sandbank plume surveys, multibeam sediment calibration and satellite wind-wave-current products.

Validated ranges. Examples span beaches, tidal bays, coral and barrier reefs, shallow sandbanks and regional-to-global satellite coverage, from intra-wave sensing through tidal and storm campaigns to multi-year morphology; each dataset remains bounded by its platform and site regime.

Recent advances. Recent advances integrate satellite winds, waves and currents; autonomous underwater and surface vehicles; calibrated multibeam concentration mapping; dense cross-shore arrays; rapid repeat topobathymetry; cloud-ready standards; and observation-model fusion with explicit uncertainty.

Disagreements. Point instruments resolve time but may miss spatial structure; remote sensors resolve area but infer rather than directly sample variables; Eulerian and Lagrangian views differ; proxy shorelines and concentrations are algorithm-dependent; denser sampling can still be biased if placement or duration misses the governing process.

Limitations. Energetic events cause sensor loss and saturation; biofouling, bubbles, turbidity and bed mobility corrupt signals; datums and clocks drift; platforms disturb or move with the flow; calibration may not span field conditions; short records alias extremes and morphology; access and maintenance constrain continuity.

Open questions. Priorities include event-resilient and low-cost sustained arrays, seamless land–water topobathymetry, direct sediment-flux sensing, uncertainty covariance across platforms, compound-hazard measurements, adaptive sampling, autonomous calibration, reproducible retrievals and equitable long-term observing networks.

Seminal papers. Early coastal field programs established pressure and current arrays, sediment concentration sampling, profile surveys, spectral analysis and wave–current boundary-layer observations; satellite altimetry, imaging, acoustics and autonomous platforms later expanded coverage and repeatability.

Claims

1. A review shows that autonomous underwater vehicles extend marine geoscience observations through repeatable, high-resolution, spatially continuous surveys while imposing navigation, endurance, payload, calibration, and data-management constraints. *Regime: Autonomous Underwater Vehicles (AUVs): Their past, present and future contributions to the advancement of marine geoscience* review statement, review [1538]
2. A two-month nine-station cross-shore array jointly measured waves, currents, and seabed position to resolve natural sandbar evolution rather than inferring morphology from hydrodynamics alone. *Regime: Observations of sand bar evolution on a natural beach* direct finding, field [472]
3. A coordinated Atlantic European dataset links the exceptional 2013/2014 winter wave climate to spatially variable beach morphological impacts across multiple sites. *Regime: Extreme wave activity during 2013/2014 winter and morphological impacts along the Atlantic coast of Europe* direct finding, field [930]
4. Field observations in Willapa Bay resolve wave–current interaction in a tidal coastal system and demonstrate that wave and circulation measurements must be interpreted jointly. *Regime: Wave-current interaction in Willapa Bay* direct finding, field [1047]
5. A reef-to-island observational transect connects offshore water levels and low-frequency wave transformation to runup, overwash, and coastal flooding. *Regime: Observations of wave transformation over a fringing coral reef and the importance of low-frequency waves and offshore water levels to runup, overwash, and coastal flooding* direct finding, field [273]

6. A review of satellite surface winds, waves, and currents identifies complementary observables, retrieval limitations, calibration needs, and scientific and operational uses. *Regime: Satellite Remote Sensing of Surface Winds, Waves, and Currents: Where are we Now?* review statement, review [575]
7. A cross-shore field-instrument network combined with numerical analysis resolves wave transformation over a barrier reef and provides a process-consistent observational benchmark. *Regime: Wave transformation over a barrier reef* direct finding, mixed [1327]
8. Wave-glider monitoring demonstrates an autonomous approach for mapping sediment transport and dredge plumes in a shallow marine sandbank environment. *Regime: Wave Glider Monitoring of Sediment Transport and Dredge Plumes in a Shallow Marine Sandbank Environment* direct finding, field [763]
9. Calibration of multibeam acoustic backscatter provides spatially resolved suspended-sediment concentration fields while making inversion and calibration assumptions explicit. *Regime: Suspended sediment concentration field quantified from a calibrated MultiBeam EchoSounder* direct finding, mixed [464]
10. A coastal-observing review provides ten recommendations for sustaining mooring networks, including stakeholder-led requirements, complementary technologies, standardized open data, performance metrics and routine network-design assessment. *Regime: Coastal Mooring Observing Networks and Their Data Products: Recommendations for the Next Decade* review statement, review [103]

30.2 Extreme-event observations

Topic field.extreme. Storm and tsunami field measurements.

Papers screened: 10. Claims: 7. Equations: 5. Shown in the Coastal Design Bench beside: Design conditions.

Well established. Extreme coastal events require direct observations because storm and tsunami response depends on time-varying waves, water levels, currents, sediment, bathymetry, structures and ecosystems in combinations not captured by single bulk forcing metrics.

Governing physics. Observed response is governed by tsunami momentum and Froude regime, wave-current interaction, spectral period and bimodality, breaking and setup, runup and overwash, vegetation drag, structure diversion, undertow, longshore currents, sediment suspension and antecedent morphology.

Dimensionless parameters. Important controls include Froude number, wave steepness, relative freeboard, barrier inertia, Shields mobility, relative depth, directional angle, spectral bimodality, structure or vegetation density, normalized surge, and model skill or uncertainty metrics.

Major equations. Field interpretation couples nonlinear shallow-water or surfbeat equations, wave-action models, sediment-flux integration, Froude scaling, overtopping and runup thresholds, vegetation drag and moment, XBeach-G wave-by-wave physics, and nonlinear data-driven waveform transfer.

Typical methods. Methods include post-event transects, video velocimetry, pressure and tide gauges, offshore pressure sensors, GPS buoys, movable instrument sleds, laser scanning, repeated profiles, sediment concentration arrays, remote sensing, numerical reconstruction and real-time assimilation.

Numerical models. Models include modified depth-integrated tsunami equations, regularized extreme learning machines, XBeach-G, SWAN and XBeach surfbeat configurations, direct suspended-flux integration, and event-specific shoreline or nourishment analyses.

Experimental datasets. Key datasets include a 340 km post-Tohoku survey, multi-instrument Tohoku waveforms, five Duck storm sediment transects, two Lake Michigan storm-current campaigns, 2 Hz Loe Bar overwash scans, Upham hurricane nourishment surveys and Ningaloo cyclone reef-lagoon observations.

Validated ranges. Explicit regimes include Sendai $U=6.2$ m/s and $Fr=1.14-1.4$; Duck H_{m0} to 3.5 m and transport to 1780 m³/h; Loe Bar $H_s=4$ m, $H_{max}=8$ m and $T_p=10-15$ s with under-5% hydrodynamic error; and barred-beach currents to 1.8 m/s.

Recent advances. High-frequency laser scanning now resolves overwash hydraulics and morphology, nonlinear machine learning accelerates near-field tsunami waveform forecasts, and coupled observations reveal when local wind waves, vegetation, structures or antecedent morphology dominate expected hazard response.

Disagreements. The CERC longshore formula alternated between over- and underprediction during storms. A surfbeat-resolving model was not superior during TC Olwyn because locally generated wind waves dominated. Hard structures can shield locally yet divert and concentrate damaging flow elsewhere.

Limitations. Extreme measurements are sparse, dangerous and often opportunistic; instruments fail or perturb flow, post-event marks mix processes, bedload may be unmeasured, pier effects and changing morphology bias transects, and event-calibrated models may not transfer to different spectra or sites.

Open questions. Priorities include redundant interoperable sensors, rapid bathymetry updates, bedload measurement during storms, uncertainty-aware real-time assimilation, compound-event observations, standardized post-event protocols, and validation across events larger than the calibration record.

Seminal papers. The 1981 barred-beach storms and 1999 Duck transport campaign established quasi-synoptic and flux-resolving storm observations; the 2011 Tohoku event enabled unprecedented regional surveys and multi-gauge tsunami reconstruction.

Equations

- *barred-beach longshore-current field dataset* [34]

$$y(t, \mathbf{x}) = \mathcal{M}[\mathbf{o}(t), \mathbf{b}, \mathbf{p}]$$

Regime: Profiles every about 3 h for 24 and 28 h; incident angle about 30 degrees; H_{rms} breaker height ≤ 1.5 m; bar about 30 m offshore and 0.75 m deep in calm conditions. Variables: y predicted or derived coastal response; $\mathbf{o}$ event observations; $\mathbf{b}$ bathymetry or beach state; $\mathbf{p}$ model or instrument parameters; normalized observation-model mapping

- *modified 2-D tsunami shallow-water model* [1010]

$$y(t, \mathbf{x}) = \mathcal{M}[\mathbf{o}(t), \mathbf{b}, \mathbf{p}]$$

Regime: Twenty-five locations over about 340 km; detailed Misawa vegetated-dune and Hachinohe seawall/forest cases; Sendai video evidence. Variables: y predicted or derived coastal response; $\mathbf{o}$ event observations; $\mathbf{b}$ bathymetry or beach state; $\mathbf{p}$ model or instrument parameters; normalized observation-model mapping

- *regularized extreme learning machine tsunami forecaster* [997]

$$y(t, \mathbf{x}) = \mathcal{M}[\mathbf{o}(t), \mathbf{b}, \mathbf{p}]$$

Regime: 2011 Tohoku waveforms from four OBP gauges plus GPS buoys, wave gauges and tide gauges; example inversion window 35 min. Variables: y predicted or derived coastal response; $\mathbf{o}$ event observations; $\mathbf{b}$ bathymetry or beach state; $\mathbf{p}$ model or instrument parameters; normalized observation-model mapping

- *XBeach-G gravel-barrier overwash model* [38]

$$y(t, \mathbf{x}) = \mathcal{M}[\mathbf{o}(t), \mathbf{b}, \mathbf{p}]$$

Regime: Loe Bar D50=2-4 mm; offshore Hs=4 m, Hmax=8 m, Tp=10-15 s, spring high tide 2.45 m and surge about 0.15 m. Variables: y predicted or derived coastal response; $\mathbf{o}$ event observations; $\mathbf{b}$ bathymetry or beach state; $\mathbf{p}$ model or instrument parameters; normalized observation-model mapping

- *SIS storm longshore-transport measurement framework* [963]

$$y(t, \mathbf{x}) = \mathcal{M}[\mathbf{o}(t), \mathbf{b}, \mathbf{p}]$$

Regime: At least nine cross-shore positions over five storms; Hm0 up to 3.5 m; SIS operable above Hs=3 m, wind 20 m/s, and currents 2 m/s. Variables: y predicted or derived coastal response; $\mathbf{o}$ event observations; $\mathbf{b}$ bathymetry or beach state; $\mathbf{p}$ model or instrument parameters; normalized observation-model mapping

Claims

1. For 2011 Great East Japan tsunami evidence along about 340 km, modeling identified the vegetated dune as the primary mitigation at Misawa and showed that straight versus crooked access roads changed velocity; at Hachinohe, seawall diversion kept vegetation moments below 0.2 kNm behind the wall but produced 2-4 kNm in exposed areas, while Sendai video implied 6.2 m/s flow 1 km inland and Fr about 1.14-1.4. *Regime: Twenty-five locations over about 340 km; detailed Misawa vegetated-dune and Hachinohe seawall/forest cases; Sendai video evidence.* direct finding, mixed [1010]
2. Across five Duck storm transects with waves up to Hm0=3.5 m, directly integrated suspended-load longshore transport reached 1780 m3/h and usually peaked near breaking at the bar and beach; CERC estimates were sometimes high and sometimes low, indicating missing short-term storm controls. *Regime: At least nine cross-shore positions over five storms; Hm0 up to 3.5 m; SIS operable above Hs=3 m, wind 20 m/s, and currents 2 m/s.* direct finding, field [963]
3. For 2011 Tohoku records from ocean-bottom pressure, GPS buoy, wave, and tide gauges, a regularized nonlinear extreme-learning-machine forecast outperformed standard linear waveform inversion without materially increasing computing time, and consecutive random-weight runs showed insignificant variability. *Regime: 2011 Tohoku waveforms from four OBP gauges plus GPS buoys, wave gauges and tide gauges; example inversion window 35 min.* direct finding, numerical [997]
4. During the 1 February 2014 Loe Bar storm (Hs=4 m, Hmax=8 m, Tp=10-15 s), 2 Hz laser observations validated XBeach-G overwash hydrodynamics to less than 5% error; simulations showed that long-period or strongly bimodal spectra reduce overtopping and overwash thresholds by enhancing runup. *Regime: Loe Bar D50=2-4 mm; offshore Hs=4 m, Hmax=8 m, Tp=10-15 s, spring high tide 2.45 m and surge about 0.15 m.* direct finding, mixed [38]
5. During two Lake Michigan storms sampled every approximately 3 h for 24 and 28 h, incident angles near 30 degrees and Hrms breaker heights no greater than 1.5 m produced time-averaged longshore currents up to 1.8 m/s that were often nearly uniform across the surf zone despite barred bathymetry. *Regime: Profiles every about 3 h for 24 and 28 h; incident angle about 30 degrees; Hrms breaker height <=1.5 m; bar about 30 m offshore and 0.75 m deep in calm conditions.* direct finding, field [34]
6. Force-balanced dynamically scaled tank experiments showed that storm waves can form the characteristic features of imbricated coastal boulder deposits, so even megagravel deposits cannot be treated as de facto tsunami indicators. *Regime: Dynamically scaled storm-wave interaction with imbricated coastal boulders.* direct finding, experimental [301]
7. During 6-8 m Atlantic storms, cliff-top vertical ground motion exceeded 50-100 μ m and two-week cliff loss was two orders of magnitude above the long-term rate, showing strongly episodic erosion. *Regime: Coastal cliff ground motions and response to extreme storm waves* direct finding, field [391]

30.3 Sediment and morphology observations

Topic field. sediment. Sampling and seabed surveys.

Papers screened: 6. Claims: 3. Equations: 1.

Well established. Field sediment transport is intermittent, vertically structured and often strongly two-way; concentration or delivery alone does not determine net accumulation because advection, resuspension, settling and export must be measured within a sediment budget.

Governing physics. Waves mobilize grains, asymmetric orbital motion, undertow and currents advect them, turbulence suspends sediment and settling plus bed exchange controls concentration. At inlets and reefs, persistent directional forcing and local trapping or flushing integrate these processes over larger scales.

Dimensionless parameters. Controls include Shields and mobility numbers, Rouse number, sediment fall velocity relative to orbital velocity, wave-current ratio, suspension height relative to boundary-layer thickness, sampling duration relative to waves and events, gross-to-net flux ratio and closure error relative to measured transport.

Major equations. Suspended flux is the time and depth integral of concentration times velocity, bed change follows sediment continuity, and settling-diffusion balances describe mean concentration profiles. Trap or tray mass rates measure gross directional exchange, while surveys close net volume budgets.

Typical methods. Campaigns pair concentration samplers, optical or acoustic sensors, pumps or pressure-difference collectors, velocity profilers, bed-level surveys, sediment traps/trays and tracers; synchronize sampling, calibrate grain and turbidity response, resolve vertical profiles and waves, estimate uncertainty and close gross and net budgets.

Numerical models. The reviewed set is observation-centered. Interpretation uses suspension-profile, flux and sediment-continuity concepts; direct process-model calibration is not established for this node, and instrument-comparison and fluorescent-tracer papers await full-text review.

Experimental datasets. Reviewed evidence includes 65 seven-level concentration profiles in about 1.5 m depth, paired two-way flux trays at four sites on two turbid Great Barrier Reef systems and seasonal entrance-bar observations at Wilson Inlet.

Validated ranges. The pressure-difference sampler resolved roughly three-minute profiles from 1 cm above bed upward; reef trays measured gross flux of 34 to over 640 g/m²/d while net sedimentation stayed below 122 g/m²/d. Results are method- and site-specific.

Recent advances. Recent advances combine acoustic/optical arrays, fluorescent and RFID tracers, sediment-imaging velocimetry, autonomous platforms, UAV and sonar morphology, Bayesian sensor fusion and real-time sediment-budget assimilation.

Disagreements. Optical sensors offer high frequency but depend on grain size and bubbles; pumped samples are direct but intrusive and sparse; traps can disturb flow and mix deposition with resuspension. Gross flux, suspended concentration and morphological change answer different questions.

Limitations. Spatial aliasing, sensor calibration drift, biofouling, bubbles, uncertain sampling efficiency, unmeasured bedload, short deployments, storm gaps, tracer recovery, changing bathymetry and open control-volume boundaries can prevent sediment-budget closure.

Open questions. Priorities include co-located bedload and suspension, event-resolving vertical flux, tracer recovery and mixing depth, autonomous calibration, cohesive-sand mixtures, reef and vegetation interactions, cross-shore/alongshore integration and open standardized field benchmarks.

Seminal papers. Pump and bottle sampling established concentration profiles; optical and acoustic backscatter enabled high-frequency suspension records, tracers revealed pathways, and synchronized velocity-concentration measurements advanced direct flux estimation.

Equations

- *pressure-difference suction sediment sampler* [1024]

$$y(t, \mathbf{x}) = \mathcal{M}[\mathbf{o}(t), \mathbf{b}, \mathbf{p}]$$

Regime: Sixty-five profiles; seven simultaneous elevations starting about 1 cm above bed; approximately 3 min samples in about 1.5 m water depth. Variables: y predicted or derived coastal response; $\mathbf{o}$ event observations; $\mathbf{b}$ bathymetry or beach state; $\mathbf{p}$ model or instrument parameters; normalized observation-model mapping

Claims

1. At four sites on two turbid Great Barrier Reef systems, paired trays measured two-way sediment flux from 34 to more than 640 g/m²/d while mean net sedimentation remained below 122 g/m²/d, demonstrating that high delivery need not imply high accumulation. *Regime: Shallow turbid inshore reefs; tray and site-specific limitations apply.* direct finding, field [1628]
2. A pump-free pressure-difference sampler collected seven suspended-sand samples simultaneously from about 1 cm above the bed upward during approximately 3 min intervals in roughly 1.5 m depth, yielding 65 field concentration profiles with hydraulic and sediment metadata. *Regime: Sixty-five profiles; seven simultaneous elevations starting about 1 cm above bed; approximately 3 min samples in about 1.5 m water depth.* direct finding, field [1024]
3. A quantitative Venice Lagoon assessment maps cumulative human disturbance of shallow coastal seabeds and associated geomorphic footprints. *Regime: Assessing the human footprint on the sea-floor of coastal systems: the case of the Venice Lagoon, Italy* direct finding, field [897]

30.4 Tracer and drifter studies

Topic field. tracers. Lagrangian and scalar transport observations.

Papers screened: 13. Claims: 12. Equations: 0.

Well established. Tracers and drifters reveal coastal advection, dispersion, retention, and exchange, but scalar parcels, surface drifters, subsurface drogues, organisms, and numerical particles can follow different pathways.

Governing physics. Observed transport reflects mean currents, tides, wind and Ekman flow, waves and surf-zone currents, fronts, vertical shear, turbulence, bathymetry, shoreline interaction, and particle buoyancy or behavior.

Dimensionless parameters. Relevant controls include excursion relative to domain scale, release duration relative to tide, inertial and windage response, horizontal/vertical diffusivity, sampling interval relative to decorrelation time, and observation duration relative to residence time.

Major equations. Core tools include advection-diffusion, single- and pair-particle statistics, residence/flushing/age/transit definitions, trajectory integration, coherent structures, mass balance, and model-observation skill metrics.

Typical methods. Methods include soluble fluorescent or SF₆ releases, GPS/RTK surface and subsurface drifters, introduced biological cohorts, conductivity plume probes, UAV concentration imaging, Eulerian co-observations, particle models, and Lagrangian coherent structures.

Numerical models. Trajectory and tracer models range from local lagoon and shoreline-release simulations to coupled wave-current shelf models and climatological coherent structures; model particles require observational checks against scalar and drifter behavior.

Experimental datasets. Accessible observations include an eight-day lagoon cohort/SF₆/drifter comparison, centimetre-accuracy 10 Hz shallow-water drifters, three UAV dye campaigns, and three nine-probe estuary plume releases.

Validated ranges. Reported capabilities include 10 Hz centimetre positioning, motion resolved to 1 Hz, UAV concentration grids of 2×2 m, three releases of nine probes, and basin-scale attraction attribution of 44% to the Black Sea Rim Current.

Recent advances. Recent advances include centimetre RTK positioning, low-cost open electronics, UAV tracer concentration maps, multiscale wave-current tracer models, and climatological coherent pathways for persistent transport diagnosis.

Disagreements. SF6/tracer models and drifter/particle models predicted different larval fate in the same lagoon experiment, showing that apparent validation depends on tracer type, vertical sampling, biological spread, and the transport quantity compared.

Limitations. Surface windage, drogue slip, positioning error, dye photochemistry, finite release duration, sparse recovery, vertical motion, shoreline loss, imperfect concentration calibration, model resolution, and ambiguous timescale definitions bias inference.

Open questions. Needs include vertical and three-dimensional dispersion, intermittent fronts and rip exchange, storm sampling, shoreline/beaching processes, multi-tracer intercomparison, source attribution, adaptive releases, and uncertainty propagation into residence and connectivity metrics.

Seminal papers. The branch connects classical dye and inert-gas dispersion, Lagrangian drifters, diagnostic timescales, biological cohort tracking, particle trajectories, and coherent-structure analysis.

Claims

1. Coastal residence, flushing, age, and transit times must be explicitly defined because they answer different transport questions and depend on the chosen Lagrangian or Eulerian framework. *Regime: Lagrangian, Eulerian, field, model, and algebraic timescales are not interchangeable and require explicit definitions and process scope.* review statement, review [870]
2. An eight-day lagoon experiment found SF6/tracer-model and drifter/particle predictions disagreed and observed larval spreading along the advective path that the models omitted. *Regime: SF6/tracer-model fate disagreed with drifter/particle predictions and observed larvae spread along the advective path, exposing missing dispersion mechanisms.* direct finding, mixed [80]
3. Gulf transport synthesis identifies coupled river, wind, wave, tide, front, inlet, shelf, and open-ocean processes as controls on tracer fate, with storm fluxes and multiscale coupling unresolved. *Regime: Winds, waves, tides, river inputs, fronts, shelf-deep-ocean exchange, and multiscale advection jointly control constituent fate; storm fluxes and scale coupling remain uncertain.* review statement, review [676]
4. A 10 Hz centimetre-accuracy RTK-GNSS shallow-water drifter resolved motion to 1 Hz and yielded dispersion coefficients comparable to dye-tracer studies. *Regime: The low-windage drifter resolved motion to 1 Hz, captured tidal elevation, and produced shallow-estuary dispersion estimates comparable to dye studies.* direct finding, field [1363]
5. A shoreline-dye simulation attributed alongshore transport to wave breaking and pressure gradients and offshore exchange to wind-driven Ekman flow and submesoscale frontogenesis rather than tides. *Regime: Wave breaking and pressure gradients drove alongshore transport; wind-driven Ekman and submesoscale frontogenesis contributed cross-shelf exchange, while tides did not explain it.* direct finding, numerical [1534]
6. UAV visible imagery calibrated against a fluorometer mapped fluorescent-tracer concentration on a 2×2 m grid across three measurement campaigns. *Regime: Visible-camera imagery produced georeferenced fluorescein and Rhodamine WT concentration maps on a 2×2 m grid.* direct finding, field [216]
7. Three nine-probe estuary releases showed open-source GPS/conductivity drifters can track plume attenuation and parameterize mean velocity and salinity-proxy concentration. *Regime: Open-source GPS probes tracked plume position, temperature, and conductivity until attenuation and supported an analytical mean-flow/salinity fit.* direct finding, field [347]
8. Black Sea coherent structures identified persistent aggregation, barriers, and cross-shelf pathways, with the Rim Current accounting for 44% of climatological attraction strength. *Regime: Persistent squeezelines, jets, barriers, and shelf pathways organized aggregation and cross-shelf export; the Rim Current contributed 44% of climatological attraction strength.* direct finding, numerical [521]
9. HF-radar Lagrangian diagnostics at 3.5–25 km scales link attracting fronts and negative divergence to phytoplankton clustering, while positive-divergence filaments indicate upward nutrient injection and high chlorophyll. *Regime: Effect of small scale transport processes on phytoplankton distribution in coastal seas* direct finding, mixed [593]
10. A 250 m triangular oxygen-control volume on a Hawaiian reef produced Eulerian net-production estimates agreeing with dye-based Lagrangian estimates at $r^2=0.81$; drifters moved 30–100% faster than dye and depth-mean currents. *Regime: Continuous measurements of net production over a shallow reef community using a modified Eulerian approach* direct finding, field [422]
11. Validation with 4,130 drifter–radar pairs gave X-band radar near-surface current RMSE of 4 cm/s and direction error of 12°, resolving roughly 500 m currents to 3 km range. *Regime: Near-Surface Current Mapping by Shipboard Marine X-Band Radar: A Validation* direct finding, field [876]
12. Marine population connectivity cannot be inferred from circulation alone: resolving larval pathways requires coordinated Lagrangian physical observations and biological evidence because fronts, stratification, vertical behavior, tides, mortality, and spawning timing jointly shape dispersal. *Regime: Coastal, shelf, bay and estuarine transport of weakly swimming marine early life stages, with source-specific hydrodynamic and biological controls.* review statement, review [485]

30.5 Wave and current observations

Topic field.waves. Fixed and mobile hydrodynamic sensors.

Papers screened: 21. Claims: 2. Equations: 0.

Well established. Field wave observations require complementary in situ and remote sensors because no platform simultaneously resolves nearshore transformation, directional spectra, extremes and broad spatial coverage; uncertainty grows near land, shallow water, breaking and vegetation or structures.

Governing physics. Wind-sea and swell propagate, refract, shoal, break, reflect and interact with currents, bottom stress, vegetation and structures; resulting spectra, setup, runup and attenuation depend on bathymetry, water level, directional partitions and local morphology.

Dimensionless parameters. Controls include relative depth, wave steepness, surf-similarity, directional spread, transmission coefficient, vegetation or reef submergence, fetch, current-to-wave celerity ratio, sensor separation relative to wavelength and error normalized by wave height.

Major equations. Wave observations are summarized by spectral moments, significant height, periods and directions; energy-flux and radiation-stress balances connect transformation to setup, while transmission ratios compare incident and lee energy. Validation uses bias, RMSE, correlation, scatter and distributional or spectral scores.

Typical methods. Campaigns collocate buoys, pressure sensors, ADCPs, wave staffs, HF radar, altimetry and shoreline measurements; synchronize clocks and datums, apply response and depth corrections, partition spectra, quality-control breaking and land contamination, quantify sampling uncertainty and validate derived quantities across sea states.

Numerical models. Field evidence evaluates empirical setup/runup parameterizations, roller and bottom-stress setup models, porous-transmission theory, vegetation-aware MIKE SW, harbor spectral reconstruction and remote-sensing retrieval algorithms. Models remain conditional on sensor and site coverage.

Experimental datasets. Reviewed evidence spans ten natural-beach setup/runup campaigns, 90 days of SandyDuck setup, living-shoreline and oyster sites, tropical-storm marsh records, a seasonal saltmarsh dataset, 40 years of harbor spectra, Iberian HF-radar/buoy/altimetry comparisons and northeastern US multi-mission altimetry.

Validated ranges. Iberian Sentinel-3 comparisons achieved correlations above 0.94, bias below 0.19 m and RMSE 0.17-0.42 m; SandyDuck setup agreed within about 30% to 6 m depth; other results cover their stated beaches, structures, vegetation and harbor regimes only.

Recent advances. Recent advances combine new coastal altimeters and SAR, compact directional buoys, distributed acoustic sensing, video and radar, autonomous vehicles, cloud-based spectral partitioning and assimilation of heterogeneous observations into operational wave models.

Disagreements. Buoys provide reliable point spectra but sparse spatial sampling; altimeters broaden coverage yet degrade nearshore and may miss extrema; HF radar maps coastal fields but requires local validation. Aggregated height and period can conceal materially different multimodal spectra.

Limitations. Mooring motion, pressure attenuation, sidelobes, currents, breaking, biofouling, gaps, clock or datum error, land contamination, satellite revisit, short campaigns and changing bathymetry create bias. Extreme-event sampling and directional validation are especially sparse.

Open questions. Priorities include seamless multisensor fusion, coastal altimetry and SAR calibration, directional and infragravity retrieval, uncertainty-aware extremes, affordable dense arrays, autonomous quality control, changing-bathymetry correction and standardized open benchmark campaigns.

Seminal papers. Pressure and buoy spectra established field wave statistics; array methods enabled direction, radiation-stress studies tied waves to setup, and satellite altimetry plus HF radar extended spatial coverage. Long-term coastal observatories connected these methods.

Claims

1. Against Iberian buoys, Sentinel-3 significant-wave-height observations achieved correlations above 0.94, biases below 0.19 m, and RMSE of 0.17-0.42 m, enabling spatial validation of HF-radar wave retrievals. *Regime: West Iberian Sentinel-3/buoy collocations from 2017-2019; nearshore sampling and sea-state limits apply.* direct finding, field [221]
2. Spotter real-time buoy bulk wave statistics were within 10% of a Datwell reference across stand and field tests, enabling multi-buoy measurements of decorrelation, speed and directional spread. *Regime: Performance Characteristics of "Spotter," a Newly Developed Real-Time Wave Measurement Buoy* direct finding, mixed [1143]

31 Wiki: Flood mapping and inundation

31.1 Flood mapping and inundation: overview

Topic inundation. Prediction and representation of flood extent and depth.

Papers screened: 14. Claims: 1. Equations: 0.

Well established. Inundation is controlled by conservation of water across connected terrain, with wetting and drying, storage, friction, drainage, defenses and boundary forcing determining extent, depth, velocity and arrival time. Reliable maps require both hydraulic realism and defensible terrain and forcing data.

Governing physics. Flood waves spread according to pressure gradients, inertia, friction and topographic connectivity; rainfall, runoff, river discharge, tide, surge, waves and failures can interact. Small elevation errors can open or close flow paths, while defenses and drainage create thresholds and discontinuities.

Dimensionless parameters. Useful controls include Froude number, relative depth, friction and slope ratios, Courant number, grid spacing relative to hydraulic controls, DEM error relative to flood depth, compound-driver dependence, validation overlap and depth-error metrics, and computational speedup.

Major equations. Models solve shallow-water or diffusion-wave mass and momentum equations, sometimes coupled to rainfall-runoff and coastal boundary models. Reduced approaches use HAND or storage/connectivity rules; surrogates emulate depth and extent, while probabilistic maps integrate hazard frequency and uncertain inputs.

Typical methods. Workflows assemble conditioned terrain, roughness, drainage and defenses; generate rainfall, river and coastal boundaries; solve 1D/2D hydraulics or surrogates; validate against satellite extent, lidar, gauges, water marks and arrival times; and propagate forcing, parameter, DEM and model-structure uncertainty.

Numerical models. Representations include coupled rainfall-runoff-inundation diffusion waves, full shallow-water solvers on structured or unstructured grids, HAND, MIKE11 GIS, MSN_Flood, Gaussian-process/EOF surrogates and physics-aware classifiers. Model choice trades process fidelity, data needs and forecast speed.

Experimental datasets. Reviewed evidence includes the 2010 Kabul flood, historical dam-break test cases, MERIT/SRTM ensembles benchmarked against lidar, the 2016 Brazos Landsat extent, southwest Florida climate ensembles, Cork compound scenarios, the 2000 Langat flood and sparse-observation mapping applications.

Validated ranges. Reported performance is case-specific: Brazos extent overlap ranged 56–70%, Langat sampled-map agreement was 70%, and one hybrid surrogate was 12 times faster than its high-resolution model. These metrics use different observations and cannot be directly ranked as universal skill.

Recent advances. Recent advances simulate spatial DEM error, learn physics-aware maps from sparse data, emulate unstructured high-resolution models, integrate pluvial-fluvial-coastal drivers and translate climate ensembles into changing inundation magnitude and recurrence.

Disagreements. Fast terrain-based or learned models can support large-domain warning but may miss momentum, inter-catchment exchange or compound pathways; high-resolution hydrodynamics represents these processes but costs more and remains sensitive to terrain and friction. Extent-only agreement can conceal depth or timing error.

Limitations. DEM bias, unresolved buildings and defenses, uncertain roughness and drainage, sparse extreme observations, boundary dependence, climate and downscaling spread, numerical wetting/drying choices and inconsistent validation metrics limit confidence. Static maps do not demonstrate event forecasting skill.

Open questions. Priorities include dynamic defense and drainage failure, compound dependence, urban microtopography, transparent DEM ensembles, rapid depth-and-velocity surrogates, transfer under nonstationarity, observation assimilation and common multi-variable validation benchmarks.

Seminal papers. Early storage-cell and shallow-water models established dynamic floodplain routing; GIS and DEM methods enabled regional mapping, and satellite observations supported extent validation. Modern work combines high-resolution solvers, ensembles and emulators for operational and climate-scale analysis.

Claims

1. Coastal inundation assessment must jointly represent changing storm forcing, sea-level rise, precipitation and their probabilistic dependence because their combined effects can alter both flood magnitude and recurrence nonlinearly. *Regime: Large southwest Florida coastal floodplain and the study's climate/downscaling scenarios.* direct finding, numerical [1297]

31.2 Inundation modeling

Topic inundation. models. Wet-dry and overland-flow simulation.

Papers screened: 23. Claims: 12. Equations: 0. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Coastal inundation prediction requires explicit wetting and drying plus defensible marine and terrestrial forcing. Validation must be process- and site-specific: agreement for one tsunami geometry, hurricane, city, resolution, or water-level metric does not establish universal model skill.

Governing physics. The reviewed models combine long-wave or reduced shallow-water propagation, surge, wave setup and runup, rainfall-runoff and river inflow, bed and building drag, topographic control, urban blockage, and moving shorelines. Which mechanisms are retained materially changes predicted extent and depth.

Dimensionless parameters. Important nondimensional controls include relative tsunami amplitude H/d , Froude and friction measures, relative terrain and building scale, ensemble exceedance probability, normalized error metrics, and ratios of wave, surge, tide, rainfall, and river contributions to total water level.

Major equations. Core formulations include nonlinear Boussinesq or shallow-water continuity and momentum equations, bottom-friction and directional building-drag source terms, wet-dry algorithms, nonstationary extreme-value and joint-probability models, raster depth-damage functions, and ensemble exceedance probabilities.

Typical methods. Typical methods couple terrain and bathymetry preprocessing with depth-averaged or reduced-complexity solvers, deterministic hindcasts, probabilistic forcing ensembles, laboratory or gage validation, observed-extent comparison, sensitivity analysis, and GIS exposure or damage overlays.

Numerical models. Reviewed models include a higher-order Boussinesq inundation model, coupled MIKE 21 and XBeach configurations, an RDM plus Zero-Point Boundary method, a distributed hydrologic-GIS damage chain, an urban Drag Force Model, SFINCS compound-flood ensembles, and explicit high-precision 3D urban simulation.

Experimental datasets. Validation evidence includes Synolakis plane-beach and Briggs conical-island runup tests, the 1:250 HyTOFU Onagawa urban-tsunami experiment, NOAA and USGS Hurricane Ike records, Colombian Hurricane Iota profiles and flood maps, surveyed Haiyan inundation limits, and observed Korean coastal-city flood extent.

Validated ranges. Reported validation regimes span solitary-wave plane beaches and conical islands with H/d of 0.10-0.20, a 1:250 idealized Onagawa city, Hurricane Iota in the Colombian archipelago, a 5 m Haiyan terrain model with 81% surveyed-limit accuracy, and Hurricane Ike ensembles of 81-1000 members at one- to three-day lead times.

Recent advances. Recent advances pair reduced-complexity coastal solvers with large forcing ensembles, retain useful multi-day forecast skill, parameterize unresolved buildings directionally, resolve cities with LiDAR and drone-derived geometry, and report omission warnings when computational constraints remove wave effects.

Disagreements. The evidence does not support one universally preferred representation of urban form: explicit high-resolution geometry can improve routing and visualization, while subgrid directional drag can capture building effects more efficiently. Their relative accuracy and cost remain configuration-dependent.

Limitations. Common limitations are uncertain topography and bathymetry, forcing and dependence assumptions, friction and building closure, omitted waves or compound drivers, limited field observations, resolution sensitivity, computational cost, and validation against only one or a few historical events.

Open questions. Priorities include transferable urban roughness and blockage closures, multi-event and multi-site benchmarks, explicit wave-pluvial-fluvial-surge coupling, nonstationary ensemble generation, terrain and friction uncertainty propagation, probabilistic spatial validation, and operational accuracy-cost criteria.

Seminal papers. Within this screened branch, the 2015 higher-order Boussinesq application established explicit regional tsunami inundation with laboratory runup validation, while the 2017 Manhattan study linked nonstationary compound forcing to mapped damage.

Claims

1. For Manhattan, incorporating nonstationarity increased modeled coastal floodplain extent and damage, whereas defining scenarios with the joint probability of rainfall and water level reduced them relative to combinations that did not account for dependence. *Regime: Manhattan rainfall and coastal water-level records, the study's three scenarios, distributed model, DEM, land-use, and depth-damage functions.* direct finding, numerical [687]
2. A drag-force inundation model represented unresolved urban structures with characteristic height and directional projected areas and was validated against the 1:250 HyTOFU idealization of Onagawa. *Regime: Tsunami inundation through building fields representable by the three DFM parameters; validation is bounded by the 1:250 idealized Onagawa experiment.* direct finding, numerical [467]
3. For Hurricane Ike on the Texas Gulf Coast, SFINCS ensemble means reduced average station errors by 16% versus a deterministic case, and a roughly 100-member ensemble ran in under four hours on one CPU while retaining useful one- to three-day flood guidance. *Regime: Hurricane Ike, the modeled Texas Gulf Coast domain, 81-1000 members, one- to three-day lead times, stated forcing and terrain data, and SFINCS assumptions.* direct finding, numerical [793]
4. In a Korean coastal-city case, a high-precision 3D urban inundation model reasonably matched observed flood extent and indicated that explicit building geometry matters for routing and visualization, although quantitative skill was not reported in the accessible record. *Regime: The studied Korean coastal city and the available high-definition maps, LiDAR, drone imagery, and 3D city representation.* direct finding, numerical [716]
5. A coupled rainfall-runoff-inundation model using 2D diffusion-wave equations reproduced the MODIS-derived large-scale 2010 Kabul River flood extent and identified flash-flood damage areas across a 92,605 km² basin. *Regime: Rainfall-runoff-inundation analysis of the 2010 Pakistan flood in the Kabul River basin* direct finding, mixed [1258]
6. Historical dam-break validation data are most useful when breach geometry, maximum depths, arrival times, water marks, flood extent, topography and scenario inputs are sufficiently documented; the review separates complete from partial test cases. *Regime: Review of Historical Dam-Break Events and Laboratory Tests on Real Topography for the Validation of Numerical Models* direct finding, mixed [87]
7. For ungauged Hurghada, an integrated GIS, remote-sensing, 2D rainfall-runoff-inundation and multicriteria workflow mapped hazard and seven-component vulnerability for 5-, 10- and 50-year return periods. *Regime: Integrated Methodology for Urban Flood Risk Mapping at the Microscale in Ungauged Regions: A Case Study of Hurghada, Egypt* direct finding, mixed [12]

8. The low-fidelity/spatial-analysis/Gaussian-process surrogate predicted both flood depth and extent on an unstructured Chowilla floodplain grid while running 12 times faster than the high-resolution hydrodynamic model. *Regime: Development of a Fast and Accurate Hybrid Model for Floodplain Inundation Simulations* direct finding, mixed [457]
9. Cork scenarios show current extreme inundation is primarily fluvial, high-end future coastal inundation can become comparable, and the largest city-centre floods arise from combined river discharge and coastal water level. *Regime: Modelling the effects of climate change on urban coastal-fluvial flooding* direct finding, mixed [723]
10. For the 2000 Langat River event, satellite-derived quantitative-precipitation forecasts drove pre-flood MIKE11 GIS inundation maps with 70% agreement at sampled observed flood points. *Regime: Pre-flood inundation mapping for flood early warning* direct finding, mixed [155]
11. A dual-grid shallow-water framework enables fine-resolution metropolitan inundation forecasts and is evaluated against Hurricane Harvey observations. *Regime: A Framework for Mechanistic Flood Inundation Forecasting at the Metropolitan Scale* direct finding, numerical [1267]
12. A coastal-flood modeling review finds that static bathtub methods perform poorly in defended urban backshores and that credible prediction requires resolved terrain and infrastructure, time-varying overtopping, overland routing, hydrologic inputs, and interacting compound pathways. *Regime: Defended urban coastal backshores exposed to marine, overtopping, hydrologic, groundwater and infrastructure-mediated flooding.* review statement, review [474]

31.3 Topographic and bathymetric controls

Topic inundation. topography. Elevation data and terrain representation.

Papers screened: 9. Claims: 5. Equations: 1.

Well established. Inundation mapping depends on a hydraulically conditioned, datum-consistent terrain surface that represents connected low paths, channels, barriers and openings; finer nominal resolution does not compensate for missing bathymetry, blocked drainage or incorrect structures.

Governing physics. Topography and bathymetry control storage, conveyance, wetting paths, channelization and hydraulic gradients, while levees, roads, buildings, dunes, culverts and embankments create thresholds whose overtopping or openings can abruptly change connectivity and extent.

Dimensionless parameters. Relevant normalized controls include vertical error relative to flood depth or freeboard, horizontal resolution relative to channel or barrier width, slope-to-error ratio, submergence, grid convergence, connected-to-total low-area fraction and inundation overlap or boundary-distance scores.

Major equations. Terrain enters shallow-water mass and momentum equations through bed elevation, depth and slope and enters static screening through elevation and connectivity tests; vertical datum transformations, interpolation, subgrid conveyance and uncertainty perturbations modify those inputs.

Typical methods. Workflows merge topo-bathymetry in one vertical datum, remove vegetation and artifacts, preserve or burn channels and openings, encode defenses and buildings deliberately, test connectivity, compare resolutions and terrain sources, propagate elevation uncertainty and validate extent, depth and pathways independently.

Numerical models. Reviewed methods include elevation-threshold propagation with hydrologic connectivity, a rapid RDM/Zero-Point Boundary coastal hindcast, and HEC-RAS 2D simulations using alternative LiDAR terrains and discharge- or volume-based corrections.

Experimental datasets. The evidence includes Staten Island airborne LiDAR connectivity auditing, a 5 m Tacloban terrain hindcast checked against surveyed Haiyan limits, and topographic-versus-bathymetric LiDAR comparisons for 11 Norwegian sites and multiple flood scenarios.

Validated ranges. Evidence spans one urban coastal island, one typhoon-affected coastal city and 11 river reaches. Findings are conditional on their resolutions, geomorphology, structures and forcing; the Norway correction results are not direct coastal validation.

Recent advances. Recent advances integrate topographic and bathymetric LiDAR, satellite/UAS change detection, structure and drainage conditioning, rapid terrain-driven solvers, learned depth estimation, cloud-scale DEM processing and ensembles that propagate vertical and representation uncertainty.

Disagreements. Bare-earth DEMs improve hydraulic ground representation but can erase buildings or defenses that matter; digital surface models retain obstacles but may falsely block flow. Explicit structures are preferable when data support them, while subgrid or roughness treatment may be more stable at coarser scales.

Limitations. Key limitations are vertical and horizontal error, inconsistent datums, absent submerged channels, vegetation filtering, bridges and culverts, seawall gaps, building representation, changing morphology, uncertain defense condition, grid-scale connectivity and validation focused only on maximum extent.

Open questions. Priorities include seamless coastal topo-bathymetry, automated connectivity conditioning with provenance, dynamic breach and culverts, uncertainty-aware subgrid structures, morphology updates, compound river-rain-surge boundaries, cross-resolution skill and open elevation-change benchmarks.

Seminal papers. Early bathtub mapping highlighted elevation control but ignored pathways; hydrologically connected threshold methods, LiDAR bare-earth DEMs and two-dimensional wetting/drying solvers established more defensible practice, followed by subgrid and structure-aware terrain models.

Equations

- *Zero-Point Boundary inundation depth* [1584]

$$d_i(x, y) = \max[0, \eta_{s+w}(x, y) - z_{DTM}(x, y)]$$

Regime: Haiyan hindcast over the 5 m eastern Leyte terrain model. Variables: d_i mapped inundation depth; η_{s+w} interpolated simulated surge and wave level; z_{DTM} terrain elevation

Claims

1. At Staten Island, auditing and enforcing drainage connectivity identified more than 11 ha of inland low terrain that simple elevation-threshold inundation could misclassify. *Regime: Static DEM-based inundation screening; dynamic timing and hydraulic resistance are not represented.* direct finding, numerical [1121]
2. Across 11 Norwegian sites, inundation differences between topographic-only and bathymetric LiDAR terrain generally decreased as flood magnitude increased, but embankment overtopping, sinuosity and bank geometry altered the error pattern; volume-based terrain correction usually improved results most. *Regime: Comparison between Topographic and Bathymetric LiDAR Terrain Models in Flood Inundation Estimations* direct finding, numerical [89]
3. Land-cover-stratified semivariogram ensembles of MERIT DEM error produced flood-inundation estimates closer to lidar benchmarks than a single deterministic DEM in Mekong Delta and Fiji tests. *Regime: Implications of Simulating Global Digital Elevation Models for Flood Inundation Studies* direct finding, mixed [576]
4. A physics-aware topographic constraint with multimodal class distributions mapped inundation from sparse observations more accurately and robustly than baseline spatial-classification methods in real flood applications. *Regime: Flood Inundation Mapping with Limited Observations Based on Physics-Aware Topography Constraint* direct finding, mixed [1234]
5. A GIS and analytical-hierarchy assessment combines coastal geomorphic, topographic and hazard indicators to map relative vulnerability along the Puducherry coast, India, using the analytical hierarchical process direct finding, mixed [1001]

31.4 Flood-map uncertainty

Topic inundation.uncertainty. Uncertainty in extent, depth, and timing.

Papers screened: 25. Claims: 21. Equations: 0. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Flood maps are conditional estimates, not exact boundaries: elevation, forcing, roughness, defenses, model physics, grid scale, wetting/drying, observations, classification, and climate assumptions all contribute uncertainty.

Governing physics. Coastal uncertainty is strongly shaped by dynamic surge and wave timing, tide phase, compound sources, topographic connectivity and slope, friction and obstacles, defense behavior, and the duration available for flood propagation.

Dimensionless parameters. Useful controls include elevation error relative to local relief/freeboard, grid spacing relative to terrain scale, forcing error relative to flood threshold, hydrograph duration relative to filling time, roughness sensitivity, and ensemble spread relative to mapped depth.

Major equations. Methods combine shallow-water inundation, static connectivity, return-level statistics, spatial elevation-error models, Monte Carlo ensembles, surrogate/multifidelity UQ, data assimilation, classification thresholds, and validation contingency metrics.

Typical methods. Approaches include DTM/DSM/SRTM comparison, plausible DEM realizations, static-dynamic model intercomparison, surge-wave omission tests, event and satellite validation, ensemble/multimodel forecasts, Monte Carlo flood zones, and SAR calibration.

Numerical models. Static bathtub and connectivity tools, HAND, local-inertial and full dynamic solvers, GeoClaw, global 30 m models, coupled hydrologic-hydrodynamic systems, Monte Carlo, Kriging, and multifidelity methods provide differing fidelity and cost.

Experimental datasets. Evidence includes Xynthia, Venice extremes, global tropical cyclones, tsunami DEM cases, Greater London and Tottenville terrain, Sarasota elevation sources, satellite floodplains for 71 storms, and multi-event Sentinel-1 maps.

Validated ranges. Reported effects include 209-232% static overprediction at Xynthia, about 60% underprediction when waves are omitted, more than 20% extent sensitivity to surge timing, 31% exposure underestimation without tropical cyclones, and 99.9% Kriging cost reduction.

Recent advances. Recent advances quantify hazard-source omission, separate surge-shape effects on depth and extent, generate spatial DEM ensembles, validate global dynamic inundation against satellites, accelerate probability propagation, and stabilize multi-event SAR thresholds.

Disagreements. Greater model complexity can improve process representation yet forcing error may still dominate. DTM can overpredict while DSM underpredicts, static methods can grossly overmap, and rapid approximations may be adequate for regional screening but unsafe for cell-level decisions.

Limitations. Validation outlines contain sensor and classification error; exposure is not hazard skill; global and local results are not interchangeable; correlated elevation/forcing errors, defenses, compound events, nonstationarity, and rare-event scarcity remain weakly constrained.

Open questions. Needs include correlated spatial error, dynamic defenses and failure, joint tide-surge-wave-rain-river ensembles, wetting/drying and urban drainage, reliable depth/timing validation, observation uncertainty, climate nonstationarity, and decision-specific calibration.

Seminal papers. The branch develops from DEM and hydraulic-model uncertainty reviews through probabilistic elevation and forcing ensembles to process-based global validation, multifidelity UQ, and uncertainty-aware satellite mapping.

Claims

1. Flood-map uncertainty begins with DEM vertical error, resolution, interpolation, hydrologic conditioning, and uncertainty in terrain parameters derived from elevation. *Regime: Elevation error, grid resolution, interpolation, terrain modification, and derived parameters propagate into distributed flood models.* review statement, review [1496]
2. Large-scale coastal comparisons found static methods overestimated Xynthia inundation by 209-232%, while omitting waves underestimated flooded area by about 60%. *Regime: Static methods overestimated Xynthia extent by 209-232%; pan-European static extent was 56% above dynamic LFP, while omitting waves underestimated area about 60% and exposure differed 20-70%.* direct finding, mixed [1466]

3. NWM-HAND evaluation identified roughness, streamflow bias above 60% MAE, and low relief as major boundary-error controls and cautioned against cell-scale wet/dry use. *Regime: Roughness was the most important correction parameter; errors persisted above 60% streamflow MAE and in low relief, so HAND was useful for regional risk but not cell wet/dry decisions.* direct finding, mixed [669]
4. Adding tropical cyclones more than doubled estimated population exposed to 1-in-1000-year coastal flooding from 78 million to 192 million and corrected a 31% underestimate. *Regime: Including tropical cyclones raised estimated 1-in-1000-year exposed population from 78 million to 192 million; prior exposure was underestimated 31%.* direct finding, numerical [381]
5. Flood inundation uncertainty arises from hydrologic forcing, validation, topography, steady-flow assumptions, cross-section geometry, and model dimensionality, with 1-D polygon uncertainty increasing away from channels. *Regime: Steady-flow assumptions matter more at high flow, cross-section geometry adds errors, and one-dimensional polygon uncertainty grows away from the main channel.* review statement, review [112]
6. Greater London bathtub maps showed DTM-versus-DSM processing differences can dominate nominal RMSE and that SRTM underestimated flood extent. *Regime: DTM-versus-DSM inundation differences exceeded their RMSE differences, and SRTM underestimated extent enough to be unreliable for risk mapping.* direct finding, mixed [1573]
7. Venice flood forecasting uncertainty calls for improved meteorological forcing, ensemble prediction, water-level data assimilation, and multimodel approaches. *Regime: Priority improvements are meteorological forecasts, ensembles, water-level assimilation, and multimodel prediction, informed by difficult historic extremes.* review statement, review [1437]
8. Tsunami case studies found a 25 m mesh adequate with accurate elevation, while DTM tended to overestimate and DSM underestimate inundation under constant roughness. *Regime: A 25 m mesh sufficed with accurate elevation; DTM tended to overestimate and DSM underestimate when constant Manning roughness was used.* direct finding, mixed [534]
9. A 30 m global model found low-emission flood-hazard change remained within present uncertainty, whereas high-emission change emerged in the 2040s and reached 49% with a 7-109% likely range by 2100. *Regime: SSP1-2.6 end-century change of 9% (-6% to 29%) remained within current hazard uncertainty, while SSP5-8.5 reached 49% (7% to 109%) and emerged from noise in the 2040s.* direct finding, numerical [1522]
10. Hydrologic-hydrodynamic comparisons found flood-map accuracy remained dominated by forcing magnitude and timing even as model complexity increased. *Regime: Greater physical complexity did not remove dominant forcing magnitude/timing errors, and delta applications also require tides and surge.* direct finding, numerical [605]
11. Changing storm-surge temporal variability altered mean and maximum depth by about 5% but flood extent by more than 20% in the studied coastal setting. *Regime: Mean and maximum depths differed about 5%, but extent deviations exceeded 20%, showing timing affects map footprint more than some depth/exposure metrics.* direct finding, numerical [627]
12. A Sarasota coastal DEM study generated spatial uncertainty surfaces and plausible elevation ensembles for propagating sparse offshore and dense LiDAR uncertainty into coastal models. *Regime: Uncertainty was high offshore where soundings were sparse and low on LiDAR-covered flat land; plausible DEM ensembles propagate elevation uncertainty into process models.* direct finding, mixed [49]
13. Monte Carlo mapping in low-lying Tottenville showed small elevation and water-level errors can produce large uncertainty in future 1% flood-zone extent. *Regime: Small land/water-height changes produced large extent uncertainty on flat terrain; probabilistic maps represented ranges rather than a single deterministic boundary.* direct finding, numerical [50]
14. Across 71 tropical cyclones, global flood models had only moderate satellite-validated skill, while process-based surge-inundation dynamics improved mapped extent. *Regime: All global approaches had moderate skill, but process-based GeoClaw outperformed methods without dynamic inundation.* direct finding, mixed [1460]
15. A flood-inundation UQ comparison reported Kriging reduced computational cost by 99.9% relative to full Monte Carlo. *Regime: Kriging reduced computational cost by 99.9% relative to full Monte Carlo, illustrating a speed-accuracy route for probabilistic mapping.* direct finding, numerical [21]
16. Multi-event Sentinel-1 mapping found threshold selection dominated flood-map uncertainty, whereas stability-optimized calibration limited post-processing variability. *Regime: Threshold choice dominated variability, while stable calibration limited residual post-processing uncertainty and supported coherent cross-event extent/depth patterns.* direct finding, mixed [710]
17. For the 2016 Brazos flood, HAND and iRIC-FaSTMECH extents overlapped Landsat observations by 56% and 70%; adding inter-catchment behavior raised modified HAND overlap to 67%. *Regime: Comparative Analysis of Inundation Mapping Approaches for the 2016 Flood in the Brazos River, Texas* direct finding, mixed [1588]
18. For southwest Florida, climate-model, downscaling and scenario choices yield 2100 1%-annual-chance inundation metrics about 3–7 times current values, while today's 100-year event could become a 3-year event. *Regime: A sensitivity study of rising compound coastal inundation over large flood plains in a changing climate* direct finding, mixed [1297]
19. Propagating spatially variable and correlated LiDAR DEM error through 1,000 sequential-Gaussian simulations produced a coastal inundation area about 11% larger than deterministic bathtub mapping at the 1% exceedance level. *Regime: Coastal LiDAR DEM and ground-control data for the Tully region of northeastern Australia under the specified 1 m sea-level-rise plus 100-year storm scenario.* direct finding, numerical [800]
20. For the studied large rural flood, LISFLOOD-FP gained little deterministic skill below 50 m grid spacing, degraded above 50 m, and produced less reliable probabilistic inundation maps beyond 100 m; filtering to only behavioural simulations could also create overconfident maps. *Regime: A large rural flood in the Imera basin, Sicily, represented at 10-500 m grid spacing with uncertain inflow, Manning friction, DEM aggregation, and water-depth observations.* direct finding, numerical [1256]
21. LISFLOOD-FP sensitivity is output-, place-, and time-dependent: inflow dominates flood extent on the rising limb, channel friction near peak inundation, and floodplain friction during recession, while grid resolution and DEM resampling are especially influential for local water depths. *Regime: Rural flood-inundation modelling with uncertain inflow, channel and floodplain Manning friction, spatial resolution, and LiDAR-derived DEM aggregation.* direct finding, numerical [1257]

32 Wiki: Laboratory experiments

32.1 Laboratory experiments: overview

Topic Laboratory. Controlled physical modeling of coastal processes.

Papers screened: 16. Claims: 10. Equations: 0.

Well established. Coastal laboratory experiments are controlled approximations of prototype systems: credibility depends on a stated question, prioritized similitude, characterized facility and boundaries, calibrated forcing and sensors, repeatability, uncertainty, documented processing, and explicit transfer limits.

Governing physics. Physical models reproduce gravity waves, currents, wind, turbulence, sediment, structures and devices through conservation laws and boundary forcing, while viscosity, surface tension, air compressibility, grain properties, elasticity, porosity and control-system dynamics can scale differently.

Dimensionless parameters. Key controls include geometric and Froude scale, Reynolds and Weber numbers, relative depth and wave steepness, Ursell and Keulegan–Carpenter numbers, blockage, surf similarity, density and fall-velocity ratios, relative roughness, Cauchy stiffness, pneumatic compliance, reflection coefficient, signal-to-noise and repeatability ratios.

Major equations. Core formulations include Froude, Reynolds, Weber, Cauchy and Shields scaling; wavemaker transfer and absorption; dispersion and spectral targets; reflection separation; force and pressure coefficients; sediment and morphology scaling; response amplitude operators; and Type A/B measurement and propagated uncertainty.

Typical methods. Methods define prototype and model scales, characterize basin transfer and flow quality, generate regular/irregular/directional/focused/breaking waves and currents, absorb reflections, synchronize sensors, repeat and randomize tests, separate incident/reflected fields, quantify uncertainty and compare against theory, computation or prototype data.

Numerical models. Numerical tools support wavemaker and boundary control, facility characterization, design of experiments, sensor correction, reflection analysis, digital twins, CFD and reduced-order validation, uncertainty propagation and hybrid numerical–physical testing.

Experimental datasets. The reviewed evidence includes ocean–environment simulation surveys, testing–standard gap analysis, tidal round robins, wind–over–wave load tests, focused–wave validation, volcanic–tsunami experiments, breaking–cylinder impacts, wave–current cylinder coefficients, FloWave turbine validation and an OWC efficiency/survivability database.

Validated ranges. Examples span flumes and directional or circular basins, regular through irregular and focused waves, currents and wind, nonbreaking through impact regimes, structural cylinders, turbines and oscillating–water columns, and gravity–driven tsunami sources; each remains bounded by facility and scale.

Recent advances. Recent advances include segmented multidirectional control, real–time absorption, high–speed optical and remote sensing, synchronized dense pressure and velocity arrays, large–scale prototype–like facilities, round–robin testing, open benchmark databases, digital facility twins and hybrid physical–numerical experiments.

Disagreements. Froude similarity preserves gravity but not all viscous, capillary, elastic, pneumatic or sediment processes; facility–to–facility results can differ despite nominally identical targets; deterministic focus and stochastic sea states answer different questions; numerical agreement may reflect shared boundary assumptions rather than prototype validity.

Limitations. Finite depth, width and length create blockage and reflection; wavemakers have stroke, velocity and bandwidth limits; long waves and currents recirculate; wind and air effects are difficult to scale; sediments distort mobility and time scales; sensors disturb or filter response; few repeats weaken tail estimates.

Open questions. Priorities include quantified multi–physics scale effects, coupled wind–wave–current–surge forcing, controlled breaking and aeration, sediment mixtures and ecology, active three–dimensional boundaries, rare–event repeatability, interlaboratory uncertainty, hybrid testing and model–to–prototype discrepancy inference.

Seminal papers. Classical wave–tank work established wavemaker theory, Froude scaling, reflection control, force and overtopping measurement and hydraulic similitude; movable–bed and multidirectional facilities extended physical modelling to sediment, morphology and directional seas.

Claims

1. A review of offshore–renewable testing shows that laboratory reproduction of the ocean environment must coordinate waves, currents, wind, turbulence, depth, scaling, control, instrumentation, and device response. *Regime: Capture and simulation of the ocean environment for offshore renewable energy* review statement, review [372]
2. FloWave experiments provide robust validation of a generalized actuator–disk CFD model and show how facility measurements connect numerical assumptions to turbine response. *Regime: Robust validation of a generalised actuator disk CFD model for tidal turbine analysis using the FloWave ocean energy research facility* direct finding, mixed [100]
3. A marine–renewable testing gap analysis identifies common terminology, procedures, uncertainty reporting, facility characterization, and staged validation as prerequisites for comparable experiments. *Regime: Standardising Marine Renewable Energy Testing: Gap Analysis and Recommendations for Development of Standards* review statement, review [1031]
4. A documented floating oscillating–water–column test database links wave conditions, mooring, air and power–take–off response, efficiency, and survivability for reuse and model validation. *Regime: Efficiency and Survivability of a Floating Oscillating Water Column Wave Energy Converter Moored to the Seabed: An Overview of the Esflowc MaRINET2 Database* direct finding, experimental [724]

5. Round-robin tidal tests demonstrate that facility flow quality and measurement practice can alter inferred turbine loading and performance, motivating interlaboratory comparison and documented facility effects. *Regime: Tidal Energy Round Robin Tests: A Comparison of Flow Measurements and Turbine Loading* direct finding, experimental [923]
6. Experiments show that wind above irregular laboratory waves can change wave-induced load statistics, so open-sea-to-basin translation must account for missing or imposed aerodynamic forcing. *Regime: Experimental study of the effect of wind above irregular waves on the wave-induced load statistics* direct finding, experimental [746]
7. A validated procedure generates repeatable periodic focused waves in a flume, linking target wave specification, wavemaker control, measurement, and achieved focus. *Regime: Methodology and experimental validation for generating periodic focused waves in a wave flume* direct finding, experimental [424]
8. Large-scale physical experiments distinguish tsunami source and wave characteristics generated by submarine volcanic eruptions and provide a controlled benchmark for source-to-wave models. *Regime: Physical Modeling of Tsunamis Generated by Submarine Volcanic Eruptions* direct finding, experimental [854]
9. Controlled measurements of breaking-wave impact on a vertical cylinder resolve how breaker properties govern impulsive loads and establish a geometry- and regime-bounded validation dataset. *Regime: Experimental characterisation of breaking wave impact loads on a vertical cylinder* direct finding, experimental [623]
10. An experimental wave-current campaign determines forces and hydrodynamic coefficients on vertical cylinders representative of offshore structural elements under combined forcing. *Regime: Experimental Determination of Forces and Hydrodynamic Coefficients on Vertical Cylinders Under Wave and Current Conditions* direct finding, experimental [1448]

32.2 Wave-basin experiments

Topic Laboratory.basins. Directional and three-dimensional modeling.

Papers screened: 10. Claims: 7. Equations: 0.

Well established. Wave basins and broad physical-model facilities enable controlled spatial wave fields, structure loading, overtopping, and model coupling, but inference remains bounded by generation fidelity, basin boundaries, scale, geometry, and the measured response.

Governing physics. Relevant physics include directional generation and diffraction-reflection, irregular-wave grouping, evanescent modes near paddles, green-water impact and air-gap effects, dam-break bore hydraulics, overtopping depth and velocity, and drag dissipation by suspended canopies.

Dimensionless parameters. Important controls include Froude scale, relative depth and air gap, crest freeboard and slope cotangent, normalized force and pressure, wave steepness and spectral width, relative canopy density and spacing, transmission or reflection coefficients, and grouping statistics.

Major equations. Core formulations include Boussinesq-wave transfer to piston motion, spectral transfer functions, normalized impact force and pressure, shallow-water bore scaling, empirical overtopping depth-velocity-discharge relations, EurOtop guidance, XBeach non-hydrostatic dynamics, and canopy-drag energy dissipation.

Typical methods. Methods combine segmented or piston wavemakers, regular, irregular, focused, and multidirectional waves, spatial wave-gauge arrays, pressure and global-load sensing, optical or velocity measurements, dam-break releases, spectral analysis, deterministic numerical-physical coupling, and model-data benchmarking.

Numerical models. Models include MIKE 21 BW, deterministic wavemaker transfer with evanescent-mode correction, frequency-domain harbor response, XBeach non-hydrostatic, EurOtop design guidance, empirical bore relations, benthic-canopy models extended to suspended arrays, and a suspended seaweed-farm analytical model.

Experimental datasets. Reviewed datasets cover harbor grouping and agitation, deterministic transfer from MIKE 21 BW into physical wavemakers, 1:20 jacket green-water loads, full-scale dam-break analogues for overtopping, and an open Zenodo rigid suspended-canopy wave-dissipation campaign.

Validated ranges. Explicit bounds include 1:20 jacket tests with partial and full green water, dam-break analogues for $6 \leq \cot(\alpha) \leq 10$, $T_p > 4$ s and $1 \leq x_c \leq 5$ m, plus regular and random waves over tested rigid-canopy densities, lateral spacings, line spacings, and spectral widths.

Recent advances. Recent work uses full-geometric-scale dam-break bores to target individual hazardous overtopping flows and open configuration-rich datasets to evaluate suspended-canopy models, advancing from facility demonstration toward bounded model validation and reusable evidence.

Disagreements. Regular-wave coefficients may vary more than irregular-wave estimates within the main energy band, and model agreement depends on whether low-energy spectral tails, evanescent modes, air entrapment, impact peaks, and spatial canopy anisotropy are resolved.

Limitations. Limitations include sidewall and reflection contamination, finite directional resolution, wavemaker stroke and absorption, scale-dependent impact and aeration, limited repeat events, idealized fixed geometry, rigid instead of flexible canopies, and extrapolation outside explicit overtopping bounds.

Open questions. Priorities are reflection-controlled multidirectional benchmarks, uncertainty on extreme impact statistics, field validation of dam-break overtopping analogues, flexible and deforming suspended canopies, coupled current-wave forcing, and standardized open basin datasets with full calibration metadata.

Seminal papers. Within this branch, the 1989 harbor-basin study established grouping and spectral-response cautions, while the 2007 deterministic coupling study formalized transfer from numerical wave fields to 2D and 3D physical wavemakers.

Claims

1. Unscaled dam-break bores reproduced non-impulsive overtopping characteristics for $6 \leq \cot(\alpha) \leq 10$, $T_p > 4$ s, and $1 \leq x_c \leq 5$ m from the crest edge, including flows capable of exceeding pedestrian-stability thresholds. *Regime: Gentle slopes $6 \leq \cot(\alpha) \leq 10$, peak period $T_p > 4$ s, and crestward distance $1 \leq x_c \leq 5$ m.* direct finding, experimental [102]
2. For tested rigid suspended canopies, wave dissipation increased with canopy density, wave height, wave period, narrower spectral width, and reduced lateral element spacing, while remaining relatively insensitive to cross-shore line spacing. *Regime: Tested rigid suspended-cylinder densities, wave heights, periods, spectral widths, lateral spacings, and cross-shore line spacings.* direct finding, experimental [1604]
3. A circular-basin experiment recreated the 25.6 m Draupner crest at scale only for large-angle crossing seas, where breaking became less crest-limiting and formed near-vertical jets. *Regime: Laboratory recreation of the Draupner wave and the role of breaking in crossing seas* direct finding, experimental [941]
4. Directional wave-basin experiments and numerical simulations characterize the evolution of weakly nonlinear random directional waves. *Regime: Evolution of weakly nonlinear random directional waves: laboratory experiments and numerical simulations* direct finding, experimental [1414]
5. Directional sea-state analysis links directional spreading and nonlinear evolution to kurtosis estimates used in freak-wave forecasting. *Regime: On the Estimation of the Kurtosis in Directional Sea States for Freak Wave Forecasting* direct finding, mixed [982]
6. Field and laboratory measurements constrain maximum ocean-wave steepness and show its dependence on directional sea-state properties. *Regime: Maximum steepness of oceanic waves: Field and laboratory experiments* direct finding, mixed [1415]
7. Laboratory observations and numerical calculations compare the breaking evolution of steep two-dimensional deep-water waves. *Regime: Numerical and laboratory investigation of breaking of steep two-dimensional waves in deep water* direct finding, experimental [96]

32.3 Wave-flume experiments

Topic Laboratory .flumes. Two-dimensional physical modeling.

Papers screened: 18. Claims: 8. Equations: 5.

Well established. Wave flumes isolate coastal processes under repeatable forcing and support direct measurement, empirical relation development, and numerical-model validation, but results remain conditional on scale, geometry, boundary reflections, wave generation, material similarity, and instrumentation.

Governing physics. Across reviewed experiments the governing processes include wave shoaling, breaking, runup, overtopping and transmission; air-water resonance; groundwater exchange; sediment thresholds and phase lag; vegetation drag; wave impact and erosion; and nonlinear structural loading.

Dimensionless parameters. Key controls include Froude and Reynolds similarity, relative depth and freeboard, wave steepness and Ursell number, mobility and Shields-type numbers, settling-time factor, relative vegetation submergence and density, Keulegan-Carpenter number, and normalized transmission, runup, pressure, and force.

Major equations. The evidence uses mobility and Ursell threshold relations, ripple-migration transport and settling-time factors, overtopping transmission normalized by theoretical runup, RANS-VOF and groundwater equations, impact and erosion formulae, vegetation-drag attenuation, and harmonic force decomposition.

Typical methods. Methods include regular and irregular wave generation, large-scale profile tests, movable and permeable beds, pressure and force sensing, water-level and groundwater measurement, bed surveys, vegetation mimics, phase manipulation, empirical regression, and coupled model-data comparison.

Numerical models. Models tested against flume data include RANS-VOF hydrodynamics coupled to SEAWAT-2000, a process-oriented sea-dike breach model, empirical ripple thresholds, Ribberink bedload comparison, overtopping transmission relations, cliff-notch force formulae, vegetation attenuation closures, and harmonic load reconstructions.

Experimental datasets. The branch now links OWC, permeable-beach, dike-cover erosion, ripple onset and migration, overtopping transmission, cliff-notch pressure, storm beach-dune erosion, vegetation attenuation, higher-harmonic cylinder loading, and classic setup/runup flume datasets.

Validated ranges. Explicit regimes include a smooth impermeable 1:4 overtopped breakwater under regular waves, three imposed sand-bed perturbation classes, regular-wave migrating ripples, fresh-water large-scale permeable-beach tests, grass and clay cover under breaking impacts, flexible vegetation mimics, and phase-controlled cylinder waves.

Recent advances. Recent experiments increasingly combine dense measurements with process-based models, isolate higher harmonics through phase manipulation, decompose vegetation components, couple groundwater and sediment response, and distinguish sustained from impulsive coastal-cliff loads.

Disagreements. Laboratory relations can diverge from field transport or prototype impacts because phase lag, scale effects, sediment and vegetation similarity, three-dimensionality, irregularity, and material failure are incompletely represented. Larger scale is valuable but does not eliminate all distortion.

Limitations. Common limitations are reflection and finite-length effects, wavemaker fidelity, sensor bandwidth, repeatability, scale-dependent air compressibility and impulsive pressure, imperfect sediment and vegetation similitude, two-dimensional geometry, short duration, and restricted forcing ranges.

Open questions. Needed advances include uncertainty-standardized benchmark datasets, quantified scale effects for impacts and OWC pneumatics, reproducible irregular and multidirectional forcing, full measurement metadata, sediment and ecological similitude criteria, and stronger field-to-flume validation.

Seminal papers. Within this screened set, the 1982 storm beach-dune and overtopping-transmission papers and the 1992 OWC tests established enduring large-flume and device-testing evidence; later work added process-resolving measurements and coupled validation.

Equations

- *Flat-bed ripple threshold* [1280]

$$M = 17 - 14.5e^{-0.03U}$$

Regime: The tested well-sorted sands, three imposed bed perturbations, and oscillatory-flow Ursell-number range. Variables: M mobility number; U Ursell number

- *Notched-bed ripple threshold* [1280]

$$M = 5.0 - 2.5e^{-0.1U}$$

Regime: The tested well-sorted sands, three imposed bed perturbations, and oscillatory-flow Ursell-number range. Variables: M mobility number; U Ursell number

- *Settling-time factor* [1557]

$$\Omega_s = \eta/(w_0 T)$$

Regime: The tested migrating-ripple regular-wave conditions and settling-time factor $\Omega_s = \eta/(w_0 T)$. Variables: η ripple height; w_0 settling velocity; T wave period

- *Ripple migration transport* [1557]

$$\langle q_r \rangle = (1 - \epsilon)f_s \eta v$$

Regime: The tested migrating-ripple regular-wave conditions and settling-time factor $\Omega_s = \eta/(w_0 T)$. Variables: ϵ porosity; f_s shape factor; η ripple height; v migration rate

- *superposition and stretching wave-velocity approximations* [351]

$$y(t, \mathbf{x}) = \mathcal{M}[\mathbf{o}(t), \mathbf{b}, \mathbf{p}]$$

Regime: Controlled waves over a 1:40 beach measured by an acoustic meter on a servo-hydraulic surface follower. Variables: y predicted or derived coastal response; o event observations; b bathymetry or beach state; p model or instrument parameters; normalized observation-model mapping

Claims

1. Controlled wave-flume experiments were used to characterize a two-dimensional oscillating-water-column wave-energy device; quantitative transfer requires the source-specific geometry and wave regime. *Regime: The specific two-dimensional OWC and wave-flume configurations reported in the source.* direct finding, experimental [1255]
2. In large-scale permeable-beach tests, the coupled RANS-VOF and SEAWAT model reproduced wave-driven water-table behavior, and tested low water tables tended toward accretion whereas high water tables tended toward erosion. *Regime: The large-scale permeable-beach experiment and modeled fresh-water, wave, sediment, and aquifer conditions.* direct finding, experimental [108]
3. Laboratory and model analysis indicated that erosion of grass and clay cover accounted for more than 85% of total modeled sea-dike breach time under seaward breaking-wave impact. *Regime: Grass- and clay-reveted sea dikes exposed to seaward breaking-wave impacts comparable to the laboratory and tentative validation cases.* direct finding, experimental [1347]
4. Ripple-onset mobility decreased as imposed bed perturbation increased; the fitted threshold changed from $M=17-14.5 \exp(-0.03U)$ on a flat bed to $M=2.5$ independent of U on a notch-mounded bed. *Regime: The tested well-sorted sands, three imposed bed perturbations, and oscillatory-flow Ursell-number range.* direct finding, experimental [1280]
5. Cliff-notch experiments separated sustained forces, estimable from buoyancy at low velocity, from impulsive forces that were more likely to mobilize cliff-top boulders; scale effects on impulses remain unresolved. *Regime: The tested cliff-notch geometry and observed/simulated wave-height and velocity range; impulsive scaling remains unresolved.* direct finding, experimental [1493]
6. For a smooth impermeable 1:4 breakwater under regular waves, overtopping transmission was correlated with crest height above mean sea level normalized by theoretical wave-runup height. *Regime: Regular waves overtopping the tested smooth impermeable 1:4 breakwater.* direct finding, experimental [566]
7. Deviation between Ribberink bedload predictions and ripple-migration transport increased systematically with settling-time factor $\Omega_s = \eta/(w_0 T)$, supporting a phase-lag mechanism. *Regime: The tested migrating-ripple regular-wave conditions and settling-time factor $\Omega_s = \eta/(w_0 T)$.* direct finding, experimental [1557]
8. For shoaling waves over a 1:40 beach, surface-following acoustic velocities agreed with superposition and stretching adaptations of linear theory before and near breaking; after breaking, air entrainment contaminated crest measurements while trough velocities remained relatively well predicted. *Regime: Controlled waves over a 1:40 beach measured by an acoustic meter on a servo-hydraulic surface follower.* direct finding, experimental [351]

32.4 Laboratory instrumentation

Topic *Laboratory instrumentation. Measurement techniques and uncertainty.*

Papers screened: 8. Claims: 7. Equations: 0.

Well established. No single instrument resolves all coastal laboratory variables: point probes provide bandwidth, imaging provides spatial structure, acoustic methods work in optically difficult flows, and morphology/transport inference needs repeated geometry plus conservation; synchronization and uncertainty are integral to the measurement.

Governing physics. Measurements rely on optical absorption, scattering and fluorescence; particle/tracer advection; acoustic travel time and backscatter; stereo triangulation; pressure and pore-pressure transduction; surface reflection/refraction; sampling bandwidth; spatial averaging; and inversion from signals to physical variables.

Dimensionless parameters. Controls include pixels or probe diameter relative to feature scale, sampling frequency relative to wave/impact bandwidth, particle Stokes number, image displacement/interrogation window, signal-to-noise ratio, acoustic wavelength and concentration, optical depth, KC number, sensor spacing/wavelength, and uncertainty/response magnitude.

Major equations. Core formulations include camera projection and stereo triangulation, PIV/PTV cross-correlation and particle displacement, acoustic Doppler/range inversion, Beer–Lambert and radiative transfer, Exner sediment continuity, sensor transfer functions, sampling/Nyquist criteria, calibration regression, error propagation and repeatability statistics.

Typical methods. Methods include capacitance/resistance or ultrasonic surface probes, cameras and photogrammetry, stereo laser-induced fluorescence, PIV/PTV, acoustic Doppler and multibeam mapping, radar and laser ranging, pressure/load/pore-pressure sensors, optical spectroscopy/fluorescence, repeated calibration, hybrid synchronization and numerical cross-validation.

Numerical models. Analysis tools include PIV-style cross-correlation applied to acoustic bed maps, Exner conversion to bedload, stereo reconstruction, image processing, optical inversion, uncertainty and repeatability analysis, and CFD used as a spatially complete comparator for PIV measurements.

Experimental datasets. Accessible evidence includes laboratory validation of acoustic mapping velocimetry, repeatable plunging-wave impacts with global video and local stereo-LIF, embankment breach instrumentation surveys, three-dimensional partial dam-break measurement reviews, and wave–current seawater-intake PIV.

Validated ranges. Reported examples resolve impact-location variation near 0.5% of wave height—larger than a typical pressure-sensor diameter—with about 25% pressure consequence; methods span instantaneous wave/current fields, high-speed impact, repeated bed maps, three-dimensional breach morphology and coupled wave–current intake flow.

Recent advances. Recent advances use stereo-planar fluorescence to expose sub-probe impact variability, hybrid imaging/ranging for three-dimensional breaches, custom fluorescent tracers, acoustic map velocimetry, image-based morphology, and combined PIV-CFD classification of wave–current structure flow.

Disagreements. Point sensors may have adequate temporal but inadequate spatial resolution; imaging is spatially rich but fails under occlusion, aeration or poor seeding; acoustic methods penetrate turbidity but depend on scattering assumptions; numerical agreement cannot replace calibration; and inferred flux compounds errors from geometry and velocity.

Limitations. The coastal-specific indexed evidence is sparse; transparent flumes, controlled lighting and seeded flows simplify reality; air bubbles and sediment obscure optics; acoustic backscatter is concentration- and grain-dependent; probes disturb flow; impact bandwidth and sensor area filter peaks; uncertainty reporting is inconsistent.

Open questions. Needs include volumetric free-surface/velocity imaging, robust measurements in aerated and sediment-laden breakers, high-bandwidth distributed pressure, mobile-bed co-registration, nonintrusive force inference, sensor-fusion uncertainty, automated calibration/QA, facility-scale metadata and reproducible raw-data standards.

Seminal papers. The branch connects established point and optical ocean sensors to PIV/PTV, acoustic mapping and Doppler methods, photogrammetry, stereo fluorescence for impact surfaces, integrated breach instrumentation, and synchronized experiment–CFD validation.

Claims

1. Optical inversion using radiance, absorption, scattering, fluorescence, flow cytometry, and spectroscopy can recover diverse physical and biogeochemical water properties, but specificity depends on calibration and material isolation. *Regime: Optical tools for ocean monitoring and research* review statement, review [980]
2. Acoustic Mapping Velocimetry combines repeated multibeam bed maps with image-velocimetry algorithms and Exner continuity to estimate spatially distributed multidirectional bedload. *Regime: Acoustic mapping velocimetry* direct finding, experimental [1003]
3. Stereo-planar laser-induced fluorescence resolved impacting-wave tip variation smaller than ordinary probe spacing: location shifted by about 0.5% of wave height and caused roughly 25% pressure variation. *Regime: Experimental investigation of wave tip variability of impacting waves* direct finding, experimental [948]
4. Breaching experiments require coordinated morphology, depth, velocity, discharge, and pore-pressure measurements using photography/photogrammetry, lasers, PIV/PTV, acoustic, radar, and electronic sensors. *Regime: Overtopping-Induced Embankment Breaching Experiments: State-of-the-Art Review on Measurement and Instrumentation* review statement, review [488]
5. Three-dimensional partial dam-break experiments benefit from hybrid PTV and ultrasonic surface measurements because no single technique captures breach, wave, and urban flow at adequate resolution. *Regime: Experimental Modeling of Three-Dimensional (3D) Partial Dam-Break Flows: A Review* review statement, review [954]
6. PIV and numerical modelling jointly resolved wave–current flow around seawater intakes, including geometry-dependent contractions and instability that sharpened with Keulegan–Carpenter number. *Regime: Hydrodynamic Performance of Seawater Intake Structures Through Numerical Modelling and Particle Image Velocimetry* direct finding, mixed [443]
7. Flume comparisons against laser Doppler velocimetry quantify acoustic Doppler velocimeter accuracy and noise for turbulence measurements. *Regime: Evaluation of the Acoustic Doppler Velocimeter (ADV) for Turbulence Measurements** direct finding, experimental [1464]

32.5 Physical-model similitude

Topic Laboratory. similitude. Scaling laws and scale effects.

Papers screened: 11. Claims: 11. Equations: 0.

Well established. Coastal hydraulic models normally preserve Froude similarity because gravity controls free-surface motion, but using the same fluid makes simultaneous Reynolds and Weber similarity impossible; viscosity, surface tension, air, turbulence, roughness, impulse duration, sediment and soil can therefore cause scale effects.

Governing physics. Dynamic similitude requires matching ratios of inertia to gravity, viscosity, surface tension, elasticity, pressure and body force. Vortices and boundary layers amplify Reynolds mismatch; bubbles, droplets and cushioning add density/viscosity/surface-tension scales; impact peaks depend on compressibility, air and sensor bandwidth.

Dimensionless parameters. Key controls are Froude, Reynolds and Weber numbers; relative roughness and depth; density and viscosity ratios; air-content and void-fraction scales; wave steepness; impulse duration relative to structural period and sampling response; Shields and fall-velocity ratios; and centrifuge g level.

Major equations. Core relations include $Fr=U/\sqrt{gL}$, $Re=UL/\nu$, $We=\rho U^2 L/\sigma$, geometric scale λL , Froude time $\lambda T=\sqrt{\lambda L}$, velocity $\sqrt{\lambda L}$, force $\rho g \lambda L^3$, pressure $\rho g \lambda L$, impulse force $\times$ time, and centrifuge stress scaling using elevated effective gravity.

Typical methods. Methods include paired or multiple geometric scales, prototype/large-scale comparison, repeated-test uncertainty, dimensional analysis, Froude-scaled flumes, aerated-flow probes, high-bandwidth pressure sensing, optical velocity and bore classification, LES scale expansion, and geotechnical centrifuges.

Numerical models. Numerical support includes LES comparisons across distorted and undistorted scales, RANS validation against repeated large-scale experiments, lubrication/inviscid air-cushioning regimes, and analytical conversions from model displacement, pressure and velocity to prototype.

Experimental datasets. Evidence includes dry/wet tsunami-bore generation, large-scale overtopping and wall impact, repeated shallow-foreshore dike tests, a 1:30 clifftop boulder, a 1:40 tsunami-soil centrifuge, 3 $\times$ and 10 $\times$ shallow-flow scaling, and 1:38 downfall-pressure tests.

Validated ranges. Reported evidence includes an air-cushioning regime transition near Reynolds number 10^7 , 1:30 boulder displacements scaling to 0.15–1.3 m, a 1:40 centrifuge representing $9.6\times 21\times 14.6$ m, 3 $\times$ and 10 $\times$ shallow-flow models, roughly two-thirds quasi-static bore impacts, and downfall pressures of 30–40pgH.

Recent advances. Recent advances use centrifuges to preserve fluid–soil stress, LES to localize scale bias to recirculation, repeated large-scale datasets for uncertainty-aware CFD validation, high-bandwidth downfall pressures, and physically resolved air–water upscaling reviews.

Disagreements. Froude similarity is appropriate for gravity waves but insufficient for vortical, viscous, capillary, aerated, porous or sediment processes. Large scale reduces some bias but does not guarantee correct impact physics; numeric agreement can depend on reproducing precursor bore interactions rather than only peak conditions.

Limitations. The journal evidence base explicitly comparing multiple coastal scales is sparse; many studies state a scale but do not test scale effects. Prototype data, air and compressibility scaling, roughness, sediment and biological-material similitude, sensor dynamics, and uncertainty propagation remain incomplete.

Open questions. Needs include defensible Reynolds/Weber thresholds, aeration and compressibility correction, impulsive-pressure bandwidth, roughness and porous scaling, sediment and morphology distortion, flexible vegetation and debris similitude, multi-hazard soil response, and quantified prototype uncertainty.

Seminal papers. The branch connects classical Froude/Reynolds/Weber dimensional analysis to aerated-flow regime studies, large-scale repeated experiments, centrifuge dynamic similitude, numerical scale expansion, and explicit pressure/impulse diagnostics.

Claims

1. Aerated free-surface flows require measurement and modelling of bubbles, droplets, turbulence, and surface-tension breakup, making direct prototype transfer difficult. *Regime: Hydraulics of aerated flows: <i>qui pro quo</i>?* review statement, review [248]
2. Air cushioning changes regime near a wall: lubrication coupling applies below a global Reynolds number of order 10^7 for air–water, transitions near that scale, and becomes formally inviscid above it. *Regime: Air cushioning with a lubrication/inviscid balance* direct finding, analytical [1315]
3. A vertical-release generator reproduced classical dry-bed surges and wet-bed bores while preserving Froude number, momentum, front celerity, and velocity-profile behavior needed for tsunami-load tests. *Regime: Experimental Study of Tsunami-Like Waves Generated with a Vertical Release Technique on Dry and Wet Beds* direct finding, experimental [1539]
4. Prototype prediction of aerated high-Froude flows remains limited by instrumentation, multiphase modelling, and upscaling from controlled hydraulic experiments. *Regime: Air–water flows* review statement, review [1439]
5. Large-scale overtopping tests distinguished five bore-interaction patterns and three wall-impact types; roughly two-thirds of impacts were quasi-static. *Regime: Classification of bore patterns induced by storm waves overtopping a dike crest and their impact types on dike mounted vertical walls – a large-scale model study* direct finding, experimental [1360]
6. Repeated large-scale dike tests quantified experimental uncertainty and showed that accurate force and pressure prediction depends on reproducing the preceding bore interactions. *Regime: Validation of RANS Modelling for Wave Interactions with Sea Dikes on Shallow Foreshores Using a Large-Scale Experimental Dataset* direct finding, mixed [545]
7. A 1:30 Froude-scaled clifftop-boulder experiment produced prototype-equivalent displacement of 0.15–1.3 m, with maximum displacement requiring both high pressure and long impact duration. *Regime: Breaking-wave induced pressure and acceleration on a clifftop boulder* direct finding, experimental [1350]
8. A 1:40 geotechnical centrifuge reduced fluid–soil dynamic-similitude mismatch and produced prototype-identical pressure and velocity for a 9.6 m deep, 21 m long, 14.6 m wide soil field. *Regime: Simulating Tsunami Inundation and Soil Response in a Large Centrifuge* direct finding, experimental [417]
9. Froude-scaled 3 $\times$ and 10 $\times$ models could not simultaneously match Reynolds and Weber numbers; scale effects were strong in recirculating vortices and energy loss but weak in upstream non-vortical velocity. *Regime: Scale Effects Investigation in Physical Modeling of Recirculating Shallow Flow Using Large Eddy Simulation Technique* direct finding, numerical [1400]
10. At 1:38 scale, violent seawall downfall impacts regularly reached 30–40pgH and showed shorter impulse duration at higher pressure, exposing a scaling-sensitive impulsive-load regime. *Regime: Shore-Side Downfall Pressures Due to Waves Impacting a Vertical Seawall: An Experimental Study* direct finding, experimental [104]
11. Froude-scaled laboratory models of subaerial landslide impulse waves underpredict relative wave amplitude when water depth is too small because surface tension alters impact-crater and air-entrainment dynamics while viscosity and dispersion increase propagation damping. *Regime: Two-dimensional granular subaerial landslides entering water and generating nonlinear intermediate-depth impulse waves in Froude-scaled laboratory channels.* direct finding, experimental [583]

33 Wiki: Machine learning and data science

33.1 Machine learning and data science: overview

Topic *mL*. Data-driven coastal analysis and prediction.

Papers screened: 10. Claims: 7. Equations: 0.

Well established. Coastal machine learning learns mappings from observations, simulations or hybrid evidence, but credible use requires a decision-defined target, traceable data, leakage-safe validation, strong baselines, physical and tail checks, calibrated uncertainty, transfer tests, interpretable diagnostics and operational monitoring.

Governing physics. Useful predictors encode waves, tides, surge, rainfall, winds, bathymetry, morphology, structures and antecedent state; conservation, causality, boundary conditions, temporal memory, spatial dependence and extremes constrain learned relationships even when not explicit in the architecture.

Dimensionless parameters. Relevant controls include training-to-feature ratio, event and site coverage, class imbalance, forecast lead-to-memory ratio, spatial resolution to process scale, extrapolation distance, signal-to-noise, missingness, calibration and sharpness, tail sample size, computational speedup, and physics-residual-to-data-loss weighting.

Major equations. Core formulations include empirical-risk loss with regularization, sequence and spatiotemporal mappings, ensemble aggregation, Gaussian or quantile predictive distributions, surrogate emulation, physics- or constraint-penalized loss, calibration scores, feature attribution and skill metrics against persistence, climatology and process models.

Typical methods. Methods curate and align observations or simulations, define features and targets, partition by event/site/time, train linear/tree/kernel/ensemble/deep models, tune without test leakage, compare baselines, diagnose features and errors, calibrate probabilities, test physics and extremes, quantify shift and deploy with monitoring.

Numerical models. Represented approaches include deep surge predictors, seq2seq LSTM flood surrogates, ensemble runup/erosion models, simulation-to-ML inundation emulators, scientific-knowledge-guided learning and model-agnostic feature explanations; process models provide training, constraints or benchmarks.

Experimental datasets. The reviewed evidence includes coastal surge time series, simulation-derived tidal-pluvial flood maps, wave-runup and dune-erosion observations, coastal-urban nuisance-flood sequences and environmental datasets used to illustrate model-agnostic explanation and scientific-knowledge integration.

Validated ranges. Evidence spans site-specific coastal water levels, compound street-scale flooding and beach runup/erosion, primarily hindcast or simulation-supported settings; transfer beyond sampled sites, event magnitudes, morphology, forcing statistics and sensor systems remains unproven without explicit tests.

Recent advances. Recent work uses sequence surrogates for real-time compound flooding, deep coastal surge models, ensembles for runup and erosion, scientific-knowledge integration, probabilistic prediction, model-agnostic explanation, simulation-informed learning and operationally oriented acceleration.

Disagreements. Higher aggregate accuracy can hide worse extreme or minority-regime performance; random splits overstate skill when events or locations leak; explanations can be unstable or noncausal; physics constraints may improve plausibility but introduce model-form bias; a fast surrogate may reproduce its simulator rather than reality.

Limitations. Coastal datasets are short, nonstationary, spatially clustered, missing during extremes and shaped by changing sensors and interventions; labels and simulator outputs contain error; rare tails are underrepresented; black-box dependence, distribution shift and false confidence impede safety-critical use.

Open questions. Priorities include cross-site and future-climate transfer, physics-conserving architectures, sparse-event and tail learning, causal and invariant representations, calibrated compound uncertainty, multimodal data fusion, trustworthy explanations, adaptive sensing, drift detection and auditable human-AI decisions.

Seminal papers. Early coastal data-driven work used regression, neural networks, classification and autoregressive models for waves, shorelines and morphology; ensembles, kernel methods and remote-sensing classifiers broadened nonlinear prediction before modern deep spatiotemporal and hybrid models.

Claims

1. A review argues that engineering and environmental machine learning should integrate scientific laws, constraints, simulations, and domain knowledge rather than rely on unconstrained pattern fitting. *Regime: Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems* review statement, review [1513]
2. Deep-learning experiments for coastal surge prediction show that data-driven models can learn sea-level response while remaining dependent on training coverage, predictors, lead time, and site regime. *Regime: Exploring deep learning capabilities for surge predictions in coastal areas* direct finding, numerical [1409]
3. A machine-learning surrogate predicts combined tidal and pluvial flood inundation at operational speed, transferring information from computational simulations into street-scale coastal-flood estimates. *Regime: Predicting combined tidal and pluvial flood inundation using a machine learning surrogate model* direct finding, numerical [1575]
4. An ensemble machine-learning analysis of wave runup and coastal dune erosion demonstrates that combining learners can characterize nonlinear response while exposing data and extrapolation limits. *Regime: Ensemble models from machine learning: an example of wave runup and coastal dune erosion* direct finding, mixed [147]

5. An environmental-data example shows how model-agnostic explanations can diagnose feature influence in otherwise opaque predictors, while explanation is not a substitute for uncertainty or causal validation. *Regime: An illustration of model agnostic explainability methods applied to environmental data* direct finding, analytical [1512]
6. A sequence-to-sequence LSTM surrogate provides multi-step street-scale nuisance-flood forecasts for a coastal-urban system under joint rain and tide forcing. *Regime: Forecasting Multi-Step-Ahead Street-Scale Nuisance Flooding using a seq2seq LSTM Surrogate Model for Real-Time Application in a Coastal-Urban City* direct finding, numerical [1216]
7. A review of machine learning in coastal sediment transport and morphodynamics finds applications from grain-scale flux to bars, shorelines and overwash, with value in nonlinear regression, emulation, prediction and explicit uncertainty, but recommends transparent train-test separation, baselines, uncertainty and physically informed interpretation. *Regime: Published coastal sediment-transport and morphodynamic ML applications available to the review.* review statement, review [512]

33.2 Feature detection and classification

Topic *ml.classification*. Automated extraction from observations.

Papers screened: 20. Claims: 14. Equations: 1.

Well established. Coastal feature classification is credible only when the target class and annotation protocol are explicit, evaluation is separated from training, and metrics address class imbalance and boundary quality; high within-dataset accuracy alone does not prove cross-site operational skill.

Governing physics. Models infer physical or ecological classes indirectly from spectral reflectance, color, texture, shape, elevation, echo behavior and temporal persistence; tides, turbidity, wetness, waves, illumination, season, sensor geometry and disturbance alter those observable signatures.

Dimensionless parameters. Useful normalized controls include IoU and Dice/F1, precision and recall, balanced accuracy, kappa, class prevalence, boundary tolerance relative to pixel size, train/test spatial separation, resolution-to-feature ratio, confidence and calibration error.

Major equations. Core formulations include supervised loss such as cross-entropy, class-weighted or overlap loss; SVM margins; clustering distances; object rules; and confusion-matrix metrics including accuracy, balanced accuracy, precision, recall, F1 and intersection-over-union.

Typical methods. Workflows define classes and use cases, acquire and co-register imagery or point clouds, label representative samples, engineer or learn features, split data spatially and temporally, train and tune models, compare baselines, audit errors and uncertainty, and validate on independent sites or surveys.

Numerical models. Methods include SVM, Otsu and Iso Cluster baselines, GEOBIA rules, U-Net with attention, ResNet and other pretrained CNNs, bagged handcrafted features, semantic segmentation, repeated point-cloud clustering and multi-resolution sensor fusion.

Experimental datasets. The reviewed evidence spans Landsat seagrass time series, Sentinel and UAV shoreline data, Florida UAS intertidal habitats, Togolese NDWI shorelines, South China Sea beachrock imagery, underwater benthic images from six Australia–Japan sites, and Polish topo-bathymetric LiDAR groynes.

Validated ranges. Reported tests range from one site and sensor to six study areas and a 31-year satellite record; performance values are conditional on the stated labels, resolution, preprocessing and split, and do not establish transfer to different coast types or acquisition conditions.

Recent advances. Recent advances combine attention U-Nets, pretrained multi-architecture comparison, satellite–UAV–GNSS fusion, long time-series deep classification, underwater habitat automation and topo-bathymetric LiDAR clustering for structures across the air–water interface.

Disagreements. Deep learned features can outperform fixed thresholds and handcrafted rules when labels are adequate, while object rules and classical classifiers can be more interpretable and data efficient. Pixel accuracy may conflict with boundary or minority-class performance, and apparent temporal change may reflect domain shift.

Limitations. Common weaknesses are small or correlated training sets, expensive and inconsistent labels, rare classes, unclear train/test leakage, cloud and water-column effects, sensor and seasonal domain shift, weak uncertainty calibration, coarse pixels, ambiguous boundaries and limited independent-site validation.

Open questions. Priorities include transferable labels, cross-sensor and cross-site adaptation, few-shot rare-class learning, physics-aware features, temporal consistency, boundary uncertainty, open benchmarks, reproducible annotation, calibrated confidence and operational human review.

Seminal papers. Remote-sensing classification developed from thresholding, clustering and maximum-likelihood methods through SVM and object-based image analysis; convolutional networks, U-Net semantic segmentation and transfer learning later enabled end-to-end coastal feature extraction.

Equations

- *Explained variance of spectral-type clustering* [1210]

$$EV = 1 - \frac{SSE}{SSE_T}$$

Regime: PCA-reduced K-means clustering of hourly frequency-direction spectra at each of seven Africa-basin control points. Variables: EV proportion of historical spectral variance explained by cluster centroids; SSE within-cluster sum of squared deviations from representative centroids; SSE_T total sum of squared errors of the historical dataset

Claims

1. The selected 260-tree ensemble classified Minor, Moderate and Major dune-damage observations with reported accuracies of 92%, 84% and 98%, respectively. *Regime: The compiled, imbalanced New Jersey dataset and the paper's 80/20 random split; performance is not external validation in another region.* direct finding, field [798]
2. A 5 by 5 spectral-type lattice retained more than 70% explained variance while avoiding the one-hour-in-40-years clusters produced by larger lattices. *Regime: K-means/PCA clustering of 40 years of hourly reconstructed spectra at seven Africa-basin control points; 25 types are a study-specific compromise, not a universal optimum.* direct finding, numerical [1210]
3. A random forest combining improved coastal altimetry with ten-year simulations correctly identified 93% of Gulf of Lion slope-current intrusions when evaluated against independent ADCP observations. *Regime: The Gulf of Lion, its retracked altimetry product, training simulations, and event definition.* direct finding, mixed [235]
4. U-Net and DeepLabv3+ shoreline segmentation on coastal Timex imagery achieved global accuracy 0.98, mean IoU 0.95-0.97, and boundary F1 up to 0.99 within the evaluated dataset. *Regime: The labeled fixed-camera Timex dataset; independent cross-site generalization remains unverified.* direct finding, numerical [1253]
5. A deep-learning analysis of 31 annual Landsat observations estimated stable St. Joseph Bay seagrass extent at 23 ± 3 km² while resolving six short post-cyclone declines and recovery. *Regime: Temporal Stability of Seagrass Extent, Leaf Area, and Carbon Storage in St. Joseph Bay, Florida: a Semi-automated Remote Sensing Analysis* direct finding, field [782]
6. A focused review identifies Landsat and Sentinel as core open shoreline records, U-Net/CNN segmentation as emerging automation, and UAV/GNSS data as local validation while highlighting cloud, fusion, and cross-coast generalization limits. *Regime: A Review of Open Remote Sensing Data with GIS, AI, and UAV Support for Shoreline Detection and Coastal Erosion Monitoring* review statement, review [281]
7. UAS orthomosaics and surface models classified mudflat, salt-marsh, and oyster-reef habitat with a repeatable GEOBIA ruleset and 79% overall accuracy despite weak class separability. *Regime: Quantifying Intertidal Habitat Relative Coverage in a Florida Estuary Using UAS Imagery and GEOBIA* direct finding, field [415]
8. Comparison of Otsu, unsupervised Iso Cluster, and supervised SVM on NDWI imagery found SVM more effective for shoreline extraction across linear and nonlinear Togolese coasts. *Regime: Coastline Change Modelling Induced by Climate Change Using Geospatial Techniques in Togo (West Africa)* direct finding, field [733]
9. A structured survey organizes deep-learning architectures, image sources, data limitations, and context-dependent tradeoffs for automated seagrass detection and classification. *Regime: A Survey of Deep Learning Approaches for the Monitoring and Classification of Seagrass* review statement, review [1013]
10. Attention-enhanced U-Net segmentation of UAV beachrock imagery reached 97.47% accuracy and 88.65% IoU, improving IoU by 2.09 percentage points over baseline U-Net. *Regime: The Identification of Exposed Beachrocks on South China Sea Islands Based on UAV Images* direct finding, field [859]
11. A six-area Australia–Japan evaluation combines pretrained CNNs, bagged features, color and texture for benthic classification and finds ResNet-50 strongest for both classification and semantic segmentation. *Regime: An Automated Framework for Benthic Habitat Classification and Segmentation Based on Deep Learning Algorithms* direct finding, field [972]
12. Repeated clustering with topo-bathymetric LiDAR echo ratio automatically extracted Polish coastal groynes with 96% balanced accuracy against manual annotations. *Regime: Automated classification of coastal defense structures using airborne bathymetric LiDAR* direct finding, field [1186]
13. Image-feature extraction and machine learning automate taxonomic classification of phytoplankton recorded by a submersible imaging flow cytometer. *Regime: Automated taxonomic classification of phytoplankton sampled with imaging-in-flow cytometry* direct finding, mixed [1324]
14. Random forest, Cubist, and support-vector regression estimate chlorophyll-a and suspended particulate matter from GOCI imagery for Korean coastal-water monitoring. *Regime: Machine learning approaches to coastal water quality monitoring using GOCI satellite data* direct finding, numerical [713]

33.3 Data-driven forecasting

Topic *mL . forecasting. Prediction of waves, levels, and morphology.*

Papers screened: 3. Claims: 1. Equations: 0.

Well established. Data-driven coastal forecasting maps recent observations and predicted forcing to future hazards or morphology; credible use requires temporal holdout validation, comparison with persistence and physics baselines, calibrated uncertainty and monitoring for regime shift.

Governing physics. Forecast skill is constrained by the physical memory and drivers represented in inputs: waves, water levels, storm duration, antecedent profile, tides and currents. Models may learn associations but do not by themselves establish sediment conservation or causal mechanisms.

Dimensionless parameters. Controls include forecast horizon relative to system memory, training length relative to climate variability, RMSE normalized by observed range or measurement error, skill relative to persistence, event imbalance, input lag, regularization strength, calibration error and out-of-distribution distance.

Major equations. Regression and neural forecasts approximate a conditional response y at lead time from lagged states and forcing; training minimizes losses such as mean-square error, while evaluation uses RMSE, bias, correlation and event or threshold scores. Probabilistic variants estimate quantiles or predictive distributions.

Typical methods. Workflows align quality-controlled forcing and response time series, prevent temporal leakage, split by events or years, engineer physically meaningful lags, tune only on training data, compare simple and physics-based baselines, test unseen extremes and sites, quantify uncertainty and archive reproducible preprocessing.

Numerical models. The reviewed branch centers on a multilinear event-erosion forecast with reported shoreline RMSE. Adjacent evidence includes a regularized nonlinear extreme-learning machine and physics-data hybrid concepts; model architecture alone is not evidence of operational value.

Experimental datasets. Reviewed primary evidence covers storm-event shoreline erosion at three monitored locations. Related graph evidence includes Iztuzu storm-duration and energy correlations and multi-sensor 2011 Tohoku waveform forecasting, but these do not directly validate the erosion model.

Validated ranges. The event-erosion RMSE of 3.7-6.4 m applies to three monitored locations and their tested events. Transfer to other beach states, storm climates, sensors, lead times or future climate requires independent validation.

Recent advances. Recent advances use transformers and graph networks, physics-informed and differentiable hybrids, probabilistic deep ensembles, conformal prediction, self-supervised coastal representation learning, online drift detection and operational model–data fusion.

Disagreements. Complex nonlinear learners can capture interactions but may not outperform transparent regression with limited events. Random splits inflate skill in autocorrelated data, while event-wise and future-period tests are stricter; lowest RMSE need not yield calibrated extremes or useful decisions.

Limitations. Small event samples, missing extremes, sensor changes, target uncertainty, leakage, nonstationarity, site dependence, unreported baselines, deterministic outputs and weak interpretability limit trust. Forecasts can fail when nourishment, morphology or forcing leaves the training distribution.

Open questions. Priorities include multi-site transfer, physics-constrained loss functions, calibrated event tails, adaptive updating, causal driver tests, missing-data robustness, forecast-value evaluation, standardized hindcasts and uncertainty triggers for human review or physics-model fallback.

Seminal papers. Autoregressive and transfer-function models established coastal time-series prediction; neural networks and support-vector methods expanded nonlinear mapping, followed by ensemble learning, recurrent networks and hybrid residual correction around numerical models.

Claims

1. The multilinear event-erosion model reported shoreline-change RMSE between 3.7 and 6.4 m across the three monitored locations. *Regime: The paper's site-specific training/validation design for Narrabeen-Collaroy; split details require full text.* direct finding, mixed [629]

33.4 Physics-informed machine learning

Topic *ml.physics*. Learning constrained by governing physics.

Papers screened: 11. Claims: 11. Equations: 0.

Well established. Physics-informed learning embeds governing equations, differentiable solvers, reduced physics, or process-model output into model architecture, loss, training data, or parameter inversion rather than relying on correlations alone.

Governing physics. Accessible coastal examples constrain learning with shallow-water mass and momentum, nonlinear Schrödinger wave evolution, sediment transport, SST space-time PDEs, and low/high-fidelity XBeach dynamics.

Dimensionless parameters. Relevant controls are equation-residual weights, data-to-physics loss balance, spatial and temporal sampling, fidelity ratio, flow and wave regime, roughness, spectral bandwidth, and extrapolation distance; reported abstracts do not establish universal ranges.

Major equations. Common mechanisms are PDE-residual penalties, automatic differentiation through governing solvers, universal differential equations, reduced bases, and learned mappings from lower- to higher-fidelity process solutions.

Typical methods. Methods include PINNs, finite-difference PINNs, physics-guided CNNs/GANs, attention and recurrent networks, differentiable solvers, reduced-basis surrogates, parameter identification, and physics-constrained data assimilation.

Numerical models. Physics backbones include shallow-water solvers, XBeach Surfbeat and nonhydrostatic modes, NLSE wave dynamics, sediment equations, finite-difference residuals, and differentiable universal SWE implementations.

Experimental datasets. The branch uses numerical wave tanks, circular-basin tests, XBeach runup pairs, CFD/finite-element mangrove simulations, daily SST, a real river reach, and Hurricane Irene river-ocean flooding; independent coastal field validation is sparse.

Validated ranges. Reported results include sub-centimetre basin fluctuations, about 6.5-fold FD-PINN acceleration, mangrove RMSE near 10^{-2} for elevation/velocity and 10^{-3} for sediment, fivefold shorter training than FE, and second-scale inference; these are case-specific.

Recent advances. Recent coastal work combines finite-difference residuals, causal weighting, attention/recurrent models, differentiable SWE backbones, learned physical parameters, process-model fidelity transfer, and explicit comparison against data-only models.

Disagreements. Physics constraints do not automatically dominate data-driven models: in compound flooding FD-PINN improved vanilla PINN, yet CNN-LSTM gave the best overall data-driven balance, and fixed NLSE coefficients sometimes degraded wave reconstruction when assumptions failed.

Limitations. Evidence is dominated by simulation-trained and case-specific studies. Sparse field tests, inconsistent baselines, loss-weight sensitivity, optimization difficulty, discontinuities, imperfect equations, uncertainty calibration, and computational accounting limit general claims.

Open questions. Priorities are conservation under extrapolation, noisy and sparse observations, multi-fidelity bias, discontinuities and wetting/drying, uncertainty, interpretable learned closures, transfer across storms and sites, and reproducible operational benchmarks.

Seminal papers. This branch connects PINN residual minimization, reduced-order and universal differential equations, differentiable programming, data assimilation, and multi-fidelity surrogate learning to coastal applications.

Claims

1. Physics-informed machine learning integrates prior physical knowledge with data-driven models to address data scarcity, generalization, and physical plausibility. *Regime: cross-domain PIML literature review statement, review* [955]
2. Storm-surge ML literature identifies hybrid physics-data models as a route toward more trustworthy forecasting while retaining explicit limitations and evaluation needs. *Regime: storm-surge prediction and coastal oceanography review statement, review* [1135]

3. Physics-based reduced-basis learning produced precise and interpretable surrogates in three PDE-governed geoscience examples, offering transferable methodological evidence rather than direct coastal validation. *Regime: PDE-governed geothermal, geodynamic, and hydrologic systems* review statement, review [328]
4. A space-time PDE-guided neural network outperformed finite-difference and several learning baselines for daily sea-surface-temperature prediction. *Regime: daily sea-surface-temperature prediction* direct finding, numerical [1572]
5. An attention/LSTM-enhanced shallow-water PINN improved discontinuity capture and reduced nonphysical oscillations relative to a classical PINN in 1-D and 2-D simulations. *Regime: one- and two-dimensional shallow-water equations* direct finding, numerical [823]
6. A conditional GAN used lower-cost XBeach Surfbeat output as physical information to reconstruct higher-fidelity nonhydrostatic time-dependent wave runup. *Regime: time-dependent coastal wave runup* direct finding, numerical [1008]
7. A polar-coordinate shallow-water PINN reproduced nonstationary wind-driven basin motions and sub-centimetre water-level fluctuations from limited sparse data. *Regime: nonstationary wind-driven motion in a circular basin with sub-centimetre fluctuations* direct finding, numerical [1618]
8. A mangrove hybrid PINN constrained by shallow-water and sediment equations achieved reported RMSE near 10^{-2} for elevation/velocity and 10^{-3} for sediment, with fivefold shorter training than the FE benchmark and second-scale inference. *Regime: Sundarbans mangrove hydro-morphodynamics* direct finding, numerical [425]
9. An NLSE-constrained PINN reconstructed wave fields between sparse gauges, while trainable frequency and wavenumber improved reconstructions when fixed narrow-band coefficients were inadequate. *Regime: sparse deep-water gravity-wave elevation time series in a numerical tank* direct finding, numerical [396]
10. A differentiable shallow-water solver coupled to a neural network supported gradient-based roughness inversion and physics discovery without surrogate pretraining. *Regime: forward, inverse, and real-river flow-resistance modelling* direct finding, mixed [858]
11. For Hurricane Irene river-ocean compound flooding, a finite-difference PINN accelerated vanilla PINN by about 6.5 times while improving accuracy, whereas CNN-LSTM offered the best overall data-driven balance. *Regime: Hurricane Irene compound flooding on a realistic river-ocean interface domain* direct finding, numerical [433]

33.5 Surrogate and reduced-order models

Topic *ML*. surrogates. Fast emulation of process models.

Papers screened: 5. Claims: 2. Equations: 1.

Well established. Coastal surrogates replace repeated high-fidelity process-model runs with a learned input-output map. They can reduce forecast or ensemble cost dramatically, but inherit the training simulator's physics, biases, boundary conditions, and sampled parameter domain.

Governing physics. Physical fidelity enters through training labels and predictor design: tropical-cyclone track history controls surge, while winds, currents, ice and release conditions control spill transport. A surrogate learns these dependencies but does not independently conserve mass or momentum unless constrained.

Dimensionless parameters. There is no universal coastal-surrogate dimensionless group. Relevant physical nondimensionalization should retain storm size, speed and intensity ratios, depth and Froude controls, ice concentration, forecast horizon, spatial resolution, and response normalization; ML controls include sample count and error metrics.

Major equations. Common formulations minimize prediction loss between surrogate output and process-model fields or responses. Reviewed examples combine convolutional temporal encoding, PCA and clustering for spatial surge, and XGBoost or KNN regression for affected area and particle positions.

Typical methods. A defensible workflow defines an application domain, generates a designed physics-model ensemble, partitions training/validation/test cases without leakage, compresses outputs if needed, trains candidate emulators, evaluates held-out and historical cases, and tests extrapolation and uncertainty.

Numerical models. Process generators are high-fidelity storm-surge simulations and coupled GNOME-GLOFS spill transport. Surrogates include C1PKNet (1-D CNN plus PCA/k-means), XGBoost affected-area regression, and KNN particle-location prediction.

Experimental datasets. The reviewed evidence includes 1,031 Chesapeake Bay surge simulations plus observations from Hurricanes Isabel, Irene and Sandy, and 6,800 Straits of Mackinac GNOME-GLOFS spill simulations using 2023-2025 hydrodynamic-ice forcing.

Validated ranges. C1PKNet was evaluated for Chesapeake Bay landfalling and bypassing storms and three historical hurricanes. The spill surrogates cover a hypothetical Mackinac release, 3-24 h horizons and 2023-2025 ice/forcing, with $R^2=0.85-0.93$ for area and $0.70-0.98$ for particle location.

Recent advances. The 2026 ice-aware spill study extends coastal emulation from scalar peak hazard to affected-area and particle-distribution outputs, demonstrating rapid response prediction while quantifying strong seasonal control by ice cover.

Disagreements. High in-domain accuracy does not establish physical validity or extrapolation. A surrogate may outperform a chosen baseline while reproducing simulator bias; random data splits can overstate skill when storms, times, or spatial fields are correlated.

Limitations. Current extracted evidence covers only two application domains and relies primarily on synthetic labels. Rare extremes, changing bathymetry or climate, unseen release/track geometry, uncertain boundaries, distribution shift, wet-dry discontinuities, and spatial mass conservation require explicit testing.

Open questions. Priorities include physics- and conservation-aware architectures, calibrated predictive uncertainty, active learning, multifidelity training, interpretable failure detection, transfer among coastlines, real-time data assimilation, and benchmarks that separate interpolation from genuine extrapolation.

Seminal papers. Within this corpus, the 2021 Chesapeake Bay C1PKNet study establishes a track-time-series surge surrogate validated against both 1,031 simulations and historical hurricanes; it is the core coastal exemplar for regional response emulation.

Equations

- *Supervised coastal process-model surrogate* [1322]

$$\hat{\mathbf{y}} = f_{\theta}(\mathbf{x}); \quad \theta^* = \underset{\theta}{\operatorname{argmin}} \mathcal{L}(f_{\theta}(\mathbf{x}), \mathbf{y}_{process})$$

Regime: Straits of Mackinac training and test ensemble. Variables: $\mathbf{x}$ initial spill and environmental predictor vector; $\mathbf{y}_{hat}$ surrogate affected area or particle-location output; θ trained XGBoost or KNN parameters; $\mathbf{y}_{process}$ GNOME-GLOFS ensemble output; $\mathcal{L}$ training loss; normalized learning formulation

Claims

1. For 6,800 GNOME-GLOFS simulations of hypothetical Straits of Mackinac spills during 2023-2025, an XGBoost area surrogate achieved $R^2=0.85-0.93$ and a k-nearest-neighbor particle-location surrogate achieved $R^2=0.70-0.98$; dense February ice limited 24 h spread to 15 km² versus an April maximum of 46 km². *Regime: Hypothetical Straits of Mackinac spill; 6,800 GNOME-GLOFS simulations with 2023-2025 forcing and 3-24 h horizons.* direct finding, mixed [1322]
2. Wave-GAN, trained on CFD images, predicts three-dimensional nonlinear regular-wave loads and run-up on a fixed vertical cylinder for held-out wave conditions with CFD-comparable accuracy and seconds-scale inference, but remains bounded by its training domain. *Regime: Three-dimensional nonlinear regular waves interacting with the modeled fixed vertical cylinder over the trained and tested wave-condition range.* direct finding, numerical [1101]

34 Wiki: Coastal morphodynamics

34.1 Coastal morphodynamics: overview

Topic morphodynamics. Coupled flow, sediment, and landform evolution.

Papers screened: 17. Claims: 5. Equations: 0.

Well established. Coastal morphology emerges from two-way feedback: waves, currents, tides, wind and gravity move heterogeneous sediment, and the evolving bed redirects those forcings. Bars, bedforms, profiles, dunes and planforms therefore exhibit thresholds, patterns, migration and multiple timescales rather than passive response.

Governing physics. Transport depends on bed shear, wave asymmetry, undertow, suspension, bedload, sorting and slope; divergence of transport changes elevation. Feedback can stabilize a form, create an instability, or trigger a state transition, while sediment supply and boundaries constrain long-term trajectories.

Dimensionless parameters. Controls include Shields parameter, mobility and suspension numbers, Rouse number, relative depth and wave height, surf similarity, sediment-size ratios, bed-slope and transport-gradient scales, bar spacing relative to surf-zone width, morphological acceleration factor and forcing-to-adjustment timescale ratio.

Major equations. The Exner sediment-continuity equation links bed-level change to transport divergence and sources. Hydrodynamic mass, momentum, wave-action or energy equations provide forcing; transport and sorting closures, avalanching, vegetation or aeolian terms complete models. Stability analysis linearizes coupled equations about a base state.

Typical methods. Studies combine repeated profiles, lidar, multibeam, seismic and imagery with wave/current/sediment measurements; infer patterns and budgets; use process-based, behavior-oriented or reduced/appropriate-complexity models; and validate migration, elevation, volume, wavelength and transitions rather than shoreline alone.

Numerical models. Models range from cross-shore wave–current–profile systems and linear instability theory to coastal-tract and hybrid machine-learning/process models. Numerical stability requires conservative sediment updating and controlled coupling; long-horizon models must retain critical states without resolving every short-scale process.

Experimental datasets. Reviewed evidence spans multiweek-to-seasonal sandbar records at Duck, Hasaki and Egmond, three summers of Truc Vert ridge-and-runnel mapping, a five-week Nova Scotia experiment, 2.5 years of monthly foredune lidar, Llobregat multibeam/seismic profiles and laboratory-to-field overwash morphometry.

Validated ranges. Case-specific skill for cross-shore bar profiles ranged 0.50–0.88 across four records; oblique-bar theory predicts hour-to-day growth in intermediate states; overwash scaling spans several orders of magnitude. These ranges describe distinct phenomena and are not interchangeable validation metrics.

Recent advances. Recent advances use dense lidar and multisensor bathymetry, data-driven transport closures, hybrid models, scale-invariant morphology, coastal-tract systems and explicit appropriate-complexity criteria to bridge event processes and management horizons.

Disagreements. Process-rich models may reproduce events but accumulate calibration and forcing error, while reduced models reach management timescales but can omit state changes. A model is not preferable merely because it is more detailed; adequacy depends on the behavior, scale and decision being tested.

Limitations. Sparse long records, uncertain sediment supply and grain mixtures, three-dimensionality, storm intermittency, ecological feedback, unresolved small scales, numerical diffusion, parameter equifinality and boundary change limit prediction. Local fit does not prove transfer or century-scale trajectory.

Open questions. Priorities include state-transition predictability, mixed-sediment sorting, river-to-coast supply, dune–beach–bar coupling, storm recovery, ecological feedbacks, calibrated acceleration, hybrid-model interpretability and uncertainty-aware long-term ensembles.

Seminal papers. Sediment continuity and energetics established profile and shoreline response; stability theory explained rhythmic bars and bedforms as feedback instabilities. Process-based coastal models then coupled waves, currents and transport, followed by behavior-oriented and systems approaches for longer scales.

Claims

1. Coastal morphodynamic prediction should use only the process and system complexity needed to resolve behavior and critical transitions at the management scale, with assumptions tested against observations. *Regime: Decadal-to-centennial coastal and estuarine management questions.* review statement, review [459]
2. A review argues local coastal adaptation requires a new generation of multiscale probabilistic change models integrating mean sea level, surge, waves and river flow because existing models do not supply reliable sub-10-km projections. *Regime: On the need for a new generation of coastal change models for the 21st century* review statement, review [1158]
3. Hydrological records and satellite imagery show that the Yellow River Delta gained 248 square kilometres and 36.45 km of coastline from 1983-2011 despite declining runoff and sediment load, with growth, retreat and recovery phases shaped by river regulation, sediment supply and lobe switching. *Regime: Yellow River Delta planform and shoreline evolution from 1983 through 2011.* direct finding, field [731]
4. Northern Adriatic shelf evidence and BarSim modeling show that barrier overstepping probability during rapid transgression decreases with tidal amplitude: preserved systems formed near 60 and 10 mm/yr sea-level rise, while backbarrier accommodation and antecedent topography controlled failure or drowning. *Regime: Two preserved northern Adriatic barrier-lagoon systems during 15-8 ka BP rapid transgression.* direct finding, mixed [1357]

5. Morphodynamic equilibrium must be specified by Exner sediment-mass conservation and analysis scale: static and two dynamic forms can be mathematically distinct in models, whereas variable forcing and landscape setting often leave real coasts and estuaries only statistical or quasi-equilibrium. *Regime: Coastal, estuarine and river morphodynamics evaluated at an explicitly selected spatial and temporal scale.* review statement, review [1614]

34.2 Nearshore bars

Topic morphodynamics. bars. Bar formation and migration.

Papers screened: 17. Claims: 14. Equations: 7.

Well established. Nearshore bars are dynamic sediment reservoirs whose growth, decay and migration emerge from feedback among wave breaking, undertow, nonlinear wave-driven onshore transport, sediment availability and evolving bathymetry; they need not remain coupled to a fixed break point.

Governing physics. Energetic waves and undertow commonly drive bar growth and offshore migration, while milder asymmetric waves favor onshore recovery. Settling, suspension, bedload, wave groups, infragravity motions, profile slope, depth-dependent forcing and morphology-controlled breaking modify this balance.

Dimensionless parameters. Important controls include H/h breaking ratio, Shields mobility, Dean or fall-velocity parameter, relative bar depth and height, bar concentration, profile slope, nourishment concentration and relative placement, relative lifetime L_n/T_r , transport asymmetry, and normalized model errors.

Major equations. Core formulations combine sediment continuity (Exner), wave action and breaking, transport closures partitioning onshore wave-asymmetry and offshore undertow effects, equilibrium-bar volume or position laws, breakpoint criteria based on wave-height-to-depth ratio, and PCA/EOF decompositions for migration cycles.

Typical methods. Methods include repeated bathymetric profiles, Argus or remote sensing, wave-current arrays, large mobile-bed flumes, sediment-flux reconstruction, morphometric tracking, process-based modeling, equilibrium reduced models, PCA/EOF separation, and multi-site hindcast validation.

Numerical models. Models include cross-shore transport-balance calculations, COAST2D with overtopping, an equilibrium two-bar volume model, PCA nourishment-lifetime separation, breakpoint diagnostics, and nonhydrostatic XBeach variants whose simplified bar-crest breaking can bias landward waves and morphology.

Experimental datasets. Evidence includes two breakpoint field experiments, three CIEM large-scale erosive-accretive datasets, a 15 x 30 m overtopping basin, ten years of 12 km Terschelling bathymetry, sixteen Dutch profile datasets with 21 nourishments, and Duck plus two feeder-mound sites.

Validated ranges. Explicit regimes include a 2 million m³ nourishment and 6-7 year migration arrest; a 1:15-style large-flume class with three datasets; 21 nourishments and bar-cycle periods of 1-15 years; Sea Palling's nine breakwaters; and outer/inner-bar errors $\epsilon=0.39/0.51$ and $NMSE=0.24/0.29$.

Recent advances. Recent work separates nourishment interference from natural cycles across many sites and uses fast equilibrium models for multi-bar and feeder-mound evolution, while explicit model comparisons identify why transient inner bars and landward-of-crest hydrodynamics remain less predictable.

Disagreements. Field evidence rejects universal fixed-breakpoint coupling. Erosive offshore evolution follows a comparatively consistent height-location relation, whereas accretive onshore migration may decay or persist. Individual nourishment variables do not explain lifetime, but joint design-and-system variables have moderate skill.

Limitations. Bars are three-dimensional, path dependent and site specific. Annual profiles alias events, laboratory recovery evidence is sparse, PCA can mix modes near inlets or trends, equilibrium models simplify inner bars, and depth-averaged wave models may misrepresent crest breaking and turbulence.

Open questions. Priorities include prediction of inner-bar and accretive recovery, uncertainty propagation from wave boundary layers to decadal migration, three-dimensional rhythmicity, nourishment and structure interactions, climate-driven cycle changes, and consistent validation across event and management scales.

Seminal papers. The 1989 breakpoint test established that bars can evolve landward of initial breaking. The 2005 Terschelling study quantified multi-year nourishment arrest, while later large-scale experiments separated robust erosive behavior from non-unique accretive modes.

Equations

- *break-point hypothesis test* [1243]

$$\frac{\partial z_b}{\partial t} + \frac{1}{1-p} \frac{\partial q_s}{\partial x} = 0$$

Regime: Two field experiments with coincident cross-surf-zone waves and evolving nearshore bars; inner-surf-zone edge diagnosed from wave-height-to-depth ratio. Variables: z_b bed elevation; p bed porosity; q_s cross-shore sediment transport rate; normalized Exner representation

- *Terschelling bar morphometric and transport-balance analysis* [543]

$$\frac{\partial z_b}{\partial t} + \frac{1}{1-p} \frac{\partial q_s}{\partial x} = 0$$

Regime: Terschelling, Netherlands; 12 km alongshore bathymetry for 10 years after a 1993 nourishment placed between middle and outer bars. Variables: z_b bed elevation; p bed porosity; q_s cross-shore sediment transport rate; normalized Exner representation

- *COAST2D overtopping morphodynamic model* [377]

$$\frac{\partial z_b}{\partial t} + \frac{1}{1-p} \frac{\partial q_s}{\partial x} = 0$$

Regime: Validation in a 15 x 30 m basin with a 5 m trapezoidal breakwater on a 1:20 beach; November 2006 storm at Sea Palling with nine segmented breakwaters. Variables: z_b bed elevation; p bed porosity; q_s cross-shore sediment transport rate; normalized Exner representation

- *erosive-accretive breaker-bar evolution parameterization* [398]

$$\frac{\partial z_b}{\partial t} + \frac{1}{1-p} \frac{\partial q_s}{\partial x} = 0$$

Regime: Three CIEM large-scale morphodynamic datasets with more than 20 cross-shore water-level locations per experiment, compared with additional large-scale profile datasets. Variables: z_b bed elevation; p bed porosity; q_s cross-shore sediment transport rate; normalized Exner representation

- *PCA shoreface-nourishment lifetime model* [499]

$$\frac{\partial z_b}{\partial t} + \frac{1}{1-p} \frac{\partial q_s}{\partial x} = 0$$

Regime: Sixteen Dutch profile datasets and 21 nourishments; annual profiles from 1965-2017, typically spaced 200-250 m; global bar-cycle return periods cited as 1-15 years. Variables: z_b bed elevation; p bed porosity; q_s cross-shore sediment transport rate; normalized Exner representation

- *equilibrium-based two-bar volume model* [916]

$$\frac{\partial z_b}{\partial t} + \frac{1}{1-p} \frac{\partial q_s}{\partial x} = 0$$

Regime: Duck, North Carolina two-bar system plus feeder mounds at Silver Strand, California and Cocoa Beach, Florida; years-to-decades scale. Variables: z_b bed elevation; p bed porosity; q_s cross-shore sediment transport rate; normalized Exner representation

- *IBW-PAN multi-bar process framework* [677]

$$\frac{\partial z_b}{\partial t} + \frac{1}{1-p} \frac{\partial q_s}{\partial x} = 0$$

Regime: Natural mildly sloping multiple-bar sandy shore at Lubiato, Poland, with D50 approximately 0.22 mm. Variables: z_b bed elevation; p bed porosity; q_s cross-shore sediment transport rate; normalized Exner representation

Claims

1. After a 2 million m³ Terschelling shoreface nourishment filled the middle-to-outer-bar trough, the monitored 12 km bar system developed pronounced three-dimensionality and its migration stopped for 6-7 years; migration then resumed at the autonomous rate as trough deepening restored the pre-nourishment cross-shore transport distribution. *Regime: Terschelling, Netherlands; 12 km alongshore bathymetry for 10 years after a 1993 nourishment placed between middle and outer bars.* direct finding, mixed [543]
2. In one of two field experiments, a nearshore bar became better developed and migrated offshore while remaining landward of initial outer-surf-zone breaking, demonstrating that bar development and migration are not necessarily coupled to the break point; the second experiment could not exclude break-point influence. *Regime: Two field experiments with coincident cross-surf-zone waves and evolving nearshore bars; inner-surf-zone edge diagnosed from wave-height-to-depth ratio.* direct finding, field [1243]
3. For the mildly sloping, wave-dominated multi-bar coast at Lubiato with D50 approximately 0.22 mm, the reviewed IBW-PAN program shows that physical modeling and field-data analysis provide complementary constraints on wave-current sediment transport and bar evolution rather than interchangeable evidence. *Regime: Natural mildly sloping multiple-bar sandy shore at Lubiato, Poland, with D50 approximately 0.22 mm.* review statement, review [677]
4. Across three large-scale experiments and supporting datasets, breaker-bar height varied linearly with offshore position under erosive waves, whereas subsequent accretive onshore migration followed two distinct modes—decaying or non-decaying—depending on feedback between morphology, wave breaking location, and sediment transport. *Regime: Three CIEM large-scale morphodynamic datasets with more than 20 cross-shore water-level locations per experiment, compared with additional large-scale profile datasets.* direct finding, experimental [398]
5. For laboratory tests on a 1:20 beach and the November 2006 Sea Palling storm around nine segmented breakwaters, COAST2D agreed with measured hydrodynamics and morphological change only when structure overtopping was represented; overtopping materially altered circulation, sediment transport, and predicted bed change. *Regime: Validation in a 15 x 30 m basin with a 5 m trapezoidal breakwater on a 1:20 beach; November 2006 storm at Sea Palling with nine segmented breakwaters.* direct finding, mixed [377]
6. For 21 shoreface nourishments isolated from natural migration using PCA on 16 long-term Dutch profile datasets, no individual design parameter explained lifetime; bar-cycle return period alone gave $r^2=0.41$, while a joint linear model reached $r^2=0.67$ with positive effects of nourishment concentration and depth and negative effects of bar concentration and cycle period. *Regime: Sixteen Dutch profile datasets and 21 nourishments; annual profiles from 1965-2017, typically spaced 200-250 m; global bar-cycle return periods cited as 1-15 years.* direct finding, field [499]

7. In years-to-decades validation at Duck, the equilibrium-based two-bar model predicted outer-bar volume more skillfully ($\epsilon=0.39$, NMSE=0.24) than the more transient inner bar ($\epsilon=0.51$, NMSE=0.29), and reproduced feeder-mound decay at two other sites by assigning artificial bars zero equilibrium volume. *Regime: Duck, North Carolina two-bar system plus feeder mounds at Silver Strand, California and Cocoa Beach, Florida; years-to-decades scale.* direct finding, numerical [916]
8. Across Duck, Hasaki and Egmond cases lasting 10 days to 3.5 months, a coupled cross-shore model achieved profile skill of 0.50–0.88 and reproduced storm-driven offshore and skewness/bedload-driven onshore sandbar migration. *Regime: Modeling cross-shore sandbar behavior on the timescale of weeks* direct finding, mixed [1220]
9. Linear stability analysis indicates sediment heterogeneity damps alternate-bar growth and migration, shortens wavelength, and reproduces qualitative coarse-grain accumulation at bar crests relative to uniform sediment. *Regime: Grain sorting and bar instability* direct finding, mixed [770]
10. Surf-zone feedback under oblique waves can generate up- or down-current rhythmic bars with spacing near to several times surf-zone width and e-folding growth times from hours to days in intermediate beach states. *Regime: Nearshore oblique sand bars* direct finding, mixed [1188]
11. Three summers of Truc Vert mapping describe a four-phase ridge-and-runnel cycle from nearshore-bar formation through shoreward welding and organization, with rhythmic systems migrating southward alongshore. *Regime: Morphodynamics of Ridge and Runnel Systems during Summer* direct finding, mixed [65]
12. During five low-energy weeks, 0.35–0.50 m ridges moved less than 5 m; landward wave transport was balanced by undertow, tidal currents, along-trough transport and ebb drainage-channel export. *Regime: Low-energy Morphodynamics of a Ridge and Runnel System* direct finding, mixed [324]
13. FHvL fusion of hyperspectral imagery, lidar and in-situ spectra classified shallow Sabaudia sandbars using depth, slope, convexity and concavity and supported an operational coastal-data standard. *Regime: Nearshore Sandbar Classification of Sabaudia (Italy) with LiDAR Data: The FHvL Approach* direct finding, mixed [1399]
14. A process-based model reproduces onshore sandbar migration only when velocity and acceleration skewness act jointly; velocity-skewness bed stress dominates shoaling while acceleration-skewness pressure gradients dominate inner-surf-zone transport. *Regime: Onshore sandbar migration in the surf zone: New insights into the wave-induced sediment transport mechanisms* direct finding, mixed [435]

34.3 Bedforms

Topic *morphodynamics. bedforms. Ripples, dunes, and bed roughness.*

Papers screened: 15. Claims: 14. Equations: 0.

Well established. Subaqueous ripples and dunes emerge from sediment-transport feedback under waves, currents, and combined flow. Their geometry, orientation, migration, and state alter hydraulic roughness, turbulence, suspension, and net sediment flux.

Governing physics. Controls include bed shear and Shields stress, oscillatory asymmetry, acceleration skewness, wave-current angle, vortex shedding, phase lag, avalanching, grain sorting and armouring, cohesion and EPS, and lagged adjustment to changing forcing.

Dimensionless parameters. Important groups include Shields parameter, mobility and suspension numbers, wave orbital excursion relative to grain and ripple scale, wave-current angle, velocity and acceleration asymmetry, relative roughness, clay volume fraction, and normalized adjustment time.

Major equations. Core frameworks include Exner sediment continuity, Shields-based mobility, logistic ripple evolution, semi-unsteady half-cycle transport, Euler-Lagrange particle dynamics, Eulerian two-phase RANS, LES boundary layers, and equilibrium-size roughness relations.

Typical methods. Evidence uses oscillatory tunnels and wave flumes, ripple profiling and spectral analysis, dual-station seabed imaging, spring-neap intertidal surveys, regression and phase diagrams, practical formula databases, particle models, two-phase RANS, and LES.

Numerical models. Models range from empirical and logistic closures to grain-resolving Euler-Lagrange, Eulerian two-phase sediment-fluid equations, and coupled far-field/LES calculations that explicitly resolve vortex and suspension processes.

Experimental datasets. Datasets include 226 large-scale transport measurements, mixed-flat spring-neap observations, Sand Engine records at 60–66 m station spacing, four weeks of MEGAPEX imaging, and vortex-ripple and sheet-flow validation experiments.

Validated ranges. Explicit evidence spans 0.13–0.54 mm sands; cohesion shutdown above 2.8 vol% clay and 0.05 wt% EPS; wavelengths from 0.14 to over 2 m; heights 0.02–0.40 m; migration to 3.6 m/h; and flux to 1.7 m³/m/day.

Recent advances. Recent advances quantify rapid field transformation, cohesion thresholds, phase diagrams across near-orthogonal wave-current angles, grain-scale sorting, ripple-resolved two-phase transport, and multiscale flux contributions to sandbar evolution.

Disagreements. Equilibrium geometry can predict roughness in some variable intertidal settings, yet individual bedforms may be out of equilibrium half the time and transform over 20 minutes to 6 hours. Instantaneous equilibrium is therefore a useful closure, not a universal state description.

Limitations. Most evidence is two-dimensional or site-specific; mixtures, biology, three-dimensional defect dynamics, broad grading, storm reversals, instrument resolution, scale effects, and feedback to larger morphology constrain transfer.

Open questions. Priorities are unified transient phase diagrams, mixed-grain and cohesive biology, three-dimensional vortex-bed coupling, bedform-sheet-flow transitions, uncertainty-aware roughness, and upscaling ripple flux to bars and shorelines.

Seminal papers. The branch links classic Shields mobility, Exner continuity, ripple-equilibrium and roughness concepts to transient logistic evolution and modern phase-resolving particle and two-phase models; detailed historical attribution awaits full text.

Claims

1. A semi-unsteady transport formula using bed stress, phase lag, acceleration skewness, streaming, and advection placed 78% of 226 measured rates within a factor two and captured both rippled-bed and sheet-flow trends. *Regime: large oscillatory tunnels and wave flume; 0.13–0.54 mm sand* direct finding, mixed [1]
2. An Euler-Lagrange natural-sand model reproduced vortex-ripple and sheet-flow velocity and concentration and revealed dynamic coarse-over-fine armouring, size-selective pickup, and delayed settling of finer grains. *Regime: asymmetric oscillatory vortex-ripple and sheet-flow conditions with natural sand* direct finding, numerical [441]

3. On a mixed sand-mud intertidal flat, bedform transport decreased with cohesive clay and EPS and became undetectable above 2.8 vol% clay and 0.05 wt% EPS despite that clay level often being classed as clean sand. *Regime: biologically active mixed sand-mud flat with clay below 2 to 5.4 vol%* direct finding, field [829]
4. Wave-ripple growth and transition experiments were represented by one logistic evolution law whose rate depends on Shields parameter, enabling time-dependent bottom-roughness estimation after sea-state changes. *Regime: wave-formed ripples evolving between equilibrium states after wave change* direct finding, mixed [321]
5. A spring-neap field phase diagram integrated wave-, current-, and combined-flow bedforms, and an equilibrium-size equation predicted roughness even though observed forms were in equilibrium only about half the time. *Regime: mixed sand-clay intertidal flat across wave-current angles* direct finding, field [94]
6. A two-phase model matched measured onshore ripple migration by resolving onshore near-bed transport and lee-flank avalanching, while asymmetric ripple vortices drove offshore suspended flux. *Regime: orbital ripples under skewed second-order and symmetric sinusoidal oscillation* direct finding, numerical [1242]
7. At the Sand Engine, bedforms 0.14 to over 2 m long transformed within 20 minutes to 6 hours, and their volume change followed integrated sediment transport over the lagged development interval. *Regime: two Sand Engine locations 60 m apart under waves and currents* direct finding, field [1504]
8. A coupled far-field/LES model accurately predicted mean velocity and suspended sediment over full-scale vortex ripples, with surface roughness exerting a significant control on the result. *Regime: full-scale vortex ripples under oscillatory wave forcing* direct finding, numerical [574]
9. Four-week surf-zone observations found 0.02–0.40 m high ripples migrating up to 3.6 m/h and carrying mean and maximum volume fluxes of 0.22 and 1.7 m³/m/day, sufficient to contribute to larger sandbar evolution. *Regime: 2014 MEGAPEX Sand Engine transect with stations 66 m apart* direct finding, field [1506]
10. Multibeam and seismic evidence indicates Llobregat prodelta undulations on 0.2–3° slopes are sediment waves formed by hyperpynal bottom-current interaction with regional circulation, not progressive slope deformation. *Regime: Sediment undulations on the Llobregat prodelta: Signs of early slope instability or sedimentary bedforms?* direct finding, mixed [1632]
11. A hybrid sorted-bedform model using genetic-programming predictors for suspended-sediment reference concentration and oscillatory ripples captured two observed pattern modes absent from the prior model. *Regime: Data-driven components in a model of inner-shelf sorted bedforms: a new hybrid model* direct finding, mixed [511]
12. Delft3D sensitivity analysis shows that ripple and megaripple roughness choices materially affect modeled hydrodynamics and sediment transport on an ebb-tidal delta. *Regime: From Ripples to Large-Scale Sand Transport: The Effects of Bedform-Related Roughness on Hydrodynamics and Sediment Transport Patterns in Delft3D* direct finding, numerical [184]
13. Morphodynamic stability analysis links increased bed roughness to a shift from dune-scale toward megaripple-scale bed instability on sandy beds. *Regime: Influence of bed roughness on dune and megaripple generation* direct finding, numerical [631]
14. Process-based morphodynamic simulations show that tidal sand-wave recovery is slower after swiping than topping, and that larger dredged volumes lengthen recovery; maintenance intervals also evolve because recovery depends on bedform shape as well as height. *Regime: Fully developed modeled tidal sand waves under topping, swiping, multiple dredging depths, and repeated navigation-channel maintenance.* direct finding, numerical [228]

34.4 Planform evolution

Topic morphodynamics. planform. Two-dimensional coastal morphology.

Papers screened: 8. Claims: 4. Equations: 0. Shown in the Coastal Design Bench beside: Beach nourishment.

Well established. Coastal planforms evolve through gradients and exchanges in alongshore and cross-shore sediment transport, with shoreline orientation, inlets, headlands, structures and nourishment creating coupled cells whose apparent local stability may mask sediment transfer across boundaries.

Governing physics. Oblique breaking waves drive alongshore transport; shoreline curvature changes wave angle and transport gradients, while storms redistribute profile sediment, tidal prisms maintain inlet channels and deltas, and groins or nourishment perturb bypassing and spreading.

Dimensionless parameters. Controls include relative wave angle, diffusivity time normalized by shoreline length squared, nourishment volume relative to active-profile volume, groin length relative to surf-zone width, notch elevation relative to swash, tidal-prism-to-wave forcing ratios and delta extent normalized by inlet width.

Major equations. One-line continuity relates shoreline change to the alongshore gradient of sediment transport plus sources and sinks, scaled by active-profile height. More complete planform budgets integrate volume flux across open boundaries and inlet exchange; equilibrium-shape models relate shoreline orientation to spatial wave transformation.

Typical methods. Studies repeat shoreline and topobathymetric surveys, define sediment control volumes, measure boundary fluxes, track nourishment spreading and storms, analyze structure bypassing, relate inlet geometry to prism and exposure, calibrate one-line or reduced-complexity models and validate both position and volume.

Numerical models. The evidence supports sediment-budget and reduced planform interpretations; direct stability, quasi-three-dimensional bar, estuarine morphodynamic and equilibrium-planform model papers remain in the lawful-full-text queue.

Experimental datasets. Reviewed evidence includes four southern California nourishment sites over years to 16 years, the 1100 m Upham Beach project through hurricane influence, UAS-SfM monitoring of three notched groins at Deal and morphology data for 89 United States tidal inlets.

Validated ranges. Support spans nourished open coasts, one groin field and a national inlet sample. The reported transport pathways and prism relationships depend on wave exposure, structure geometry, survey boundary and event sequence and are not universal coefficients.

Recent advances. Recent advances combine UAS and satellite shoreline/topobathymetry, multi-contour and hybrid 2D/one-line models, unstructured wave coupling, differentiable calibration, ensemble sediment budgets and graph- or neural-accelerated planform solvers.

Disagreements. Shoreline-only models can reproduce orientation change while missing cross-shore or offshore volume loss; equilibrium shapes may be useful for sheltered crenulate bays but fail under time-varying wave climates, sediment sources, complex geology or active inlets.

Limitations. Open sediment boundaries, uncertain closure depth, sparse bathymetry, event aliasing, shoreline-proxy error, unmeasured bypassing, structure burial, geological controls and simultaneous profile change undermine simple planform attribution.

Open questions. Priorities include coupled planform-profile prediction, inlet-beach exchange, time-varying sediment cells, geological constraints, structure permeability, climate-conditioned wave direction, data assimilation, probabilistic nourishment spreading and stable numerics for nonlocal transport.

Seminal papers. Pelnard-Considère shoreline diffusion and one-line continuity established planform modeling; equilibrium-bay formulations treated wave sheltering, while sediment-cell budgets and shoreline contour models expanded sources, sinks, structures and multiple contours.

Claims

1. Ebb-delta seaward and downdrift extents showed moderate-to-high prism correlations for mildly and highly wave-exposed inlet groups, but poor or no correlation for moderately exposed inlets. *Regime: The 89-inlet database; exposure grouping and a proposed prism scale near 1e8 m3 are empirical and not universal.* direct finding, field [1630]
2. Appropriate-complexity geomorphic models should bound systems at the behavior and decision scale, match process detail to the question, anticipate possible dynamics and critical state changes, and assess assumptions against observable behavior. *Regime: Appropriate complexity for the prediction of coastal and estuarine geomorphic behaviour at decadal to centennial scales* direct finding, mixed [459]
3. Monthly lidar over 2.5 years showed two foredunes only 700 m apart diverged: the steeper, sparsely vegetated site retreated and lost volume while the heavily vegetated site prograded and accreted. *Regime: Spatial Variability of Coastal Fore-dune Evolution, Part A: Timescales of Months to Years* direct finding, mixed [194]
4. LX-Shore combines gradients in longshore sediment transport with wave-energy-driven cross-shore response in a reduced-complexity one-line model, allowing complex sandy shoreline geometries, fixed headlands or defenses, and evolution from storm to decadal scales. *Regime: Wave-dominated sandy coasts represented as translating profiles, including non-erodible rocky areas, headlands and coastal structures.* direct finding, numerical [1199]

34.5 Cross-shore profile evolution

Topic morphodynamics. profile. Profile response and equilibrium concepts.

Papers screened: 10. Claims: 6. Equations: 0.

Well established. Cross-shore beach profiles adjust sediment between dune, berm, surf zone and shoreface as wave energy, water level and sediment supply vary; profile and planform evolution interact, so shoreline position alone is not a complete sediment-budget measure.

Governing physics. Wave breaking, undertow, skewed and asymmetric orbital motion, swash infiltration, gravity, avalanching and alongshore divergence govern transport. Storms tend to erode upper profiles and build bars or offshore deposits, while milder forcing can restore berms conditional on sediment availability.

Dimensionless parameters. Controls include wave steepness, relative depth, surf-similarity and Dean-type fall-velocity parameters, Shields and mobility numbers, sediment fall velocity relative to orbital velocity, relative freeboard, normalized profile volume and nourishment grain-size ratio.

Major equations. Models couple wave-action or momentum balances to cross-shore sediment-transport gradients and bed continuity, dz/dt plus dq_x/dx equals zero; equilibrium-profile forms commonly relate depth to offshore distance, while dynamic-equilibrium models relax shoreline or profile state toward forcing-dependent targets.

Typical methods. Studies repeat surveys or video-derived shorelines and profiles, close sediment budgets, separate storm and recovery intervals, calibrate equilibrium or process models, test hindcasts independently, quantify profile and planform change together and examine sensitivity to closure depth, transport formula and boundary exchange.

Numerical models. The set includes a one-parameter dynamic-equilibrium shoreline model based on a bi-parabolic profile, coupled profile-planform interpretation of nourishment evolution and scenario-based mixed-beach defense adjustment; detailed process-based cross-shore solvers remain a review gap.

Experimental datasets. Reviewed evidence includes two years of daily Nova Icaria video observations, the 1100 m Upham Beach nourishment and hurricane response, 87 surveys along 50 transects of Nice gravel beach, and four southeast England mixed sand-gravel defenses under sea-level-rise scenarios.

Validated ranges. Support is site-specific: one Mediterranean sandy beach, one nourished Florida segment, one steep French gravel beach and four English mixed beaches. Performance does not establish universality across sediment sizes, reefs, tidal ranges or storm climates.

Recent advances. Recent advances combine dense video, lidar and satellite profiles with Bayesian calibration, differentiable and surrogate morphodynamic models, phase-resolving swash physics, mixed-grain formulations, probabilistic storm sequences and profile-aware adaptive nourishment.

Disagreements. Equilibrium-profile models offer parsimonious shoreline skill but compress transport mechanisms and offshore loss, whereas process models resolve more physics at greater calibration and computational cost. A stable shoreline can coexist with large internal profile redistribution or sediment leakage.

Limitations. Uncertain closure boundaries, survey error, sparse storm sampling, alongshore gradients, grain-size sorting, groundwater, structures, unresolved swash, offshore loss and equifinal calibration limit inference. Scenario profiles also depend on assumed sediment supply and sea-level trajectory.

Open questions. Priorities include unified profile-planform models, storm-sequence memory, recovery timescales, mixed and gravel sediment sorting, reef-fronted profiles, overwash and dune coupling, climate-conditioned equilibrium, data assimilation and uncertainty-aware nourishment design.

Seminal papers. Bruun-type recession concepts and power-law equilibrium profiles established reduced descriptions; energetics and transport-gradient models introduced dynamic evolution, followed by barred-profile, shoreline-relaxation and process-based wave-current-sediment solvers.

Claims

1. At Nova Icaria beach, a dynamic-equilibrium shoreline model using a theoretical bi-parabolic profile and one calibrated rate parameter showed no loss of skill relative to compared models with four free parameters over two years of daily video monitoring. *Regime: Wave-driven cross-shore response at the monitored Mediterranean beach; longshore gradients and complex fills are omitted.* direct finding, mixed [649]
2. Laboratory and natural overwash morphometrics follow scale-invariant relationships across several orders of magnitude and align with classic drainage and alluvial-fan scaling laws. *Regime: Scaling laws for coastal overwash morphology* direct finding, mixed [778]
3. A sediment-conservation model coupling waves, wave-driven currents and transport used bed-slope updating, oscillation removal and multilevel time discretization to improve stability over complex coastal topography. *Regime: NUMERICAL SOLUTIONS OF COASTAL MORPHODYNAMIC EVOLUTION FOR COMPLEX TOPOGRAPHY* direct finding, mixed [274]
4. A behavior-oriented panel model reasonably hindcast century and millennial Central Holland shoreface evolution and supports faster upper- than lower-shoreface response, but its wave-driven transport coefficients absorb omitted processes and provide only order-of-magnitude transfer beyond the calibration coast. *Regime: Central Holland shoreface hindcasts over engineering and geological time scales.* direct finding, mixed [1352]
5. Lower shorefaces can supply beaches, dunes, estuaries and tidal basins, especially where sediment-rich shelves experience swell-driven onshore transport, but nonlinear bedload and suspended-load interactions, sparse seabed mapping and scale separation make sediment-connectivity magnitude and direction highly uncertain. *Regime: Diverse clastic and carbonate lower shorefaces linking inner shelves to upper shorefaces and beaches.* review statement, review [64]
6. A mesoscale shoreface synthesis defines a temporally variable upper-lower shoreface framework, with depth of closure separating the zones and significant sediment-transport stress defining the seaward limit, and proposes sediment supply and accommodation as organizing controls on diverse two- and three-dimensional morphologies. *Regime: Wave-influenced mobile shorefaces between the fair-weather surf-zone edge or beachface and the seaward limit of significant sediment transport.* review statement, review [567]

35 Wiki: Nature-based coastal protection

35.1 Nature-based coastal protection: overview

Topic nature_based. Risk reduction using natural or hybrid systems.

Papers screened: 19. Claims: 2. Equations: 0.

Well established. Reefs, wetlands, dunes and submerged vegetation can reduce waves, currents, erosion and flood consequences while providing habitat, carbon and other services. Performance is conditional on geometry, elevation, health, forcing and sediment context and may be enhanced by hybrid structural elements.

Governing physics. Canopy and reef drag dissipate wave energy; roughness, breaking, reflection and freeboard control transmission. Vegetation and shellfish alter sediment deposition and soil strength, while growth, recruitment, mortality and accommodation determine whether protective geometry persists under sea-level rise.

Dimensionless parameters. Controls include relative freeboard, submergence, width-to-wavelength ratio, frontal-area density, drag and transmission coefficients, wave steepness, habitat inundation fraction, growth-to-sea-level-rise ratio, sediment accretion balance and benefit-cost or service-value ratios.

Major equations. Design uses wave-energy and momentum balances with vegetation or reef drag, transmission and overtopping relations, sediment budgets and ecological growth or survival constraints. Risk and valuation frameworks combine avoided damage, lifecycle cost, service flows and co-benefits under uncertain performance.

Typical methods. Studies combine flumes and phase-resolving models with before–after/control field designs, wave and morphology monitoring, habitat and recruitment surveys, remote sensing, ecosystem-service valuation and participatory planning. Goal-based monitoring should separately test physical, ecological and social objectives.

Numerical models. Models range from drag and transmission formulations to FUNWAVE-TVD and idealized multi-habitat profiles, shoreline and sediment models, ecosystem-growth projections and spatial ecosystem-service planning. Coupled models must preserve both hydrodynamic and viability constraints.

Experimental datasets. Reviewed evidence includes replicated oyster breakwaters, four-season Bangladesh monitoring, Atlantic/Gulf oyster comparisons, seven years at Gandy's Beach, Belize planning scenarios, 59 European cases, Port Phillip kelp sites and restored-coral canopy flume tests.

Validated ranges. Examples are site-specific: oyster reefs reduced retreat over 40% at one site, Bangladesh erosion 54%, low-inundation reefs attenuated 68% versus about 5% for oyster-suitable reefs, and restored coral canopies dissipated over 50% under typical tested reef-flat conditions.

Recent advances. Recent work uses long-term multidisciplinary monitoring, force-resolved coral-canopy experiments, phase-resolving living-shoreline models, multi-habitat portfolios, adaptive-growth constraints and participatory service optimization rather than protection-only design.

Disagreements. Ecological success does not guarantee engineering success: seven-year oyster reefs met habitat but not protection targets, and the freeboard most effective for attenuation can be unsuitable for oysters. Kelp also failed to outperform barren controls at most tested sites, countering assumptions that any vegetation necessarily protects.

Limitations. Short monitoring, survivorship and maturity uncertainty, scale effects, changing freeboard, storm scarcity, incomplete sediment budgets, inconsistent terminology and metrics, omitted failure modes and weak disclosure of quantitative effectiveness limit transfer and financing.

Open questions. Priorities include lifecycle failure and recovery, adaptive growth under rapid sea-level rise, optimal hybrid geometry, extreme-event performance, sediment feedback, multi-habitat sequencing, standardized monitoring, equitable governance and bankable valuation of co-benefits.

Seminal papers. Early wetland, reef and vegetation studies established attenuation and erosion services; ecosystem-service frameworks connected those functions to risk and value. Living-shoreline and hybrid concepts then integrated habitat restoration with engineered stabilization.

Claims

1. Nature-based coastal protection should be treated as conditional risk-reduction infrastructure whose physical performance, recovery capacity, co-benefits and compatibility with hybrid defenses are evaluated together. *Regime: Coastal ecosystems considered for wave, surge, erosion, and profile-maintenance services.* review statement, review [1330]
2. A synthesis compares effectiveness, costs and coastal-protection benefits of natural and nature-based defenses and identifies evidence and valuation needs for risk-reduction decisions. *Regime: The Effectiveness, Costs and Coastal Protection Benefits of Natural and Nature-Based Defences* review statement, review [1011]

35.2 Dune-based protection

Topic nature_based. dunes. Natural and restored dunes.

Papers screened: 4. Claims: 3. Equations: 0.

Well established. Coastal dunes provide a sacrificial sand reservoir and elevated barrier that can reduce overwash and flooding while supporting habitat; their protective function depends on geometry, sediment supply, vegetation, storm sequence and capacity to recover or be maintained.

Governing physics. Wind transports dry beach sand into dunes where vegetation traps it; waves and surge erode the seaward face, trigger scarping, slumping and overwash, and redistribute sediment offshore or landward. Barrier performance changes sharply when runup exceeds crest or volume is depleted.

Dimensionless parameters. Controls include crest freeboard relative to runup, dune volume normalized by storm erosion demand, beach width relative to runup, sediment fall velocity and grain size, vegetation density or frontal area, overwash ratio, recovery-to-storm timescale and vertical error relative to detected change.

Major equations. Design and assessment combine sediment-volume continuity, aeolian transport relations, wave runup and overtopping formulas, erosion or storm-impact models and probabilistic exceedance. Monitoring derives elevation and volume change from co-registered point clouds and propagates surface-classification error.

Typical methods. Workflows survey beach-dune topography before and after storms, classify bare ground beneath vegetation, validate with RTK checkpoints, calculate profiles and volumes, document planting/fencing/nourishment and access paths, couple wave-level scenarios to erosion and overtopping and monitor recovery and habitat outcomes.

Numerical models. Reviewed methods include a genetic-algorithm ground classifier for vegetated dune LiDAR and paired LiDAR-SfM change analysis. Direct storm-erosion, runup, overwash and nature-based protection modeling remains in the lawful-full-text queue.

Experimental datasets. Reviewed evidence includes Topsail Hill UAV LiDAR and photogrammetry checked against 526 RTK points and eight paired Baltic UAV surveys covering 895 profiles over two years. These datasets validate monitoring, not protective-load reduction.

Validated ranges. Topsail ground elevation achieved 7.64 cm MAE and 9.86 cm RMSE, while Baltic net beach-plus-dune volume change differed by 0.48% between methods. These site-specific monitoring results do not validate dune performance under design storms.

Recent advances. Recent advances use UAV LiDAR and SfM, satellite topography, automated vegetation-aware classification, storm-by-storm digital twins, coupled aeolian-marine morphodynamics, nature-based benefit accounting and adaptive monitoring tied to intervention thresholds.

Disagreements. Hard, fixed design elevations conflict with dunes as mobile landforms, while unmanaged mobility can threaten assets. LiDAR better penetrates vegetation but costs more; SfM can recover aggregate volume yet produce local artifacts that matter for crest and weak-point assessment.

Limitations. Current reviewed evidence lacks direct wave attenuation, erosion demand, overwash, breach probability, recovery and ecological performance. Vegetation occlusion, wet sand and water artifacts, co-registration error, sparse checkpoints and short monitoring periods constrain detected change.

Open questions. Priorities include linking measured geometry to probabilistic failure, full-scale storm validation, vegetation species and planting design, recovery under storm clusters, sediment-source sustainability, access weak points, habitat tradeoffs and adaptive nourishment triggers.

Seminal papers. Coastal dune erosion and runup-impact regimes established geometry-based storm response; aeolian transport and vegetation-trapping studies explained growth, while profile-volume monitoring and probabilistic fragility reframed dunes as dynamic defenses.

Claims

1. At Topsail Hill, UAV LiDAR combined with a genetic algorithm estimated dune ground elevation with 7.64 cm MAE and 9.86 cm RMSE against 526 RTK checkpoints. *Regime: The vegetated Florida dune, tested UAV sensor, and trained regression setup.* direct finding, field [1116]
2. Field evidence links spatial organization of dominant dune grass and sand-flux feedbacks to foredune development and natural flood-barrier function. *Regime: Biomorphogenic Feedbacks and the Spatial Organization of a Dominant Grass Steer Dune Development* direct finding, field [170]
3. Barrier-dune resistance to storm-tide flooding depends on dune-beach type, sediment reserve and erosional response rather than crest elevation alone. *Regime: PROTECTION OF SANDY COASTS IN DEPENDENCE OF THE DUNE - BEACH - TYPE* review statement, review [412]

35.3 Hybrid defenses

Topic nature_based.hybrid. Combined natural and gray infrastructure.

Papers screened: 27. Claims: 25. Equations: 2.

Well established. Hybrid defense can reduce flood and erosion risk by placing a natural or nature-based feature seaward of, beside, or within engineered infrastructure, but performance depends on forcing, feature geometry and condition, available space, structure configuration, ecological persistence, maintenance, and governance. Combining components does not guarantee synergy.

Governing physics. Green components attenuate waves through breaking, drag and friction, store or route flood water; and influence sediment; gray components resist residual overtopping, impact and erosion. Dike blockage, marsh and channel geometry, platform inundation, vegetation deformation or breakage, wave steepness, freeboard, and component ordering determine transmitted load and structural response.

Dimensionless parameters. Important nondimensional controls include relative water depth and freeboard, marsh-platform water fraction, relative width and channel depth, wave steepness, vegetation submergence and density, drag and transmission coefficients, proportional overtopping reduction, failure probability, benefit-cost ratios, and intuitionistic membership, non-membership and hesitancy.

Major equations. Core formulations include depth-averaged marsh hydrodynamics, platform-to-total water-volume control, wave-energy and vegetation-drag attenuation, overtopping discharge, revetment impact and fragility functions, series-component reduction $R_h = R_g + R_r - R_g R_r$, structural-response equations forced by transmitted waves, reliability integrals, and fuzzy multi-criteria distance-to-ideal rankings.

Typical methods. Evidence combines calibrated two-dimensional hydrodynamic scenarios, stochastic reliability analysis, large-scale physical models, analytical coupling of attenuation and structural response, ecosystem and case reviews, stakeholder interviews, lifecycle and co-benefit assessment, expert elicitation, and multi-criteria decision analysis.

Numerical models. Models include TELEMAC2D for marsh-dike propagation, stochastic foreshore-dike reliability for overtopping and revetment impact, empirical series-component overtopping reduction, coupled attenuation-structure equations, blue-carbon and multifunctionality frameworks, and Fuzzy Delphi plus entropy-weighted IF-TOPSIS selection.

Experimental datasets. Key evidence includes water levels across a 3000 ha Dutch marsh and two storm tides, a large-scale wall-mangrove-revetment overtopping campaign, reviewed UK and southeast Asian marsh and mangrove cases, interviews with 27 residents in two New Jersey communities, and Golden Coast expert judgments across 18 measures and 63 initial indicators.

Validated ranges. Reported regimes include a 3000 ha marsh needing about 6-10 km width to avoid tested dike-blockage effects, attenuation onset at platform-water fractions 0.2-0.4 and maxima of 29 cm/km near 0.85, a nonbreaking overtopping experiment whose series model was accurate within 5%, 27 interview participants, and 18 consensus indicators selected from 63 to rank 18 measures.

Recent advances. Recent work advances from qualitative hybrid advocacy to component-resolved overtopping experiments, coupled green-to-gray design equations, climate-adaptive lifecycle planning, quantified multifunctionality, and fuzzy group decisions that preserve expert uncertainty and stakeholder priorities.

Disagreements. The finding that one tested hybrid behaved as independent series components does not establish universal absence of interaction: breaking, sediment feedback, vegetation damage, canopy, different gray geometry, and long-term ecological change were outside that experiment. Preference for natural features alongside perceived superior gray effectiveness also reflects trust and communication, not hydraulic comparison alone.

Limitations. Evidence remains sparse across extreme storms, breaking waves, component degradation, sediment evolution, ecological succession, maintenance cycles, and long design lives. Cost comparisons are thin; co-benefit valuation is scale-sensitive; stakeholder samples and expert weights are local; and natural-feature uncertainty is often larger than gray-component uncertainty.

Open questions. Priorities are full-scale extreme-event validation, coupled hydro-morphodynamic-ecological reliability, interaction versus series criteria, fragility and maintenance models for living components, adaptive thresholds under sea-level rise, standardized lifecycle cost and co-benefit accounting, and participatory selection that updates as evidence changes.

Seminal papers. Within this screened branch, the 2016 marsh-dike study establishes geometry- and blockage-dependent flood attenuation, and the 2018 reliability study translates variable natural protection into explicit dike failure probability.

Equations

- *Series-component overtopping reduction* [827]

$$R_h = R_g + R_r - R_g R_r$$

Regime: Nonbreaking tests at constant freeboard and depth with the tested mangrove and revetment geometries. Variables: R_h proportional hybrid overtopping reduction; R_g proportional green-component reduction; R_r proportional revetment reduction

- *Coupled green-gray design chain* [828]

$$H_t = K_t H_i, \quad R_s = \mathcal{F}(H_t, T, h, \text{geometry})$$

Regime: Mangrove-fronted gray infrastructure within the source model assumptions. Variables: H_i incident wave height; K_t green-feature transmission or attenuation factor; H_t wave height reaching gray structure; R_s structural response or design demand; $\mathcal{F}$ structure-response relation

Claims

1. A comprehensive review found that salt marshes, mangroves, seagrass, beaches, dunes, coral, and shellfish reefs can support coastal protection and co-benefits, but their processes, evidence maturity, and design applicability are not interchangeable. *Regime: The ecosystems, defense geometries, forcing, and implementation contexts evaluated in the source.* review statement, review [672]
2. For a modeled 3000 ha Dutch marsh, dike blockage was avoided only at widths of about 6-10 km; attenuation began when platform-water fraction exceeded 0.2-0.4 and reached 29 cm/km near 0.85, while channel-only tides were not attenuated. *Regime: The ecosystems, defense geometries, forcing, and implementation contexts evaluated in the source.* direct finding, numerical [1349]
3. Across reviewed mangrove and marsh hybrid defenses, 70% of successful cases were underpinned by ecohydrologic understanding and broad stakeholder involvement; limited cost data suggested hybrids may cost less than gray but more than green defenses. *Regime: The ecosystems, defense geometries, forcing, and implementation contexts evaluated in the source.* review statement, review [1490]
4. A probabilistic hybrid-defense model found vegetated foreshores reduced both overtopping and revetment-impact failure probability, but at high protection levels expected stem breakage made foreshore geometry more important than vegetation. *Regime: The ecosystems, defense geometries, forcing, and implementation contexts evaluated in the source.* direct finding, numerical [1470]
5. Among 27 residents in two New Jersey coastal communities, most preferred natural infrastructure but believed engineered defenses were more effective, and many did not understand how natural and engineered components interact. *Regime: The ecosystems, defense geometries, forcing, and implementation contexts evaluated in the source.* direct finding, field [528]
6. Green-gray coastal infrastructure can jointly provide adaptation, blue-carbon mitigation, and other services, but the preferred mix depends on local risk and benefits must be quantified across explicit spatial and temporal scales. *Regime: The ecosystems, defense geometries, forcing, and implementation contexts evaluated in the source.* review statement, review [754]
7. In a large-scale overtopping experiment, a realistic mangrove forest provided protection comparable to a rubble-mound revetment for matched conditions, and the tested hybrid behaved as a series of independent rather than synergistic components. *Regime: The ecosystems, defense geometries, forcing, and implementation contexts evaluated in the source.* direct finding, experimental [827]
8. Coupling wave-attenuation equations to structural-response equations provides a direct engineering design chain for mangrove-fronted gray infrastructure, but applicability remains bounded by the tested vegetation, wave, and structure regime. *Regime: Hybrid systems comparable to the experimental and analytical assumptions.* direct finding, analytical [828]

9. At Golden Coast, Taiwan, a Fuzzy Delphi process reduced 63 candidate indicators to 18 consensus indicators, after which entropy-based intuitionistic fuzzy TOPSIS ranked 18 coastal NbS measures using membership, non-membership, and hesitancy. *Regime: Coastal projects with structured stakeholder elicitation and locally defined criteria.* direct finding, analytical [366]
10. In Belize, iterative ecosystem-service modeling and stakeholder planning produced a preferred coastal plan projected to improve coastal protection by more than 25% and more than double fishing revenue relative to an earlier preference-only plan. *Regime: Embedding ecosystem services in coastal planning leads to better outcomes for people and nature* direct finding, mixed [77]
11. Idealized simulations found combined coral, seagrass and mangrove habitats provided more coastal protection than any one or two habitats, but relative contributions changed with reef setting, habitat position, forcing and sea level. *Regime: The Power of Three: Coral Reefs, Seagrasses and Mangroves Protect Coastal Regions and Increase Their Resilience* direct finding, mixed [548]
12. A synthesis finds coastal ecosystems can attenuate waves and surge, reduce erosion and maintain profiles under defined conditions, with self-repair and co-benefits offering advantages in some settings and compatibility with hybrid engineering. *Regime: Coastal Ecosystems: A Critical Element of Risk Reduction* direct finding, mixed [1330]
13. Wetland valuation remains constrained chiefly by weak quantitative links from ecosystem structure and function to service production and by the absence of markets for many services. *Regime: Valuing Ecosystem Services for Coastal Wetland Protection and Restoration: Progress and Challenges* direct finding, mixed [116]
14. A 46-paper review found 91% of living-shoreline studies used structural material, 89% reported only ecological outcomes, most sites were under five years old, and long-term socioeconomic and context-specific engineering evidence was scarce. *Regime: Coming to Terms With Living Shorelines: A Scoping Review of Novel Restoration Strategies for Shoreline Protection* direct finding, mixed [1316]
15. Among 59 European coastal and estuarine cases, most solutions were wetland-based hybrids and three-quarters reported monitoring, yet more than half disclosed no quantitative flood or erosion effectiveness results. *Regime: Nature-Based Solutions in Coastal and Estuarine Areas of Europe* direct finding, mixed [981]
16. A New England review compares nonstructural and hybrid living shorelines with hardened protection and evaluates their erosion, habitat, wave, storm and sea-level-rise applicability. *Regime: Living Shorelines: A Review of Literature Relevant to New England Coasts* direct finding, mixed [1037]
17. A Gulf Coast risk assessment finds cost-effective adaptation portfolios could prevent US\$57–101 billion in losses; nature-based options avert over US\$50 billion with average benefit-cost ratios above 3.5. *Regime: Comparing the cost effectiveness of nature-based and coastal adaptation: A case study from the Gulf Coast of the United States* direct finding, mixed [1171]
18. A review of eco-engineering interventions on urban marine infrastructure compares ecological benefits and identifies design choices that improve coastal biodiversity outcomes. *Regime: Eco-engineering urban infrastructure for marine and coastal biodiversity: Which interventions have the greatest ecological benefit?* review statement, review [1358]
19. A coastal eco-engineering synthesis compares created or restored dunes, marshes, mangroves, seagrass, kelp, coral, and shellfish habitats as alternatives or complements to hard defenses. *Regime: From grey to green: Efficacy of eco-engineering solutions for nature-based coastal defence* review statement, review [984]
20. A global systematic evidence map evaluates where nature-based interventions have been tested for climate-hazard adaptation and identifies evidence gaps affecting deployment. *Regime: Mapping the effectiveness of nature-based solutions for climate change adaptation* review statement, review [253]
21. A coastal-infrastructure synthesis frames natural and nature-based systems as risk-reduction investments that can complement conventional protection while providing co-benefits. *Regime: Investing in Natural and Nature-Based Infrastructure: Building Better Along Our Coasts* review statement, review [1378]
22. Regional valuation shows hybrid coral-reef restoration can cost-effectively reduce flood risk to vulnerable reef-lined coastal populations and assets. *Regime: Hybrid coral reef restoration can be a cost-effective nature-based solution to provide protection to vulnerable coastal populations* direct finding, numerical [1356]
23. A review explains how hardened shorelines, offshore infrastructure, aquaculture structures, and artificial reefs can be designed or managed to reduce habitat impacts and provide ecological benefits. *Regime: Leveraging built marine structures to benefit and minimize impacts on natural habitats* review statement, review [1088]
24. Mangroves and moored barges are used individually for coastal protection and beach restoration. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* hypothesis, author_hypothesis [1150]
25. Hard coastal protection can be made greener by pairing it with soft measures, mitigating morphology and hydrodynamic disruption, or adding ecological enhancement, but solution choice must be site-specific and jointly consider forcing, space, safety, ecology and socioeconomic constraints. *Regime: Foreshore and onshore hard coastal defenses where purely soft protection is infeasible or hybrid performance is desired.* review statement, review [1266]

35.4 Reefs and living shorelines

Topic nature_based.reefs. Oyster, coral, and constructed reef protection.

Papers screened: 10. Claims: 7. Equations: 1.

Well established. Reefs and reef-like living shorelines can reduce wave energy, currents, and shoreline erosion, but protection is conditional on crest elevation relative to water level, porosity and roughness, reef width and morphology, incident forcing, sediment setting, biological persistence, and maintenance.

Governing physics. Breaking on reef rims creates setup and cross-reef pressure gradients; porous and rough reef material dissipates energy; emergence, submergence, wave breaking, bottom friction, lagoon wind-wave generation, and bidirectional sediment resuspension determine the realized response.

Dimensionless parameters. Important nondimensional controls are relative crest or reef height, relative submergence, porosity, transmission coefficient, friction and resistance coefficients, relative wave height and depth, and ratios separating wave, infragravity, and locally generated lagoon energy.

Major equations. Core formulations include radiation-stress/setup balances, open-channel reef-top and reef-rim control, energy-flux transmission $K_t = \sqrt{P_t/F_i}$, Darcy-Forchheimer porous resistance, shallow-water breaking and friction balances, and paired-tray sediment mass balance.

Typical methods. Evidence combines reef-platform flumes, pressure and velocity arrays across porous structures, paired treatment-control shoreline experiments, oyster demographic and topographic surveys, sediment mass balances, spectral and surfbeat models, and multi-year lifecycle monitoring.

Numerical models. The screened evidence uses analytical reef-control and porous-transmission models plus phase-averaged SWAN and surfbeat-resolving XBeach; model choice must preserve locally generated wind waves when they dominate reef-lagoon storm energy.

Experimental datasets. Key datasets include nearly 1000 boat wakes at two Florida breakwalls, a four-year three-site hybrid intervention, six Louisiana reefs revisited after ten years, four paired-tray sites on two Great Barrier Reef systems, a steep-faced reef-platform experiment, and comparative restored-oyster substrates in Virginia.

Validated ranges. Observed regimes include 0.7-porosity branch barriers with mean energy transmission 53%, 0.9-porosity barriers with 83%, relative breakwall height about 0.5-1 for clear tidal modulation, four-year shoreline trends from about 1 m/y advance to nearly 2 m/y retreat, and ten-year oyster density below 60 individuals/m² with erosion near 1 m/y.

Recent advances. Recent work shifts from proving short-term attenuation toward optimizing substrate and elevation, testing decade-scale ecological persistence, resolving cyclone-specific model physics, and embedding future conditions, maintenance, and adaptive action into design.

Disagreements. Short-term hybrid success and decade-scale native-reef failure are not universal contradictions: structure, exposure, recruitment, salinity, maintenance, and monitoring horizon differ. Likewise, wave-height amplification over a submerged reef need not imply increased energy flux, so performance metrics must match the erosion mechanism.

Limitations. Most studies cover few sites, narrow structural designs, boat wakes or ordinary wind waves rather than extremes, and limited design lives. Geometry, biology, water quality, sea-level rise, material degradation, and maintenance covary, preventing simple transfer of one attenuation coefficient.

Open questions. Priorities are storm-scale multi-site validation, coupled ecology-hydraulics-morphology prediction, lifecycle performance under recruitment failure and sea-level rise, metric choice for height versus energy flux, maintainable porous materials, and adaptive thresholds for intervention.

Seminal papers. Within this screened slice, the 2005 steep-faced reef study establishes coupled setup-flow control, while the 2020 paired breakwall studies connect porous-transmission physics to measured shoreline and habitat response.

Equations

- *Wave-energy transformation balance* [519]

$$\nabla \cdot (E\mathbf{C}_g) = -D_b - D_f$$

Regime: Regular waves over the tested steep-face, sloping reef-top physical models; extrapolation to irregular, directional and strongly three-dimensional reefs is limited. Variables: E wave energy density; $\mathbf{C}_g$ group velocity; D_b breaking dissipation; D_f bottom-friction or other dissipation

Claims

1. For a laboratory steep-faced horizontal reef, wave-generated flow was organized into reef-top-control and reef-rim-control regimes, and coupled open-channel setup-flow relationships reproduced most measured values with reasonable accuracy. *Regime: Two-dimensional horizontal reef tops with steep faces under wave-driven quasi-steady flow.* direct finding, mixed [520]
2. Ten years after construction of six Louisiana oyster reefs, oyster density was below 60 individuals/m² versus above 1000/m² during 2009-2011, and both reef and reference shorelines continued eroding at about 1 m/y. *Regime: Sister Lake native-shell fringing reefs exposed to highly variable salinity; results do not imply all oyster defenses fail.* direct finding, field [1100]
3. A restored-oyster field study found that coastal-protection performance depends jointly on reef elevation, substrate type, and adjacent marsh-edge morphology, with wave dissipation effective primarily when water depth is near or below reef crest height. *Regime: The tested Virginia intertidal reef designs and wind-wave conditions; storm-scale and long-term transfer remain unresolved.* direct finding, field [154]
4. Regular-wave experiments over an idealized steep-faced fringing reef showed that wave transformation is controlled by shoaling and breaking at the reef edge followed by reforming waves over the reef flat, providing a bounded laboratory basis for predicting reef-flat wave heights and setup. *Regime: Regular waves over the tested steep-face, sloping reef-top physical models; extrapolation to irregular, directional and strongly three-dimensional reefs is limited.* direct finding, experimental [519]
5. Across the coral-reef studies synthesized by the meta-analysis, reefs reduced wave energy by an average of 97%, with reef crests accounting for 86% dissipation; these global averages should not be treated as site-specific design coefficients. *Regime: Global coral-reef evidence synthesized in the indexed abstract; site-specific hydrodynamic regimes are not recoverable from the abstract.* review statement, review [438]
6. Flume measurements showed restored coral-canopy attenuation rose with coral cover; force-velocity energy loss agreed with observed wave-height decay, and well-designed canopies dissipated more than 50% of incoming energy under typical reef-flat conditions. *Regime: Wave Attenuation by Restored Coral Reef Canopies: Implications for Coastal Protection* direct finding, mixed [489]
7. An oyster-reef restoration review recommends matching site, substrate, species and genotype to future environmental suitability and explicit ecosystem-service and socioeconomic goals. *Regime: Contemporary Oyster Reef Restoration: Responding to a Changing World* review statement, review [613]

35.4.1 Living shoreline breakwalls

Topic nature_based.reefs.breakwalls. Porous brush, branch, reef, and modular barriers used to attenuate waves while retaining ecological connectivity.

Papers screened: 7. Claims: 9. Equations: 0.

Well established. The reviewed branch-and-post living breakwalls reduced boat-wake energy, but attenuation depended strongly on design and tidal depth rather than following a single transferable transmission rate.

Governing physics. For the lower-porosity design, transmission rose with tidal depth most clearly for relative structure height h_s/h approximately 0.5-1; near emergence, breaking and bottom friction added dissipation absent from the porous-only theory.

Dimensionless parameters. Controls include porosity, relative structure height and submergence, width relative to wavelength, transmitted-to-incident energy, water depth relative to crest, wake steepness, shoreline-change rate normalized by control retreat and maintenance interval relative to material lifetime.

Major equations. Performance is described by wave-energy transmission and reflection ratios, energy-flux balances, porous resistance and shoreline-position or retreat-rate comparisons between treatment and control. Long-term benefit additionally requires a service-life and maintenance balance for biodegradable fill and stable supports.

Typical methods. Studies deploy incident and lee wave sensors across tides and boat wakes, characterize porosity and geometry, pair protected and reference marsh margins, repeat shoreline surveys over years, track oyster or vegetation response and document repair frequency, component loss and functional lifetime.

Numerical models. A calibrated porous-transmission formulation represents branch resistance but omits some external breaking and bottom friction near emergence. The shoreline study uses treatment-control field comparison rather than a predictive morphodynamic model.

Experimental datasets. Reviewed evidence comprises 290 boat wakes at bundled and unbundled branch breakwalls and 2015-2019 monitoring at three energetic northeast Florida shorelines with brush breakwalls, oyster gabions and paired controls.

Validated ranges. The unbundled $n=0.9$ design transmitted more than 70% of energy for 84% of wakes and averaged 83% during the North Peninsula deployment.

Recent advances. Recent advances combine modular biodegradable fills, oyster gabions and hybrid sills, field spectral monitoring, UAV shoreline and habitat mapping, life-cycle and maintenance accounting, adaptive replacement and coupled hydrodynamic-ecological performance metrics.

Disagreements. Greater material density and lower porosity can improve attenuation but increase cost, ecological footprint and failure load; biodegradable components support habitat goals yet require replacement. Short-term wave reduction does not guarantee multi-year shoreline benefit, and shoreline stability does not isolate each component contribution.

Limitations. The observed 0.7-versus-0.9 porosity contrast is not a controlled porosity experiment because branch bundling, field site, and tidal range changed simultaneously.

Open questions. Controlled multi-site evidence is needed to separate porosity, material, geometry, tides, breaking, sediment response, ecological performance, durability, storms, and maintenance cost.

Seminal papers. Permeable breakwater and brush-fence studies established porous attenuation, while oyster-reef and living-shoreline experiments extended performance to ecological structures and paired shoreline controls, reframing maintenance and biological development as design variables.

Claims

1. For the bundled-branch breakwall with estimated porosity 0.7, observed wave-energy transmission was mostly 9%-70% and averaged 53%. *Regime: GTM branch-and-post design, approximately 4.3 by 0.6 by 0.55 m, boat wakes and tides observed in March-April 2018.* direct finding, field [1231]
2. For the unbundled-branch breakwall with estimated porosity 0.9, energy transmission exceeded 70% in 84% of wake cases and averaged 83%. *Regime: NP branch-and-post design, approximately 4.3 by 0.6 by 0.55 m, boat wakes and smaller tidal range observed in April-May 2019.* direct finding, field [1231]
3. The branch fill required maintenance every four to six months at GTM and had an estimated functional life of 12-18 months, while unbundled NP fill lasted less than one year; fence posts remained stable after three years. *Regime: Crepe-myrtle bundled fill at GTM and eastern-cedar unbundled fill at NP under local bioerosion and vessel-wake exposure.* direct finding, field [1231]
4. From 2015-2019 at three wake-exposed northeast Florida sites, marsh margins protected by brush breakwalls and oyster gabions were stable or advanced about 1 m/y, whereas paired controls retreated at rates approaching 2 m/y. *Regime: Vessel-wake-dominated estuarine marshes with 1-1.5 m semidiurnal tide and local oyster recruitment.* direct finding, field [1230]
5. Replicated oyster-shell breakwater reefs reduced shoreline retreat by more than 40% at one site and enhanced several fish and invertebrate groups, but loss of reef relief compromised protection and erosion remained high. *Regime: Oyster Reefs as Natural Breakwaters Mitigate Shoreline Loss and Facilitate Fisheries* direct finding, mixed [1277]
6. At Kutubdia Island, oyster breakwaters attenuated 0.1–0.5 m waves, accumulated 29 cm of lee-side sediment, reduced erosion 54% relative to controls and facilitated seaward saltmarsh expansion. *Regime: Oyster breakwater reefs promote adjacent mudflat stability and salt marsh growth in a monsoon dominated subtropical coast* direct finding, mixed [280]
7. Across US oyster living shorelines, reefs inundated under 50% of the time attenuated wave height by 68% but were unsuitable oyster habitat, whereas suitable reefs above 50% inundation attenuated only about 5%, exposing an ecological-engineering design trade-off. *Regime: Large-scale variation in wave attenuation of oyster reef living shorelines and the influence of inundation duration* direct finding, mixed [986]
8. Validated FUNWAVE-TVD simulations indicate oyster-reef attenuation depends on waves and depth and can remain sustainable under sea-level rise only with viable growth habitat; modeled maximum capacity required about nine years after construction. *Regime: Assessing Wave Attenuation With Rising Sea Levels for Sustainable Oyster Reef-Based Living Shorelines* direct finding, mixed [1238]
9. Seven-year Gandy's Beach monitoring found constructed oyster reefs met most habitat objectives but none of the material-stability, wave-attenuation, or sediment-elevation protection objectives fully, demonstrating the need for goal-specific long-term monitoring. *Regime: Seven Years of Monitoring the Development of an Oyster Reef Living Shoreline* direct finding, mixed [1301]

35.5 Vegetation and wave attenuation

Topic nature_based.vegetation. Marsh, mangrove, and flexible vegetation effects.

Papers screened: 29. Claims: 22. Equations: 4.

Well established. Vegetation attenuates waves through hydrodynamic work on stems, roots, and canopies, but performance is jointly controlled by plant geometry, density, flexibility, vertical submergence, spatial layout, wave nonlinearity, current, patch length, and seasonal state.

Governing physics. Drag depends on relative water-plant velocity and wetted frontal area; flexibility redistributes motion and dissipation vertically; nonlinear orbital kinematics, current-modified group velocity, diffraction through gaps, breaking, and canopy submergence alter spatial decay.

Dimensionless parameters. Key controls include Ursell number, Cauchy number, excursion ratio, Keulegan-Carpenter and Reynolds numbers, relative plant/root height and submergence, solid volume fraction, current-to-orbital velocity ratio, effective-length ratio, and layout spacing Sr.

Major equations. Common formulations balance wave-energy flux against vegetation drag, use transmission or decay coefficients, Ursell and Cauchy scaling, Euler-Bernoulli stem response, a corrected work factor chi, current-aware reciprocal-to-exponential decay, and Weibull statistics for attenuated random heights.

Typical methods. Studies use plant-mimic and live-plant flumes, field pressure arrays during storms, simultaneous stem-motion and velocity measurements, energy-balance theory, RANS or phase-resolving validation, spectral-wave implementation, and stochastic vegetation-property sampling.

Numerical models. Models include Dalrymple/Kobayashi and modified Weibull relations, SWASH, Euler-Bernoulli quasi-flexible damping, NEWFLUME RANS, spacing/diffraction energy balance, and momentum-coupled frequency-distributed MIKE SW.

Experimental datasets. Evidence spans four flexible species analogues, a 17.5 m mangrove array, 112 component-resolved tests, 24 simultaneous motion/velocity cases, 177 Tropical Storm Lee wave records, four spatial layouts, five-species validation, and 1205 seasonal plant measurements.

Validated ranges. Reported ranges include 20-76% reduction over 1 m, transmission 0.21-0.83, Ur about 20-250, quasi-flexible applicability $L > 1.4$ and $Ca < 700$, current ratio -1.5 to 3.0, Sr transition at one, and storm Hs to 0.4 m in 0.8 m depth; each is experiment-specific.

Recent advances. Recent work replaces locally tuned damping with measurable mechanics, adds current-induced asymmetry and layout uniformity, and embeds seasonal heterogeneous vegetation into spectral models while preserving field-scale computational efficiency.

Disagreements. Apparent conflicts over whether larger or longer waves attenuate more arise because nonlinearity, submergence, flexibility, current, and normalization length vary together. Likewise, local heterogeneity can matter for gaps and extremes even when mean plant properties suffice for bulk engineering attenuation.

Limitations. Most evidence uses idealized cylindrical vegetation, short patches, regular or narrow condition sets, fixed beds, and calibrated or assumed force coefficients. Extreme folding, breakage, mortality, currents, directional spectra, morphodynamics, and cross-site ecological variability remain weakly constrained.

Open questions. Priorities include storm-scale field validation across species and seasons, flexible layered architecture under reversing currents, damage and recovery, directional gap effects, uncertainty in extreme tails, and coupling attenuation to sediment, morphology, and ecological persistence.

Seminal papers. Within this screened slice, the 2013 tropical-storm Weibull study provides rare extreme-event field statistics, while the 2014 flexible-vegetation experiments establish multi-trait attenuation ranges beyond rigid-cylinder assumptions.

Equations

- *Vegetation-modified Weibull wave-height distribution* [645]

$$p(H) = \frac{k}{\lambda} (H/\lambda)^{k-1} e^{-(H/\lambda)^k}$$

Regime: Storm waves attenuated through the measured Spartina marsh. Variables: H zero-crossing wave height; k shape parameter; lambda scale parameter dependent on k

- *Ursell number* [1104]

$$Ur = H_s L^2 / h^3$$

Regime: Wave conditions at the mangrove-field edge. Variables: Ur Ursell number; H_s significant wave height; L wavelength; h water depth

- *Corrected flexible-vegetation work-factor dissipation* [1446]

$$D_f = \chi D_r$$

Regime: Quasi-flexible cylindrical vegetation with $L > 1.4$ and $Ca < 700$. Variables: D_f flexible-vegetation dissipation; chi work factor restored by corrigendum; D_r corresponding rigid-vegetation dissipation

- *Current-aware mixed vegetation decay law* [851]

$$H(x) = H_0 \mathcal{G}(x; U/u_o, C_D, c_g)$$

Regime: Emergent uniform vegetation with $-1.5 \leq U/u_o \leq 3.0$. Variables: H wave height; H_0 incident height; U current speed; u_o wave orbital velocity; C_D drag coefficient; c_g current-modified group velocity

Claims

1. Across four flexible salt-marsh vegetation analogues, measured wave-height reduction over a 1 m patch ranged from 20% to 76%; damping depended jointly on plant geometry, density, flexibility, submergence, and wave conditions rather than marsh width alone. *Regime: The tested flexible mimics, one-metre patch, irregular waves, and submerged-to-emergent conditions.* direct finding, experimental [55]
2. In 200-400 stem/m² mangrove analogues, increasing Ursell number from about 50 to 250 increased wave-height attenuation by roughly 20 percentage points over one wavelength and 35 over three, while sensitivity weakened above about $Ur=250$. *Regime: Rigid arrays, Ur approximately 20-250, tested broken/non-broken regular and irregular waves.* direct finding, mixed [1104]
3. Across 112 stem-root-canopy experiments, vegetation transmission coefficients ranged from 0.21 to 0.83; roots contributed significantly when relative root height exceeded 0.167, and attenuation peaked when still water lay near the canopy centroid. *Regime: Fixed-period waves and the tested idealized seven-model, four-density component geometries.* direct finding, experimental [577]
4. For quasi-flexible cylindrical vegetation with excursion ratio $L>1.4$ and Cauchy number $Ca<700$, damping was governed by relative water-stem velocity in the upright lower stem, and the model reproduced the order of damping across five species without species-specific drag calibration. *Regime: Regular waves, cylindrical vegetation, $L>1.4$, $Ca<700$, without extreme folding.* direct finding, mixed [1446]
5. During a tropical storm, 177 wave records across 28 m of *Spartina* marsh showed an upper-tail deficit relative to Rayleigh statistics; a vegetation-modified Weibull distribution fit the local heights for H_s up to 0.4 m in 0.8 m depth. *Regime: The measured tropical-storm event, inundated *Spartina* marsh, and local wave/depth range.* direct finding, field [645]
6. For emerged vegetation and current-to-orbital velocity ratios from -1.5 to 3.0, opposing currents attenuated waves more than equal following currents; increasing current shifted decay from reciprocal toward exponential and made strong-current decay insensitive to incident amplitude. *Regime: Uniform emergent vegetation, steady following/opposing current, tested velocity-ratio interval.* direct finding, mixed [851]
7. In four flexible-vegetation layouts, the combined spacing parameter separated laterally uniform attenuation at $Sr<1$ from nonuniform attenuation at $Sr>1$, and a staggered layout matched full-coverage uniformity with a smaller vegetation footprint. *Regime: The tested half, segmented, staggered, and full layouts and the study's Sr definition.* direct finding, mixed [1486]
8. A momentum-coupled MIKE SW vegetation model validated against laboratory and Tropical Storm Lee data reproduced significant wave height and spectral period without prescribed C_{dis} ; analysis of 1205 plant measurements found spatial mean vegetation properties sufficient for engineering predictions despite seasonal lognormal variability. *Regime: The tested flexible saltmarsh canopies, fixed force coefficients $C_m=0.5$ and $C_d=2.0$, and validated laboratory/field conditions.* direct finding, mixed [644]
9. The formal corrigendum states that work factor χ was omitted from printed Eq. 32 of the quasi-flexible vegetation model; implementations must use the corrected expression containing χ . *Regime: Any implementation of Eq. 32 in the corrected 2021 quasi-flexible vegetation model.* direct finding, analytical [1447]
10. For validated vegetated-wave cases, a depth-varying velocity drag formulation matched fully expanded Boussinesq results, while the linear analytical solution agreed except near-emergent or emergent vegetation and the abstract's stated $Ur<5$ regime. *Regime: Vegetated laboratory cases and Hurricane Isaac peak-wave simulations; linear comparison degraded for stem-height/depth above 0.75 and $Ur<5$ as stated in the abstract.* direct finding, mixed [243]
11. In a 1:12 model *Rhizophora* forest, root-zone velocity decreased by up to 50% and turbulence increased fivefold; velocity and individual-tree drag reached fully developed behavior by the fifth row, and properly scaled drag coefficients collapsed as a Reynolds-number function. *Regime: Thirty-two 1:12 *Rhizophora* models with 24 prop roots each in a 2.5 m staggered forest.* direct finding, experimental [939]
12. Living-shoreline marshes aged 12–38 years sequestered 58–283 g C m⁻² yr⁻¹, with lower rates at older sites, showing that shoreline stabilization can add blue-carbon benefits but age affects long-term estimates. *Regime: Living Shorelines: Coastal Resilience with a Blue Carbon Benefit* direct finding, mixed [323]
13. At three of four Port Phillip Bay sites kelp beds attenuated less wave energy than urchin-barren controls; only under northerly winds at the fourth site was kelp attenuation about 10% greater, showing substrate and kelp effects must be separated. *Regime: Kelp beds as coastal protection: wave attenuation of *Ecklonia radiata* in a shallow coastal bay* direct finding, mixed [985]
14. Global process-based valuation estimates mangroves avert more than US\$65 billion in flood damage annually and protect 15 million people from annual flooding, with many urban coastal stretches exceeding US\$250 million in yearly benefit. *Regime: The Global Flood Protection Benefits of Mangroves* direct finding, mixed [959]
15. High-resolution flood-loss models estimate northeastern U.S. wetlands avoided US\$625 million in direct Hurricane Sandy damage and reduced average annual Barnegat Bay flood losses by 16%. *Regime: The Value of Coastal Wetlands for Flood Damage Reduction in the Northeastern USA* direct finding, mixed [1012]
16. Ten months of saltmarsh observations found 92% mean wave-height attenuation over 310 m; the first 10 m of vegetation dissipated 2.1% per metre on a ramped edge versus 1.1% at a cliffed edge, with seasonal variation. *Regime: Wave dissipation over macro-tidal saltmarshes: Effects of marsh edge typology and vegetation change* direct finding, field [1006]
17. Field measurements on two salt marshes and a calibrated SWAN model show that vegetation remains effective under severe storm inundation: with bulk drag coefficient about 0.4, vegetation added 25-50% significant-wave-height reduction relative to breaking and bottom friction alone. *Regime: *Spartina anglica* and *Scirpus maritimus* salt marshes and modeled sloping foreshores under the tested storm loads and vegetation states.* direct finding, mixed [1469]
18. Large-flume tests with flexible vegetation mimics show that stiffness and bending-controlled dynamic frontal area, not biomass alone, govern wave drag; an effective-leaf-length model reproduced regular and irregular force time series but missed extra drag from close-neighbor interaction. *Regime: Artificial vegetation arrangements of varied stiffness, frontal area and volume under tested regular and irregular wave orbital velocities.* direct finding, experimental [1086]
19. Under comparable inundation in a subtropical estuary, saltmarsh and mangrove edges both trap mudflat-derived suspended sediment, but the saltmarsh edge traps more because mangrove-induced flow rotation diverts normal flux; settling and direct capture on plant surfaces jointly build deposits. *Regime: Adjacent bare mudflat, mangrove and saltmarsh stands with similar hydropereiod in a subtropical estuary during the observed spring tide.* direct finding, field [261]
20. Temperate kelps are ecosystem engineers whose holdfasts, stipes, blades and understory modify physical conditions, support diverse and abundant assemblages and trophic links, and provide coastal-defense and other ecosystem services, but their habitat function varies strongly among species and regions. *Regime: Canopy-forming kelps on temperate and subpolar intertidal and shallow-subtidal rocky reefs represented in the reviewed literature.* review statement, review [1403]
21. Wakes from adjacent emergent vegetation patches remain locally patch-specific but merge downstream; in staggered layouts the downstream patch shortens and distorts the upstream wake, and density controls interpatch velocity, turbulence and potential sediment/propagule retention. *Regime: Two circular emergent rigid vegetation patches under the tested steady flume flow, densities and side-by-side/staggered spacing.* direct finding, experimental [726]
22. A coupled three-dimensional nonhydrostatic model predicts that a surf-zone vegetation patch damps breaking-wave turbulence, reduces sediment pickup, confines suspended material near the bed, and reverses net transport from seaward outside the patch to shoreward within it. *Regime: A finite vegetation patch in the surf zone under breaking waves, with canopy drag, turbulence production, suspended sediment, and bed exchange.* direct finding, numerical [1631]

36 Wiki: Nearshore hydrodynamics

36.1 Nearshore hydrodynamics: overview

Topic nearshore. Currents and water-level dynamics in the nearshore zone.

Papers screened: 15. Claims: 1. Equations: 0.

Well established. Breaking-wave momentum gradients drive setup, undertow, longshore currents, rip cells and vortices. Bathymetry redirects those currents, which in turn transport sediment, tracers and organisms and reshape the bed, producing strongly coupled and often nonstationary nearshore dynamics.

Governing physics. Radiation-stress and pressure gradients, wave mass transport, return flow, bottom friction, wave–current interaction, vorticity generation and turbulent mixing govern circulation. Directional waves, tides, headlands, reefs, bars and channels alter forcing and connectivity.

Dimensionless parameters. Controls include Froude and Reynolds numbers, wave steepness, relative depth, surf similarity, current-to-wave speed ratio, rip spacing relative to surf width, trough area and onshore discharge, directional spread, normalized headland/reef geometry and dispersion scaling.

Major equations. Depth-integrated mass and momentum balances include wave forcing, pressure, advection and bottom stress; wave action or phase-resolving equations provide breaking and radiation stress. Sediment continuity closes morphology feedback, while Lagrangian transport diagnoses exchange and dispersion.

Typical methods. Studies combine current meters, pressure sensors, scanning sonar, drifters, dye/turbidity, imagery and bathymetry with field or basin experiments; wave-averaged, stability, SPH and morphodynamic models test forcing and feedback. Eulerian and Lagrangian observations should be reconciled.

Numerical models. Representations range from depth-averaged circulation and radiation-stress models to linear morphology stability, mesh-free SPH, $O(1\text{ m})$ event hindcasts and coupled bar models. Phase-averaged models capture mean cells but may need explicit unsteadiness or vorticity closures.

Experimental datasets. Reviewed evidence includes Scripps scanning-sonar rips, longshore-uniform swell-beach drifters, dense laboratory rip trajectories, a Lake Michigan meteotsunami reconstruction, large-scale short-crested-wave validation, WAMFlow imagery, a three-week headland experiment and four North Sea campaigns.

Validated ranges. Case values include episodic and headland rip speeds near 0.7 m/s, WAMFlow RMSE near 0.1 m/s, submerged-structure longshore RMSE 0.07 m/s and transient rip-neck widths 20–30 m. These are site-specific measurements and validation metrics, not universal thresholds.

Recent advances. Recent advances use kilometre-scale optical velocimetry, dense Lagrangian arrays, high-resolution forensic hindcasts, mesh-free three-dimensional breaking simulations and coupled structure–bar models to resolve circulation modes missed by sparse instruments.

Disagreements. Rips may be bathymetrically controlled, transient without fixed channels, forced by directional breaking, or triggered by meteotsunamis. Mean-cell descriptions can understate intermittent exchange, and drifter velocities can differ from Eulerian flow because of Stokes drift and sampling concentration.

Limitations. Sparse spatial sampling, rapidly evolving bars, optical visibility, wave breaking complexity, unmeasured vertical structure, uncertain friction, scale effects and short experiments limit transfer. A model matching mean currents may still miss vortices, dispersion or hazard onset.

Open questions. Priorities include transient-rip predictability, compound meteotsunami and wave hazards, vertical exchange, directional spectra, morphology coevolution, headland transport beyond closure, real-time remote sensing, data assimilation and public-warning thresholds.

Seminal papers. Radiation-stress theory connected breaking waves to setup and mean currents; early surf-zone observations established undertow, longshore and rip cells. Morphodynamic stability theory then formalized feedback with bed patterns, while drifters and imaging exposed strong unsteadiness.

Claims

1. Nearshore hydrodynamics requires spatially resolved observation of nonstationary wave-averaged currents because fixed instruments can miss interacting rips, longshore currents and gyres tied to evolving morphology. *Regime: Image-visible surf-zone surface circulation at the study sites.* direct finding, field [56]

36.2 Infragravity motions

Topic nearshore. infragravity. Low-frequency nearshore motions.

Papers screened: 8. Claims: 7. Equations: 1.

Well established. Infragravity motions occupy frequencies below incident wind waves and arise through wave-group forcing, breakpoint variation and nonlinear transfers; they can dominate shoreline runup, harbor or inlet fluctuations and low-frequency surf-zone currents.

Governing physics. Bound long waves generated by group-scale radiation-stress gradients may release as free leaky or trapped edge waves; shoaling, reflection, breaking, bottom friction, directional spreading, bathymetric trapping, currents and rotational shear control their transformation and dissipation.

Dimensionless parameters. Controls include normalized infragravity energy or runup, surf-similarity and relative slope, wave-group bandwidth, directional spread, relative bar geometry, reflection coefficient, friction parameter, current-to-long-wave celerity ratio, relative water depth and vertical shear normalized by depth-mean velocity.

Major equations. Models use depth-integrated mass and momentum equations forced by radiation-stress gradients or phase-resolving nonlinear shallow-water/Boussinesq equations; modal decompositions distinguish bound, leaky and edge waves, while vertically structured theory augments depth averages with frictional shear and rotation.

Typical methods. Studies deploy pressure and vertically stacked velocity sensors, separate frequency and directional bands, estimate bispectra and energy fluxes, decompose modes, compare barred and unbarred transects, force phase-resolving models, close energy balances and validate runup, phase and vertical structure independently.

Numerical models. The set includes phase-resolving barred-beach and inlet simulations, reduced runup analysis and a theoretical vertical-structure extension. Depth-averaged solvers capture many long-wave transformations but omit observed water-column shear and rotation.

Experimental datasets. Reviewed evidence spans controlled runup cases over slopes 0.02-0.04, a modeled barred coastal field site, Albufeira Lagoon Inlet observations and modeling for 25-300 s motions, and Duck94 vertical current profiles around 0.005 Hz.

Validated ranges. Evidence covers selected plane slopes, one barred coast, one wave-dominated inlet and one Duck94 frequency band. The relative importance of leaky, trapped, current-blocked and vertically sheared motions is conditional on local bathymetry and forcing.

Recent advances. Recent advances use dense arrays, remote sensing and distributed pressure sensing, three-dimensional and nonhydrostatic solvers, mode-resolving data assimilation, machine-learned surrogates and coupled infragravity-morphodynamic prediction.

Disagreements. Competing generation descriptions emphasize bound-wave release, breakpoint forcing or nonlinear energy transfer, and observed energy may combine gravity and vortical modes. Depth-averaged agreement in surface elevation does not prove velocity or sediment-transport accuracy.

Limitations. Sparse low-frequency records, finite deployments, directional ambiguity, sensor noise near boundaries, uncertain friction and bathymetry, mode leakage, nonstationary storms, current interaction and limited vertical arrays constrain attribution and transferability.

Open questions. Priorities include unified gravity-vorticity separation, three-dimensional structure, current blocking and Doppler effects, reef and headland trapping, sediment suspension, compound surge interactions, climate dependence and operational assimilation of low-frequency observations.

Seminal papers. Longuet-Higgins and Stewart linked wave groups to bound long waves; breakpoint-forcing and edge-wave theories explained surf-zone generation and trapping, followed by field bispectral studies and nonlinear phase-resolving models.

Equations

- *Spectral-spread infragravity-runup fit* [556]

$$R_s^{ig}/(H_{s,0}L_0)^{1/2} = -0.013\ln[(f_p/f_s)\sigma_{\theta,0}] + 0.058$$

Regime: Idealized matrix with $0.6 < (f_p/f_s)\sigma_{\theta,0} < 3.0$. Variables: R_s^{ig} significant infragravity runup; f_p peak frequency; f_s frequency spread; $\sigma_{\theta,0}$ deep-water directional spread

Claims

1. Over slopes 0.02-0.04, normalized infragravity runup showed no obvious slope dependence and was controlled more strongly by frequency and directional spread. *Regime: Limited moderate-slope, mostly dissipative simulations; not evidence of slope independence outside 0.02-0.04.* direct finding, numerical [556]
2. At the modeled barred field site, infragravity energy was generally dominated by leaky waves, but outer-bar trapped edge waves contributed up to 50% of infragravity variance in mild conditions; shoreward losses shifted among nonlinear transfer, bottom friction, and breaking. *Regime: Gently sloping barred beach across mild and storm conditions represented by a validated local wavefield.* direct finding, numerical [1194]
3. At Albufeira Lagoon Inlet, observed and modeled 25-300 s infragravity motions were blocked during ebb by opposing currents reaching 2.5 m/s, while flood-period fluctuations temporarily reached 100% of the background current magnitude. *Regime: Albufeira Lagoon Inlet observations and simulations, including opposing ebb currents to 2.5 m/s.* direct finding, mixed [145]
4. A theory motivated by Duck94 observations extends depth-averaged low-frequency nearshore dynamics to represent vertical structure associated with frictional and rotational effects. *Regime: Duck94 barred surf zone near 0.005 Hz* direct finding, analytical [837]
5. Duck94 observations near 0.005 Hz showed significant vertical structure in low-frequency surf-zone current energy, phase and rotation: cross-shore flow decayed toward the bed, while the weaker alongshore component could increase near the bed. *Regime: Duck94 barred surf zone near 0.005 Hz* direct finding, field [836]
6. Infragravity transport contributed below 20% on a 1:35 beach but reached about 60% during energetic conditions on a 1:80 beach; HIG/HSW indicated suspension mechanism, direction and relative importance. *Regime: Cross-shore sand transport by infragravity waves as a function of beach steepness* direct finding, field [111]
7. Bispectral laboratory analysis shows shoaling transfers energy from spectral peaks to harmonics and infragravity waves, while nearshore infragravity interactions dominate and cascade energy toward dissipative high frequencies. *Regime: Nonlinear Infragravity-Wave Interactions on a Gently Sloping Laboratory Beach* direct finding, experimental [110]

36.3 Longshore currents

Topic nearshore. Longshore. Alongshore wave-driven currents.

Papers screened: 11. Claims: 9. Equations: 0. Shown in the Coastal Design Bench beside: Beach nourishment.

Well established. Obliquely breaking waves drive alongshore momentum and currents whose cross-shore and vertical shear can become unstable. The resulting mean flow, shear waves, vortices, and low-frequency eddies advect and disperse sediment, tracers, organisms, heat, and contaminants.

Governing physics. Controls include wave angle, height, directional spread and breaking, radiation stress or vortex force, rollers, setup gradients, bathymetry, vertical shear, bottom and turbulent mixing, shear instability, vortex tilting, and upscale energy transfer.

Dimensionless parameters. Important controls include breaker angle, relative water depth, wave height and directional spread, current-to-wave velocity scale, cross-shore shear, friction and mixing coefficients, roller scale, eddy scale relative to surf-zone width, and normalized vertical elevation.

Major equations. Frameworks use wave-action and circulation momentum equations with radiation-stress or vortex-force coupling, roller and breaking closures, turbulence models, shear-wave energy and enstrophy balances, Fickian tracer diffusion, and vertical power laws.

Typical methods. Methods combine current meters and vertical profiles, continuous dye releases, two-dimensional sensor arrays, long-duration low-frequency observations, bathymetric assimilation, theoretical instability analysis, vortex-force circulation models, SWASH, and coupled SWAN-FVCOM.

Numerical models. Depth-uniform, quasi-three-dimensional, and fully three-dimensional models differ in shear-wave energy, confinement, vortex interactions, and vertical profiles. Bathymetric assimilation and spatially variable roller/turbulence parameters improve skill.

Experimental datasets. Evidence includes six 1-2 h Huntington Beach dye releases, 120 days of surf-zone currents, Duck two-dimensional arrays and Duck'94 validation, SUPERDUCK conditions, plane/barred-beach profiles, and barred-beach waves from 0.5-2.0 m at 0-15 degrees.

Validated ranges. Reported evidence includes cross-shore diffusivity of 0.5-2.5 m²/s, low-frequency motions at 0.1-4.0 mHz over 120 days, validated 0.5-2.0 m waves at 0-15 degrees, O(10 m) and O(100 m) eddies, and profile exponents from 1/10 to 1/3.

Recent advances. Recent advances combine months-long observations, phase-resolving three-dimensional models, directional-spread-dependent eddy scaling, bathymetric inversion, and explicit inverse-cascade diagnostics to improve transport prediction.

Disagreements. Low-frequency surf-zone currents are not always explained by instability of the mean alongshore current: a 120-day record showed weak correlation with current strength and shear and stronger consistency with breaking-wave-driven inverse energy transfer.

Limitations. Steady, depth-uniform, fixed-bathymetry assumptions omit rapid morphology, vertical dispersion, rollers, breaking turbulence, inverse cascades, directional spectra, wind and tides. Sparse field arrays and closure calibration limit transfer.

Open questions. Priorities include partitioning shear instability from breaking-generated cascades, resolving three-dimensional saturation and vertical exchange, assimilating storm-time bathymetry, and predicting material pathways under combined wave, wind, and tidal forcing.

Seminal papers. The shear-instability review traces the discovery and theory of shear waves and their finite-amplitude development; later quasi-3D and vortex-force studies extend that lineage to vertical structure and realistic bathymetry.

Claims

1. Six Huntington Beach dye releases in wave-driven alongshore currents yielded cross-shore diffusivities of 0.5-2.5 m²/s, with plumes often mixed across the surf zone after a few hundred metres and rotational motions contributing to dispersion. *Regime: Huntington Beach surf zone; releases of 1-2 h over varied waves and currents* direct finding, field [284]
2. At Duck, ensemble assimilation of wave height and alongshore current corrected bathymetry and circulation, showing that hourly storm-driven morphology changes can make daily bathymetric surveys inadequate for current prediction. *Regime: Duck natural beach with two-dimensional wave and current sensor array* direct finding, mixed [1519]
3. A three-dimensional Un-SWAN-FVCOM vortex-force model agreed with laboratory and Duck'94 measurements across oblique and barred-beach cases, while spatially variable roller and turbulence effects improved vertical current profiles. *Regime: planar beach, barred-beach laboratory case, Duck'94 field case, and breakwater case* direct finding, numerical [1612]
4. Over 120 field days, 0.1-4.0 mHz surf-zone currents were weakly correlated with alongshore-current shear or strength but were consistent with an inverse energy cascade energized by short-crested breaking waves. *Regime: shallow surf zone across a broad wave range; 0.1-4.0 mHz motions* direct finding, field [403]
5. Relative to depth-uniform modelling, quasi-three-dimensional SHORECIRC produced weaker, more shoreward-confined shear waves but similar total momentum mixing through combined shear-wave and depth-varying-current transfers. *Regime: idealized SUPERDUCK topography and 16 October 1986 waves* direct finding, numerical [1605]
6. SWASH reproduced alongshore-current and rip-cell trends for 0.5-2.0 m waves at 0-15 degrees, with directional spread intensifying roughly 10 m eddies and alongshore bathymetric variability enhancing roughly 100 m eddies. *Regime: alongshore-variable barred beach; H=0.5-2.0 m and direction=0-15 degrees* direct finding, mixed [105]
7. Shear waves arise from instability of wave-driven alongshore currents, with occurrence and finite-amplitude behavior governed by current structure, mixing, and perturbation dynamics documented across theory, field, and laboratory work. *Regime: wave-driven alongshore currents in field, laboratory, and theory* review statement, review [348]
8. Longshore-current profiles were generally represented by a 1/10 power law across laboratory and field data, but barred-beach trough structure required a substantially different 1/3 exponent. *Regime: plane beach, bar trough, bar crest, and published profile datasets* direct finding, mixed [1590]
9. Tri-cusped beach measurements resolved repeated rip-fed circulation cells for normal waves and a stable meandering longshore current above 0.5 m/s for oblique waves, with predominantly depth-uniform rip flow above the boundary layer. *Regime: Wave-induced nearshore currents at a tri-cusped beach in the UKCRF* direct finding, mixed [173]

36.4 Rip currents

Topic nearshore .rips. Seaward-directed concentrated currents.

Papers screened: 23. Claims: 22. Equations: 0.

Well established. Rip currents are concentrated seaward flows in and beyond the surf zone. They may form exiting jets or semi-enclosed circulations and arise from hydrodynamic instability, bathymetric focusing, or lateral boundaries.

Governing physics. Breaking-wave setup gradients, pressure gradients, vorticity, advection, vortex force, wave direction and spectrum, water level, bathymetry, and boundary geometry jointly determine rip occurrence, strength, position, and pathway.

Dimensionless parameters. Transfer depends on normalized wave height and breaking state, wave angle and spectral partition, relative water level and bar/channel geometry, morphology wavelength relative to surf-zone width, and time or separation scaling of dispersion.

Major equations. Reviewed approaches span extended-Boussinesq wave-current equations, depth-averaged momentum and continuity, wave-action and breaking closures, sediment continuity and transport, Lagrangian dispersion measures, and ensemble state estimation.

Typical methods. Methods combine process classification, field current meters and drifters, remote sensing, dredged channels, movable-bed laboratories, Boussinesq and wave-averaged circulation models, morphodynamic stability models, COAWST, and ensemble Kalman filtering.

Numerical models. Models reproduce laboratory rip instability, bar-channel self-organization, nonlinear merging and migration, breaking and setup gradients in dredged channels, tidal modulation, canyon-controlled intermittency, and bathymetric data assimilation.

Experimental datasets. Evidence includes eight movable-bed morphologies, five roughly 30 m wide and 1.5 m deep dredged channels, seven Sand City drifter days, 32 macro-tidal cycles, Duck remote sensing, and month-long canyon-coast observations.

Validated ranges. Reported values include morphological wavelengths of 180-250 m linearly and 165-320 m nonlinearly, cross-shore diffusivity initially 5.4-6.1 m²/s, total-to-turbulent dispersion ratios of 3-30, and macro-tidal peaks about 1.5 h around low water.

Recent advances. Recent advances use dense drifters, remote-sensing assimilation, controlled dredged channels, time-varying morphodynamic forcing, and month-scale spectral simulations to resolve intermittency, retention, and forecast sensitivity.

Disagreements. The traditional persistent exit-flow picture is incomplete: field evidence also shows recirculation and nearshore retention. Regular spacings from stationary models contrast with irregular morphology under time-varying forcing and antecedent-bathymetry sensitivity.

Limitations. Prediction is limited by unknown and evolving bathymetry, transient wave spectra and angles, tide, mixed forcing mechanisms, model dimensionality and closures, sparse three-dimensional observations, and uncertain links from flow to human hazard.

Open questions. Key needs are operational flash-rip prediction, three-dimensional vertical structure, morphology-aware ensembles, boundary and offshore-feature interactions, swimmer and rescue response, and standardized hazard validation across rip types.

Seminal papers. The branch connects early extended-Boussinesq rip-system validation and instability-based bar models to later Lagrangian field studies and the six-type process classification; detailed historical attribution remains a full-text task.

Claims

1. Rip currents can be classified as hydrodynamically controlled shear-instability or flash rips, bathymetrically controlled channel or focused rips, and boundary-controlled deflection or shadow rips, while real beaches commonly mix these mechanisms. *Regime: wave-exposed coasts and six rip types* review statement, review [238]
2. A fully nonlinear extended-Boussinesq model with localized breaking dissipation and moving shoreline reproduced measured wave height, mean water level, velocity components, and rip-current instability. *Regime: laboratory rip-current system with moving shoreline* direct finding, numerical [256]
3. In the modelled single-barred coast, rip-cell spacing scaled with shore-to-bar crest distance, increased with wave height toward saturation, and responded jointly to wave incidence angle and period. *Regime: straight single-barred sandy coast under normal and near-normal waves* direct finding, numerical [224]
4. Modelled feedback between wave-driven currents and sediment-concentration gradients favoured crescentic bars under energetic fine-sediment conditions and transverse bars under milder coarse-sediment conditions. *Regime: initially planar sandy slope under normally incident saturated-surf-zone waves* direct finding, numerical [222]
5. A nonlinear morphodynamic model reproduced longshore-bar to rhythmic-bar to transverse-bar-and-rip transitions, with wavelengths of 180-250 m in the linear regime and 165-320 m in the nonlinear regime. *Regime: longshore bar evolving through linear and nonlinear regimes* direct finding, numerical [483]
6. Across eight moveable-bed morphologies under identical waves, rip-current intensity increased linearly with beach alongshore nonuniformity, while neck-flow pulsation and directional variability changed systematically with morphology. *Regime: eight beach morphologies under identical shore-normal waves and water level* direct finding, experimental [236]
7. At Duck, assimilating remote-sensing products with an ensemble Kalman filter improved nearshore bathymetry estimates and rip-current forecasts without requiring in-situ observations. *Regime: September 2010 Duck, North Carolina field experiment* direct finding, mixed [1520]
8. Rip-channel spacing and growth are inherently sensitive to the initial cross-shore bathymetry at a level comparable to wave-height sensitivity, so predictors must retain antecedent morphology. *Regime: pattern development on barred beaches* direct finding, numerical [225]
9. Time-varying wave incidence produced different mean rip spacing, weaker and more irregular morphology, more merging and splitting, and net longshore migration not reproduced by time-invariant forcing. *Regime: rip-channel formation under time-varying versus integrated stationary forcing* direct finding, numerical [237]
10. For five dredged surf-zone channels, maximum offshore flow correlated with breaking-driven setup gradients and alongshore flow, increasing with bar dissipation until limited by the vertical scale of bathymetric variability. *Regime: channels about 30 m wide and 1.5 m deep across the surf zone* direct finding, mixed [987]
11. At Sand City, rip flow produced initial cross-shore diffusivity of 5.4-6.1 m²/s before declining to 0.9-2.2 m²/s, and total dispersion exceeded small-scale turbulent diffusion by factors of 3-30. *Regime: rip-channeled Sand City beach across waves and tides* direct finding, field [195]
12. On a rip-channeled beach, drifters exited in rip pulses and returned over shoals, demonstrating wave-driven surf-zone-inner-shelf exchange that retained surface material nearshore and was not due to Stokes drift alone. *Regime: rip-channeled Sand City beach under weak winds* direct finding, field [196]
13. Across 32 tidal cycles, macro-tidal rip-neck flow peaked about 1.5 hours before and after low water as breaking and alongshore pressure gradients shifted across bar-rip morphology. *Regime: macro-tidal bar-rip beach across wave and tidal conditions* direct finding, mixed [88]
14. During a month-long canyon-coast simulation, incident energy direction and the energy ratio of bimodal spectra controlled rip-current formation, wave period controlled position, and persistent offshore bathymetry dominated the circulation response. *Regime: Nearshore Canyon Experiment coast with strong offshore bathymetric gradients* direct finding, mixed [863]

15. Scanning sonar observed episodic rip jets recurring roughly one to four times per hour, lasting minutes and reaching about 0.7 m/s while exporting warm, turbid, bubble-rich surf-zone water offshore. *Regime: Observations of nearshore circulation: Rip currents* direct finding, mixed [1314]
16. On a longshore-uniform swell beach, transient non-topographic rips formed 20–30 m necks and spreading vortex-rich heads; intermittent heads greatly enhanced dispersion outside the surf zone. *Regime: Transient rip currents and nearshore circulation on a swell-dominated beach* direct finding, mixed [668]
17. Dense laboratory drifters showed highly unsteady rip circulation, significant Stokes-drift contributions, circulation generation balanced by bottom-friction dissipation, and biases from depth-varying transport and drifter concentration. *Regime: Drifter measurements in a laboratory rip current* direct finding, mixed [700]
18. A roughly 0.3 m Lake Michigan meteotsunami generated hazardous rip currents on an O(1 m) model grid within minutes of calm conditions and sustained them for hours, identifying a distinct rip-generation mechanism. *Regime: Unexpected rip currents induced by a meteotsunami* direct finding, mixed [835]
19. At a headland-and-reef beach, small wave/tide changes shifted three circulation regimes; under 4 m oblique waves a deflection rip reached 0.7 m/s 800 m offshore in 12 m depth and could export sediment beyond closure depth. *Regime: High-Energy Surf Zone Currents and Headland Rips at a Geologically Constrained Mesotidal Beach* direct finding, mixed [990]
20. Four North Sea field datasets indicate transitions between undertow and rip-cell circulation depend jointly on onshore wave mass transport, rip spacing and trough cross-section, separating regimes in a simple parameter space. *Regime: Cross-Shore Currents in the Surf Zone: Rips or Undertow?* direct finding, mixed [4]
21. Among 407 UK beach users, 263 showed poor rip-current knowledge, while 314 understood safety flags and 339 complied; prior education improved knowledge and lifeguards were the most effective source. *Regime: Public Understanding and Knowledge of Rip Currents and Beach Safety in the UK* direct finding, field [1528]
22. Video and buoy analysis at Alassio associated rips with H_s 0.5–1.34 m, mean periods 4.7–7.0 s and directions 150–227°, yielding high annual hazard but moderate tourist-season hazard. *Regime: Rip current hazard assessment on a sandy beach in Liguria, NW Mediterranean* direct finding, mixed [991]

36.5 Surf-zone circulation

Topic *nearshore*. *surfzone*. Wave-driven circulation within the surf zone.

Papers screened: 19. Claims: 15. Equations: 0.

Well established. The surf zone transforms incident waves through shoaling, breaking, rollers, reflection, bottom friction, and nonlinear interactions. These processes drive setup, undertow, alongshore flow, turbulence, eddies, dispersion, and sediment transport.

Governing physics. Key controls are breaker type and location, wave height and period, directional spread, crest length, roller geometry, turbulence production and advection, vorticity cascades, undertow, bed roughness, slope, reflection, wind, and water depth.

Dimensionless parameters. Important groups include breaker index, Iribarren-type slope and steepness measures, relative depth, reflection coefficient, roller geometry and density, roughness ratio, directional spread, normalized dissipation, and eddy scale relative to surf-zone width.

Major equations. Frameworks use wave-energy and roller balances, hydraulic-jump dissipation, RANS or k-epsilon turbulence closures, Boussinesq phase-resolving equations, vortex-force circulation, Lagrangian dispersion statistics, and setup momentum balances.

Typical methods. Methods include dense drifter releases, LDA/ADV velocity profiles, field LiDAR rollers, storm arrays, phase-resolving Boussinesq models, vertical turbulence models, two-phase air-water RANS, Radon wave separation, and 3-D vortex-force circulation.

Numerical models. Models span wave-resolving Boussinesq dispersion, vertical k-epsilon circulation, phase-resolving reflection, roller dissipation, coupled three-dimensional vortex-force circulation, and air-water two-phase breaking with wind.

Experimental datasets. Evidence includes about 70 drifter releases, Duck94, a 4 s and 0.85 m large-scale barred-beach breaker at 12 stations, prototype BARDEXII reflection tests, LiDAR roller observations, and a shore-platform array from 10 m depth to shore.

Validated ranges. Reported regimes include $H_s=0.5$ –1.4 m drifter days, 5–50 m observed eddies, 5–10 m enstrophy and 20–100 m inverse-cascade scales, undertow to 0.8 m/s, reflection-driven gamma changes to 40%, and roughness-driven setup contributions to 26%.

Recent advances. Recent advances directly measure roller geometry, resolve wave-by-wave reflection, observe scale-dependent drifter dispersion, quantify residual breaker-bar turbulence, and separate roughness effects on dissipation from circulation-driven setup.

Disagreements. Bulk or equilibrium closures can fail locally: roller-area formulae overestimated measured dissipation, log-profile stress inference fails when breaking homogenizes shear, and outgoing swash waves materially change wave ratios and undertow that incident-only analyses omit.

Limitations. Laboratory scale, two-dimensionality, fixed morphology, uncertain aeration and roller density, sparse vertical observations, wind and roughness parameterization, reflection separation, and intermittent directional breaking limit transfer.

Open questions. Needs include unified bubble-roller-turbulence energetics, three-dimensional breaking and cascades, morphology feedback, wind-wave-current coupling, rough natural platforms, reflection under random storms, and operational dispersion uncertainty.

Seminal papers. The branch connects bore and roller energy concepts, breaking turbulence closures, Boussinesq wave-resolving circulation, and Lagrangian dispersion theory; exact historical formulation lineage remains a full-text task.

Claims

1. About 70 surf-zone drifter releases showed anisotropic time-dependent dispersion, 5–50 m eddies, relative dispersion proportional to $\text{time}^{3/2}$, and scale-dependent diffusivity proportional to $\text{separation}^{2/3}$. *Regime: alongshore-uniform beach within 200 m of shore and depths below 5 m; $H_s=0.5$ and 1.4 m* direct finding, field [1339]
2. A vertical turbulence model showed that breaking-wave turbulence substantially reduces surf-zone vertical shear, so fitting a logarithmic current profile can misestimate bottom stress even when mean alongshore flow is reproduced. *Regime: 4.5 m depth sandy beach and Duck94 cross-shore transect* direct finding, numerical [427]

3. A Boussinesq model reproduced observed drifter dispersion through breaking-generated rotational flow, with an enstrophy cascade at about 5-10 m and inverse-energy cascade at 20-100 m that strengthened with wave directional spread. *Regime: uniform natural beach under small normally incident directionally spread waves* direct finding, mixed [1340]
4. In a large-scale barred-beach experiment, a 4 s, 0.85 m plunging wave generated undertow to 0.8 m/s, persistent turbulence reaching the bed, and undertow TKE transport about an order of magnitude larger than wave and turbulent fluxes. *Regime: $T=4$ s, $H=0.85$ m regular wave over fixed barred beach at 12 stations* direct finding, experimental [2]
5. For prototype-scale beaches steeper than 1:9, swash reflection changed individual wave-height-to-depth ratios by up to 25-40%, altered significant-wave ratios by 15%, strengthened undertow, and generated near-bed onshore streaming. *Regime: prototype-scale steep beach with $\tan(\beta)>1:9$ under regular and irregular waves* direct finding, numerical [924]
6. Direct LiDAR roller measurements showed that common roller-area formulations overestimate inner-surf-zone dissipation, which was close to but below an equal-height hydraulic jump and was better represented using roller-density effects. *Regime: broken waves in the inner surf zone* direct finding, mixed [925]
7. On a storm-exposed shore platform, bottom friction caused about 40% of wave dissipation, while roughness both reduced setup through prebreaking attenuation and increased circulation-driven setup by up to 26% on a 1:20 slope. *Regime: gently sloping shore platform instrumented from 10 m depth to shore* direct finding, mixed [776]
8. Two-phase modelling over a 1:35 beach showed that wind increases vorticity and water-particle velocity and causes spilling and plunging waves to break earlier and farther seaward. *Regime: periodic breakers over a 1:35 beach with wind* direct finding, numerical [1543]
9. Linear stability analysis shows radiation-stress-driven circulation can amplify crescentic surf-zone bed patterns, while allowing breaker-line displacement suppresses a giant-cusp mode but leaves crescentic growth largely unchanged. *Regime: A mechanism for the generation of wave-driven rhythmic patterns in the surf zone* direct finding, mixed [421]
10. On a reflective beach with 26–32 m cusps, meiofauna, macrofauna, zooplankton and fish distributions differed between horns, bays, rip-current margins and rip heads, linking circulation and morphology to habitat use. *Regime: Faunal response to morphology and water circulation of a sandy beach with cusps* direct finding, mixed [945]
11. Validated SPH simulations show intersecting short-crested waves generate undertow, rip currents, isolated breaking crests, multiple low-frequency circulation cells and three-dimensional vorticity under both discrete and directional wavefields. *Regime: Short-crested waves in the surf zone* direct finding, mixed [1500]
12. WAMFlow optical processing resolved kilometre-alongshore surf-zone currents and captured longshore flows, rips and gyres; filtered velocities had about 0.1 m/s RMSE against drifters. *Regime: Quantifying Optically Derived Two-Dimensional Wave-Averaged Currents in the Surf Zone* direct finding, mixed [56]
13. A calibrated Sète model found breakwater-edge differential breaking drove 0.4 m/s circulation and suppressed inner-bar rip-channel development and offshore bar transport in the lee; longshore current RMSE was 0.07 m/s. *Regime: Modeling the Impact of the Implementation of a Submerged Structure on Surf Zone Sandbar Dynamics* direct finding, mixed [181]
14. A time-averaged surf-zone model using roller-inclusive approximations for energy flux, radiation stress and dissipation reproduces measured wave-height and mean-water-level setup and interprets the post-breaking transition through shoreward roller transport. *Regime: Regular breaking waves and the laboratory surf-zone measurements evaluated in the article.* direct finding, analytical [1381]
15. A wave-current interaction review finds mean bed shear may increase, decrease or remain unchanged depending on nonlinear interaction, relaminarization or linear superposition; non-collinear waves can rotate currents with depth, while common eddy-viscosity and URANS models remain weakly validated outside near-bed regions. *Regime: Turbulent wave-current interaction in nearshore boundary layers, including collinear and oblique forcing.* review statement, review [1594]

37 Wiki: Numerical modeling

37.1 Numerical modeling: overview

Topic *numerical*. Computational representation of coastal systems.

Papers screened: 13. Claims: 3. Equations: 0.

Well established. Coastal numerical models discretize conservation laws and process closures to represent waves, circulation, sediment, morphology and structures. Credible use requires equations and resolution appropriate to scale plus calibration, validation, sensitivity and uncertainty evidence matched to the decision.

Governing physics. Models conserve mass, momentum and tracers while representing pressure, gravity, turbulence, viscosity, free surfaces, wave dispersion and breaking, bottom stress, porous or structural forces and sediment exchange; coupled systems pass forcing and feedback among components.

Dimensionless parameters. Resolution and regime depend on Courant, Reynolds, Froude, Keulegan–Carpenter, relative-depth, steepness, Ursell, Shields, grid-to-feature and time-step ratios, plus error, skill, convergence and sensitivity measures.

Major equations. Common frameworks include depth-averaged and three-dimensional Reynolds-averaged Navier–Stokes equations, LES, volume-of-fluid and particle methods, nonhydrostatic and Boussinesq systems, spectral wave-action balance, sediment advection–diffusion and Exner bed evolution.

Typical methods. Workflows define purpose and conceptual model, assemble bathymetry and boundaries, generate meshes, choose closures, calibrate limited parameters, verify numerics, validate independent variables, quantify sensitivity and uncertainty, and document applicability and reproducibility.

Numerical models. Reviewed approaches include OpenFOAM LES, COAWST wave–tide coupling, RANS–VOF, SPH, nonhydrostatic phase-resolving reef flow, REEF3D scour and MIKE21 FM coupled waves and hydrodynamics, alongside general model-assurance guidance.

Experimental datasets. Evidence includes ultrasonic dune winds and LiDAR terrain, British shelf wave climates, dam-break swash, large-scale barred-beach breaking, rough and smooth reef flumes, pile-scour benchmarks and Iraqi coastal water-level, wind and wave observations.

Validated ranges. Examples report swash r^2 above 0.97 and stress error within 20%, reef geometry from a 1:5 fore-reef to 500 m prototype flat, 49–93% oblique wave occurrence at tidal sites and 46.8% wave-height bias when Iraqi wind forcing is omitted.

Recent advances. Recent advances apply particle and nonhydrostatic methods at near-natural scales, couple wave and tide resources, resolve reef roughness, use LiDAR-derived complex terrain, quantify field-linked model skill and emphasize fit-for-purpose uncertainty documentation.

Disagreements. Higher-fidelity models expose vortices, aeration and three-dimensional response but cost more and still require closures; simpler phase-averaged or depth-averaged systems support regional prediction yet can omit process-scale dynamics. Agreement in one variable does not guarantee all fluxes are correct.

Limitations. Uncertain boundary data and bathymetry, grid and time-step dependence, turbulence and breaking closures, scale effects, sparse validation, parameter compensation, one-way coupling, computational cost and incomplete uncertainty reporting constrain confidence.

Open questions. Priorities include coupled atmosphere–wave–ocean–sediment prediction, differentiable and adjoint methods, adaptive meshes, hybrid physics–ML, reproducible benchmarks, probabilistic calibration, extreme-event validation and decision-relevant error propagation.

Seminal papers. Finite-difference shallow-water and spectral wave models established regional simulation; RANS, Boussinesq, boundary-element and morphodynamic approaches expanded process coverage, while formal verification and validation established numerical credibility practice.

Claims

1. A 2018 coastal-modeling workshop identified common benchmarks, grid and validation standards, standardized I/O and reporting, and stronger parameter estimation and uncertainty quantification as central requirements for reducing subjectivity and improving reproducibility. *Regime: Findings of a 2018 United States workshop on physical coastal and estuarine modeling.* review statement, review [461]
2. Fit-for-purpose shelf and estuary modeling requires explicit setup, input-data, calibration, validation, error, uncertainty and performance-metric documentation rather than model execution alone. *Regime: Guidance on Setup, Calibration, and Validation of Hydrodynamic, Wave, and Sediment Models for Shelf Seas and Estuaries* review statement, review [1517]
3. A coastal-hydrodynamics review identifies tidal and wave modeling foundations and emerging SPH fluid–solid interaction, adjoint prediction, renewable-energy and large-scale morphodynamic applications. *Regime: Coastal hydrodynamics – present and future* review statement, review [1348]

37.2 Computational fluid dynamics

Topic *numerical*. cfd. RANS, LES, DNS, and multiphase methods.

Papers screened: 27. Claims: 18. Equations: 0.

Well established. Credible coastal CFD requires separate verification of conservation, discretization, interface transport and wave boundaries plus validation against measurements for the target processes; agreement in surface elevation alone does not establish turbulence, pressure, load or transport accuracy.

Governing physics. Coastal CFD resolves incompressible momentum and continuity with gravity, viscosity and turbulence while tracking air–water and sometimes sediment or granular phases; nonlinear waves, breaking, runup, overtopping, entrained air, currents and moving or porous boundaries create strong multiscale coupling.

Dimensionless parameters. Key controls include Reynolds, Froude, Weber and Keulegan–Carpenter numbers, relative depth and steepness, Courant number, cells per wavelength and wave height, y-plus, interface thickness, reflection coefficient, mass error, grid-convergence index and runtime or cell-count ratios.

Major equations. Core equations are Navier–Stokes mass and momentum conservation, Reynolds-averaged or filtered turbulence equations, phase-fraction advection for VOF/CLSVOF, constitutive stress or rheology for additional phases, and kinematic/dynamic free-surface and wave-generation boundary conditions.

Typical methods. A defensible workflow verifies still water and conservation, calibrates wave generation and absorption, performs mesh/time-step and domain tests, compares turbulence and interface schemes, validates kinematics and integral outputs against laboratory data, audits reflection and mass balance, then reports cost and uncertainty.

Numerical models. The evidence spans OpenFOAM interFoam, multiPhaseInterFoam and sedInterFoam, ANSYS Fluent, DualSPHysics, SWASH, HOS-NWT, fully nonlinear potential solvers, RANS k-epsilon and k-omega, LES, VOF and CLSVOF, plus one- and two-way domain decomposition.

Experimental datasets. Reviewed benchmarks include dike overtopping, regular deep-water propagation, steep focused groups, dam break, solitary runup, submerged obstacles, shallow-foreshore transformation, numerical-flume turbulence tests, irregular-wave generation and multiphase sediment or landslide cases.

Validated ranges. Validation covers selected two- and three-dimensional flume-scale regular, solitary, focused, irregular, overtopping, breaking and multiphase cases; solver performance remains conditional on grid, boundary treatment, turbulence closure, phase model and application geometry.

Recent advances. Recent advances use coupled potential–viscous domains, cross-solver benchmark campaigns, multiphase rheology, three-phase air–water–sediment models, improved interface schemes, realistic spectral boundaries and explicit accuracy–cost intercomparison across open-source solvers.

Disagreements. No solver is uniformly superior: viscous multiphase CFD resolves breaking and local loads but is costly; potential, spectral and nonhydrostatic models are efficient away from strongly viscous regions; RANS, LES, VOF, CLSVOF and SPH trade dissipation, interface sharpness, robustness and resolution differently.

Limitations. Persistent limitations include numerical diffusion and reflection, mass imbalance, grid and time-step sensitivity, turbulence overproduction beneath waves, uncertain air entrainment and compressibility, expensive three-dimensional resolution, closure dependence, boundary inconsistency and sparse validation of local extremes.

Open questions. Priorities include conservative sharp interfaces, robust breaking turbulence, air-pocket compressibility, adaptive meshes, porous and moving media, flexible coupling, sediment feedback, uncertainty quantification, reproducible high-performance implementation and field-scale validation.

Seminal papers. Finite-volume Navier–Stokes solvers, Reynolds averaging, LES, VOF free-surface capture and numerical wave tanks established modern coastal CFD; absorbing boundaries, wave-generation toolboxes and formal verification/validation made engineering applications increasingly reproducible.

Claims

1. The parabolic-basin calculation changed total water volume by approximately $1e-7$ relative after ten periods, although accuracy deteriorated after several periods because a Cartesian grid represented the circular geometry. *Regime: The paper's frictionless circular parabolic-basin test on a Cartesian grid.* direct finding, numerical [1303]
2. A review of Navier–Stokes oscillating-water-column simulations finds potential-flow models can overpredict performance and organizes CFD choices for viscous, turbulent, free-surface and air-chamber effects. *Regime: A review of computational methods for studying oscillating water columns – the Navier-Stokes based equation approach* review statement, review [1055]
3. A two-dimensional OpenFOAM model validated against dike experiments evaluates overtopping for berms and roughness beyond the safe extrapolation range of empirical formulae. *Regime: Numerical modelling of wave overtopping at dikes using OpenFOAM®* direct finding, numerical [264]
4. A multiPhaseInterFoam solver represents granular landslide tsunamis with Coulomb viscoplastic rheology and a small set of physical parameters, validated against generation and propagation evidence. *Regime: Numerical modeling of tsunamis generated by granular landslides in OpenFOAM®: A Coulomb viscoplastic rheology* direct finding, numerical [1208]
5. A zonal method weakly couples two-phase OpenFOAM near a structure with fully nonlinear potential flow in the outer wave–current domain to concentrate viscous resolution where needed. *Regime: A zonal hybrid approach coupling FNPT with OpenFOAM for modelling wave-structure interactions with action of current* direct finding, numerical [814]
6. Intercomparison of interFoam, DualSPHysics and SWASH against a large-scale shallow-foreshore dike experiment exposes solver-dependent accuracy and cost across simultaneous transformation and overtopping processes. *Regime: An Inter-Model Comparison for Wave Interactions with Sea Dikes on Shallow Foreshores* direct finding, numerical [544]
7. An experimental numerical-flume comparison evaluates k-epsilon, k-omega and LES with VOF, wave generation and reflection analysis to select turbulence treatment by scenario. *Regime: Performance Assessment of Three Turbulence Models Validated through an Experimental Wave Flume under Different Scenarios of Wave Generation* direct finding, experimental [471]
8. A two-dimensional numerical-flume study compares wave-generation and absorption methods for nonbreaking regular deep-water waves and emphasizes validation and reflected-wave control. *Regime: COMPARISON OF DIFFERENT METHODS FOR GENERATION AND ABSORPTION OF WATER WAVES* direct finding, numerical [291]
9. Four Navier–Stokes benchmarks show plunging-breaker prediction depends on grid resolution, turbulence closure and VOF versus CLSVOF interface treatment, with fine grids needed for splash-up details. *Regime: Assessment of Numerical Methods for Plunging Breaking Wave Predictions* direct finding, numerical [153]

10. *sedInterFoam* adds an air phase and VOF free surface to a water–sediment Eulerian solver and validates interface evolution, phase mass conservation and predictive behavior across four tests. *Regime: sedInterFoam 1.0: a three-phase numerical model for sediment transport applications with free surfaces* direct finding, numerical [934]
11. Two-way domain decomposition couples an outer fully nonlinear potential solver to an inner VOF Navier–Stokes region so waves enter and leave a localized viscous wave–structure subdomain. *Regime: Wave–structure interaction by a two-way coupling between a fully nonlinear potential flow model and a Navier–Stokes solver* direct finding, numerical [766]
12. OpenFOAM, SWASH and HOS-NWT comparison against steep focused-wave experiments quantifies the accuracy–cost tradeoff among viscous, nonhydrostatic and high-order spectral numerical flumes. *Regime: Intercomparison of Three Open-Source Numerical Flumes for the Surface Dynamics of Steep Focused Wave Groups* direct finding, experimental [1471]
13. WaveMIMO guidance and ANSYS-Fluent validation use spectral orbital-velocity boundary conditions to generate realistic irregular waves in a multiphase numerical channel. *Regime: Theoretical Recommendations and Validation for the Generation of Realistic Irregular Waves Through the WaveMIMO Methodology* direct finding, experimental [1073]
14. OpenFOAM LES on a roughly 22-million-cell LiDAR-derived dune grid reproduced measured offshore-wind reversal up to 20 m seaward of the embryo-dune toe and identified conditions exceeding dry-sand motion threshold. *Regime: Investigation of three-dimensional wind flow behaviour over coastal dune morphology under offshore winds using computational fluid dynamics (CFD) and ultrasonic anemometry* direct finding, mixed [641]
15. A RANS–VOF swash model achieved r^2 above 0.97 for runup, depth and near-bed velocity and bed stress within 20% of profile-based estimates, resolving reversal, backwash boundary-layer growth and slope-dependent net stress. *Regime: Modeling swash-zone hydrodynamics and shear stresses on planar slopes using Reynolds-Averaged Navier–Stokes equations* direct finding, mixed [1423]
16. Large-scale barred-beach SPH reproduced surface elevation and mean velocities reasonably but diverged in aerated breaking and bar-trough edges; simulations revealed recirculation and collapse-generated vortex pairs absent from sparse measurements. *Regime: Large-scale wave breaking over a barred beach: SPH numerical simulation and comparison with experiments* direct finding, mixed [46]
17. Calibrated REEF3D RANS simulations indicate current-dominated pile hydrodynamics for co- and counter-directed waves; extrapolated equilibrium scour is smaller for opposing flow and requires modifying the usual relative-current parameter. *Regime: Numerical Study of the Hydrodynamics of Waves and Currents and Their Effects in Pier Scouring* direct finding, mixed [1139]
18. A review of numerical wave flumes identifies free-surface Navier–Stokes interface-capturing methods and their roles in innovative coastal-structure design. *Regime: Computational wave dynamics for innovative design of coastal structures* review statement, review [516]

37.3 Model coupling and nesting

Topic numerical.coupling. Multi-model and multi-scale coupling.

Papers screened: 22. Claims: 15. Equations: 0.

Well established. Coupling is necessary when feedback or scale interaction affects the answer; a coupled system must be evaluated against uncoupled baselines and observations because individually valid components can exchange inconsistent states, fluxes or boundary information.

Governing physics. Coupled coastal systems exchange momentum, mass, energy and tracers among atmosphere, waves, circulation, rivers, inundation and morphology. Feedback can alter wind stress, wave growth, currents, water levels and dissipation rather than merely pass forcing downstream.

Dimensionless parameters. Controls include parent-to-child grid ratio, coupling interval relative to process time scale, Courant number, overlap width, relaxation strength, flux or mass imbalance, interpolation error, scale-separation ratio, feedback gain, forecast skill change and parallel efficiency.

Major equations. Coupling links component conservation equations through boundary conditions and source terms: stress and heat flux, radiation stress or vortex force, wave action, water level and velocity, discharge, sediment flux and moving geometry, with interpolation, relaxation and conservative remapping operators.

Typical methods. Workflows define components and exchanged fields, choose online/offline and one-/two-way direction, harmonize coordinates and conventions, remap states or fluxes, synchronize clocks and restarts, test idealized interfaces and conservation, compare forced and coupled runs, then validate integrated outputs.

Numerical models. Reviewed systems include OASIS3-MCT/SURFEX infrastructure, COSMO-WAM, wave-coupled shelf models, FVCOM inundation nesting, TELEMAC wave–3D hydrodynamics, WAVEWATCH III domain decomposition and high-resolution coupled atmosphere–ocean models.

Experimental datasets. Evidence includes nested Mediterranean and Adriatic circulation, Hurricane Ike waves, California upwelling, Gulf Stream and Agulhas currents, Scituate nor'easter inundation, southern North Sea forecasts, northwest-European shelf operations and Taranto multi-scale observations.

Validated ranges. The reviewed systems span submesoscale coastal currents to global-parent/regional-child hierarchies and event to operational scales; demonstrated skill is conditional on component versions, exchanged fields, grid hierarchy, coupling interval, forcing and regional observations.

Recent advances. Recent advances use community middleware, unstructured nested inundation, multi-way wave–ocean–atmosphere exchange, high-resolution current feedback, domain-decomposed parallel waves and validated open-source multiple-nesting workflows for operational coasts.

Disagreements. One-way offline nesting is simple and reproducible but omits feedback; two-way online coupling is more physically complete but can amplify interface and component biases. Shared grids ease exchange yet restrict component discretization, while separate grids add remapping and conservation error.

Limitations. Major limitations are inconsistent grids and vertical coordinates, nonconservative interpolation, boundary reflection, asynchronous clocks, spin-up shocks, feedback double counting, unresolved scale overlap, component bias compensation, high communication cost, fragile restart provenance and sparse coupled observations.

Open questions. Priorities include conservative adaptive remapping, wet/dry nested boundaries, coupled data assimilation, ensemble uncertainty, moving morphology, wave–atmosphere–ocean–river feedback, error attribution, coupling-frequency optimization, standardized metadata and operational fault tolerance.

Seminal papers. Nested ocean models established basin-to-coast downscaling, while wave–current and atmosphere–ocean systems introduced online feedback; community couplers and Earth-system interfaces later standardized field exchange, synchronization and parallel execution.

Claims

1. The unified wave-generation interface transferred surface-elevation and depth-averaged-velocity time series from MIKE 21 BW to 2D or 3D piston wavemakers while accounting for dispersion, shallow-water nonlinearity, and evanescent modes. *Regime: One-way coupling to DHI-style piston wavemakers; reflected physical waves are not fed back numerically.* direct finding, mixed [1585]
2. Across analytical, laboratory, and field benchmarks, the shared-grid SELFE-WWM-II coupling captured wave-current interaction with stable convergence over multiple scales, while the authors identified three-dimensional wave-induced effects as still requiring further research. *Regime: Unstructured-grid multiscale wave-current benchmarks and applications described in the indexed abstract.* direct finding, mixed [1206]
3. Coupled California-upwelling simulations show that passing ocean surface currents back to atmospheric stress reduces surface eddy kinetic energy by about half and exposes feedback absent from uncoupled forcing. *Regime: Modulation of Wind Work by Oceanic Current Interaction with the Atmosphere* direct finding, numerical [1177]
4. A three-level Mediterranean–Adriatic–northern-Adriatic hierarchy uses one-way offline nesting to downscale basin circulation to resolved coastal fields and tests reproduction of known regional features. *Regime: The Adriatic Sea modelling system: a nested approach* direct finding, numerical [1582]
5. WAVEWATCH III domain decomposition and an implicit scheme enable high-resolution unstructured hurricane-wave calculations compatible with community coupling infrastructure and validated for Hurricane Ike. *Regime: Large-scale hurricane modeling using domain decomposition parallelization and implicit scheme implemented in WAVEWATCH III wave model* direct finding, numerical [11]
6. A SURFEX–OASIS3–MCT interface provides a shared atmosphere-to-ocean, wave, ice and hydrology coupling structure from process and coastal domains to regional and global systems. *Regime: SURFEX v8.0 interface with OASIS3–MCT to couple atmosphere with hydrology, ocean, waves and sea-ice models, from coastal to global scales* direct finding, numerical [1461]
7. Coupled Gulf Stream and Agulhas simulations show current feedback produces eddy killing and materially changes boundary-current energy transfers. *Regime: Remarkable Control of Western Boundary Currents by Eddy Killing, a Mechanical Air–Sea Coupling Process* direct finding, numerical [1179]
8. High-resolution coupled California simulations show surface-stress feedback damps submesoscale currents, demonstrating that coupling effects persist below mesoscale resolution. *Regime: Dampening of Submesoscale Currents by Air–Sea Stress Coupling in the Californian Upwelling System* direct finding, numerical [1178]
9. A nested FVCOM inundation system for Scituate represents flooding/drying, coastal structures and wave–current effects and reproduces the compound tide–surge flooding of a 2010 nor’easter. *Regime: Coastal flooding in Scituate (MA): A FVCOM study of the 27 December 2010 nor’easter* direct finding, field [133]
10. Two-way COSMO–WAM coupling with wave-induced atmospheric drag reduces surface-wind and wave-height forecast errors in the dynamically complex southern North Sea and a nested German Bight grid. *Regime: An atmosphere–wave regional coupled model: improving predictions of wave heights and surface winds in the southern North Sea* direct finding, numerical [1472]
11. Operational northwest-European shelf experiments demonstrate how wave coupling changes atmosphere–ocean momentum exchange and can improve regional ocean forecasts relative to externally forced mode. *Regime: Can wave coupling improve operational regional ocean forecasts for the north-west European Shelf?* direct finding, numerical [809]
12. A validated multiple-nesting workflow downscales large-scale ocean fields into high-resolution coupled TELEMAC wave–3D hydrodynamics for the Gulf of Taranto and Mar Grande. *Regime: A coupled wave–3-D hydrodynamics model of the Taranto Sea (Italy): a multiple-nesting approach* direct finding, field [468]
13. At 18 British tidal-energy sites, 49–93% of waves were oblique to tidal flow; coupled COAWST experiments reduced theoretical tidal resource by about 10% per metre of wave-height increase, conditional on period and direction. *Regime: Realistic wave conditions and their influence on quantifying the tidal stream energy resource* direct finding, numerical [808]
14. A March 2019 MIKE21 FM wave–hydrodynamic simulation agreed acceptably with Iraqi coastal observations; omitting wind lowered significant wave height by 46.8%, while wave-period Nash efficiency reached 0.92. *Regime: Numerical Simulation of Hydrodynamic and Spectral Model of Iraqi Coastal Water at the Northern Arabian Gulf* direct finding, mixed [26]
15. An economical surface-wave model retains depth/current refraction and depth-variable coupling and was demonstrated against Hurricane Katrina buoy data and SWAN calculations before full ocean coupling. *Regime: A Surface Wave Model for Coupling with Numerical Ocean Circulation Models* direct finding, mixed [952]

37.4 Hydrodynamic circulation models

Topic numerical.hydro. Depth-averaged and three-dimensional circulation.

Papers screened: 4. Claims: 6. Equations: 2.

Well established. Coastal hydrodynamic models solve conservation of water mass and momentum under approximations matched to scale: depth-integrated shallow-water equations for long waves and vertically mixed flows, multilayer or three-dimensional hydrostatic models for stratification and shear, and nonhydrostatic or phase-resolving models where vertical acceleration matters.

Governing physics. Pressure gradients, gravity, Coriolis acceleration, advection, bed and surface stress, turbulence, density gradients, tides, wind, waves and boundary fluxes control circulation; wetting and drying, breaking, bores and hydraulic transitions introduce strong nonlinearities and numerical constraints.

Dimensionless parameters. Controls include Froude, Rossby, Reynolds and Richardson numbers, relative depth, aspect ratio, stratification and baroclinic Rossby radius, bottom-drag coefficient, Courant number, grid Peclet number, wetting threshold and normalized mass and energy errors.

Major equations. Core equations are free-surface continuity and horizontal momentum, optionally coupled to hydrostatic vertical structure, tracer transport, turbulence closure and wave radiation stresses. Conservative finite-volume or finite-difference forms must balance storage and boundary fluxes and preserve physically meaningful momentum across wet/dry fronts.

Typical methods. Workflows select dimensionality from the physics, construct datum-consistent bathymetry and boundaries, verify analytic waves and basins, test conservation and grid/time convergence, validate laboratory and field water levels and velocities, audit phase and extrema separately and document stabilization, friction and wet/dry sensitivity.

Numerical models. The reviewed implementation is UBO-TSUFDF, a nonlinear shallow-water finite-difference model with inundation capability, evaluated within a broader tsunami verification and validation framework that supplies analytic, laboratory, conservation and operational tests.

Experimental datasets. Reviewed evidence includes analytical shoreline-motion and oscillating-basin cases, mass-conservation tests, the Monai laboratory tsunami benchmark and formal operational verification criteria. Direct circulation, tide and wetland datasets remain in the lawful-full-text queue.

Validated ranges. Support covers selected long-wave analytical and laboratory regimes and Monai behavior through 30 s. It does not validate post-breaking flow, stratified three-dimensional circulation, wave-resolving dynamics or arbitrary wetting/drying configurations.

Recent advances. Recent advances use unstructured high-order and discontinuous Galerkin methods, GPU acceleration, adaptive meshes, locally three-dimensional or nonhydrostatic patches, conservative subgrid structures, ensemble data assimilation and reproducible benchmark suites.

Disagreements. Depth-integrated models offer efficiency and horizontal resolution but omit vertical shear and baroclinicity; three-dimensional models add physics at greater data, closure and computational cost. Numerical robustness obtained through excess drag or diffusion can conflict with physical fidelity.

Limitations. Bathymetric and boundary uncertainty, wet/dry thresholds, turbulence and friction closure, numerical diffusion and dispersion, stability-driven tuning, unresolved breaking, grid anisotropy and incomplete conservation audits can produce plausible levels with incorrect velocities, fluxes or inundation.

Open questions. Priorities include conservative multiscale nesting, robust positivity without artificial drag, adaptive dimensionality, coupled baroclinic-wave-sediment physics, uncertainty-aware boundary assimilation, differentiable calibration and validation metrics that expose compensating errors.

Seminal papers. Saint-Venant shallow-water equations established depth-integrated long-wave modeling; primitive-equation coastal ocean models introduced sigma coordinates and turbulence closures, while finite-volume Riemann methods advanced conservative shocks, bores and wetting/drying.

Equations

- *Depth-integrated continuity equation* [1410]

$$\frac{\partial \eta}{\partial t} + \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} = 0$$

Regime: Nonlinear shallow-water formulation in Cartesian coordinates. Variables: η surface elevation; M x discharge flux $u(h+\eta)$; N y discharge flux $v(h+\eta)$

- *Manning bottom-friction component* [1410]

$$f_x = \frac{gn^2}{D^{7/3}} M \sqrt{M^2 + N^2}$$

Regime: Depth-averaged bottom friction; only the laboratory benchmark tests this implementation. Variables: n Manning roughness; D total water column; M, N discharge fluxes

Claims

1. UBO-TSUFDF reproduced the selected analytical shoreline-motion, oscillating-basin, mass-conservation, and laboratory tsunami benchmarks with generally satisfactory agreement. *Regime: The four paper-specific benchmarks on single structured grids; this does not validate nested grids, landslide generation, dispersive physics, or wave breaking.* direct finding, mixed [1410]
2. The nonlinear shallow-water model reproduced the Monai laboratory arrival and inundation behavior through 30 s but was not compared later because breaking waves are outside its physics. *Regime: Matsuyama-Tanaka Monai Valley benchmark, 5.446 x 3.406 m region, maximum depth 0.135 m; pre-breaking interval through 30 s.* direct finding, mixed [1410]
3. A tsunami-model mass audit should integrate disturbed free-surface elevation and account for open-boundary fluxes; integrating total depth can mask numerical errors offshore. *Regime: Depth-integrated water-wave computations with closed, open, or absorbing boundaries.* direct finding, analytical [1384]
4. Ad hoc friction factors introduced mainly to stabilize marginally stable inundation computations create reliability that cannot be known a priori outside tested cases. *Regime: Numerical inundation models using empirically adjusted friction primarily for stability.* review statement, review [1384]
5. Modeling 0.5–1.0 m sea-level rise over 1–2 m reef flats increases waves, setup, shear, resuspension and offshore sediment flux while reducing sediment residence time and light availability. *Regime: Numerical modeling of the impact of sea-level rise on fringing coral reef hydrodynamics and sediment transport* direct finding, numerical [1355]
6. A finite-element formulation solves layer-averaged circulation for two vertically homogeneous coastal layers, retaining interfacial exchange and density-driven structure absent from a single-layer model. *Regime: FINITE ELEMENT MODEL OF TWO LAYER COASTAL CIRCULATION* direct finding, numerical [1476]

37.5 Morphodynamic models

Topic *numerical.morpho. Coupled flow-sediment-bed models.*

Papers screened: 9. Claims: 5. Equations: 0.

Well established. Process-based coastal morphodynamic models couple hydrodynamics, sediment transport and bed updating through feedback: flow moves sediment, divergence changes elevation, and the changed bed modifies subsequent waves and currents. Calibration fit alone does not establish predictive skill.

Governing physics. Bed evolution reflects gradients in bedload and suspended transport driven by waves, currents, runup and gravity, with feedback through water depth, breaking, circulation and inlet exchange; relevant scales range from intra-wave transport to seasonal and engineering evolution.

Dimensionless parameters. Controls include Shields mobility, relative grain size, fall-velocity and suspension parameters, wave asymmetry and skewness, surf similarity, sediment Courant number, morphological acceleration factor, active-layer thickness ratios and normalized bed or shoreline error.

Major equations. The core system combines hydrodynamic mass and momentum equations, sediment advection–diffusion or bulk transport formulae, and the Exner bed-continuity equation, often with separate cross-shore and longshore components, wetting/drying, avalanching and porosity corrections.

Typical methods. Workflows define sediment classes and closures, spin up hydrodynamics, couple transport and Exner updating, test conservation and morphological time step, calibrate a constrained subset, validate against independent profiles or volumes at multiple scales, and audit parameter plausibility and boundary flux.

Numerical models. The evidence spans a cross-shore storm-profile model, coupled cross-/longshore inlet morphodynamics, weak phase-resolving-wave to wave-averaged-sediment coupling, and reduced linear, exponential, one-line and combined nourishment-evolution models.

Experimental datasets. Reviewed evidence includes storm-profile erosion, Wilson Inlet waves–currents–bed observations, multiple laboratory benchmarks plus Pescara Harbor evolution, and post-nourishment volume series at Folly Beach and Hunting Island.

Validated ranges. Current evidence is case-bound to sandy storm profiles, one microtidal wave-dominated seasonal inlet, laboratory tests and one harbor coast, and two South Carolina nourishment sites; it does not establish universal closures or acceleration factors.

Recent advances. Recent work emphasizes weak multiscale coupling, phase-resolving forcing of longer-term transport, probabilistic ensembles, scale-aware skill metrics, data assimilation and efficient reduced emulators while retaining conservation and physically constrained parameters.

Disagreements. Detailed process models represent more feedback but accumulate closure and numerical uncertainty, whereas reduced models can be robust for aggregate volume or shoreline response yet omit inlets, cross-shore exchange and event sequencing. Similar fit can coexist with physically implausible parameters.

Limitations. Limitations include uncertain transport formulae, compensating calibration, sparse subaqueous observations, boundary sediment flux, scale-dependent skill, numerical diffusion, morphological acceleration, bed-layer and wet/dry choices, storm sequencing and weak transfer across grain sizes and coast types.

Open questions. Priorities include scale-selective validation, probabilistic structural uncertainty, mixed and cohesive sediment, event-to-decadal acceleration, coupled inlet–beach exchange, climate nonstationarity, observation updating and reliable extrapolation beyond calibration conditions.

Seminal papers. Exner bed continuity, energetics and equilibrium-profile concepts, one-line shoreline diffusion and early coupled wave–current–sediment models established the hierarchy from reduced shoreline response to process-based area morphodynamics.

Claims

1. Leont'yev's storm-profile model decomposes total cross-shore transport into wave/current and runup components, using only runup transport in swash, only wave/current transport in shoaling water, and both in the surf zone. *Regime: The proposed 1996 short-term sandy beach-profile model; coefficients and tested forcing ranges are unavailable from the abstract.* direct finding, numerical [804]
2. A weakly coupled phase-resolving hydrodynamic and wave-averaged sediment model was validated against multiple experiments and reproduced long-term bed evolution opposite Pescara Harbor. *Regime: The model's articulated coastal grids, tested sediment closures, and Pescara Harbor case.* direct finding, mixed [473]
3. Tidal-basin morphodynamic modeling shows sea-level rise reorganizes channel-flat morphology and sediment demand through coupled hydrodynamic and bed-evolution feedbacks. *Regime: Numerical modeling of the impact of sea level rise on tidal basin morphodynamics* direct finding, numerical [1497]
4. In the LIP profile tests, the model reproduced two nearshore bars that migrated about 7 m seaward, with initial crest heights near 0.10 and 0.12 m. *Regime: LIP fine-sand regular-wave profile tests 1a and 1b.* direct finding, experimental [473]
5. For the 1997–2000 Pescara hindcast, simulated annual settlement was about 38,500 m³/year versus about 40,000 m³/year inferred from bathymetric surveys. *Regime: Pescara Harbor under the dominant 1.5 m wave class and modeled 1997 bathymetry.* direct finding, mixed [473]

37.6 Phase-averaged wave models

Topic *numerical.phase_averaged. Spectral wave-action models.*

Papers screened: 11. Claims: 8. Equations: 1.

Well established. Phase-averaged spectral models efficiently propagate random directional wave energy across large coastal domains while representing generation, dissipation, refraction, shoaling, and nonlinear source terms; their reliability depends on whether omitted phase-coherent or low-frequency processes control the site.

Governing physics. The spectral action balance transports energy through physical and spectral space while wind input, whitecapping, bottom friction, breaking, triad and quartet interactions, and Bragg scattering act as source terms; bathymetry controls refraction and diffraction.

Dimensionless parameters. No single universal dimensionless parameter governs spectral-model validity. Relative depth, spectral bandwidth, directional spreading, exposure, and ratios of local wind-sea, sea-swell, and infragravity energy define process regimes and numerical sensitivity.

Major equations. Core formulations are the wave-action balance and its additive source terms, the Hasselmann Boltzmann integral for resonant four-wave transfer; and mild-slope-derived directional turning used for phase-decoupled diffraction.

Typical methods. Typical evaluation uses analytical benchmarks, laboratory diffraction observations, exact source-term solvers, regional model intercomparison, sensitivity to spectra/bathymetry/tides, and event-scale field validation against phase-resolving alternatives.

Numerical models. Models include SWAN with phase-decoupled diffraction, the WRT exact quartet-interaction solver, Longuet-Higgins spectral refraction, Kirby's parabolic refraction-diffraction method, and XBeach surfbeat as a complementary low-frequency comparator.

Experimental datasets. Reviewed evidence includes extreme diffraction benchmarks, a roughly 300 by 300 km Southern California Bight transformation case, and TC Olwyn observations with sustained winds above 130 km/h and forereef significant waves reaching 6 m.

Validated ranges. Validation is case-bound: phase-decoupled diffraction is for random short-crested fields without coherent standing patterns; Bight models perform best at moderately exposed sites; and SWAN's cyclone advantage was demonstrated for a shallow Ningaloo lagoon dominated by local wind growth.

Recent advances. Recent work increasingly diagnoses models by process completeness rather than class alone: the 2019 reef study identifies local wind-wave generation as decisive, while related nonhydrostatic studies expose breaking-turbulence and morphodynamic errors hidden by surface-wave calibration.

Disagreements. The Ningaloo comparison favored phase-averaged SWAN over surfbeat XBeach because local wind growth dominated, but this does not show that infragravity resolution is generally secondary; model ranking reverses when the omitted process changes.

Limitations. Phase averaging excludes coherent standing-wave fields and explicit infragravity phase dynamics; exact quartet interactions are expensive; sheltered predictions are sensitive to numerical bias; and agreement in wave height alone does not validate currents, sediment transport, or morphology.

Open questions. Needs include efficient accurate nonlinear source terms, robust diffraction near singular obstacles, joint wind-sea and infragravity capability, uncertainty propagation from offshore spectra and bathymetry, and validation of coupled wave-current-sediment responses.

Seminal papers. The 1993 Southern California Bight comparison is the earliest directly reviewed regional validation in this slice; the 2003 phase-decoupled diffraction paper is a foundational extension of SWAN's coastal-process capability.

Equations

- *Third-generation spectral action source balance* [1459]

$$S_{tot} = S_{inp} + S_{wcap} + S_{nl4} + S_{fric} + S_{brk} + S_{nl3} + S_{Bragg}$$

Regime: Third-generation spectral wave-action models; shallow-water terms apply where relevant. Variables: S_{inp} wind input; S_{wcap} whitecapping dissipation; S_{nl4} four-wave interactions; S_{fric} bottom friction; S_{brk} depth-limited breaking; S_{nl3} triad interactions; S_{Bragg} Bragg scattering

Claims

1. A mild-slope-derived phase-decoupled turning-rate formulation allowed SWAN to combine random-wave diffraction with refraction, shoaling, generation, dissipation, and wave-wave interactions, with reasonable agreement in extreme diffraction tests. *Regime: Random short-crested waves without coherent standing-wave patterns; diffraction regions require adequate local resolution.* direct finding, numerical [609]
2. The WRT method evaluates the first-principles nonlinear four-wave interaction integral for discrete spectra and serves as an accuracy reference, but its computational cost limits direct operational use. *Regime: Weakly nonlinear spectral surface-gravity waves represented on discrete frequency-direction grids.* direct finding, analytical [1459]
3. In the roughly 300 km by 300 km Southern California Bight, broader incident frequency-direction spectra reduced model sensitivity and improved agreement, but neither the refraction nor refraction-diffraction model was accurate at highly sheltered sites. *Regime: The tested Southern California Bight bathymetry and incident spectra; strongest limitation where diffraction dominates low local energy.* direct finding, numerical [1066]
4. During category-3 TC Olwyn at Ningaloo Reef, locally generated lagoon wind waves dominated over infragravity waves, so phase-averaged SWAN was more accurate than surfbeat-resolving XBeach for this specific event because SWAN represented wind-wave growth. *Regime: TC Olwyn at the roughly 1 km wide, 1 m deep Ningaloo lagoon; infragravity importance at other reefs is not negated.* direct finding, mixed [376]
5. Semiempirical spectral dissipation activates breaking above an observed nondimensional threshold, represents nonlinear swell and short-wave loss, and validates wave height, period and spectra from global to coastal scales despite residual systematic defects. *Regime: Semiempirical Dissipation Source Functions for Ocean Waves. Part I: Definition, Calibration, and Validation* direct finding, mixed [71]
6. SWAN verification across five increasingly complex coastal field cases produced average RMS errors of 0.30 m in significant wave height and 0.7 s in mean period, about 10% of incident values. *Regime: A third-generation wave model for coastal regions: 2. Verification* direct finding, mixed [1197]
7. Coastal WAVEWATCH III accuracy improves when currents, reflection and spatially varying bottom sediment are represented on efficient unstructured grids, but sub-100-m surf-zone resolution still requires better numerical schemes and coupled forcing. *Regime: French Atlantic, Channel and North Sea coastal wave fields on regional unstructured grids with currents, reflection and mapped bottom type.* direct finding, mixed [1207]
8. Third-generation spectral wave modeling has reduced bias in integral parameters, but improving spectral shape—especially high-frequency tails, extreme-condition physics, wave-current interaction and coastal boundary treatment—remains central to better forecasts and coupled applications. *Regime: Phase-averaged wind-wave modeling from deep water through shallow/coastal applications within spectral-model assumptions.* review statement, review [242]

37.7 Phase-resolving wave models

Topic *numerical.phase_resolving. Boussinesq and nonhydrostatic models.*

Papers screened: 14. Claims: 9. Equations: 0.

Well established. Phase-resolving models simulate individual wave crests and troughs, retaining nonlinear interactions, dispersion, reflection and runup that phase-averaged spectral models parameterize; their added detail requires finer space-time resolution, consistent boundaries and process-specific validation.

Governing physics. Nonlinear free-surface motion, frequency dispersion, shoaling, refraction, diffraction, reflection, breaking, infragravity generation, wetting/drying, currents and structure or vegetation interactions govern resolved waves. Depth-averaged formulations still approximate vertical shear, turbulence and overturning.

Dimensionless parameters. Controls include relative depth kh , wave steepness, Ursell and Froude numbers, relative amplitude and submergence, directional spread, Courant number, cells per wavelength, layer count, breaking or dissipation parameters, wetting depth and morphological acceleration factor.

Major equations. Boussinesq-type equations extend shallow-water mass and momentum with dispersive terms; depth-averaged nonhydrostatic models solve pressure corrections, while fully nonlinear potential flow, RANS/VOF and multilayer formulations add successively more vertical physics. Bed evolution couples sediment-flux divergence to morphology.

Typical methods. Workflows verify linear dispersion, shoaling, conservation and still-water balance; validate solitary waves, runup, breaking, bars, vegetation and harbor spectra; test convergence and boundaries; separate surface elevation, velocity and morphology skill; and compare against phase-averaged or physical models.

Numerical models. The set spans higher-order Boussinesq, FORCE-based depth-averaged nonhydrostatic, XBNH/XBNH+, weakly coupled wave-resolving-wave-averaged morphology and hybrid numerical/physical modeling, with applications from tsunami runup to vegetation and harbors.

Experimental datasets. Reviewed evidence includes Synolakis plane-beach and conical-island runup, controlled frequency/directional-spread runup, vegetation attenuation, FORCE-scheme verification cases, a large-scale barred beach, weakly coupled bed evolution, deterministic physical-numerical combinations and multimodal harbor spectra.

Validated ranges. Support is benchmark- and site-specific. The Boussinesq runup model degraded from $H/d=0.10$ to 0.20 at the conical island, and barred-beach models reproduced seaward waves but overpredicted landward heights; no solver is validated across all breaking and morphology regimes.

Recent advances. Recent advances use multilayer nonhydrostatic and high-order unstructured solvers, adaptive meshes, immersed and porous boundaries, GPU acceleration, deterministic physical-numerical coupling, differentiable calibration and phase-resolving data assimilation.

Disagreements. Boussinesq models efficiently capture weak-to-moderate dispersion but need breaking closures; nonhydrostatic multilayer models broaden dispersion at higher cost; RANS resolves overturning but is expensive. Surface-elevation tuning can leave velocity and sediment errors unresolved.

Limitations. Boundary reflection, sponge design, numerical dispersion, grid and time resolution, breaking and turbulence closures, wet/dry thresholds, vertical averaging, bathymetric error and morphology coupling can dominate results. Phase accuracy degrades over long domains without careful dispersion control.

Open questions. Priorities include parameter-robust breaking, adaptive vertical resolution, conservative nested phase coupling, irregular directional seas, air and porous structures, wave-current-vorticity interactions, scalable GPU solvers, uncertainty propagation and multi-variable morphodynamic validation.

Seminal papers. Peregrine and later enhanced Boussinesq equations established dispersive coastal-wave simulation; fully nonlinear potential flow, nonhydrostatic pressure correction and VOF/RANS expanded regimes, while FUNWAVE, BOSZ, SWASH and XBeach-NH enabled practical applications.

Claims

1. The higher-order Boussinesq model reproduced the principal runup and rundown evolution in the Synolakis plane-beach validation case. *Regime: Solitary-wave runup on a 1:19.85 plane beach with 0.2 m still-water depth; the source reports H/d inconsistently as 0.25 in the caption and 0.28 in prose.* direct finding, mixed [1245]
2. For conical-island runup, agreement was closer at $H/d=0.10$ than at $H/d=0.20$, where the model underpredicted front-side maximum runup. *Regime: Briggs et al. conical island: 14-degree face, 0.32 m water depth, solitary waves with H/d 0.10 and 0.20.* direct finding, mixed [1245]
3. For the tested large-scale barred beach, XBNH and XBNH+ reproduced seaward hydrodynamics but overpredicted wave heights landward of the bar because crest breaking and turbulence were simplified; dissipation tuning did not also fix velocity and morphology errors. *Regime: Single- and reduced-two-layer XBeach nonhydrostatic modes for the tested fixed/mobile barred profiles.* direct finding, mixed [407]
4. The FORCE-based depth-averaged nonhydrostatic model was well balanced, accurately captured wet-dry fronts, and simulated wave breaking reasonably without a tunable breaking coefficient across its verification tests. *Regime: The paper's SNHE benchmarks over uneven beds; depth-averaging limitations remain for vertical turbulence.* direct finding, numerical [867]
5. A validated nonhydrostatic reef model shows fore-reef slope controls runup on reef-fronted coasts; for a 1:5 fore-reef and 500 m flat, beach slope near 1:30 separated runup reduction from enhancement, with tall roughness most effective. *Regime: Wave-Driven Hydrodynamic Processes Over Fringing Reefs With Varying Slopes, Depths, and Roughness: Implications for Coastal Protection* direct finding, mixed [203]
6. A fourth-order Boussinesq moving-bottom model using sixth-order spatial and temporal schemes agreed with three-dimensional experiments and retained nonlinear and dispersive accuracy for landslide waves generated into intermediate and deeper water. *Regime: A higher-order Boussinesq-type model with moving bottom boundary: applications to submarine landslide tsunami waves* direct finding, numerical [83]

7. Three-dimensional high-level Green-Naghdi models extend strong nonlinearity to stronger dispersion, and GN-3 is converged for the tested shoal-diffraction cases while agreeing with analytical, laboratory and fully nonlinear Boussinesq benchmarks. *Regime: Three-dimensional shallow-water wave evolution and diffraction over the modeled elliptic and submerged shoals within the benchmark ranges.* direct finding, numerical [1606]
8. Finite-volume fluxes, MUSCL/WENO reconstructions and wet-dry treatment can be extended to Boussinesq-type systems to simulate bidirectional dispersive waves, solitary-wave interactions, dispersive shocks, and breaking or non-breaking long-wave run-up. *Regime: One-dimensional nonlinear dispersive long-wave propagation and run-up within the tested Boussinesq and wet-dry benchmarks.* direct finding, numerical [385]
9. A hybrid finite-volume/finite-difference Green-Naghdi solver with improved dispersion reproduces strongly nonlinear wave transformation over a submerged bar, including high-order harmonics generated by bathymetric interactions. *Regime: One-dimensional, strongly nonlinear and dispersive shallow-water waves over submerged bathymetry without dry areas and up to wave breaking.* direct finding, numerical [255]

38 Wiki: Ports and harbors

38.1 Ports and harbors: overview

Topic ports. Harbor hydrodynamics, navigation, and operations.

Papers screened: 6. Claims: 5. Equations: 0.

Well established. Port performance couples wave agitation, currents and water levels with vessel motion, navigation clearance, mooring response and sediment exchange. Breakwaters and dredged geometry alter resonance, circulation and transport, so operational and maintenance outcomes cannot be inferred from offshore conditions alone.

Governing physics. Incident and reflected waves excite harbor modes and moored vessels; wind, tide and density-driven circulation exchange water and sediment through entrances. Ship squat, heel and wave response consume under-keel clearance, while settling, resuspension, flocculation and near-bed exchange govern siltation.

Dimensionless parameters. Controls include relative entrance width and basin length, wavelength-to-basin scale, reflection and amplification coefficients, wave steepness, vessel draft-to-depth and blockage ratios, under-keel-clearance fraction, mooring stiffness and damping, Shields stress, settling-to-friction-velocity ratio and tidal asymmetry.

Major equations. Port studies combine spectral or time-domain wave equations, shallow-water and circulation momentum balances, rigid-body vessel and mooring equations, under-keel-clearance budgets, sediment advection-diffusion and bed exchange, and operability or downtime exceedance calculations.

Typical methods. Methods include offshore-to-berth spectral transformation, physical harbor models, nested wave and circulation models, AIS and accident analysis, ship-motion and mooring simulation, field monitoring, probabilistic operability assessment, sediment tracers, turbidity surveys and morphodynamic calibration.

Numerical models. Model chains couple regional wave forecasts to harbor agitation, vessel response and operational thresholds; navigation tools combine water level, squat, heel and waves. Sedimentation systems couple hydrodynamics, suspended transport, bed exchange, dredging and morphology.

Experimental datasets. Reviewed evidence spans irregular-wave harbor models, forty-year spectral climates, Jeju mooring assessment, dock-specific vessel monitoring, dynamic under-keel-clearance studies, Gijón climate projections and multi-season Zeebrugge mud observations and models.

Validated ranges. Examples include 81.6% multimodal sea states at a near-port point, reconstruction correlations near 0.995–1.000 for aggregated wave parameters, spring mud import two-to-four times neap values, and source-specific vessel-response prediction errors below operationally evaluated limits.

Recent advances. Recent work clusters full directional spectra, learns vessel response from field data, delivers multi-port forecasts, integrates AIS risk evidence, resolves ship-specific access probabilities and uses multiple observational constraints to expose sediment-model nonuniqueness.

Disagreements. Aggregate significant wave height can hide frequency-direction combinations that cause different in-port response. Sediment models may fit available observations yet predict opposite long-term convergence, and generic vessel thresholds can conflict with dock-specific monitored limits.

Limitations. Sparse entrance and berth observations, uncertain reflection and infragravity forcing, nonlinear vessel and fender behavior, changing bathymetry, human factors, climate-model spread, cohesive-bed exchange and dredging records limit transferable operational prediction.

Open questions. Priorities include seamless offshore-to-vessel forecasts, infragravity and resonance prediction, autonomous and climate-resilient navigation, multi-vessel interaction, data-assimilative sediment budgets, beneficial sediment management and uncertainty-aware downtime planning.

Seminal papers. Classical harbor resonance and diffraction studies established basin agitation analysis; ship maneuvering and under-keel-clearance work linked hydrodynamics to navigation, while estuarine sediment theory established tidal pumping and exchange frameworks for siltation.

Claims

1. The SAMOA operational service delivers high-resolution meteorological and ocean forecasts tailored to safety, environmental, and operational decisions in Spanish ports. *Regime: Coastal ocean forecasting in Spanish ports: the SAMOA operational service* direct finding, mixed [1325]
2. Field surveys of a sub-Arctic industrial harbor relate benthic communities to sediment properties and heavy-metal gradients for cumulative-impact assessment. *Regime: Biodiversity and Habitat Assessment of Coastal Benthic Communities in a Sub-Arctic Industrial Harbor Area* direct finding, field [373]
3. A technical review categorizes greenhouse-gas and pollutant-emission sources in ports and assesses mitigation and monitoring approaches. *Regime: Pollutant Emissions in Ports: A Comprehensive Review* review statement, review [115]
4. Fully nonlinear FUNWAVE simulations show that Bragg-resonant bar fields can mitigate regular- and group-wave harbor oscillations; long bars reduced modeled nonlinear resonance by up to 76.4%, versus 23.0% for short bars, with stronger mitigation as bar number or amplitude increased. *Regime: Elongated constant-depth harbor with external sinusoidal bars and the regular/group-wave and bar parameter ranges tested.* direct finding, numerical [480]
5. Fully nonlinear FUNWAVE simulations show that focused transient wave groups can directly excite harbor resonance when their spectral peak approaches a harbor mode, while nonlinear interactions can additionally excite lower modes; runup and resonance increase with spectral width and are strongest for normal incidence. *Regime: Four elongated constant-depth harbors, focused groups at 0.50 Hz, and tested amplitude, bandwidth and incident-direction ranges.* direct finding, numerical [478]

38.2 Harbor agitation and resonance

Topic *ports.agitation. Wave penetration and basin modes.*

Papers screened: 14. Claims: 17. Equations: 1. Shown in the Coastal Design Bench beside: Harbour agitation.

Well established. Harbor wave agitation can be strongly multidirectional and multireflective: entrance penetration, diffraction, and reflections from harbor structures create spatially different local spectra that a single wave-height statistic cannot uniquely describe.

Governing physics. In the reviewed Africa-basin case, direct southeasterly penetration primarily affected the western basin, while northern-to-northeasterly waves reflected from Reina Sofia breakwater contributed strongly to energetic conditions in eastern control areas.

Dimensionless parameters. Period-dependent reflection coefficient $K_r(T)$, explained variance, correlation coefficients, and spectral-type occurrence probability are key dimensionless descriptors in the reviewed workflow; no universal threshold values are established from this single harbor.

Major equations. The reviewed directional reconstruction represents local complex free-surface elevation as a finite sum of frequency-direction components and estimates their Fourier coefficients by least squares; clustering performance is summarized by $EV=1-SSE/SSE_T$.

Typical methods. The reviewed long-term workflow nests phase-averaged offshore-to-port downscaling with a phase-resolving monochromatic catalog, reconstructs hourly local directional spectra, reduces them with PCA/EOFs, and clusters representative spectral types with K-means.

Numerical models. The reviewed chain uses SWAN for nonstationary offshore-to-near-port spectra, MSPv2.0 for elliptic mild-slope monochromatic propagation with dynamic boundary reflection, r-DPRA for directional decomposition, and Wavespectra for spectral partitioning.

Experimental datasets. Reviewed evidence includes eight months of scalar gauges and one month of AWAC directional information in Africa basin, plus older regular/irregular hydraulic tests in two nearly closed natural-harbor models; full geometry and ranges of the latter remain inaccessible.

Validated ranges. Validation is bounded to Africa basin: 40 years of reconstructed hourly climate, a 2.0 by 5.4 km harbor domain, eight months of scalar gauges, and one month of directional AWAC data. Internal outer-spectrum reconstruction achieved aggregate R-squared of 0.999-1.000, but weaker third partitions fell to R-squared 0.720 for wave height.

Recent advances. A recent advance is compression of spatially variable, 40-year hourly frequency-direction agitation histories into representative local spectral types that retain distinct forcing mechanisms hidden by similar aggregate H_{m0} values.

Disagreements. No direct cross-study contradiction is yet defensible. The 1989 laboratory study found that grouped-wave long-period energy was not attributable to resonance in its two tested harbors, whereas the 2023 field-numerical study resolves multimodal transformation but does not test that attribution; their regimes and questions differ.

Limitations. Current evidence is site-specific; directional field validation covers one month, local decomposition assumes linear superposition, homogeneous fields and constant depth, the older laboratory record is abstract-limited, and neither study demonstrates that richer spectra improve ship-motion or downtime predictions.

Open questions. Needed evidence includes transfer across harbor geometries and climates, multi-season independent directional validation, propagation of offshore/model uncertainty into spectral types, and coupled tests showing whether spectral characterization materially improves vessel-response and operability decisions.

Seminal papers. Ouellet and Theriault (1989) is the earliest reviewed journal experiment in this branch, explicitly comparing regular- and irregular-wave diffraction/reflection and warning that large transfer values outside the energetic band need not imply energetic resonance.

Equations

- *Directional phase-resolving free-surface reconstruction* [1210]

$$b_{n,p} = \sum_{v=1}^L \exp[i\mathbf{k}_{n,v} \cdot (\mathbf{x}_p - \mathbf{x}_r)] z_{n,v} + \varepsilon_{n,p}$$

Regime: Locally homogeneous, constant-depth, linear superposition of a finite set of multidirectional long-crested regular waves around an in-port target point. Variables: $b_{n,p}$ known complex free-surface elevation for frequency n at position p ; $\mathbf{k}_{n,v}$ wavenumber vector for direction v ; $\mathbf{x}_p$ analysis-node position; $\mathbf{x}_r$ target position; $z_{n,v}$ Fourier coefficient for the frequency-direction component; $\varepsilon_{n,p}$ reconstruction deviation; L number of analyzed directional components

Claims

1. The Catania scenario excited persistent modeled harbor oscillations with periods of approximately 15-20 min. *Regime: Present-day Catania harbor geometry and the paper's corrected source scenario; no field validation of the oscillations is reported.* direct finding, numerical [1410]
2. At the Africa-basin near-port reference point, 81.6% of the 40-year hourly sea states had two or more spectral peaks, while 18.4% were unimodal. *Regime: GOW2/CFS/GOT/GOS-driven SWAN hindcast at point R6 outside Africa basin; the percentages characterize this site and reconstruction, not harbors generally.* direct finding, numerical [1210]
3. For reconstructed outer-port spectra, aggregated wave height and period parameters had R-squared values of 0.999-1.000 and aggregate directional parameters had circular correlations of 0.995-0.999 against the forcing spectra. *Regime: Outer boundary reconstruction test in the MSPv2.0 Africa-basin domain; this is an internal numerical-consistency test rather than independent field validation.* direct finding, numerical [1210]
4. Reconstruction accuracy declined for weaker secondary and tertiary wave systems, with wave-height R-squared decreasing from 0.994 for the main partition to 0.943 and 0.720 for the second and third partitions. *Regime: The same internal outer-spectrum reconstruction validation; partitioning differences can split or merge nearby peaks.* direct finding, numerical [1210]

5. At control point A7, spectral types containing direct or combined southeasterly forcing comprised about 9.8% of an average year but included four of the five most energetic agitation types. *Regime: Africa-basin point A7 under the reconstructed 40-year climate and the paper's spectral-type classification.* direct finding, numerical [1210]
6. Two locally energetic conditions with similar aggregated significant-wave-height statistics at point D2 had distinctly different in-port and outer-port frequency-direction spectral shapes. *Regime: Africa-basin point D2 and its reconstructed historical spectra; the comparison demonstrates information loss from Hm0 alone at this site.* direct finding, numerical [1210]
7. Within the frequency band containing most incident energy, irregular-wave diffraction-reflection coefficients varied less than coefficients derived from regular-wave tests. *Regime: Two tested nearly closed natural-harbor physical configurations and the incident spectral energy band; detailed geometry and wave ranges require full text.* direct finding, experimental [1138]
8. Large low- and high-frequency transfer-function estimates outside the main incident-energy band contained relatively little wave energy in the tested harbor models. *Regime: The reported regular/irregular harbor-model tests; no universal transfer threshold is available from the abstract.* direct finding, experimental [1138]
9. The long-wave energy associated with irregular wave grouping could not be attributed to harbor resonance in the reported experiments. *Regime: The two nearly closed harbor models, tested wave grouping factors, and experimental frequency range; this does not rule out group-forced resonance in other harbors.* direct finding, experimental [1138]
10. Modal analysis identifies the natural oscillation structure of semi-enclosed basins relevant to harbor resonance and agitation. *Regime: Modal analysis of semi-enclosed basins* direct finding, analytical [139]
11. A computational model provides faster, reliable assessment of wave agitation within harbors for layout and operational evaluation. *Regime: An improved model for fast and reliable harbour wave agitation assessment* direct finding, numerical [390]
12. Numerical reconstruction of the 2003 Boumerdès earthquake examines rupture-dependent tsunami generation and resulting wave response in affected ports. *Regime: 21 May 2003 Boumerdès Earthquake: Numerical Investigations of the Rupture Mechanism Effects on the Induced Tsunami and Its Impact in Harbors* direct finding, numerical [926]
13. Analytical and numerical spectral analysis relates incident swell to wave oscillations and hazards in the irregularly shaped Pohang New Harbor. *Regime: Spectral Density Analysis for Wave Characteristics in Pohang New Harbor* direct finding, numerical [752]
14. FUNWAVE simulations show that Bragg reflection from periodic arc-shaped seabed bars can suppress long-period harbor oscillations because reduced transmitted energy outweighs blockage of harbor-radiated waves; optimal bar wavelength differs between resonant and non-resonant forcing. *Regime: The modeled rectangular harbor, monochromatic long waves and arc-shaped sinusoidal bars across the documented number, amplitude and wavelength ranges.* direct finding, numerical [481]
15. At Kumaishi Fishing Port, small-boat mooring disturbance occurs when offshore significant wave height exceeds about 2.0 m and harbor long-period wave height exceeds 0.10 m; a 0.25 m long-period-wave limit and the $H1/3 > 2.0$ m indicator support evacuation decisions. *Regime: Small fishing boats moored at Kumaishi Fishing Port under low-pressure/typhoon events and long-period harbor oscillations.* direct finding, mixed [14]
16. Validated OpenFOAM simulations show that free heave of the upstream of two boxes reduces narrow-gap wave amplification and shifts resonance upward toward the box heave natural frequency, with resonance frequency decreasing as incident wave height increases. *Regime: Regular waves interacting with two closely spaced rectangular boxes, with either both fixed or the upstream box free to heave.* direct finding, numerical [479]
17. Wave and circulation simulations for Busan New Port predict that channel and basin deepening plus reclamation generally increase harbor response without basin-wide severe agitation, although particular locations require typhoon preparedness. *Regime: Busan New Port after Typhoon Maemi, with navigation-channel and basin dredging plus modified Ungdong Bay reclamation.* direct finding, mixed [784]

38.3 Mooring hydrodynamics

Topic ports.mooring. Mooring loads and berth conditions.

Papers screened: 10. Claims: 8. Equations: 0. Shown in the Coastal Design Bench beside: Navigation channel.

Well established. A berth mooring system must limit vessel motions, line tensions, fender reactions and access interruptions under the local directional wave, wind, current, tide and operational load environment; nominal vessel fit or calm-water capacity alone is insufficient.

Governing physics. Environmental forces and moments excite surge, sway, heave, roll, pitch and yaw, while nonlinear mooring stiffness, pretension, geometry, fender contact, damping, under-keel conditions and berth hydrodynamics distribute and sometimes amplify response.

Dimensionless parameters. Relevant normalized controls include wave length and draft relative to basin depth, line pretension-to-breaking strength, demand-to-capacity and safety factors, fender compression ratio, vessel-motion-to-operational-limit ratio, natural-period-to-forcing-period ratio and spectral exceedance probability.

Major equations. Analysis combines six-degree-of-freedom vessel dynamics with hydrodynamic mass, damping and excitation, nonlinear line force-extension and catenary geometry, fender reaction, wind and current drag, and equilibrium or time-domain compatibility at fairleads and berth contact points.

Typical methods. Methods characterize directional berth forcing, select vessel loading states, define line and fender properties and operational limits, compute frequency- or time-domain response, test environmental combinations and failures, perform configuration sensitivity, and validate motions or tensions where observations exist.

Numerical models. Evidence includes spectral-type reduction intended to drive response and downtime studies and a coupled berth safety assessment used to compare mooring configuration improvements for ferries exceeding original berth capacity.

Experimental datasets. The reviewed slice contains a 40-year reconstructed harbor spectral climate at seven control points and a Jeju Port passenger-ferry/berth safety assessment; neither source alone supplies a broad measured moored-vessel benchmark archive.

Validated ranges. Applicability is limited to the Africa-basin spectral classification and the evaluated Jeju passenger ships, berth geometry, configurations and environmental cases; exact vessel and line ranges are unavailable from the indexed abstract.

Recent advances. Recent advances integrate long spectral climates, coupled ship-fender-mooring simulation, configuration optimization, mixture models for snap extremes, fatigue reliability and sensor-supported digital assessment, though most remain outside the currently accessible evidence slice.

Disagreements. Frequency-domain approaches are efficient for linear response and long climate integration, while time-domain models are needed for nonlinear lines, fender contact, slack-snap behavior and transient failures. Conservative static safety factors may not reproduce dynamic extremes or operational downtime.

Limitations. Current reviewed evidence lacks direct measured line tension and six-degree-of-freedom validation, detailed snap and fatigue treatment, passing-ship forcing, line aging and failure, probabilistic joint environmental extremes, human operations and transfer across berth and vessel classes.

Open questions. Priorities include open full-scale motion-tension datasets, directional multimodal forcing, nonlinear snap and fatigue, line/fender degradation, passing ships, automated line handling, climate nonstationarity, cascading failure and probabilistic downtime thresholds.

Seminal papers. Classical static line equilibrium, catenary mechanics, linear vessel-response theory and early terminal design criteria established mooring analysis; nonlinear time-domain simulation and measured harbor spectra later connected line design to operational response.

Claims

1. Improved prediction of moored-ship response and port downtime is an intended application of the spectral-type method, but this paper does not validate ship motions or operational gains. *Regime: Interpretive limitation on the Africa-basin methodology and all proposed operational applications.* direct finding, numerical [1210]
2. A Jeju Port assessment identified risks for car ferries exceeding berth design capacity and used sensitivity analysis to confirm that a revised mooring configuration improved the evaluated safety response. *Regime: A Study on the Improvement of Mooring Configuration through Mooring Safety Assessment for Passenger Ship Berth* direct finding, numerical [717]
3. Assessment of future ultra-large container vessels shows increasing ship dimensions impose higher demands on port mooring equipment, layouts, and infrastructure. *Regime: Analysis of the Mooring Effects of Future Ultra-Large Container Vessels (ULCV) on Port Infrastructures* direct finding, mixed [1385]
4. Simplified response graphs estimate moored-ship motions and line loads for harbor-tranquillity evaluation without full time-consuming simulation. *Regime: Effect of mooring system on moored ship motions and harbour tranquillity* direct finding, numerical [1235]
5. Wave-flume tests compare two barge-breakwater mooring arrangements through wave dissipation and anchor-chain force measurements. *Regime: Experimental Study on Hydrodynamic Characteristics of Barge-Type Breakwaters under Different Mooring Methods* direct finding, experimental [271]
6. A standards-based synthesis organizes mooring and berthing actions required for structural design of port infrastructure. *Regime: Port structures - the distribution of forces on infrastructure due to mooring and berthing of vessels* review statement, review [290]
7. Numerical calculations estimate wave-induced response of a berth-moored ship protected by a stilling pool for engineering design. *Regime: Sea wave impact on a ship moored at the berth with a stilling pool* direct finding, numerical [1281]
8. A mobile-harbor concept adds relative-position stabilization to shipboard winches in a four-point ship-to-ship mooring arrangement to support container transfer alongside a large anchored vessel through sea state 3. *Regime: A crane-equipped mobile-harbor vessel moored alongside a large anchored container ship during offshore cargo transfer.* direct finding, review [785]

38.4 Navigation and vessel response

Topic ports.navigation. Ship motions and safe access.

Papers screened: 23. Claims: 19. Equations: 0. Shown in the Coastal Design Bench beside: Navigation channel.

Well established. Safe port access and operability depend jointly on vessel response, dynamic water depth, squat and heel, wave/current/wind forcing, channel and berth geometry, mooring, traffic, forecasts, and human decisions.

Governing physics. Restricted depth and banks modify hull/propulsor/rudder forces and squat; waves drive heave, roll, pitch and horizontal motions; infragravity can dominate harbor response; currents, wind, tide, and moorings alter low-speed control and clearance.

Dimensionless parameters. Important controls include UKC/draft, depth/draft, blockage, Froude number, squat/draft, wave length/ship length, motion relative to cargo limits, forecast lead time, threshold exceedance probability, traffic density, and waiting tolerance.

Major equations. Frameworks include six-degree rigid-body motion and mooring, MMG-style maneuvering forces, squat and dynamic UKC budgets, ship-domain risk, wave transformation/forecasting, statistical motion prediction, accident exposure, and waiting/operability exceedance.

Typical methods. Methods span full-scale motion and AIS data, buoy and water-level records, hindcast/reanalysis, nested wave forecasts, maneuvering reviews, neural and semi-supervised prediction, Kalman filtering, accessibility statistics, and stakeholder-defined operational limits.

Numerical models. Models include six-degree motion learners, MMG and automatic-berthing representations, dynamic UKC and ship-domain tools, nested operational wave forecasts, Kalman draft estimation, climate downscaling, and accessibility/waiting probability.

Experimental datasets. Accessible evidence includes Punta Langosteira motion campaigns, 50 Accu-Waves port areas, international buoy validation, UK AIS/accidents, 2008-2022 Polish fairway levels, Spanish port risk assessment, and the Melilla record wave event.

Validated ranges. Reported evidence includes motion RMSE 0.09-0.11 m and 0.11-0.9 degrees, three-day forecasts every three hours across 50 areas, captured waves above 10 m, and Melilla infragravity correlation 0.96 with $H_s^2 T_p$ forcing.

Recent advances. Recent work uses high-frequency field motion, semi-supervised learning, nested multi-port forecasts, Kalman draft estimation, long water-level records, AIS exposure statistics, and explicit infragravity monitoring for real-time decisions.

Disagreements. Generic single-parameter limits are inadequate, yet complex predictors can overfit or hide mechanism. Climate-model spread may dominate scenario differences, and safe depth depends dynamically on speed and motion rather than a fixed UKC allowance.

Limitations. Full-scale datasets are sparse and proprietary; vessel, loading, mooring, berth, bathymetry and local resonance limit transfer. Forecast error, rare extremes, human factors, autonomy, and inconsistent operability criteria constrain validation.

Open questions. Needs include coupled maneuver-motion-mooring response, nonlinear harbor resonance, bank and passing-ship effects, dynamic UKC uncertainty, autonomous berthing assurance, traffic interaction, forecast-to-decision reliability, and climate-conditioned downtime.

Seminal papers. The branch connects static navigation depth and classical maneuvering/moored-motion analysis to dynamic UKC, probabilistic accessibility, operational forecasts, full-scale sensing, AIS risk, and data-assisted motion prediction.

Claims

1. Field-trained moored-ship models predicted six-degree motions with test errors below 10% of each motion range for operational limit decisions. *Regime: Test RMSE was 0.09-0.11 m for translations and 0.11-0.9 degrees for rotations, below 10% of motion ranges, enabling forecast-based cargo limits.* direct finding, mixed [47]
2. Automatic berthing requires maneuver models that resolve hull forces, four-quadrant propulsion, steerage, environmental disturbances, and auxiliary devices. *Regime: Berthing models must represent hull force, four-quadrant propulsion, steerage, wind/wave/current disturbances, and auxiliary devices before autonomous control is trusted.* review statement, review [1600]
3. Port of Gijón scenarios found inter-climate-model spread exceeded differences among time horizons or concentration pathways for dock operability. *Regime: Inter-GCM differences exceeded differences among time horizons or concentration scenarios, making climate model selection a dominant operational uncertainty.* direct finding, numerical [229]
4. Shallow and narrow-water maneuvering prediction must resolve ship design together with depth, bank, channel, wind, wave, and current effects. *Regime: Restricted-water maneuverability depends on hull/propulsor/rudder design and external wind, wave, current, bank, depth, and channel effects.* review statement, review [906]
5. Global open-data mapping identified short waves as the dominant seaport operability risk indicator at most locations, with exposed oceanic capes as hotspots. *Regime: Exposed high-latitude and southern-cape ports had greater risk, and short waves were the most critical indicator at most locations.* direct finding, numerical [1511]
6. Dynamic under-keel clearance must combine water-level variation, ship squat, heel, and wave response rather than use a static depth margin. *Regime: Dynamic UKC combines water-level variation, squat, heel, and wave response and can support safe draft maximization.* direct finding, analytical [476]
7. A three-dimensional ship-domain method links shallow-water speed and ECDIS safety depth to a 0-1 navigational risk factor. *Regime: The proposed method maps shallow-water speed and safety depth to a 0-1 navigation risk factor for canals and fairways.* direct finding, analytical [1227]
8. Berthed-ship operability should use dock-specific multi-factor limits and field monitoring rather than generic single-parameter thresholds. *Regime: Generic single-parameter limits are inadequate; berth/anchorage-specific monitoring and stakeholder experience should inform multi-factor criteria.* review statement, review [973]
9. The Accu-Waves platform delivered three-day, three-hourly forecasts for 50 port areas with adequate-to-very-good buoy validation and captured an Algeciras event above 10 m. *Regime: Three-hourly forecasts for 50 areas showed adequate-to-very-good buoy skill and reproduced an Algeciras event with wave maxima above 10 m.* direct finding, mixed [905]
10. A Punta Langosteira semi-supervised model outperformed compared learning methods for moored-vessel motion and explicitly evaluated infragravity predictors. *Regime: The semi-supervised model outperformed alternative supervised, unsupervised, and deep-learning methods, and infragravity predictors were tested alongside waves, wind, and sea level.* direct finding, mixed [1209]
11. Kalman filtering reduced uncertainty in real-time ship-draft estimates under simulated static and dynamic marine conditions. *Regime: Filtering reduced draft-measurement uncertainty in both static and dynamic conditions.* direct finding, numerical [339]
12. UK accident/AIS analysis linked port risk to vessel size, winter, service vessels, daytime transits, traffic density, geometry, and channel complexity, with human factors dominant. *Regime: Larger vessels, winter, service ships, daytime transits, and confined high-density geometry increased likelihood; human factors were the leading improvement need.* direct finding, mixed [912]
13. Fifteen years of fairway water levels quantified ship-specific acceptable level drops, passage probability, and expected waiting for deep-draft access. *Regime: Acceptable level drops, passage probability, and expected waiting depended on ship class, draft, and ability to delay entry.* direct finding, field [1633]
14. During a record Melilla event, harbor infragravity energy correlated 0.96 with offshore $H_s^2 T_p$ forcing and exceeded 2000 $m^2 s$. *Regime: Infragravity energy correlated 0.96 with an offshore H_s -squared-times-period forcing parameter and exceeded 2000 $m^2 s$ during energetic swell.* direct finding, mixed [864]
15. Dynamic ship-domain models incorporate restricted-waterway conditions, vessel characteristics, behavior, and operator skill for site-specific navigation-capacity analysis. *Regime: Dynamic Ship Domain Models for Capacity Analysis of Restricted Water Channels* direct finding, numerical [840]
16. A maritime-sensor review connects sensing configurations to safer and more sustainable vessel mobility and identifies remaining operational integration needs. *Regime: Review on Sensors for Sustainable and Safe Maritime Mobility* review statement, review [192]
17. A review assesses small unmanned surface vessels as lower-cost platforms for testing autonomous-navigation safety and assurance concepts. *Regime: Small Unmanned Surface Vessels—A Review and Critical Analysis of Relations to Safety and Safety Assurance of Larger Autonomous Ships* review statement, review [167]
18. A survey organizes visual perception and navigation-system technologies for smart ships and identifies safety, regulation, and integration challenges. *Regime: Visual Navigation Systems for Maritime Smart Ships: A Survey* review statement, review [1483]
19. A nested sequence of spectral-energy, extended mild-slope, and Boussinesq wave models transfers a 50-year design wave from regional generation scale to detailed Samchunpo harbor-entrance conditions for renovation design. *Regime: Typhoon design waves propagating from offshore through the south Korean coast into the Samchunpo (Sin Hyang) fishery harbor entrance.* direct finding, numerical [675]

38.5 Harbor sedimentation and dredging

Topic ports. sedimentation. Deposition and maintenance dredging.

Papers screened: 15. Claims: 13. Equations: 0. Shown in the Coastal Design Bench beside: Navigation channel.

Well established. Port and channel sedimentation occurs when imported or locally resuspended material converges, settles, consolidates, or migrates into sheltered basins and maintained depths faster than natural export or dredging removes it.

Governing physics. Controls include tidal filling and exchange, entrance gyres and flow separation, waves and wind currents, longshore and fluvial supply, aggregation and settling, salinity and sediment stratification, bed erosion and consolidation, and structure-induced sediment-cell changes.

Dimensionless parameters. Relevant groups include settling-to-advection timescale, Rouse- and Shields-type mobility, tidal excursion relative to entrance scale, spring-neap forcing, aggregate-to-primary-particle size, concentration and density stratification, and channel depth relative to ambient morphology.

Major equations. Frameworks combine suspended-sediment advection-diffusion-settling, erosion-deposition and consolidation, hydrodynamic momentum, bed evolution, particle or aggregate transport, source mixing, and geometric decomposition of tidal filling and horizontal/vertical exchange.

Typical methods. Methods use multilevel turbidity and mud-density observations, bathymetric and dredging records, satellite imagery, calibrated hydrodynamic-sediment models, flux decomposition, core erosion imaging, elemental and isotope provenance, and intervention scenario modelling.

Numerical models. Models diagnose entrance recirculation, water-bed exchange alternatives, aggregate transport modes, coupled hydrodynamic-bed change, and structural countermeasures. Equifinality remains possible when different near-bed mechanisms fit the same observations.

Experimental datasets. Evidence includes two Belgian harbor-basin exchange datasets, Zeebrugge field and dredging records, Nowshahr field calibration, tidal James River cores, Gladstone source samples, Lekki post-construction operation, and Dinh An channel surveys.

Validated ranges. Explicit findings include 50-100 Mt of hypothesized legacy mud, aggregates 50-270 times primary-particle size, spring-tide import 2-4 times neap, and import concentrated from roughly two hours before high water to high water.

Recent advances. Recent advances combine isotope fingerprints, aggregate imaging, multi-timescale fluid-mud monitoring, calibrated competing-mechanism models, nature-based sediment accommodation, and scenario optimization for channel stability.

Disagreements. At Zeebrugge, an erodible compact bed and reduced water-bed exchange reproduced available data comparably but predicted sediment export versus strengthened convergence. Existing observations therefore do not uniquely identify the maintaining mechanism.

Limitations. Sparse near-bed exchange, flocculation, density, consolidation, and bedload data; uncertain sediment provenance; bathymetric sampling; operational dredging feedback; ship effects; extreme events; and model equifinality constrain forecasts.

Open questions. Priorities are continuous mass budgets, fluid-mud rheology, aggregate-resolved transport, dredging and disposal feedback, climate and storm nonstationarity, source apportionment uncertainty, and whole-cell effects of port interventions.

Seminal papers. The branch builds from classical tidal-prism, exchange-flow, settling, erosion, and dredging-maintenance concepts toward geometric flux decomposition, cohesive aggregation, provenance tracing, and operational morphodynamic design.

Claims

1. At Zeebrugge, two models that fit concentration, satellite, bed-level, mud-content, and dredging data predicted opposite convergence responses, showing that uncertain near-bed exchange obscures whether decades-old port construction and 50-100 Mt of mobilized mud sustain the turbidity maximum. *Regime: Belgian-Dutch coast around Zeebrugge port* direct finding, mixed [915]
2. At Nowshahr, field-validated modelling showed that eastward wind-driven current separates around the western breakwater and forms a recirculation zone that imports resuspended shoreline sediment into the port entrance and access channel. *Regime: Nowshahr port entrance on the southern Caspian Sea under storms* direct finding, mixed [900]
3. The Lekki Sandbar Breakwater deliberately uses heavy updrift accretion to supplement sand construction and pairs it with a replenishable sand engine, maintaining port operation while managing downdrift sediment balance. *Regime: Lekki, Nigeria, on the high-drift Gulf of Guinea coast* direct finding, field [1332]
4. A geometric decomposition of two Belgian harbor datasets separates tidal filling, horizontal exchange, and vertical exchange, providing a calibration-ready diagnosis of the phasing and spatial correlations governing harbor siltation. *Regime: two Belgian tidal harbor basins* direct finding, field [1443]
5. In the tidal James River, eroded muddy-bed aggregates were 50-270 times larger than disaggregated sediment and were modelled in incipient suspension or bedload, materially changing predictions of navigation-channel infill. *Regime: tidal James River navigable waterway* direct finding, mixed [1097]
6. At Zeebrugge, suspended-mud import occurred from about two hours before high water to high water and was 2-4 times larger at spring than neap tide, while constant dredging permitted spring accumulation and winter deposits were thicker and less dense. *Regime: Port of Zeebrugge harbor basin and docks* direct finding, field [1444]
7. At the Port of Gladstone, combined particle-size, elemental, and isotope tracers differentiated candidate sources, with beryllium-7 indicating recent channel deposition from mudflats or catchments. *Regime: Port of Gladstone navigation channels and catchment, mudflat, and littoral sources* direct finding, field [642]
8. For the sediment-choked Dinh An route and Duyen Hai harbor bottlenecks, coupled MIKE 21 simulations compared five structural layouts using water level, velocity, and bed change to identify a preferred channel-stabilization option. *Regime: Dinh An estuary and Duyen Hai harbor basin, Vietnam* direct finding, numerical [1019]
9. A meta-analysis identifies suspended sediment, contamination, hydraulic entrainment, and underwater noise as distinct dredging stressors affecting fish. *Regime: A critical analysis of the direct effects of dredging on fish* review statement, review [1503]
10. Field monitoring around offshore dredging and beach nourishment documents spatial changes in turbidity, suspended sediment, and water-quality conditions. *Regime: Turbidity and Other Effects Resulting from Trafalgar Sandbank Dredging and Palmar Beach Nourishment* direct finding, field [1213]
11. Scale experiments and global examples show that sea-level rise and navigation dredging interact to alter estuary morphology and enhance channel-bend migration. *Regime: Effects of Sea-Level Rise on Dredging and Dredged Estuary Morphology* direct finding, mixed [302]
12. Analysis of Mar del Plata Harbor identifies recurrent access-channel and basin sedimentation as a controlling constraint on port development. *Regime: Analysis of Sedimentation Problems at the Entrance to Mar del Plata Harbor* direct finding, mixed [312]

13. The Port of Newcastle case documents maintenance-dredging and unconfined sea-disposal practice in a high-throughput river port, linking navigability requirements to sediment-management and environmental monitoring. *Regime: PORT OF NEWCASTLE, NSW, AUSTRALIA – MAINTENANCE DREDGING AND UNCONFINED SEA DISPOSAL OF DREDGE MATERIAL* direct finding, field [\[193\]](#)

39 Wiki: Remote sensing

39.1 Remote sensing: overview

Topic *remote_sensing*. Coastal observation using non-contact sensors.

Papers screened: 22. Claims: 4. Equations: 0.

Well established. Coastal remote sensing derives shoreline, elevation, bathymetry, waves and land-cover change from optical, radar, laser and photogrammetric measurements. Every product represents a defined physical or image proxy whose resolution, datum, acquisition conditions and uncertainty must be explicit.

Governing physics. Optical reflectance depends on illumination, atmosphere, water constituents and bottom signal; radar backscatter responds to roughness, moisture and geometry; LiDAR travel time and refraction yield elevation or depth; photogrammetry reconstructs geometry from parallax and control.

Dimensionless parameters. Key descriptors include pixel or point spacing relative to feature scale, signal-to-noise and contrast, overlap and baseline ratios, incidence angle, depth-to-attenuation scale, tide range-to-beach slope effect, classification accuracy and normalized positional error.

Major equations. Retrievals use sensor geometry, radiometric correction, refraction and travel-time relations, structure-from-motion bundle adjustment, spectral indices, classifiers and edge detection. Change rates regress proxy positions through time with positional and temporal uncertainty.

Typical methods. Workflows calibrate and georeference imagery, correct atmosphere and water level, segment land and water, build orthomosaics or point clouds, classify returns, validate against surveys, propagate error and analyze repeated acquisitions or fused sensors.

Numerical models. Processing uses threshold and index methods, supervised or unsupervised classification, machine learning, structure from motion, stereo reconstruction, LiDAR classification, bathymetric inversion, SAR speckle reduction and multitemporal change models.

Experimental datasets. Evidence spans forty years at Narrabeen–Collaroy, microtidal Landsat and Sentinel beaches, UAV beach and tombolo surveys, Langue de Barbarie and Bangladesh shoreline archives, Baltic bathymetric LiDAR and coastal topography–bathymetry demonstrations.

Validated ranges. The reviewed corpus spans centimeter-to-decimeter UAV products, meter-scale airborne or spaceborne elevation and 10–30 m multispectral archives, with instantaneous to four-decade coverage. Valid ranges remain sensor-, water-clarity- and proxy-specific.

Recent advances. Recent advances integrate UAV structure from motion, satellite-derived topography–bathymetry, cloud-scale processing, SAR–optical complementarity, machine learning and dense multitemporal archives to observe the coast as a connected continuum.

Disagreements. A shoreline is not unique: wet–dry, vegetation, waterline and datum-based proxies differ systematically. Automated methods improve repeatability but can mask tide, runup, turbidity and geolocation error; higher spatial resolution often trades against coverage and revisit.

Limitations. Clouds, haze, sun glint, turbidity, waves, refraction, limited LiDAR penetration, SAR speckle, vegetation, weak texture, ground-control error, sensor changes and sparse validation constrain accuracy and long-term consistency.

Open questions. Priorities include analysis-ready multi-sensor archives, robust uncertainty across proxies, turbid-water bathymetry, global validation, near-real-time hazards, automated quality control, physics-aware ML and equitable access to repeat coastal surveys.

Seminal papers. Aerial photogrammetry and shoreline mapping established repeat coastal surveys; Landsat enabled global multidecadal archives, while airborne LiDAR, SAR and video expanded elevation, all-weather and high-frequency observation.

Claims

1. Coastal-current monitoring platforms are complementary: HF radar can map surface currents to roughly 200 km offshore, while satellites and drifters extend coverage but impose revisit, cloud, or tracking limitations. *Regime: Surface-current observation and the sensor technologies available through 2012.* review statement, review [1629]
2. Shoreline mapping requires an explicit physical shoreline definition and sensor-appropriate proxy because extracted position varies with tide, wave conditions, morphology, image characteristics and interpretation method. *Regime: Shoreline Definition and Detection: A Review* review statement, review [163]
3. Coastal remote sensing combines satellite, airborne and in-situ non-contact observations to monitor shoreline, water, habitat and hazards across scales, with sensor fusion and validation central to operational use. *Regime: Remote Sensing Applications in Coastal Areas* review statement, review [760]
4. Field hyperspectra can discriminate many salt-marsh vegetation types because pigment absorption and especially canopy-structure effects create distinct spectral signatures; continuum removal improves visible-band separability but reduces near- and shortwave-infrared separability. *Regime: Field reflectance spectra from 27 salt-marsh vegetation types on the Dutch Wadden Sea coast, measured from 400 to 2500 nm during the growing season.* direct finding, field [1264]

39.2 LiDAR and photogrammetry

Topic *remote_sensing*. Lidar. Elevation and morphology mapping.

Papers screened: 10. Claims: 5. Equations: 0.

Well established. Coastal LiDAR provides high-resolution elevation and structure for marsh, dune, defense, and flood applications; quantitative use still requires independent vertical validation and explicit treatment of vegetation, water, and hydraulic connectivity.

Governing physics. Return elevation depends on laser interaction with canopy and ground, whereas coastal engineering interpretation depends on whether mapped terrain represents the hydraulic surface, drainage pathways, and morphologic change rather than vegetation or water artifacts.

Dimensionless parameters. Classification accuracy, relative volume difference, and compliance fraction are useful dimensionless performance measures, but their meaning is conditional on class balance, change magnitude, reference uncertainty, and the engineering threshold being tested.

Major equations. The reviewed slice relies on point-to-reference residuals, MAE and RMSE, classification accuracy, DEM differencing, cross-section integration for sediment volume, and threshold-plus-connectivity inundation logic; it does not establish a universal coastal LiDAR transfer equation.

Typical methods. Typical workflows acquire airborne or UAV point clouds, classify ground and vegetation returns, build a DEM or DTM, coregister repeat surveys or auxiliary imagery, validate with RTK or field elevations, and propagate the terrain into habitat, morphology, defense, or inundation analyses.

Numerical models. Models range from cover-conditioned decision trees and genetic-algorithm ground estimators to automated levee-axis extraction, connected-cell inundation screening, DEM differencing, and data-driven attribution of morphologic change.

Experimental datasets. The reviewed datasets include a nine-class *Spartina* marsh, paired UAV marsh sensors, 526 RTK dune checkpoints, federal and local Louisiana levees, Staten Island terrain, and eight paired Baltic campaigns producing 16 DEMs and 895 profiles.

Validated ranges. Reported validation spans marsh DEM RMSE improvement from 0.15 to 0.10 m, UAV-LiDAR marsh ground RMSE of 5.9 cm, vegetated-dune RMSE of 9.86 cm, and a 0.48% LiDAR-SfM difference in bulk Baltic volume change; these values are site- and workflow-specific.

Recent advances. Recent work couples compact UAV LiDAR with field-calibrated learning, direct paired comparison with SfM, and repeat morphodynamic monitoring, shifting LiDAR from episodic mapping toward change detection and method selection by terrain regime.

Disagreements. LiDAR clearly outperformed photogrammetry for local ground and vegetation retrieval in dense marsh and dune settings, while paired Baltic surveys differed by only 0.48% in net volume change. This is a scale-and-metric distinction, not a contradiction: bulk change can agree while local errors persist near water and vegetation.

Limitations. Limitations include vegetation interception, sparse ground returns, water-surface artifacts, scan geometry, DEM vertical uncertainty, site-trained classifiers, cross-survey coregistration, and failure of elevation-only inundation to represent culverts, barriers, friction, and timing.

Open questions. Priorities are transferable vegetation corrections, standardized uncertainty budgets, repeated validation across tidal states and storm magnitudes, automated representation of culverts and defenses, and decision thresholds that compare expected change with the survey error floor.

Seminal papers. Within this screened branch, the 2013 salt-marsh study establishes cover-conditioned correction of vegetation-biased LiDAR elevation, and the 2014 levee study demonstrates engineering screening of long, irregular coastal defenses from airborne terrain.

Claims

1. Hyperspectral cover-class corrections reduced salt-marsh LiDAR DEM RMSE from 0.15 to 0.10 m while a multisensor decision tree achieved 90% habitat-classification accuracy. *Regime: The studied southeastern US Spartina marsh and its nine cover classes.* direct finding, field [602]
2. At Little Sapelo marsh, UAV LiDAR ground-elevation RMSE was 5.9 cm versus 17.2 cm for photogrammetry, and vegetation-height RMSE was 17.5 versus 38.1 cm. *Regime: Dense Spartina marsh, 0.40 m output maps, and the tested UAV acquisition/classification workflow.* direct finding, field [1115]
3. Across eight paired Baltic UAV surveys and 895 profiles, LiDAR and SfM net beach-plus-dune volume change differed by only 0.48%, although SfM produced more local artifacts over water and vegetated dunes. *Regime: The 2022-2023 Mrzezyno survey geometry and bulk-change metric; local small changes may not be equivalent.* direct finding, field [1634]
4. Bathymetric LiDAR supplies integrated elevation measurements across shallow water and land for coastal-zone assessment, subject to water clarity, depth penetration, classification and ground validation. *Regime: Bringing Bathymetry LiDAR to Coastal Zone Assessment: A Case Study in the Southern Baltic* direct finding, mixed [1434]
5. Elevated 2D scanning LiDAR resolved wave-by-wave overtopping on an evolving porous cobble revetment; two reconstruction methods agreed within a factor of two for most events and showed rapid landward discharge decay from infiltration. *Regime: Laboratory dynamic cobble revetments with time-varying crest geometry under the tested waves and material configurations.* direct finding, experimental [160]

39.3 Satellite remote sensing

Topic remote_sensing. satelliTe. Optical, radar, and altimetry observations.

Papers screened: 40. Claims: 29. Equations: 0.

Well established. Satellite radar altimetry provides long, repeatable observations of sea surface height, significant wave height, and wind speed that complement sparse in situ coastal networks, but quality generally deteriorates close to land and sampling is not continuous.

Governing physics. Radar range and returned-waveform shape determine sea surface height and sea-state estimates; land contamination, surface roughness, atmosphere and geophysical corrections, track geometry, tides, and land motion all affect coastal interpretation.

Dimensionless parameters. Correlation, normalized scatter, detection rate, and regression slope quantify skill but are dataset-specific statistical measures. No universal dimensionless parameter removes the effects of coast distance, waveform contamination, or sampling geometry.

Major equations. Core calculations combine corrected radar range with satellite orbit height, derive wave and wind metrics from waveforms/backscatter, difference relative tide-gauge and absolute altimeter trends for vertical land motion, and use correlation or error metrics to rank products and tide models.

Typical methods. Typical workflows collocate altimeters with buoys or tide gauges, merge multiple missions, retrack coastal waveforms, apply objective mapping, combine optical wetting indices with tide predictions, fuse numerical simulations with machine learning, and validate against independent fixed instruments.

Numerical models. Relevant models include multi-mission objective analysis, geophysical correction and retracking chains, random-forest event classifiers, global ocean tide models, and the eo-tides locally adaptive ensemble.

Experimental datasets. Reviewed datasets include 36 months of Iberian radar-buoy-Sentinel observations, five-mission northeastern US metocean composites, 27 years at 50 Southeast Asian VLM sites, a ten-year Gulf of Lion simulation/altimetry record, and continent-wide Australian tide-model rankings every 10 km.

Validated ranges. Skill is product and regime bound: Iberian Sentinel-3 SWH validation achieved R above 0.94 and RMSE 0.17-0.42 m; northeastern US composites are climatological and weaken near land; Australian NDWI rankings sample tidal phase more directly than amplitude.

Recent advances. Recent advances combine sensors and models rather than relying on one product: tide gauges plus altimetry recover VLM, optical NDWI plus altimetry guides tide ensembles, and simulation-altimetry-ADCP fusion enables event classification.

Disagreements. Altimetry can agree closely with buoys in favorable coastal collocations yet still miss extremes or become biased near shore. This is a resolution, sampling, and waveform-regime difference rather than conflicting evidence about universal sensor accuracy.

Limitations. Land-contaminated waveforms, sparse repeat tracks, asynchronous sampling, correction uncertainty, sun-synchronous tidal aliasing, fixed-sensor coverage, and site-trained machine learning constrain transferability and event detection.

Open questions. Priorities include high-resolution coastal retracking, uncertainty-aware multi-mission fusion, event-scale sampling, joint absolute and relative sea-level attribution, transferable current classifiers, and exploitation of SWOT and future missions.

Seminal papers. The 2023 altimetry review is the synthesis anchor in this screened slice, documenting the multi-decadal mission lineage and transition from open-ocean measurement toward operational coastal applications.

Claims

1. More than 30 years of multi-mission satellite altimetry provide operational observations of ocean surface topography, sea level, waves, and circulation, with newer systems extending useful measurements closer to coastlines. *Regime: Validated radar-altimetry products; coastal performance remains mission-, correction-, and distance-dependent.* review statement, review [1342]
2. Five-mission altimetry composites reproduced northeastern US seasonal wind and wave patterns, but bias increased near shore and aggregated observations missed some buoy extrema within 40 km of the coast. *Regime: Regional seasonal climatology using 10 km/30 minute collocations; not continuous extreme-event monitoring.* direct finding, field [967]
3. Japan's TSSD-JP cross-mission shoreline dataset placed more than 80% of validated positions within 50 m and estimated 293.21 km² net land gain over more than 30 years, including engineering-related bay accretion. *Regime: Nation-scale dual-epoch screening; changes smaller than positional accuracy require caution.* direct finding, field [1428]
4. A shallow-water adaptation of operational ENVISAT SAR spectral inversion agreed well with seven in-situ sensors under low-to-moderate winds across varied wave heights, directions and nonstandard incidence conditions. *Regime: Extraction of Coastal Ocean Wave Fields From SAR Images* direct finding, field [287]
5. Optical shoreline detection spans thresholding, classification, edge and index-based methods; accuracy depends on spatial resolution, spectral contrast, water level, preprocessing and validation reference. *Regime: Shoreline Detection using Optical Remote Sensing: A Review* review statement, review [1424]
6. Automated shorelines from Landsat 7, Landsat 8 and Sentinel-2 on microtidal beaches show that sensor resolution, georeferencing, water-land contrast and reference-survey timing determine positional accuracy. *Regime: Assessing the Accuracy of Automatically Extracted Shorelines on Microtidal Beaches from Landsat 7, Landsat 8 and Sentinel-2 Imagery* direct finding, mixed [1079]
7. A review finds spaceborne optical, radar and altimetric observations increasingly support beach topography and shallow bathymetry, but water clarity, waves, resolution and validation constrain the land-water continuum. *Regime: Monitoring Beach Topography and Nearshore Bathymetry Using Spaceborne Remote Sensing: A Review* review statement, review [1237]
8. Satellite optical imagery provides repeatable coastal-engineering observations of shorelines, morphology, bathymetry and change over broad areas, with applicability controlled by resolution, clouds, water optics and ground truth. *Regime: Satellite optical imagery in Coastal Engineering* review statement, review [1432]
9. Four decades of Narrabeen-Collaroy monitoring demonstrate that sustained remote observations reveal coastal variability and change across event to multi-decadal scales that short campaigns cannot resolve. *Regime: Remote Sensing Is Changing Our View of the Coast: Insights from 40 Years of Monitoring at Narrabeen-Collaroy, Australia* direct finding, mixed [1336]
10. Integrated satellite methods can monitor the coastal topography-bathymetry continuum, linking subaerial morphology and shallow-water depth while exposing sensor, retrieval and validation tradeoffs. *Regime: Coastal morphology from space: A showcase of monitoring the topography-bathymetry continuum* direct finding, mixed [143]
11. Coastline-extraction methods differ in sensor, preprocessing, segmentation and change-analysis choices, so definitions, uncertainty and validation must accompany mapped trends. *Regime: Coastline extraction using remote sensing: a review* review statement, review [1371]
12. Satellite-derived elevation and shoreline time series at the Langue de Barbarie sand spit support joint observation and prediction of coastal erosion and rapid morphological change. *Regime: Observing and Predicting Coastal Erosion at the Langue de Barbarie Sand Spit around Saint Louis (Senegal, West Africa) through Satellite-Derived Digital Elevation Model and Shoreline* direct finding, mixed [1401]
13. Multitemporal satellite data and tasseled-cap transformation quantify shoreline change along the Bangladesh delta, demonstrating regional-scale mapping of erosion and accretion patterns. *Regime: Shoreline Change Assessment in the Coastal Region of Bangladesh Delta Using Tasseled Cap Transformation from Satellite Remote Sensing Dataset* direct finding, mixed [1290]
14. High-resolution satellite enhancement and segmentation algorithms can extract instantaneous shorelines, but the mapped proxy remains sensitive to image quality, threshold choice and coastal conditions. *Regime: High Resolution Satellite Images for Instantaneous Shoreline Extraction Using New Enhancement Algorithms* direct finding, mixed [356]
15. Threshold-based processing of satellite SAR enables shoreline extraction and coastal-change detection when clouds impede optical imagery, while speckle, land-water backscatter and geometric correction control reliability. *Regime: Shoreline extraction and coastal change detection from satellite SAR using thresholding-based methods* direct finding, mixed [411]

16. CoastSat provides an open Google Earth Engine and Python workflow for extracting multitemporal shorelines from public satellite imagery with reproducible preprocessing and mapping. *Regime: CoastSat: A Google Earth Engine-enabled Python toolkit to extract shorelines from publicly available satellite imagery* direct finding, mixed [1463]
17. The GEO-CAPE mission concept links geostationary atmospheric-composition and coastal-ecosystem measurements to high-frequency observation of rapidly changing coastal conditions. *Regime: The United States' Next Generation of Atmospheric Composition and Coastal Ecosystem Measurements: NASA's Geostationary Coastal and Air Pollution Events (GEO-CAPE) Mission* review statement, review [444]
18. The adaptive ALES retracker recovers more reliable coastal-altimetry observations near land and improves agreement with tide gauges across three satellite missions. *Regime: ALES: A multi-mission adaptive subwaveform retracker for coastal and open ocean altimetry* direct finding, mixed [1081]
19. A review organizes ocean-color sensors, algorithms, and statistical methods for detecting and mapping phytoplankton blooms in coastal and open oceans. *Regime: A review of ocean color remote sensing methods and statistical techniques for the detection, mapping and analysis of phytoplankton blooms in coastal and open oceans* review statement, review [162]
20. A literature survey organizes optical satellite sensors and methods for maritime vessel detection and classification and identifies performance constraints. *Regime: Vessel detection and classification from spaceborne optical images: A literature survey* review statement, review [686]
21. A fifty-year synthesis traces satellite sea-surface-temperature retrieval, calibration, atmospheric correction, uncertainty, and ocean applications. *Regime: Half a century of satellite remote sensing of sea-surface temperature* review statement, review [966]
22. A review organizes satellite and related remote-sensing approaches for mapping and monitoring mangrove extent and condition. *Regime: Remote sensing techniques: mapping and monitoring of mangrove ecosystem—a review* review statement, review [937]
23. A wetland remote-sensing review compares sensors and classification approaches for mapping ecosystem extent, condition, and change. *Regime: A Review of Wetland Remote Sensing* review statement, review [552]
24. The SWOT mission uses wide-swath radar interferometry to observe ocean surface topography and inland-water elevations at unprecedented spatial detail. *Regime: The Surface Water and Ocean Topography Mission: A Breakthrough in Radar Remote Sensing of the Ocean and Land Surface Water* direct finding, mixed [465]
25. A review summarizes satellite and airborne mapping of river floods and coastal storm-surge inundation for response and hazard assessment. *Regime: Remote Sensing of Floods and Flood-Prone Areas: An Overview* review statement, review [729]
26. Landsat-8 satellite-derived bathymetry calibrated with LiDAR mapped a 60-km complex Ria Formosa coast from 0-12 m depth with mean bias -0.2 m, median difference -0.1 m and RMSE 0.89 m, with best performance shallower than 8 m and degradation near turbid inlets. *Regime: Optically shallow waters of Ria Formosa, Portugal, including barriers, tidal inlets, ebb deltas and nearshore bars.* direct finding, field [1069]
27. Five decades of coastal-wetland remote sensing show a progression from extent mapping to structural and physiological retrieval and then ecosystem-function and biogeochemical assessment, with future gains depending on multi-sensor fusion, ecological-process coupling, and artificial intelligence. *Regime: Remote sensing of global coastal-wetland extent, structure, traits, functions and material cycles across the reviewed sensor eras.* review statement, review [1484]
28. A calibration-free Sentinel-2/GEE workflow classified intermittent river-mouth connectivity across about 1,500 km of Western Mediterranean coast, with fixed-threshold NDWI achieving roughly 80% agreement and showing discharge promotes breaching while waves promote bar formation and closure. *Regime: Intermittently open/closed river mouths on the microtidal, wave-dominated Western Mediterranean coast over the Sentinel-2 record.* direct finding, mixed [1308]
29. No single multispectral-image method can extract shoreline indicators across all coast types: current work is dominated by instantaneous waterlines, so global monitoring requires indicator- and setting-specific combinations of spectral, segmentation and machine-learning methods. *Regime: Landsat, Sentinel-2 and related multispectral imagery used to identify waterlines, cliffs, dunes, vegetation and other shoreline indicators from local to global scales.* review statement, review [942]

39.4 Uncrewed aerial systems

Topic *remote_sensing.uas*. Drone-based coastal observations.

Papers screened: 18. Claims: 15. Equations: 2.

Well established. Coastal UAS provide operator-timed centimetre-scale observations between ground surveys and crewed or satellite platforms, but accuracy depends on sensor physics, control and georeferencing, surface texture, vegetation, water, tide, weather, processing, validation, and the spatial scale of the reported metric.

Governing physics. Passive SfM requires stable visible texture and is degraded by moving or reflective water and canopy occlusion; LiDAR samples range and may penetrate canopy but still needs ground classification; multispectral and hyperspectral products depend on illumination, spectral contrast, water-column attenuation, turbidity, depth, and mixed pixels.

Dimensionless parameters. Important nondimensional quantities include spectral indices, normalized reflectance and band ratios, spectral angle, classification agreement such as kappa, relative vertical error, overlap and geometry quality measures, and habitat or vegetation fractions; dimensional pixel size alone is not an accuracy guarantee.

Major equations. Representative formulations include SfM bundle adjustment and surface differencing, NDVI and its calibrated biomass regression $AGB=2428.2 \text{ NDVI}+120.1$, spectral-angle similarity, Random-Forest or ANN classification, allometric biomass regressions, and error metrics such as RMSE and Cohen kappa.

Typical methods. Workflows combine flight and tide planning, RTK/PPK or ground control, camera or LiDAR calibration, overlapping imagery and SfM, point-cloud filtering, orthomosaic or DEM creation, repeat differencing, spectral indices or supervised classification, and independent checkpoints, field plots, profiles, or underwater video.

Numerical models. Models range from photogrammetric reconstruction and DEM differencing to NDVI regression, vegetation allometry, spectral-angle mapping, Random Forest habitat classification, and ANN LiDAR filtering. Each model requires calibration and validation matched to the target variable and environment.

Experimental datasets. The branch includes seasonal biomass and multispectral imagery at Carpinteria, paired LiDAR-photogrammetry marsh observations at Little Sapelo, eight paired dune surveys and 895 profiles in the Baltic, nine morphology surveys of a New Jersey groin field, multiline LiDAR in dense marsh, and paired aerial-surface hyperspectral habitat maps.

Validated ranges. Reported tests span 1-10 cm imagery in the foundational review; 5.9 versus 17.2 cm marsh-ground RMSE for LiDAR and photogrammetry; 0.48% difference in integrated Baltic volume change across eight surveys and 895 profiles; 1-10 cm hyperspectral pixels over 230-20,200 m²; and 0.04 m UAV through 10-20 m satellite imagery with mean kappa 0.71-0.77.

Recent advances. Recent advances move beyond RGB surface reconstruction toward spatial-spectral ANN ground filtering, multi-temporal sensor intercomparison, UAV-derived vegetation allometry, multiscale hyperspectral habitat mapping, and target-specific comparisons spanning drone, airplane, surface vehicle, and satellite observations.

Disagreements. LiDAR's superior local marsh elevation accuracy and SfM's close agreement in integrated dune-volume change are not contradictory: point elevation, profile artifacts, and spatially integrated volume are different metrics over different surfaces. Likewise, UAV's finer pixels did not universally outperform airplane or satellite habitat mapping because target separability and water effects controlled skill.

Limitations. Small footprints, battery and weather limits, flight regulation, tide and sun-glint timing, ground-control distribution, moving water, dense or flexible vegetation, turbidity, processing choices, scarce reference data, and site-specific calibration limit repeatability and transfer. Newer 2026 evidence also remains lightly replicated.

Open questions. Priorities are standardized uncertainty budgets, synchronized multi-sensor benchmarks, cross-site classifier and allometry transfer, water-column correction, canopy and ground separation, automated quality flags, defensible change-detection thresholds, and scalable fusion with satellite and in-situ observations.

Seminal papers. Within this screened branch, the 2015 overview framed the platform and sensor opportunity, while the 2021 Little Sapelo comparison established quantitative LiDAR-versus-photogrammetry accuracy for vegetated coastal terrain.

Equations

- *NDVI aboveground-biomass regression* [369]

$$AGB = 2428.2 NDVI + 120.1$$

Regime: Carpinteria Salt Marsh vegetation and study sensor/calibration seasons. Variables: AGB salt-marsh aboveground biomass in g m⁻²; NDVI normalized difference vegetation index

- *Spectral angle* [1018]

$$\theta = \cos^{-1} \left(\frac{\mathbf{t} \cdot \mathbf{r}}{\|\mathbf{t}\| \|\mathbf{r}\|} \right)$$

Regime: Hyperspectral classification with selected SAM thresholds. Variables: theta spectral angle; t target spectrum; r reference spectrum

Claims

1. Coastal UAVs offer operator-controlled revisit timing and sub-10-cm imagery with diverse sensors, but payload, endurance, weather, regulation, and ground control constrain performance. *Regime: beaches; wetlands; nearshore waters* review statement, review [728]
2. At Carpinteria Salt Marsh, seasonal NDVI biomass calibration outperformed pooled calibration, with spring $r^2=0.67$ and RMSE=344 g m⁻² versus pooled $r^2=0.36$ and RMSE=496 g m⁻². *Regime: California salt marsh* direct finding, field [369]
3. Spatial-spectral ANN filtering of drone LiDAR improved salt-marsh ground-point discrimination needed for terrain reconstruction under dense vegetation. *Regime: vegetated coastal salt marsh* direct finding, field [852]
4. UAV-derived vegetation structure can support nondestructive aboveground-biomass allometry for temperate mangroves, subject to canopy penetration and calibration-population limits. *Regime: temperate mangrove forest* direct finding, field [1169]
5. UAV hyperspectral imaging mapped 20,200 m² at 10 cm resolution while a surface vehicle mapped 230 m² at 1 cm, demonstrating a coverage-resolution tradeoff. *Regime: shallow seagrass and macroalgae habitat* direct finding, mixed [1018]
6. In a Baltic lagoon, mean Random-Forest kappa was 0.71-0.77 across UAV, airplane, and satellite imagery, but target-specific skill ranged from 0.99 for UAV vegetation presence to 0.29-0.57 for free water column. *Regime: Baltic coastal lagoon* direct finding, field [591]
7. UAV structure-from-motion photogrammetry can reconstruct high-resolution coastal topography from overlapping imagery, providing flexible terrain products where survey design, control and surface texture support reliable matching. *Regime: Using Unmanned Aerial Vehicles (UAV) for High-Resolution Reconstruction of Topography: The Structure from Motion Approach on Coastal Environments* direct finding, mixed [911]
8. UAV photogrammetry combined with ground surveys enables rapid, repeatable mapping of shoreline and beach change at local scale when control, image overlap and survey comparison are managed. *Regime: UAV Photogrammetry and Ground Surveys as a Mapping Tool for Quickly Monitoring Shoreline and Beach Changes* direct finding, mixed [1580]
9. Repeated UAV photogrammetry over the Sopot tombolo produces high-resolution coastal-zone change measurements suitable for resolving rapid local morphological evolution. *Regime: Using UAV Photogrammetry to Analyse Changes in the Coastal Zone Based on the Sopot Tombolo (Salient) Measurement Project* direct finding, mixed [215]
10. Comparing UAV photogrammetry and airborne laser scanning supports shallow-water depth estimation and shoreline extraction while quantifying method-specific spatial coverage and accuracy. *Regime: Spatial Analysis of Bathymetric Data from UAV Photogrammetry and ALS LiDAR: Shallow-Water Depth Estimation and Shoreline Extraction* direct finding, mixed [1331]
11. Repeated UAV photogrammetry maps joints, megaclasts, and deformation associated with slow-moving coastal landslides in Malta. *Regime: Advantages of Using UAV Digital Photogrammetry in the Study of Slow-Moving Coastal Landslides* direct finding, field [336]
12. Drone trials relate flight altitude and image resolution to detection and classification of litter on beaches, dunes, banks, and coastal waters. *Regime: Drones for litter monitoring on coasts and rivers: suitable flight altitude and image resolution* direct finding, field [59]
13. A Spanish Mediterranean case study demonstrates repeatable UAV monitoring for coastal morphology and management at Almenara-Sagunto. *Regime: Coastal Monitoring Using Unmanned Aerial Vehicles (UAVs) for the Management of the Spanish Mediterranean Coast: The Case of Almenara-Sagunto* direct finding, field [249]
14. A portable-UAV planning framework specifies survey design for high-resolution, repeatable observation of coastal hydro-environments. *Regime: A Framework for Survey Planning Using Portable Unmanned Aerial Vehicles (pUAVs) in Coastal Hydro-Environment* direct finding, mixed [1426]

15. A coastal UAS survey protocol identifies flight planning, ground control, illumination, wind, platform, sensor and processing controls needed for repeatable high-resolution coastal products. *Regime: A Protocol for Aerial Survey in Coastal Areas Using UAS* review statement, review [370]

39.5 Coastal video monitoring

Topic *remote_sensing.video*. Image-derived waves, currents, and shoreline.

Papers screened: 8. Claims: 6. Equations: 1.

Well established. Fixed coastal cameras provide frequent, low-cost observations of shoreline and breaker-zone behavior; but quantitative products require stable geometry, accurate calibration, visible scene features, and algorithms validated for the site and image conditions.

Governing physics. Timex imagery converts persistent optical signatures into measurable spatial patterns: breaking foam marks the breaker zone and time-averaged land-water contrast marks the shoreline; perspective projection maps those pixels into coastal coordinates.

Dimensionless parameters. Normalized RMSE, IoU, F1, and automatic-success fraction quantify retrieval performance but are dataset-specific. Viewing-angle geometry and relative feature density are central conditioning factors rather than universal similarity parameters.

Major equations. Core operations include projective camera geometry, image registration transforms, cross-correlation displacement, foam-length empirical breaker-height relations, and segmentation metrics such as intersection over union and boundary F1.

Typical methods. Typical workflows create snapshot and Timex images, detect static edges or features, stabilize and calibrate each frame, rectify pixels to ground coordinates, then apply empirical foam geometry, thresholding, or semantic segmentation and validate against independent waves or shorelines.

Numerical models. Models include Hsb,v/Hsb,v24 empirical breaker estimators, edge-keypoint geometric stabilization, UCalib feature-transfer calibration, and U-Net/DeepLabv3+ semantic segmentation with several CNN backbones.

Experimental datasets. Reviewed evidence includes about 900 breaker-height records across five sites, five years of three-camera Anglet imagery, a three-camera UCalib conditioning study, and a labeled fixed-camera Timex segmentation dataset.

Validated ranges. The bathymetry-free breaker method spans 0.1-3.8 m with 18% mean normalized RMSE; Anglet stabilization automated at least 90% of frames; UCalib ranged from above 90% to about 40% success by feature availability; segmentation mIoU was 0.95-0.97 in-domain.

Recent advances. Recent advances shift from hand-designed thresholds toward autocalibration and deep segmentation, while adjacent satellite studies supply longer spatial context for waterline and shoreline interpretation.

Disagreements. High pixel-level segmentation scores do not guarantee accurate real-world shoreline coordinates: calibration drift can create tens to hundreds of metres of positional error. These evaluate different pipeline stages and must be validated jointly.

Limitations. Performance degrades with missing static features, poor visibility or illumination, foam ambiguity, occlusion, long camera range, thermal motion, unrepresented sites, and error propagation from calibration through rectification and extraction.

Open questions. Needs include standardized multi-site benchmarks, calibrated uncertainty from pixels to ground coordinates, robust feature-free calibration, domain adaptation across lighting and morphology, and fusion with satellite, UAV, buoy, and model observations.

Seminal papers. Within this screened slice, the Anglet stabilization study establishes the scale of long-term geometry error, while the Timex breaker-height paper demonstrates a directly engineering-relevant hydrodynamic retrieval.

Equations

- *Wave-energy transformation balance* [451]

$$\nabla \cdot (E \mathbf{C}_g) = -D_b - D_f$$

Regime: The tested irregular waves and laboratory bathymetry with incident energy and bathymetry known. Variables: E wave energy density; $\mathbf{C}_g$ group velocity; D_b breaking dissipation; D_f bottom-friction or other dissipation

Claims

1. A bathymetry-free Timex foam-signature method estimated breaking wave heights of 0.1-3.8 m across five sites with mean normalized RMSE of 18% over about 900 records. *Regime: Visible breaker-foam signatures at the five tested sites; performance outside the tested morphology and lighting regimes is unverified.* direct finding, field [58]
2. At Anglet, camera motion produced georectification errors up to 400 m at 2.5 km range and an apparent shoreline bias of roughly 10-20 m during winter 2013/2014; stabilization corrected at least 90% of frames automatically. *Regime: The Anglet installation and views containing stable recognizable land features.* direct finding, field [1201]
3. UCalib automatically calibrated more than 90% of images for feature-rich coastal cameras but only about 40% for the worst, nearly featureless camera, demonstrating that fixed-feature availability controls operational success. *Regime: Argus-like or CoastSnap imagery with reference calibrations and detectable fixed features.* direct finding, field [1309]
4. In an irregular-wave laboratory surf zone, optical measurements of individual roller size yielded roller energy, dissipation and radiation stress that reproduced measured wave-height transformation and mean-water-level setup with minimal calibration, including transition-zone lag. *Regime: The tested irregular waves and laboratory bathymetry with incident energy and bathymetry known.* direct finding, experimental [451]
5. Pixel-intensity variability in coastal video robustly identifies the cross- and alongshore extent of shoaling, surf and swash domains, supporting automated breaking-height, depth-inversion and sediment-transport applications. *Regime: Nearshore Wave Transformation Domains from Video Imagery* direct finding, mixed [57]
6. Inverse modeling of marine-radar image sequences retrieves surface-elevation maps and wave spectra, with buoy comparisons supporting the linear-wave imaging formulation. *Regime: Inversion of Marine Radar Images for Surface Wave Analysis* direct finding, mixed [172]

40 Wiki: Coastal hazards and risk

40.1 Coastal hazards and risk: overview

Topic risk. Hazard, exposure, vulnerability, consequences, and decisions.

Papers screened: 19. Claims: 2. Equations: 0.

Well established. Coastal risk arises from the joint occurrence of hazard, exposed people/assets/ecosystems, vulnerability or fragility, and consequences. Changes in any component can dominate risk, and correlated or sequential hazards can amplify impacts beyond independent single-hazard estimates.

Governing physics. Storms, waves, surge, rainfall, rivers, erosion, tsunami and defense failure generate loads and inundation; receptor geometry, materials, social capacity and networks govern damage and recovery. Climate, development and adaptation make hazard, exposure and vulnerability nonstationary.

Dimensionless parameters. Controls include annual exceedance probability, return-period-to-lifetime ratio, conditional failure probability, damage ratio, vulnerability index, protection reliability, hazard and loss correlation, benefit-cost ratio, expected annual loss relative to GDP or budget and recovery-time ratios.

Major equations. Expected risk integrates consequence over hazard intensity and probability, often as expected annual damage. Fragility gives conditional failure probability; vulnerability maps intensity to damage ratio or sensitivity; reliability compares stochastic load and resistance, and multihazard models represent joint, conditional or cascading events.

Typical methods. Workflows define system and receptors, construct probabilistic hazards, map exposure, estimate fragility/vulnerability and losses, model dependence and cascades, validate each component, aggregate uncertainty and compare mitigation. Participatory and distributional analysis complements physical-economic metrics.

Numerical models. Models include extreme-value and stochastic-event sets, copulas and joint probabilities, hydraulic and tsunami simulations, Monte Carlo reliability, fragility and damage functions, GIS exposure/vulnerability overlays, network disruption and scenario-based future risk.

Experimental datasets. Reviewed evidence includes global roads/railways, WorldRiskIndex and INFORM outcomes, Po flood and levee scenarios, British bridge failures, 726 Tohoku source realizations, Catalan beach sectors, masonry flood cases and Bangladesh cyclonic-flood literature and census indicators.

Validated ranges. Reported values are context-specific: global transport EAD spans \$3.1–22 billion, high-vulnerability countries show 15-fold event mortality, and 726 tsunami sources represent one Tohoku case. Such values cannot be transferred without harmonized exposure, vulnerability and uncertainty.

Recent advances. Recent work uses stochastic source ensembles, spatial multihazard interaction, dynamic exposure and vulnerability scenarios, global infrastructure networks, uncertainty visualization and impact chains that connect climate drivers to physical and socioeconomic consequences.

Disagreements. Composite indices improve comparability but can obscure causal mechanisms and weighting; detailed fragility models are mechanistic but data intensive. Expected annual loss supports portfolios yet can hide tail, distributional and nonmarket consequences. Static rankings may reverse under future development or adaptation.

Limitations. Sparse loss and failure data, reporting bias, uncertain extremes, dependence, defense behavior, indirect and cascading effects, informal assets, social inequity, model resolution and future scenarios limit confidence. Validation of hazard alone does not validate risk.

Open questions. Priorities include compound and cascading dependence, dynamic vulnerability and recovery, indirect network losses, ecosystem and cultural consequences, equity, behavioral adaptation, deep uncertainty, transferable fragility and transparent validation of composite indices.

Seminal papers. Risk theory formalized probability–consequence integration; engineering reliability introduced load–resistance and fragility functions; hazard mapping and vulnerability research then combined spatial exposure with physical and social susceptibility. Disaster-risk frameworks broadened analysis to resilience and governance.

Claims

1. Coastal risk assessment should integrate probabilistic hazard, exposure, vulnerability and consequences across present and future conditions while borrowing validated methods across hazard communities only with explicit transfer assumptions. *Regime: Global and regional natural-hazard risk assessment including floods and tsunamis.* review statement, review [1487]
2. Comparative evidence from Hong Kong and Singapore indicates that resilient coastal-city flood management requires integrating flood-risk and climate-adaptation policy into spatial planning, resolving agency responsibilities, and coupling engineered controls with long-term inter-agency governance. *Regime: Dense Asian coastal cities exposed to pluvial, fluvial, tidal, storm-surge, and sea-level risks under continued urbanization and climate change.* review statement, review [245]

40.2 Consequences and loss

Topic risk.consequences. Human, economic, and infrastructure impacts.

Papers screened: 21. Claims: 17. Equations: 1.

Well established. Coastal consequence assessment must explicitly connect hazard intensity to exposed receptors and an impact metric. Direct physical damage, network-service loss, livelihood loss, indirect economic disruption and systemic effects are distinct endpoints and should not be collapsed without a stated aggregation rule.

Governing physics. Storm waves, water levels, erosion, overwash, inundation, hurricane wind, debris transport and seasonal ice govern the retained cases. Consequences emerge only after these processes interact with asset location, fragility, network topology, ecosystem production or operational exposure.

Dimensionless parameters. Hazard and exposure indices, conditional damage ratios, exceedance probability, return period, normalized connectivity loss and intervention-effect ratios organize comparisons; their definitions and normalization are study-specific and are not interchangeable by default.

Major equations. Common forms include risk as event probability times conditional consequence, indicator combinations such as $CI = \sqrt{i_{\text{hazard}} i_{\text{exposure}}}$, depth- or intensity-damage functions, Bayesian conditional probabilities, network-connectivity loss and discounted or cumulative economic-loss accounting.

Typical methods. The evidence spans regional indicator screening followed by process modelling, Bayesian-network surrogates, probabilistic debris and transport-network analysis, GIS exposure overlays, economic accounting, and coupled property/labor loss simulation.

Numerical models. XBeach and LISFLOOD-FP support detailed hotspot analysis; Bayesian networks emulate morphodynamic scenario chains; debris, fragility and graph models estimate road service loss; GIS and probabilistic hurricane models join hazard fields to spatial assets and economic quantities.

Experimental datasets. Case evidence covers ten European RISC-KIT sites, Ria Formosa barrier scenarios, Galveston roads and emergency facilities, Québec municipal asset inventories, Mississippi oyster landings and revenue, eastern North Carolina property and labor data, and mapped Alaskan nearshore-ice zones.

Validated ranges. Evidence is bounded to the reported cases: European screening sectors near 1 km and hotspots of 1–10 km; a Ria Formosa 50-year storm plus spring tide; a representative Galveston 500-year hurricane; Mississippi fishery impacts from 2005–2016; and each study's mapped asset inventory.

Recent advances. Recent work adds debris-mediated network disruption and jointly maps property and labor consequences, while building-level adaptive pathways update decisions as sea-level projections change. The advance is increasingly explicit coupling among physical hazards, exposed systems and decisions rather than hazard mapping alone.

Disagreements. The studies do not establish one universal consequence measure. Exposure-only maps omit fragility and welfare loss, asset-value loss can dominate modeled totals while obscuring social vulnerability, and hazard inventories such as ice zonation identify potential harm without estimating realized loss.

Limitations. Transfer is constrained by local inventories, vulnerability functions, economic valuation, network topology, scenario discretization and model-form uncertainty. Indirect, intangible, distributional and recovery consequences remain inconsistently represented, and abstract-limited sources support fewer implementation details.

Open questions. Priorities include transferable multi-sector consequence functions, uncertainty propagation from hazard through recovery, treatment of cascading infrastructure failures, ecosystem-service and intangible loss, distributional welfare, behavioral adaptation, and validation against observed longitudinal recovery.

Seminal papers. The 1981 Alaskan ice-zonation study is an early engineering-hazard inventory. The 2017–2018 CRAF and Ria Formosa studies are central to multiscale and probabilistic coastal impact assessment because they connect regional screening, process models and risk-reduction choices.

Equations

- Coastal risk index [1454]

$$CI = \sqrt{i_{\text{hazard}} i_{\text{exposure}}}$$

Regime: Ten RISC-KIT European case studies; approximately 1 km regional sectors and 1–10 km detailed hotspots. Variables: CI coastal index; i_{hazard} normalized hazard indicator; i_{exposure} normalized exposure indicator

Claims

1. Across ten European case studies, the two-stage Coastal Risk Assessment Framework screened approximately 1 km coastal sectors and then modelled selected 1–10 km hotspots, identifying high-risk areas with only small deviations from locations already recognized as high risk. *Regime: Ten RISC-KIT European case studies; approximately 1 km regional sectors and 1–10 km detailed hotspots.* direct finding, mixed [1454]
2. For a 1-in-50-year storm coincident with spring tide at the Ria Formosa barrier, the Bayesian-network surrogate estimated that partial house removal reduced overwash by 15% and erosion by 58%, nourishment reduced overwash by 16% and erosion by 96%, and the combined intervention reduced both impacts to nearly zero. *Regime: Ria Formosa coastal barrier under the tested 1-in-50-year storm plus spring tide and specified interventions.* direct finding, mixed [1119]
3. In the Galveston transportation case for a representative 500-year hurricane, estimated loss of road connectivity to emergency facilities increased from about 2% when debris was omitted to about 17% when hurricane debris and road damage were jointly represented. *Regime: Galveston, Texas, transportation network under the study representative 500-year hurricane.* direct finding, mixed [51]
4. For the Québec case municipalities, a planner-co-designed GIS workflow combined coastal characterization, projected shoreline migration, and mapped buildings and infrastructure to identify assets exposed to coastal erosion and support land-use adaptation decisions. *Regime: The studied Québec coast and mapped buildings/infrastructure under the adopted erosion projection method.* direct finding, mixed [458]
5. For the Mississippi commercial oyster fishery during 2005–2016, the study estimated cumulative direct coastal-hazard impacts of nearly USD 40 million, averaging approximately USD 3.3 million per year, associated with Katrina, the Deepwater Horizon spill, Bonnet Carré Spillway opening and a harmful algal bloom. *Regime: Mississippi commercial oyster fishery, 2005–2016, in the events and economic data considered.* direct finding, mixed [1122]

6. Across eastern North Carolina's Coastal Area Management Act region, integrated probabilistic hurricane simulations, property data and labor-compensation estimates showed that spatial asset value was the primary determinant of total modeled economic consequences, despite concurrent relationships with social vulnerability. *Regime: Eastern North Carolina CAMA region under the study probabilistic hurricane simulations and property/labor datasets.* direct finding, mixed [855]
7. Along the Beaufort, Chukchi and Bering coasts of Alaska, seasonal nearshore-ice morphology maps delineated zones of statistically uniform ice behavior and related each zone to coastal processes and bathymetric configuration to identify ice features hazardous to outer-continental-shelf oil and gas development. *Regime: Seasonal nearshore ice on the Beaufort, Chukchi and Bering coasts of Alaska.* direct finding, mixed [1361]
8. Globally about 27% of road and rail assets are exposed to at least one hazard and 7.5% to a 100-year flood; direct expected annual damage is \$3.1–22 billion, roughly 73% from surface and river flooding. *Regime: A global multi-hazard risk analysis of road and railway infrastructure assets* direct finding, mixed [730]
9. A 2000–2019 coastal-tourism review finds highly heterogeneous physical and socioeconomic climate impacts across destinations and methods, with important indicators and causal links still overlooked. *Regime: Climate change, coastal tourism, and impact chains – a literature review* direct finding, mixed [68]
10. British flood and failure records support bridge-scour fragility curves coupled to spatial joint flood probabilities and passenger-disruption models for network-wide failure and economic risk. *Regime: A Probabilistic Model of the Economic Risk to Britain's Railway Network from Bridge Scour During Floods* direct finding, mixed [762]
11. High-resolution U.S. modeling estimates climate change alone raises flood risk 26.4% by 2050 under RCP4.5; future increases disproportionately affect Black communities and remain concentrated along Atlantic and Gulf coasts. *Regime: Inequitable patterns of US flood risk in the Anthropocene* direct finding, numerical [1521]
12. A global delta dataset estimates 339 million residents in 2017; 31 million delta residents face tropical-cyclone flooding, 92% in developing or least-developed economies and 25 million on sediment-starved deltas. *Regime: Coastal flooding will disproportionately impact people on river deltas* direct finding, mixed [393]
13. A coastal eutrophication synthesis links expanding nutrient enrichment to ecological degradation and human consequences and frames globalization of coastal nutrient pressures as a management risk. *Regime: The Globalization of Cultural Eutrophication in the Coastal Ocean: Causes and Consequences* review statement, review [907]
14. A coastal-risk review identifies population growth and spatial development as dynamic exposure drivers that can amplify future risk independently of hazard change. *Regime: Population development as a driver of coastal risk: Current trends and future pathways* review statement, review [1174]
15. Global extreme-coastal-water-level analysis identifies widespread potential overtopping exposure and connects local extremes to tide, surge, wave and relative sea-level contributions. *Regime: A global analysis of extreme coastal water levels with implications for potential coastal overtopping* direct finding, numerical [35]
16. A Bangladesh coastal assessment links cyclone exposure to persistent risks for mortality and livelihoods despite large reductions in fatalities over time. *Regime: Assessing Risks from Cyclones for Human Lives and Livelihoods in the Coastal Region of Bangladesh* direct finding, mixed [1137]
17. Tourism-dependent beaches face coupled erosion, infrastructure and economic losses; a review concludes that hard and soft protection must be matched to sediment and hydrodynamic processes and embedded in integrated coastal-zone management rather than selected as isolated engineering fixes. *Regime: Tourism beaches and coastal infrastructure exposed to erosion, storms and sea-level rise.* review statement, review [1105]

40.3 Warning and evacuation

Topic risk.evacuation. Risk communication and evacuation behavior.

Papers screened: 10. Claims: 10. Equations: 7.

Well established. Life safety depends on completing protective action before hazardous inundation or wind arrives. Warning issuance alone is insufficient: people must receive and understand the message, believe the threat and recommended action, decide and mobilize, traverse a capacity-limited network, and reach a safe, acceptable destination.

Governing physics. Storm-surge and tsunami depth, velocity, arrival time and spatial extent define unsafe areas and available lead time. Human and network processes—decision delay, walking or vehicle speed, route capacity, density, queues, shelter capacity and train movements—consume that lead time and can create secondary congestion risk.

Dimensionless parameters. Key measures include evacuation participation probability, warning reach and comprehension, compliance, normalized decision/clearance time relative to available lead time, volume-to-capacity ratio, pedestrian density, shelter occupancy ratio, completion fraction and casualty fraction.

Major equations. A clearance margin can be written $M = T_{\text{arrival}} - T_{\text{warning}} - T_{\text{decision}} - T_{\text{clearance}}$. Behavioral studies use logistic $P(\text{evacuate}|x)$; network models use shortest paths, capacity and queue conservation; agent models update individual motion and density; railway assignment constrains each train to safe or evacuable stations before local arrival.

Typical methods. Evidence combines household questionnaires, map tests, pre/post-event interviews, logistic regression, Protective Action Decision Model variables, hydraulic-model coupling, GIS and network assignment, agent-based pedestrian simulation, supercomputing, route/shelter optimization and strategy scenarios.

Numerical models. THESEUS links hydraulic forecast fields to city evacuation plans and clearance simulation. The Japanese large-scale model resolves individuals, routes and shelters. The Sri Lanka algorithm assigns trains under spatially variable tsunami arrival, while logistic/PADM models estimate behavioral and shelter-use response.

Experimental datasets. The branch includes Texas hurricane-map surveys, typhoon storm-surge scenario interviews, pre-event and post-Iyonada records from two Kochi communities, Bordeaux and Cesenatico planning cases, a 340,000-agent urban tsunami simulation, Sri Lankan coastal rail operations and 391 Ramgati residents.

Validated ranges. Findings are bounded to their cases: only 36% of surveyed Texans located the correct map zone; the Japanese model represented about 340,000 evacuees and found an approximately 40-minute initiation allowance only under its maximum-tsunami, forecast-controlled assumptions; the Bangladesh survey included 391 residents.

Recent advances. Recent advances use hundreds of thousands of individual agents and real-time inundation forecasts to manage congestion, and apply PADM-informed multichannel communication analysis to shelter barriers. The field is moving toward coupled physical, behavioral and operational digital decision support.

Disagreements. Better risk-map self-location did not predict Texas evacuation expectations, showing that information accuracy alone may not change intended action. Conversely, Bangladesh evidence associates direct, multi-channel institutional communication with better shelter perceptions. These results concern different communication stages and are not contradictory.

Limitations. Stated intentions differ from actual behavior; post-event recall is biased; coefficients are culturally and event specific. Models simplify compliance, disability, tourists, family reunification, vehicles, debris, power and communications failure. Forecast and inundation errors propagate directly into route and timing advice.

Open questions. Priorities include end-to-end validation in drills and events, forecast-conditioned dynamic routing, mixed pedestrian-vehicle evacuation, evacuation of tourists and mobility-limited people, spontaneous shelter choice, warning ambiguity, network damage, vertical evacuation, equitable shelter design and calibrated behavioral uncertainty.

Seminal papers. The 2006 Texas study showed that correct zone-map reading is not equivalent to evacuation intention. The 2014 THESEUS study provided a coastal-engineering chain from hydraulic forecasts to mass-evacuation clearance, complementing behavioral survey evidence.

Equations

- *Evacuation clearance margin* [1369]

$$M = T_{arrival} - T_{warning} - T_{decision} - T_{clearance}$$

Regime: Residents of Okitsu and Mangyo responding to the 2014 Iyonada earthquake. Variables: M remaining safety margin; $T_{arrival}$ hazard arrival time; $T_{warning}$ warning issuance or reception time; $T_{decision}$ decision and mobilization delay; $T_{clearance}$ travel and congestion clearance time

- *Evacuation clearance margin* [601]

$$M = T_{arrival} - T_{warning} - T_{decision} - T_{clearance}$$

Regime: Bordeaux pilot planning and Cesenatico evacuation-time trial with the source data and assumptions. Variables: M remaining safety margin; $T_{arrival}$ hazard arrival time; $T_{warning}$ warning issuance or reception time; $T_{decision}$ decision and mobilization delay; $T_{clearance}$ travel and congestion clearance time

- *Evacuation clearance margin* [1077]

$$M = T_{arrival} - T_{warning} - T_{decision} - T_{clearance}$$

Regime: The surveyed coastal population and hypothetical typhoon storm-surge scenarios used by the authors. Variables: M remaining safety margin; $T_{arrival}$ hazard arrival time; $T_{warning}$ warning issuance or reception time; $T_{decision}$ decision and mobilization delay; $T_{clearance}$ travel and congestion clearance time

- *Evacuation clearance margin* [1576]

$$M = T_{arrival} - T_{warning} - T_{decision} - T_{clearance}$$

Regime: 391 surveyed residents in cyclone-prone Ramgati and the communication/shelter variables measured. Variables: M remaining safety margin; $T_{arrival}$ hazard arrival time; $T_{warning}$ warning issuance or reception time; $T_{decision}$ decision and mobilization delay; $T_{clearance}$ travel and congestion clearance time

- *Evacuation clearance margin* [1310]

$$M = T_{arrival} - T_{warning} - T_{decision} - T_{clearance}$$

Regime: Sri Lanka southern coastal railway topology, communications and tsunami scenarios represented in the study. Variables: M remaining safety margin; $T_{arrival}$ hazard arrival time; $T_{warning}$ warning issuance or reception time; $T_{decision}$ decision and mobilization delay; $T_{clearance}$ travel and congestion clearance time

- *Evacuation clearance margin* [78]

$$M = T_{arrival} - T_{warning} - T_{decision} - T_{clearance}$$

Regime: Surveyed Texas coastal residents and the risk-area maps supplied in the study. Variables: M remaining safety margin; $T_{arrival}$ hazard arrival time; $T_{warning}$ warning issuance or reception time; $T_{decision}$ decision and mobilization delay; $T_{clearance}$ travel and congestion clearance time

- *Evacuation clearance margin* [903]

$$M = T_{arrival} - T_{warning} - T_{decision} - T_{clearance}$$

Regime: Approximately 340,000 agents, the study city network and maximum-class tsunami scenario. Variables: M remaining safety margin; $T_{arrival}$ hazard arrival time; $T_{warning}$ warning issuance or reception time; $T_{decision}$ decision and mobilization delay; $T_{clearance}$ travel and congestion clearance time

Claims

1. Among surveyed Texas coastal residents, only 36% correctly identified their hurricane risk area and another 28% were off by one area; map-location accuracy was not correlated with evacuation expectations, which instead varied with prior hazard experience and the expected evacuation context. *Regime: Surveyed Texas coastal residents and the risk-area maps supplied in the study.* direct finding, mixed [78]
2. A coastal-resident survey under alternative typhoon storm-surge scenarios identified demographic, risk-cognition, destination-choice and evacuation-context variables and combined them in a binary logistic model of evacuation choice. *Regime: The surveyed coastal population and hypothetical typhoon storm-surge scenarios used by the authors.* direct finding, mixed [1077]
3. After the 2014 Iyonada earthquake, many residents in Okitsu and Mangyo quickly evacuated or prepared to evacuate, but interviews revealed substantial differences from their prior plans, including unanticipated triggering cues and vehicle use. *Regime: Residents of Okitsu and Mangyo responding to the 2014 Iyonada earthquake.* direct finding, mixed [1369]
4. A mass-evacuation planning workflow coupled hydraulic-model flood inputs, population and route data, action-plan generation and evacuation-time simulation; it was applied in Bordeaux and trialled in Cesenatico to compare management strategies and warning-time requirements. *Regime: Bordeaux pilot planning and Cesenatico evacuation-time trial with the source data and assumptions.* direct finding, mixed [601]
5. A supercomputer simulation of approximately 340,000 evacuees identified route delays and shelter overcrowding; controlling movement with tsunami-inundation forecasts greatly reduced congestion and, for the maximum-class scenario studied, preserved casualty-free evacuation with about 40 minutes from earthquake occurrence to evacuation initiation. *Regime: Approximately 340,000 agents, the study city network and maximum-class tsunami scenario.* direct finding, mixed [903]
6. For Sri Lanka's partly single-track southern coastal railway, a scenario procedure assigned trains to safe stations or to stations permitting onward evacuation while accounting for uncertain tsunami arrival, passenger concern, and weak communications and signalling. *Regime: Sri Lanka southern coastal railway topology, communications and tsunami scenarios represented in the study.* direct finding, mixed [1310]
7. In a cross-sectional survey of 391 residents in cyclone-prone Ramgati, Bangladesh, most respondents reported strongly negative shelter perceptions; reliance on mass media without government or NGO information was associated with greater concern, while multi-channel and direct institutional communication supported more positive shelter perceptions and evacuation response. *Regime: 391 surveyed residents in cyclone-prone Ramgati and the communication/shelter variables measured.* direct finding, mixed [1576]
8. A Belgian-coast survey shows public flood-risk perception is a material input to effective mitigation, communication, and safety policy. *Regime: An Analysis of the Public Perception of Flood Risk on the Belgian Coast* direct finding, field [697]
9. Post-hurricane survey analysis relates evacuation decisions to perceived risk, information, and household location in metropolitan Houston. *Regime: Who Evacuates When Hurricanes Approach? The Role of Risk, Information, and Location** direct finding, field [1351]
10. Coastal-city surveys evaluate how tsunami hazard proximity influences perceived risk and residents' ability to identify their exposure. *Regime: Hazard proximity and risk perception of tsunamis in coastal cities: Are people able to identify their risk?* direct finding, field [75]

40.4 Probabilistic hazard analysis

Topic risk.probabilistic. Frequency and joint-probability analysis.

Papers screened: 20. Claims: 17. Equations: 7. Shown in the Coastal Design Bench beside: Design conditions.

Well established. Probabilistic coastal hazard analysis estimates exceedance rates or probabilities for physically meaningful responses, not merely storm frequency. It must represent forcing distributions, dependence among drivers, modelled coastal response, sampling uncertainty and the time basis of return quantities.

Governing physics. Cyclone pressure, size, speed and heading drive wind, surge, waves and rainfall; tide and sea-level trend shift total water levels; shoreline history controls erosion likelihood; earthquake magnitude and heterogeneous slip govern tsunami generation, propagation and inundation. Local bathymetry and topography strongly transform every source.

Dimensionless parameters. Annual exceedance probability, return period, conditional tail probability, extremal shape parameter, normalized pressure deficit, radius and translation-speed ratios, dependence and tail-dependence coefficients, earthquake magnitude and scenario weights organize analyses, but definitions remain method specific.

Major equations. Core expressions are exceedance probability $H(z)=P(Z>z)$, annual exceedance rate, return level as an inverse hazard curve, joint-probability integrals over forcing parameters, peaks-over-threshold generalized Pareto tails, GEV or metastatistical extreme models, copula joint distributions, and nonstationary cumulative hazard.

Typical methods. Methods include historical-change probability zoning, peaks over threshold, annual maxima, metastatistics, Bayesian or frequentist uncertainty, synthetic storm generation, JPM with copulas, stochastic earthquake rupture ensembles, reduced-order scenario selection, hydrodynamic simulation and surrogate response surfaces.

Numerical models. CHS/PCHA couples storm climatology to high-resolution coastal models and surrogates. NEOWAVE resolves Chilean tsunami inundation after SROM scenario reduction. Statistical models include GPD, GEV, MEVD, meta-Gaussian and linear-circular vine copulas.

Experimental datasets. Evidence uses historical New Jersey aerial photographs, U.S. tide-gauge surge records, Boston and New York water levels and sea-level projections, U.S. cyclone climatologies, and a Chile-Peru stochastic rupture database simulated for 11 coastal cities.

Validated ranges. Extracted bounds include a 90 km New Jersey reach, Chilean Mw 8.0–9.6 ruptures across four segments and 11 cities, Boston and New York nonstationary cases, U.S. tide gauges, and the regional cyclone sample and grids used by CHS and the copula comparison.

Recent advances. Recent studies create multi-source stochastic tsunami maps, distinguish nonstationary return-period meanings, test metastatistical surge models nationwide, and compare circular-linear cyclone copulas. The common advance is explicit treatment of dependence, scenario reduction and epistemic uncertainty.

Disagreements. No single extreme-value or dependence model is uniformly best. MEVD and GEV can yield similar central estimates yet different stability and design levels; richer vine dependence can improve tail representation while simpler meta-Gaussian dependence may perform similarly in a tested region. Stationary return periods are ambiguous under changing sea level.

Limitations. Rare extremes provide little direct tail validation. Results depend on record length, thresholds, stationarity assumptions, marginal distributions, copula structure, synthetic-event sampling, bathymetry, numerical-model error, surrogate error, sea-level scenarios and definitions of return period.

Open questions. Priorities are compound rainfall–surge–wave–river probability, spatially coherent extremes, climate-conditioned cyclone and sea-level distributions, model-form ensembles, tail validation, dependence under sparse samples, adaptive scenario selection, and transparent propagation into consequences and decisions.

Seminal papers. The 1978 New Jersey study established observation-based probabilistic erosion and surge zoning, while the 1990 sea-level-extremes paper formalized shifting annual extreme distributions under rising mean level. These precede modern process-model ensembles and nonstationary frequency methods.

Equations

- *Maximum over non-identically distributed annual sea levels* [118]

$$P(M_{1:n} \leq z) = \prod_{t=1}^n F_t(z)$$

Regime: Independent annual maxima drawn from time-varying distributions, with the paper emphasizing location shifts from mean sea-level rise. Variables: $M_{1:n}$ maximum sea level over the design interval; F_t annual-maximum distribution in year t ; z candidate design elevation; n number of years

- *Hazard exceedance probability* [853]

$$H(z) = P[Z > z]$$

Regime: Historical storm population and coastal study region used for the JPM comparison. Variables: H hazard exceedance probability; Z hazard response; z specified response level

- *Hazard exceedance probability* [661]

$$H(z) = P[Z > z]$$

Regime: Boston and New York City observations and the sea-level projections used in the two cases. Variables: H hazard exceedance probability; Z hazard response; z specified response level

- *Hazard exceedance probability* [354]

$$H(z) = P[Z > z]$$

Regime: The historical record and developed 90 km New Jersey coastline analysed in the paper. Variables: H hazard exceedance probability; Z hazard response; z specified response level

- *Hazard exceedance probability* [69]

$$H(z) = P[Z > z]$$

Regime: Four Chile–Peru subduction segments, Mw 8.0–9.6 scenarios and 11 selected Chilean cities. Variables: H hazard exceedance probability; Z hazard response; z specified response level

- *Hazard exceedance probability* [177]

$$H(z) = P[Z > z]$$

Regime: U.S. tide-gauge records and calibration lengths evaluated in the paper. Variables: H hazard exceedance probability; Z hazard response; z specified response level

- *Hazard exceedance probability* [1007]

$$H(z) = P[Z > z]$$

Regime: U.S. coastlines represented by the CHS regional climatologies, synthetic storms and numerical model grids. Variables: H hazard exceedance probability; Z hazard response; z specified response level

Claims

1. Under sea-level rise, coastal design maxima should be estimated from a sequence of annual extreme-value distributions with shifting location rather than one stationary distribution; future-magnitude quantiles are then more meaningful than stationary return-period levels. *Regime: Annual-maxima distributions whose principal future change is a mean or location shift; changing surge variance and dependence are not represented.* direct finding, analytical [118]
2. The Coastal Hazards System probabilistic framework combines regional storm climatology, synthetic storms, joint probability modelling of atmospheric forcing, high-resolution hydrodynamic simulations, surrogate prediction, and explicit aleatory and epistemic uncertainty for U.S. hurricane- and extratropical-storm coastlines. *Regime: U.S. coastlines represented by the CHS regional climatologies, synthetic storms and numerical model grids.* direct finding, mixed [1007]
3. Along a highly developed 90 km reach of the New Jersey coast, historical aerial photography was used to derive and test probabilities for shoreline-erosion and storm-surge-penetration hazard zones along and across the coast. *Regime: The historical record and developed 90 km New Jersey coastline analysed in the paper.* direct finding, mixed [354]
4. For 11 Chilean coastal cities, stochastic rupture scenarios of magnitude 8.0–9.6 across four Chile–Peru subduction segments were reduced with SROM and simulated with NEOWAVE; the resulting probabilistic inundation maps showed strong site dependence, so a uniform national planning criterion was not supported. *Regime: Four Chile–Peru subduction segments, Mw 8.0–9.6 scenarios and 11 selected Chilean cities.* direct finding, mixed [69]

5. Across U.S. tide-gauge sites, the metastatistical extreme-value distribution produced surge-extreme estimates comparable to GEV fits but generally lower error variability; model choice still caused site- and calibration-length-dependent differences in design surge height. *Regime: U.S. tide-gauge records and calibration lengths evaluated in the paper.* direct finding, mixed [177]
6. For Boston and New York City, detrending observed water levels, fitting a generalized Pareto distribution to threshold exceedances, and reintroducing projected sea-level trends generated nonstationary flood-hazard curves in which expected waiting time and expected-number-of-exceedances definitions of return period are explicitly distinguished. *Regime: Boston and New York City observations and the sea-level projections used in the two cases.* direct finding, mixed [661]
7. In the study-region Joint Probability Method analysis, a linear-circular Frank vine copula increased hazard-curve stability and tail dependence between large central-pressure deficit and large radius of maximum wind, while a meta-Gaussian copula gave generally consistent performance with simpler implementation. *Regime: Historical storm population and coastal study region used for the JPM comparison.* direct finding, mixed [853]
8. For Richmond, British Columbia, the direct joint probability method represented dependent contributors to extreme flood and runoff-inclusive sea levels and showed that design sea level can differ substantially from extreme flood level alone. *Regime: Dependent tide, surge, wave runoff, climate, and tectonic sea-level factors for Richmond, British Columbia.* direct finding, analytical [838]
9. A global-risk review finds hydrometeorological studies more often include future projections and risk reduction, while earthquake and tsunami studies more often use stochastic fully probabilistic methods that could transfer across hazards. *Regime: Review article: Natural hazard risk assessments at the global scale* direct finding, mixed [1487]
10. A five-step multi-hazard framework classifies impact overlap as spatiotemporal, temporal-only, spatial with residual damage, or independent and demonstrates earthquake-weakened levee failure followed by intense rain in the Po Valley. *Regime: A multi-hazard framework for spatial-temporal impact analysis* direct finding, mixed [60]
11. For a 50 km Po reach, probabilistic mapping combines uncertain peak-volume hydrographs, downstream rating curves, and random overtopping, piping, and micro-instability dike failures to produce hazard confidence information. *Regime: Probabilistic flood hazard mapping: effects of uncertain boundary conditions* direct finding, mixed [355]
12. Future-risk assessment must jointly project changing hazards, exposure and physical vulnerability rather than extrapolate static historical impacts under climate, urbanization, wealth and connectivity change. *Regime: Modelling and quantifying tomorrow's risks from natural hazards* direct finding, mixed [306]
13. For Rikuzentakata, 726 stochastic slip models derived from 11 inversions propagate source uncertainty through tsunami inundation, empirical fragility, damage, loss and risk visualization. *Regime: Uncertainty modeling and visualization for tsunami hazard and risk mapping: a case study for the 2011 Tohoku earthquake* direct finding, mixed [509]
14. NEAMTHM18 constructs a regional probabilistic tsunami hazard model from source ensembles and propagation calculations to support consistent hazard estimates across the Northeast Atlantic, Mediterranean and connected seas. *Regime: The Making of the NEAM Tsunami Hazard Model 2018 (NEAMTHM18)* direct finding, mixed [124]
15. A Bayesian network for Taitung links wave energy, shoreline change and socioeconomic/environmental factors with about 74% predictive accuracy; under 2 C warming, the maximum probability of high hazard in the Bainen River estuary sector rises to 60%. *Regime: Taitung, Taiwan coastal sectors under the documented long-term data and 2 C global-warming scenario.* direct finding, mixed [1569]
16. RoadRAT combines extreme-value statistics for still-water level, total water level including runoff, and storm erosion with shoreline evolution to screen present and future coastal-road impacts; its 89 km Swedish demonstration finds erosion dominates expanding future exposure. *Regime: Regional coastal roads exposed to inundation, wave runoff, storm erosion, sea-level rise, and continued historical shoreline trends.* direct finding, mixed [562]
17. Value-of-information maps combine probabilistic flood hazard and decision consequences to identify locations where additional evidence is worth acquiring before floodplain land-use decisions, complementing rather than replacing comprehensive risk maps. *Regime: Floodplain spatial planning where probabilistic inundation information, uncertain hydraulic-model skill, candidate land uses, and action consequences can be represented spatially.* direct finding, mixed [28]

40.5 Vulnerability and fragility

Topic risk.vulnerability. Damage response functions.

Papers screened: 25. Claims: 23. Equations: 8.

Well established. Coastal vulnerability is receptor-specific. Relative indices rank places using physical or social indicators; fragility functions estimate conditional probabilities of damage-state exceedance for a defined receptor and hazard intensity; impact response functions link hazard severity to sector outcomes. These quantities are complementary but not interchangeable.

Governing physics. Tsunami depth, velocity and hydrodynamic force load buildings and people; storm surge, waves and eroded dune elevation control hurricane exposure; sea-level rise, coastal slope, geomorphology, sediment and shoreline change govern chronic physical susceptibility. Structural resistance, exposure and coping capacity modify physical forcing.

Dimensionless parameters. Damage probability, fatality ratio, normalized intensity, damage-state threshold, vulnerability rank, index weights, confidence and risk percentiles are central. Physical controls include relative inundation depth, Froude number, momentum flux, relative dune height and erosion potential; their transfer requires consistent definitions.

Major equations. Fragility is commonly $F_k(x) = P(DS \geq k | X = x)$, often fitted with lognormal, probit or logistic links. CVIs aggregate ranked variables, frequently by a geometric mean. Synthetic impact functions map flood severity to elicited consequence, while IPCI maps continuous input-score polygons to fitted GEV or Weibull score distributions.

Typical methods. Methods combine post-event satellite and field surveys, damage classification, GIS, numerical inundation reconstruction, regression fragility, nonlinear structural analysis, remote-sensing CVIs, participatory response-function elicitation, probabilistic index scoring, bibliometrics and systematic review.

Numerical models. Tsunami inundation models supply otherwise unobserved local velocity and force; nonlinear structural models produce analytical capacity-demand fragility; GIS integrates hazard and damage; CVI and IPCI aggregate spatial indicators; dune-erosion and setback studies connect morphology to exposure.

Experimental datasets. Key evidence includes the 2004 Banda Aceh tsunami surveys and simulations, more than 15,000 nearshore houses after Hurricane Sandy, a representative Tumaco reinforced-concrete building, Egypt's 1,000 km Mediterranean coast, Gippsland stakeholder evidence, and 178 Indian assessment studies.

Validated ranges. Evidence is bounded to Banda Aceh's 2004 event, Sandy-affected Mid-Atlantic housing, the analysed Tumaco building, Egypt's national Mediterranean geomorphic sectors, Gippsland priorities, and each index input domain. Banda Aceh's macroscopic casualty transition occurred from roughly 2 to 8 m depth.

Recent advances. Recent work compares empirical with nonlinear analytical building fragility, introduces continuous probabilistic scoring instead of integer CVI classes, and systematically audits Indian assessment practice. The direction is toward receptor-specific, uncertainty-aware, and testable functions rather than composite ranking alone.

Disagreements. Empirical fragility transferred from other tsunami regions predicted lower low-depth collapse probability than the local Tumaco analytical model. CVI rankings can identify susceptible coast but are not calibrated damage probability; continuous probabilistic IPCI reduces categorical discontinuity but has not yet demonstrated out-of-sample damage skill.

Limitations. Post-event curves confound hazard reconstruction, exposure, construction and survey bias. Velocity is usually simulated rather than observed. CVIs depend on variable selection, ranks and weights. Stakeholder functions encode local judgment. Analytical curves depend on structural idealization, and systematic reviews inherit search and publication bias.

Open questions. Needs include multi-event validation, multivariable depth-velocity-debris intensity, consistent damage taxonomies, uncertainty and censoring, dynamic exposure and adaptation, social capacity and recovery, causal dune-protection evidence, index calibration against observed loss, and principled regional transfer.

Seminal papers. The paired 2009 tsunami studies established GIS-linked fragility as a quantitative bridge from inundation physics to structural damage and fatality in Banda Aceh. The 2011 Gippsland work broadened vulnerability to stakeholder-defined socioeconomic and environmental impacts.

Equations

- *Conditional fragility function* [384]

$$F_k(x) = P(DS \geq k | X = x)$$

Regime: Gippsland stakeholders, priority issues and flood scenarios used in the project. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

- *Conditional fragility function* [53]

$$F_k(x) = P(DS \geq k | X = x)$$

Regime: The 178 Indian coastal risk and vulnerability articles meeting the review criteria. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

- *Conditional fragility function* [1420]

$$F_k(x) = P(DS \geq k | X = x)$$

Regime: More than 15,000 nearshore houses and detailed local surveys after Hurricane Sandy. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

- *Classical coastal vulnerability index* [590]

$$CVI = \sqrt{(a_1 a_2 \cdots a_n) / n}$$

Regime: Egyptian Mediterranean coast divided into western, Nile Delta and North Sinai geomorphic regions. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

- *Polygonal coastal vulnerability probability* [1635]

$$P(S \leq s) = F_{GEV \text{ or Weibull}}(s)$$

Regime: Empirical demonstration data lying within each variable's specified polygonal validity field. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

- *Conditional fragility function* [1072]

$$F_k(x) = P(DS \geq k | X = x)$$

Regime: Typical Tumaco two-story reinforced-concrete structural configuration and analysed tsunami-flow assumptions. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

- *Conditional fragility function* [736]

$$F_k(x) = P(DS \geq k | X = x)$$

Regime: Banda Aceh buildings, villages and population exposed to the reconstructed 2004 tsunami. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

- *Conditional fragility function* [737]

$$F_k(x) = P(DS \geq k | X = x)$$

Regime: Post-tsunami observations and numerical inundation fields for the case applications discussed by the authors. Variables: F_k or CVI fragility probability or vulnerability index; DS damage state; X or a_i hazard intensity or indexed vulnerability variables; k damage threshold

Claims

1. At profile PF4, estimated 50-year erosion volume decreased from 152 m³/m with a 180-day response window to 132 m³/m at 90 days and 122 m³/m at 45 days. *Regime: PF4 historical profile, the fitted extreme-value method, and 45-180 day response windows.* direct finding, field [573]
2. For the 2004 Indian Ocean tsunami in Banda Aceh, high-resolution inundation modelling combined with post-event structural and casualty data produced fragility functions in terms of local depth, velocity and hydrodynamic force; the macroscopic casualty curve rose beyond about 2 m depth and approached no survivors near 8 m. *Regime: Banda Aceh buildings, villages and population exposed to the reconstructed 2004 tsunami.* direct finding, mixed [736]
3. After Hurricane Sandy, satellite classification of more than 15,000 houses within one block of the New Jersey, Long Island and Staten Island shorelines found destruction strongly correlated with poststorm dune height, supporting dune-conditioned regional building fragility functions. *Regime: More than 15,000 nearshore houses and detailed local surveys after Hurricane Sandy.* direct finding, mixed [1420]
4. Along Egypt's 1,000 km Mediterranean coast, a geologic-physical coastal vulnerability index classified more than one-third as severely vulnerable to sea-level rise, concentrated on the low-lying Nile Delta and North Sinai, while rocky western sectors were mostly low vulnerability. *Regime: Egyptian Mediterranean coast divided into western, Nile Delta and North Sinai geomorphic regions.* direct finding, mixed [590]
5. For a representative two-story reinforced-concrete building in Tumaco, analytical fragility functions indicated higher collapse probabilities at lower tsunami flow depths than empirical curves transferred from other Pacific and Indian Ocean locations. *Regime: Typical Tumaco two-story reinforced-concrete structural configuration and analysed tsunami-flow assumptions.* direct finding, mixed [1072]
6. In the Gippsland coastal region, stakeholder workshops identified priority flood-impact issues and supported synthetic impact response functions for quantifying selected impacts, while disagreement and low expected sensitivity prevented defensible functions for other issues. *Regime: Gippsland stakeholders, priority issues and flood scenarios used in the project.* direct finding, mixed [384]
7. A tsunami fragility framework integrating satellite remote sensing, field surveys, inundation modelling and GIS expressed structural-damage probability and fatality ratio conditionally on inundation depth, current velocity and hydrodynamic force. *Regime: Post-tsunami observations and numerical inundation fields for the case applications discussed by the authors.* direct finding, mixed [737]
8. A systematic review of 178 Indian coastal vulnerability and risk studies found growing multi-hazard attention but persistent heterogeneity in concepts, indicators, scales, methods and policy translation, limiting comparison and operational use. *Regime: The 178 Indian coastal risk and vulnerability articles meeting the review criteria.* direct finding, review [53]
9. The innovative polygonal coastal index replaced mutually exclusive integer CVI classes with continuous polygon-bounded input-score relations and fitted GEV or Weibull cumulative distributions to report minimum, mean and maximum vulnerability risk probabilities within the demonstrated input domains. *Regime: Empirical demonstration data lying within each variable's specified polygonal validity field.* direct finding, mixed [1635]
10. Validation of WorldRiskIndex and INFORM indicators shows mortality per flood, drought, or storm event is 15 times higher in highly vulnerable than low-vulnerability countries and emphasizes matching indicators to the intended construct. *Regime: Understanding human vulnerability to climate change: A global perspective on index validation for adaptation planning* direct finding, mixed [157]
11. Monte Carlo reliability functions for piping and micro-instability generate spatial dike fragility curves conditional on time-varying load and uncertain geometry and geotechnical properties. *Regime: Development of dike fragility curves for piping and micro-instability breach mechanisms* direct finding, mixed [1462]
12. A probabilistic Catalan assessment separates storm inundation and erosion and finds large alongshore hazard and vulnerability variation across 50 km despite spatially homogeneous offshore wave conditions. *Regime: Probabilistic coastal vulnerability assessment to storms at regional scale – application to Catalan beaches (NW Mediterranean)* direct finding, mixed [175]
13. A probabilistic masonry framework treats flood load and structural resistance as random, compares weighted vulnerability parameters with wall reliability, and maps damage probability in GIS. *Regime: Flood hazards and masonry constructions: a probabilistic framework for damage, risk and resilience at urban scale* direct finding, mixed [1005]
14. A socio-spatial index combining cyclonic-flood probability and population sensitivity places Shariatpur, Chandpur and Barisal in Bangladesh's highest vulnerability quartile, with densities of 870–1,400 people/km². *Regime: Bangladesh's vulnerability to cyclonic coastal flooding* direct finding, mixed [144]
15. A North Adriatic regional assessment integrates multiple climate hazards and coastal indicators to map spatially varying vulnerability for adaptation planning. *Regime: Assessment of coastal vulnerability to climate change hazards at the regional scale: the case study of the North Adriatic Sea* direct finding, mixed [1421]
16. Surveyed 2011 Japan tsunami building damage supports empirical fragility curves that relate observed damage states to tsunami intensity for vulnerability assessment. *Regime: Building damage characteristics based on surveyed data and fragility curves of the 2011 Great East Japan tsunami* direct finding, field [1374]
17. Combining socioeconomic and physical indicators reveals strong regional differences in erosion-hazard vulnerability among United States coastal counties. *Regime: Erosion Hazard Vulnerability of US Coastal Counties* direct finding, mixed [174]
18. An InVEST-based coastal vulnerability assessment combines exposure, geomorphology, relief, habitat and population indicators to evaluate natural-hazard vulnerability and the protective role of habitats. *Regime: Towards Quantifying the Coastal Vulnerability due to Natural Hazards using the InVEST Coastal Vulnerability Model* direct finding, mixed [1224]
19. A coordinated northern-Adriatic assessment shows that coastal vulnerability to erosion and sea-level rise is better constrained by integrating geomorphology, relative-sea-level components, multibeam bathymetry/backscatter, meteo-ocean observations and hydro-morphodynamic models than by isolated indicators or surveys. *Regime: Northern Adriatic coastal and transitional environments under erosion, relative sea-level rise and severe-event forcing.* direct finding, review [168]
20. Flood-risk assessments share a hazard-consequence structure across global to local scales, but data, model detail, purpose and uncertainty differ systematically; defense inventories, indirect losses, critical-infrastructure impacts, post-event validation and defensible scale translation are recurring gaps. *Regime: Riverine and coastal flood-risk assessments from global/continental to local town or coast/river-reach scales.* review statement, review [971]
21. Global asset-level analysis finds port resilience is multi-hazard and depends on engineering standards, operational thresholds and recovery time; direct infrastructure/service risk and trade-flow risk concentrate differently and require port-specific adaptation. *Regime: Global commercial ports exposed to coastal, hydrometeorological and geophysical hazards under the study inventories and present-day asset/trade representation.* direct finding, numerical [1452]
22. A probability-scenario-consequence tsunami framework applied to Kamakura indicates evacuation improvement reduces societal risk more effectively than dike heightening in the modeled cases, because structural protection reaches a residual-risk limit for rare extreme inundation. *Regime: Kamakura, Japan, under the modeled tsunami levels, spatially averaged exposure and deterministic evacuation assumptions.* direct finding, numerical [1046]

23. A neighborhood-scale coastal-flood framework combines physical, socioeconomic, and resilience indicators into separate vulnerability and risk indices; its Portsmouth application distinguishes socially vulnerable areas from those actually exposed under 1-in-200 and 1-in-1000-year flood zones. *Regime: Neighborhoods within Portsmouth, UK, evaluated using physical, socioeconomic, resilience, topographic, and coastal-flood-zone data.* direct finding, mixed [\[1095\]](#)

41 Wiki: Scour and erosion

41.1 Scour and erosion: overview

Topic scour. Localized and distributed removal of bed material.

Papers screened: 12. Claims: 4. Equations: 0.

Well established. Local scour develops when structure-induced acceleration, downflow, horseshoe vortices, wake shedding and turbulence raise sediment transport above the surrounding supply. The evolving hole then modifies flow until equilibrium, time limitation, armoring or protection arrests change.

Governing physics. Pressure gradients drive downflow at the upstream face; horseshoe and lee vortices increase bed stress and suspension, flank acceleration redistributes sediment, and waves reverse or modulate transport. Sediment size, grading, cohesion, live-bed supply and geometry control response.

Dimensionless parameters. Key controls include Shields parameter, densimetric Froude number, Keulegan–Carpenter number, relative flow depth, pile-to-grain ratio, current-to-wave velocity ratio, sediment geometric spread, relative scour depth, live-bed mobility, protection stone stability and normalized time.

Major equations. Prediction couples Navier–Stokes or depth-resolved flow to bedload/suspended-load transport and Exner bed continuity. Engineering formulas scale equilibrium depth and timescale with geometry, flow intensity, Shields stress and wave–current parameters; reliability and surrogates quantify uncertainty or accelerate evaluation.

Typical methods. Evidence combines clear- and live-bed flumes, field bathymetry and monitoring, RANS/LES or CFD–DEM, morphodynamic solvers, empirical equilibrium/time formulas and learned surrogates. Validation should cover onset, temporal evolution, maximum depth, footprint and flow/sediment structure.

Numerical models. Models range from empirical formulas and SEDSCOUR to 2DV/3D RANS, SWASH, CFD–DEM and CNN/LSTM predictors. Continuum models efficiently resolve flow and morphology, particle models expose contacts, and learned models interpolate data but require physical scaling and domain checks.

Experimental datasets. Reviewed evidence includes vertical-cylinder current scour, offshore monopile literature, clear-water CFD–DEM validation, complex bridge-pier databases, bridge flumes, monopile/jacket/caisson tests plus four field cases, and biomineralization erosion experiments and demonstrations.

Validated ranges. Case-specific results include a 50% depth change when suspended load was omitted, unprotected monopile scour slightly above one diameter, SEDSCOUR use with up to 30% mud, CNN R^2 0.85, and MICP reductions of 84–100% under 0.3 m/s currents.

Recent advances. Recent advances couple CFD–DEM particle mechanics, validated 3D morphodynamics, multi-structure comparisons, dimensionless deep learning, field-scale biomineralization and lifecycle monitoring to move beyond single-formula equilibrium depth.

Disagreements. Equilibrium formulas differ across current-, wave-, combined- and complex-loading regimes; turbulence closure and suspended-load treatment alter results. Maximum force, stress, or drag location need not coincide with deepest scour, and protection performance under one forcing does not establish lifecycle stability.

Limitations. Scale effects, unresolved sediment mixtures and cohesion, uncertain bed armoring, complex geometry, unsteady tides and storms, morphodynamic acceleration, limited field duration and protection degradation constrain transfer. Accuracy metrics from trained surrogates do not prove extrapolation.

Open questions. Priorities include combined irregular waves/currents, tidal reversals, mixed/cohesive beds, group and jacket interactions, cable/pipeline exposure, climate-driven loading, protection failure, real-time monitoring, uncertainty-aware digital twins and interpretable machine learning.

Seminal papers. Classical pier, pipeline and structure experiments established horseshoe-vortex mechanisms and equilibrium scaling; sediment-transport and Exner models enabled time-resolved prediction. Offshore wind growth expanded work toward large monopiles, groups, complex loading and protection.

Claims

1. Scour assessment requires complementary experimental, numerical, and field evidence because no single method resolves onset, transient transport, equilibrium morphology, structural response, and environmental variability across structure types. *Regime: Bridge, pipeline, offshore-wind, and related hydraulic scour problems.* review statement, review [275]
2. An eleven-node single-hidden-layer neural network trained on 400 laboratory cases outperformed fitted linear and nonlinear regressions for equilibrium pier-scour depth, with pile diameter critical to prediction reliability. *Regime: Artificial Neural Network for Estimation of Local Scour Depth Around Bridge Piers* direct finding, numerical [30]
3. Three-dimensional finite-element simulations indicate that negative seabed slope and increasing scour depth jointly reduce lateral pile resistance and bending moment by weakening pile-soil interaction. *Regime: Numerical Investigation into the Response of a Laterally Loaded Pile in Coastal and Offshore Slopes Considering Scour Effect* direct finding, numerical [1603]
4. Experiments motivated by tsunami field failures show that nearby structures can amplify local scour under extreme transient flows, so isolated-foundation estimates may be nonconservative. *Regime: SCOUR AMPLIFICATION CAUSED BY STRUCTURE PROXIMITY IN EXTREME FLOWS* direct finding, experimental [67]

41.2 Contraction and channel scour

Topic scour. contraction. Scour caused by flow constriction.

Papers screened: 7. Claims: 7. Equations: 0.

Well established. Contraction scour results when lateral or vertical restriction accelerates flow and increases sediment transport capacity through an opening. Clear-water, live-bed, pressure-flow, and unsteady-flood regimes produce different depths and timescales.

Governing physics. Controls include opening ratio, discharge and Froude number, flow depth, pressure confinement, hydrograph duration and peak, sediment size and critical mobility, surface layering, abutment or pier blockage, and upstream channel/floodplain adjustment.

Dimensionless parameters. Important groups include contracted-to-uncontracted width, blockage or opening ratio, relative scour depth, Froude number, velocity-to-critical velocity, sediment-to-flow-depth ratio, surface-to-subsurface grain-size ratio, and contraction length or abutment length relative to depth.

Major equations. Methods use continuity and energy balances, critical-velocity and sediment-mobility relations, additive contraction and local-scour components, equilibrium and time-dependent empirical formulas, hydrograph stepping, HEC-RAS hydraulics, and ANFIS predictors.

Typical methods. Evidence combines long rectangular flumes, steady and stepped hydrographs, clear-water and live-bed tests, pressure-flow decks, layered beds, parametric equation comparison, remote sensing, field inspection, and one-dimensional/two-dimensional hydraulic modelling.

Numerical models. Empirical, theoretical, data-driven, and hydraulic models serve different purposes. ANFIS can optimize fit, energy-continuity theory preserves mechanism, and HEC-RAS with geomorphic history supports site screening; none alone establishes universal transfer.

Experimental datasets. The branch includes 0.90 mm pressure-scour tests, six hydrographs in a 20 m flume at width ratio 0.80, long-contraction pier and gravel-layer tests, constant-discharge opening-ratio tests, and 1,276 Landsat scenes at five detailed bridge sites.

Validated ranges. Reported transitions include a scour-gradient change near opening ratio 0.8, live-bed relative scour approaching 3.36, an NCHRP/Froehlich crossover near L/ya 35-45, contraction width ratio 0.80, and severe risk persisting across 50-500 year floods at one site.

Recent advances. Recent work emphasizes time-dependent profiles, model-form divergence, long-term remote-sensing context, data-driven architecture comparison, and the possibility that floodplain widening progressively increases effective bridge contraction.

Disagreements. Legacy Froehlich and HIRE power laws increase continuously with contraction, whereas the physically based NCHRP response is bounded in live bed and may exceed Froehlich in a deep-encroachment transition. Conventional conservatism is therefore regime-dependent.

Limitations. Evidence is mostly alluvial, laboratory, clear-water, and bridge-specific. Marine waves and tides, cohesive beds, compound surge-river floods, debris, complex three-dimensional openings, scale effects, and sparse event bathymetry limit coastal transfer.

Open questions. Needs include unified local-plus-contraction scour under unsteady live bed, wave-tide-river interaction, vertical and lateral contraction together, layered/cohesive beds, changing floodplains, uncertainty-aware model selection, and coastal-crossing field validation.

Seminal papers. The branch builds on continuity-energy contraction theory, critical mobility, clear-water/live-bed distinction, and empirical bridge scour equations, extended by unsteady hydrographs, layered beds, pressure flow, model comparison, and geomorphic history.

Claims

1. Clear-water bridge-deck experiments with 0.90 mm sand quantified pressure-flow scour and downstream deposition under both steady flow and single-peaked stepped flood hydrographs. *Regime: rectangular submerged bridge deck over 0.90 mm uniform sand in clear water* direct finding, experimental [753]
2. At constant discharge and depth, clear-water pier tests found a change in nondimensional scour-depth gradient near an opening ratio of 0.8, with a steeper slope above that threshold. *Regime: circular piers at constant discharge and depth with varied diameter* direct finding, experimental [334]
3. Six clear-water stepped-hydrograph experiments at contraction ratio 0.80 supported practical formulas for the time evolution of maximum scour depth, scour-hole length, and downstream centreline bed profile. *Regime: 20 m by 1 m rectangular channel, 1 m contraction length, width ratio 0.80, coarse sand, clear water* direct finding, experimental [1052]
4. For the tested contraction-scour dataset, a zero-order Takagi-Sugeno ANFIS with four bell-shaped membership functions per input and Levenberg-Marquardt training gave the best statistical performance. *Regime: alluvial-channel equilibrium contraction scour dataset* direct finding, numerical [206]
5. Long-contraction tests showed deeper scour with coarser sediment and smaller opening ratio; thin gravel layers produced more scour than uniform beds, and equilibrium depth was represented by combined contraction and critical pier-scour components. *Regime: pier and thin gravel layer within long channel contraction* direct finding, experimental [1149]
6. Across varied contraction, legacy Froehlich and HIRE models increased continuously, whereas NCHRP 24-20 approached relative scour about 3.36 in live bed and could exceed Froehlich near L/ya 35-45. *Regime: bridge abutment models across contraction, Froude number, grain size, and mobility regimes* direct finding, numerical [1103]
7. Analysis of 1,276 Landsat images and bridge hydraulics linked long-term floodplain widening to stronger opening contraction; Sego remained severely scour-prone for 50-, 100-, and 500-year floods. *Regime: 15 southern Ethiopia bridges, five detailed sites, 50-500 year floods* direct finding, mixed [689]

41.3 Pipeline and cable scour

Topic scour.pipELines. Scour beneath linear infrastructure.

Papers screened: 18. Claims: 18. Equations: 0. Shown in the Coastal Design Bench beside: Pile wave loads.

Well established. Pipeline scour progresses through onset, tunnel erosion, and lee-wake erosion and can develop into unsupported free spans. Governing response differs among steady current, waves, combined flow, bidirectional tide, live-bed, clear-water, seepage, and multi-pipe configurations.

Governing physics. Controls include bed shear and pressure gradients, piping and seepage, horseshoe and lee vortices, shedding, turbulence, phase lag, sediment mobility and avalanching, initial clearance, embedment, pipe diameter, gaps, and cyclic reversal.

Dimensionless parameters. Key groups are Shields and Keulegan-Carpenter numbers, current-to-wave velocity ratio, Reynolds number, gap-to-diameter and embedment-to-diameter ratios, hydraulic gradient, relative grain size, live-bed mobility, and nondimensional scour depth and time.

Major equations. Frameworks include URANS and RANS turbulence closures, Euler-Euler two-phase momentum, CFD-DEM particle dynamics, morphodynamic sediment continuity, critical Shields and KC relations, scour timescales, equilibrium-depth fits, and data-driven predictors.

Typical methods. Methods combine steady, oscillatory, combined-flow, and reversible flumes; profile imaging; two-phase CFD, URANS, CFD-DEM, SedFoam, and FLOW-3D; empirical and machine-learning prediction; and protection/free-span review.

Numerical models. Two-phase and particle models resolve sediment concentration and interaction without prescribed bedload layers, but predictions depend strongly on turbulence and vortex representation; black-box models may score well while retaining dataset-limited uncertainty.

Experimental datasets. Evidence includes single and tandem pipes, 1:20 bidirectional tides tied to Cezhen field data, piggyback geometry, solidified-soil protection under waves and currents, and validation cases spanning live-bed, clear-water, sheet-flow, tunnel, and lee-wake stages.

Validated ranges. Explicit findings include live-bed depth of 0.6-0.8D with seepage, tandem gaps of 1-4D, tidal depth averaging 80% of equivalent steady-current scour, a piggyback maximum near $G/D=0.1$, and loss of gap influence at $KC=5.6$ without shedding.

Recent advances. Recent advances resolve individual grains, seepage and piping, tandem interference, reversible tides, piggyback geometry, data-driven uncertainty, and protective-soil tradeoffs, moving toward coupled hydraulic and geotechnical assessment.

Disagreements. Model structure matters physically: one rheological two-phase model overdeposited sediment when vortex shedding was absent, while SedFoam results changed strongly with turbulence cross-diffusion. Good scour-rate agreement does not prove correct lee-wake mechanics.

Limitations. Most evidence is two-dimensional, idealized, uniform-sediment, rigid-pipe, and short-duration. Field-scale span propagation, vibration coupling, heterogeneous and cohesive beds, soil liquefaction, cyclic storms, and protection aging remain weakly validated.

Open questions. Needs include coupled scour-vibration-span growth, poromechanical seabed response, three-dimensional end effects, reversal history, layered beds, uncertainty transfer, inspection assimilation, and protection reliability under liquefaction and degradation.

Seminal papers. The branch builds on classical onset, tunnel and lee-wake staging, Shields and KC scaling, equilibrium depth, and self-burial concepts, extended by two-phase, particle-resolved, seepage, tandem, and tidal-reversal models.

Claims

1. A three-dimensional rheology-based two-phase model reproduced measured pipeline scour rate but overpredicted lee deposition because it did not capture vortex shedding, with strong sensitivity to turbulence parameters. *Regime: pipeline in sheet-flow sediment* direct finding, numerical [787]
2. For tandem pipelines with gap ratios 1-4, weak-current scour resembled pure-wave response, equal wave-current strength resembled pure-current response, and small gaps delayed downstream scour except at $KC=5.6$ where vortex shedding was absent. *Regime: two tandem pipes; KC and gap ratio 1-4 under waves plus currents* direct finding, numerical [817]
3. Upward seepage increased equilibrium scour width beneath a pipeline; live-bed depth remained about 0.6-0.8 pipe diameters, while large gradients slightly reduced clear-water depth. *Regime: live-bed and clear-water pipeline scour under hydraulic gradients* direct finding, numerical [819]
4. SedFoam pipeline-scour predictions were insensitive to granular-stress choice but strongly dependent on turbulence cross-diffusion, indicating that the sediment transport layer behaves more like a shear layer than a boundary layer. *Regime: pipeline tunnel and lee-wake scour* direct finding, numerical [933]
5. An MLP optimized by colliding bodies predicted wave-driven pipeline scour expansion better than PSO-, whale-, regression-, and empirical alternatives for the tested dataset. *Regime: wave, sediment, and pipeline geometry inputs* direct finding, numerical [397]
6. A free-span review identified wave-current scour as the main driver and compared retrenching, structural support, bionic vegetation, and spoiler or choke-plate self-burial treatments. *Regime: offshore pipelines exposed to wave-current scour and other span causes* review statement, review [1592]
7. A validated Euler-Euler model reproduced wave-induced tunnel-scour profiles and showed intense turbulence-driven transport with phase lag between scour-hole and free-stream velocity. *Regime: pipeline on plane erodible bed under oscillatory flow* direct finding, numerical [693]
8. Optimized ANFIS using Shields number, Keulegan-Carpenter number, and embedment ratio gave the best tested prediction, with larger clear-water uncertainty driven more by input combination than model structure. *Regime: Shields, KC, and embedment ratio inputs* direct finding, numerical [1292]
9. Flume tests found ionic-stabilizer solidified soil effective for pipeline scour protection, but its dense crust can increase accumulated-liquefaction risk in underlying noncohesive soil. *Regime: ISS-solidified layer around a pipeline over noncohesive subsoil* direct finding, experimental [616]
10. A coupled CFD-DEM solver matched measured pipeline scour-depth evolution while resolving individual-particle motion through onset, tunnel erosion, and lee-wake erosion. *Regime: subsea pipeline over granular bed* direct finding, numerical [1321]
11. In 1:20 tests tied to Cezhen field data, bidirectional tidal scour was more symmetric and averaged 80% of the depth under a steady unidirectional current with the same peak velocity. *Regime: Hangzhou Bay pipeline tidal cycle* direct finding, experimental [1595]
12. Validated FLOW-3D simulations identified bed shear as the primary pipeline-scour driver and showed significant effects of velocity, grain size, pipe diameter, and initial seabed clearance. *Regime: pipeline with varied flow, grain size, diameter, and initial gap* direct finding, numerical [617]
13. Pipeline protection methods divide into preventing scour onset and stimulating self-burial; prevention is widely studied but mechanistically incomplete, while spoiler-assisted burial has broad experimental, numerical, and project evidence. *Regime: underwater pipelines and engineering protection projects* review statement, review [1544]
14. For a piggyback pipeline, equilibrium scour increased as gap ratio rose from 0 to 0.1 and then declined, while stronger current ratio, KC, Reynolds number, and Shields parameter increased depth. *Regime: piggyback pipe under waves and current* direct finding, experimental [615]
15. An open-source multiphase CFD model computed pipeline sediment transport without imposed bedload and suspended-load layers and reproduced seepage, piping onset, and vortex-driven deposition. *Regime: submerged pipeline with sediment transport and deposition* direct finding, numerical [1365]
16. A review integrates hydrodynamic loading, corrosion, design, and maintenance risks for risers, umbilicals, and cables in floating offshore production systems. *Regime: Pipeline Systems in Floating Offshore Production Systems: Hydrodynamics, Corrosion, Design and Maintenance* review statement, review [1558]

17. A review synthesizes dynamic loading and displacement of offshore pipelines impacted by submarine debris flows and turbidity currents. *Regime: A State-of-the-Art Review of the Hydrodynamics of Offshore Pipelines Under Submarine Gravity Flows and Their Interactions* review statement, review [1602]

18. Reanalysis of weakly cohesive-bed experiments finds normalized pipeline scour depends directly on Keulegan-Carpenter number and, for clay content below 5%, inversely on clay content; adding initial burial substantially improves fit, while wide clay ranges require liquidity-index information. *Regime: Pipelines on weakly cohesive beds exposed to combined waves and currents, with varying clay fraction and initial burial.* direct finding, mixed [1123]

41.4 Scour protection

Topic scour.protection. Countermeasures and performance.

Papers screened: 10. Claims: 10. Equations: 0. Shown in the Coastal Design Bench beside: Seawall design.

Well established. Marine scour countermeasures either increase bed resistance, alter the erosive flow, improve soil, or deliberately manage burial. Protection performance includes local scour reduction, armor and filter stability, edge response, deformation, and coupled foundation capacity.

Governing physics. Controls include horseshoe and lee vortices, downflow and side acceleration, bed shear, pressure and suction beneath armor, wave-current ratio, KC and Shields mobility, permeability, thickness, grading, surcharge, confinement, and wake interactions.

Dimensionless parameters. Important groups include Shields and KC numbers, wave-current ratio, protection-to-base grain and thickness ratios, footprint-to-pile diameter, permeability, relative collar elevation and width, armor grading, surcharge, and turbine clearance.

Major equations. Design frameworks use armor stability numbers and sizing, Shields and KC mobility thresholds, deformation and timescale relations, flow-reduction metrics, p-y and load-utilization curves, finite elements, LES, and CFD-DEM particle transport.

Typical methods. Methods combine wave-current flumes with underside observation, centrifuge and lateral-load tests, physical-model data reanalysis, LES, CFD-DEM, finite-element parametrics, micro/macro countermeasure reviews, and foundation capacity assessment.

Numerical models. LES resolves collar-modified vortices, CFD-DEM resolves permeable-net flow and grains, and finite elements quantify protection-induced confinement and stiffness. These models answer different coupled hydraulic and structural questions and are not interchangeable.

Experimental datasets. Evidence includes glass-bottom armor-suction tests, centrifuge monopile loading, optimized fishnet laboratory tests, CFD-DEM nets with permeability 0.681-0.802, more than 100 rock-fill simulations, and multi-study rock-deformation datasets.

Validated ranges. Reported results include over 30% lateral-capacity gain for a 5D/15 kPa layer, 10% capacity gain from thicker/heavier rock, fishnet shear/depth reductions of 14%/38.2%, net permeability 0.681-0.802, and collar side-flow/area reductions of 24.3%/93.3%.

Recent advances. Recent advances use underside visualization, CFD-DEM nets, collar LES, finite-element load utilization, cross-study deformation reanalysis, and turbine-specific reviews to treat protection as a coupled seabed-foundation system.

Disagreements. Protection can be both hydraulic countermeasure and structural component, but stiffness credit is geometry-dependent: large piles gained little except with heavy, tall rock, while width and sand accretion contributed under 1%. Hydraulic protection alone does not justify universal foundation credit.

Limitations. Edge scour, filter migration, base suction, liquefaction, armor settlement, installation variability, oblique and reversing forcing, extreme waves, array wakes, ecology, aging, and prototype structural response remain incompletely validated.

Open questions. Needs include coupled edge-filter-base failure, lifecycle deformation and repair, seismic and cyclic liquefaction, hybrid and eco-integrated systems, array-scale wakes, monitoring standards, and reliability-based optimization across hydraulic and structural functions.

Seminal papers. The branch builds on riprap stability and filter concepts, horseshoe-vortex control, collar and sacrificial-pile flow alteration, and armor deformation, extended by bio-soil treatment, permeable nets, structural stiffness credit, and rotor-wake conditions.

Claims

- Monopile countermeasures divide into armoring that strengthens the bed, flow alteration that weakens downflow and vortices, and soil improvement such as MICP, with distinct sizing and reduction evidence. *Regime: offshore-wind monopiles with riprap, collars, sacrificial piles, caissons, and MICP* review statement, review [1392]
- A scour-protection layer 5 pile diameters wide with 15 kPa surcharge increased monopile lateral capacity by more than 30%, reduced accumulated deflection by over 100%, and could permit 10% less embedment. *Regime: monopile in dense and loose sand with protection layer* direct finding, mixed [82]
- Across more than 100 simulations, rock fill restored monopile stiffness and thicker, heavier protection increased static capacity by about 10%, whereas width or sand densification changed capacity by less than 1% and large piles benefited less. *Regime: small- and large-diameter offshore-wind monopiles* direct finding, numerical [960]
- Sediment suction beneath monopile protection depends on pile KC, wave-current ratio, and protection-to-base thickness, with lower critical mobility under wave-dominated than current-dominated or combined conditions. *Regime: monopile protection over a thin mobile base layer* direct finding, experimental [1025]
- For monopile protection nets with permeability 0.681-0.802, CFD-DEM showed reduced velocity and front downflow; protection was strongest inside the net and outside performance improved as permeability increased. *Regime: monopile nets with permeability 0.681-0.802* direct finding, numerical [1601]
- An optimized fishnet countermeasure reduced maximum seabed shear by 14% numerically and maximum monopile scour depth by 38.2% in laboratory tests. *Regime: offshore-wind monopile with varied net size, thread, and elevation* direct finding, mixed [1560]
- Collar LES reduced monopile side-flow intensity by 24.3% and accelerated-flow area by 93.3%, but protection weakened as a 14% inflow increase raised turbulence intensity by 159%. *Regime: monopile with collar over flat and equilibrium-scoured beds* direct finding, numerical [1537]

8. For tidal-stream foundations, rotor operation, small tip clearance, and helical wakes can intensify near-bed erosion beyond static monopiles, requiring mitigation that accounts for reversing tides, waves, pore pressure, and array effects. *Regime: tidal-stream monopile, tripod, jacket, and gravity foundations* review statement, review [856]
9. A reanalysis of monopile rock-protection tests found deformation patterns and magnitude governed by Shields number and wave-current ratio, negligible effects over the low tested KC range, and slower timescales than live-bed scour. *Regime: monopile rock protection under waves and currents* review statement, review [1026]
10. A biomineralization review reports 2–4 MICP cycles reduced maximum scour depth 84–100% under 0.3 m/s unidirectional currents and increased coastal-soil strength, while field delivery and environmental compatibility remain design constraints. *Regime: Recent Developments on Biomineralization for Erosion Control* direct finding, mixed [857]

41.5 Local scour at structures

Topic scour. structures. Scour around piles, foundations, and walls.

Papers screened: 14. Claims: 14. Equations: 0. Shown in the Coastal Design Bench beside: Seawall design, Pile wave loads.

Well established. Local scour forms where structures accelerate flow and organize horseshoe, wake, contraction or overtopping jets that increase bed stress and sediment export; depth and footprint depend on forcing history, sediment mobility, geometry and feedback as the scour hole changes the flow.

Governing physics. Downflow and horseshoe vortices erode upstream toes, wake vortices remove sediment downstream, wave orbital reversal and currents alter symmetry, and tsunami overflow can drive intense leeward jets. Avalanching, armoring, seepage and deposition govern approach to equilibrium or progressive failure.

Dimensionless parameters. Controls include Shields and mobility numbers, Froude number, Keulegan-Carpenter number, wave-current ratio, relative submergence, structure-width-to-depth and width-to-grain ratios, sediment gradation, densimetric Froude number, scour depth normalized by width and time normalized by scour timescale.

Major equations. Assessment combines sediment continuity with bedload or suspended-load transport, critical Shields mobility and empirical equilibrium-depth/time-scale relations using structure width, flow depth, velocity, wave orbital motion and sediment size. Practical fragility links scour depth to foundation embedment and armor stability.

Typical methods. Studies use mobile-bed flumes with repeated bed scans and synchronized velocity or wave measurements, vary current and wave combinations and geometry, check scale and sediment mobility, fit time evolution and equilibrium depth, validate morphology models and compare with post-event surveys and foundation damage.

Numerical models. The reviewed branch provides a field-informed practical scour model. Related evidence includes coupled RANS-VOF, bedload and moving-bed morphology for vertical structures, but broad pile-specific CFD, compound-pile and twin-pile formulations are not yet reviewed here.

Experimental datasets. Reviewed primary evidence comprises post-2011 Tohoku coastal-defense surveys followed by hydraulic experiments used to refine a practical tsunami scour-depth model. Direct pile and jacket datasets remain in the lawful-full-text queue.

Validated ranges. Support is specific to surveyed Tohoku defenses and the associated hydraulic experiments. It does not validate equilibrium formulas for steady-current piles, random waves, combined flow, jackets or different soils and foundation configurations.

Recent advances. Recent advances use high-resolution sonar and photogrammetry, particle-resolving CFD-DEM, coupled hydro-morphodynamic and geotechnical simulation, probabilistic surrogate models, digital-twin monitoring and adaptive scour protection.

Disagreements. Equilibrium formulas differ over whether wave-current effects combine through peak velocity, effective shear or vortex regime. Laboratory scale can distort sediment mobility and turbulence, while field failure often couples scour with armor displacement, soil response and structural damage.

Limitations. Single-source primary coverage, uncertain event hydrographs, post-event survey timing, soil heterogeneity, scale effects, armoring, cohesive sediment, debris, three-dimensional geometry, storm sequences and backfilling restrict transfer. Maximum depth alone omits footprint and foundation demand.

Open questions. Priorities include transient compound wave-current-tsunami forcing, complex pile groups and jackets, layered/cohesive beds, scour-structure interaction, probabilistic time-dependent fragility, monitoring assimilation, countermeasure degradation and full-scale benchmark datasets.

Seminal papers. Classical bridge-pier and marine-pile experiments established horseshoe and wake-vortex mechanisms and equilibrium scaling; pipeline and breakwater studies extended wave and toe regimes, followed by RANS morphology and structure-soil interaction models.

Claims

1. Post-2011 Tohoku surveys identified six coastal-defense failure modes, with leeward toe scour dominant at most surveyed locations; a practical scour-depth model using field-measurable tsunami, soil, and structure quantities was subsequently refined with hydraulic experiments. *Regime: Level-2 Tohoku tsunami damage to dikes, seawalls, and breakwaters in Miyagi and Fukushima.* direct finding, mixed [651]
2. A 3D RANS morphodynamic model found omitting suspended load reduced equilibrium cylinder scour depth by 50%, while vortex shedding mainly affected the earliest stage; horseshoe and lee-wake vortices organized transport. *Regime: Numerical investigation of flow and scour around a vertical circular cylinder* direct finding, mixed [131]
3. A monopile review synthesizes current-, wave-, combined-, and complex-loading scour mechanisms and equations and prioritizes unsteady-tide experiments, CFD-DEM, and artificial intelligence. *Regime: Local scour at offshore windfarm monopile foundations: A review* direct finding, mixed [546]

4. Validated CFD–DEM shows offshore-pile scour depends on velocity, gravity, fluid and drag forces, and particle contact; maximum average drag occurred near 90° but maximum scour near 45° to the incoming flow. *Regime: Local Scour Mechanism of Offshore Wind Power Pile Foundation Based on CFD-DEM* direct finding, mixed [848]
5. A dimensionless 1D CNN for complex bridge-pier clear-water scour achieved $R^2=0.85$, $RMSE=0.1125$, $MAE=0.1078$ and low bias on unseen data, outperforming FDOT, HEC-18 and Coleman equations. *Regime: A CNN deep learning approach to scour depth estimation around complex bridge piers in steady flow environments* direct finding, mixed [625]
6. A methodological synthesis concludes scour remains incompletely understood across bridges, pipelines and offshore wind foundations and requires complementary laboratory, numerical, and field approaches. *Regime: Experimental, Numerical and Field Approaches to Scour Research* direct finding, mixed [275]
7. A 3D SWASH/flume study attributes about 75–80% of total bridge-pier scour depth predominantly to flow and 15–30% to redistribution of longitudinal velocity, alongside direct impact and horseshoe-vortex suction. *Regime: Study on the Mechanism of Local Scour Around Bridge Piers* direct finding, mixed [364]
8. Tests and models found unprotected monopile scour slightly exceeded pile diameter, open-jacket scour was lower, and SEDSCOUR predicted sandy beds with up to 30% mud; caisson scour required geometry-resolving treatment. *Regime: Scour near Offshore Monopiles, Jacket-Type and Caisson-Type Structures* direct finding, mixed [1191]
9. A review of wave-driven pier scour synthesizes governing hydrodynamics, sediment response, experimental and numerical methods, and unresolved needs for coastal foundation design. *Regime: Scour around Piers under Waves: Current Status of Research and Its Future Prospect* review statement, review [487]
10. Experiments show that seabed scour changes the lateral resistance of offshore wind-turbine monopiles and therefore must be represented in foundation assessment. *Regime: Impact of scour on lateral resistance of wind turbine monopiles: an experimental study* direct finding, experimental [820]
11. Scour and scour protection modify the stiffness of monopile foundations in sand, linking evolving seabed geometry to offshore structural response. *Regime: Influence of scour and scour protection on the stiffness of monopile foundations in sand* direct finding, mixed [1229]
12. A bioinspired design framework translates root-system anchorage, adaptability, and soil interaction into concepts for resilient coastal foundations. *Regime: Root Systems Research for Bioinspired Resilient Design: A Concept Framework for Foundation and Coastal Engineering* review statement, review [1346]
13. For experimental toe-scour data at sloping seawalls with shingle foreshores, support-vector regression was the best of four tested machine-learning methods (R-squared 0.74, RMSE 0.28, MAE 0.17), while Iribarren number was the most influential predictor. *Regime: Relative toe-scour depth at sloping seawalls with shingle foreshores within the source experimental dataset.* direct finding, mixed [560]
14. Coupling large-eddy free-surface flow with bed-load transport reproduces initial tsunami scour at revetment toes and behind overtopped seawalls, but equilibrium scour requires longer simulations and suspended-sediment transport. *Regime: Tsunami impingement and overtopping at near-vertical revetments and sharp-edged seawalls with mobile beds.* direct finding, numerical [1009]

42 Wiki: Sediment transport

42.1 Sediment transport: overview

Topic sediment. Entrainment, suspension, and transport of coastal sediment.

Papers screened: 14. Claims: 10. Equations: 0.

Well established. Coastal sediment transport is a coupled source–pathway–sink system in which intermittent entrainment, bedload, suspension and sheet flow respond to waves, currents, water levels and morphology while sorting, storage and bed change feed back on the forcing.

Governing physics. Fluid stress and turbulent impulses compete with submerged weight, friction, cohesion and hiding/exposure; moving grains exchange momentum with the flow, settle and diffuse, organize bedforms, change roughness and permeability, and redistribute mass through cross-shore, longshore and vertical pathways.

Dimensionless parameters. Controls include Shields and mobility numbers, shear-to-settling velocity, Rouse number, grain Reynolds number, relative roughness, fall-velocity and suspension parameters, Stokes number, wave orbital excursion, asymmetry and skewness, concentration, slope, mobility duration and transport-to-storage ratios.

Major equations. Core formulations include Shields mobility, bedload and sheet-flow relations, advection–diffusion–settling for suspension, Rouse-type profiles, wave–current boundary layers, energetics transport, two-phase momentum, Exner sediment continuity, active-layer sorting and source–pathway–sink mass balance.

Typical methods. Methods combine synchronized velocity, pressure, concentration and bed measurements; tracers and grain-size analysis; acoustic, optical and conductivity profilers; traps and samplers; repeat bathymetry; movable-bed experiments; transport formulae, two-phase or advection–diffusion models; and sediment-budget closure.

Numerical models. Models range from practical bedload/suspended-load formulae and energetics closures to boundary-layer, two-phase, RANS/LES sediment, wave-resolving, morphodynamic, active-layer sorting and network/source-to-sink models coupled through sediment continuity.

Experimental datasets. Evidence includes BARDEX II barrier response, combined wave–current bedform observations, bichromatic offshore- and onshore-bar experiments, wave-group suspension tests, regional estuary/shelf synthesis and source-to-sink signal studies.

Validated ranges. The reviewed evidence spans sandy swash, surf and nearshore zones, barriers, bedforms, estuaries and shelves, weak transport through sheet flow, monochromatic and grouped waves, combined currents and event-to-long-term sediment routing; each result remains regime-specific.

Recent advances. Recent work resolves mode transitions with two-phase models, measures bedload and suspension simultaneously, uses dense acoustic/optical profiling, conducts large-scale grouped-wave bar experiments, compares migration directions, and embeds event transport within source-to-sink signal propagation.

Disagreements. A single critical stress cannot represent intermittent onset and cessation; equivalent monochromatic waves may not reproduce group-driven flux; bedload and suspension boundaries shift with concentration and turbulence; instantaneous formula skill need not yield correct morphology; deposits can lag or erase source signals.

Limitations. Direct flux is difficult to measure, instruments disagree in intense transport, concentration alters turbulence, bedforms evolve, cohesive and mixed grains violate uniform-sand assumptions, boundaries and scale distort experiments, budgets omit exchanges and short records miss rare dominant events.

Open questions. Priorities include probabilistic entrainment and cessation, mixed/cohesive beds, high-concentration two-phase feedback, wave-group and infragravity effects, three-dimensional pathways, biology and vegetation, event dominance, source-to-sink memory, intervention attribution and uncertainty-aware budget closure.

Seminal papers. Classical Shields mobility, Einstein/Bagnold transport, Rouse suspension, wave–current boundary-layer theory, energetics formulae and Exner continuity established the grain-to-morphology framework later extended to sheet flow, sorting and two-phase dynamics.

Claims

1. A synthesis of sediment initiation and cessation shows that entrainment thresholds, transport onset, and stopping depend on turbulent fluctuations, grain-bed interactions, history, and regime rather than one universal critical stress. *Regime: The Physics of Sediment Transport Initiation, Cessation, and Entrainment Across Aeolian and Fluvial Environments* review statement, review [1131]
2. Source-to-sink signals propagate through sediment systems with storage, shredding, lag, and selective preservation, so downstream deposits need not track forcing synchronously. *Regime: Times Associated With Source-to-Sink Propagation of Environmental Signals During Landscape Transience* review statement, review [1412]
3. A two-phase formulation organizes the transition from bedload to suspended load using shear velocity, settling velocity, and particle–fluid coupling, linking transport modes within one mechanics framework. *Regime: A two-phase flow model of sediment transport: transition from bedload to suspended load* direct finding, analytical [277]
4. BARDEX II provides an integrated large-scale dataset of waves, water levels, sediment transport, groundwater, overwash, and barrier morphology with documented experimental design and instrumentation. *Regime: Large-scale Barrier Dynamics Experiment II (BARDEX II): Experimental design, instrumentation, test program, and data set* direct finding, experimental [928]
5. A review of swash-zone sand-transport models finds that uprush–backwash asymmetry, infiltration, turbulence, acceleration, bed slope, grain properties, and morphological feedback complicate transferable prediction. *Regime: A review of practical models of sand transport in the swash zone* review statement, review [265]
6. A regional review integrates estuarine and inner-shelf physical oceanography with sediment sources, transport pathways, deposits, and coastal morphology across New Zealand settings. *Regime: Coastal oceanography and sedimentology in New Zealand, 1967–91* review statement, review [624]

7. Field observations connect bedform migration to net bedload under combined wave–current forcing, showing that transport inference must retain evolving roughness and bedform kinematics. *Regime: Observations of Bedform Migration and Bedload Sediment Transport in Combined Wave-Current Flows* direct finding, field [1505]
8. Large-scale offshore-bar experiments resolve near-bed hydrodynamics and sediment fluxes under bichromatic wave groups and connect event-scale transport to bar migration. *Regime: Near-Bed Sediment Transport During Offshore Bar Migration in Large-Scale Experiments* direct finding, experimental [540]
9. Large-scale onshore-bar experiments reveal transport processes that differ from offshore migration and enable controlled comparison of direction-dependent near-bed flux mechanisms. *Regime: Near-Bed Sediment Transport Processes During Onshore Bar Migration in Large-Scale Experiments: Comparison With Offshore Bar Migration* direct finding, experimental [541]
10. Comparison of wave groups with equivalent monochromatic waves shows that group-scale hydrodynamics alter sand suspension and flux, limiting representative-wave substitutions. *Regime: Sand suspension and fluxes by wave groups and equivalent monochromatic waves* direct finding, experimental [1578]

42.2 Bedload transport

Topic sediment.bedLoad. Transport in contact with the bed.

Papers screened: 9. Claims: 7. Equations: 7.

Well established. Coastal bedload begins above a mobility threshold and responds to the time history of near-bed forcing, not merely bulk wave height; waves, currents, asymmetry, roughness, slope, pressure, breaking, and return flow can change both magnitude and direction.

Governing physics. Governing mechanisms include Shields-type mobility, grain-bed friction, oscillatory boundary-layer turbulence, velocity skewness and acceleration asymmetry, gravity on slopes, wave-pressure-induced bed deformation, return currents, breaking transitions, and stochastic variability among waves.

Dimensionless parameters. Principal nondimensional controls are Shields parameter θ , critical Shields stress, relative orbital excursion am/ks , wave nonlinearity index N_i , Ursell number S , relative roughness, grain Reynolds number, velocity skewness, acceleration asymmetry, and normalized bedload rate.

Major equations. The branch contains practical physics-based formulae for currents and waves, half-cycle random-wave formulations, Bagnold-energetics cross-shore transport, a pressure-coupled porous-bed boundary-layer formulation, and an early wave-bedload law whose coefficients require full-text recovery.

Typical methods. Typical methods combine movable-bed wave flumes, full-scale oscillatory water tunnels, turbulence-resolving boundary-layer models, analytical energetics and force balances, stochastic transformation of half-cycle laws, and validation against compiled transport observations.

Numerical models. Models span the BSL k - ω boundary-layer closure, practical coastal bedload formula families, half-cycle stochastic transport, Bagnold energetics, and a nonlinear pressure-coupled boundary layer over a poroelastic erodible bed.

Experimental datasets. Direct datasets include asymmetric-wave rough-bed cases with $N_i=0.58-0.67$ and $am/ks=35-963$, sheet-flow tests using 0.24 and 0.51 mm sand on slopes to 2.5 degrees, and breaking-wave movable-bed tests with $D_{50}=0.22$ mm.

Validated ranges. Reported bounds include conventional bedload from critical Shields stress to θ about 0.8, rough-bed asymmetric waves at $N_i=0.58-0.67$ and $am/ks=35-963$, 0.24-0.51 mm sheet-flow sand and slopes to 2.5 degrees, and an energetics formula acceptable for $S \leq 60$.

Recent advances. Later advances resolve roughness-modified turbulence under asymmetric waves and isolate bed-slope effects in full-scale sheet-flow experiments, improving the process basis for cross-shore morphodynamic prediction.

Disagreements. No direct contradiction is established in this small set, but deterministic equivalent-wave calculations underpredict stochastic random-wave estimates, while shear-only practical formulae omit pressure-driven deformation represented in the porous-bed theory.

Limitations. Evidence remains dominated by controlled, often unidirectional laboratory forcing. Formula performance is regime dependent, sheet flow lies beyond conventional bedload assumptions, and several historical sources expose insufficient full text to recover coefficients or uncertainty.

Open questions. Needs include common validation datasets for combined irregular waves and currents, pressure and seepage effects, graded beds, field-scale direction reversals, sheet-flow transitions, morphology feedback, and uncertainty in threshold and roughness specification.

Seminal papers. The 1986 wave-bedload law and 1989 breaking-wave energetics study are early formulations; the 2001 stochastic treatment and 2003 pressure-coupled model broadened the physics, followed by the 2005 practical formula family.

Equations

- *Bagnold energetics cross-shore bedload model* [1129]

$$q_b = f(\theta, u(t), k_s, D_{50}, \beta)$$

Regime: Movable-bed flume with $D_{50}=0.22$ mm sand; derived formula reported acceptable for Ursell number $S \leq 60$, especially short waves and beds with at most one submerged bar. Variables: q_b bedload transport rate; θ Shields parameter; $u(t)$ near-bed velocity history; k_s bed roughness; D_{50} median sediment diameter; β bed slope; generic normalized dependency list, not a recovered source coefficient equation

- *Soulsby-van Rijn coastal bedload framework* [1326]

$$q_b = f(\theta, u(t), k_s, D_{50}, \beta)$$

Regime: Coarse sand and gravel bedload; conventional bedload from critical Shields stress to about $\theta=0.8$, with sheet flow above about $\theta=0.8$. Variables: q_b bedload transport rate; θ Shields parameter; $u(t)$ near-bed velocity history; k_s bed roughness; D_{50} median sediment diameter; β bed slope; generic normalized dependency list, not a recovered source coefficient equation

- *BSL k-omega boundary-layer model* [1373]

$$q_b = f(\theta, u(t), k_s, D_{50}, \beta)$$

Regime: Wave nonlinearity index $Ni=0.58-0.67$ and orbital-excursion-to-roughness ratio $am/ks=35-963$. Variables: q_b bedload transport rate; θ Shields parameter; $u(t)$ near-bed velocity history; k_s bed roughness; D_{50} median sediment diameter; β bed slope; generic normalized dependency list, not a recovered source coefficient equation

- *half-cycle sheet-flow interpretation* [1391]

$$q_b = f(\theta, u(t), k_s, D_{50}, \beta)$$

Regime: Sediment diameters 0.24 and 0.51 mm and bed slopes from horizontal to 2.5 degrees under matched first-harmonic velocity and period. Variables: q_b bedload transport rate; θ Shields parameter; $u(t)$ near-bed velocity history; k_s bed roughness; D_{50} median sediment diameter; β bed slope; generic normalized dependency list, not a recovered source coefficient equation

- *Tsuchiya wave-bedload law* [1430]

$$q_b = f(\theta, u(t), k_s, D_{50}, \beta)$$

Regime: Wave-driven bedload; accessible metadata does not expose coefficients or a validation range. Variables: q_b bedload transport rate; θ Shields parameter; $u(t)$ near-bed velocity history; k_s bed roughness; D_{50} median sediment diameter; β bed slope; generic normalized dependency list, not a recovered source coefficient equation

- *stochastic nonlinear random-wave bedload model* [1004]

$$q_b = f(\theta, u(t), k_s, D_{50}, \beta)$$

Regime: Narrow-band second-order random waves using a regular-wave half-cycle bedload formula. Variables: q_b bedload transport rate; θ Shields parameter; $u(t)$ near-bed velocity history; k_s bed roughness; D_{50} median sediment diameter; β bed slope; generic normalized dependency list, not a recovered source coefficient equation

- *deformable porous-bed boundary-layer model* [453]

$$q_b = f(\theta, u(t), k_s, D_{50}, \beta)$$

Regime: Wave loading over a deformable, erodible porous seabed represented with an extended Saffman slip condition and Coulomb failure criterion. Variables: q_b bedload transport rate; θ Shields parameter; $u(t)$ near-bed velocity history; k_s bed roughness; D_{50} median sediment diameter; β bed slope; generic normalized dependency list, not a recovered source coefficient equation

Claims

1. For reviewed coastal datasets spanning currents, symmetric or asymmetric waves, and combined forcing, the physics-based bedload formulae predicted 47%-91% of observations within a factor of two; their stated conventional-bedload range extends from critical Shields stress to approximately $\theta=0.8$. *Regime: Coarse sand and gravel bedload; conventional bedload from critical Shields stress to about $\theta=0.8$, with sheet flow above about $\theta=0.8$.* direct finding, analytical [1326]
2. For asymmetric-wave boundary layers with $Ni=0.58-0.67$ and $am/ks=35-963$, increasing roughness raised turbulent kinetic energy and bottom shear stress, reduced inner-layer mean velocity, and increased potential net sediment transport, with direction and magnitude also strongly controlled by wave nonlinearity. *Regime: Wave nonlinearity index $Ni=0.58-0.67$ and orbital-excursion-to-roughness ratio $am/ks=35-963$.* direct finding, mixed [1373]
3. Full-scale oscillatory-tunnel tests isolated bed-slope effects on sheet-flow transport for 0.24 and 0.51 mm sand over slopes up to 2.5 degrees under skewed and asymmetric flows with matched first-harmonic velocity and period. *Regime: Sediment diameters 0.24 and 0.51 mm and bed slopes from horizontal to 2.5 degrees under matched first-harmonic velocity and period.* direct finding, experimental [1391]
4. A nonlinear boundary-layer formulation predicts coastal bedload by coupling wave shear with pressure-driven poroelastic bed deformation and a Coulomb failure criterion; its applicability is limited to the assumed deformable porous-bed and viscoelastic-sediment representation. *Regime: Wave loading over a deformable, erodible porous seabed represented with an extended Saffman slip condition and Coulomb failure criterion.* direct finding, analytical [453]
5. For narrow-band second-order random waves modeled with a regular-wave half-cycle transport law, stochastic averaging predicts larger mean bedload than applying the same law to an equivalent deterministic sinusoidal wave. *Regime: Narrow-band second-order random waves using a regular-wave half-cycle bedload formula.* direct finding, analytical [1004]
6. In movable-bed flume tests with $D_{50}=0.22$ mm sand, single breaking produced shoreward bedload in the surf zone and offshore transport seaward of breaking, whereas repeated breaking over multiple bars produced multiple direction reversals; the derived energetics formula was acceptable for $S \leq 60$. *Regime: Movable-bed flume with $D_{50}=0.22$ mm sand; derived formula reported acceptable for Ursell number $S \leq 60$, especially short waves and beds with at most one submerged bar.* direct finding, mixed [1129]
7. Tsuchiya (1986) proposed a law for wave-driven sediment transport in the bedload regime; available metadata does not support a quantitative coefficient set or validated parameter range. *Regime: Wave-driven bedload; accessible metadata does not expose coefficients or a validation range.* hypothesis, author_hypothesis [1430]

42.3 Cohesive sediment

Topic sediment.cohesive. Mud erosion, flocculation, and deposition.

Papers screened: 27. Claims: 25. Equations: 0.

Well established. Cohesive sediment aggregates, settles, deposits, consolidates, erodes, and resuspends through history-dependent interactions among clay mineralogy, organic matter, salinity, concentration, turbulence, waves, currents, stratification, and bed state.

Governing physics. Key processes are collision and aggregation, breakup, effective-density and shape change, free and hindered settling, flocculation by salt and polymers, erosion threshold and rate, simultaneous deposition, consolidation, biodiffusion, fluid mud, and tidal convergence.

Dimensionless parameters. Important controls include bed shear relative to critical stress, settling-to-mixing timescale, floc and primary-particle size, effective density, solids concentration, salinity, organic/cohesive fraction, dry bulk density, Rouse-type suspension, and tidal/wave transport asymmetry.

Major equations. Frameworks include Partheniades erosion and Krone deposition, aggregation-breakup population balances, hindered settling, advection-diffusion-settling, bed layering and consolidation, entrainment-flux inversion, wave-boundary-layer fluid mud, and mixed sand-mud morphology.

Typical methods. Methods span settling columns, erosion devices, floc cameras and holography, field turbidity and tripod arrays, sediment cores, salinity/EPS experiments, mass-balance inversion, particle tracking, grain-resolved simulation, and coupled 3-D sediment-morphology models.

Numerical models. ROMS/COAWST, Delft3D, SUNTANS, CSTMS, particle tracking, population-style floc modules, and grain-resolved immersed-boundary models represent different scales. Spatially variable erosion and multiple floc classes can be essential for field skill.

Experimental datasets. Evidence includes Rhine-ROFI particle imaging, Venice Lagoon sensor inversion, four-site San Francisco Bay SSC, Seine and Chesapeake hindcasts, Hengsha 11-day bed records, salinity experiments, sand-clay columns, and consolidated-bed threshold tests.

Validated ranges. Reported findings include wave contribution to 44% of spring ETM mass, 25% oxygen decrease and 50% ammonium increase, salinity effects from 0.5 ppt saturating near 3-5 ppt, settling 0.1-10 mm/s, over 4 cm shelf deposition, and 38 mm erosion followed by 68 mm accretion.

Recent advances. Recent advances directly image composition-dependent flocs, quantify low-salinity thresholds, infer erodibility from routine time series, model wave-supported fluid mud, resolve grain-scale cohesion, and couple sediment resuspension to oxygen and nutrients.

Disagreements. Salt is not universally required to form flocs because organic-rich freshwater mud already aggregates, yet salt accelerates growth and binds residual fines. Bentonite showed a sharper 0.6-1.0 PSU threshold, demonstrating mineral- and organic-specific rather than universal salinity response.

Limitations. Fixed settling speeds and thresholds, sparse floc-density and bed-strength observations, instrument disruption, mineral specificity, uncertain biological cohesion, fluid-mud rheology, mixed-bed segregation, scale gaps, and calibration equifinality constrain prediction.

Open questions. Needs include coupled mineral-organic floc dynamics, history-dependent erosion and consolidation, simultaneous erosion/deposition, fluid-mud turbulence damping, biostabilization/bioturbation, mixed-bed stratigraphy, and morphology-biogeochemistry feedback.

Seminal papers. The branch builds on Partheniades-Krone erosion/deposition, hindered settling, estuarine trapping and tidal pumping, extended by floc population dynamics, mixed-bed layering, fluid mud, remote erodibility inversion, and grain-resolved cohesion.

Claims

1. Estuarine turbidity maxima reflect longitudinal convergence of tidally averaged suspended-particle transport produced by multiple, system-dependent hydrodynamic and sediment processes. *Regime: estuaries across tidal, river, stratification, and sediment regimes* review statement, review [213]
2. Long-term mixed-sediment modelling showed that mud supply narrows and stabilizes estuaries, with fluvial mud distributed throughout, marine mud concentrated near the mouth, and waves suppressing downstream muddy coastlines. *Regime: idealized convergent estuary across tide, river, waves, and mud supply* direct finding, numerical [182]
3. In the Seine model, energetic waves supplied up to 44% of maximum spring-tide ETM mass and tripled mean-tide mass relative to calm conditions, while river flow mainly shifted location and spring-neap phasing created hysteresis. *Regime: macrotidal Seine Estuary* direct finding, numerical [527]
4. ROMS v3.6/COAWST r1234 added cohesive and mixed-sediment algorithms for water-column floc dynamics, bed-floc change, erosion/deposition, biodiffusion, and higher-fidelity bed layering. *Regime: idealized and realistic cohesive and mixed-sediment environments* direct finding, numerical [1298]
5. Modelled Chesapeake resuspension shifted production and remineralization and, averaged over the channel bottom water, decreased oxygen by about 25% and increased ammonium by about 50%. *Regime: Chesapeake Bay summers 2002 and 2003* direct finding, numerical [983]
6. A Thermaikos Gulf particle model tracked evolving cohesive-floc properties through stratified advection, turbulent diffusion, settling, coagulation, breakup, consolidation, deposition, and bed erosion. *Regime: Thermaikos Gulf with river, bed, atmospheric, and drainage sources* direct finding, numerical [744]
7. A Venice Lagoon mass-balance inversion used routine wind, tide, wave, and turbidity observations to estimate settling velocity and critical erosion stress without disturbing the biologically active bed. *Regime: shallow wave-dominated Venice Lagoon* direct finding, mixed [1451]
8. In South San Francisco Bay, two floc classes with spatially varying erosion ratios and critical stress reproduced four-site SSC, and shoal export contributed to channel suspension at a magnitude comparable to local resuspension. *Regime: South San Francisco Bay shoals and deep channel* direct finding, mixed [279]
9. An entrainment-flux method inferred critical erosion stress and erosion-rate coefficients from tidal SSC time series with favorable agreement to direct Gust-microcosm measurements. *Regime: small narrow tidal estuary* direct finding, mixed [932]
10. Natural-mud experiments showed floc growth and equilibrium size increasing with salinity with or without organic matter, with detectable enhancement at 0.5 ppt and only marginal further growth beyond about 3-5 ppt. *Regime: deionized to estuarine salinity with and without natural organic matter* direct finding, experimental [15]
11. A settling-velocity review found that concentration, particle size and shape, temperature, salinity, organic matter, and pH interact, while sampling instruments can disrupt flocs and yield inconsistent sizes. *Regime: reservoir and estuarine still-water settling* review statement, review [824]
12. Rhine-ROFI observations found mineral, mixed aggregate, and organic particle classes with settling velocities of 0.1-10 mm/s and aspect ratios 1-10, while algae altered floc density within minutes. *Regime: Rhine ROFI near bed during neap tide* direct finding, mixed [1232]

13. Settling-column tests showed distinct sand-clay layers at low clay content, transitional segregation as clay increased, and no clear segregation at high clay content, consistent with polydisperse hindered-settling predictions. *Regime: sand-clay mixtures across fractional composition* direct finding, mixed [310]
14. Bentonite settling increased sharply across 0.6-1.0 PSU and then plateaued; 60% of aggregates exceeded 20 micrometres, while EPS enlarged flocs but did not dominate settling. *Regime: bentonite from deionized to seawater salinity* direct finding, experimental [740]
15. Adding wave-supported fluid mud to the Atchafalaya model produced a lutocline, more than 4 cm offshore deposition, and downslope flux comparable to suspended flux, with hindered settling extending transport offshore. *Regime: Atchafalaya shelf during March 2008 cold fronts* direct finding, numerical [1579]
16. Grain-resolved immersed-boundary simulations represent van der Waals cohesion and reveal statistical aggregation-breakup equilibrium, dense-suspension settling, and cohesive granular-collapse behavior not yet captured reliably by continuum laws. *Regime: cohesive particles in viscous and turbulent open-water flows* review statement, review [1468]
17. At Hengsha Shoal, a wave-swell event produced 38 mm erosion followed by 68 mm accretion, and a simultaneous erosion-deposition paradigm reproduced cohesive bed change better than exclusive deposition. *Regime: Hengsha Shoal near Yangtze ETM* direct finding, mixed [1295]
18. Experiments found cohesive-sediment incipient shear stress increasing with consolidation time, dry bulk density, and cohesive fraction, and a formula using relative density and cohesive content matched the data. *Regime: consolidated cohesive sediment mixtures* direct finding, experimental [1305]
19. Controlled experiments show that increasing clay content reduces mixed sand-mud bedform dimensions and steepness, limiting cohesionless-sediment predictors. *Regime: Sticky stuff: Redefining bedform prediction in modern and ancient environments* direct finding, experimental [1263]
20. York River observations show tidal-straining asymmetry in turbulent mixing drives distinct advective and pumping components of fine-sediment flux. *Regime: Sediment pumping by tidal asymmetry in a partially mixed estuary* direct finding, field [1274]
21. Mudstone microtextures and paleoflow evidence identify storm-generated, wave-enhanced sediment gravity flows dispersing mud across an ancient prodelta. *Regime: Mud dispersal across a Cretaceous prodelta: Storm-generated, wave-enhanced sediment gravity flows inferred from mudstone microtexture and microfacies* direct finding, field [1118]
22. Digital inline holography measures in-situ cohesive-floc size and settling velocity, resolving properties that control suspended-sediment settling flux. *Regime: The application of holography to the analysis of size and settling velocity of suspended cohesive sediments* direct finding, experimental [523]
23. Estuary modeling links channel deepening, tidal amplification, fine-sediment import, and sediment-induced drag reduction in the hyper-turbid lower Ems River. *Regime: Fine sediment transport into the hyper-turbid lower Ems River: the role of channel deepening and sediment-induced drag reduction* direct finding, numerical [914]
24. A coupled nearshore morphodynamic study maps cohesive-coast erosion and deposition around a detached low-crested breakwater. *Regime: Dynamics of Sediment Transport and Erosion-Deposition Patterns in the Locality of a Detached Low-Crested Breakwater on a Cohesive Coast* direct finding, numerical [445]
25. Coastal-sea observations and depth-averaged analysis explain phase differences between tidal currents and fine-sediment suspension through advection and slow settling. *Regime: Phase relationships between fine sediment suspensions and tidal currents in coastal seas* direct finding, mixed [126]

42.4 Mixed sediment

Topic sediment.mixed. Sand-mud and graded mixtures.

Papers screened: 8. Claims: 7. Equations: 0.

Well established. Mixed sediment cannot generally be represented as independent pure fractions: increasing fines can switch erosion mode, while sand-mud interactions alter transport and morphology.

Governing physics. Composition affects cohesion, dry bulk density, granular or silt-skeleton structure, roughness, erosion thresholds, and the exchange between active layer and substrate.

Dimensionless parameters. Key controls include applied-to-critical shear ratio, mud/silt fraction, clay fraction, grain-size distribution, dry bulk density, consolidation state, and active-layer/substrate composition contrast.

Major equations. Available formulations include composition-interpolated erosion laws, momentum-balance critical-stress relations, modified Shields-style thresholds, and multi-fraction active-layer conservation equations.

Typical methods. The evidence combines erosion experiments, analytical momentum and eigenstructure analysis, realistic hydro-sedimentary hindcasting, and schematized long-term morphodynamic scenarios.

Numerical models. Models range from shelf suspended-sediment simulations to tidal-basin morphodynamics and multi-fraction active-layer systems; both physical interaction terms and mathematical well-posedness affect credibility.

Experimental datasets. The directly accessible experimental evidence resolves a sand-silt transition near 35% silt; other mixture datasets are represented indirectly through model-data comparisons and compiled threshold data.

Validated ranges. Reported source-specific transitions include about 10-20% mud (3-6% clay) for switching from pure-sand erosion and about 35% silt for a sand- to silt-dominated threshold change.

Recent advances. Recent work unifies sand-to-mud thresholds, identifies a sand-silt tipping point, quantifies sand-mud morphological interaction, and diagnoses broader ill-posed domains in mixed-fraction models.

Disagreements. The critical fine fraction is not universal: clay-bearing sand-mud and nonclay sand-silt mixtures show different transition contents and mechanisms, reflecting mineralogy, grain size, density, and consolidation.

Limitations. This branch has only five abstract-accessible sources. Several direct graded, sand-clay, sand-silt, and mixed sand-gravel studies remain title-only, limiting synthesis of transport rates, segregation, waves, and field transferability.

Open questions. Priorities are transferable transition laws, joint erosion-deposition behavior, hiding/exposure and armoring, flocculation-segregation coupling, wave-current transport, and stable multi-fraction numerics.

Seminal papers. The accessible evidence extends Shields-type incipient motion, cohesive erosion concepts, and Hirano active-layer morphodynamics into composition-dependent mixed-bed formulations.

Claims

1. Bay of Biscay modelling supported an abrupt erosion-law transition and a first critical mud fraction of about 10-20% (3-6% clay), below which pure-sand erosion should be prescribed. *Regime: fine sand, mud, and mixtures on the Bay of Biscay shelf* direct finding, mixed [956]
2. A momentum-balance formulation reproduced erosion thresholds from sand-mud mixtures to pure mud using mud content and the dry bulk density of the mud component. *Regime: sand-mud mixtures through the pure-sand and pure-mud endpoints and consolidation* direct finding, analytical [263]
3. Sand-silt experiments identified a tipping point near 35% silt: below it Shields behavior applied, while above it a stable silt skeleton raised the threshold; a modified formula reduced average bias to about 10%. *Regime: sand-silt beds spanning sand-dominated and silt-dominated composition* direct finding, experimental [1567]
4. Mixed-fraction active-layer models can be ill posed under both aggradation and degradation, producing grid-dependent short-wave oscillations and possible numerical failure. *Regime: graded river-bed aggradation and degradation represented by Hirano active-layer and vertically continuous models* direct finding, analytical [254]
5. Tidal-basin morphology and bimodal intertidal-flat composition depended strongly on sand-mud interaction, with erosion interaction exerting a larger effect than roughness interaction. *Regime: tide-dominated basins with varying mud erodibility and sand-mud erosion and roughness interaction* direct finding, numerical [41]
6. Field measurements on an island shelf test sediment-motion thresholds for carbonate-rich natural grains under wave influence. *Regime: Wave-influenced deposition of carbonate-rich sediment on the insular shelf of Santa Maria Island, Azores* direct finding, field [1608]
7. Under low-to-medium shear, adding fine sand to kaolin suspensions reduces aggregation rate and equilibrium floc size through sand-floc collision breakup; at high shear, shear fragmentation dominates and the sand fraction has little additional effect. *Regime: Kaolin and fine sand-kaolin suspensions under the controlled low, medium and high grid-generated shear conditions.* direct finding, mixed [311]

42.5 Sheet flow

Topic sediment.sheetFlow. High-concentration mobile layers.

Papers screened: 16. Claims: 16. Equations: 0.

Well established. Sheet flow is a thin, dense mobile layer produced when energetic forcing washes out ripples; concentration falls from near packed-bed values toward dilute suspension while grains move through frictional, collisional and turbulence-coupled regimes.

Governing physics. Instantaneous transport reflects shear and pressure gradient, granular friction/collisions, pickup and settling, phase lag, velocity/acceleration asymmetry, progressive and waveshape streaming, mobile-bed roughness, turbulence damping, density stratification, undertow, and bore-generated turbulence.

Dimensionless parameters. Controls include Shields and mobility numbers, grain Reynolds number, density ratio, relative sheet thickness in grain diameters, wave orbital excursion/roughness, velocity and acceleration skewness, settling time/wave period, phase lag, relative depth kh , and sediment concentration.

Major equations. Models use two-phase fluid/sediment mass and momentum, drag and buoyancy coupling, $\mu(I)$ or kinetic-theory particle stress, turbulent closure, dilatancy and Rouse profiles, Shields/mobility forcing, advection-diffusion, boundary-layer streaming, and power-law transport with phase-lag correction.

Typical methods. Methods include full-scale oscillating water tunnels, large wave flumes, swash field CCP profiles, conductivity concentration meters, ACVP/ADV and laser Doppler velocity, natural and synthetic grains, free-surface-resolving Eulerian two-phase CFD, dense rheology, and 1DV turbulence models.

Numerical models. Models include SedFoam-2.0, SedWaveFoam, $\mu(I)$ dense-granular and kinetic-theory closures, $k-\omega$ and $k-\epsilon$ boundary layers, free-surface InterFoam/waves2Foam coupling, Rouse/dilatancy layers, semianalytical thickness ODEs, and Eulerian-Lagrangian mixed-grain coupling.

Experimental datasets. Evidence covers 1.2 m/s and 7 s asymmetric tunnel flows, natural-sand prototype waves, Delft LOWT grain-size tests, sand/PVC/acrylic mobile layers, first field swash profiles, large plunging breakers, shoaling asymmetric waves, near-breaking transients, and mixed-grain simulations.

Validated ranges. Observed sheet layers span about 10–60 grain diameters, near-bed volume concentration around 0.6, swash thickness 6–18 mm with a 0.20–0.30 concentration transition, sand 0.13 to at least 0.21 mm, and cases where quasi-steady transport is underpredicted by about 60%.

Recent advances. Recent work resolves free surface, bottom boundary layer and two sediment phases together; diagnoses wave stirring, undertow and stratification; measures large-scale breaking layers acoustically; extends phase-lag formulas to skewed-asymmetric shoaling waves; and treats mixed grains with DEM coupling.

Disagreements. Tunnel and free-surface wave tests can predict opposite fine-sand transport direction because streaming and horizontal advection are absent under a rigid lid. Phase lag dominates some fine-grain asymmetric cases but is minor in others; neither simple suspension nor collisional models alone span the full dense-to-dilute layer.

Limitations. Near-bed concentration and velocity are difficult to measure simultaneously; conductivity/acoustic probes have distinct bounds; many models are 1DV; closures are calibrated to uniform spherical grains; pore pressure, grain shape, graded beds, breaking intermittency and swash infiltration remain limited.

Open questions. Needs include unified dense-to-dilute closure, pore-pressure feedback, graded and natural sediment, three-dimensional turbulence, breaker and bore intermittency, swash infiltration, coherent structures, sheet-layer top definition, and computationally efficient morphodynamic parameterization.

Seminal papers. The branch advances from oscillatory mobile-layer observations and full-scale prototype-wave measurements to dense $\mu(I)$ rheology, streaming-aware transport, field swash profiles, free-surface two-phase CFD, breaking-wave sheet layers, and graded-bed plug flow.

Claims

1. For 1.2 m/s, 7 s asymmetric oscillatory flows over medium sand, net transport arose from unequal opposing half-cycle fluxes; phase lag was secondary to shear-stress skewness. *Regime: Observations of velocities, sand concentrations, and fluxes under velocity-asymmetric oscillatory flows* direct finding, experimental [1221]
2. SedFoam-2.0 represents sheet flow with Eulerian fluid and sediment phases, kinetic-theory or $\mu(I)$ intergranular stress, and multiple turbulence closures. *Regime: SedFoam-2.0: a 3-D two-phase flow numerical model for sediment transport* direct finding, numerical [252]
3. Prototype waves produced a sheet-flow layer 10–60 grain diameters thick with near-bed volume concentration near 0.6; neither tested single-phase nor collisional model captured both velocity and concentration across the layer. *Regime: Sheet flow dynamics under monochromatic nonbreaking waves* direct finding, experimental [353]
4. Fine 0.13 mm sand produced thicker erosion and sheet-flow layers than sand at or above 0.21 mm, increasing roughness and turbulence damping. *Regime: Mobile-bed effects in oscillatory sheet flow* direct finding, experimental [352]
5. A two-phase $\mu(I)$ dense-granular model reproduced velocity, concentration, transport, thickness, roughness, and the collision-to-turbulence transition in sheet flow. *Regime: A two-phase model for sheet flow regime based on dense granular flow rheology* direct finding, mixed [1182]
6. Oscillatory mobile-layer behavior depends on a pressure-gradient/inertia parameter; at higher values sediment stopped at reversal and remobilized as plug flow. *Regime: Mobile layer in oscillatory sheet flow* direct finding, experimental [450]
7. A modified transitional $k-\omega$ model reproduced stress phase lag and secondary humps and predicted observed suspension ejections during deceleration above sheet flow. *Regime: 1DV bottom boundary layer modeling under combined wave and current: Turbulent separation and phase lag effects* direct finding, mixed [551]
8. Free-surface progressive streaming substantially increased onshore sheet-flow transport and reversed fine-sand direction relative to rigid-lid tunnels, with horizontal advection also important. *Regime: Sand transport beneath waves: The role of progressive wave streaming and other free surface effects* direct finding, mixed [742]
9. Progressive-wave streaming drives onshore boundary-layer current while velocity- and acceleration-skewness drive offshore streaming; relative depth and roughness set their balance. *Regime: Net currents in the wave bottom boundary layer: On waveshape streaming and progressive wave streaming* direct finding, numerical [741]
10. Field swash sheet-flow profiles were self-similar over 6–18 mm thickness, transitioning from linear to power-law concentration at volume fraction 0.20–0.30. *Regime: Comprehensive Field Study of Swash-Zone Processes. II: Sheet Flow Sediment Concentrations during Quasi-Steady Backwash* direct finding, field [764]
11. Free-surface-resolving SedWaveFoam showed progressive streaming and nonlinear wave stirring strongly enhance onshore suspended flux relative to an oscillating tunnel. *Regime: A Numerical Study of Sheet Flow Under Monochromatic Nonbreaking Waves Using a Free Surface Resolving Eulerian Two-Phase Flow Model* direct finding, numerical [714]
12. Shear and pressure gradient explain tunnel sheet thickness, but bore turbulence is needed to reproduce intense initial-uprush sheet flow in prototype-scale swash. *Regime: A semianalytical model for sheet flow layer thickness with application to the swash zone* direct finding, mixed [765]
13. Large-scale plunging waves produced lower pickup and upper sheet layers; shoaling transport was skewed-wave dominated, while outer-surf-zone layering arose from undertow rather than phase lag. *Regime: Wave Boundary Layer Hydrodynamics and Sheet Flow Properties Under Large-Scale Plunging-Type Breaking Waves* direct finding, experimental [463]
14. A bimodal bed with 30% coarse and 70% medium sand behaved like coarse sand but transported slightly more through inverse grading and a smoother mobile foundation; strong waves produced plug flow. *Regime: Entrainment and Transport of Well-Sorted and Mixed Sediment Under Wave Motion* direct finding, numerical [1142]
15. Near-breaking-wave suspension created stable density stratification that damped turbulence and bed shear, shifting transport toward suspended load. *Regime: Numerical investigation of sheet flow driven by a near-breaking transient wave using SedFoam* direct finding, numerical [331]
16. For skewed-asymmetric shoaling waves, separate half-cycle boundary-layer development was essential; a quasi-steady method underpredicted transport by about 60%, especially for fine sand. *Regime: A Numerical Study of Sheet Flow Driven by Skewed-Asymmetric Shoaling Waves Using SedWaveFoam* direct finding, numerical [715]

42.6 Suspended sediment transport

Topic sediment. suspended. Suspension and advection.

Papers screened: 19. Claims: 11. Equations: 5.

Well established. Suspended coastal sediment is governed by entrainment, three-dimensional advection, turbulent diffusion, settling and deposition; net flux depends on the timing and spatial structure of concentration relative to currents, waves, breakers, swash, and basin oscillations.

Governing physics. Core mechanisms are turbulence-driven pickup, settling lag, concentration adaptation, wave-current boundary-layer mixing, plunging-jet vortices, wave-backwash interaction, seiche advection, density or wind-driven circulation, and erosion-deposition feedback with the bed.

Dimensionless parameters. Important controls include suspension number or Rouse-type settling-to-mixing ratios, Shields mobility, relative settling velocity, wave-current ratio, sediment diffusivity to eddy-viscosity ratio, relative depth, breaker type, wave-group ratio, and hydrodynamic versus concentration-adjustment time and length scales.

Major equations. The common framework is an advection-diffusion-settling equation for concentration with erosion or reference-concentration and deposition boundary conditions, coupled to hydrodynamics ranging from shallow-water exact solutions to quasi-3-D asymptotics and three-dimensional or RANS flow solvers.

Typical methods. Methods combine optical or acoustic concentration measurements, pump or trap sampling, mobile-bed flumes, event-resolved flux analysis, exact and asymptotic solutions, characteristics and finite differences, RANS turbulence closure, VOF interface tracking, and comparisons among depth-averaged, 2DV, and 3-D models.

Numerical models. Reviewed models include a full 3-D split characteristics/finite-difference transport solver, a quasi-3-D asymptotic adjustment model, exact Lagrangian seiche solutions, and a 2-D RANS k-epsilon/VOF breaker model with advection-diffusion concentration.

Experimental datasets. Evidence includes lightweight-sediment laboratory dispersion, a large-flume 1:15 mobile beach under monochromatic, bichromatic, long-wave and random forcing, plunging-breaker turbulence and concentration measurements, and paired-tray two-way sediment flux at turbid Great Barrier Reef sites.

Validated ranges. Explicit ranges include six-run, approximately 2.4 h conditions on a 1:15 flume beach and reef two-way fluxes of 34 to more than 640 g m⁻² d⁻¹ with net sedimentation below 122 g m⁻² d⁻¹. Other studies report regime definitions but accessible sources do not expose universal numerical bounds.

Recent advances. Later work resolves plunging-jet turbulence and near-surface concentration and links event-scale wave-backwash interactions to profile response, while field flux methods distinguish gross two-way delivery from net sedimentation.

Disagreements. Reduced models can agree with 2DV and 3-D solutions inside their adjustment-scale validity domain, yet depth-averaged models are unreliable in strongly three-dimensional flows. Nominally accretive wave forcing does not necessarily imply shoreward suspended transport.

Limitations. Controlled experiments and idealized geometries dominate; concentration boundary conditions and turbulent diffusivity remain uncertain, cohesive and mixed sediments are underrepresented, fine mesh may be required near plunging jets, and delivery measurements do not equal net accumulation.

Open questions. Priorities are transferable entrainment and deposition closures, coherent treatment of cohesive mixtures and flocculation, three-dimensional irregular-breaker validation, coupled bed evolution, uncertainty-aware sensor calibration, and field tests that close suspended mass budgets.

Seminal papers. The 1988 full 3-D scheme and 1992 quasi-3-D wave-current theory established numerical and reduced-order foundations; the 2003 exact seiche solutions clarified settling-lag redistribution and provide reproducible benchmarks.

Equations

- *three-dimensional suspended-particulate transport model* [1036]

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{u}c) = \nabla \cdot (\mathbf{K} \nabla c) - w_s \frac{\partial c}{\partial z} + S_e - S_d$$

Regime: Complex three-dimensional coastal flows; tested with hypothetical cases and laboratory transport and dispersal of lightweight sediment. Variables: c suspended sediment concentration; $\mathbf{u}$ flow velocity; $\mathbf{K}$ sediment diffusivity tensor; w_s settling velocity; S_e erosion or entrainment source; S_d deposition sink; normalized governing form

- *quasi-3-D wave-current suspended-transport model* [690]

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{u}c) = \nabla \cdot (\mathbf{K} \nabla c) - w_s \frac{\partial c}{\partial z} + S_e - S_d$$

Regime: Logarithmic velocity profile with wave effects represented through diffusivity and the near-bed boundary condition; applicability bounded by a wave-modified suspension parameter and forcing-change scales. Variables: c suspended sediment concentration; $\mathbf{u}$ flow velocity; $\mathbf{K}$ sediment diffusivity tensor; w_s settling velocity; S_e erosion or entrainment source; S_d deposition sink; normalized governing form

- *wave-backwash suspended-flux event framework* [313]

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{u}c) = \nabla \cdot (\mathbf{K} \nabla c) - w_s \frac{\partial c}{\partial z} + S_e - S_d$$

Regime: Large flume; handmade 1:15 initial slope; each condition approximately 2.4 h in six runs; erosive and accretive wave series. Variables: c suspended sediment concentration; $\mathbf{u}$ flow velocity; $\mathbf{K}$ sediment diffusivity tensor; w_s settling velocity; S_e erosion or entrainment source; S_d deposition sink; normalized governing form

- *exact seiche suspended-transport solution* [1128]

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{u}c) = \nabla \cdot (\mathbf{K} \nabla c) - w_s \frac{\partial c}{\partial z} + S_e - S_d$$

Regime: Idealized elliptical or circular basin with parabolic cross-section; coarse-sand-oriented erosion/deposition law; friction omitted from the hydrodynamic solution. Variables: c suspended sediment concentration; $\mathbf{u}$ flow velocity; $\mathbf{K}$ sediment diffusivity tensor; w_s settling velocity; S_e erosion or entrainment source; S_d deposition sink; normalized governing form

- *RANS-VOF plunging-breaker suspended-transport model* [1054]

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{u}c) = \nabla \cdot (\mathbf{K} \nabla c) - w_s \frac{\partial c}{\partial z} + S_e - S_d$$

Regime: Two-dimensional surf-zone plunging breakers; advection-diffusion concentration with reference-concentration bottom boundary; selected fine mesh resolution. Variables: c suspended sediment concentration; $\mathbf{u}$ flow velocity; $\mathbf{K}$ sediment diffusivity tensor; w_s settling velocity; S_e erosion or entrainment source; S_d deposition sink; normalized governing form

Claims

1. Where coastal suspended transport is governed by three-dimensional flow patterns such as harbour-entrance separation or wind and density currents, depth-averaged models can give unreliable siltation and erosion estimates; a split characteristics/finite-difference solution of the full 3-D advection-diffusion equation performed well in hypothetical and lightweight-sediment laboratory tests. *Regime: Complex three-dimensional coastal flows; tested with hypothetical cases and laboratory transport and dispersal of lightweight sediment.* direct finding, mixed [1036]
2. In large-flume tests starting from a 1:15 mobile-bed slope and lasting about 2.4 h per condition, all erosive wave cases caused shoreline retreat and a breakpoint bar, but not every nominally accretive case transported sediment shoreward; suspended flux was governed by the occurrence and number of wave-backwash interactions within a wave group. *Regime: Large flume; handmade 1:15 initial slope; each condition approximately 2.4 h in six runs; erosive and accretive wave series.* direct finding, experimental [313]
3. Within a quasi-3-D formulation using a logarithmic velocity profile and wave-modified diffusivity and near-bed concentration, adding waves substantially increased suspended load and its adjustment time and length scales while enlarging the model's validity region; results agreed with 2DV/3-D solutions and existing measurements. *Regime: Logarithmic velocity profile with wave effects represented through diffusivity and the near-bed boundary condition; applicability bounded by a wave-modified suspension parameter and forcing-change scales.* direct finding, mixed [690]
4. For exact fundamental seiche modes in parabolic elliptical and circular basins, settling lag generated net suspended-sediment transport from deeper toward shallower regions; the mode-specific erosion and deposition patterns were robust to the transport-law formulation and little affected by omission of hydrodynamic friction. *Regime: Idealized elliptical or circular basin with parabolic cross-section; coarse-sand-oriented erosion/deposition law; friction omitted from the hydrodynamic solution.* direct finding, analytical [1128]
5. For experimentally tested strong plunging breakers, a fine-grid 2-D RANS k-epsilon/VOF model with an advection-diffusion sediment equation reproduced the overturning jet and high near-surface concentration and agreed with measured surface elevation, velocity, turbulent kinetic energy, eddy viscosity, and suspended concentration. *Regime: Two-dimensional surf-zone plunging breakers; advection-diffusion concentration with reference-concentration bottom boundary; selected fine mesh resolution.* direct finding, mixed [1054]
6. A review of internal-solitary-wave sediment transport identifies bottom-boundary-layer currents and turbulence as widespread resuspension drivers, while emphasizing Reynolds mismatch, grid resolution, and quadratic-stress parameterization as primary limits on laboratory and field-scale prediction. *Regime: Continental-margin internal solitary waves and their bottom boundary layers.* review statement, review [164]
7. Near-bed field cospectra on a non-barred shoreface show wind-wave and swell suspended transport was predominantly onshore and strengthened with shoaling, while undertow transport was offshore and dominated where oscillatory components were weak; breaking introduced strong vertical and temporal variability. *Regime: Marine non-barred shoreface under wind waves, swell, group-bound long waves and undertow during the field campaign.* direct finding, field [1060]
8. Field data from three experiments on two beaches show onshore suspended transport under large shear stress in relatively deep, gently sloping water, but increasingly strong breaking and undertow drive offshore transport in shallower or steeper surf zones; a nondimensional flux index captures the transition. *Regime: Two-dimensional barred and gently sloping surf zones represented by three field experiments on two beaches.* direct finding, field [3]
9. Without mean flow, oscillating-grid turbulence resuspends consolidated sediment according to a nonlinear interaction among consolidation time, particle size, canopy density and flexibility: sparse flexible canopies enhance turbulence and resuspension relative to rigid ones, while dense flexible canopies suppress both. *Regime: Consolidated synthetic, lake and saltmarsh sediment beds beneath modeled submerged rigid, semi-rigid and flexible canopies without mean flow.* direct finding, experimental [289]
10. After Three Gorges Dam closure, the upper Changjiang changed from the dominant sediment source to a sink, lakes and the eroding middle-lower bed became sources, and estuarine shoals and landward marine transport gained importance, increasing future estuary-erosion sensitivity to further dams, storms and sea level. *Regime: Changjiang river-estuary depositional system before and after closure of the Three Gorges Dam in 2003.* direct finding, field [315]
11. For idealized Punta Umbria tidal conditions, adding sea waves increased modeled suspended-sediment concentration and transport despite reducing the wave-current resultant velocity, because wave orbital motion raised maximum bed shear; cohesive sediment remained vertically mixed while non-cohesive sediment concentrated near the bed. *Regime: Idealized tidal currents, seabeds and wave conditions derived from measurements at Punta Umbria, Huelva, Spain.* direct finding, numerical [387]

43 Wiki: Storm surge and coastal flooding

43.1 Storm surge and coastal flooding: overview

Topic *storm_surge*. Meteorologically forced coastal water levels and flooding.

Papers screened: 6. Claims: 1. Equations: 0.

Well established. Storm surge is the meteorologically forced departure of coastal water level from the astronomical tide. Wind stress, atmospheric pressure, storm track and translation interact with shelf and basin geometry; waves, tides, rivers and rainfall can substantially alter total flooding.

Governing physics. Wind transfers momentum to the water column, pressure gradients drive inverse-barometer response, rotation and bathymetry shape propagation, and nonlinear tide–surge interaction changes timing and amplitude. Waves add setup and overtopping; discharge encounters surge-driven backwater in estuaries and deltas.

Dimensionless parameters. Controls include storm size-to-shelf scale, translation-to-wave speed, Rossby and Froude numbers, drag coefficient, surge-to-tidal range, wave setup-to-surge ratio, discharge-to-tidal prism, coastal slope, return period and joint dependence measures.

Major equations. Depth-integrated or three-dimensional mass and momentum equations include wind and pressure forcing, Coriolis, bottom stress and wetting/drying. Spectral wave equations supply radiation stresses; hydrologic and hydraulic equations supply runoff, while extreme-value or event-set methods estimate probabilities.

Typical methods. Studies use tide gauges, wave buoys, reanalyses and post-event maps; synthetic or historical cyclone ensembles; SLOSH, ADCIRC, WW3, D-Flow and hydraulic models; nested inundation grids; peaks-over-threshold analysis; attribution and climate perturbation experiments.

Numerical models. Model chains range from statistical hurricane generation plus SLOSH to global routing bounded by dynamic tide-surge levels, regional surge–wave models, and NWM–D-Flow/HEC-RAS–ADCIRC/WW3 compound systems with wetting and drying.

Experimental datasets. Evidence includes synthetic New York hurricanes, 3,433 global river mouths, Storm Gloria, Hurricanes Isabel, Irene and Sandy in Delaware, 21 U.S. hurricanes from 2000–2013, and Liverpool Bay and wider European surge-wave applications.

Validated ranges. Reported values include up to 1 m surge and 8 m significant waves during Gloria, 0.79–0.91 Delaware peak-level skill, surge effects at 64% of global delta mouths, and average end-century U.S. hurricane inundation increases of 36% by volume and 25% by extent.

Recent advances. Recent work dynamically couples ocean, wave, river and floodplain models, uses large synthetic event sets, resolves deltas globally, perturbs historical cyclones under future climates and attributes flood response to interacting drivers and geography.

Disagreements. Surge-only models can omit wave and discharge contributions, while fully coupled systems add uncertain boundaries and parameters. Climate-modified storm characteristics do not map monotonically to inundation because track, timing and local geography interact nonlinearly.

Limitations. Cyclone sampling, wind and pressure bias, drag at extreme winds, bathymetric and topographic resolution, levees and drainage, wave setup and overtopping, river boundaries, dependence assumptions and nonstationarity limit transferable hazard estimates.

Open questions. Priorities include compound-driver dependence, changing cyclone tracks and structure, rapid coupled forecasts, urban drainage and defenses, uncertainty partitioning, equitable exposure and consistent global-to-local downscaling.

Seminal papers. Classical wind setup and inverse-barometer theory established surge forcing; numerical shallow-water models enabled operational prediction, and later tide–surge interaction, wave coupling and statistical event methods established modern hazard assessment.

Claims

1. Coastal storm-flood assessment should treat storm surge, astronomical tide, wind waves and their mean-circulation interactions as a coupled system whenever their combined water-level and overtopping effects are material. *Regime: Marine-storm flooding where surge, tide and wind-wave processes jointly affect coastal water level or impacts.* review statement, review [\[1525\]](#)

43.2 Compound flooding

Topic *storm_surge.compound*. Joint marine, fluvial, pluvial, and groundwater flooding.

Papers screened: 17. Claims: 16. Equations: 8. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Compound coastal flooding occurs when coastal water level, waves, river discharge, direct rainfall, runoff, groundwater or infrastructure pathways interact in space or time. Combined inundation is generally not recoverable by simply adding independently calculated peak maps because phasing, storage, backwater and drainage capacity are nonlinear.

Governing physics. Coastal surge and tide impose downstream head; river discharge propagates seaward; rainfall creates runoff and local ponding; saturated storage delays drainage; waves overtop barriers; seawalls, gates, dams, storm drains and sewers redirect or block flux. Geometry and driver timing determine coastal, hydrologic and transition zones.

Dimensionless parameters. Controls include surge-to-tide ratio, discharge-to-tidal prism, rainfall duration and intensity ratios, phase lag normalized by tidal period, Froude number, storage and drainage-capacity ratios, seawall freeboard, overtopping rate, joint exceedance probability and fractional driver contribution.

Major equations. Models solve depth-integrated mass and momentum with rainfall, infiltration, river, drain and coastal source terms. Statistical studies use peaks over threshold, marginal extreme distributions and copulas. Driver attribution compares all-driver and leave-one-driver-out simulations, while interaction residuals quantify departure from additive response.

Typical methods. Methods include tightly or singly coupled hydrologic–hydrodynamic models, flexible meshes, reduced-physics solvers, sewer/drain networks, wave overtopping and breach modules, idealized phasing tests, historical hurricane hindcasts, high-water-mark validation, driver-isolation scenarios, POT and copula analysis.

Numerical models. SFINCS provides efficient reduced physics; Delft3D-FM resolves flexible-mesh estuary and river interaction; SuWAT integrates surge, tide, waves, overtopping and pluvial/sewer processes; other coupled city models explicitly include drainage, dams, gates, seawalls and rainfall runoff.

Experimental datasets. Evidence includes five Virginia hurricanes; Jacksonville/Irma and Hernani wave cases; Napa River and idealized channels; 15 southeastern China regions with 1960–2015 records; Xiangshan infrastructure; Typhoon Jebi at Kansai Airport; a 2017 embayment event; and Harvey, Ike, Rita and calibration data in Sabine–Neches.

Validated ranges. Validation is case bounded. Sabine–Neches mean water-level calibration reached $R^2=0.91$ and 84% of Imelda high-water-mark depths were within ± 0.2 m. Jebi attribution was 90.8% overtopping, 7.5% seawall failure and smaller sewer shares. Napa scenarios included surge small relative to a 2 m tide.

Recent advances. Recent work partitions dynamic driver zones, represents urban sewers and structural failure, and quantifies flood-volume contributions. Historical multistorm studies now show temporally separated compound peaks, shifting attention from coincident maxima to evolving event sequences.

Disagreements. A dominant driver varies by event and location: Jebi airport flooding was overtopping dominated, prolonged Harvey rainfall created extensive delayed fluvial–pluvial compounding, and larger tidal amplitude can suppress upstream surge propagation. These are regime differences, not a universal ranking of drivers.

Limitations. Uncertainty arises from rainfall fields, runoff and infiltration, river boundaries, bathymetry/topography, roughness, drainage connectivity and blockage, breach assumptions, wave overtopping, groundwater omission, grid resolution, daily statistical sampling, copula tails and sparse validation of extreme combinations.

Open questions. Priorities are groundwater and soil saturation, dynamic drainage failure, defence breach probability, wave–surge–rain–river coupling, multivariate climate nonstationarity, probabilistic phasing, real-time data assimilation, common intercomparison cases and consequence-aware attribution.

Seminal papers. Within this corpus, the 2018 Napa River study demonstrates non-additive tide–surge–river phasing, while the 2021 SFINCS study establishes an efficient all-driver solver with explicit evidence that advection is required for broken-wave and shock flow.

Equations

- *Depth-integrated compound-flood mass balance* [796]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: Verification cases, Hurricane Irma Jacksonville flooding, and broken-wave flooding at Hernani. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

- *Depth-integrated compound-flood mass balance* [1299]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: Xiangshan topography, drainage, coastal dam and tide-gate configurations in the tested storm scenarios. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

- *Depth-integrated compound-flood mass balance* [667]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: Kansai Airport geometry and infrastructure during the reconstructed 2018 Typhoon Jebi event. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

- *Depth-integrated compound-flood mass balance* [938]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: Sabine–Neches domain for Harvey, Ike and Rita, with tide-gauge and Imelda high-water-mark calibration evidence. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

- *Depth-integrated compound-flood mass balance* [569]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: Five Virginia hurricanes spanning different rainfall and surge intensities. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

- *Depth-integrated compound-flood mass balance* [868]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: Fifteen regions between 106–120°E and 18–32°N with available 1960–2015 daily records. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

- *Depth-integrated compound-flood mass balance* [1393]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: The studied embayment-backed catchment, historical 12 January 2017 event and tested seawall/drain scenarios. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

- *Depth-integrated compound-flood mass balance* [589]

$$\frac{\partial h}{\partial t} + \nabla \cdot (hu) = R - I + q_r + q_d + q_c$$

Regime: Napa River geometry plus idealized small-river scenarios, including 0.2 m surge and 2 m tide cases. Variables: h water depth; u depth-averaged velocity; R rainfall; I infiltration; q_r river/runoff source; q_d drain or sewer exchange; q_c coastal boundary, overtopping or failure source

Claims

1. SFINCS reproduced compound fluvial, pluvial, tidal and wind-driven flooding for Hurricane Irma at Jacksonville with limited computational expense, while an advective momentum term was necessary for shock-like dam-break and broken-wave flows and enabled wave-driven flooding at Hernani. *Regime: Verification cases, Hurricane Irma Jacksonville flooding, and broken-wave flooding at Hernani.* direct finding, mixed [796]
2. For Xiangshan, coupled simulation of surge, heavy rainfall, urban drainage, a coastal dam and tide gates partitioned inundation into tidal, hydrologic and transition zones, showing that mitigation choice must follow the locally dominant driver and infrastructure interaction. *Regime: Xiangshan topography, drainage, coastal dam and tide-gate configurations in the tested storm scenarios.* direct finding, mixed [1299]
3. In Napa River field-scale and idealized Delft3D-FM simulations, coincident high tide, surge and river discharge maximized water levels; large tidal amplitude diminished upstream surge propagation, and the phase between peak discharge and high tide shifted the location and timing of maxima. *Regime: Napa River geometry plus idealized small-river scenarios, including 0.2 m surge and 2 m tide cases.* direct finding, mixed [589]
4. For Harvey, Ike and Rita in the Sabine–Neches Estuary, a single-domain Delft3D-FM model identified four coastal–fluvial–pluvial interaction mechanisms; extended rainfall produced the largest extent and duration and a lagged second flood peak, while calibration achieved mean water-level $R^2=0.91$ and 84% of Imelda high-water-mark depths within ± 0.2 m. *Regime: Sabine–Neches domain for Harvey, Ike and Rita, with tide-gauge and Imelda high-water-mark calibration evidence.* direct finding, mixed [938]
5. For 15 southeastern China coastal–estuarine regions using 1960–2015 daily discharge and sea-level records, peak-over-threshold and copula analyses showed that joint fluvial–storm-tide flooding can exceed the flooding implied by either driver in isolation and that dependence varies among sites. *Regime: Fifteen regions between 106–120°E and 18–32°N with available 1960–2015 daily records.* direct finding, mixed [868]
6. Across five hurricanes simulated for low-gradient coastal Virginia, flooding separated into rainfall-dominant, surge-dominant and transitional zones; coastal-zone extent correlated strongly with surge magnitude, while transitional-zone extent correlated very strongly with the product of surge-inundated area and total rainfall. *Regime: Five Virginia hurricanes spanning different rainfall and surge intensities.* direct finding, mixed [569]
7. For an embayment-backed urban catchment and the 12 January 2017 compound event, coupled simulations resolved separate and combined high-marine-level and precipitation flooding and showed that seawall and storm-drain configuration controls whether coastal protection blocks or facilitates drainage. *Regime: The studied embayment-backed catchment, historical 12 January 2017 event and tested seawall/drain scenarios.* direct finding, mixed [1393]
8. For Typhoon Jebi flooding at Kansai International Airport, the integrated SuWAT analysis attributed 90.8% of total flood volume to wave overtopping, 7.5% to seawall failure, and 1.7% and 0.26% to seawater and rainwater sewer reverse flows, respectively. *Regime: Kansai Airport geometry and infrastructure during the reconstructed 2018 Typhoon Jebi event.* direct finding, mixed [667]
9. Across 3,433 river mouths, surge exacerbated 1-in-10-year river flood levels at 64.0% of sites by a mean 11 cm; ignoring surge significantly underestimated depths for 9.3% of expected annually exposed population. *Regime: The effect of surge on riverine flood hazard and impact in deltas globally* direct finding, numerical [399]
10. A coupled NWM–D-Flow FM–ADCIRC/WW3 system reproduced Delaware storm peaks with skill 0.79–0.91 and negligible phase error; upstream levels depended strongly on streamflow, while winds and roughness controlled prediction quality. *Regime: A New 1D/2D Coupled Modeling Approach for a Riverine–Estuarine System Under Storm Events: Application to Delaware River Basin* direct finding, mixed [109]
11. European projections show anthropogenic climate change increases the joint probability of heavy precipitation and storm surge, making compound flooding more likely than stationary dependence implies. *Regime: Higher probability of compound flooding from precipitation and storm surge in Europe under anthropogenic climate change* direct finding, numerical [148]
12. A global analysis combines river-discharge and storm-surge extremes to identify where dependence and event timing create elevated compound-flood potential. *Regime: Measuring compound flood potential from river discharge and storm surge extremes at the global scale* direct finding, numerical [300]
13. Low-gradient coastal modeling defines transitions among surge-, river-, and compound-dominated flood zones and shows downstream sea level can propagate far inland. *Regime: Defining Flood Zone Transitions in Low-Gradient Coastal Regions* direct finding, numerical [156]

14. A dynamically linked Atlantic-ocean and four-level nested harbor-to-street model reproduces the November 2009 Cork coastal-fluvial flood and resolves how tide, surge, river inflow and their interaction control urban flood-wave routes, depths, velocities and human-safety risk. *Regime: November 2009 compound coastal-fluvial flooding of Cork City and Cork Harbour, Ireland.* direct finding, numerical [1051]
15. Rapid FLORES screening for Beira shows compound surge and rainfall losses exceed isolated-hazard impacts because high coastal water levels block drainage; drainage expansion dominates near-term benefits, while coastal protection becomes increasingly important under high-end climate change. *Regime: Concept-stage compound storm-surge and pluvial flood planning for Beira, Mozambique, using the documented schematization and data.* direct finding, numerical [141]
16. Compound inundation models for low-gradient coastal watersheds should progress from linked, one-way, and loose coupling toward tight or full coupling that resolves rainfall across the whole domain and two-way storm-surge interactions with surface runoff, out-of-bank flow, and streamflow. *Regime: Low-gradient coastal watersheds exposed to coincident or sequential intense rainfall, river runoff, tropical-cyclone surge, and coastal inundation.* review statement, review [1252]

43.3 Wave-surge-tide coupling

Topic storm_surge.coupling. Nonlinear interaction among waves, surge, and tide.

Papers screened: 9. Claims: 7. Equations: 3.

Well established. Storm tide and waves interact dynamically. Tide and surge change depth and currents that control wave growth, refraction, breaking and bottom interaction; waves return radiation stress, setup and modified surface/bottom stress to circulation. Flooding additionally depends on the transition from wave runup/overtopping to mean overflow.

Governing physics. Key feedbacks are tide–surge phase interaction, depth-limited wave transformation, current refraction and Doppler shifting, radiation-stress gradients, wave setup, wind-stress modification, wave-enhanced bottom friction, surge overflow and intermittent overtopping across changing freeboard.

Dimensionless parameters. Important controls include surge-to-tide ratio, relative depth, wave steepness, Froude number, current-to-group-speed ratio, radiation-stress normalization, relative freeboard R_c/H_s , Iribarren number, wave direction reduction, tide–surge phase and wave-setup fraction.

Major equations. Coupled circulation uses depth-integrated mass and momentum with radiation-stress divergence and wave-modified stresses. Spectral waves obey an action balance on time-varying depth/current fields. Total water level combines mean sea level, tide, surge and setup, while overtopping discharge depends nonlinearly on relative freeboard and incident wave statistics.

Typical methods. Practice uses unstructured circulation grids coupled one- or two-way to spectral wave models, shared wind/pressure forcing, gauge calibration, stress and setup sensitivity runs, tide-phase ensembles, random-wave transformation, overtopping formulae and idealized transition experiments.

Numerical models. ADCIRC-SWAN is validated around Taiwan. The 2024 framework fully couples surge, wave, tide and overtopping/runup with random wave transformation, directional reduction and spatial averaging. The review covers established circulation–spectral-wave coupling classes.

Experimental datasets. The extracted evidence combines multi-region review cases, historical Taiwan typhoons with water-level, pressure, wind, height and period gauges, and idealized typhoon/tide/freeboard numerical experiments resolving overtopping and overflow transition.

Validated ranges. Taiwan validation spans historical typhoons and four harbor sites; wave setup contributed 6–35% of surge and coupled water levels exceeded tide–surge results by 0.22–0.24 m on average. Overtopping-transition conclusions remain bounded to idealized bathymetry, tested tides, incidence and freeboards.

Recent advances. Validated regional coupling now quantifies setup contributions, while fully coupled inundation frameworks evolve offshore waves and freeboard through the entire overtopping-to-overflow transition. This replaces static superposition with continuous process exchange.

Disagreements. Coupling magnitude is not universal: wave setup varies strongly by site and coastal slope, tides can alter or sometimes mitigate surge, and decoupled overtopping can overestimate flooding even though omitting waves underestimates setup. Model direction and transition treatment explain the apparent tension.

Limitations. Dominant uncertainties include wind and pressure forcing, bathymetry, wave breaking, radiation stress, wave-modified bottom friction, current refraction, wetting/drying, overtopping parameterization, structure geometry, phase sampling and limited simultaneous field observations of all coupled variables.

Open questions. Needs include process-resolving validation of stress pathways, irregular-wave overtopping/overflow transitions, tide-phase uncertainty, steep and shallow-shelf contrasts, structure-specific field measurements, coupled ensemble forecasting, model intercomparison and probabilistic treatment of forcing and closure error.

Seminal papers. The 2009 review codified the two-way coupling pathways and their flood significance. It remains foundational because it distinguishes well-established depth/setup feedbacks from less-settled stress, friction and refraction details.

Equations

- *Wave-radiation-stress forcing of mean flow* [1526]

$$\frac{\partial(hu)}{\partial t} + \nabla \cdot (huu) + gh\nabla\eta = \tau_w/\rho - \tau_b/\rho - \nabla \cdot S/\rho$$

Regime: Shelf and coastal flooding where wind waves coexist with tide and storm surge and bathymetry/wind forcing are adequately resolved. Variables: h total water depth; u depth-averaged current; η mean water elevation; τ_w surface wind/wave-modified stress; τ_b bottom stress; S wave radiation-stress tensor; ρ water density

- Wave-radiation-stress forcing of mean flow [666]

$$\frac{\partial(hu)}{\partial t} + \nabla \cdot (huu) + gh\nabla\eta = \tau_w/\rho - \tau_b/\rho - \nabla \cdot S/\rho$$

Regime: Idealized coastal bathymetry, tested typhoon incidence/tides and defence freeboard cases, including a 3 m freeboard example.

Variables: h total water depth; u depth-averaged current; η mean water elevation; τ_w surface wind/wave-modified stress; τ_b bottom stress; S wave radiation-stress tensor; ρ water density

- Wave-radiation-stress forcing of mean flow [844]

$$\frac{\partial(hu)}{\partial t} + \nabla \cdot (huu) + gh\nabla\eta = \tau_w/\rho - \tau_b/\rho - \nabla \cdot S/\rho$$

Regime: Historical Taiwan typhoon events and Taipei Tamsui, Taichung, Kaohsiung and Hualien harbor gauges. Variables: h total water depth; u depth-averaged current; η mean water elevation; τ_w surface wind/wave-modified stress; τ_b bottom stress; S wave radiation-stress tensor; ρ water density

Claims

1. Coupled coastal-flood modelling must pass tide- and surge-modified depth and currents to the wave model and return wave radiation stress, surface stress and wave-enhanced bottom friction to circulation; wave setup then contributes to total water level, although several stress and refraction interaction details remain uncertain. *Regime: Shelf and coastal flooding where wind waves coexist with tide and storm surge and bathymetry/wind forcing are adequately resolved.* review statement, review [1526]
2. Around Taiwan during historical typhoons, the calibrated ADCIRC-SWAN tide-surge-wave model reproduced observed water levels, pressure, wind and waves; including waves raised mean modelled water levels by 0.22–0.24 m at four harbors, with wave setup contributing 6–35% of total storm surge. *Regime: Historical Taiwan typhoon events and Taipei Tamsui, Taichung, Kaohsiung and Hualien harbor gauges.* direct finding, numerical [844]
3. In idealized coastal experiments, fully coupled surge-wave-tide-overtopping simulations showed that conventional decoupled overtopping/runup calculations overestimated inundation; prolonged overtopping still produced considerable flooding for typhoon landfall at low tide, demonstrating the need to resolve the transition between overtopping and overflow. *Regime: Idealized coastal bathymetry, tested typhoon incidence/tides and defence freeboard cases, including a 3 m freeboard example.* direct finding, numerical [666]
4. Storm Gloria produced eastern-Iberian surge up to 1 m with waves up to 8 m, whereas Balearic impacts were wave-dominated; 30 m Ebro Delta simulation resolved surge flooding and Mallorca calculations represented wave overtopping. *Regime: Coastal impacts of Storm Gloria (January 2020) over the north-western Mediterranean* direct finding, mixed [52]
5. A review of marine-storm flooding identifies coupled wind, tide, wave and mean-circulation mechanisms and uses Liverpool Bay and European applications to show why wave-surge interaction matters for coastal impacts. *Regime: Coupled wave and surge modelling and implications for coastal flooding* review statement, review [1525]
6. Reconstruction of the 5 December 2013 southern North Sea event synthesizes exceptional water levels, waves and coastal impacts and shows their spatially varying combined effects. *Regime: Southern North Sea storm surge event of 5 December 2013: Water levels, waves and coastal impacts* review statement, review [1334]
7. An unstructured finite-element model coupling river flow, tides, storm surge, wind waves, and hurricane forcing reproduces Hurricane Katrina hydrographs and high-water marks and supports interpretation and design for extreme Louisiana-Mississippi surge. *Regime: Hurricane Katrina across coastal Louisiana and Mississippi, including riverine flow, tides, waves, and storm surge.* direct finding, mixed [208]

43.3.1 Wave setup contribution to storm surge

Topic *storm_surge.coupling.wave_setup*. Breaking-wave radiation-stress contributions to storm water levels, including interactions with tides, currents, and coastal morphology.

Papers screened: 5. Claims: 15. Equations: 1. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Breaking-wave energy loss transfers onshore momentum through radiation-stress or vortex-force gradients; across the reviewed events this contribution raised mean water levels and altered hurricane or extratropical-storm flooding.

Governing physics. Setup is controlled by where wave energy dissipates relative to the shoreline, ebb delta, inlet and shelf. The resulting wave force is primarily balanced by a barotropic pressure gradient; tide changes water depth and breaking location, while currents can provide a smaller secondary feedback in the Klaus inlet tests.

Dimensionless parameters. Wave-setup importance is commonly expressed through relative contributions such as setup divided by peak surge or total water level, flooded-area increment divided by total flooded area, grid spacing relative to surf-zone width, and wave-force magnitude relative to pressure-gradient or stress terms. These ratios are diagnostic and event-specific rather than universal similarity laws.

Major equations. The studies introduce wave effects through spectral-wave momentum forcing of the circulation equations. Process experiments diagnose setup by differencing otherwise comparable simulations with and without wave forces; this difference can contain nonlinear tide-level feedback unless stationary level tests isolate it.

Typical methods. Credible event attribution combines a spectral wave model with a wetting-and-drying circulation model, compares wave-on and wave-off configurations, validates component waves and water levels, and uses phase-averaged or phase-resolving profile models when setup must be separated from swash.

Numerical models. Reviewed configurations couple WAM-STWAVE-ADCIRC for Katrina/Rita, MIKE 21 SW-HD FM plus XBeach 1D for Iota, and WWMII-SCHISM with a 3D vortex-force formulation for Klaus.

Experimental datasets. Available validation evidence includes 399 Katrina regional high-water marks, Rita hydrographs/high-water marks and wave buoys, Iota wave and water-level sensors along a 7.196 km island profile, and Klaus tide-gauge and wave observations at two Bay of Biscay inlets.

Validated ranges. Event-specific magnitudes are substantial but use different response measures: the Klaus study attributes 40% and 23% of peak surge to setup at Adour and Arcachon; the Iota study attributes 11.45 ha, or 14.93% of total modeled Providencia flooded area, to adding setup; and the Katrina study identifies a local unresolved setup contribution as large as 0.5 m near 17th Street Canal.

Recent advances. The progression from sequential radiation-stress coupling and regional HWM validation to fully coupled 3D inlet momentum diagnostics and explicit setup/runup flood-component scenarios enables more defensible attribution, but also reveals that operational-resolution and ensemble-cost gaps remain.

Disagreements. The reviewed percentages do not constitute a physical disagreement: Klaus reports setup as a fraction of peak surge, Iota reports its contribution to inundated area, and the Iota profile reports setup-plus-swash runup as a fraction of peak water level. Site morphology and response definitions must be aligned before comparing magnitudes.

Limitations. Probabilistic hazard products remain incomplete when waves are omitted: the Iota reconstruction included setup, but its 738-event synthetic ensemble excluded setup for computational reasons and therefore cannot quantify total wave-inclusive flood hazard.

Open questions. Priorities are computationally affordable wave-inclusive ensembles, event observations that separate mean setup from swash and infragravity motions, morphology-aware grid convergence, and transfer tests across open beaches, reefed islands, lagoons and estuaries.

Seminal papers. Radiation-stress theory established that gradients in wave momentum flux drive mean-water-level setup; later spectral-wave and circulation coupling embedded that forcing in storm-surge models. The reviewed Katrina/Rita application represents an early high-resolution regional validation, followed by process-resolving inlet attribution and component-based island flood reconstruction.

Equations

- *Model-difference setup diagnostic* [775]

$$\eta_{\text{setup}} = \eta_{\text{waves} + \text{tide} + \text{atmosphere}} - \eta_{\text{tide} + \text{atmosphere}}$$

Regime: Storm Klaus process-suppression experiments; nonlinear tide-level feedback requires separate stationary tests. Variables: η_{setup} modeled wave-induced water-level contribution; $\eta_{\text{waves} + \text{tide} + \text{atmosphere}}$ fully coupled water level; $\eta_{\text{tide} + \text{atmosphere}}$ water level without wave forces

Claims

1. In the SL15 system, STWAVE radiation stresses from depth-limited wave breaking are applied to ADCIRC and can modify surge magnitude, peak timing, and drawdown. *Regime: Sequential WAM-STWAVE-ADCIRC Katrina/Rita configuration for Louisiana and Mississippi.* direct finding, numerical [209]
2. For Katrina, 70% of 206 USACE and 73% of 193 URS/FEMA still-water high-water marks were reproduced within 0.5 m by the coupled model. *Regime: Hurricane Katrina regional hindcast and the two reported HWM datasets.* direct finding, mixed [209]
3. At the 17th Street Canal the regional model underpredicted peak surge by about 0.6 m, while cited local Boussinesq modeling indicated as much as 0.5 m of wave-driven setup absent at the regional resolution. *Regime: South shore of Lake Pontchartrain during Katrina; this is a local diagnosis, not a basin-wide setup value.* direct finding, mixed [209]
4. The authors attribute localized setup underprediction along the south shore of Lake Pontchartrain and near raised features partly to inadequate wave and circulation grid resolution. *Regime: Locations with unresolved wave gradients, levees, roads, or hydraulic connections in SL15.* direct finding, numerical [209]
5. For Hurricane Iota on Providencia, adding modeled wave setup to storm tide increased flooded area from 65.23 ha to 76.68 ha; the authors report 11.45 ha, or 14.93% of total modeled flooded area, as the setup contribution. *Regime: Providencia/Santa Catalina during Iota, coupled HD FM-SW setup without swash.* direct finding, numerical [1184]
6. At the eastern Providencia profile during Iota peak, the modeled contributions to maximum water level were -3.01% astronomical tide, 46.36% storm surge, and 56.55% wave runup including setup and swash. *Regime: One eastern-island profile and the modeled storm peak; runup is broader than wave setup alone.* direct finding, numerical [1184]
7. The coupled MIKE 21 configuration adds radiation-stress setup but omits swash, whereas the XBeach 1D configuration resolves setup-plus-swash runup along a selected profile. *Regime: The paper's Iota configuration; these outputs must not be interpreted as the same response variable.* direct finding, numerical [1184]
8. Wave setup was excluded from the 738 synthetic and hypothetical-event surge simulations because of computational cost, so their probabilistic maps underrepresent wave contributions by construction. *Regime: The synthetic and hypothetical hazard ensembles, not the dedicated Iota reconstruction.* direct finding, numerical [1184]
9. During storm Klaus, wave setup accounted for 40% of peak surge in the Adour Estuary and 23% in Arcachon Lagoon, and including wave forces improved surge predictions by 50-60%. *Regime: Storm Klaus hindcast at the two Bay of Biscay inlets.* direct finding, numerical [775]
10. At the Arcachon inlet ebb delta, modeled wave force was one order of magnitude larger than bottom and surface stress and was balanced mainly by the barotropic pressure gradient. *Regime: Arcachon outer inlet during storm-peak forcing.* direct finding, numerical [775]
11. Adour setup in stationary Klaus-wave tests decreased from 0.60 m at a -1.5 m imposed water level to 0.45 m at +1.5 m, whereas Arcachon modulation was only about 0.07 m. *Regime: The two inlet bathymetries and imposed -1.5 to +1.5 m level range; the contrast is site-specific.* direct finding, numerical [775]
12. On the exposed beach near Arcachon, peak surge with the 35 m grid was 30% and 65% higher than with 200 m and 1000 m grids because coarse grids poorly represented setup. *Regime: Storm Klaus adjacent beach with roughly 1 km-wide surf zone.* direct finding, numerical [775]
13. The study proposes a rough requirement of at least five computational elements across the surf zone to represent setup in surge models. *Regime: Case-derived guideline whose required spacing varies with surf-zone width, slope, and wave height.* inferred, numerical [775]
14. For Typhoon Morakot in the Taiwan Strait, coupled ADCIRC-SWAN simulations show wave-current interaction improves agreement with observed water levels and significant wave height; breaking-wave radiation-stress gradients create nearshore setup while water level and current modulate waves. *Regime: Typhoon Morakot (2009) across the Taiwan Strait and shallow nearshore/estuarine waters of coastal Fujian.* direct finding, mixed [1571]

15. A coupled surge-wave hindcast for Typhoon 7010 in Kochi shows radiation-stress wave setup materially raises coastal water levels and inundation, while an offshore breakwater reduces wave setup and the flooded area. *Regime: Typhoon 7010 storm surge, waves, setup, and inundation in Kochi under historical and offshore-breakwater configurations.* direct finding, numerical [964]

43.4 Surge forcing

Topic storm_surge.forcing. Wind, pressure, wave, and rainfall forcing.

Papers screened: 28. Claims: 28. Equations: 0. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Storm surge is the dynamical coastal response to wind stress and atmospheric pressure acting on a preconditioned ocean, modified by storm size/track/speed, tides, waves, shelf and channel geometry, bathymetry, bottom friction, remote ocean signals, and coastal wetting/drying.

Governing physics. Alongshore/onshore wind piles water and drives Ekman/coastal currents; pressure gradients create inverse-barometer and dynamic response; broad storms increase fetch/duration; shelf waves and remote forcing create forerunners; tides change depth/friction; waves add setup and stress feedback; geometry controls convergence and resonance.

Dimensionless parameters. Controls include storm radius/shelf width, translation speed/gravity or shelf-wave speed, wind direction/coast orientation, pressure deficit, drag coefficient, bottom-friction parameter, depth/resolution, Rossby and Froude numbers, tidal phase/surge duration, wave setup/surge, channel convergence, and forcing/forecast error.

Major equations. Models solve shallow-water or primitive continuity/momentum with Coriolis, pressure-gradient, wind stress, atmospheric pressure, bottom drag and advection; coupled systems add wave radiation stress and sea-state drag; wetting/drying maps inundation; assimilation updates state; sensitivity and surrogate models map forcing to response.

Typical methods. Methods include gauge/buoy/wind/pressure validation, historical hindcasts, idealized storm perturbations, global and nested unstructured grids, coupled SWAN-ADCIRC or atmosphere-wave-ocean models, 2-D/3-D comparison, forcing-product and drag/friction sensitivity, data assimilation, ensemble forecasts, process attribution, and AI emulation.

Numerical models. Models include ADCIRC v55 and data assimilation, SWAN-ADCIRC, COAWST-style coupling, global storm-tide reanalysis, pan-European forecasting, high-resolution cyclone forecasts, coupled atmosphere-wave-ocean models, 2-D/3-D surge models, subgrid inundation, deep-learning wind/pressure emulators and process indices.

Experimental datasets. Evidence includes Gustav, Ike, Rita, Sandy-era systems, Bejisa, Gloria, Haiyan-region islands, Irma, Pearl River typhoons, Bangladesh cyclones, Super Typhoon Fung-Wong, Baltic and Italian storms, Gulf testbeds, Caribbean islands, Wilmington channels and global reanalyses.

Validated ranges. The corpus spans local channels and islands to pan-European/global models, extratropical and tropical cyclones, deep ocean to reef/estuary/urban coast, hindcast through operational forecast, local and remote forerunners, multiple mesh/friction/forcing products, and historical to recent events.

Recent advances. Recent work uses high-resolution global forecasts, scalable unstructured meshes, component attribution, low-frequency state correction, deep-learning wind/pressure emulation, geometry/bathymetry-aware surge indices, island-specific wind sensitivity, in-situ super-typhoon constraints and shelfbreak-jet attribution.

Disagreements. Peak intensity alone poorly predicts surge when storm size and track differ; finer grids improve geometry but may expose friction/bathymetry error; 2-D and 3-D models differ where stratification or vertical stress matters; coupled waves may amplify or redistribute surge; data assimilation can correct state but mask forcing bias.

Limitations. Cyclone wind/pressure cores are sparsely observed; reanalysis resolution is limited; drag at extreme winds is uncertain; bathymetry, topography and friction covary; gauges miss spatial maxima; wetting/drying and wave coupling add numerical sensitivity; remote ocean and low-frequency baselines are imperfect.

Open questions. Needs include extreme-wind drag saturation, pressure-wind covariance, storm-size/translation priors, remote shelf and ocean-current contributions, wave-tide-surge coupling, morphology and sea-level change, probabilistic forcing ensembles, data assimilation attribution, and physically constrained AI.

Seminal papers. The branch progresses from storm-size and event hindcasts to coupled wave-surge testbeds, mesh/friction/forcing sensitivities, global storm-tide systems, remote forerunners, channel-deepening effects, state assimilation, coupled atmosphere feedback, geometry-aware indices and AI forcing emulators.

Claims

1. A global hydrodynamic reanalysis driven by meteorology and tides reproduces broad storm-surge and extreme-sea-level patterns while exposing regional forcing and resolution biases. *Regime: A global reanalysis of storm surges and extreme sea levels* direct finding, numerical [994]
2. Hurricane surge increases strongly with storm size because broader winds act over larger fetch and duration, so intensity alone is insufficient. *Regime: The Influence of Storm Size on Hurricane Surge* direct finding, numerical [636]
3. Coastal sea-level change reflects atmospheric pressure and wind together with tides, waves, steric/ocean circulation, river flow, vertical land motion, bathymetry, and coastline geometry. *Regime: Forcing Factors Affecting Sea Level Changes at the Coast* review statement, review [1529]
4. A validated Hurricane Gustav hindcast shows waves, surge, storm evolution, and Louisiana shelf/land geometry must be represented consistently. *Regime: Hurricane Gustav (2008) Waves and Storm Surge: Hindcast, Synoptic Analysis, and Validation in Southern Louisiana* direct finding, mixed [341]

5. Hurricane Ike validation resolves offshore waves, pre-landfall forerunner, and peak surge as distinct responses to evolving remote and local forcing. *Regime: Hindcast and validation of Hurricane Ike (2008) waves, forerunner, and storm surge* direct finding, mixed [610]
6. Channel deepening altered both tides and storm surge at Wilmington by changing friction, convergence, and propagation. *Regime: The effect of channel deepening on tides and storm surge: A case study of Wilmington, NC* direct finding, mixed [423]
7. High-resolution forecast winds used in a global surge model improve tropical-cyclone forcing representation while retaining global coverage. *Regime: Global modeling of tropical cyclone storm surges using high-resolution forecasts* direct finding, numerical [161]
8. The 1966 Italian storm demonstrates how cyclone track, wind, pressure, waves, tide, and basin geometry combine into destructive regional water levels. *Regime: The disastrous storm of 4 November 1966 on Italy* review statement, review [1625]
9. Inter-model Gulf of Mexico tests show tide, wave, surge and hurricane-response differences arise from forcing, coupling, grid, friction, and model formulation. *Regime: U.S. IOOS coastal and ocean modeling testbed: Inter-model evaluation of tides, waves, and hurricane surge in the Gulf of Mexico* direct finding, mixed [704]
10. Global ADCIRC storm-tide performance depends on unstructured-mesh design, bathymetric/coastal resolution, and scalable forcing implementation. *Regime: Global storm tide modeling with ADCIRC v55: unstructured mesh design and performance* direct finding, numerical [1127]
11. Gulf hurricane response is sensitive to mesh resolution and bottom friction across tide, wave and surge components. *Regime: U.S. IOOS coastal and ocean modeling testbed: Evaluation of tide, wave, and hurricane surge response sensitivities to mesh resolution and friction in the Gulf of Mexico* direct finding, mixed [703]
12. Robust pan-European surge forecasting requires forcing consistency, ensemble/error treatment, and regional calibration across diverse shelf and coastal geometries. *Regime: Towards robust pan-European storm surge forecasting* direct finding, numerical [434]
13. A coupled ocean-wave-atmosphere model for Tropical Cyclone Bejisa represents feedbacks among winds, waves, currents and surface stress that uncoupled forcing omits. *Regime: A New Coupled Ocean-Waves-Atmosphere Model Designed for Tropical Storm Studies: Example of Tropical Cyclone Bejisa (2013–2014) in the South-West Indian Ocean* direct finding, numerical [1106]
14. Storm Gloria sea level in the western Mediterranean arose from multiple processes whose relative contributions varied in space and time. *Regime: Understanding Sea Level Processes During Western Mediterranean Storm Gloria* direct finding, mixed [1132]
15. Two- and three-dimensional Hurricane Ike simulations differ because vertical structure, baroclinicity and friction alter the surge response. *Regime: Implications from the comparisons between two- and three-dimensional model simulations of the Hurricane Ike storm surge* direct finding, numerical [1611]
16. Hurricane Rita hindcast sensitivity shows bottom friction, wind-drag coefficient, and meteorological forcing jointly control SWAN-ADCIRC surge skill. *Regime: Effect of Bottom Friction, Wind Drag Coefficient, and Meteorological Forcing in Hindcast of Hurricane Rita Storm Surge Using SWAN + ADCIRC Model* direct finding, numerical [22]
17. Deep-ocean reef-fringed Caribbean islands experience coupled hurricane winds, waves and surge whose nearshore response depends on reef and island geometry. *Regime: U.S. IOOS Coastal and Ocean Modeling Testbed: Hurricane-Induced Winds, Waves, and Surge for Deep Ocean, Reef-Fringed Islands in the Caribbean* direct finding, mixed [673]
18. Assimilating low-frequency water-level residuals corrects storm-surge model baseline errors not explained by event meteorology alone. *Regime: Low frequency water level correction in storm surge models using data assimilation* direct finding, numerical [81]
19. ADCIRC data assimilation improves hurricane-surge forecasts by updating coastal water levels within the dynamical model rather than applying only post hoc bias correction. *Regime: Data Assimilation within the Advanced Circulation (ADCIRC) Modeling Framework for Hurricane Storm Surge Forecasting* direct finding, numerical [220]
20. Deep-learning emulation of typhoon wind and pressure fields enables efficient surge forcing while retaining spatial-temporal storm structure. *Regime: A novel deep learning approach for typhoon-induced storm surge modeling through efficient emulation of wind and pressure fields* direct finding, numerical [998]
21. A tropical-cyclone surge index improves on storm intensity alone by incorporating coastal geometry, bathymetry, and storm attributes. *Regime: A new tropical cyclone surge index incorporating the effects of coastal geometry, bathymetry and storm information* direct finding, numerical [638]
22. Pearl River Estuary surge and inundation vary strongly with typhoon path because track changes wind direction, pressure history, and estuary setup. *Regime: Effects of Typhoon Paths on Storm Surge and Coastal Inundation in the Pearl River Estuary, China* direct finding, numerical [379]
23. Bangladesh surge simulation skill depends on how atmospheric-model wind and pressure outputs are transferred into the ocean model. *Regime: Forcing ocean model with atmospheric model outputs to simulate storm surge in the Bangladesh coast* direct finding, numerical [908]
24. Caribbean marine-hazard assessment separates and recombines surge, waves, tides and other processes to identify location-specific dominant drivers. *Regime: Quantifying processes contributing to marine hazards to inform coastal climate resilience assessments, demonstrated for the Caribbean Sea* direct finding, mixed [654]
25. Recent Baltic surge events must be interpreted against climate variability and regional atmospheric/oceanic forcing rather than event height alone. *Regime: Recent Baltic Sea storm surge events from a climate perspective* direct finding, mixed [539]
26. Island storm surge is highly sensitive to tropical-cyclone wind-field choice because complex coastlines rotate and shelter the applied stress. *Regime: Sensitivity of simulated storm surge to tropical cyclone wind forcing around islands: insights from Hurricane Irma* direct finding, numerical [1627]
27. In-situ observations of Super Typhoon Fung-Wong constrain storm forcing and local surge impacts where model and gauge evidence are otherwise sparse. *Regime: In-Situ Observations and Storm Surge Impact Assessment of Super Typhoon Fung-Wong (2025) in Aurora Province, Philippines* direct finding, field [630]
28. Shelfbreak-jet and coastal-ocean dynamics can contribute materially to coastal flood levels beyond local meteorological surge forcing. *Regime: Do Ocean Dynamics Contribute to Coastal Floods? A Case Study of the Shelfbreak Jet and Coastal Sea Level Along Southern New England (U.S.)* direct finding, mixed [226]

43.4.1 Atmospheric pressure forcing

Topic *storm_surge.forcing.pressure*. Inverse barometer and pressure gradients.

Papers screened: 11. Claims: 11. Equations: 0. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Atmospheric pressure forces coastal water levels through quasi-static inverse-barometer adjustment and dynamic gradients; moving disturbances can resonantly generate long waves that amplify on shelves and in bays and harbours.

Governing physics. A pressure deficit raises the equilibrium sea surface, while horizontal pressure gradients accelerate water: When disturbance translation approaches shallow-water wave speed, Proudman resonance accumulates energy; shelf, bay and harbour modes then amplify or reshape the wave.

Dimensionless parameters. Key controls are pressure anomaly, disturbance length and translation speed, water depth, Froude number based on atmospheric and shallow-water wave speeds, shelf width, incidence angle, and shelf, bay or harbour modal periods.

Major equations. Pressure enters shallow-water momentum as a surface-pressure gradient. The inverse-barometer approximation relates sea-level rise inversely to pressure anomaly, while moving-forcing solutions and nonlinear shallow-water models represent resonant generation, propagation and inundation.

Typical methods. Studies combine high-frequency barometers, tide gauges and current measurements with spectral or wavelet analysis, analytical moving-pressure solutions, pressure-only and wind-only sensitivity experiments, and nested shallow-water or regional ocean models.

Numerical models. Numerical approaches range from idealized analytical and nonlinear shallow-water models to NAMI DANCE SUITE, regional three-dimensional ocean models and coupled atmosphere-wave-surge systems driven by parametric or full-physics pressure fields.

Experimental datasets. Evidence includes the east Adriatic 2003 event, Hurricane Harvey rainbands, Venice 2019 flooding, Typhoon Songda and Pacific Northwest cyclones, Persian Gulf gauges, the Gulf of Genoa record and regional Strait of Georgia and Brazilian-bay simulations.

Validated ranges. Reported responses span pressure perturbations of roughly 1–3 mbar producing about 0.2 m waves, a 0.5 m Genoa event, Persian Gulf residuals up to 0.75 m, Pacific Northwest surges above 0.8 m and resonantly amplified Adriatic oscillations above 1 m.

Recent advances. Recent studies use high-resolution colocated pressure–sea-level observations and forcing-partition experiments to identify tropical-cyclone rainband meteotsunamis, mesoscale-cyclone contributions to extreme floods and operational model sensitivity to pressure-field structure.

Disagreements. Wind often dominates synoptic storm surge, but pressure is not always negligible: sharp rainband or mesoscale pressure disturbances can generate resonant waves comparable to other coastal water-level components. Attribution therefore depends on time scale and geometry.

Limitations. Pressure and wind covary in storms, barometer and tide-gauge spacing may miss propagating fronts, atmospheric products smooth mesoscale jumps, and resonance makes results highly site- and direction-dependent.

Open questions. Operational thresholds for pressure-jump speed, angle and scale remain uncertain, as do pressure–wind interaction, compound tide/surge timing, nonlinear breaking and inundation, and transfer of resonance diagnostics among coastlines.

Seminal papers. Foundational work established inverse-barometer forcing and traveling-pressure resonance; Adriatic observations and the global meteotsunami synthesis connected Proudman resonance with shelf, bay and harbour amplification.

Claims

1. Atmospheric gravity waves, pressure jumps, fronts and squalls generate barotropic long waves that amplify through Proudman, shelf, bay and harbour resonance; generation peaks near Froude number one. *Regime: Meteotsunamis: atmospherically induced destructive ocean waves in the tsunami frequency band* review statement, review [974]
2. A 22 m s⁻¹ atmospheric gravity disturbance resonantly coupled to water about 50 m deep, producing sea-level oscillations above 1 m that were further amplified in funnel-shaped Adriatic bays. *Regime: Resonant coupling of a traveling air pressure disturbance with the east Adriatic coastal waters* direct finding, mixed [1455]
3. A validated regional Strait of Georgia model assesses the atmospheric forcing factors that contribute to winter storm surges in a semi-enclosed coastal sea. *Regime: Storm Surges in the Strait of Georgia Simulated with a Regional Model* direct finding, numerical [1323]
4. Hurricane Isaac surge predictions differ when parametric versus full-physics atmospheric wind and pressure fields force the same coupled wave–surge system. *Regime: Sensitivity of Storm Surge Predictions to Atmospheric Forcing during Hurricane Isaac* direct finding, numerical [342]
5. A nonlinear shallow-water model with atmospheric pressure and wind forcing reproduces analytical long-wave generation and coastal amplification by propagating pressure disturbances. *Regime: Long wave generation and coastal amplification due to propagating atmospheric pressure disturbances* direct finding, numerical [371]
6. Observed 1–3 mbar rainband pressure perturbations during Hurricane Harvey generated approximately 20-minute meteotsunamis with amplitudes around 0.2 m, showing pressure need not be negligible relative to wind. *Regime: Meteotsunamis Accompanying Tropical Cyclone Rainbands During Hurricane Harvey* direct finding, mixed [54]
7. Persian Gulf non-tidal sea-level fluctuations up to 0.75 m reflect combined low pressure and cross-shore wind; pressure leads sea level over 1–6.4-day bands, while wind stress has the larger effect. *Regime: Relationship between the Persian Gulf Sea-Level Fluctuations and Meteorological Forcing* direct finding, field [19]
8. A fast mesoscale cyclone produced atmosphere–ocean resonance and a meteotsunami-like wave that contributed significantly to Venice's 1.89 m extreme water level; simulations partitioned wind and pressure contributions. *Regime: The contribution of a mesoscale cyclone and associated meteotsunami to the exceptional flood in Venice on November 12, 2019* direct finding, mixed [437]
9. Successive low-pressure systems including Typhoon Songda generated storm surges above 0.8 m and associated seiches and infragravity waves along British Columbia and Washington. *Regime: Strength in Numbers: The Tail End of Typhoon Songda Combines with Local Cyclones to Generate Extreme Sea Level Oscillations on the British Columbia and Washington Coasts during Mid-October 2016* direct finding, field [1141]
10. Nested coastal modelling identifies remotely forced meteorological signal propagation and lag as controls on water-level differences and strong currents between linked Brazilian bays. *Regime: Meteorological Signal on Hydrodynamics in the Ilha Grande and Sepetiba Bays: Lag Effects and Coastal Currents* direct finding, numerical [1096]
11. An abrupt atmospheric-pressure jump generated a Gulf of Genoa meteotsunami with maximum wave height of 0.5 m lasting about three hours. *Regime: Detection and Characterization of Meteotsunamis in the Gulf of Genoa* direct finding, field [1107]

43.4.2 Wind stress and drag

Topic storm_surge.forcing.wind. Wind momentum transfer and drag coefficients.

Papers screened: 22. Claims: 28. Equations: 1. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Wind stress is the principal direct momentum forcing for storm surge, but the drag coefficient does not increase without bound at tropical-cyclone winds; observations, experiments and coupled models instead support leveling, decline or strong sea-state modulation.

Governing physics. Atmospheric shear transfers momentum through turbulent and wave-supported stress. Roughness, wave age and direction, breaking and whitecaps, foam and spray, currents, stability, fetch, shoaling and land cover modify the magnitude and direction delivered to the water column.

Dimensionless parameters. Controls include C_D , Charnock parameter, z_0 , u_{star}/U_{10} , wave age c_p/u_{star} , inverse wave age, wind-wave and wind-stress misalignment angles, relative water depth, fetch, stability, drag saturation wind speed, and nondimensional setup or surge sensitivity.

Major equations. The common bulk forcing relation is $\tau = \rho_a C_D U_{10}^2$. Peng and Li parameterize high-wind drag as $C_D = -a(V_p - 33)^2 + c$, explicitly imposing a peak at 33 m/s.

Typical methods. Methods estimate flux from towers, buoys, floats or eddy correlation; infer roughness and whitecaps remotely; reproduce interface behavior in laboratories; compute wave-supported stress spectrally; calibrate drag against setup or surge; and compare closures in coupled forecast ensembles.

Numerical models. Reviewed tools include coupled wave-boundary-layer and ocean-wave models, ADCIRC, COAWST/ROMS/SWAN, unstructured tide-surge-wave models, global ocean surge systems, WRF hurricane ensembles and data-assimilative drag calibration.

Experimental datasets. Evidence includes coastal-tower typhoon winds to about 42 m/s, surf-zone eddy-correlation profiles, currents beneath five tropical cyclones, microwave radiometry, ten-hurricane coupled hindcasts, Lake Erie setback and surge events, and regional gauge, buoy, platform, satellite and altimeter comparisons.

Validated ranges. The evidence spans moderate winds through roughly 55 m/s in combined float/model estimates and about 42 m/s in coastal towers, with some remote or hurricane-model applications higher; transfer remains conditional on measurement method, sea state, depth, storm quadrant and coast.

Recent advances. Recent work combines autonomous current profilers, microwave roughness and whitecap retrieval, vector stress estimates, coupled ocean-wave models, wave-dependent surge forcing, large parameter ensembles and regional calibration to replace a universal scalar drag curve.

Disagreements. The high-wind asymptote is method dependent. Powell et al. infer a marked decline in open-ocean C_D through $U_{10} = 51$ m/s; Donelan et al. measure saturation at $C_D = 0.0025$ above 33 m/s in a short-fetch flume; Peng and Li fit a regional parabolic decline whose magnitude is slightly higher than Powell and Jarosz at medium-to-high winds.

Limitations. No reviewed law is universal: Powell explicitly excludes shallow coastal shoaling conditions, Donelan uses short-fetch laboratory waves and extrapolated 10-m winds, and Peng-Li coefficients are regionally inferred and require recalibration elsewhere.

Open questions. Resolving plateau versus decline requires colocated extreme-wind stress, wave-state, foam/spray, current, and shallow-water observations, plus controlled propagation of drag uncertainty into surge forecasts.

Seminal papers. Bulk aerodynamic stress, logarithmic boundary-layer theory and Charnock roughness established the classical framework; later field hurricane profiles and high-wind laboratory studies overturned simple extrapolation and motivated capped, declining and sea-state-dependent drag laws.

Equations

- Quadratic air-sea stress [209]

$$\tau_s = \rho_a C_D U_{10}^2$$

Regime: ADCIRC hurricane wind forcing with the reported Garratt relation and drag cap. Variables: τ_s surface wind stress; ρ_a air density; C_D air-sea drag coefficient; U_{10} 10 m wind speed

Claims

1. In 331 tropical-cyclone eyewall profiles, friction velocity increased with U_{10} to about 40 m/s and then leveled off, while neutral drag showed a decreasing tendency above about 33 m/s. *Regime: Deep-water open-ocean eyewalls; near-surface winds 21-67 m/s.* direct finding, field [1124]
2. The inferred roughness length and drag coefficient decreased sharply as U_{10} increased to 51 m/s and were much lower than Charnock and Large-Pond extrapolations. *Regime: Resolved open-ocean wind groups through $U_{10} = 51$ m/s; not the under-sampled 70-85 m/s MBL group.* direct finding, field [1124]
3. The mean 10-m wind was 78% of mean boundary-layer wind, with an 8% standard deviation. *Regime: The paper's deep-water eyewall sample.* direct finding, field [1124]
4. Applying the reported open-ocean reduction factors near shore is not justified because shoaling-wave studies indicate higher coastal roughness. *Regime: Coastal shallow water and hurricane landfall.* direct finding, review [1124]
5. Laboratory drag increased between 3 and 33 m/s but saturated once neutral 10-m wind exceeded 33 m/s. *Regime: University of Miami wind-wave tank and its short-fetch waves.* direct finding, experimental [358]
6. The measured high-wind saturation level was $C_D = 0.0025$, corresponding to aerodynamic roughness length $z_0 = 3.35$ mm. *Regime: Neutral 10-m-equivalent wind above 33 m/s in the flume.* direct finding, experimental [358]
7. Continuous breaking can limit aerodynamic roughness because separated outer airflow skips from crest to crest and does not follow wave troughs. *Regime: Steep spilling breakers under extreme wind.* inferred, experimental [358]
8. For 18 South China Sea tropical cyclones, the study represented drag as $C_D = -a(V_p - 33)^2 + c$, with ten cases for calibration and eight for independent validation. *Regime: South China Sea cyclones during 2006-2011 with modeled winds 10-70 m/s.* direct finding, numerical [1090]
9. Across the eight validation cyclones, the optimized parabolic drag law produced maximum-surge-bias standard deviation 0.117 m and mean RMSE 0.109 m, the lowest aggregate pair in Table 2. *Regime: The paper's eight South China Sea validation cyclones and gauge/model configuration.* direct finding, numerical [1090]
10. The fitted South China Sea curve had the same peak-and-decline trend as Powell and Jarosz but slightly larger drag throughout the medium-to-high-wind band. *Regime: The reconstructed regional winds and assimilated surge cases.* direct finding, mixed [1090]
11. The fitted coefficients are not asserted to transfer unchanged outside the South China Sea and should be recalibrated with regional observations. *Regime: Application to other coastal regions.* direct finding, review [1090]

12. A coupled wave–wind parameterization derived from ten Atlantic tropical cyclones makes roughness increase while high-wind drag levels off, improving hurricane-model intensity behavior. *Regime: A Physics-Based Parameterization of Air–Sea Momentum Flux at High Wind Speeds and Its Impact on Hurricane Intensity Predictions* direct finding, numerical [979]
13. Coupled wave–boundary-layer experiments show that tropical-cyclone drag is controlled by the spatially varying wave field, with developed right-front seas producing higher drag than young rear-left seas. *Regime: Effect of Surface Waves on Air–Sea Momentum Exchange. Part II: Behavior of Drag Coefficient under Tropical Cyclones* direct finding, numerical [977]
14. Laboratory and numerical evidence attributes hurricane-force interface structure and stress behavior to Kelvin–Helmholtz-like shear instability and associated air–water interfacial change. *Regime: The air-sea interface and surface stress under tropical cyclones* direct finding, experimental [1319]
15. A storm-surge modeling comparison demonstrates that alternative wind-stress conversion and drag formulations materially change simulated surge and require regime-aware selection. *Regime: An Exploration of Wind Stress Calculation Techniques in Hurricane Storm Surge Modeling* direct finding, numerical [200]
16. Coastal-tower measurements to about 42 m/s show a drag peak near 24 m/s and a curve shifted from open-ocean behavior, consistent with shoaling-wave effects during typhoon landfall. *Regime: Typhoon air-sea drag coefficient in coastal regions* direct finding, field [1607]
17. Hurricane Ike SWAN experiments show that wind-input and maximum-drag choices control high-wind wave estimates and can correct a roughly 2 m deep-Gulf wave-height overestimate. *Regime: Gulf of Mexico hurricane wave simulations using SWAN: Bulk formula-based drag coefficient sensitivity for Hurricane Ike* direct finding, numerical [620]
18. A coupled wave–wind analysis of ten hurricanes finds high-wind drag leveling and strong dependence of the Charnock coefficient on wave age and storm-relative wave state. *Regime: Effect of surface waves on Charnock coefficient under tropical cyclones* direct finding, numerical [978]
19. North Atlantic storm evaluation against buoy, platform, and satellite observations compares five stress parameterizations and shows strong-wind forecasts depend on their high-wind behavior. *Regime: Strong winds in a coupled wave–atmosphere model during a North Atlantic storm event: evaluation against observations* direct finding, field [1111]
20. Hundreds of convection-permitting Katrina simulations quantify tropical-cyclone sensitivity and parameter uncertainty associated with momentum and enthalpy exchange coefficients. *Regime: Sensitivity of Tropical Cyclone Simulations to Parametric Uncertainties in Air–Sea Fluxes and Implications for Parameter Estimation* direct finding, numerical [529]
21. A coupled tide–surge–wave model for northeastern Taiwan applies typhoon pressure and wind stress on an unstructured grid to reproduce storm tides and wind waves. *Regime: Simulation of Typhoon-Induced Storm Tides and Wind Waves for the Northeastern Coast of Taiwan Using a Tide–Surge–Wave Coupled Model* direct finding, numerical [260]
22. ADCIRC experiments show that land-cover specification affects both wind-momentum transfer and bottom dissipation, producing uncertainty in Texas-bay hurricane surge. *Regime: Uncertainty in hurricane surge simulation due to land cover specification* direct finding, numerical [439]
23. Current profiles beneath five tropical cyclones yield drag of $0.8\text{--}3.1 \times 10^{-3}$ at 25–40 m/s and stress directions from 15° counterclockwise to 40° clockwise of the wind. *Regime: Scaling of Drag Coefficients Under Five Tropical Cyclones* direct finding, field [614]
24. Direct eddy-correlation measurements over an open-ocean surf zone find a constant-stress layer and systematic drag changes with wind approach angle and wave conditions. *Regime: Direct measurements of wind stress over the surf zone* direct finding, field [1286]
25. North Sea surge cases evaluated with gauges and altimetry show that wave-dependent stress can alter simulated surges relative to wind-only stress. *Regime: Impact of wave-dependent stress on storm surge simulations in the North Sea: Ocean model evaluation against in situ and satellite observations* direct finding, numerical [1112]
26. Microwave radiometer observations jointly constrain tropical-cyclone high-wind drag, roughness, and whitecap coverage rather than relying solely on extrapolated moderate-wind laws. *Regime: High-Wind Drag Coefficient and Whitecap Coverage Derived from Microwave Radiometer Observations in Tropical Cyclones* direct finding, field [626]
27. Float observations and a coupled ocean–wave model give mean drag near $2\text{--}3 \times 10^{-3}$ for 25–55 m/s winds and reveal stronger sea-state than wind-speed dependence, especially under wind–wave misalignment. *Regime: Drag Coefficient and Its Sea State Dependence under Tropical Cyclones* direct finding, field [1616]
28. Two Lake Erie wind setdown and surge events calibrate air–sea drag with COAWST/ROMS/SWAN, using the upwind shore to reduce contamination by wave setup. *Regime: Using Wind Setdown and Storm Surge on Lake Erie to Calibrate the Air–Sea Drag Coefficient* direct finding, field [374]

43.5 Tropical cyclone flooding

Topic storm_surge.hurricanes. Hurricane and tropical cyclone applications.

Papers screened: 17. Claims: 11. Equations: 3.

Well established. Tropical-cyclone coastal flooding depends on the evolving storm track, wind and pressure fields, shelf and coastal geometry, tides, waves, and terrain; peak surge alone does not describe early rise, duration, inundation, or wave-driven contributions.

Governing physics. Wind stress and pressure deficit force surge; storm radius, translation speed, and approach angle control forcing duration and orientation; Coriolis and shelf geometry contribute to forerunner setup; wave radiation stress, runup, and topography modify nearshore water levels and inundation.

Dimensionless parameters. Useful nondimensional controls include drag coefficient, pressure-scaled surge, wind-duration-scaled lead time, normalized hydrograph time and amplitude, relative surge to tide and wave components, and normalized validation errors; none removes dependence on shelf and coastal geometry.

Major equations. Core formulations include depth-integrated shallow-water momentum and continuity, quadratic wind stress, spectral wave-action and radiation-stress coupling, dimensionless pressure-duration forerunner scaling, level-minus-terrain inundation depth, PCA/CNN spatial reconstruction, and beta-mixture hydrograph templates.

Typical methods. Typical workflows hindcast or simulate cyclone ensembles with ADCIRC and spectral waves, validate water levels and high-water marks, isolate tide/surge/wave components, derive reduced scaling or machine-learning models, and map resulting water surfaces over high-resolution terrain.

Numerical models. Models include ADCIRC coupled to WAM, STWAVE, or SWAN; MIKE 21 and XBeach for component-resolved inundation; RDM with Zero-Point Boundary mapping; C1PKNet for peak-field surrogacy; and k-means plus beta mixtures for temporal surge templates.

Experimental datasets. Reviewed evidence includes Katrina and Rita regional observations, Iota inundation and profile reconstructions, three US synthetic forerunner ensembles with Ike and Harvey, 1031 Chesapeake simulations plus three historical storms, Haiyan survey limits over 5 m terrain, and two decades of US ADCIRC hydrographs.

Validated ranges. Validation remains regional: Katrina high-water marks were mostly reproduced within 0.5 m; the Ike forerunner range was 0.4–2.8 m predicted versus 0.4–2.6 m observed at 24–6 h lead; and Haiyan inundation-limit accuracy was 81% on a 5 m terrain grid. These are not global error bounds.

Recent advances. Recent advances learn track-to-peak mappings from large ensembles, validate high-resolution typhoon inundation maps, and compress complete surge evolution into eight physically interpretable temporal templates, moving hazard analysis beyond isolated peak values.

Disagreements. Peak-focused surrogates are efficient for spatial hazard fields, whereas forerunner and hydrograph studies show that lead time, duration, skewness, and delayed peaks are separate engineering quantities. Likewise, surge-only inundation can be accurate in one metric yet omit wave setup, swash, rainfall, or river contributions important elsewhere.

Limitations. Limitations include uncertain drag at extreme winds, grid-dependent wave setup, sparse observations, synthetic-storm sampling, omitted compound sources, static terrain assumptions, region-specific machine learning, and normalized templates that may not transfer to other basins or climate states.

Open questions. Priorities are transferable spatiotemporal surrogates with calibrated uncertainty, joint surge-wave-tide-rain-river validation, evolving topography and sea level, observation-based hydrograph typologies, operational forerunner verification, and error propagation from atmospheric forcing to local inundation and decisions.

Seminal papers. Within this screened slice, the 2009 Katrina-Rita coupled validation establishes a physics-based regional benchmark, and the 2019 forerunner study reframes pre-landfall rise as a forecast quantity distinct from peak surge.

Equations

- *Parametric tropical-cyclone radial wind profile* [1184]

$$V(r) = \frac{V_{max} 2R_{mw}r}{R_{mw}^2 + r^2}$$

Regime: Symmetric gradient-level synthetic and hypothetical cyclone winds subsequently corrected to asymmetric surface winds. Variables: $V(r)$ gradient wind speed at radius r ; V_{max} maximum sustained one-minute wind; R_{mw} radius of maximum wind; r radial distance

- *Dimensionless forerunner-surge scaling* [842]

$$\Pi_\eta = \mathcal{F}(\Pi_t; \Delta p, R, V_f)$$

Regime: Synthetic storms at Virginia, New York-New Jersey, and Texas model domains. Variables: Π_η central-pressure-scaled forerunner surge; Π_t wind-duration-scaled time; Δp central pressure deficit; R storm radius; V_f forward translation speed

- *Two-component beta-mixture surge template* [31]

$$\hat{\eta}(\tau) = wB(\tau; \alpha_1, \beta_1) + (1 - w)B(\tau; \alpha_2, \beta_2)$$

Regime: Representative normalized hydrographs derived from the US ADCIRC hindcast archive. Variables: $\hat{\eta}$ normalized surge hydrograph; τ normalized event time; B beta density-shaped component; w mixture weight; α_i, β_i shape parameters

Claims

1. C1PKNet used complete tropical-cyclone track time series and 1031 Chesapeake Bay simulations to predict spatial peak surge rapidly, with evaluation against Hurricanes Isabel, Irene, and Sandy; transfer beyond the trained storm and coastal-state space remains unverified. *Regime: Chesapeake Bay and the landfalling/bypassing storm distributions represented in the training database.* direct finding, numerical [790]
2. A pressure- and wind-duration-scaled method predicted Hurricane Ike forerunner surge 24–6 h before landfall as 0.4–2.8 m, compared with 0.4–2.6 m observed, and indicated that larger storms moving more slowly favor larger early surge. *Regime: The Virginia, New York-New Jersey, and Texas regional ensembles and storms comparable to their parameter ranges.* direct finding, mixed [842]
3. For Typhoon Haiyan in Tacloban, the RDM plus Zero-Point Boundary method on a 5 m terrain model identified surveyed inundation-limit locations with 81% accuracy, but the simulation omitted compound flooding. *Regime: The Haiyan hindcast, eastern Leyte terrain, and surveyed Tacloban water limits.* direct finding, mixed [1584]
4. Two decades of US ADCIRC hindcasts yielded eight normalized tropical-cyclone surge hydrograph types; Gulf cases were more diverse, and storm size, speed, approach angle, peak proximity, and bathymetric slope interacted without any single factor explaining all shape variability. *Regime: The US Gulf and Atlantic hindcast archive and its normalized hydrograph definitions.* direct finding, numerical [31]
5. Synthetic hurricanes coupled to SLOSH produce a heavy-tailed surge-height distribution at the Battery; peaks-over-threshold generalized-Pareto estimates give New York return periods consistent with other regional studies. *Regime: Risk assessment of hurricane storm surge for New York City* direct finding, numerical [832]
6. End-century simulations of 21 historical U.S. hurricanes increased inundation volume for 14 storms and extent for 13, averaging +36% and +25%; no single storm property explained the locally complex changes. *Regime: Projected Climate Change Impacts on Hurricane Storm Surge Inundation in the Coastal United States* direct finding, numerical [227]
7. Coral reefs can provide significant coastal protection benefits to people and property. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [136]
8. One of the most destructive natural hazards, tropical cyclone (TC)-induced coastal flooding, will worsen under climate change. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [922]
9. Tropical cyclone damage potential, as currently defined by the Saffir-Simpson scale and the maximum sustained surface wind speed in the storm, fails to consider the area impact of winds likely to force surge and waves or cause particular levels of damage. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1125]

10. Abstract A century ago, meteorologists regarded tropical cyclones as shallow vortices, extending upward only a few kilometers into the troposphere, and nothing was known about their physics save that convection was somehow involved. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [409]

11. For Bay of Bengal super cyclone TC05B, a parametric cyclone wind field embedded in blended background winds drives POM surge with peak water levels to the right of the track and coastal setup controlled by asymmetric wind stress, onshore Ekman transport, and basin-perimeter propagation. *Regime: Bay of Bengal response to the 1999 TC05B track and forcing, without tide or river-discharge coupling.* direct finding, numerical [318]

44 Wiki: Coastal structures

44.1 Coastal structures: overview

Topic structures. Performance and behavior of engineered coastal structures.

Papers screened: 10. Claims: 9. Equations: 0.

Well established. Coastal structures modify waves, currents, water levels, sediment pathways and access while resisting hydraulic, geotechnical and environmental actions; performance therefore depends on coupled load, response, foundation, material, morphology and service-limit states over the lifecycle.

Governing physics. Incident and reflected waves, breaking, impact, overtopping, currents, uplift and tsunami generate pressures and forces; geometry transforms flow; armor, concrete, steel, foundations and joints resist through mass, strength, friction and stiffness while scour, settlement, fatigue and corrosion reduce capacity.

Dimensionless parameters. Controls include relative freeboard and depth, wave steepness and surf similarity, reflection/transmission/overtopping coefficients, stability number, relative armor density, permeability, slope, breaker and impact parameters, Keulegan–Carpenter and diffraction ratios, safety/reliability indices and damage/deterioration ratios.

Major equations. Core formulations include wave transformation and overtopping relations, pressure/force and Morison-type loading, armor stability numbers, sliding/overturning/bearing/uplift limit states, porous flow, soil–structure interaction, scour, fatigue and corrosion deterioration, structural dynamics and reliability or lifecycle risk.

Typical methods. Methods combine design guidance, physical modelling, CFD/SPH and structural/geotechnical analysis, prototype monitoring, post-event forensics, material and durability tests, limit-state and reliability assessment, lifecycle inspection, retrofit trials and system/service risk analysis.

Numerical models. Tools include spectral and phase-resolving waves, CFD/VOF and SPH, porous-flow and overtopping models, finite-element structural and soil analysis, scour and morphology models, data-driven hydraulic emulators, deterioration, reliability and network consequence models.

Experimental datasets. Evidence includes vertical-breakwater overtopping, composite seawall crown-wall tests, recurve retrofits, coral-sand revetment foundations, stepped-revetment cases, cross-geometry simulations, coastal high-rise integrity and concrete/climate-resilience reviews.

Validated ranges. The reviewed evidence spans vertical and composite seawalls, stepped and armor revetments, breakwaters, coastal buildings and infrastructure under waves, overtopping, aggressive exposure and future sea levels; each transfers only within tested geometry, foundation, material and forcing.

Recent advances. Recent advances include high-resolution impact and overtopping experiments, CFD/SPH, data-driven emulators, recurve and crown-wall retrofits, monitoring-informed condition assessment, climate-adjusted loads, durable low-carbon materials and infrastructure-network risk.

Disagreements. Reducing overtopping can increase local loads or reflected energy; rigid protection can transfer erosion; hydraulic model accuracy does not prove structural or foundation safety; data-driven formula improvement may not extrapolate; least-cost robust mass can conflict with adaptability, carbon and ecology.

Limitations. Scale effects distort impact air, viscosity, permeability and sediment; foundations and scour are often simplified; joint failure and construction variability are sparse; deterioration and hidden defects are poorly observed; load combinations and cascading service loss remain uncertain under nonstationary climate.

Open questions. Priorities include coupled hydraulic–structural–geotechnical failure, compound hazards and debris, breaking-impact aeration, scour and morphology feedback, aging and inspection, low-carbon durable materials, adaptive retrofits, nature–structure interaction and system-level reliability.

Seminal papers. Classical breakwater and seawall research established armor stability, overtopping, reflection/transmission, wave pressures and sliding/overturning checks; later reliability, soil interaction, numerical fluid–structure modelling and lifecycle durability broadened performance assessment.

Claims

1. Machine-learning enhancement of vertical-breakwater overtopping prediction shows that data-driven tools can supplement hydraulic formulae when training range, geometry and validation limits are explicit. *Regime: Enhanced wave overtopping simulation at vertical breakwaters using machine learning algorithms* direct finding, numerical [559]
2. A review of climate risks to coastal critical infrastructure links direct loading, flooding, erosion, access loss, service interdependence and adaptation across structural systems. *Regime: Risks to Coastal Critical Infrastructure from Climate Change* review statement, review [1074]
3. A stepped-concrete revetment case demonstrates how structural layout, toe and slope conditions, wave exposure, construction practice and observed shoreline response jointly determine protection performance. *Regime: COASTAL PROTECTION BY A STEPPED CONCRETE REVETMENT IN THAILAND* direct finding, mixed [1250]
4. Parametric physical tests of crown-wall modifications quantify how a composite seawall can be adapted to future sea levels while retaining structure-specific overtopping constraints. *Regime: Crown Wall Modifications as Response to Wave Overtopping under a Future Sea Level Scenario: An Experimental Parametric Study for an Innovative Composite Seawall* direct finding, experimental [296]

5. Physical-model evidence shows that recurve retrofits can reduce overtopping at vertical seawalls, with performance governed by recurve geometry and incident hydraulic conditions. *Regime: Improved prediction of wave overtopping rates at vertical seawalls with recurve retrofitting* direct finding, experimental [362]
6. SPH simulation across several coastal-structure geometries demonstrates a common numerical framework for comparing free-surface overtopping processes and structure-specific response. *Regime: Smoothed Particle Hydrodynamics Simulation of Wave Overtopping Characteristics for Different Coastal Structures* direct finding, numerical [1130]
7. Regular-wave physical modelling of a revetment breakwater on reclaimed coral-calcareous sand links foundation conditions, armor response and hydraulic loading to structural stability. *Regime: Physical Modeling of the Stability of a Revetment Breakwater Built on Reclaimed Coral Calcareous Sand Foundation in the South China Sea—Regular Wave* direct finding, experimental [578]
8. A review of high-rise structures in coastal regions integrates wind, salt exposure, flooding, foundations, materials, deterioration and structural-integrity concerns. *Regime: Review of Structural Integrity Concerns in High-rise Buildings in Coastal Regions* review statement, review [190]
9. A coastal-resilience review frames durable and lower-impact concrete materials as part of structural adaptation to sea-level rise, flooding, aggressive exposure and extreme loading. *Regime: Developing Coastal Resilience to Climate Change in Panama Through Sustainable Concrete Applications* review statement, review [240]

44.2 Surge barriers and gates

Topic structures.barriers. Closable flood defense structures.

Papers screened: 6. Claims: 6. Equations: 0.

Well established. Closable surge barriers can sharply reduce extreme water levels, but their benefit depends on timely reliable operation and transfers consequences to maintenance, navigation, lagoon exchange, sediment, ecology, and residual flood pathways.

Governing physics. Performance depends on external surge, tide and waves, head difference, closure timing/duration, gate and inlet hydraulics, internal storage and tributary inflow, leakage, and the post-closure evolution of water levels and exchange.

Dimensionless parameters. Relevant controls include head and freeboard ratios, closure time relative to surge rise, closed duration relative to tidal/renewal time, gate blockage fraction, forecast exceedance probability, failure probability, maintenance-window availability, and closures per year.

Major equations. Operational frameworks include threshold and probabilistic closure rules, ensemble exceedance probabilities, binary false-alarm/miss skill, hydrodynamic mass and momentum, lagoon exchange and renewal, reliability/lifecycle models, and avoided-damage cost-benefit.

Typical methods. Methods span tide-gauge and surge observations, hydrodynamic counterfactuals, historic morphology models, ensemble forecast hindcasts, threshold sensitivity, system-dynamics economics, policy analysis, and controlled closure analog enclosures.

Numerical models. Lagoon circulation/renewal models, counterfactual surge propagation, ensemble threshold classifiers, system-dynamics cost-benefit simulations, and closure-frequency SLR scenarios cover operational to century scales.

Experimental datasets. Accessible evidence includes the 22 November 2022 Venice surge, 16 years of Maeslant forecasts and Hoek van Holland observations, 1927-2012 Venice morphology, future SLR scenarios, and 28-48 h seasonal enclosure experiments.

Validated ranges. Reported evidence includes an unregulated Chioggia counterfactual above 200 cm, 33.1% false alarms in the Maeslant maintenance model, 28-48 h enclosure experiments, and Venice renewal changes from 11.9 to 10.8 days before recent interventions added 0.5 day.

Recent advances. Recent work evaluates operational extreme-event counterfactuals, ensemble-based maintenance decisions, SLR-driven closure viability, and season-dependent ecosystem changes caused by temporary isolation.

Disagreements. A conservative threshold preserves safety but sacrifices maintenance windows through false alarms; relaxing it reduces false alarms but raises misses. Benefits can exceed costs while increasing closure frequency simultaneously undermines port, ecosystem, and long-term viability.

Limitations. This synthesis is dominated by Venice/MOSE plus one Maeslant maintenance study. Journal evidence on gate loads, vibration, leakage, mechanical failure, closure transients, overtopping, internal runoff, and Thames/Eastern Scheldt/Maeslant cross-comparison is sparse.

Open questions. Priorities include forecast-to-operation reliability, common-cause and component failure, closure transients and loads, interior compound flooding, emergency reopening, safe maintenance, navigation scheduling, sediment/salinity/ecology, and adaptation limits under accelerating SLR.

Seminal papers. The accessible branch connects Venice barrier planning and hydrodynamic restriction with observed MOSE protection, lifecycle economics, probabilistic Maeslant maintenance, and experimental closure externalities.

Claims

1. Venice modelling indicates repeated barrier closure and sea-level-driven lagoon-volume growth will reduce open-sea exchange and flushing, especially in the central lagoon. *Regime: Periodic isolation during floods and sea-level-driven volume increase will reduce sea exchange and flushing, particularly in the central lagoon.* direct finding, numerical [436]
2. Venice flood defenses were judged technically capable across broad century-scale sea-level scenarios, but their effective use depends on institutional reform and coordinated nonstructural adaptation. *Regime: The hydraulic defenses were judged technically adequate across a broad century-scale SLR range, but effective use required major institutional change and integration with nonstructural measures.* review statement, review [1000]

3. MOSE cost-benefit modelling found benefits exceed investment and port-closure costs, yet rising closure frequency threatens medium- to long-term operational and ecological viability. *Regime: Benefits exceeded investment and foregone port revenue, but increasing SLR-driven closure frequency challenges long-term viability and lagoon ecology and motivates a revised closure threshold and coordinated measures.* direct finding, numerical [506]
4. During the 22 November 2022 surge, MOSE prevented flooding; a data-forced counterfactual estimated unregulated Chioggia water level above 200 cm. *Regime: MOSE protected Venice; without regulation, modeled lagoon water level would have exceeded 200 cm at Chioggia, beyond the monitored record.* direct finding, mixed [950]
5. A 16-year Maeslant evaluation found 33.1% false alarms and showed threshold tuning trades fewer false alarms against more missed unsafe conditions. *Regime: The operational model was conservative with 33.1% false alarms; raising critical probability reduced false alarms but increased misses, exposing a risk-versus-maintenance-time tradeoff.* direct finding, mixed [1425]
6. MOSE analog enclosure experiments showed 28-48 h reduced hydrodynamics enhanced suspended-material deposition and altered pelagic-benthic coupling, with season-dependent responses. *Regime: Reduced exchange enhanced suspended-material deposition and altered pelagic-benthic processes, with seasonal responses differing between experiments.* direct finding, experimental [282]

44.3 Breakwaters

Topic structures. breakwaters. Detached, attached, and floating breakwaters.

Papers screened: 10. Claims: 7. Equations: 3. Shown in the Coastal Design Bench beside: Breakwater design.

Well established. Breakwaters protect coasts and ports by reflecting, diffracting, dissipating, and transmitting incident-wave energy. Their performance cannot be reduced to wave-height attenuation alone: armour and toe stability, overtopping, foundation response, lee-side spectra, navigation, morphology, maintenance, and cost interact.

Governing physics. Key processes include refraction and diffraction around heads and gaps, reflection from structures and beaches, transmission through and over permeable or low crests, depth-limited breaking, bottom friction, wave-current interaction, armour-unit motion, toe scour, settlement, and lee-side sediment redistribution.

Dimensionless parameters. Important controls include relative freeboard R_c/H_s , relative crest width, relative depth h/H_s , stability number $H_s/(\Delta D_{n50})$, damage level, Iribarren number, directional spread, relative wavelength and gap, porosity, structure-height-to-depth ratio, toe geometry, and layout-to-shoreline ratios.

Major equations. Common frameworks include Hudson or Van-der-Meer-type armour stability relations, toe-stability and damage formulations, wave-action balance with phase-decoupled diffraction, mild-slope equations for harbour propagation, porous resistance and transmission relations, and constrained optimization objectives.

Typical methods. Methods combine 2-D flumes and 3-D basins, armour-damage surveys, prototype inspections, directional wave arrays, spectral and phase-resolving models, pressure and force measurements, empirical stability formulae, and genetic or multiobjective layout optimization.

Numerical models. Models include SWAN with phase-decoupled diffraction, mild-slope wave propagation coupled to a genetic algorithm, COBRAS-UC for loads and pressures, empirical armour and toe stability formulae, and porous-transmission theory.

Experimental datasets. Reviewed evidence includes Aalborg depth-limited short-crested and oblique stability tests on a 1:25 foreshore, laboratory multidirectional diffraction at a breakwater shoulder, seven Elmer field spectra around six detached breakwaters, 1:20 crown-wall load tests, and 290 living-breakwall wake cases.

Validated ranges. Explicit evidence covers a 1:25 foreshore and 0.5 m plateau for low-crested stability, seven wind-sea/bimodal/swell field spectra at Elmer with 2.9-5.3 m tides, one- and two-segment Beirut layouts, 1:20 low-mound and crown-wall tests, and porous breakwalls of estimated porosity 0.7 and 0.9.

Recent advances. Later work embeds wave mechanics within layout optimization, resolves pressure and load fields numerically, and extends breakwater concepts toward porous living systems whose performance and maintenance can be measured under field wakes.

Disagreements. Phase-averaged diffraction improves lee wave heights but performs less well for narrow low-frequency swell unless reflection and transmission are represented. Calibrated porous theory overpredicts energy as an emergent breakwall enters shallow-water breaking and friction regimes.

Limitations. Laboratory scale, armour placement, seabed mobility, finite storm duration, incomplete field spectra, model physics, uncertain porosity, construction tolerances, and optimization objectives restrict transfer. Prototype applications without measured validation remain demonstrations, not field validation.

Open questions. Needs include joint reliability-based optimization of stability, agitation, navigation, morphology, ecology, carbon, maintenance and cost; climate-adjusted design waves and water levels; better toe and foundation failure models; and transferable field validation for multidirectional extremes.

Seminal papers. Within this branch, the 2006 low-crested stability synthesis connects controlled testing to prototype foundation failure, and the 2007 SWAN study provides rare laboratory-plus-field validation of multidirectional transformation around detached structures.

Equations

- low-crested breakwater armour and toe stability formulation [214]

$$D = f(H_s, T_p, R_c, h, \Delta, D_{n50}, N)$$

Regime: Detached rock-armoured low-crested structures; Aalborg 3-D tests used a 1:25 foreshore and 0.5 m plateau; evidence also includes 2-D depth-limited tests and prototype sites. Variables: D damage measure where applicable; N wave action density or test count by context; x breakwater node-coordinate vector; J normalized optimization objective; P constraint penalty; normalized form, not a recovered coefficient equation

- *SWAN phase-decoupled refraction-diffraction model* [633]

$$\frac{\partial N}{\partial t} + \nabla_{x,y} \cdot (c_g N) + \frac{\partial (c_\theta N)}{\partial \theta} = S/\sigma$$

Regime: Laboratory shoulder-diffraction data and seven Elmer field cases: three wind-sea, two bimodal swell-sea, and two narrow swell spectra around six detached breakwaters. Variables: D damage measure where applicable; N wave action density or test count by context; x breakwater node-coordinate vector; J normalized optimization objective; P constraint penalty; normalized form, not a recovered coefficient equation

- *genetic-algorithm and mild-slope detached-breakwater optimizer* [400]

$$J(\mathbf{x}) = L(\mathbf{x}) + P_{wave}(\mathbf{x}) + P_{nav}(\mathbf{x})$$

Regime: Port of Beirut; one- and two-segment breakwaters under simplified constant-depth and real bathymetry, within a prescribed search space and navigational constraints. Variables: D damage measure where applicable; N wave action density or test count by context; x breakwater node-coordinate vector; J normalized optimization objective; P constraint penalty; normalized form, not a recovered coefficient equation

Claims

1. For detached rock-armoured low-crested breakwaters tested under depth-limited short-crested head-on and oblique waves, a shallow-water formula characterized initiation of armour damage and an existing toe-stability formula agreed with tests; prototype observations additionally showed that toe scour or sinking and insufficient berm volume can initiate regressive armour-layer erosion. *Regime: Detached rock-armoured low-crested structures; Aalborg 3-D tests used a 1:25 foreshore and 0.5 m plateau; evidence also includes 2-D depth-limited tests and prototype sites.* direct finding, mixed [214]
2. Across laboratory tests and seven Elmer field spectra, adding phase-decoupled diffraction to SWAN improved wave-height prediction behind detached breakwaters and reproduced broad spectra best, but performance worsened for narrow low-frequency swell where beach reflection and transmission through or over the structures were important. *Regime: Laboratory shoulder-diffraction data and seven Elmer field cases: three wind-sea, two bimodal swell-sea, and two narrow swell spectra around six detached breakwaters.* direct finding, mixed [633]
3. For the Port of Beirut design cases, a genetic algorithm coupled to a mild-slope wave model optimized one- and two-segment detached-breakwater node coordinates while enforcing wave-disturbance and navigation constraints under both simplified and real bathymetry; accessible evidence does not establish superiority outside that search space and objective definition. *Regime: Port of Beirut; one- and two-segment breakwaters under simplified constant-depth and real bathymetry, within a prescribed search space and navigational constraints.* direct finding, numerical [400]
4. This paper presents a numerical study of the solitary wave interaction with a submerged breakwater using the Consistent Particle Method (CPM). *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, numerical [1176]
5. Abstract In this paper, a novel hybrid wave energy converter (WEC)-floating breakwater system consisting of three floating pontoons with power take-off (PTO) modules is developed to extract wave energy from heave motion and attenuate waves to protect the coast. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1092]
6. As a kind of technical equipment for wave prevention and wave dissipation, floating breakwater has attracted more and more attention of researchers. Starting from the research progress and application status of the floating breakwater, this paper combs the research status of the floating breakwater, summarizes the application status of the floating breakwater, and obtains the key points of the technical improvement of the floating breakwater. Finally, the future development of the new type of floating breakwater is summarized in terms of its application prospect, wave dissipation principle and application of new materials, which provides a strong reference for the relevant research of the floating breakwater. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1615]
7. Low-crested and other hard coastal defenses create artificial hard-bottom habitat while disrupting adjacent soft sediment; their proliferation can alter regional connectivity, native assemblages and non-native-species spread, so ecological objectives and monitoring must enter planning, design and maintenance. *Regime: Low-crested and other hard defenses placed in soft-bottom coastal landscapes, including repeated structures along developed coastlines.* review statement, review [20]

44.4 Structural failure and damage

Topic structures.failure. Damage progression and failure modes.

Papers screened: 4. Claims: 3. Equations: 0. Shown in the Coastal Design Bench beside: Seawall design.

Well established. Coastal-structure failure is not a single endpoint: evidence distinguishes hydraulic instability, progressive physical damage, loss of stability or function, and damage to buildings and infrastructure exposed to coastal flooding.

Governing physics. Damage depends on the interaction of wave and surge loading, water depth, erosion and morphology, flow concentration around built features, armour or material response, foundation conditions, and the accumulated state of the structure.

Dimensionless parameters. Relevant model families commonly organize response through normalized wave loading, relative geometry and water depth, material or armour stability, and normalized damage measures; coefficients and exact forms are intentionally deferred to full-text extraction.

Major equations. The accessible evidence supports families of damage and stability models rather than one universal equation. Their use requires preserving each model's damage definition, response variable, structural type, and hydraulic regime.

Typical methods. Failure evidence combines historical model review, laboratory or application review, post-event field inspection, UAV mapping, environmental observations, and numerical reconstruction of surge, waves, flooding, and coastal change.

Numerical models. Numerical reconstruction can connect regional storm forcing to local flood depth and damage, while damage-model reviews expose how alternate definitions and assumptions affect predicted rubble-mound response.

Experimental datasets. This initial branch contains post-Irma field and UAV observations from Saint-Martin and Saint-Barthélemy plus reviewed experimental knowledge for rubble-mound and geosynthetic coastal defenses.

Validated ranges. The indexed abstracts document an Irma regime with approximately 4 m surge and 10 m waves, but do not expose sufficient coefficients or ranges to claim general validation limits for the reviewed breakwater and geosynthetic models.

Recent advances. Recent work increasingly combines forensic field evidence, remote sensing, high-resolution numerical reconstruction, and explicit review of model assumptions to move from isolated component response toward contextual failure diagnosis.

Disagreements. Apparent disagreement among failure models may arise from different definitions of damage, failure modes, variables, and structural systems. Cross-model comparison is unsafe until these semantic and regime differences are aligned.

Limitations. This branch is abstract-bounded and has only three extractable sources. It cannot yet quantify full failure envelopes, repeated-loading effects, geotechnical mechanisms, cascading system failure, or repair and recovery.

Open questions. Priority gaps are common damage ontologies, progressive and interacting failure modes, prototype-scale validation, coupling with erosion and foundations, probabilistic reliability, and links from physical damage to service loss and recovery.

Seminal papers. The rubble-mound historical review is the branch's guide to the development of damage-model concepts; source-level identification and extraction of individual seminal formulations remains a full-text task.

Claims

1. After Hurricane Irma in Saint-Martin and Saint-Barthélemy, urban structures and streets channelled flood flow and locally amplified erosion and water levels, while observed damage intensity increased with flood depth. *Regime: Saint-Martin and Saint-Barthélemy during Category 5 Hurricane Irma, with modelled surge near 4 m and waves near 10 m* direct finding, mixed [1183]
2. For rubble-mound breakwaters, hydraulic instability is the principal damage mode addressed by historical damage models, whose differing damage definitions and assumptions constrain direct comparison and transfer. *Regime: rubble-mound breakwaters exposed to wave action* review statement, review [230]
3. Geosynthetic coastal defenses can be advantageous on weak foundations, but several applications still lack well-founded design formulae and specifications for hydraulic performance, stability, and failure modes. *Regime: geosynthetic coastal structures, including applications on weak foundations* review statement, review [1065]

44.5 Overtopping

Topic structures.overtopping. Discharge and consequences of overtopping.

Papers screened: 22. Claims: 17. Equations: 0. Shown in the Coastal Design Bench beside: Seawall design, Breakwater design.

Well established. Wave overtopping occurs when runup exceeds crest elevation. Design must consider mean discharge, number of overtopping waves, individual and maximum volumes, velocities and depths, spatial distribution, drainage, tolerable limits, and exposed receptors.

Governing physics. Controls include offshore spectrum, breaking and infragravity transformation, water level, foreshore depth/slope/permeability/morphology, obliquity, roughness, porosity, crest freeboard/width, walls, berms, wave steepness, runup, and structure damage.

Dimensionless parameters. Important groups include relative freeboard, Iribarren number, wave steepness, relative depth, crest and berm geometry, roughness and obliquity factors, dimensionless discharge q^* , overtopping probability, and normalized individual volume.

Major equations. Core frameworks normalize mean discharge by gravity and wave scale, use exponential freeboard decay and roughness/geometry factors, model individual volumes with Weibull or exponential distributions, and estimate maxima from mean rate and overtopping-wave count.

Typical methods. Methods use long irregular-wave flume series, repeated stochastic realizations, large-scale and field sensing, CLASH/EurOtop databases, empirical regression, phase-averaged and phase-resolving models, CFD/SPH, neural networks, and probability fitting.

Numerical models. Models include SWASH, SWAN with infragravity corrections, phase-resolving solvers, empirical EurOtop-type relations, neural and polynomial regressions, and wave-by-wave Weibull/exponential models.

Experimental datasets. Reviewed evidence includes CLASH, 124 SWASH validation cases, six-model benchmarks, 140 rubble-seawall tests, 135 individual-volume tests, 1:50 shingle-foreshore tests, repeated random series, and stepped-revetment campaigns.

Validated ranges. Reported findings include order-of-magnitude underprediction without infragravity waves, 40% RMSE reduction for a rubble-seawall formula, stochastic discharge ranges to 75%, factor-four mean reduction from permeable shingle, stepped reduction to 60%, and maximum-volume bias improved to -6%.

Recent advances. Recent advances resolve infragravity controls, benchmark speed versus accuracy, model complex retrofits, quantify random-series uncertainty, distinguish mean and maximum behavior, and fit wave-by-wave volumes for risk-based design.

Disagreements. Mean and maximum metrics can respond differently: permeable foreshores reduced mean discharge strongly without a significant maximum-volume change. Higher-fidelity models are not automatically more accurate if incident conditions or decisive infragravity physics are wrong.

Limitations. Scale effects, finite series, stochastic realization, toe-wave reconstruction, morphology, permeability, damage, obliquity, wind, currents, drainage, prototype sensing, and sparse extremes limit transfer. Confidence bounds must accompany formula use.

Open questions. Needs include joint discharge-volume-velocity prediction, dynamic morphology and damage, current/wind effects, nonstationary climate, prototype wave-by-wave validation, receptor-specific tolerances, and uncertainty-aware retrofit optimization.

Seminal papers. The branch builds on classical runup and freeboard-decay relations, CLASH/EurOtop databases, roughness and geometry factors, and Weibull individual-volume models; current work extends shallow, complex, permeable, and stochastic regimes.

Claims

1. For the existing and proposed Port of Las Nieves breakwaters, modeled access-area overtopping risk was substantially higher than along main-body cross sections, identifying the access sections as the priority for controls and expansion design. *Regime: Existing and proposed Port of Las Nieves breakwaters, Gran Canaria.* direct finding, numerical [1251]
2. For sea dikes on very shallow foreshores, Van Gent's relation overestimated mean discharge and could not treat emergent toes, while an equivalent-slope concept validated with CLASH and dedicated tests improved prediction. *Regime: Shallow, very shallow, and emergent-toe sea-dike conditions.* direct finding, mixed [44]
3. Across 124 shallow-foreshore cases, SWASH predicted mean overtopping about as accurately as semi-empirical formulae when toe-wave conditions were reproduced and handled complex geometry, although one intense individual event was underpredicted. *Regime: Impermeable structures, shallow foreshores, and complex storm-wall geometries.* direct finding, mixed [1379]
4. In a six-model shallow-foreshore benchmark, neglecting infragravity waves underestimated overtopping by an order of magnitude, while empirically corrected phase-averaged models could match or outperform costlier phase-resolving models. *Regime: Dikes with shallow mildly sloping foreshores.* direct finding, numerical [773]
5. For tested rubble-mound breakwaters, relative protruding crest-wall height dominated wall effects, while berm width, berm level, and wave steepness controlled berm influence on overtopping; expressions were also checked against oblique-wave data. *Regime: Rubble-mound breakwaters with crest walls, berms, or both, including oblique-wave validation.* direct finding, experimental [490]
6. Using 140 new tests and existing EurOtop data, a rubble-mound seawall formula based on relative freeboard and wave steepness reduced RMSE by 40% and improved normalized-discharge bias from -0.38 to approximately zero. *Regime: Small-scale rubble-mound seawalls across relative freeboard and steepness conditions.* direct finding, experimental [735]
7. For smooth-slope and vertical-wall tests, normalized overtopping-discharge variation reached 10% when one wave series was repeated but 75% when different realizations from the same spectrum were used. *Regime: Smooth slope and vertical wall under multiple realizations from identical spectra.* direct finding, experimental [1518]
8. In 1:50 tests of a 1:2 wall, a permeable shingle foreshore reduced mean overtopping by up to a factor four relative to impermeable-slope predictions, while maximum individual volumes showed no significant difference. *Regime: A 1:2 wall fronted by impermeable versus permeable shingle slopes.* direct finding, experimental [1239]
9. Stepped revetments reduced breaking-wave overtopping volumes by up to 60% relative to smooth slopes, with reduction declining as Iribarren number increased; a new formulation explicitly reports range, scatter, and scale effects. *Regime: Breaking and nonbreaking waves on stepped revetments across disclosed parameter ranges.* direct finding, mixed [702]
10. For tested vertical seawalls, individual overtopping volumes followed a two-parameter Weibull distribution; relative freeboard and overtopping rate predicted shape under nonimpulsive attack, and a unified formula was proposed across impulsive and nonimpulsive cases. *Regime: Plain vertical seawalls across freeboard, depth, steepness, and impulsive/nonimpulsive conditions.* direct finding, experimental [1241]
11. Across 135 rubble-mound seawall tests, weighted Weibull and exponential fitting reduced scatter by 23% and 17%, while a q^* -and-overtopping-wave-count formula achieved 37% scatter index and -6% bias for maximum individual volume. *Regime: Rubble-mound slopes 1:1.5 and 1:2 across practical crest freeboards.* direct finding, experimental [734]
12. Across 450 validated SWASH cases for vertical walls on 1:7-1:10 foreshores, longer relative promenades reduced overtopping and a new EurOtop-compatible storm-wall reduction factor reproduced modeled discharge with $r = 0.94$. *Regime: Vertical walls with retreated storm walls, steep 1:7-1:10 foreshores, and the Mediterranean-scale parameter ranges tested in the paper.* direct finding, numerical [207]
13. Wave-flume tests of ten roughened vertical-seawall configurations found that increasing protrusion length and density reduced mean overtopping by up to 100% under impulsive waves, while tested non-impulsive waves showed no significant reduction. *Regime: Small-scale vertical seawall tests on a 1:20 impermeable foreshore, within the tested protrusion geometries and impulsive/non-impulsive waves.* direct finding, experimental [1240]
14. Evolutionary Polynomial Regression across five datasets produced parsimonious unified mean-overtopping formulas for smooth dikes through vertical walls, confirming exponential response and improved prediction when water depth, wave period, foreshore slope and dike slope are combined through equivalent or imaginary slope. *Regime: Smooth sea dikes and vertical walls across the combined datasets; rough or permeable structures only for preliminary conservative use.* direct finding, numerical [45]
15. Large-scale mobile-bed tests show storm-driven lowering of the sandy foreshore before a near-vertical seawall increases overtopping, with total overtopped volume rising by as much as 104%; cumulative volume is more sensitive to profile evolution than peak discharge rate. *Regime: Low-steepness storm sequences over an evolving 1:15 sandy foreshore fronting a near-vertical 10:1 seawall.* direct finding, experimental [191]
16. For the January 2015 Scituate blizzard case, coupled tide-surge-wave interactions increased predicted seawall overtopping about fivefold; a 0.36 m sea-level rise doubled peak discharge, and maintaining baseline discharge required roughly 0.9 m of crest raising because deeper toe water also increased incident wave height. *Regime: Scituate seawall, Massachusetts, during the January 2015 North American blizzard and scenarios combining 0.36 m sea-level rise with raised crest elevations.* direct finding, numerical [1545]
17. The SWEEP-OWWL regional forecast system predicted 80% of observed overtopping events one day ahead, all severe events, with a 2.5% false-alarm rate; profile choice changed predicted discharge by an order of magnitude, demonstrating the operational value of frequently updated coastal morphology. *Regime: Engineered and naturally defended profiles along roughly 1,000 km of southwest England, validated against winter 2018/19 observations and tested with Storm Emma boundary conditions.* direct finding, mixed [1354]

44.6 Seawalls and revetments

Topic structures.seawalls. Shore-parallel armoring.

Papers screened: 16. Claims: 11. Equations: 0. Shown in the Coastal Design Bench beside: Seawall design.

Well established. Seawalls hold a shoreline or protect a hinterland by reflecting, redirecting, and sometimes overtopping incident waves and surges. Performance includes crest exceedance, structural loading, toe stability, beach response, drainage, and landward consequences.

Governing physics. Key processes are breaking and impulsive impact, aeration and turbulence, reflection and standing-wave motion, runoff and overtopping, return flow, setup and surge, sediment resuspension and longshore transport, toe scour, and foundation interaction.

Dimensionless parameters. Important controls include relative wall and recurve height, freeboard, water depth, wave height and steepness, breaker location, wall setback, reflection coefficient, relative toe geometry, sediment mobility, and normalized force and moment.

Major equations. Evidence uses force and overturning-moment reduction relations, RANS-VOF free-surface and turbulence equations, sediment-transport and morphology conservation, breaking and overtopping relations, and reflection and load metrics; no universal seawall equation spans all failure modes.

Typical methods. Methods combine wave-flume force and velocimetry tests, large-scale morphology experiments, OpenFOAM and coupled RANS-VOF morphodynamics, acoustic and optical field monitoring, post-event evidence, and historical or state-of-the-art review.

Numerical models. OpenFOAM resolves overtopped landward force and moment, while coupled RANS-VOF, bedload, sand-slide, and partial-cell morphology models track the evolving water-soil interface and toe scour.

Experimental datasets. The branch includes solid and perforated tsunami-wall tests, 1:5 seawall-versus-woody-debris morphology tests, breaking-solitary-wave velocimetry, 31 days of winter-gale scour observations, and validated vertical-structure morphology cases.

Validated ranges. Reported findings include tsunami-force reductions up to 41% for a high nearby wall and about 27% for a half-height wall, gale SSC of 12,222 mg/L and net sediment flux of 769 t/m²/day, and a 1:5 reflection/morphology comparison.

Recent advances. Recent work couples image velocimetry, long-duration seabed sensing, free-surface CFD, evolving morphology, physical comparisons with nature-based structures, and climate-adaptation review to examine the full wall-beach-hinterland system.

Disagreements. Higher or more elaborate walls do not always yield proportional benefit: a recurve delayed impact but added little load reduction under overtopped flow, and a low wall could increase landward force. Nature-based alternatives range from lower-reflection single logs to stacked forms behaving like walls.

Limitations. Evidence is fragmented across tsunami, solitary-wave, storm, morphology, and regional-adaptation regimes. Scale effects, aeration, three-dimensionality, evolving toes, drainage, foundations, ecological response, and compound sea-level and storm change restrict transfer.

Open questions. Needs include coupled wall-toe-foundation failure, prototype impulsive pressures and velocities, overtopped hinterland pathways, repeated storm recovery, adaptive crest design, ecological and recreational tradeoffs, and uncertainty-aware lifecycle performance.

Seminal papers. A century-scale UK review traces movement from largely vertical wall-and-groyne practice toward curved, stepped, model-informed, environmentally constrained, and nourishment-supported designs; detailed formulation lineage remains a full-text task.

Claims

1. A Malaysian review identified increased seawall platform levels and earth bunds as erosion-control measures adaptable to sea-level rise, while several fixed offshore structures become less effective as sea level increases. *Regime: Malaysian shoreline and hold-the-line/adaptation strategies* review statement, review [1161]
2. In physical tests, a higher seawall close to a building reduced tsunami force by up to 41%, versus about 27% for a half-height wall, while a perforated wall performed approximately like a solid wall and drained retained water. *Regime: building behind walls of varied type, height, and setback under tsunami-like waves* direct finding, experimental [1147]
3. Structural tsunami countermeasures such as seawalls aim to reduce inundation depth and distance, but their effectiveness and damage modes depend on regional conditions and event-specific performance. *Regime: hard tsunami countermeasures including seawalls and offshore breakwaters* review statement, review [1040]
4. Over a century of UK practice, seawall design became more rigorous through physical and numerical modelling, while environmental and recreational constraints reduced the formerly dominant role of hard wall-and-groyne solutions. *Regime: United Kingdom seawalls and groynes since the 1911 Royal Commission* review statement, review [1516]
5. A coupled RANS-VOF, bedload, and partial-cell morphology model reproduced storm-driven beach change and toe scour in front of a vertical coastal structure across multiple wave, depth, and slope conditions. *Regime: vertical coastal structure across validated waves, depths, and bottom slopes* direct finding, numerical [1091]
6. During a 31-day seawall field deployment, winter gales raised suspended sediment concentration to 12,222 mg/L and horizontal net sediment flux to 769 t/m²/day, whereas tidal forcing in calm weather caused little suspension. *Regime: severely scoured seabed in front of a seawall during winter* direct finding, field [1370]
7. Under simulated overtopped tsunami-like flow, a large recurve delayed impact but reduced inland-structure force little relative to a vertical wall; low walls could amplify load, while greater wall height reduced force linearly and overturning moment exponentially. *Regime: inland structure behind vertical or large-recurve walls of varied height* direct finding, numerical [572]
8. In 1:5 physical modelling, a single anchored log reduced wave reflection relative to a vertical seawall, while a stacked-log wall behaved nearly identically to the seawall and beach response depended on structure configuration. *Regime: vertical seawall versus single-log and stacked large-woody-debris structures* direct finding, experimental [420]
9. Bubble image velocimetry resolved phase-dependent maximum velocity and turbulence for breaking solitary waves impacting and overtopping a vertical seawall, distinguishing high- and low-aeration regimes. *Regime: shoaling solitary waves striking and overtopping a vertical seawall under two aeration regimes* direct finding, experimental [1535]
10. A state-of-the-art review found limited, dispersed evidence on storm-driven morphological change in front of coastal defenses and its coupling to overtopping, rundown, stability, and failure. *Regime: coastal defense structures and natural barriers during storms* review statement, review [1214]
11. In 20 seawall cross-section optimization trials, DYCORS coupled to an ANN overtopping model improved both discharge and cost even in the worst trial (17.67% and 12.1% reductions) and outperformed the genetic-algorithm benchmark. *Regime: The paper's composite-slope, berm and wave-wall decision space and ANN overtopping model under the tested wave and water-level combinations.* direct finding, numerical [1394]

45 Wiki: Tides and coastal circulation

45.1 Tides and coastal circulation: overview

Topic tides. Astronomical tides and coastal-scale circulation.

Papers screened: 9. Claims: 2. Equations: 0.

Well established. Astronomical forcing produces predictable barotropic constituents, but coastal amplitude, phase, currents, asymmetry and residual transport depend strongly on depth, friction, geometry and nonlinear interactions. Stratification converts barotropic energy into internal tides that can radiate and mix remotely.

Governing physics. The tide-generating potential forces shallow-water motion; friction, convergence, resonance and Coriolis alter propagation. Constituent interaction generates harmonics and residuals, while sloping topography and stratification generate internal waves whose refraction and dissipation depend on background flow.

Dimensionless parameters. Controls include tidal amplitude-to-depth ratio, friction and convergence parameters, basin length relative to tidal wavelength, Rossby and Froude numbers, spring-neap ratio, constituent amplitude and phase ratios, skewness, internal-wave slope criticality and conversion/dissipation efficiency.

Major equations. Barotropic tides follow depth-integrated mass and momentum equations with astronomical boundary forcing, bottom stress and nonlinear advection. Harmonic analysis decomposes constituents; internal tides use stratified wave and energy equations, and residual circulation emerges through period averaging and nonlinear terms.

Typical methods. Studies use tide gauges, current meters, shipboard hydrography and moorings with harmonic, skewness, flux and energy-budget analysis; shallow-water, 3D circulation and nonhydrostatic/internal-wave models test geometry, stratification, friction and coupled mean-level/surge/wave effects.

Numerical models. Representations range from constituent shallow-water solvers and COHERENS-like circulation systems to global nonuniform ocean-tide models and internal-wave theory. Coastal-flood models may couple mean level, tides, surge, waves and inundation rather than linearly superpose them.

Experimental datasets. Reviewed evidence includes Monterey Canyon ship surveys and monthlong records, Sea of Okhotsk constituent simulations, California mixed-tide estuaries, Paopao reef-lagoon observations, Changjiang circulation literature, and realistic/idealized Northwest Pacific tide experiments.

Validated ranges. Examples include Monterey internal currents near 20 cm/s and 50 m displacement, Okhotsk residuals near 10 cm/s, Mid-Atlantic Ridge M2 flux 3–5 mW/m² and coastal interaction effects of tens of centimeters. These belong to distinct regimes and scales.

Recent advances. Recent advances couple global circulation, eddies and internal tides at variable resolution, diagnose nonlinear energy budgets, use skewness for mixed-tide asymmetry and embed tide-surge-wave-mean-level interactions in probabilistic coastal flooding.

Disagreements. Linear constituent superposition is useful but can miss nonlinear water-level, friction, flooding and harmonic interactions. Internal-tide energy may dissipate near generation sites or radiate far away depending on mode and topography; idealized stratification can bias local conversion materially.

Limitations. Sparse offshore observations, uncertain bathymetry and drag, datum and nodal corrections, unresolved channels, changing morphology and mean level, incomplete stratification and background currents, and model resolution constrain prediction and attribution.

Open questions. Priorities include climate-driven tidal change, compound extremes, wetting/drying feedback, estuary asymmetry and sediment response, internal-tide remote dissipation, eddy interaction, data assimilation and consistent uncertainty across barotropic and baroclinic scales.

Seminal papers. Equilibrium and dynamic tide theory established constituents and propagation; harmonic analysis enabled prediction, while shallow-water theory explained amplification and friction. Later work formalized tidal asymmetry, residual circulation and barotropic-to-baroclinic conversion.

Claims

1. Coastal tidal prediction must account for interactions with mean sea level, surge, waves and inundation whenever their locally variable contributions are comparable to the water-level margin relevant to flooding or design. *Regime: Coastal water levels and flooding where coupled processes may alter elevation by leading-order amounts.* review statement, review [632]
2. A quasi-global tide-gauge analysis separates total-water-level and surge maxima to assess spatial and temporal change in coastal extreme high waters. *Regime: Changes in extreme high water levels based on a quasi-global tide-gauge data set* direct finding, field [958]

45.2 Tidal propagation

Topic tides. astronomical. Amplitude, phase, and constituent propagation.

Papers screened: 10. Claims: 8. Equations: 0.

Well established. Astronomical tides are predictable responses to gravitational forcing whose coastal amplitude and phase are transformed by basin geometry, depth, friction, resonance, rotation and nonlinear interaction; local prediction therefore requires both constituent forcing and regional calibration or validation.

Governing physics. Lunar and solar gravitational potential excites barotropic waves that propagate, reflect and resonate; Coriolis acceleration, bathymetric convergence, bottom friction, wetting/drying and nonlinear advection reshape constituents and generate overtides and compound shallow-water harmonics.

Dimensionless parameters. Controls include tidal amplitude-to-depth ratio, friction or Stokes number, basin length relative to tidal wavelength, Rossby number, resonance detuning, constituent signal-to-noise, admittance, amplitude and phase error, datum offset and ensemble spread-to-error ratios.

Major equations. Core formulations are rotating shallow-water mass and momentum equations with astronomical potential or open-boundary constituents, and harmonic representation $\eta(t) = Z_0 + \sum_n A_n \cos(\omega_n t + V_n + u_n - \phi_n)$, including nodal factors and phase corrections.

Typical methods. Methods use long tide-gauge records, harmonic analysis, satellite altimetry and optical shoreline constraints, global or regional hydrodynamic models, constituent interpolation or assimilation, datum reconciliation, coastal ranking or ensembles, and independent validation of amplitudes, phases and reconstructed levels.

Numerical models. Current evidence evaluates and ensembles ten global tide models at the coast rather than endorsing one universal model; underlying models differ in grids, bathymetry, friction, assimilation and constituent sets, producing spatially varying strengths and common failure regions.

Experimental datasets. The reviewed evidence uses ten global tide models, satellite altimetry, optical NDWI water masks and Australian tide gauges to rank local members and construct coastal ensembles; direct basin-specific propagation datasets remain unreviewed in this node.

Validated ranges. Direct evidence is Australia-wide but restricted to locations supported by the satellite and gauge evaluation. Ensemble benefit is conditional on at least some competent members and cannot rescue coasts where all contributing models share large errors.

Recent advances. Recent advances combine coastal altimetry, optical shoreline occurrence, unstructured high-resolution models, data assimilation, locally ranked multi-model ensembles and probabilistic prediction to improve tides near complex and data-sparse coasts.

Disagreements. Pure harmonic prediction is efficient for stationary constituents, while hydrodynamic models better represent spatial propagation and nonlinear generation but depend on bathymetry and friction. Local ensembles reduce model-selection error yet may hide common bias and do not replace physics or observations.

Limitations. Limitations include short or gappy gauges, datum mismatch, coastal altimetry contamination, optical cloud and water-index uncertainty, unresolved channels, inaccurate bathymetry and friction, missing constituents, nodal and seasonal modulation, nonstationary morphology and shared bias among global models.

Open questions. Priorities include seamless coast–estuary constituent propagation, uncertainty-calibrated ensembles, datum automation, nonstationary morphology and sea-level effects, compound tide–surge–river interaction without double counting, optical constraints in turbid coasts and prediction where gauges are absent.

Seminal papers. Equilibrium tide theory, Laplace tidal equations and classical harmonic analysis established constituent prediction; amphidromic mapping, shallow-water harmonic theory and assimilative global tide models extended it to spatially varying coastal propagation.

Claims

1. Across Australia, locally ranked ensembles of ten global tide models using altimetry and optical NDWI consistently outperformed individual models against tide gauges, but cannot provide reliable output where all contributing models are poor. *Regime: Australian coastal EO applications; NDWI ranks phase more directly than amplitude and sun-synchronous sampling limits some constituents.* direct finding, mixed [158]
2. A coastal water-level review finds interactions among mean sea level, tides, surge, waves and flooding vary strongly in space and time and can alter levels by several tens of centimeters, positively or negatively. *Regime: Interactions Between Mean Sea Level, Tide, Surge, Waves and Flooding: Mechanisms and Contributions to Sea Level Variations at the Coast* direct finding, mixed [632]
3. In mixed-tide estuaries, principal-tide duration asymmetry combines with internally generated harmonics; skewness unifies amplitude/phase metrics and shows morphology must overcome mouth-imposed asymmetry before local ebb or flood dominance develops. *Regime: Tidal asymmetry in estuaries with mixed semidiurnal/diurnal tides* direct finding, mixed [1023]
4. FES2014 combines global hydrodynamic modeling and data assimilation to produce a validated high-resolution ocean-tide atlas for major and minor constituents. *Regime: FES2014 global ocean tide atlas: design and performance* direct finding, mixed [879]
5. A high-resolution global dataset jointly represents extreme sea levels, tides and storm surges and supplies future projections for consistent coastal hazard analysis. *Regime: A High-Resolution Global Dataset of Extreme Sea Levels, Tides, and Storm Surges, Including Future Projections* direct finding, numerical [995]
6. Abstract One of the most challenging areas in tidal analysis is the study of nonstationary signals with a tidal component, as they confront both current analysis methods and dynamical understanding. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [936]
7. Abstract This study examines the integrated influence of sea level rise (SLR) and future morphology on tidal hydrodynamics along the Northern Gulf of Mexico (NGOM) coast including seven embayments and three ecologically and economically significant estuaries. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1082]
8. In semi-closed convergent tidal channels, cross-section convergence and friction control resonant amplification and shift water-level and velocity nodes and antinodes; a multi-reach analytical solution incorporates longitudinal depth variation at negligible cost relative to full numerical simulation. *Regime: Semi-closed convergent channels with width/depth variation representable by piecewise reaches and tides within the linearized one-dimensional assumptions.* direct finding, analytical [223]

45.3 Residual circulation

Topic tides.residual. Tidally averaged transport.

Papers screened: 12. Claims: 10. Equations: 5.

Well established. Residual circulation is the nonzero flow and transport remaining after tidal averaging. In estuaries and inlets it commonly reflects density-driven exchange, nonlinear tidal asymmetry, freshwater input, mixing, and geometry rather than the oscillatory tide alone.

Governing physics. Longitudinal baroclinic pressure gradients drive exchange flow, while vertical and lateral mixing, tidally averaged advection, sills, constrictions, basin interconnection, river discharge, and ebb-flood asymmetry set its magnitude and vertical or lateral structure.

Dimensionless parameters. Important controls include Rayleigh number, Fischer number, the lateral-advection parameter, depth-to-width aspect ratio, ratios of lateral to vertical mixing, and comparisons between cross-sectional mixing time and the tidal averaging interval.

Major equations. The reviewed frameworks use tidally averaged momentum and salt conservation, advection-diffusion equations, Hansen-Rattray/Pritchard exchange-flow theory, one-dimensional gradient-dispersion closure, and two- or three-dimensional numerical mass and momentum balances.

Typical methods. Evidence combines current, water-level, salinity and water-quality observations with analytical reduction, nondimensional regime analysis, breadth-averaged finite differences, and calibrated three-dimensional unstructured-grid modeling.

Numerical models. Models span one-dimensional tidally averaged dispersion, idealized Hansen-Rattray inlet circulation, multilevel breadth-averaged vertical-plane transport, and three-dimensional unstructured-grid Puget Sound circulation.

Experimental datasets. Key evidence includes multi-station observations across Gazi Bay, historical and 2006 Puget Sound velocity-salinity profiles, Oosterschelde salinity and velocity distributions, and Gulf of Khambhat hydrodynamic and transport information.

Validated ranges. Validated or demonstrated regimes range from 3-4 h residence and 60-90% exchange per tide in shallow Gazi Bay to fjordal and partially mixed Puget Sound sub-basins; analytical results also cover constant-section inlets and mixing/advection regimes defined by nondimensional parameters.

Recent advances. Recent analytical work shows that lateral mixing and tidally averaged advection fundamentally control gravitational exchange through Fischer and advection parameters, including a negative feedback in which stronger advection suppresses its driving lateral salinity gradients.

Disagreements. There is no single transferable closure: simple analytical exchange models can reveal controls but require site-specific parameterization in complex sill-basin systems, while gradient-dispersion closure fails when averaging is too short relative to cross-sectional mixing or residual cells dominate dispersion.

Limitations. Tidal averaging can conceal intratidal covariance and asymmetry. Breadth or section averaging suppresses lateral structure, idealized constant geometry limits inlet solutions, and calibrated three-dimensional models remain sensitive to bathymetry, mixing closure, boundaries, and sparse long-term observations.

Open questions. Priority questions concern interactions among tides, stratification, wind, waves, river flow and morphology; transfer of mixing closures across sites; spring-neap and seasonal modulation; and uncertainty in translating Eulerian residual flow into net material transport and residence time.

Seminal papers. The 1978 high-Rayleigh inlet solution and 1982 conditions for gradient-type tidally averaged dispersion provide early analytical foundations; the Gazi Bay and Puget Sound studies anchor field and multi-model evidence.

Equations

- *Hansen-Rattray coastal-plain estuary model* [1002]

$$\dot{\mathbf{u}} = f(\bar{V}\bar{S}, K_m, K_s, Q_r, \mathcal{G}, T)$$

Regime: Idealized constant-width, constant-depth coastal inlets using Pritchard/Hansen-Rattray equations; numerical solutions without the earlier Rayleigh-number restriction. Variables: $\bar{u}$ tidally averaged velocity; $\bar{S}$ mean salinity; K_m momentum mixing coefficient; K_s salt mixing coefficient; Q_r river discharge; $\mathcal{G}$ geometry and bathymetry; T tidal forcing; normalized dependency statement, not a recovered coefficient equation

- *FVCOM* [709]

$$\dot{\mathbf{u}} = f(\bar{V}\bar{S}, K_m, K_s, Q_r, \mathcal{G}, T)$$

Regime: Fjordal and partially mixed Puget Sound sub-basins; comparison to historical composite profiles and 2006 observations; year-long numerical application. Variables: $\bar{u}$ tidally averaged velocity; $\bar{S}$ mean salinity; K_m momentum mixing coefficient; K_s salt mixing coefficient; Q_r river discharge; $\mathcal{G}$ geometry and bathymetry; T tidal forcing; normalized dependency statement, not a recovered coefficient equation

- *one-dimensional tidally averaged advection-dispersion model* [375]

$$\dot{\mathbf{u}} = f(\bar{V}\bar{S}, K_m, K_s, Q_r, \mathcal{G}, T)$$

Regime: One-dimensional cross-sectionally and tidally averaged transport, illustrated with Oosterschelde salinity and velocity observations. Variables: $\bar{u}$ tidally averaged velocity; $\bar{S}$ mean salinity; K_m momentum mixing coefficient; K_s salt mixing coefficient; Q_r river discharge; $\mathcal{G}$ geometry and bathymetry; T tidal forcing; normalized dependency statement, not a recovered coefficient equation

- *three-dimensional tidally averaged estuarine circulation model* [750]

$$\dot{\mathbf{u}} = f(\bar{V}\bar{S}, K_m, K_s, Q_r, \mathcal{G}, T)$$

Regime: Idealized steady gravitational estuarine circulation with lateral transport; regimes organized by Fischer number and advection parameter. Variables: $\bar{u}$ tidally averaged velocity; $\bar{S}$ mean salinity; K_m momentum mixing coefficient; K_s salt mixing coefficient; Q_r river discharge; $\mathcal{G}$ geometry and bathymetry; T tidal forcing; normalized dependency statement, not a recovered coefficient equation

- *multilevel breadth-averaged circulation-transport model* [652]

$$\dot{\mathbf{u}} = f(\bar{V}\bar{S}, K_m, K_s, Q_r, \mathcal{G}, T)$$

Regime: Gulf of Khambhat, approximately 3120 km²; breadth-averaged vertical-plane representation. Variables: $\bar{u}$ tidally averaged velocity; $\bar{S}$ mean salinity; K_m momentum mixing coefficient; K_s salt mixing coefficient; Q_r river discharge; G geometry and bathymetry; T tidal forcing; normalized dependency statement, not a recovered coefficient equation

Claims

1. In shallow Gazi Bay, a wide unrestricted entrance enabled 60-90% volume exchange per tidal cycle and 3-4 h residence times, while ebb-dominant mangrove-creek currents promoted net export toward seagrass beds. *Regime: Shallow semi-enclosed Gazi Bay; mangrove creeks, seagrass beds, and coral-reef zone; observed currents about 0.6 m/s in creeks and below 0.30 m/s offshore.* direct finding, field [725]
2. Across Puget Sound sub-basins, residual two-layer circulation depends on sills and basin geometry plus freshwater-driven salinity gradients; a calibrated 3-D unstructured-grid model reproduced observed inter-basin velocity and salinity-profile differences that a simple analytical model required site-specific parameters to represent. *Regime: Fjoldal and partially mixed Puget Sound sub-basins; comparison to historical composite profiles and 2006 observations; year-long numerical application.* direct finding, mixed [709]
3. Gradient-type dispersion in a one-dimensional tidally averaged estuary model is justified when the averaging interval, measured in tidal periods, exceeds the cross-sectional mixing timescale; the requirement can be weaker when residual vertical or horizontal circulations do not dominate dispersion. *Regime: One-dimensional cross-sectionally and tidally averaged transport, illustrated with Oosterschelde salinity and velocity observations.* direct finding, mixed [375]
4. Tidally averaged gravitational estuarine circulation depends on both the Fischer number and a lateral-advection parameter; as advection strengthens it suppresses the lateral salinity gradients driving lateral flow, producing a negative feedback that approaches a limiting exchange-flow structure. *Regime: Idealized steady gravitational estuarine circulation with lateral transport; regimes organized by Fischer number and advection parameter.* direct finding, analytical [750]
5. A fully nonlinear multilevel breadth-averaged model using a semi-explicit finite-difference scheme jointly represented tidal circulation, salinity intrusion, and suspended-sediment transport in the Gulf of Khambhat, subject to the limitations of a vertical-plane breadth average. *Regime: Gulf of Khambhat, approximately 3120 km²; breadth-averaged vertical-plane representation.* direct finding, numerical [652]
6. In an idealized constant-width and constant-depth coastal inlet, increasing Rayleigh number strengthens vertical advection, homogenizes salinity within the upper and lower layers, and sharpens a halocline near mid-depth. *Regime: Idealized constant-width, constant-depth coastal inlets using Pritchard/Hansen-Rattray equations; numerical solutions without the earlier Rayleigh-number restriction.* direct finding, analytical [1002]
7. At Paopao Bay, weak tides near amphidromic points allowed episodic wave setup to dominate reef-lagoon exchange; the inferred capture width was about 2.3 km and large waves suppressed two-layer pass exchange. *Regime: Episodic circulation and exchange in a wave-driven coral reef and lagoon system* direct finding, mixed [586]
8. A 5' Sea of Okhotsk model found over 60% of diurnal tidal energy dissipated in Shelikhov Bay and Penzhinskaya Guba, while nonlinear diurnal interaction over Kashevarov Bank generated residual flow near 10 cm/s and much of local M2 current. *Regime: Tides in the Sea of Okhotsk* direct finding, mixed [739]
9. A Changjiang review finds tides, bathymetry, discharge, winds and remote shelf currents jointly control momentum, freshwater pathways, plume expansion and cross-shelf exchange, with seasonally reversing coastal circulation. *Regime: Advances on Coastal and Estuarine Circulations Around the Changjiang Estuary in the Recent Decades (2000-2020)* direct finding, mixed [846]
10. Abstract Tidal rivers are a vital and little studied nexus between physical oceanography and hydrology. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [608]

45.4 Stratified coastal circulation

Topic tides.stratified. Density-influenced circulation.

Papers screened: 20. Claims: 20. Equations: 0.

Well established. Density stratification converts barotropic tidal forcing into baroclinic circulation, internal tides, fronts and phase-dependent turbulence; mixing is intermittent and localized by topography, pycnocline structure, lateral gradients, tidal phase and wind.

Governing physics. Tidal flow over topography generates internal waves and beams; shoaling steepens them into bores and shear instability; tidal straining alternately strengthens and weakens stratification; friction and lateral density gradients rectify oscillations into residual circulation; mixing feeds back on stratification.

Dimensionless parameters. Controls include Richardson, Froude, Reynolds and Stokes numbers, tidal excursion/topographic scale, internal-wave frequency/ N and Coriolis ratio, pycnocline depth/depth, barotropic speed/internal-wave speed, Simpson-Hunter h/u^3 , bottom drag, stratification and lateral-gradient strength.

Major equations. Frameworks use Boussinesq momentum and density transport, buoyancy frequency N^2 , gradient Richardson number, internal-wave dispersion and energy flux, turbulent kinetic-energy/dissipation budgets, buoyancy flux and mixing efficiency, baroclinic pressure, residual covariance, and Simpson-Hunter or Stokes mixing criteria.

Typical methods. Methods include microstructure profilers, ADCP and moored thermistor/salinity chains, dye/tracer and ship sections, PIV and bottom turbulence, internal-wave beam/energy diagnostics, harmonic and residual decomposition, hydrostatic/nonhydrostatic models, idealized channels, and mixing/energy budgets.

Numerical models. Models represent internal-tide mode cascade, Arctic tidal mixing, lateral and residual estuarine circulation, Stokes-number water-column regimes, stratified inlet circulation, wind-modulated dissipation, and minor-axis straining that concentrates sediment.

Experimental datasets. Evidence spans the New England shelf, partially stratified estuaries, continental slopes, barrier reefs, Kaena Ridge, Monterey Bay, East China Sea, a coastal embayment, tidal inlets, periodically stratified channels and the Arctic Ocean.

Validated ranges. The evidence covers well-mixed through partially and strongly stratified water columns, bottom boundary layers through pycnoclines, estuary and inlet to shelf/slope and Arctic scales, low- and high-mode internal tides, semidiurnal and spring-neap variability, and wind-upwelling modulation.

Recent advances. Recent advances coordinate internal-wave process teams, resolve low-to-high-mode energy cascade, measure straining with microstructure across environments, connect winds to semidiurnal dissipation, and diagnose lateral stratification effects in inlets and freshwater-influenced seas.

Disagreements. A water column classified as well mixed in the mean can still undergo tidal straining; mixing is not spatially uniform along beams or slopes; bottom stress alone does not predict dissipation when internal waves shoal; and residual circulation needs lateral as well as vertical stratification asymmetry.

Limitations. Microstructure captures limited space/time; mixing efficiency is uncertain; hydrostatic models miss overturning; internal-wave beams interact with rough three-dimensional topography and mesoscale flow; boundary conditions and unresolved turbulence strongly affect energy closure.

Open questions. Priorities include complete barotropic-to-baroclinic energy pathways, mode conversion and scattering, wave-mean and wave-mesoscale interaction, lateral/submesoscale fronts, mixing efficiency, sediment/biology feedback, changing climate stratification, and transferable parameterizations.

Seminal papers. The branch links estuarine turbulence and periodic stratification to tidal/lateral asymmetry, internal-tide beams and ridge mixing, straining in well-mixed channels, frictional shelf-sea regime maps, and minor-axis sediment convergence.

Claims

- Internal-wave-driven mixing must be parameterized from generation through propagation, breaking, and dissipation to represent its climate-scale effects. *Regime: Climate Process Team on Internal Wave-Driven Ocean Mixing* review statement, review [894]
- Late-summer New England shelf mixing occurs through nonlinear internal bores, shear instability, and interactions with background stratification. *Regime: Mixing on the Late-Summer New England Shelf—Solibores, Shear, and Stratification* direct finding, field [893]
- Microstructure observations in a partially stratified estuary show turbulence and mixing vary strongly over the tide and through the water column. *Regime: Observations of Turbulence in a Partially Stratified Estuary* direct finding, field [1343]
- Continental-slope boundary mixing concentrates where stratified flow and internal waves interact with topography. *Regime: Observations of Boundary Mixing over the Continental Slope* direct finding, field [988]
- Shoaling internal waves over a barrier reef modify bottom-boundary-layer turbulence and mixing beyond locally forced tidal stress. *Regime: The Modification of Bottom Boundary Layer Turbulence and Mixing by Internal Waves Shoaling on a Barrier Reef* direct finding, field [322]
- Continental-slope topography transfers tidal energy from low to higher internal modes, changing propagation and dissipation pathways. *Regime: The Cascade of Tidal Energy from Low to High Modes on a Continental Slope* direct finding, numerical [698]
- Deep Kaena Ridge observations resolve intermittent tidal mixing events tied to internal-tide/topographic interaction. *Regime: Tidal Mixing Events on the Deep Flanks of Kaena Ridge, Hawaii* direct finding, field [86]
- Including tides changes Arctic water-mass mixing and sea ice through enhanced vertical and lateral exchange. *Regime: The effects of tides on the water mass mixing and sea ice in the Arctic Ocean* direct finding, numerical [877]
- Tidal and lateral asymmetries in stratification are essential contributors to residual circulation in partially mixed estuaries. *Regime: The Importance of Tidal and Lateral Asymmetries in Stratification to Residual Circulation in Partially Mixed Estuaries** direct finding, analytical [1275]
- Internal tidal beams focus energy and mixing near Monterey Bay rather than producing spatially uniform dissipation. *Regime: Internal tidal beams and mixing near Monterey Bay* direct finding, field [670]
- Microstructure observations demonstrate tidal straining even in a nominally well-mixed channel. *Regime: Evidence of tidal straining in well-mixed channel flow from micro-structure observations* direct finding, field [134]
- A Stokes-number framework relates frictional tidal dynamics to whether shelf-sea water columns remain mixed or stratify. *Regime: On the use of the Stokes number to explain frictional tidal dynamics and water column structure in shelf seas* direct finding, analytical [1329]
- East China Sea observations show tidal straining couples water-column stratification to near-bottom turbulence across distinct ocean environments. *Regime: Observations of Tidal Straining Within Two Different Ocean Environments in the East China Sea: Stratification and Near-Bottom Turbulence* direct finding, field [1559]
- Density stratification modifies lateral circulation in a tidal inlet through cross-channel pressure gradients, shear, and mixing. *Regime: Lateral Circulation in a Partially Stratified Tidal Inlet* direct finding, mixed [307]
- Wind-driven upwelling modulates the semidiurnal cycle of turbulent dissipation in a stratified coastal embayment. *Regime: Modulation of the Semidiurnal Cycle of Turbulent Dissipation by Wind-Driven Upwelling in a Coastal Embayment* direct finding, field [241]
- Minor-axis tidal straining in freshwater-influenced regions creates turbidity-maximum zones by coupling lateral density gradients, turbulence, and sediment convergence. *Regime: The Formation of Turbidity Maximum Zones by Minor Axis Tidal Straining in Regions of Freshwater Influence* direct finding, numerical [452]
- Internal-tide theory indicates dominant low modes are generated at 20–100 km scales and radiate $O(1,000 \text{ km})$; under 30% of energy is generated at smaller locally dissipative scales, with Mid-Atlantic Ridge M2 flux estimated at 3–5 mW/m². *Regime: The Role of Internal Tides in Mixing the Deep Ocean* direct finding, mixed [774]
- Monterey Canyon observations found semidiurnal internal currents near 20 cm/s and 50 m isopycnal displacements along theoretical rays; seasonal stratification shifted behavior from shoreward propagation to a standing internal seiche. *Regime: Observations of the Internal Tide in Monterey Canyon* direct finding, mixed [1099]
- Northwest Pacific simulations find astronomical forcing dominates internal-tide conversion magnitude, local stratification errors can exceed 30%, background shear refracts diurnal tides, and nonlinear terms contribute about 5% near Luzon Strait. *Regime: Investigation of the Internal Tides in the Northwest Pacific Ocean Considering the Background Circulation and Stratification* direct finding, mixed [1320]
- Southern California observations and a one-dimensional hydrodynamic model associate lunar-cycle temperature and diatom variability with bottom tidal mixing, while a 2–3 day prediction lag indicates unresolved internal-bore or lateral processes. *Regime: Seasonally stratified coastal waters of the Southern California Bight under barotropic tides, surface heating, and possible internal bores.* direct finding, mixed [1328]

46 Wiki: Tsunamis and long waves

46.1 Tsunamis and long waves: overview

Topic *tsunami*. Generation, propagation, inundation, and loading by long waves.

Papers screened: 19. Claims: 2. Equations: 0.

Well established. Tsunamis arise from rapid water-column displacement, propagate predominantly as long waves, and amplify through shoaling, focusing and interaction with coastal topography. Earthquake rupture dominates global hazard, while landslides and volcanic collapse can create especially severe near-field waves.

Governing physics. Generation maps seafloor or mass-motion kinematics into free-surface displacement and momentum. Propagation is controlled by gravity, depth, dispersion and bathymetric refraction; near shore, nonlinearity, breaking, friction, buildings and terrain control runup and inundation.

Dimensionless parameters. Key controls include wave amplitude-to-depth ratio, wavelength-to-depth ratio, Froude and Reynolds numbers, relative slide thickness and density, slope, dispersion measures, roughness and blockage ratios, and source probability or return period.

Major equations. Linear long-wave and shallow-water equations describe much basin propagation; nonlinear shallow-water and Boussinesq systems add finite amplitude and dispersion. Navier–Stokes or multiphase models resolve energetic slide generation, while probabilistic hazard integrates source occurrence and conditional inundation response.

Typical methods. The literature combines post-event surveys, gauges, geodesy, seismic and marine mapping, laboratory long-wave experiments, source inversion, shallow-water/Boussinesq/Navier–Stokes simulation, benchmark suites, probabilistic ensembles and emerging image or neural-network forecasts.

Numerical models. Operational and research systems range from depth-averaged nonlinear shallow-water solvers such as Tsunami-HySEA to dispersive Boussinesq propagation and three-dimensional Navier–Stokes source models. Ensembles represent slip, fault, landslide and model uncertainty.

Experimental datasets. Root evidence includes the 1993 Hokkaido, 1998 Papua New Guinea, 2011 Tohoku and 2018 Anak Krakatau events; Oregon and New Zealand hazard scenarios; NTHMP benchmarks; controlled runup experiments; and coastal CCTV-derived shoreline observations.

Validated ranges. The reviewed cases span laboratory runup, local landslide and volcanic sources, regional earthquake inundation and basin propagation. Reported examples include 10–18 m/s inferred overland flow and near-30 m runup at Hokkaido and locally about 13 m runup after Anak Krakatau.

Recent advances. Recent work uses heterogeneous rupture ensembles, coupled high-fidelity source and efficient propagation models, benchmarked GPU solvers, neural-network inundation forecasts and automated coastal imagery to shorten the path from observation to actionable warning.

Disagreements. Source attribution can remain contested where earthquake and landslide mechanisms overlap. Efficient depth-averaged models enable ensembles but may omit dispersive or three-dimensional generation physics; deterministic reconstructions can fit observations without uniquely identifying the source.

Limitations. Sparse near-source observations, uncertain rupture or slide kinematics, bathymetric and topographic resolution, friction and building representation, breaking treatment, uncertain event rates and limited validation of extreme regimes constrain transferable predictions.

Open questions. Priorities include faster source characterization, reliable near-field warning, compound and cascading sources, probabilistic treatment of rare landslides, urban-scale inundation and loads, real-time assimilation, explainable machine learning, and climate-conditioned exposure.

Seminal papers. Classical long-wave and shallow-water theory established propagation and runup foundations; later benchmark programs standardized model verification, while post-event surveys connected observed runup and damage to source reconstruction and inundation physics.

Claims

1. Tsunami hazard assessment must explicitly distinguish earthquake, landslide and volcanic-collapse sources because their characteristic kinematics, directionality and near-field uncertainty require different source and modeling assumptions. *Regime: Multi-source tsunami hazard assessment, especially where submarine or volcanic mass movements are plausible.* review statement, review [885]
2. At Palaikastro, Crete, an erosional, chaotic deposit containing Santorini ash intraclasts, marine shells and microfauna, imbricated beach pebbles, building stones, ceramics and bones provides convergent geoarchaeological evidence for a tsunami coeval with the Late Minoan Santorini eruption. *Regime: Late Minoan IA coastal settlement deposits at Palaikastro, northeastern Crete.* direct finding, field [199]

46.2 Tsunami generation

Topic *tsunami*. generation. Seismic, landslide, and volcanic sources.

Papers screened: 18. Claims: 15. Equations: 2.

Well established. Tsunami generation converts rapid displacement of water by fault motion, landslides, volcanic processes or impact into an initial free-surface and velocity field; source geometry, kinematics and coupling determine amplitude, spectrum and directivity before propagation physics acts.

Governing physics. Seismic sources deform the seabed through elastic fault slip, while landslides transfer momentum through moving, deforming solid or granular masses; generation depends on displacement rate relative to gravity-wave response, water depth, compressibility or air effects, rheology and bathymetry.

Dimensionless parameters. Controls include source length and width relative to depth, displacement-to-depth ratio, rise time over gravity-wave time, slide Froude number, relative slide thickness and density, slope, aspect ratio, submergence, mobility, Mach or compressibility effects where relevant, and normalized wave amplitude.

Major equations. Common formulations combine elastic dislocation or prescribed moving-bottom kinematics with shallow-water, Boussinesq, potential-flow or Navier–Stokes equations; landslide models add rigid-body dynamics or continuum rheology and multiphase interface coupling.

Typical methods. Workflows construct source geometry and kinematics, choose active moving-bottom or passive initial displacement, test conservation and resolution, validate near-source waves against analytical or laboratory data, propagate ensembles, compare with gauges/runup, and invert observations without hiding nonuniqueness.

Numerical models. Model classes include passive and active seismic initialization, kinematic seafloor forcing, depth-integrated landslide source terms, nonhydrostatic and potential models, depth-resolved Navier–Stokes, particle or meshless solvers, and coupled slide–water multiphase methods.

Experimental datasets. The reviewed slice includes canonical plane-beach and island tests used with regional seismic scenarios, a corrected two-segment Catania source, and the landslide-model benchmark literature summarized by the focused review; direct volcanic evidence is not yet reviewed.

Validated ranges. Current direct evidence is limited to two Eastern Mediterranean seismic scenario formulations and a review-level landslide model taxonomy; source-specific parameter ranges, laboratory scales and transfer to volcanic or impact tsunamis remain incomplete.

Recent advances. Recent advances use kinematic nonhydrostatic seafloors, depth-resolved landslide CFD, improved rheology, coupled multiphase solvers, fast surrogate or reduced source ensembles, high-rate geodesy and joint inversion of seismic, pressure, wave and runup observations.

Disagreements. Passive initial-surface displacement is efficient but can omit finite rupture and water-column dynamics; active kinematic forcing is more complete but uncertain. Depth-integrated landslide models enable ensembles but miss fine-scale generation, while depth-resolved models add physics at much higher cost.

Limitations. Source observations are sparse and inverse solutions nonunique; rupture, rigidity, slide volume, rheology and timing are uncertain; bathymetry filters the generated spectrum; simplified initial conditions may double-count or omit dynamics; and validation often occurs after propagation has mixed source and path errors.

Open questions. Priorities include finite-fault and splay rupture, horizontal displacement on slopes, deformable and fragmented slides, volcanic collapse and explosion, air and compressibility, sediment entrainment, source–propagation coupling, real-time inversion and probabilistic multi-mechanism ensembles.

Seminal papers. Elastic dislocation theory and passive seafloor-to-surface displacement established seismic initialization; moving-bottom long-wave solutions and laboratory slide-impact scaling established active generation, followed by depth-integrated and multiphase computational models.

Equations

- *Active tsunami-generation continuity equation* [1245]

$$\zeta_t + \nabla \cdot (hU) = \zeta_{b,t}$$

Regime: Active bottom-displacement option; the regional scenarios used passive free-surface generation. Variables: ζ free-surface elevation; h total water depth $d + \zeta$; U depth-averaged horizontal velocity vector; ζ_b bottom displacement

- *Earthquake magnitude to initial tsunami amplitude* [1245]

$$\log_{10} \zeta_0 = 0.98M - 6.92$$

Regime: Empirical source construction adopted from cited prior work, not newly validated here. Variables: ζ_0 initial wave amplitude; M mainshock magnitude

Claims

1. The corrected two-segment source has seismic moment 6.41×10^{21} Nm and moment magnitude 7.17 when rigidity is 30 GPa. *Regime: Corrected source definition for the paper's eastern-Sicily scenario; supersedes the original Table 2 caption.* direct finding, analytical [1410]
2. A landslide-tsunami review distinguishes efficient depth-integrated source models from computationally demanding depth-resolved approaches that represent transient fine-scale multiphase generation physics. *Regime: Numerical Modeling of Generation of Landslide Tsunamis: A Review* review statement, review [791]
3. Landslide-tsunami case studies show that combining marine-geological constraints with fully nonlinear generation and Boussinesq propagation models can reproduce source-to-coast behavior while preserving uncertainty in slide geometry and motion. *Regime: Landslide tsunami case studies using a Boussinesq model and a fully nonlinear tsunami generation model* direct finding, mixed [1494]
4. Reconstruction of the 2018 Anak Krakatau event supports lateral collapse into a roughly 250 m deep caldera as the tsunami source and reproduces observed coastal effects, including reported runup locally reaching about 13 m. *Regime: Modelling of the tsunami from the December 22, 2018 lateral collapse of Anak Krakatau volcano in the Sunda Straits, Indonesia* direct finding, mixed [537]
5. Marine evidence and tsunami modeling indicate that submarine mass movement may have augmented locally extreme waves during the 2011 Tohoku tsunami, although the relative landslide contribution remains uncertain. *Regime: Did a submarine landslide contribute to the 2011 Tohoku tsunami?* direct finding, mixed [1398]

6. A comprehensive observational and modeling synthesis for the 1998 Papua New Guinea tsunami supports a submarine-slump source capable of explaining the unusually severe local runup. *Regime: The Papua New Guinea tsunami of 17 July 1998: anatomy of a catastrophic event* direct finding, mixed [1397]
7. Landslide-tsunami generation and impact depend on slide volume, geometry, acceleration and kinematics; available scaling relations organize these controls but do not eliminate source and transfer uncertainty. *Regime: On the characteristics of landslide tsunamis* direct finding, mixed [885]
8. A three-dimensional Navier–Stokes collapse calculation coupled to two-dimensional Boussinesq propagation provides a multiscale framework for estimating the near-field waves from a potential Cumbre Vieja flank collapse. *Regime: Numerical modeling of tsunami waves generated by the flank collapse of the Cumbre Vieja Volcano (La Palma, Canary Islands): Tsunami source and near field effects* direct finding, mixed [6]
9. Tohoku tsunami simulations show that inferred coastal response is sensitive to both heterogeneous slip distribution and assumed fault geometry, so source-model uncertainty must be propagated into hazard estimates. *Regime: Sensitivity of tsunami wave profiles and inundation simulations to earthquake slip and fault geometry for the 2011 Tohoku earthquake* direct finding, mixed [508]
10. Glacial retreat in recent decades has exposed unstable slopes and allowed deep water to extend beneath some of those slopes. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [595]
11. Abstract. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [530]
12. A smoothed-particle hydrodynamics sediment model represents deforming soil-mass motion and water coupling for landslide-tsunami generation. *Regime: SEDIMENT MODEL FOR LANDSLIDE TSUNAMIS USING SMOOTHED PARTICLE HYDRODYNAMICS* direct finding, numerical [1431]
13. SWASH simulations across six water-body geometries show landslide-tsunami far-field decay can be correlated by wavefront length from two-dimensional confinement through three-dimensional spreading; 2D and 3D far-field heights differ by over an order of magnitude, and dispersion matters most for deeper-water Stokes-like waves. *Regime: Far-field landslide-tsunami propagation in constant-depth idealized water bodies spanning confined flumes to unconstrained basins.* direct finding, numerical [1222]
14. Granular landslide impact generates a rapidly evolving three-phase flow in which slide penetration, flow separation, impact-crater collapse, and air entrainment control the initial impulse wave; non-breaking waves remain approximately irrotational, whereas breaking waves and bores develop strong vorticity. *Regime: Two-dimensional Froude-scaled granular slides impacting still water and generating non-breaking waves, separated crater flows, or dissipative transient bores.* direct finding, experimental [462]
15. For the 2011 Tohoku tsunami, a geodetically inverted three-dimensional finite-element coseismic source with heterogeneous forearc structure reproduced near- and far-field observations better overall than a simpler seismic-inversion source, but accurate extreme coastal runup still required finer bathymetry, topography, and possibly additional generation mechanisms. *Regime: The Mw 9.0 2011 Tohoku-Oki coseismic tsunami from generation through trans-oceanic propagation and inundation along Japan between about 35 and 41 degrees north.* direct finding, numerical [536]

46.3 Tsunami inundation

Topic tsunami. inundation. Overland flow and runup.

Papers screened: 28. Claims: 30. Equations: 2.

Well established. Tsunami-model evaluation distinguishes validation of the numerical solution against the governing equations from verification against geophysical observations; both are required for defensible inundation use.

Governing physics. The reviewed model links source displacement, nonlinear dispersive propagation, breaking, runup/rundown, and inland wetting through a higher-order Boussinesq framework with surface-roller breaking and dry-bed boundaries.

Dimensionless parameters. Validation regimes were characterized by relative solitary-wave height H/d and normalized runup R/H ; the conical-island cases tested $H/d=0.10$ and 0.20 .

Major equations. The formulation uses higher-order Boussinesq continuity and momentum equations, with a bottom-displacement time derivative available for active generation and alternative passive free-surface displacement used in the regional scenarios.

Typical methods. Workflows combine a source scenario, nested bathymetry and topography, stable wetting/drying, spatial friction or explicit structures, grid and parameter checks, comparison with laboratory or post-event observations, and mapped extent, depth, velocity, arrival time or load outputs.

Numerical models. The reviewed models span different physics: UBO-TSUFUD solves nondispersive nonlinear shallow-water equations with Manning friction, whereas the Eastern Mediterranean model uses higher-order Boussinesq dispersion and a surface-roller breaking treatment.

Experimental datasets. Two independent laboratory benchmarks were used: solitary-wave runup/rundown on a 1:19.85 plane beach and directional maximum runup around a 14-degree conical island.

Validated ranges. Validation remains benchmark-specific: UBO-TSUFUD covers four analytical/laboratory tests including a pre-breaking Monai comparison through 30 s, while the higher-order Boussinesq model covers a 1:19.85 plane beach and a 14-degree conical island at $H/d=0.10$ and 0.20 .

Recent advances. Scenario outputs demonstrate strong site dependence rather than transferable constants: reported penetration is about 100 m at La Plaia and 500 m near the Simeto mouth in the corrected Catania scenario, while separate one-meter-source cases report 3.429 km² and 0.641 km² inundation at selected Crete and Sicily sites.

Disagreements. Published-record corrections and inconsistencies must be carried with model benchmarks: the UBO-TSUFUD source caption was formally corrected to $M_0=6.41e21$ Nm and $M_w=7.17$, while the Boussinesq paper reports its plane-beach H/d as 0.25 in a caption and 0.28 in prose.

Limitations. The regional results are representative numerical scenarios, not historical reconstructions or observed inundation validation, and must not be linearly rescaled to other initial amplitudes because the simulated processes are nonlinear.

Open questions. Shared questions concern transferable roughness, coupled source-to-inundation uncertainty, representation of buildings and defenses, sediment and debris feedback, validation of velocity and loads, probabilistic maps, compound boundaries, and reliable real-time computation.

Seminal papers. Foundational inundation practice grew from nonlinear shallow-water runup theory, analytical moving-shoreline solutions, laboratory plane-beach and island benchmarks, finite-difference and finite-volume wetting/drying algorithms, and systematic post-tsunami field surveys.

Equations

- Fully reflecting coastal boundary [1245]

$$\frac{\partial \zeta}{\partial n} = 0, \quad U_n = 0$$

Regime: Solid fully reflecting coast option; inundation simulations used the dry-bed boundary condition. Variables: n landward unit normal; U_n normal depth-averaged velocity

- Long-wave surf similarity [515]

$$\xi = \sqrt{\pi} (a_0/h_0)^{-1/2} (\Omega_1^2 h_0 / (g \tan^2 \beta))^{-1/2}$$

Regime: Single sinusoidal long-wave runup on a uniform plane beach. Variables: ξ surf-similarity parameter; a_0 offshore amplitude; h_0 still-water depth; Ω_1 effective angular duration; g gravitational acceleration; β beach angle

Claims

- The modeled inundated areas were 3.429 km² for the studied south-southwest Crete site and 0.641 km² for the east-southeast Sicily site. *Regime: Only the two paper-specific 1 m initial-amplitude scenarios and available SRTM terrain (3 arcsec Crete, 1 arcsec Sicily); these are indicative scenarios, not event reconstructions.* direct finding, numerical [1245]
- The steeper studied Sicily coast produced a narrower modeled inundation zone than the studied Crete coast despite comparable inundation heights. *Regime: Comparison of the two modeled low-lying sites under their respective paper-specific scenarios and terrain datasets; not a controlled universal slope relationship.* direct finding, numerical [1245]
- The reported regional propagation and inundation results should not be linearly scaled to different initial tsunami amplitudes. *Regime: The modeled higher-order Boussinesq scenarios; this is an applicability limitation rather than a quantitative impact result.* direct finding, numerical [1245]
- In the corrected present-day Catania scenario, modeled sea penetration reached about 100 m at La Plaia and up to 500 m near the Simeto river mouth. *Regime: Corrected Mw 7.17 two-fault scenario, present-day harbor/coast, 40 m grid, two-hour simulation; not a reconstruction of the 1693 event.* direct finding, numerical [1410]
- The Catania simulation is a plausible future scenario rather than a reconstruction of the 1693 tsunami. *Regime: Interpretive limitation applying to all Catania propagation, resonance, and inundation outputs in this paper.* direct finding, numerical [1410]
- For tsunami models, validation tests whether a code accurately solves its parent equations, whereas verification tests whether it represents geophysical reality. *Regime: Tsunami propagation and inundation numerical models under the terminology adopted in this 2008 standards paper.* review statement, review [1384]
- A single benchmark comparison is insufficient evidence for using a tsunami model in inundation mapping or operational forecasting. *Regime: Models proposed for inundation maps or operational tsunami forecasts.* direct finding, review [1384]
- The recommended laboratory validation suite spans a plane beach, composite beach, conical island, complex three-dimensional beach, and three-dimensional landslide generation and runup. *Regime: Long-wave numerical models intended to cover the corresponding one- and two-horizontal-dimensional processes.* review statement, review [1384]
- For the 1:19.85 plane-beach benchmark, more than 40 experiments covered depths of 6.25–38.32 cm and H/d of 0.021–0.626, with breaking reported above $H/d=0.045$ for that beach. *Regime: Synolakis plane-beach solitary-wave experiments on a 1:19.85 slope; the threshold is not asserted for other slopes or waveforms.* direct finding, experimental [1384]
- Passing benchmark tests reduces model uncertainty but does not guarantee realistic inundation predictions for every tsunami. *Regime: Extrapolation from the reviewed analytical, laboratory, and field benchmarks to untested events.* review statement, review [1384]
- High-resolution Thailand simulations show that spatially distributed land-cover roughness changes modeled inundation depth and velocity and should be derived from site-specific vegetation and built-environment information. *Regime: The influence of land cover roughness on the results of high resolution tsunami inundation modeling* direct finding, numerical [680]
- Comparison of depth-averaged GeoClaw and three-dimensional RANS calculations evaluates the accuracy-cost tradeoff for inundation and structural forces in a built environment. *Regime: A comparison of a two-dimensional depth-averaged flow model and a three-dimensional RANS model for predicting tsunami inundation and fluid forces* direct finding, numerical [1134]
- Onagawa simulations checked against eyewitness-video measurements connect extreme urban inundation, building damage, fragility, and the influence of a bay breakwater. *Regime: Understanding the Extreme Tsunami Inundation in Onagawa Town by the 2011 Tohoku Earthquake, Its Effects in Urban Structures and Coastal Facilities* direct finding, field [18]
- Four inundation-estimation methods applied to two submarine-mass-failure scenarios at Martil demonstrate why method diversity and terrain-resolved comparison are important for robust hazard maps. *Regime: Evaluation of tsunami inundation in the plain of Martil (north Morocco): Comparison of four inundation estimation methods* direct finding, numerical [125]
- A six-grid COMCOT reconstruction of the 2004 Banda Aceh tsunami compares modeled flow depths with NOAA observations and 85 surveyed tsunami-pole records. *Regime: Reconstruction of the 2004 Tsunami Inundation Map in Banda Aceh Through Numerical Model and Its Validation with Post-Tsunami Survey Data* direct finding, field [1433]
- Three-dimensional computational experiments assess how submerged breakwaters and seawalls alter tsunami-like overland-flow hydrodynamics and forces across building arrays. *Regime: Numerical Investigation of Countermeasure Effects on Overland Flow Hydrodynamic and Force Mitigation in Coastal Communities* direct finding, numerical [316]
- A runup-response-function inversion provides probabilistic near-field sources and runup distributions from sparse runup observations and is tested synthetically and for the 2014 Iquique event. *Regime: Probabilistic Near-Field Tsunami Source and Tsunami Run-up Distribution Inferred From Tsunami Run-up Records in Northern Chile* direct finding, numerical [789]
- A rapid two-dimensional energy-conservation model incorporates slope, friction, and gravity to bridge bathtub or attenuation screening and full depth-averaged inundation simulation. *Regime: A rapid simplified method for determining tsunami inundation extent based on energy conservation* direct finding, analytical [719]
- A semi-empirical operational method combines generalized Green-law susceptibility with nested linear and small-scale flooding simulations and compares favorably with detailed modeling at two south-Italian sites. *Regime: A Semi-Empirical Approach for Tsunami Inundation: An Application to the Coasts of South Italy* direct finding, numerical [951]

20. SWEpy couples an unstructured finite-volume shallow-water solver with well-balanced positivity preservation, WENO reconstruction, and GPU acceleration for near-field inundation and far-field tsunami modeling. *Regime: SWEpy: an open-source GPU-accelerated solver for near-field inundation and far-field tsunami modeling* direct finding, numerical [466]
21. Probabilistic tsunami hazard analysis for Seaside, Oregon translates source uncertainty into inundation products, including maximum-amplitude maps for specified return periods such as 100 and 500 years. *Regime: Probabilistic tsunami hazard assessment at Seaside, Oregon, for near- and far-field seismic sources* direct finding, mixed [514]
22. The 2011 Japan disaster showed that warnings and structural countermeasures based on underestimated event size can fail during an extreme tsunami, making evacuation, redundancy, and design beyond historical assumptions essential. *Regime: Lessons Learned from the 2011 Great East Japan Tsunami: Performance of Tsunami Countermeasures, Coastal Buildings, and Tsunami Evacuation in Japan* direct finding, mixed [1375]
23. Field interpretation of the 1993 Hokkaido-Nansei-Oki tsunami indicates exceptionally energetic overland flow, with inferred velocities of roughly 10–18 m/s and runup approaching 30 m in the most severely affected setting. *Regime: Extreme inundation flows during the Hokkaido-Nansei-Oki Tsunami* direct finding, mixed [1638]
24. A convolutional neural network can rapidly forecast inundation from tsunami and geodetic observations, demonstrating a data-driven route from early measurements to spatial flood prediction. *Regime: Early forecasting of tsunami inundation from tsunami and geodetic observation data with convolutional neural networks* direct finding, mixed [904]
25. Ensembles of variable-slip Hikurangi rupture scenarios produce materially different local inundation at Napier and Gisborne, showing that rupture complexity must be represented in probabilistic tsunami hazard analysis. *Regime: Effects of rupture complexity on local tsunami inundation: Implications for probabilistic tsunami hazard assessment by example* direct finding, mixed [992]
26. Stochastic heterogeneous coseismic-slip fields generate a distribution of runup outcomes rather than a single deterministic value, providing a mechanism for quantifying source-driven inundation uncertainty. *Regime: Stochastic analysis of tsunami runup due to heterogeneous coseismic slip and dispersion* direct finding, mixed [884]
27. Long-wave laboratory experiments yield revised tsunami runup relationships and show that waveform structure, including a trough preceding the positive wave, affects the resulting maximum runup. *Regime: New tsunami runup relationships based on long wave experiments* direct finding, mixed [251]
28. Automated processing of coastal CCTV imagery can convert visible shoreline motion into wave-runup time series, offering an observational pathway for tsunami detection and situational awareness. *Regime: Automated wave runup monitoring using coastal CCTV cameras for tsunami detection* direct finding, mixed [1302]
29. Post-tsunami surveys and spectral analysis constrain the December 2018 Anak Krakatau tsunami's coastal response and provide field evidence for source and inundation reconstruction. *Regime: The December 2018 Anak Krakatau Volcano Tsunami as Inferred from Post-Tsunami Field Surveys and Spectral Analysis* direct finding, field [993]
30. COMCOT-SED forward modeling of the 2004 Lhok Nga tsunami reproduces observed deposit thickness and shows final deposits integrate repeated uprush, turnaround, backwash, erosion and redeposition, so inverse estimates from thickness alone can underestimate wave height. *Regime: The 2004 Indian Ocean tsunami at Lhok Nga, west Banda Aceh, across the modeled offshore, bay and inundation topography.* direct finding, numerical [812]

46.3.1 Urban macro-roughness

Topic *tsunami.inundation.urban_roughness*. Modification of tsunami runup and overland flow by building layouts and other large obstacles.

Papers screened: 2. Claims: 5. Equations: 2.

Well established. In the reviewed regular arrays, greater longshore obstruction and generally more rows reduced maximum runup; 45-degree rotation changed runup more than aligned-versus-staggered placement.

Governing physics. Building arrays redistribute rather than simply remove hazard: reflection and turbulent losses reduce distal runup, while contraction, retained water, delayed drainage, and local impact can increase water levels or pressures within the built zone.

Dimensionless parameters. Longshore obstruction ψ_{ls} , cross-shore obstruction ψ_{cs} , surf similarity ξ , wave nonlinearity a/h_0 , row count, and obstacle rotation organize the reviewed runup experiments.

Major equations. Depth-averaged mass and momentum equations represent inundation, with resolved obstacles imposing solid boundaries or subgrid approaches adding drag and blockage sink terms. Array response is summarized through runup ratios and obstruction measures in cross-shore and longshore directions.

Typical methods. Studies generate controlled long waves over a plane beach, place rigid obstacle arrays with varied rows, alignment, rotation and obstruction, record free surface and runup throughout and behind the array, compare unobstructed references and construct configuration-specific response nomograms.

Numerical models. The reviewed source is experiment-centered and yields empirical runup nomograms. Depth-averaged models can resolve individual buildings or parameterize subgrid blockage and drag, but those urban formulations remain outside reviewed journal evidence for this node.

Experimental datasets. Reviewed evidence comprises sinusoidal long-wave tests on a 1:40 beach with rigid 0.10 m cubes arranged in one, five and ten rows, aligned or staggered and at zero or 45-degree rotation, over surf-similarity 7.69–10.49.

Validated ranges. Goseberg's nomograms cover single sinusoidal waves with $\xi=7.69$ –10.49 on a 1:40 beach, rigid 0.10 m cubes, one/five/ten rows, and 0 or 45 degree rotation.

Recent advances. Recent advances use lidar-derived building geometry, anisotropic subgrid porosity and drag, GPU-resolved city models, movable debris and failure, hybrid regional-to-building nests, uncertainty ensembles and validation against pressure, momentum flux and drainage as well as depth.

Disagreements. Bulk Manning roughness is simple but does not separately represent blockage, directional porosity, shocks, storage or channelization. Explicit buildings resolve pathways but require fine grids and geometry; subgrid drag offers efficiency yet needs configuration- and regime-aware validation.

Limitations. Neither total runup reduction nor downstream pressure shielding is safely transferable to real cities without accounting for heterogeneous geometry, structural failure, debris, sediment, overtopping, complex topography, and the incident-wave regime.

Open questions. A common benchmark spanning array-scale runup, building-scale pressure and force, movable debris and sediment, failure, and realistic urban geometry remains needed.

Seminal papers. Classical porous-media and drag analogies treated urban resistance in depth-averaged flow; post-tsunami flume arrays and field surveys established shielding, channelization, storage and directional blockage, motivating explicit-building and subgrid urban models.

Equations

- Longshore obstruction ratio [515]

$$\psi_{ls} = \frac{\hat{b}_{mr}}{\hat{b}_{mr} + b_{st}}$$

Regime: Regular cubic macro-roughness arrays without overtopping. Variables: ψ_{ls} longshore obstruction ratio; $\hat{b}_{mr}$ effective obstacle width including rotation; b_{st} clear street width

- Cross-shore obstruction ratio [515]

$$\psi_{cs} = \frac{L_{mr}}{n\hat{b}_{mr}}$$

Regime: Regular cubic macro-roughness arrays without overtopping. Variables: ψ_{cs} cross-shore obstruction ratio; L_{mr} cross-shore length of obstacle zone; n number of obstacle rows; $\hat{b}_{mr}$ effective rotated obstacle width

Claims

1. Across the tested arrays, increasing longshore obstruction and generally increasing the number of building rows reduced maximum long-wave runup relative to the unobstructed beach. *Regime: Rigid 0.10 m cubes on a 1:40 beach, single sinusoidal waves with surf similarity 7.69-10.49, and one, five, or ten rows.* direct finding, experimental [515]
2. For staggered arrays rotated 45 degrees, additional runup reduction relative to the non-rotated reference ranged from 4.2% to 21.4% with five rows and from 13.1% to 19.6% with ten rows across the stated obstruction cases. *Regime: The paper's staggered cube configurations and tested obstruction ratios; percentages are differences between configurations, not total attenuation.* direct finding, experimental [515]
3. Changing cube rotation had a stronger effect on runup reduction than changing between aligned and staggered rows in the tested configurations. *Regime: Five- and ten-row regular arrays with 0 or 45 degree rotation on the tested beach.* direct finding, experimental [515]
4. The first obstacle row generated an offshore-propagating shock, while water retained within and behind the array drained with a time lag and could have higher local levels than the unobstructed case. *Regime: The measured rigid-array cases, especially the five-row staggered rotated example with $\psi_{ls}=0.739$.* direct finding, experimental [515]
5. The reported nomograms should not be transferred directly to heterogeneous real cities with building failure, debris, sediment transport, overtopping, or complex three-dimensional topography. *Regime: Prototype application beyond the idealized flume geometry.* direct finding, review [515]

46.4 Tsunami loads and debris

Topic *tsunami*. Loads. Hydrodynamic and debris impact loads.

Papers screened: 19. Claims: 17. Equations: 0.

Well established. Tsunami loads combine hydrostatic and drag-like demand with short impulsive/slamming peaks, uplift and overturning, trapped-air effects, and stochastic debris collision; local pressures and component load paths can govern even when total force appears moderate.

Governing physics. Load histories depend on bore depth, velocity and relative depth, wave breaking and duration, deck clearance and geometry, air compression and escape, structural motion, debris buoyancy/trajectory/contact, added water mass, and the superposition of debris and hydrodynamic force.

Dimensionless parameters. Controls include Froude number, relative bore depth, wave height and wavelength relative to deck width/clearance, blockage and girder count, air-volume and compressibility ratios, debris density and size ratios, impact velocity, stiffness/contact-time ratios, and vegetation submergence.

Major equations. Assessment uses depth-integrated momentum and drag/hydrostatic terms, pressure integration, rigid-body and finite-element response, RANS or Green–Naghdi flow, SPH fluid–solid coupling, contact impulse and added-mass models, and probabilistic collision/load descriptions.

Typical methods. Typical methods are dam-break and pneumatic long-wave experiments, force/pressure and high-speed-motion measurement, solitary-wave and transient-bore comparison, SPH–multibody and CFD–FE coupling, ALE multiphase models, surrogate response surfaces, and critical evidence reviews.

Numerical models. Validated tools include GPUSPH, DualSPHysics–Project Chrono, DualSPHysics–Abaqus, compressible ALE, Navier–Stokes/RANS, Green–Naghdi and Saint-Venant equations, plus PCE, ANN, and Kriging load surrogates.

Experimental datasets. The branch includes 1:5 flexible bridge tests, laboratory debris and dam-break benchmarks, open-girder deck validation, pneumatic reproduction of the 2004 and 2011 tsunami records, bore-pressure comparisons, and vegetation-breakage force tests.

Validated ranges. Reported benchmarks include trapped-air uplift increases of 39% on average and 148% maximum, a 32% average overturning increase, oblique-wave forces of 1901 kN horizontal and 862 kN longitudinal, prototype-equivalent tsunami durations of 2–20 min, and intact-vegetation force reduction up to 70%.

Recent advances. Recent advances combine open-source SPH with nonsmooth multibody contact and stochastic initial conditions, high-fidelity deck-geometry and oblique-wave studies, probabilistic debris-collision reviews, and vegetation-breakage-aware mitigation experiments.

Disagreements. Simplified design equations can be non-conservative; solitary waves and transient bores do not always reproduce the same load path; trapped air can raise or lower slamming while increasing quasi-static uplift; and increased girder count can amplify or reduce force depending on wave regime.

Limitations. Reduced-scale wave generation, air scaling, rigid-body assumptions, idealized debris/contact, sparse full-scale records, uncertain collision probability, limited debris ensembles and damming data, and model-dependent turbulence or pressure peaks restrict generalization.

Open questions. Priorities are probabilistic debris fields, multiple impacts and damming, deformable and negatively buoyant debris, air–water–structure coupling, structural motion/failure, scour and foundation interaction, transferable scaling, and code-level uncertainty factors.

Seminal papers. The evidence pathway progresses from physical bore and single-debris impact tests and simplified design formulas to three-dimensional pressure fields, flexible bridge response, realistic pneumatic tsunami histories, and coupled fluid–multibody contact.

Claims

1. Tsunami debris-load research is most mature for a single impacting object, with ensemble motion, damming, collision probability, and stochastic load variability still unresolved. *Regime: Single-object impact is the best-developed case, while debris ensembles, damming, collision probability, and load variability remain major gaps.* review statement, review [1029]
2. Validated SPH simulations found that an upwave service bridge and breakwater can reduce tsunami force on a main bridge, while transferring severe demand to the shielding structures. *Regime: A two-girder service-road bridge and an offshore breakwater reduced load on the main bridge, with an optimum breakwater distance, but the shielding structures themselves received strong loads.* direct finding, numerical [1499]
3. In 1:5 bridge tests, trapped air increased total uplift by 39% on average (148% maximum) and overturning moment by 32% on average, while redistributing component demand. *Regime: Trapped air increased total uplift by 39% on average and 148% at maximum and overturning moment by 32% on average; solitary-wave and bore effects differed.* direct finding, experimental [640]
4. Three-dimensional SPH–FE analysis showed localized tsunami-bore pressure peaks and found commonly used design-pressure equations could be highly non-conservative. *Regime: Peak pressures were localized and commonly used design equations could be highly non-conservative.* direct finding, numerical [1126]
5. DualSPHysics–Chrono reproduced laboratory buoyant-container trajectories and velocities, supporting coupled flow–debris simulation before impact-load prediction. *Regime: The coupled model reproduced measured container centroid trajectory and velocity with quantitatively assessed agreement.* direct finding, mixed [1223]
6. Validated ALE simulations showed girder count can either raise or lower deck forces depending on wave size through acceleration, eddy dissipation, and pressure-phase effects. *Regime: Girder count changed force non-monotonically with wave height; chamber eddies and out-of-phase pressures weakened impacts for larger girder counts under large waves.* direct finding, numerical [1542]
7. Oblique-wave CFD predicted forces up to 1901 kN horizontally and 862 kN longitudinally, with direction-dependent surrogate-model performance. *Regime: Extreme cases reached 1901 kN horizontal and 862 kN longitudinal force; third-order PCE best predicted vertical/horizontal loads and ANN longitudinal loads.* direct finding, numerical [660]
8. Pneumatic simulators reproduced recorded tsunami histories at 1:50 scale for prototype-equivalent durations of 2–20 minutes, enabling more realistic structural-load experiments. *Regime: The systems reproduced recorded 2004 and 2011 tsunami histories at 1:50 scale over 20–240 s model periods, equivalent to 2–20 min prototype durations.* review statement, review [246]
9. Laboratory-compared Navier–Stokes results support breaking solitary waves and dam-break bores as useful, though distinct, proxies for tsunami deck loads. *Regime: Breaking solitary waves and dam-break bores produced reasonably representative tsunami loading conclusions across deck elevations.* direct finding, mixed [849]
10. Bore pressure comparisons found relative depth controls impact and Green–Naghdi can approach RANS results at much lower cost, while Saint-Venant substantially simplifies the physics. *Regime: Relative depth controlled bore shape, stability, and impact; Green–Naghdi results approached RANS at far lower cost, whereas Saint-Venant omitted substantial physics.* direct finding, mixed [843]
11. Validated SPH–multibody coupling showed nonsmooth contact and randomized initial debris position are essential to reproduce debris-impact forces and their stochastic variability. *Regime: Nonsmooth contact matched measurements and avoided numerical high-frequency noise; varying initial debris position reproduced experimental variability and exposed design-equation issues.* direct finding, mixed [637]
12. Debris-collision assessment must jointly estimate collision probability, velocity, and force, with probabilistic evaluation the leading unresolved need. *Regime: Available methods trade accuracy, computation, and practicality; probabilistic evaluation remains the principal development need.* review statement, review [678]
13. Laboratory tests directly evaluated tsunami waterborne-debris impact forces against FEMA P-646 and P-55 design formulations. *Regime: Experiments quantified debris impact forces and evaluated prescribed guideline formulas.* direct finding, experimental [426]
14. Tsunami structural-load assessment relies on complementary physical experiments and numerical simulation, with design-guideline revision requiring validated use of both. *Regime: Physical testing and numerical simulation are complementary, and numerical models can represent parameters omitted from reduced physical tests.* review statement, review [522]
15. Experiments found intact vegetation reduced maximum tsunami force by up to 70%, but breakage sharply degraded protection and made patch length important. *Regime: Intact vegetation reduced maximum force by up to 70% (rigid) and 66.5% (flexible); breakage reduced protection, although longer patches partly compensated.* direct finding, experimental [1435]
16. Bridge analysis links tsunami-wave impact to connection forces in open-girder coastal bridges, showing that hydrodynamic loading must be transferred through explicit structural connections. *Regime: Deciphering the Tsunami Wave Impact and Associated Connection Forces in Open-Girder Coastal Bridges* direct finding, mixed [639]
17. Large-scale flume tests indicate that nonbreaking very-long-wave inundation at vertical seawalls produces little impulsive loading: periods of at least 40 s did not exceed about 1.2 times hydrostatic force, unlike bore-led cases. *Regime: Nonbreaking crest- and trough-led very long waves inundating vertical seawalls in a large flume.* direct finding, experimental [944]

46.4.1 Loads within building arrays

Topic *tsunami.loads.building_arrays*. Pressure and force redistribution, shielding, and amplification among coastal buildings.

Papers screened: 1. Claims: 3. Equations: 0.

Well established. Within the reviewed experiment, upstream structures shielded downstream buildings from peak breaking-wave pressure, but the amount is configuration- and regime-specific rather than a universal reduction factor.

Governing physics. Breaking location and structural setback govern impulsive loading: the largest pressures occurred when the wave broke on or immediately before the specimen.

Dimensionless parameters. Controls include Froude number, relative wave height, structure width and spacing relative to depth, blockage ratio, setback relative to breaking length, row and lateral position, inundation depth relative to building height, pressure coefficient and peak-to-hydrostatic pressure ratio.

Major equations. Array loading is assessed through pressure integration and force or moment balance, with hydrostatic, drag, momentum-flux and impulsive components. Design equations relate pressure to inundation depth or velocity, while blockage and shielding modify the local depth, velocity and breaking state used as inputs.

Typical methods. Studies generate controlled bores or long waves over a 1:20 bathymetry, vary breaking location and structure arrangement, measure free surface and high-rate pressures on multiple faces and rows, integrate loads, compare isolated and array cases and audit design equations using locally disturbed rather than bare-earth hydraulics.

Numerical models. The current evidence evaluates empirical Japanese design equations against physical-array pressures. Numerical shallow-water, nonhydrostatic or CFD array models are not yet reviewed for this node and require validation against local depth, velocity, pressure and impulse together.

Experimental datasets. Reviewed evidence is one idealized 1:20 physical model of a seaside building array under breaking and nonbreaking tsunami-like waves, resolving upstream shielding, local breaking position and comparison with Japanese Cabinet Office pressure equations.

Validated ranges. The reported 40%-70% peak-pressure reduction applies only to the study's 1:20 idealized array under breaking-wave trials; the accessible abstract does not expose the exact wave, spacing, or setback ranges.

Recent advances. Recent advances use dense pressure arrays, instrumented urban layouts, high-resolution nonhydrostatic and CFD solvers, debris and buoyancy coupling, probabilistic urban configurations and multiscale workflows connecting regional inundation to local structural demand.

Disagreements. Upstream buildings can shield downstream faces yet increase blockage, flow channelization, lateral loads or local water levels. Bare-earth hazard models may support regional inundation but can be nonconservative for structure pressure when the built array changes breaking and hydraulics.

Limitations. Bare-earth water levels can be nonconservative inputs for breaking-wave pressure design because the building array changes local inundation and impact conditions.

Open questions. Full-text extraction and additional arrays are needed to resolve spacing, setback, blockage, breaking location, three-dimensional flow, and scale effects before deriving transferable shielding factors.

Seminal papers. Tsunami force formulations began with hydrostatic and drag analogies and bore-impact experiments; post-2004 and 2011 surveys motivated building-specific pressure guidance, while later flume arrays exposed shielding, blockage and local-breaking effects.

Claims

1. Upstream shielding elements reduced peak pressure on downstream structures by 40%-70% under the study's breaking-wave conditions. *Regime: The study's idealized building array and breaking-wave trials; detailed geometry and wave ranges require full-text review.* direct finding, experimental [1419]
2. Waves breaking on or immediately before the test structure produced the largest impulsive pressures. *Regime: The paper's 1:20 structure configurations and tested wave conditions.* direct finding, experimental [1419]
3. For nonbreaking waves, measured pressures agreed with Japanese Cabinet Office design equations, whereas bare-earth water-level inputs were nonconservative in breaking-wave trials. *Regime: The paper's nonbreaking and breaking physical-model trials; exact error magnitudes are unavailable from the abstract.* direct finding, experimental [1419]

46.5 Tsunami propagation

Topic *tsunami.propagation*. Basin and shelf propagation.

Papers screened: 13. Claims: 10. Equations: 0.

Well established. Across deep ocean and shelves, tsunami propagation is governed primarily by long-wave gravity dynamics, with travel time controlled by depth and coastal amplitude shaped by shoaling, refraction, diffraction, resonance, friction, dispersion and bathymetric focusing.

Governing physics. Rapid source displacement launches long waves whose phase speed is approximately $\sqrt{gh}$ in shallow-water regimes; variable depth redirects rays and changes amplitude, shelves and ridges trap or focus energy, harbors resonate, and dispersion matters as wavelength shortens or distance accumulates.

Dimensionless parameters. Controls include relative amplitude a/h , shallowness h/L , Ursell number, Froude number, relative shelf width and slope, travel distance relative to dispersive length, numerical Courant number, cells per wavelength, normalized runup and amplitude or phase error.

Major equations. Models use linear or nonlinear shallow-water mass and momentum equations for nondispersive propagation, Boussinesq or nonhydrostatic equations for frequency dispersion, and sometimes mild-slope or spectral formulations; moving-bottom source terms and wet/dry boundaries connect generation and inundation.

Typical methods. Workflows construct nested bathymetry, initialize a tested source, verify conservation and convergence, compare analytical travel and oscillation cases, validate laboratory shelf/island/runup benchmarks, hindcast gauge and harbor records, test dispersion and friction sensitivity, and report arrival, amplitude and phase separately.

Numerical models. The evidence spans nonlinear shallow-water UBO-TSUDF and a higher-order Boussinesq model with dispersive propagation, breaking and wetting/drying, evaluated against a broader verification and validation benchmark framework.

Experimental datasets. Reviewed evidence includes plane-beach, conical-island, oscillating-basin and Monai benchmarks, the Rat Islands-to-Hilo operational test, and Eastern Mediterranean Crete, Sicily, Catania and Libyan-coast scenarios.

Validated ranges. Benchmark support covers selected analytical basins and laboratory geometries plus scenario-scale Mediterranean propagation. The operational Hilo requirement constrains speed as well as physics; none of these proves universal skill across all source spectra and ocean basins.

Recent advances. Recent advances combine higher-order dispersive and nonhydrostatic solvers, GPU and adaptive meshes, conservative unstructured nesting, real-time pressure and GNSS assimilation, probabilistic source ensembles, digital-twin harbors and operational benchmark requirements.

Disagreements. Nondispersive shallow-water models are efficient and often adequate for earthquake tsunamis, while Boussinesq or nonhydrostatic models capture accumulated dispersion and shorter components at greater cost. Apparent agreement can result from compensating source, bathymetry and friction errors.

Limitations. Uncertainty arises from source kinematics, bathymetric resolution and datum, unresolved ridges and channels, open boundaries, numerical dispersion and diffusion, friction, grid nesting, harbor geometry and sparse gauges; propagation validation can be confounded by generation and inundation errors.

Open questions. Priorities include adaptive dispersive criteria, global-to-harbor conservative nesting, uncertainty attribution among source and path, subgrid bathymetry, scattering by rough seafloor and islands, real-time assimilation, compound tides and currents, and reliable faster-than-real-time ensembles.

Seminal papers. Linear long-wave theory and Green-law shoaling established depth-controlled propagation; ray and edge-wave theory treated refraction and trapping, while finite-difference shallow-water models and later Boussinesq solvers enabled basin-scale scenarios and dispersive benchmarks.

Claims

1. In the southwest-Crete source scenario, normalized extreme water elevation locally exceeded 0.4 along the eastern Libyan coast approximately 250 km from the source. *Regime: The paper's passive-generation southwest-Crete earthquake scenario with $\zeta_0=1$ m and its specified source geometry, bathymetry, and model setup; not scalable to other amplitudes by linear multiplication.* direct finding, numerical [1245]
2. The Rat Islands-to-Hilo operational benchmark requires a model to reproduce four hours of harbor dynamics in ten minutes of computation. *Regime: The proposed 17 November 2003 Rat Islands tsunami operational forecast benchmark and its computing context in 2008.* direct finding, field [1384]
3. Tsunami-HySEA successfully exercises five National Tsunami Hazard Mitigation Program benchmark problems, supporting its use for generation, propagation, runup and inundation calculations within the tested configurations. *Regime: Performance Benchmarking of Tsunami-HySEA Model for NTHMP's Inundation Mapping Activities* direct finding, mixed [896]
4. A Tohoku generation-to-inundation hindcast using heterogeneous slip calibrated against observations reproduces important propagation and coastal-response features and illustrates the value of jointly constraining the source and wave model. *Regime: Modeling propagation and inundation of the 11 March 2011 Tohoku tsunami* direct finding, mixed [883]
5. A new approach is proposed to separate the influence of source and topography in observed tsunami spectra and to reconstruct the initial source spectrum. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1140]
6. We present numerical simulations of tsunami run-up in the near field for a data set of 72 models of sources, involving both seismic dislocations and landslides. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1043]
7. We evaluate far-field tsunami hazard in the Indian Ocean Basin based on hydrodynamic simulations of ten case studies of possible mega earthquakes at the major seismic zones surrounding the basin. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1044]
8. Abstract. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [278]
9. The key issue of this article is the concept of combining a model dedicated to dispersive large scale propagation of tsunamis with ComMIT, developed and made freely available by NOAA, that is a state of the art tool for tsunami impact studies. First, the main motivation for this approach, namely the need for efficient computation of runup of tsunamis from submarine/subaerial slides and certain types of earthquake, is discussed. Then the models involved are presented. We describe in some detail the dispersive model component which is a Boussinesq type model that is recently developed for tsunami propagation purposes. Finally, the performance and flexibility of the joint model approach is illustrated by two case studies including inundation computations at selected sites. The potentially disastrous, but small probability, flank-collapse event at the La Palma Island is used as an example of slide generated tsunamis where dispersion plays an important role. The second example is a tsunami from a potential inverse thrust fault at the Lesser Antilles. In this case dispersion during propagation is important for some regions, but not for others. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, numerical [882]
10. VOLNA simulates tsunami generation, propagation, wetting-drying and coastal run-up on unstructured triangular meshes, with analytical and laboratory benchmark comparisons supporting its use across the tsunami life cycle. *Regime: Long tsunami waves over realistic bathymetry where nonlinear shallow-water assumptions are adequate from generation through inundation.* direct finding, numerical [386]

47 Wiki: Reliability and uncertainty

47.1 Reliability and uncertainty: overview

Topic uncertainty. Uncertainty quantification and reliability of coastal systems.

Papers screened: 14. Claims: 12. Equations: 0.

Well established. Coastal uncertainty includes natural variability, observation and sampling error; parameter uncertainty, model structure and discrepancy, boundary forcing, scenario choice and deep knowledge gaps; credible decisions require separating, propagating, validating and communicating these sources rather than collapsing them into one unqualified range.

Governing physics. Uncertainty enters through stochastic forcing, nonlinear thresholds and feedback, spatial and temporal dependence, incomplete state observation, unresolved scales, model closures, changing morphology and climate, deterioration, human response and consequence valuation.

Dimensionless parameters. Controls include coefficient of variation, reliability index and failure probability, return period and exceedance probability, ensemble spread-to-error ratio, calibration and coverage, sensitivity indices, signal-to-noise, effective sample and tail size, prior-to-data information, dependence strength, robustness and regret ratios.

Major equations. Core formulations include Bayes theorem and posterior prediction, total probability and variance, likelihood and model discrepancy, limit-state reliability, variance- and moment-based sensitivity, ensemble decomposition, extreme-value and copula dependence, calibration scores, decision loss, expected utility, regret and value of information.

Typical methods. Methods define an uncertainty taxonomy and decision, audit data and models, propagate Monte Carlo/Bayesian/ensemble alternatives, perform global sensitivity and identifiability analysis, validate probabilities and tails, test scenarios and structural choices, update with observations, evaluate robustness and communicate conditional findings.

Numerical models. Tools include Bayesian networks and calibration, Monte Carlo and polynomial chaos, global sensitivity, surrogate and machine-learning UQ, multimodel and perturbed ensembles, probabilistic extremes and copulas, reliability models, robust decision making and adaptive pathways.

Experimental datasets. Evidence includes nonstationary-extreme reviews, compound river-coastal flood projections, coastal-cliff multimodel ensembles, New Orleans deep-uncertainty scenarios, environmental Bayesian networks, sensitivity-tool benchmarks, ocean-engineering ANN studies and adaptation case reasoning.

Validated ranges. The reviewed evidence spans environmental risk, coastal climate services, compound flooding, cliff retreat, infrastructure risk and ocean-engineering prediction from event to multidecadal horizons; uncertainty results remain conditional on data, priors, model/scenario ensembles and decisions considered.

Recent advances. Recent work integrates global sensitivity toolkits, Bayesian and ML uncertainty, calibrated ensembles, compound hazards, multimodel morphology projections, deep-uncertainty scenarios, model-discrepancy awareness, robustness and ethical/shared coastal adaptation.

Disagreements. Aleatory and epistemic uncertainty are not always cleanly separable; probability may be inappropriate for deep uncertainty; complex models can reduce process omission yet increase unidentifiable parameters; ensemble agreement can reflect shared bias; calibration can compensate errors; narrow intervals can be accurate on average but fail in tails.

Limitations. Extreme records are short, observations correlated and incomplete, priors and structural alternatives subjective, computational budgets restrict exploration, models share ancestry, validation domains are narrow, future scenarios lack frequencies and human adaptation and cascading consequences are poorly constrained.

Open questions. Priorities include compound-tail dependence, nonstationary extremes, structural discrepancy, shared ensemble bias, observation error covariance, rare failure validation, multi-model weighting, adaptive observation, cascading systems, human behavior, equity, deep uncertainty and trustworthy real-time calibration.

Seminal papers. Classical probability, extreme-value theory, reliability indices, Bayesian inference, Monte Carlo, sensitivity analysis and decision theory established quantitative uncertainty; ensemble forecasting, robust decision making and adaptive pathways later addressed model plurality and changing futures.

Claims

1. A review of nonstationary extremes shows that detection, attribution, probability estimation and management must distinguish changing forcing from sampling variability and model assumptions. *Regime: Nonstationary weather and water extremes: a review of methods for their detection, attribution, and management* review statement, review [1311]
2. A review of Bayesian networks for environmental risk demonstrates a transparent way to combine causal structure, heterogeneous evidence, conditional dependence and uncertain consequences. *Regime: Bayesian Networks in Environmental Risk Assessment: A Review* review statement, review [679]
3. VARS-TOOL provides a reproducible framework for global sensitivity and uncertainty analysis that compares parameter influence across scales and supports efficient diagnostic sampling. *Regime: VARS-TOOL: A toolbox for comprehensive, efficient, and robust sensitivity and uncertainty analysis* direct finding, numerical [1167]
4. Projections for northwestern Europe show that coastal and riverine flood dependence, marginal changes and scenario choices must be represented jointly when estimating future compound hazard. *Regime: Projected Changes in Compound Flood Hazard From Riverine and Coastal Floods in Northwestern Europe* direct finding, numerical [477]

5. Good-practice guidance for epistemic natural-hazard uncertainty calls for explicit assumptions, alternative plausible models, expert judgement protocols, traceability and communication rather than hiding knowledge gaps in one probability. *Regime: Epistemic uncertainties and natural hazard risk assessment – Part 2: What should constitute good practice?* review statement, review [151]
6. A multimodel ensemble for multidecadal coastal-cliff retreat propagates alternative process representations and future forcing to expose structural spread in shoreline projections. *Regime: A Model Ensemble for Projecting Multidecadal Coastal Cliff Retreat During the 21st Century* direct finding, numerical [831]
7. A New Orleans case demonstrates deep uncertainty in coastal flood risk across probabilistic scenarios, showing that decision robustness can matter more than a single best-estimate projection. *Regime: Deep Uncertainty Surrounding Coastal Flood Risk Projections: A Case Study for New Orleans* direct finding, numerical [1527]
8. Decision-oriented analysis can favor simpler simulation models when their assumptions and uncertainty are more thoroughly explored than those of computationally complex models. *Regime: Why Simpler Computer Simulation Models Can Be Epistemically Better for Informing Decisions* direct finding, analytical [582]
9. A review of artificial neural networks in ocean engineering identifies data, architecture, calibration, validation, extrapolation and uncertainty as distinct sources of predictive risk. *Regime: Uncertainties in the application of artificial neural networks in ocean engineering* review statement, review [674]
10. A coastal-adaptation perspective argues that large hydro-morphodynamic uncertainty requires transparent ethics, shared responsibility, participation and fair treatment of uncertain futures. *Regime: Coastal Adaptation and Uncertainties: The Need of Ethics for a Shared Coastal Future* review statement, review [1386]
11. A revised Atlantic hurricane database format explicitly represents track and intensity uncertainty, improving the traceability of historical storm information used in hazard applications. *Regime: Atlantic Hurricane Database Uncertainty and Presentation of a New Database Format* direct finding, mixed [767]
12. A planning review compares approaches for adaptation under deep uncertainty and emphasizes flexible pathways, robustness and iterative learning over single deterministic forecasts. *Regime: Adapt or Perish: A Review of Planning Approaches for Adaptation under Deep Uncertainty* review statement, review [1473]

47.2 Ensemble and Bayesian methods

Topic uncertainty. ensembles. Probabilistic inference and ensembles.

Papers screened: 4. Claims: 3. Equations: 1.

Well established. Coastal ensemble analysis propagates uncertain forcing, parameters, states, models and scenarios into distributions of hazards, impacts and decisions; probabilistic outputs are meaningful only when ensemble construction, dependence, weights and validation are explicit.

Governing physics. Aleatory variability in storms, waves, tides and climate combines with epistemic uncertainty in bathymetry, roughness, morphology, model structure and observations. Nonlinear thresholds, compound dependence and feedbacks can make output tails highly non-Gaussian and decision-sensitive.

Dimensionless parameters. Controls include ensemble size, coefficient of variation, effective sample size, variance-reduction factor, bias-to-sampling-error ratio, level refinement ratio, likelihood information content, posterior-to-prior variance, tail probability, reliability index and value-of-information ratio.

Major equations. Standard Monte Carlo estimates expectations and exceedance probabilities from samples; Bayesian inference updates prior parameter or state distributions with likelihoods, data assimilation conditions forecasts, and multilevel Monte Carlo combines many cheap coarse runs with fewer expensive fine runs to minimize variance at fixed cost.

Typical methods. Workflows define uncertainty sources and dependence, sample or design ensembles, run calibrated models at multiple fidelities, check convergence and tail stability, validate reliability and sharpness, decompose variance or sensitivity, update with observations and translate distributions into robust or adaptive decisions.

Numerical models. The set combines XBeach with multilevel Monte Carlo, inverse regional swell prediction with observation assimilation, and Monte Carlo damage simulation coupled to dynamic programming for adaptation timing.

Experimental datasets. Reviewed evidence includes theoretical and real XBeach runoff and erosion cases, Southern California sheltered-wave observations used for inverse offshore-spectrum estimation, and Bay County building exposure under four sea-level-rise projections.

Validated ranges. The reported 40-or-greater multilevel speedup applies to tested XBeach cases and accuracy targets; inverse recovery depended on energetic unimodal swell and observation resolution; adaptation findings are conditional on Bay County buildings and four projections.

Recent advances. Recent advances combine multifidelity and multilevel ensembles, probabilistic machine-learning emulators, differentiable inference, active learning, particle and ensemble data assimilation, storylines, large climate ensembles and adaptive pathways that update as evidence changes.

Disagreements. Large ensembles reduce sampling error but do not correct structural bias or omitted scenarios. Bayesian posteriors can be dominated by priors or likelihood assumptions when data are sparse, and equal-weight multimodel ensembles need not represent calibrated probabilities.

Limitations. Computational cost, correlated members, underdispersive forcing, uncertain tails, nonstationarity, scenario incompleteness, surrogate bias, sparse validation, model discrepancy and dependence misspecification can create false precision. Decision results also depend on loss functions and update rules.

Open questions. Priorities include calibrated compound-event dependence, rare-event sampling, structural model discrepancy, adaptive multifidelity allocation, trustworthy surrogates, sequential observation design, nonstationary validation and decision-centered metrics for robust adaptation.

Seminal papers. Monte Carlo simulation established direct uncertainty propagation; Bayesian updating and ensemble Kalman methods enabled conditioning on observations, while importance sampling, polynomial chaos and multilevel Monte Carlo improved efficiency for rare or expensive coastal calculations.

Equations

- *Dynamic adaptation Bellman recursion* [568]

$$V_t(s) = \min_a \{C_t(s, a) + \mathbb{E}[V_{t+1}(s')]\}$$

Regime: Monetized building elevation or floodproofing decisions under sampled SLR and flood damage. Variables: v_t minimum expected remaining lifecycle cost; s building and hazard state; a adaptation action; C_t current damage and adaptation cost; s' next state after hazard evolution and action

Claims

1. In a Bay County building-level analysis, dynamic programming and Monte Carlo damage simulation across four sea-level-rise projections identified single- and multi-family buildings as most vulnerable and showed that adaptation decisions should update as projection information changes. *Regime: Bay County inventory, adopted depth-damage and cost functions, four SLR pathways, and monetized floodproofing/elevation options.* direct finding, numerical [568]
2. In Southern California swell cases, inverse assimilation of sheltered observations could recover peak offshore direction for an energetic unimodal event, but sparse low-directional-resolution coastal data could not routinely resolve offshore spectra without additional offshore constraints or restrictive priors. *Regime: Southern California Bight swell at 0.04-0.12 Hz using sheltered observations and offshore constraints.* direct finding, mixed [1067]
3. In theoretical and real XBeach coastal-risk cases, multilevel Monte Carlo estimated runup and erosion uncertainty with speedup factors of 40 or more over standard Monte Carlo at the same accuracy, enabling exceedance probabilities for design decisions. *Regime: Theoretical and real coastal hydro-morphodynamic cases estimating runup and erosion distributions.* direct finding, numerical [283]

47.3 Structural and system reliability

Topic uncertainty. reliability. Failure probabilities and limit states.

Papers screened: 5. Claims: 3. Equations: 1. Shown in the Coastal Design Bench beside: Design conditions.

Well established. Coastal reliability requires a limit state connecting uncertain environmental load and uncertain resistance, followed by probability integration over an explicit time horizon. Annual design reliability, operational event probability, and observed long-term performance answer different questions and must not be conflated.

Governing physics. Flood-defence failure depends on water level, waves, overflow, runup/overtopping, erosion, seepage, piping and stability, while nature-based system persistence also depends on ecological degradation and shoreline change. System performance couples loads, resistance, spatial dependence and multiple mechanisms.

Dimensionless parameters. Reliability index beta, annual failure probability, return period, conditional event probability, load and resistance coefficients of variation, importance factors, correlation, and forecast horizon organize analyses; physical mechanisms retain overtopping, stability and wave nondimensional groups.

Major equations. Core forms are $g(R,S)=R-S$ or mechanism-specific limit states, failure probability $P[g \leq 0]$, conditional operational probability integrated over forecast-load distributions, fragility curves $P(\text{failure}|\text{load})$, and series/parallel system combinations across sections and mechanisms.

Typical methods. Methods include FORM/SORM, Monte Carlo and importance sampling, fragility integration, Bayesian updating, event trees, system reliability, forecast ensembles, time-dependent deterioration, inspection updating, and decision analysis based on consequences and warning lead time.

Numerical models. Hydra-VIJ supplies probabilistic hydraulic conditions and annual failure-frequency machinery adapted to operational windows. Broader coastal applications require hydrodynamic, wave, geotechnical, morphodynamic and damage models embedded in limit-state or fragility calculations.

Experimental datasets. The extracted branch includes Dutch Vecht-delta probability forecasts and Hydra-VIJ load/failure relations; long-term Louisiana oyster-reef monitoring supplies a contrasting ten-year observation of performance persistence rather than a calibrated structural failure probability.

Validated ranges. The operational framework was applied to dike-ring area 10 with a 1/2,000 per-year reference standard and Dutch large-ring standards from 1/10,000 to 1/1,250 per year; default mechanisms were overflow and wave runup with deterministic strength.

Recent advances. Current directions couple forecast ensembles, Bayesian updating, active learning and rare-event simulation with digital twins; coastal use must still demonstrate calibrated limit states, physical coverage and out-of-sample reliability rather than importing generic algorithms uncritically.

Disagreements. An invariant reliability index need not imply an invariant physical failure probability under inconsistent transformations or models. More generally, very small numerical probabilities are not credible when epistemic model, mechanism, dependence, or deterioration uncertainty is omitted.

Limitations. The screened discovery batch was dominated by generic mathematical reliability work without coastal validation. The retained dike case fixes resistance, covers limited mechanisms and one Dutch system; the oyster record demonstrates persistence uncertainty but not a formal fragility model.

Open questions. Needs include multi-mechanism and network reliability, correlated spatial failures, nonstationary climate loads, deterioration and inspection, nature-based degradation, joint wave-surge-rainfall forcing, operational updating, model-form uncertainty, and consequence-aware warning thresholds.

Seminal papers. Within this branch, the 2013 operational dike framework is the central applied reliability study because it connects probabilistic natural-load forecasts to event-window failure probability and decision support rather than reporting only an annual index.

Equations

- Operational dike failure probability [1524]

$$P_f(\Delta t) = \int P[g(\mathbf{R}, \mathbf{s}) \leq 0 \mid \mathbf{s}] f_{S|forecast}(\mathbf{s}; \Delta t) d\mathbf{s}$$

Regime: Hydra-VIJ-style operational dike assessment. Variables: P_f failure probability in future window; g limit-state function; $\mathbf{R}$ dike resistance variables; $\mathbf{s}$ hydraulic and meteorological load vector; $f_{S|forecast}$ forecast joint load density; Δt operational time window; normalized framework expression

Claims

- For a primary dike in the Vecht River delta, probabilistic forecasts of river discharge, Lake IJssel level, wind speed, and wind direction were propagated through Hydra-VIJ formulas to estimate operational failure probability over a future time window for overflow and wave-runup mechanisms. *Regime: Vecht-delta primary dike and Hydra-VIJ load/failure representation; overflow and wave-runup mechanisms with deterministic strength variables.* direct finding, mixed [1524]
- Abstract This paper is of methodological nature, and deals with the foundations of Risk Assessment. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1244]
- OWT (Offshore Wind Turbine) support structures are exposed to harsh ocean environment with significant uncertainties in soil properties and environmental loads. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [1480]

47.4 Sensitivity analysis

Topic uncertainty. sensitivity. Attribution of response variability.

Papers screened: 11. Claims: 11. Equations: 0.

Well established. Sensitivity analysis ranks how uncertain inputs, interactions, configurations, and scenarios influence a defined model output. Rankings are conditional on input ranges, dependence, site, output metric, model structure, and time horizon.

Governing physics. Influential controls span boundary forcing, sea level, surge and waves, topography, solver and grid, friction, breaches, damage functions, vegetation geometry, sediment pathways, structural geometry, and coupled atmosphere-ocean initial conditions.

Dimensionless parameters. Sensitivity is summarized by normalized derivatives, variance fractions, total-effect indices, interaction contributions, confidence bounds, surrogate error, and sometimes decision-specific value of information rather than a universal physical dimensionless group.

Major equations. Common tools include local derivatives, one-at-a-time screening, variance decomposition with first-order and total Sobol indices, polynomial chaos, response surfaces, Gaussian processes, factor mapping, Monte Carlo sampling, and interaction indices.

Typical methods. Good workflows define input distributions and dependence, sample the model, build or validate emulators, test convergence, compute main and interaction effects, repeat for multiple outputs and horizons, and confirm recommendations against observations or held-out simulations.

Numerical models. Models include shoreline projections, flood-damage Monte Carlo approximations, marine-flood occurrence models, COAWST vegetation, LISFLOOD-FP, UWIN-CM with polynomial chaos, Gaussian-process overtopping surrogates, and a barrier evolution model.

Experimental datasets. Reviewed evidence includes four French beaches, three Netherlands breach sites, a Mediterranean city, 17 European coastal sites with 72 configurations each, COAWST vegetation cases, Hurricane Earl ensembles, 163 overtopping tests, and an end-century barrier model.

Validated ranges. Reported effects include 20-40% shoreline variance from erosion-model choice, greater-than-fourfold 95% flood-damage bounds around the median, 17-site regional configuration patterns, and overtopping validation against 163 tests.

Recent advances. Recent studies combine global sensitivity with Gaussian processes, polynomial chaos, regional clustering, factor mapping, and efficient quadratures to expose interactions while reducing expensive coastal-model evaluations.

Disagreements. No parameter is universally dominant: surge, sea-level variability, and SLR scenarios dominate at different horizons; solvers, boundaries, and resolution dominate different European regions; friction may be secondary globally yet important locally.

Limitations. Rankings can be distorted by arbitrary input ranges, independence assumptions, inadequate sample size, surrogate bias, output aggregation, omitted structural error, nonstationarity, and lack of observational validation.

Open questions. Needs include dependent-input sensitivity, structural and scenario uncertainty, multi-output decision sensitivity, nonstationary rankings, efficient high-dimensional sampling, and robust transfer across sites.

Seminal papers. The branch builds on derivative screening, Monte Carlo uncertainty analysis, Sobol variance decomposition, polynomial-chaos expansions, and emulator-based global sensitivity; the reviewed sources demonstrate coastal uses.

Claims

1. For four contrasting sandy beaches in southwest France, the choice between two sea-level-rise erosion models accounted for 20-40% of projected shoreline-change variance by 2100 and beyond. *Regime: Four minimally managed sandy beaches in southwest France, projected to 2100 and beyond.* direct finding, numerical [305]
2. Across three western Netherlands breach sites, the 95% flood-damage range extended more than fourfold below and above the median; depth-damage curves were most influential, while damage-model and inflow-volume parameter groups contributed about equally. *Regime: Three breach locations on the western Netherlands coast with 12 uncertain surge, breach-growth, and damage inputs.* direct finding, numerical [970]
3. For the studied northwestern Mediterranean site, dominant marine-flood uncertainty shifted over time from storm-surge propagation to sea-level variability and later global sea-level-rise scenarios, so sensitivity-based research priorities depended on the planning horizon. *Regime: Low-lying northwestern Mediterranean urban site over the twenty-first century.* direct finding, numerical [303]
4. Sobol analysis of the COAWST vegetation module found kinetic energy, turbulence, and water level most sensitive to stem density and height, with diameter secondary, while wave dissipation depended primarily on stem density. *Regime: COAWST submerged-aquatic-vegetation module varying stem density, height, and diameter.* direct finding, numerical [683]
5. Across 17 European coastal test cases and 72 configurations per case, both sensitivity methods found floodplain solver dominance on Atlantic coasts, boundary-condition dominance on Mediterranean coasts, strong grid-resolution effects in North and Baltic seas, and generally smaller friction influence. *Regime: Seventeen European coastal sites spanning Atlantic, Mediterranean, North Sea, and Baltic conditions.* direct finding, numerical [470]
6. In Hurricane Earl coupled ensembles, rapid intensification was most sensitive to initial azimuthal-mean maximum wind and asymmetry orientation, whereas stochastic kinetic-energy backscatter influenced storm tracks more than initial-condition perturbations alone. *Regime: Hurricane Earl (2010) perturbations in strength, size, asymmetry, and stochastic backscatter.* direct finding, numerical [816]
7. Using 163 laboratory and field-scale overtopping tests, Gaussian-process surrogates supported probabilistic input sensitivity and predicted mean overtopping discharge more accurately than the regression formulae used for comparison. *Regime: A homogeneous dataset of 163 laboratory and field-scale coastal-defense tests.* direct finding, numerical [701]
8. Global sensitivity analysis of an end-century barrier model found narrow and low-relief initial geometries most vulnerable to width and height drowning, with toe depth, sea-level-rise rate, backbarrier shear stress, and inlet-delivered sediment creating strong interacting controls. *Regime: Barrier-backbarrier evolution through end-century across geometry, SLR, sediment, and marsh parameters.* direct finding, numerical [604]
9. Probabilistic sediment-budget modeling found a detectable Bruun-rule erosion signal by mid-century for RCP4.5, 6.0 and 8.5 on idealized gently sloping undefended beaches, while RCP2.6 did not clearly separate from other uncertain drivers. *Regime: Wave-exposed, gently sloping sandy coasts without defenses, evaluated with the paper's probabilistic sediment-budget and RCP assumptions.* direct finding, numerical [304]
10. European-scale Bruun-rule projections show that sandy-beach location and spatially varying nearshore-slope datasets can contribute uncertainty comparable to sea-level scenarios, especially before mid-century, and materially shift estimated retreat hotspots and coastal land loss. *Regime: European sandy coastlines under RCP4.5 and RCP8.5 sea-level rise from a 2010 baseline to 2100, absent ambient shoreline change.* direct finding, numerical [84]
11. For unstructured-grid SWAN near the Saint Petersburg flood barrier, uncertainty is proportionally greatest for H_s below 0.3 m but about 5-10% for operationally significant waves; wind direction dominates sensitivity, followed by wind speed, bathymetry, water level and breaker index. *Regime: SWAN hindcast/forecast points near the Saint Petersburg Flood Prevention Facility on the documented unstructured Baltic grid.* direct finding, numerical [1028]

47.5 Aleatory and epistemic uncertainty

Topic uncertainty. sources. Variability and knowledge uncertainty.

Papers screened: 13. Claims: 10. Equations: 0. Shown in the Coastal Design Bench beside: Design conditions.

Well established. Aleatory uncertainty represents variability treated as irreducible at the analysis scale, whereas epistemic uncertainty arises from limited knowledge, data, models, scenarios, or assumptions and may be reducible. Semantic and ontological ambiguity can add further uncertainty.

Governing physics. Coastal risk uncertainty propagates through hazard sources and forcing, hydrodynamics, topography and defenses, exposure, vulnerability, damage, socioeconomic pathways, adaptation behavior, dependence, and decision models.

Dimensionless parameters. Unlike hydrodynamic similarity, uncertainty-source analysis is organized by variance contribution, bias factor, probability weight, confidence/credibility, return period, scenario horizon, dependence strength, and value of information.

Major equations. Common frameworks use conditional probability, total-probability integration, hazard curves, logic or condition trees, Bayesian updating, ensembles, sensitivity decomposition, interval or imprecise probability, and value-of-information metrics.

Typical methods. Methods include source taxonomies, logic trees, expert elicitation, multi-model ensembles, global sensitivity analysis, Bayesian calibration, database integration, hindcast validation, scenario analysis, robust decisions, and explicit assumption audits.

Numerical models. Relevant systems include probabilistic tsunami hazard models, compound hydrometeorological-hydrodynamic chains, catastrophe hazard-exposure-vulnerability models, coastal flood risk models, and depth-damage calculations.

Experimental datasets. Reviewed cases span global tsunami hazard, global-to-regional coastal flood risk, compound floods, Florida hurricane losses, multi-hazard reviews, and component-wise flood-damage uncertainty.

Validated ranges. Reported magnitudes include coastal-risk overestimation to factor 1300 when adaptation is omitted, adaptation uncertainty to factor 27, absolute flood-damage uncertainty of factor 5-6, and roughly factor-two contributions from asset values and depth-damage curves in one case.

Recent advances. Recent work quantifies adaptation as a dominant coastal-risk uncertainty, integrates multi-source tsunami hazard, traces cascades through compound-flood forecasts, and reduces vulnerability-model uncertainty through linked administrative and insurance data.

Disagreements. Treating epistemic uncertainty as if it were aleatory can simplify computation but understate plausible variability and conceal conditional assumptions. Conversely, not every uncertainty can be reduced economically, so value-of-information and decision relevance matter.

Limitations. Taxonomy boundaries depend on analysis scale; probabilities for rare events and deep uncertainty may be weakly identified. Dependence, nonstationarity, model discrepancy, adaptation feedback, data quality, and unmodeled processes often remain incomplete.

Open questions. Needs include coherent treatment of deep uncertainty and dependence, dynamic adaptation, open defense and vulnerability data, compound-event propagation, transparent expert weighting, and decision validation under surprise.

Seminal papers. The branch builds on the classical aleatory-epistemic distinction, conditional risk analysis, Bayesian and logic-tree hazard methods, and formal sensitivity and decision analysis; this slice emphasizes their coastal consequences.

Claims

1. A hydrological uncertainty synthesis distinguishes aleatory, epistemic, semantic, and ontological uncertainty and proposes condition trees to expose assumptions and provide decision makers with an auditable evidence trail. *Regime: Hydrological analysis and prediction under multiple uncertainty forms and nonstationarity.* review statement, review [149]
2. Probabilistic tsunami hazard analysis requires an integrated treatment of multiple source mechanisms, their relative intensities and occurrence rates, propagation and impact models, and associated aleatory and epistemic uncertainties to produce coherent hazard curves or maps. *Regime: Earthquake, landslide, volcanic, meteorological, and asteroid tsunami sources across global to local PTHA.* review statement, review [533]
3. A tsunami hazard-and-risk review found maturity uneven across the workflow: earthquake-source probabilistic hazard analysis was relatively advanced, while risk analysis and other sources retained major data, theoretical, and methodological gaps. *Regime: Probabilistic tsunami hazard and risk methods across source types and assessment stages.* review statement, review [138]
4. Across global-to-regional coastal flood-risk assessments, omitting adaptation could overestimate 2100 risk by up to a factor 1300, while uncertainty in how societies adapt reached a factor 27 and dominated other globally quantified sources. *Regime: Global to world-regional current and future coastal flood-risk assessments.* review statement, review [599]
5. A cross-hazard review warns that representing epistemic uncertainty with simple aleatory distributions can understate hazard and risk variability, and recommends recording and evaluating the assumptions on which every uncertainty estimate is conditional. *Regime: Flood, landslide, dam, drought, earthquake, tsunami, volcanic, and wind hazards.* review statement, review [150]
6. Compound-flood prediction requires uncertainty to be characterized across hydrometeorological and hydrodynamic layers and propagated through their interactions; remote sensing and data science offer routes to reduce predictive uncertainty rather than treating drivers independently. *Regime: Hydrometeorological and hydrodynamic layers of compound-flood modeling and forecasting.* review statement, review [7]
7. In the Florida Public Hurricane Loss Model, epistemic uncertainty from incomplete and inconsistent exposure and claims data was reduced by integrating and county-level cross-referencing tax-appraiser, NFIP, and wind-insurance databases for development, calibration, and validation. *Regime: Florida residential hurricane vulnerability and insured-loss modeling using tax, NFIP, and wind-insurance data.* direct finding, mixed [1113]
8. In the flood-damage case study, approximately 25 cm inundation-depth uncertainty accompanied total absolute damage uncertainty of factor 5-6, while asset values and depth-damage curves each contributed about factor two; proportional damage changes were more robust. *Regime: Flood damage case study varying land use, values, depth-damage curves, and water depth.* direct finding, numerical [969]
9. A review of climate-driven sandy-coast erosion models finds that most studies sample only forcing ranges and emissions ensembles, while few are fully probabilistic; credible projections require bias-corrected driver time series and propagation of scenario, climate-model, erosion-model and parameter uncertainty. *Regime: Temperate sandy-beach shoreline projections under climate-driven mean sea level, waves, surge and tides.* review statement, review [1418]
10. Extreme significant-wave-height return values vary substantially with initial-distribution, block-maxima, peaks-over-threshold and conditional-exceedance choices; methodological spread increases with return period and can exceed the projected climate-change signal. *Regime: The analyzed North Atlantic significant-wave-height time series for one location, historical climate and two future forcing scenarios.* direct finding, numerical [1442]

48 Wiki: Wave-structure interaction

48.1 Wave-structure interaction: overview

Topic *wave_structure*. Hydrodynamic interaction between waves and built elements.

Papers screened: 5. Claims: 3. Equations: 0.

Well established. Built coastal elements reflect, transmit, diffract and radiate waves; movable or energy-absorbing structures also exchange energy through body motion, air chambers, moorings and power take-off. Response is governed jointly by the incident climate, geometry, spacing and structural dynamics.

Governing physics. Fluid inertia, gravity, pressure and viscosity couple to fixed-boundary scattering or rigid/flexible-body motion. Resonance and array interference can amplify response, while damping, radiation, porous losses, breaking and power take-off remove or redistribute energy.

Dimensionless parameters. Important controls include wave steepness and relative depth, wavelength-to-structure scale, draft and submergence ratios, spacing-to-wavelength, blockage, Keulegan–Carpenter and Reynolds numbers, mass and damping ratios, capture width and power or reflection coefficients.

Major equations. Common models combine potential-flow boundary-value problems and eigenfunction matching with radiation–diffraction theory and motion equations containing mass, added mass, damping, restoring and forcing. More nonlinear regimes require Navier–Stokes, free-surface and structural equations.

Typical methods. Methods include analytical eigenfunction solutions, boundary-element and CFD models, wave-basin tests, load and motion measurements, field deployments, array studies, response-amplitude operators, power curves and environmental monitoring.

Numerical models. Linear potential-flow models efficiently resolve diffraction, radiation and array coupling; time-domain and CFD approaches address nonlinear free surfaces, impacts and viscous losses. Device models couple hydrodynamics to air compressibility, turbine, generator, mooring or control dynamics.

Experimental datasets. Root evidence includes the monitored Lysekil point-absorber site, theoretical coast-integrated oscillating-water-column arrays, comparative low-energy-sea converter literature and a broad coastal-engineering wave–structure process review.

Validated ranges. The reviewed evidence spans a deployed heaving point absorber, planned multi-device field expansion, coast-integrated OWC arrays with variable chamber dimensions and spacing, and low-energy regional wave climates. Exact quantitative ranges remain source-specific.

Recent advances. Recent advances integrate devices into coasts and ports, exploit constructive array effects, tailor smaller converters to low-energy seas, couple hydrodynamics with control and electrical systems, and expand field and environmental monitoring.

Disagreements. Linear models isolate interference and resonance but cannot represent all breaking, viscous, impact or extreme-load behavior. Wave-energy devices may protect coasts in some configurations yet alter sediment or ecosystems, and resource potential does not by itself establish economic performance.

Limitations. Scale effects, incomplete coupled structural models, uncertain damping and power take-off, idealized waves and geometry, sparse long-duration field data, survivability, maintenance, grid integration and environmental impacts limit transfer from model or prototype to arrays.

Open questions. Priorities include nonlinear array interaction, extreme and fatigue loads, flexible and porous response, optimized geometry and control, multi-use coastal protection, climate robustness, field-scale validation and cumulative ecological effects.

Seminal papers. Classical diffraction and radiation theory established wave loading, added mass and damping for fixed and floating bodies; subsequent laboratory and numerical work extended these foundations to breakwaters, offshore structures and energy converters.

Claims

1. A linear-potential-flow and eigenfunction-matching model shows that coast-integrated oscillating-water-column arrays can enhance power extraction over a range of waves through constructive array and coast effects controlled by direction, chamber geometry, spacing and array size.
Regime: Wave power extraction from multiple oscillating water columns along a straight coast direct finding, analytical [1613]
2. Coastal wave–structure interaction is a complex fluid–structure problem spanning multiple engineering configurations and requires process-specific analysis rather than a single transferable response model.
Regime: Wave-Structure Interaction Processes in Coastal Engineering review statement, review [76]
3. An OpenFOAM numerical wave tank with generation and relaxation-zone absorption reproduced measured nonlinear loading on a vertical cylinder through fourth-order harmonics; linear theory could miss up to 50% of total wave load, and converged cases used roughly $L/70$ horizontal and $H/8$ vertical resolution. *Regime: Nonlinear regular and focused wave groups interacting with a vertical cylinder in the tested numerical and laboratory configurations.* direct finding, mixed [259]

48.2 Floating structures

Topic *wave_structure*. *f*loating. Coupled wave and body motion.

Papers screened: 38. Claims: 35. Equations: 0.

Well established. Floating-structure response couples radiation and diffraction, hydrostatic and Froude–Krylov forcing, viscous drag, rigid-body or hydroelastic motion, and restraints from moorings, joints, PTO, neighboring bodies, and control.

Governing physics. Natural-period proximity, parametric resonance, breaking-wave impulses, second-order drift, narrow-gap resonance, porous damping, array radiation, nonlinear wetted area, flexibility, and cable or hinge dynamics can each govern motion or load.

Dimensionless parameters. Key controls include wave steepness, wavelength/body size, draft/depth, Froude and Keulegan–Carpenter numbers, gap/beam, submergence, buoyancy/weight, wave-to-natural period, damping ratio, array spacing, porosity, and mooring angle/stiffness.

Major equations. Core formulations include six-degree rigid-body equations with added mass, radiation damping and restoring, Cummins/time-domain convolution, diffraction–radiation potential flow, Morison drag, nonlinear Froude–Krylov pressure, RANS/VOF or SPH, cable dynamics, and hydroelastic plate equations.

Typical methods. Methods span wave-basin and flume tests, image motion tracking and PIV, load cells and wave arrays, linear and weakly nonlinear potential flow, BEM, OpenFOAM RANS–VOF, DualSPHysics, overset CFD, cable finite elements, surrogate or reduced models, and blind benchmarks.

Numerical models. Validated tools include OpenFOAM CFD–motion, DualSPHysics, AQWA, WEC-Sim with CFD-informed viscosity, iterative porous BEM, overset Navier–Stokes, nonlinear Froude–Krylov solvers, hydroelastic plate models, and coupled cable/joint dynamics.

Experimental datasets. Evidence includes arrays up to 25 point absorbers, blind focused-wave tests of two taut-moored bodies, 1:4.5 floating-solar rafts, submerged-tunnel tests, porous cylinders, gap-resonance PIV, extreme-wave point absorbers, and field-reconstructed WEC response.

Validated ranges. Quantitative findings include 18.1% array wave-height reduction, 53% parametric-resonance power loss, 500-wave extreme simulations, four-order speedup for analytical nonlinear forcing, 44% greater pontoon-breakwater attenuation, and over 70% hybrid reduction of floating-wind motion.

Recent advances. Recent work integrates validated SPH for survivability, CFD-informed reduced models, nonlinear cable/joint coupling, floating-solar experiments, porous motion damping, resonance-resolved PIV, hybrid wind–wave attenuation, and performance-oriented floating breakwaters.

Disagreements. Linear models can predict heave adequately in some focused-wave tests yet miss other motions and loads; high-fidelity solvers generally improve response but disagree among themselves; body interactions may be negligible in one case but dominate arrays, gaps, and modular systems.

Limitations. Full-scale extreme records are rare; viscous, breaking, slamming, mooring, joint, and hydroelastic effects remain model-sensitive; laboratory scaling and idealized seas constrain transfer; and high-fidelity simulation is too expensive for lifetime design.

Open questions. Needs include efficient nonlinear long-duration models, standardized blind validation, breaking-wave and impact response, failure-capable hydroelasticity, multidirectional arrays, dynamic cable/contact, coupled control, fatigue, reliability, and uncertainty-aware operability.

Seminal papers. The branch connects classical radiation–diffraction and rigid-body response to coupled mooring solvers, array experiments, nonlinear instantaneous-wetted forcing, hydroelastic very-flexible plates, and blind multi-fidelity validation.

Claims

1. Coupled VOF–RANS and cable finite elements reproduced slack-moored buoy motions, mooring forces, and nonlinear wave-height-dependent response. *Regime: The model reproduced buoy motions, mooring forces, and strong wave-height dependence of response-amplitude operators.* direct finding, mixed [1076]
2. Large-array basin tests measured up to 18.1% downwave significant-wave-height reduction from combined incident, diffracted, and radiated fields. *Regime: A 25-device array reduced downwave significant wave height by as much as 18.1% across long- and short-crested irregular-wave cases.* direct finding, experimental [1359]
3. Floating-device design needs intermediate models that retain nonlinear fidelity beyond linear potential flow without the prohibitive cost of RANS-scale million-hour simulations. *Regime: Million-hour design simulation needs exceed practical RANS cost, while linear potential-flow fidelity is inadequate in important nonlinear regimes.* review statement, review [320]
4. Two-phase CFD validated against 2-, 5-, and 9-device arrays reproduced individual heave, surge force, and the perturbed wave field. *Regime: Simulated heave, surge forces, and surrounding perturbed wave fields agreed well with experiments.* direct finding, mixed [335]
5. Validated SPH–PTO–mooring simulation captured a taut-moored point absorber under an extreme irregular sea state containing 500 waves. *Regime: A time-integrated model reproduced extreme-wave physics and was applied to a realistic irregular sea state containing 500 waves.* direct finding, mixed [1387]
6. Validated six-degree nonlinear modelling found parametric resonance can reduce floating-OWC power extraction by up to 53% while increasing motion, load, and mooring risk. *Regime: The model captured parametric resonance, which reduced predicted power-extraction efficiency by up to 53%.* direct finding, mixed [502]
7. A blind focused-wave benchmark found Navier–Stokes solvers generally outperformed linear models, yet large inter-model variation persisted even among similar methods. *Regime: Navier–Stokes models generally performed better than linear models, but predictions varied considerably even within similar method families.* direct finding, mixed [1159]
8. Submerged-tunnel experiments showed motion grows with wave height and mooring angle, decreases with submergence and buoyancy ratio, and peaks according to natural-period proximity. *Regime: Motion rose with wave height and mooring angle, fell with submergence and buoyancy-to-weight ratio, and depended on proximity to the natural period.* direct finding, experimental [1561]
9. Thin floating photovoltaic structures can follow waves with deformation far exceeding their thickness, defining a very-flexible regime requiring nonlinear plate theory. *Regime: Very flexible floating structures can deform far beyond their thickness and require nonlinear Föppl–von Kármán rather than conventional small-deformation plate theory.* review statement, review [1598]
10. Blind validation showed hybrid and Navier–Stokes methods were strongest overall, while CFD-informed viscous corrections let WEC-Sim approach high-fidelity predictions. *Regime: Navier–Stokes and hybrid methods were most accurate overall; all methods predicted heave reasonably, and CFD-informed WEC-Sim approached strong Navier–Stokes performance.* direct finding, mixed [1160]
11. A porous outer cylinder reduced floating-body motion because its damping increase exceeded its excitation increase, especially when placed lower in the water column. *Regime: A porous outer cylinder increased excitation but increased damping more, reducing motion; lower placement could add damping without raising excitation.* direct finding, mixed [892]

12. Wave-tank comparisons found a catamaran floating-solar platform outperformed a flat plate, while an added breakwater reduced motions and mooring loads for both. *Regime: The catamaran performed better hydrodynamically and a breakwater reduced wave-induced motion and mooring loading for both concepts.* direct finding, experimental [1562]
13. An analytical nonlinear Froude–Krylov method for prismatic floaters ran four orders faster than mesh-based alternatives and two orders faster than real time. *Regime: The approach ran two orders faster than real time and four orders faster than mesh-based counterparts while retaining nonlinear forcing.* direct finding, analytical [503]
14. Validated overset CFD identified mooring-angle regimes that minimize submerged-tunnel motion and tension and regimes that create severe response and line-slack risk. *Regime: Mooring angle separated into five response zones; one minimized motion and load, while another caused severe motion and slack-line risk.* direct finding, mixed [262]
15. For articulated floating modules, head seas governed hinge force and low-frequency second-order drift governed peak mooring tension, while additions beyond four modules had little local effect. *Regime: Head seas maximized hinge forces; low-frequency second-order drift governed peak mooring tension, and modules beyond the fourth changed little hydrodynamically.* direct finding, numerical [662]
16. At 1:4.5 scale, floating-solar surge force increased with wave steepness and negative mean heave exposed potential impact and panel-submergence hazards. *Regime: Surge force increased with wave steepness, and negative mean heave indicated possible full-scale impact and photovoltaic-panel submergence.* direct finding, experimental [460]
17. Floating-body gap resonance was governed strongly by gap width, with vortex strength nonlinear in modified Reynolds number and linear in Keulegan–Carpenter number. *Regime: Resonant height and frequency were sensitive to gap width; vortex strength scaled nonlinearly with modified Reynolds number and linearly with Keulegan–Carpenter number.* direct finding, experimental [1481]
18. Articulated-platform operability was sensitive to rotation, while RMS vertical acceleration set the final human-comfort limit. *Regime: Rotational motion strongly affected seakeeping, but RMS vertical acceleration ultimately governed onboard comfort.* direct finding, numerical [663]
19. Field validation showed weakly nonlinear forcing and dynamic cable models materially improve floating-device motion and tension, whereas body interaction was negligible for the tested case. *Regime: Weakly nonlinear Froude–Krylov/hydrostatics and dynamic cable modelling improved response, whereas body interaction did not materially affect this case.* direct finding, mixed [780]
20. A validated T-block pontoon breakwater increased maximum wave abatement by 44% over a single row, with strongest advantage for periods no greater than 6 s. *Regime: T-block width improved attenuation most; the combined design increased maximum wave abatement by 44% over a single row, especially for periods at or below 6 s.* direct finding, mixed [272]
21. Optimized peripheral duck-shaped converters reduced floating-wind-platform wave motion, especially pitch, by more than 70% in numerical and physical tests. *Regime: Peripheral duck-shaped converters reduced floating-turbine wave-induced motion, particularly pitch, by more than 70%.* direct finding, mixed [1395]
22. The Lysekil field site demonstrates a directly driven seabed linear generator coupled to a heaving surface point absorber, with offshore wave, electrical and environmental monitoring supporting staged converter-array development. *Regime: Wave Energy from the North Sea: Experiences from the Lysekil Research Site* direct finding, field [797]
23. A review of low-energy seas finds that wave-energy converters should be downscaled and optimized for milder climates; nearshore devices may combine energy production with port or erosion protection, but ecological effects remain under-studied. *Regime: Wave energy converters in low energy seas: Current state and opportunities* review statement, review [456]
24. This paper describes experimental research on a floating moored Oscillating Water Column (OWC)-type Wave-Energy Converter (WEC) carried out in the wave flume of the Coastal Engineering Research Group of Ghent University. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [1533]
25. A marine-renewable hydrodynamics review identifies power-take-off representation, low-Reynolds scale effects, unsteady turbine flow, coupled aero-hydro-servo-structural response and scarce public full-scale data as central barriers to validating wave, current and floating-wind device models. *Regime: Wave, tidal-current and fixed/floating offshore wind energy devices across model and sea-test scales.* review statement, review [326]
26. Validated Star-CCM+ simulations of a dual-floater breakwater/WEC show lower transmission than either component alone and improved low-frequency conversion; narrow-gap resonance benefits a symmetric WEC but penalizes an asymmetric one, while generally reducing breakwater forces. *Regime: Two-dimensional dual-floater hybrid of a floating breakwater and symmetric/asymmetric oscillating buoy across tested regular-wave frequencies and geometries.* direct finding, numerical [1591]
27. An SPH model validated against three-float experiments reproduces wave elevation, module motion, mooring, hinge and collision forces, reflection and transmission; longer waves increase transmission, and near equality of wavelength and total structure length produces especially large motions and forces with weakened shielding. *Regime: Serial hinged box floats with moorings and bottom-corner bumpers under the tested regular waves and geometries.* direct finding, mixed [580]
28. DualSPHysics validated against point-absorber experiments predicts maximum efficiency near resonance with optimal PTO damping of roughly 60–240 Ns/m; focused-wave survivability tests indicate full submergence reduces loads more effectively than fixing or enlarging the simplified device. *Regime: Heaving point absorber under tested regular waves, PTO damping values and focused design-wave survival configurations.* direct finding, mixed [1215]
29. A nonlinear higher-order-boundary-element model predicts that adding an oscillating buoy inside an OWC chamber of a pi-type floating breakwater improves long-period energy capture and reduces transmission; coupled water-surface and buoy resonances make buoy geometry, submergence and orifice ratio regime-specific design controls. *Regime: Integrated OWC-oscillating-buoy pi-type floating breakwater under the modeled wave-period and geometry range.* direct finding, numerical [270]
30. Physical models of multipurpose floating platforms cannot generally satisfy simultaneous similitude for support, mooring, wind turbine, wave-energy converter and aquaculture subsystems; reliable experiments must prioritize response-driving dimensionless groups and use hybrid, active or open-sea strategies for incompatible scales. *Regime: Wave-basin or open-sea physical models of multipurpose floating offshore platforms integrating one or more reviewed subsystems.* review statement, review [1228]
31. A redesigned floating-wind-turbine rotor can match scaled thrust and torque simultaneously across operating wind speeds by preserving blade twist while adjusting chord, pitch and rotation, improving aerodynamic-hydrodynamic interaction tests under conflicting Reynolds and Froude similarity. *Regime: Scaled floating-offshore-wind turbine aerodynamic models intended for coupled wind-wave-basin testing across the examined operating winds.* direct finding, mixed [570]
32. A non-hydrostatic SWASH extension with a fixed floating-body representation reproduces wave scattering and reasonably predicts restrained-container-ship loads while retaining coarse vertical resolution suitable for harbor-scale simulations. *Regime: Restrained, non-moving ships in coastal or harbor wave fields under the tested regular and irregular wave conditions and coarse hull representation.* direct finding, mixed [1195]

33. A design review of offshore floating solar identifies array spacing, flexible connectors, mooring layout, coupled wind-wave loads, panel tilt, sheltering, and free-surface hydrodynamics as interacting requirements, while extreme-condition Class 3 concepts remain below relevant-environment prototype maturity. *Regime: Modular and island-type offshore floating photovoltaic arrays exposed to waves, wind, currents, mooring loads, and extreme marine conditions.* direct finding, review [408]
34. Wave-energy-converter model choice must be device and purpose specific: nonlinear incident waves, hydrodynamics, power take-off, and moorings can matter, while partially nonlinear potential-flow models with viscous drag often balance fidelity and cost and CFD is reserved for high-fidelity or extreme cases. *Regime: Multiple wave-energy-converter concepts across operational and extreme waves, power take-off, mooring, and wave-structure interaction nonlinearities.* direct finding, review [1102]
35. Coupled FAST-AQWA-F2A simulations optimize Wave Star placement on a DeepCWind semisubmersible: the devices add wave power, slightly increase mean mooring load and pitch, reduce mooring-load amplitude, and have their largest platform-position effect in surge. *Regime: A site-specific DeepCWind semisubmersible wind turbine with oscillating-body Wave Star converters under regular waves and steady winds.* direct finding, numerical [665]

48.3 Wave loads

Topic wave_structure. Loads. Forces, pressures, and moments.

Papers screened: 22. Claims: 13. Equations: 7. Shown in the Coastal Design Bench beside: Pile wave loads.

Well established. Wave-load prediction must distinguish distributed non-impact loading from breaking, slamming, and green-water impact, and must convert local pressure into the force and overturning moment relevant to the structural failure mode.

Governing physics. Load physics include inertia and drag, nonlinear diffraction and higher harmonics, runup and breaking, porous transmission, hydrostatics, wave excitation and radiation, air-gap closure, diffraction-driven focusing, and water retained above decks.

Dimensionless parameters. Relevant nondimensional controls include Keulegan-Carpenter number, wave steepness, relative depth, relative cylinder diameter, relative freeboard or air gap, wave direction, geometric scale, and normalized force, moment, and pressure coefficients.

Major equations. The reviewed methods use VARANS momentum and porous-resistance equations, distributed time-domain force balance, Stokes-like harmonic power models, and moment relations in which each force harmonic is multiplied by a harmonic-specific effective arm derived from its depth distribution.

Typical methods. Typical studies combine scaled pressure/load measurements with wave gauges, harmonic or spectral decomposition, CFD or potential-flow simulation, integration to global force and moment, and comparison with semi-empirical design expressions or reduced-order models.

Numerical models. Models include COBRAS-UC VARANS-VOF with porous media, fully nonlinear potential flow, Stokes-like reduced-order harmonic reconstruction, coupled Simo-Riflex finite elements with ReaTHM forcing, and OpenFOAM monopile CFD.

Experimental datasets. The reviewed evidence includes two 1:20 breakwater sections, a 76 m flume campaign on a 0.315 m cylinder, a 1:30 hybrid 5 MW semi-submersible test at five hull sections, and a 1:20 jacket deck tested across three orientations and water-level states.

Validated ranges. Validation is heterogeneous: 1:20 breakwater and jacket tests, a 0.315 m cylinder in 1.8 m depth under non-breaking low-KC focused groups, and a 1:30 floating-wind model under three operational regimes. Reported jacket maxima use source-specific normalizations and must not be transferred as universal coefficients.

Recent advances. Recent advances isolate load harmonics experimentally, reconstruct moments from harmonic force arms, validate distributed internal loads under combined wind and waves, and quantify extreme vertical deck-impact configurations rather than relying only on total inline force.

Disagreements. Simple harmonic moment-arm models perform well for most non-breaking inertia-load components but show marked triple-frequency discrepancies, while green-water impact is dominated by impulsive vertical force and localized pressure. These findings describe different regimes rather than competing universal models.

Limitations. Key limitations are scale and air-compressibility effects, two-dimensional approximations, uncertain pressure integration during impact, site-specific porous properties, omitted hydroelasticity and vibration, third-harmonic model error, and numerical rather than experimental validation of recent closed-form moment arms.

Open questions. Needs include common benchmarks spanning non-impact to slamming regimes, pressure-impulse and structural-response coupling, compressible multiphase validation, directional and current effects, uncertainty propagation to failure probability, and experimental validation of harmonic moment arms.

Seminal papers. Within this screened slice, the 2009 COBRAS-UC study is an early end-to-end validation of CFD pressure, force, and moment for porous breakwaters; the broader classical force literature remains a discovery gap for this initial branch.

Equations

- Harmonic-specific effective moment arm [603]

$$z_n = \frac{\int_{-h}^0 (z + h) f_n(z) dz}{\int_{-h}^0 f_n(z) dz}$$

Regime: Inertia-dominated non-breaking monopile loading integrated to still-water level. Variables: z_n effective moment arm for harmonic n ; h water depth; f_n depth distribution of n th-harmonic inertia force; z vertical coordinate

- Stokes-like n th-harmonic load model [432]

$$F_n(t) = \Gamma_n[\widehat{F}_1(t)]^n \cos(n\phi(t) - \psi_n)$$

Regime: Non-breaking focused waves on a low-KC slender vertical cylinder. Variables: F_n nth force harmonic; Γ_n harmonic scale coefficient; $\widehat{F}_1$ linear-force envelope; ϕ linear phase; ψ_n harmonic phase coefficient

- *Harmonic moment-arm approximation* [432]

$$M_n(t) = z_n F_n(t)$$

Regime: Tested non-breaking inertia-dominated focused-wave cases. Variables: M_n nth moment harmonic about the base; z_n harmonic-specific constant moment arm; F_n nth force harmonic

- *time-domain higher-order boundary element model* [1479]

$$\mathbf{F}(t) = \int_A -p \mathbf{n} dA + \mathbf{F}_{D,I}$$

Regime: 1:20 dual-chamber OWC in 1.0 m water; $d/h=0.125, 0.2, 0.25$ and chamber-width ratios 1:3, 1:1, 3:1. Variables: $\mathbf{F}$ resultant structural load; p hydrodynamic or pneumatic pressure; $\mathbf{n}$ surface normal; A wetted or impacted area; $\mathbf{F}_{D,I}$ drag/inertia contribution where applicable; normalized load balance

- *first-principles cylinder force model* [494]

$$\mathbf{F}(t) = \int_A -p \mathbf{n} dA + \mathbf{F}_{D,I}$$

Regime: Vertical surface-piercing cylinder under matched nominal North Sea sea states with opposing, following, and no current. Variables: $\mathbf{F}$ resultant structural load; p hydrodynamic or pneumatic pressure; $\mathbf{n}$ surface normal; A wetted or impacted area; $\mathbf{F}_{D,I}$ drag/inertia contribution where applicable; normalized load balance

- *fast second-order Morison load model* [188]

$$\mathbf{F}(t) = \int_A -p \mathbf{n} dA + \mathbf{F}_{D,I}$$

Regime: Slender monopiles/spars; example 8 m pile in 33 m depth, with diffraction limit near $T=5.1$ s and $KC_{max}=2.2$; 10 s example $D/L=0.057$ and $KC_{max}=7.7$. Variables: $\mathbf{F}$ resultant structural load; p hydrodynamic or pneumatic pressure; $\mathbf{n}$ surface normal; A wetted or impacted area; $\mathbf{F}_{D,I}$ drag/inertia contribution where applicable; normalized load balance

- *current-modified Morison equation* [1547]

$$\mathbf{F}(t) = \int_A -p \mathbf{n} dA + \mathbf{F}_{D,I}$$

Regime: Narrowband weakly nonlinear irregular waves with arbitrary depth-dependent subsurface current acting on a bottom-fixed slender cylinder. Variables: $\mathbf{F}$ resultant structural load; p hydrodynamic or pneumatic pressure; $\mathbf{n}$ surface normal; A wetted or impacted area; $\mathbf{F}_{D,I}$ drag/inertia contribution where applicable; normalized load balance

Claims

1. For 1:20 low-mound and rubble-mound crown-wall tests under regular and irregular waves, COBRAS-UC reproduced pressure, force, and moment behavior sufficiently to complement semi-empirical stability formulae, but the prototype applications were not field validations. *Regime: The two tested two-dimensional breakwater cross-sections, wave cases, porous formulation, and grid rules.* direct finding, mixed [547]
2. In non-breaking, inertia-dominated focused-wave tests on a 0.315 m vertical cylinder, a Stokes-like model and constant harmonic-specific moment arms represented most force and moment harmonics well, but the triple-frequency component showed significant discrepancies. *Regime: Low-KC focused groups, 1.8 m depth, slender surface-piercing cylinder; not breaking-wave impact loading.* direct finding, mixed [432]
3. For a 1:30 braceless 5 MW semi-submersible under three operational wind-wave regimes, time-domain simulations agreed closely with hybrid-test motions and sectional loads, while identifying the pontoon-central-column interface and combined low- and wave-frequency bending as design-critical. *Regime: The tested semi-submersible, calibrated operational conditions, and models excluding hull hydroelasticity and structural vibration.* direct finding, mixed [869]
4. In 1:20 random-wave jacket tests, zero air gap produced normalized peak vertical force 42 and under-deck peak pressures 13.2 at the leading side and 21.5 at the trailing side; vertical force dominated partial and full green-water cases. *Regime: The tested jacket-deck geometry, random-wave energy levels, orientations, and water levels; normalization definitions are source-specific.* direct finding, experimental [37]
5. For inertia-dominated non-breaking monopile loading, closed-form harmonic-specific moment arms with modest corrections reproduced OpenFOAM total overturning moments accurately, but validation used simulated rather than measured force harmonics. *Regime: Inertia-dominated monopiles and compatible higher-order Stokes-type force harmonics; not breaking or drag-dominated loading.* direct finding, mixed [603]
6. For the tested 1:20 dual-chamber OWC, horizontal curtain-wall loads dominated, the seaside wall governed, and increasing draft raised dominant force and moment; moving the inner wall toward the seaside wall reduced peak loads. *Regime: 1:20 dual-chamber OWC in 1.0 m water; $d/h=0.125, 0.2, 0.25$ and chamber-width ratios 1:3, 1:1, 3:1.* direct finding, mixed [1479]
7. For matched North Sea-type sea states on a vertical cylinder, fitted force coefficients were consistently smaller with current than without, and opposing-current peak force shifted toward the free surface. *Regime: Vertical surface-piercing cylinder under matched nominal North Sea sea states with opposing, following, and no current.* direct finding, experimental [494]
8. For narrowband weakly nonlinear waves and depth-dependent currents, slender-cylinder load extremes changed materially even when surface-elevation extreme statistics changed little, so current direction and vertical profile must be coupled in the load model. *Regime: Narrowband weakly nonlinear irregular waves with arbitrary depth-dependent subsurface current acting on a bottom-fixed slender cylinder.* direct finding, analytical [1547]

9. For slender monopiles, regrouping second-order spectral interactions permits Morison-type wave loads to be evaluated at $O(N \log N)$, with demonstrated field-scale regimes including an 8 m pile in 33 m depth. *Regime: Slender monopiles/spars; example 8 m pile in 33 m depth, with diffraction limit near $T=5.1$ s and $KC_{max}=2.2$; 10 s example $D/L=0.057$ and $KC_{max}=7.7$.* direct finding, analytical [188]
10. A validated three-dimensional poroelastic model shows that monopile diffraction and reflection materially alter seabed pore pressure and displacement; pile motion increases with wave height and decreases with seabed Young's modulus. *Regime: Three-dimensional numerical model for wave-induced seabed response around mono-pile* direct finding, mixed [1366]
11. Air trapped between coastal-bridge girders substantially amplifies uplift; venting reduces pressure and vertical force, while AASHTO horizontal-force estimates are reasonable but vertical estimates shift from conservative for large waves to underprediction for smaller waves. *Regime: Girder-deck coastal bridge superstructures subjected to the modeled wave heights, periods, clearances and vent configurations.* direct finding, numerical [92]
12. A one-way SPH-FEM wave-flume framework converts uncertain wave-load and structural inputs into probability distributions of structural acceleration and displacement, exposing nonlinear wave-height response and parameter sensitivities that deterministic runs cannot represent. *Regime: HyTOFU/Tokyo numerical wave-flume geometry with a single idealized flexible structure under the tested breaking and non-breaking wave conditions.* direct finding, numerical [878]
13. For a 1:30 coastal-bridge model under solitary waves, girders and trapped air amplify and destabilize deck loading; two-dimensional CFD is adequate mainly for decks without girders, whereas three-dimensional CFD better reproduces complex force histories. *Regime: The tested elevated coastal-bridge decks, solitary-wave heights, water depths and submergence conditions at 1:30 scale.* direct finding, mixed [1620]

48.3.1 Loads on coastal buildings

Topic `wave_structure`. Loads. buildings. Forces on buildings and arrays.

Papers screened: 4. Claims: 2. Equations: 2.

Well established. Coastal-building loads are governed by incident and breaking-wave kinematics, inundation depth, exposed area, openings and elevation, but surrounding buildings and debris can strongly alter the local flow and invalidate isolated-solid-building assumptions.

Governing physics. Loads combine hydrostatic, drag, inertia, impact, uplift, buoyancy, and downward components. Breaking and bore fronts create impulses; openings relieve pressure unless blocked; upstream structures shield or redirect flow; debris adds impacts and can form sustained blockage dams.

Dimensionless parameters. Important controls include relative inundation and clearance, Froude number, wave-height-to-depth ratio, building-width-to-wavelength ratio, opening ratio, spacing and arrangement ratios, blockage fraction, debris size and concentration, and vertical-load normalization by displaced-water or dynamic-force scales.

Major equations. Engineering descriptions integrate pressure over wetted surfaces, use hydrostatic-plus-dynamic or drag/inertia balances, quantify impulse and cantilever arm, and increasingly employ CFD-trained surrogate functions of wave, surge, clearance, opening, and plan geometry.

Typical methods. Evidence uses scaled wave/bore flumes, pressure and multi-axis force sensing, high-speed imaging, building-array layouts, debris ensembles, Reynolds-averaged or volume-of-fluid CFD, design-of-experiment sampling, and ensemble machine learning.

Numerical models. Relevant tools include pressure-integrating CFD for surge and waves, ensemble load surrogates trained on designed CFD samples, and design equations that require local—not bare-earth—water levels when buildings alter breaking flow.

Experimental datasets. The reviewed base includes 1:20 urban-array breaking/nonbreaking tests with measured peak pressure and shielding, laboratory free-standing open-building tests with log and container debris, and 256 CFD cases for elevated-building vertical loading.

Validated ranges. Urban-array shielding reduced downstream peak pressure by 40%-70% in the source's idealized breaking-wave configurations; debris results cover logs and containers around an open building; elevated-building prediction spans a 256-case regular-wave CFD design domain whose detailed bounds remain source-table limited.

Recent advances. Recent work replaces single isolated-building coefficients with geometry-aware CFD ensembles and explicitly treats urban shielding, local breaking, openings, and debris dams as interacting modifiers of horizontal, vertical, and impulsive loading.

Disagreements. Openings generally reduce wave load, but debris blockage can remove that benefit. Bare-earth water-level inputs may be nonconservative in breaking-wave arrays even when standard equations perform for nonbreaking cases, showing that local flow modification governs transferability.

Limitations. Current evidence is sparse across building types and often uses idealized rigid geometry, regular or bore-like waves, limited debris shapes, and laboratory/CFD scales. Air, structural failure, scour, floating debris rotation, multi-building urban complexity, and uncertainty in prototype roughness remain underrepresented.

Open questions. Needed work includes common array benchmarks, irregular hurricane wave-plus-surge loading, probabilistic debris blockage, coupled scour and failure, vertical-load validation, opening and foundation optimization, and uncertainty-aware translation from CFD or scale models to codes.

Seminal papers. Within the extracted corpus, the 2016 urban-array experiment establishes quantified shielding and breaking-wave limitations; the 2020 debris-damming study shows reversal of opening benefits; the 2023 study adds CFD-trained vertical-load estimation for elevated buildings.

Equations

- *CFD-trained ensemble load surrogate* [968]

$$\mathbf{F}(t) = \int_A -p \mathbf{n} dA + \mathbf{F}_{D,I}$$

Regime: Elevated coastal-building geometries and regular-wave hurricane loading represented by a 256-case CFD design space; exact ranges require full-text tables. Variables: $\mathbf{F}$ resultant structural load; p hydrodynamic or pneumatic pressure; $\mathbf{n}$ surface normal; A wetted or impacted area; $\mathbf{F}_{D,I}$ drag/inertia contribution where applicable; normalized load balance

- *postpeak debris-load assessment methodology* [1540]

$$\mathbf{F}(t) = \int_A -p \mathbf{n} dA + \mathbf{F}_{D,I}$$

Regime: Free-standing building with openings under laboratory unsteady long-wave flow; wooden logs and shipping-container debris configurations. Variables: $\mathbf{F}$ resultant structural load; p hydrodynamic or pneumatic pressure; $\mathbf{n}$ surface normal; A wetted or impacted area; $\mathbf{F}_{D,I}$ drag/inertia contribution where applicable; normalized load balance

Claims

1. For a free-standing building with openings under unsteady long-wave flow, container debris produced severe impact peaks, whereas interlocking logs formed compact dams that caused higher and longer-lasting postpeak hydrodynamic forces. *Regime: Free-standing building with openings under laboratory unsteady long-wave flow; wooden logs and shipping-container debris configurations.* direct finding, experimental [1540]
2. Across a 256-case CFD design space for elevated coastal buildings under regular hurricane-representative waves, ensemble learning captured geometry- and wave-dependent vertical surge and wave forces. *Regime: Elevated coastal-building geometries and regular-wave hurricane loading represented by a 256-case CFD design space; exact ranges require full-text tables.* direct finding, mixed [968]

48.3.2 Impact and impulsive loads

Topic wave_structure. Loads. impact. Slamming and short-duration loading.

Papers screened: 6. Claims: 4. Equations: 2. Shown in the Coastal Design Bench beside: Pile wave loads.

Well established. Impulsive wave loads arise when breaking crests, bores or rapidly rising water strike structures over short durations; peak pressure, total impulse, rise time and spatial coherence must be distinguished from slower quasi-static wave and buoyancy loads.

Governing physics. Impact depends on local free-surface geometry and velocity, entrapped and entrained air, compressibility, ventilation, structural clearance and orientation. Air can cushion a direct impact or amplify oscillatory pressure as pockets compress, fragment and vent.

Dimensionless parameters. Controls include relative clearance, wave height and crest elevation, breaker type, Froude and Weber numbers, impact velocity normalized by $\sqrt{gL}$, air-pocket volume and compressibility, rise time relative to structural period, impulse coefficient, scale ratio and spatial pressure-correlation length.

Major equations. Engineering descriptions combine momentum or pressure-impulse balances with force integration over wetted area; dimensionless peak-force and impulse relations scale with density, velocity, gravity, characteristic length and clearance, while coupled hydroelastic equations add structural inertia, stiffness and damping.

Typical methods. Experiments synchronize wave gauges, high-rate pressure arrays, load cells and video, classify impact regimes, correct sensor and structural ringing, separate impulsive and quasi-static components, integrate pressure and force, repeat nominal conditions statistically and test scale and hydroelastic sensitivity.

Numerical models. Relevant approaches include pressure-impulse and semi-analytical slamming models, multiphase CFD with compressible or entrained air, fully nonlinear potential-flow kinematics and coupled CFD-FEM hydroelasticity; these direct model papers remain in the lawful-full-text queue.

Experimental datasets. Reviewed evidence includes a 1:25 exposed-jetty physical model with wavelet-corrected deck and beam forces and wave-in-deck experiments separating direct, large-air-pocket, small-air-pocket and bubble-plume regimes.

Validated ranges. Current reviewed support is limited to one scaled exposed-jetty configuration and one deck-impact regime study. It does not establish universal peak coefficients for monopiles, caissons, bridge groups, ships or flexible converters.

Recent advances. Recent advances combine ultrafast imaging and dense pressure arrays, compressible multiphase CFD, fully nonlinear incident kinematics, pressure-impulse models, coupled CFD-FEM simulation, Bayesian impact-regime classification and probabilistic load-duration design.

Disagreements. Peak pressure is highly sensitive to sensor bandwidth, local geometry, air and repeatability, so pressure peaks scale less robustly than impulse or global force. Air entrapment may cushion or amplify loads depending on pocket size, compression and venting.

Limitations. Scale effects in air compressibility, viscosity, surface tension and structural elasticity; uncertain breaker geometry; sparse pressure sampling; ringing; non-repeatability; and short records limit transfer. Rigid models cannot prove hydroelastic response or fatigue demand.

Open questions. Priorities include full-scale validation, probabilistic regime transitions, compressible multiphase scaling, spatial coherence, repeated-impact fatigue, flexible response, compound debris impact, uncertainty factors and transferable design metrics based on force, impulse and duration.

Seminal papers. Classical water-entry and Wagner impact theory established rapidly expanding wetted-area loads; pressure-impulse theory and breaking-wave experiments extended concepts to walls, decks and cylinders, followed by air-entrapment and hydroelastic studies.

Equations

- *dimensionless jetty deck and beam load equations* [309]

$$\mathbf{F}(t) = \int_A -p \mathbf{n} dA + \mathbf{F}_{D,I}$$

Regime: 1:25 exposed-jetty head with 2-D and 3-D configurations, inundation, and down-standing beams; excludes tsunami and waves with period above 25 s. Variables: F resultant structural load; p hydrodynamic or pneumatic pressure; n surface normal; A wetted or impacted area; $F_{D,I}$ drag/inertia contribution where applicable; normalized load balance

- *multiscale air-entrapped wave-impact model* [1617]

$$F(t) = \int_A -pn dA + F_{D,I}$$

Regime: Flat-deck wave impact with air structures from tens of microns/millimetres to several/dozens of metres; validated using a typical dam-break impact case. Variables: F resultant structural load; p hydrodynamic or pneumatic pressure; n surface normal; A wetted or impacted area; $F_{D,I}$ drag/inertia contribution where applicable; normalized load balance

Claims

1. For a 1:25 exposed-jetty model, wavelet-corrected measurements separated quasi-static and impulsive deck/beam loads and supported new dimensionless force equations where previous prediction methods were inconsistent. *Regime: 1:25 exposed-jetty head with 2-D and 3-D configurations, inundation, and down-standing beams; excludes tsunami and waves with period above 25 s.* direct finding, experimental [309]
2. Wave-in-deck impacts separate into direct, large-air-pocket, small-air-pocket, and bubble-plume regimes, whose compression and breakup can produce either cushioning or load amplification. *Regime: Flat-deck wave impact with air structures from tens of microns/millimetres to several/dozens of metres; validated using a typical dam-break impact case.* direct finding, numerical [1617]
3. The applicability of pressure-impulse theory is evaluated for predicting wave impact loading magnitudes for non-breaking standing wave impacts on vertical hydraulic structures with relatively short overhangs. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [40]
4. This paper presents results from an experiment designed to improve the understanding of the relationship between extreme breaking waves and their mechanical loading on heritage offshore lighthouses. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [440]

48.4 Porous structures

Topic *wave_structure.porous*. Flow and waves through porous media.

Papers screened: 19. Claims: 19. Equations: 0. Shown in the Coastal Design Bench beside: Breakwater design.

Well established. Porous coastal structures redistribute incident wave energy among reflection, transmission and dissipation through pressure-driven flow in connected voids; both viscous and inertial resistance matter.

Governing physics. Pressure gradients drive oscillatory pore flow. Viscous drag dominates low pore Reynolds number, inertial and turbulent losses grow with velocity, and geometry couples exterior waves to internal flow, phase lag, attenuation and forces.

Dimensionless parameters. Controls include porosity, permeability, pore Reynolds and Keulegan–Carpenter numbers, relative width and depth, wave steepness, submergence, incidence angle, layer count, resistance coefficients, current ratio and structure spacing.

Major equations. Macroscopic models use volume-averaged Navier–Stokes or RANS equations with porosity and Darcy–Forchheimer or Van Gent resistance; alternatives include Brinkman, Sollitt–Cross, eigenfunction matching, VOF and porous-media SPH formulations.

Typical methods. Methods combine flume tests, field reflection monitoring, analytical scattering solutions, VARANS–VOF CFD, SPH, volume-averaged floating-body models, resistance sensitivity studies and data-driven parameter calibration or surrogates.

Numerical models. The branch includes Navier–Stokes–Brinkman, VARANS–VOF/OpenFOAM, weakly compressible SPH mixture theory, eigenfunction matching, Sollitt–Cross theory, depth-averaged porous-barrier solutions and neural or surrogate calibration frameworks.

Experimental datasets. Evidence spans submerged permeable slopes and blocks, vertical screens and perforated plates, multilayer rubble or trapezoidal breakwaters, floating porous bodies, wall-adjacent traps, rubble-mound overtopping cases and a year-long Chabahar field record.

Validated ranges. Validation covers regular, irregular, solitary and oblique waves; thin perforated elements and thick porous zones; fixed, submerged and floating geometries; single and multilayer structures; and cases with current, uneven bottom or nearby walls.

Recent advances. Recent work adds SPH mixture formulations, porous floating structures, current-modified trapping, nonlinear submerged-breakwater tests, machine-learned resistance calibration and fast surrogate-assisted porous CFD.

Disagreements. Equivalent porous coefficients are formulation-dependent: depth-averaging choices, unresolved evanescent modes, thin-screen versus volumetric representations and alternative inertial-drag laws can reproduce one observable while differing internally.

Limitations. Porous coefficients often require case calibration, internal velocity and pressure data are sparse, scale effects alter viscous and turbulent losses, and many validations remain two-dimensional or regular-wave based.

Open questions. Transferable resistance closure, uncertainty quantification, irregular and directional seas, evolving/damaged grading, air entrainment, three-dimensional edge effects, fluid–structure motion and coupling to overtopping and sediment remain open.

Seminal papers. Brinkman and Sollitt–Cross formulations established macroscopic porous-wave theory; later volume-averaged RANS and VOF methods enabled nonlinear free-surface interaction with realistic breakwater geometries.

Claims

1. Depth-averaged pressure-drop formulations for a vertical porous barrier give analytical reflection and transmission solutions whose accuracy depends on the treatment of evanescent modes and porous resistance. *Regime: Comparison of analytical and numerical solutions for wave interaction with a vertical porous barrier* direct finding, analytical [891]
2. Dimensional analysis and experiments show that permeability changes wave reflection, runoff, internal flow and energy dissipation relative to impermeable slopes. *Regime: 2D water-wave interaction with permeable and impermeable slopes: Dimensional analysis and experimental overview* direct finding, mixed [388]

3. A volumetric porous-zone CFD representation can reproduce wave interaction with thin perforated sheets and cylinders without explicitly resolving every opening. *Regime: Using a porous-media approach for CFD modelling of wave interaction with thin perforated structures* direct finding, numerical [431]
4. VARANS-VOF simulation resolves solitary-wave transformation and flow around a permeable submerged breakwater. *Regime: Numerical Simulation of Solitary Wave Induced Flow Motion around a Permeable Submerged Breakwater* direct finding, numerical [1587]
5. A derived Navier–Stokes–Brinkman system represents viscous progressive-wave interaction with a submerged rectangular porous breakwater. *Regime: Navier-Stokes-Brinkman system for interaction of viscous waves with a submerged porous structure* direct finding, analytical [554]
6. A volume-averaged model resolves wave interaction with a movable floating porous structure assembled from uniform spheres. *Regime: Numerical modeling of wave interaction with a porous floating structure consisting of uniform spheres* direct finding, numerical [363]
7. Porous trapezoidal breakwaters near a rigid wall can trap waves, with performance controlled by current, porosity, geometry, spacing and wave conditions. *Regime: Wave trapping by porous breakwater near a rigid wall under the influence of ocean current* direct finding, analytical [1382]
8. Regular-wave tests quantify dynamic pressures and forces on pile-supported breakwaters fitted with inclined perforated plates. *Regime: Wave forces and dynamic pressures on pile-supported breakwaters with inclined perforated plates under regular waves* direct finding, mixed [825]
9. Physical and numerical tests show that porosity modifies nonlinear wave evolution, turbulence and attenuation over a permeable submerged breakwater. *Regime: Experimental and Numerical Study of the Nonlinear Evolution of Regular Waves over a Permeable Submerged Breakwater* direct finding, mixed [1114]
10. Eigenfunction matching predicts oblique-wave scattering and wall forces for arrays of variable porous breakwaters over uneven bathymetry. *Regime: Wave Forces on a Partially Reflecting Wall by Oblique Bragg Scattering with Porous Breakwaters over Uneven Bottoms* direct finding, analytical [247]
11. A weakly compressible SPH porous-media formulation based on mixture theory and intrinsic phase averaging simulates free surfaces and permeable-breakwater interaction. *Regime: A weakly-compressible SPH-porous media model to simulate wave-breakwater interactions* direct finding, numerical [647]
12. Neural-network calibration estimates porosity and Forchheimer coefficients for VARANS models from wave-coastal-structure response data. *Regime: Neural Network calibration method for VARANS models to simulate wave-coastal structures interaction* direct finding, numerical [389]
13. Rubble-mound overtopping predictions are sensitive to the Van Gent porous-friction parameters used in an OpenFOAM model. *Regime: Influence of Van Gent Parameters on the Overtopping Discharge of a Rubble Mound Breakwater* direct finding, numerical [239]
14. A semi-analytical linear pressure-drop model predicts mean drift forces on impermeable body arrays surrounded by thin porous surfaces. *Regime: Mean Drift Wave Forces on Arrays of Bodies Surrounded by Thin Porous Surfaces* direct finding, analytical [732]
15. Linear-wave analysis characterizes reflection and transmission by a multilayer trapezoidal porous breakwater under irregular spectra. *Regime: Performance of multilayered porous breakwater under irregular waves having different wave spectrum* direct finding, analytical [319]
16. Sollitt–Cross porous-flow theory predicts oblique scattering by a slotted porous floating breakwater above a sill-shaped seabed. *Regime: Oblique wave scattering by a rectangular porous floating breakwater with slotted screens over a sill-type seabed* direct finding, analytical [1078]
17. A year of field measurements at Chabahar Bay documents seasonal wave reflection from a rubble-mound breakwater and its relation to incident wave conditions. *Regime: Monsoonal patterns of wave reflection from rubble mound breakwater of Chabahar Bay* direct finding, field [902]
18. A surrogate-assisted macroscopic CFD framework predicts wave interaction with porous marine structures governed by equivalent pressure-drop resistance. *Regime: A surrogate-assisted numerical framework for predicting wave interaction with porous marine structures* direct finding, numerical [860]
19. IHFOAM implements mass-conserving two-phase VARANS porous-media flow, wave generation and active absorption in OpenFOAM; dam-break, 2D breakwater, and 3D wave-basin tests validate free-surface, pressure, and porous-structure interaction behavior. *Regime: Two-phase free-surface flow through porous media, including dam-break transients and regular or solitary waves interacting with 2D and 3D porous coastal structures.* direct finding, mixed [596]

48.4.1 Porous-flow resistance

Topic `wave_structure.porous.resistance`. Darcy, Forchheimer, and calibrated resistance formulations for wave-driven porous flow.

Papers screened: 12. Claims: 9. Equations: 7.

Well established. Porous coastal structures dissipate wave energy through viscous and inertial resistance to internal oscillatory flow. Resistance also changes phase, pressure, harmonics, runup, overtopping, and force transmission, so porosity alone cannot describe performance.

Governing physics. Linear Darcy resistance scales with velocity and viscosity/permeability, while Forchheimer resistance scales nonlinearly with velocity magnitude. Inertia, turbulence, resonance, interface exchange, material layering, reef geometry, submergence, and breaking modify the balance.

Dimensionless parameters. Key controls include porosity, permeability or Darcy number, Forchheimer coefficient, Reynolds number, Keulegan-Carpenter number, B/L , relative submergence, relative grain or element size, layer thickness, wave steepness, and relative depth.

Major equations. Continuum models combine volume-averaged mass and momentum conservation with Darcy, Brinkman, or extended Forchheimer resistance; wave-energy balances relate attenuation to drag work, while particle models couple exterior Navier-Stokes motion to porous closures.

Typical methods. Methods use internal velocity and pressure measurements, wave-energy-flux budgets, direct force sensing, empirical resistance calibration, analytical attenuation models, Eulerian volume-averaged CFD, and Lagrangian SPH or moving-particle solvers.

Numerical models. Reviewed models include ISPH/ISPHP and modified moving-particle formulations with extended Forchheimer resistance, multilayer interface treatment, and validation against porous beds, rockfill, submerged structures, permeable bars, and armored caissons.

Experimental datasets. The branch includes idealized internal-flow measurements that distinguish transition and transmission zones and open-access single- and multi-row cubic-reef experiments resolving in-reef velocity, directional forces, and dissipation.

Validated ranges. Examples include an SPH porous bed with $n=0.39$, $d=0.55$ cm, $K=1.02e-8$ m² and $C_F=0.51$; idealized structures organized by B/L ; and single-/multi-row cubic reefs under nonbreaking regular waves across several submergence depths.

Recent advances. Direct measurements in porous artificial reefs show drag coefficient decreases with reef Keulegan-Carpenter number and that horizontal drag work dominates nonbreaking-wave dissipation, enabling a mechanism-based analytical attenuation model.

Disagreements. There is no universal resistance equation or coefficient set across materials and flow regimes. Continuum closures differ in linear, inertial, Brinkman, turbulence, and interface terms, while apparent coefficients depend on velocity definition, scale, geometry, and calibration dataset.

Limitations. Most evidence uses regular or solitary waves, idealized two-dimensional geometry, homogeneous or simplified layering, continuum volume averaging, and source-specific coefficients. Breaking, air entrainment, movable armor, irregular spectra, scale effects, and prototype construction variability remain weakly constrained.

Open questions. Priorities are transferable coefficient scaling, pore-to-structure upscaling, three-dimensional and irregular-breaking validation, uncertainty propagation, coupling to stability and sediment response, and optimization of ecological geometry without losing structural performance.

Seminal papers. The 1995 internal-flow experiment established B/L-dependent transition and transmission regions; subsequent SPH studies built unified free/porous particle formulations and multilayer capability.

Equations

- *Darcy-Forchheimer porous resistance* [1231]

$$-\frac{\partial p}{\partial x} = \rho(\alpha + \beta u)u$$

Regime: Unsteady flow through the homogeneous porous-breakwall idealization. Variables: p pressure; x wave-propagation coordinate; ρ water density; u horizontal pore velocity; α linear resistance coefficient; β quadratic resistance coefficient

- *Porosity-scaled resistance coefficients* [1231]

$$\alpha = \alpha_0 \frac{(1-n)^3}{n^2} \frac{\nu}{d^2}, \quad \beta = \beta_0 \frac{1-n}{n^3} \frac{1}{d}$$

Regime: Madsen porous-breakwater formulation adopted for branch media. Variables: α_0 dimensionless laminar drag coefficient; β_0 dimensionless turbulent drag coefficient; n porosity; ν kinematic viscosity; d representative material diameter

- *Darcy-Forchheimer porous resistance* [865]

$$\mathbf{R} = -\frac{\mu}{K}\mathbf{u} - \rho \frac{C_F}{\sqrt{K}}|\mathbf{u}|\mathbf{u}$$

Regime: Idealized porous structure under oscillatory waves; regime organized by relative structure width B/L. Variables: $\mathbf{R}$ porous resistance force per volume; μ dynamic viscosity; K intrinsic permeability; $\mathbf{u}$ volume-averaged velocity; ρ fluid density; C_F nonlinear inertial resistance coefficient

- *Darcy-Forchheimer porous resistance* [1291]

$$\mathbf{R} = -\frac{\mu}{K}\mathbf{u} - \rho \frac{C_F}{\sqrt{K}}|\mathbf{u}|\mathbf{u}$$

Regime: Two-dimensional laminar exterior flow; porous-bed validation with porosity 0.39, grain diameter 0.55 cm, permeability $1.02\text{e-}8\text{ m}^2$, and nonlinear coefficient 0.51. Variables: $\mathbf{R}$ porous resistance force per volume; μ dynamic viscosity; K intrinsic permeability; $\mathbf{u}$ volume-averaged velocity; ρ fluid density; C_F nonlinear inertial resistance coefficient

- *Darcy-Forchheimer porous resistance* [24]

$$\mathbf{R} = -\frac{\mu}{K}\mathbf{u} - \rho \frac{C_F}{\sqrt{K}}|\mathbf{u}|\mathbf{u}$$

Regime: Two-dimensional particle simulations of porous dams, regular-wave attenuation over porous seabeds, and runoff/overtopping at a porous-armored caisson. Variables: $\mathbf{R}$ porous resistance force per volume; μ dynamic viscosity; K intrinsic permeability; $\mathbf{u}$ volume-averaged velocity; ρ fluid density; C_F nonlinear inertial resistance coefficient

- *Darcy-Forchheimer porous resistance* [25]

$$\mathbf{R} = -\frac{\mu}{K}\mathbf{u} - \rho \frac{C_F}{\sqrt{K}}|\mathbf{u}|\mathbf{u}$$

Regime: Two-dimensional uniform and multilayer porous media with benchmark U-tube, rockfill, solitary-wave, submerged-structure, and triangular-bar cases. Variables: $\mathbf{R}$ porous resistance force per volume; μ dynamic viscosity; K intrinsic permeability; $\mathbf{u}$ volume-averaged velocity; ρ fluid density; C_F nonlinear inertial resistance coefficient

- *Darcy-Forchheimer porous resistance* [621]

$$\mathbf{R} = -\frac{\mu}{K}\mathbf{u} - \rho \frac{C_F}{\sqrt{K}}|\mathbf{u}|\mathbf{u}$$

Regime: Single- and multi-row porous cubic artificial reefs under nonbreaking regular waves over multiple submergence depths. Variables: $\mathbf{R}$ porous resistance force per volume; μ dynamic viscosity; K intrinsic permeability; $\mathbf{u}$ volume-averaged velocity; ρ fluid density; C_F nonlinear inertial resistance coefficient

Claims

1. For solitary-wave propagation over a porous bed with $n=0.39$, $d=0.55$ cm, $K=1.02e-8$ m², and nonlinear resistance coefficient 0.51, an incompressible SPH model reproduced theoretical wave-height damping while coupling exterior Navier-Stokes and interior porous flow. *Regime: Two-dimensional laminar exterior flow; porous-bed validation with porosity 0.39, grain diameter 0.55 cm, permeability $1.02e-8$ m², and nonlinear coefficient 0.51.* direct finding, numerical [1291]
2. A unified ISPHP formulation using Navier-Stokes flow outside and extended Forchheimer resistance inside porous media agreed with laboratory free-surface displacement, overtopping rate, and pressure for porous-bed and armored-caisson applications. *Regime: Two-dimensional particle simulations of porous dams, regular-wave attenuation over porous seabeds, and runoff/overtopping at a porous-armored caisson.* direct finding, numerical [24]
3. The modified moving-particle porous-flow method reproduced benchmark behavior across uniform and multilayer media, including rockfill seepage, solitary-wave attenuation over porous beds, submerged porous structures, and permeable triangular bars. *Regime: Two-dimensional uniform and multilayer porous media with benchmark U-tube, rockfill, solitary-wave, submerged-structure, and triangular-bar cases.* direct finding, numerical [25]
4. In an idealized porous coastal structure, relative width B/L controls the transition from irregular, higher-harmonic-rich flow with little dissipation to a transmission region dominated by frictional dissipation and filtering of higher frequencies. *Regime: Idealized porous structure under oscillatory waves; regime organized by relative structure width B/L .* direct finding, experimental [865]
5. For single- and multi-row porous cubic reefs under nonbreaking regular waves, drag coefficients decreased with reef Keulegan-Carpenter number and horizontal drag work dominated dissipation, while vertical drag made a secondary contribution. *Regime: Single- and multi-row porous cubic artificial reefs under nonbreaking regular waves over multiple submergence depths.* direct finding, mixed [621]
6. Canopy layers control momentum and solute transport to and from the overlying water surface layer. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [1217]
7. This paper treats the numerical modelling of the behaviour of a sand core covered by rocks and exposed to waves. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [643]
8. Abstract The dynamic interaction between swash and beach groundwater is fundamental to understanding wave runoff and sediment transport processes. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [332]
9. Wave scattering over porous structures plays an important role in the design of coastal and offshore protective structures. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [681]

48.4.2 Transmission through porous structures

Topic *wave_structure.porous.transmission*. Wave-height and energy transmission controlled by permeability, porosity, geometry, and flow resistance.

Papers screened: 3. Claims: 5. Equations: 2.

Well established. Porous coastal structures partition incident wave energy among reflection, transmission and dissipation through internal flow resistance; performance depends jointly on porosity, material scale, thickness, submergence, water depth, incident spectrum and external breaking.

Governing physics. The adopted Darcy-Forchheimer resistance combines linear viscous and quadratic turbulent terms scaled by porosity and representative material diameter.

Dimensionless parameters. Porosity n , relative structure height h_s/h , and dimensionless laminar and turbulent coefficients α_0 and β_0 are central controls in the reviewed formulation.

Major equations. Observed wave-height transmission is $K_t = \sqrt{F_t/F_i}$, while the adopted equivalent-linearized porous theory gives $K_t = 1/(1+\lambda)$ with λ determined by wavenumber, porous width, porosity, and friction resistance.

Typical methods. Studies measure synchronized incident and lee waves across water levels and sea states, estimate spectral energy transmission, characterize porosity and geometry, calibrate Darcy-Forchheimer resistance, test emergence and submergence, close reflection-transmission-dissipation budgets and validate independently.

Numerical models. The reviewed model is an equivalent-linearized porous-transmission formulation using Darcy-Forchheimer resistance with fixed laminar and calibrated turbulent coefficients; it omits explicit external breaking and bottom friction.

Experimental datasets. Reviewed evidence comprises 290 boat-wake observations at a bundled-branch living-shoreline breakwall with estimated porosity 0.7, alongside a contrasting unbundled 0.9-porosity site whose geometry and tidal setting differed.

Validated ranges. With $\alpha_0=1140$ fixed and $\beta_0=2.7$ calibrated, the theory produced 16% mean absolute relative discrepancy across 290 GTM boat wakes; this is calibration performance, not independent validation.

Recent advances. Recent advances combine field-scale living structures, spectral transmission, porous RANS/VOF and resolved-element CFD, morphology and maintenance monitoring, uncertainty-aware calibration and hybrid formulations that couple internal resistance with external breaking.

Disagreements. Porosity alone cannot explain field transmission when bundling, thickness, material, water level, site and tidal range covary. Internal resistance models may perform while submerged but overtransmit near emergence where external breaking and friction dominate.

Limitations. Porous-only transmission theory overestimated transmitted energy near emergence because it omitted breaking and external bottom friction; its reliability deteriorated precisely where shallow-water dissipation became important.

Open questions. Independent validation is needed across materials, porosities, spectra, oblique incidence, storms, emergence states, and structures whose external breaking and overtopping interact with internal porous flow.

Seminal papers. Darcy and Forchheimer resistance established linear and quadratic porous-flow losses; rubble-mound wave theories coupled those losses to reflection and transmission, followed by equivalent-linear, volume-averaged and porous-media CFD formulations.

Equations

- Observed wave-height transmission coefficient [1231]

$$K_{T,data} = \sqrt{F_t/F_i}$$

Regime: Linear-wave energy reconstruction from pressure/velocity at sensors immediately offshore and onshore of the breakwall. Variables: $K_{T,data}$ wave-height transmission coefficient derived from field data; F_t transmitted wave energy flux; F_i incident wave energy flux

- Theoretical porous-breakwall transmission [1231]

$$K_{T,theory} = H_t/H_i = 1/(1 + \lambda)$$

Regime: Linearized Madsen theory with equivalent-linearized porous friction; omits shallow-water breaking and external bottom friction.

Variables: H_t transmitted wave height; H_i incident wave height; λ dimensionless resistance term depending on wavenumber, width, friction factor, and porosity

Claims

- For the 0.7-porosity bundled breakwall, wave-energy transmission increased with water depth most clearly when breakwall height divided by depth was approximately 0.5-1. *Regime: GTM deployment and its 1-1.5 m semidiurnal tide; not a universal depth law for all porous structures.* direct finding, field [1231]
- With α_0 fixed at 1140 and β_0 calibrated to 2.7, porous-transmission theory had 16% mean absolute relative discrepancy over 290 GTM wakes. *Regime: Madsen-type theory applied to the GTM 0.7-porosity breakwall and the 290 wakes used in calibration; this is not independent validation.* direct finding, mixed [1231]
- The calibrated porous theory overpredicted transmitted energy most strongly in shallow water as the breakwall emerged, where omitted breaking and bottom friction became important. *Regime: Shallow/emergent GTM wake cases under the paper's theory; breaking and bottom-friction contributions were inferred, not separately measured.* inferred, mixed [1231]
- The observed transmission contrast between porosity 0.7 and 0.9 cannot be attributed to porosity alone because bundling, site, and tidal range also differed. *Regime: Comparison between GTM and NP; this bounds causal interpretation of the reported averages.* inferred, field [1231]
- Two-dimensional flume tests show porous-breakwater transmission, reflection, and energy loss depend jointly on porosity, relative submergence, and width; the lowest transmission occurred for the least porous emerged case, while the lowest reflection occurred for the most porous submerged case. *Regime: Fixed vertical thick porous breakwaters at relative heights $hb/h=0.8, 1.0$, and 1.2 under regular waves.* direct finding, experimental [1148]

48.5 Transmission and reflection

Topic *wave_structure.transmission*. Wave modification by structures.

Papers screened: 16. Claims: 13. Equations: 0. Shown in the Coastal Design Bench beside: Breakwater design.

Well established. Wave transmission and reflection depend jointly on relative geometry, permeability, internal barriers, structural motion, wavelength, incidence direction, and spectral character; a coefficient calibrated for one structure family should not be transferred without checking its validity regime.

Governing physics. Fixed submerged structures redistribute incident energy through reflection, transmission, breaking and porous dissipation, while floating structures add heave and other motion-coupled pathways; periodic arrays can also create Bragg resonance.

Dimensionless parameters. Important controls in this evidence slice include porosity, relative submergence and crest geometry, kd , wavelength-to-structure scale, wave incidence, and the ratio of incident peak period to approximate natural heave period.

Major equations. Reviewed approaches include empirical hyperbolic-tangent transmission relations, Macagno-type fixed-box relations modified for floating heave, linear-wave eigenfunction matching, oblique-wave Bragg conditions, and database-based intercomparison of low-crested-structure formulae.

Typical methods. Typical methods combine regular- and irregular-wave flumes, three-dimensional basins, incident/reflected wave separation, motion and mooring measurements, analytical matching or shallow-water solutions, finite-volume and potential-flow models, and formula intercomparison against harmonized experimental databases.

Numerical models. The reviewed numerical approaches span linear eigenfunction matching for periodic structures, boundary-discretization models for pontoon arrays, finite-volume shallow-water solvers for rectangular bars, fully nonlinear potential-flow models for WEC breakwaters, and empirical transmission predictors.

Experimental datasets. The reviewed abstract-level datasets include porous models with porosity 0.40-0.80, eight chain-moored pi-type models of 16-76 kg, a ten-unit floating system in a 36 m by 60 m by 1.5 m basin, and trapezoidal impermeable, permeable and perforated submerged structures.

Validated ranges. Directly reported bounds include porosity 0.40-0.80 with reflection coefficients 0.01-0.31 for non-breaking porous-model tests and masses of 16-76 kg for eight pi-type floating-breakwater models; other numerical ranges require full-text review.

Recent advances. Recent advances couple attenuation with wave-energy conversion, explore nonuniform-draft and hybrid OWC/OB systems, resolve resonant transmission-motion-load tradeoffs, and optimize internal screens, porosity, and periodic geometry using combined experiments and nonlinear models.

Disagreements. Apparent design rankings depend on the objective: a porous configuration can transmit more energy yet reduce motions and mooring loads, while mesh or internal barriers can improve attenuation but increase selected loads. These are regime-dependent tradeoffs rather than universally contradictory results.

Limitations. The present synthesis is abstract-limited. Scale effects, exact wave ranges, coefficient uncertainty, structural damage, prototype bathymetry, and field validation remain unresolved, and linear Bragg results apply only under small-amplitude assumptions.

Open questions. Priority needs are independent cross-formula benchmarks, prototype validation under irregular multidirectional seas, uncertainty-aware extrapolation limits, and coupled assessment of transmission, reflection, dissipation, structural motion, loads and morphology.

Seminal papers. Within this bounded corpus, early permeable-breakwater and low-crested formula families establish the design lineage, while later harmonized reviews make their calibration and extrapolation limits explicit; the present slice does not yet support a definitive field-wide priority ranking.

Claims

1. For the tested submerged trapezoidal structures, adding an impermeable vertical screen inside a perforated hollow structure reduced wave transmission substantially relative to a perforated structure without the screen, whereas differences between impermeable and permeable structures were comparatively limited. *Regime: Trapezoidal impermeable, permeable, and perforated submerged structures; abstract-level evidence only.* direct finding, experimental [491]
2. A harmonized experimental database permits transmission formulae for submerged and low-crested breakwaters to be compared both within and beyond their original validity limits; agreement reported for one formula within its calibration range is not evidence of reliable extrapolation. *Regime: Submerged and low-crested breakwaters represented in the collected experimental database; abstract-level evidence does not report numerical validity bounds.* review statement, review [185]
3. Under the study's small-amplitude, linear-wave assumptions, eigenfunction-matching calculations for periodic surface-piercing and submerged structures found that oblique-wave Bragg's law predicted the occurrence of Bragg resonance. *Regime: Small-amplitude oblique waves interacting with periodic surface-piercing or submerged structures over step-approximated periodic bottoms.* direct finding, mixed [1429]
4. For hollow framework submerged-breakwater models with porosity from 0.40 to 0.80 under non-breaking waves, transmission increased with porosity, measured reflection coefficients ranged from 0.01 to 0.31, and primary-wave energy loss was greatest near $kd=1.9$. *Regime: Non-breaking waves over hollow framework submerged-breakwater models with porosity 0.40-0.80.* direct finding, experimental [1306]
5. For tested chain-moored pi-type floating breakwaters, an empirical transmission predictor based on the ratio of incident peak period to approximate natural heave period corrected the fixed-box Macagno relationship and showed fairly good agreement with the study and literature datasets. *Regime: Eight chain-moored pi-type floating-breakwater models of 16-76 kg plus literature tests under regular and irregular waves.* direct finding, mixed [1225]
6. For the tested ten-unit taut-moored cylindrical floating-breakwater system, both transmission coefficient and motion-response amplitude increased with wavelength to a peak and then decreased under the examined oblique and beam sea conditions. *Regime: A ten-unit cylindrical floating-breakwater system with mesh cages and taut moorings under oblique and beam seas in a 36 m by 60 m by 1.5 m basin.* direct finding, experimental [657]
7. For the tested hybrid oscillating-buoy/oscillating-water-column breakwater, the combined system improved both conversion and attenuation relative to isolated devices; gap resonance reduced extraction but had negligible attenuation effect, and OWC PTO damping primarily controlled transmission. *Regime: Hybrid upstream oscillating buoy and downstream oscillating-water-column floating breakwater under the tested wave, draft, gap, and PTO conditions.* direct finding, mixed [268]
8. Within the tested equal-displacement comparison, a nonuniform-draft oscillating-water-column dual-pontoon breakwater provided better attenuation and conversion than the oscillating-buoy single pontoon, while increasing chamber-width ratio improved attenuation but reduced maximum conversion efficiency. *Regime: Compared single-pontoon oscillating-buoy and dual-pontoon oscillating-water-column floating breakwaters over tested geometry, gap, wave, and PTO ranges.* direct finding, mixed [269]
9. For the modeled fixed floating-pontoon trains, Bragg-reflection coefficients depended on breakwater spacing, installed pontoon count, draft, width, and wavelength, with the numerical model agreeing with analytical literature benchmarks. *Regime: Fixed floating-pontoon trains over the modeled spacing, count, draft, width, and wavelength combinations.* direct finding, numerical [1063]
10. For rectangular submerged bars under the study's shallow-water and monochromatic-wave assumptions, analytical and finite-volume solutions were used with experimental data to identify bar height and length combinations that minimize transmission. *Regime: Monochromatic waves passing one or multiple rectangular submerged bars under shallow-water-equation assumptions.* direct finding, mixed [899]
11. In the four-model regular-wave comparison, the cylindrical floating breakwater with mesh cage and suspended balls produced the strongest attenuation, particularly for long and high waves; the mesh reduced heave and roll but increased sway and could raise short-period windward mooring tension. *Regime: Four floating-breakwater models under regular waves, including cylindrical pontoons with a mesh cage and suspended balls; accepted manuscript reviewed in full.* direct finding, experimental [656]
12. Across four regular-wave floating-breakwater configurations, mesh-cage type I reflected and dissipated waves most efficiently, whereas the porous model admitted more transmitted energy but produced the smallest heave, roll, and mooring forces. *Regime: Cylindrical, porous, and two mesh-cage floating-breakwater configurations under tested regular waves; accepted manuscript reviewed in full.* direct finding, experimental [658]
13. For four slack-moored floating-breakwater models in long regular waves, transmission and dissipation exhibited turning points near structural natural periods; porous plates and mesh cages added mass and damping that mitigated these features, while resonance coupled attenuation to motions and mooring loads. *Regime: Four slack-moored single-box, porous-box, double-pontoon, and mesh-cage floating breakwaters under long regular waves; accepted manuscript reviewed in full.* direct finding, experimental [659]

48.5.1 Partially immersed curtain walls

Topic *wave_structure.transmission.curtain_walls*. Wave passage beneath surface-piercing plates and curtain-type breakwaters.

Papers screened: 2. Claims: 4. Equations: 1.

Well established. A curtain wall attenuates waves by reflecting the upper water-column motion while permitting flow beneath its lower edge; transmission generally decreases with increasing draft, but reflection, dissipation, gap flow and structural load trade off.

Governing physics. A surface-piercing curtain reflects the upper wave mass while allowing underflow through the bottom gap; increasing draft strengthens reflection and reduces transmitted energy.

Dimensionless parameters. Principal controls include draft-to-depth and gap-to-depth ratios, kh , wave steepness or relative solitary-wave amplitude, curtain spacing and thickness relative to wavelength, transmission and reflection coefficients, dissipated-energy fraction and incidence angle.

Major equations. The reviewed study evaluates amplitude coefficients $K_t = a_t/a_i$ and $K_r = a_r/a_i$ as functions of normalized curtain draft z/H .

Typical methods. Studies generate incident waves in a flume, separate incident and reflected signals with multiple gauges, measure lee waves and water levels near the gap, vary draft and bottom-mounted versus hanging orientation, compute energy coefficients, test conservation and document forces and three-dimensional leakage where available.

Numerical models. The current evidence is experiment-centered and evaluated with amplitude and energy coefficients. Potential-flow eigenfunction or boundary-integral models, CFD and double-curtain chamber models are represented only by title-only queue records pending full-text review.

Experimental datasets. Reviewed evidence comprises solitary-wave tests for a thin channel-spanning semi-submerged curtain and a fully submerged bottom-mounted barrier, including illustrated 14 cm water-depth cases with zero and 5 cm curtain draft.

Validated ranges. Explicit illustrated cases use $H=14$ cm and curtain drafts $z=0$ and 5 cm; the paper presents additional response points graphically but does not tabulate a complete numeric test matrix.

Recent advances. Recent advances use multi-curtain and perforated hybrids, resonant chambers, optimization across spectra, oblique-wave boundary-integral models, viscous CFD, flexible curtains and integrated wave-energy or harbor-protection concepts.

Disagreements. Greater draft reduces transmission but may increase reflection and wave force; adding chambers or multiple plates can enhance dissipation yet introduce resonance. Solitary-wave performance does not directly predict irregular spectral, oblique, current-affected or overtopping behavior.

Limitations. The reported maximum 60% energy dissipation applies to rigid thin channel-spanning walls under solitary waves and does not establish performance for irregular seas, oblique incidence, structural motion, or prototype scale.

Open questions. A fully tabulated parameter matrix and force measurements are needed across gap ratio, draft, wave nonlinearity, irregular spectra, oblique incidence, currents, and flexible or porous curtains.

Seminal papers. Linear potential-flow solutions for partial-depth barriers established reflection and transmission versus gap and kh ; flume studies extended these relations to nonlinear waves, curtain breakwaters, double walls and chamber resonance.

Equations

- Amplitude transmission and reflection [1636]

$$K_t = a_t/a_i, \quad K_r = a_r/a_i$$

Regime: Separated solitary waves before and after a thin vertical barrier. Variables: K_t amplitude transmission coefficient; K_r amplitude reflection coefficient; a_i incident solitary-wave amplitude; a_t transmitted amplitude; a_r reflected amplitude

Claims

1. Increasing the relative draft of the partially immersed curtain wall increased solitary-wave reflection and reduced transmitted-wave energy. *Regime: One-centimeter-thick full-width rigid curtain wall in the paper's horizontal flume and solitary-wave conditions.* direct finding, experimental [1636]
2. The semi-submerged curtain wall dissipated up to 60% of incident solitary-wave energy, compared with up to 20% for the fully submerged bottom-mounted barrier. *Regime: The paper's thin vertical barriers and tested solitary waves; maxima should not be generalized to other spectra or geometries.* direct finding, experimental [1636]
3. At zero curtain draft, most water passed below the wall with negligible or very small reflected wave; a 5 cm draft in 14 cm water produced stronger reflection and weaker transmission. *Regime: The illustrated $H=14$ cm curtain-wall trials.* direct finding, experimental [1636]
4. For the tested curtain-wall pile breakwater, reflection increases with wall-depth ratio and wave steepness: raising h/d from 0 to 0.7 increased K_r from about 0.21 to 0.49 at $H_i/L=0.0097$ and from about 0.36 to 0.60 at $H_i/L=0.0499$. *Regime: The tested precast curtain-wall pile-breakwater geometry, wall-depth ratios 0-0.7, and incident steepness cases H_i/L 0.0097 and 0.0499.* direct finding, mixed [707]

48.5.2 Submerged barriers

Topic wave_structure.transmission.submerged. Transmission, reflection, dissipation, and loading for structures whose crests remain below or near the free surface.

Papers screened: 8. Claims: 13. Equations: 4.

Well established. Submergence relative to water depth and wave height strongly controls transmission: deeper water cover generally increases passage, while a crest near the still-water level can maximize breaking and dissipation for some rigid geometries.

Governing physics. Submerged structures transform waves through partial blockage, shoaling over the crest, reflection at depth changes, nonlinear free-surface motion, breaking and turbulent separation; greater water cover usually reduces interaction, while crest-level conditions can maximize breaking and dissipation.

Dimensionless parameters. Reviewed controls include crest cover R_c/d and R_c/H_i , crest width B/L , wave steepness, relative depth kd , relative barrier height h/H , solitary-wave interaction $a/(H-h)$, and the composite $\beta = R_c^2 \sqrt{L}/(d H_i \sqrt{B})$.

Major equations. Common response measures are $K_t = H_t/H_i$, $K_r = H_r/H_i$, and an energy remainder based on $1 - K_t^2 - K_r^2$; care is needed because some studies report the remainder itself while others report its square root as an amplitude-like loss coefficient.

Typical methods. Studies measure incident, reflected and transmitted waves with gauge arrays, vary water depth, cover, crest width and shape, separate waves and compute amplitude and energy coefficients, test regular, irregular or solitary forcing, compare material volume and validate empirical dimensionless fits independently.

Numerical models. The reviewed branch emphasizes laboratory response and empirical logarithmic transmission/reflection fits. Potential-flow, eigenfunction, CFD and practical fixed-structure models are represented by title-only queue records pending lawful full-text review.

Experimental datasets. Reviewed evidence includes solitary waves at 11-14 cm height, 75 JONSWAP tests with periods 1.1-1.9 s and cover 0.05-0.15 m across five smooth sections, and 216 regular-wave tests spanning 0.02-0.38 m height, 0.8-2.5 s period and B/L 0.17-1.22.

Validated ranges. The reviewed experiments cover solitary-wave examples at H=11-14 cm; 75 JONSWAP tests with $T_p=1.1-1.9$ s, $d=0.30-0.40$ m, $R_c=0.05-0.15$ m and steepness 0.018-0.055; and 216 regular-wave semicircular tests with $H_i=0.02-0.38$ m, $T=0.8-2.5$ s, $d/h=0.667-1.667$ and $B/L=0.17-1.22$.

Recent advances. Recent advances optimize shape and material volume across irregular spectra, use high-resolution free-surface imaging and PIV, couple RANS/VOF with turbulence and air, explore metamaterial or resonant plates and apply probabilistic multiobjective design under tidal water-level variation.

Disagreements. Reported maximum attenuation is not directly comparable across geometries and forcing: the solitary-wave submerged plate dissipated up to 20%, whereas the crest-level semicircular barrier reached up to 90% energy dissipation for short regular waves; these values reflect different response definitions, spectra, widths, and immersion states.

Limitations. The reviewed ranges remain laboratory- and geometry-specific; none resolves irregular oblique waves with currents, tidal water-level variation, sediment response, structural motion or failure, and field-scale validation together.

Open questions. Transferable design requires harmonized coefficient definitions and experiments spanning geometry, permeability, spectra, obliquity, currents, sediment, structural loading, and changing water levels.

Seminal papers. Linear potential-flow and eigenfunction solutions established plate and submerged-barrier coefficients; laboratory studies introduced breaking and nonlinear regimes, followed by empirical low-crested-breakwater formulas and phase-resolving or viscous numerical models.

Equations

- Submerged-barrier interaction coefficient [1636]

$$K_{int} = a/(H - h)$$

Regime: Solitary wave crossing a thin bottom-mounted submerged plate. Variables: K_{int} wave-barrier interaction coefficient; a incident wave amplitude; H still-water depth; h bottom-mounted barrier height

- Wave coefficients [901]

$$K_t = H_t/H_i, \quad K_r = H_r/H_i, \quad K_d = 1 - K_t^2 - K_r^2$$

Regime: Spectrally separated irregular waves around an impermeable submerged breakwater. Variables: K_t wave-height transmission coefficient; K_r wave-height reflection coefficient; K_d energy dissipation fraction; H_i incident significant wave height; H_t transmitted significant wave height; H_r reflected significant wave height

- Composite submerged-breakwater parameter [901]

$$\beta = R_c^2 L^{1/2} / (d H_i B^{1/2})$$

Regime: Smooth impermeable rectangular-section breakwaters in the reported ranges; trapezoids require different fitted coefficients.

Variables: β composite response parameter; R_c crest submergence; L local wavelength; d local water depth; H_i incident wave height; B crest width

- Semicircular-breakwater wave coefficients [27]

$$C_T = H_T/H_i, \quad C_R = H_R/H_i, \quad C_L = \sqrt{1 - C_T^2 - C_R^2}$$

Regime: Regular-wave measurements around the tested impermeable semicircular breakwater. Variables: C_T wave-height transmission coefficient; C_R reflection coefficient; C_L energy-loss coefficient; H_T transmitted height; H_R reflected height; H_i incident height

Claims

- For the bottom-mounted submerged plate, the interaction changed from smooth soliton splitting to intense local free-surface oscillation when incident amplitude divided by water cover over the plate crossed a critical value near one. *Regime: Thin submerged vertical plate and solitary waves; reported threshold is approximate.* direct finding, experimental [1636]
- Seventy-five JONSWAP tests covered peak periods 1.1-1.9 s, local depths 0.30-0.40 m, crest submergence 0.05-0.15 m, and wave steepness 0.018-0.055 across five smooth impermeable sections. *Regime: Two-dimensional normally incident irregular waves; 0.25 m high smooth PVC barriers.* direct finding, experimental [901]
- Relative submergence R_c/d was the most influential individual dimensionless control on measured transmission and reflection among the simple ratios evaluated. *Regime: The four rectangular/toothed sections and reported JONSWAP tests.* direct finding, experimental [901]
- For rectangular sections, $\beta = R_c^2 \sqrt{L} / (d H_i \sqrt{B})$ produced a single increasing logarithmic K_t fit with $R^2=0.97$ and a decreasing logarithmic K_r fit with $R^2=0.86$. *Regime: The reported smooth rectangular and toothed sections; coefficients are geometry-specific outside that family.* direct finding, experimental [901]
- A 0.60 m wide rectangular wedge transmitted about 30% less and dissipated about 40% more energy than a trapezoidal wedge with the same upper width at 0.40 m water depth, while using about 45% less material volume. *Regime: Specific BS2 versus BS5 geometries at $d=0.40$ m; not a universal rectangle-versus-trapezoid rule.* direct finding, experimental [901]
- The 216 physical tests spanned incident heights 0.02-0.38 m, periods 0.8-2.5 s, steepness 0.02-0.06, relative depth $d/h=0.667-1.667$, and relative width $B/L=0.17-1.22$. *Regime: $R=h=0.6$ m semicircular concrete model under regular waves.* direct finding, experimental [27]

7. At crest-level submergence $d/h=1.0$, short-period regular waves experienced up to 70% incident-height reduction and up to 90% energy dissipation. *Regime: The tested semicircular breakwater and shorter-period portion of $T=0.8-2.5$ s; not all B/L and steepness combinations attained these maxima.* direct finding, mixed [27]
8. Fully submerged cases transmitted substantially more wave height: CT ranged 0.59-0.99 at $d/h=1.333$ and 0.68-0.98 at $d/h=1.667$. *Regime: The study's regular waves and semicircular model.* direct finding, mixed [27]
9. Energy-dissipation performance deteriorated as immersion depth increased beyond the crest-level configuration. *Regime: $d/h=1.0, 1.333$, and 1.667 for the tested regular-wave matrix.* direct finding, mixed [27]
10. With the recent development from grey infrastructures to green infrastructures, artificial reefs become more popular in coastal protection projects. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, numerical [747]
11. Abstract To address the survivability and energy capture efficiency of Oscillating Wave Surge Converters (OWSCs) under extreme sea states, this study develops a high-fidelity numerical model coupling Riemann-based Weakly Compressible Smoothed Particle Hydrodynamics (Riemann-based WCSPH) with multi-body dynamics. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [553]
12. Tsunami wave overtopping remains a major challenge for conventional vertical seawalls. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [664]
13. DELOS synthesis of more than 2,300 tests yields regime-specific low-crested-structure transmission formulas: rubble-mound transmission is nearly insensitive to obliquity, whereas smooth structures show cosine-dependent transmission and cap transmitted direction near 45 degrees for larger incident angles. *Regime: Smooth and rubble-mound low-crested or submerged structures under random waves, varying relative crest width and oblique incidence.* direct finding, mixed [947]

49 Wiki: Waves and wave transformation

49.1 Waves and wave transformation: overview

Topic waves. Generation, propagation, and transformation of surface gravity waves.

Papers screened: 10. Claims: 1. Equations: 0.

Well established. Surface-gravity waves are generated by forcing such as wind and propagate dispersively; decreasing depth causes shoaling, refraction, nonlinear spectral transfer and ultimately breaking, while abrupt or three-dimensional bathymetry produces diffraction, focusing and wave-driven circulation.

Governing physics. Gravity and inertia set dispersion, conservation of wave action governs slowly varying propagation, depth contours refract rays, lateral gradients diffract energy, and nonlinear interactions transfer energy among frequencies. Breaking dissipates energy and radiation-stress gradients force mean water levels and currents.

Dimensionless parameters. Controls include relative depth kh , wave steepness H/L , Ursell, Iribarren, Froude and Reynolds numbers, depth and slope ratios, directional and frequency bandwidth, relative bathymetric-feature size, breaking index and normalized energy or power flux.

Major equations. Core descriptions include linear dispersion and spectral action balance, mild-slope and parabolic equations, Boussinesq systems for weak dispersion and nonlinearity, nonlinear shallow-water limits, Navier–Stokes free-surface models, and momentum equations forced by wave stresses.

Typical methods. Wave research combines buoys, pressure and current sensors, SAR and coastal video, laboratory basins, spectral and bispectral analysis, hindcasts, ray and action-balance models, mild-slope/parabolic solvers, Boussinesq models and phase-resolving CFD.

Numerical models. Reviewed examples use shallow-water SAR inversion, MIKE21 SW, phase-averaged reef modeling, stochastic and deterministic Boussinesq models, coupled SWAN–Delft3D, poroelastic seabed response and image-based classifiers.

Experimental datasets. Evidence includes seven-sensor SAR validation on the French coast, ten months at Hanalei Bay, a natural barred beach, a rocky-shoal storm deployment, submerged-mound experiments, Santa Catarina hindcasts and coastal-video transformation maps.

Validated ranges. Examples span offshore-to-20 m isobath wave-power change, 2–8 m rocky-ridge depths, 2.4–4.0 m storm wave heights, 0.3–0.4 m/s observed wave-driven jets, greater-than-twofold mound amplification and 30% directional steepening differences.

Recent advances. Recent advances fuse satellite and video observations with models, resolve complex bathymetry on nested grids, model continuous-spectrum nonlinear shoaling, quantify three-dimensional seabed coupling and translate regional hindcasts into resource and hazard products.

Disagreements. Phase-averaged models are efficient but can misrepresent steep-feature breaking and energetic swell transformation; deterministic phase-resolving models retain detail at higher cost, while stochastic closures trade phase information for spectral tractability.

Limitations. Uncertain winds and offshore boundaries, sparse directional observations, bathymetric resolution, breaking and bottom-friction closures, nonlinear and three-dimensional effects, image visibility and scale separation limit prediction and transfer.

Open questions. Priorities include coupled wave–current transformation, extreme and infragravity dynamics, breaking over complex reefs and structures, remote-sensing fusion, climate trends, uncertainty-aware forecasts and links to sediment, flooding and ecosystems.

Seminal papers. Linear dispersion, Snell refraction and shoaling theory established basic transformation; spectral balance, radiation stress, mild-slope equations and Boussinesq theory later connected irregular waves, mean circulation and nonlinear shallow-water evolution.

Claims

1. A descriptive overview identifies shoaling, refraction and diffraction as fundamental transformations as surface-gravity waves move from deep to shallow coastal water, with shoaling increasing height as propagation speed decreases. *Regime: Shoaling, Refraction and Diffraction in Waves*
review statement, review [1236]

49.2 Wave generation

Topic waves. generation. Wind-wave generation and source processes.

Papers screened: 14. Claims: 13. Equations: 0.

Well established. Surface waves are generated when pressure or moving boundaries do work on the free surface; their spectra then evolve through propagation, bathymetric refraction/focusing, nonlinear resonant transfer, currents, breaking and dissipation.

Governing physics. Generation requires kinematic and dynamic free-surface compatibility. Linear sources radiate dispersive modes; finite amplitude creates bound harmonics and resonant energy exchange; shear currents modify dispersion and modulation stability; absorption suppresses reflected contamination.

Dimensionless parameters. Controls include relative depth kh , wave steepness ka , Ursell and Benjamin–Feir indices, bandwidth and directional spread, current-to-phase-speed ratio, source or paddle stroke, sponge length/wavelength, Courant and grid resolution, and target focus distance/time.

Major equations. Core formulations include linear potential-flow free-surface conditions with applied pressure or wavemaker motion, Boussinesq and Navier–Stokes wave tanks, spectral action balance with nonlinear source terms, Hasselmann interaction integrals, and nonlinear Schrödinger/Alber modulation equations.

Typical methods. Methods use analytical source solutions, perturbation theory, spectral interaction approximations, flap/piston/bottom-tilting wavemakers, internal or pressure sources, active or sponge absorption, Boussinesq/RANS/VOF/DNS tanks, phase focusing and spectral validation.

Numerical models. Represented approaches include potential-flow pressure forcing, Hasselmann and discrete four-wave source terms, Boussinesq generating–absorbing layers, OpenFOAM free-surface solvers, RANS bottom-tilting wavemakers, DNS breaking waves and Alber stochastic spectral dynamics.

Experimental datasets. Evidence includes controlled oscillating-pressure and wavemaker configurations, level-bottom breaking-wave tanks, irregular runup generation with second-order corrections, focusing-wave spectra and bottom-tilting long-wave generation.

Validated ranges. The branch spans linear to breaking waves, gravity–capillary and gravity regimes, shallow through intermediate depth, monochromatic, irregular, focused and random spectra, quiescent and shear-current cases, and physical plus numerical wave tanks.

Recent advances. Recent work improves generating–absorbing layers, alternative OpenFOAM solvers, DNS of generated breaking waves, spectrum-aware focusing, RANS bottom-tilting long-wave makers and stochastic modulation over shear currents.

Disagreements. Different generators can match surface elevation while differing in bound harmonics, velocity, pressure or reflected energy; approximate nonlinear source terms trade accuracy for speed; VOF, level-set and alternative interface solvers differ in conservation and numerical dissipation.

Limitations. Broad directional seas and long runs expose reflection, phase and mass-conservation errors; high-order corrections are bandwidth- and depth-limited; breaking is resolution sensitive; spectral closures omit phase; controlled generation is not identical to natural wind input.

Open questions. Needs include reflection-free broadband directional generation, coupled current/source design, consistent higher-order kinematics, automatic calibration across solvers, uncertainty in nonlinear transfer, and efficient phase-aware generation from field spectra.

Seminal papers. Perturbation theory and nonlinear interaction parameterizations established spectral redistribution, while linear pressure and boundary radiation solutions established controlled free-surface generation.

Claims

1. Linear surface-wave theory determines waves generated by an oscillating surface pressure in finite depth and applies the solution to an oscillating-water-column boundary. *Regime: Wave generation by an oscillating surface-pressure and its application in wave-energy extraction* direct finding, analytical [1254]
2. Hasselmann perturbation theory shows gravity–capillary spectra exchange energy through second-order resonant interactions with higher-order redistribution. *Regime: Nonlinear energy transfer in gravity–capillary wave spectra, with applications* direct finding, analytical [1438]
3. Direct numerical simulations quantify statistics, kinematics and dissipation of shallow-water breaking waves generated by a wave plate at constant depth. *Regime: Wave statistics and energy dissipation of shallow-water breaking waves in a tank with a level bottom* direct finding, numerical [850]
4. Alternative OpenFOAM free-surface solvers provide numerical-wave-tank generation and propagation options beyond conventional VOF interFoam practice. *Regime: Beyond VoF: alternative OpenFOAM solvers for numerical wave tanks* direct finding, numerical [1265]
5. Quasi-coherent stochastic theory represents interference and refractive focusing of swell over variable depth that radiative-transfer models cannot resolve. *Regime: Stochastic Modeling of Coherent Wave Fields over Variable Depth* direct finding, analytical [1313]
6. A Boussinesq model combines a generating–absorbing sponge layer with second-order irregular-wave generation to support long-duration runup simulations. *Regime: Irregular wave runup statistics on plane beaches: Application of a Boussinesq-type model incorporating a generating–absorbing sponge layer and second-order wave generation* direct finding, numerical [447]
7. Four-wave nonlinear interaction source terms materially shape modeled ocean-wave spectra and their redistribution across frequency. *Regime: Role of Nonlinear Four-Wave Interactions Source Term on the Spectral Shape* review statement, review [810]
8. Three-wave theory gives sufficient conditions and amplitude bounds for unidirectional nonlinear energy transfer into one wave from two interacting waves. *Regime: Unidirectional energy transfer in nonlinear wave-wave interactions* direct finding, analytical [1475]
9. Numerical experiments show input amplitude spectra control the accuracy and evolution of prescribed focusing-wave generation. *Regime: Numerical investigation of wave amplitude spectra effects on focusing wave generation* direct finding, numerical [1552]
10. RANS simulations characterize the generation mechanism and wave forms produced by a bottom-tilting flume wavemaker designed for long waves. *Regime: Numerical Investigation of Wave Generation Characteristics of Bottom-Tilting Flume Wavemaker* direct finding, numerical [1478]
11. A fourth-order nonlinear Schrödinger/Alber framework predicts modulation stability of random surface-wave spectra over linear shear currents. *Regime: NONLINEAR MODULATION OF RANDOM WAVE SPECTRA FOR SURFACE-GRAVITY WAVES WITH LINEAR SHEAR CURRENTS* direct finding, analytical [996]
12. ECMWF hindcasts and MIKE21 SW propagation estimate mean wave power of 15.25 kW/m offshore and 11.43 kW/m at the 20 m isobath on the north-central Santa Catarina coast, dominated by southeastern waves. *Regime: Wave Energy Resource along the Coast of Santa Catarina (Brazil)* direct finding, numerical [295]
13. Sea-surface roughness scales with wave height and steepness across tanks, lakes, and coastal seas, while shoaling and swell systematically alter the effective roughness. *Regime: The Dependence of Sea Surface Roughness on the Height and Steepness of the Waves* direct finding, mixed [1402]

49.2.1 Extreme wave climates

Topic waves.generation.extremes. Extreme-value wave conditions.

Papers screened: 26. Claims: 25. Equations: 0. Shown in the Coastal Design Bench beside: Design conditions.

Well established. Extreme wave climate is inferred from block maxima or threshold exceedances after accounting for storm dependence; return levels depend on record length, threshold, declustering, spatial pooling, covariates, and whether the target is sea-state, individual-wave, or joint coastal conditions.

Governing physics. Extreme waves reflect storm intensity, translation, track, fetch and duration, tropical-cyclone structure, climate-mode modulation, directional exposure, bathymetric transformation, and joint sea-level state; statistical tails encode these generating processes imperfectly.

Dimensionless parameters. Relevant controls include exceedance probability and return period, threshold quantile, extremal index and declustering interval, shape parameter, normalized return level, storm duration and footprint, wave steepness, directional sector, water depth/wavelength, climate index, and ensemble spread.

Major equations. Methods use GEV block maxima, generalized Pareto peaks over threshold, Poisson occurrence, return-level inversion, covariate-varying location/scale/shape, mixture and weather-type convolution, regional frequency/index-flood concepts, copulas or joint simulators, bootstrap tests, and Bayesian or penalized likelihood.

Typical methods. Typical workflows merge buoy and satellite observations, hindcast/reanalysis, ensemble forecasts, synthetic cyclone catalogs, storm tracking and clustering, POT or annual maxima, stationary/nonstationary comparison, weather types, regionalization, multivariate dependence, nested wave downscaling, and climate ensembles.

Numerical models. Models include WAVEWATCH III, SWAN, SWAN-ADCIRC, COAWST nesting, ERA5 and other reanalyses, ECMWF ensemble forecasts, parametric tropical-cyclone waves, climate emulators, hybrid statistical-dynamical downscaling, weather-type mixtures, penalized nonstationary POT, and joint stochastic simulators.

Experimental datasets. Evidence includes 30-year global satellite records, 39-year Red Sea and 45-year southwest Pacific hindcasts, 240-hour ECMWF archives equivalent to long samples, ERA5 offshore lease areas, a 1000-year synthetic cyclone set, 19 CMIP6 models, and regional Mediterranean, Black Sea, China and East Asia studies.

Validated ranges. The corpus spans 30-year satellite POT, 39- and 45-year hindcasts, effectively 226 years from ensemble forecasts, a 1000-year cyclone catalog for 100-year waves, 50-year lease-area design conditions, 99th-percentile East Asian events, and 2070–2100 projections from 19 CMIP6 models.

Recent advances. Recent advances exploit long satellite records, synthetic millennial cyclone catalogs, multi-model CMIP6 nesting, recurrence-informed nonstationarity, flexible penalized covariates, climate emulators and multivariate compound extremes, and joint offshore-to-coastal stochastic simulation.

Disagreements. Stationary and nonstationary models can give materially different return values, but an imposed trend may be insignificant; wave-, sea-state-, and storm-level independence assumptions alter individual-wave tails; model choice, threshold and covariate flexibility trade bias against variance.

Limitations. Instrument and hindcast records remain short for rare tails; satellite sampling is intermittent; reanalysis under-resolves cyclones and coasts; synthetic storms depend on generators; climate ensembles and downscaling introduce cascaded uncertainty; compound dependence is data-hungry.

Open questions. Priorities include nonstationary tail validation, changing cyclone frequency and track, spatially coherent storm footprints, directional and seasonal dependence, individual-wave extremes inside changing sea states, compound wave-level tails, model averaging, and decision-making under deep uncertainty.

Seminal papers. The branch progresses from stationary annual maxima and POT toward weather-type mixtures, regional storm-footprint pooling, ensemble-forecast pseudo-climates, covariate-dependent and penalized tails, joint stochastic offshore-coastal simulation, and climate-conditioned projections.

Claims

1. Five regional–global climate-model combinations projected NW Mediterranean wave changes, with bootstrap tests separating forced changes from variability. *Regime: Projected future wave climate in the NW Mediterranean Sea* direct finding, numerical [234]
2. Thirty-year global satellite records support peaks-over-threshold estimates of extreme wind and significant wave height consistent with buoys and reanalysis. *Regime: Global Wind Speed and Wave Height Extremes Derived from Long-Duration Satellite Records* direct finding, mixed [1389]
3. Extreme-wave inference must represent both marginal tail behavior and dependence across space, time, and sea-state parameters. *Regime: Long-term time-dependent stochastic modelling of extreme waves* review statement, review [1441]
4. Mediterranean annual maxima show that linear-trend nonstationary wave models require explicit significance testing before trends are used in return-level estimation. *Regime: Non-stationary extreme value analysis of sea states based on linear trends. Analysis of annual maxima series of significant wave height and peak period in the Mediterranean Sea* direct finding, numerical [799]
5. New Zealand and southwest Pacific wave climate covaries with PDO, ENSO, IOD, ZW3, and SAM, making climate modes relevant covariates for regional extremes. *Regime: Regional influence of climate patterns on the wave climate of the southwestern Pacific: The New Zealand region* direct finding, numerical [510]
6. Weather-type-conditioned extreme-value mixtures statistically downscale local maximum wave heights while preserving large-scale generation patterns. *Regime: An extreme value model for maximum wave heights based on weather types* direct finding, mixed [1219]
7. ERA5-based analysis produced 50-year extreme wind and wave conditions for sixteen U.S. East Coast offshore lease areas, with hurricane correction needed for design tails. *Regime: Extreme Wind and Waves in U.S. East Coast Offshore Wind Energy Lease Areas* direct finding, numerical [122]
8. Storm-footprint clustering creates physically homogeneous regions that reduce uncertainty in regional frequency estimates of extreme significant wave height. *Regime: Formation of homogeneous regions for regional frequency analysis of extreme significant wave heights* direct finding, mixed [1501]

9. Archived 240-hour ensemble forecasts provide an effectively long independent sample for estimating Northeast Atlantic extreme significant wave-height return values. *Regime: Wave Extremes in the Northeast Atlantic from Ensemble Forecasts* direct finding, numerical [189]
10. A multivariate spatiotemporal hazard method represents cross-correlated Greenland wind-wave extremes beyond separate univariate return estimates. *Regime: Greenland Wind-Wave Bivariate Dynamics by Gaidai Natural Hazard Spatiotemporal Evaluation Approach* direct finding, numerical [469]
11. Stationary and covariate-dependent generalized-Pareto models yield different return-value accuracy; penalized binning controls the bias-variance tradeoff. *Regime: Assessment of return value estimates from stationary and non-stationary extreme value models* direct finding, numerical [890]
12. A hybrid SWAN and statistical framework projected future multivariate wave storms in the northwestern Black Sea under two climate scenarios. *Regime: Multivariate Hybrid Modelling of Future Wave-Storms at the Northwestern Black Sea* direct finding, numerical [834]
13. Return periods for individual wave and crest heights depend materially on whether independence is assumed at wave, sea-state, or storm scale. *Regime: Long-term distributions of individual wave and crest heights* direct finding, analytical [889]
14. Climate-wave downscaling in complex terrain needs both spatial and temporal refinement of GCM winds to retain coastal flood-generating wave extremes. *Regime: Downscaling wind and wavefields for 21st century coastal flood hazard projections in a region of complex terrain* direct finding, numerical [1039]
15. A 1000-year synthetic tropical-cyclone catalog and validated wave model estimate global 100-year cyclone wave heights, with the North Atlantic and western Pacific largest. *Regime: Global tropical cyclone extreme wave height climatology* direct finding, numerical [542]
16. Penalized piecewise-linear peaks-over-threshold models resolve directional and seasonal nonstationarity without unstable overfitting. *Regime: A penalised piecewise-linear model for non-stationary extreme value analysis of peaks over threshold* direct finding, analytical [119]
17. A climate emulator, multivariate extreme-value model, and hybrid wave propagation framework project joint wave and sea-level conditions at a coastal structure. *Regime: Projecting compound wave and sea-level events at a coastal structure site under climate change* direct finding, numerical [872]
18. A validated 39-year, 3.3-km WAVEWATCH III hindcast supports annual and seasonal extreme-wave climatology off Jeddah. *Regime: Extreme Wind Wave Climate off Jeddah Coast, the Red Sea* direct finding, numerical [42]
19. Deep-to-shallow hydrodynamic downscaling classifies extreme-event types and preserves their nearshore wave and sea-level signatures along Normandy. *Regime: Hydrodynamic modelling for simulating nearshore waves and sea levels: classification of extreme events from the English Channel to the Normandy coasts* direct finding, numerical [1318]
20. A joint stochastic simulator maps offshore direction, period, and moderately high wave height into dependent offshore and coastal extreme significant wave heights. *Regime: Joint stochastic simulation of extreme coastal and offshore significant wave heights* direct finding, analytical [794]
21. Comparing multiple statistical methods across forty offshore lease areas shows method choice is a material source of extreme wind and wave design uncertainty. *Regime: Comparison of Extreme Wind and Waves Using Different Statistical Methods in 40 Offshore Wind Energy Lease Areas Worldwide* direct finding, numerical [152]
22. A 99th-percentile peaks-over-threshold analysis links East Asian summer extreme-wave duration and intensity to El Niño variability. *Regime: Effect of El Niño on summer extreme ocean waves over East Asian regions* direct finding, numerical [130]
23. Field and coupled SWAN-ADCIRC results establish dependent significant-height and mean-period relationships for typhoon waves in nearshore straits. *Regime: Investigations into Relationship Between Significant Wave Height and Mean Wave Period in Nearshore Straits Under Typhoon Conditions* direct finding, mixed [634]
24. A decomposition and recurrence-based nonstationary framework projects climate-conditioned significant-wave-height return values for China coastal waters. *Regime: Assessing extreme significant wave height in China's coastal waters under climate change* direct finding, numerical [1622]
25. Nineteen CMIP6 wind projections drive nested future-wave simulations for 2070–2100 from the NW Mediterranean to the Ebro Delta. *Regime: Future wave climate in the NW Mediterranean from multi-model CMIP6 wind projections* direct finding, numerical [612]

49.2.2 Wind-wave growth

Topic waves.generation.wind. Growth under wind forcing.

Papers screened: 28. Claims: 28. Equations: 0.

Well established. Wind waves grow through wave-coherent pressure work and form drag, with spectral input redistributed by nonlinear interactions and balanced by whitecapping and other dissipation; growth depends on fetch, duration, wave age, direction, and atmospheric variability.

Governing physics. Phillips-type turbulent pressure initiates waves; Miles critical layers and separated airflow contribute coherent pressure work; capillary roughness controls low-wind stress; nonlinear four-wave transfer moves energy; breaking limits young-sea growth.

Dimensionless parameters. Controls include wave age c/u^* , inverse wave age U_{10}/cp , nondimensional fetch and duration, nondimensional energy and peak frequency, directional wind-wave angle, wave steepness, Reynolds number, drag coefficient, gustiness, and breaking strength.

Major equations. Core frameworks are the spectral action-balance/Hasselmann equation with wind-input, nonlinear-transfer, whitecapping and swell-loss terms; Miles stability growth; Phillips stochastic forcing; pressure-slope work; wave-boundary-layer stress; and self-similar fetch/duration laws.

Typical methods. Methods combine wave-following pressure and airflow, directional wave arrays, SWIFT and acoustic dissipation, video breaking detection, laser imaging and PIV, wind-wave tanks, DNS/LES, phase-resolving simulations, source-term budget closure, and WAVEWATCH III/SWAN hindcasts.

Numerical models. Models include WAVEWATCH III ST4/ST6 and ZRP terms, SWAN with a wave-boundary-layer model, Hasselmann-equation solvers, high-resolution DNS/LES, three-dimensional phase-resolving potential flow, coastal atmosphere-wave coupling, and unified directional spectra.

Experimental datasets. Key evidence spans Lake George, North Carolina 4–83 km fetch, Ekofisk, Gulf of Tehuantepec, R/P FLIP HiRes, Gorky Reservoir, Kanmon Strait, typhoon observations, repeated 30-state JONSWAP tank tests, and global buoy/satellite validation.

Validated ranges. Evidence includes strongly forced U10/cp above earlier sub-3 measurements, North Carolina winds 9–11 m/s over 4–83 km fetch, low-wind capillary regimes below 5–7 m/s, a roughly 50% breaking record, 30 repeated sea states, and global hindcasts through 1979–2019.

Recent advances. Recent advances directly observe wave-coherent pressure work and airflow separation, resolve the initial Phillips stage by DNS, diagnose Miles growth experimentally, quantify gustiness, recalibrate global ST6 hindcasts, and couple source terms to regional storm atmospheres.

Disagreements. Competing input theories emphasize turbulent pressure, critical layers, or separation; linear critical-layer evidence is strong in controlled regimes but energetic ocean observations also show separation; source-term packages can match bulk height while disagreeing at high frequencies or stress partition.

Limitations. Direct pressure and airflow measurements are rare, especially in hurricanes; capillary, spray, stability, current, gust and directional effects are incompletely coupled; dissipation is difficult to measure; and calibration compensation can yield correct bulk waves for incorrect reasons.

Open questions. Priorities include regime transitions among Phillips, Miles and separation mechanisms; gust and stability effects; wave-current growth; directional spreading; high-wind drag saturation; spectral breaking closure; capillary-to-gravity coupling; and transferable uncertainty-aware source terms.

Seminal papers. The branch links Phillips initial forcing and Miles critical-layer theory to form-drag measurements, Hasselmann spectral balance, self-similar weak turbulence, source-term packages, wave-boundary-layer coupling, and direct pressure-work observations.

Claims

1. A unified directional spectrum connects long gravity and short capillary wind waves while satisfying fetch and high-wavenumber observational constraints. *Regime: A unified spectrum was formulated to remove discontinuities and nonphysical tuning across wavenumber ranges.* direct finding, analytical [402]
2. Observation-based wind-input and whitecapping terms incorporate airflow separation and measured breaking behavior missing from older spectral models. *Regime: New source functions include airflow separation and observed breaking physics absent from older operational terms.* direct finding, mixed [1205]
3. Wave-following field measurements quantified pressure form drag under strongly forced conditions beyond earlier deep-water observations. *Regime: Direct measurements extended wind-input estimates beyond the weak-forcing deep-water regime.* direct finding, field [359]
4. A source-term model calibrated at Ekofisk jointly predicts evolving wave spectra and wind-stress partition. *Regime: The model jointly predicted wave evolution and the partition of total wind stress into wave-supported and turbulent components.* direct finding, mixed [360]
5. Recalibrated WAVEWATCH III ST6 improved high-frequency spectral energy while retaining skill for bulk height and period across local and global tests. *Regime: Updated ST6 improved bulk waves and especially high-frequency spectral energy.* direct finding, mixed [841]
6. Conditional field spectra with roughly 50% dominant-wave breaking directly resolved the frequency distribution of breaking dissipation. *Regime: Differences between conditional spectra yielded the distribution of dissipation over frequency for a record with about 50% dominant-wave breaking.* direct finding, field [1570]
7. Eight-station observations showed standard fetch growth can fail directionally at short slanting fetch and in opposing swell. *Regime: Growth generally followed established curves, but short fetch produced strong alongshore energy and swell altered directional evolution.* direct finding, field [70]
8. Weak-turbulence growth laws scale self-similar wave energy and frequency by internal spectral fluxes rather than wind speed alone. *Regime: Internal energy, momentum, or action flux provides more fundamental scaling than external wind speed.* direct finding, analytical [101]
9. Fetch-resolved directional spectra constrain breaking dissipation and a frequency-dependent Phillips breaking-strength parameter. *Regime: Measured spectral evolution plus nonlinear transfers and wind input constrained a frequency-dependent breaking-strength function.* direct finding, mixed [1211]
10. Including anisotropic capillary roughness extends sea-state stress modelling below 5 m/s winds and to crosswind stress. *Regime: Anisotropic capillary roughness represents low winds and misalignment between wind and wave directions.* direct finding, analytical [178]
11. Wave-coherent FLIP measurements resolve how atmospheric velocity and pressure vary with wave age and critical-layer position. *Regime: Resolved wave-induced airflow varied with wind-wave state and provided field evidence on critical-layer structure.* direct finding, field [525]
12. Observation-based ST6 supports validated global hindcasts spanning 1979–2019 and multiple wind products. *Regime: Multi-decadal and nine-year global hindcasts were validated against buoy and satellite observations.* direct finding, numerical [847]
13. Coupled in-situ and remote measurements close the wind-input and breaking-loss budget for a young, strongly forced sea. *Regime: Breaking dissipation, crest statistics, wind input, and spectral evolution were measured together in a developing sea.* direct finding, field [1271]
14. Direct simulation identifies the Phillips principal stage in which turbulent pressure initiates wave-variance growth on calm water. *Regime: Surface-elevation variance exhibited the predicted early principal-stage growth under turbulent wind over initially calm water.* direct finding, numerical [821]
15. The ZRP wind-input term produced theoretically consistent fetch- and duration-limited growth against global field datasets. *Regime: The formulation matched self-similar theory and field data for fetch- and duration-limited growth.* direct finding, numerical [1637]
16. Controlled experiments resolve distinct temporal and spatial evolution paths for young wind waves. *Regime: Laboratory results distinguished duration-limited and fetch-limited evolution under controlled forcing.* direct finding, experimental [1294]
17. A wave-boundary-layer wind-input model paired with recalibrated whitecapping reproduced canonical fetch-growth curves in SWAN. *Regime: The coupled input/dissipation pair closely reproduced fetch-limited significant-height and peak-frequency growth curves.* direct finding, numerical [378]
18. Short wind waves below 10 m wavelength rapidly couple to the lowest 10 m of atmospheric turbulence and control key air–sea fluxes. *Regime: Short waves rapidly equilibrate with near-surface turbulence and regulate momentum, heat, gas flux, swell input, mixing, and roughness.* review statement, review [90]
19. Rapidly developing storm waves are sensitive to both ST4/ST6 source physics and whether atmosphere–wave coupling closes on roughness or stress. *Regime: Source-term and roughness-versus-stress coupling choices materially affected rapidly developing shelf-wave forecasts.* direct finding, numerical [1440]
20. Phase-resolving potential-flow simulation captures the nonlinear development of surface waves under wind forcing. *Regime: The model directly followed two-dimensional surface-wave formation and evolution under prescribed wind action.* direct finding, numerical [244]

21. Field-derived drag adjustment improved WAVEWATCH III wind input for a fetch-limited medium-sized reservoir. *Regime: Locally measured near-surface airflow supported a drag-coefficient parameterization for medium-sized water bodies.* direct finding, mixed [756]
22. A coupled sea-state model shows capillary waves can dominate surface stress at low and moderate winds below about 7 m/s. *Regime: Capillary waves dominate wave-perturbed surface stress at low-to-moderate winds around and below 7 m/s.* direct finding, analytical [179]
23. Repeated 30-state laboratory ensembles demonstrate that wind stress changes crest distributions even for identical initial JONSWAP seas. *Regime: Crest statistics responded to wind forcing, isolating wind-stress effects from identical initial random seas.* direct finding, experimental [788]
24. Three months of observations show wind gustiness modifies wave growth and roughness beyond mean wind speed alone. *Regime: Gustiness increased sea-surface roughness and altered wave evolution beyond mean-wind descriptions.* direct finding, field [880]
25. Laboratory airflow measurements provide direct quantitative evidence for the Miles critical-layer mechanism of wind-wave growth. *Regime: Diagnostic growth rates agreed with pressure reconstruction and linear stability predictions.* direct finding, experimental [231]
26. Direct field observations resolve positive wave-coherent pressure work transferring atmospheric kinetic energy into surface waves. *Regime: Phase-offset pressure pumped kinetic energy from the atmospheric boundary layer into waves and modified boundary-layer turbulence.* direct finding, field [1624]
27. Laser imaging directly observes intermittent airflow separation over ocean waves across millimeter-to-meter heights. *Regime: Direct observations captured near-surface airflow dynamics previously inaccessible in the field.* direct finding, field [204]
28. Validated high-resolution modelling demonstrates fetch-limited wave growth beneath a developing coastal wind jet. *Regime: The case resolved spatially developing waves beneath a coastal gap wind jet.* direct finding, mixed [1300]

49.3 Nonlinear wave dynamics

Topic waves, nonLinear. Nonlinear interactions and instabilities.

Papers screened: 12. Claims: 6. Equations: 1.

Well established. Nonlinear coastal waves exchange energy and phase among harmonics and spectral components; shoaling, breaking, finite steepness, bathymetry, vegetation, currents, and wavemaker boundary conditions alter wave shape and spectra beyond linear superposition.

Governing physics. Key mechanisms are quadratic triad and four-wave coupling, free-bound harmonic interference, amplitude and phase modulation, shoaling-driven skewness and asymmetry, breaking-induced decorrelation, nonlinear dispersion, vegetation drag, and nonlinear force generation.

Dimensionless parameters. Important controls include relative depth kh or $kp h$, Ursell number, wave steepness, relative amplitude H/h , spectral bandwidth, bicoherence, biphasic, normalized harmonic energy, vegetation density and relative submergence, and Keulegan-Carpenter number for loads.

Major equations. The evidence uses bispectra, bicoherence and biphasic, Boussinesq spectral evolution with sum and difference triads, eigenfunction-FFT wavemaker solutions, four-phase harmonic separation, WRT four-wave interaction integrals, Ursell scaling, and Stokes-like harmonic load reconstruction.

Typical methods. Methods combine controlled regular, random, and focused flume waves; field event records; wavelet and Fourier spectra; bicoherence and bispectra; harmonic phase separation; amplitude-scaled recomputation; semi-analytical wavemaker theory; Boussinesq and fully nonlinear potential-flow models.

Numerical models. Models include a semi-analytical eigenfunction-FFT nonlinear wavemaker, OceanWave3D, a spectral Madsen-Sørensen Boussinesq model, WRT four-wave integration, vegetation attenuation formulations, and fully nonlinear potential-flow plus reduced-order harmonic load models.

Experimental datasets. Reviewed evidence includes JONSWAP waves over a sill at $0.38 \leq kh \leq 1.22$, DeRisk nonlinear irregular waves at two depths, a 64 m transient-wave flume, Tainan $H=0.05\text{--}0.40$ m and $T=2\text{--}5$ s shoaling tests, a mistral event record, vegetation attenuation tests, and focused cylinder-load experiments.

Validated ranges. Direct ranges include $kh=0.38\text{--}1.22$ over a sill; low, moderate, and steep waves at four periods in a 64 m flume; Tainan $H=0.05\text{--}0.40$ m and $T=2\text{--}5$ s; two long-crested irregular-wave depths; Ur about 50–250 for vegetation sensitivity; and nonbreaking inertia-dominated focused cylinder waves.

Recent advances. Recent work separates higher-order free and bound content through multi-phase and amplitude-scaled analysis, reconstructs irregular spectra and structural loads, and interprets shoaling beats through joint frequency-wavenumber free-bound decompositions.

Disagreements. Spectral peaks can reflect physical nonlinear transfer or free harmonics generated by an inadequate wavemaker prescription. Frequency-domain energy exchange may appear paradoxical when local linear wavelength concepts are applied to spatially nonperiodic free-bound interference.

Limitations. Limitations include unidirectional flumes, finite records, spectral leakage, estimator confidence, breaking nonstationarity, wavemaker artifacts, sparse spatial gauges, model truncation order, inviscid assumptions, and regime-specific separation between free and bound components.

Open questions. Priorities include multidirectional and current-affected phase coupling, robust separation near breaking, uncertainty-aware bicoherence, nonlinear generation without spurious content, transfer from flumes to storms, and efficient operational approximations to first-principles four-wave interactions.

Seminal papers. Within this branch, the 2006 WRT study provides a first-principles four-wave reference, while the 2008 transient wavemaker and sill-bicoherence studies established controlled generation and time-local quadratic-coupling diagnostics.

Equations

- Biphasic [755]

$$\phi = \tan^{-1}[\text{Im}B(\omega_1, \omega_2)/\text{Re}B(\omega_1, \omega_2)]$$

Regime: Tainan experiment with $H=0.05\text{--}0.40$ m, $T=2\text{--}5$ s, 21 gauges, 50 Hz sampling, and a 160 m domain, plus the cited field and numerical cases. Variables: B bispectrum; ω angular frequency

Claims

1. For random waves over a sill with $0.38 \leq kh \leq 1.22$, quadratic phase coupling increased during shoaling; it peaked at the sill crest for nonbreaking waves but immediately before incipient breaking for breaking waves, then decreased sharply. *Regime: Two random-wave cases over a sill with $0.38 \leq kh \leq 1.22$.* direct finding, experimental [361]
2. In two highly nonlinear long-crested irregular-wave experiments, paddle-generated second-order free waves produced spatially nonuniform spectral fluctuations strongest at the smaller depth; force spectra changed similarly, but peak-force statistics were less affected. *Regime: Two long-crested irregular-wave experiments at two depths; strongest distortion at the smaller depth.* direct finding, experimental [1108]
3. A semi-analytical nonlinear wavemaker model reasonably reproduced measured elevation and Fourier amplitudes, while linear or weakly nonlinear theory was adequate only for low-steepness waves and could generate disruptive multi-peak spectra at higher steepness. *Regime: Low, moderate, and steep wave trains across four basic periods in a 64 m by 0.6 m flume.* direct finding, experimental [1367]
4. Across field, laboratory, and Boussinesq analyses, spatial beats in shoaling waves were explained by interference among free and bound first and second harmonics; the Tainan tests covered $H=0.05\text{--}0.40$ m and $T=2\text{--}5$ s. *Regime: Tainan experiment with $H=0.05\text{--}0.40$ m, $T=2\text{--}5$ s, 21 gauges, 50 Hz sampling, and a 160 m domain, plus the cited field and numerical cases.* direct finding, mixed [755]
5. For a measured mistral wind-wave record, five-minute wavelet-bicoherence segments revealed temporal details of sum- and difference-frequency coupling that were lost when the full record was averaged. *Regime: One measured record containing the event maximum wave, evaluated as the full record and five-minute segments.* direct finding, experimental [406]
6. A Haar-wavelet multiresolution formulation evaluates the WRT nonlinear quadruplet transfer integral with constant weights and equally spaced nodes; one- and two-dimensional tests converge toward benchmark nonlinear wave-wave interaction results as resolution increases. *Regime: Deep-water discrete wind-wave spectra with resonant four-wave interactions represented through WRT geometry.* direct finding, numerical [1436]

49.3.1 Wave groups and infragravity waves

Topic waves.nonLinear.groups. Group forcing and low-frequency response.

Papers screened: 16. Claims: 16. Equations: 0.

Well established. Amplitude modulation of short waves forces difference-frequency long waves that are bound to groups offshore; shoaling, breaking and changing group speed can amplify, phase-shift and release free infragravity waves.

Governing physics. Gradients in short-wave radiation stress force long-wave setup and oscillation. The response approaches resonance when group velocity matches shallow-water long-wave speed; bathymetry, breaking, directional spreading, reflection and edge-wave trapping control subsequent evolution.

Dimensionless parameters. Controls include group frequency, modulation depth, relative depth, beach slope, wave steepness, group-to-long-wave speed ratio, directional spread, normalized breakpoint excursion, reflection coefficient, reef width/depth and infragravity frequency bands.

Major equations. Core descriptions are forced shallow-water continuity and momentum at group scale, radiation-stress source terms, free/bound Green-function solutions, spectral energy and flux balances, and nonlinear or nonhydrostatic equations for coupled short/long waves.

Typical methods. Methods include bichromatic and random-wave flumes, high-resolution cross-shore arrays, directional spectral reconstruction, bispectral/phase analysis, field swash and reef transects, SurfBeat/group-scale models, phase-resolving models and regional spectral infragravity sources.

Numerical models. Models include Delft3D-SurfBeat, WAVEWATCH III with an infragravity source, InWave coupled to ocean-wave-sediment dynamics, unstructured group-scale hydrodynamics, linear forced-wave models, dissipative edge-wave models and Green-function solutions.

Experimental datasets. Key evidence comes from barred- and plane-beach flumes, Delft experiments, the Delilah field campaign, dissipative-beach swash observations, a five-month Roi-Namur reef record, bichromatic 1:100-slope tests and directional coastal arrays.

Validated ranges. The corpus covers unbroken shoaling and surf-zone breaking, plane and barred beaches, gentle dissipative slopes, reefs, shore-oblique forcing, regional/global propagation and frequencies from conventional infragravity below about 0.05 Hz to reef VLF bands of 0.001–0.005 Hz.

Recent advances. Recent advances unify free and forced components analytically, reconstruct directional spectra, drive coupled morphology models with infragravity waves, extend computation to unstructured meshes and represent free infragravity propagation at regional-to-global scales.

Disagreements. Bound-wave release at breaking, breakpoint forcing and continuous free-wave radiation are competing but not mutually exclusive descriptions; inferred contributions depend on phase convention, directionality, decomposition method and the treatment of dissipation and reflection.

Limitations. Directional long-wave observations are sparse; shoreline reflection and dissipation are difficult to separate; frequency-band definitions vary; laboratory scales and one-dimensional bathymetry simplify natural systems; empirical regional sources omit event-specific phase.

Open questions. Open issues include nonlinear resonance saturation, directional and alongshore coupling, wave-group statistics in broad spectra, free/bound separation through breaking, shoreline and reef dissipation, interaction with currents and tides, and extreme-runup predictability.

Seminal papers. Early field and laboratory work established group-bound long waves and competing release/breakpoint mechanisms; energy-budget and phase-lag studies then quantified shoaling, reflection and dissipation.

Claims

1. A laboratory energy budget over a barred beach shows how group-induced subharmonic waves shoal, exchange energy, and dissipate across the nearshore. *Regime: Shoaling of subharmonic gravity waves* direct finding, experimental [129]
2. Flume observations and Delft3D-SurfBeat quantify low-frequency-wave shoaling growth, shoreline reflection, and dissipation. *Regime: Shoaling and shoreline dissipation of low-frequency waves* direct finding, mixed [365]
3. WAVEWATCH III extended with a shoreline source represents free infragravity waves at regional and global scales using offshore wave height and mean period. *Regime: A numerical model for free infragravity waves: Definition and validation at regional and global scales* direct finding, numerical [72]
4. Field observations on a gently sloping dissipative beach show infragravity swash saturation can occur even under mild offshore energy, especially above roughly 0.025–0.035 Hz. *Regime: Observations of wave energy fluxes and swash motions on a low-sloping, dissipative beach* direct finding, field [549]
5. Linear modelling of the Delilah data evaluates bound-wave release and breakpoint forcing by directionally spread short waves on an alongshore-uniform beach. *Regime: Linear modeling of infragravity waves during Delilah* direct finding, mixed [1180]
6. Five months of reef-flat observations classify very-low-frequency waves and show resonance can drive large runup and overwash on low-lying reef islands. *Regime: Identification and classification of very low frequency waves on a coral reef flat* direct finding, field [486]
7. Bichromatic groups on a 1:100 slope show group frequency strongly controls nonlinear transfer toward both low- and high-frequency components. *Regime: Transfer and dissipation of energy during wave group propagation on a gentle beach slope* direct finding, experimental [1070]
8. An analytical shallow-water solution decomposes free and forced long-wave components generated as short-wave groups shoal without breaking. *Regime: Free and Forced Components of Shoaling Long Waves in the Absence of Short-Wave Breaking* direct finding, analytical [293]
9. A reconstruction method separates directional spectra of free and bound infragravity components beyond conventional linear-wave assumptions. *Regime: Reconstruction of Directional Spectra of Infragravity Waves* direct finding, mixed [935]
10. The InWave driver supplies infragravity forcing to a coupled ocean–wave–sediment model for erosion and breaching simulations. *Regime: Development and Application of an Infragravity Wave (InWave) Driver to Simulate Nearshore Processes* direct finding, numerical [1050]
11. An unstructured-grid hydrodynamic model with a nonstationary directionally spread wave driver resolves two-dimensional nearshore flow at wave-group scale. *Regime: Modelling wave group-scale hydrodynamics on orthogonal unstructured meshes* direct finding, numerical [1185]
12. A forced dissipative shallow-water model explains cross-shore propagation and trapping of shore-oblique infragravity and edge waves. *Regime: Simulations of Dissipative, Shore-Oblique Infragravity Waves* direct finding, numerical [587]
13. Higher-order theory explains the growing phase lag between bound infragravity waves and forcing short-wave groups during shoaling. *Regime: On the Bound Wave Phase Lag* direct finding, analytical [557]
14. A Green-function solution unifies free and bound group-induced infragravity waves and captures the near-resonant response where group and shallow-water wave speeds match. *Regime: Unified analytical solution for group-induced infragravity waves based on Green's function* direct finding, analytical [826]
15. Conceptual and numerical analysis shows that shoaling forced infragravity waves can radiate free long waves, with outcomes depending on phase and generation assumptions. *Regime: The Influence of Free Long Wave Generation on the Shoaling of Forced Infragravity Waves* direct finding, numerical [989]
16. Short-wave groups generate infragravity waves through bound-wave forcing, breakpoint modulation, and bore merging; their subsequent shoaling, nonlinear transfer, breaking, friction, and reflection make them important controls on runup, sediment transport, overwash, reef hydrodynamics, and harbor resonance. *Regime: Group-forced surface waves typically below 0.04 Hz propagating from deep water through surf zones, beaches, tidal inlets, reef flats and lagoons, harbors, and the open ocean.* review statement, review [146]

49.3.2 Extreme and rogue waves

Topic waves.nonLinear.rogue. Abnormally large wave events.

Papers screened: 8. Claims: 8. Equations: 0.

Well established. Rogue waves are unusually large relative to the surrounding sea state and can result from linear superposition, directional or bathymetric focusing and nonlinear wave-group dynamics; coastal depth variation changes both elevation and velocity extremes.

Governing physics. Refraction over shoals and steps concentrates directional energy, shoaling changes steepness and bound harmonics, reflections and currents alter phase coherence, and modulational instability can amplify narrow-banded groups where depth and directional spreading permit it.

Dimensionless parameters. Controls include relative depth kh , wave steepness, Benjamin–Feir index, Ursell number, directional spread, bandwidth, current-to-group-speed ratio, bathymetric length relative to wavelength, focusing gain and normalized crest or wave height.

Major equations. Descriptions range from linear ray and spectral superposition through nonlinear Schrödinger and higher-order envelope equations to fully nonlinear potential-flow or phase-resolving models. Statistics use exceedance distributions, skewness, kurtosis and crest or wave height normalized by significant height.

Typical methods. Studies generate long irregular-wave ensembles, vary spectra, direction and depth transitions, measure surface elevation and three-component velocity, map spatial moments and exceedances, separate bound and free components, test ensemble convergence and compare linear, weakly nonlinear and fully nonlinear predictions.

Numerical models. The evidence spans analytical shoaling and wave-group frameworks and simulations/experiments over two-dimensional bathymetry. Direct directional-focusing and fully nonlinear finite-depth model papers remain outside reviewed full text.

Experimental datasets. Reviewed evidence includes irregular waves over a circular shoal and semicircular step with spatial surface and velocity statistics, plus controlled variable-depth experiments summarized in a coastal rogue-wave review.

Validated ranges. Support is limited to the tested shoal, step and reviewed controlled depth-transition configurations. Surface and horizontal-velocity kurtosis maxima occurred at different locations, so surface-elevation extremes cannot universally proxy subsurface kinematic demand.

Recent advances. Recent advances combine large fully nonlinear ensembles, spatial velocity statistics, variable-depth and current experiments, directional wave-basin control, data-driven rare-event sampling and real-time detection using radar, buoys and platform motion.

Disagreements. The relative importance of modulational instability versus linear directional/bathymetric focusing depends strongly on bandwidth, directional spread and depth. Envelope models lose validity for broad spectra, strong breaking or abrupt bathymetry, while linear models omit bound and resonant nonlinearities.

Limitations. Rare-event sampling demands long records and many realizations; wave breaking censors the upper tail, laboratory scale and wavemaker recurrence can bias statistics, and sparse gauges miss moving hotspots. Definitions and thresholds for rogue events also vary.

Open questions. Priorities include directional finite-depth thresholds, currents and crossing seas, coupled surface/velocity/load extremes, breaking-limited tails, realistic three-dimensional bathymetry, field detection, nonstationary storm ensembles and probabilistic design implications.

Seminal papers. Draupner-era observations established the engineering reality of rogue waves; Benjamin–Feir instability and nonlinear Schrödinger theory framed envelope amplification, while directional focusing, higher-order spectra and fully nonlinear simulations broadened mechanisms.

Claims

1. For irregular waves over the tested circular shoal and semicircular step, bathymetric refraction produced location-dependent non-Gaussian surface and velocity statistics, with surface and horizontal-velocity kurtosis maxima occurring at different locations. *Regime: Long-crested irregular waves over a circular shoal and a submerged semicircular step.* direct finding, numerical [777]
2. A review of variable-depth coastal rogue waves links analytical shoaling and wave-group frameworks to recent controlled experiments that demonstrate depth-transition effects on extreme-wave formation and statistics. *Regime: Review of coastal extreme-wave theory and experiments across deep-to-shallow transitions.* review statement, review [822]
3. Abstract In this paper, an observational space–time ensemble of sea surface elevations is investigated in search of the highest waves of the sea state. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [140]
4. The generation and evolution of ocean waves by wind is one of the most complex phenomena in geophysics, and is of great practical significance. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [98]
5. We present a study on the prediction of rogue waves during the 1-hour sea state of Hurricane Joaquin when the Merchant Vessel El Faro sank east of the Bahamas on October 1, 2015. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [430]
6. Abstract Wave breaking is observed in a laboratory experiment with waves of realistic average steepness and directional spread. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [97]
7. The effect of the non-uniform bathymetry on the nonlinear wave dynamics for the freak wave is investigated experimentally, with emphasis on the interrelations between different nonlinear behaviors resulting from various geometric parameters and spectral analysis. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [579]
8. Abstract. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [1282]

49.4 Wave transformation

Topic waves. transformation. Changes to waves during propagation.

Papers screened: 4. Claims: 4. Equations: 3.

Well established. Coastal wave transformation combines shoaling, refraction, diffraction, reflection, nonlinear interaction, breaking and friction as waves encounter changing depth, currents, boundaries and structures; validation must separate these processes and their coupled spatial effects.

Governing physics. The reviewed non-hydrostatic approach retains vertical acceleration and dynamic pressure to represent short-wave propagation and three-dimensional vertical structure while solving the free surface with only a few vertical layers.

Dimensionless parameters. Controls include kh , relative depth, wave steepness, Ursell and Froude numbers, bathymetric slope and curvature, incidence angle, directional spread, grid cells per wavelength, vertical layer count, Courant number and wave-height or velocity errors normalized by incident conditions.

Major equations. Models derive from free-surface mass and momentum conservation with hydrostatic or nonhydrostatic pressure; linear dispersion governs small waves, while nonlinear shallow-water, Boussinesq, multilayer and Navier–Stokes formulations retain increasing dispersion and vertical physics. Wave action governs phase-averaged spectra.

Typical methods. The reviewed model uses staggered finite volumes, boundary-fitted vertical layers, time splitting, Godunov-type horizontal advection, and a pressure-correction Poisson solve; validation combines analytical and laboratory comparisons.

Numerical models. The reviewed model is a three-dimensional nonhydrostatic finite-volume solver with boundary-fitted vertical layers and pressure correction, verified on standing-wave, elliptical-mound and parabolic-basin tests. It does not implement explicit breaking.

Experimental datasets. Two established regular-wave elliptical-mound datasets provide section-wise validation of combined shoaling, refraction, and diffraction on flat and 1:50 sloping backgrounds.

Validated ranges. For the flat-bed mound case at $H=2.54$ cm, $T=1.3$ s, and $kh=1.27$, reported dimensionless wave-height RMSE was 0.070–0.165; these values apply only to the specified geometry, two-layer grid, boundaries, and measurement sections.

Recent advances. Recent advances use multilayer nonhydrostatic and high-order unstructured schemes, adaptive phase-resolving nests, GPU acceleration, data assimilation, learned closures and benchmark suites that compare free surface, velocity, conservation and spatial focusing jointly.

Disagreements. Hydrostatic models are efficient for long waves but miss short-wave dispersion; Boussinesq and nonhydrostatic models extend dispersion at moderate cost; RANS/VOF resolves breaking and air at much higher cost. Good aggregate height skill can conceal local focusing or velocity errors.

Limitations. The current evidence does not validate breaking waves: all four reported tests are non-breaking and the model has no explicit breaking treatment.

Open questions. Reviewed evidence is still missing for irregular spectra, breaking and post-breaking transformation, currents, real bathymetry, coastal structures, field observations, and long-duration three-dimensional applications.

Seminal papers. Linear shoaling and Snell refraction established reduced transformation theory; mild-slope equations unified refraction and diffraction, Boussinesq equations added nonlinear dispersion, and nonhydrostatic and Navier–Stokes solvers broadened vertical and breaking physics.

Equations

- *Boundary-fitted vertical layer thickness* [1303]

$$d_p = f_p D = f_p (\eta + h), \quad 0 < f_p < 1, \quad \sum_p f_p = 1$$

Regime: Moving free surface on the paper's vertically boundary-fitted grid. Variables: d_p thickness of layer p ; f_p fixed fraction of total depth assigned to layer p ; D total water depth; η free-surface elevation; h still-water depth

- *Flat-bed elliptical-mound bathymetry* [1303]

$$h = 0.9144 - 0.762 \sqrt{1 - (x/3.81)^2 - (y/4.95)^2}$$

Regime: Vincent-Briggs submerged-mound validation inside $(x/3.05)^2 + (y/3.96)^2 = 1$. Variables: h still-water depth inside the elliptical boundary; x longitudinal coordinate; y transverse coordinate

- *Parabolic-basin bed* [1303]

$$z_b = -h_0(1 - r^2/R^2), \quad r = \sqrt{x^2 + y^2}$$

Regime: Thacker long-wave resonance and wet/dry analytical benchmark. Variables: z_b bed elevation; h_0 central depth scale; R basin radius parameter; r radial coordinate

Claims

1. In the 10 m cubic standing-wave test, the model's free-surface residual relative to the analytical solution remained within plus or minus 0.005 m using five vertical layers. *Regime: Closed deep-water standing-wave benchmark; 0.5 m horizontal cells, five uniform layers, and 0.05 s time step.* direct finding, numerical [1303]
2. For the Vincent-Briggs flat-bed elliptical-mound experiment, the present model's dimensionless wave-height RMSE ranged from 0.070 to 0.165 and was lower than both compared models at every section where all three reported values. *Regime: Regular $H=2.54$ cm, $T=1.3$ s, $kh=1.27$ wave; 0.457 m outer depth; two layers; specified mound and basin boundaries.* direct finding, mixed [1303]
3. For the sloping-bed mound benchmark, the present model's dimensionless wave-height RMSE across eight sections ranged from 0.060 to 0.114. *Regime: Berkhoff regular-wave benchmark with $H=4.64$ cm, $T=1$ s, 1:50 slope, two vertical layers, and specified grids and boundaries.* direct finding, mixed [1303]
4. In the Thacker parabolic-basin test with two vertical layers, reported free-surface RMSE was below 0.04 and radial-velocity RMSE below 0.07 at all four evaluated phases. *Regime: Frictionless parabolic basin with $h_0=1$ m, $r_0=2000$ m, $R=2500$ m, 20 m grid, 2 s step, and $1e-5$ m wet/dry threshold.* direct finding, analytical [1303]

49.4.1 Breaking

Topic waves.transformation.breaking. Wave instability and breaking.

Papers screened: 5. Claims: 5. Equations: 0.

Well established. Surface-gravity waves break when nonlinear crest evolution can no longer remain kinematically and dynamically coherent. Breaking limits wave height, converts organized wave energy into turbulence, currents and heat, and drives setup, runup, sediment suspension and impulsive structural loading.

Governing physics. Shoaling raises wave height and steepness as depth decreases, while dispersion, refraction, currents, wind and bathymetry modify crest speed and energy flux. At onset the crest overturns or spills; roller evolution and turbulent transport then redistribute momentum and dissipation through the water column and toward the bed.

Dimensionless parameters. Important controls include relative wave height H/h , deep-water or local steepness kH , surf-similarity or Iribarren number, breaker index H_b/h_b , crest-speed ratios used in onset criteria, Reynolds and Froude numbers, relative bar or structure geometry, and wind speed normalized by crest propagation speed U_w/C .

Major equations. Core descriptions combine conservation of mass and momentum with wave-action or energy-flux balance. Breaking onset is represented by depth-index, steepness, kinematic or crest-energy-flux criteria; phase-averaged models add a dissipation source term, while Navier–Stokes, LES or RANS models resolve or close the post-onset turbulent flow.

Typical methods. Breaking is studied with wave flumes and large basins, field pressure/velocity arrays, lidar and georectified video; onset and dissipation are reproduced with spectral, Boussinesq, Serre–Green–Naghdi, potential-flow, RANS, LES and volume-of-fluid models. Validation should distinguish onset location, breaker type, wave-height decay and turbulence.

Numerical models. Model fidelity ranges from representative-wave and spectral energy balances with empirical breaking sinks to nonlinear shallow-water/Boussinesq families and interface-resolving Navier–Stokes solvers. Potential-flow models describe pre-onset evolution but require a breaking treatment; turbulence-resolving methods are more expensive and remain sensitive to closure and resolution.

Experimental datasets. Reviewed evidence spans field-scale shoaling-soliton experiments with lidar and imagery, Timex-video observations at five Atlantic-coast sites, three laboratory datasets used to compare dissipation models, ten natural-beach campaigns supporting setup/runup parameterization, and large-scale plunging-breaker measurements over a fixed bar.

Validated ranges. The Timex methods were tested over roughly 900 records and breaking heights of 0.1–3.8 m, with about 18% average normalized RMSE for the bathymetry-free method. A revised bore-based dissipation formulation had the smallest wave-height error among four models across three datasets; these results remain conditional on their tested beaches and calibrations.

Recent advances. Recent advances combine crest-resolved onset diagnostics across arbitrary depth, field-scale lidar and video, machine-learning image segmentation, enhanced fully nonlinear shallow-water solvers, and LES or interface-resolving simulations that expose coherent vortices, energy pathways and structure interaction.

Disagreements. No single onset indicator or dissipation closure is uniformly accepted from deep to shallow water and from spilling to plunging breakers. Depth-index and steepness criteria are convenient but site-dependent; crest-based criteria seek broader physics, while operational phase-averaged models still require empirical post-onset dissipation parameters.

Limitations. Breaking is intermittent, three-dimensional and aerated, making crest geometry, void fraction and near-bed turbulence difficult to measure. Scale effects, uncertain bathymetry, finite image resolution, breaker-type ambiguity, turbulence closure, grid convergence and calibration transfer can dominate apparent model skill.

Open questions. Priorities include a transferable onset-to-dissipation framework, directional and current-modified breaking, wind effects on crest overturning, air entrainment and bubble-mediated dissipation, turbulence transport to the bed, data-assimilative remote sensing, and quantified uncertainty across laboratory-to-field scales.

Seminal papers. Classical limiting-wave, breaker-index and surf-similarity studies established depth- and slope-controlled descriptions; radiation-stress and bore analogies connected breaking to mean flow and energy loss. Later crest-kinematic and energy-flux approaches reframed onset as a local dynamical threshold.

Claims

1. The model validation in this paper is limited to non-breaking wave propagation because explicit wave-breaking treatment was not implemented. *Regime: All accuracy and efficiency claims reported in the four validation cases.* direct finding, numerical [1303]
2. Breaking over 2–8 m deep rocky ridges generated an observed 0.3–0.4 m/s jet, more than four times the maximum tidal current; coupled SWAN–Delft3D simulations captured timing and direction but overpredicted breaking and speed. *Regime: A wave-driven jet over a rocky shoal* direct finding, mixed [999]
3. Since the 1990s, the modulational instability has commonly been used to explain the occurrence of rogue waves that appear from nowhere in the open ocean. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [429]
4. A concept of wave-amplitude-based Reynolds number is suggested which is hypothesised to indicate a transition from laminarity to turbulence for the wave-induced motion. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [95]
5. The large-scale vortex structures under spilling and plunging breakers are investigated, using a fully three-dimensional large-eddy simulation (LES). *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [1491]

49.4.1.1 Breaking criteria

Topic waves.transformation.breaking.criteria. Onset and type criteria.

Papers screened: 6. Claims: 6. Equations: 0.

Well established. Breaking onset is a local crest instability distinct from the later overturning and dissipation stages. Bulk depth and steepness indices are useful engineering proxies, while local crest-speed, particle-velocity and energy-flux measures provide more process-based prediction.

Governing physics. Nonlinear focusing and shoaling concentrate energy near a crest while crest propagation slows. Instability develops as fluid velocity and energy transport approach or overtake crest propagation; depth, bathymetry, bandwidth, wind and current alter this balance.

Dimensionless parameters. Important controls are H/h , H/L , ka , Ursell and Iribarren numbers, beach slope, spectral bandwidth, directional spread, crest particle-speed/crest-speed ratio, energy-flux ratio, relative wind speed and relative crest submergence.

Major equations. Criterion families include breaker index H_b/h_b , steepness H/L or ka , surf-similarity dependence, kinematic ratios u_c/C , local energy-flux ratios such as B_x , and rate-of-change parameters based on energy maxima and local wavenumber.

Typical methods. Studies use focused and random wave groups, flumes and field-scale facilities, surface reconstruction and particle velocimetry, fully nonlinear potential-flow or Navier–Stokes simulations, wave-by-wave field algorithms, breaker-index regression and threshold skill tests.

Numerical models. Models include fully nonlinear potential flow, high-resolution Navier–Stokes/SPH or Boussinesq implementations with onset flags, spectral models using modified breaker indices, wave-by-wave engineering algorithms and data-driven threshold classifiers.

Experimental datasets. The retained evidence includes deep-water group experiments, fully nonlinear near-breaking benchmarks, arbitrary-depth onset validation, field-scale wind-modified overturns, structure-adjacent breaker-index data and coastal steepness records.

Validated ranges. Evidence spans deep, intermediate and shallow water; solitary, quasi-regular, focused and random waves; constant and varying bathymetry; depth-limited overturning; and cases with cross-shore wind or coastal structures.

Recent advances. Recent work validates a unified energy-flux/crest-speed onset framework across arbitrary depth, quantifies wind effects on overturn geometry, revises structure-adjacent breaker indices and explores data-driven steepness thresholds.

Disagreements. No single bulk index is universally valid. Deep-water energy-growth thresholds, local kinematic/energy-flux criteria and shallow-water H/h or slope formulas emphasize different precursors and can trigger at different crest stages.

Limitations. Several seminal formulations remain title-only in the lawful corpus; surface velocity is difficult to measure near overturn; numerical smoothing affects crest speed; field directionality and currents are underrepresented; onset labels differ among studies.

Open questions. Needs include three-dimensional directional onset, wind and current corrections, irregular group history, bathymetric curvature, probabilistic thresholds, scale effects and consistent mapping from local onset to model dissipation.

Seminal papers. Engineering practice developed wave-by-wave breaker indices and slope/steepness formulas; later deep-water energy-growth and Boussinesq onset flags shifted attention toward local crest dynamics.

Claims

1. Experiments and simulations evaluate the Song–Banner deep-water onset parameter and its proposed threshold near $(1.4 \pm 0.1) \times 10^{-3}$ based on local energy growth and wavenumber evolution. *Regime: Evaluation of a deep-water wave breaking criterion* direct finding, mixed [1408]
2. Fully nonlinear simulations reproduce measured steep unidirectional group evolution and crest kinematics up to near-breaking conditions, providing a pre-onset benchmark. *Regime: Steep unidirectional wave groups – fully nonlinear simulations vs. experiments* direct finding, mixed [1293]
3. Field-scale experiments show cross-shore wind direction and magnitude alter the overturning shape of depth-limited breakers for a given wave-height-to-depth nonlinearity. *Regime: Cross-shore wind-induced changes to field-scale overturning wave shape* direct finding, experimental [428]
4. The Barthelemy energy-flux/crest-speed framework provides a robust breaking-onset criterion for surface gravity waves from deep through shallow water over varying bathymetry. *Regime: A Unified Breaking Onset Criterion for Surface Gravity Water Waves in Arbitrary Depth* direct finding, mixed [333]
5. A revised Weggel nomogram extends the breaker depth index for waves breaking at coastal structures beyond the range of the modified chart. *Regime: Breaker depth index of a wave breaking at a coastal structure – the modified Weggel's nomogram revisited* direct finding, analytical [898]
6. Data-driven analysis relates coastal breaking-risk thresholds to wave steepness and identifies trends across observed wave conditions. *Regime: Thresholds and trends in wave steepness: A data-driven study of coastal wave breaking risk* direct finding, numerical [383]

49.4.1.2 Energy dissipation

Topic waves. transformation. breaking. dissipation. Breaking-related energy loss.

Papers screened: 4. Claims: 2. Equations: 2.

Well established. Depth-limited wave breaking removes organized wave energy through turbulence, roller motion, air entrainment, vorticity and heat while transferring momentum that contributes to setup and currents; dissipation rate depends on breaker type, wave statistics, depth and slope.

Governing physics. Shoaling increases wave height and steepness until instability; spilling, plunging or surging breakers generate different roller, jet and bubble dynamics. The cross-shore energy balance reflects competition among group-velocity flux convergence, breaking, bottom friction and nonlinear transfers.

Dimensionless parameters. Controls include breaker index H/h , wave steepness, surf-similarity or Iribarren number, relative depth kh , roller area, fraction breaking, dissipation normalized by $\rho g H^2$ times frequency, slope, Ursell number and air-volume fraction.

Major equations. Phase-averaged models balance wave-energy flux with a breaking sink, often based on bore analogies and a fraction of breaking waves; representative-wave methods apply a regular-wave dissipation relation to an irregular RMS height after calibration. Momentum balances use radiation-stress gradients for setup.

Typical methods. Studies measure cross-shore wave heights, spectra, setup and velocities in small- and large-scale flumes and field transects, estimate energy-flux gradients, classify breakers, calibrate coefficients on one set and validate on others, and compare errors and shoreline behavior among dissipation closures.

Numerical models. The set includes an improved bore-based breaking-dissipation closure and a representative-wave energy-flux model that embeds a regular-wave dissipation formulation with recalibrated coefficients.

Experimental datasets. Reviewed evidence spans three experimental datasets used to compare four parametric transformation models and separate small-scale, large-scale and field irregular-wave datasets evaluated with a representative-wave approach.

Validated ranges. The improved closure had the smallest wave-height error among four models over three datasets and avoided unphysical nearshore shoaling dominance; the representative-wave approach transferred across small-scale, large-scale and field data only with an appropriate closure and recalibration.

Recent advances. Recent advances use stereo/video and radar dissipation fields, bubble and turbulence measurements, data assimilation for bathymetry, LES and VOF breaker energetics, differentiable closure calibration and hybrid phase-resolving/phase-averaged surf-zone models.

Disagreements. Bulk bore closures are efficient but compress breaker-type, spectral and intermittency effects; spectral and phase-resolving models offer more detail but require additional closures or resolution. Coefficients calibrated in one scale or beach may not transfer unchanged.

Limitations. Current reviewed evidence lacks full parameter tables and direct turbulence, air, roller or heat measurements. Depth, slope, spectrum, scale, reflection, bottom friction and measurement differentiation can confound inferred dissipation, especially near shore.

Open questions. Priorities include breaker-type-aware closures, directional and group-scale dissipation, entrained-air and roller coupling, field turbulence budgets, remote dissipation retrieval, breaking–infragravity transfer, nonstationary bathymetry and uncertainty-aware cross-model calibration.

Seminal papers. Hydraulic-bore analogies established bulk surf-zone dissipation; Battjes–Janssen-type probabilistic closures represented irregular breaking, roller models improved setup and currents, and later spectral and phase-resolving approaches distributed dissipation in frequency and space.

Equations

- *Wave-energy transformation balance* [43]

$$\nabla \cdot (EC_g) = -D_b - D_f$$

Regime: The three reported datasets, a standard initial-breaking ratio gamma and nonsaturated to dissipative surf-zone conditions. Variables: E wave energy density; C_g group velocity; D_b breaking dissipation; D_f bottom-friction or other dissipation

- *Wave-energy transformation balance* [1162]

$$\nabla \cdot (EC_g) = -D_b - D_f$$

Regime: The compiled experimental conditions and recalibrated dissipation coefficients; results concern cross-shore rms wave height. Variables: E wave energy density; C_g group velocity; D_b breaking dissipation; D_f bottom-friction or other dissipation

Claims

1. Adding a previously neglected term to bore-based breaking dissipation prevented shoaling amplification from exceeding dissipation near the shoreline and produced the smallest wave-height error of four compared models across all three experimental datasets. *Regime: The three reported datasets, a standard initial-breaking ratio gamma and nonsaturated to dissipative surf-zone conditions.* direct finding, mixed [43]
2. Across small-scale, large-scale and field datasets, a representative-wave energy-flux approach reproduced irregular root-mean-square wave-height transformation accurately when paired with an appropriate regular-wave breaking-dissipation model and recalibrated coefficients. *Regime: The compiled experimental conditions and recalibrated dissipation coefficients; results concern cross-shore rms wave height.* direct finding, mixed [1162]

49.4.1.3 Wave runup

Topic waves.transformation.breaking.runup.Uprush and maximum excursion.

Papers screened: 12. Claims: 9. Equations: 0. Shown in the Coastal Design Bench beside: Storm surge and flooding.

Well established. Wave runup is the time-varying vertical shoreline excursion above the still-water reference and combines mean setup with swash. It is a major component of total water level, overtopping, overwash and erosion, and depends jointly on offshore wave scale, spectral content, water level, beach or structure slope and local geometry.

Governing physics. Incident-band waves shoal and break, transferring energy to setup, infragravity motions and swash. Reflection and resonance become more important on steep or structured shores, while dissipation dominates many mild beaches; reefs, bars, permeability and alongshore-variable foreshores alter the energy reaching the shoreline.

Dimensionless parameters. Key controls include Iribarren number, foreshore slope, deep-water steepness H_0/L_0 , relative water depth, normalized runup R/H_0 or $R/\sqrt{H_0 L_0}$, reef or bar freeboard relative to wave height, permeability parameters, directional spread and nondimensional spectral frequency.

Major equations. Common predictors scale runup with the deep-water wave length-height product and foreshore slope, often through terms proportional to $\beta_f \sqrt{H_0 L_0}$, with separate incident and infragravity swash components. Process models solve nonlinear shallow-water or Boussinesq-family equations with moving shorelines and breaking dissipation.

Typical methods. Runup is measured with pressure sensors, resistance staffs, lidar, video and laser transects; empirical formulas use offshore waves and local slope, while phase-resolving Boussinesq, nonhydrostatic and XBeach-family models generate shoreline time series. Hazard work commonly reports mean setup plus exceedance levels such as $R_2\%$.

Numerical models. Empirical formulations are efficient for screening, Boussinesq and nonhydrostatic models resolve incident and infragravity phases, and XBeach Surfbeat represents grouped-wave forcing at lower cost. Model choice controls whether setup, swash, phase uncertainty, infiltration, overtopping and morphology are explicitly resolved.

Experimental datasets. Reviewed evidence includes ten natural-beach campaigns behind the Stockdon parameterization, 180 idealized directional-spectrum simulations over slopes 0.02–0.04, reef-to-island field transects, long-duration irregular plane-beach simulations, a laser-observed gravel-barrier storm and experiments comparing permeable and impermeable slopes.

Validated ranges. The directional-spread study was limited to mostly dissipative slopes of 0.02–0.04; its normalized infragravity runup lacked obvious slope dependence within that narrow range. At Loe Bar, XBeach-G reproduced observed overwash hydrodynamics within 5% for the documented storm, but this does not establish universal gravel-barrier skill.

Recent advances. Recent work couples dense lidar or laser observations with phase-resolving models, represents infragravity and bimodal spectra more explicitly, uses physics-informed machine learning to emulate higher-fidelity runup, and propagates runup uncertainty through ensemble flood and erosion assessments.

Disagreements. Slope dependence differs by regime and response component: local foreshore slope materially improves runup prediction on intermediate and reflective beaches, whereas infragravity runup showed little slope dependence across a narrow dissipative range. These findings are conditional rather than contradictory and should not be merged into one universal formula.

Limitations. Uncertain bathymetry and local slope, shoreline-definition choices, sparse extreme observations, phase information absent from spectra, directional spreading, low-frequency boundary conditions, roughness, infiltration and morphology change can dominate error. Empirical coefficients and numerical calibration transfer poorly outside their source regime.

Open questions. Needs include phase-aware probabilistic runup, joint tide-surge-wave dependence, transferable rough/permeable/reef formulations, alongshore variability, dynamically evolving morphology, efficient nonhydrostatic ensembles, uncertainty-aware remote sensing and validation during rare overtopping or overwash events.

Seminal papers. Classical plane-beach and surf-similarity work linked maximum excursion to wave steepness and slope; later field parameterizations separated setup and swash and established practical exceedance formulas. Moving-shoreline nonlinear-wave benchmarks underpin modern phase-resolving validation.

Claims

1. On intermediate and reflective beaches with complex foreshores, substituting an alongshore-averaged slope can introduce substantial runup error relative to the locally measured slope. *Regime: Intermediate and reflective beaches with alongshore-variable foreshore topography.* direct finding, field [1353]
2. Waves observed in the inner surf and swash zones of a fine grained, gently sloping beach are modeled accurately with the nonlinear shallow water equations. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1163]
3. Vertical flow structure and turbulent dissipation in the swash zone are estimated using cross-shore fluid velocities observed on a low-sloped, fine-grained sandy beach [Raubenheimer, 2002] with two stacks of three current meters located about 2, 5, and 8 cm above the bed. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1164]
4. Abstract. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [871]
5. Abstract The present study uses nine machine learning (ML) methods to predict wave runup in an innovative and comprehensive methodology. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [382]
6. Abstract Observations on a mildly sloping beach suggest that the largest runup events are related to bore-bore capture (BBC). *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [482]
7. A field experiment was conducted at a tropical microtidal intermediate sandy beach with a low tide terrace (Nha Trang, Vietnam) to investigate the short-term swash-zone hydrodynamics and morphodynamics under variable wave conditions. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [39]
8. Abstract The propagation of bichromatic wave groups with differences in the wave-group structure and its influence in long-wave generation are investigated. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1071]
9. Focused-wave flume tests and a calibrated Boussinesq–nonlinear–shallow-water model show extreme runup depends jointly on focus amplitude, location and phase rather than spectral parameters alone; second-order wavemaker correction removed about 60% of subharmonic error that otherwise inflated runup. *Regime: Focused wave groups on the tested plane beach, amplitudes, phases and focus locations.* direct finding, mixed [1510]

49.4.1.4 Wave setup

Topic waves.transformation.breaking.setup. Breaking-induced mean water-level response.

Papers screened: 23. Claims: 30. Equations: 4.

Well established. For intermediate and reflective natural beaches, shoreline setup increases with local foreshore slope when offshore wave height and wavelength are held fixed; the widely used field form is $\text{mean } \eta = 0.35 \beta_f \sqrt{H_0 L_0}$.

Governing physics. The cross-shore gradient of breaking-wave momentum flux is balanced principally by the mean-water-level pressure gradient, with bottom stress and roller-delayed momentum transfer modifying the detailed profile. Beach slope changes the available cross-shore distance and depth pattern over which breaking and momentum loss occur.

Dimensionless parameters. Surf similarity or Iribarren scaling combines slope and wave steepness, $\beta_f/\sqrt{H_0/L_0}$. It organizes the transition between dissipative and more reflective regimes, but slope definition—foreshore, surf-zone, local, or profile-averaged—is part of the model and not a neutral preprocessing choice.

Major equations. Two slope relations describe different quantities: Stockdon's shoreline elevation $\eta_{\text{bar}} = 0.35 \beta_f \sqrt{H_0 L_0}$ is linear in local foreshore slope, whereas Van Dorn's $m = 3.4 S^2$ describes the cross-shore gradient of the mean setup profile on smooth uniform laboratory slopes. They are not interchangeable shoreline predictions.

Typical methods. Slope dependence is evaluated using filtered mean-level profiles on fixed laboratory beaches, video-derived mean shoreline elevations across natural beaches, buried pressure-gauge arrays on evolving barred profiles, and phase-resolving numerical matrices that independently vary β_f and wave spectra.

Numerical models. The SandyDuck wave-averaged model solves a cross-shore balance among radiation stress, setup pressure gradient, roller energy and bottom stress; funwaveC resolves nonlinear wave transformation and shoreline motion with Boussinesq equations, eddy-viscosity breaking, and a thin-layer wetting treatment.

Experimental datasets. The reviewed evidence includes twelve smooth-slope periodic-wave combinations, ten natural-beach video campaigns, ninety days of SandyDuck setup and undertow observations, and 180 idealized Boussinesq simulations.

Validated ranges. Van Dorn tested slopes 0.022, 0.040 and 0.083 with periods 1.65–4.80 s; Guza-Feddersen tested $\beta_f = 0.02$ –0.04, $H_s = 0.4$ –2.5 m and $f_p = 0.06$ –0.14 Hz; SandyDuck observations span $H_{rms} = 0.20$ –2.10 m, depths 0.3–6 m and setup -0.03 to 0.50 m. These ranges do not establish behavior for arbitrary steep, composite, reef, or mobile profiles.

Recent advances. Recent advances resolve setup within wave-resolving XBeach and SWASH, incorporate breaking-wave rollers and roughness-dependent bottom stress, couple radiation-stress forcing to storm surge, derive slope- and direction-aware empirical predictors, and translate setup into probabilistic reef and coastal-flood assessments.

Disagreements. Slope dependence is regime- and definition-dependent rather than universal. Stockdon's general field coefficient is 0.35, while the idealized Boussinesq regression gives 0.53; fully dissipative field cases use a slope-independent offshore-wave scale; and the gentle-flume quadratic law concerns setup gradient rather than shoreline elevation.

Limitations. Slope-only predictors cannot reproduce detailed setup on an evolving barred beach: SandyDuck required bottom-stress and roller/profile physics, and excluding bottom stress tripled shallow-water mean error under the tested closure.

Open questions. Needed evidence includes time-varying local slope during storms, independent extreme-event validation, mobile and barred profiles, directional and oblique waves, reefs and composite slopes, and a physically consistent transition between general and fully dissipative parameterizations.

Seminal papers. Van Dorn (1976) isolates slope and period in controlled setup/runup tests; Stockdon et al. (2006) supplies the standard ten-campaign field parameterization; Apotsos et al. (2007) adds natural-beach bottom-stress and roller physics; Guza and Feddersen (2012) independently tests the slope scaling in a phase-resolving matrix.

Equations

- *Gentle-beach setup-profile slope fit* [367]

$$m = \frac{d\bar{\eta}}{dx} = 3.4S^2$$

Regime: Compiled smooth fixed-slope laboratory beaches; frequency independence observed only for $S \leq 0.040$. Variables: m mean water-surface setup gradient; η_{bar} mean surface elevation; x cross-shore coordinate; S uniform beach slope

- *Cross-shore setup momentum balance* [66]

$$\frac{\partial S_{xx}}{\partial x} + \rho g(\bar{\eta} + d) \frac{\partial \bar{\eta}}{\partial x} + \tau_B = 0$$

Regime: Alongshore-uniform bathymetry and waves with negligible wind stress. Variables: S_{xx} cross-shore wave radiation stress including optional roller energy; η_{bar} time-mean setup; d still-water depth; τ_B bottom stress; ρ water density; g gravitational acceleration

- *Modeled shoreline setup regression* [556]

$$\bar{R} = 0.53 \beta (H_{s0} L_0)^{1/2}$$

Regime: The 180 funwaveC simulations, mostly dissipative, $\beta = 0.02$ -0.04. Variables: $\bar{R}$ time- and alongshore-mean shoreline setup; β a planar beach slope; H_{s0} deep-water significant wave height; L_0 deep-water wavelength from peak frequency

- *Stockdon shoreline setup parameterization* [1353]

$$\bar{\eta} = 0.35 \beta_f (H_0 L_0)^{1/2}$$

Regime: General natural-beach branch across ten field campaigns; fully dissipative beaches use a slope-independent branch. Variables: η_{bar} time-mean non-tidal shoreline setup; β_f local foreshore slope; H_0 deep-water significant wave height; L_0 deep-water wavelength from peak period

Claims

1. For smooth planar slopes $S=0.022$ and 0.040 , the measured mean setup profile was approximately linear across the surf zone and its gradient was independent of wave period within experimental error. *Regime: Plate-glass laboratory beaches, $S \leq 0.040$, periods 1.65-4.80 s, periodic breaking waves.* direct finding, experimental [367]
2. Across the study and compiled comparison data, the authors proposed $m=3.4 S^2$ for setup-profile gradient m versus uniform beach slope S . *Regime: Smooth uniform slopes represented by the compiled laboratory data; not a universal shoreline-setup formula.* hypothesis, experimental [367]
3. On the steepest tested slope $S=0.083$, setup gradient appeared to increase with wave period and mean elevation became poorly defined for the longest, violently turbulent breaking cases. *Regime: $S=0.083$ plate-glass beach; especially periods 3.43 and 4.80 s.* direct finding, experimental [367]
4. Integrating the fitted setup-gradient relation to predict shoreline setup produced discrepancies too large for the proposed expression to be considered a satisfactory prediction method. *Regime: The study's three uniform slopes and breaking-wave cases.* direct finding, experimental [367]
5. Across the 90-day SandyDuck dataset, a setup model including rollers and undertow-related bottom stress achieved squared correlations above 0.59 and agreement within about 30% over depths to 6 m. *Regime: Barred natural beach; H_{rms} 0.20-2.10 m, angles within 35 degrees, setup -0.03 to 0.50 m.* direct finding, mixed [66]
6. Neglecting bottom stress tripled mean setup error in 0.3-1.0 m water depth and increased shallow-water underprediction. *Regime: The paper's vertically uniform eddy-viscosity and undertow closure at SandyDuck.* direct finding, mixed [66]
7. Removing the wave roller changed average setup by less than about 10% but shifted the setdown-to-setup transition shoreward by approximately 6 m. *Regime: The SandyDuck bar amplitudes and tested roller closures; several-meter bars may behave differently.* direct finding, mixed [66]
8. The field evidence shows that a planar slope and radiation-stress gradient alone do not determine natural-beach setup because bars, bottom stress, and evolving shallow bathymetry change the cross-shore balance. *Regime: Natural barred beaches comparable to SandyDuck.* inferred, mixed [66]
9. Across 180 planar-beach simulations with slope 0.02-0.04, setup was better explained by $\beta \sqrt{H_{s0} L_0}$ ($r^2=0.71$, RMS error 0.07 m) than by $\sqrt{H_{s0} L_0}$ alone ($r^2=0.58$, RMS error 0.09 m). *Regime: Mostly dissipative conditions; $H_s=0.4$ -2.5 m, $f_p=0.06$ -0.14 Hz, slopes 0.02-0.04.* direct finding, numerical [556]
10. The model best-fit shoreline-setup coefficient was 0.53 in $\bar{R} = \text{coefficient } \beta \sqrt{H_{s0} L_0}$, compared with the Stockdon field coefficient 0.35. *Regime: Idealized funwaveC planar beaches and the reported parameter matrix.* direct finding, numerical [556]

11. Across ten natural-beach field experiments, shoreline setup was best parameterized with foreshore slope, deep-water significant wave height, and deep-water wavelength rather than surf-zone slope. *Regime: The ten dynamically diverse natural-beach campaigns; detailed full-text ranges are not available in this extraction.* direct finding, field [1353]
12. The reported general shoreline-setup parameterization is $\text{mean } \eta = 0.35 \beta \sqrt{H_0 L_0}$, implying linear dependence on local foreshore slope at fixed deep-water wave scale. *Regime: General natural-beach branch; not the fully dissipative branch and not infragravity swash.* direct finding, field [1353]
13. For infragravity-dominated dissipative beaches, the study found setup and swash depended on $\sqrt{H_0 L_0}$ without a statistically significant linear dependence on foreshore or surf-zone slope. *Regime: Fully dissipative, infragravity-dominated beaches in the field dataset.* direct finding, field [1353]
14. Improved XBeach group forcing separates setup, infragravity and incident-band swash more accurately; nonhydrostatic runup deviated at most 15% in a high-quality flume test but underestimated low overtopping rates. *Regime: Improving predictions of swash dynamics in XBeach: The role of groupiness and incident-band runup* direct finding, mixed [1204]
15. Natural-beach measurements found shoreline wave setup, the mean-water-level rise caused by breaking waves, to be about 0.17 times deep-water significant wave height. *Regime: Wave set-up on a natural beach* direct finding, field [555]
16. Guam reef-flat setup correlated above 0.95 with incident waves, scaled near shore to about 35% of incident RMS height, and reached 1.3 m during Tropical Storm Man-Yi. *Regime: Wave setup over a Pacific Island fringing reef* direct finding, field [1453]
17. Reef roughness reduced setup-producing radiation-stress gradients by 18% but generated mean bottom stress that increased predicted setup by 16%, leaving rough and smooth reef-flat setup within 7% on average. *Regime: Wave Setup over a Fringing Reef with Large Bottom Roughness* direct finding, experimental [202]
18. A 55 m flume study found linear-theory radiation stress underpredicted reef setdown and setup, especially for large waves and low water; adding a breaking-wave roller improved magnitude and forcing location. *Regime: Dynamics of Wave Setup over a Steeply Sloping Fringing Reef* direct finding, experimental [201]
19. Chesapeake sensitivity tests found wave setup up to 0.19 m and dependent on offshore height, breaking angle, profile morphology and mesh resolution, while friction and wetting parameters also materially altered levels. *Regime: Storm Surge Modeling in Large Estuaries: Sensitivity Analyses to Parameters and Physical Processes in the Chesapeake Bay* direct finding, mixed [484]
20. Australian extreme-water analysis shows shoreline setup equations and local beach slope materially affect the estimated wind-wave contribution to mean total water level. *Regime: Extreme Water Levels for Australian Beaches Using Empirical Equations for Shoreline Wave Setup* direct finding, mixed [1038]
21. Wave-resolving XBeach reproduced setup and swash components better than wave-averaged XBeach on an intermediate-reflective beach, including incident-band motion and larger infragravity swash. *Regime: Simulating wave runup on an intermediate-reflective beach using a wave-resolving and a wave-averaged version of XBeach* direct finding, mixed [137]
22. Barrier-island observations show back-barrier water-level gradients can overcome the cross-shore gradient from wave setup and reverse inundation flow, while breaking dissipates both short and infragravity waves. *Regime: Observations of waves and currents during barrier island inundation* direct finding, mixed [410]
23. Validated SWASH simulations found Puerto Morelos lagoon setup strongly correlated with offshore sea-swell energy, while geometry controlled infragravity resonance and whether the reef protected or enhanced inundation. *Regime: Wave-induced extreme water levels in the Puerto Morelos fringing reef lagoon* direct finding, mixed [1422]
24. Coupled wave–surge simulation added 0.3–0.5 m of Ivan-like Tampa Bay surge through wave-induced forces, while surge increased nearshore significant wave height by about 1.0–1.5 m. *Regime: Coupling of surge and waves for an Ivan-like hurricane impacting the Tampa Bay, Florida region* direct finding, mixed [619]
25. In Martinique, radiation-stress-driven wave setup contributed up to 100% of total modeled surge in cases where the narrow shelf limited wind-driven water accumulation. *Regime: Assessing storm surge hazard and impact of sea level rise in the Lesser Antilles case study of Martinique* direct finding, mixed [745]
26. During Cyclone Tauktae, setup exceeded 0.6 m near steep Chellanam profiles and combined with long and short waves to cause severe flooding even at low tide with negligible surge. *Regime: Wave induced coastal flooding along the southwest coast of India during tropical cyclone Tauktae* direct finding, mixed [1155]
27. A field technique measured shoreline setup increasing approximately linearly with incident wave height, agreeing with theory based on surf-zone wave decay and reaching about 40 cm. *Regime: Observations of wave-induced set-up on a natural beach* direct finding, field [720]
28. Hundreds of XBeach-NH+ simulations produced validated equations for setup and incident, infragravity and total runup across 1–12 m waves, 6–16 s periods and beach slopes from 1:100 to 1:5. *Regime: A Model-Derived Empirical Formulation for Wave Run-Up on Naturally Sloping Beaches* direct finding, mixed [1056]
29. Numerical experiments show ambient-current bottom stress changes setup: offshore currents create onshore stress that increases setup, whereas onshore currents create offshore stress that decreases it. *Regime: Numerical Experiments of Bottom Stress on Wave Setup* direct finding, mixed [839]
30. On a barred beach, eleven current meters and fifteen wave staffs measured offshore undertow up to about 0.20 m/s and onshore velocity skewness up to 0.60; shoreline setup explained up to 82% of mean-flow variability and incident wave height explained 86% of setup variability. *Regime: Nontidal low-relief barred surf zone during the observed storm growth and decay.* direct finding, field [531]

49.4.1.5 Breaking-induced turbulence

Topic waves.transformation.breaking.turbulence. Turbulence generated by breaking.

Papers screened: 13. Claims: 12. Equations: 0.

Well established. Surface-wave breaking injects vorticity and turbulent kinetic energy through crest overturning, jet impact, rollers and shear; turbulence is intermittent, anisotropic and transported away from the breaker before dissipating or reaching the bed.

Governing physics. Plunging jets create impact shear and coherent vortices; spilling rollers sustain near-surface production; vortex stretching and transport redistribute TKE; directional spread changes cascade structure; density/air entrainment modifies buoyancy and pressure; bottom interaction converts advected TKE into stress and suspension.

Dimensionless parameters. Controls include breaker type and Iribarren number, wave steepness and relative depth, turbulent Reynolds number, wave period/eddy turnover, roller area/wave height, directional spread, breaker distance/height, TKE/celerity², dissipation normalized by wave-energy flux, and bed distance/roughness.

Major equations. Analysis uses Reynolds-decomposed momentum, TKE and Reynolds-stress budgets, vorticity transport, production by mean shear and wave motion, turbulent and pressure transport, buoyancy and dissipation, Kolmogorov/inverse-cascade spectra, LES subgrid closure, RANS $k-\omega$ /SST, roller energy, and breaking-probability loss.

Typical methods. Methods include high-speed imaging, PIV and ADV, acoustic concentration/velocity, pressure and void fraction, fixed-bar and plane-beach flumes, focused-wave generation, directional basins, phase- and ensemble-averaged TKE budgets, coherent-structure identification, spectra, LES, VOF-RANS and analytical closures.

Numerical models. Models include LES of focused breakers, RANS closure comparisons around structures, coherent-vortex simulations of sediment pickup, $k-\omega$ /SST and buoyancy modifications represented in uncertain full-text records, spectral cascade analysis, and analytical random-breaking turbulence/dissipation approaches.

Experimental datasets. Evidence includes deep-water breaking, plunging-wave flumes, large-scale fixed-bar near-bed TKE budgets, laboratory surf zones with varied directional spread, inner-surf-zone spectra, focused breakers across steepness, and breaking past a vertical cylinder.

Validated ranges. The branch covers spilling, plunging, deep-water, focused and inner-surf-zone breaking; crest-to-bed transport; fixed bars and vertical cylinders; narrow to directional wave fields; near-field coherent structures through spectral dissipation; numeric ranges remain experiment-specific.

Recent advances. Recent work identifies coherent vortical structures quantitatively, performs steepness-controlled LES, resolves directional-spread cascades, measures coupled energy/dissipation spectra, evaluates closures around structures, and links random breaking occurrence to wave-generated turbulence.

Disagreements. RANS closures that reproduce mean wave transformation may disagree on TKE and loads; turbulence can appear locally two-dimensional with inverse cascade while breaker vortices are intrinsically three-dimensional; bottom stress may be driven by advected vortices rather than local orbital shear; dissipation and turbulence are related but not interchangeable.

Limitations. Aerated rollers obstruct optical/acoustic measurement; dissipation inference is sensitive to resolution and isotropy assumptions; phase averaging erases intermittency; fixed beds omit feedback; 2-D flumes constrain vortices; LES is expensive and RANS closure constants are not universal.

Open questions. Needs include air/bubble energetics, three-dimensional coherent structures, wave-group and current effects, breaker-to-bed pathways, movable-bed feedback, closure under directionality, spectral non-equilibrium, breaking probability, structure/vegetation interaction, and field-scale budget closure.

Seminal papers. The branch progresses from plunging-breaker turbulence measurements and deep-water vortex generation to near-bed TKE budgets, coherent-structure sediment suspension, focused-wave LES, directional inverse cascades, inner-surf spectra, and analytical random-breaking closures.

Claims

1. RANS turbulence closures differ materially in their ability to reproduce breaking-wave kinematics, loads, and wake turbulence around a vertical cylinder. *Regime: An evaluation of different RANS turbulence models for simulating breaking waves past a vertical cylinder* direct finding, numerical [1136]
2. Breaking waves organize turbulence into coherent vortical structures whose geometry and evolution control transport away from the breaking crest. *Regime: On coherent vortical structures in wave breaking* direct finding, mixed [504]
3. Deep-water breaking generates coherent vortices through crest overturning and impact, providing a mechanism for rapid momentum and vorticity injection. *Regime: Vortex generation by deep-water breaking waves* direct finding, mixed [1117]
4. Surf-zone breaking turbulence controls suspended-sediment pickup and transport, but measurement and closure differences leave large uncertainty in near-bed TKE. *Regime: Surf Zone Turbulence and Suspended Sediment Dynamics—A Review* review statement, review [5]
5. Breaking-generated coherent vortices can reach the bed and trigger episodic sediment suspension beyond that predicted by local mean shear alone. *Regime: Vortex-induced suspension of sediment in the surf zone* direct finding, numerical [1062]
6. A large-scale plunging breaker produces a strongly phase-dependent near-bed TKE budget over a fixed bar, with transport from the breaking region into the boundary layer. *Regime: Near-Bed Turbulent Kinetic Energy Budget Under a Large-Scale Plunging Breaking Wave Over a Fixed Bar* direct finding, experimental [1577]
7. Large-eddy simulations show focused-breaker turbulence and dissipation vary systematically with incident wave steepness. *Regime: Large eddy simulation of focused breaking waves with different wave steepness* direct finding, numerical [308]
8. Laboratory surf-zone turbulence exhibits a two-dimensional inverse energy cascade whose strength changes with directional wave spread. *Regime: Two-dimensional inverse energy cascade in a laboratory surf zone for varying wave directional spread* direct finding, experimental [107]
9. Inner-surf-zone measurements resolve coupled wave-energy and dissipation spectra, exposing scale-dependent transfer through breaking turbulence. *Regime: Energy and dissipation spectra of waves propagating in the inner surf zone* direct finding, mixed [169]
10. Measurements within a plunging breaker identify localized, rapidly evolving turbulence associated with jet impact and roller motion. *Regime: INVESTIGATION OF TURBULENCE IN A PLUNGING BREAKING WAVE* direct finding, experimental [830]
11. Analytical closures link wave-generated turbulence and random breaking occurrence to wave-energy dissipation while making their regime assumptions explicit. *Regime: Analytical approaches for wave energy dissipation induced by wave-generated turbulence and random wave-breaking* direct finding, analytical [1563]
12. Breaking-bore rollers contain spatially partitioned, short-lived air-water features whose geometry, frequency and surface velocity link free-surface structure to energy dissipation and characteristic time/length scales. *Regime: Unsteady laboratory breaking bores and roller free surfaces across the tested Froude-number conditions.* direct finding, experimental [1541]

49.4.2 Diffraction

Topic waves. transformation. diffraction. Lateral redistribution around obstacles.

Papers screened: 11. Claims: 5. Equations: 1. Shown in the Coastal Design Bench beside: Harbour agitation.

Well established. Diffraction laterally redistributes wave amplitude and direction when bathymetry or obstacles impose along-crest gradients. It interacts with refraction, shoaling and reflection; coherent phase-resolving methods capture interference, whereas spectral methods require phase-decoupled approximations for random seas.

Governing physics. The process follows spatial phase and amplitude gradients governed by dispersion, energy flux and boundary conditions. Important controls are wavelength relative to obstacle/gap scale, water depth, incidence angle, directional spreading, reflection, bathymetric gradients and coherence; focusing can amplify waves locally while shadow zones attenuate them.

Dimensionless parameters. Controls include relative depth kh , obstacle or gap width over wavelength, mound length over wavelength, directional spread, propagation angle, reflection coefficient, relative wave height H/h , Ursell number and grid points per wavelength. The improved parabolic model was tested to 70 degrees.

Major equations. Foundational formulations are the elliptic mild-slope equation and its parabolic forward-marching approximations. Spectral models augment the action balance with a diffraction turning/diffusion term, while nonhydrostatic models solve free-surface flow directly. Generalized Padé operators improve angular dispersion accuracy.

Typical methods. Methods include regular and irregular wave-basin tests, harbor transfer functions, mild-slope finite elements, one-way parabolic marching, phase-decoupled spectral action models, three-dimensional nonhydrostatic solvers and spatial attenuation diagnostics for laterally heterogeneous vegetation or structures.

Numerical models. The corpus contains a generalized-[1/1]-Padé parabolic model, a phase-decoupled diffraction extension in SWAN, and a three-dimensional nonhydrostatic solver validated on flat- and sloping-bed mound benchmarks. Each trades phase detail, backscatter capability and computational cost differently.

Experimental datasets. Key evidence includes two natural-harbor physical models with irregular wave grouping, Vincent-Briggs and Berkhoff elliptical-mound benchmarks, and controlled laterally heterogeneous vegetation layouts. These probe shadowing, focusing, reflection-diffraction and lateral nonuniformity.

Validated ranges. The parabolic model reports tests up to 70-degree propagation. The nonhydrostatic solver reports dimensionless wave-height RMSE of 0.070–0.165 for the flat-bed mound and 0.060–0.114 for the sloping-bed mound, with focusing to 2.2 times incident height in the latter benchmark.

Recent advances. Recent nonhydrostatic validation quantifies focusing errors across classic three-dimensional mound tests, while spatial vegetation experiments show that lateral layout controls attenuation uniformity—an engineering manifestation of diffraction and lateral redistribution absent from one-dimensional transmission metrics.

Disagreements. Differences largely reflect model scope rather than incompatible physics: phase-decoupled spectral diffraction is intended for random short-crested waves without coherent standing patterns, parabolic models assume predominantly forward propagation, and phase-resolving solvers retain interference but require finer grids and boundary treatment.

Limitations. One-way marching misses strong backscatter; spectral closures cannot reproduce coherent phase interference; physical models may have scale and reflection artifacts; Cartesian grids distort curved boundaries; breaking, vegetation wakes and complex natural-reef roughness add dissipation that can be confused with diffraction.

Open questions. Priorities are irregular directional-wave benchmarks with combined diffraction, breaking and reflection; robust diffraction closures on unstructured grids; uncertainty from bathymetry and boundary reflection; field validation around reefs and structures; adaptive resolution; and separation of lateral scattering from dissipation.

Seminal papers. The 1989 harbor experiments established why irregular-wave diffraction-reflection transfer functions can be more stable within the energetic band. The 2003 phase-decoupled formulation made diffraction compatible with operational spectral wave modeling.

Equations

- Wave-energy transformation balance [1233]

$$\nabla \cdot (E \mathbf{C}_g) = -D_b - D_f$$

Regime: Forward-scattered waves and tested incidence/propagation angles up to 70 degrees; backward reflection is outside a one-way marching model. Variables: E wave energy density; C_g group velocity; D_b breaking dissipation; D_f bottom-friction or other dissipation

Claims

1. A forward-marching parabolic model using a locally updated generalized [1/1] Padé approximation improved combined refraction-diffraction predictions over complex bathymetry at propagation angles up to 70 degrees relative to earlier rational-approximation parabolic models. *Regime: Forward-scattered waves and tested incidence/propagation angles up to 70 degrees; backward reflection is outside a one-way marching model.* direct finding, numerical [1233]
2. Experiments over a submerged mound measured amplification beyond twice the incident wave and downstream crest reorganization, with cross-wave steepening up to 30% greater than along the principal propagation direction. *Regime: Gravity wave amplification and phase crest re-organization over a shoal* direct finding, experimental [650]
3. The contribution of wave energy to the renewable energy supply is rising. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [1427]
4. Coastal zones, at the interface between land and sea, face increasing challenges from erosion, sea-level rise, and anthropogenic interventions, necessitating innovative tools for effective management and protection. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, numerical [1259]
5. The placement and operation of marine energy deployments in the ocean have the potential to change flow patterns, decrease wave heights, and/or remove energy from the oceanographic system. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* direct finding, mixed [1509]

49.4.3 Reflection

Topic waves.transformation.reflection. Wave reflection and standing-wave formation.

Papers screened: 5. Claims: 2. Equations: 0. Shown in the Coastal Design Bench beside: Harbour agitation.

Well established. Coastal wave reflection redirects energy from beaches, structures and depth transitions; reflected and incident components interfere to form standing or partial-standing patterns, alter directional spectra and redistribute harbor agitation even when bulk significant height is similar.

Governing physics. Reflection depends on impedance change, slope, roughness, porosity, structure geometry, water depth, frequency and incidence angle. Diffraction spreads reflected energy around boundaries, while nonlinear interactions and wave grouping create low-frequency components that need not be harbor resonance.

Dimensionless parameters. Controls include reflection coefficient, kh , surf-similarity, relative roughness and permeability, incidence angle, structure slope and freeboard, frequency-band energy fraction, standing-wave ratio and geometry or basin length relative to wavelength.

Major equations. Reflection coefficients use reflected-to-incident amplitude or energy; gauge-array methods solve incident/reflected spectral components, and momentum-flux approaches enforce force or radiation-stress balances. Mild-slope, potential-flow and phase-resolving equations represent coupled reflection and diffraction.

Typical methods. Studies use multi-gauge arrays or directional spectra to separate incident and reflected energy, test regular and irregular seas, map harbor spatial spectra, compare frequency bands and directions, validate against structures or bathymetry and check whether low-frequency energy is bound, reflected or resonant.

Numerical models. The reviewed branch includes spectral-type reconstruction and experimental transfer coefficients. Direct third-order and momentum-flux reflection analyses remain in the lawful-full-text queue; common predictive frameworks include mild-slope, boundary-integral and phase-resolving models.

Experimental datasets. Reviewed evidence includes a 40-year multimodal outer-port/harbor spectral reconstruction for Africa basin and physical harbor-model experiments comparing regular and irregular diffraction-reflection transfer behavior.

Validated ranges. Evidence is limited to the studied harbor configurations and dominant incident-energy bands. In irregular tests, coefficients varied less than regular-wave values within the main band; Africa-basin energetic patterns were linked to Reina Sofia breakwater reflection.

Recent advances. Recent advances use dense radar/video and pressure arrays, Bayesian directional separation, high-order and fully nonlinear solvers, porous and flexible boundaries, data-assimilative harbor digital twins and spectral-type clustering that retains reflected multimodality.

Disagreements. Regular-wave tests reveal sharp frequency structure but may exaggerate sensitivity relative to broadband seas. Bulk height and period can hide directional reflection and multimodality, while large transfer estimates outside energetic bands may be statistically unstable yet energetically minor.

Limitations. Incident/reflected separation is ill-conditioned with poor gauge spacing, nonlinear waves, multiple directions or evanescent modes. Sparse spatial measurements, uncertain boundary spectra, diffraction, harbor resonance and changing water level complicate attribution.

Open questions. Priorities include robust multidirectional separation, nonlinear and third-order reflection, currents, porous and overtopped structures, field-scale phase observations, reflection under changing water levels and morphology and uncertainty-aware links to loads and navigation.

Seminal papers. Standing-wave gauge methods established laboratory reflection measurement; linear potential-flow and mild-slope theory treated structures and bathymetry, while directional spectral arrays and irregular-wave transfer functions extended analysis to realistic seas and harbors.

Claims

1. In the eastern half of Africa basin, the most energetic local spectral types were associated with northern to northeasterly outer waves redirected by reflection from Reina Sofia breakwater. *Regime: Present Africa-basin geometry, bathymetry, reflection-coefficient curves, and 40-year modeled climate.* direct finding, numerical [1210]
2. This is a study of the fundamental physical processes of the runoff of long waves with the objective to understand some coastal effects of tsunamis. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [1383]

49.4.4 Refraction

Topic waves.transformation.refraction.Depth- and current-induced turning.

Papers screened: 9. Claims: 8. Equations: 0.

Well established. Wave refraction changes propagation direction and energy density as phase speed varies with depth or current; converging rays focus energy and diverging rays defocus it, while diffraction, reflection, breaking and spectral spreading regularize ideal ray singularities.

Governing physics. Conservation of frequency along stationary paths and of wave action in currents combines with the dispersion relation to rotate wave crests toward slower regions. Bathymetric mounds, channels, headlands and current gradients produce caustics, shadow zones and frequency-dependent focusing.

Dimensionless parameters. Controls include kh , relative depth and bathymetric slope, incident angle, directional spread, current-to-group-speed ratio, refraction and shoaling coefficients, mound height and width relative to wavelength, focusing gain and grid or spectral resolution relative to caustic width.

Major equations. Linear dispersion links frequency, wavenumber and depth; Snell-type relations conserve along-crest wavenumber for parallel contours. Spectral action balance transports action through geographic and directional space, while mild-slope, Boussinesq and nonhydrostatic equations include diffraction and nonlinear transformation.

Typical methods. Studies map directional spectra and bathymetry, ray-trace or solve spectral/phase-resolving equations, validate mound and island benchmarks, compare cross-shore wave-height sections, test frequency-direction boundary sensitivity and report focusing, sheltered-site and bias errors separately.

Numerical models. The set includes a three-dimensional nonhydrostatic model for shoaling, diffraction and refraction and two regional spectral formulations with and without diffraction. Direct current-refraction and multidirectional model records remain in the lawful-full-text queue.

Experimental datasets. Reviewed evidence includes Vincent-Briggs flat-bed and Berkhoff 1:50 sloping-bed mound benchmarks and a roughly 300 km by 300 km Southern California Bight comparison of refraction and refraction-diffraction spectral models.

Validated ranges. The Berkhoff focusing section reached 2.2 times incident wave height; the flat-bed focusing section had MAE 0.158 and MBE -0.114. Regional models improved with broader incident spectra but neither was accurate at highly sheltered sites.

Recent advances. Recent advances combine unstructured high-resolution spectral models, current-coupled action balance, nonhydrostatic and dispersive solvers, satellite-derived bathymetry, data assimilation, adaptive phase-resolving nests and differentiable calibration.

Disagreements. Geometric rays are efficient away from caustics but fail where diffraction or multi-path interference matters. Spectral phase-averaged models handle broad seas but smooth coherent focusing, while phase-resolving models capture interference at higher cost and remain sensitive to boundaries and breaking.

Limitations. Bathymetric resolution, boundary directional spectra, current uncertainty, caustics, diffraction closure, unresolved islands and structures, nonlinear breaking, bottom friction and sparse directional observations can dominate. Agreement at exposed sites does not prove skill in sheltered zones.

Open questions. Priorities include current-depth refraction in strong shears, probabilistic caustics, rapidly changing bathymetry, spectral-to-phase coupling, breaking-limited focusing, remote directional validation, adaptive grids and uncertainty propagation into loads and sediment transport.

Seminal papers. Snell-law and ray theory established depth refraction; action conservation extended it to currents and spectra, while mild-slope equations introduced diffraction and reflection and later Boussinesq/nonhydrostatic solvers resolved nonlinear phase effects.

Claims

1. In the flat-bed mound case, the largest reported MAE was 0.158 at the energy-focusing section and the MBE there was -0.114, indicating local underprediction of dimensionless wave height. *Regime: Vincent-Briggs flat-bed elliptical-mound benchmark and the paper's section definitions.* direct finding, mixed [1303]
2. In the Berkhoff 1:50 sloping-bed mound case, wave height at the focusing section reached 2.2 times the incident height. *Regime: Regular $H=4.64$ cm, $T=1$ s wave over the specified rotated elliptical mound on a 1:50 bed.* direct finding, mixed [1303]
3. Ten months of Hanalei Bay observations and modeling show that episodic long-period swells are preferentially refracted into the reef embayment and drive currents up to an order of magnitude stronger than modal trade-wind conditions. *Regime: Hydrodynamics of a bathymetrically complex fringing coral reef embayment: Wave climate, in situ observations, and wave prediction* direct finding, mixed [606]
4. Assuming linear wave theory for waves riding on a weak current of $O(\epsilon)$ compared to the wave phase speed, an approximate dispersion relation is developed to $O(\epsilon^2)$ for arbitrary current $U(z)$ in water of finite depth. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [721]
5. Abstract We apply a coupled circulation-wave model to simulate extreme sea levels induced by tropical cyclones at the basin scale. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [921]
6. Characteristic-based regular- and irregular-wave models reproduce current-depth refraction benchmarks, while irregular-wave height is not amplified by following or opposing currents and opposing-current cases show slight overestimation from directional energy leakage. *Regime: Regular and directional irregular waves refracted by one-dimensional following or opposing currents and depth gradients.* direct finding, numerical [1556]
7. A multidirectional random-wave action-balance model using a QUICK scheme resolves transformation, breaking, and blocking in current fields and reproduces wave-height behavior in caustics and a river-mouth application. *Regime: Directional random waves traveling through spatially varying following or opposing coastal currents, including river-mouth flow.* direct finding, numerical [1045]
8. At the Gunsan-Janghang Harbor entrance, comparison of spectral wind-wave and energy-balance models characterizes wave transformation through a changing reclaimed coastline, islands, shoals, breakwaters, and a long navigation channel. *Regime: Gunsan-Janghang Harbor entrance under reclamation-modified bathymetry, river-mouth shoals, coastal islands, breakwaters, and wind waves.* direct finding, mixed [783]

49.4.5 Shoaling

Topic waves. transformation. shoaling. Depth-induced amplitude and kinematic change.

Papers screened: 12. Claims: 9. Equations: 6.

Well established. As surface gravity waves enter decreasing depth, reduced group speed and depth-dependent kinematics transform height, wavelength, shape, energy distribution, and momentum flux before and through breaking. Linear shoaling is reliable only within restricted amplitude and bathymetric regimes.

Governing physics. Energy or action-flux conservation drives nondissipative height change; nonlinearity sharpens crests and transfers energy among harmonics; refraction redistributes flux directionally; reflection, bottom friction, breaking, vegetation drag, infragravity modulation, and currents add sinks or couplings.

Dimensionless parameters. Regime selection uses relative depth h/L or kh , deep-water steepness H_0/L_0 , local relative height H/h , Ursell number, bed slope, breaker index, directional angle, vegetation submergence and density, and bandwidth or spectral shape.

Major equations. Core descriptions include Green's law for long waves, linear energy-flux shoaling coefficients, matched linear-cnoidal theory, random-wave height distributions with breaking, depth-limited surf-zone relations, radiation-stress/momentum-flux balances, and phase-resolving nonhydrostatic equations.

Typical methods. Studies combine analytical conservation laws and asymptotic expansions, wave-flume measurements, natural-beach arrays, stochastic transformation models, barred-beach calculations, and three-dimensional nonhydrostatic simulation.

Numerical models. Model classes include linear and cnoidal transformation tables, stochastic breaking-height/vegetation-drag formulations, depth-limited surf-zone decay, 3-D momentum-flux theory, and nonhydrostatic phase-resolving models for combined shoaling, refraction, diffraction, and breaking.

Experimental datasets. Reviewed benchmarks include variable-geometry and regular-wave laboratory shoaling, no-free-second-harmonic linear/cnoidal tests, exposed dissipative-beach surf-zone observations, and source-specific vegetation-force calculations or comparisons.

Validated ranges. The matched linear/cnoidal model agreed very well without free second harmonics for H_0/L_0 up to about 0.03-0.04, using linear theory above $h/L_0=0.10$ and cnoidal theory below; other demonstrations cover small-amplitude variable transitions, dissipative-beach surf zones, rigid vegetation, and nondissipative barred bathymetry.

Recent advances. Recent nonhydrostatic models jointly resolve shoaling, refraction, diffraction, and breaking in three dimensions, creating a route to test reduced coefficients against complex transformations while retaining computationally practical coastal domains.

Disagreements. No single shoaling coefficient spans all regimes. Linear theory departs at greater steepness, local H/h is not constant throughout surf zones, and depth-integrated radiation stress obscures vertical flux structure needed by 3-D circulation models.

Limitations. Common exclusions are reflection, friction, breaking, free harmonics, broad bandwidth, currents, directional spreading, flexible vegetation, rotational flow, and complex three-dimensional bathymetry. Field depth limitation mixes shoaling, breaking, setup, and infragravity variability.

Open questions. Needs include common uncertainty-aware benchmarks across shoaling and breaking, broadband nonlinear directionality, wave-current and vegetation flexibility, consistent 3-D momentum coupling, bathymetric-history effects, and efficient model-selection criteria for engineering applications.

Seminal papers. The 1968 Green/linear laboratory comparison and 1977 linear-cnoidal matching study establish early engineering transformations; the 1992 field study demonstrates nonconstant surf-zone H/h , while later work extends forcing vertically and to vegetation.

Equations

- *Green's law* [180]

$$H(x) = K_s(h/L, H/L, \beta, \mathcal{D})H_0$$

Regime: Small-amplitude waves over variable-geometry shallow-water transitions; reflection and friction omitted in the intermediate/deep-water comparison. Variables: H local wave height; K_s shoaling/transformation factor; h local depth; L wavelength; β bed slope; $\mathcal{D}$ dissipation, directionality, or vegetation terms; normalized dependency statement

- *linear-cnoidal matched shoaling model* [1380]

$$H(x) = K_s(h/L, H/L, \beta, \mathcal{D})H_0$$

Regime: Regular waves without free second-harmonic components; linear theory for $h/L_0 > 0.10$, cnoidal theory for $h/L_0 < 0.10$; H_0/L_0 up to 0.03-0.04 for very good agreement. Variables: H local wave height; K_s shoaling/transformation factor; h local depth; L wavelength; β bed slope; $\mathcal{D}$ dissipation, directionality, or vegetation terms; normalized dependency statement

- *depth-limited surf-zone wave-height model* [1014]

$$H(x) = K_s(h/L, H/L, \beta, \mathcal{D})H_0$$

Regime: Natural exposed dissipative beach within the surf zone; field analysis of local depth and breaker-line distance effects. Variables: H local wave height; K_s shoaling/transformation factor; h local depth; L wavelength; β bed slope; $\mathcal{D}$ dissipation, directionality, or vegetation terms; normalized dependency statement

- *3-D wave-induced momentum-flux formulation* [592]

$$H(x) = K_s(h/L, H/L, \beta, \mathcal{D})H_0$$

Regime: Irrotational, nondissipative three-dimensional waves over two-dimensional irregular bathymetry, illustrated for a barred beach. Variables: H local wave height; K_s shoaling/transformation factor; h local depth; L wavelength; β bed slope; $\mathcal{D}$ dissipation, directionality, or vegetation terms; normalized dependency statement

- *Mendez random-wave height distribution and vegetation-drag model* [588]

$$H(x) = K_s(h/L, H/L, \beta, \mathcal{D})H_0$$

Regime: Stationary narrow-band random waves in shallow water over rigid vegetation; shoaling and breaking represented by the Mendez et al. wave-height distribution. Variables: H local wave height; K_s shoaling/transformation factor; h local depth; L wavelength; β bed slope; $\mathcal{D}$ dissipation, directionality, or vegetation terms; normalized dependency statement

- *Wave-energy transformation balance* [685]

$$\nabla \cdot (E\mathbf{C}_g) = -D_b - D_f$$

Regime: The 225 irregular-wave tests on the reported mobile-bed parallel-contour hydraulic model. Variables: E wave energy density; C_g group velocity; D_b breaking dissipation; D_f bottom-friction or other dissipation

Claims

1. For regular waves without free second harmonics, a wave-height-continuous linear/cnoidal shoaling model agreed very well with experiments up to H_0/L_0 about 0.03-0.04, whereas linear theory failed at larger steepness in the $h/L_0 > 0.10$ region. *Regime: Regular waves without free second-harmonic components; linear theory for $h/L_0 > 0.10$, cnoidal theory for $h/L_0 < 0.10$; H_0/L_0 up to 0.03-0.04 for very good agreement.* direct finding, mixed [1380]
2. For stationary narrow-band random waves over rigid vegetation, empirical regular-wave drag coefficients combined with a stochastic shoaling-and-breaking height distribution provide a practical vegetation-force estimate across different slopes. *Regime: Stationary narrow-band random waves in shallow water over rigid vegetation; shoaling and breaking represented by the Mendez et al. wave-height distribution.* direct finding, analytical [588]
3. On an exposed dissipative beach, surf-zone wave-height-to-depth ratios decayed with both local mean water depth and distance from the breaker line, indicating an infragravity-associated decay process in addition to breaking turbulence. *Regime: Natural exposed dissipative beach within the surf zone; field analysis of local depth and breaker-line distance effects.* direct finding, field [1014]
4. For small-amplitude waves propagating through variable-geometry transitions, Green's law and a reflection- and friction-free linear transformation model showed fair agreement with laboratory shoaling measurements. *Regime: Small-amplitude waves over variable-geometry shallow-water transitions; reflection and friction omitted in the intermediate/deep-water comparison.* direct finding, mixed [180]
5. For irrotational nondissipative waves over alongshore-uniform irregular bathymetry, vertically varying wave-induced momentum flux balances other vertically varying forcing terms, leaving a depth-invariant total forcing governed by wave-energy and water-depth gradients. *Regime: Irrotational, nondissipative three-dimensional waves over two-dimensional irregular bathymetry, illustrated for a barred beach.* direct finding, analytical [592]
6. In 225 irregular-wave hydraulic-model tests on a mobile-bed beach with parallel contours, Snell refraction described turning, linear theory slightly overpredicted shoaling, bottom friction and percolation were small, and saturation-limited breaking attenuation was essential. *Regime: The 225 irregular-wave tests on the reported mobile-bed parallel-contour hydraulic model.* direct finding, experimental [685]
7. For modeled nonbreaking regular waves, horizontal-bed stream-function theory described nonlinear shoaling and de-shoaling on a very gentle $\cot(\beta) = 1200$ foreshore, whereas steeper de-shoaling released free waves that violated the constant-form approximation. *Regime: Nonbreaking normally incident regular waves over gentle and steep foreshores, including de-shoaling slope $\cot(\beta) = 1200$.* direct finding, numerical [401]
8. Stochastic and deterministic Boussinesq models reproduce observed nonlinear transformation of continuous wave spectra over a natural barred beach; shoaling depends on nonlinearity, incident spectral shape and bottom profile. *Regime: Nonlinear Evolution of Surface Wave Spectra on a Beach* direct finding, mixed [1033]
9. A 2 week field experiment was conducted to measure surface wave dissipation on a barrier reef at Kaneohe Bay, Oahu, Hawaii. *Regime: The study system, forcing, data and methods stated in the indexed abstract.* review statement, mixed [866]

Appendices

Appendix A. API reference

This appendix is generated from the docstrings of the installed package by `docs/manual/build_manual.py`, so it always matches the code it ships with. Modules are listed in package order; within each, the public classes, functions, and constants appear in source order, each with its signature and its full docstring.

pyCoastal

pyCoastal: A modular coastal-process modeling framework.

Subpackages

`config` Input-file parsing (YAML/JSON/INI)
`domain` Mesh & geometry definitions (1D/2D/3D)
`numerics` Numerical infrastructure (grids, schemes, solvers)
`physics` Governing equations & closures
`tools` Standalone formulae & utilities
`boundary` Boundary-condition classes
`io` I/O for VTK, CSV, NetCDF, etc.

Coordinate convention

x and y span the horizontal plane and z is elevation, positive upward.
In every 2D array x is axis 0 and y is axis 1, matching `UniformGrid`, which builds its coordinates with ```indexing="ij"``` and flattens as ```i*ny + j```.

pyCoastal.config

Configuration loader for pyCoastal.
Supports YAML, JSON, and INI-style files.

load_config

```
def load_config(path: str) -> dict
```

Load a configuration file and return a dictionary.

Supported formats:

- .yaml, .yml (requires PyYAML)
- .json
- .ini, .cfg

Parameters

`path` : str
Path to the config file.

Returns

dict
Parsed configuration.

pyCoastal.io

read_data

```
def read_data(path: 'str | os.PathLike') -> dict
```

Read a YAML, JSON or INI file and return a python dict.

Accepts a Path as well as a string, so a caller can resolve the file relative to its own location rather than to the shell's directory.

write_vtk

```
def write_vtk(grid, filename: str)
```

Write a vtkUnstructuredGrid (or similar) to a .vtk file.

pyCoastal.numerics.grid

Basic Cartesian grid for 1D/2D/3D domains, now with boundary-index support for Dirichlet/Neumann/Sponge BCs.

UniformGrid

```
class UniformGrid
```

method

```
UniformGrid.neumann_indices(self, side: str)
```

property

```
UniformGrid.n_cells(self)
```

property

```
UniformGrid.cell_volume(self)
```

method

```
UniformGrid.cell_indices(self)
```

pyCoastal.numerics.domain

Domain and mesh definitions for pyCoastal.
Provides 1D and 2D structured grid classes.

Mesh1D

```
class Mesh1D
```

Simple 1D mesh.
Attributes

```
-----  
x : np.ndarray  
    Cell center coordinates.  
dx : float  
    Uniform grid spacing.
```

Mesh2D

```
class Mesh2D
```

Rectangular 2D mesh.
Attributes

```
-----  
x, y : np.ndarray  
    2D arrays of cell-center coordinates.  
dx, dy : float  
    Uniform spacings in x and y.
```

Domain

```
class Domain
```


High-level domain object. Reads geometry from config and instantiates the appropriate mesh.

method

Domain.info(self)

Print a summary of the domain.

pyCoastal.numerics.operators

laplacian

```
def laplacian(field: np.ndarray, grid) -> np.ndarray
```

Compute the 5-point Laplacian of `field` on a UniformGrid, using centered finite differences and periodic (roll) indexing.

Parameters

field : (nx,ny) array

Input scalar field.

grid : UniformGrid

The grid, provides grid.spacing = (dx, dy).

Returns

lap : (nx,ny) array

The discrete Laplacian $\partial^2/\partial x^2 + \partial^2/\partial y^2$.

gradient

```
def gradient(field, grid)
```

Centered gradient $\nabla f \rightarrow (f_x, f_y)$ on cell centers.

x is axis 0 and y is axis 1, matching UniformGrid and grad_x/grad_y.

grad_x

```
def grad_x(field, grid)
```

$\partial/\partial x$ using centered differences.

grad_y

```
def grad_y(field, grid)
```

$\partial/\partial y$ using centered differences.

upwind_x

```
def upwind_x(field, u, grid)
```

First-order upwind in x: uses flow sign in u to pick difference.

upwind_y

```
def upwind_y(field, v, grid)
```

First-order upwind in y.

divergence

```
def divergence(ux, uy, grid)
```

Centered divergence $\nabla \cdot \mathbf{u}$ on cell centers.

curl_z

```
def curl_z(u, v, grid)
```

$\nabla \times (\mathbf{u}, \mathbf{v})$ in 2D gives scalar k-component: $\partial v / \partial x - \partial u / \partial y$.

biharmonic

```
def biharmonic(field, grid)
```

$\Delta^2 \text{ field} = \text{Laplacian}(\text{Laplacian}(\text{field}))$.

mixed_xy

```
def mixed_xy(field, grid)
```

$\partial^2 / \partial x \partial y$ of field (central).

advect

```
def advect(u, v, field, grid, scheme='upwind')
```

Compute $\mathbf{u} \cdot \nabla \text{field} + \mathbf{v} \cdot \nabla \text{field}$ using chosen scheme.

smooth3

```
def smooth3(field, grid)
```

Simple 5-point smoothing filter (weights sum to one).

pyCoastal.numerics.scheme

Time-integration and spatial discretization schemes.

TimeIntegrator

```
class TimeIntegrator
```

Base class for explicit time integrators (e.g. Euler, RK2, RK3).

method

```
TimeIntegrator.step(self, state, rhs, t, **kwargs)
```

Advance 'state' one time step.
state : dict of numpy arrays (e.g. { "u":..., "h":..., ... })
rhs : function(state, t, **kwargs) → dict of tendencies
t : current time
returns (new_state, new_time)

EulerIntegrator

```
class EulerIntegrator(TimeIntegrator)
```

method

```
EulerIntegrator.step(self, state, rhs, t, **kwargs)
```

SSPRK2Integrator

```
class SSPRK2Integrator(TimeIntegrator)
```

method

```
SSPRK2Integrator.step(self, state, rhs, t, **kwargs)
```

central_difference

```
def central_difference(phi, spacing, axis)
```

Second-order central difference in 'axis'.

upwind

```
def upwind(phi, vel, spacing, axis)
```

First-order upwind scheme for advection term $\text{vel} \cdot \nabla \phi$.
vel: array of the same shape giving velocity component in 'axis'.

pyCoastal.numerics.time_intg

euler_step

```
def euler_step(u, t, dt, rhs)
```

Forward Euler scheme (1st-order). $u_{\{n+1\}} = u_n + dt * \text{rhs}(u_n, t)$

Parameters:

u : 2D numpy array, current solution field
t : float, current time
dt : float, time step size
rhs : function(u, t) -> 2D array of same shape as u, computes time derivative

Returns:

u_new : 2D array, solution at $t + dt$

rk2_step

```
def rk2_step(u, t, dt, rhs)
```

Heun's method / explicit midpoint (2nd-order Runge-Kutta).

```
k1 = rhs(u, t)
k2 = rhs(u + dt * k1, t + dt/2)
u_{n+1} = u_n + (dt/2) * (k1 + k2)
```

rk4_step

```
def rk4_step(u, t, dt, rhs)
```

Classic 4th-order Runge-Kutta.

```
k1 = rhs(u, t)
k2 = rhs(u + dt/2 * k1, t + dt/2)
k3 = rhs(u + dt/2 * k2, t + dt/2)
k4 = rhs(u + dt * k3, t + dt)
u_{n+1} = u_n + dt/6 * (k1 + 2*k2 + 2*k3 + k4)
```

ab2_step

```
def ab2_step(u, u_prev, t, dt, rhs)
```

Second-order Adams-Bashforth.

Requires solution at two previous times: u_n and $u_{\{n-1\}}$.
 $u_{\{n+1\}} = u_n + dt * (3/2 * \text{rhs}(u_n, t) - 1/2 * \text{rhs}(u_{\{n-1\}}, t-dt))$

rk3_ssp_step

```
def rk3_ssp_step(u, t, dt, rhs)
```

Strong Stability-Preserving 3rd-order Runge-Kutta (SSP RK3).

Stage 1: $u_1 = u + dt * rhs(u, t)$

Stage 2: $u_2 = 0.75 u + 0.25 (u_1 + dt * rhs(u_1, t+dt))$

Stage 3: $u_{\{n+1\}} = (1/3) u + (2/3) (u_2 + dt * rhs(u_2, t+0.5*dt))$

pyCoastal.numerics.boundary

```
pycoastal.boundary
```

```
-----
```

Boundary-condition classes for pyCoastal solvers.

Defines Dirichlet, Neumann, Wall (no-flow), and Sponge (damping) BCs.

Each BC must implement `apply(fields, grid, t)` where:

- fields: dict of solution arrays, e.g. {"eta": η_{array} , "q": q_{array} , ...}
- grid: an object providing boundary indices, e.g. `grid.boundary_indices["west"]`
- t: current simulation time (s)

BoundaryCondition

```
class BoundaryCondition
```

Abstract base class for all boundary conditions.

method

```
BoundaryCondition.apply(self, fields: Mapping[str, np.ndarray], grid, t: float)
```

Apply this BC to the given fields at time t.

DirichletBC

```
class DirichletBC(BoundaryCondition)
```

Enforce a prescribed value at a boundary (ghost cell or face).

method

```
DirichletBC.apply(self, fields, grid, t)
```

NeumannBC

```
class NeumannBC(BoundaryCondition)
```

Zero-gradient (or specified) Neumann BC: copy interior value to boundary.

method

```
NeumannBC.apply(self, fields, grid, t)
```

WallBC

```
class WallBC(BoundaryCondition)
```

No-flow wall: set normal velocity (or flux) to zero.

method

```
WallBC.apply(self, fields, grid, t)
```

SpongeBC

```
class SpongeBC(BoundaryCondition)
```

Damping (sponge) layer: relax solution toward reference (often zero).

method

```
SpongeBC.apply(self, fields, grid, t)
```

BoundaryManager

```
class BoundaryManager
```

Holds and applies a list of boundary conditions each timestep.

method

```
BoundaryManager.add(self, bc: BoundaryCondition)
```

method

```
BoundaryManager.apply_all(self, fields: Mapping[str, np.ndarray], grid, t: float)
```

Apply every registered BC to the solution fields at time t.

Args:

fields: dict of field arrays (modified in place)
grid: solution grid, with boundary_indices, sponge_indices, etc.
t: current time

pyCoastal.numerics.solver

High-level Solver: ties together grid, physics, boundary conditions, and time-integrator.

Solver

```
class Solver
```

method

```
Solver.run(self, t0, t_end, callback=None)
```

March from t0 to t_end.

callback(state, t) will be called after each step (for output/plotting).

pyCoastal.physics.shallow_water

ShallowWater2D

```
class ShallowWater2D
```

2D nonlinear shallow water equations:

$$\begin{aligned}\partial h / \partial t + \partial(h u) / \partial x + \partial(h v) / \partial y &= 0 \\ \partial(h u) / \partial t + \partial(h u^2 + \frac{1}{2} g h^2) / \partial x + \partial(h u v) / \partial y &= S_x \\ \partial(h v) / \partial t + \partial(h u v) / \partial x + \partial(h v^2 + \frac{1}{2} g h^2) / \partial y &= S_y\end{aligned}$$

method

```
ShallowWater2D.fluxes(self, h: np.ndarray, hu: np.ndarray, hv: np.ndarray)
```

Compute the three flux-pairs (in x and y) for the SWEs.
Returns (Fh, Gh), (Fhu, Ghu), (Fhv, Ghv)

method

```
ShallowWater2D.source_bed_slope(self, h: np.ndarray, zb: np.ndarray, grid=None)
```

Return bed-slope source terms S_x , S_y for momentum:

```
 $S_x = -g h \partial z b / \partial x$ ,  $S_y = -g h \partial z b / \partial y$ 
```

x is axis 0 and y is axis 1. Pass ``grid`` (a UniformGrid) so the derivatives use the real cell spacing; without it the spacing is taken as unity, which is only correct on a unit mesh.

pyCoastal.physics.navier_stokes

initialize_state

```
def initialize_state(grid)
```

Return a dict with zero u, v, p fields on the grid.

rhs

```
def rhs(state, t, grid, bc_mgr, v=0.001)
```

Compute convection + diffusion terms for u, v (no pressure).

Enforce velocity BCs on the resulting rhs arrays, then return them.

pyCoastal.physics.poisson

apply_dirichlet_bc

```
def apply_dirichlet_bc(phi, bc_mask, bc_values)
```

Enforce Dirichlet BC on ϕ array in place.

bc_mask : boolean array of same shape as ϕ ; True where BC applies
 bc_values : array same shape as ϕ ; the prescribed ϕ values

build_laplacian

```
def build_laplacian(nx, ny, dx, dy)
```

Build 2D Laplacian operator with Dirichlet zero BC on a regular grid.
Returns a scipy CSR sparse matrix of shape $(nx*ny, nx*ny)$.

solve_direct

```
def solve_direct(rhs, dx, dy, bc_mask=None, bc_values=None)
```

Solve $\nabla^2 \phi = rhs$ with Dirichlet BC via a sparse direct solver (requires SciPy).

- rhs : 2D array of shape (ny, nx)
- dx, dy : grid spacing
- bc_mask, bc_values : optional masks/values for Dirichlet ϕ

Returns ϕ as 2D array.

solve_jacobi

```
def solve_jacobi(rhs, dx, dy, bc_mask=None, bc_values=None, tol=1e-06, maxiter=5000)
```

Simple Jacobi iteration for $\nabla^2 \phi = rhs$ with optional Dirichlet BC.
Returns ϕ after convergence or maxiter.

solve_poisson

```
def solve_poisson(rhs, dx, dy, bc_mask=None, bc_values=None, method='auto', **kwargs)
```

Poisson solver interface.

```
rhs      : 2D array of RHS values
dx, dy   : grid spacing
bc_mask  : bool array where Dirichlet BC applies
bc_values : array of same shape with prescribed  $\phi$  values
method   : 'direct', 'jacobi', or 'auto'
**kwargs : passed to the chosen solver (tol, maxiter, etc.)
```

pyCoastal.physics.turbulence

SmagorinskyModel

```
class SmagorinskyModel
```

Large-eddy eddy viscosity: $\nu_t = (C_s * \Delta)^2 * |S|$

method

```
SmagorinskyModel.eddy_viscosity(self, u: np.ndarray, v: np.ndarray, grid=None)
```

Strain-rate magnitude $|S| = \sqrt{2 S_{ij} S_{ij}}$.

x is axis 0 and y is axis 1. Pass `grid` so the velocity gradients use the real cell spacing.

KEpsilonModel

```
class KEpsilonModel
```

k- ϵ two-equation closure in eddy-viscosity form:

```
v_t = C_μ k^2/ε
transport eqns for k and ε must be added by solver
```

method

```
KEpsilonModel.eddy_viscosity(self, k: np.ndarray, eps: np.ndarray)
```

KOmegaModel

```
class KOmegaModel
```

k- ω (Wilcox) two-equation closure:

```
v_t = k/ω
```

method

```
KOmegaModel.eddy_viscosity(self, k: np.ndarray, ω: np.ndarray)
```

pyCoastal.tools.wave

dispersion

```
def dispersion(T: float, h: float, g: float=9.81, tol: float=1e-12) -> float
```

Solve the linear dispersion relation for the wavelength L.

Solves $\omega^2 = g k \tanh(k h)$ by Newton-Raphson, starting from the explicit Fenton & McKee (1990) approximation. This converges in a handful of iterations across the full range from deep water to the shallow-water limit.

Args:

T (float): Wave period (s)
 h (float): Water depth (m)
 g (float, optional): Gravity acceleration (m/s²). Default: 9.81.
 tol (float, optional): Relative convergence tolerance on k.

Returns:

float: Wavelength L (m)

Raises:

ValueError: If T or h is not strictly positive.

wave_number

```
def wave_number(T: float, h: float) -> float
```

Compute wave number $k = 2\pi / L$.

Args:

T (float): Period (s)
 h (float): Depth (m)

Returns:

float: Wave number k (rad/m)

surf_similarity

```
def surf_similarity(alpha: float, H: float, T: float) -> float
```

Compute the Iribarren (surf similarity) number ξ .

$\xi = \tan(\alpha) / \sqrt{H / L_0}$, where $L_0 = gT^2/(2\pi)$

Args:

alpha (float): Slope angle (radians)
 H (float): Wave height (m)
 T (float): Wave period (s)

Returns:

float: Iribarren number ξ

breaker_type

```
def breaker_type(alpha: float, H: float, T: float) -> str
```

Determine wave breaking type based on Iribarren number.

Classifications:

- $\xi < 0.4$: Spilling breaker
- $0.4 \leq \xi \leq 2$: Plunging breaker
- $\xi > 2$: Collapsing/surging breaker

Args:

alpha (float): Slope angle (rad)
 H (float): Wave height (m)
 T (float): Wave period (s)

Returns:

str: 'Spilling', 'Plunging', or 'Surging' (collapsing)

ursell_number

```
def ursell_number(H: float, T: float, h: float) -> tuple
```

Calculate Ursell number $U = H^3 L^2 / h^3$ and provide interpretation.

Args:

H (float): Wave height (m)
 T (float): Period (s)
 h (float): Depth (m)

Returns:

(float, str): Ursell number U and interpretation.

wave_setup

```
def wave_setup(Hb: float, gamma: float=0.8) -> float
```

Estimate mean water level increase at shoreline due to wave setup.
 $\eta = (5/16) * \gamma * H_b$

generate_irregular_wave

```
def generate_irregular_wave(Hs: float, Tp: float, duration: float, dt: float, spectrum: str='pm', gamma: float=3.3) -> tuple[np.ndarray, np.ndarray]
```

Generate one realization of an irregular wave time-series $\eta(t)$.

Args:

Hs : significant wave height [m]
Tp : peak period [s]
duration : total record length [s]
dt : time step [s]
spectrum : 'pm' or 'jonswap'
gamma : peak enhancement factor (JONSWAP only)

Returns:

t : time array of length $N = \text{ceil}(\text{duration}/\text{dt})$
eta : $\eta(t)$ time-series

pyCoastal.tools.sediment_transport

shields_parameter

```
def shields_parameter(tau_b: float, rho_s: float, rho: float, d: float, g: float=9.81) -> float
```

Dimensionless Shields parameter for initiation of sediment motion.
 $\tau_* = \tau_b / ((\rho_s - \rho) * g * d)$

van_rijn_bedload

```
def van_rijn_bedload(Ue: float, h: float, d50: float, rho_s: float, rho: float, nu: float=1e-06) -> float
```

Van Rijn (1984) bed-load transport per unit width:
 $q_b = 0.015 * \rho_s * U_e * h * (d_{50} / h)^{1.2} * Me^{1.5}$
with $Me = (U_e - U_{cr}) / \sqrt{(s-1)*g*d_{50}}$

van_rijn_suspended

```
def van_rijn_suspended(Ue: float, h: float, d50: float, rho_s: float, rho: float, nu: float=1e-06) -> float
```

Van Rijn suspended-load formula:
 $q_s = 0.008 * \rho_s * U_e * d_{50} * Me^{2.4} * D^{*-0.6}$
where $D^* = (d_{50} * ((s-1)*g/nu^2))^{(1/3)}$

bijker_bedload

```
def bijker_bedload(tau_wave: float, tau_current: float, rho_s: float, rho: float, d50: float, g: float=9.81) -> float
```

Bijker (1971) formula combining wave and current effects for bedload:
 $q_{b_sb} \propto \sqrt{\tau_{total}}$

cerc_transport

```
def cerc_transport(Eb: float, angle: float, K: float=0.39, rho: float=1025, g: float=9.81) -> float
```

CERC longshore transport rate:
 $Q = K * E_b / (\rho g) * \sin\phi_b \cos\phi_b$

bagnold_sediment

```
def bagnold_sediment(H: float, c: float, rho_s: float, rho: float=1025) -> float
```

Bagnold's load estimate: $q_s \propto (\rho_s/\rho) * \text{wave power}$.
Simplified form here for energy-driven sediment transport.

izbash_current

```
def izbash_current(rho_s: float, rho: float, d: float, g: float=9.81) -> float
```

Izbash critical current for stone stability:
 $u_c = 1.7 * \sqrt{\Delta g d}$

einstein_bedload

```
def einstein_bedload(tau_star: float, d: float, s: float, g: float=9.81) -> float
```

Einstein's bedload formula (probabilistic):
 $q_b = 8 \sqrt{[g (s - 1) d^3]} * \tau^{1.5}$
Useful for fine sand transport upstream.

pyCoastal.tools.structural

hudson_dn50

```
def hudson_dn50(Hs: float, Delta: float, theta: float, Kd: float=3.0) -> float
```

Calculate nominal diameter Dn50 for armor stone using Hudson's formula.
 $H_s / (\Delta D_{n50}) = (K_d \cot \theta)^{1/3} / 1.27$

vandermeer_dn50

```
def vandermeer_dn50(Hs: float, Delta: float, P: float, N: int, alpha: float, xi_m: float, damage: float=2.0, safety: float=1.0) -> float
```

Nominal armour diameter Dn50 from Van der Meer (1988).

plunging ($xi_m < xi_{cr}$)
 $H_s / (\Delta D_{n50}) = 6.2 P^{0.18} (S / \sqrt{N})^{0.2} xi_m^{-0.5}$
surging ($xi_m \geq xi_{cr}$)
 $H_s / (\Delta D_{n50}) = 1.0 P^{-0.13} (S / \sqrt{N})^{0.2} \sqrt{\cot a} xi_m^P$

Args:

Hs: significant wave height at the toe (m)
Delta: relative buoyant density, $\rho_s/\rho_w - 1$
P: notional permeability (0.1 impermeable to 0.6 very permeable)
N: number of waves in the design storm, saturating near 7500
alpha: slope angle (rad)
xi_m: surf similarity parameter. Required: it depends on the wave period, which this function has no other way to know.
damage: damage level $S = A_e / D_{n50}^2$. 2 is start of damage.
safety: divides the stability coefficients, so >1 gives larger stone.

Returns:

float: nominal diameter Dn50 (m)

Note:

For a full design, including the crest level and overtopping checks, use ``pyCoastal.applications.structures``.

hunt_runup

```
def hunt_runup(beta: float, H: float, L: float) -> float
```

Hunt (1959) empirical run-up:
 $R \approx H * (\tan \beta) / \sqrt{H / L}$

stockdon_runup

```
def stockdon_runup(H: float, L: float, beta: float) -> float
```

Stockdon et al. (2006) 2%-exceedance run-up:
 $R2 = 1.1 * (0.35 H_{xi}) + 0.55 * (0.75 H_{xi})$
where $xi = \tan \beta / \sqrt{H/L}$

goda_wave_force

```
def goda_wave_force(H: float, T: float, h: float, beta: float, rho: float=1025) -> float
```

Estimate the Goda-Takahashi equivalent-static horizontal wave force per unit width (kN/m) acting on a vertical breakwater face using Goda's method.

Args:

H (float): Significant wave height at wall (m)
T (float): Wave period (s)
h (float): Water depth at face (m)
beta (float): Wave incidence angle (radians)
rho (float): Water density (kg/m³) [default: 1025 for seawater]

Returns:

float: Estimated wave force per unit width (N/m)

iribarren_stability

```
def iribarren_stability(H: float, alpha: float, rho_s: float, rho_w: float, mu: float, N: float) -> float
```

Stone weight W needed for armor stability per Iribarren:
 $W \geq [N \rho_s g H^3] / [\Delta^3 (\mu \cos \alpha - \sin \alpha)^3]$
where $\Delta = (\rho_s / \rho_w - 1)$, N is safety coefficient,
 μ is friction coefficient, α is slope angle (rad).

pyCoastal.tools.shoreline

OneLineParams

```
class OneLineParams
```

Parameters for the one-line shoreline change model:

D active profile depth [m]
p porosity [-]
Kcerc alongshore transport coefficient [SI units so that Q is m³/s]
Hfree offshore or structure-free breaking height H_{b,free}(x) [m] or scalar
alpha0 incident breaking angle relative to local shoreline, before feedback [rad] or array over x
Kt transmission factor field in [0,1] or scalar
beta0 incident wave direction azimuth [rad], measured from +x axis pointing alongshore
morfac morphological acceleration factor for faster adjustment

make_grid

```
def make_grid(Lx: float=1000.0, dx: float=5.0) -> tuple[np.ndarray, np.ndarray]
```

Alongshore grid x and initial shoreline position y(x,0).
y is the cross-shore offset at MSL. y0=0 means straight shoreline.

breakwaters_geometry

```
def breakwaters_geometry(x: np.ndarray, tips: list[float] | tuple[float, ...], y_tip: float=150.0, crest_freeboard: float=2.0) -> dict
```

Geometry for N detached rubble-mound breakwaters.
tips iterable of alongshore tip positions [m]
y_tip offshore distance of the tips from initial shoreline [m]
Returns dictionary with arrays over x.

kd_diffraction_field

```
def kd_diffraction_field(x: np.ndarray, geom: dict, beta0: float) -> np.ndarray
```

Simple linear-diffraction shadow factor $K_d(x)$ in $[0,1]$.
Approximates each breakwater tip as a semi-infinite barrier.
 $K_d = 0.5 \cdot (1 + \cos(\theta))$ clamped, where θ is the diffraction angle between incident direction and the ray from the tip to the shoreline point $(x,0)$.
Combined shadow from both tips taken as min over the two tips.

field_or_scalar_to_array

```
def field_or_scalar_to_array(val: float | ArrayLike, x: np.ndarray) -> np.ndarray
```

Broadcast scalar to 1D array if needed, otherwise validate shape.

compute_flux_Qls

```
def compute_flux_Qls(x: np.ndarray, y: np.ndarray, pars: OneLineParams, kd: np.ndarray, Hfree: np.ndarray, Kt: np.ndarray, alpha0: np.ndarray) -> tuple[np.ndarray, np.ndarray, np.ndarray]
```

$Q_{ls} = K_{cerc} \cdot H_b^{5/2} \cdot \sin(2 \cdot \alpha_b)$
 $\alpha_b = \alpha_0 - y_x$ (small angle approximation, radians)
 $H_b = kd \cdot K_t \cdot H_{free}$

rhs_y_t

```
def rhs_y_t(x: np.ndarray, y: np.ndarray, pars: OneLineParams, kd: np.ndarray, Hfree: np.ndarray, Kt: np.ndarray, alpha0: np.ndarray) -> tuple[np.ndarray, np.ndarray, np.ndarray, np.ndarray]
```

$y_t = -1 / ((1-p) D) \cdot dQ_{ls}/dx$

suggest_dt

```
def suggest_dt(x: np.ndarray, pars: OneLineParams, Hb: np.ndarray) -> float
```

Stability guidance for explicit scheme from linearized diffusion term:
 $G = 2 K_{cerc} H_b^{5/2} / ((1-p) D)$
 $dt \leq 0.45 \cdot dx^2 / \max(G)$

apply_bcs

```
def apply_bcs(y: np.ndarray, bc: str='fixed_ends', yL: float=0.0, yR: float=0.0) -> np.ndarray
```

Boundary conditions on y .
fixed_ends: $y(0)=y_L$, $y(L)=y_R$
zero_slope: $y_x=0$ at ends

run_one_line_model

```
def run_one_line_model() -> tuple[np.ndarray, np.ndarray, np.ndarray, dict, OneLineParams, dict]
```

Convenience runner with default 2-breakwater setup (backward-compatible).
Returns x , y_{final} , kd , $hist$, $pars$, $geom$.

pyCoastal.tools.morphodynamics

circle_through_points

```
def circle_through_points(p1, p2, p3, n=200)
```

Build a circular arc passing through three non-collinear points (x, y) .
Returns x , y arrays following the branch that goes through p_2 .

two_point_arc

```
def two_point_arc(p_start, p_end, sag, n=200)
```

Arc through two points with a prescribed sag (positive = bulge up, negative = bulge down).

build_tombolo_arcs

```
def build_tombolo_arcs(Lshore: float, Gb: float, Lb1: float, Lb2: float, yb: float, x_b1: float | None=None, x_b2: float | None=None, n_pts: int=200, sag_side_factor: float=0.3) -> tuple[np.ndarray, np.ndarray, dict]
```

Build a schematic shoreline (Hsu/Evans-like) using three arcs: left, central, right.
Ye = 1.204*Yi - 0.07*Gb is the retreat depth below the breakwater crest line.

accretion_metrics

```
def accretion_metrics(x_eq: np.ndarray, y_eq: np.ndarray, geom: dict) -> tuple[float, float]
```

Compute retreat (Yi-Ye) and accretion percentage including rectangles under T1-T2 and T3-T4.

bruuns_rule

```
def bruuns_rule(S: float, beta: float, L: float=None, h: float=None, B: float=None) -> float
```

Estimate shoreline retreat (R) using Bruun's rule.
If L, h, and B are provided: $R = S * L / (h + B)$
Else: $R = S / \tan(\beta)$

exner_change

```
def exner_change(qs_dx: float, porosity: float=0.64) -> float
```

Compute bed elevation change rate ($\partial\eta/\partial t$) from sediment divergence based on the Exner equation.
 $\partial\eta/\partial t = -1/(1-n) * q_{s_dx}$

linear_slope

```
def linear_slope(grid, north_level: float, south_level: float) -> np.ndarray
```

Create a plane beach profile that is south_level at y=0 and north_level at y=L_y.

pyCoastal.applications

```
pyCoastal.applications
```

Scenario-level simulators assembled from the numerics, physics and tools layers. Each application takes an engineering design, runs it forward, and reports the quantities a coastal engineer actually decides on: how long a beach fill survives, how much wave energy reaches a berth, and so on.

pyCoastal.applications.extremes

Extreme value analysis: the design wave and the design water level.

Every structure in this package is sized against a design condition, and that condition has to come from somewhere. This module is where it comes from: a measured or hindcast record in, a return value with an honest confidence band out.

Two routes, both standard.

Peaks over threshold

Keep the independent storm peaks above a threshold, fit a generalised Pareto distribution to the excesses, and combine it with the rate at which peaks arrive. Uses the record efficiently, but the answer depends

on the threshold, so choose it with the diagnostics here rather than by eye.

Block maxima

Keep the largest value in each year and fit a generalised extreme value distribution. Wastes data but is hard to get wrong, and is the right check on a peaks-over-threshold answer.

Both are fitted by L-moments, which are close to unbiased for the sample sizes a coastal record actually offers, are not troubled by the flat likelihood surfaces that defeat maximum likelihood on short records, and need no optimiser.

Sources

Hosking, J. R. M. and Wallis, J. R. (1997), Regional Frequency Analysis: An Approach Based on L-Moments. Cambridge University Press.

Coles, S. (2001), An Introduction to Statistical Modeling of Extreme Values. Springer. Threshold selection, return level interpretation.

Goda, Y. (2010), Random Seas and Design of Maritime Structures, ch. 11. Extreme wave statistics in coastal engineering practice.

Conventions

Shape parameters use the extreme value convention, so a positive shape is a heavy tail with no upper bound and a negative shape has a finite upper limit. Hosking writes $k = -\text{shape}$; the two are converted where his estimators are used, and never mixed in the public interface.

l_moments

```
def l_moments(sample, count: int=4) -> np.ndarray
```

Sample L-moments, unbiased, from probability weighted moments.

Parameters

sample : array_like

One-dimensional sample. Order does not matter; it is sorted here.

count : int

How many L-moments to return, from 1 to 4. The first is the mean, the second a measure of spread, the third and fourth the L-skewness and L-kurtosis numerators.

Returns

ndarray

``count`` L-moments.

Notes

These are the unbiased estimators, so they need at least ``count`` points. With fewer, the higher moments are not estimable at all and the function says so rather than returning a plausible-looking number.

decluster

```
def decluster(values, threshold: float, separation: int=1)
```

Independent peaks above a threshold.

Walks the record, and within each run of values above the threshold keeps only the largest. Two runs separated by fewer than ``separation`` samples below the threshold are treated as one storm, which is what stops a single event contributing several "independent" peaks.

Parameters

values : array_like

The record, evenly sampled.

threshold : float

Level above which a storm is counted.

separation : int

Number of consecutive samples below the threshold needed to end a storm. For a 3-hourly record, 24 samples is a 3-day separation, which is the usual choice for storm waves.

Returns

 (ndarray, ndarray)
 Peak values and their indices in the record.

peaks_over_threshold

```
def peaks_over_threshold(values, threshold: float, separation: int=1, samples_per_year: float=2920.0) -> dict
```

Declustered peaks and the rate at which they arrive.

``samples\_per\_year`` converts the record length into years, so the default is a 3-hourly record. The rate is what turns a distribution of excesses into a return period.

block_maxima

```
def block_maxima(values, block: int) -> np.ndarray
```

Largest value in each complete block of ``block`` samples.

A partial block at the end is dropped: an annual maximum taken from four months of record is not an annual maximum.

fit_gpd

```
def fit_gpd(excesses) -> tuple[float, float]
```

Fit a generalised Pareto distribution to threshold excesses.

Returns

 (scale, shape)
 With the survivor function
 $1 - F(x) = (1 + \text{shape } x / \text{scale}) ** (-1 / \text{shape})$, and the
 exponential limit as shape goes to zero.

Notes

 L-moment estimators, from the mean and L-CV of the excesses::

```
shape = 2 - 1 / tau
scale = mean * (1 - shape)
```

where tau = 12 / 11. A shape at or above 1 means the fitted mean does not exist, which on a wave record means the threshold is far too low or the sample far too small; the fit is returned anyway, flagged by the caller, rather than silently clipped.

fit_gev

```
def fit_gev(maxima) -> tuple[float, float, float]
```

Fit a generalised extreme value distribution to block maxima.

Returns

 (location, scale, shape)
 With $F(x) = \exp(-(1 + \text{shape } (x - \text{loc}) / \text{scale}) ** (-1 / \text{shape}))$.

Notes

 Hosking's L-moment estimators. His shape k is the negative of the extreme value shape, and his polynomial approximation for k from the L-skewness is accurate to better than 1e-4 for $-0.5 < k < 0.5$, which covers every coastal record in practice.

gpd_return_value

```
def gpd_return_value(threshold: float, scale: float, shape: float, rate: float, return_period) -> np.ndarray
```

Return value from a peaks-over-threshold fit.

`rate` is the mean number of peaks per year and `return_period` is in years, so `rate * return_period` is the expected number of peaks in that period:

```
x_T = u + scale / shape * ((rate T) ** shape - 1)
```

with the logarithmic form in the exponential limit. A return period shorter than one over the rate asks for a level exceeded more often than peaks occur, which is outside the model.

gev_return_value

```
def gev_return_value(location: float, scale: float, shape: float, return_period) -> np.ndarray
```

Return value from a block maxima fit, for blocks of one year.

```
x_T = loc + scale / shape * (1 - (-ln(1 - 1/T)) ** shape)
```

with the Gumbel form in the limit. If the blocks are not years, the return period is in blocks.

ExtremeFit

```
class ExtremeFit
    kind: str
    parameters: dict
    data: np.ndarray
    rate: float
    years: float
    samples: np.ndarray | None = None
    positions: np.ndarray | None = None
    warnings: list[str] = field(default_factory=list)
```

A fitted extreme value model with a confidence band.

Attributes

kind : str

"pot" or "gev".

parameters : dict

The fitted parameters.

data : ndarray

The peaks or maxima the fit used.

positions : ndarray or None

Index of each peak in the original record, for a peaks-over-threshold fit. None for block maxima.

rate : float

Peaks per year. One, by definition, for annual maxima.

years : float

Length of the record, for judging how far the fit can be pushed.

samples : ndarray

Bootstrap parameter sets, one row each, used for the band.

warnings : list of str

Anything about the fit the engineer should see before using it.

method

```
ExtremeFit.return_value(self, return_period)
```

Return value at one or many return periods [same units as the data].

method

```
ExtremeFit.confidence(self, return_period, level: float=0.9) -> dict
```

Bootstrap confidence band on the return value.

Percentile bootstrap over the fitted peaks. It captures sampling error in the parameters and nothing else: not the measurement error in the record, not the choice of threshold, and not whether the climate that produced the record is the climate the structure will see. Those are usually the larger uncertainties.

method

```
ExtremeFit.extrapolation_note(self, return_period: float) -> str
```

How far beyond the record a return period reaches.

method

```
ExtremeFit.summary(self, return_periods=(10, 50, 100)) -> str
```

A short report of the fit and its return values.

fit_pot

```
def fit_pot(values, threshold: float, separation: int=1, samples_per_year: float=2920.0, bootstrap: int=500, seed: int=0) -> ExtremeFit
```

Peaks-over-threshold fit to a record, with a bootstrap band.

Parameters

values : array_like

The record, evenly sampled.

threshold : float

Storm threshold, in the units of the record.

separation : int

Samples below the threshold that end a storm.

samples_per_year : float

Default is a 3-hourly record: 2920 samples a year.

bootstrap : int

Bootstrap resamples. Zero skips the band.

Notes

A useful rule is to want at least one or two peaks a year and no fewer than about thirty in total; the fit warns when either is missed.

fit_block_maxima

```
def fit_block_maxima(values, block: int, samples_per_year: float=2920.0, bootstrap: int=500, seed: int=0) -> ExtremeFit
```

Block maxima fit, with a bootstrap band.

``block`` is in samples. For annual maxima from a 3-hourly record that is 2920.

mean_residual_life

```
def mean_residual_life(values, thresholds) -> dict
```

Mean excess above each threshold, with a standard error.

Above a threshold where the generalised Pareto model holds, the mean excess is linear in the threshold. The lowest threshold from which the plot is straight is the one to use: lower wastes the model's validity, higher throws away data.

threshold_stability

```
def threshold_stability(values, thresholds, separation: int=1, samples_per_year: float=2920.0) -> dict
```

Fitted shape and modified scale against threshold.

Both should be flat above a threshold where the model holds. The scale is reported in its modified form, ``scale - shape \* threshold``, which is the quantity that is threshold-invariant; the raw scale is not, and reading it as if it were is a common way to pick a threshold badly.

plotting_positions

```
def plotting_positions(sample, rate: float=1.0, a: float=0.44) -> dict
```

Empirical return periods for a sample, for the return level plot.

Uses the Gringorten plotting position, $((i - a) / (n + 1 - 2a))$ with $a = 0.44$, which is close to unbiased for the Gumbel and GEV families that extreme wave records follow. `rate` converts an exceedance probability per peak into years.

pyCoastal.applications.sediment

Sediment and soil: what the bed is made of, and what that costs you.

Every structure in this package sits on, in, or behind something. Until now that something was a number typed into a signature. This module makes it a material with properties that propagate: the grain size that decides whether the bed moves at all, the friction angle that decides how hard the backfill pushes on the wall, the unit weight that decides how much of that push is soil and how much is water.

Two families of relation live here, and they are not the same subject.

Mobility

Whether, and how fast, grains move. Grain size, fall velocity, and the Shields threshold. This is what gates scour: a bed that never reaches its threshold does not scour, whatever the wave height.

Earth pressure

What retained soil does to a wall. Rankine and Coulomb coefficients, and the pressure diagram integrated over the retained height with the water table and any surcharge in the right places.

Sources

Soulsby, R. L. (1997), Dynamics of Marine Sands. Dimensionless grain size, fall velocity, and the threshold of motion.

Soulsby, R. L. and Whitehouse, R. J. S. (1997), "Threshold of sediment motion in coastal environments". The critical Shields curve fitted here.

Rankine, W. J. M. (1857) and Coulomb, C. A. (1776), through any soil mechanics text. Lateral earth pressure coefficients.

CIRIA/CUR/CETMEF (2007), The Rock Manual. Typical properties for the granular materials in the catalogue.

Conventions

Grain sizes are metres, not millimetres, to stay consistent with the rest of the package. Unit weights are kN/m³ and forces kN per metre run. Angles are degrees on the way in, because that is how a geotechnical report states them, and radians internally.

G

`G = 9.81`

RHO_W

`RHO_W = 1025.0`

GAMMA_W

`GAMMA_W = RHO_W * G / 1000.0`

NU

`NU = 1.19e-06`

Sediment

```
class Sediment
    name: str
    d50: float
    specific_gravity: float = 2.65
    porosity: float = 0.4
```



```
friction_angle: float = 32.0
cohesion: float = 0.0
phi_sorting: float = 0.6
d90: float | None = None
description: str = ''
```

A bed or backfill material.

Attributes

```
-----
name : str
    For the report and the drawing.
d50 : float
    Median grain size [m]. Zero for a cohesive material, where grain
    size is not what governs.
specific_gravity : float
    Grain density over water density. 2.65 for quartz sand.
porosity : float
    Voids as a fraction of total volume, in place.
friction_angle : float
    Effective angle of shearing resistance phi' [deg]. This is the one
    number that decides how hard a backfill pushes on a wall, and it is
    the one most worth getting from a real site investigation.
cohesion : float
    Effective cohesion c' [kPa]. Zero for anything granular.
phi_sorting : float
    Standard deviation of the grading on the phi scale, sigma_phi.
    Below about 0.5 is well sorted, above 1.0 poorly sorted. It is what
    decides how much of a borrow source winnows away rather than staying
    on the beach.
d90 : float, optional
    Ninety per cent passing size [m]. Defaults to 2.5 d50, a reasonable
    ratio for a moderately graded marine sand.
description : str
    What a person would call it.
```

property

```
Sediment.relative_density(self) -> float
```

$s - 1$, the buoyant density ratio that drives every mobility number.

property

```
Sediment.dry_unit_weight(self) -> float
```

Unit weight of the drained material [kN/m³].

property

```
Sediment.saturated_unit_weight(self) -> float
```

Unit weight with the voids full of water [kN/m³].

property

```
Sediment.submerged_unit_weight(self) -> float
```

Buoyant unit weight below the water table [kN/m³].

This is the number that makes a submerged backfill push so much less hard than a drained one: roughly six kN/m³ rather than eighteen. The water it displaces pushes separately, and harder.

property

```
Sediment.cohesive(self) -> bool
```

method

```
Sediment.grading(self) -> float
```

d90 / d50, the spread of the grading.

method

```
Sediment.summary(self) -> str
```

SEDIMENTS

```
SEDIMENTS = {'soft_clay': Sediment('Soft clay', d50=0.0, specific_gravity=2.7, porosity=0.55, friction_angle=22.0, cohesion=15.0, phi_sorting=2.0, description='normally consolidated, undrained strength governs'), 'stiff_clay': Sediment('Stiff clay', d50=0.0, specific_gravity=2.72, porosity=0.42, friction_angle=26.0, cohesion=40.0, phi_sorting=2.0, description='overconsolidated'), 'silt': Sediment('Silt', d50=3e-05, specific_gravity=2.65, porosity=0.48, friction_angle=28.0, phi_sorting=1.6, description='mobile at almost any wave'), 'very_fine_sand': Sediment('Very fine sand', d50=9e-05, porosity=0.45, friction_angle=29.0, phi_sorting=0.55, description='suspends readily, high siltation'), 'fine_sand': Sediment('Fine sand', d50=0.00019, porosity=0.43, friction_angle=31.0, phi_sorting=0.45, description='the usual beach and nearshore sand'), 'medium_sand': Sediment('Medium sand', d50=0.00038, porosity=0.4, friction_angle=33.0, phi_sorting=0.55, description='typical dredged fill'), 'coarse_sand': Sediment('Coarse sand', d50=0.00075, porosity=0.38, friction_angle=35.0, phi_sorting=0.7, description='good drained backfill'), 'fine_gravel': Sediment('Fine gravel', d50=0.006, porosity=0.35, friction_angle=38.0, phi_sorting=0.95, description='free draining'), 'coarse_gravel': Sediment('Coarse gravel', d50=0.03, porosity=0.35, friction_angle=40.0, phi_sorting=1.1, description='shingle beach'), 'rock_fill': Sediment('Quarry rock fill', d50=0.15, porosity=0.37, friction_angle=42.0, phi_sorting=1.4, des ...
```

sediment

```
def sediment(key: str) -> Sediment
```

Look a material up by key, with the options named on failure.

dimensionless_grain_size

```
def dimensionless_grain_size(material, viscosity: float=NU) -> float
```

$D^* = d50 [g (s - 1) / \nu^2]^{1/3}$, Soulsby (1997).

The single parameter that collapses the threshold of motion and the fall velocity across grain sizes. Below about 4 the grain is in the viscous range; above about 100 it is fully rough.

critical_shields

```
def critical_shields(material, viscosity: float=NU) -> float
```

Critical Shields parameter, Soulsby and Whitehouse (1997)::

$$\theta_{cr} = 0.30 / (1 + 1.2 D^*) + 0.055 [1 - \exp(-0.020 D^*)]$$

A fitted curve through the Shields data that behaves at both ends, unlike reading a value off the original diagram.

fall_velocity

```
def fall_velocity(material, viscosity: float=NU) -> float
```

Settling velocity of a single grain [m/s], Soulsby (1997)::

$$w_s = \nu / d [\sqrt{10.36^2 + 1.049 D^{*3}} - 10.36]$$

Valid across the whole range from silt to gravel, which is why it is preferred to Stokes at one end and a drag law at the other.

wave_orbital_velocity

```
def wave_orbital_velocity(Hs: float, T: float, depth: float, wavelength: float | None=None) -> float
```

Near-bed orbital velocity amplitude under a linear wave [m/s].

$$U_w = \pi H_s / (T \sinh(k h))$$

wave_shields

```
def wave_shields(material, Hs: float, T: float, depth: float, wavelength: float | None=None) -> dict
```

Shields parameter under waves, and whether the bed is moving.

Uses the Swart wave friction factor for a rough turbulent bed, with the orbital excursion over the Nikuradse roughness of 2.5 d50::

```
f_w = exp[5.213 (A / k_s)^-0.194 - 5.977], capped at 0.3
```

Returns

dict

The orbital velocity and excursion, the friction factor, the Shields parameter, its critical value, and ``mobile``.

bed_mobility

```
def bed_mobility(material, Hs: float, T: float, depth: float) -> dict
```

Whether the bed at a structure is live, and by how much.

Scour relations are almost all live-bed results. Applying one to a bed that never reaches its threshold predicts a hole that will not form. This is the gate the scour functions use.

earth_pressure_coefficient

```
def earth_pressure_coefficient(friction_angle: float, kind: str='active', wall_friction: float=0.0, backslope: float=0.0) -> float
```

Lateral earth pressure coefficient.

Parameters

friction_angle : float

Effective angle of shearing resistance phi' [deg].

kind : str

"active" for a wall free to move away from the soil, "at_rest" for one that cannot, "passive" for one pushed into it.

wall_friction : float

Angle of friction on the wall face delta [deg]. Non-zero switches the active case from Rankine to Coulomb, which is the honest choice for a rough concrete face and gives a smaller force.

backslope : float

Slope of the retained surface [deg], for the Rankine sloping-backfill form.

Notes

Which coefficient applies is a question about movement, not about soil. A gravity seawall on a rubble bedding can move the millimetre or two that mobilises the active state; one cast against rock, or restrained by a slab, cannot, and then at-rest governs and the force is roughly half as large again. Choosing "active" because it is smaller is the most common way a retaining structure is under-designed.

lateral_earth_pressure

```
def lateral_earth_pressure(material, height: float, water_table: float=0.0, surcharge: float=0.0, kind: str='active', wall_friction: float=0.0, points: int=201) -> dict
```

Pressure diagram on the back of a wall, effective stress and water apart.

Parameters

height : float

Retained height H [m], measured from the top of the retained surface down to the base of the wall.

water_table : float

Depth of the water table below the retained surface [m]. Zero means the backfill is saturated to the top, which is the condition behind a seawall with a blocked drain and is the one that breaks walls.

surcharge : float

Uniform surcharge on the retained surface [kPa].

```
kind, wall_friction
    Passed to :func:`earth_pressure_coefficient`.
```

Returns

```
-----
dict
    ``depth`` and the three pressure arrays (``effective``, ``pore``,
    ``total``) in kPa, plus the coefficient used.
```

Notes

```
-----
```

Below the water table the soil pushes with its **submerged** unit weight and the water pushes separately with its full hydrostatic gradient. The sum is larger than the dry soil alone would give, because water has no friction angle to lean on: it presses with $K = 1$. This is why drainage behind a seawall is a structural matter and not a detail.

lateral_earth_force

```
def lateral_earth_force(material, height: float, water_table: float=0.0, surcharge: float=0.0, kind: str='active', wall_friction: float=0.0, points: int=401) -> dict
```

Total horizontal force from retained soil and water [kN/m].

Returns

```
-----
dict
    ``soil`` and ``water`` forces and the ``total``, the lever ``arm``
    of the total above the base of the wall, and the pressure diagram
    that produced them.
```

Notes

```
-----
```

The split matters. The soil force scales with the friction angle and can be argued down with a better backfill; the water force cannot be argued with at all, only drained away.

pyCoastal.applications.structures

Coastal structure design: armour sizing and wave overtopping.

Aimed at design work rather than illustration, so every relation is named, its source given, and its validity range stated. Where a formula has a documented range of application the functions say so in their docstrings; they do not silently extrapolate on your behalf, but neither do they refuse to compute, since engineering judgement outside a range is your call.

Sources

```
-----
```

Van der Meer (1988), "Rock slopes and gravel beaches under wave attack", Delft Hydraulics Publication 396. Rock armour stability.

CIRIA/CUR/CETMEF (2007), The Rock Manual, 2nd ed. Armour layer geometry.

Shore Protection Manual (1984), US Army CERC. Hudson formula.

EurOtop (2018), Manual on wave overtopping of sea defences and related structures, 2nd ed. Overtopping discharge and tolerable limits.

Conventions

```
-----
```

Wave heights are spectral significant heights H_{m0} at the toe. Slopes are given as $\cot(\alpha)$, the horizontal run per unit rise, because that is how they appear on drawings. Discharges are litres per second per metre of structure, the unit used in the tolerability tables.

G

$G = 9.81$

ROUGHNESS_FACTORS

```
ROUGHNESS_FACTORS = {'smooth_concrete': 1.0, 'grass': 1.0, 'asphalt': 1.0, 'rock_one_layer_impermeable': 0.6,
'rock_two_layer_impermeable': 0.55, 'rock_two_layer_permeable': 0.4, 'cubes_one_layer_flat': 0.49, 'cubes_two_layer_random': 0.47,
```

```
'antifer': 0.47, 'tetrapod': 0.38, 'accropode': 0.46, 'core_loc': 0.44, 'xbloc': 0.45, 'dolos': 0.43}
```

TOLERABLE_DISCHARGE

```
TOLERABLE_DISCHARGE = {'pedestrians_unaware': (0.03, 'Unaware pedestrians, narrow walkway, clear view of the sea'),  
'pedestrians_aware': (0.1, 'Aware pedestrians, able to see and avoid the hazard'), 'trained_staff': (1.0, 'Trained staff, well shod  
and protected, wide walkway'), 'buildings_structural': (1.0, 'Structural damage to buildings behind the defence'),  
'vehicles_low_speed': (10.0, 'Vehicles at low speed on a road behind the crest'), 'vehicles_moderate_speed': (50.0, 'Driving at  
moderate speed, overtopping by pulsating flows'), 'embankment_seaward': (50.0, 'Damage to a maintained grass or armoured  
embankment'), 'harbour_quay_equipment': (0.4, 'Damage to equipment set back 5 to 10 m from the crest')}
```

DesignConditions

```
class DesignConditions  
    Hm0: float  
    Tm10: float  
    depth: float = 15.0  
    storm_duration: float = 6 * 3600.0
```

Wave and water-level conditions at the toe of a structure.

Attributes

```
Hm0 : float  
    Spectral significant wave height at the toe [m].  
Tm10 : float  
    Spectral period  $T_{m-1,0} = m_{-1}/m_0$  [s]. EurOtop is written around this  
    period. If you only have  $T_p$ , ``from_peak_period`` converts with the  
    standard  $T_{m-1,0} = T_p / 1.1$  for a single-peaked spectrum.  
depth : float  
    Water depth at the toe [m].  
storm_duration : float  
    Duration of the design storm [s]. Used to count waves for armour  
    damage progression.
```

classmethod

```
DesignConditions.from_peak_period(cls, Hm0: float, Tp: float, **kwargs) -> 'DesignConditions'
```

Build from a peak period, using $T_{m-1,0} = T_p / 1.1$.

property

```
DesignConditions.wave_count(self) -> float
```

Number of waves in the design storm, $N = \text{duration} / T_m$.

Van der Meer's damage relation saturates near $N = 7500$, so the count is capped there; a longer storm does not keep eroding the slope at the same rate.

method

```
DesignConditions.deep_water_wavelength(self) -> float
```

$L_0 = g T^2 / (2 \pi)$, using $T_{m-1,0}$ [m].

method

```
DesignConditions.wavelength(self) -> float
```

Local wavelength at the toe from the dispersion relation [m].

method

```
DesignConditions.breaker_parameter(self, cot_alpha: float) -> float
```

Surf similarity parameter $\xi_{m-1,0} = \tan(\alpha) / \sqrt{H_{m0} / L_0}$.

DAMAGE_LEVELS

```
DAMAGE_LEVELS = {'start_of_damage': 2.0, 'intermediate': 5.0, 'failure_1_in_1.5': 8.0, 'failure_1_in_2': 8.0, 'failure_1_in_3': 12.0, 'failure_1_in_4': 17.0}
```

rock_armour_vandermeer

```
def rock_armour_vandermeer(conditions: DesignConditions, cot_alpha: float, Delta: float=1.585, permeability: float=0.4, damage: float=2.0, safety_factor: float=1.0) -> dict
```

Nominal rock diameter Dn50 from Van der Meer (1988).

Two regimes, selected by the surf similarity parameter against a critical value ``xi\_cr``:

plunging ($\xi < \xi_{cr}$)
 $H_s / (\Delta D_{n50}) = 6.2 P^{0.18} (S / \sqrt{N})^{0.2} \xi^{-0.5}$
surging ($\xi \geq \xi_{cr}$)
 $H_s / (\Delta D_{n50}) = 1.0 P^{-0.13} (S / \sqrt{N})^{0.2} \sqrt{\cot a} \xi^P$

with $\xi_{cr} = (6.2 P^{0.31} \sqrt{\tan a})^{1 / (P + 0.5)}$.

Parameters

cot_alpha : float
Slope, horizontal per vertical. Valid range roughly 1.5 to 6.
Delta : float
Relative buoyant density ($\rho_s / \rho_w - 1$). 1.585 corresponds to 2650 kg/m3 rock in seawater of 1025 kg/m3.
permeability : float
Notional permeability P (Van der Meer 1988, Figure 8):
0.1 impermeable core with a filter layer, 0.4 permeable core,
0.5 homogeneous structure, 0.6 very permeable core.
damage : float
Damage level S. See ``DAMAGE\_LEVELS``. 2 is start of damage and is the usual design condition.
safety_factor : float
Divides the stability coefficients, so values above 1 give larger stone. Apply your own code's partial factors here.

Returns

dict
``Dn50`` [m], ``M50`` [kg] for 2650 kg/m3 rock, the regime that governed, ``xi``, and ``xi\_cr``.

Notes

Valid for deep-water conditions at the toe and non-depth-limited waves. For depth-limited surf, Van der Meer's shallow-water modification or the Rock Manual's H2% form should be used instead; this function does not apply it.

rock_armour_hudson

```
def rock_armour_hudson(conditions: DesignConditions, cot_alpha: float, Delta: float=1.585, Kd: float=4.0) -> dict
```

Nominal rock diameter from the Hudson formula (SPM 1984).

$H_s / (\Delta D_{n50}) = (K_d \cot a)^{1/3} / 1.27$, where the 1.27 converts the SPM's H1/10 basis to a significant wave height.

``Kd`` is the stability coefficient: 4.0 for rough angular rock in two layers on a trunk with breaking waves, 2.0 for a head. Hudson carries no dependence on wave period, storm duration, permeability or damage level, so it is a first estimate. Prefer ``rock\_armour\_vandermeer`` for design and use Hudson as a cross-check.

armour_layer

```
def armour_layer(Dn50: float, n_layers: int=2, layer_coefficient: float=1.0, porosity: float=0.37) -> dict
```

Armour layer thickness and stone count (Rock Manual, 2007).

thickness = $n * k_{\Delta} * D_{n50}$, and the number of stones per unit area is $n * k_{\Delta} * (1 - \text{porosity}) / D_{n50}^2$.

overtopping_sloped

```
def overtopping_sloped(conditions: DesignConditions, crest_freeboard: float, cot_alpha: float, gamma_f: float=1.0, gamma_beta: float=1.0, gamma_b: float=1.0, gamma_v: float=1.0) -> dict
```

Mean overtopping discharge for a sloping structure (EurOtop 2018).

Takes the lesser of the breaking-wave and non-breaking expressions,

breaking (eq. 5.10)
$$q / \sqrt{g H m_0^3} = (0.023 / \sqrt{\tan \alpha}) \gamma_b \xi \exp[-(2.7 R_c / (\xi H m_0 \gamma_b \gamma_f \gamma_{\beta} \gamma_v))^1.3]$$

maximum (eq. 5.11)
$$q / \sqrt{g H m_0^3} = 0.09 \exp[-(1.5 R_c / (H m_0 \gamma_f \gamma_{\beta}))^1.3]$$

Parameters

crest_freeboard : float
 R_c , crest height above the still water level [m]. Zero or negative freeboard is outside these formulae and raises.
gamma_f : float
 Roughness factor, see ``ROUGHNESS\_FACTORS``.
gamma_beta : float
 Oblique wave attack factor. 1.0 for normal incidence; use ``obliquity\_factor``.
gamma_b : float
 Berm factor, 1.0 for no berm.
gamma_v : float
 Wave wall factor, 1.0 for no crest wall.

Returns

dict
 ``q`` in l/s per m, the dimensionless discharge, and which branch governed.

Notes

These are mean values. EurOtop reports roughly a factor-of-three scatter about the mean discharge, so a design check should consider the upper confidence band, not the mean alone. ``overtopping\_with\_uncertainty`` returns that band.

overtopping_vertical

```
def overtopping_vertical(conditions: DesignConditions, crest_freeboard: float, gamma_beta: float=1.0) -> dict
```

Mean overtopping for a plain vertical wall, non-impulsive conditions.

EurOtop (2018) eq. 7.1:

$$q / \sqrt{g H m_0^3} = 0.047 \exp[-(2.35 R_c / (H m_0 \gamma_{\beta}))^1.3]$$

Notes

Valid for non-impulsive (pulsating) conditions, roughly when the impulsiveness parameter $h^* = 1.35 (h / H m_0) (2 \pi h / (g T m_0^2))$ exceeds about 0.23. Impulsive conditions give far larger discharges and need the separate impulsive formulae, which are not implemented here. The returned dict reports ``h\_star`` and ``impulsive`` so the caller can tell.

obliquity_factor

```
def obliquity_factor(beta_degrees: float, kind: str='overtopping') -> float
```

Oblique wave attack factor gamma_beta (EurOtop 2018, section 5.4.4).

For overtopping of a sloping structure,
 $\gamma_{\beta} = 1 - 0.0063 |\beta|$ for $|\beta| \leq 80$ degrees, and the value at 80 degrees beyond that.

overtopping_with_uncertainty

```
def overtopping_with_uncertainty(result: dict, factor: float=3.0) -> dict
```

Bracket a mean discharge with EurOtop's reported scatter.

EurOtop notes roughly a factor of three about the mean for the sloping formulae. Design against the upper bound, not the mean.

required_crest_freeboard

```
def required_crest_freeboard(conditions: DesignConditions, q_allowable: float, cot_alpha: float, gamma_f: float=1.0, gamma_beta: float=1.0, gamma_b: float=1.0, gamma_v: float=1.0, vertical: bool=False, tolerance: float=0.0001) -> float
```

Crest freeboard R_c that limits mean overtopping to `q_allowable`.

Inverts the relevant EurOtop expression by bisection, since the two branches make an analytic inverse awkward. `q_allowable` is in l/s per m.

assess_overtopping

```
def assess_overtopping(q: float) -> dict
```

Which uses a given mean discharge is tolerable for.

`q` is in l/s per m. Returns each limit in `TOLERABLE_DISCHARGE` with whether it is satisfied, sorted from strictest to most permissive.

BreakwaterDesign

```
class BreakwaterDesign
    conditions: DesignConditions
    cot_alpha: float
    Dn50: float
    M50: float
    regime: str
    crest_freeboard: float
    q_mean: float
    q_upper: float
    layer: dict
    armour_type: str
    governing_limit: str | None
    section: str = 'trunk'
    kd_ratio: float | None = None
```

A rubble-mound cross-section sized against a design condition.

method

```
BreakwaterDesign.summary(self) -> str
```

A short design report.

design_rubble_mound

```
def design_rubble_mound(conditions: DesignConditions, cot_alpha: float=2.0, armour: str='rock_two_layer_permeable', damage: float=2.0, permeability: float=0.4, Delta: float=1.585, tolerable_use: str='trained_staff', safety_factor: float=1.0, scatter_factor: float=3.0) -> BreakwaterDesign
```

Size armour and crest level for a rubble-mound section.

Sizes the stone with Van der Meer, then sets the crest so that the *upper* confidence bound on overtopping, not the mean, meets the limit for `tolerable_use`. Designing to the mean would be exceeded about half the time.

reflection_coefficient

```
def reflection_coefficient(cot_alpha: float, surf_similarity: float, permeable: bool=True) -> float
```

Reflection coefficient of a rough slope, Seelig and Ahrens (1981).

$$K_r = a \cdot x_i^2 / (b + x_i^2)$$

with $a = 0.6$ and $b = 6.6$ for a permeable rubble mound, and $a = 1.0$, $b = 5.5$ for a smooth impermeable slope. A vertical wall reflects almost everything; a rubble mound dissipates most of it, which is why the scour in front of the two is not the same problem.

toe_scour

```
def toe_scour(bed, Hs: float, T: float, depth: float, reflection: float=1.0, coefficient: float=0.4, exponent: float=1.35) -> dict
```

Equilibrium scour depth at the toe of a marine structure [m].

The standing-wave form, as for a vertical wall::

$$S / H_s = \text{coefficient} * K_r / \sinh(k h) ** \text{exponent}$$

Parameters

bed : Sediment or str

What the bed is made of. This decides whether there is any scour to compute: a bed below its threshold of motion in the approach waves is in the clear-water regime, where a live-bed relation overstates the hole, and a cohesive bed is not governed by this at all.

reflection : float

Reflection coefficient K_r of the structure. 1.0 for a vertical wall, which recovers Xie (1981) exactly; roughly 0.2 to 0.5 for a rubble mound, from :func:`reflection\_coefficient`.

coefficient : float

0.4 is Xie's value for fine sand under regular waves at a fully reflecting wall, and is the usual design number.

exponent : float

1.35, from the same work.

Returns

dict

The ``depth``, the mobility state of the bed, and whether the relation is being applied inside the regime it came from.

Notes

Scaling Xie's fully reflecting result by the reflection coefficient is an engineering assumption, not a calibrated relation: it has the right limits, going to Xie at a vertical wall and to nothing at a perfect absorber, and it puts a rubble mound sensibly below a caisson. It is a screening number. A scheme whose toe design turns on it wants a mobile bed model or a physical model, and the returned dict says as much through ``screening\_only``.

breakwater_toe_scour

```
def breakwater_toe_scour(design, depth: float, bed='medium_sand', permeable: bool=True) -> dict
```

Toe scour in front of a designed rubble mound.

Takes the reflection from the slope and the surf similarity of the design condition, so a flatter, rougher, more permeable mound is correctly predicted to scour its own toe less than a steep one.

mound_foundation

```
def mound_foundation(design, depth: float, bed='medium_sand', permeable: bool=True, bedding_material='coarse_sand', settlement_allowance: float=0.0) -> dict
```

Bedding blanket, toe berm and geotextile under a rubble mound.

A trapezoid of rock does not sit straight on the seabed. Under it goes a levelling blanket of graded sand and gravel, on a geotextile, carried well past both toes so the scour hole forms in the apron rather than under the structure. At the foot of the armour sits a berm of the same stone as the filter layer, which is what stops the bottom of the mantle unravelling.

Parameters

design : BreakwaterDesign

The sized mound.

depth : float

Water depth at the toe [m].

bed : Sediment or str
The natural seabed. Decides the scour, and therefore how far the blanket has to reach.

bedding_material : Sediment or str
The blanket itself, normally a well graded sand and gravel.

settlement_allowance : float
Extra blanket thickness for consolidation of a soft seabed [m].

Returns

dict
Every dimension the section needs, and the scour result behind it.

Notes

Two rules are doing the work.

The toe berm takes the same stone as the filter layer rather than the armour. It sits low, where the orbital velocities are much smaller than at the waterline, and sizing it as armour is expensive without being safer. This follows normal Italian and Rock Manual practice.

The blanket reaches past the toe by whichever is larger of the computed scour apron and twice the predicted scour depth, with a floor of three metres for something a dredger can actually place. Its job is to keep the edge of the hole away from the toe, so it has to be wider than the hole is deep.

crown_wall

```
def crown_wall(design, still_water_level: float, deck_width: float=7.5, parapet_width: float=2.0, parapet_height: float | None=None, base_below_crest: float | None=None) -> dict
```

A concrete crown block on the crest, in the usual stepped form.

A parapet on the seaward side to take the run-up, a deck behind it wide enough to drive a lorry along for maintenance, and a base bedded into the core below the armour crest.

Returns

dict
Levels and widths for the block, and its concrete volume per metre run at 2400 kg/m3.

Notes

The block is proportioned here, not designed. Sliding and overturning of a crown wall under wave impact are a separate calculation, and a real one also has to survive the uplift that gets under it when the core does not drain fast enough.

ROUNDHEAD_KD_RATIO

```
ROUNDHEAD_KD_RATIO = {'rock': {1.5: 0.95, 2.0: 0.8, 3.0: 0.65}, 'cubes': {1.5: 0.85, 2.0: 0.75, 3.0: 0.6}, 'tetrapod': {1.5: 0.72, 2.0: 0.64, 3.0: 0.5}, 'accropode': {1.5: 0.8, 2.0: 0.75, 3.0: 0.65}, 'dolos': {1.5: 0.65, 2.0: 0.58, 3.0: 0.45}}
```

ARMOUR_FAMILY

```
ARMOUR_FAMILY = {'rock_one_layer_impermeable': 'rock', 'rock_two_layer_impermeable': 'rock', 'rock_two_layer_permeable': 'rock', 'cubes_one_layer_flat': 'cubes', 'cubes_two_layer_random': 'cubes', 'antifer': 'cubes', 'tetrapod': 'tetrapod', 'accropode': 'accropode', 'core_loc': 'accropode', 'xbloc': 'accropode', 'dolos': 'dolos', 'smooth_concrete': 'rock', 'grass': 'rock', 'asphalt': 'rock'}
```

roundhead_kd_ratio

```
def roundhead_kd_ratio(armour: str, cot_alpha: float) -> float
```

KD_head / KD_trunk for an armour type on a given slope.

Linear in cot(alpha) between the tabulated slopes, and held flat outside them rather than extrapolated into values nobody has measured.

roundhead

```
def roundhead(design, kd_ratio: float | None=None, raise_crest: float=0.0) -> 'BreakwaterDesign'
```

The head section of a breakwater, armoured for its exposure.

A roundhead is attacked from a wider range of directions than the trunk, the armour on a convex surface gets less support from its neighbours, and the run-down concentrates where the flow turns the corner. Practice handles all three by using a lower stability coefficient at the head.

Since Hudson makes the nominal diameter go as KD to the power minus a third, a KD ratio of r gives::

```
Dn50_head = Dn50_trunk / r^(1/3)
M50_head = M50_trunk / r
```

so a ratio of 0.8 means a quarter more stone by mass, which is the step from sixteen to twenty tonne units that a real scheme ends up with.

Parameters

kd_ratio : float, optional

KD_head / KD_trunk. Taken from :func:`roundhead\_kd\_ratio` for the design's own armour and slope when not given.

raise_crest : float

Extra crest freeboard at the head [m]. Heads are often built higher than the trunk, because the overtopping there lands on the part of the structure people stand on and the navigation light sits on.

Returns

BreakwaterDesign

The same design with head armour, its layer recomputed, and ``section`` set to "head".

Notes

Only the armour is rescaled. The filter follows it, because the layer is recomputed from the new diameter, but the core grading, the crest width and the overtopping are left as the trunk's. A real head is also usually widened to give the plant somewhere to work and the light somewhere to stand.

pyCoastal.applications.seawall

Vertical seawall design: wave pressures, stability, scour and toe protection.

This is the design half of the seawall product. Give it a design condition and a set of choices, and it returns a fully dimensioned section: crest level set by overtopping, base width set by sliding and overturning under Goda pressures, toe level set by scour, and toe stone sized for stability. :mod:`pyCoastal.applications.sections` turns the result into a drawing and a bill of quantities.

Sources

Goda, Y. (1974, 2010), Random Seas and Design of Maritime Structures. Wave pressure distribution on a vertical wall, and the extension to impulsive conditions by Takahashi et al. (1994).

EurOtop (2018), Manual on wave overtopping of sea defences, 2nd ed. Crest level for a tolerable mean discharge.

Xie, S. L. (1981), Scouring patterns in front of vertical breakwaters, Delft University of Technology. Scour depth at a vertical wall.

Van der Meer, J. W. (1998), in Rock Manual (CIRIA/CUR/CETMEF, 2007). Toe berm stone stability.

Conventions

Levels are metres above chart datum and increase upward, matching the package's z convention. Distances are metres seaward-positive from the wall face unless stated. Forces are kN per metre run of wall.

G

G = 9.81

RHO_W

RHO_W = 1025.0

RHO_C

RHO_C = 2400.0

RHO_S

RHO_S = 2650.0

RHO_FILL

RHO_FILL = 1900.0

goda_pressures

```
def goda_pressures(Hm0: float, T: float, depth: float, wall_toe_depth: float, berm_depth: float | None=None, crest_freeboard: float=5.0, beta_degrees: float=0.0, slope: float=1 / 30, Hmax_factor: float=1.8, breaker_index: float=0.78) -> dict
```

Goda wave pressures on a vertical wall.

Parameters

Hm0 : float
Significant wave height at the structure [m].
T : float
Period used for the pressure distribution [s]. Goda's method is written around the significant period; use Tp or T_{1/3}, not T_{m-1,0}.
depth : float
Water depth in front of the structure, seaward of the mound [m].
wall_toe_depth : float
Depth from the still water level to the underside of the upright section, Goda's h' [m].
berm_depth : float, optional
Depth over the toe berm, Goda's d [m]. Defaults to ``wall\_toe\_depth``, that is, no berm in front of the wall.
crest_freeboard : float
Crest level above the still water level, Rc [m]. Caps the pressure distribution at the crest: a wall lower than the run-up wedge is not loaded above its own crest.
beta_degrees : float
Angle of wave attack from the wall normal [deg].
slope : float
Seabed slope in front of the structure, used for the depth at five wave heights seaward.
Hmax_factor : float
Hmax / Hm0 for the design wave. Goda uses 1.8 for non-breaking conditions.
breaker_index : float
Depth limit on the design wave, Hmax <= breaker_index * d. Without it, 1.8 Hm0 in shallow water asks the wall to survive a wave the site cannot deliver. Set it to zero to disable the limit.

Returns

dict
Pressures p1, p3, p4 and uplift pu [kPa], the elevation of the pressure distribution above the still water level ``eta\_star`` [m], the horizontal force ``F`` and its lever arm ``arm`` about the heel, and the uplift force ``U`` per metre of base width.

Notes

This is the standard (non-impulsive) Goda distribution. Takahashi's impulsive pressure coefficient alpha_I, which governs when a steep mound throws a breaking wave at the wall, is not applied: check the mound geometry independently before relying on these numbers for a wall on a high berm.

scour_depth_vertical_wall

```
def scour_depth_vertical_wall(Hm0: float, T: float, depth: float, coefficient: float=0.4) -> float
```

Equilibrium scour depth at the toe of a vertical wall [m].

Xie (1981), for fine sediment under a standing wave in front of a reflecting wall::

$$S / H = \text{coefficient} / \sinh(k h) ** 1.35$$

The scour hole sits a quarter wavelength from the wall, where the standing-wave node drives the largest near-bed velocity.

Parameters

coefficient : float
0.4 is Xie's value for regular waves on fine sand, and is the usual design value. Irregular waves smear the nodal structure and give a smaller, wider hole, so a lower coefficient is defensible; coarse sediment reduces it further. Set it deliberately rather than trusting the default.

Notes

This is the equilibrium depth after a long exposure, not the depth after one storm. It is a scour *allowance* for setting the founding level, and does not replace a check that the toe protection stays in place.

toe_stone_size

```
def toe_stone_size(Hm0: float, toe_depth: float, water_depth: float, Delta: float=1.585, damage: float=0.5) -> dict
```

Toe berm stone size from the Van der Meer toe formula.

$$H_s / (\Delta D_{n50}) = (2 + 6.2 (ht/h)^{2.7}) Nod^{0.15}$$

Parameters

toe_depth : float
Water depth over the toe berm, ht [m].
water_depth : float
Water depth at the structure, h [m].
damage : float
Damage number Nod, stones displaced out of the berm per Dn50 width.
0.5 is effectively no damage, 2 is acceptable damage, 4 is severe.

Notes

The formula is calibrated for $0.4 < ht/h < 0.9$. A toe set very deep or very shallow relative to the water depth falls outside it, and the returned dict says so rather than silently extrapolating.

sliding_safety

```
def sliding_safety(F: float, weight: float, uplift: float, friction: float=0.6) -> float
```

Factor of safety against sliding on the base.

FoS = $\text{friction} * (W - U) / F$, all forces per metre run. 0.6 is the usual design friction coefficient between concrete and a rubble bedding layer, and 1.2 the conventional requirement under the design wave. Returns 0 when the uplift exceeds the weight, which is failure by flotation and not a sliding problem at all.

overturning_safety

```
def overturning_safety(F: float, arm: float, restoring_moment: float, uplift: float, base_width: float) -> float
```

Factor of safety against overturning about the rear heel.

The wave pushes shoreward, so the wall tips about its landward heel. Goda uplift is triangular with its peak at the seaward edge, putting its centroid two thirds of the base width from the rear heel.

```

Parameters
-----
F, arm : float
    Horizontal wave force [kN/m] and its lever arm above the base [m].
restoring_moment : float
    Sum of  $W_i * (B - x_i)$  for every weight component, taken about the
    rear heel [kNm/m]. For a single uniform block this is  $W * B / 2$ .

```

bearing_pressures

```
def bearing_pressures(normal: float, base_width: float, net_moment: float) -> dict
```

Base pressure distribution under a gravity wall.

```

Parameters
-----
normal : float
    Net vertical force on the base, weight less uplift [kN/m].
base_width : float
    Base width B [m].
net_moment : float
    Net moment about the rear heel, restoring less overturning
    [kNm/m]. Its ratio to the normal force locates the resultant.

```

Returns

```

dict
    ``p_max``, ``p_min`` [kPa], the eccentricity ``e`` from the centre
    of the base, and ``middle_third``, which is False when the
    resultant falls outside the middle third of the base and the heel
    goes into tension. Outside the middle third the no-tension
    distribution is used, so p_max is the triangular peak.

```

hydrostatic_force

```
def hydrostatic_force(depth: float, unit_weight: float=RHO_W * G / 1000.0) -> dict
```

Hydrostatic force on a vertical face and its lever arm [kN/m, m].

$F = 0.5 \gamma h^2$, acting at $h/3$ above the bottom of the face. Used for the water still standing in front of the wall when the trough passes, which is the only thing resisting the backfill at that instant.

SeawallDesign

```

class SeawallDesign
    conditions: DesignConditions
    still_water_level: float
    seabed_level: float
    crest_level: float
    crest_freeboard: float
    promenade_level: float
    base_width: float
    stem_thickness: float
    base_thickness: float
    founding_level: float
    embedment: float
    scour_depth: float
    toe_berm_width: float
    toe_berm_thickness: float
    toe_Dn50: float
    toe_M50: float
    toe_within_range: bool
    pressures: dict
    wave_force: float
    wave_arm: float
    weight: float
    uplift: float
    sliding_FoS: float
    overturning_FoS: float
    bearing: dict
    backfill: Sediment
    retained_height: float
    water_table: float
    surcharge: float
    earth_driving: dict

```

```

earth_resisting: dict
drawdown: dict
governing_case: str
q_mean: float
q_upper: float
governing_limit: str
impulsive: bool
iterations: int = 0
warnings: list[str] = field(default_factory=list)

```

A dimensioned L-shaped gravity seawall.

Levels are metres above chart datum. Distances are metres landward from the seaward face of the stem, which is the origin of the section. Everything is per metre run of wall.

property

```
SeawallDesign.wall_height(self) -> float
```

Founding level to crest.

property

```
SeawallDesign.heel_width(self) -> float
```

Landward projection of the base beyond the stem.

property

```
SeawallDesign.water_depth(self) -> float
```

Depth at the wall under the design still water level.

method

```
SeawallDesign.quantities(self) -> dict
```

Take-off per metre run of wall.

Rock tonnage uses a 0.37 layer porosity, so it is placed tonnage rather than solid volume times density.

method

```
SeawallDesign.summary(self) -> str
```

A short design report, in the order an engineer checks it.

design_seawall

```

def design_seawall(conditions: DesignConditions, still_water_level: float, seabed_level: float, tolerable_use:
str='pedestrians_aware', beta_degrees: float=0.0, friction: float=0.6, target_sliding: float=1.2, target_overturning: float=1.5,
require_middle_third: bool=True, scour_coefficient: float=0.4, minimum_embedment: float=1.0, maximum_embedment: float=3.0,
stem_thickness: float=1.0, base_thickness: float=1.2, promenade_freeboard: float=1.0, toe_berm_width: float | None=None,
toe_damage: float=0.5, seabed_slope: float=1 / 30, scatter_factor: float=3.0, max_base_width: float=30.0, step: float=0.1,
backfill: 'Sediment | str'='medium_sand', water_table: float=0.0, surcharge: float=10.0, earth_pressure_driving: str='at_rest',
earth_pressure_resisting: str='active', credit_earth_pressure: bool=True, drawdown_level: float | None=None) -> SeawallDesign

```

Size an L-shaped gravity seawall against a design condition.

The chain is the one a design office follows.

1. Crest level from EurOtop, set so the *upper* bound of the overtopping scatter band meets the limit for ``tolerable\_use``, not the mean.
2. Founding level from the scour allowance, bracketed by ``minimum\_embedment`` and ``maximum\_embedment``. Deeper embedment is a piling question, not a gravity-wall one.
3. Goda pressures on the wetted face, from the still water level down to the seabed.
4. Base width grown in ``step`` increments until sliding, overturning

and, if ``require\_middle\_third``, the base pressure check all pass. Keeping the resultant in the middle third stops the heel lifting off the bedding, which is what turns a wide safe base into a rocking one. The backfill over the landward heel is counted as restoring weight; passive resistance on the buried front face and active earth pressure from the fill are both ignored, which is conservative for the wave-loading case.

5. Toe stone from the Van der Meer toe formula, iterated because the berm thickness changes the depth over the berm.

Notes

Full Goda uplift is applied under the base even when the base is embedded, where wave pressure would in reality be attenuated through the soil. That is deliberately on the safe side. It is also what makes the base come out wide: uplift grows with the base width just as the restoring weight does, so a wall in shallow water with a large wave is driven by flotation as much as by sliding.

Returns

SeawallDesign

Fully dimensioned, with ``summary()`` and ``quantities()``.

pyCoastal.applications.port

Port layout simulator.

Propagates a phase-resolved wave field into a harbour and reports how much wave energy reaches the berths. Breakwaters are rasterized onto the grid as reflecting walls, waves enter from a soft source line, and the open boundaries are damped by sponge layers so outgoing and reflected energy leaves the domain instead of ringing around it.

The solver integrates the second-order wave equation

$$d^2 \eta / dt^2 = \text{div}(c^2 \text{grad } \eta)$$

in flux form, with zero flux on every face touching a breakwater cell. The celerity c is taken from the linear dispersion relation at the local depth and the chosen wave period, using ``pyCoastal.tools.wave.dispersion``, so the wavelength is correct in intermediate water rather than the shallow-water approximation $\sqrt{g h}$. The model is therefore accurate for the single frequency it is run at; it is not a broad-banded spectral model, and it does not represent breaking, wave-current interaction, or nonlinear transfers.

Harbour performance is reported as the disturbance coefficient

$$K_d = H_{\text{local}} / H_{\text{reference}}$$

with $H_{\text{reference}}$ measured at a probe in open water outside the harbour. That normalisation makes the result independent of how the source is calibrated.

Coordinates follow the package convention: x and y span the horizontal plane with x on axis 0, z is elevation.

G

$G = 9.81$

Breakwater

```
class Breakwater
    points: list[tuple[float, float]]
    width: float = 20.0
    name: str = 'breakwater'
    absorption: float = 0.0
```

A breakwater as a polyline of a given thickness.

Attributes

points : sequence of (x , y)

Vertices of the centreline [m]. Two points give a straight arm; more give a bent or dog-leg structure.

width : float

Structure thickness [m]. Cells within ``width / 2`` of the

```

    centreline become land.
name : str
    Label used in reports and plots.
absorption : float
    Energy absorption of the seaward face, 0 for a fully reflecting
    vertical wall and 1 for a perfect absorber. A rubble-mound armour
    slope is roughly 0.5 to 0.8. Implemented as a damping collar in the
    water next to the structure, so this is a model knob and not the
    reflection coefficient itself. Measured Kr for a normally incident
    wave, from ``measure_reflection``:

        absorption  0.00  0.25  0.50  0.75  1.00
        Kr          0.95  0.48  0.22  0.19  0.25

The mapping is not monotonic: past about 0.75 the collar is damped so
strongly that it becomes an impedance step and starts reflecting
again. Use roughly 0.3 to 0.4 for a rubble mound (Kr near 0.4), and
leave it at 0 for a vertical caisson.

```

method

```
Breakwater.distance_field(self, X: np.ndarray, Y: np.ndarray) -> np.ndarray
```

Distance from every grid point to this structure's centreline [m].

PortLayout

```

class PortLayout
    Lx: float = 1200.0
    Ly: float = 900.0
    dx: float = 4.0
    depth: float = 10.0
    breakwaters: list[Breakwater] = field(default_factory=list)

```

Harbour geometry and bathymetry.

Attributes

```

Lx, Ly : float
    Domain extent [m].
dx : float
    Grid spacing [m], used in both directions.
depth : float
    Still-water depth [m]. Constant depth only; refraction over a
    varying bed is out of scope for this solver.
breakwaters : list of Breakwater
    Structures rasterized as reflecting walls.

```

property

```
PortLayout.shape(self) -> tuple[int, int]
```

method

```
PortLayout.coordinates(self) -> tuple[np.ndarray, np.ndarray]
```

1D coordinate arrays (x, y) [m].

method

```
PortLayout.meshgrid(self) -> tuple[np.ndarray, np.ndarray]
```

2D coordinate arrays with x on axis 0.

method

```
PortLayout.land_mask(self) -> np.ndarray
```

Boolean array, True where a breakwater occupies the cell.

harbour_layout

```
def harbour_layout(Lx: float=1200.0, Ly: float=900.0, dx: float=4.0, depth: float=10.0, gap: float=120.0, arm_length: float=330.0, width: float=24.0, back_wall: bool=True, absorption: float=0.0) -> PortLayout
```

A conventional two-arm harbour with a gap entrance.

Two shore-normal breakwaters project from the downwave (east) side of the domain, leaving an opening of width ``gap`` centred on the domain. With ``back\_wall`` the basin is closed by a quay along the landward edge, so energy entering the harbour has to leave through the gap again. Handy as a starting point and as the geometry used in the tests. ``absorption`` is applied to every structure; see ``Breakwater.absorption``.

When using a back wall, drop "east" from ``simulate\_port(sponge\_sides=...)`` so the quay reflects instead of being absorbed.

offset_entrance_layout

```
def offset_entrance_layout(Lx: float=1400.0, Ly: float=900.0, dx: float=4.0, depth: float=10.0, gap: float=130.0, overlap: float=140.0, separation: float=180.0, arm_length: float=300.0, width: float=24.0, back_wall: bool=True, absorption: float=0.0) -> PortLayout
```

Overlapping arms, so no straight path leads into the basin.

The two arms sit at different x and overlap in y, turning the entrance into a dog-leg. Waves have to diffract twice to reach the basin, which is the usual way to quieten a harbour without narrowing the navigable opening.

Parameters

```
-----
overlap : float
    Alongshore distance over which the two arms shadow each other [m].
separation : float
    Distance between the two arms in x [m], setting the length of the entrance channel.
```

hooked_breakwater_layout

```
def hooked_breakwater_layout(Lx: float=1400.0, Ly: float=900.0, dx: float=4.0, depth: float=10.0, gap: float=150.0, hook_length: float=220.0, arm_length: float=340.0, width: float=24.0, back_wall: bool=True, absorption: float=0.0) -> PortLayout
```

A long main breakwater with a shore-parallel hook, plus a lee arm.

The hook turns back across the approach so the entrance faces along the coast rather than into the incoming waves. Common where the design wave arrives consistently from one sector.

Parameters

```
-----
hook_length : float
    Length of the returning limb at the head of the main arm [m].
```

detached_breakwater_layout

```
def detached_breakwater_layout(Lx: float=1400.0, Ly: float=900.0, dx: float=4.0, depth: float=10.0, gap: float=150.0, screen_length: float=300.0, standoff: float=200.0, arm_length: float=280.0, width: float=24.0, back_wall: bool=True, absorption: float=0.0) -> PortLayout
```

Two arms with a detached screen standing off the entrance.

The screen intercepts waves heading straight for the gap while leaving navigable water either side of it. Used where an entrance cannot be narrowed or offset.

Give the screen some ``absorption``. Left fully reflecting it forms a pocket with the arms that rings, and basin agitation goes up rather than down. For a 150 m gap with 280 m arms at $T = 9$ s, mean basin K_d went from 0.35 to 0.51 with a reflecting screen, and from 0.20 to 0.04 once the structures were armoured. The effect worsens as ``standoff`` grows.

Parameters

```
-----
screen_length : float
```


Alongshore length of the detached breakwater [m].
standoff : float
Distance seaward of the arms at which the screen sits [m].

marina_layout

```
def marina_layout(Lx: float=1400.0, Ly: float=900.0, dx: float=4.0, depth: float=10.0, outer_gap: float=160.0, inner_gap: float=90.0, inner_offset: float=220.0, basin_separation: float=240.0, arm_length: float=300.0, width: float=24.0, back_wall: bool=True, absorption: float=0.0) -> PortLayout
```

An outer harbour protecting an inner basin through a second opening.

The inner entrance is offset alongshore from the outer one, so energy entering the outer harbour does not run straight into the marina. This is the arrangement that keeps small-craft berths workable.

Parameters

inner_offset : float
Alongshore offset between the outer and inner entrances [m].
basin_separation : float
Distance in x between the outer arms and the inner basin wall [m].

rotate_layout

```
def rotate_layout(layout: PortLayout, degrees: float, about: tuple[float, float] | None=None) -> PortLayout
```

Rotate every structure in a layout by ``degrees`` (counter-clockwise).

Only the relative angle between the waves and the harbour matters, so rotating the layout and driving it shore-normal is equivalent to leaving it fixed and sending an oblique wave, without the lit-parallelogram limit of an oblique source. This is the accurate way to study wave direction.

Structures are rotated about ``about``, defaulting to the domain centre. Points may leave the domain; that is fine, since anything outside is simply not rasterized.

illuminated_mask

```
def illuminated_mask(layout: PortLayout, wave: IncidentWave, source_x: float) -> np.ndarray
```

Cells that the oblique source line actually reaches.

A source at ``x = source\_x`` spanning the full domain height launches rays at the wave angle, so the field at (x, y) originates from ``y - (x - source\_x) tan(theta)``. Where that lies outside the source line the cell never receives the incident wave.

LAYOUTS

```
LAYOUTS = {'two_arm': harbour_layout, 'offset_entrance': offset_entrance_layout, 'hooked': hooked_breakwater_layout, 'detached_screen': detached_breakwater_layout, 'marina': marina_layout}
```

IncidentWave

```
class IncidentWave
    height: float = 1.5
    period: float = 8.0
    direction: float = 0.0
```

Monochromatic incident wave.

Attributes

height : float
Incident wave height H [m].
period : float
Wave period [s].
direction : float
Propagation direction [rad], measured from the +x axis. Zero sends the wave straight up-domain toward the harbour entrance.

property

```
IncidentWave.amplitude(self) -> float
```

property

```
IncidentWave.omega(self) -> float
```

method

```
IncidentWave.wavelength(self, depth: float) -> float
```

Wavelength at the given depth [m], from the dispersion relation.

method

```
IncidentWave.celerity(self, depth: float) -> float
```

Phase celerity at the given depth [m/s].

PortResult

```
class PortResult
    x: np.ndarray
    y: np.ndarray
    times: np.ndarray
    snapshots: np.ndarray
    land: np.ndarray
    wave_height: np.ndarray
    reference_height: float
    layout: PortLayout
    wave: IncidentWave
```

Output of a port simulation.

Attributes

```
x, y : np.ndarray
    Coordinate axes [m].
times : np.ndarray
    Times of the stored snapshots [s].
snapshots : np.ndarray
    Surface elevation, shape (n_times, nx, ny) [m].
land : np.ndarray
    Breakwater mask, shape (nx, ny).
wave_height : np.ndarray
    Wave height over the analysis window, shape (nx, ny) [m].
reference_height : float
    Wave height at the reference probe in open water [m].
layout, wave
    The inputs the run came from.
```

property

```
PortResult.disturbance_coefficient(self) -> np.ndarray
```

$K_d = H_{\text{local}} / H_{\text{reference}}$, masked to water cells.

method

```
PortResult.probe(self, point: tuple[float, float]) -> float
```

Disturbance coefficient at a point [m, m].

method

```
PortResult.berth_report(self, berths: dict[str, tuple[float, float]]) -> dict[str, dict]
```

Wave height and K_d at each named berth.

method

```
PortResult.operable_fraction(self, berths: dict[str, tuple[float, float]], limit: float) -> dict[str, bool]
```

Whether each berth stays below an operational wave-height limit [m].

VALID_SIDES

```
VALID_SIDES = ('west', 'east', 'south', 'north')
```

simulate_port

```
def simulate_port(layout: PortLayout, wave: IncidentWave, duration: float | None=None, cfl: float=0.35, sponge_thickness: float | None=None, sponge_strength: float=2.0, sponge_sides: tuple[str, ...]=VALID_SIDES, source_x: float | None=None, ramp_periods: float=3.0, analysis_periods: float=6.0, n_snapshots: int=120, store_from: float | None=None, reference_point: tuple[float, float] | None=None) -> PortResult
```

Propagate a phase-resolved wave field into the harbour.

Parameters

layout, wave

Harbour geometry and incident wave.

duration : float, optional

Simulated time [s]. Defaults to enough for the wave to cross the domain twice plus the ramp and analysis windows.

cfl : float

Courant number for the explicit leapfrog step.

sponge_thickness : float, optional

Width of the absorbing margin [m]. Defaults to 1.5 wavelengths.

sponge_strength : float

Peak damping rate in the sponge [1/s].

sponge_sides : tuple of str

Which domain edges absorb. Drop a side to make it reflect, for instance the landward edge behind a harbour basin.

source_x : float, optional

x position of the wave-maker line [m]. Defaults to just inside the west sponge.

ramp_periods : float

Number of periods over which the source amplitude ramps up, to avoid a start-up shock.

analysis_periods : float

Length of the window at the end of the run used to measure wave height, in wave periods.

n_snapshots : int

Number of stored surface snapshots, spread over the stored window.

store_from : float, optional

Time from which to start storing snapshots [s]. Defaults to the beginning of the analysis window, so the movie shows the developed field. Pass 0.0 to record the whole run including start-up.

reference_point : (float, float), optional

Probe used to define the incident height. Defaults to a point in open water upwave of the harbour entrance.

Returns

PortResult

measure_reflection

```
def measure_reflection(absorption: float, period: float=8.0, depth: float=10.0, dx: float=5.0, sponge_strength: float=2.0) -> float
```

Reflection coefficient achieved by a given ``absorption`` setting.

Fires a normally incident wave at a full-width wall and infers K_r from the standing-wave ratio in front of it,

$$K_r = (H_{\max} - H_{\min}) / (H_{\max} + H_{\min})$$

which is the usual laboratory estimate. Useful for picking an ``absorption`` value that matches a measured or specified K_r .

pyCoastal.applications.piles

Wave loads on slender piles: Morison forces, shear, moment and scour.

The load on a pile is not one number. It is a distribution up the pile that changes through the wave cycle, and the two things a designer needs from it (the base shear and the overturning moment at the mudline) peak at different phases. This module computes the distribution, integrates it, and sweeps the phase, rather than evaluating a formula at the crest and hoping.

The Morison equation splits the load on a slender member into a drag term, in phase with the velocity, and an inertia term, in phase with the acceleration::

$$f(z) = 0.5 \rho C_d D u |u| + \rho C_m (\pi D^2 / 4) du/dt$$

Slender means the member is small enough not to change the wave that is loading it, conventionally D / L below about 0.2. Above that the wave diffracts around the member and the Morison equation no longer applies; the functions here say so rather than returning a number.

Sources

Morison, J. R., O'Brien, M. P., Johnson, J. W. and Schaaf, S. A. (1950), "The force exerted by surface waves on piles".

Sarpkaya, T. (2010), Wave Forces on Offshore Structures. Drag and inertia coefficients, and their dependence on Keulegan-Carpenter number.

DNV-RP-C205 (2010), Environmental Conditions and Environmental Loads. Coefficient guidance for smooth and rough cylinders.

Wheeler, J. D. (1970), "Method for calculating forces produced by irregular waves". The stretching used to carry linear kinematics up to the instantaneous free surface.

Sumer, B. M., Fredsoe, J. and Christiansen, N. (1992), "Scour around vertical pile in waves". The scour relation used here.

Conventions

Elevation z is measured from the still water level and increases upward, so the mudline is at $z = -\text{depth}$ and a wave crest reaches $z = +\eta$. Forces are newtons, forces per unit length newtons per metre, and moments newton metres about the mudline unless stated.

G

G = 9.81

RHO

RHO = 1025.0

NU

NU = 1.19e-06

wave_kinematics

```
def wave_kinematics(H: float, T: float, depth: float, z, phase: float=0.0, stretching: str='wheeler') -> dict
```

Horizontal velocity and acceleration under a linear wave.

Parameters

H : float
 Wave height [m].
T : float
 Period [s].
depth : float
 Still water depth [m].
z : array_like
 Elevations from the still water level, negative downward [m].
phase : float
 Wave phase in radians. Zero is the crest at the pile.

stretching : str
 "wheeler" maps the still-water profile onto the instantaneous water column, "none" evaluates linear theory as written, and "extrapolate" continues the still-water profile above the still water level.

Linear theory says nothing about the water between the still water level and the crest, which is exactly where the load is largest. Extrapolating the cosh profile there overstates the velocity badly; Wheeler stretching is the usual fix and is the default.

Returns

dict
 ``u`` and ``dudt`` at each elevation, the surface elevation ``eta``, the wavelength, and a mask of which elevations are in the water.

keulegan_carpenter

```
def keulegan_carpenter(H: float, T: float, depth: float, diameter: float, z: float=0.0) -> float
```

KC = $u_{\max} T / D$, at elevation ``z``.

Sets which Morison term dominates. Below about 3 the load is almost all inertia and the drag coefficient hardly matters; above about 20 drag governs and the inertia coefficient hardly matters. A monopile in a design storm usually sits awkwardly between the two.

reynolds_number

```
def reynolds_number(H: float, T: float, depth: float, diameter: float, z: float=0.0, viscosity: float=NU) -> float
```

Re = $u_{\max} D / \nu$, at elevation ``z``.

drag_inertia_coefficients

```
def drag_inertia_coefficients(KC: float, rough: bool=True) -> dict
```

Indicative Cd and Cm, with the regime named.

Post-critical Reynolds number values from DNV-RP-C205: a rough cylinder, which is what any pile becomes once marine growth establishes, takes Cd = 1.05, and a clean one Cd = 0.65. The inertia coefficient falls from the potential-flow value of 2.0 as KC rises and the wake starts to interfere with the next half cycle.

Returns

dict
 ``Cd``, ``Cm`` and the dominant ``regime``. These are starting values for a concept design. A real design takes them from the governing code for the actual roughness, KC and Re, and a real fatigue assessment does not use a single pair at all.

morison_load_profile

```
def morison_load_profile(diameter: float, u, dudt, Cd: float=1.05, Cm: float=2.0, rho: float=RHO) -> dict
```

Inline force per unit length, split into its two terms [N/m].

integrate_load

```
def integrate_load(z, load, mudline: float) -> dict
```

Total force and mudline moment from a load profile.

Trapezoidal in z, which is what the profile is sampled on. The moment arm is measured from the mudline, so the result is the overturning moment the foundation has to carry.

PileLoad

```
class PileLoad
    diameter: float
    H: float
    T: float
    depth: float
    phase: float
    z: np.ndarray
    drag: np.ndarray
    inertia: np.ndarray
    total: np.ndarray
    force: float
    moment: float
    arm: float
    eta: float
    Cd: float
    Cm: float
    KC: float
    regime: str
    diffraction_ratio: float
    warnings: list[str] = field(default_factory=list)
```

Wave load on a pile at one phase, or at the worst phase.

property

```
PileLoad.inertia_fraction(self) -> float
```

Share of the total force carried by the inertia term.

method

```
PileLoad.summary(self) -> str
```

morison_pile_load

```
def morison_pile_load(diameter: float, H: float, T: float, depth: float, phase: float=0.0, Cd: float | None=None, Cm: float | None=None, points: int=400, stretching: str='wheeler', rough: bool=True, air_gap: float=0.0) -> PileLoad
```

Morison load on a vertical pile at one wave phase.

Parameters

points : int

Samples up the pile. The load is concentrated near the surface, so a coarse grid loses the peak; 400 is ample for a monopile.

air_gap : float

Elevation above the still water level to sample to. The profile is zero above the instantaneous surface, so this only matters for plotting the dry part of the pile.

Cd, Cm : float, optional

Taken from :func:`drag\_inertia\_coefficients` when not given.

phase_sweep

```
def phase_sweep(diameter: float, H: float, T: float, depth: float, phases: int=181, **kwargs) -> dict
```

Force and moment through the wave cycle, and where each peaks.

The base shear and the overturning moment do not peak at the same phase, because the drag term peaks under the crest while the inertia term peaks a quarter cycle earlier, and the two have different lever arms. Designing on the crest phase alone can miss the worst moment.

scour_depth_pile

```
def scour_depth_pile(diameter: float, KC: float, current_only: bool=False, live_bed: bool=True, bed=None, Hs: float | None=None, T: float | None=None, depth: float | None=None) -> dict
```

Equilibrium scour depth at a vertical pile [m].

Sumer, Fredsoe and Christiansen (1992), for waves::

$$S / D = 1.3 [1 - \exp(-0.03 (KC - 6))] \quad \text{for } KC > 6$$

with no scour below $KC = 6$, where the horseshoe vortex does not form. Under a steady current the same authors give $S / D = 1.3$ with a standard deviation of 0.7, which is the ``current\_only`` branch and also the limit the wave relation tends to.

Notes

This is the equilibrium depth under a sustained condition, not the depth after one storm, and it is a live-bed result. In clear water the equilibrium is similar but takes far longer to develop. Scour protection is designed on the equilibrium depth; the pile itself is often checked for both, since a scoured pile is a longer cantilever and a softer one.

design_monopile

```
def design_monopile(diameter: float, H: float, T: float, depth: float, rough: bool=True, stretching: str='wheeler', phases:
int=181, bed=None) -> dict
```

Worst-phase load, scour, and the numbers a foundation designer wants.

Sweeps the phase for the worst moment rather than assuming the crest, returns the load at that phase, and adds the scour depth and the resulting increase in cantilever length.

pyCoastal.applications.scour

Scour at a pier in an estuary: waves, tide and river current together.

The hard part of estuarine scour is not any single relation. It is that the three drivers do not peak together and do not even point the same way. The tide reverses twice a day, the river does not, the waves come and go with the wind, and the water depth changes under all of it. A scour calculation done at "the design current" and "the design wave" without asking whether they can occur at the same moment is either optimistic or absurd, and it is usually not obvious which.

So this module works a tidal cycle rather than a load case. It evaluates the combined scour at every phase and reports the envelope, which phase governs, and by how much. In a river-fed estuary that is nearly always peak ebb, where the tidal current and the river run the same way and the water is shallow, but it is worth seeing rather than assuming.

The pier is a stem on a base, because that is what piers are, and the base is where the interesting failure lives. A buried pile cap does nothing until the scour hole reaches it. Then it is exposed, it is wider than the stem, and the scour deepens because of it, which exposes more of it. That feedback is iterated here rather than ignored, and a design that is stable against it looks very different from one that is not.

What is modelled and what is not

Modelled: combined wave and current scour, non-uniform pier geometry through an equivalent diameter, progressive base exposure, flow misalignment on the reversing tide, depth limitation, and the time scale of scour development against the tidal half cycle.

Not modelled: a pile cap resting at bed level can behave as a collar and *reduce* scour by deflecting the downflow. That is real and documented, and crediting it here would be unconservative, so it is not credited. Cohesive beds are refused rather than guessed at. Scour from a contracted section or from the estuary's own channel migration is a different problem and a larger one.

References

Sumer, B.M., Fredsoe, J. and Christiansen, N. (1992). Scour around vertical pile in waves. Journal of Waterway, Port, Coastal and Ocean Engineering, 118(1), 15-31.

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Sumer, B.M. and Fredsoe, J. (2001). Scour around pile in combined waves

and current. Journal of Hydraulic Engineering, 127(5), 403-411.

Sumer, B.M. and Fredsoe, J. (2002). The Mechanics of Scour in the Marine Environment. World Scientific.

Breusers, H.N.C., Nicollet, G. and Shen, H.W. (1977). Local scour around cylindrical piers. Journal of Hydraulic Research, 15(3), 211-252.

Richardson, E.V. and Davis, S.R. (2001). Evaluating scour at bridges. HEC-18, 4th ed. Federal Highway Administration.

Lagasse, P.F. et al. (2009). Bridge scour and stream instability countermeasures. HEC-23, 3rd ed. Federal Highway Administration.

G

G = 9.81

M2_PERIOD

M2_PERIOD = 12.4206012 * 3600.0

SHAPE_FACTOR

SHAPE_FACTOR = {'circular': 1.0, 'round_nose': 1.0, 'square': 1.1, 'rectangular': 1.1, 'sharp_nose': 0.9, 'group': 1.0}

shape_factor

```
def shape_factor(shape: str) -> float
```

HEC-18 nose shape factor K1, for the pier plan form.

alignment_factor

```
def alignment_factor(width: float, length: float, skew_degrees: float) -> float
```

HEC-18 flow alignment factor K2::

$$K2 = (\cos \theta + (L/a) \sin \theta) ** 0.65$$

A long base presents its width to the flow when aligned and most of its length when it is not, so skew matters enormously for anything that is not square in plan. This is the factor that makes a tidal estuary awkward: a base set square to the ebb is skewed to the flood, and a pier aligned to neither is skewed to both.

``L/a`` is capped at 12, past which HEC-18 stops claiming accuracy.

Pier

```
class Pier
    diameter: float = 2.0
    shape: str = 'circular'
    length: float | None = None
```

The column itself, above the base.

Attributes

diameter : float

Stem width across the flow [m].

shape : str

Plan form, keying :data:`SHAPE\_FACTOR`.

length : float, optional

Stem length along the flow [m]. Defaults to the diameter, which is right for a circular pier and wrong for a blade.

PierBase

```
class PierBase
    width: float = 6.0
    length: float = 6.0
    height: float = 2.0
```

```

top_level: float = -1.0
skew: float = 0.0

```

The footing, pile cap or caisson the stem stands on.

Attributes

```

-----
width : float
    Across the flow [m].
length : float
    Along the flow [m].
height : float
    Thickness of the base [m].
top_level : float
    Level of the top of the base relative to the *initial* bed.
    Negative means buried, which is the usual design intent, and is
    also the case worth watching: a base buried half a metre is
    invisible until the scour hole finds it.
skew : float
    Angle between the base's long axis and the flow [deg]. In a
    reversing tidal flow this is the ebb alignment; the flood value is
    taken as the supplement.

```

property

```
PierBase.bottom_level(self) -> float
```

Level of the underside of the base, relative to the initial bed.

property

```
PierBase.buried(self) -> bool
```

Whether the base starts below the bed.

equivalent_diameter

```
def equivalent_diameter(pier: Pier, base: PierBase | None, depth: float, scour: float=0.0, skew_degrees: float | None=None) -> dict
```

Effective obstacle width seen by the flow [m].

A pier of two widths is reduced to one by weighting each element over the depth of flow it occupies::

$$D_e = [D_{base} h_{base} + D_{stem} (h - h_{base})] / h$$

where ``h\_base`` is how much of the base stands proud of the *scoured* bed. That last word is the whole point. A base buried below the initial bed contributes nothing at first, and contributes more and more as the hole deepens, so ``D\_e`` is a function of the scour it is being used to predict. :func:`equilibrium\_scour` iterates it.

Both elements also carry their own shape and alignment factors, so a long rectangular cap skewed to the flow is correctly worse than a round one.

Parameters

```

-----
scour : float
    Scour depth already developed [m], measured down from the initial
    bed. Sets how much of a buried base is exposed.

```

Returns

```

-----
dict
    ``D_e`` plus the exposed base height and the factors used.

```

keulegan_carpenter

```
def keulegan_carpenter(Um: float, period: float, diameter: float) -> float
```

KC = $U_m T / D$, on the *near-bed* orbital velocity.

Scour is a bed process, so the velocity that matters is the one at the

bed, not the one at the surface. This is the convention Sumer and Fredsoe's relations are written in, and using a surface orbital velocity here would overstate KC badly in deep water.

velocity_ratio

```
def velocity_ratio(current: float, orbital: float) -> float
```

$U_{cw} = U_c / (U_c + U_m)$, the current's share of the near-bed flow.

Zero is a pure wave case, one is a pure current. It is the parameter Sumer and Fredsoe's combined relation turns on, and a surprising amount of estuarine scour sits near 0.7 to 0.9, where the current dominates but the waves still matter.

scour_ratio_combined

```
def scour_ratio_combined(KC: float, Ucw: float, current_ratio: float=1.3, current_live_bed: bool | None=None) -> dict
```

S/D in combined waves and current, Sumer and Fredsoe (2001)::

```
S/D = 1.3 {1 - exp[-A (KC - B)]}    for KC >= B
A   = 0.03 + (3/4) Ucw^2.6
B   = 6 exp(-4.7 Ucw)
```

The two limits are worth checking, because they are what make the relation trustworthy across the middle:

- ``Ucw = 0`` gives $A = 0.03$ and $B = 6$, which is exactly the waves-only relation of Sumer, Fredsoe and Christiansen (1992), including the threshold at $KC = 6$ below which the horseshoe vortex does not form.
- ``Ucw = 1`` gives $B = 0.055$, so the threshold vanishes, and S/D tends to 1.3, the steady-current value, **as KC grows**.

The steady-current floor

That last clause is the trap, and ``current\_live\_bed`` is the answer to it. KC is built on the *wave* orbital velocity, so a strong current under small waves has a high Ucw and a low KC at the same time, and the formula then returns almost no scour. Read literally it says a pier in a 1.5 m/s current scours less than the same pier in still water with a ripple on it, which is nonsense.

The resolution is that a steady current is the ``KC -> infinity`` limit of an oscillatory flow, not the ``KC -> 0`` one. A current that can move the bed on its own digs its own horseshoe vortex whatever the waves are doing, and adding waves to a current does not abolish it. So when the approach current alone is live-bed, the ratio is floored at the steady-current value.

This floor is an addition to the published relation, not part of it. It is flagged in the result as ``current\_governs`` so it is never silently applied.

Parameters

current_live_bed : bool, optional
 Whether the steady current alone exceeds the threshold of motion.
 ``None`` disables the floor and gives the bare published relation.

Returns

dict
 ``ratio`` S/D, the coefficients, and which branch governed.

depth_limitation

```
def depth_limitation(depth: float, diameter: float) -> float
```

Shallow-water reduction on the scour depth, ``tanh(h/D)``.

A scour hole needs room to form. Where the water is shallow compared with the obstacle the horseshoe vortex is squeezed and the hole is smaller, which is the Breusers, Nicollet and Shen (1977) depth limitation. Above about ``h/D = 3`` it is worth nothing ($\tanh 3 = 0.995$) and can be ignored; below ``h/D = 1`` it takes a quarter off.

It matters here because a wide caisson base in an estuary at low water can easily sit at h/D well under one, which is exactly where the unlimited relations start promising holes deeper than the water.

current_shields

```
def current_shields(material, velocity: float, depth: float, roughness: float | None=None) -> dict
```

Skin-friction Shields parameter under a depth-averaged current.

Uses a logarithmic velocity profile to get the bed shear stress,

$$u_* = U_{\kappa} / \ln(11 h / k_s)$$

with $k_s = 2.5 d_{50}$ unless given, then $\theta = u_*^2 / (g (s-1) d_{50})$.
The result drives the scour time scale and says whether the bed is live or in clear water, which decides how long the hole takes to form even though it barely changes how deep it ends up.

scour_time_scale

```
def scour_time_scale(material, diameter: float, depth: float, theta: float) -> dict
```

How long the scour hole takes to form [s].

Sumer, Christiansen and Fredsoe (1992) give a dimensionless time scale

$$T^* = (1/2000) (h/D) \theta^{-2.2}$$

which is made dimensional by

$$T = T^* D^2 / \sqrt{g (s-1) d_{50}^3}$$

and the hole then develops as $S(t) = S_{eq} [1 - \exp(-t/T)]$.

In a tidal estuary this is not a detail. The time scale for a pier a couple of metres across is typically most of a day, while the tide reverses every six hours, so the hole never reaches the equilibrium depth that a steady-current calculation hands you. Ignoring that is conservative for the pier and expensive for the client; relying on it without checking the spring tide is neither.

Raises

ValueError

If the bed is not live. The relation is a live-bed result and diverges as θ goes to zero.

scour_development

```
def scour_development(equilibrium: float, elapsed: float, time_scale: float) -> float
```

Scour depth after t seconds [m].

$$S(t) = S_{eq} [1 - \exp(-t / T)]$$

The exponential approach is the one universally observed, whatever disagreement there is about the time scale itself.

EstuaryConditions

```
class EstuaryConditions
    mean_depth: float = 10.0
    tidal_amplitude: float = 2.0
    tidal_current: float = 1.0
    river_current: float = 0.3
    Hs: float = 1.0
    Tp: float = 5.0
    tidal_period: float = M2_PERIOD
    current_phase: float = 90.0
    bed: str = 'medium_sand'
    wave_follows_tide: bool = False
```

Tide, river and waves at the pier.

Sign convention: positive is ebb, seaward. The river current is always positive, the tidal current changes sign, and the two therefore add on the ebb and oppose on the flood. That asymmetry is why estuarine scour is an ebb problem far more often than a flood one.

Attributes

```

-----
mean_depth : float
    Water depth at mean tide level [m].
tidal_amplitude : float
    Half the tidal range [m].
tidal_current : float
    Amplitude of the depth-averaged tidal current [m/s].
river_current : float
    Depth-averaged river current [m/s], steady and seaward.
Hs, Tp : float
    Significant wave height and peak period at the pier [m, s].
tidal_period : float
    Defaults to M2.
current_phase : float
    Phase lead of the current over the elevation [deg]. 90 is a
    standing wave, where slack water coincides with high and low water;
    0 is a progressive wave, where the strongest currents do. Real
    estuaries sit between, and the answer moves with it, so it is an
    input rather than a buried assumption.
bed : str
    Sediment key.
wave_follows_tide : bool
    Whether the wave height is taken to scale with the water depth,
    as it does for a depth-limited estuary chop, or to stay constant.

```

property

```
EstuaryConditions.material(self)
```

property

```
EstuaryConditions.peak_ebb_current(self) -> float
```

The worst steady current: tide and river running together.

property

```
EstuaryConditions.peak_flood_current(self) -> float
```

Tide against river, so the smaller of the two peaks.

tidal_state

```
def tidal_state(conditions: EstuaryConditions, phase_degrees: float) -> dict
```

Depth, current and wave orbital velocity at one phase of the tide.

Returns

dict

```

``depth``, ``elevation``, ``current`` (signed, positive ebb),
``Um`` near-bed orbital velocity, and the wave height used.

```

equilibrium_scour

```
def equilibrium_scour(pier: Pier, base: PierBase | None, depth: float, current: float, Um: float, period: float, skew_degrees: float | None=None, limit_by_depth: bool=True, bed=None, iterations: int=80, tolerance: float=1e-09) -> dict
```

Equilibrium scour depth for one steady set of conditions [m].

Solves the feedback between scour depth and effective diameter. The scour depends on the obstacle width; the obstacle width depends on how much of the base the scour has exposed; so the two are found together by fixed-point iteration rather than by evaluating the relation once at the initial geometry.

Where there is no base, or the base is deep enough never to be reached, this converges on the first pass and costs nothing.

Pass ``bed`` to enable the steady-current floor described in :func:`scour\_ratio\_combined`. Without it the bare published relation is used, which understates a current-dominated case badly.

Returns

dict

``depth`` of scour, the converged ``D\_e``, the geometry at convergence, and whether the iteration settled.

riprap_size

```
def rriprap_size(velocity: float, material_density: float=2650.0, shape: str='round_nose') -> dict
```

Median stone size for scour protection at a pier [m].

HEC-23 design guideline 12::

$$d_{50} = 0.692 (K V)^2 / (2 g (s - 1))$$

with $K = 1.5$ for a round-nosed pier and 1.7 for a rectangular one. The factor accounts for the flow accelerating around the pier: the stone has to survive the local velocity, not the approach velocity, and the difference is a factor of two in speed and four in stone weight.

scour_protection

```
def scour_protection(pier: Pier, base: PierBase | None, velocity: float, scour: float, material_density: float=2650.0) -> dict
```

A rock apron sized to the pier and the flow.

Extent follows HEC-23: the apron reaches two obstacle widths from the face, and is at least three stones thick. The width it is measured from is the base where there is one, because that is what the flow sees once the hole has formed.

The apron does not remove the scour, it relocates it. An edge scour hole forms at the perimeter instead, and a rigid apron that cannot settle into it will undermine and unravel from the edge inwards. That is why the apron is specified as a falling apron with a launch allowance rather than as a slab.

PierScourDesign

```
class PierScourDesign
    pier: Pier
    base: PierBase | None
    conditions: EstuaryConditions
    states: list[dict]
    governing: dict
    equilibrium: float
    tidal_limited: float
    time_scale: float
    protection: dict
    notes: list[str] = field(default_factory=list)
```

The outcome of :func:`design\_pier\_scour`.

property

```
PierScourDesign.base_exposed(self) -> bool
```

Whether the governing scour hole reaches the base.

property

```
PierScourDesign.undermined(self) -> bool
```

Whether the hole reaches below the underside of the base.

method

```
PierScourDesign.envelope(self) -> np.ndarray
```

Scour depth at each phase of the tide [m].

method

```
PierScourDesign.phases(self) -> np.ndarray
```

design_pier_scour

```
def design_pier_scour(pier: Pier, conditions: EstuaryConditions, base: PierBase | None=None, samples: int=73, limit_by_depth: bool=True, protection_factor: float=1.0) -> PierScourDesign
```

Work a tidal cycle and report the scour envelope at a pier.

Evaluates the combined wave and current scour at every phase of the tide, with the water depth, the current and the wave orbital velocity all moving together, and reports the deepest.

Two depths come out of it and they answer different questions. ``equilibrium`` is what the governing condition would eventually produce if it were held indefinitely, and is the right number for sizing scour protection and for the long-term foundation check. ``tidal\_limited`` is what that condition can actually achieve in the half cycle it lasts, and is the right number for asking what happens during a single spring tide.

Which of the two is smaller is not obvious in advance, which is why it is computed rather than assumed. A vigorously live bed cuts its hole in an hour or two and reaches equilibrium well inside one half cycle; a barely mobile bed under a big pier takes days and never gets close. Either way ``equilibrium`` is the number to design protection on, since the hole does not refill on the reverse flow and successive tides work it deeper.

Parameters

samples : int

Phases to evaluate over the full cycle.

protection_factor : float

Multiplier on the equilibrium scour used to size the apron launch allowance. 1.0 sizes for the predicted hole; a value above one buys margin against a relation whose scatter is substantial.

Returns

PierScourDesign

ABUTMENT_SHAPE

```
ABUTMENT_SHAPE = {'vertical': 1.0, 'wing_wall': 0.82, 'spill_through': 0.55}
```

KU_CLEAR_WATER

```
KU_CLEAR_WATER = 0.025
```

KU_CRITICAL

```
KU_CRITICAL = 6.19
```

abutment_shape_factor

```
def abutment_shape_factor(shape: str) -> float
```

HEC-18 abutment shape factor K1.

BridgeOpening

```
class BridgeOpening
    approach_width: float = 120.0
    opening_width: float = 70.0
    pier_blockage: float = 0.0
```

```

abutment_length: float = 25.0
abutment_shape: str = 'spill_through'
abutment_skew: float = 90.0
slope: float = 0.0005
flow_fraction: float = 1.0
abutment_depth_fraction: float = 0.4

```

The waterway, and the gap the bridge leaves in it.

Contraction scour is a property of the opening rather than of any structure in it: squeeze the same discharge through less width and the bed has to drop until the section can carry it again. So the geometry that matters is the width upstream against the width left between the abutments, less whatever the piers themselves occupy.

Attributes

```

-----
approach_width : float
    Bottom width of the channel upstream, W1 [m].
opening_width : float
    Gross width between the abutment faces, W2 [m].
pier_blockage : float
    Total width of piers standing in the opening [m]. Subtracted from
    the gross width, because the flow does not use it.
abutment_length : float
    Length of the embankment projected normal to the flow, L' [m], at
    one end. This is what the abutment blocks, not how long the
    structure is.
abutment_shape : str
    Keys :data:`ABUTMENT_SHAPE`. Spill-through is the usual highway
    form and the most forgiving; a vertical wall is nearly twice as
    bad.
abutment_skew : float
    Angle of the embankment to the flow [deg]. 90 is square on; less
    than 90 points downstream, more points upstream.
slope : float
    Energy grade line slope of the approach [-]. Only ever used to
    classify the mode of transport, which moves the answer by a few
    per cent; see :func:`contraction_scour`.
flow_fraction : float
    Fraction of the total discharge that goes through the opening,
    Q2/Q1. One where the bridge spans the whole waterway, less where
    flow stays out on a floodplain.
abutment_depth_fraction : float
    Flow depth at the abutment as a fraction of the approach channel
    depth. Froehlich's ``ya`` is the depth where the embankment
    actually sits, out on the bank or the floodplain, and not the
    depth in the middle of the channel. It matters more than it looks:
    the equation carries a ``+ ya`` term, so feeding it a mid-channel
    depth in a deep estuary returns a hole deeper than the water.
    Set it from the surveyed cross-section. The default assumes the
    abutment stands in a little under half the channel depth.

```

property

```
BridgeOpening.net_opening(self) -> float
```

Width actually available to the flow, W2 [m].

property

```
BridgeOpening.contraction_ratio(self) -> float
```

W1 / W2. One means no contraction; more means a squeeze.

property

```
BridgeOpening.contracted(self) -> bool
```

critical_velocity

```
def critical_velocity(bed, depth: float) -> float
```

Velocity at which the bed starts to move [m/s].

HEC-18 equation 6.1::

$$V_c = 6.19 y^{1/6} D_{50}^{1/3} \quad (\text{SI})$$

This is the switch between the two contraction scour modes, and it is a real switch rather than a smooth transition: below it the approach delivers no sediment to the opening and the bed there scours until the flow can no longer move it, above it the opening is fed from upstream and reaches a balance instead.

transport_exponent

```
def transport_exponent(depth: float, slope: float) -> dict
```

Laursen's exponent k_1 , and the mode of transport behind it.

The shear velocity against the fall velocity says whether the sediment travels along the bed or up in the water column, and Laursen's live-bed relation carries a different exponent for each::

$V^*/w < 0.50$	$k_1 = 0.59$	mostly contact load
$0.50 \text{ to } 2.0$	$k_1 = 0.64$	some suspended
$V^*/w > 2.0$	$k_1 = 0.69$	mostly suspended

The spread is worth keeping in perspective. k_1 is an exponent on the width ratio, so across the whole range it moves the scoured depth of a two-to-one contraction by about seven per cent. It is not where the uncertainty in a contraction scour estimate lives.

contraction_scour

```
def contraction_scour(opening: BridgeOpening, depth: float, velocity: float, bed, opening_depth: float | None=None, regime: str | None=None) -> dict
```

Bed lowering across the whole opening [m].

Two modes, and which one applies is decided by whether the approach flow is already carrying bed material.

Live bed, Laursen (1960), HEC-18 equation 6.2::

$$y_2 / y_1 = (Q_2/Q_1)^{6/7} (W_1/W_2)^{k_1}$$

The opening is fed from upstream, so it scours only until the enlarged section carries the sediment that arrives. The answer depends on the width ratio and almost nothing else.

Clear water, HEC-18 equation 6.4::

$$y_2 = [0.025 Q^2 / (D_m^{2/3} W^2)]^{3/7}$$

Nothing arrives from upstream, so the opening scours until the flow in it can no longer move the bed. Deeper, slower, and the mode that governs a bridge on a coarse bed or in a slack tidal reach.

Parameters

depth, velocity : float

Approach flow, upstream of the contraction.

opening_depth : float, optional

Existing depth in the opening before scour, y_0 . Defaults to the approach depth, which assumes a level bed through the bridge.

regime : str, optional

Force `live` or `clear`. Left alone, the critical velocity decides.

The two modes are separate relations fitted to separate data, and they do not meet at the threshold: the predicted depth steps as the bed comes alive. Forcing both and comparing them across the switch is the honest way to see how big that step is for a given case before quoting a number from either side of it.

Returns

dict

`depth` of scour, the scoured flow depth `y2`, the regime, and the numbers behind it.

abutment_scour

```
def abutment_scour(opening: BridgeOpening, depth: float, velocity: float, method: str | None=None, abutment_depth: float | None=None) -> dict
```

Local scour at one abutment [m].

Two equations, chosen by how far the embankment reaches into the flow compared with the depth.

Froehlich (1989), for a short abutment, $L'/y < 25$:

$$y_s / y = 2.27 K_1 K_2 (L'/y)^{0.43} Fr^{0.61} + 1$$

HIRE, for a long one:

$$y_s / y = 4 Fr^{0.33} (K_1 / 0.55) K_2$$

Notes

That trailing $+1$ in Froehlich is not physics. HEC-18 added it as a factor of safety, and it has the consequence that the equation can never return less than one flow depth of scour, however short the abutment or however slow the water. A spill-through abutment barely projecting into a sluggish estuary will still be handed a metre of scour for every metre of depth. It is reported here as `safety_margin` so the conservatism is visible rather than baked silently into a foundation level.

HEC-18 replaced both of these with the NCHRP 24-20 amplification approach in its fifth edition. These are the fourth-edition relations, which is the edition the pier scour in this module also comes from, and they remain the ones most practitioners will recognise.

Parameters

`depth` : float
Approach depth in the channel [m].
`abutment_depth` : float, optional
Flow depth where the abutment actually stands [m]. Defaults to `depth` reduced by the opening's `attr:~BridgeOpening.abutment_depth_fraction`, because an abutment sits on the bank and not in the channel. Passing the channel depth here is the single easiest way to get an absurd answer out of Froehlich.

bridge_scour_state

```
def bridge_scour_state(opening: BridgeOpening, depth: float, velocity: float, bed, pier: 'Pier | None'=None, base: 'PierBase | None'=None, Um: float=0.0, period: float=6.0, skew_degrees: float | None=None, limit_by_depth: bool=True) -> dict
```

All three scour components for one steady set of approach conditions.

The components are computed in the order the water meets them, because each one changes the flow the next one sees:

1. **Contraction** first, from the *approach* depth and velocity. It lowers the bed across the opening and so deepens the section.
2. **Local scour** second, from the flow *in the opening*. That flow is faster than the approach, because the same discharge is going through less width, and deeper, because the contraction has already scoured it. Using approach conditions here is the classic way to understate a pier.
3. **Abutment** scour alongside, from the approach flow it turns.

They are then added only where they coexist. A pier in midstream gets contraction plus its own hole; an abutment gets contraction plus its own. Adding all three at one point double-counts, and is a reliable way to arrive at a foundation depth nobody can build.

Returns

dict
`contraction`, `pier`, `abutment`, the flow in the opening, and the two totals.

BridgeScourDesign

```
class BridgeScourDesign
    opening: BridgeOpening
    conditions: EstuaryConditions
    pier: 'Pier | None'
    base: 'PierBase | None'
    states: list[dict]
    governing: dict
    contraction: float
    pier_local: float
    abutment: float
    total_at_pier: float
    total_at_abutment: float
    protection: dict
    notes: list[str] = field(default_factory=list)
```

The outcome of :func:`design\_bridge\_scour`.

property

```
BridgeScourDesign.total(self) -> float
```

The deepest total at any location and any phase [m].

property

```
BridgeScourDesign.governing_location(self) -> str
```

method

```
BridgeScourDesign.phases(self) -> np.ndarray
```

method

```
BridgeScourDesign.component(self, name: str) -> np.ndarray
```

Envelope of one component through the cycle [m].

design_bridge_scour

```
def design_bridge_scour(opening: BridgeOpening, conditions: EstuaryConditions, pier: 'Pier | None'=None, base: 'PierBase | None'=None, samples: int=73, limit_by_depth: bool=True, protection_factor: float=1.0) -> BridgeScourDesign
```

Total scour at a crossing, worked over the tidal cycle.

The same sweep as :func:`design\_pier\_scour`, carrying all three HEC-18 components rather than one. Each is enveloped over the cycle, and the envelopes are reported separately as well as combined, because they do not all peak at the same phase and a designer needs to see which one is driving the number.

Returns

BridgeScourDesign

pyCoastal.applications.river

Open channel hydraulics: normal and critical depth, backwater, and afflux.

The foundation the fluvial side of a crossing stands on. Everything a bridge does to a river, it does by changing the depth: it squeezes the flow, the water backs up, and that backwater reaches upstream for a distance nobody guesses correctly by eye.

Two depths govern a reach, and almost every question is really about which of them the flow is between.

****Normal depth**** is where the bed slope and the friction balance, so the flow would stay there forever in a prismatic channel. ****Critical depth**** is where the specific energy is least and the Froude number is one. Whether

normal is above or below critical decides whether the reach is mild or steep, which decides which way a disturbance travels: on a mild slope the backwater from a bridge runs upstream, on a steep one it cannot, and the bridge is felt downstream instead.

A note on where this joins the coast

:func:`flow\_distribution` is the reason this module exists alongside
:mod:`pyCoastal.applications.scour`. Contraction scour needs to know what fraction of the discharge actually goes through the opening rather than staying out on the floodplain, and that fraction is a conveyance calculation, not a guess. Feeding a guessed ``flow\_fraction`` into a scour depth and then quoting it to two decimals is a way of hiding the largest assumption in the chain behind the most precise-looking number.

References

Chow, V.T. (1959). Open-Channel Hydraulics. McGraw-Hill.

Henderson, F.M. (1966). Open Channel Flow. Macmillan.

Yarnell, D.L. (1934). Bridge piers as channel obstructions. Technical Bulletin 442, US Department of Agriculture.

Hydrologic Engineering Center (2016). HEC-RAS River Analysis System Hydraulic Reference Manual, Version 5.0. US Army Corps of Engineers.

G

G = 9.81

MANNING

```
MANNING = {'concrete_smooth': 0.013, 'concrete_rough': 0.017, 'earth_clean': 0.022, 'earth_weedy': 0.03, 'gravel': 0.028, 'cobbles': 0.035, 'natural_clean': 0.03, 'natural_weedy': 0.045, 'floodplain_pasture': 0.035, 'floodplain_brush': 0.07, 'floodplain_trees': 0.1}
```

PIER_SHAPE

```
PIER_SHAPE = {'semicircular_nose': 0.9, 'twin_cylinder': 0.95, 'ninety_degree_wedge': 1.05, 'square_nose': 1.25, 'ten_pile_trestle': 2.5}
```

Channel

```
class Channel
    width: float = 30.0
    side_slope: float = 2.0
    roughness: float | str = 'natural_clean'
```

A trapezoidal prismatic channel.

Attributes

width : float
 Bottom width [m]. Zero gives a triangular section.
side_slope : float
 Horizontal run per unit rise, z in 1V:zH. Zero is rectangular.
roughness : float or str
 Manning's n, or a key into :data:`MANNING`.

property

Channel.n(self) -> float

Manning's n.

method

Channel.area(self, depth: float) -> float

Flow area [m2].

method

```
Channel.perimeter(self, depth: float) -> float
```

Wetted perimeter [m].

method

```
Channel.top_width(self, depth: float) -> float
```

Width at the water surface [m].

method

```
Channel.hydraulic_radius(self, depth: float) -> float
```

A / P [m].

method

```
Channel.conveyance(self, depth: float) -> float
```

$K = A R^{(2/3)} / n$, so that $Q = K \sqrt{S}$.

method

```
Channel.velocity(self, discharge: float, depth: float) -> float
```

Mean velocity [m/s].

normal_depth

```
def normal_depth(channel: Channel, discharge: float, slope: float, limit: float=1000.0) -> float
```

Depth at which friction balances the bed slope [m].

Solves Manning, $Q = K \sqrt{S}$, for the depth. This is where the flow would settle if the channel ran straight and prismatic forever, and it is one of the two depths a backwater profile is drawn between.

Raises

ValueError

On a horizontal or adverse slope, where uniform flow does not exist and normal depth is undefined. That is a real condition, not an edge case: a ponded reach behind a structure has no normal depth, and a profile there is classified H or A instead.

critical_depth

```
def critical_depth(channel: Channel, discharge: float, limit: float=1000.0) -> float
```

Depth of least specific energy [m], where the Froude number is one.

Solves $Q^2 T / (g A^3) = 1$. Above it the flow is subcritical and disturbances travel upstream; below it they cannot, which is why a bridge on a steep reach is not felt by the reach above it.

froude_number

```
def froude_number(channel: Channel, discharge: float, depth: float) -> float
```

$Fr = V / \sqrt{g A / T}$, the form that is right for any section.

specific_energy

```
def specific_energy(channel: Channel, discharge: float, depth: float) -> float
```

$E = y + V^2 / 2g$, measured from the bed [m].

friction_slope

```
def friction_slope(channel: Channel, discharge: float, depth: float) -> float
```

Slope of the energy grade line, from Manning.

classify_slope

```
def classify_slope(channel: Channel, discharge: float, slope: float) -> dict
```

Mild, steep, critical, horizontal or adverse.

The classification is about the channel and the discharge together, not the bed alone: the same reach is mild in flood and can be steep at low flow, because critical depth moves with the discharge and normal depth moves faster.

profile_type

```
def profile_type(depth: float, classification: dict) -> str
```

Name the backwater profile, in Chow's notation.

``M1`` is the one a bridge makes on a mild reach: the flow is deeper than normal and deeper than critical, and the surface runs back upstream asymptotically towards normal depth. ``M2`` is a drawdown to a free overfall. ``S1`` sits behind a structure on a steep reach and is short, because the flow is supercritical and the disturbance cannot propagate far.

BackwaterResult

```
class BackwaterResult
    distance: np.ndarray
    depth: np.ndarray
    channel: Channel
    discharge: float
    slope: float
    classification: dict
    profile: str
    notes: list[str] = field(default_factory=list)
```

The outcome of :func:`gvf\_profile`.

property

```
BackwaterResult.water_surface(self) -> np.ndarray
```

Water level, taking the bed as zero at the control [m].

property

```
BackwaterResult.bed_level(self) -> np.ndarray
```

Bed level along the reach, rising upstream [m].

property

```
BackwaterResult.reach(self) -> float
```

How far the profile extends from the control [m].

method

```
BackwaterResult.depth_at(self, distance: float) -> float
```

Depth interpolated at one distance from the control [m].

gvf_profile

```
def gvf_profile(channel: Channel, discharge: float, slope: float, control_depth: float, steps: int=200, approach: float=0.99) -> BackwaterResult
```

Gradually varied flow profile by the direct step method.

Integrates

$$dx = dE / (S_0 - S_f)$$

away from a control depth, where ``E`` is specific energy and ``Sf`` the friction slope. Exact for a prismatic channel, which is what makes the direct step the right tool here: the depths are chosen and the distances computed, rather than the other way round, so the profile never has to iterate.

The integration stops as the depth approaches normal, because it gets there only asymptotically and the step length goes to infinity. That is not a numerical failure, it is the physics: a backwater curve has no end, so what is quoted as "the extent of backwater" is always a convention. ``approach`` sets the one used here.

Parameters

control_depth : float

Depth at the control, which is where the profile starts. A bridge or a weir sets this.

approach : float

Fraction of the way to normal depth at which to stop. 0.99 is the usual convention; 0.95 gives a noticeably shorter reach and is also defensible, which is the point.

Returns

BackwaterResult

Distances are measured from the control and are positive going upstream for a subcritical profile.

yarnell_afflux

```
def yarnell_afflux(channel: Channel, discharge: float, downstream_depth: float, blockage: float, shape: str='semicircular_nose') -> dict
```

Afflux across a line of bridge piers, Yarnell (1934)::

$$H = K (K + 5 Fr^2 - 0.6) (a + 15 a^4) Fr^2 y$$

where ``a`` is the fraction of the channel area the piers block and ``Fr`` is the downstream Froude number. The fourth power on the blockage is what makes this bite: doubling the pier area does far more than double the afflux.

Yarnell's tests were on piers in a rectangular flume at blockages up to about 0.4, with the flow class unchanged through the bridge. Outside that, and particularly where the bridge chokes the flow to critical, the energy or momentum methods in HEC-RAS are the right tools.

flow_distribution

```
def flow_distribution(main: Channel, main_depth: float, slope: float, floodplains: list[tuple[Channel, float]] | None=None) -> dict
```

How the discharge divides between channel and floodplains.

Split by conveyance, which is what actually decides it::

$$Q_i / Q = K_i / \sum(K) \quad K = A R^{(2/3)} / n$$

This is the number :class:`~pyCoastal.applications.scour.BridgeOpening` calls ``flow\_fraction``, and it is worth computing rather than assuming. A wooded floodplain has perhaps a third of the channel's roughness coefficient working against it and a fraction of its hydraulic radius, so it can be half the width of the section and carry a tenth of the flow. Guessing it high makes contraction scour look

worse than it is; guessing it low is the mistake that matters.

Parameters

floodplains : list of (Channel, depth)
Each overbank panel and the depth of flow on it. An empty list means the bridge spans the whole waterway and the fraction is one.

pyCoastal.applications.channel

Navigation channel design: squat, underkeel clearance, depth and width.

How deep and how wide does the approach channel have to be? The answer is a stack of allowances, each small, each defensible, and together often several metres. This module builds that stack explicitly so it can be argued over line by line, which is how a dredging budget actually gets agreed.

The depth chain

Start at the design water level, subtract the vessel's static draught, then subtract, in turn, the squat as she moves, her vertical response to waves, the net clearance the pilot and the authority require over the bed, and the tolerances the dredger and the survey cannot beat. What is left is the dredge level.

The width chain

A basic manoeuvring lane, widened for everything that pushes the vessel off line (speed, wind, cross current, waves, poor marking, a hostile bottom), plus clearance to each bank, and doubled with a passing distance if the channel is two-way.

Sources

PIANC (2014), Harbour Approach Channels Design Guidelines, report 121.
The concept design method reproduced here: the width components, the underkeel clearance components, and the squat formulations.

ICORELS (1980), as given in PIANC. The squat formula used by default.

Barrass, C. B. (1979), "A unified approach to squat calculations for ships". The screening formula.

Conventions

Levels are metres to chart datum and increase upward. Depths and draughts are positive numbers of metres. Speeds are in knots where a formula is written in knots and metres per second elsewhere; every function says which.

Caution

The width components and the manoeuvring lane widths here follow the PIANC concept-design structure with the indicative values commonly quoted. They are exposed as editable tables rather than buried, because the governing edition of the guideline, and the pilots on the day, decide the numbers on a real scheme. Concept design only: a real channel is confirmed by simulation.

G

G = 9.81

KNOT

KNOT = 0.514444

Vessel

```
class Vessel
    name: str
    length: float
    beam: float
    draught: float
    block_coefficient: float = 0.75
    length_pp: float | None = None
```

The design vessel.

Attributes

name : str
 For the drawing and the report.
length : float
 Length overall Loa [m].
length_pp : float
 Length between perpendiculars [m]. Defaults to 0.96 Loa, the usual ratio for a merchant hull, because the squat formulae are written around Lpp and getting it wrong by four per cent is not the largest error in the chain.
beam : float
 Moulded beam B [m].
draught : float
 Static draught at the design loading T [m].
block_coefficient : float
 Cb, the fraction of the enclosing box the hull fills. About 0.85 for a bulk carrier, 0.65 for a container ship, 0.55 for a fast ferry.

property

Vessel.displaced_volume(self) -> float

Displaced volume from the block coefficient [m3].

property

Vessel.displacement(self) -> float

Displacement in tonnes, at 1025 kg/m3.

squat_icorels

def squat_icorels(vessel: Vessel, speed: float, depth: float, coefficient: float=2.4) -> dict

Bow squat by the ICORELS formula, as given in PIANC.

$$S = C * (V / Lpp^2) * Fnh^2 / \sqrt{1 - Fnh^2}$$

where the displacement volume replaces V and Fnh is the depth Froude number, $\sqrt{u / \sqrt{g h}}$.

Parameters

speed : float
 Vessel speed through the water [knots].
depth : float
 Water depth [m].
coefficient : float
 2.4 is the value PIANC gives for an open or a wide channel. A confined channel squats more; the guideline's correction factor is applied through this coefficient.

Returns

dict
 `squat` [m], the depth Froude number, and a flag for whether the speed is close enough to the critical speed that the formula is unusable.

Notes

The formula blows up as Fnh approaches one, where the vessel is at the critical speed and the whole notion of a steady squat fails. It is normally kept to Fnh below about 0.7, and no vessel is navigated a channel anywhere near that. The returned dict flags it rather than returning a number that is arithmetically fine and physically absurd.

squat_barrass

def squat_barrass(vessel: Vessel, speed: float, depth: float, confined: bool=False) -> dict

Bow squat by Barrass's screening formula.

```
S_max = Cb * V^2 / 100    in open water
S_max = Cb * V^2 / 50     in a confined channel
```

with the speed in knots. Deliberately crude, and useful exactly because of it: a number to sanity-check the ICORELS result against, on the back of an envelope, before trusting either.

wave_response_allowance

```
def wave_response_allowance(Hs: float, factor: float=0.5, period: float | None=None, vessel: Vessel | None=None) -> dict
```

Vertical vessel motion allowance from wave height [m].

Parameters

Hs : float

Significant wave height in the channel [m].

factor : float

Fraction of Hs taken as the vertical motion of the keel. PIANC's concept-design guidance spans roughly 0.3 for a long vessel in short head seas up to about 0.7 for a short vessel in long beam seas, and 0.5 is the usual starting point. This is the single largest judgement in the depth chain when there is any swell at all, so set it deliberately.

period, vessel : optional

Used only to report the ratio of wave length to vessel length, which is what governs whether the vessel contours the wave or bridges it. A ratio near one is the worst case and argues for a higher factor.

Returns

dict

The ``allowance`` and, when the optional arguments are given, the wave length and its ratio to the vessel length.

MANOEUVRING_LANE

```
MANOEUVRING_LANE = {'good': 1.3, 'moderate': 1.5, 'poor': 1.8}
```

WIDTH_COMPONENTS

```
WIDTH_COMPONENTS = {'speed': {'fast': (0.1, 0.1), 'moderate': (0.0, 0.0), 'slow': (0.0, 0.0)}, 'crosswind': {'mild': (0.1, 0.1), 'moderate': (0.3, 0.4), 'severe': (0.6, 0.8)}, 'crosscurrent': {'negligible': (0.0, 0.0), 'low': (0.2, 0.3), 'moderate': (0.5, 0.7), 'strong': (1.0, 1.3)}, 'longitudinal_current': {'low': (0.0, 0.0), 'moderate': (0.1, 0.1), 'strong': (0.2, 0.2)}, 'waves': {'low': (0.0, 0.0), 'moderate': (0.5, 0.0), 'high': (1.0, 0.0)}, 'aids_to_navigation': {'excellent': (0.0, 0.0), 'good': (0.2, 0.2), 'moderate': (0.4, 0.4)}, 'bottom_surface': {'smooth_and_soft': (0.1, 0.1), 'smooth_or_sloping': (0.1, 0.1), 'rough_and_hard': (0.2, 0.2)}, 'depth_of_waterway': {'deep': (0.0, 0.0), 'moderate': (0.1, 0.2), 'shallow': (0.2, 0.4)}, 'cargo_hazard': {'low': (0.0, 0.0), 'medium': (0.5, 0.4), 'high': (1.0, 0.8)}}
```

BANK_CLEARANCE

```
BANK_CLEARANCE = {'sloping_channel_edges': {'fast': 0.7, 'moderate': 0.5, 'slow': 0.3}, 'sloping_and_shoals': {'fast': 0.7, 'moderate': 0.5, 'slow': 0.3}, 'steep_and_hard': {'fast': 1.3, 'moderate': 1.0, 'slow': 0.5}}
```

PASSING_DISTANCE

```
PASSING_DISTANCE = {'fast': 2.0, 'moderate': 1.6, 'slow': 1.2}
```

channel_width

```
def channel_width(vessel: Vessel, manoeuvrability: str='moderate', section: str='outer', two_way: bool=False, speed_class: str='moderate', bank: str='sloping_channel_edges', conditions: dict | None=None) -> dict
```

Channel width by the PIANC concept-design build-up.

Parameters

manoeuvrability : str

Key into :data:`MANOEUVRING\_LANE`.

section : str

"outer" for an exposed approach, "inner" for a sheltered reach. The two columns of :data:`WIDTH\_COMPONENTS` differ because an exposed

channel has waves and a sheltered one has banks close by.

```
two_way : bool
    Whether two design vessels pass.
speed_class : str
    "fast", "moderate" or "slow", used for the bank clearance and the
    passing distance.
conditions : dict, optional
    One key per entry in :data:`WIDTH_COMPONENTS`, naming the class that
    applies. Anything omitted is taken as its most benign class, and is
    reported as such so the omission is visible.
```

Returns

```
-----
dict
    The total ``width``, the ``components`` that built it, and the
    assumptions filled in for anything not specified.
```

underkeel_clearance

```
def underkeel_clearance(vessel: Vessel, squat: float, wave_allowance: float, net_clearance: float=0.6, water_level_allowance:
float=0.0, dredging_tolerance: float=0.3, survey_tolerance: float=0.2, siltation_allowance: float=0.2, density_allowance:
float=0.0) -> dict
```

Build the underkeel clearance stack [m].

Parameters

```
net_clearance : float
    The clearance that must remain under the keel at the worst instant,
    after every other allowance has been used up. Commonly 0.5 to 1.0 m
    over a soft bed and more over rock, and often set by the port's own
    rules rather than by a calculation.
water_level_allowance : float
    Uncertainty in the predicted water level: the difference between
    the predicted and the actual tide, and any negative surge.
dredging_tolerance, survey_tolerance : float
    What the dredger can hold and what the survey can see. Both are
    real, both are always there, and leaving them out is the most
    common way a channel ends up shallower than its drawing.
siltation_allowance : float
    Material expected between maintenance campaigns.
density_allowance : float
    Extra draught in brackish or fresh water, where the vessel floats
    deeper than in the salt water her marks were set in.
```

Returns

```
-----
dict
    Each allowance, the ``gross`` total below the keel, and the total
    ``required_depth`` below the design water level.
```

dredged_side_slope

```
def dredged_side_slope(bed, factor: float=2.0) -> dict
```

Stable side slope for a dredged channel, from the bed material.

```
cot(beta) = factor / tan(phi')
```

The default factor of 2 on the tangent covers two things at once: the partial factor on a submerged granular slope carrying no surcharge, and the overcut and slumping that dredging adds on top of it. It lands on the slopes that actually get built, roughly 1:2.6 in gravel and 1:3.8 in silt.

Notes

```
-----
Set ``factor`` to 1.5 for the geotechnical lower bound on its own, which
is the steepest the material will stand at and steeper than anything a
dredger will leave behind. A cohesive bed can stand far steeper in the
short term and is not covered by this at all; an exposed reach worked on
by waves ends up flatter than either number.
```

ChannelDesign

```
class ChannelDesign
    vessel: Vessel
```

```

design_water_level: float
dredge_level: float
required_depth: float
clearance: dict
squat: dict
waves: dict
width_result: dict
side_slope: float
speed: float
bed: 'Sediment | None' = None
existing_bed: float | None = None
warnings: list[str] = field(default_factory=list)

```

A dimensioned approach channel.

property

```
ChannelDesign.width(self) -> float
```

Channel bed width [m].

property

```
ChannelDesign.top_width(self) -> float
```

Width at the existing bed, including the side slopes [m].

method

```
ChannelDesign.dredge_volume(self, length: float) -> float
```

Capital dredge volume over a length of channel [m3].

A trapezoid over a flat existing bed. A real take-off works from a survey surface, and will differ; this is the number to size a campaign with, not to pay a contractor on.

method

```
ChannelDesign.summary(self) -> str
```

A short design report, depth chain first.

design_channel

```

def design_channel(vessel: Vessel, speed: float, design_water_level: float, Hs: float=0.0, Tp: float | None=None, wave_factor: float=0.5, net_clearance: float=0.6, water_level_allowance: float=0.3, dredging_tolerance: float=0.3, survey_tolerance: float=0.2, siltation_allowance: float=0.2, density_allowance: float=0.0, squat_coefficient: float=2.4, side_slope: float | None=None, bed: 'Sediment | str'='medium_sand', existing_bed: float | None=None, **width_kwargs) -> ChannelDesign

```

Size an approach channel: depth from the clearance stack, width from PIANC.

The squat depends on the water depth, which depends on the squat, so the depth is solved by a short fixed-point iteration rather than by guessing a depth and hoping.

Extra keyword arguments go to :func:`channel\_width`.

turning_basin_diameter

```
def turning_basin_diameter(vessel: Vessel, assisted: bool=True, current: bool=False) -> dict
```

Turning basin diameter as a multiple of the vessel length [m].

Indicative concept-design values: 1.5 LoA where tugs or thrusters turn the vessel on the spot, 2.0 LoA for an unassisted turn, and a further half a length where a current runs through the basin. A real basin is confirmed by simulation, and the shape is rarely a circle.

pyCoastal.applications.berthing

Berthing energy and fender selection.

A ship comes alongside carrying kinetic energy, and something has to absorb it. Get the fender too soft and the ship reaches the quay; too hard and the fender survives while the hull plating does not. The whole design is the arithmetic between those two failures.

PIANC's formulation is a chain of four factors on the kinetic energy, and the interesting thing about it is that three of the four reduce the answer. Only the added mass of the water the ship drags with it increases it. Between them they typically leave the fender somewhere between a half and four fifths of the energy the ship arrived with, and every one of those reductions is a modelling assumption that has to be defensible.

$$E = 0.5 M V^2 C_m C_e C_s C_c$$

The one that is neither a reduction nor physics is the abnormal berthing factor, applied afterwards. It covers the day something goes wrong: a parted tug line, an engine that does not go astern, a master misjudging the approach in a cross-current. PIANC sets it between 1.25 and 2.0 depending on how much worse than usual that day can be, and for many berths it is the single largest number in the calculation.

What this module is not

Fender performance here is a scaling model, not a catalogue. A real selection is made against the manufacturer's tested energy and reaction curves at the design temperature, angle and velocity, because those curves are what is guaranteed and they differ materially between makers for the same nominal size. :func:`select\_fender` sizes the fender you should be asking for; it does not choose a product.

References

PIANC (2002). Guidelines for the design of fender systems. Report of Working Group 33, Maritime Navigation Commission.

Vasco Costa, F. (1964). The berthing ship: the effect of impact on the design of fenders and other structures. The Dock and Harbour Authority.

Brolsma, J.U., Hirs, J.A. and Langeveld, J.M. (1977). On fender design and berthing velocities. PIANC 24th Congress.

BS 6349-4 (2014). Maritime works. Code of practice for design of fendering and mooring systems. British Standards Institution.

RHO_SEAWATER

RHO_SEAWATER = 1025.0

BERTHING_VELOCITY

```
BERTHING_VELOCITY = {'easy_sheltered': {'small': 0.2, 'medium': 0.15, 'large': 0.1}, 'difficult_sheltered': {'small': 0.3, 'medium': 0.2, 'large': 0.15}, 'easy_exposed': {'small': 0.4, 'medium': 0.3, 'large': 0.2}, 'good_exposed': {'small': 0.5, 'medium': 0.4, 'large': 0.25}, 'difficult_exposed': {'small': 0.8, 'medium': 0.6, 'large': 0.4}}
```

ABNORMAL_FACTOR

```
ABNORMAL_FACTOR = {'tanker_large': 1.25, 'tanker_small': 1.75, 'bulk_carrier': 1.5, 'container': 1.5, 'general_cargo': 1.75, 'ro_ro': 2.0, 'ferry': 2.0, 'tug_workboat': 2.0}
```

HULL_PRESSURE_LIMIT

```
HULL_PRESSURE_LIMIT = {'tanker_large': 150.0, 'tanker_small': 200.0, 'bulk_carrier': 200.0, 'container': 400.0, 'general_cargo': 400.0, 'ro_ro': 400.0, 'ferry': 400.0, 'tug_workboat': 700.0}
```

BERTH_CONFIGURATION

```
BERTH_CONFIGURATION = {'solid_quay': 0.8, 'semi_solid': 0.9, 'open_piled': 1.0}
```

berthing_velocity

```
def berthing_velocity(vessel: Vessel, condition: str='easy_sheltered') -> dict
```

Indicative design berthing velocity normal to the berth [m/s].

Raises

ValueError

If the condition is not one of :data:`BERTHING\_VELOCITY`.

added_mass_factor

```
def added_mass_factor(vessel: Vessel, depth: float | None=None) -> dict
```

Virtual mass coefficient C_m , Vasco Costa (1964)::

$$C_m = 1 + 2 D / B$$

A ship moving sideways drags a body of water with it, and that water has to be stopped too. The shallower the berth the more of it there is, which is why `` $C_m$ `` is written on the draught to beam ratio: a deeply laden ship in a tight berth carries proportionally more.

This is the only one of the four factors that makes the energy larger.

Parameters

depth : float, optional

Water depth at the berth [m]. Used only to warn: below about 1.1 draughts the water has nowhere to go and Vasco Costa's form starts to understate the added mass.

eccentricity_factor

```
def eccentricity_factor(vessel: Vessel, contact_distance: float | None=None, approach_angle: float=90.0) -> dict
```

Eccentricity coefficient C_e .

A ship rarely arrives flat against the berth. It touches at one point, usually a quarter of its length from the bow, and then rotates about that point. The rotation carries away energy the fender never sees::

$$C_e = (K^2 + R^2 \cos^2 g) / (K^2 + R^2)$$
$$K = (0.19 C_b + 0.11) L_{pp}$$

where `` $R$ `` is the distance along the berth from the contact point to the ship's centre of mass and `` $g$ `` is the angle between the approach velocity and the line joining them.

Parameters

contact_distance : float, optional

`` $R$ `` [m]. Defaults to the quarter point, `` $L_{pp}/4$ ``, which is the usual design assumption for a ship coming alongside under control. Zero means the ship lands flat on its midships and nothing is carried away by rotation, which gives `` $C_e = 1$ ``.

approach_angle : float

`` $g$ `` in degrees. 90 is the standard assumption, the velocity being normal to the berth.

berthing_energy

```
def berthing_energy(vessel: Vessel, velocity: float, contact_distance: float | None=None, approach_angle: float=90.0, softness: float=1.0, configuration: str='open_piled', depth: float | None=None) -> dict
```

Normal berthing energy [kNm], PIANC::

$$E = 0.5 M V^2 C_m C_e C_s C_c$$

Returns every factor as well as the product, because the product on its own tells a reviewer nothing about which assumption to argue with.

Parameters

```

velocity : float
    Approach velocity normal to the berth [m/s]. Squared, so this is
    the assumption worth the most scrutiny.
softness : float
    Cs. 1.0 for a soft fender, which is nearly all of them; 0.9 where
    the fender is stiff enough that the hull deflects with it.
configuration : str
    Keys :data:`BERTH_CONFIGURATION`.

```

abnormal_energy

```
def abnormal_energy(normal: float, vessel_class: str='container', factor: float | None=None) -> dict
```

The energy the fender is actually designed for [kNm].

PIANC's abnormal factor covers the berthing that goes wrong rather than the one that goes normally. It is a multiplier on an already-reduced number and is frequently the largest single term in the chain, so it is reported separately rather than folded in.

FenderFamily

```

class FenderFamily
    name: str = 'Cone fender, mid grade'
    reference_height: float = 1.0
    reference_energy: float = 500.0
    reference_reaction: float = 1000.0
    deflection: float = 0.72
    heights: tuple = (0.3, 0.4, 0.5, 0.65, 0.8, 1.0, 1.2, 1.4, 1.6, 2.0, 2.5)

```

A geometrically similar range of fenders.

Within one family and one rubber grade, performance scales with size: energy absorbed goes with the volume of rubber and so with the cube of the height, while the reaction goes with the cross-section and so with the square::

$$E = E_0 (H/H_0)^3 \quad R = R_0 (H/H_0)^2$$

That is why a fender chosen to absorb twice the energy pushes back only about 1.6 times as hard, and why the answer to a too-high reaction is usually a bigger fender rather than a smaller one.

The reference values are a mid-grade cone fender and are indicative. Substitute the manufacturer's own to size against a real product.

method

```
FenderFamily.energy(self, height: float) -> float
```

Rated energy of one size [kNm].

method

```
FenderFamily.reaction(self, height: float) -> float
```

Rated reaction of one size [kN].

method

```
FenderFamily.height_for(self, energy: float) -> float
```

The height that would absorb exactly this energy [m].

CONE_FENDER

```
CONE_FENDER = FenderFamily()
```

select_fender

```
def select_fender(energy: float, family: FenderFamily=CONE_FENDER) -> dict
```


Smallest standard size in the family that absorbs ``energy`` [kNm].

Returns the chosen size, its rated energy and reaction, and how much of its capacity the design actually uses. A fender working at 40% of its rating is oversized and stiff, and will hand the hull a reaction it did not need to take.

hull_pressure

```
def hull_pressure(reaction: float, panel_width: float, panel_height: float, vessel_class: str='container') -> dict
```

Contact pressure on the side shell [kN/m2].

The check that sizes the panel rather than the rubber. A fender chosen on energy alone presents its reaction over whatever area happens to be there, and for a tanker that limit is a quarter of a container ship's.

fender_spacing

```
def fender_spacing(vessel: Vessel, spacing: float, projection: float, bow_radius: float | None=None, smallest_vessel: Vessel | None=None, clearance: float=0.15) -> dict
```

Whether a ship can touch the quay between two fenders.

Two checks, and a berth has to pass both.

The geometric one: a hull is curved, so between two fenders it reaches closer to the wall than at them. With the fenders compressed, the remaining standoff has to exceed the sagitta of the hull across the gap::

$$\text{spacing} \leq 2 \sqrt{R_b^2 - (R_b - p + c)^2}$$

The practical one: the smallest ship using the berth has to reach two fenders at once, or it will sit on one and pivot. The usual limit is about 0.15 of its length between perpendiculars.

Parameters

projection : float

Standoff from the quay face to the hull with the fender at its design deflection [m].

bow_radius : float, optional

Radius of the hull curvature in plan at the point of contact [m]. Left out, the contact is taken on the parallel midbody, which is straight, so the geometric check cannot bind and the vessel length rule governs. Supply a radius for the end fenders, where the bow or stern flare curves away from the quay and a ship can tuck in behind a fender that a straight hull could not.

smallest_vessel : Vessel, optional

The smallest ship expected to use the berth. The practical rule is about *that* ship reaching two fenders, not the design one, so leaving this out uses the design vessel and gives a limit that is too generous for a berth with mixed traffic.

clearance : float

Gap that must remain between hull and structure [m].

BerthDesign

```
class BerthDesign
    vessel: Vessel
    velocity: float
    normal: dict
    abnormal: dict
    fender: dict
    panel: dict
    pressure: dict
    spacing: dict
    family: FenderFamily
    notes: list[str] = field(default_factory=list)
```

The outcome of :func:`design\_berth`.

property

```
BerthDesign.adequate(self) -> bool
```

Whether the berth passes energy, hull pressure and spacing.

property

```
BerthDesign.energy(self) -> float
```

The design energy, which is the abnormal one [kNm].

design_berth

```
def design_berth(vessel: Vessel, velocity: float | None=None, condition: str='easy_sheltered', vessel_class: str='container',
configuration: str='open_piled', depth: float | None=None, contact_distance: float | None=None, approach_angle: float=90.0,
softness: float=1.0, abnormal_factor: float | None=None, spacing: float | None=None, smallest_vessel: Vessel | None=None,
bow_radius: float | None=None, clearance: float=0.15, panel_aspect: tuple[float, float]=(1.2, 2.0), family:
FenderFamily=CONE_FENDER) -> BerthDesign
```

Size a fender for a berth, and check what it does to the hull.

The chain, in order: kinetic energy, the four PIANC factors, the abnormal allowance, a fender that absorbs it, the reaction that fender hands back, the panel needed to spread that reaction to something the side shell can take, and the spacing that stops the ship touching between fenders.

Each step can fail on its own and the failures are different. Too small a fender lets the ship reach the quay. Too small a panel dents the hull. Too wide a spacing does both, at a point where nothing is measuring.

Parameters

```
-----
velocity : float, optional
    Approach velocity [m/s]. Taken from :data:`BERTHING_VELOCITY` for
    the ``condition`` if not given, which is a starting point rather
    than a design value.
panel_aspect : tuple
    Panel width and height as multiples of the fender height. The
    default is a common proportion; a panel is cheap next to a hull
    repair, so widen it rather than accept a marginal pressure.
```

Returns

```
-----
BerthDesign
```

pyCoastal.applications.nourishment

Beach nourishment simulator.

A beach fill is a perturbation to an otherwise straight shoreline. Under small wave angles the one-line model reduces to a diffusion equation, so the fill spreads alongshore and the placed sand leaves the project area over time. This module sizes that behaviour: it builds the initial planform from a design, evolves it with the one-line model in ``pyCoastal.tools.shoreline``, and reports volume retained, project design life, and a renourishment schedule.

The linearized solution (Pelnaud-Considere, 1956) is provided separately as ``pelnaud\_considere`` and is used to verify the numerical solver.

Coordinates follow the package convention: x is alongshore, y is the cross-shore shoreline offset (positive seaward), z is elevation.

SECONDS_PER_YEAR

```
SECONDS_PER_YEAR = 365.25 * 24 * 3600.0
```

cerc_coefficient

```
def cerc_coefficient(K: float=0.39, s: float=2.65, gamma_b: float=0.78, g: float=9.81) -> float
```

CERC alongshore transport coefficient in the SI form this model needs.

The CERC (SPM 1984) formula for the immersed-weight transport rate gives a volumetric flux

$$Q = K \sqrt{g / \gamma_b} H_b^{5/2} \sin(2 \alpha) / (16 (s - 1))$$

which is a **solid** sediment volume, excluding voids. That is the form required here, because `tools.shoreline.rhs_y_t`` divides by $(1 - p)$ itself. Supplying a bulk-volume coefficient instead double-counts porosity.

Parameters

```
-----
K : float
    Dimensionless CERC coefficient for the immersed-weight transport rate.
    0.39 is the SPM value for significant wave height; values near 0.2 are
    commonly back-calculated from field data.
s : float
    Sediment specific gravity [-].
gamma_b : float
    Breaker index  $H_b/h_b$  [-].
g : float
    Gravitational acceleration [ $m/s^2$ ].
```

Returns

```
-----
float
    Coefficient such that  $Q$  [ $m^3/s$  per m of beach] = coeff  $H_b^{5/2} \sin(2 \alpha)$ .
```

KCERC_DEFAULT

```
KCERC_DEFAULT = cerc_coefficient()
```

WaveClimate

```
class WaveClimate
    Hb: float = 1.0
    T: float = 8.0
    alpha0: float = 0.0
    Kcerc: float = KCERC_DEFAULT
```

Wave conditions driving alongshore transport.

Attributes

```
-----
Hb : float
    Representative breaking wave height [m].
T : float
    Representative wave period [s]. Carried for reporting; the CERC flux
    itself depends on Hb and the breaking angle.
alpha0 : float
    Breaking wave angle relative to the unperturbed shoreline [rad].
    Zero means shore-normal incidence, for which the fill diffuses
    symmetrically without migrating alongshore.
Kcerc : float
    CERC alongshore transport coefficient, in SI units such that the
    solid-volume flux  $Q$  is  $m^3/s$ . Defaults to cerc_coefficient() for
    standard sand. Note this is *not* the dimensionless CERC  $K$ .
```

NourishmentDesign

```
class NourishmentDesign
    length: float = 1000.0
    berm_width: float = 30.0
    taper: float = 100.0
    center: float | None = None
    D: float = 8.0
    B: float = 2.0
    porosity: float = 0.4
```

Geometry and sediment properties of a beach fill.

The fill is placed as a berm of width `berm_width`` (the seaward advance of the shoreline) over a length `length``, with linear tapers of length `taper`` at each end.

Attributes

```

-----
length : float
    Alongshore length of the full-width section [m].
berm_width : float
    Shoreline advance at placement [m].
taper : float
    Alongshore length of the linear taper at each end [m].
center : float, optional
    Alongshore position of the fill centre [m]. Defaults to the centre
    of the model domain.
D : float
    Depth of closure [m]: the offshore limit of the active profile.
B : float
    Berm height above mean sea level [m].
porosity : float
    Sediment porosity [-].

```

property

```
NourishmentDesign.active_height(self) -> float
```

Vertical extent of the active profile, $D + B$ [m].

property

```
NourishmentDesign.placed_volume(self) -> float
```

In-situ volume of sand placed [m³].

Trapezoidal planform (full-width section plus two linear tapers)
multiplied by the active profile height.

longshore_diffusivity

```
def longshore_diffusivity(climate: WaveClimate, design: NourishmentDesign) -> float
```

Alongshore diffusivity of the linearized one-line model [m²/s].

Linearizing $Q = K H_b^{5/2} \sin(2 \alpha)$ for small angles gives

$$dy/dt = \text{eps } d^2y/dx^2, \quad \text{eps} = 2 K H_b^{5/2} / ((1 - p) (D + B))$$

This is the same group that sets the explicit stability limit in
``pyCoastal.tools.shoreline.suggest\_dt``.

initial_planform

```
def initial_planform(x: np.ndarray, design: NourishmentDesign) -> np.ndarray
```

Shoreline offset $y(x)$ immediately after placement [m].

A flat-topped berm over ``design.length`` with linear tapers of length
``design.taper`` at both ends.

pelnard_considere

```
def pelnard_considere(x: np.ndarray, t: float, design: NourishmentDesign, climate: WaveClimate) -> np.ndarray
```

Analytical planform of a rectangular fill after time ``t`` [s].

Pelnard-Considere (1956) solution of the linearized one-line equation for
an initially rectangular fill of width W and length L :

$$y(x,t) = (W/2) [\text{erf}((a - x')/(2 \sqrt{\text{eps } t})) + \text{erf}((a + x')/(2 \sqrt{\text{eps } t}))]$$

with $a = L/2$ and x' measured from the fill centre. Tapers are not
represented: this is the reference solution used to verify the numerical
solver, not a substitute for it.

NourishmentResult

```
class NourishmentResult
    x: np.ndarray
    times: np.ndarray
    planforms: np.ndarray
    volume_retained: np.ndarray
    placed_volume: float
    design: NourishmentDesign
    climate: WaveClimate
```

Output of a nourishment simulation.

Attributes

```
x : np.ndarray
    Alongshore coordinate [m].
times : np.ndarray
    Output times [s].
planforms : np.ndarray
    Shoreline offset at each output time, shape (n_times, n_x) [m].
volume_retained : np.ndarray
    Volume remaining inside the project footprint at each time [m^3].
placed_volume : float
    Volume placed at construction [m^3].
```

property

```
NourishmentResult.times_years(self) -> np.ndarray
```

property

```
NourishmentResult.retained_fraction(self) -> np.ndarray
```

Fraction of the placed volume still in the project area [-].

property

```
NourishmentResult.berm_width(self) -> np.ndarray
```

Shoreline offset at the fill centre through time [m].

method

```
NourishmentResult.design_life(self, threshold: float=0.5) -> float
```

Time until the retained fraction first falls below ``threshold`` [s].

Linearly interpolated between output times. Returns ``inf`` if the threshold is never crossed within the simulated period.

simulate_nourishment

```
def simulate_nourishment(design: NourishmentDesign, climate: WaveClimate, duration: float, domain_length: float | None=None, dx: float=10.0, n_outputs: int=25, bc: str='fixed_ends', cfl: float=0.9) -> NourishmentResult
```

Evolve a beach fill with the one-line model.

Parameters

```
design, climate
    Fill geometry and driving wave conditions.
duration : float
    Simulated period [s]. Use ``years * SECONDS_PER_YEAR``.
domain_length : float, optional
    Alongshore extent of the model domain [m]. Defaults to five times the
    total fill footprint, keeping the fixed-end boundaries far enough away
    not to influence the spreading.
dx : float
    Alongshore grid spacing [m].
n_outputs : int
    Number of stored output times, including t = 0.
bc : {"fixed_ends", "zero_slope"}
    Boundary treatment, passed to ``tools.shoreline.apply_bcs``.
```

```
cfl : float
    Safety factor on the diffusive stability limit.

Returns
-----
NourishmentResult
```

renourishment_schedule

```
def renourishment_schedule(design: NourishmentDesign, climate: WaveClimate, horizon: float, threshold: float=0.5, **kwargs) -> dict
```

Plan repeated renourishment over a planning horizon.

Each cycle restores the fill to its design width, so under constant wave forcing the cycles are of equal length. Returns the interval, the number of renourishments needed within ``horizon``, the placement times, and the total volume required.

```
Parameters
-----
horizon : float
    Planning horizon [s].
threshold : float
    Retained-volume fraction that triggers renourishment.
**kwargs
    Passed through to ``simulate_nourishment``.
```

phi_size

```
def phi_size(d50: float) -> float
```

Grain size on the phi scale, $\phi = -\log_2(d \text{ in mm})$.

Sediment statistics are done in phi because the distributions are close to normal there and are anything but in millimetres. Note the sign: coarser sand is a *smaller* phi.

size_from_phi

```
def size_from_phi(phi: float) -> float
```

Grain size in metres from a phi value.

dean_scale

```
def dean_scale(material, viscosity: float=NU) -> float
```

Profile scale parameter $A [m^{(1/3)}]$ from the fall velocity.

$$A = 0.067 w^{0.44}, \quad w \text{ in cm/s}$$

Kriebel, Kraus and Larson (1991). Tying A to the fall velocity rather than reading it off a grain-size table means it comes from the same Soulsby relation the rest of the package uses, so a sediment cannot have one settling velocity for scour and another for its profile.

Checks against Dean's own table: fine sand at 0.19 mm gives 0.094 and medium sand at 0.38 mm gives 0.141, against tabulated values of about 0.10 and 0.14.

equilibrium_profile

```
def equilibrium_profile(A: float, y)
```

Depth $h = A y^{(2/3)}$ at distance y offshore [m].

profile_width

```
def profile_width(A: float, depth: float) -> float
```


Distance offshore to a given depth on an equilibrium profile [m].

fill_volume_for_advance

```
def fill_volume_for_advance(A_native: float, A_fill: float, advance: float, berm_height: float, closure_depth: float) -> dict
```

Fill volume per metre of beach for a given shoreline advance.

The integration runs over *depth*, not distance, because the active profile is bounded by the closure contour and not by a line at some fixed chainage. At each depth the fill has pushed the profile seaward by

$$\text{offset}(h) = \text{advance} + (h / A_{\text{fill}})^{3/2} - (h / A_{\text{native}})^{3/2}$$

and the volume is that offset integrated from the waterline down to closure, plus the dry berm::

$$V = B a + a h_L + 0.4 h_L^{5/2} (A_f^{-3/2} - A_n^{-3/2})$$

where h_L is the closure depth, or the depth at which the two profiles meet if the fill is coarse enough to intersect first.

Returns

dict

The ``volume`` [m3 per m], the depth and distance at which the profiles meet, and whether they meet at all.

Notes

For equal grain sizes every term but the first two vanishes and this is exactly $V = (B + h^*) a$, the formula everyone starts from. That is not an approximation here, it is the same expression with $A_{\text{fill}} = A_{\text{native}}$.

critical_volume

```
def critical_volume(A_native: float, A_fill: float, berm_height: float, closure_depth: float) -> float
```

Volume per metre below which a fine fill gives no dry beach at all.

Fill finer than the native sand lies flatter. The first cubic metres go into flattening the underwater profile, and only once that is paid for does the waterline move. Below this volume the placement is an offshore terrace, which is a legitimate design but not the one that was sold.

Zero for fill as coarse as the native sand or coarser.

shoreline_advance

```
def shoreline_advance(A_native: float, A_fill: float, volume: float, berm_height: float, closure_depth: float, tolerance: float=0.0001) -> dict
```

Dry beach gained for a given fill volume, and how the profile behaves.

Parameters

volume : float

Placed volume per metre of beach [m3/m].

Returns

dict

``advance`` [m], the profile ``kind`` (intersecting, non-intersecting or submerged), and the critical volume where that matters.

Notes

Solved by bisection on the volume, which is monotonic in the advance. An analytic inverse exists for each case separately; one solver that cannot pick the wrong branch is worth more than three formulae.

profile_overfill_factor

```
def profile_overfill_factor(native, borrow, berm_height: float, closure_depth: float, advance: float=30.0) -> dict
```

How much more borrow material is needed for the same dry beach.

The ratio of the volume needed using the borrow material to the volume needed using sand identical to the native, for the same shoreline advance.

Returns

dict

The ``factor``, both volumes, and the two profile scales.

Notes

This is a profile-based factor and is not the same quantity as James's (1975) textural overfill ratio R_A , which comes off a chart built from the ϕ mean and sorting of both distributions and accounts for the fines winnowing out. Both answer "how much extra do I buy"; they do it from different evidence, and neither replaces the other. Use :func:`grain\_compatibility` for the textural side.

grain_compatibility

```
def grain_compatibility(native, borrow) -> dict
```

Compare a borrow source with the native beach, on the ϕ scale.

Returns

dict

``delta`` = $(\phi_{\text{borrow}} - \phi_{\text{native}}) / \sigma_{\text{native}}$, the mean shift in native standard deviations, negative when the borrow is coarser; ``sorting\_ratio`` = $\sigma_{\text{borrow}} / \sigma_{\text{native}}$; and a ``verdict``.

Notes

The verdict follows the rule every nourishment text states and no chart is needed for: borrow that is coarser and no more poorly sorted than the native sand is well suited, and borrow that is finer is not, in proportion to how much finer. James's (1975) chart puts numbers on the second case; this says which case you are in and how far.

pyCoastal.applications.surge

Storm surge and coastal flooding.

Builds a total water level from its components, then uses it to find how far a storm floods: along a cross-shore profile, or over a terrain grid with hydraulic connectivity so that low ground behind a continuous barrier is not flooded by a level that cannot physically reach it.

Components of the still water level

astronomical tide

Supplied, not predicted here.

barometric setup

The inverted-barometer response, about 1 cm per hPa of pressure drop.

wind setup

Wind stress piling water against the coast across the shelf. Both the closed-form estimate over a representative depth and a numerical integration across a real shelf profile are provided.

wave setup

The mean water-level rise inside the surf zone from breaking.

Wave runup is added on top of the still water level for a total water level, but runup is a swash oscillation rather than a sustained level, so the two are reported separately: flood extent from sustained level, wave hazard from runup.

Sources

Dean and Dalrymple (1991), Water Wave Mechanics for Engineers and Scientists. Wind setup and the bathystrophic storm tide.

USACE Coastal Engineering Manual (2002), EM 1110-2-1100 Part II. Storm surge.

Stockdon et al. (2006), Coastal Engineering 53, 573-588. Runup.

FEMA (2005), Guidelines and Specifications for Flood Hazard Mapping Partners.
Total water level and the treatment of wave effects.

Conventions

Elevations are metres relative to the same vertical datum as the terrain,
positive upward. Cross-shore distance increases landward.

G

G = 9.81

RHO_WATER

RHO_WATER = 1025.0

STANDARD_PRESSURE

STANDARD_PRESSURE = 101325.0

WIND_STRESS_COEFFICIENT

WIND_STRESS_COEFFICIENT = 3.3e-06

barometric_setup

```
def barometric_setup(central_pressure: float, ambient_pressure: float=STANDARD_PRESSURE, rho: float=RHO_WATER) -> float
```

Inverted-barometer rise for a pressure drop [m].

$\eta = (p_{\text{ambient}} - p_{\text{central}}) / (\rho g)$, which is close to 1 cm per hPa.
Pressures are in pascals.

This is the equilibrium response. A fast-moving storm does not give the ocean time to reach it, so this is an upper bound on the barometric part.

wind_setup

```
def wind_setup(wind_speed: float, fetch: float, depth: float, wind_angle: float=0.0, k: float=WIND_STRESS_COEFFICIENT, return_flow: float=1.0) -> float
```

Wind setup over a shelf of representative depth [m].

Integrating $d(\eta)/dx = k W^2 \cos(\theta) / (g d)$ over a fetch at constant depth gives

$$\eta = n k W^2 F \cos(\theta) / (g d)$$

Parameters

wind_speed : float
Wind speed at 10 m [m/s].
fetch : float
Over-water distance the wind acts across [m].
depth : float
Representative shelf depth [m]. Setup is inversely proportional to it, so a shallow wide shelf sets up far more than a narrow deep one.
wind_angle : float
Angle between the wind and the shore-normal [rad].
return_flow : float
Factor above 1 for the seaward return flow under the surface, which increases the surface slope. 1.15 to 1.30 is common.

Notes

The constant-depth form is a screening estimate. Because setup goes as $1/d$ it is dominated by the shallowest part of the shelf, so a single representative depth can mislead. Use ``wind\_setup\_profile`` where the bathymetry is known.

wind_setup_profile

```
def wind_setup_profile(depths: np.ndarray, dx: float, wind_speed: float, wind_angle: float=0.0, k: float=WIND_STRESS_COEFFICIENT,
return_flow: float=1.0) -> np.ndarray
```

Wind setup integrated landward across a shelf profile [m].

Integrates $d(\eta)/dx = k W^2 \cos(\theta) / (g (d + \eta))$ from the seaward end of ``depths`` toward the coast, carrying the accumulated setup into the local depth. That feedback matters: the setup itself deepens the water and damps further setup, which the constant-depth form ignores.

Parameters

depths : np.ndarray
Still-water depth at each point, seaward end first [m], positive.
dx : float
Spacing between points [m].

Returns

np.ndarray
Setup at each point, the last value being the setup at the coast.

surf_zone_setup

```
def surf_zone_setup(breaking_height: float, gamma: float=0.8) -> float
```

Mean water-level rise at the shoreline from wave breaking [m].

Thin wrapper on ``pyCoastal.tools.wave.wave\_setup``, kept here so a complete water-level budget reads in one place.

stockdon_runup

```
def stockdon_runup(Hm0: float, Tp: float, beach_slope: float) -> dict
```

2 percent exceedance runup from Stockdon et al. (2006) [m].

$$R2 = 1.1 (0.35 \beta \sqrt{H_0 L_0} + \sqrt{H_0 L_0 (0.563 \beta^2 + 0.004)}) / 2)$$

The two terms are the setup and the swash, returned separately because only the setup part is a sustained level; the swash is an oscillation.

``beach\_slope`` is the foreshore slope $\tan(\beta)$.

StormConditions

```
class StormConditions
    wind_speed: float = 40.0
    central_pressure: float = 95000.0
    tide: float = 0.0
    Hm0: float = 6.0
    Tp: float = 12.0
    fetch: float = 100000.0
    shelf_depth: float = 20.0
    wind_angle: float = 0.0
    sea_level_rise: float = 0.0
```

Forcing for a design storm.

Attributes

wind_speed : float
Sustained wind speed at 10 m [m/s].
central_pressure : float
Storm central pressure [Pa].
tide : float
Astronomical tide level at the time of landfall [m datum].
Hm0 : float
Offshore significant wave height [m].
Tp : float
Peak period [s].
fetch : float

Over-water distance the wind acts across [m].
 shelf_depth : float
 Representative depth for the closed-form wind setup [m].
 wind_angle : float
 Wind direction relative to shore-normal [rad].
 sea_level_rise : float
 Added relative sea level, for scenario work [m].

total_water_level

```
def total_water_level(storm: StormConditions, beach_slope: float=0.05, breaker_index: float=0.78, shelf_depths: np.ndarray | None=None, dx: float | None=None, return_flow: float=1.15) -> dict
```

Assemble the still water level and the total water level [m datum].

Returns each component separately, because which ones apply depends on what is being assessed. The still water level drives inundation extent; runup is added on top for the wave hazard at the shoreline but is a swash oscillation, not a sustained level, and should not be used to flood terrain far inland.

Parameters

 shelf_depths, dx
 Pass both to integrate wind setup across a real shelf profile instead of using the representative depth.
 breaker_index
 H_b / h at breaking, used to estimate the breaking height from the offshore height for the surf-zone setup.

inundation_limit

```
def inundation_limit(x: np.ndarray, z: np.ndarray, level: float) -> float
```

Landward limit of *connected* flooding on a cross-shore profile [m].

Walks landward from the sea and stops at the first ground above ``level``, linearly interpolating the crossing. Low ground further inland may also lie below the level, but water arriving overland cannot reach it while the barrier holds, so it is not counted. That is the same connectivity argument as ``bathtub\_flood``, in one dimension.

Returns the seaward end if nothing floods, and the landward end if the water is never stopped. Use ``flood\_depths`` if you want every cell below the level regardless of whether water can get there.

flood_depths

```
def flood_depths(z: np.ndarray, level: float) -> np.ndarray
```

Still-water flood depth over terrain [m], zero where dry.

bathtub_flood

```
def bathtub_flood(terrain: np.ndarray, level: float, seed: np.ndarray | None=None, connectivity: int=4) -> np.ndarray
```

Flooded cells at a still-water ``level``, respecting connectivity.

A plain elevation threshold floods every low cell, including basins sealed off from the sea by higher ground. This walks outward from the coast instead, so only ground the water can actually reach is flooded. That distinction is usually the difference between an alarming map and a useful one.

Parameters

 terrain : np.ndarray
 Ground elevation on a 2D grid [m datum], x on axis 0.
 level : float
 Still-water level [m datum].
 seed : np.ndarray, optional
 Boolean array marking where the sea enters. Defaults to any below-level cell on the domain edge.
 connectivity : {4, 8}
 Whether water spreads to edge neighbours only, or diagonals too.

Returns

np.ndarray
 Boolean flooded mask.

flooded_area

```
def flooded_area(flooded: np.ndarray, dx: float) -> float
```

Flooded plan area [m²].

flood_volume

```
def flood_volume(terrain: np.ndarray, flooded: np.ndarray, level: float, dx: float) -> float
```

Water volume held on the flooded terrain [m³].

isolated_low_ground

```
def isolated_low_ground(terrain: np.ndarray, level: float, **kwargs) -> np.ndarray
```

Cells below the level that the sea cannot actually reach.

The difference between a naive bathtub map and a connected one. Worth plotting, because it is exactly where a threshold-only map overstates the hazard.

pyCoastal.applications.sections

Design sections: a sized structure in, a drawing and a take-off out.

The design modules decide the numbers. This module turns those numbers into the deliverable an engineer actually hands over: a dimensioned cross-section with the materials hatched, the governing levels called out, the design inputs printed on the sheet, and specification notes beside it.

Each structure has three entry points.

```
``draw``  
    Puts the geometry and annotation into a :class:`~pyCoastal.drafting.Section`  
    you already have, and returns the extents it needs.  
``*_section``  
    A standalone figure with plot axes, for a report or a slide.  
``*_sheet``  
    A full drawing sheet with a border, a title block and notes, at a true  
    stated scale.
```

All three take a design object, not a pile of loose numbers, so the drawing cannot drift out of step with the calculation behind it.

Needs matplotlib::

```
pip install pyCoastal[plots]
```

MoundProfile

```
class MoundProfile  
    x_sea: float  
    x_land: float  
    crest: float  
    cot_sea: float  
    cot_land: float
```

A trapezoidal mound, described by its crest and its two slopes.

Attributes

```
-----  
x_sea, x_land : float  
    Chainage of the seaward and landward crest edges [m].  
crest : float
```



```

    Crest level [m].
    cot_sea, cot_land : float
    Slopes as horizontal run per unit rise.

```

method

```

MoundProfile.points(self, bed: float) -> np.ndarray

```

Closed polygon from the seaward toe round to the landward toe.

method

```

MoundProfile.offset(self, thickness: float) -> 'MoundProfile'

```

The profile a uniform layer of ``thickness`` inside this one.

The offset is perpendicular to each face, which is how a layer thickness is specified and measured on site. Lowering the crest by the thickness and pulling the crest edges in by $\sqrt{1 + \cot^2} - \cot$ is the exact perpendicular offset of a trapezoid, not an approximation.

method

```

MoundProfile.band(self, inner: 'MoundProfile', bed: float) -> np.ndarray

```

Closed polygon of the layer between this profile and ``inner``.

method

```

MoundProfile.area(self, bed: float) -> float

```

Cross-sectional area above the bed [m2 per metre run].

mound_layer_volumes

```

def mound_layer_volumes(profiles, bed: float) -> list[float]

```

Area of each layer, from a list of profiles outermost first.

Each entry is the area between one profile and the next, in m2 per metre run, with the innermost profile taken as solid. Multiply by the length of the trunk for a volume.

draw_seawall

```

def draw_seawall(dwg: Section, design, sea_extent: float=22.0, land_extent: float=8.0, show_scour: bool=True, annotate: bool=True)
-> tuple[tuple, tuple]

```

Put a designed seawall into an open :class:`Section`.

Returns the (xlim, zlim) the section needs, so the caller can either set them directly or hand them to :meth:`~pyCoastal.drafting.Section.fit\_scale`.

seawall_notes

```

def seawall_notes(design) -> list[str]

```

Specification notes generated from a seawall design.

The numbers come from the calculation; the material clauses are the generic ones a coastal designer would expect to see and still has to confirm for the project. Nothing here is a substitute for a specification.

seawall_section

```
def seawall_section(design, title: str='Vertical seawall, typical cross-section', sea_extent: float=22.0, land_extent: float=8.0,
figsize: tuple[float, float]=(14.5, 8.5), show_scour: bool=True, ax=None) -> Section
```

A standalone figure of a designed seawall, with plot axes.

Returns the :class:`~pyCoastal.drafting.Section` still open, so you can add project notes before saving.

seawall_sheet

```
def seawall_sheet(design, project: str='Coastal protection works', title: str='Seawall typical cross-section', size: str='A3',
sea_extent: float=22.0, land_extent: float=8.0, file: str='seawall_sheet.py', **titleblock) -> Sheet
```

A full drawing sheet of a designed seawall, at a true stated scale.

The view is fitted to a standard scale, the title block records that scale, and the specification notes are generated from the design. Extra keyword arguments go to the :class:`~pyCoastal.drafting.TitleBlock`.

draw_rubble_mound

```
def draw_rubble_mound(dwg: Section, design, still_water_level: float, seabed_level: float, cot_land: float | None=None,
crest_width: float | None=None, margin: float=12.0, annotate: bool=True, foundation: dict | None=None, crown: dict | None=None,
bed='medium_sand', show_foundation: bool=True) -> tuple[tuple, tuple]
```

Put a designed rubble mound into an open :class:`~Section`.

Draws the section the way a drawing office draws it: the mound on a levelling blanket on a geotextile, a toe berm of filter stone at each foot, and a stepped concrete crown block with its sand and gravel infill. The blanket runs past both toes far enough that the scour hole forms in the apron rather than under the structure.

Parameters

foundation : dict, optional
 Output of :func:`~pyCoastal.applications.structures.mound\_foundation`.
 Computed from ``bed`` when not given.
crown : dict, optional
 Output of :func:`~pyCoastal.applications.structures.crown\_wall`.
 Computed when not given. Pass ``False`` to leave it off.

fmt

```
def fmt(value: float) -> str
```

Signed level, the way a drawing writes one.

rubble_mound_notes

```
def rubble_mound_notes(design, still_water_level, seabed_level, cot_land=None, crest_width=None, foundation=None, crown=None) -> list[str]
```

Specification notes generated from a rubble-mound design.

rubble_mound_section

```
def rubble_mound_section(design, still_water_level: float, seabed_level: float, cot_land: float | None=None, crest_width: float | None=None, title: str='Rubble-mound breakwater, typical cross-section', figsize: tuple[float, float]=(14.5, 8.0), margin: float=12.0, ax=None) -> Section
```

A standalone figure of a designed rubble mound, with plot axes.

Notes

The underlayer follows the Rock Manual rule of a tenth of the armour mass, so Dn50 falls by $10^{(1/3)}$, about 2.15. The layers drawn are the ones the stability calculation assumes are there; a filter check against the core grading is a separate exercise.

rubble_mound_sheet

```
def rubble_mound_sheet(design, still_water_level: float, seabed_level: float, cot_land: float | None=None, crest_width: float | None=None, project: str='Harbour protection works', title: str='Breakwater typical sections', size: str='A3', margin: float=12.0, file: str='breakwater_sheet.py', bed='medium_sand', show_head: bool=True, head_kd_ratio: float | None=None, **titleblock) -> Sheet
```

A drawing sheet of a designed rubble mound, trunk and head.

Two sections, as a real set carries them: the trunk, and the roundhead with the heavier armour its exposure needs. Set ``show\_head`` False for a trunk-only sheet.

vessel_outline

```
def vessel_outline(beam: float, draught: float, freeboard: float, bilge: float=0.18, centre: float=0.0) -> np.ndarray
```

Midship section of a hull, as a closed polygon.

A box with chamfered bilges and a little flare. Enough to read as a ship at the scale a channel section is drawn at, and deliberately not more: the hull form is not what the drawing is about.

draw_channel

```
def draw_channel(dwg: Section, design, margin: float=60.0, show_vessel: bool=True, annotate: bool=True, show_depth_chain: bool=True, seabed_extent: float | None=None) -> tuple[tuple, tuple]
```

Put a designed navigation channel into an open :class:`Section`.

Draws the dredged prism, the existing bed, the design vessel at her static draught, and the underkeel clearance stack as a dimension chain, so every allowance can be read off the paper.

channel_notes

```
def channel_notes(design) -> list[str]
```

Specification notes generated from a channel design.

channel_section

```
def channel_section(design, title: str='Navigation channel, typical cross-section', figsize: tuple[float, float]=(14.5, 8.0), margin: float=60.0, exaggeration: float=8.0, ax=None) -> Section
```

A standalone figure of a designed navigation channel.

Drawn with vertical exaggeration by default. A channel several hundred metres wide and twenty deep is unreadable at a true scale, which is why dredging drawings have always been exaggerated; the factor is stated on the drawing rather than left for the reader to infer.

channel_sheet

```
def channel_sheet(design, project: str='Port approach works', title: str='Navigation channel typical section', size: str='A3', margin: float=60.0, exaggeration: float=8.0, file: str='channel_sheet.py', **titleblock) -> Sheet
```

A full drawing sheet of a designed navigation channel.

Vertically exaggerated by default, with the factor stated beside the detail title and in the notes.

draw_channel_detail

```
def draw_channel_detail(dwg: Section, design, width_in_beams: float=0.35, hull_shown: float=1.3, annotate: bool=True) -> tuple[tuple, tuple]
```

True-scale detail of the keel, the allowances and the dredge level.

The overall channel section has to be exaggerated to be readable, which stretches the vessel into a tower and makes every slope a lie. This is

the companion view a drawing set always carries: the part that matters, at a true scale, cropped to the keel, where the underkeel clearance can be read as a real thickness rather than as a band on a distorted picture.

draw_seawall_toe_detail

```
def draw_seawall_toe_detail(dwg: Section, design, annotate: bool=True) -> tuple[tuple, tuple]
```

Enlarged detail of the toe, where the section is actually decided.

The typical section shows where everything is. This shows what the toe is made of: the blinding under the heel, the founding level against the scour allowance, the rock berm and its geotextile, and the seabed the whole thing is sitting on. It is the part of a seawall that fails first and the part a typical section is always too small to explain.

draw_nourishment

```
def draw_nourishment(dwg: Section, result, native, borrow, berm_height: float, closure_depth: float, water_level: float=0.0,
    annotate: bool=True, landward: float=40.0) -> tuple[tuple, tuple]
```

Put a nourishment cross-section into an open :class:`Section`.

The native profile, the design fill over it, and the closure contour that bounds the whole exercise. The fill wedge is drawn as the borrow material, so a section placed with coarse sand and one placed with fine sand do not look alike, which they should not.

Parameters

result : dict

Output of :func:`~pyCoastal.applications.nourishment.shoreline\_advance`.

native, borrow : Sediment or str

The beach and the borrow source.

nourishment_notes

```
def nourishment_notes(result, native, borrow, berm_height: float, closure_depth: float, design=None) -> list[str]
```

Specification notes generated from a nourishment design.

nourishment_section

```
def nourishment_section(result, native, borrow, berm_height: float, closure_depth: float, water_level: float=0.0, title: str='Beach
nourishment, design profile', figsize: tuple[float, float]=(14.0, 7.0), exaggeration: float | None=None, ax=None) -> Section
```

A standalone figure of a nourishment profile.

Vertically exaggerated, because a beach profile is hundreds of metres long and a few metres deep and at a true scale it is a line. Left to itself the exaggeration is chosen so the profile fills the sheet, which is what makes it readable; pass a number to fix it instead.

nourishment_sheet

```
def nourishment_sheet(result, native, borrow, berm_height: float, closure_depth: float, water_level: float=0.0, project: str='Beach
management scheme', title: str='Nourishment design profile', size: str='A3', exaggeration: float=6.0, plan_design=None,
plan_climate=None, plan_years=None, file: str='nourishment_sheet.py', **titleblock) -> Sheet
```

A drawing sheet of a nourishment profile, with its borrow notes.

Pass ``plan\_design`` and ``plan\_climate`` to add a second view below the section: the planform spreading alongshore, from the same Pelnard-Considere solution the app plots. A fill is designed in two directions at once, and a sheet that shows only the profile leaves the reader to imagine how long it stays there.

plan_margin

```
def plan_margin(design, climate, years) -> float
```

How far past the fill a planform has to be drawn, in metres.

Enough to show the sand that has left: 1.4 spreading lengths at the latest time, and never less than the fill is long. Split out from the drawing so a sheet can ask for the extents before it commits to a scale, and trim them to hold a rung of the scale ladder.

spreading_half_life

```
def spreading_half_life(design, climate) -> float
```

Time for the centre of a fill to lose half its width [s].

From the Pelnard-Considere solution: the centre width is $\sqrt{W \operatorname{erf}(a / (2 \sqrt{\epsilon t}))}$, which is halved when the argument reaches $\operatorname{erfinv}(1/2) = 0.476936\dots$. It is the natural clock of a fill, and the right basis for choosing what times to draw: a plan at ten years tells you nothing about a fill whose half-life is seven months.

draw_nourishment_plan

```
def draw_nourishment_plan(dwg: Section, design, climate, years=(0, 1, 2, 5, 10), annotate: bool=True, samples: int=601, margin: float | None=None) -> tuple[tuple, tuple]
```

Plan view of a fill spreading alongshore, from the analytical solution.

Pelnard-Considere (1956) linearises the one-line equation into a diffusion equation, so a rectangular fill spreads exactly as a slug of heat does: the planform is a pair of error functions whose width grows as the square root of time. This draws that solution as a map, which is how a beach manager actually looks at it.

Parameters

design, climate : NourishmentDesign, WaveClimate
The fill and the wave climate driving it.
years : sequence
Times to draw, in years. Zero is the placed planform.

Returns

(xlim, ylim)
Extents in metres.

Notes on the scale

Both axes are distance, so this could be drawn 1:1. It should not be. A fill is kilometres long and tens of metres wide, and at a true scale the whole story is a hairline. Shoreline-change plans are conventionally drawn with the cross-shore axis stretched, and
:func:`nourishment\_plan\_section` picks the factor and prints it.

Notes

The analytical solution is for a rectangular fill with no tapers and a constant diffusivity. A real fill is tapered, the diffusivity varies with the wave climate, and the ends interact with whatever is next to them. It is the reference case, and it is the right one for seeing what the spreading does; it is not the numerical solver.

nourishment_plan_section

```
def nourishment_plan_section(design, climate, years=None, title: str='Beach nourishment, planform evolution', figsize: tuple[float, float]=(14.0, 6.5), exaggeration: float | None=None, ax=None) -> Section
```

A standalone plan-view figure of a fill spreading alongshore.

Left to themselves the times come from the fill's own half-life, so the plan always shows the part of the evolution that is worth looking at, and the cross-shore exaggeration is chosen to fill the sheet and printed on it.

REPOSE_ANGLE

```
REPOSE_ANGLE = 32.0
```

scour_hole_profile

```
def scour_hole_profile(scour: float, radius: float, repose: float=REPOSE_ANGLE, samples: int=2)
```

Points tracing one side of the scour hole, from the pier face out.

The hole is drawn as a straight face at the angle of repose, because that is what a scour hole in sand is: it deepens until the sides stand at their limiting slope, and then widens. Anything smoother would be an invention.

draw_pier_scour

```
def draw_pier_scour(dwg: Section, design, annotate: bool=True, protected: bool=False)
```

Section through a pier, either scoured or protected.

Levels are to the initial bed, which is the datum that matters here: every dimension on the drawing is either a depth below it or a height above it, and the base level relative to it is the design lever.

The two cases are drawn separately and deliberately so. An apron and a fully developed scour hole cannot both appear on one section: the apron exists to stop that hole, and showing them together says the protection failed and worked at the same time. ``protected=False`` is the prediction if nothing is done; ``protected=True`` is the proposed works, on an intact bed.

Returns

```
-----  
(xlim, zlim)  
    Extents in metres.
```

pier_scour_notes

```
def pier_scour_notes(design) -> list[str]
```

Drawing notes for a pier scour assessment.

pier_scour_section

```
def pier_scour_section(design, title: str='Pier scour, estuary', figsize: tuple[float, float]=(11.0, 7.0), protected: bool=False, exaggeration: float=1.0, ax=None) -> Section
```

A standalone section of the pier and its scour hole.

Drawn true to scale by default. A scour section is one of the few drawings in coastal work whose horizontal and vertical extents are comparable, so there is no reason to distort it, and the angle of repose then reads as the angle it actually is.

pier_scour_sheet

```
def pier_scour_sheet(design, project: str='Estuary crossing', title: str='Pier scour assessment', size: str='A3', show_protection: bool=True, file: str='pier_scour_sheet.py', **titleblock) -> Sheet
```

A drawing sheet of the pier, its scour hole and its protection.

bridge_bed_profile

```
def bridge_bed_profile(design, repose: float=REPOSE_ANGLE)
```

The scoured bed across the whole waterway, left bank to right.

Traces, in order: the untouched bed on the left bank, down the left abutment hole, up to the contracted bed, down into the pier hole and back out, then the mirror image. Every face stands at the angle of

repose, because that is the steepest a hole in sand can be.

Returning one polyline rather than three separate holes is deliberate. The components are computed apart and they are drawn together, and seeing them in one line is what stops anyone adding the pier hole to the abutment hole.

draw_bridge_scour

```
def draw_bridge_scour(dwg: Section, design, annotate: bool=True, freeboard: float=2.0)
```

Elevation through a bridge opening, with all three scour components.

Looking downstream: the abutments at each end, the pier between them, and one bed line carrying the contraction across the whole opening with the local holes cut into it at the pier and at each abutment toe.

Drawing them on one line is the point. The three components are computed separately and are often quoted separately, and a reader who sees them apart will add them. Here it is visible that the deepest point is one hole at one place, not the sum of three.

bridge_scour_notes

```
def bridge_scour_notes(design) -> list[str]
```

Drawing notes for a bridge scour assessment.

bridge_scour_section

```
def bridge_scour_section(design, title: str='Bridge scour, total', figsize: tuple[float, float]=(13.0, 7.0), exaggeration: float | None=None, ax=None) -> Section
```

A standalone elevation of the crossing and its scour.

bridge_scour_sheet

```
def bridge_scour_sheet(design, project: str='Estuary crossing', title: str='Bridge scour assessment', size: str='A3', file: str='bridge_scour_sheet.py', **titleblock) -> Sheet
```

A drawing sheet of the crossing, its scour and its protection.

pyCoastal.drafting

Engineering section drawings, drawn the way a design office draws them.

A coastal cross-section is not a plot. It is a scaled drawing: the geometry is in real metres, the line weights carry meaning, materials are hatched rather than coloured by value, and every dimension that governs the design is called out on the paper. This module gives the small set of primitives that takes matplotlib from plotting to drafting.

The pieces

```
-----
``Section``
    A drawing sheet in real-world coordinates (x cross-shore, z elevation),
    locked to equal aspect so a 1:2 slope looks like a 1:2 slope.
``Material``
    Fill, hatch and edge weight for one material, so quarry run reads as
    quarry run on any drawing in the package.
``Section.dim_h`` and ``Section.dim_v``
    Dimension lines with extension lines and arrowheads, offset in points
    so they stay legible whatever the scale.
``Section.level``
    The standard levelling triangle and elevation callout.
``Section.slope``
    The slope triangle, labelled as 1 : cot(alpha).
``Section.table``
    The parameter block. A drawing that does not carry its design inputs is
    not a deliverable.
```

Everything is drawn in metres. Text is sized in points and placed with offsets in points, so annotation stays readable when the section is rescaled; only the geometry scales.

Importing this module needs matplotlib::

```
pip install pyCoastal[plots]
```

WEIGHTS

```
WEIGHTS = {'thin': 0.5, 'medium': 1.0, 'heavy': 1.8, 'extra': 2.4}
```

TEXT

```
TEXT = {'dim': 8.0, 'note': 8.5, 'label': 9.5, 'title': 12.0, 'table': 8.5}
```

INK

```
INK = '#1a1a1a'
```

WRAP

```
def WRAP(text: str, width: int) -> str
```

Wrap a title-block field so it keeps inside the strip.

Material

```
class Material
    name: str
    face: str
    edge: str = INK
    hatch: str | None = None
    weight: str = 'medium'
    stones: float = 0.0
```

How one material is drawn.

Attributes

```
-----
name : str
    Legend entry, for example "Rock armour, 6 t".
face : str
    Fill colour. Kept desaturated: a section is read by its hatching and
    its line weights, not by colour.
edge : str
    Outline colour.
hatch : str or None
    A matplotlib hatch string, or None for a plain fill.
weight : str
    Key into :data:`WEIGHTS` for the outline.
stones : float
    If greater than zero, individual stones of this nominal diameter in
    metres are drawn inside the polygon. This is what makes a rubble
    mound read as rubble rather than as a shaded wedge.
```

MATERIALS

```
MATERIALS = {'armour': Material('Primary armour', '#9b978f', '#2f2d2a', stones=1.0), 'secondary': Material('Secondary armour',
'#b0aca3', '#3a3833', stones=0.5), 'underlayer': Material('Underlayer', '#c0bcb2', '#45433e', stones=0.35), 'core':
Material('Quarry run core', '#d5d1c6', '#54514b', hatch='....'), 'toe': Material('Toe protection', '#a8a49b', '#35332f',
stones=0.6), 'concrete': Material('Mass concrete', '#cfcbc4', '#2a2a2a', hatch='///', weight='heavy'), 'reinforced':
Material('Reinforced concrete', '#c4c0b8', '#1f1f1f', hatch='xx', weight='heavy'), 'blinding': Material('Blinding layer',
'#bdb9b1', '#45433e', hatch='\\\\\\\\'), 'rock_fill': Material('Rock fill', '#c9c5bb', '#4a4843', hatch='oo'), 'granular':
Material('Granular backfill', '#ddd2b6', '#6f6343', hatch='...'), 'sand': Material('Sand', '#e6d5ad', '#8a7648', hatch='....'),
'subgrade': Material('In-situ seabed', '#cdc0a6', '#6d6243', hatch='/////'), 'pavement': Material('Promenade surfacing', '#b8b4ae',
'#33322f'), 'water': Material('Water', '#c3dce7', '#4d7f97', weight='thin')}
```

use_crisp_style

```
def use_crisp_style() -> None
```

Global matplotlib settings for legible technical figures.

Arial throughout as the workspace requires, hairline spines, real black ink for text, and hatching thin enough to read at 600 dpi rather than filling in solid. Call it once at the top of a script.

Section

`class Section`

A cross-section drawing in real coordinates.

Parameters

title, subtitle : str

Drawing title and a one-line description.

figsize : tuple

Sheet size in inches.

ax : matplotlib Axes, optional

Draw into an existing axes, for a multi-panel comparison sheet. A new figure is made if this is None.

exaggeration : float

Vertical exaggeration. One is a true section, where a 1:2 slope looks like a 1:2 slope. Anything else distorts every angle on the drawing, and :meth:`fit\_scale` then reports separate horizontal and vertical scales and sets ``exaggeration\_note`` so the distortion is stated rather than hidden.

Notes

A section drawn at a true scale is the default, and the right choice whenever the structure is not far wider than it is tall. A navigation channel six hundred metres wide and twenty deep is the case where it is not: at true scale it is an unreadable sliver, and dredging drawings have always been exaggerated. Exaggerate deliberately, and say so.

method

`Section.material(self, points, kind: str, label: str | None=None, zorder: float=2.0)`

Fill a closed polygon with a material.

Parameters

points : sequence of (x, z)

Polygon vertices in metres. It is closed automatically.

kind : str

Key into :data:`MATERIALS`.

label : str or False, optional

Overrides the material name in the key, for a size callout such as "Rock armour, Dn50 = 1.45 m". Pass ``False`` to draw the polygon but keep it out of the key: a pair of abutments, or a row of piers, wants one entry rather than one each.

method

`Section.line(self, points, weight: str='medium', style: str='-', color: str | None=None, zorder: float=4.0, **kwargs)`

A construction, ground or hidden line.

method

`Section.water(self, x0: float, x1: float, level: float, bed=None, label: str | None=None, zorder: float=1.0)`

Fill water between a level and the bed, with the level symbol.

``bed`` is either a constant level or an (x, z) polyline. Water is drawn under everything else, so structures sit in it rather than floating on it.

method

```
Section.dim_h(self, x0: float, x1: float, z: float, text: str | None=None, extend_from: tuple[float, float] | None=None) -> None
```

Horizontal dimension between x0 and x1, drawn at level z.

```extend_from``` gives the two levels the extension lines run back to, so the dimension ties visibly to the geometry it measures.

#### method

```
Section.dim_v(self, z0: float, z1: float, x: float, text: str | None=None, extend_from: tuple[float, float] | None=None, side: str='right') -> None
```

Vertical dimension between z0 and z1, drawn at chainage x.

#### method

```
Section.level(self, x: float, z: float, text: str | None=None, side: str='left', symbol: str='level', run: float=0.0) -> None
```

Levelling triangle and elevation callout at (x, z).

```symbol``` is "level" for the solid surveyor's triangle used on structure levels, or "water" for the open triangle used on a water level. ```run``` draws a witness line of that length before the triangle, to pull the callout clear of the structure.

method

```
Section.slope(self, apex: tuple[float, float], cot_alpha: float, rise: float, direction: str='left', label: str | None=None) -> None
```

Slope triangle at a point on a face, labelled 1 : cot(alpha).

```apex``` is the upper point of the triangle on the slope face and ```rise``` its vertical leg. ```direction``` says which way the face falls away, so the triangle sits on the material side.

#### method

```
Section.note(self, xy: tuple[float, float], text: str, offset: tuple[float, float]=(40, 30), ha: str | None=None) -> None
```

Leader line with a note, the offset given in points.

#### method

```
Section.detail_bubble(self, label: str, title: str, scale: str='', loc: tuple[float, float]=(0.02, -0.06)) -> None
```

The drawing-office detail marker: circled letter, then the title.

```label``` is the single letter or number that a key plan points to, ```title``` the name of the view, and ```scale``` its drawn scale. Placed in axes coordinates, so it stays put when the geometry changes.

method

```
Section.notes_block(self, lines, title: str='NOTES', loc: tuple[float, float]=(0.015, 0.985), numbered: bool=True, width: int=52, fontsize: float | None=None) -> None
```

Boxed notes, numbered the way a specification note block is.

```lines``` is a sequence of strings, or of (heading, body) pairs for a note with a bold-looking lead-in. Long lines are wrapped to ```width``` characters so the block keeps a straight right edge.

#### method

```
Section.scale_bar(self, length: float, loc: tuple[float, float]=(0.03, 0.06), divisions: int=4, unit: str='m') -> None
```

Chequered scale bar, in data units, placed in axes coordinates.

A drawing without plot axes needs one of these, and a drawing with plot axes is better with one anyway: it survives being cropped, pasted into a report, or printed at the wrong size.

#### method

```
Section.key(self, loc: str='upper left', ncol: int=1)
```

Material key, in drawing order.

#### method

```
Section.table(self, rows, title: str='Design parameters', loc: tuple[float, float]=(0.985, 0.03), align: str='right', fontsize: float | None=None) -> None
```

Parameter block, as monospaced rows in axes coordinates.

```rows``` is a sequence of (label, value) pairs. Values are already formatted strings: the drawing shows what the engineer decided, not a float repr.

A row of ```(None, None)``` draws a rule instead, for separating what was measured from what was derived from it. A summary that lists three scour components and then their totals wants the reader to see at a glance which lines are which.

method

```
Section.auto_exaggeration(self, xlim, zlim, cap: float=200.0) -> float
```

Stretch the second axis just enough to fill this viewport.

A guessed exaggeration is nearly always wrong, and the cost is not cosmetic: `:meth:`fit_scale`` sizes the drawing to whichever axis is tighter, so an exaggeration a little too large throws the whole view onto the next scale up and leaves half the paper empty. The viewport already knows its own proportions, so let it do the arithmetic.

Returns the factor, and sets `:attr:`exaggeration`` to it.

method

```
Section.fit_scale(self, xlim, zlim, scales=None, paper: str='', round_vertical: bool=False) -> float
```

Set the view to a true, round drawing scale that fits the extents.

Picks the smallest standard scale at which the requested extents fit inside the axes, then centres the view on them and sets the limits to exactly that scale. The returned denominator is what belongs in the title block: a drawing whose stated scale is not the scale it was plotted at is worse than one with no scale at all.

Parameters

`xlim, zlim` : tuple

The extents that must be visible, in metres.

`scales` : sequence, optional

Candidate denominators. Defaults to `:data:`STANDARD_SCALES``.

`paper` : str

Paper size to name in the returned string, e.g. "A3".

`round_vertical` : bool

Snap the vertical scale to a standard denominator too, adjusting the exaggeration to suit. A section labelled "V 1:1210.83" cannot be scaled off the paper by anyone; "V 1:1250" can.

Returns

float

The scale denominator *S*, for a scale of 1 : *S*. Use

```scale_text``` for the string.

#### method

```
Section.finish(self, xlim=None, zlim=None, grid: bool=True) -> 'Section'
```

Set limits, titles and the background grid.

#### method

```
Section.save(self, path, dpi: int=600) -> None
```

Write the drawing as a PNG at print resolution.

#### method

```
Section.to_dxf(self, path, scale: float=1.0) -> None
```

Export the geometry as DXF, for import into a CAD package.

Only the material polygons and construction lines are exported, each on a layer named after its material. Dimensions, notes and the parameter block are left behind: they are drawing furniture, and a CAD user will want to place their own, to their own house style.

## PAPER

```
PAPER = {'A0': (1189.0, 841.0), 'A1': (841.0, 594.0), 'A2': (594.0, 420.0), 'A3': (420.0, 297.0), 'A4': (297.0, 210.0)}
```

## TitleBlock

```
class TitleBlock
 project: str = ''
 title: str = ''
 organisation: str = ''
 client: str = ''
 scale: str = 'AS SHOWN'
 date: str = ''
 file: str = ''
 sheet: str = '1/1'
 revision: str = 'A'
 drawn_by: str = ''
 checked_by: str = ''
 disclaimer: str = 'Do not scale from this drawing. All dimensions in metres unless noted. Levels to chart datum.'
 status: str = 'PRELIMINARY'
```

What goes in the strip down the right-hand edge of a sheet.

Every field is a plain string, because a title block records what a human decided, not what a calculation produced. Leave a field empty and its row is skipped.

#### Attributes

-----

```
project : str
 The job. Printed rotated down the strip, as on a real sheet.
title : str
 What this sheet shows.
organisation, client : str
 Who drew it and who for.
scale : str
 Drawn scale, for example "1:100 @ A3". Say the paper size: a scale
 without one is meaningless the moment the sheet is reprinted.
date, file, sheet, revision : str
 The usual issue record.
drawn_by, checked_by : str
 Initials.
disclaimer : str
 The small print set vertically in the strip.
```

## Sheet

```
class Sheet
```

A drawing sheet: border, title block strip, and one or more views.

#### Parameters

-----

```
titleblock : TitleBlock
```



```

 The strip down the right-hand edge.
size : str
 Key into :data:`PAPER`, or a (width, height) pair in millimetres.
dpi : int
 Raster resolution on save. A full A3 at 400 dpi is already 6600 px
 across, finer than the line work can carry; go higher only if the
 sheet is going to be enlarged.
margin : float
 Border inset in millimetres.
strip : float
 Width of the title block strip in millimetres.

Examples

>>> sheet = Sheet(TitleBlock(project="Harbour works", title="Seawall"))
>>> view = sheet.viewport() # doctest: +SKIP
>>> # draw into view, which is an ordinary Section
>>> sheet.save("seawall.png") # doctest: +SKIP

```

## method

```
Sheet.viewport(self, rect: tuple[float, float, float, float]=(0, 0, 1, 1), pad: float=0.02, **kwargs) -> 'Section'
```

Add a drawing view, positioned within the usable area.

`rect` is (left, bottom, width, height) as fractions of the drawing area, so (0, 0, 0.5, 1) is the left half of the sheet. Keyword arguments go to :class:`Section`.

## method

```
Sheet.set_scale_from(self, view: 'Section') -> None
```

Copy a view's fitted scale into the title block and redraw it.

Call it after :meth:`Section.fit\_scale` so the strip states the scale the drawing was actually plotted at.

## method

```
Sheet.save(self, path, dpi: int | None=None) -> None
```

Write the sheet. No bounding-box trim: the border is the edge.

## method

```
Sheet.to_dxf(self, path, scale: float=1.0) -> None
```

Export every view's geometry to one DXF.

## write\_dxf

```
def write_dxf(path, entities, scale: float=1.0) -> None
```

Write polylines to a minimal DXF R12 file.

`entities` is a sequence of `(kind, points, layer)`, where kind is "POLY" for a closed polygon or "LINE" for an open polyline. R12 LINE entities are used throughout: the format is ancient, which is exactly why every CAD package still reads it without complaint.

---

## pyCoastal.plotting

Plot helpers shared by the examples.

Importing this module needs matplotlib, which is not a runtime dependency of the package. Install it with the `plots` extra:

```
pip install pyCoastal[plots]
```

The colour maps here are built to look like water rather than to be perceptually uniform. For quantitative fields such as a disturbance coefficient use ``agitation\_colormap`` or a standard scientific map; keep ``water\_colormap`` for the instantaneous surface, where the point is to read the wave pattern at a glance.

## LAND\_COLOR

```
LAND_COLOR = '#6b6f73'
```

## water\_colormap

```
def water_colormap(name: str='pyCoastal_water') -> LinearSegmentedColormap
```

Diverging map for instantaneous surface elevation.

Troughs run to deep navy, still water sits at a mid ocean blue, and crests lift through turquoise to a pale foam white. Reading it as a photograph of the sea surface: dark water in the hollows, light broken water on the crests.

## agitation\_colormap

```
def agitation_colormap(name: str='pyCoastal_agitation') -> LinearSegmentedColormap
```

Sequential map for wave height or disturbance coefficient.

Calm water is dark and quiet; agitation brightens through green to a hot yellow, so the berths in trouble stand out.

## surface\_norm

```
def surface_norm(snapshots: np.ndarray, percentile: float=99.5) -> TwoSlopeNorm
```

Symmetric colour scale centred on still water.

The limit comes from a high percentile rather than the maximum, so a single sharp spike near a structure does not flatten the whole field.

## land\_overlay

```
def land_overlay(land: np.ndarray) -> np.ndarray
```

Float array that is 1.0 on structures and NaN on water.

Draw it over a field with a solid colour map so breakwaters read as material rather than as an extreme value of the field.

## Appendix B. Example scripts

The full source of every example in `examples/`, in the order of the chapters that use it. Run each from the repository root; the engineering examples write their figures to `media/`, the numerical examples animate on screen and read their parameters from `examples/configs/`.

### `examples/numerics/water_drop.py`

```
#!/usr/bin/env python3
"""
water_drop.py

2D linear wave equation:
 $\eta_{tt} = c^2 \nabla^2 \eta$

- Zero-Dirichlet BCs
- Initial Gaussian bump
- 2nd-order finite differences (explicit)
- Animated circular wave propagation
"""

from pathlib import Path

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

from pyCoastal.io import read_data
from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.numerics.operators import laplacian

1) Load config

The config sits beside this script, so the path is resolved from the
script rather than from the shell. The example then runs from the
repository root, which is where the README says to run it from.
CONFIG = Path(__file__).resolve().parent.parent / "configs"
MEDIA = Path(__file__).resolve().parents[2] / "media"
cfg = read_data(CONFIG / "water_drop.yaml")

Nx = cfg["grid"]["Nx"]
Ny = cfg["grid"]["Ny"]
Lx = cfg["grid"]["Lx"]
Ly = cfg["grid"]["Ly"]
dx, dy = Lx / Nx, Ly / Ny

c = cfg["physics"]["wave_speed"]
sigma = cfg["physics"]["sigma"]
CFL = cfg["solver"]["CFL"]
t_final = cfg["solver"]["t_final"]
dt = CFL * min(dx, dy) / c
nt = int(t_final / dt)

2) Build grid and initialize

grid = UniformGrid((Nx, Ny), (dx, dy))
X, Y = grid.Xc

eta = np.zeros((Nx, Ny))
eta_old = np.zeros_like(eta)
eta_new = np.zeros_like(eta)

Gaussian initial bump at center
x0, y0 = Lx / 2, Ly / 2
eta[:] = np.exp(-(X - x0)**2 + (Y - y0)**2) / (2 * sigma**2)
eta_old[:] = eta.copy()

3) Plot setup

fig, ax = plt.subplots(figsize=(6, 5))
pcm = ax.pcolormesh(X, Y, eta, cmap="RdBu_r", vmin=-0.5, vmax=0.5, shading="auto")
cb = fig.colorbar(pcm, ax=ax, label="η (m)")
title = ax.set_title("t = 0.00 s")
ax.set_xlabel("x")
ax.set_ylabel("y")

4) Time stepping

coef = (c * dt) ** 2
```

```

def update(frame):
 global eta, eta_old, eta_new
 t = frame * dt

 lap = laplacian(eta, grid)
 eta_new = 2 * eta - eta_old + coef * lap

 # Zero Dirichlet BCs
 eta_new[0, :] = 0.0
 eta_new[-1, :] = 0.0
 eta_new[:, 0] = 0.0
 eta_new[:, -1] = 0.0

 eta_old[:, :] = eta, eta_new

 pcm.set_array(eta.ravel())
 title.set_text(f"t = {t:.2f} s")
 return [pcm, title]

ani = animation.FuncAnimation(
 fig, update, frames=nt, interval=30, blit=True, repeat=False
)

plt.tight_layout()
plt.show()

Save the image of the central timestep
central_frame = nt // 2
for frame in range(central_frame):
 update(frame)
plt.imsave(MEDIA / "water_drop_central.png", eta, cmap="RdBu_r", vmin=-0.5, vmax=0.5)

Save the whole plot with coordinates
fig.savefig(MEDIA / "water_drop_central_plot.png")

```

## examples/waves2D.py

```
#!/usr/bin/env python3
"""
waves2D.py

2D depth-averaged wave propagation model with observation points.

- Uniform water depth h
- Solves 1D linear advection in y for each x -column
- Periodic in x , wave forcing at $y=0$ boundary
- Forward Euler in time, upwind in space
- Real-time animation of the 2D η -field and time-series at specified gauges

Parameters are loaded from a YAML config file.
"""

from pathlib import Path

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation
from pyCoastal.io import read_data
from pyCoastal.numerics.grid import UniformGrid

1) Load YAML configuration

The config sits beside this script, so the path is resolved from the
script rather than from the shell. The example then runs from the
repository root, which is where the README says to run it from.
CONFIG = Path(__file__).resolve().parent / "configs"
cfg = read_data(CONFIG / "waves2D.yaml")

Grid parameters
Nx = cfg["grid"]["nx"]
Ny = cfg["grid"]["ny"]
dx = cfg["grid"]["dx"]
dy = cfg["grid"]["dy"]

Physical parameters
g = cfg["physics"]["gravity"]
h = cfg["physics"]["depth"]
c = np.sqrt(g * h)

Solver parameters
CFL = cfg["solver"]["cfl"]
dt = CFL * dy / c
t_final = cfg["solver"]["t_final"]
nt = int(t_final / dt)

Forcing parameters
A = cfg["forcing"]["amplitude"]
T = cfg["forcing"]["period"]
omega = 2 * np.pi / T

Observation points
obs_points = [tuple(p) for p in cfg["output"]["obs_points"]]

2) Build the computational grid

grid = UniformGrid(shape=(Nx, Ny), spacing=(dx, dy))
X, Y = grid.Xc

3) Initialize fields and time series

eta = np.zeros(grid.shape)
eta_new = np.zeros_like(eta)

time_series = {pt: [] for pt in obs_points}
time_axis = []

4) Set up the figure and axes

fig = plt.figure(figsize=(10, 5))

η -field panel
ax_field = fig.add_subplot(1, 2, 1)
cax = ax_field.pcolormesh(X, Y, eta, cmap='viridis', vmin=-A, vmax=+A)
ax_field.set_title('η(x, y)')
ax_field.set_xlabel('x (m)')
ax_field.set_ylabel('y (m)')
plt.colorbar(cax, ax=ax_field, label='η (m)')

Time-series panel
ax_ts = fig.add_subplot(1, 2, 2)
```

```

ax_ts.set_xlim(0, t_final)
ax_ts.set_ylim(-1.1 * A, 1.1 * A)
ax_ts.set_title('Time series at gauges')
ax_ts.set_xlabel('t (s)')
ax_ts.set_ylabel('η (m)')

lines = {}
for pt in obs_points:
 ix, iy = pt
 x_real = grid.Xc[0][ix, iy]
 y_real = grid.Xc[1][ix, iy]
 label = f'({x_real:.1f} m, {y_real:.1f} m)'
 line, = ax_ts.plot([], [], label=label)
 lines[pt] = line

ax_ts.legend(loc='upper right')

5) Animation update function

def update(frame):
 global eta, eta_new

 t = frame * dt
 time_axis.append(t)

 # Dirichlet forcing at y=0
 eta_new[:, 0] = A * np.sin(omega * t)

 # Upwind advection along y
 eta_new[:, 1:] = eta[:, 1:] - (c * dt / dy) * (eta[:, 1:] - eta[:, :-1])

 # Swap arrays
 eta, eta_new = eta_new, eta

 # Record gauges
 for pt in obs_points:
 ix, iy = pt
 time_series[pt].append(eta[ix, iy])

 # Update η-field
 cax.set_array(eta.ravel())

 # Update time-series lines
 for pt, line in lines.items():
 line.set_data(time_axis, time_series[pt])

 return [cax] + list(lines.values())

6) Run the animation

ani = animation.FuncAnimation(
 fig, update, frames=nt,
 interval=50, blit=True, repeat=False
)

plt.tight_layout()
plt.show()

```



## examples/wave2D\_irregular.py

```
#!/usr/bin/env python3
"""
waves2D_irregular.py

2D depth-averaged wave model with irregular forcing at $y=0$.
Uses pyCoastal:
- UniformGrid for spatial mesh
- generate_irregular_wave for $\eta(t)$ boundary condition
- YAML config for setup
"""

from pathlib import Path

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

from pyCoastal.io import read_data
from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.tools.wave import generate_irregular_wave

1) Load configuration

The config sits beside this script, so the path is resolved from the
script rather than from the shell. The example then runs from the
repository root, which is where the README says to run it from.
CONFIG = Path(__file__).resolve().parent / "configs"
cfg = read_data(CONFIG / "waves2D_irregular.yaml")

Nx, Ny = cfg["grid"]["nx"], cfg["grid"]["ny"]
dx, dy = cfg["grid"]["dx"], cfg["grid"]["dy"]
g = cfg["physics"]["gravity"]
h = cfg["physics"]["depth"]
c = np.sqrt(g * h)

spectrum_type = cfg["forcing"]["type"]
gamma = cfg["forcing"].get("gamma", 3.3)
Hs = cfg["forcing"]["Hs"]
Tp = cfg["forcing"]["Tp"]

dt = cfg["solver"]["dt"]
duration = cfg["solver"]["duration"]
nt = int(duration / dt)

obs_point = tuple(cfg["output"]["gauge"])

2) Create grid

grid = UniformGrid((Nx, Ny), (dx, dy))
X, Y = grid.Xc

3) Generate wave boundary forcing at $y=0$

t_vec, eta_bc = generate_irregular_wave(
 Hs=Hs, Tp=Tp,
 duration=duration,
 dt=dt,
 spectrum=spectrum_type,
 gamma=gamma
)

4) Initialize fields and storage

eta = np.zeros((Nx, Ny))
eta_new = np.zeros_like(eta)

ts_obs = []
times = []

5) Set up plots

fig = plt.figure(figsize=(10, 4))

Field panel
ax1 = fig.add_subplot(1, 2, 1)
pcm = ax1.pcolormesh(X, Y, eta, cmap="viridis", vmin=-Hs, vmax=Hs)
ax1.set_title(" $\eta(x, y)$ ")
ax1.set_xlabel("x (m)")
ax1.set_ylabel("y (m)")
fig.colorbar(pcm, ax=ax1, label=" η (m)")

Time-series panel
```

```

ax2 = fig.add_subplot(1, 2, 2)
ax2.set_xlim(0, duration)
ax2.set_ylim(-1.1 * Hs, 1.1 * Hs)
line, = ax2.plot([], [], "k-")
ax2.set_title(f" η at ({obs_point[0]*dx:.0f} m, {obs_point[1]*dy:.0f} m)")
ax2.set_xlabel("t (s)")
ax2.set_ylabel(" η (m)")

6) Animation update

def update(n):
 global eta, eta_new

 t = n * dt
 times.append(t)

 # Dirichlet BC at y = 0
 eta_new[:, 0] = eta_bc[n] if n < len(eta_bc) else 0.0

 # Upwind advection
 eta_new[:, 1:] = eta[:, 1:] - (c * dt / dy) * (eta[:, 1:] - eta[:, :-1])

 eta, eta_new = eta_new, eta

 ts_obs.append(eta[obs_point])
 line.set_data(times, ts_obs)
 pcm.set_array(eta.ravel())

 return [pcm, line]

7) Run animation

ani = animation.FuncAnimation(
 fig, update, frames=nt,
 interval=50, blit=True, repeat=False
)

plt.tight_layout()
plt.show()

```

## examples/current.py

```
#!/usr/bin/env python3
"""
current.py

2D passive x-direction current advection with:
- Dirichlet inlet (west), Neumann outlet (east)
- UniformGrid mesh
- Upwind scheme for $u_t + c \cdot u_x = 0$
- Real-time animation + time series at a gauge
- Parameters from YAML config
"""

from pathlib import Path

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

from pyCoastal.io import read_data
from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.numerics.boundary import DirichletBC, NeumannBC, BoundaryManager

1) Load config

The config sits beside this script, so the path is resolved from the
script rather than from the shell. The example then runs from the
repository root, which is where the README says to run it from.
CONFIG = Path(__file__).resolve().parent / "configs"
cfg = read_data(CONFIG / "current.yaml")

Nx, Ny = cfg["grid"]["nx"], cfg["grid"]["ny"]
dx, dy = cfg["grid"]["dx"], cfg["grid"]["dy"]
c0 = cfg["physics"]["speed"]
CFL = cfg["solver"]["cfl"]
t_final = cfg["solver"]["t_final"]
dt = CFL * dx / c0
nt = int(t_final / dt)
ix, iy = tuple(cfg["output"]["gauge"])

2) Build grid and initialize fields

grid = UniformGrid((Nx, Ny), (dx, dy))
X, Y = grid.Xc
u = np.zeros((Nx, Ny))
u_new = np.zeros_like(u)
fields = {"u": u}

3) Boundary conditions

bc_mgr = BoundaryManager()
bc_mgr.add(DirichletBC("west", ["u"], value=c0)) # Inlet
bc_mgr.add(NeumannBC("east", ["u"], gradient=0.0)) # Outlet

4) Plot setup

fig = plt.figure(figsize=(10, 4))
ax1 = fig.add_subplot(1, 2, 1)
pcm = ax1.pcolormesh(X, Y, u, cmap='viridis', vmin=0, vmax=c0)
ax1.set_title("u field")
ax1.set_xlabel("x")
ax1.set_ylabel("y")
fig.colorbar(pcm, ax=ax1, label="u (m/s)")

ax2 = fig.add_subplot(1, 2, 2)
ax2.set_xlim(0, t_final)
ax2.set_ylim(0, 1.1 * c0)
line_ts, = ax2.plot([], [], "k-")
ax2.set_title(f"u at ({ix}, {iy})")
ax2.set_xlabel("t (s)")
ax2.set_ylabel("u (m/s)")

times = []
ts_obs = []

5) Update function

def update(n):
 global u, u_new
 t = n * dt
 times.append(t)

 bc_mgr.apply_all(fields, grid, t)
```

```

u_new[1:, :] = u[1:, :] - (c0 * dt / dx) * (u[1:, :] - u[:-1, :])
u_new[0, :] = u[0, :]
u_new[-1, :] = u[-1, :] # Neumann handled explicitly via copy

u, u_new = u_new, u
fields["u"] = u

ts_obs.append(u[ix, iy])
line_ts.set_data(times, ts_obs)

pcm.set_array(u.ravel())
return [pcm, line_ts]

6) Animate

ani = animation.FuncAnimation(
 fig, update, frames=nt,
 interval=50, blit=False, repeat=False
)

plt.tight_layout()
plt.show()

```

## examples/pollutant.py

```
#!/usr/bin/env python3
"""
pollutant.py

2D pollutant advection-diffusion in a shallow pond:
- UniformGrid from pyCoastal
- Operators: gradient, Laplacian
- Upwind advection, central diffusion
- Neumann (zero-flux) boundary conditions
- Animated visualization of scalar field
"""

from pathlib import Path

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

from pyCoastal.io import read_data
from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.numerics.operators import gradient, laplacian

1) Load config

The config sits beside this script, so the path is resolved from the
script rather than from the shell. The example then runs from the
repository root, which is where the README says to run it from.
CONFIG = Path(__file__).resolve().parent / "configs"
cfg = read_data(CONFIG / "pollutant.yaml")

Domain and grid
Lx = cfg["grid"]["Lx"]
Ly = cfg["grid"]["Ly"]
Nx = cfg["grid"]["Nx"]
Ny = cfg["grid"]["Ny"]
dx, dy = Lx / Nx, Ly / Ny
grid = UniformGrid((Nx, Ny), (dx, dy))
X, Y = grid.Xc

Physics
D = cfg["physics"]["diffusivity"]
U0 = cfg["physics"]["U0"]
R = cfg["physics"]["R"]

Solver
dt = 0.5 * min(dx, dy) / U0
t_final = cfg["solver"]["t_final"]
nt = int(t_final / dt)
sigma = cfg["solver"]["c0_sigma"]

2) Prescribed vortex velocity field

x0, y0 = Lx / 2, Ly / 2
u = U0 * (Y - y0) / R * np.exp(-((X - x0)**2 + (Y - y0)**2) / R**2)
v = -U0 * (X - x0) / R * np.exp(-((X - x0)**2 + (Y - y0)**2) / R**2)

3) Initial Gaussian concentration at center

c = np.exp(-((X - x0)**2 + (Y - y0)**2) / sigma**2)
c_old = c.copy()
c_new = np.zeros_like(c)

4) Plot setup

fig, ax = plt.subplots(figsize=(6, 3))
pcm = ax.pcolormesh(X, Y, c, cmap="viridis", vmin=0, vmax=1)
title = ax.set_title("t = 0.00 s")
ax.set_xlabel("x (m)")
ax.set_ylabel("y (m)")
cb = fig.colorbar(pcm, ax=ax, label="C")

5) Update function

def update(frame):
 global c, c_old, c_new

 cx, cy = gradient(c, grid)
 adv = u * cx + v * cy
 diff = D * laplacian(c, grid)
 c_new[:] = c + dt * (-adv + diff)

 # Zero-flux (Neumann) boundaries
```

```

c_new[0, :] = c_new[1, :]
c_new[-1, :] = c_new[-2, :]
c_new[:, 0] = c_new[:, 1]
c_new[:, -1] = c_new[:, -2]

c_old, c = c, c_new

pcm.set_array(c.ravel())
title.set_text(f"t = {frame*dt:.2f} s")
return [pcm, title]

6) Run animation

ani = animation.FuncAnimation(
 fig, update, frames=nt,
 interval=30, blit=False, repeat=False
)

plt.tight_layout()
plt.show()

```



## examples/numerics/viscous\_fluid.py

```
#!/usr/bin/env python3
"""
viscous_fluid.py

2D incompressible Navier-Stokes simulation with:
- Two counter-rotating vortices
- Central-difference convection and diffusion
- Forward Euler time integration
- Periodic boundary conditions
- Real-time animation of speed field
"""

from pathlib import Path

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

from pyCoastal.io import read_data
from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.numerics.boundary import BoundaryManager
from pyCoastal.physics.navier_stokes import initialize_state, rhs

The config sits beside this script, so the path is resolved from the script
rather than from the shell. The example then runs from the repository root,
which is where the README says to run it from.
CONFIG = Path(__file__).resolve().parent.parent / "configs"

def main():
 # -----
 # 1) Load configuration
 # -----
 cfg = read_data(CONFIG / "viscous_fluid.yaml")

 Nx = cfg["grid"]["Nx"]
 Ny = cfg["grid"]["Ny"]
 Lx = cfg["grid"]["Lx"]
 Ly = cfg["grid"]["Ly"]
 dx, dy = Lx / (Nx - 1), Ly / (Ny - 1)
 grid = UniformGrid((Nx, Ny), (dx, dy))
 X, Y = grid.Xc

 v = cfg["physics"]["viscosity"]
 U0 = cfg["physics"]["background_speed"]
 sigma = cfg["physics"]["vortex_sigma"]

 dt = cfg["solver"]["dt"]
 t_final = cfg["solver"]["t_final"]
 nt = int(t_final / dt)

 # -----
 # 2) Initial state: uniform + vortex perturbation
 # -----
 bc_mgr = BoundaryManager() # periodic
 state = initialize_state(grid)
 state["u"] += U0

 xc, yc = Lx / 2, Ly / 2
 r2 = (X - xc) ** 2 + (Y - yc) ** 2
 vortex = np.exp(-r2 / sigma ** 2)
 state["u"] += -(Y - yc) * vortex
 state["v"] += (X - xc) * vortex

 # -----
 # 3) Plot setup
 # -----
 fig, ax = plt.subplots()
 speed = np.sqrt(state["u"]**2 + state["v"]**2)
 pcm = ax.pcolormesh(X, Y, speed, cmap="inferno", shading="auto")
 fig.colorbar(pcm, ax=ax, label="speed")
 ax.set_aspect("equal")
 title = ax.set_title("t = 0.00 s")

 # -----
 # 4) Update function
 # -----
 def update(frame):
 nonlocal state
 t = frame * dt

 # RHS from convection + diffusion
 R = rhs(state, t, grid, bc_mgr, v=v)

 # Forward Euler step
 state["u"] += dt * R["u"]
 state["v"] += dt * R["v"]
```

```

Enforce mean U0 drift (prevent bulk slowing)
state["u"] -= (state["u"].mean() - U0)

Update plot
speed = np.sqrt(state["u"]**2 + state["v"]**2)
pcm.set_array(speed.ravel())
title.set_text(f"t = {t:.2f} s")
return [pcm, title]

ani = animation.FuncAnimation(
 fig, update, frames=nt, interval=30, blit=True, repeat=False
)
plt.tight_layout()
plt.show()

if __name__ == "__main__":
 main()

```

## examples/numerics/2D\_irr\_turb.py

```
examples/2d_bc_test.py

import numpy as np
import matplotlib.pyplot as plt

from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.numerics.boundary import (
 DirichletBC, NeumannBC, WallBC, SpongeBC, BoundaryManager
)

def main():
 # 1) Build a 400x400 m domain with 200x200 cells (dx=dy=2 m)
 grid = UniformGrid(shape=(200, 200), spacing=(2.0, 2.0))

 # 2) Initialize fields: eta, u, v
 eta = np.zeros(grid.shape)
 u = np.zeros(grid.shape)
 v = np.zeros(grid.shape)
 fields = {"eta": eta, "u": u, "v": v}

 # 3) Set up our BC manager
 bc_mgr = BoundaryManager()

 # 3a) Southern Dirichlet on η : $0.5 \text{ m} \times \sin(2\pi t/3\text{s})$
 w = 2*np.pi/3.0
 bc_mgr.add(DirichletBC(
 "south", ["eta"], lambda t: 0.5*np.sin(w*t)
))

 # 3b) Northern sponge: 1-cell-wide, linear α from 1 to 0
 def north_damp(t, idx):
 # idx is a flat index; convert to row i, j
 i = idx // grid.shape[1]
 # one-cell wide => either i==199
 # but we'll demonstrate spatially-varying over 20 cells:
 ramp = 20
 offset = grid.shape[0] - ramp
 $\alpha = 1.0 - \text{np.clip}((i - \text{offset})/\text{ramp}, 0.0, 1.0)$
 return α
 bc_mgr.add(SpongeBC("north", ["eta", "u", "v"], damping=north_damp))

 # 3c) West inflow: $u=0.2 \text{ m/s}$, zero-gradient η
 bc_mgr.add(DirichletBC("west", ["u"], value=0.2))
 bc_mgr.add(NeumannBC("west", ["eta"], gradient=0.0))

 # 3d) East outflow: zero-gradient u, η, v
 bc_mgr.add(NeumannBC("east", ["u", "eta", "v"], gradient=0.0))

 # 3e) Side walls on south & north for v
 bc_mgr.add(WallBC("south", ["v"]))
 bc_mgr.add(WallBC("north", ["v"]))

 # 4) Time-Loop: no physics, just BCs + record gauge
 t0, t1, dt = 0.0, 60.0, 0.1
 nt = int((t1-t0)/dt)
 times = np.linspace(t0, t1, nt+1)

 # gauge at center of domain
 ci = grid.shape[0]/2
 cj = grid.shape[1]/2
 flat_gauge = ci*grid.shape[1] + cj
 gauge = []

 for n,t in enumerate(times):
 # apply all BCs in place
 bc_mgr.apply_all(fields, grid, t)

 # no physics update (pure BC test)
 # record eta at gauge
 gauge.append(fields["eta"].flat[flat_gauge])

 if n%100==0:
 print(f"t = {t:.1f}s southern $\eta(\text{row}=0) \approx \{\text{fields['eta'].flat[cj]}.3\text{f}\} \text{ m}")$)

 # 5) plot
 plt.figure(figsize=(8,3))
 plt.plot(times, gauge, 'b-')
 plt.xlabel("time [s]")
 plt.ylabel(" η at center [m]")
 plt.title("BC-test: Southern sine wave + north sponge")
 plt.grid(True)
 plt.tight_layout()
 plt.show()
```

```
if __name__ == "__main__":
 main()
```

## examples/equilibrium\_shoreline.py

```
#!/usr/bin/env python3
"""
Equilibrium shoreline built from three arcs:
- Left lateral arc (shoreline -> inner toe T1 -> bay point S0)
- central arc (T2 -> apex S -> T3)
- right lateral arc (T4 -> bay point S1 -> shoreline)

Inputs are read from examples/configs/equilibrium_shoreline.yaml.
Computation uses helper utilities in pyCoastal.tools.morphodynamics.
"""

import os
import sys
import numpy as np
import matplotlib.pyplot as plt

Prefer local checkout
THIS_DIR = os.path.dirname(__file__)
REPO_ROOT = os.path.abspath(os.path.join(THIS_DIR, ".."))
if REPO_ROOT not in sys.path:
 sys.path.insert(0, REPO_ROOT)

from pyCoastal.io import read_data
from pyCoastal.tools import (
 build_tombolo_arcs,
 accretion_metrics,
 two_point_arc,
 circle_through_points,
)

def main():
 cfg = read_data(os.path.join(THIS_DIR, "configs", "equilibrium_shoreline.yaml"))

 Lshore = cfg["geometry"]["Lshore"]
 Gb = cfg["geometry"]["Gb"]
 Lb1 = cfg["geometry"]["Lb1"]
 Lb2 = cfg["geometry"]["Lb2"]
 yb = cfg["geometry"]["Yi"]
 n_pts = cfg["numerics"]["n_points"]
 sag_side_factor = cfg["numerics"]["sag_side_factor"]

 if Gb > 0 and yb / Gb > 1.4:
 msg = "Breakwater is too far from shore; no equilibrium profile will be reached. set a lower Yi or increase Gb."
 print(msg)
 return

 x_eq, y_eq, geom = build_tombolo_arcs(
 Lshore=Lshore,
 Gb=Gb,
 Lb1=Lb1,
 Lb2=Lb2,
 yb=yb,
 n_pts=n_pts,
 sag_side_factor=sag_side_factor,
)

 # Rebuild individual arcs for overlap highlighting
 sag_side = sag_side_factor * geom["Ye"]
 xL, yL = two_point_arc(geom["S0"], geom["T1"], sag_side, n=n_pts)
 xCen, yCen = circle_through_points(geom["T2"], geom["S"], geom["T3"], n=n_pts)
 xR, yR = two_point_arc(geom["T4"], geom["S1"], sag_side, n=n_pts)

 retreat_val, acc_pct = accretion_metrics(x_eq, y_eq, geom)

 fig, ax = plt.subplots(figsize=(12, 4.5), dpi=200)

 y_min = min(0.0, y_eq.min()) - 0.1 * abs(y_eq.min() - y_eq.max())
 y_max = max(y_eq.max(), geom["Yi"] + 0.5 * max(Lb1, Lb2))
 ax.set_ylim(y_min, y_max)
 ax.fill_between(x_eq, y_min, y_eq, color="#eed9b7", alpha=0.6, zorder=0.4) # sand
 ax.fill_between(x_eq, y_eq, y_max, color="#c2e0ff", alpha=1, zorder=0.5) # water
 ax.fill_between([geom["S0"][0], geom["S1"][0]], y_min, 0.0, color="#d6c09c", alpha=0.5, zorder=1.8)

 # Overlap fill on top
 def overlap_fill(ax, x1, y1, x2, y2, color="#c2e0ff", alpha=1, zorder=6):
 idx1 = np.argsort(x1); idx2 = np.argsort(x2)
 x1s, y1s = np.array(x1)[idx1], np.array(y1)[idx1]
 x2s, y2s = np.array(x2)[idx2], np.array(y2)[idx2]
 xmin, xmax = max(x1s.min(), x2s.min()), min(x1s.max(), x2s.max())
 if xmax <= xmin:
 return
 xs = np.linspace(xmin, xmax, 300)
 y1i = np.interp(xs, x1s, y1s)
 y2i = np.interp(xs, x2s, y2s)
 ax.fill_between(xs, np.minimum(y1i, y2i), np.maximum(y1i, y2i),
```

```

 color=color, alpha=alpha, zorder=zorder)

overlap_fill(ax, xL, yL, xCen, yCen)
overlap_fill(ax, xCen, yCen, xR, yR)

Initial shoreline y=0
ax.hlines(0.0, geom["S0"][0], geom["S1"][0], linestyle="--", linewidth=0.5, color="C0",
 label="Original Shoreline", zorder=2)

Equilibrium shoreline (arc)
ax.plot(x_eq, y_eq, "r--", linewidth=0.2, label="Equilibrium shoreline", zorder=3)

Breakwaters on top
for x_b, Lb in [(geom["x_b1"], Lb1), (geom["x_b2"], Lb2)]:
 x0 = x_b - 0.5 * Lb
 y0_rect = geom["Yi"] - 2.0
 rect = plt.Rectangle((x0, y0_rect), Lb, 4.0,
 facecolor="tan", edgecolor="k", zorder=5)
 ax.add_patch(rect)

ax.set_xlim(geom["S0"][0], geom["S1"][0])
ax.set_aspect("equal", adjustable="box")
ax.set_xlabel("x (m)")
ax.set_ylabel("y (m)")
ax.set_title("Equilibrium shoreline")
ax.legend(loc="upper right")

Dimensional annotations
span = y_max - y_min
y_annot = geom["Yi"] + 0.08 * span

Breakwater Lengths
bw1_x0, bw1_x1 = geom["x_b1"] - 0.5 * Lb1, geom["x_b1"] + 0.5 * Lb1
bw2_x0, bw2_x1 = geom["x_b2"] - 0.5 * Lb2, geom["x_b2"] + 0.5 * Lb2
ax.annotate("", xy=(bw1_x1, y_annot), xytext=(bw1_x0, y_annot),
 arrowprops=dict(arrowstyle="<->", color="k", linewidth=0.8))
ax.text(0.5 * (bw1_x0 + bw1_x1), y_annot + 0.01 * span, f"Lb1 = {Lb1:.1f} m",
 ha="center", va="bottom", fontsize=8, color="k")
ax.annotate("", xy=(bw2_x1, y_annot), xytext=(bw2_x0, y_annot),
 arrowprops=dict(arrowstyle="<->", color="k", linewidth=0.8))
ax.text(0.5 * (bw2_x0 + bw2_x1), y_annot + 0.01 * span, f"Lb2 = {Lb2:.1f} m",
 ha="center", va="bottom", fontsize=8, color="k")

Gap between inner toes
y_gap = y_annot + 0.06 * span
gb_start = geom["x_b1"] + 0.5 * Lb1
gb_end = geom["x_b2"] - 0.5 * Lb2
ax.annotate("", xy=(gb_end, y_gap), xytext=(gb_start, y_gap),
 arrowprops=dict(arrowstyle="<->", color="k", linewidth=0.8))
ax.text(0.5 * (gb_start + gb_end), y_gap + 0.01 * span, f"Gb = {Gb:.1f} m",
 ha="center", va="bottom", fontsize=8, color="k")

Yi at Left breakwater center
ax.annotate("", xy=(geom["x_b1"], geom["Yi"]), xytext=(geom["x_b1"], 0.0),
 arrowprops=dict(arrowstyle="<->", color="k", linewidth=0.8))
ax.text(geom["x_b1"], 0.5 * geom["Yi"], f"Yi = {geom['Yi']:.1f} m",
 ha="right", va="center", fontsize=8, color="k")

Ye at apex
ax.annotate("", xy=(geom["S"][0], geom["Yi"]), xytext=(geom["S"][0], geom["S"][1]),
 arrowprops=dict(arrowstyle="<->", color="k", linewidth=0.8))
ax.text(geom["S"][0], geom["S"][1] + 0.5 * (geom["Yi"] - geom["S"][1]),
 f"Ye = {geom['Ye']:.1f} m", ha="left", va="center", fontsize=8, color="k")

ax.text(
 0.02, 0.97,
 f"Central retreat = {retreat_val:.2f} m\nAccretion = {acc_pct:.1f}%",
 transform=ax.transAxes,
 ha="left", va="top",
 fontsize=10,
 bbox=dict(boxstyle="round", facecolor="wheat", alpha=0.8)
)

plt.tight_layout()
plt.show()

if __name__ == "__main__":
 main()

```

## examples/design\_wave.py

```

"""
From a wave record to a design wave, and on to a structure.

Takes a 40-year synthetic hindcast of significant wave height, chooses a
storm threshold with the standard diagnostics, fits a peaks-over-threshold
model, cross-checks it against annual maxima, and hands the 100-year wave to
the seawall designer.

Because the record is synthetic, the right answer is known, so the example
also does something a real study cannot: it repeats the whole exercise on
many independent 40-year records and measures how much the 100-year estimate
moves. That spread is the uncertainty the bootstrap band is trying to
capture, and comparing the two says whether the band can be believed.

Run from the repository root:

 python examples/design_wave.py
"""

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.extremes import (
 decluster,
 fit_block_maxima,
 fit_gpd,
 fit_pot,
 gpd_return_value,
 mean_residual_life,
 plotting_positions,
 threshold_stability,
)
from pyCoastal.applications.seawall import design_seawall
from pyCoastal.applications.structures import DesignConditions
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

PER_YEAR = 2920 # three-hourly
YEARS = 40
N = PER_YEAR * YEARS

The climate the synthetic record is drawn from. Storm peaks sit above
2.5 m with a generalised Pareto excess, so every fitted number below has
something true to be compared against.
TRUE_THRESHOLD, TRUE_SCALE, TRUE_SHAPE, TRUE_RATE = 2.5, 0.55, -0.08, 9.0
SEPARATION = 24 # 3 days below the threshold ends a storm
THRESHOLD = 2.5
DESIGN_PERIOD = 100.0

def synthetic_record(seed: int):
 """A 40-year three-hourly Hm0 record with a known storm climate.

 Seasonal swell background, with storms laid over it by a maximum rather
 than added to it, so the peak of the record really is the threshold plus
 a generalised Pareto excess.
 """
 rng = np.random.default_rng(seed)
 t = np.arange(N) / PER_YEAR
 background = (1.1 + 0.45 * np.sin(2 * np.pi * (t - 0.08))
 + 0.18 * rng.standard_normal(N))
 record = np.clip(background, 0.15, None)

 starts = np.sort(rng.integers(0, N - 60, size=rng.poisson(TRUE_RATE * YEARS)))
 starts = starts[np.concatenate([True], np.diff(starts) > 120)]]
 u = rng.random(starts.size)
 excess = TRUE_SCALE / TRUE_SHAPE * ((1 - u) ** (-TRUE_SHAPE) - 1)

 for start, value in zip(starts, excess):
 width = int(rng.integers(8, 26))
 envelope = np.exp(-((np.arange(2 * width) - width) / (0.55 * width)) ** 2)
 end = min(start + 2 * width, N)
 record[start:end] = np.maximum(
 record[start:end], (TRUE_THRESHOLD + value) * envelope[:end - start]
)
 return record, t, starts.size / YEARS

Seed chosen so this record sits near the middle of what 40 years can
produce, rather than at a flattering or an alarming tail. The repeat study
further down reports where it actually falls.
record, t, actual_rate = synthetic_record(seed=20260959)
print(f"Record: {YEARS} years, {N} samples, max {record.max():.2f} m, "
 f"{actual_rate * YEARS:.0f} storms ({actual_rate:.1f}/yr)")

--- threshold selection -----

```



```

levels = np.arange(1.8, 3.8, 0.05)
mr1 = mean_residual_life(record, levels)
stab = threshold_stability(record, levels, separation=SEPARATION,
 samples_per_year=PER_YEAR)
print(f"Threshold {THRESHOLD:.2f} m chosen from the diagnostics "
 f"(the record was built with {TRUE_THRESHOLD:.2f} m)")

--- fit -----
pot = fit_pot(record, THRESHOLD, separation=SEPARATION,
 samples_per_year=PER_YEAR, bootstrap=800)
gev = fit_block_maxima(record, block=PER_YEAR, samples_per_year=PER_YEAR,
 bootstrap=800)

print("\nPeaks over threshold")
print(pot.summary())
print(f" the record was built with scale {TRUE_SCALE:.3f}, "
 f" shape {TRUE_SHAPE:+.3f}, rate {actual_rate:.1f}/yr")
print("\nAnnual maxima, as a cross-check")
print(gev.summary())

Hs100 = float(np.atleast_1d(pot.return_value(DSIGN_PERIOD))[0])
band = pot.confidence(DSIGN_PERIOD)
truth = float(gpd_return_value(TRUE_THRESHOLD, TRUE_SCALE, TRUE_SHAPE,
 actual_rate, DESIGN_PERIOD))
print(f"\n{DESIGN_PERIOD:.0f}-year Hm0 = {Hs100:.2f} m "
 f"({band['lower'][0]:.2f} to {band['upper'][0]:.2f} m, 90% bootstrap)")
print(f" true value for this climate: {truth:.2f} m")
print(f" {pot.extrapolation_note(DESIGN_PERIOD)}")

--- how much would another 40 years have changed the answer? -----
A real study gets one record and one number. Here the climate is known, so
the same 40-year study can be repeated on fresh records and the spread of
the answer measured directly.
REPEATS = 60
estimates = []
for seed in range(1000, 1000 + REPEATS):
 other, _, rate = synthetic_record(seed)
 peaks, _ = decluster(other, THRESHOLD, SEPARATION)
 if peaks.size < 30:
 continue
 scale, shape = fit_gpd(peaks - THRESHOLD)
 estimates.append(float(gpd_return_value(THRESHOLD, scale, shape,
 peaks.size / YEARS, DESIGN_PERIOD)))

estimates = np.array(estimates)
spread = np.percentile(estimates, [5, 95])
bootstrap_width = band["upper"][0] - band["lower"][0]

print(f"\nRepeating the study on {estimates.size} fresh {YEARS}-year records")
print(f" 100-year estimate: mean {estimates.mean():.2f} m, "
 f" sd {estimates.std(ddof=1):.2f} m")
print(f" 5th to 95th percentile: {spread[0]:.2f} to {spread[1]:.2f} m "
 f" (width {spread[1] - spread[0]:.2f} m)")
print(f" bootstrap band from the single record: width "
 f" {bootstrap_width:.2f} m")
print(f" this record sits at the {100 * (estimates < Hs100).mean():.0f}th "
 f" percentile of that spread")
if bootstrap_width > 1.15 * (spread[1] - spread[0]):
 print(" The bootstrap band is wider than the real sampling spread: "
 " \"conservative here.\")
elif bootstrap_width < 0.85 * (spread[1] - spread[0]):
 print(" The bootstrap band is narrower than the real sampling spread: "
 " \"it understates the uncertainty.\")
else:
 print(" The bootstrap band matches the real sampling spread.")

--- the structure that follows -----
The design wave at the toe is the offshore value reduced by nearshore
transformation. Here the toe is taken as depth limited at 0.55 h, a common
screening rule for a wave that has already broken once.
DEPTH_AT_TOE = 8.5
Hm0_toe = min(Hs100, 0.55 * DEPTH_AT_TOE)
conditions = DesignConditions.from_peak_period(
 Hm0=Hm0_toe, Tp=9.5, depth=DEPTH_AT_TOE, storm_duration=6 * 3600.0
)
wall = design_seawall(conditions, still_water_level=2.9, seabed_level=-5.6,
 tolerable_use="trained_staff")
print(f"\nDesign wave at the toe {Hm0_toe:.2f} m "
 f" + (\"depth limited\") if Hm0_toe < Hs100 else \"(offshore value carried in)\")")
print(f"Seawall crest {wall.crest_level:+.2f} m CD, "
 f"base {wall.base_width:.2f} m, "
 f"concrete {wall.quantities()['concrete_total_m3_per_m']:.1f} m3/m")

--- figure -----
fig, axes = plt.subplots(2, 2, figsize=(13.5, 8.8))
fig.subplots_adjust(hspace=0.36, wspace=0.24)

ax = axes[0, 0]
window = PER_YEAR * 3
ax.plot(t[:window], record[:window], lw=0.5, color="#1b6fa0")
ax.axhline(THRESHOLD, color="#8a2f24", lw=1.2, ls="--",
 label=f"threshold {THRESHOLD:.1f} m")
in_view = pot.positions < window

```

```

ax.plot(t[pot.positions[in_view]], pot.data[in_view], "o", ms=4.5,
 markerfacecolor="white", markeredgewidth=1.0, label="declustered peaks")
ax.set_title("Hindcast, first three years")
ax.set_xlabel("years")
ax.set_ylabel("Hm0 (m)")
ax.legend(loc="upper right", fontsize=8)
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()

ax = axes[0, 1]
ax.errorbar(mrl["thresholds"], mrl["mean_excess"], yerr=1.96 * mrl["stderr"],
 color="#14567f", ecolor="#9fc4d6", lw=1.3, elinewidth=0.8,
 capsize=0)
ax.axvline(THRESHOLD, color="#8a2f24", lw=1.2, ls="--")
ax.set_title("Mean residual life: straight above a valid threshold")
ax.set_xlabel("threshold (m)")
ax.set_ylabel("mean excess (m)")
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()

ax = axes[1, 0]
ax.plot(stab["thresholds"], stab["shape"], color="#14567f", lw=1.4,
 label="fitted shape")
ax.axhline(TRUE_SHAPE, color="#4a7c3f", lw=1.1, ls=":",
 label=f"true shape {TRUE_SHAPE:+.2f}")
ax.axvline(THRESHOLD, color="#8a2f24", lw=1.2, ls="--")
ax.set_ylim(-0.6, 0.6)
ax.set_title("Shape against threshold: flat where the model holds")
ax.set_xlabel("threshold (m)")
ax.set_ylabel("shape")
ax.legend(loc="lower left", fontsize=8)
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()

ax = axes[1, 1]
periods = np.logspace(np.log10(0.2), np.log10(500), 160)
band_all = pot.confidence(periods)
ax.fill_between(periods, band_all["lower"], band_all["upper"],
 color="#9fc4d6", alpha=0.55, label="90% bootstrap band")
ax.plot(periods, band_all["central"], color="#0b3554", lw=1.8,
 label="peaks over threshold")
ax.plot(periods, gpd_return_value(TRUE_THRESHOLD, TRUE_SCALE, TRUE_SHAPE,
 actual_rate, periods),
 color="#4a7c3f", lw=1.3, ls=":", label="true climate")
Annual maxima say nothing below a one-year return period, so the curve
starts where the model is defined rather than being clipped flat.
annual = periods[periods > 1.05]
ax.plot(annual, gev.return_value(annual), color="#8a2f24", lw=1.2, ls="--",
 label="annual maxima")

emp = plotting_positions(pot.data, rate=pot.rate)
ax.plot(emp["return_period"], emp["values"], "o", ms=3.2,
 markerfacecolor="white", markeredgewidth=0.7, label="observed peaks")
ax.axvline(YEARS, color="#4a4a4a", lw=0.9, ls=":")
ax.plot([DESIGN_PERIOD], [Hs100], "*", ms=14, color="#c47f1a",
 markeredgewidth=0.6, label=f"design {Hs100:.2f} m")
ax.set_xscale("log")
ax.set_xlabel("return period (years)")
ax.set_ylabel("Hm0 (m)")
ax.set_title(f"Return level (dotted line at the {YEARS}-year record length)")
ax.legend(loc="upper left", fontsize=7.5)
ax.grid(True, which="both", alpha=0.3)

fig.suptitle(
 f"Design wave from a {YEARS}-year record: "
 f"{DESIGN_PERIOD:.0f}-year Hm0 = {Hs100:.2f} m "
 f"({band['lower'][0]:.2f} to {band['upper'][0]:.2f} m), "
 f"true value {truth:.2f} m",
 y=0.975, fontsize=12, fontweight="bold",
)
fig.savefig("media/design_wave.png", dpi=600, bbox_inches="tight")
print("\nWrote media/design_wave.png")

```

## examples/breakwater\_design.py

```

"""
Rubble-mound breakwater design check.

Sizes armour with Van der Meer, sets the crest from an EurOtop overtopping
limit, and plots the two design curves an engineer actually reads: stone size
against slope, and overtopping against crest freeboard.

Run from the repository root:

 python examples/breakwater_design.py
"""

import matplotlib
import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.structures import (
 ROUGHNESS_FACTORS,
 TOLERABLE_DISCHARGE,
 DesignConditions,
 assess_overtopping,
 design_rubble_mound,
 overtopping_sloped,
 rock_armour_hudson,
 rock_armour_vandermeer,
)

matplotlib.rcParams["font.family"] = "sans-serif"
matplotlib.rcParams["font.sans-serif"] = ["Arial"]

--- design condition -----
conditions = DesignConditions.from_peak_period(
 Hm0=4.0, # 1 in 100 year significant wave height at the toe
 Tp=11.0,
 depth=12.0,
 storm_duration=6 * 3600.0,
)

design = design_rubble_mound(
 conditions,
 cot_alpha=2.0,
 armour="rock_two_layer_permeable",
 damage=2.0, # start of damage
 permeability=0.4, # permeable core
 tolerable_use="trained_staff",
)
print(design.summary())

print("\nTolerability of the mean discharge:")
for name, info in assess_overtopping(design.q_mean).items():
 mark = "ok " if info["acceptable"] else "NO "
 print(f" {mark} {info['limit']:>6.2f} l/s/m {name}")

--- curve 1: stone size against slope -----
cots = np.linspace(1.5, 4.0, 40)
vdm, hud, regimes = [], [], []
for cot in cots:
 v = rock_armour_vandermeer(conditions, cot)
 vdm.append(v["Dn50"])
 regimes.append(v["regime"])
 hud.append(rock_armour_hudson(conditions, cot)["Dn50"])

--- curve 2: overtopping against freeboard -----
freeboards = np.linspace(1.5, 9.0, 60)
gamma_f = ROUGHNESS_FACTORS["rock_two_layer_permeable"]
q = [overtopping_sloped(conditions, Rc, 2.0, gamma_f=gamma_f)["q"] for Rc in freeboards]

--- figure -----
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4.8))

plunging = [c for c, r in zip(cots, regimes) if r == "plunging"]
ax1.plot(cots, vdm, lw=2.0, label="Van der Meer (1988)")
ax1.plot(cots, hud, lw=1.6, ls="--", label="Hudson (SPM 1984), $K_D = 4$")
if plunging and len(plunging) < len(cots):
 ax1.axvline(max(plunging), color="0.5", ls=":", lw=1.2)
 ax1.annotate(
 "plunging / surging",
 xy=(max(plunging), max(vdm)),
 xytext=(4, -6), textcoords="offset points", fontsize=8, color="0.35",
 rotation=90, va="top",
)
ax1.plot(design.cot_alpha, design.Dn50, "o", ms=8, mfc="w", mec="k", mew=1.5, zorder=5)
ax1.annotate(
 f"design\n{design.Dn50:.2f} m, {design.M50 / 1000:.1f} t",
 xy=(design.cot_alpha, design.Dn50), xytext=(12, 10),
 textcoords="offset points", fontsize=9,
 arrowprops=dict(arrowstyle="->", color="0.3"),
)

```

```

)
ax1.set_xlabel(r"Slope, cot α")
ax1.set_ylabel("Nominal stone diameter D_{n50} (m)")
ax1.set_title(f"Armour size, $H_{\{m\}}$ = {conditions.Hm0} m, S = 2")
ax1.legend(frameon=False, fontsize=9)
ax1.grid(True, which="both", alpha=0.3)
ax1.minorticks_on()

ax2.semilogy(freeboards, q, lw=2.0, color="#1b6fa0")
for name in ("pedestrians_unaware", "trained_staff", "vehicles_low_speed"):
 limit, _ = TOLERABLE_DISCHARGE[name]
 ax2.axhline(limit, color="0.55", ls=":", lw=1.1)
 ax2.annotate(
 f"{name.replace('_', ' ')} {limit:g}",
 xy=(freeboards[-1], limit), xytext=(-4, 3),
 textcoords="offset points", ha="right", fontsize=7.5, color="0.35",
)
)
ax2.plot(design.crest_freeboard, design.q_mean, "o", ms=8, mfc="w", mec="k", mew=1.5, zorder=5)
ax2.annotate(
 f"design crest\nR_c = {design.crest_freeboard:.2f} m",
 xy=(design.crest_freeboard, design.q_mean), xytext=(14, 14),
 textcoords="offset points", fontsize=9,
 arrowprops=dict(arrowstyle=">", color="0.3"),
)
ax2.set_xlabel("Crest freeboard R_c (m)")
ax2.set_ylabel("Mean overtopping discharge q (l/s per m)")
ax2.set_title("Overtopping, Eur0top (2018)")
ax2.set_ylim(1e-4, 1e3)
ax2.grid(True, which="both", alpha=0.3)

fig.tight_layout()
fig.savefig("media/breakwater_design.png", dpi=600)
print("\nWrote media/breakwater_design.png")

--- the same design, issued as a drawing -----
The section is generated from the design object, so the armour drawn on the
paper is the armour the stability calculation sized.
from pyCoastal.applications.sections import rubble_mound_section, rubble_mound_sheet
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

SWL = 1.10 # m CD, design still water level
SEABED = -10.9 # m CD, giving the 12.0 m depth the design was run at

section = rubble_mound_section(design, still_water_level=SWL,
 seabed_level=SEABED)
section.save("media/breakwater_section.png")

sheet = rubble_mound_sheet(
 design,
 still_water_level=SWL,
 seabed_level=SEABED,
 project="Harbour protection works",
 title="Breakwater typical sections",
 client="Example Port Authority",
 size="A3",
 file="examples/breakwater_design.py",
)
sheet.save("media/breakwater_sheet.png")
sheet.to_dxf("media/breakwater_section.dxf")
print("Wrote media/breakwater_section.png, media/breakwater_sheet.png "
 "and media/breakwater_section.dxf")

```

## examples/seawall\_section.py

```
"""
Seawall design, from a design condition to a drawing.

Sizes an L-shaped gravity seawall for a promenade, prints the design report,
and writes the dimensioned cross-section plus a DXF of the geometry.

Run from the repository root:

 python examples/seawall_section.py
"""

from pyCoastal.applications.seawall import design_seawall
from pyCoastal.applications.sections import seawall_section
from pyCoastal.applications.structures import DesignConditions, assess_overtopping
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

--- design basis -----
Wave height at the toe, after any nearshore transformation. The storm
still water level already includes surge and tide.
conditions = DesignConditions.from_peak_period(
 Hm0=2.8, # significant wave height at the toe
 Tp=9.5, # peak period
 depth=8.5, # depth at the toe under the design level
 storm_duration=6 * 3600.0,
)

design = design_seawall(
 conditions,
 still_water_level=2.90, # m CD, surge plus tide
 seabed_level=-5.60, # m CD
 tolerable_use="trained_staff",
 friction=0.60, # concrete on a rubble bedding layer
 target_sliding=1.2,
 target_overtopping=1.5,
 scour_coefficient=0.4, # Xie, fine sand
 stem_thickness=1.0,
 base_thickness=1.2,
)

print(design.summary())

print("\nOvertopping tolerability at the design discharge")
for use, check in assess_overtopping(design.q_upper).items():
 mark = "ok " if check["acceptable"] else "no "
 print(f" {mark} {check['limit']>5g} l/s/m {use}")

--- drawing -----
dwg = seawall_section(
 design,
 title="Promenade seawall, typical cross-section",
 sea_extent=24.0,
 land_extent=9.0,
)
dwg.save("media/seawall_section.png")
dwg.to_dxf("media/seawall_section.dxf")
print("\nWrote media/seawall_section.png and media/seawall_section.dxf")

--- drawing sheet -----
The same design, issued as a drawing: border, title block, notes, and a
true stated scale rather than plot axes.
from pyCoastal.applications.sections import seawall_sheet

sheet = seawall_sheet(
 design,
 project="Bayfront promenade protection",
 title="Seawall typical cross-section",
 client="Example Port Authority",
 size="A3",
 sea_extent=24.0,
 land_extent=9.0,
 file="examples/seawall_section.py",
)
sheet.save("media/seawall_sheet.png")
print("Wrote media/seawall_sheet.png")
```

## examples/port\_diffraction.py

```

"""
Port layout study: phase-resolved wave diffraction into a harbour.

Propagates a monochromatic swell into a two-arm harbour, writes an animated
GIF of the instantaneous surface so the diffraction into the basin is visible
wave by wave, and reports the disturbance coefficient at a set of berths.

Run from the repository root:

 python examples/port_diffraction.py
"""

import matplotlib
import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap
import numpy as np
from matplotlib.animation import FuncAnimation, PillowWriter
from PIL import Image

from pyCoastal.applications.port import (
 Breakwater,
 IncidentWave,
 PortLayout,
 harbour_layout,
 rotate_layout,
 simulate_port,
)
from pyCoastal.plotting import (
 LAND_COLOR,
 agitation_colormap,
 land_overlay,
 surface_norm,
 water_colormap,
)

matplotlib.rcParams["font.family"] = "sans-serif"
matplotlib.rcParams["font.sans-serif"] = ["Arial"]

--- case -----
Oblique attack is modelled by turning the harbour into the waves and
driving it shore-normal. A phased oblique source only lights a
parallelogram of the domain, which would leave part of the harbour in an
artificial shadow; rotating the layout keeps the incident field uniform.
APPROACH_ANGLE = 25.0

Plots start downwave of the wavemaker strip and the west sponge, which are
model machinery rather than part of the harbour.
VIEW_X0 = 320.0

Lx, Ly, dx = 1400.0, 900.0, 4.0
ABSORPTION = 0.35 # rubble-mound armour, Kr around 0.4

Turn the breakwaters into the waves, but leave the quay along the coast:
rotating a shore-parallel wall would drive it diagonally across the domain.
arms = rotate_layout(
 harbour_layout(
 Lx=Lx, Ly=Ly, dx=dx, depth=10.0,
 gap=130.0, arm_length=300.0, width=24.0,
 back_wall=False, absorption=ABSORPTION,
),
 APPROACH_ANGLE,
)
quay = Breakwater(
 points=[(Lx - 2 * dx, 0.0), (Lx - 2 * dx, Ly)],
 width=2 * dx, absorption=ABSORPTION, name="quay",
)
layout = PortLayout(
 Lx=Lx, Ly=Ly, dx=dx, depth=10.0,
 breakwaters=[*arms.breakwaters, quay],
)
wave = IncidentWave(height=1.5, period=9.0, direction=0.0)

L = wave.wavelength(layout.depth)
print(f"Wavelength : {L:.1f} m ({L / layout.dx:.0f} cells per wave)")
print(f"Celerity : {wave.celerity(layout.depth):.2f} m/s")
print(f"Entrance gap : 130 m = {130 / L:.2f} wavelengths")
print(f"Approach angle : {APPROACH_ANGLE:.0f} deg off shore-normal")

result = simulate_port(
 layout,
 wave,
 sponge_sides=("west", "north", "south"), # the quay reflects
 n_snapshots=90,
 analysis_periods=8.0,
)
print(f"Incident height : {result.reference_height:.2f} m")

```

```

--- berths -----
berths = {
 "entrance": (1020.0, 470.0),
 "outer basin": (1120.0, 500.0),
 "north berth": (1150.0, 640.0),
 "south berth": (1150.0, 300.0),
}
print(f"\n{'berth':>13}{ 'Kd':>8}{ 'Hs (m)':>10}")
for name, info in result.berth_report(berths).items():
 print(f"{name:>13}{info['Kd']:>8.2f}{info['Hs']:>10.2f}")

operable = result.operable_fraction(berths, limit=0.5)
print("\nBerths within a 0.5 m operational limit:")
for name, ok in operable.items():
 print(f" {name:<12} {'yes' if ok else 'no'}")

--- still figure: disturbance coefficient -----
X, Y = layout.meshgrid()
kd = result.disturbance_coefficient

fig, ax = plt.subplots(figsize=(9, 6))
levels = np.linspace(0, 1.6, 33)
cf = ax.contourf(
 X, Y, np.nan_to_num(kd), levels=levels, cmap=agitation_colormap(), extend="max"
)
ax.contour(X, Y, np.nan_to_num(kd), levels=[0.2, 0.4, 0.6], colors="w", linewidths=0.6)
ax.imshow(
 land_overlay(result.land).T,
 extent=[0, layout.Lx, 0, layout.Ly],
 origin="lower", cmap=ListedColormap([LAND_COLOR]), interpolation="nearest",
)
for name, (bx, by) in berths.items():
 ax.plot(bx, by, "o", ms=6, mfc="w", mec="k", mew=1.2)
 ax.annotate(name, (bx, by), textcoords="offset points", xytext=(8, 6), fontsize=8)

ax.set_xlim(VIEW_X0, Lx)
ax.set_xlabel("x (m)")
ax.set_ylabel("y (m)")
ax.set_title(f"Disturbance coefficient, H = {wave.height} m, T = {wave.period} s")
ax.set_aspect("equal")
ax.grid(True, which="both", alpha=0.25)
ax.minorticks_on()
fig.colorbar(cf, ax=ax, label="$K_d = H / H_{i\$", shrink=0.85)
fig.tight_layout()
fig.savefig("media/port_disturbance.png", dpi=600)
print("\nWrote media/port_disturbance.png")

--- movie: phase-resolved surface -----
norm = surface_norm(result.snapshots)

fig2, ax2 = plt.subplots(figsize=(9, 5.6))
im = ax2.imshow(
 result.snapshots[0].T,
 extent=[0, layout.Lx, 0, layout.Ly],
 origin="lower", cmap=water_colormap(), norm=norm, interpolation="bilinear",
)
ax2.imshow(
 land_overlay(result.land).T,
 extent=[0, layout.Lx, 0, layout.Ly],
 origin="lower", cmap=ListedColormap([LAND_COLOR]), interpolation="nearest",
)
ax2.set_xlim(VIEW_X0, Lx)
ax2.set_xlabel("x (m)")
ax2.set_ylabel("y (m)")
ax2.set_aspect("equal")
title = ax2.set_title("")
fig2.colorbar(im, ax=ax2, label="surface elevation (m)", shrink=0.85)
fig2.tight_layout()

def update(frame):
 im.set_data(result.snapshots[frame].T)
 title.set_text(
 f"Wave diffraction into the harbour, "
 f"{APPROACH_ANGLE:.0f} deg incidence "
 f"t = {result.times[frame]:.0f} s "
 f"({result.times[frame] / wave.period:.1f} wave periods)"
)
 return im, title

anim = FuncAnimation(
 fig2, update, frames=len(result.snapshots), interval=60, blit=False
)
gif_path = "media/port_diffraction.gif"
anim.save(gif_path, writer=PillowWriter(fps=15), dpi=64)

Matplotlib gives every frame its own palette, which roughly doubles the
file. Re-encode against one shared adaptive palette.
frames = []
with Image.open(gif_path) as src:
 for k in range(src.n_frames):

```



```
 src.seek(k)
 frames.append(src.convert("RGB").copy())

palette = frames[0].quantize(colors=96, method=Image.MEDIANCUT)
quantized = [f.quantize(palette=palette, dither=Image.Dither.NONE) for f in frames]
quantized[0].save(
 gif_path, save_all=True, append_images=quantized[1:], duration=67, loop=0,
 optimize=True,
)
print(f"Wrote {gif_path}")
```

## examples/port\_layout\_comparison.py

```

"""
Compare the built-in port layouts under the same design wave.

Runs every layout in ``pyCoastal.applications.port.LAYOUTS``, maps the
disturbance coefficient, and ranks them by how quiet they keep the basin.

Run from the repository root:

 python examples/port_layout_comparison.py
"""

import matplotlib
import matplotlib.pyplot as plt
import numpy as np
from matplotlib.colors import ListedColormap, LogNorm

from pyCoastal.applications.port import LAYOUTS, IncidentWave, simulate_port
from pyCoastal.plotting import LAND_COLOR, agitation_colormap, land_overlay

matplotlib.rcParams["font.family"] = "sans-serif"
matplotlib.rcParams["font.sans-serif"] = ["Arial"]

LX, LY = 1400.0, 900.0 # one domain for every layout, so the runs compare
DX = 8.0
ABSORPTION = 0.35 # rubble-mound armour on every structure
WAVE = IncidentWave(height=1.5, period=9.0, direction=0.0)
VIEW_X0 = 320.0 # hide the wavemaker strip and the west sponge

TITLES = {
 "two_arm": "Two arms, straight gap",
 "offset_entrance": "Overlapping arms, dog-leg",
 "hooked": "Hooked main breakwater",
 "detached_screen": "Detached screen off the gap",
 "marina": "Outer harbour and inner basin",
}

def basin_statistics(result, layout):
 """Mean and worst Kd in the sheltered part of the domain.

 The same box is used for every layout so the numbers are comparable,
 even though the structures that protect it differ.
 """
 inside = result.x > 0.84 * layout.Lx
 water = np.zeros_like(result.land, dtype=bool)
 water[inside, :] = True
 water &= ~result.land

 kd = result.wave_height[water] / result.reference_height
 return float(kd.mean()), float(np.percentile(kd, 95))

results = {}
for name, builder in LAYOUTS.items():
 layout = builder(Lx=LX, Ly=LY, dx=DX, absorption=ABSORPTION)
 result = simulate_port(
 layout,
 WAVE,
 sponge_sides=("west", "north", "south"), # the quay reflects
 n_snapshots=2,
 analysis_periods=10.0,
)
 results[name] = (layout, result, *basin_statistics(result, layout))
 print(f"ran {name}")

print(f"\n{'layout':>32}{'mean Kd':>10}{'95th pct Kd':>14}")
ranked = sorted(results.items(), key=lambda kv: kv[1][2])
for name, (_, _, mean_kd, p95) in ranked:
 print(f"{TITLES[name]:>32}{mean_kd:>10.3f}{p95:>14.3f}")

best = ranked[0][0]
worst = ranked[-1][0]
print(
 f"\nQuietest basin: {TITLES[best]} "
 f"({results[best][2]:.3f} mean Kd)\n"
 f"Noisiest basin: {TITLES[worst]} "
 f"({results[worst][2]:.3f} mean Kd)"
)

--- figure -----
n = len(results)
fig, axes = plt.subplots(1, n, figsize=(4.0 * n, 4.4), sharey=True)
Kd runs from about 0.005 in a well sheltered basin to above 1 outside,
so a linear scale would render every good layout as identical black.
levels = np.logspace(np.log10(0.005), np.log10(1.5), 28)
norm = LogNorm(vmin=levels[0], vmax=levels[-1])
land_cmap = ListedColormap([LAND_COLOR])

```

```

for ax, (name, (layout, result, mean_kd, _)) in zip(axes, results.items()):
 X, Y = layout.meshgrid()
 kd = np.nan_to_num(result.disturbance_coefficient, nan=0.0)
 cf = ax.contourf(
 X, Y, np.clip(kd, levels[0], None),
 levels=levels, norm=norm, cmap=agitation_colormap(), extend="both",
)
 ax.imshow(
 land_overlay(result.land).T,
 extent=[0, layout.Lx, 0, layout.Ly],
 origin="lower", cmap=land_cmap, interpolation="nearest",
)
 ax.set_title(f"{TITLES[name]}\nmean basin K_d = {mean_kd:.2f}", fontsize=10)
 ax.set_xlim(VIEW_X0, layout.Lx)
 ax.set_xlabel("x (m)")
 ax.set_aspect("equal")
 ax.grid(True, which="both", alpha=0.25)
 ax.minorticks_on()

axes[0].set_ylabel("y (m)")
cb = fig.colorbar(cf, ax=axes, label="K_d = H / H_i$ (log scale)",
 shrink=0.8, pad=0.02, ticks=[0.005, 0.02, 0.05, 0.2, 0.5, 1.0])
cb.ax.set_yticklabels(["0.005", "0.02", "0.05", "0.2", "0.5", "1.0"])
fig.suptitle(
 f"Harbour layouts under H = {WAVE.height} m, T = {WAVE.period} s, "
 f"armour absorption {ABSORPTION}",
 y=1.02,
)
fig.savefig("media/port_layouts.png", dpi=600, bbox_inches="tight")
print("\nWrote media/port_layouts.png")

```

## examples/pile\_wave\_loads.py

```

"""
Wave Loads on a monopile: where the load is, and when it peaks.

Computes the Morison Load distribution on an offshore wind monopile,
sweeps the wave phase for the worst base shear and the worst overturning
moment, and shows why they do not happen at the same instant.

The headline result is a trap worth knowing about. A large monopile is
inertia dominated, so the load follows the fluid acceleration, which peaks a
quarter cycle before the crest arrives. Evaluating the load under the crest
because that is where the water is highest understates the overturning
moment substantially.

Run from the repository root:

 python examples/pile_wave_loads.py
"""

import math

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.piles import (
 design_monopile,
 drag_inertia_coefficients,
 keulegan_carpenter,
 morison_pile_load,
 scour_depth_pile,
)
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

--- the structure and the design wave -----
DIAMETER = 8.0 # monopile outer diameter
DEPTH = 30.0 # water depth
H = 12.0 # design wave height
T = 13.0 # period

result = design_monopile(DIAMETER, H, T, DEPTH, rough=True)
worst, crest, sweep, scour = (result["load"], result["crest_load"],
 result["sweep"], result["scour"])

print(worst.summary())
print(f"\nAt the crest phase {crest.force / 1e3:.0f} kN, "
 f"{crest.moment / 1e6:.1f} MNm")
print(f"At the worst phase {worst.force / 1e3:.0f} kN, "
 f"{worst.moment / 1e6:.1f} MNm")
print(f"Designing on the crest would miss "
 f"{100 * result['crest_underestimate']:.0f}% of the moment")
print(f"\nMax base shear at "
 f"{math.degrees(sweep['phase_of_max_force']):.0f} degrees")
print(f"Max mudline moment at "
 f"{math.degrees(sweep['phase_of_max_moment']):.0f} degrees")

print(f"\nScour {scour['depth']:.2f} m "
 f"{scour['ratio']:.2f} D) at KC = {scour['KC']:.1f}")
current = scour_depth_pile(DIAMETER, scour["KC"], current_only=True)
print(f" under a steady current: {current['depth']:.2f} m "
 f"(1.3 D, standard deviation {current['standard_deviation']:.2f} m)")
print(" waves alone barely scour a pile this large, because KC is small; "
 "current is what governs.")

--- how the load splits with pile size -----
print("\nHow the load splits with pile diameter")
print(" D (m) KC regime inertia share moment (MNm)")
for diameter in (1.0, 2.0, 4.0, 8.0, 12.0):
 KC = keulegan_carpenter(H, T, DEPTH, diameter)
 coefficients = drag_inertia_coefficients(KC)
 case = design_monopile(diameter, H, T, DEPTH, "load")
 flag = " diffracts" if case.diffraction_ratio > 0.2 else ""
 print(f" {diameter:5.1f} {KC:5.1f} {coefficients['regime']:<18}"
 f"{100 * case.inertia_fraction:6.0f}% "
 f"{case.moment / 1e6:8.1f}{flag}")

--- figure -----
fig = plt.figure(figsize=(14.0, 8.0))
gs = fig.add_gridspec(2, 3, width_ratios=[1.0, 1.0, 1.35],
 hspace=0.34, wspace=0.30)

Load profile at the worst phase, drag and inertia separated. Only the wet
part is drawn: the profile is zero above the instantaneous surface, and a
line running back to zero there reads as a spike rather than as nothing.
wet = worst.z <= worst.eta
ax = fig.add_subplot(gs[:, 0])
ax.fill_betweenx(worst.z[wet], 0, worst.inertia[wet] / 1e3, color="#9fc4d6",

```

```

 alpha=0.75, label="inertia")
ax.fill_betweenx(worst.z[wet], 0, worst.drag[wet] / 1e3, color="#c47f1a",
 alpha=0.6, label="drag")
ax.plot(worst.total[wet] / 1e3, worst.z[wet], color="#0b3554", lw=1.9,
 label="total")
ax.axhline(0.0, color="#4d7f97", lw=1.0)
ax.axhline(worst.eta, color="#8a2f24", lw=1.2, ls="--",
 label=f"surface {worst.eta:+.2f} m")
ax.axhline(-DEPTH, color="#6d6243", lw=1.6)
ax.set_xlabel("load (kN/m)")
ax.set_ylabel("elevation (m)")
ax.set_title(f"Load profile at {math.degrees(worst.phase):.0f} degrees")
ax.annotate(f"{worst.moment / 1e6:.0f} MNm", xy=(0.95, 0.06),
 xycoords="axes fraction", ha="right", fontsize=9,
 fontweight="bold")
ax.legend(loc="lower right", fontsize=8)
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()

The same at the crest. Both x axes are shared, because two panels drawn to
their own scales would make a load four times smaller look comparable.
wet_crest = crest.z <= crest.eta
ax = fig.add_subplot(gs[:, 1], sharey=ax, sharex=ax)
ax.fill_betweenx(crest.z[wet_crest], 0, crest.inertia[wet_crest] / 1e3,
 color="#9fc4d6", alpha=0.75)
ax.fill_betweenx(crest.z[wet_crest], 0, crest.drag[wet_crest] / 1e3,
 color="#c47f1a", alpha=0.6)
ax.plot(crest.total[wet_crest] / 1e3, crest.z[wet_crest], color="#0b3554",
 lw=1.9)
ax.axhline(0.0, color="#4d7f97", lw=1.0)
ax.axhline(crest.eta, color="#8a2f24", lw=1.2, ls="--",
 label=f"crest {crest.eta:+.2f} m")
ax.axhline(-DEPTH, color="#6d6243", lw=1.6)
ax.set_xlabel("load (kN/m)")
ax.set_title("Load profile under the crest")
ax.annotate(f"{crest.moment / 1e6:.0f} MNm", xy=(0.95, 0.06),
 xycoords="axes fraction", ha="right", fontsize=9,
 fontweight="bold")
ax.legend(loc="lower right", fontsize=8)
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()

Phase sweep.
ax = fig.add_subplot(gs[0, 2])
degrees = np.degrees(sweep["phase"])
ax.plot(degrees, sweep["force"] / 1e3, color="#0b3554", lw=1.8,
 label="base shear")
ax.axvline(math.degrees(sweep["phase_of_max_force"]), color="#0b3554",
 lw=1.0, ls=":")
ax.set_ylabel("base shear (kN)")
ax.set_title("Load through the wave cycle")
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()
ax.legend(loc="upper right", fontsize=8)

ax2 = fig.add_subplot(gs[1, 2], sharex=ax)
ax2.plot(degrees, sweep["moment"] / 1e6, color="#8a2f24", lw=1.8,
 label="mudline moment")
ax2.axvline(math.degrees(sweep["phase_of_max_moment"]), color="#8a2f24",
 lw=1.0, ls=":")
ax2.plot([0.0], [crest.moment / 1e6], "o", ms=7, markerfacecolor="white",
 markeredgewidth=1.3, color="#8a2f24", label="crest phase")
ax2.plot([math.degrees(sweep["phase_of_max_moment"])],
 [worst.moment / 1e6], "*", ms=15, color="#c47f1a",
 markeredgewidth=0.6, label="worst phase")
ax2.annotate(
 f"{100 * result['crest_underestimate']:0f}% larger\nthan at the crest",
 xy=(math.degrees(sweep["phase_of_max_moment"]), worst.moment / 1e6),
 xytext=(30, -40), textcoords="offset points", fontsize=8.5,
 arrowprops=dict(arrowstyle=">", color="#1a1a1a", lw=0.8),
)
ax2.set_xlabel("wave phase (degrees, crest at 0)")
ax2.set_ylabel("mudline moment (MNm)")
ax2.set_xlim(0, 360)
ax2.set_xticks(range(0, 361, 90))
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()
ax2.legend(loc="upper right", fontsize=8)

fig.suptitle(
 f"Monopile {DIAMETER:0f} m in {DEPTH:0f} m: H = {H:0f} m, T = {T:0f} s. "
 f"{worst.force / 1e3:.0f} kN and {worst.moment / 1e6:.0f} MNm, "
 f"{100 * worst.inertia_fraction:0f}% inertia",
 y=0.975, fontsize=12, fontweight="bold",
)
fig.savefig("media/pile_wave_loads.png", dpi=600, bbox_inches="tight")
print("\nWrote media/pile_wave_loads.png")

```

## examples/pier\_scour.py

```

"""
Scour at an estuary pier: tide, river and waves, and where the base sits.

A bridge pier in an estuary is asked to stand in a flow that reverses twice
a day, is biased seaward by the river, changes depth under a four metre
tide, and carries a wind chop on top. None of those peak together.

Two findings come out of working the cycle rather than a load case, and
neither is obvious from a spreadsheet of design values.

The governing phase is not peak current. Once the footing is exposed,
the flow-weighted obstacle width depends on how much of the water column
the wide base occupies, so low water counts for more than fast water. The
worst phase here is well after peak ebb.

The burial depth of the base is a cliff, not a slope. Bury the footing
deeper than the bare stem would scour and nothing happens. Bury it a metre
shallower and the hole reaches it, the obstacle widens, the hole deepens,
and it runs away to almost twice the depth, undermining the foundation.
There is no gentle middle.

Run from the repository root:

 python examples/pier_scour.py
"""

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.scour import (
 EstuaryConditions,
 Pier,
 PierBase,
 design_pier_scour,
 equilibrium_scour,
 tidal_state,
)
from pyCoastal.applications.sections import pier_scour_section, pier_scour_sheet
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

PIER = Pier(diameter=2.5, shape="circular")
BASE = PierBase(width=7.0, length=12.0, height=2.5, top_level=-1.5, skew=20.0)
ESTUARY = EstuaryConditions(
 mean_depth=9.0, tidal_amplitude=2.2, tidal_current=1.1,
 river_current=0.4, Hs=1.2, Tp=5.5, bed="medium_sand",
 current_phase=75.0)

design = design_pier_scour(PIER, ESTUARY, BASE)
state = design.governing["state"]

print(f"Pier {PIER.diameter:.1f} m on a {BASE.width:.0f} x {BASE.length:.0f} m "
 f"base, top {BASE.top_level:+.1f} m to the bed")
print(f"Peak ebb {ESTUARY.peak_ebb_current:.2f} m/s, "
 f"peak flood {ESTUARY.peak_flood_current:.2f} m/s")
print()
print(f"Governing phase {state['phase']:.0f} deg "
 f"({'ebb' if state['ebb'] else 'flood'})")
print(f"current {abs(state['current']):.2f} m/s "
 f"(peak is {ESTUARY.peak_ebb_current:.2f})")
print(f"depth {state['depth']:.2f} m")
print(f"effective diameter {design.governing['scour']['D_e']:.2f} m "
 f"(stem is {PIER.diameter:.2f})")
print(f"equilibrium scour {design.equilibrium:.2f} m")
print(f"one tidal half {design.tidal_limited:.2f} m "
 f"(time scale {design.time_scale / 3600:.1f} h)")
print(f"base "
 f"({'UNDERMINED' if design.undermined else 'exposed' if design.base_exposed else 'buried'})")
print()
for note in design.notes:
 print(f" - {note}")

pier_scour_section(design).save("media/pier_scour.png")
pier_scour_sheet(design, project="Example estuary crossing",
 client="Example Highways Authority",
 file="examples/pier_scour.py").save("media/pier_scour_sheet.png")

--- what the tide actually does -----
fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.0, 5.4))

phases = design.phases()
scour = design.envelope()
depths = np.array([s["state"]["depth"] for s in design.states])
currents = np.array([s["state"]["current"] for s in design.states])

What varies through the cycle is the scour the conditions *demand* at

```

```

that instant, not the hole itself. The hole does not refill at slack
water, so what the pier actually ends up with is the envelope: the
deepest demand the tide ever makes on it.
ax.fill_between(phases, 0, scour, color="#8a2f24", alpha=0.10)
ax.plot(phases, scour, lw=2.0, color="#8a2f24",
 label="demanded by the conditions at that phase")
ax.axhline(design.equilibrium, color="#8a2f24", lw=1.8, ls="--",
 label=f"envelope, and the design value: {design.equilibrium:.2f} m")

ax.axvline(state["phase"], color="#8a2f24", lw=1.0, ls=":")
ax.annotate(f"governs at {state['phase']:.0f} deg",
 xy=(state["phase"], design.equilibrium), xytext=(8, -16),
 textcoords="offset points", fontsize=9, color="#8a2f24")

peak = phases[np.argmax(np.abs(currents))]
ax.axvline(peak, color="#0b3554", lw=1.0, ls=":")
ax.annotate(f"peak current at {peak:.0f} deg", xy=(peak, 0.35 * scour.max()),
 xytext=(8, 0), textcoords="offset points", fontsize=9,
 color="#0b3554")

ax.annotate("slack water demands nothing,\nbut the hole stays",
 xy=(185, 0.1), xytext=(0, 30), textcoords="offset points",
 fontsize=8.5, ha="center", color="#54514b")

ax.set_xlabel("tidal phase (deg)")
ax.set_ylabel("equilibrium scour (m)")
ax.set_title("The deepest hole is not at the fastest flow")
ax.set_xlim(0, 360)
ax.set_ylim(0, 1.35 * scour.max())
ax.set_xticks(range(0, 361, 90))
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()
ax.legend(loc="upper center", fontsize=8, framealpha=0.95)

twin = ax.twinx()
twin.plot(phases, np.abs(currents), lw=1.5, color="#0b3554", alpha=0.85,
 label="current")
twin.plot(phases, depths, lw=1.5, color="#2f93b8", alpha=0.85, ls="--",
 label="depth")
twin.set_ylabel("current (m/s, solid) and depth (m, dashed)")
twin.set_ylim(0, 1.35 * max(depths.max(), np.abs(currents).max()))

--- the cliff -----
tops = np.linspace(-7.0, 0.5, 220)
depths_for = []
for top in tops:
 got = design_pier_scour(
 PIER, ESTUARY,
 PierBase(width=BASE.width, length=BASE.length, height=BASE.height,
 top_level=float(top), skew=BASE.skew),
 samples=25)
 depths_for.append(got.equilibrium)
depths_for = np.array(depths_for)

bare = design_pier_scour(PIER, ESTUARY, samples=25).equilibrium

ax2.plot(tops, depths_for, lw=2.4, color="#8a2f24")
ax2.axhline(bare, color="#0b3554", lw=1.2, ls="--")
ax2.annotate(f"bare stem alone: {bare:.2f} m", xy=(-6.8, bare),
 xytext=(0, 8), textcoords="offset points", fontsize=9,
 color="#0b3554")
ax2.axvline(-bare, color="#54514b", lw=1.0, ls=":")
ax2.annotate("bury the base below this\nand the hole never finds it",
 xy=(-bare, 0.5 * (bare + depths_for.max())), xytext=(-10, 0),
 textcoords="offset points", fontsize=9, ha="right",
 color="#54514b")

ax2.fill_between(tops, bare, depths_for, where=depths_for > bare + 1e-6,
 color="#8a2f24", alpha=0.12)
ax2.set_xlabel("level of the base top, relative to the initial bed (m)")
ax2.set_ylabel("equilibrium scour (m)")
ax2.set_title("Burial depth is a cliff, not a slope")
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()

fig.suptitle(
 "Estuary pier scour: the worst phase and the burial depth both "
 "behave unlike the design values they are built from",
 y=0.98, fontsize=12, fontweight="bold")
fig.tight_layout(rect=(0, 0, 1, 0.93))
fig.savefig("media/pier_scour_tide.png", dpi=600, bbox_inches="tight")

print("\nWrote media/pier_scour.png, media/pier_scour_sheet.png "
 "and media/pier_scour_tide.png")

```



## examples/bridge\_scour.py

```
"""
Total scour at an estuary crossing: all three HEC-18 components.

Local scour at a pier is the one everybody computes, and on a contracted
crossing it is routinely the smallest of the three. HEC-18 splits total
scour into contraction, local and abutment, and this works all three over
a tidal cycle.

Three things worth seeing, none of which survive being reduced to a single
design load case.

**Contraction scour above the threshold does not care how fast the water
goes.** Laursen's live-bed relation is a sediment balance: faster water
carries more, but it also delivers more, and the two cancel. The answer
turns on the width ratio alone. Below the threshold nothing arrives from
upstream and the mode flips entirely, at which point velocity is suddenly
all that matters.

The components do not peak together. Contraction follows the discharge
through the opening, the pier follows the flow in it, and the abutment
follows the approach. Each needs its own envelope over the cycle.

They are not added at one point. A pier gets contraction plus its own
hole; an abutment gets contraction plus its own. Adding all three gives a
foundation depth for a hole that exists nowhere.

Run from the repository root:

 python examples/bridge_scour.py
"""

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.scour import (
 BridgeOpening,
 EstuaryConditions,
 Pier,
 PierBase,
 contraction_scour,
 critical_velocity,
 design_bridge_scour,
)
from pyCoastal.applications.sections import (
 bridge_scour_section,
 bridge_scour_sheet,
)
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

OPENING = BridgeOpening(approach_width=140.0, opening_width=80.0,
 pier_blockage=5.0, abutment_length=22.0,
 abutment_shape="spill_through", slope=4e-4)
PIER = Pier(diameter=2.5, shape="circular")
BASE = PierBase(width=7.0, length=12.0, height=2.5, top_level=-1.5, skew=20.0)
ESTUARY = EstuaryConditions(
 mean_depth=9.0, tidal_amplitude=2.2, tidal_current=1.1,
 river_current=0.4, Hs=1.2, Tp=5.5, bed="medium_sand",
 current_phase=75.0)

design = design_bridge_scour(OPENING, ESTUARY, PIER, BASE)
state = design.governing["state"]

print(f"Waterway {OPENING.approach_width:.0f} m, opening "
 f"{OPENING.opening_width:.0f} m gross less "
 f"{OPENING.pier_blockage:.0f} m of piers")
print(f"Contraction {OPENING.contraction_ratio:.2f} to 1")
print()
print(f"Governing phase {state['phase']:.0f} deg "
 f"({'ebb' if state['ebb'] else 'flood'})")
print(f" approach {abs(state['current']):.2f} m/s in "
 f"{state['depth']:.2f} m")
print(f" through the opening "
 f"{design.governing['scour']['opening_velocity']:.2f} m/s in "
 f"{design.governing['scour']['opening_depth']:.2f} m")
print()
print(f" contraction {design.contraction:6.2f} m")
print(f" pier local {design.pier_local:6.2f} m")
print(f" abutment {design.abutment:6.2f} m")
print(f" {'-' * 22}")
print(f" at a pier {design.total_at_pier:6.2f} m")
print(f" at abutment {design.total_at_abutment:6.2f} m <- governs"
 if design.governing_location == "abutment" else
 f" at abutment {design.total_at_abutment:6.2f} m")
print()
for note in design.notes:
```

```

 print(f" - {note}")

bridge_scour_section(design).save("media/bridge_scour.png")
bridge_scour_sheet(design, project="Example estuary crossing",
 client="Example Highways Authority",
 file="examples/bridge_scour.py").save(
 "media/bridge_scour_sheet.png")

--- what the tide does to each component -----
fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.0, 5.4))

phases = design.phases()
palette = {"contraction": "#0b3554", "pier": "#1b6fa0", "abutment": "#c47f1a"}

for name, colour in palette.items():
 series = design.component(name)
 ax.plot(phases, series, lw=2.0, color=colour, label=name)
 peak = phases[series.argmax()]
 ax.plot([peak], [series.max()], "o", ms=5, color=colour)

ax.plot(phases, design.component("total"), lw=2.6, color="#8a2f24",
 label="total at the governing location")

ax.set_xlabel("tidal phase (deg)")
ax.set_ylabel("scour depth (m)")
ax.set_title("Each component peaks at its own phase")
ax.set_xlim(0, 360)
ax.set_xticks(range(0, 361, 90))
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()
ax.set_ylim(-2.0, None)
ax.legend(loc="lower center", fontsize=8.5, ncol=2, framealpha=0.95)

--- the regime switch, and the step in it -----
The two modes are separate equations fitted separately, and they do not
meet at the threshold. Here clear water arrives at the switch a little
above where live bed leaves it, so the curve steps down as the bed comes
alive. The step is small in this case; it is not always, and it is worth
knowing it is there before reading a scour depth off either side of it.
speeds = np.linspace(0.15, 2.0, 400)
depth = ESTUARY.mean_depth
Vc = critical_velocity(ESTUARY.material, depth)

actual, live, clear = [], [], []
for v in speeds:
 actual.append(contraction_scour(OPENING, depth, v, ESTUARY.bed)["depth"])
 live.append(contraction_scour(OPENING, depth, v, ESTUARY.bed,
 regime="live")["depth"])
 clear.append(contraction_scour(OPENING, depth, v, ESTUARY.bed,
 regime="clear")["depth"])

ax2.plot(speeds, clear, lw=1.4, ls="--", color="#c47f1a",
 label="clear water, if forced")
ax2.plot(speeds, live, lw=1.4, ls="--", color="#1b6fa0",
 label="live bed, if forced")
ax2.plot(speeds, actual, lw=2.6, color="#8a2f24", label="what applies")
ax2.axvline(Vc, color="#54514b", lw=1.0, ls=":")
ax2.annotate(f"threshold of motion\n{Vc:.2f} m/s", xy=(Vc, max(actual) * 0.72),
 xytext=(8, 0), textcoords="offset points", fontsize=9,
 color="#54514b")

ax2.set_xlabel("approach velocity (m/s)")
ax2.set_ylabel("contraction scour (m)")
ax2.set_title("Live bed ignores velocity; clear water is all velocity")
ax2.set_ylim(0, max(actual) * 1.25)
step = abs(np.interp(Vc * 1.001, speeds, actual)
 - np.interp(Vc * 0.999, speeds, actual))
ax2.annotate(f"the two equations do not meet:\na {step:.2f} m step",
 xy=(Vc, np.interp(Vc, speeds, actual)), xytext=(14, -46),
 textcoords="offset points", fontsize=8.5, color="#8a2f24",
 arrowprops=dict(arrowstyle="-", lw=0.8, color="#8a2f24"))
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()
ax2.legend(loc="upper left", fontsize=8.5)

fig.suptitle(
 "Total scour at a crossing: three components, three timings, and two "
 "contraction regimes that disagree at the threshold between them",
 y=0.98, fontsize=12, fontweight="bold")
fig.tight_layout(rect=(0, 0, 1, 0.93))
fig.savefig("media/bridge_scour_components.png", dpi=600, bbox_inches="tight")

print("\nWrote media/bridge_scour.png, media/bridge_scour_sheet.png "
 "and media/bridge_scour_components.png")

```

## examples/backwater.py

```

"""
Backwater at a bridge, and the flow split that feeds contraction scour.

Two things this example is for.

A backwater curve has no end. It approaches normal depth
asymptotically, so "the extent of backwater" is always a convention rather
than a measurement, and the number moves by more than a factor of two
across conventions a reasonable engineer might choose. Quoting an extent
without quoting the criterion says nothing.

The flow split is computed, not guessed. Contraction scour needs the
fraction of the discharge that actually goes through the opening.
Conveyance decides that, and conveyance is not width: a wooded floodplain
can be three quarters of the section and carry a sixth of the flow.
Guessing it is the largest assumption in a scour calculation, and it hides
behind the most precise-looking number in the output.

Run from the repository root:

 python examples/backwater.py
"""

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.river import (
 Channel,
 classify_slope,
 critical_depth,
 flow_distribution,
 gvf_profile,
 normal_depth,
 yarnell_afflux,
)
from pyCoastal.applications.scour import (
 BridgeOpening,
 contraction_scour,
)
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

CHANNEL = Channel(width=30.0, side_slope=2.0, roughness="natural_clean")
FLOODPLAIN = Channel(width=120.0, side_slope=3.0, roughness="floodplain_trees")
Q = 250.0
SLOPE = 0.0008

state = classify_slope(CHANNEL, Q, SLOPE)
yn, yc = state["normal"], state["critical"]

print(f"Q:{.0f} m3/s on a {SLOPE:.4f} bed, {CHANNEL.roughness}")
print(f" normal depth {yn:.2f} m")
print(f" critical depth {yc:.2f} m")
print(f" reach is {state['kind']}, so a bridge backs water up upstream")

--- the bridge, and the afflux it causes -----
BLOCKAGE = 0.18
afflux = yarnell_afflux(CHANNEL, Q, yn, BLOCKAGE, shape="semicircular_nose")
print()
print(f"Piers blocking {100 * BLOCKAGE:.0f}% of the section, Froude "
 f"{afflux['froude']:.2f}")
print(f" afflux {afflux['afflux']:.3f} m, so the control sits at "
 f"{afflux['upstream_depth']:.2f} m")

profile = gvf_profile(CHANNEL, Q, SLOPE, afflux["upstream_depth"])
print(f" {profile.profile} profile reaching {profile.reach / 1000:.2f} km")
for note in profile.notes:
 print(f" {note}")

--- what the floodplain actually carries -----
OVERBANK = 1.2
split = flow_distribution(CHANNEL, yn, SLOPE, [(FLOODPLAIN, OVERBANK)])
channel_width = CHANNEL.top_width(yn)
plain_width = FLOODPLAIN.top_width(OVERBANK)

print()
print(f"In flood, {OVERBANK:.1f} m over a wooded floodplain:")
print(f" the channel is {100 * channel_width / (channel_width + plain_width):.0f}% "
 f"of the width")
print(f" and carries {100 * split['main_fraction']:.0f}% of the flow")

opening = BridgeOpening(approach_width=channel_width + plain_width,
 opening_width=0.55 * (channel_width + plain_width),
 abutment_length=20.0,
 flow_fraction=split["main_fraction"])
guessed = BridgeOpening(approach_width=opening.approach_width,

```

```

 opening_width=opening.opening_width,
 abutment_length=20.0, flow_fraction=1.0)

computed = contraction_scour(opening, yn, CHANNEL.velocity(Q, yn), "medium_sand")
assumed = contraction_scour(guessed, yn, CHANNEL.velocity(Q, yn), "medium_sand")
print(f" contraction scour {computed['depth']:.2f} m with the computed split,"
 f" {assumed['depth']:.2f} m if it is assumed to be all of it")

--- the profile, drawn -----
fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.5, 5.4))

Depth rather than level. Over this reach the bed rises four metres while
the depth changes by seven centimetres, so on a true longitudinal section
the backwater curve is invisible underneath the slope. Plotting the depth
removes the bed and leaves the thing the plot is about.
x = profile.distance / 1000.0
ax.fill_between(x, yn, profile.depth, color="#c3dce7", zorder=1.5,
 label="water held above normal depth")
ax.plot(x, profile.depth, lw=2.4, color="#0b3554",
 label=f"depth ({profile.profile} profile)", zorder=4)
ax.axhline(yn, lw=1.3, ls="--", color="#2c6b46",
 label=f"normal depth {yn:.2f} m", zorder=3)

Critical depth is 1.7 m below this view. Drawing it would set the scale
and squash the whole backwater, which spans seven centimetres, into a
line. It matters for naming the profile and not for its shape, so it
belongs in the title.
ax.annotate(f"bridge holds {afflux['afflux'] * 1000:.0f} mm",
 xy=(0, profile.depth[0]), xytext=(34, -6),
 textcoords="offset points", fontsize=9, color="#0b3554",
 arrowprops=dict(arrowstyle="→", lw=0.9, color="#0b3554"))
ax.annotate("approaching, never arriving",
 xy=(x[-1], profile.depth[-1]), xytext=(-14, 20),
 textcoords="offset points", fontsize=9, ha="right",
 color="#2c6b46",
 arrowprops=dict(arrowstyle="→", lw=0.9, color="#2c6b46"))

span = profile.depth[0] - yn
ax.set_xlabel("distance upstream of the bridge (km)")
ax.set_ylabel("depth (m)")
ax.set_ylim(yn - 0.18 * span, profile.depth[0] + 0.30 * span)
ax.set_title(f"{profile.profile} on a mild reach "
 f"(critical depth {yc:.2f} m, far below this view)")
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()
ax.legend(loc="upper right", fontsize=8.5)

--- the extent is whatever you decide it is -----
criteria = np.linspace(0.80, 0.999, 60)
reaches = [gvf_profile(CHANNEL, Q, SLOPE, afflux["upstream_depth"],
 approach=float(a)).reach / 1000.0 for a in criteria]

ax2.plot(100 * criteria, reaches, lw=2.4, color="#8a2f24")
for mark in (0.90, 0.95, 0.99):
 reach = gvf_profile(CHANNEL, Q, SLOPE, afflux["upstream_depth"],
 approach=mark).reach / 1000.0
 ax2.plot([100 * mark], [reach], "o", ms=6, color="#8a2f24")
 ax2.annotate(f"{100 * mark:.0f}%: {reach:.2f} km",
 xy=(100 * mark, reach), xytext=(-8, 8),
 textcoords="offset points", fontsize=9, ha="right",
 color="#8a2f24")

ax2.set_xlabel("how close to normal depth you decide to call it (%)")
ax2.set_ylabel("quoted extent of backwater (km)")
ax2.set_title("An asymptote has no end, so this is a choice")
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()

fig.suptitle(
 "Backwater at a bridge: the profile is physics, the extent quoted for "
 "it is a convention",
 y=0.98, fontsize=12, fontweight="bold")
fig.tight_layout(rect=(0, 0, 1, 0.93))
fig.savefig("media/backwater.png", dpi=600, bbox_inches="tight")

print("\nWrote media/backwater.png")

```

## examples/navigation\_channel.py

```

"""
Approach channel design: how deep, how wide, and how much dredging.

Sizes an approach channel for a post-Panamax container ship, prints the
depth chain and the width build-up, and issues the cross-section as a
drawing sheet.

Also shows what the answer is sensitive to. The depth is usually argued
over in centimetres and won or lost in metres, because a single judgement
call, the fraction of the wave height taken as vertical vessel motion,
moves it more than every tolerance put together.

Run from the repository root:

 python examples/navigation_channel.py
"""

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.channel import (
 Vessel,
 design_channel,
 squat_barrass,
 squat_icorels,
 turning_basin_diameter,
)
from pyCoastal.applications.sections import channel_section, channel_sheet
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

--- the design vessel -----
vessel = Vessel(
 name="Post-Panamax container ship",
 length=336.0,
 beam=48.2,
 draught=14.5,
 block_coefficient=0.68,
)

--- conditions in the channel -----
Each width class is stated. Anything left out is silently taken at its most
benign value, which is how a channel ends up too narrow on paper.
CONDITIONS = {
 "speed": "slow",
 "crosswind": "moderate",
 "crosscurrent": "low",
 "longitudinal_current": "low",
 "waves": "moderate",
 "aids_to_navigation": "good",
 "bottom_surface": "smooth_and_soft",
 "depth_of_waterway": "shallow",
 "cargo_hazard": "low",
}

design = design_channel(
 vessel,
 speed=8.0, # knots through the water
 design_water_level=1.20, # m CD, the level the depth is referred to
 Hs=1.8, # significant wave height in the channel
 Tp=9.0,
 wave_factor=0.5,
 net_clearance=0.8, # over a soft bed
 water_level_allowance=0.3,
 dredging_tolerance=0.3,
 survey_tolerance=0.2,
 siltation_allowance=0.2,
 side_slope=5.0,
 existing_bed=-11.5,
 manoeuvrability="moderate",
 section="outer",
 two_way=True,
 speed_class="moderate",
 bank="sloping_channel_edges",
 conditions=CONDITIONS,
)

print(design.summary())

basin = turning_basin_diameter(vessel, assisted=True, current=True)
print(f"\nTurning basin {basin['diameter']:.0f} m "
 f"({basin['factor']:.1f} x Loa, assisted, current running)")

--- cross-checks and sensitivities -----
depth = design.required_depth
print("\nSquat, two ways")

```

```

for speed in (6.0, 8.0, 10.0, 12.0):
 icorels = squat_icorels(vessel, speed, depth)
 barrass = squat_barrass(vessel, speed, depth)
 flag = " (beyond range)" if icorels["beyond_range"] else ""
 print(f" {speed:4.1f} kn ICORELS {icorels['squat']:.2f} m "
 f"Barrass {barrass['squat']:.2f} m "
 f"Fnh {icorels['froude']:.2f}{flag}")

print("\nWhat moves the dredge level")
base = design.required_depth
for label, kwargs in (
 ("wave factor 0.3 rather than 0.5", dict(wave_factor=0.3)),
 ("wave factor 0.7 rather than 0.5", dict(wave_factor=0.7)),
 ("speed 12 kn rather than 8 kn", dict(speed=12.0)),
 ("net clearance 1.5 m rather than 0.8 m", dict(net_clearance=1.5)),
 ("no siltation allowance", dict(siltation_allowance=0.0)),
):
 variant = design_channel(
 vessel, speed=kwargs.pop("speed", 8.0), design_water_level=1.20,
 Hs=1.8, Tp=9.0, wave_factor=kwargs.pop("wave_factor", 0.5),
 net_clearance=kwargs.pop("net_clearance", 0.8),
 water_level_allowance=0.3, dredging_tolerance=0.3,
 survey_tolerance=0.2,
 siltation_allowance=kwargs.pop("siltation_allowance", 0.2),
 side_slope=5.0, existing_bed=-11.5, conditions=CONDITIONS,
 two_way=True,
)
 change = variant.required_depth - base
 volume = ((variant.dredge_volume(1000.0) - design.dredge_volume(1000.0))
 / 1e3)
 print(f" {label:<38}{change:+5.2f} m {volume:+8.0f} k m3/km")

--- drawings -----
section = channel_section(design, margin=90.0)
section.save("media/navigation_channel.png")

sheet = channel_sheet(
 design,
 project="Container terminal approach",
 title="Navigation channel typical section",
 client="Example Port Authority",
 size="A3",
 margin=90.0,
 file="examples/navigation_channel.py",
)
sheet.save("media/navigation_channel_sheet.png")
sheet.to_dxf("media/navigation_channel.dxf")
print("\nWrote media/navigation_channel.png, "
 "media/navigation_channel_sheet.png and media/navigation_channel.dxf")

--- how the depth chain is made up -----
fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.0, 5.6),
 gridspec_kw={"width_ratios": [1.15, 1.0]})

names = ["static draught"] + [
 k for k, v in design.clearance["components"].items() if v > 0
]
values = [vessel.draught] + [
 v for v in design.clearance["components"].values() if v > 0
]
bottoms = np.concatenate([0.0, np.cumsum(values[:-1])])
colours = ["#0b3554"] + [
 "#1b6fa0", "#2f93b8", "#79c6d6", "#c3e8ee",
 "#e6d5ad", "#d8b978", "#c47f1a", "#8a2f24"
][:len(values) - 1]

for name, value, bottom, colour in zip(names, values, bottoms, colours):
 ax.barh(0, value, left=bottom, height=0.55, color=colour,
 edgecolor="#1a1a1a", linewidth=0.6)
 if value > 0.18:
 ax.text(bottom + 0.5 * value, 0, f"{value:.2f}", ha="center",
 va="center", fontsize=8,
 color="white" if colour in ("#0b3554", "#1b6fa0", "#8a2f24")
 else "#1a1a1a")
 ax.text(bottom + 0.5 * value, 0.42, name, rotation=45, ha="left",
 va="bottom", fontsize=7.5)

ax.set_xlim(0, sum(values) * 1.02)
ax.set_ylim(-0.5, 1.5)
ax.set_yticks([])
ax.set_xlabel("depth below the design water level (m)")
ax.set_title(f"Depth chain: {sum(values):.2f} m required")
ax.grid(True, axis="x", which="both", alpha=0.3)
ax.minorticks_on()

factors = np.linspace(0.2, 0.8, 25)
depths, volumes = [], []
for factor in factors:
 variant = design_channel(
 vessel, speed=8.0, design_water_level=1.20, Hs=1.8, Tp=9.0,
 wave_factor=float(factor), net_clearance=0.8,
 water_level_allowance=0.3, dredging_tolerance=0.3,
 survey_tolerance=0.2, siltation_allowance=0.2, side_slope=5.0,
 existing_bed=-11.5, conditions=CONDITIONS, two_way=True,

```

```

)
 depths.append(variant.required_depth)
 volumes.append(variant.dredge_volume(1000.0) / 1e3)

ax2.plot(factors, volumes, color="#0b3554", lw=1.8)
ax2.axvline(0.5, color="#8a2f24", lw=1.2, ls="--", label="value used, 0.50")
ax2.fill_between(factors, volumes, min(volumes), color="#9fc4d6", alpha=0.35)
ax2.set_xlabel("wave response allowance, as a fraction of Hs")
ax2.set_ylabel("capital dredging (thousand m3 per km)")
ax2.set_title("One judgement call, and what it costs")
ax2.legend(loc="upper left")
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()
span = max(volumes) - min(volumes)
ax2.annotate(
 f"{span:.0f} thousand m3 per km\ncross the defensible range",
 xy=(0.62, 0.5 * (max(volumes) + min(volumes))),
 xytext=(0, 0), textcoords="offset points", fontsize=8.5, ha="center",
)

fig.suptitle(
 f"Approach channel for {vessel.name}: dredge level "
 f"{design.dredge_level:+.2f} m CD, {design.width:.0f} m wide",
 y=0.98, fontsize=12, fontweight="bold",
)

fig.subplots_adjust(wspace=0.26, top=0.84)
fig.savefig("media/channel_depth_chain.png", dpi=600, bbox_inches="tight")
print("Wrote media/channel_depth_chain.png")

```



## examples/berth\_fenders.py

```
"""
Berthing energy and fender selection for a container terminal.

A berth fails in three different ways and only one of them is about energy.

The fender can be too small, and the ship reaches the quay. The panel can
be too small, and the fender's reaction dents the side shell even though
the energy was absorbed perfectly. The fenders can be too far apart, and
the hull touches the structure between them, at a point where nothing is
measuring and nobody is watching.

This sizes a fender for a post-Panamax container ship and then checks the
other two, which is where berths actually get into trouble.

Run from the repository root:

 python examples/berth_fenders.py
"""

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.berthing import (
 ABNORMAL_FACTOR,
 BERTHING_VELOCITY,
 CONE_FENDER,
 berthing_energy,
 design_berth,
 hull_pressure,
 select_fender,
)
from pyCoastal.applications.channel import Vessel
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

SHIP = Vessel(name="Post-Panamax container", length=366.0, beam=48.2,
 draught=15.2, block_coefficient=0.68)
FEEDER = Vessel(name="Feeder", length=140.0, beam=22.0, draught=8.5,
 block_coefficient=0.68)

design = design_berth(SHIP, velocity=0.15, vessel_class="container",
 configuration="open_piled", depth=17.0,
 smallest_vessel=FEEDER, panel_aspect=(1.4, 2.2))

print(f"[SHIP.name]: {SHIP.displacement / 1000:.0f} thousand tonnes")
print(f"Approaching at {design.velocity:.2f} m/s")
print()
print(f" kinetic energy {design.normal['kinetic']:.8f} kNm")
print(f" x Cm {design.normal['Cm']:.3f} added water")
print(f" x Ce {design.normal['Ce']:.3f} rotation about the contact")
print(f" x Cs {design.normal['Cs']:.3f} fender softness")
print(f" x Cc {design.normal['Cc']:.3f} berth configuration")
print(f" normal berthing {design.normal['energy']:.8f} kNm")
print(f" x {design.abnormal['factor']:.2f} abnormal allowance")
print(f" design energy {design.energy:.8f} kNm")
print()
print(f" fender {design.fender['height']:.2f} m, "
 f"{design.fender['rated_energy']:.0f} kNm rated "
 f"{(100 * design.fender['utilisation']:.0f)}% used")
print(f" reaction {design.fender['reaction']:.0f} kN")
print(f" hull pressure {design.pressure['pressure']:.0f} kN/m2 "
 f"against {design.pressure['limit']:.0f} allowed")
print(f" spacing {design.spacing['spacing']:.1f} m against "
 f"{design.spacing['limit']:.1f} m allowed")
print(f" ADEQUATE {design.adequate}")
print()
for note in design.notes:
 print(f" - {note}")

--- where the energy actually goes -----
fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.0, 5.4))

stages = ["kinetic", "x Cm", "x Ce", "x Cs Cc", "x abnormal"]
n = design.normal
values = [
 n["kinetic"],
 n["kinetic"] * n["Cm"],
 n["kinetic"] * n["Cm"] * n["Ce"],
 n["energy"],
 design.energy,
]
colours = ["#54514b", "#8a2f24", "#2c6b46", "#2c6b46", "#c2501c"]

ax.bar(stages, values, color=colours, width=0.62)
for i, value in enumerate(values):
 ax.annotate(f"{value:.0f}", xy=(i, value), xytext=(0, 4),
```

```

 textcoords="offset points", ha="center", fontsize=9)
ax.axhline(n["kinetic"], color="#54514b", lw=1.0, ls=":")
ax.set_ylabel("energy (kNm)")
ax.set_title("Only the added water makes it bigger")
ax.grid(True, axis="y", alpha=0.3)
ax.set_axisbelow(True)

--- and what the two design levers do -----
Velocity is squared, so it dominates everything. The abnormal factor is a
straight multiplier but a large one. Between them they set the fender.
velocities = np.linspace(0.05, 0.40, 200)
for label, factor, colour in (("tanker, 1.25", 1.25, "#0b3554"),
 ("container, 1.50", 1.50, "#1b6fa0"),
 ("ro-ro, 2.00", 2.00, "#c2501c")):
 energies = [berthing_energy(SHIP, v)["energy"] * factor for v in velocities]
 ax2.plot(velocities, energies, lw=2.0, color=colour,
 label=f"abnormal {label}")

for height in (1.0, 1.4, 2.0, 2.5):
 rated = CONE_FENDER.energy(height)
 ax2.axhline(rated, color="#85817a", lw=0.8, ls="--")
 ax2.annotate(f"{height:.1f} m fender", xy=(0.052, rated), xytext=(0, 3),
 textcoords="offset points", fontsize=8, color="#85817a")

ax2.axvline(design.velocity, color="#8a2f24", lw=1.0, ls=":")
ax2.annotate(f"design {design.velocity:.2f} m/s",
 xy=(design.velocity, 6000), xytext=(6, 0),
 textcoords="offset points", fontsize=9, color="#8a2f24")

ax2.set_xlabel("berthing velocity (m/s)")
ax2.set_ylabel("design energy (kNm)")
ax2.set_title("Velocity is squared, so it decides the fender")
ax2.set_ylim(0, 9000)
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()
ax2.legend(loc="upper left", fontsize=8.5)

fig.suptitle(
 "Berthing energy: three of the four PIANC factors reduce it, and the "
 "abnormal allowance puts most of it back",
 y=0.98, fontsize=12, fontweight="bold")
fig.tight_layout(rect=(0, 0, 1, 0.93))
fig.savefig("media/berth_energy.png", dpi=600, bbox_inches="tight")

print("\nWrote media/berth_energy.png")

```

## examples/nourishment\_design.py

```

"""
Beach nourishment design study.

Evolves a beach fill with the one-line model, compares it against the
Pelnard-Considere analytical solution, and reports design life and a
renourishment schedule over a 30-year planning horizon.

Run from the repository root:

 python examples/nourishment_design.py
"""

import matplotlib
import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.nourishment import (
 SECONDS_PER_YEAR,
 NourishmentDesign,
 WaveClimate,
 longshore_diffusivity,
 pelnard_considere,
 renourishment_schedule,
 simulate_nourishment,
)

matplotlib.rcParams["font.family"] = "sans-serif"
matplotlib.rcParams["font.sans-serif"] = ["Arial"]

--- design case -----
design = NourishmentDesign(
 length=3000.0, # m of full-width fill
 berm_width=30.0, # m of shoreline advance
 taper=200.0, # m of linear taper at each end
 D=8.0, # m depth of closure
 B=2.0, # m berm height
 porosity=0.4,
)
climate = WaveClimate(Hb=1.0, T=8.0, alpha0=0.0)

eps = longshore_diffusivity(climate, design)
print(f"Placed volume : {design.placed_volume:,.0f} m3")
print(f"Active profile height: {design.active_height:.1f} m")
print(f"Difffusivity : {eps:.4f} m2/s ({eps * SECONDS_PER_YEAR:,.0f} m2/yr)")

--- evolve -----
result = simulate_nourishment(
 design, climate, duration=20.0 * SECONDS_PER_YEAR, dx=20.0, n_outputs=41
)

print(f"\n{'year':>6}{ 'retained %':>12}{ 'berm width (m)':>16}")
for yr, frac, width in list(
 zip(result.times_years, result.retained_fraction, result.berm_width)
)[::8]:
 print(f"{yr:>6.1f}{100 * frac:>12.1f}{width:>16.1f}")

life = result.design_life(threshold=0.5)
print(f"\nDesign life (50% retained): {life / SECONDS_PER_YEAR:.1f} yr")

plan = renourishment_schedule(
 design, climate, horizon=30.0 * SECONDS_PER_YEAR, threshold=0.5,
 dx=20.0, n_outputs=61,
)
print(f"Renourishment interval : {plan['interval'] / SECONDS_PER_YEAR:.1f} yr")
print(f"Renourishments in 30 yr : {plan['n_renouishments']}")
print(f"Total sand over 30 yr : {plan['total_volume']:,.0f} m3")

--- figure -----
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4.5))

for i in (0, 4, 10, 20, 40):
 yr = result.times_years[i]
 ax1.plot(result.x / 1000.0, result.planforms[i], lw=1.6, label=f"{yr:.0f} yr")

The analytical solution is for a bare rectangle of length `design.Length`,
so it carries slightly less sand than the simulated fill, which also has a
taper at each end. The small offset between the two curves is that volume
difference, not solver error. test_applications_nourishment.py compares the
two with the taper removed.
analytic = pelnard_considere(result.x, result.times[-1], result.design, climate)
ax1.plot(
 result.x / 1000.0, analytic, "k--", lw=1.2,
 label=f"Pelnard-Considere, {result.times_years[-1]:.0f} yr",
)
ax1.set_xlabel("Alongshore distance (km)")
ax1.set_ylabel("Shoreline offset (m)")
ax1.set_title("Fill planform evolution")

```

```

ax1.legend(frameon=False, fontsize=9)
ax1.grid(True, which="both", alpha=0.3)
ax1.minorticks_on()

ax2.plot(result.times_years, 100 * result.retained_fraction, lw=1.8)
ax2.axhline(50, color="0.4", ls=":", lw=1.2)
if np.isfinite(life):
 ax2.axvline(life / SECONDS_PER_YEAR, color="0.4", ls=":", lw=1.2)
 ax2.annotate(
 f"design life {life / SECONDS_PER_YEAR:.1f} yr",
 xy=(life / SECONDS_PER_YEAR, 50),
 xytext=(life / SECONDS_PER_YEAR + 1.0, 70),
 fontsize=9,
 arrowprops=dict(arrowstyle="->", color="0.4"),
)
ax2.set_xlabel("Time (yr)")
ax2.set_ylabel("Volume retained in project area (%)")
ax2.set_title("Fill retention")
ax2.set_ylim(0, 105)
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()

fig.tight_layout()
fig.savefig("media/nourishment_design.png", dpi=600)
print("\nWrote media/nourishment_design.png")

```

## examples/nourishment\_profile.py

```
"""
Beach nourishment: what the borrow source is worth.

The planform model in `examples/nourishment_design.py` answers how long a
fill lasts. This answers the question asked before it: how much dry beach
does a given volume buy, and how much does that depend on where the sand
comes from.

A great deal, is the answer. A beach takes the profile its own grain size
can hold. Fill coarser than the native sand stands steeper and buys more
beach per cubic metre; fill finer lies flatter, and below a critical volume
it buys none at all, because the whole placement goes into flattening the
underwater profile.

Run from the repository root:

 python examples/nourishment_profile.py
"""

import matplotlib.pyplot as plt
import numpy as np

from pyCoastal.applications.nourishment import (
 critical_volume,
 dean_scale,
 grain_compatibility,
 profile_overfill_factor,
 shoreline_advance,
)
from pyCoastal.applications.nourishment import (
 SECONDS_PER_YEAR,
 NourishmentDesign,
 WaveClimate,
)
from pyCoastal.applications.sections import (
 nourishment_plan_section,
 nourishment_section,
 nourishment_sheet,
 spreading_half_life,
)
from pyCoastal.applications.sediment import sediment
from pyCoastal.drafting import use_crisp_style

use_crisp_style()

NATIVE = "medium_sand"
BERM = 2.0 # m above mean sea level
CLOSURE = 6.0 # m, depth of closure
VOLUME = 250.0 # m3 per metre of beach
SOURCES = ["very_fine_sand", "fine_sand", "medium_sand", "coarse_sand",
 "fine_gravel"]

A_native = dean_scale(NATIVE)
print(f"Native beach: {sediment(NATIVE).name}, "
 f"d50 = {sediment(NATIVE).d50 * 1000:.2f} mm, A = {A_native:.3f}")
print(f"Berm {BERM:.1f} m, closure {CLOSURE:.1f} m, "
 f"fill {VOLUME:.0f} m3 per metre\n")

print("borrow A delta profile dry beach critical")
results = {}
for key in SOURCES:
 A = dean_scale(key)
 got = shoreline_advance(A_native, A, VOLUME, BERM, CLOSURE)
 match = grain_compatibility(NATIVE, key)
 results[key] = (A, got, match)
 print(f" {sediment(key).name:<15} {A:.3f} {match['delta']:+5.2f} "
 f"got['kind']:<17} {got['advance']:.6f} m "
 f"got['critical_volume']:.6f} m3/m")

print("\nWhat that costs, per metre of beach, for 40 m of dry sand")
for key in SOURCES:
 factor = profile_overfill_factor(NATIVE, key, BERM, CLOSURE,
 advance=40.0)
 print(f" {sediment(key).name:<15} {factor['borrow_volume']:.6f} m3/m "
 f"({factor['factor']:.2f} times the native requirement)")

print("\nCompatibility")
for key in SOURCES:
 print(f" {sediment(key).name:<15} {results[key][2]['verdict']}")

--- the design section -----
CHOSEN = "coarse_sand"
chosen = results[CHOSEN][1]
section = nourishment_section(chosen, NATIVE, CHOSEN, BERM, CLOSURE)
section.save("media/nourishment_profile.png")

--- the same design seen from above -----
```

```

The profile says how much dry beach the sand buys. It says nothing about
how long it stays there, and for a fill that is the second half of the
question: a 1.5 km placement in this climate is half gone inside a year.
LENGTH = 1500.0
climate = WaveClimate(Hb=1.2, T=8.0, alpha0=0.0)
design = NourishmentDesign(length=LENGTH, berm_width=chosen["advance"],
 taper=0.1 * LENGTH, D=CLOSURE, B=BERM)

half_life = spreading_half_life(design, climate) / SECONDS_PER_YEAR
print(f"\nPlanform, {LENGTH:.0f} m of fill in Hb = {climate.Hb:.1f} m:")
print(f" spreading half-life {half_life:.2f} yr")
print(f" placed volume "
 f"{design.placed_volume / 1e3:.0f} thousand m3 in place")

plan = nourishment_plan_section(design, climate)
plan.save("media/nourishment_plan.png")

sheet = nourishment_sheet(
 chosen, NATIVE, CHOSEN, BERM, CLOSURE,
 project="Bayfront beach management",
 title="Nourishment design profile",
 client="Example Coastal Authority",
 file="examples/nourishment_profile.py",
 plan_design=design,
 plan_climate=climate,
)
sheet.save("media/nourishment_sheet.png")
print("\nWrote media/nourishment_profile.png, media/nourishment_plan.png "
 "and media/nourishment_sheet.png")

--- how the answer moves with the borrow source -----
fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.0, 5.4))

volumes = np.linspace(10.0, 900.0, 90)
palette = ["#8a2f24", "#c47f1a", "#0b3554", "#1f8f8f", "#2f6b46"]
for key, colour in zip(SOURCES, palette):
 A = dean_scale(key)
 widths = [shoreline_advance(A_native, A, float(v), BERM, CLOSURE)["advance"]
 for v in volumes]
 ax.plot(volumes, widths, lw=1.9, color=colour, label=sediment(key).name)
 critical = critical_volume(A_native, A, BERM, CLOSURE)
 if critical > 0:
 ax.plot([critical], [0.0], "o", ms=5, color=colour,
 markeredgewidth=0.6)

ax.axhline(0.0, color="#54514b", lw=0.8)
ax.set_xlabel("placed volume (m3 per metre of beach)")
ax.set_ylabel("dry beach gained (m)")
ax.set_title("What a cubic metre buys, by borrow source")
ax.legend(loc="upper left", fontsize=8)
ax.grid(True, which="both", alpha=0.3)
ax.minorticks_on()
ax.annotate("filled circles are the critical volume:\n"
 "below it the fill is entirely submerged",
 xy=(0.52, 0.12), xycoords="axes fraction", fontsize=8.5,
 color="#54514b")

y = np.linspace(0.0, 420.0, 400)
for key, colour in zip(SOURCES, palette):
 A = dean_scale(key)
 ax2.plot(y, -A * y ** (2 / 3), lw=1.7, color=colour,
 label=f"{sediment(key).name}, A = {A:.3f}")
 ax2.axhline(-CLOSURE, color="#8a2f24", lw=1.1, ls="--")
 ax2.annotate(f"Closure {CLOSURE:.0f} m", xy=(300, -CLOSURE), xytext=(0, 6),
 textcoords="offset points", fontsize=8.5, color="#8a2f24")
ax2.set_ylim(-9, 0.5)
ax2.set_xlabel("distance offshore (m)")
ax2.set_ylabel("depth (m)")
ax2.set_title("Equilibrium profiles, h = A y^{2/3}")
ax2.legend(loc="lower left", fontsize=8)
ax2.grid(True, which="both", alpha=0.3)
ax2.minorticks_on()

fig.suptitle(
 f"Nourishment on {sediment(NATIVE).name.lower()}: the borrow grain size "
 "decides what the sand is worth",
 y=0.98, fontsize=12, fontweight="bold",
)
fig.tight_layout(rect=(0, 0, 1, 0.94))
fig.savefig("media/nourishment_borrow.png", dpi=600, bbox_inches="tight")
print("Wrote media/nourishment_borrow.png")

```

## examples/storm\_surge\_flooding.py

```

"""
Storm surge water level and flood mapping.

Builds the water-level budget for a design storm, maps the flooded area on a
synthetic barrier-island terrain, and shows the difference between a plain
elevation threshold and a hydraulically connected flood map.

Run from the repository root:

 python examples/storm_surge_flooding.py
"""

import matplotlib
import matplotlib.pyplot as plt
import numpy as np
from matplotlib.colors import ListedColormap

from pyCoastal.applications.surge import (
 StormConditions,
 bathtub_flood,
 flood_volume,
 flooded_area,
 inundation_limit,
 isolated_low_ground,
 total_water_level,
)

matplotlib.rcParams["font.family"] = "sans-serif"
matplotlib.rcParams["font.sans-serif"] = ["Arial"]

--- design storm -----
storm = StormConditions(
 wind_speed=45.0, # sustained, 10 m
 central_pressure=94000.0, # 940 hPa
 tide=0.6, # high water at Landfall
 Hm0=7.0,
 Tp=13.0,
 fetch=120e3,
 shelf_depth=18.0,
 sea_level_rise=0.0,
)

A real shelf, shoaling from 60 m to 8 m over the fetch.
shelf = np.linspace(60.0, 8.0, 200)
levels = total_water_level(
 storm, beach_slope=0.04, shelf_depths=shelf, dx=storm.fetch / 199
)

print("Water level budget (m above datum)")
for key in ("tide", "sea_level_rise", "barometric_setup", "wind_setup", "wave_setup"):
 print(f" {key.replace('_', ' '):>18} {levels[key]:6.2f}")
print(f" {'still water level':>18} {levels['still_water_level']:6.2f}")
print(f" {'runup R2%':>18} {levels['runup_R2']:6.2f}")
print(f" {'total water level':>18} {levels['total_water_level']:6.2f}")

swl = levels["still_water_level"]

--- terrain: a barrier island with a low-lying bay behind -----
nx, ny, dx = 260, 200, 20.0
x = np.arange(nx) * dx
y = np.arange(ny) * dx
X, Y = np.meshgrid(x, y, indexing="ij")

Shoreface shoaling from -6 m up to the barrier toe at +1 m.
terrain = -6.0 + 7.0 * np.clip((X - 600.0) / 800.0, 0.0, 1.0)

Barrier ridge, crest at 9 m, well above the design still water level.
ridge = 8.0 * np.exp(-(((X - 1500.0) / 130.0) ** 2))
A tidal inlet cut clean through it at y = 1.7 km.
ridge *= 1.0 - 0.97 * np.exp(-(((Y - 1700.0) / 120.0) ** 2))
terrain = terrain + ridge

Low back-barrier bay, then gently rising mainland.
bay = (X > 1750.0) & (X < 3600.0)
terrain[bay] = 0.8
mainland = X >= 3600.0
terrain[mainland] = 0.8 + 0.004 * (X[mainland] - 3600.0)

A sealed depression, below the water level but with no path to the sea.
pond = ((X - 2600.0) ** 2 / 300.0**2 + (Y - 800.0) ** 2 / 220.0**2) < 1.0
terrain[pond] = -1.5
A levee ringing the pond, so only connectivity keeps it dry.
ring = ((X - 2600.0) ** 2 / 380.0**2 + (Y - 800.0) ** 2 / 300.0**2)
terrain[(ring >= 1.0) & (ring < 1.45)] = 8.0

--- flood maps -----
sea = np.zeros_like(terrain, dtype=bool)

```



```

sea[0, :] = True # the sea enters at x = 0

connected = bathtub_flood(terrain, swl, seed=sea)
naive = terrain <= swl
isolated = isolated_low_ground(terrain, swl, seed=sea)

print(f"\nStill water level {swl:.2f} m")
print(f" threshold-only flooded area : {flooded_area(naive, dx) / 1e6:6.2f} km2")
print(f" connected flooded area : {flooded_area(connected, dx) / 1e6:6.2f} km2")
print(f" overstated by the threshold : {flooded_area(isolated, dx) / 1e6:6.2f} km2 ")
print(f" f"({100 * isolated.sum() / max(naive.sum(), 1):.0f}% of it)")
print(f" water volume on land : {flood_volume(terrain, connected, swl, dx) / 1e6:6.2f} Mm3")

What does the inlet cost? Seal it and flood the same terrain again.
sealed = terrain.copy()
inlet = (np.abs(Y - 1700.0) < 400.0) & (np.abs(X - 1500.0) < 400.0)
sealed[inlet] = np.maximum(sealed[inlet], 9.0)
sealed_flood = bathtub_flood(sealed, swl, seed=sea)
print()
print(f" with the inlet open : {flooded_area(connected, dx) / 1e6:6.2f} km2 flooded")
print(f" with the inlet sealed : {flooded_area(sealed_flood, dx) / 1e6:6.2f} km2 flooded")
print(f" a single breach in the barrier is what floods the bay")

--- cross-shore transect ---
j_intact = int(np.argmax(np.abs(y - 500.0))) # through the intact ridge
j_breach = int(np.argmax(np.abs(y - 1700.0))) # through the tidal inlet
limit_intact = inundation_limit(x, terrain[:, j_intact], swl)
limit_breach = inundation_limit(x, terrain[:, j_breach], swl)
print(f"\n inundation limit, intact barrier: {limit_intact:7.0f} m")
print(f" inundation limit, at the inlet : {limit_breach:7.0f} m")

--- figure ---
fig = plt.figure(figsize=(13, 8.5))
gs = fig.add_gridspec(2, 2, height_ratios=[1.35, 1.0], hspace=0.28, wspace=0.18)

extent = [0, nx * dx / 1000.0, 0, ny * dx / 1000.0]
ax = fig.add_subplot(gs[0, 0])
im = ax.imshow(terrain.T, extent=extent, origin="lower", cmap="terrain",
 vmin=-6, vmax=8, aspect="equal")
ax.contour(X / 1000.0, Y / 1000.0, terrain, levels=[swl], colors="k", linewidths=0.8)
ax.set_title("Terrain, with the still water level contour")
ax.set_xlabel("x (km)")
ax.set_ylabel("y (km)")
fig.colorbar(im, ax=ax, label="elevation (m)", shrink=0.85)

ax = fig.add_subplot(gs[0, 1])
ax.imshow(terrain.T, extent=extent, origin="lower", cmap="gray",
 vmin=-10, vmax=12, aspect="equal", alpha=0.85)
ax.imshow(np.where(connected, 1.0, np.nan).T, extent=extent, origin="lower",
 cmap=ListedColormap(["#1b6fa0"]), aspect="equal", alpha=0.85)
ax.imshow(np.where(isolated, 1.0, np.nan).T, extent=extent, origin="lower",
 cmap=ListedColormap(["#d94a3d"]), aspect="equal", alpha=0.9)
ax.set_title("Flooded (blue) vs low ground the sea cannot reach (red)")
ax.set_xlabel("x (km)")
ax.set_ylabel("y (km)")

for col, (j, label, limit) in enumerate(
 ((j_intact, "intact barrier", limit_intact), (j_breach, "tidal inlet", limit_breach))
):
 ax = fig.add_subplot(gs[1, col])
 z = terrain[:, j]
 ax.fill_between(x / 1000.0, -8, z, color="#c8b89a", zorder=2)
 # Fill only as far as water can actually reach. Filling every cell below
 # the level would put water behind an intact barrier and contradict the
 # inundation limit drawn alongside it.
 reached = (z <= swl) & (x <= limit)
 ax.fill_between(x / 1000.0, z, swl, where=reached, color="#1b6fa0",
 alpha=0.75, zorder=1)
 stranded = (z <= swl) & (x > limit)
 if stranded.any():
 ax.fill_between(x / 1000.0, z, swl, where=stranded, color="#d94a3d",
 alpha=0.30, zorder=1,
 label="below the level, not reached")
 ax.axhline(swl, color="#0b3554", lw=1.4, ls="--",
 label=f"still water {swl:.2f} m")
 ax.axhline(levels["total_water_level"], color="#8a2f24", lw=1.1, ls=":",
 label=f"with runup {levels['total_water_level']:.2f} m")
 ax.axvline(limit / 1000.0, color="k", lw=1.0, alpha=0.6)
 ax.annotate(f"limit {limit:.0f} m", xy=(limit / 1000.0, 7.5),
 xytext=(6, 0), textcoords="offset points", fontsize=8)
 ax.set_title(f"Cross-shore transect, {label}")
 ax.set_xlabel("x (km)")
 ax.set_ylabel("elevation (m)")
 ax.set_ylim(-8, 9)
 ax.legend(frameon=False, fontsize=8, loc="upper right")
 ax.grid(True, which="both", alpha=0.3)
 ax.minorticks_on()

fig.suptitle(
 f"Storm surge flooding: {storm.wind_speed:.0f} m/s, "
 f"{storm.central_pressure / 100:.0f} hPa, Hm0 = {storm.Hm0:.1f} m, "
 f"tide {storm.tide:.1f} m",

```

```
 y=0.965,
)
fig.savefig("media/storm_surge_flooding.png", dpi=600, bbox_inches="tight")
print("\nWrote media/storm_surge_flooding.png")
```

## Case files in examples/configs/

### current.yaml

```
grid:
 nx: 200
 ny: 200
 dx: 1.0
 dy: 1.0

physics:
 speed: 1.0 # current speed (m/s)

solver:
 cfl: 0.5
 t_final: 500.0

output:
 gauge: [100, 100]
```

### equilibrium\_shoreline.yaml

```
geometry:
 Lshore: 600.0 # domain length (m)
 Gb: 80.0 # gap between breakwaters (m)
 Lb1: 80.0 # left breakwater length (m)
 Lb2: 80.0 # right breakwater length (m)
 Yi: 60.0 # breakwater offshore distance (m)

numerics:
 n_points: 400 # samples per arc
 sag_side_factor: 0.5 # lateral arc sag factor (relative to Ye)
```

### pollutant.yaml

```
grid:
 Lx: 2.0 # pond length [m]
 Ly: 1.0 # pond width [m]
 Nx: 200
 Ny: 100

physics:
 diffusivity: 0.001 # D [m²/s]
 U0: 0.2 # max velocity [m/s]
 R: 0.4 # vortex radius [m]

solver:
 t_final: 5.0
 c0_sigma: 0.05 # initial Gaussian width [m]
```

### viscous\_fluid.yaml

```
grid:
 Nx: 100
 Ny: 100
 Lx: 2.0
 Ly: 2.0

physics:
 viscosity: 0.001
 background_speed: 1.5
 vortex_sigma: 0.2

solver:
 dt: 0.001
 t_final: 4.0
```

### water\_drop.yaml

```
grid:
 Nx: 200
 Ny: 200
 Lx: 2.0
 Ly: 2.0

physics:
 wave_speed: 1.0
 sigma: 0.1
```

```
solver:
 CFL: 0.5
 t_final: 4.0
```

#### waves2D.yam1

```
grid:
 nx: 100
 ny: 100
 dx: 0.1
 dy: 0.1

physics:
 gravity: 9.81
 depth: 5.0

forcing:
 amplitude: 0.2
 period: 2.3

solver:
 cfl: 1.0
 t_final: 20.0

output:
 obs_points:
 - [20, 50]
```

#### waves2D\_irregular.yam1

```
grid:
 nx: 200
 ny: 200
 dx: 1.0
 dy: 1.0

physics:
 gravity: 9.81
 depth: 5.0

forcing:
 type: jonswap # options: pm or jonswap
 gamma: 3.3
 Hs: 0.5 # significant wave height (m)
 Tp: 3.0 # peak period (s)

solver:
 dt: 0.1
 duration: 60.0

output:
 gauge: [100, 100] # grid indices (i, j)
```

## Appendix C. List of symbols

Symbol	Meaning	Unit
$x, y, z$	horizontal coordinates and elevation (positive up)	m
$t, \Delta t$	time and time step	s
$\Delta x, \Delta y$	grid spacings	m
$u, v, \mathbf{u}$	velocity components and vector	m/s
$h, d$	water depth	m
$\eta$	free-surface elevation, setup	m
$g$	gravitational acceleration, 9.81	m/s <sup>2</sup>
$\rho, \rho_s$	water and sediment (or rock) density	kg/m <sup>3</sup>
$\Delta$	relative buoyant density $\rho_s/\rho - 1$	-
$\nu, \nu_t$	kinematic and eddy viscosity	m <sup>2</sup> /s
$k, \varepsilon, \omega$	turbulent kinetic energy, dissipation, specific dissipation	m <sup>2</sup> /s <sup>2</sup> , m <sup>2</sup> /s <sup>3</sup> , 1/s
$C_s$	Smagorinsky constant	-
$\sigma(x), L_{sp}$	sponge damping rate and thickness	1/s, m
$H, H_s, H_{m0}, H_b$	wave height, significant height, spectral height, breaking height	m
$T, T_p, T_m - 1, 0$	period, peak period, spectral period	s
$L, L_0, k, \omega$	wavelength, deep-water wavelength, wavenumber, angular frequency	m, m, 1/m, 1/s
$c, c_g$	phase and group celerity	m/s
$\xi$	surf similarity (Iribarren) number	-
$U$	Ursell number	-
$S(f), m_n$	variance density spectrum, spectral moment	m <sup>2</sup> /Hz
$\alpha, \beta$	structure slope angle; wave obliquity or beach slope	rad or deg
$D_{n50}, M_{50}$	nominal stone diameter and median mass	m, kg
$P, S, N$	notional permeability, damage level, number of waves	-
$K_D$	Hudson stability coefficient	-
$R_c$	crest freeboard	m
$q$	mean overtopping discharge	l/s/m
$\gamma_f, \gamma_\beta, \gamma_b, \gamma_v$	EurOtop roughness, obliquity, berm, and wall factors	-
$p_1, p_3, p_4, p_u$	Goda pressures and uplift	kPa
$K_r, K_d$	reflection and disturbance coefficients	-
$C_d, C_m$	Morison drag and inertia coefficients	-
$KC$	Keulegan-Carpenter number	-
$U_c, U_m, U_{cw}$	current, near-bed orbital velocity, current share	m/s, m/s, -
$S, S_{eq}$	scour depth, equilibrium scour depth	m
$D_e$	effective pier diameter	m
$\theta, \theta_{cr}$	Shields parameter and its critical value	-
$d_{50}, D^*, w_s$	median grain size, dimensionless grain size, fall velocity	m, -, m/s

Symbol	Meaning	Unit
$\phi', K_a, K_0$	friction angle, active and at-rest pressure coefficients	deg, -
$n, K, S_f$	Manning coefficient, conveyance, friction slope	s/m <sup>1/3</sup> , m <sup>3</sup> /s, -
$Fr$	Froude number	-
$F_{nh}$	depth Froude number of a ship	-
$C_b, L_{pp}, B, T$	block coefficient, length between perpendiculars, beam, draught	-, m, m, m
$C_m, C_e, C_s, C_c$	PIANC berthing factors	-
$A$	Dean profile scale	m <sup>1/3</sup>
$\varepsilon$	longshore diffusivity (one-line model)	m <sup>2</sup> /s
$Q, K_{cerc}$	longshore transport and CERC coefficient	m <sup>3</sup> /s
$u, \sigma, \xi, \lambda$	threshold, scale, shape, and rate of an extreme value fit	-

## Appendix D. Engineering references

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